

TECHNICAL MANUAL

AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR

**ARMY MODELS UH-60A, UH-60L,
EH-60A, HH-60A AND HH-60L
HELICOPTERS**

**CHAPTER 2
TROUBLESHOOTING PROCEDURES**

*This manual together with TM 1-1520-237-23-1, TM 1-1520-237-23-3, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-6, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10, TM 1-1520-237-23-11, TM 1-1520-237-23-12, TM 1-1520-237-23-13, TM 1-1520-237-23-14, TM 1-1520-237-23-15, TM 1-1520-237-23-16, TM 1-1520-237-23-17, TM 1-1520-237-23-18, TM 1-1520-237-23-19 and TM 1-1520-237-23-20, dated 17 April 2006, supersedes TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-3, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-6, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10-1, TM 1-1520-237-23-10-2, TM 1-1520-237-23-10-3, TM 1-1520-237-23-11, dated 1 May 2003 including all changes.

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HEADQUARTERS, DEPARTMENT OF THE ARMY

17 April 2006

WARNING SUMMARY

Personnel performing operations, procedures, and practices which are included or implied in this technical manual shall observe the following warnings. To disregard these warnings and precautionary information can cause serious injury, death, or an aborted mission.

WARNING

FIRST AID - refer to FM 4-25-11.

AC POWER - before applying ac power to the helicopter, make sure that the area around the stabilator is clear of personnel and equipment. Should the stabilator be in any position other than trailing edge fully down, it may reposition automatically to fully down when ac power is applied.

CARBON MONOXIDE - during cold weather operations when preheating is necessary, all personnel shall become acquainted with the various types of heaters and where they are to be used. Be careful of accumulations of carbon monoxide and damage to heat-sensitive equipment.

CONSUMABLE MATERIALS - observe all cautions and warnings on the containers when using consumables. When applicable wear necessary protective gear during handling and use. If a consumable is flammable or explosive, MAKE SURE consumable and its vapors are kept away from heat, spark, and flame. MAKE SURE helicopter is properly grounded and firefighting equipment is readily available for use.

SAFETY PINS - make sure all safety pins are installed in ejector racks prior to performing any maintenance. To disregard this warning can cause serious injury, death, or an aborted mission.

ELECTRICAL GROUNDING OF THE HELICOPTER - army regulations require the grounding of the helicopter when doing all fueling and defueling operations. Do not operate helicopter electrical switches, except those essential for servicing during fueling and defueling. Do not smoke or use flame during fueling and defueling operations.

HIGH VOLTAGE - there are dangerous voltages in the helicopter. Use extreme care when working with equipment having these voltages.

HYDRAULIC SYSTEMS - there are dangerous high hydraulic system pressures in the helicopter. Use extreme care when working on systems having this pressure.

USE OF FIRE EXTINGUISHER - when fire extinguishers are used in a confined area, ventilate immediately. Serious injury or death could result if the area is not ventilated or the user does not use a self-contained breathing device.

MAIN ROTOR PYLON SLIDING COVER - when opening and closing main pylon sliding cover, keep hands away from sharp edges near track, mating end, and especially ventilation blower inlet hole.

TECHNICAL ACETONE - is flammable and toxic to eyes, skin, and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well ventilated areas (or use approved respirator as determined by local safety/industrial hygiene personnel). Keep away from open flames, sparks, hot surfaces or other sources of ignition.

ALKALINE AQUEOUS CLEANER - alkaline cleaner is harmful to skin and eyes. Use adequate ventilation. Operators should wear protective gloves, aprons, and goggles. Upon contact with cleaning compound, wash affected area for at least 20 minutes with clean water. Seek medical attention.

CLEANING COMPOUND SOLVENT - cleaning Compound Solvent is combustible and toxic to eyes, skin, and respiratory tract. Wear protective gloves and goggles/face shield. Avoid repeated or prolonged contact. Use only in well-ventilated areas (or use approved respirator). Keep away from open flames or other sources of ignition.

ELECTRON - use electron in well ventilated area only. Avoid prolonged breathing of vapors. Avoid bodily contact. The use of chemical gloves and chemical splash goggles are required. Do not use near heat, spark or flame. This solvent is

reactive with acids and oxidizers; do not mix or cross-apply with other chemicals. Organic vapor respirator with dust and mist filter is recommended when solvent is spray applied. Keep containers closed between applications.

- Provide mechanical ventilation if used in confined spaces. Coordinate the use of this material with your supporting industrial hygiene and safety offices. Ensure you read and understand the material safety data sheet (MSDS) for electron prior to use.

ISOPROPYL ALCOHOL - isopropyl alcohol is a volatile and flammable liquid which must be kept away from flame and other ignition sources. It must be used only in well ventilated area. Personnel must wear protective gloves and eye protection. Excessive inhalation of vapors or contact with skin must be avoided. If contact occurs with eyes or skin, flush area thoroughly with water. If ingestion occurs, induce vomiting and call a doctor. Remove to fresh air if overexposed to vapor.

DS-108 - use in well ventilated area. Avoid breathing vapors. Organic vapor respirator with dust and mist filter is recommended. May be harmful by inhalation, ingestion, or skin absorption. Avoid bodily contact. Chemical safety glasses/goggles and chemical resistant gloves are required. Combustible, keep away from heat, spark, or flame. Vapors are heavier than air and may travel over a distance to an ignition source and then flash back. Do not mix or cross apply with other chemicals.

LOCKWIRE - in critical applications use lockwire instead of safety cable which is specifically prohibited from use in critical applications. The use of safety cable is restricted in these specific applications with a warning.

PURPOSE OF A WARNING - alerts personal to operation, procedure, practice, etc., which if not strictly observed, could result in personal injury or loss of life.

CHANGE
NO. 3

HEADQUARTERS DEPARTMENT OF THE ARMY
WASHINGTON, D.C., 20 June 2008

TECHNICAL MANUAL
AVIATION UNIT AND INTERMEDIATE MAINTENANCE
FOR
ARMY MODELS
UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS
CHAPTER 2 TROUBLESHOOTING PROCEDURES

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GEORGE W. CASEY, JR.
General, United States Army
Chief of Staff

Official:

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JOYCE E. MORROW
Administrative Assistant to the
Secretary of the Army
0815612

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TECHNICAL MANUAL

**AVIATION UNIT AND INTERMEDIATE MAINTENANCE
FOR
ARMY MODELS
UH-60A, UH-60L, EH-60A, HH-60A AND HH-60L HELICOPTERS**

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JOYCE E. MORROW
*Administrative Assistant to the
Secretary of the Army*
0733115

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TECHNICAL MANUAL

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i and ii
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Cover

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WP 0070 00
WP 0073 00
WP 0074 00
WP 0075 00
WP 0076 00

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WP 0077 00
WP 0078 00
WP 0079 00
WP 0084 00
WP 0085 00
WP 0086 00
WP 0087 00
WP 0088 00
WP 0089 00
WP 0090 00
WP 0097 00
WP 0099 00
WP 0100 00

7. Add the following new work packages.

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WP 0097 01

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Dates of issue for original and changed pages / work packages are:

Original	0	17 April 2006
Change	1	21 March 2007
Change	2	15 December 2007
Change	3	20 June 2008

TOTAL NUMBER OF PAGES FOR FRONT AND REAR MATTER IS 6 AND TOTAL NUMBER OF WORK PACKAGES IS 33 CONSISTING OF THE FOLLOWING:

Page / WP No.	*Change No.	Page / WP No.	*Change No.	Page / WP No.	*Change No.
Volume 2 Title	1	WP 0077 00 (12 pgs)	1	WP 0091 00 (14 pgs)	2
a - b	3	WP 0078 00 (24 pgs)	1	WP 0092 00 (26 pgs)	0
A	3	WP 0079 00 (56 pgs)	1	WP 0093 00 (24 pgs)	0
B Blank	3	WP 0080 00 (10 pgs)	0	WP 0094 00 (14 pgs)	0
i - ii	2	WP 0081 00 (8 pgs)	0	WP 0095 00 (34 pgs)	0
Chapter 2 Title Page	1	WP 0082 00 (2 pgs)	1	WP 0096 00 (36 pgs)	0
WP 0069 00 (6 pgs)	0	WP 0083 00 (8 pgs)	0	WP 0097 00 (32 pgs)	1
WP 0070 00 (14 pgs)	1	WP 0084 00 (6 pgs)	1	WP 0097 01 (4 pgs)	1
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WP 0073 00 (22 pgs)	1	WP 0087 00 (22 pgs)	1	WP 0100 00 (64 pgs)	1
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AVIATION UNIT AND INTERMEDIATE MAINTENANCE FOR

Army Models UH-60A, UH-60L, EH-60A, HH-60A, and HH-60L Helicopters

CHAPTER 2 TROUBLESHOOTING PROCEDURES

REPORTING ERRORS AND RECOMMENDING IMPROVEMENTS.

You can help improve this manual. If you find any mistakes or if you know of a way to improve these procedures, please let us know. Mail your letter or DA Form 2028 (Recommended Changes to Publications and Blank Forms) located in the back of this manual directly to: Commander, U.S. Army Aviation and Missile Command, ATTN: AMSAM-MMC-MA-NP, Redstone Arsenal, AL 35898-5000. A reply will be furnished to you. You may also provide DA Form 2028 information to AMCOM via e-mail, fax, or the World Wide Web. Our fax number is: DSN 788-6546 or Commercial 256-842-6546. Our e-mail address is: 2028@redstone.army.mil. Instructions for sending an electronic 2028 may be found at the back of this manual immediately preceding the hard copy 2028. For the World Wide Web use: <https://amcom2028.redstone.army.mil>.

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*This manual together with TM 1-1520-237-23-1, TM 1-1520-237-23-3, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-6, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10, TM 1-1520-237-23-11, TM 1-1520-237-23-12, TM 1-1520-237-23-13, TM 1-1520-237-23-14, TM 1-1520-237-23-15, TM 1-1520-237-23-16, TM 1-1520-237-23-17, TM 1-1520-237-23-18, TM 1-1520-237-23-19, and TM 1-1520-237-23-20, dated 17 April 2006, supersedes TM 1-1520-237-23-1, TM 1-1520-237-23-2, TM 1-1520-237-23-3, TM 1-1520-237-23-4, TM 1-1520-237-23-5, TM 1-1520-237-23-6, TM 1-1520-237-23-7, TM 1-1520-237-23-8, TM 1-1520-237-23-9, TM 1-1520-237-23-10, TM 1-1520-237-23-10-1, TM 1-1520-237-23-10-2, TM 1-1520-237-23-10-3, TM 1-1520-237-23-11, dated 1 May 2003 including all changes.

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CHAPTER 2
TROUBLESHOOTING PROCEDURES
FOR
ARMY MODELS UH-60A, UH-60L, EH-60A, HH-60A,
AND HH-60L
HELICOPTERS



UNIT LEVEL

AIRFRAME

POSITION TRANSMITTER/LIMIT SWITCH ASSEMBLY (AVIM)

INITIAL SETUP:

Test Equipment

Angle Position Indicator (API), 8810-F1110
 Stabilator Indicators (2 Required), ID-2279/A

Towel, Machinery Wiping, Item 323, WP 1803 00

Tools and Special Tools

Electrical Repairer Toolkit, SC 5180-99-B06
 Locally-Made Stabilator Position Sensor/Limit Switch
 Assembly Lockpin Figure 179, WP 1805 00
 Stabilator Position Sensor/Limit Switch Test Unit
 Figure 154, WP 1805 00
 Torque Wrench, 0 - 30 in. lbs, A-A-2411

References

WP 0371 00
 WP 0372 00
 WP 1803 00
 WP 1805 00

Equipment Condition

Between 27 and 28 vdc, 115 vac, 400 Hz, Single-Phase
 Bench Power Available

Material/Parts

Cloth, Cheesecloth Item 85, WP 1803 00
 Isopropyl Alcohol, Technical, Item 170, WP 1803 00

POSITION TRANSMITTER/LIMIT SWITCH ASSEMBLY (AVIM) OPERATIONAL/TROUBLESHOOTING
PROCEDURE

PROCEDURE

Step

NOTE

See Figure 1 for component location diagram, and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Remove all dirt, dust, oil, grease, or other foreign matter from position transmitter, limit switches, wiring, and connectors using cheesecloth, Item 85, WP 1803 00, dampened with solvent compound cleaning, Item 170, WP 1803 00.
2. Dry surfaces using machinery wiping towel, Item 344, WP 1803 00.
3. Inspect for broken or burnt wires, bent or broken connector pins, security of components, and general condition.

Indication/Condition

- a. Wires shall not be broken or burnt.
- b. Connector pins shall not be broken or bent.
- c. Components shall be secure and in good condition.

Corrective Action

If Indication/Conditions are not as specified, repair/replace wiring, as required.

POSITION TRANSMITTER/LIMIT SWITCH ASSEMBLY (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Step

4. Place locally-made stabilator position sensor/limit switch test unit (test unit) (WP 1805 00) 28 VDC and 115 VAC ON-OFF switches to OFF.
5. Connect transmitter/limit switch assembly electrical connector P604 to test unit J604, and electrical connector P605 to test unit J605.
6. Connect stabilator indicators to test unit harness connectors P116 and P471R.
7. Connect test unit harness 28 vdc and 115 vac to bench power source.
8. Connect API to corresponding tipjacks as follows:

API	TEST UNIT
S1	S1
S2	S3
S3	S2
REF HI	REF HI
REF LO	REF LO

9. Turn API power on by pressing SYN button.
10. Place test unit 28 VDC and 115 VAC ON-OFF switches to ON.
11. Press test unit LAMP TEST button and observe that all lamps go on.
12. While observing stabilator indicators, move pivot arm on limit switch assembly throughout its range of travel.

Indication/Condition

Stabilator indicator pointers shall move smoothly through a range of -8° up to +40° down as pivot assembly is moved.

Corrective Action

If Indication/Condition is not as specified, replace position transmitter (WP 0372 00).

Step

13. Turn pivot arm to +10° down position and lock arm in place using locally-made stabilator position sensor/limit switch assembly lockpin (lockpin) (WP 1805 00) through slot in base and hole in pivot arm.

Indication/Condition

API shall indicate between 350.5° and 352.5°.

Corrective Action

1. If Indication/Condition is not as specified, adjust transmitter by removing cotter pins, and loosen nuts enough to allow transmitter to turn.
 - a. Turn transmitter until 10° indication is obtained.

**POSITION TRANSMITTER/LIMIT SWITCH ASSEMBLY (AVIM) OPERATIONAL/TROUBLESHOOTING
PROCEDURE - Continued**

- (1) If adjustment cannot be obtained, replace transmitter (WP 0372 00).
- b. **TIGHTEN NUTS AND TORQUE TO 19 - 21 INCH-POUNDS.**
- c. Install cotter pins (WP 0372 00).

Step**CAUTION**

Use care when positioning the pivot assembly in the extreme -8° and +40° slot positions. Too much force through the alignment pin may damage the position transmitter synchro shaft.

- 14. Turn pivot arm to -8° up position and lock arm in place using lockpin.

Indication/Condition

#1 UP and #2 UP lights on test unit remain off.

Corrective Action

- 1. If Indication/Condition is not as specified, adjust switch until light just goes off.
 - a. If trouble remains, replace switch (WP 0371 00).
 - b. **TIGHTEN NUT AND TORQUE TO 19 - 21 INCH-POUNDS.**

Step

- 15. With pin still installed, turn pivot arm counterclockwise toward up-limit switches.

Indication/Condition

Both #1 UP and #2 UP lights on test unit shall go on.

Corrective Action

- 1. If Indication/Condition is not as specified, adjust switch until light just goes on.
 - a. If trouble remains, replace limit switch (WP 0371 00).
 - b. **TIGHTEN NUTS AND TORQUE TO 19 - 21 INCH-POUNDS.**

Step

- 16. Gently move pivot arm back and forth within notch.

Indication/Condition

The API shall indicate between 7° and 9°.

Corrective Action

If Indication/Condition is not as specified, replace baseplate and/or retainer (WP 0372 00).

Step

- 17. Remove lockpin from pivot arm and baseplate.
- 18. Position the pivot arm towards the opposite limit switch to obtain an API indication of 324° to 326°.
- 19. Install lockpin in the +40 alignment slot.
- 20. Ensure pivot arm stays in contact with the switch plungers and gently move pivot arm back and forth.

**POSITION TRANSMITTER/LIMIT SWITCH ASSEMBLY (AVIM) OPERATIONAL/TROUBLESHOOTING
PROCEDURE - Continued****Indication/Condition**

- a. There shall be a deflection of 37° to 39° on the stabilator indicator.
- b. Both #1 DN and #2 DN lights on test unit shall remain off.

Corrective Action

1. If Indication/Condition a. is not as specified, ensure that pressure is only applied to the hole end of the pivot arm to hold against limit switch spring force.
2. If Indication/Condition b. is not as specified, adjust switch until light just goes off.
 - a. If trouble remains, replace switch (WP 0371 00).
 - b. **TIGHTEN NUT AND TORQUE TO 19 - 21 INCH-POUNDS.**

Step

21. With pin still installed, turn pivot arm clockwise toward down-limit switches.

Indication/Condition

Both #1 DN and #2 DN lights on test unit shall go on.

Corrective Action

1. If Indication/Condition is not as specified, adjust switch until light just goes on.
 - a. If trouble remains, replace switch (WP 0371 00).
 - b. **TIGHTEN NUTS AND TORQUE TO 19-21 INCH-POUNDS.**

Step

22. Gently move pivot arm back and forth within notch.

Indication/Condition

The API shall indicate between 300.5° and 323.5°.

Corrective Action

If Indication/Condition is not as specified, replace baseplate and/or retainer (WP 0372 00).

Step

23. Remove lockpin from pivot arm and baseplate.
24. Place test unit 28 VDC and 115 VAC ON-OFF switches OFF.
25. Disconnect all test equipment from position transmitter/limit switch.

Indication/Condition

Test unit shall be off and all test equipment disconnected.

Corrective Action

None Required

LOCATION

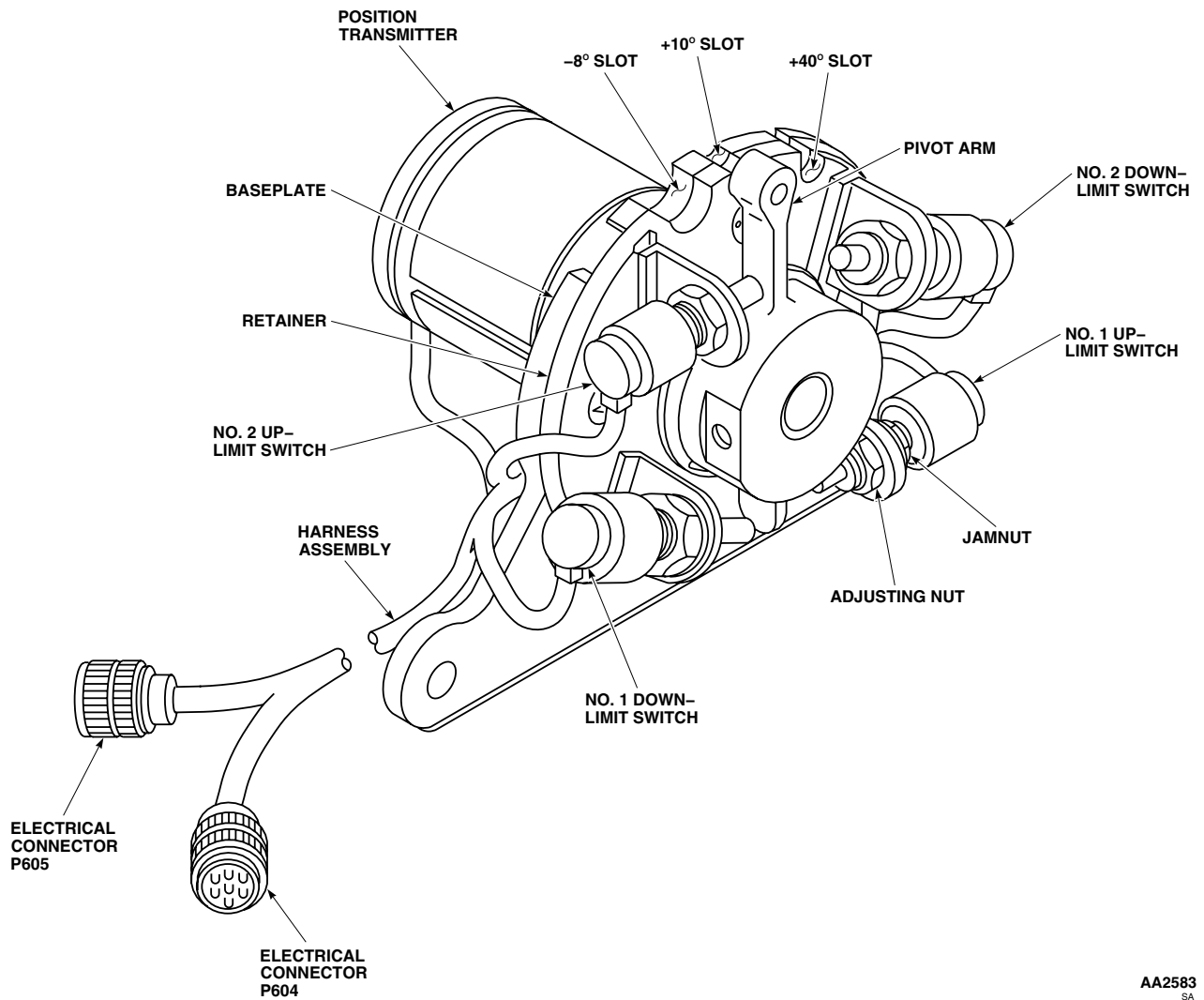
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Figure 1. Position Transmitter/Limit Switch Assembly Parts Location.

SCHEMATICS

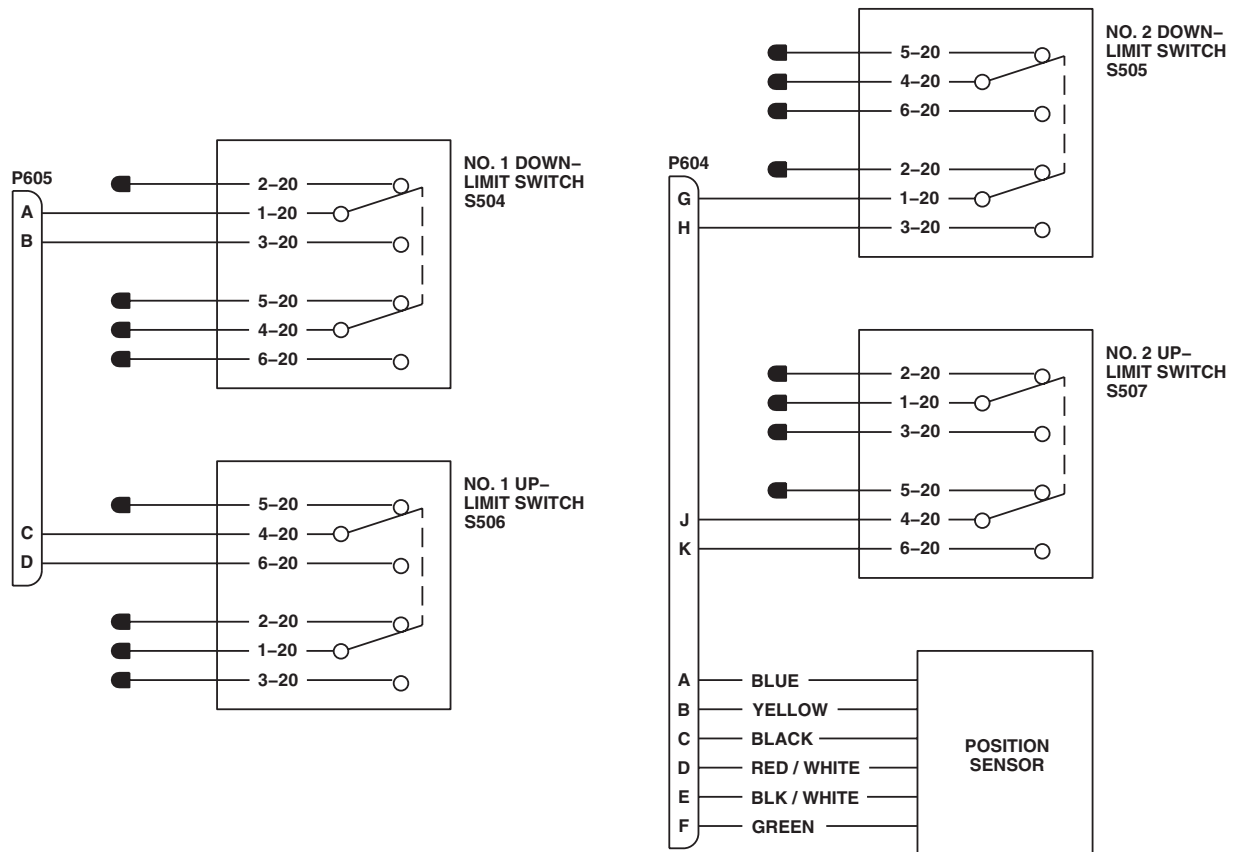
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Figure 2. Position Transmitter/Limit Switch Assembly Schematic Diagram.

END OF WORK PACKAGE

UNIT LEVEL

LANDING GEAR

PARKING BRAKE SYSTEM

This WP supersedes WP 0070 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment

Multimeter, AN/PSM-45A

WP 0451 00

WP 0452 00

WP 0454 00

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Electrical Repairer Toolkit, SC 5180-99-B06

WP 0455 00

WP 0457 00

WP 0458 00

Personnel Required

Aircraft Electrician MOS 15F (1)

UH-60 Helicopter Repairer MOS 15T (1)

WP 0824 00

WP 0925 00

WP 1685 00

References

TM 1-1520-237-10

WP 0090 00

WP 0100 00

WP 0104 00

Equipment Condition

External Power Available, WP 1685 00

or APU Operational, TM 1-1520-237-10

PARKING BRAKE SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER

CIRCUIT BREAKER PANEL

CAUT/ADVSY PNL

Upper Console

LIGHTS ADVSY

Copilot's

LIGHTS CAUT/ADVSY

Copilot's

PARKING BRAKE SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. Turn on electrical power. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed). ■
3. Momentarily press top of pilot's then copilot's directional pedals.

Indication/Condition

Pedals shall have little movement and feel firm.

Corrective Action

1. If Indication/Condition is not as specified, check pilot's or copilot's brakes for air in system or leaks.
 - a. If air is present, bleed system (WP 0457 00).
 - b. If leaks are present, repair/replace leaking component.
2. If pilot's or copilot's pedals bottom out (no resistance felt), go to BRAKE PEDALS BOTTOM OUT (NO RESISTANCE FELT), in this work package.

Step**NOTE**

Make sure that helicopter is on level ground before releasing tailwheel lock.

4. Manually disengage tailwheel lock. With pilot's or copilot's brake pedals pressed, pull PARKING brake handle up. Take both feet off pedals. Push side of tail cone near tailwheel.

Indication/Condition

- a. PARKING BRAKE ON capsule on caution/advisory panel shall go on.
- b. PARKING BRAKE handle shall stay up.
- c. Helicopter shall not move when pushed at tail cone.

Corrective Action

1. If Indication/Condition a. is not as specified, go to PARKING BRAKE ON CAPSULE DOES NOT GO ON, in this work package.
2. If Indication/Condition b. is not as specified, check linkage between handle and valve.
 - a. If linkage is binding or damaged, repair/replace as required (WP 0458 00).
 - (1) If trouble remains, check valve for leakage, and repair/replace as required (WP 0454 00).
3. If Indication/Condition c. is not as specified, go to BRAKES DO NOT HOLD, in this work package.

Step

5. Momentarily press pilot's, then copilot's, left brake pedal.

Indication/Condition

- a. PARKING BRAKE ON capsule on caution/advisory panel shall go off.
- b. PARKING BRAKE handle shall move down.
- c. Brakes shall release.

PARKING BRAKE SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition a. is not as specified, replace parking brake switch (WP 0455 00).
2. If Indication/Condition b. is not as specified, check that linkage is intact and does not bind at cockpit floor and console; repair/replace, as necessary (WP 0458 00).
 - a. If trouble remains, replace parking brake valve (WP 0454 00).
3. If Indication/Condition c. is not as specified, check pilot's or copilot's brakes for air in system or leaks.
 - a. If air is present, bleed system (WP 0457 00).
 - b. If leaks are present, repair/replace leaking component.

Step

6. Lock tailwheel.
7. Turn MFD switch OFF.
8. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

BRAKE PEDALS BOTTOM OUT (NO RESISTANCE FELT)**SYMPTOM**

Pedals shall have little movement and feel firm.

MALFUNCTION

Brake pedals bottom out (no resistance felt).

CORRECTIVE ACTION**1. Check lines and system components for leaks.**

- a. If leaks are found, repair/replace components as necessary. Go to **3**.
- b. If leaks are not found, go to **2**.

2. Check master cylinders for proper fluid level.

- a. If level is proper, replace master cylinders as required (WP 0452 00). Go to **3**.
- b. If level is not proper, service master cylinders as required (WP 0452 00). Go to **3**.

3. Procedure completed.

PARKING BRAKE ON CAPSULE DOES NOT GO ON**SYMPTOM**

PARKING BRAKE ON capsule on caution/advisory panel shall go on.

MALFUNCTION

PARKING BRAKE ON capsule does not go on.

CORRECTIVE ACTION

1. **Make sure linkage between brake handle and parking brake valve is installed.**
 - a. If linkage installed, go to **2**.
 - b. If linkage not installed, connect linkage as required (WP 0458 00). Go to **10**.
2. **Hold caution/advisory panel HH-60A HH-60L instrument panel INDICATOR LTS BRT/DIM-TEST switch to TEST. Go to 3.**
3. **Check that PARKING BRAKE ON capsule goes on.**
 - a. If capsule goes on, go to **4**.
 - b. If capsule does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **10**.
4. **Check helicopter configuration.**
 - a. **EMEP** , go to **5**.
 - b. **W/O EMEP** , go to **7**.
5. **EMEP Remove and check pin filtered adapter P118 or HH-60A HH-60L P1019R or P1020R (WP 0925 00).**
 - a. If pin filtered adapter is good, go to **6**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **10**.
6. **EMEP Install pin filtered adapter (WP 0925 00). Go to 7.**
7. **Check for 28 vdc between P118-H or HH-60A HH-60L P1019R-33 or P1020R-33 and ground.**
 - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **10**.
 - b. If voltage is not as specified, go to **8**.
8. **Check for 28 vdc between LIGHTS ADVSY circuit breaker terminal 1 and ground.**
 - a. If voltage is as specified, go to **9**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **10**.
9. **Check continuity between:**
 - **P118-H and S42-NC**
 - **P249-L and S42-COM**
 - a. If continuity is present, replace parking brake switch (WP 0455 00). Go to **10**.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **10**.

PARKING BRAKE ON CAPSULE DOES NOT GO ON - Continued

10. Procedure completed.

BRAKES DO NOT HOLD**SYMPTOM**

Brakes do not hold.

MALFUNCTION

Helicopter shall not move when pushed at tail cone.

CORRECTIVE ACTION

1. Check lines and system components for leaks.

- a. If leaks are found, repair/replace components as necessary. Go to **4**.
- b. If leaks are not found, go to **2**.

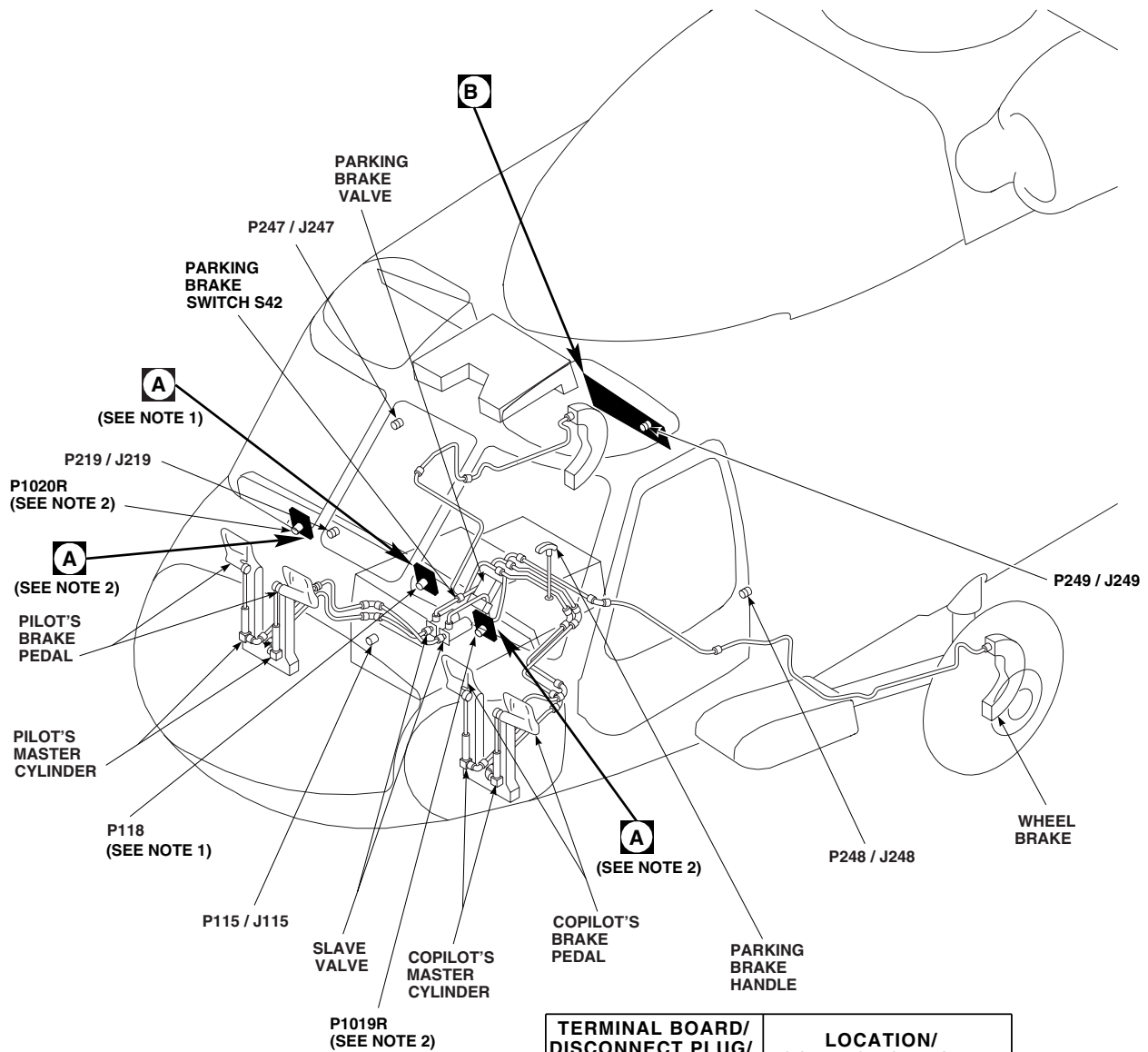
2. Push wear indicator pin in on wheel brake housing. Go to 3.

3. Is wear pin flush with housing?

- a. If wear pin is flush with housing, replace brake pads (WP 0451 00). Go to **4**.
- b. If wear pin is not flush with housing, replace parking brake valve (WP 0454 00). Go to **4**.

4. Procedure completed.

LOCATION



NOTES

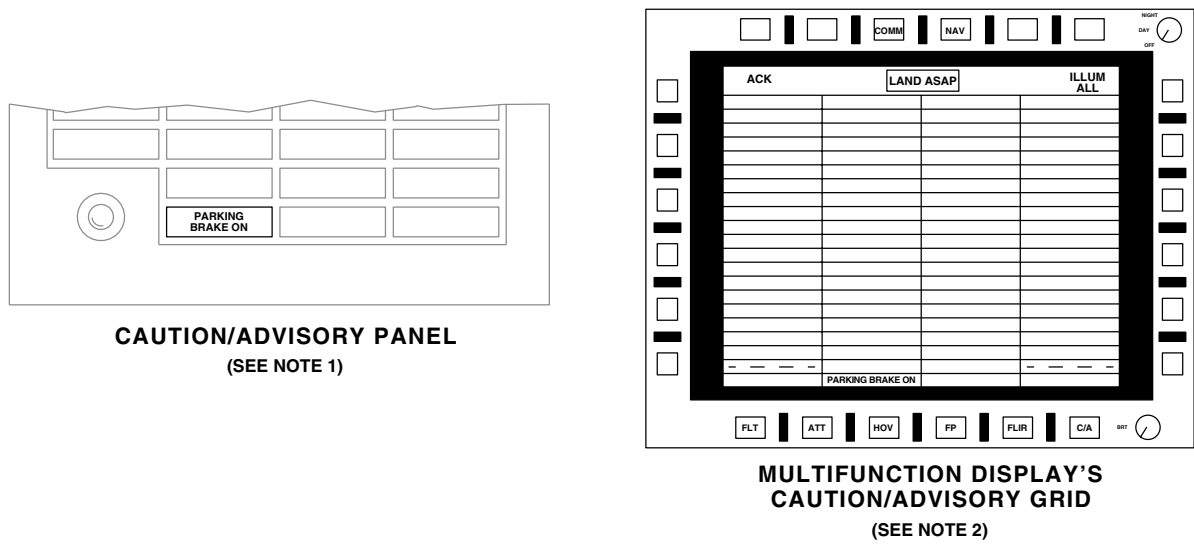
1. UH-60A UH-60L EH-60A
2. HH-60A HH-60L

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P115 / J115	COCKPIT CONSOLE BL 7.5 RH, STA 210
P118 / J118	CAUTION / ADVISORY PANEL (SEE NOTE 1)
P219 / J219	CABIN FLOOR BL 25 RH, STA 242
P247 / J247	CABIN, BL 40 RH, STA 248
P248 / J248	CABIN, BL 40 LH, STA 248
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23.0 LH
P1019R / J3	COPILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 2)
P1020R / J3	PILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 2)

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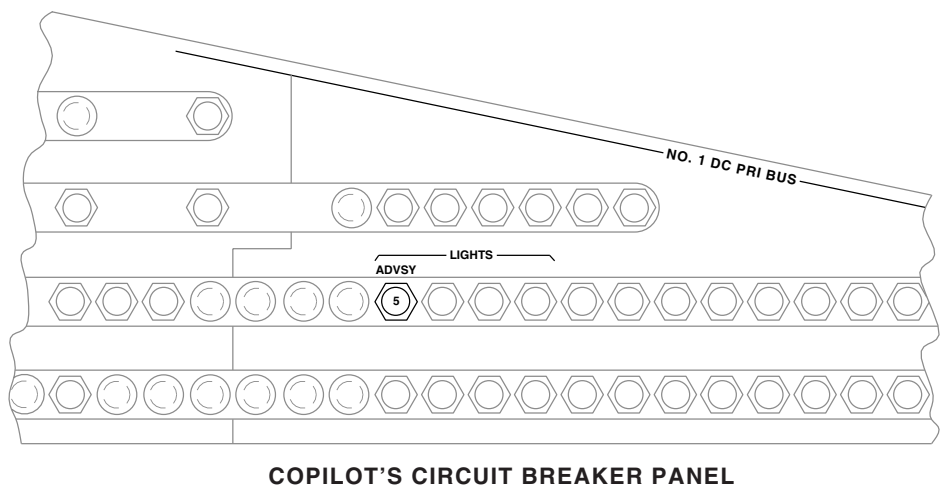
Figure 1. Parking Brake System Location Diagram. (Sheet 1 of 2)

LOCATION - Continued



A

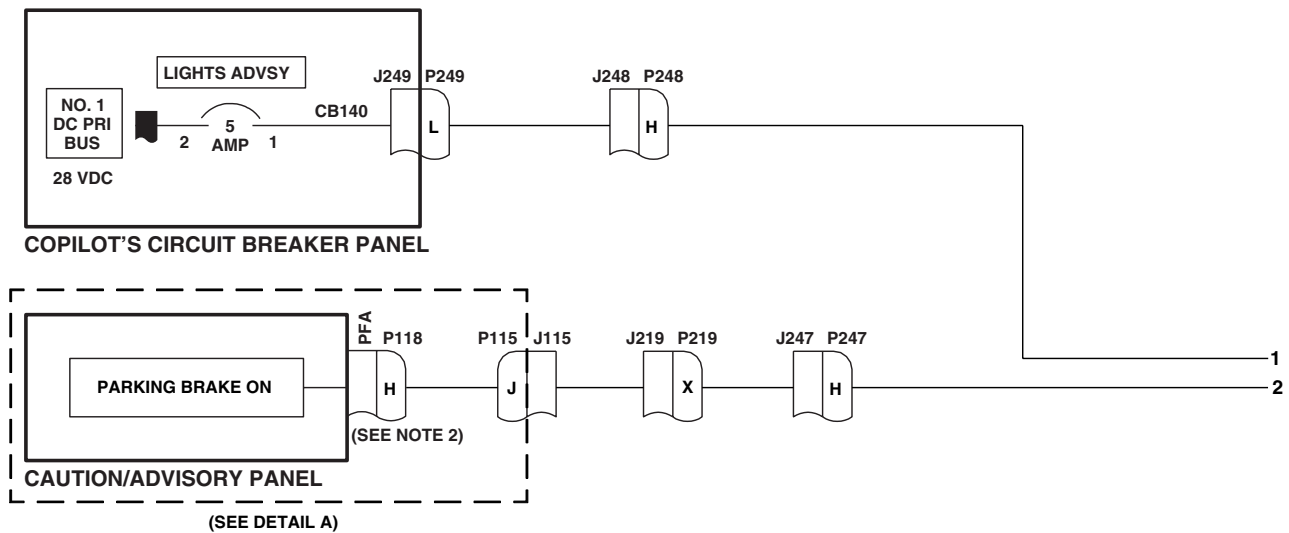
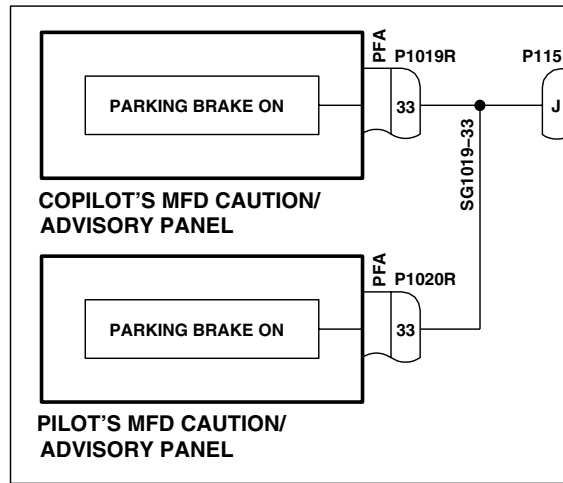
B



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SA

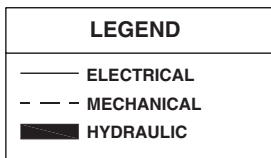
Figure 1. Parking Brake System Location Diagram. (Sheet 2 of 2)

SCHEMATICS



NOTES

1. LINES ARE PRESSURIZED ONLY WHEN PEDALS ARE PRESSED.
2. **W/O EMEP** PIN FILTERED ADAPTER (PFA) IS NOT INSTALLED.
3. **HH-60A HH-60L**



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SA

Figure 2. Parking Brake System Schematic Diagram. (Sheet 1 of 6)

SCHEMATICS - Continued

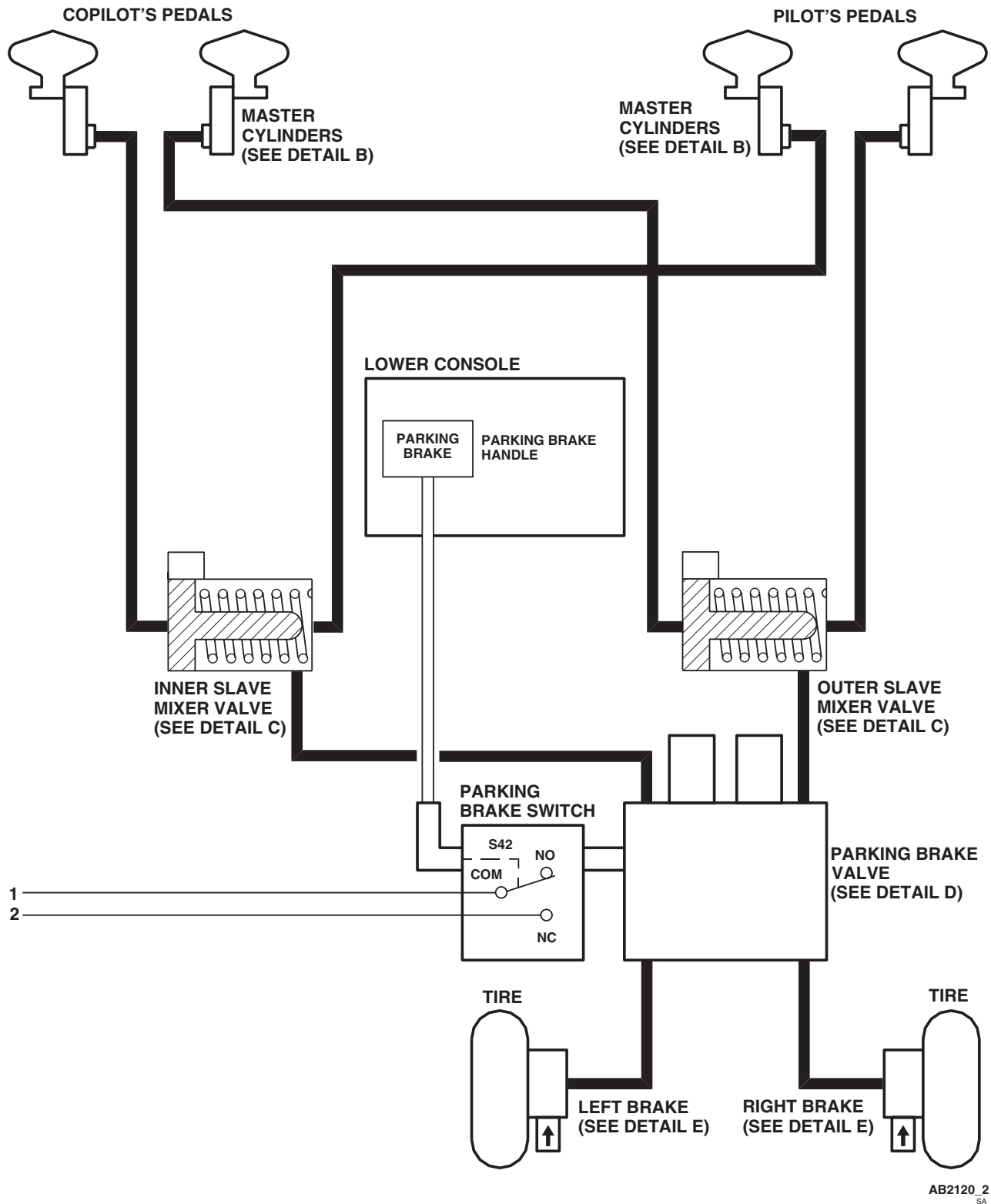
AB2120_2
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Figure 2. Parking Brake System Schematic Diagram. (Sheet 2 of 6)

SCHEMATICS - Continued

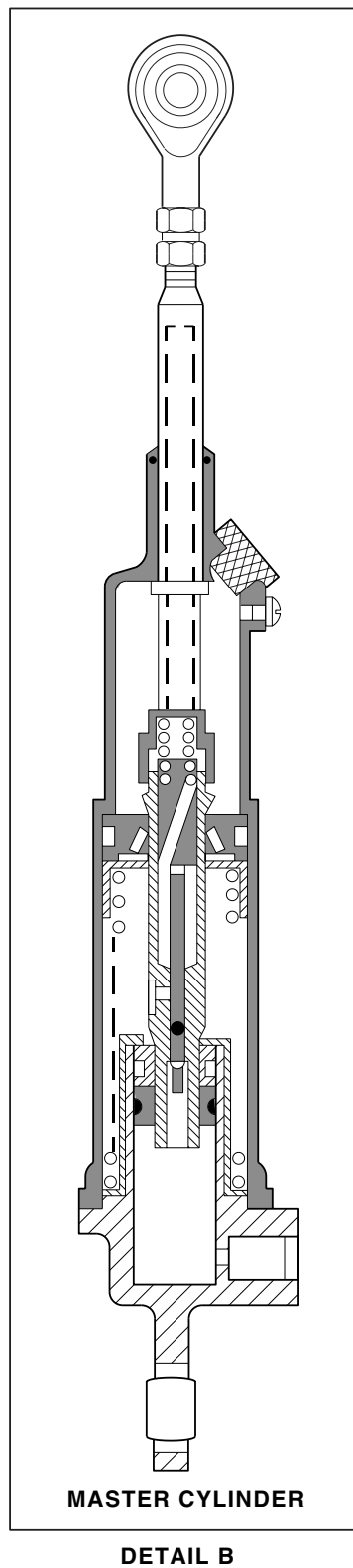
AB2120 3
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Figure 2. Parking Brake System Schematic Diagram. (Sheet 3 of 6)

SCHEMATICS - Continued

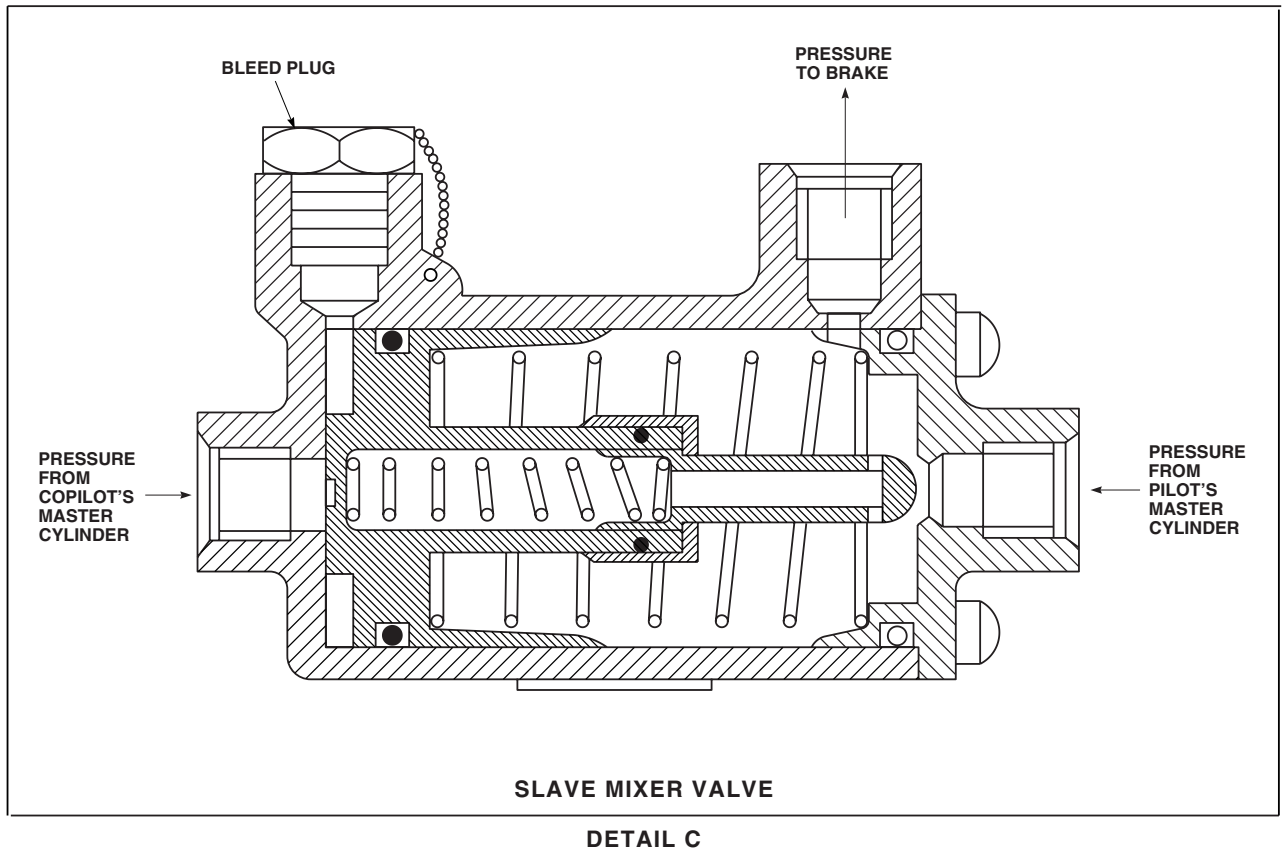
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Figure 2. Parking Brake System Schematic Diagram. (Sheet 4 of 6)

SCHEMATICS - Continued

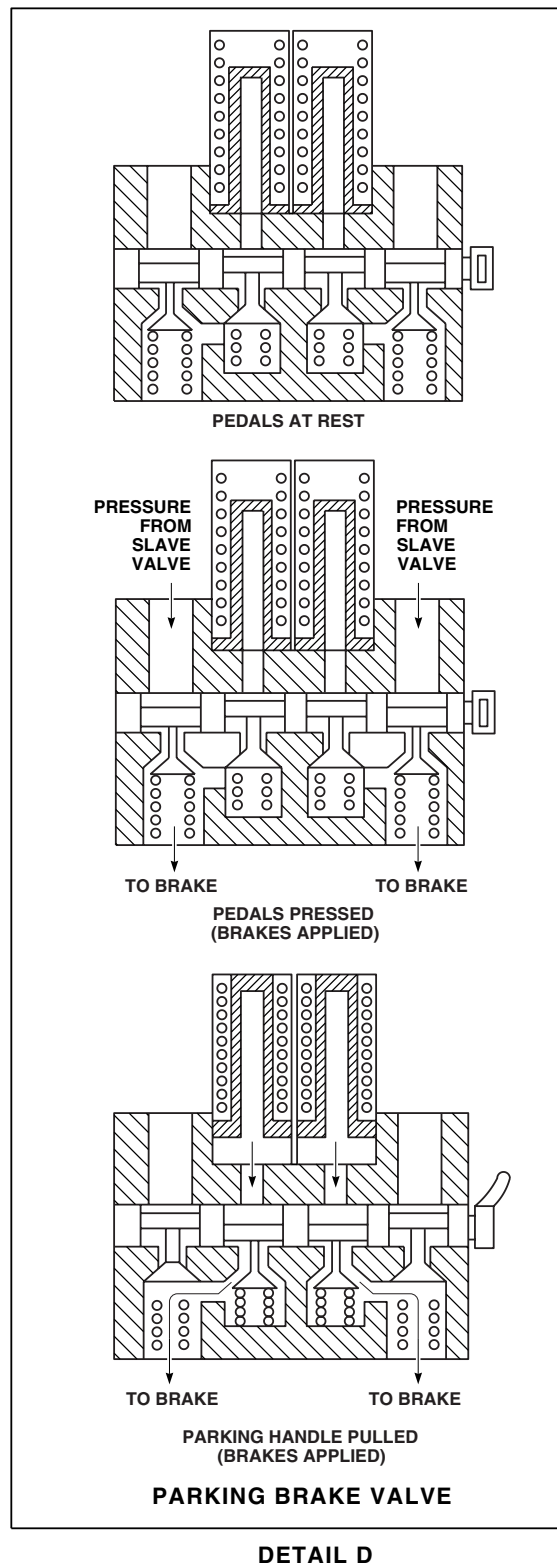
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Figure 2. Parking Brake System Schematic Diagram. (Sheet 5 of 6)

SCHEMATICS - Continued

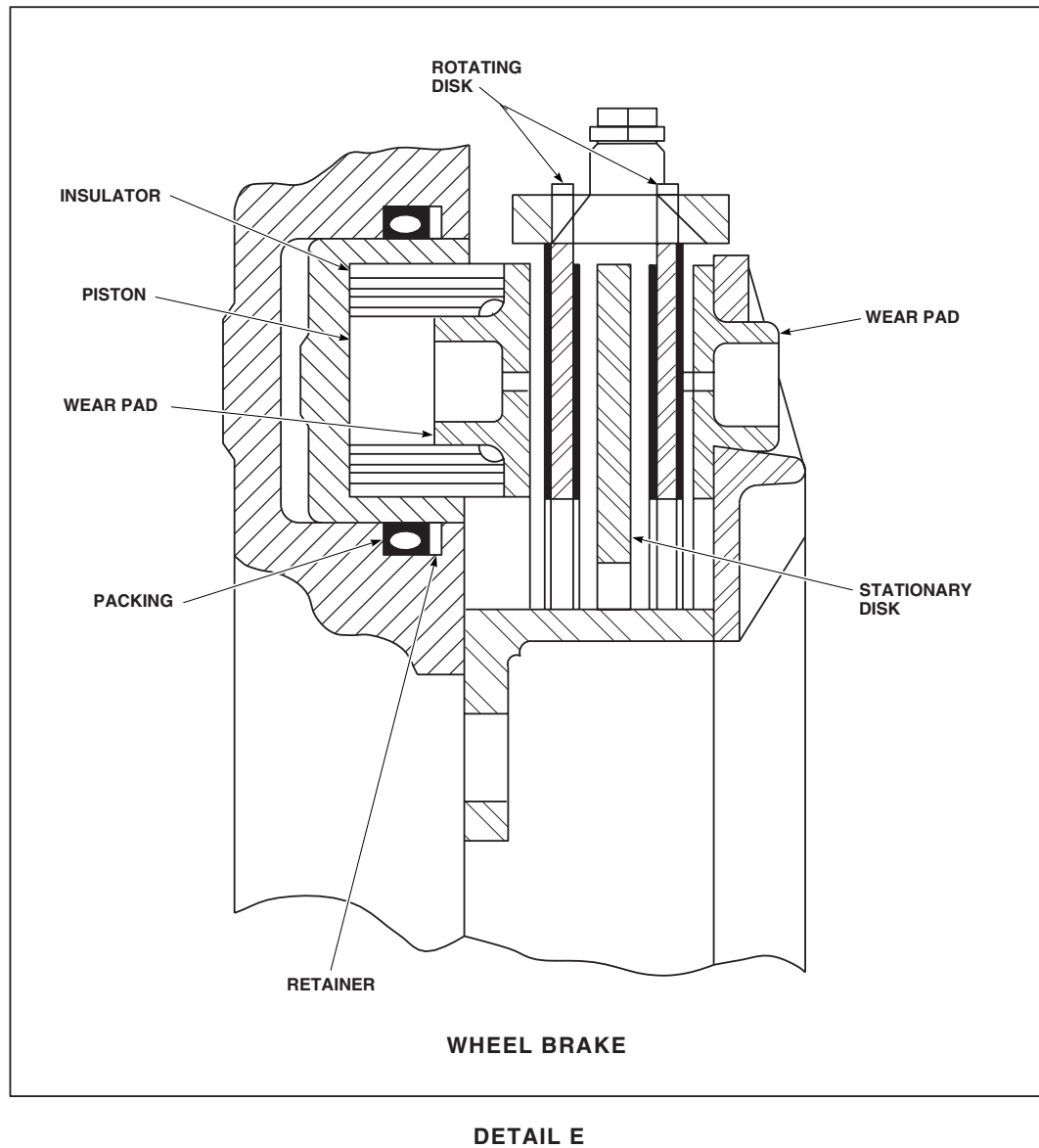
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Figure 2. Parking Brake System Schematic Diagram. (Sheet 6 of 6)

END OF WORK PACKAGE

UNIT LEVEL
LANDING GEAR
TAIL WHEEL LOCK SYSTEM

INITIAL SETUP:**Test Equipment**

Multimeter, AN/PSM-45A

WP 0104 00

WP 0111 00

WP 0464 00

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 0824 00

Electrical Repairer Toolkit, SC 5180-99-B06

WP 0830 00

WP 1685 00

WP 1747 00

Personnel Required

Aircraft Electrician MOS 15F (1)

Tactical Transport Helicopter Repairer MOS 15T (1)

Equipment Condition

External Electrical Power Available, WP 1685 00
 or APU Operational, TM 1-1520-237-10

References

TM 1-1520-237-10

WP 0099 00

TAIL WHEEL LOCK SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER**CIRCUIT BREAKER PANEL**

CAUT/ADVSY PNL

Upper Console

LIGHTS CAUT/ADVSY

Copilot's

TAIL WHEEL LOCK

Upper Console

2. Visually check to see if lockpin is lined up with hole.

Indication/Condition

Lockpin should be lined up with hole.

TAIL WHEEL LOCK SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If Indication/Condition is not as specified, push manual lever down and push side of tail cone as required to line up hole and pin.

Step**NOTE**

- Make sure unlock lever is left in position as shown in Figure 1, Sheet 2, Detail C or pin will not go up or down.
- Battery power (BATT switch ON) will only operate tail wheel lock actuator. TAIL WHEEL switch light will not go on.

3. Turn on electrical power.
4. Check MISC SW panel and position of tail wheel lockpin for agreement.
5. Push TAIL WHEEL switch on MISC SW panel.

Indication/Condition

- a. Tail wheel lockpin shall engage or disengage.
- b. TAIL WHEEL switch shall likewise indicate LOCK or UNLK.

Corrective Action

1. If tail wheel lockpin does not disengage and TAIL WHEEL switch indicates LOCK, go to TAIL WHEEL LOCKPIN DOES NOT DISENGAGE AND TAIL WHEEL SWITCH INDICATES LOCK, in this work package.
2. If tail wheel lockpin disengages, but TAIL WHEEL switch still indicates LOCK, go to TAIL WHEEL LOCKPIN DISENGAGES, BUT TAIL WHEEL SWITCH INDICATES LOCK, in this work package.
3. If tail wheel lockpin does not engage and TAIL WHEEL switch indicates UNLK, go to TAIL WHEEL LOCKPIN DOES NOT ENGAGE AND TAIL WHEEL SWITCH INDICATES UNLK, in this work package.
4. If tail wheel lockpin engages, but TAIL WHEEL switch still indicates UNLK, go to TAIL WHEEL LOCKPIN ENGAGES, BUT TAIL WHEEL SWITCH INDICATES UNLK, in this work package.

Step

6. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

TAIL WHEEL LOCKPIN DOES NOT DISENGAGE AND TAIL WHEEL SWITCH INDICATES LOCK**SYMPTOM**

Tail wheel lockpin shall engage or disengage. TAIL WHEEL switch shall likewise indicate LOCK or UNLK.

MALFUNCTION

Tail wheel lockpin shall engage or disengage. TAIL WHEEL switch shall likewise indicate LOCK or UNLK.

CORRECTIVE ACTION**1. Check actuator for proper position (retracted).**

- a. If actuator is in proper position, go to **2**.
- b. If actuator is not in proper position, go to **6**.

2. Check linkage from actuator to lockpin for proper installation or freedom from damage.

- a. If linkage is good, go to **3**.
- b. If linkage is not good, replace/install linkage or spring, as required (WP 0464 00). Go to **11**.

3. Check that lockpin becomes disengaged using manual release.

- a. If lockpin becomes disengaged, check lockpin for damage or hole for signs of corrosion. Repair/replace as necessary (WP 0464 00). Go to **11**.
- b. If lockpin does not become disengaged, go to **4**.

4. Push side of tail cone. Go to 5.**5. Check that lockpin disengages.**

- a. If lockpin disengages, system is operational. Go to **11**.
- b. If lockpin does not disengage, replace lockpin (WP 0464 00). Go to **11**.

6. Check for 28 vdc between P507-B and P507-C.

- a. If voltage is as specified, replace tail wheel lock actuator (WP 0464 00). Go to **11**.
- b. If voltage is not as specified, go to **7**.

7. Check for 28 vdc between P507-B and ground.

- a. If voltage is as specified, repair/replace wiring between P507-C and GND500-1 (WP 1747 00). Go to **11**.
- b. If voltage is not as specified, go to **8**.

8. Check continuity between J135-A and J135-D.

- a. If continuity is present, go to **9**.
- b. If continuity is not present, replace MISC SW panel (WP 0830 00). Go to **11**.

9. Check for 28 vdc between P135-A and ground.

- a. If voltage is as specified, repair/replace wiring between, as required (WP 1747 00), then go to **11**:
 - P135-D and P507-B
 - P135-F and P507-A

TAIL WHEEL LOCKPIN DOES NOT DISENGAGE AND TAIL WHEEL SWITCH INDICATES LOCK - Continued

b. If voltage is not as specified, go to **10**.

10. Check for 28 vdc between TAIL WHEEL LOCK circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring between terminal 1 and P135-A (WP 1747 00). Go to **11**.

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **11**.

11. Procedure completed.

TAIL WHEEL LOCKPIN DISENGAGES, BUT TAIL WHEEL SWITCH INDICATES LOCK**SYMPTOM**

Tail wheel lockpin shall engage or disengage. TAIL WHEEL switch shall likewise indicate LOCK or UNLK.

MALFUNCTION

Tail wheel lockpin shall engage or disengage. TAIL WHEEL switch shall likewise indicate LOCK or UNLK.

CORRECTIVE ACTION

1. Check for 28 vdc between P135-G and P659R-H.

a. If voltage is as specified, replace MISC SW panel (WP 0830 00). Go to **6**.

b. If voltage is not as specified, go to **2**.

2. Check for 28 vdc between P135-G and ground.

a. If voltage is as specified, repair/replace wiring between P659R-H and GG12-5 (WP 1747 00). Go to **6**.

b. If voltage is not as specified, go to **3**.

3. Check continuity between P135-G and J500-6.

a. If continuity is present, go to **4**.

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.

4. Check for 28 vdc between J500-4 and ground.

a. If voltage is as specified, replace harness assembly. Go to **6**.

b. If voltage is not as specified, go to **5**.

5. Check for 28 vdc between P902-X and ground.

a. If voltage is as specified, repair/replace wiring between (WP 1747 00), then go to **6**:

- J500-4 and P902-X
- J500-12 and P902-X

b. If voltage is not as specified, troubleshoot instrument panel and consoles indicator lights dimming system (WP 0111 00). Go to **6**.

6. Procedure completed.

TAIL WHEEL LOCKPIN DOES NOT ENGAGE AND TAIL WHEEL SWITCH INDICATES UNLK**SYMPTOM**

Tail wheel lockpin shall engage or disengage. TAIL WHEEL switch shall likewise indicate LOCK or UNLK.

MALFUNCTION

Tail wheel lockpin shall engage or disengage. TAIL WHEEL switch shall likewise indicate LOCK or UNLK.

CORRECTIVE ACTION**1. Check actuator for proper position (extended).**

- a. If actuator is extended, go to **2**.
- b. If actuator is not extended, go to **5**.

2. Check linkage from actuator to lockpin for proper installation or freedom from damage.

- a. If linkage is good, go to **3**.
- b. If linkage is not good, repair/replace parts, as required (WP 0464 00). Go to **8**.

3. Push side of tail cone to line up lockpin with hole. Go to 4.**4. Check that lockpin engages.**

- a. If lockpin engages, system is operational. Go to **8**.
- b. If lockpin does not engage, check lockpin hole for damage. Replace lockpin if necessary (WP 0464 00). Go to **8**.

5. Check for 28 vdc between P507-A and P507-C.

- a. If voltage is as specified, replace tail wheel lock actuator (WP 0464 00). Go to **8**.
- b. If voltage is not as specified, go to **6**.

6. Check continuity between P507-C and ground.

- a. If continuity is present, go to **7**.
- b. If continuity is not present, repair/replace wiring between J500-9 and GND500-1 (WP 1747 00). Go to **8**.
- c. If trouble remains, replace harness assembly. Go to **8**.

7. Check continuity between P135-F and P507-A.

- a. If continuity is present, troubleshoot miscellaneous switch panel (WP 0099 00). Go to **8**.
- b. If continuity is not present, repair/replace wiring between P135-F and J500-7 (WP 1747 00). Go to **8**.
- c. If trouble remains, replace harness assembly. Go to **8**.

8. Procedure completed.

TAIL WHEEL LOCKPIN ENGAGES, BUT TAIL WHEEL SWITCH INDICATES UNLK**SYMPTOM**

Tail wheel lockpin shall engage or disengage. TAIL WHEEL switch shall likewise indicate LOCK or UNLK.

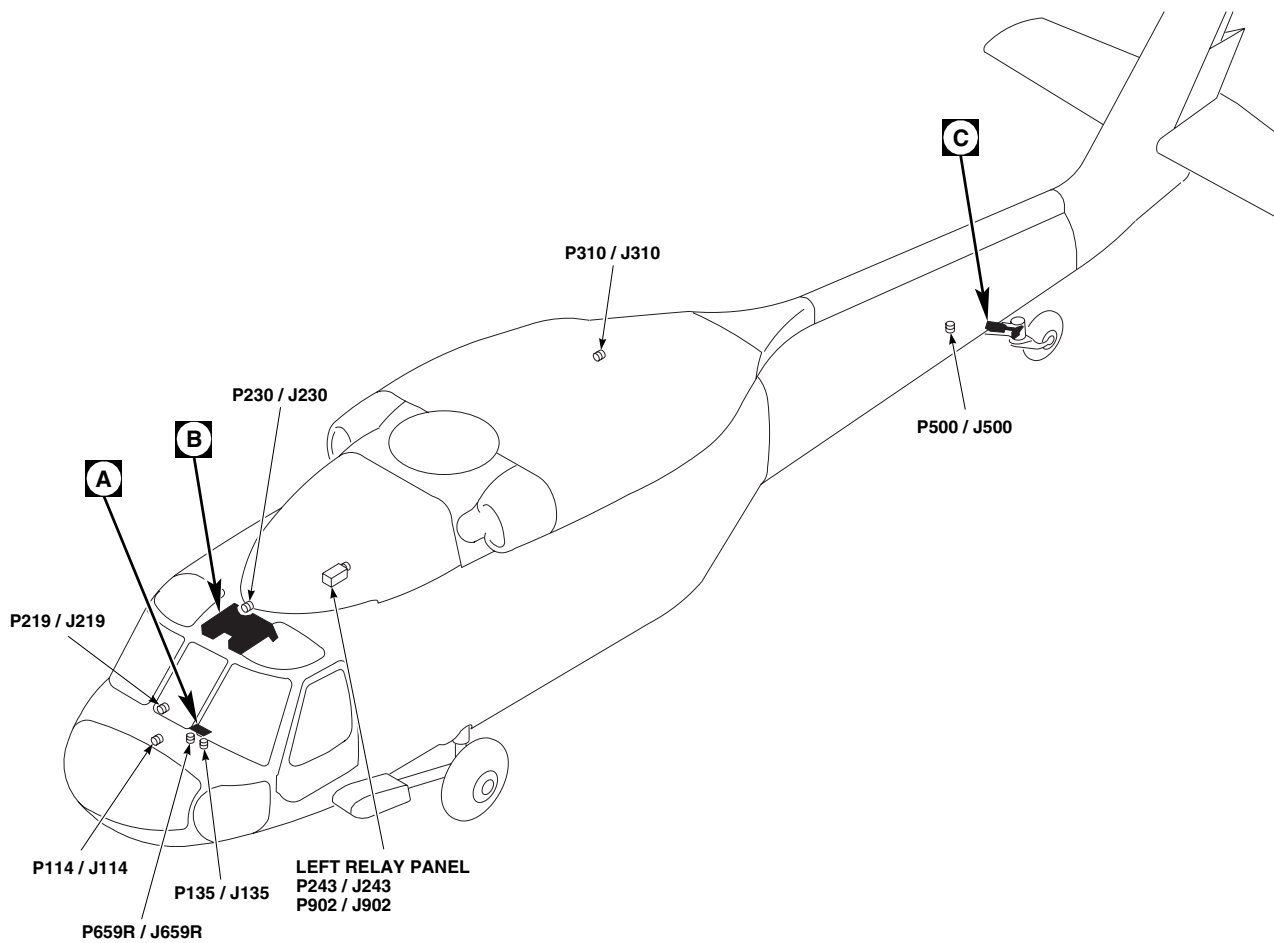
MALFUNCTION

Tail wheel lockpin shall engage or disengage. TAIL WHEEL switch shall likewise indicate LOCK or UNLK.
Tail wheel lockpin engages, but tail wheel switch indicates UNLK.

CORRECTIVE ACTION

- 1. Check continuity between J507-J and J507-H.**
 - a. If continuity is present, go to **2**.
 - b. If continuity is not present, replace tail wheel lock actuator (WP 0464 00). Go to **4**.
- 2. Check for 28 vdc between P507-J and ground.**
 - a. If voltage is as specified, go to **3**.
 - b. If voltage is not as specified, go to TAIL WHEEL LOCKPIN DISENGAGES, BUT TAIL WHEEL SWITCH INDICATES LOCK, in this work package. Go to **4**.
- 3. Check continuity between P500-1 and P500-3.**
 - a. If continuity is present, repair/replace wiring between J500-1 and J500-11 (WP 1747 00). Go to **4**.
 - b. If continuity is not present, replace harness assembly. Go to **4**.
- 4. Procedure completed.**

LOCATION

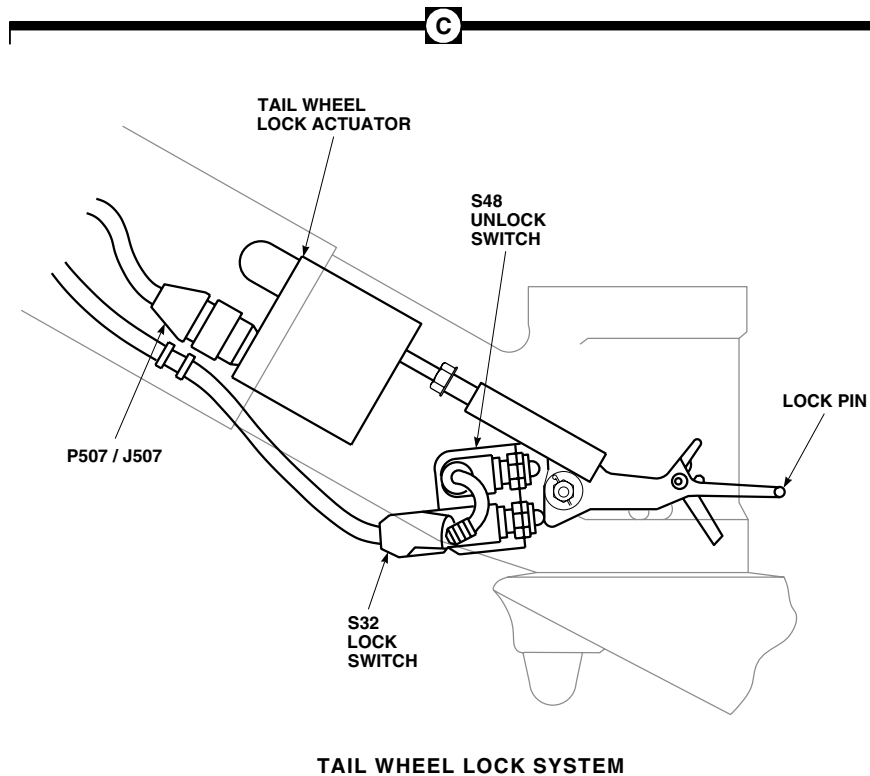
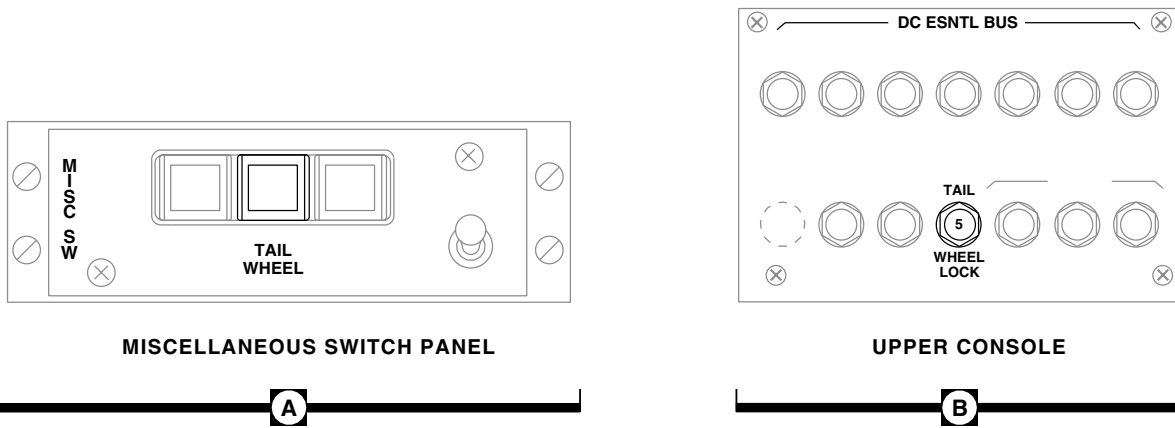


TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P114 / J114	COCKPIT BL 8 RH, STA 200
P135 / J135	MISCELLANEOUS SWITCH PANEL
P219 / J219	CABIN FLOOR BL 25 RH, STA 242
P230 / J230	CABIN CEILING BL 8 RH, STA 247
P243 / J243	LEFT RELAY PANEL
P310 / J310	CABIN CEILING BL 16.7 RH, STA 390
P500 / J500	TAIL CONE BL 0, STA 609
P507 / J507	TAIL WHEEL LOCK ACTUATOR
P659R / J659R	MISCELLANEOUS SWITCH PANEL
P902 / J902	LEFT RELAY PANEL

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Figure 1. Tail Wheel Lock System Location Diagram. (Sheet 1 of 2)

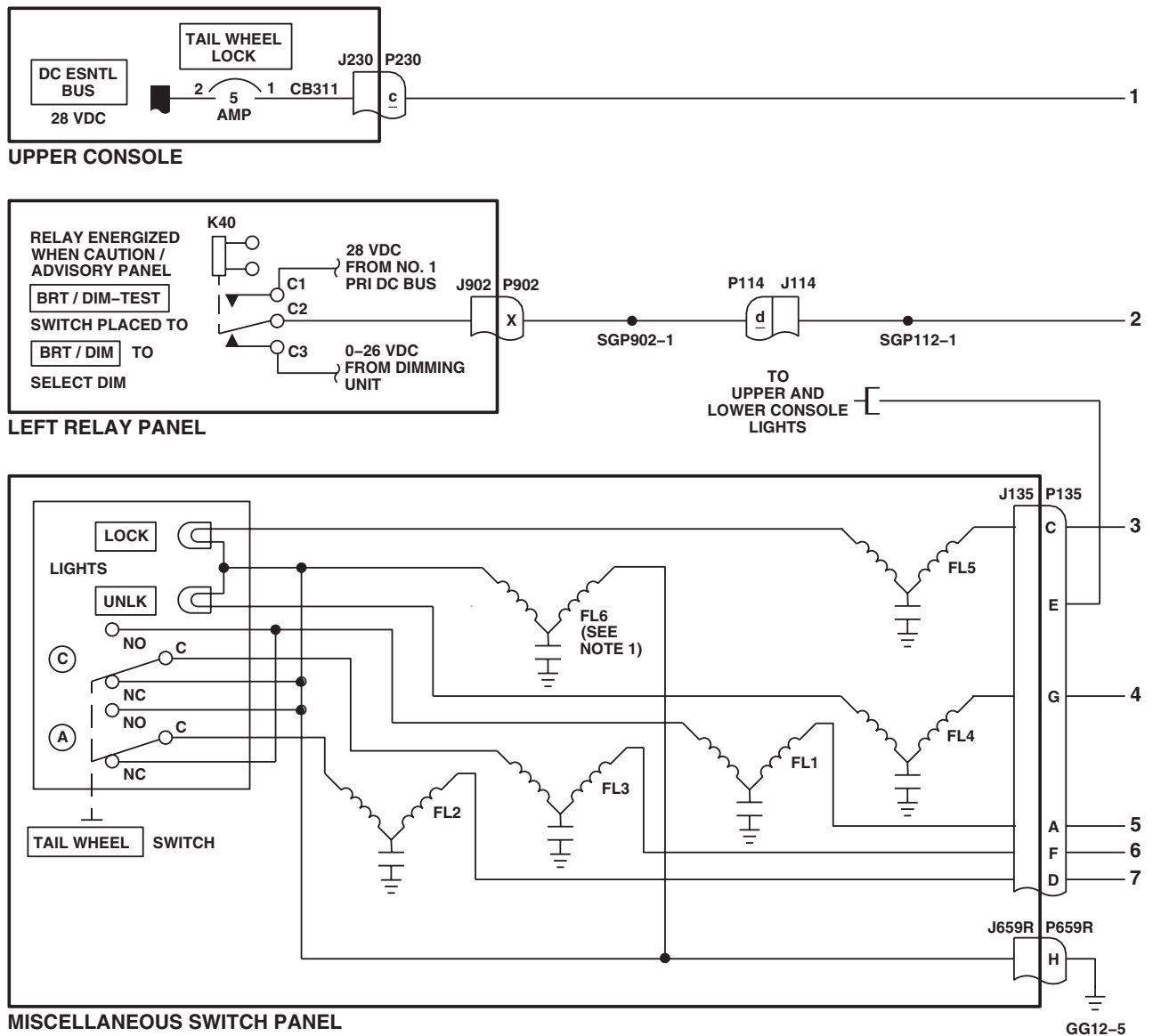
LOCATION - Continued



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Figure 1. Tail Wheel Lock System Location Diagram. (Sheet 2 of 2)

SCHEMATICS



NOTES

1. **W/O EMEP** FILTERS FL1 THROUGH FL6 NOT INSTALLED.
2. ALL SWITCHES CHANGE POSITION WHEN ACTUATOR REACHES EXTENDED OR RETRACTED POSITION.

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Figure 2. Tail Wheel Lock System Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued

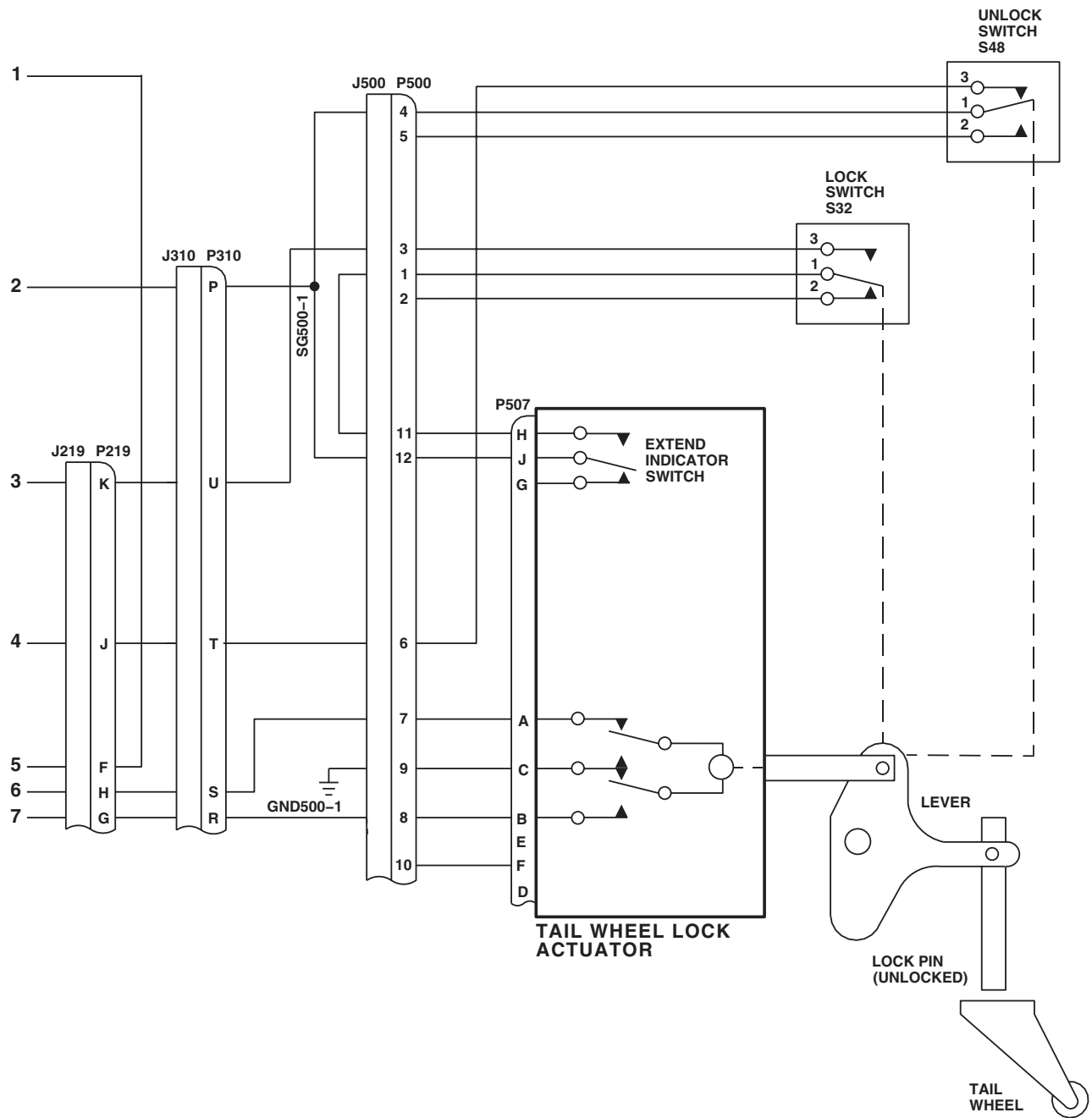
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Figure 2. Tail Wheel Lock System Schematic Diagram. (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL
ENGINE SYSTEM
ENGINE AND ENGINE INTERFACE

INITIAL SETUP:**References**

TM 1-1520-237-10
 TM 1-2840-248-23
 WP 0075 00
 WP 0076 00
 WP 0089 00
 WP 0102 00
 WP 0129 00
 WP 0498 00
 WP 0499 00

WP 0500 00
 WP 0506 00
 WP 0507 00
 WP 0508 00
 WP 0511 00
 WP 0512 00
 WP 0799 00
 WP 0979 00
 WP 1090 00
 WP 1747 00

ENGINE AND ENGINE INTERFACE INTRODUCTION OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE

Step

1. This operational/troubleshooting procedure provides fault isolation of engine malfunction indication as reported by the pilot. An engine malfunction indication may be due to an airframe or an engine component malfunction. All troubleshooting tables here relate to airframe items first. If it has been determined that the airframe is not the cause, troubleshoot the engine (TM 1-2840-248-23).

NOTE

- As a general rule, if only one parameter of the engine is malfunctioning and all other indications are normal, the general cause is a malfunctioning airframe indicating system. Troubleshoot instrument display system (WP 0089 00).
 - See Figure 1 for component location diagram, and Figure 2 for schematic diagram as an aid in operational/troubleshooting.
2. **UH60L** The following table lists the self-test BITE indications on the appropriate Torque Meter and references to troubleshooting procedures. ■

Table 1. Self-test BITE Indications.

BITE INDICATION	TROUBLE	TROUBLESHOOTING PROCEDURES
15%	DEC	TM 1-2840-248-23
25%	Np Demand	TM 1-2840-248-23
35%	Load Share	TM 1-2840-248-23

ENGINE AND ENGINE INTERFACE INTRODUCTION OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Table 1. Self-test BITE Indications. - Continued

BITE INDICATION	TROUBLE	TROUBLESHOOTING PROCEDURES
45%	TGT	WP 0089 00/TM 1-2840-248-23
55%	Alternator Power	TM 1-2840-248-23
65%	Ng	WP 0089 00/TM 1-2840-248-23
75%	Np	WP 0089 00/TM 1-2840-248-23
85%	Torque	WP 0089 00/TM 1-2840-248-23(Torque)
85%	Overspeed	WP 0075 00/TM 1-2840-248-23 (Overspeed)
95%	Hot Start Prevention	TM 1-2840-248-23
105%	Helicopter 400 Hz Power	WP 0102 00/TM 1-2840-248-23 (Torque)
115%	Collective Channel	TGT SPIKES ON EITHER ENGINE, in this work package
125%	Nr	BITE READS 115% OR DURING RECOVERY, NP DOES NOT INCREASE TO MATCH NR, in this work package/TM 1-2840-248-23 (Torque)

Indication/Condition

None Required

Corrective Action

None Required

ENGINE NG LOW AT GROUND IDLE SPEED (IDLE SPEED BELOW LIMITS)

SYMPTOM

Engine Ng low.

MALFUNCTION

Engine Ng low at ground idle speed (idle speed below limits).

CORRECTIVE ACTION

1. **Troubleshoot instrument display system, %Ng 1 and 2 scales for proper operation (WP 0089 00).**
 - a. If instrument display system operates as specified, troubleshoot engine for low ground idle speed (TM 1-2840-248-23). Go to **2**.
2. **Procedure completed.**

ENGINE NG HIGH AT GROUND IDLE SPEED (IDLE SPEED ABOVE LIMITS)**SYMPTOM**

Engine Ng high.

MALFUNCTION

Engine Ng high at ground idle speed (idle speed above limits).

CORRECTIVE ACTION

1. **Does helicopter have HMU 6068T77P03 or 6068T97P03?**
 - a. If helicopter is as specified, go to **2**.
 - b. If helicopter is not as specified, go to **4**.
2. **Check power available spindle system for proper rigging and operation (WP 0512 00).**
 - a. If rigging operation is proper, go to **3**.
 - b. If rigging operation is not proper, rig power available spindle controls (WP 0512 00) or replace components as required (WP 0499 00, WP 0507 00, or WP 0508 00). Go to **6**.
3. **Troubleshoot instrument display system %Ng 1 and 2 scales for proper operation (WP 0089 00).**
 - a. If scales show proper operation, troubleshoot engine for high Ng during ground idle (TM 1-2840-248-23). Go to **6**.
 - b. If scales do not show proper operation, repair/replace components as required. Go to **6**.
4. **Check power available spindle system for proper rigging and operation (WP 0512 00).**
 - a. If rigging shows proper operation, go to **5**.
 - b. If rigging does not show proper operation, rig power available spindle controls (WP 0512 00) or replace components as required (WP 0499 00, WP 0507 00, or WP 0508 00). Go to **6**.
5. **Perform load demand spindle verification (LOAD DEMAND SPINDLE VERIFICATION, in this work package) to check for proper rigging and operation.**
 - a. If load demand spindle system operates as specified, troubleshoot engine for high Ng during ground idle (TM 1-2840-248-23). Go to **6**.
 - b. If load demand spindle system is not correct, repair/replace components, as required. Go to **6**.
6. **Procedure completed.**

ENGINE FLAMES OUT**SYMPTOM**

Engine flames out.

MALFUNCTION

Engine flames out.

CORRECTIVE ACTION

1. **Move control quadrant ENG POWER CONT lever in cockpit to LOCKOUT. Go to 2.**

ENGINE FLAMES OUT - Continued

2. **Move control quadrant ENG FUEL SYS SELECTOR LEVER to detent. Go to 3.**
3. **Place FUEL PUMP switch to FUEL PRIME. Check that fuel is coming from overboard drain.**
 - a. If fuel is coming from overboard drain, go to **4.**
 - b. If fuel is not coming from overboard drain, troubleshoot fuel prime boost system (WP 0129 00). Go to **5.**
4. **Visually check self sealing breakaway main fuel valve on top of tank for a yellow stripe.**
 - a. If yellow stripe is present, replace fuel valve (WP 0979 00). Go to **5.**
 - b. If yellow stripe is not present, troubleshoot engine for flame out (TM 1-2840-248-23). Go to **5.**
5. **Procedure completed.**

ENGINE TGT FLUCTUATES AT GROUND IDLE SPEED**SYMPTOM**

Engine TGT fluctuates.

MALFUNCTION

Engine TGT fluctuates at ground idle speed.

CORRECTIVE ACTION

1. **Troubleshoot %Ng 1 or %Ng 2 scale on instrument display unit (WP 0089 00).**
 - a. If instrument display system operates as specified, troubleshoot engine for fluctuating TGT at ground idle speed (TM 1-2840-248-23). Go to **2.**
2. **Procedure completed.**

ENGINE TGT EXCEEDS LIMITER SETTING**SYMPTOM**

Engine TGT exceeded.

MALFUNCTION

Engine TGT exceeds limiter setting.

CORRECTIVE ACTION

1. **Troubleshoot TGT 1 or 2 scale on instrument display unit (WP 0089 00).**
 - a. If instrument display system operates as specified, troubleshoot engine for exceeding TGT limiter (TM 1-2840-248-23). Go to **2.**
2. **Procedure completed.**

ENGINE HAS UNCONTROLLED ACCELERATION (NG)**SYMPTOM**

Engine unstable.

MALFUNCTION

Engine has uncontrolled acceleration (Ng).

CORRECTIVE ACTION

1. **Check ENG POWER CONT level for proper operation.**
 - a. If ENG POWER CONT level is normal, troubleshoot engine for uncontrolled acceleration (TM 1-2840-248-23). Go to **2**.
 - b. If ENG POWER CONT level is not normal, repair/replace components in the PAS system, as required (WP 0499 00, WP 0507 00, or WP 0508 00). Go to **2**.
2. **Procedure completed.**

ENGINE HAS UNSTABLE OPERATION (NG, NP, AND TORQUE)**SYMPTOM**

Engine unstable operation.

MALFUNCTION

Engine has unstable operation (Ng, Np, and torque).

CORRECTIVE ACTION

1. **Check continuity between:**
 - **P137-N and P800-20**
 - **P138-N and P801-20**
 - a. If continuity is present, troubleshoot engine for unstable operation (TM 1-2840-248-23). Go to **2**.
 - b. If continuity is not present, repair/replace wiring, as required (WP 1747 00). Go to **2**.
2. **Procedure completed.**

TORQUE SPLIT GREATER THAN 5% OR ERRATIC**SYMPTOM**

Torque split exceeded.

MALFUNCTION

Torque split greater than 5% or erratic.

CORRECTIVE ACTION

1. **Check electrical connectors between DEC of both engines for proper connections.**
 - a. If connectors have proper connections, go to **2**.
 - b. If connectors do not have proper connections, connect electrical connectors. Go to **10**.
2. **Check load demand cable spring pin (for each engine) for sheared condition.**

TORQUE SPLIT GREATER THAN 5% OR ERRATIC - Continued

- a. If sheared conditions is present, go to **3**.
- b. If sheared condition is not present, go to **9**.
3. **Disconnect collective bias tube from mixing unit lever (WP 0506 00). Push cable rod end in and out. Go to 4.**
4. **Check that cable movement is smooth (no binding).**
 - a. If cable movement is smooth, go to **5**.
 - b. If cable movement is not smooth, go to **6**.
5. **Replace collective bias spring pin (WP 0511 00). Go to 10.**
6. **Disconnect load demand rotary input from HMU (WP 0498 00). Go to 7.**
7. **Push cable rod end in and out. Go to 8.**
8. **Check that cable movement is smooth (no binding).**
 - a. If cable movement is smooth, replace spring pin in LDS cable (WP 0511 00). Go to **10**.
 - b. If cable movement is not smooth, replace LDS cable (WP 0506 00). Go to **10**.
 - c. If trouble remains, replace HMU (TM 1-2840-248-23). Go to **10**.
9. **Check continuity between:**
 - **P800-3 and P801-3**
 - **P800-22 and P801-22**
 - **P800-23 and P801-24**
 - **P800-24 and P801-23**
 - a. If continuity is present, troubleshoot engine for excessive torque split for erratic torque indication (TM 1-2840-248-23). Go to **10**.
 - b. If continuity is not present, repair/replace wiring between P800 and P801, as required (WP 1747 00). Go to **10**.
10. **Procedure completed.**

BITE READS 125%**SYMPTOM**

BITE reads 125%.

MALFUNCTION

BITE reads 125%.

CORRECTIVE ACTION

1. **Does only No. 2 engine BITE read 125%?**
 - a. If only No. 2 engine BITE reads 125%, go to **2**.
 - b. If both engine BITEs read 125%, go to **3**.
2. **Check continuity between:**
 - **P407-1 and P800-26**

BITE READS 125% - Continued

- **P407-2 and P800-25**
 - a. If continuity is present, replace main rotor speed sensor (WP 0500 00). Go to **6**.
 - b. If trouble remains, troubleshoot No. 2 engine (TM 1-2840-248-23). Go to **6**.
 - c. If continuity is not present, repair/replace wiring, as required (WP 1747 00). Go to **6**.
- 3. **Do No. 1 and No. 2 engines BITE read 125%?**
 - a. If both engine BITEs are as specified, go, to **4**.
 - b. If both engine BITEs are not as specified, go to **5**.
- 4. **Check continuity between:**
 - **P407-1 and P800-26**
 - **P407-2 and P800-25**
 - a. If continuity is present, replace main rotor speed sensor (WP 0500 00). Go to **6**.
 - b. If continuity is not present, repair/replace wiring, as required (WP 1747 00). Go to **6**.
- 5. **Check continuity between:**
 - **P407-1 and P800-26**
 - **P407-2 and P800-25**
 - a. If continuity is present, replace main rotor speed sensor (WP 0500 00). Go to **6**.
 - b. If trouble remains, troubleshoot No. 1 engine (TM 1-2840-248-23). Go to **6**.
 - c. If continuity is not present, repair/replace wiring, as required (WP 1747 00). Go to **6**.
- 6. **Procedure completed.**

BITE READS 115% OR DURING POWER RECOVERY, NP DOES NOT INCREASE TO MATCH NR**SYMPTOM**

BITE reads 115%.

MALFUNCTION

BITE reads 115% or during power recovery, Np does not increase to match Nr.

CORRECTIVE ACTION

1. **Does only No. 2 engine BITE read 115%?**
 - a. If only No. 2 engine BITE reads 115%, go to **2**.
 - b. If both engine BITEs read 115%, go to **4**.
2. **Check continuity between:**
 - **J351-4 and P800-29**
 - **J351-5 and P800-30**
 - **J351-6 and P800-31**
 - a. If continuity is present, go to **3**.
 - b. If continuity is not present, repair/replace wiring, as required (WP 1747 00). Go to **7**.

BITE READS 115% OR DURING POWER RECOVERY, NP DOES NOT INCREASE TO MATCH NR -
Continued

3. **Is No. 3 collective stick position sensor adjusted properly?**
 - a. If sensor is adjusted properly, troubleshoot No. 2 engine (TM 1-2840-248-23). Go to **7**.
 - b. If sensor is not adjusted properly, adjust sensor (WP 1090 00). Go to **7**.
 - c. If sensor cannot be adjusted, replace sensor (WP 1090 00). Go to **7**.
4. **Do No. 1 and No. 2 engine BITEs read 115%?**
 - a. If both engine BITEs read 115%, replace No. 3 collective stick position sensor (WP 1090 00). Go to **7**.
 - b. If both engines do not read 115%, go to **5**.
5. **Check continuity between:**
 - **J351-1 and P801-29**
 - **J351-2 and P801-30**
 - **J351- 3 and P801-31**
 - a. If continuity is present, go to **6**.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **7**.
6. **Is No. 3 collective stick position sensor adjusted properly?**
 - a. If sensor is adjusted properly, troubleshoot No. 1 engine (TM 1-2840-248-23). Go to **7**.
 - b. If sensor is not adjusted properly, adjust sensor (WP 1090 00). Go to **7**.
 - c. If sensor cannot be adjusted, replace sensor (WP 1090 00). Go to **7**.
7. **Procedure completed.**

TGT SPIKES ON EITHER ENGINE**SYMPTOM**

TGT spikes.

MALFUNCTION

TGT spikes on either engine.

CORRECTIVE ACTION

1. **Check that maintenance has been performed in the following areas:**
 - **No. 1 or No. 2 engines**
 - **No. 1 or No. 2 SDCs**
 - **Flight control harness to cockpit feed through connectors.**
 - a. If maintenance has been performed go to **2**.
 - b. If maintenance has not been performed, go to **11**.
2. **Check that all bonded and grounded connections are secure and free of corrosion and that electrical harnesses and connectors have not been damaged and are free of moisture.**
 - a. If connectors are as specified, go to **3**.

TGT SPIKES ON EITHER ENGINE - Continued

- b. If connections are not as specified, repair/replace wiring, as required (WP 1747 00). Go to **41**.
3. **Disconnect yellow harness connector W4P5 from alternator. Check for specified resistance between:**
 - **J2-1 and J2-1 (Infinity)**
 - **J2-2 and J2-2 (Infinity)**
 - **J2-3 and J2-3 (0.6 - 1.0)**
 - **J2-4 and J2-4 (0.6 - 1.0)**
 - **J2-5 and J2-5 (1.2 - 2.0)**
 - a. If resistance is as specified, go to **4**.
 - b. If resistance is not as specified, replace alternator (TM 1-2840-248-23). Go to **41**.
4. **Check circuit at S39 connector on ECU (TM 1-2840-248-23).**
 - a. If circuit is good, go to **5**.
 - b. If circuit is not good, repair/replace wiring as required (WP 1747 00). Go to **41**.
5. **Disconnect connectors P814 and P824. Run No. 2 engine (TM 1-1520-237-10). Is TGT still spiking?**
 - a. If TGT still spikes, go to **6**.
 - b. If TGT is not spiking, go to **26**.
6. **Check all associated engine compartment harnesses and connectors, SDC harness and connectors, and interconnecting wiring for damage and moisture intrusion.**
 - a. If harness and connectors are good, go to **7**.
 - b. If harness and connectors are not good, repair/replace wiring, as required (WP 1747 00). Go to **41**.
7. **Is related spiking indication also associated with torque?**
 - a. If spiking is as specified, go to **8**.
 - b. If spiking is not as specified, go to LOAD DEMAND SPINDLE VERIFICATION, in this work package.
8. **Is indication of torque spiking associated with No. 2 engine?**
 - a. If indication of torque is as specified, go to **9**.
 - b. If indication of torque is not as specified, go to **10**.
9. **Check continuity between:**
 - **P800-10 and P814-15**
 - **P800-11 and P814-14**
 - **P800-20 and P814-30**
 - a. If continuity is present, go to LOAD DEMAND SPINDLE VERIFICATION, in this work package.

TGT SPIKES ON EITHER ENGINE - Continued

- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **41**.
- 10. **Check continuity between:**
 - **P801-10 and P815-15**
 - **P801-11 and P815-16**
 - a. If continuity is present, go to LOAD DEMAND SPINDLE VERIFICATION, in this package.
 - b. If continuity is not present, repair/replace wiring, as required (WP 1747 00). Go to **41**.
- 11. **Disconnect yellow harness connector W4P5 from alternator. Check for specified resistance between:**
 - **J2-1 and J2-1 (Infinity)**
 - **J2-2 and J2- 2 (Infinity)**
 - **J2-3 and J2-3 (0.6 - 1.0)**
 - **J2-4 and J2-4 (0.6 - 1.0)**
 - **J2-3 and J2-3 (1.2 - 2.0)**
 - a. If resistance is as specified, go to **12**.
 - b. If resistance is not as specified, replace alternator (TM 1-2840-248-23). Go to **41**.
- 12. **Do circuit checks at S39 connector on ECU (TM 1-2840-248-23).**
 - a. If circuit checks are as specified, go to **13**.
 - b. If circuit checks are not as specified, repair/replace wiring, as required (WP 1747 00). Go to **41**.
- 13. **Disconnect connectors P814 and P824. Run No. 2 engine (TM 1-2840-248-23). Is TGT still spiking?**
 - a. If TGT still spikes, go to **14**.
 - b. If TGT does not spike, go to **19**.
- 14. **Check all associated engine compartment harnesses and connectors, SDC harness and connectors, and interconnecting wiring for damage and moisture intrusion.**
 - a. If harness and connectors are good, go to **15**.
 - b. If harness and connectors are not good, repair/replace wiring, as required (WP 1747 00). Go to **41**.
- 15. **Is related spiking indication also associated with torque?**
 - a. If spiking is as specified, go to **16**.
 - b. If spiking is not as specified, go to LOAD DEMAND SPINDLE VERIFICATION, in this work package.
- 16. **Is indication of torque spiking associated with No. 2 engine?**
 - a. If spiking is associated with No. 2 engine, go to **17**.
 - b. If spiking is not associated with No. 2 engine, go to **18**.
- 17. **Check continuity between:**

TGT SPIKES ON EITHER ENGINE - Continued

- **P800-10 and P814-15**
 - **P800-11 and P814-14**
 - **P800-20 and P814-30**
- a. If continuity is present, go to LOAD DEMAND SPINDLE VERIFICATION, in this work package.
- b. If continuity is not present, repair/replace wiring, as required (WP 1747 00). Go to **41**.
18. **Check continuity between:**
- **P801-10 and P815-15**
 - **P801-11 and P815-16**
- a. If continuity is present, go to LOAD DEMAND SPINDLE VERIFICATION, in this work package.
- b. If continuity is not present, repair/replace wiring, as required (WP 1747 00). Go to **41**.
19. **Reconnect connectors P814 and P824. Disconnect connectors P815 and P825. Run No. 1 engine (TM 1-1520-237-23). Is TGT still spiking?**
- a. If TGT still spikes, go to **20**.
- b. If TGT does not spike, go to **25**.
20. **Check all associated engine compartment harnesses and connectors, SDC harness and connectors, and interconnecting wiring for damage and moisture intrusion.**
- a. If harness and connectors are good, go to **21**.
- b. If harness and connectors are not good, repair/replace wiring as required (WP 1747 00). Go to **41**.
21. **Is related spiking indication also associated with torque?**
- a. If spiking is related with torque, go to **22**.
- b. If spiking is not related with torque, go to LOAD DEMAND SPINDLE VERIFICATION, in this work package.
22. **Is indication of torque spiking associated with No. 1 engine?**
- a. If spiking is associated with No. 1 engine, go to **23**.
- b. If spiking is not associated with No. 1 engine, go to **24**.
23. **Check continuity between:**
- **P800-10 and P814-15**
 - **P800-11 and P814-14**
 - **P800-20 and P814-30**
- a. If continuity is present, go to LOAD DEMAND SPINDLE VERIFICATION, in this work package.
- b. If continuity is not present, repair/replace wiring, as required (WP 1747 00). Go to **41**.
24. **Check continuity between:**

TGT SPIKES ON EITHER ENGINE - Continued

- **P801-10 and P815-15**
 - **P801-11 and P815-16**
- a. If continuity is present, go to LOAD DEMAND SPINDLE VERIFICATION, in this work package.
- b. If continuity is not present, repair/replace wiring, as required (WP 1747 00). Go to **41**.
25. **Troubleshoot engine speed trim system (WP 0076 00). Is TGT still spiking?**
- a. If TGT still spikes, go to **37**.
- b. If TGT does not spike, system is operational. Go to **41**.
26. **Reconnect connectors P814 and P824. Disconnect connectors P815 and P825. Run No. 1 engine (TM 1-1520-237-23). Is TGT still spiking?**
- a. If TGT still spikes, go to **27**.
- b. If TGT does not spike, go to **32**.
27. **Check all associated engine compartment harnesses and connectors, SDC harness and connectors, and interconnecting wiring for damage and moisture intrusion.**
- a. If harness and connectors are good, go to **28**.
- b. If harness and connectors are not good, repair/replace wiring, as required (WP 1747 00). Go to **41**.
28. **Is related spiking indication also associated with torque?**
- a. If related spiking is associated with torque, go to **29**.
- b. If related spiking is not associated with torque, go to LOAD DEMAND SPINDLE VERIFICATION, in this work package.
29. **Is indication of torque spiking associated with No. 2 engine?**
- a. If spiking is associated with No. 2 engine, go to **30**.
- b. If spiking is not associated with No. 2 engine, go to **31**.
30. **Check continuity between:**
- **P800-10 and P814-15**
 - **P800-11 and P814-14**
 - **P800-20 and P814-30**
- a. If continuity is present, go to LOAD DEMAND SPINDLE VERIFICATION, in this work package.
- b. If continuity is not present, repair/replace wiring, as required (WP 1747 00). Go to **41**.
31. **Check continuity between:**
- **P801-10 and P815-15**
 - **P801-11 and P815-16**
- a. If continuity is present, go to LOAD DEMAND SPINDLE VERIFICATION, in this work package.

TGT SPIKES ON EITHER ENGINE - Continued

- b. If continuity is not present, repair/replace wiring, as required (WP 1747 00). Go to **41**.
- 32. **Troubleshoot engine speed trim system (WP 0076 00). Is TGT still spiking?**
 - a. If TGT still spikes, go to **33**.
 - b. If TGT does not spike, system is operational. Go to **41**.
- 33. **Is related spiking indication also associated with torque?**
 - a. If spiking is associated with torque, go to **34**.
 - b. If spiking is not associated with torque, go to LOAD DEMAND SPINDLE VERIFICATION, in this work package.
- 34. **Is indication of torque spiking associated with No. 2 engine?**
 - a. If torque is associated with No. 2 engine, go to **35**.
 - b. If torque is not associated with No. 2 engine, go to **36**.
- 35. **Check continuity between:**
 - **P800-10 and P814-15**
 - **P800-11 and P814-14**
 - **P800-20 and P814-30**
 - a. If continuity is present, go to LOAD DEMAND SPINDLE VERIFICATION, in this work package.
 - b. If continuity is not present repair/replace wiring, as required (WP 1747 00). Go to **41**.
- 36. **Check continuity between:**
 - **P801-10 and P815-15**
 - **P801-11 and P815-16**
 - a. If continuity is present, go to LOAD DEMAND SPINDLE VERIFICATION, in this work package.
 - b. If continuity is not present, repair/replace wiring, as required (WP 1747 00). Go to **41**.
- 37. **Is related spiking also associated with torque?**
 - a. If spiking is associated with torque, go to **38**.
 - b. If spiking is not associated with torque, go to LOAD DEMAND SPINDLE VERIFICATION, in this work package.
- 38. **Is indication of torque spiking associated with No. 2 engine?**
 - a. If spiking is associated with No. 2 engine, go to **39**.
 - b. If spiking is not associated with No. 2 engine, go to **40**.
- 39. **Check continuity between:**
 - **P800-10 and P814-15**
 - **P800-11 and P814-14**
 - **P800-20 and P814-30**

TGT SPIKES ON EITHER ENGINE - Continued

- a. If continuity is present, go to LOAD DEMAND SPINDLE VERIFICATION, in this work package.
 - b. If continuity is not present, repair/replace wiring, as required (WP 1747 00). Go to **41**.
40. **Check continuity between:**
- **P801-10 and P815-15**
 - **P801-11 and P815-16**
- a. If continuity is present, go to LOAD DEMAND SPINDLE VERIFICATION, in this work package.
 - b. If continuity is not present, repair/replace wiring, as required (WP 1747 00). Go to **41**.
41. **Procedure completed.**

TGT AND/OR TORQUE SPIKING APPEAR ON PDUS WITHOUT ASSOCIATED CHANGE IN ENGINE POWER**SYMPTOM**

TGT and/or torque spiking appears.

MALFUNCTION

TGT and/or torque spiking appear on PDUs without associated change in engine power.

CORRECTIVE ACTION

1. **Troubleshoot vertical indicator display system (WP 0089 00). Is system malfunctioning?**
 - a. If vertical indicator display system is malfunctioning, repair, is required (WP 0799 00). Go to **14**.
 - b. If vertical indicator display system is not malfunctioning, go to **2**.
2. **Check No. 1 and No. 2 engine wiring harness continuity.**
 - a. If continuity is present, go to **3**.
 - b. If continuity is not present, replace wiring harness, as required (TM 1-2840-248-23). Go to **14**.
3. **Disconnect No. 2 engine history recorder. Run engines (TM 1-1520-237-23). Check that TGT and/or torque spiking indications do not appear on PDU's without change in engine power.**
 - a. If TGT and/or torque spiking appear on PDU, go to **4**.
 - b. If TGT and/or torque spiking does not appear on PDU, go to **5**.
4. **Check connectors between recorder and ECU for contamination.**
 - a. If contamination is present, clean connectors (TM 1-2840-248-23). Go to **14**.
 - b. If contamination is not present, replace history recorder (TM 1-2840-248-23). Go to **14**.
 - c. If trouble remains, replace yellow electrical cable (TM 1-2840-248-23). Go to **14**.
5. **Shutdown engines and disconnect No. 1 engine history recorder. Run engines (TM 1-1520-237-23). Has trouble been remedied?**

TGT AND/OR TORQUE SPIKING APPEAR ON PDUS WITHOUT ASSOCIATED CHANGE IN ENGINE POWER - Continued

- a. If trouble has been remedied, go to **6**.
- b. If trouble has not been remedied, go to **7**.
- 6. Check connectors between recorder and ECU for contamination.**
 - a. If contamination is present, clean connectors (TM 1-2840-248-23). Go to **14**.
 - b. If contamination is not present, replace history recorder (TM 1-2840-248-23). Go to **14**.
 - c. If trouble remains, replace yellow electrical cable (TM 1-2840-248-23). Go to **14**.
- 7. Check yellow electrical cable for contamination.**
 - a. If contamination is present, clean connectors (TM 1-2840-248-23). Go to **14**.
 - b. If contamination is not present, go to **8**.
- 8. Check continuity of yellow electrical cable.**
 - a. If continuity is present, go to **9**.
 - b. If continuity is not present, replace cable (TM 1-2840-248-23). Go to **14**.
- 9. Check blue electrical cable for contamination.**
 - a. If contamination is present, clean connectors (TM 1-2840-248-23). Go to **14**.
 - b. If contamination is not present, go to **10**.
- 10. Check continuity of blue electrical cable.**
 - a. If continuity is present, go to **11**.
 - b. If continuity is not present, replace cable (TM 1-2840-248-23). Go to **14**.
- 11. Check green electrical cable for contamination.**
 - a. If contamination is present, clean connectors (TM 1-2840-248-23). Go to **14**.
 - b. If contamination is not present, go to **12**.
- 12. Check continuity of green electrical cable.**
 - a. If continuity is present, go to **13**.
 - b. If continuity is not present, replace cable (TM 1-2840-248-23). Go to **14**.
- 13. Troubleshoot Np sensor system (TM 1-2840-248-23). Is Np sensor system malfunctioning?**
 - a. If Np sensor system is malfunctioning, replace Np sensor system (TM 1-2840-248-23). Go to **14**.
 - b. If Np system is not malfunctioning, replace engine as required (TM 1-2840-248-23). Go to **14**.
- 14. Procedure completed.**

LOAD DEMAND SPINDLE VERIFICATION**SYMPTOM**

Load demand.

MALFUNCTION

Load demand spindle verification.

CORRECTIVE ACTION

1. **Disconnect LDS cable from mixer unit lever assembly. Go to 2.**
2. **Measure load demand system force (T1) (WP 0506 00).**
 - a. If force is greater than 45 pounds, go to **3**.
 - b. If force is less than or equal to 45 pounds, reconnect LDS cable to mixer unit lever assembly. Go to **10**.
3. **Disconnect load demand rotary input from HMU (WP 0498 00). Go to 4.**
4. **Measure load demand system force (T2) (WP 0506 00).**
 - a. If force (T2) is greater than T1 minus 12 pounds, go to **5**.
 - b. If force (T2) is less or equal to T1 minus 12 pounds, go to **7**.
5. **Disconnect LDS cable from rotary input assembly (WP 0498 00). Go to 6.**
6. **Measure cable friction force (T3) (WP 0506 00).**
 - a. If force (T3) is greater than 3 pounds, replace LDS cable (WP 0506 00). Go to **10**.
 - b. If force (T3) is less than or equal to 3 pounds, replace rotary input assembly (WP 0498 00). Go to **10**.
7. **Measure HMU spindle torque (Q) (TM 1-2840-248-23).**
 - a. If spindle torque is greater than 15 inch-pounds, replace HMU (TM 1-2840-248-23). Go to **10**.
 - b. If spindle torque is less or equal to 15 inch-pounds, go to **8**.
8. **Disconnect LDS cable from rotary input assembly (WP 0498 00). Go to 9.**
9. **Measure cable friction force (T3) (WP 0506 00).**
 - a. If force (T3) is greater than 3 pounds, replace LDS cable (WP 0506 00). Go to **10**.
 - b. If force (T3) is less or equal to 3 pounds, replace rotary input assembly (WP 0498 00). Go to **10**.
10. **Procedure completed.**

LOCATION

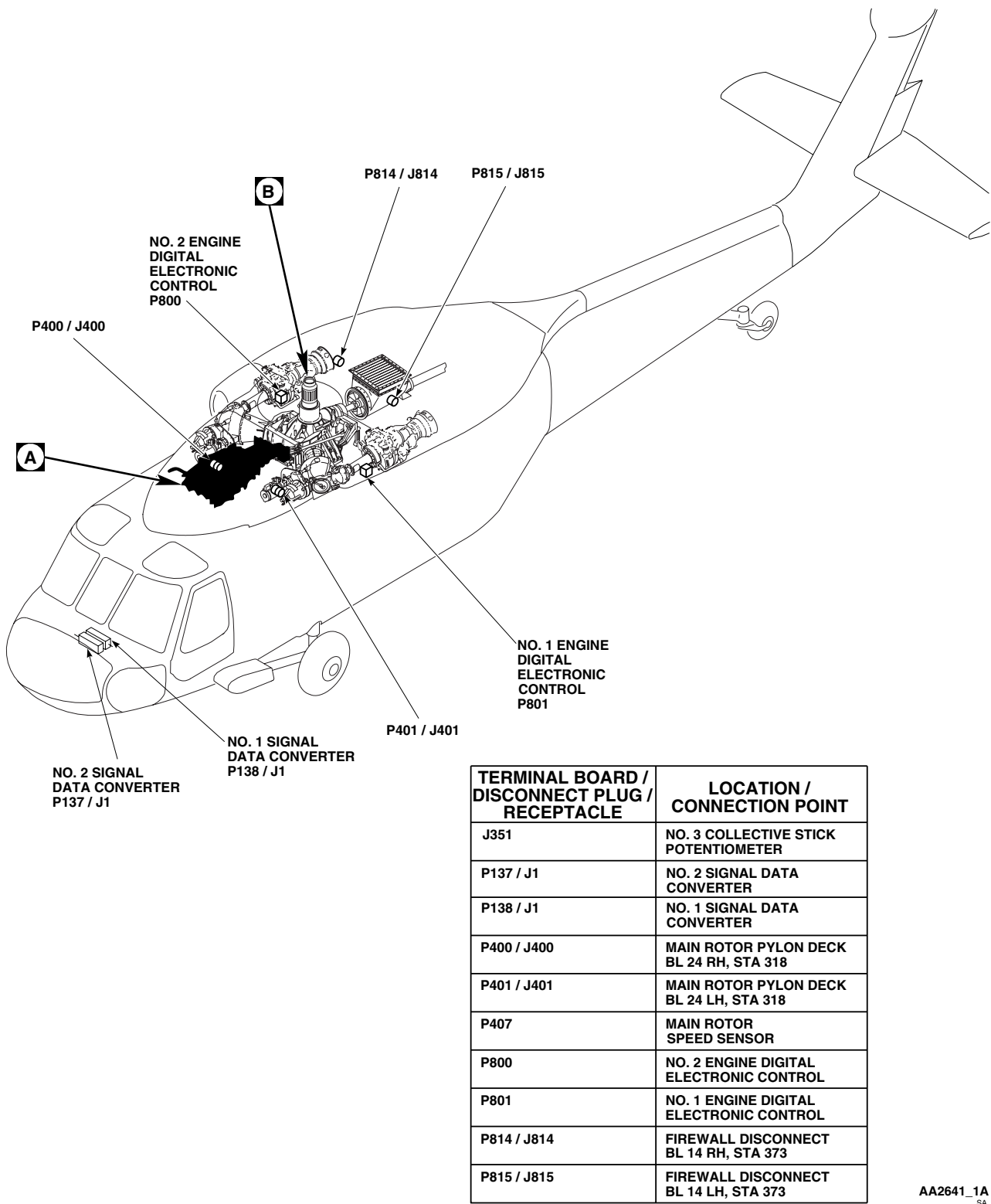
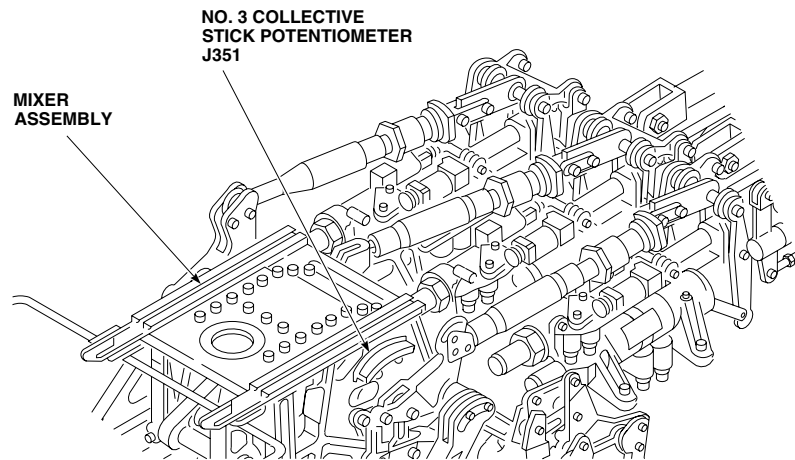


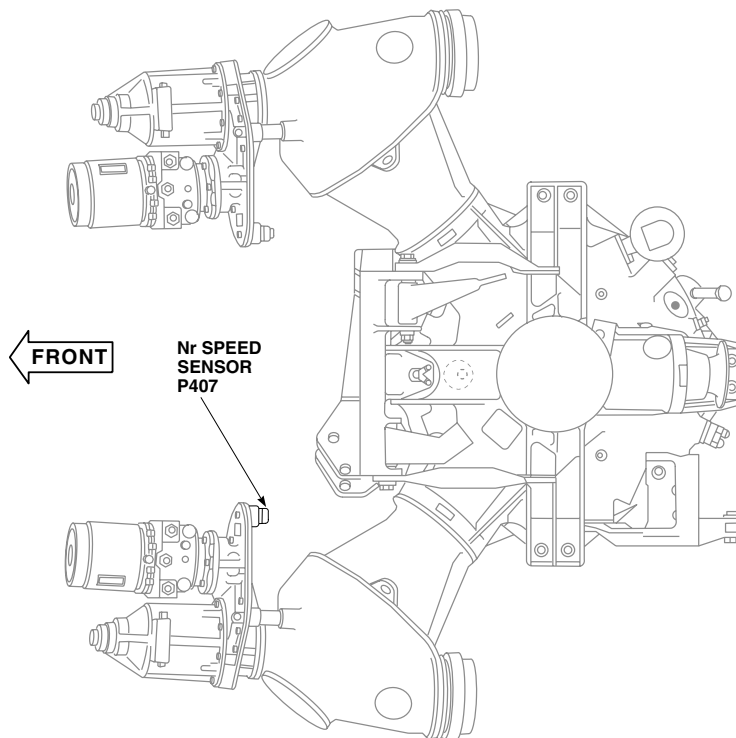
Figure 1. Engine and Engine Interface Location Diagram **UH60L** . (Sheet 1 of 2)

LOCATION - Continued



A

B

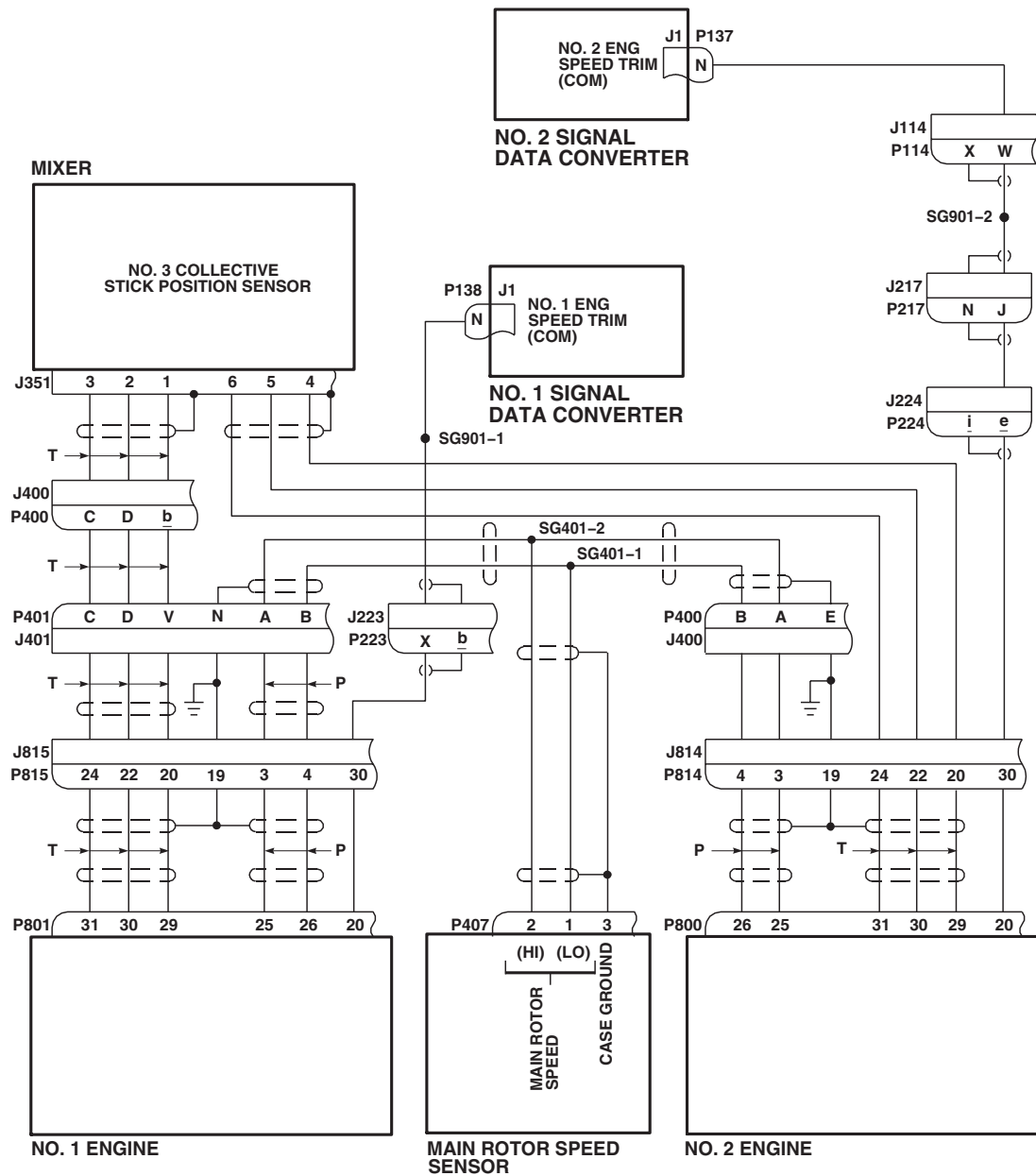


MAIN TRANSMISSION

AA2641_2
SA

Figure 1. Engine and Engine Interface Location Diagram **UH60L** . (Sheet 2 of 2)

SCHEMATICS

AB1453B
SAFigure 2. Engine and Engine Interface Schematic Diagram **UH60L**.

END OF WORK PACKAGE

UNIT LEVEL

ENGINE SYSTEM

NO. 1 ENGINE INSTRUMENTS AND WARNING LIGHTS SYSTEM

This WP supersedes WP 0073 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment

Multimeter, AN/PSM-45A

WP 0075 00

WP 0089 00

WP 0090 00

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Electrical Repairer Toolkit, SC 5180-99-B06

Engine Repairman's Toolkit, SC 5180-99-B07

Insulated Jumper Wire

WP 0100 00

WP 0104 00

WP 0481 00

WP 0482 00

WP 0824 00

Personnel Required

Aircraft Electrician MOS 15F (1)

Aircraft Powerplant Repairer MOS 15B (1)

UH-60 Helicopter Repairer MOS 15T (1)

WP 0925 00

WP 0976 00

WP 1685 00

WP 1747 00

References

TM 1-1520-237-MTF

TM 1-1520-237-10

TM 1-2840-248-23

Equipment Condition

External Electrical Power Available, WP 1685 00 00
or APU Operational, TM 1-1520-237-10

No. 1 Fuel Tank Serviced, TM 1-1520-237-10

NO. 1 ENGINE INSTRUMENTS AND WARNING LIGHTS SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for system location diagram, Figure 2 for system schematic diagram, **UH-60A UH-60L EH-60A** Figure 3 and **HH-60A HH-60L** Figure 4 for warning lights schematic diagrams as an aid in operational/troubleshooting.

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Turn on electrical power. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed).

Indication/Condition

- a. #1 FUEL PRESS capsule on caution/advisory panel or legend MFD/caution/advisory panel shall go on.

NO. 1 ENGINE INSTRUMENTS AND WARNING LIGHTS SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- b. #1 ENGINE OIL PRESS capsule on caution/advisory panel or legend on MFD/caution/advisory panel shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, go to #1 FUEL PRESS CAPSULE OR LEGEND DOES NOT GO ON, in this work package.
2. If Indication/Condition b. is not as specified, go to #1 ENGINE OIL PRESS CAPSULE OR LEGEND DOES NOT GO ON, in this work package.

Step

3. Install an insulated jumper wire between P803-13 and P803-14.

Indication/Condition

#1 OIL FLTR BYPASS capsule on caution/advisory panel or legend on MFD/caution/advisory panel shall go on.

Corrective Action

If Indication/Condition is not as specified, go to #1 OIL FLTR BYPASS CAPSULE OR LEGEND DOES NOT GO ON, in this work package.

Step

4. Remove jumper wire from P803.
5. Install an insulated jumper wire between P803-15 and P803-16.

Indication/Condition

#1 FUEL FLTR BYPASS capsule on caution/advisory panel or legend on MFD/caution/advisory panel shall go on.

Corrective Action

If Indication/Condition is not as specified, go to #1 FUEL FLTR BYPASS CAPSULE OR LEGEND DOES NOT GO ON, in this work package.

Step

6. Remove jumper wire from P803.
7. Connect P803 to No. 1 engine (WP 0481 00).
8. Turn off electrical power.
9. With qualified personnel at controls, run No. 1 engine up to idle speed (TM 1-1520-237-10).

Indication/Condition

#1 FUEL PRESS capsule shall be off.

Corrective Action

If Indication/Condition is not as specified, go to #1 FUEL PRESS CAPSULE OR LEGEND REMAINS ON WHEN ENGINE IS AT IDLE SPEED, in this work package.

Step

10. With qualified personnel at controls, run No. 1 engine with main rotor turning at 100% Nr until engine instruments stabilize.

NO. 1 ENGINE INSTRUMENTS AND WARNING LIGHTS SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

Engine status shall be as follows:

CAUTION/ADVISORY PANEL OR MFD/CAUTION/ADVISORY PANEL	CAPSULE STATUS
#1 FUEL PRESS	OFF
#1 ENGINE OIL PRESS	OFF
#1 ENGINE OIL TEMP	OFF
#1 FUEL FLTR BYPASS	OFF
#1 OIL FLTR BYPASS	OFF

CENTRAL DISPLAY UNIT	INDICATION
Ng SPEED 1	(Refer to TM 1-1520-237-MTF for normal operating limits)
TGT TEMP 1	
ENG OIL TEMP 1	
ENG OIL PRESS 1	

PILOT'S DISPLAY UNIT	INDICATION
% RPM 1	(Refer to TM 1-1520-237-MTF for normal operating limits)
% TRQ 1	

Corrective Action

1. If any of these caution/advisory panel capsules or MFD/caution/advisory panel legends are on:

CAPSULE	GO TO
#1 ENGINE OIL PRESS	NO. 1 ENGINE OIL PRESSURE (ENG OIL PRESS 1) IS BELOW MINIMUM LIMITS, in this work package.
#1 ENGINE OIL TEMP	NO. 1 ENGINE OIL TEMPERATURE (ENG OIL TEMP 1) IS TOO HIGH OR HAS NO INDICATION, in this work package.

2. If #1 FUEL PRESS capsule is on, troubleshoot engine for fuel low pressure caution light on during flight (TM 1-2840-248-23).

NO. 1 ENGINE INSTRUMENTS AND WARNING LIGHTS SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

3. If #1 FUEL FLTR BYPASS or #1 OIL FLTR BYPASS capsule is on, troubleshoot engine (TM 1-2840-248-23).
4. If No. 1 engine gas generator speed (% Ng 1) has no indication, go to NO. 1 ENGINE HAS NO GAS GENERATOR (% NG 1) INDICATION, in this work package.
5. If No. 1 engine turbine inlet temperature (TGT TEMP 1) has no indication, go to NO. 1 ENGINE TURBINE INLET TEMPERATURE (TGT TEMP 1) HAS NO INDICATION, in this work package.
6. If No. 1 engine oil temperature (ENG OIL TEMP 1) has no indication, go to NO. 1 ENGINE OIL TEMPERATURE (ENG OIL TEMP 1) IS TOO HIGH OR HAS NO INDICATION, in this work package.
7. If No. 1 engine oil pressure (ENG OIL PRESS 1):
 - Is too high (10 psi above normal), go to NO. 1 ENGINE OIL FLUCTUATES GREATER THAN 5 PSI OR IS TOO HIGH (10 PSI ABOVE NORMAL OPERATING PRESSURE), in this work package.
 - Has no indication, go to NO. 1 ENGINE OIL HAS NO OIL PRESSURE INDICATION, in this work package.
 - Fluctuates greater than 5 psi, go to NO. 1 ENGINE OIL FLUCTUATES GREATER THAN 5 PSI OR IS TOO HIGH (10 PSI ABOVE NORMAL OPERATING PRESSURE).
 - Is below minimum limits (45 psi), go to NO. 1 ENGINE OIL PRESSURE (ENG OIL PRESS 1) IS BELOW MINIMUM LIMITS, in this work package.
8. If No. 1 engine power turbine speed (% RPM 1) has no indication, go to NO. 1 ENGINE HAS NO POWER TURBINE SPEED (% RPM ENG 1) INDICATION, in this work package.
9. If No. 1 engine has no torque indication (% TRQ 1), go to NO. 1 ENGINE HAS NO TORQUE (% TRQ 1) INDICATION, in this work package.

Step

11. Turn MFD switch to OFF.
12. Shut down No. 1 engine (TM 1-1520-237-10).

Indication/Condition

No. 1 engine shall be shutdown.

Corrective Action

None Required

#1 FUEL PRESS CAPSULE OR LEGEND DOES NOT GO ON**SYMPTOM**

#1 FUEL PRESS capsule on caution/advisory panel or legend on MFD/caution/advisory panel shall go on.

MALFUNCTION

#1 FUEL PRESS capsule or legend does not go on.

CORRECTIVE ACTION**1. Check helicopter configuration.**

- a. **EMEP** , go to **2**.
- b. **W/O EMEP** , go to **4**.

#1 FUEL PRESS CAPSULE OR LEGEND DOES NOT GO ON - Continued

2. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R (WP 0925 00).
 - a. If pin filtered adapter is good, go to 3.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 9.
3. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 4.
4. Hold caution/advisory panel **HH-60A HH-60L** INDICATOR LTS BRT/DIM-TEST switch to TEST.
 - a. If #1 FUEL PRESS capsule goes on, go to 5.
 - b. If #1 FUEL PRESS capsule does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 9.
5. Check for 28 vdc between P117-2 or **HH-60A HH-60L** P1019R-71 or P1020R-71 and ground.
 - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 9.
 - b. If voltage is not as specified, go to 6.
6. Check for 28 vdc between P803-23 and ground.
 - a. If voltage is as specified, go to 7.
 - b. If voltage is not as specified, go to 8.
7. Check continuity between P117-2 or **HH-60A HH-60L** P1019R-71 or P1020R-71 and P803-24.
 - a. If continuity is present, replace fuel pressure switch (WP 0976 00). Go to 9.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 9.
8. Check for 28 vdc between NO. 1 ENG WARN LTS circuit breaker terminal 1 and ground.
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P803-23 (WP 1747 00). Go to 9.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to 9.
9. Procedure completed.

#1 ENGINE OIL PRESS CAPSULE OR LEGEND DOES NOT GO ON**SYMPTOM**

#1 ENGINE OIL PRESS capsule on caution/advisory panel or legend on MFD/caution/advisory panel shall go on.

MALFUNCTION

#1 ENGINE OIL PRESS capsule or legend does not go on.

CORRECTIVE ACTION

1. Check helicopter configuration.
 - a. **EMEP** , go to 2.
 - b. **W/O EMEP** , go to 4.

#1 ENGINE OIL PRESS CAPSULE OR LEGEND DOES NOT GO ON - Continued

2. **EMEP** Remove and check pin filtered adapters (WP 0925 00).
 - a. If pin filtered adapters are good, go to 3.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 7.
3. **EMEP** Install pin filtered adapters (WP 0925 00). Go to 4.
4. Hold caution/advisory panel **HH-60A HH-60L** instrument panel INDICATOR LTS **BRT/DIM-TEST** switch to TEST.
 - a. If #1 ENGINE OIL PRESS capsule goes on, go to 5.
 - b. If #1 ENGINE OIL PRESS capsule does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 7.
5. Check for 28 vdc between P117-3 or **HH-60A HH-60L** P1019R-72 or P1020R-72 and ground.
 - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 7.
 - b. If voltage is not as specified, go to 6.
6. Check continuity between P117-3 or **HH-60A HH-60L** P1019R-72 or P1020R-72 and P138-X.
 - a. If continuity is present, troubleshoot instrument display system (WP 0089 00). Go to 7.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 7.
7. Procedure completed.

#1 OIL FLTR BYPASS CAPSULE OR LEGEND DOES NOT GO ON**SYMPTOM**

#1 OIL FLTR BYPASS capsule on caution/advisory panel or legend on MFD/caution/advisory panel shall go on.

MALFUNCTION

#1 OIL FLTR BYPASS capsule or legend does not go on.

CORRECTIVE ACTION

1. Hold caution/advisory panel **HH-60A HH-60L** instrument panel INDICATOR LTS **BRT/DIM-TEST** switch to TEST.
 - a. If #1 OIL FLTR BYPASS capsule goes on, go to 2.
 - b. If #1 OIL FLTR BYPASS capsule does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 9.
2. Remove jumper wire from P803. Go to 3.
3. Check for 28 vdc between P803-13 and P803-14.
 - a. If voltage is as specified, troubleshoot engine oil filter bypass system (TM 1-2840-248-23). Go to 9.
 - b. If voltage is not as specified, go to 4.
4. Check for 28 vdc between P803-13 and ground.
 - a. If voltage is as specified, go to 5.

#1 OIL FLTR BYPASS CAPSULE OR LEGEND DOES NOT GO ON - Continued

- b. If voltage is not as specified, repair/replace wiring between NO. 1 ENG WARN LTS circuit breaker terminal 1 and P803-13 (WP 1747 00). Go to **9**.

5. Check helicopter configuration.

- a. **EMEP** , go to **6**.
- b. **W/O EMEP** , go to **8**.

6. EMEP Remove and check pin filtered adapter P117 or HH-60A HH-60L P1019R or P1020R (WP 0925 00).

- a. If pin filtered adapter is good, go to **7**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **9**.

7. EMEP Install pin filtered adapter (WP 0925 00). Go to 8.**8. Check for 28 vdc between P117-23 or HH-60A HH-60L P1019R-22 or P1020R-22 and ground.**

- a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **9**.
- b. If voltage is not as specified, repair/replace wiring between P803-14 and P117-23 or HH-60A HH-60L P1019R-22 or P1020R-22 (WP 1747 00). Go to **9**.

9. Procedure completed.**#1 FUEL FLTR BYPASS CAPSULE OR LEGEND DOES NOT GO ON****SYMPTOM**

#1 FUEL FLTR BYPASS capsule on caution/advisory panel or legend on MFD/caution/advisory panel shall go on.

MALFUNCTION

#1 FUEL FLTR BYPASS capsule or legend does not go on.

CORRECTIVE ACTION**1. Check helicopter configuration.**

- a. **EMEP** , go to **2**.
- b. **W/O EMEP** , go to **4**.

2. EMEP Remove and check pin filtered adapter P117 or HH-60A HH-60L P1019R or P1020R (WP 0925 00).

- a. If pin filtered adapter is good, go to **3**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **8**.

3. EMEP Install pin filtered adapter (WP 0925 00). Go to 4.**4. Hold caution/advisory panel HH-60A HH-60L instrument panel INDICATOR LTS BRT/DIM-TEST switch to TEST.**

- a. If #1 FUEL FLTR BYPASS capsule goes on, go to **5**.
- b. If #1 FUEL FLTR BYPASS capsule does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **8**.

#1 FUEL FLTR BYPASS CAPSULE OR LEGEND DOES NOT GO ON - Continued

5. Check for 28 vdc between P117-6 or **HH-60A HH-60L** P1019R-74 or P1020R-74 and ground.
 - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 8.
 - b. If voltage is not as specified, go to 6.
6. Remove jumper wire from P803. Check for 28 vdc between P803-15 and ground.
 - a. If voltage is as specified, go to 7.
 - b. If voltage is not as specified, repair/replace wiring between NO. 1 ENG WARN LTS circuit breaker terminal 1 and P803-15 (WP 1747 00). Go to 8.
7. Check continuity between P117-6 or **HH-60A HH-60L** P1019R-74 or P1020R-74 and P803-16.
 - a. If continuity is present, troubleshoot engine fuel filter bypass system (TM 1-2840-248-23). Go to 8.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 8.
8. Procedure completed.

#1 FUEL PRESS CAPSULE OR LEGEND REMAINS ON WHEN ENGINE IS AT IDLE SPEED**SYMPTOM**

#1 FUEL PRESS capsule shall be off.

MALFUNCTION

#1 FUEL PRESS capsule or legend remains on when engine is at idle speed.

CORRECTIVE ACTION

1. With No. 1 engine operating at ground idle speed, place upper console FUEL PUMP switch to FUEL PRIME. Go to 2.
2. Check that #1 FUEL PRESS capsule goes off.
 - a. If capsule goes off, go to 3.
 - b. If capsule does not go off, troubleshoot engine for FUEL LOW PRESSURE caution light on below flight idle speed (TM 1-2840-248-23). Go to 6.
3. Shut down No. 1 engine (TM 1-1520-237-10). Go to 4.
4. Disconnect P803 from No. 1 engine fuel pressure switch. Go to 5.
5. Check that #1 FUEL PRESS capsule goes off.
 - a. If capsule goes off, troubleshoot airframe fuel system for possible leaks between engine and fuel tanks. Repair/replace components as required. Go to 6.
 - b. If capsule does not go off, repair/replace wiring between (WP 1747 00):
 - P803-23 and P249-V
 - P803-24 and P117-2
 - **HH-60A HH-60L** P1019R-71 or P1020R-71

#1 FUEL PRESS CAPSULE OR LEGEND REMAINS ON WHEN ENGINE IS AT IDLE SPEED - Continued

- c. If trouble remains, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **6**.

6. Procedure completed.**NO. 1 ENGINE OIL PRESSURE (ENG OIL PRESS 1) IS BELOW MINIMUM LIMITS****SYMPTOM**

If No. 1 engine oil pressure (ENG OIL PRESS 1) below minimum limits (45 psi).

MALFUNCTION

No. 1 engine oil pressure (ENG OIL PRESS 1) is below minimum limits.

CORRECTIVE ACTION**CAUTION**

If engine ran for more than 1 minute while oil pressure was below minimum limits, and indicating system was operating properly, replace engine.

1. Visually check engine for external oil leaks.

- a. If leaks are found, repair/replace items causing leak. Check oil tank level and service as necessary (TM 1-2840-248-23). Go to **2**.
- b. If no leaks are found, go to NO.1 ENGINE OIL FLUCTUATES GREATER THAN 5 PSI OR IS TOO HIGH (10 PSI ABOVE NORMAL OPERATING PRESSURE), in this work package.

2. Procedure completed.**NO. 1 ENGINE OIL TEMPERATURE (ENG OIL TEMP 1) IS TOO HIGH OR HAS NO INDICATION****SYMPTOM**

No. 1 engine oil temperature (ENG OIL TEMP 1) has no indication.

MALFUNCTION

No. 1 Engine oil temperature (ENG OIL TEMP 1) is too high or has no indication.

CORRECTIVE ACTION**1. Check engine oil tank for proper level.**

- a. If tank has proper oil level, go to **2**.
- b. If tank does not have proper oil level, troubleshoot engine oil system (TM 1-2840-248-23). Go to **16**.

2. Troubleshoot instrument display system, ENG OIL TEMP 1 scale (WP 0089 00 00).

- a. If instrument display system is good, go to **3**.

3. Install an insulated jumper wire between P803-9 and P803-10. Go to 4.**4. Check helicopter configuration.**

- a. **EMEP** , go to **5**.
- b. **W/O EMEP** , go to **7**.

5. **EMEP Remove and check pin filtered adapter P138 (WP 0925 00).**

NO. 1 ENGINE OIL TEMPERATURE (ENG OIL TEMP 1) IS TOO HIGH OR HAS NO INDICATION - Continued

- a. If pin filtered adapter is good, go to **6**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **16**.
- 6. EMEP Install pin filtered adapter (WP 0925 00). Go to 7.**
- 7. Check continuity between P138-g and P138-h.**
 - a. If continuity is present, go to **8**.
 - b. If continuity is not present, go to **14**.
- 8. Remove jumper wire from P803. Go to 9.**
- 9. Install No. 2 engine oil temperature sensor into No. 1 engine. Go to 10.**
- 10. Operate No. 1 engine (TM 1-1520-237-10). Go to 11.**
- 11. Check that No. 1 engine temperature is normal.**
 - a. If No. 1 engine temperature is normal, go to **12**.
 - b. If No. 1 engine temperature is not normal, troubleshoot engine oil system (TM 1-2840-248-23). Go to **16**.
- 12. Reinstall No. 2 engine oil temperature sensor (WP 0482 00). Go to 13.**
- 13. Install a new oil temperature sensor in No. 1 engine (WP 0482 00). Go to 16.**
- 14. Remove jumper wire from P803. Go to 15.**
- 15. Repair/replace wiring, as required, between (WP 1747 00), then go to 16:**
 - **P138- g and P803-10**
 - **P138-h and P803-9**
- 16. Procedure completed.**

NO. 1 ENGINE HAS NO GAS GENERATOR (% NG 1) INDICATION**SYMPTOM**

No. 1 engine gas generator speed (% Ng 1) has no indication.

MALFUNCTION

No. 1 engine has no gas generator (% Ng 1) indication.

CORRECTIVE ACTION

- 1. Troubleshoot engine Ng indicating system (TM 1-2840-248-23).**
 - a. If indicating system is good, go to **2**.
- 2. Troubleshoot instrument display system, % Ng 1 scale (WP 0089 00).**
 - a. If display system is good, repair/replace wiring, as required, between (WP 1747 00), then go to **3**:
 - **P138- c and P803-20**
 - **P138-d and P803-21**
- 3. Procedure completed.**

NO. 1 ENGINE TURBINE INLET TEMPERATURE (TGT TEMP 1) HAS NO INDICATION**SYMPTOM**

No. 1 engine turbine inlet temperature (TGT TEMP 1) has no indication.

MALFUNCTION

No. 1 engine turbine inlet temperature (TGT TEMP 1) has no indication.

CORRECTIVE ACTION

1. Troubleshoot instrument display system, TGT TEMP 1 scale (WP 0089 00).
 - a. If instrument display system is good, go to 2.
2. Install an insulated jumper wire between P801-16 and P801-17. Go to 3.
3. Check helicopter configuration.
 - a. **EMEP** , go to 4.
 - b. **W/O EMEP** , go to 6.
4. **EMEP** Remove and check pin filtered adapter P138 (WP 0925 00).
 - a. If pin filtered adapter is good, go to 5.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 11.
5. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 6.
6. Check continuity between P138-e and P138-f.
 - a. If continuity is present, go to 7.
 - b. If continuity is not present go to 9.
7. Remove jumper wire. Go to 8.
8. Troubleshoot engine for TGT not indicating (TM 1-2840-248-23). Go to 11.
9. Remove jumper wire. Go to 10.
10. Repair/replace wiring, as required, between (WP 1747 00), then go to 11:
 - P138-f and P801-16
 - P138-e and P801-17
11. Procedure completed.

NO. 1 ENGINE OIL FLUCTUATES GREATER THAN 5 PSI OR IS TOO HIGH (10 PSI ABOVE NORMAL OPERATING PRESSURE)**SYMPTOM**

No. 1 engine oil fluctuates.

MALFUNCTION

No. 1 engine oil fluctuates greater than 5 psi or is too high (10 psi above normal operating pressure).

CORRECTIVE ACTION

1. Check engine oil tank for proper level.
 - a. If engine oil is at proper level, go to 2.

NO. 1 ENGINE OIL FLUCTUATES GREATER THAN 5 PSI OR IS TOO HIGH (10 PSI ABOVE NORMAL OPERATING PRESSURE) - Continued

- b. If engine oil is not at proper level, troubleshoot engine oil system (TM 1-2840-248-23). Go to **20**.
- 2. Troubleshoot instrument display system, ENG OIL PRESS 1 scale (WP 0089 00).**
 - a. If instrument display system is good, go to **3**.
- 3. Install an insulated jumper wire between P803-6 and P803-7. Go to 4.**
- 4. Check helicopter configuration.**
 - a. **EMEP** , go to **5**.
 - b. **W/O EMEP** , go to **7**.
- 5. EMEP Remove and check pin filtered adapter P138 (WP 0925 00).**
 - a. If pin filtered adapter is good, go to **6**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **20**.
- 6. EMEP Install pin filtered adapter (WP 0925 00). Go to 7.**
- 7. Check continuity between P138-i and P138-j.**
 - a. If continuity is present, go to **8**.
 - b. If continuity is not present, go to **18**.
- 8. Install insulated jumper wire between P803-5 and P803-7. Go to 9.**
- 9. Check continuity between P138-j and P138-k.**
 - a. If continuity is present, go to **10**.
 - b. If continuity is not present, go to **16**.
- 10. Remove jumper wire from P803. Go to 11.**
- 11. Remove No. 2 engine oil pressure sensor and install in No. 1 engine (WP 0482 00). Go to 12.**
- 12. With qualified personnel at controls, motor No. 1 engine (TM 1-1520-237-10). Go to 13.**
- 13. If a positive oil pressure indication appears during motoring of engine, do a normal start and check for normal oil pressure. Go to 14.**
- 14. Check that oil pressure indication is within limits.**
 - a. If oil pressure is within limits, go to **15**.
 - b. If oil pressure is not within limits, troubleshoot engine oil system (TM 1-2840-248-23). Go to **20**.
- 15. No. 1 engine oil system is operational. Install a new oil pressure sensor in No. 2 engine (WP 0482 00 and TM 1-2840-248-23). Go to 20.**
- 16. Remove jumper wire. Go to 17.**
- 17. Repair/replace wiring between P138-k and P803-5 (WP 1747 00). Go to 20.**
- 18. Remove jumper wire. Go to 19.**
- 19. Repair/replace wiring, as required, between (WP 1747 00), then go to 20:**
 - **P138-i and P803-6**

NO. 1 ENGINE OIL FLUCTUATES GREATER THAN 5 PSI OR IS TOO HIGH (10 PSI ABOVE NORMAL OPERATING PRESSURE) - Continued

- **P138-j and P803-7**

20. Procedure completed.

NO. 1 ENGINE OIL HAS NO OIL PRESSURE INDICATION**SYMPTOM**

No. 1 engine oil has no oil pressure indication.

MALFUNCTION

No. 1 engine oil has no oil pressure indication.

CORRECTIVE ACTION**CAUTION**

If engine ran for more than 1 minute while oil pressure was below minimum limits, and indicating system was operating properly, replace engine.

1. Make sure engine green harness is connected to sensor.

- If harness is connected, go to NO. 1 ENGINE OIL FLUCTUATES GREATER THAN 5 PSI OR IS TOO HIGH (10 PSI ABOVE NORMAL OPERATING PRESSURE), in this work package.
- If harness is not connected, reconnect harness to sensor. Go to **2**.

2. Procedure completed.**NO. 1 ENGINE HAS NO POWER TURBINE SPEED (%RPM ENG 1) INDICATION****SYMPTOM**

No. 1 engine has no power turbine speed indication.

MALFUNCTION

No. 1 engine has no power turbine speed (%RPM ENG 1) indication.

CORRECTIVE ACTION**1. Troubleshoot instrument display system, %RPM ENG 1 scale (WP 0089 00).**

- If display system is good, go to **2**.

2. Install an insulated jumper wire between P801-8 and P801-9. Go to 3.**3. Check helicopter configuration.**

- EMEP**, go to **4**.
- W/O EMEP**, go to **6**.

4. EMEP Remove and check pin filtered adapter P138 (WP 0925 00).

- If pin filtered adapter is good, go to **5**.
- If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **11**.

5. EMEP Install pin filtered adapter (WP 0925 00). Go to 6.**6. Check continuity between P138-C and P138-D.**

- If continuity is present, go to **7**.

NO. 1 ENGINE HAS NO POWER TURBINE SPEED (%RPM ENG 1) INDICATION - Continued

- b. If continuity is not present, go to **9**.
- 7. Remove jumper wire. Go to 8.**
- 8. Troubleshoot engine Np indicating system (TM 1-2840-248-23). Go to 11.**
- 9. Remove jumper wire. Go to 10.**
- 10. Repair/replace wiring, as required, between (WP 1747 00), then go to 11:**
 - **P138-C and P801-8**
 - **P138-D and 8019-9**
- 11. Procedure completed.**

NO. 1 ENGINE HAS NO TORQUE (%TRQ 1) INDICATION**SYMPTOM**

No. 1 engine has no torque indication.

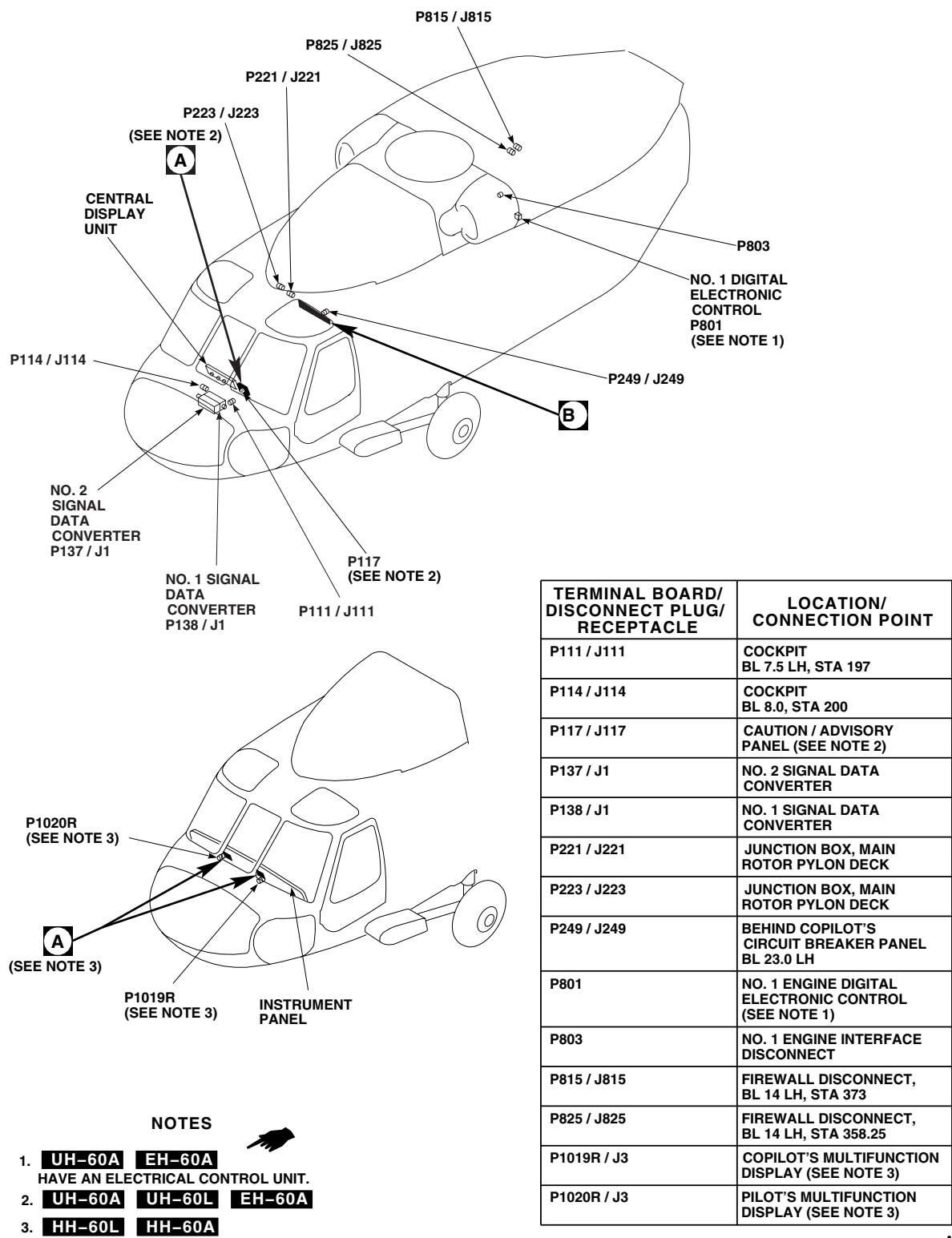
MALFUNCTION

No. 1 engine has no torque (%TRQ 1) indication.

CORRECTIVE ACTION

- 1. Troubleshoot engine overspeed protection system (WP 0075 00).**
 - a. If engine overspeed protection system is good, go to **2**.
- 2. Troubleshoot instrument display system, %TRQ 1 scale (WP 0089 00).**
 - a. If instrument display system is good, go to **3**.
- 3. Repair/replace wiring, as required, between (WP 1747 00), then go to 4:**
 - **P138-E and P801-10**
 - **P138-F and P801-11**
- 4. Procedure completed.**

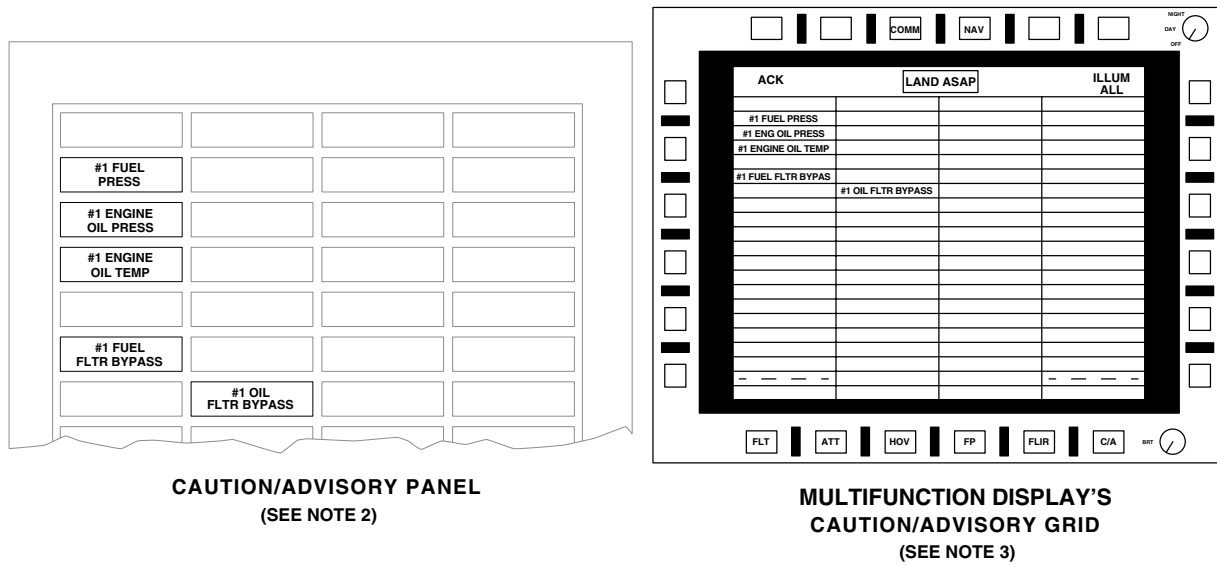
LOCATION



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SA

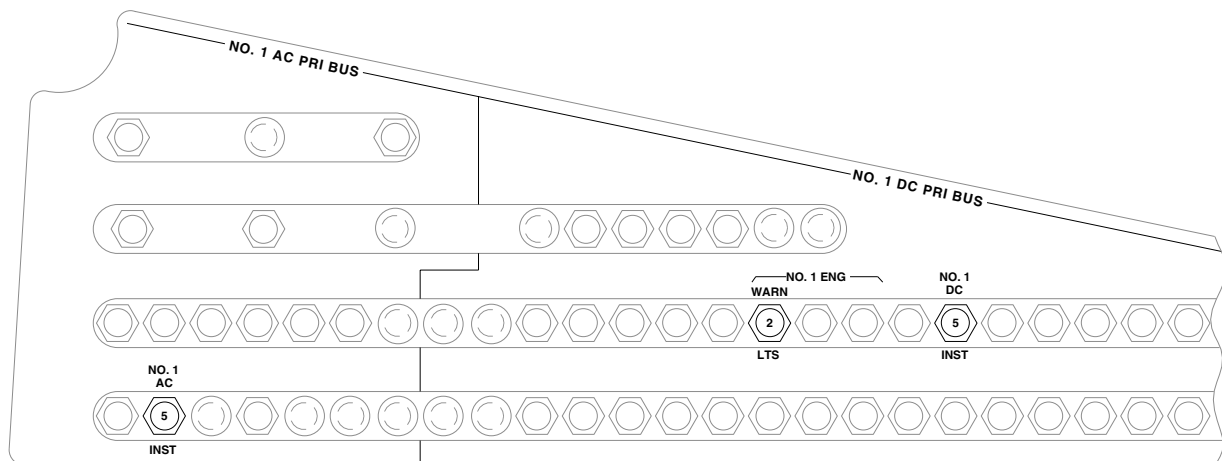
Figure 1. No. 1 Engine Instruments and Warning Lights System Location Diagram. (Sheet 1 of 2)

LOCATION - Continued



A

B



COPILOT'S CIRCUIT
BREAKER PANEL

AB2157_2
SA

Figure 1. No. 1 Engine Instruments and Warning Lights System Location Diagram. (Sheet 2 of 2)

SCHEMATICS

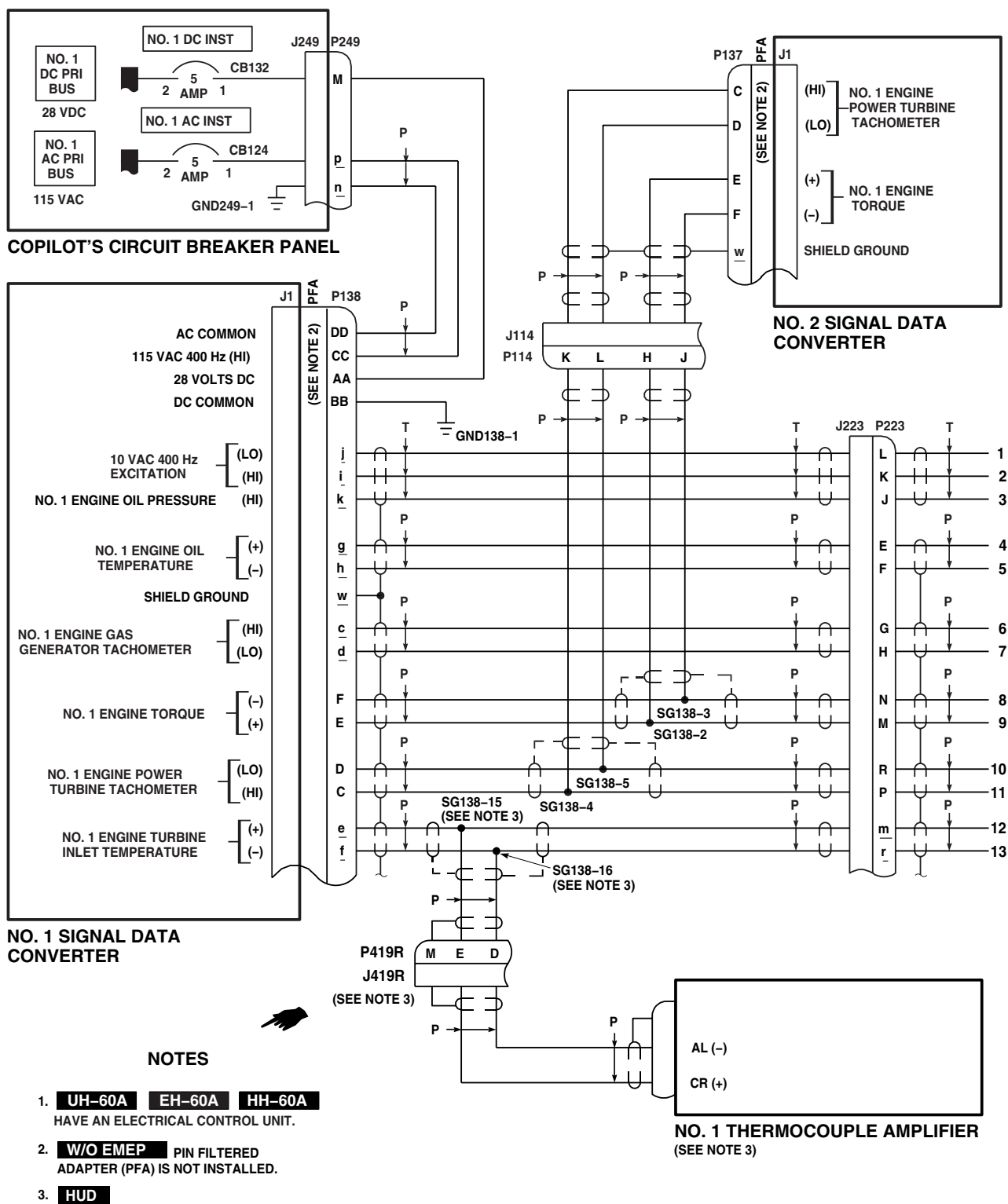
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Figure 2. No. 1 Engine Instrument System Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued

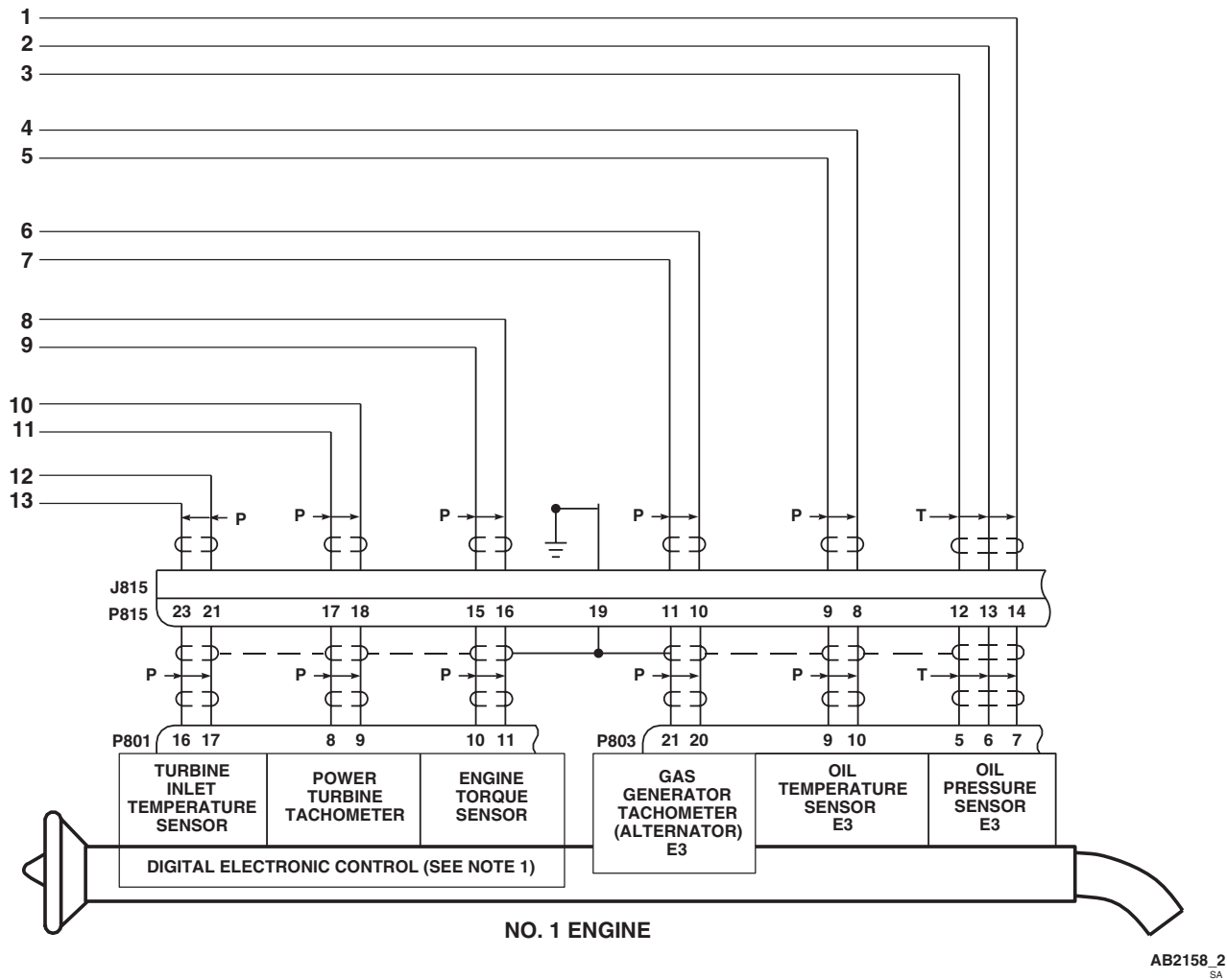
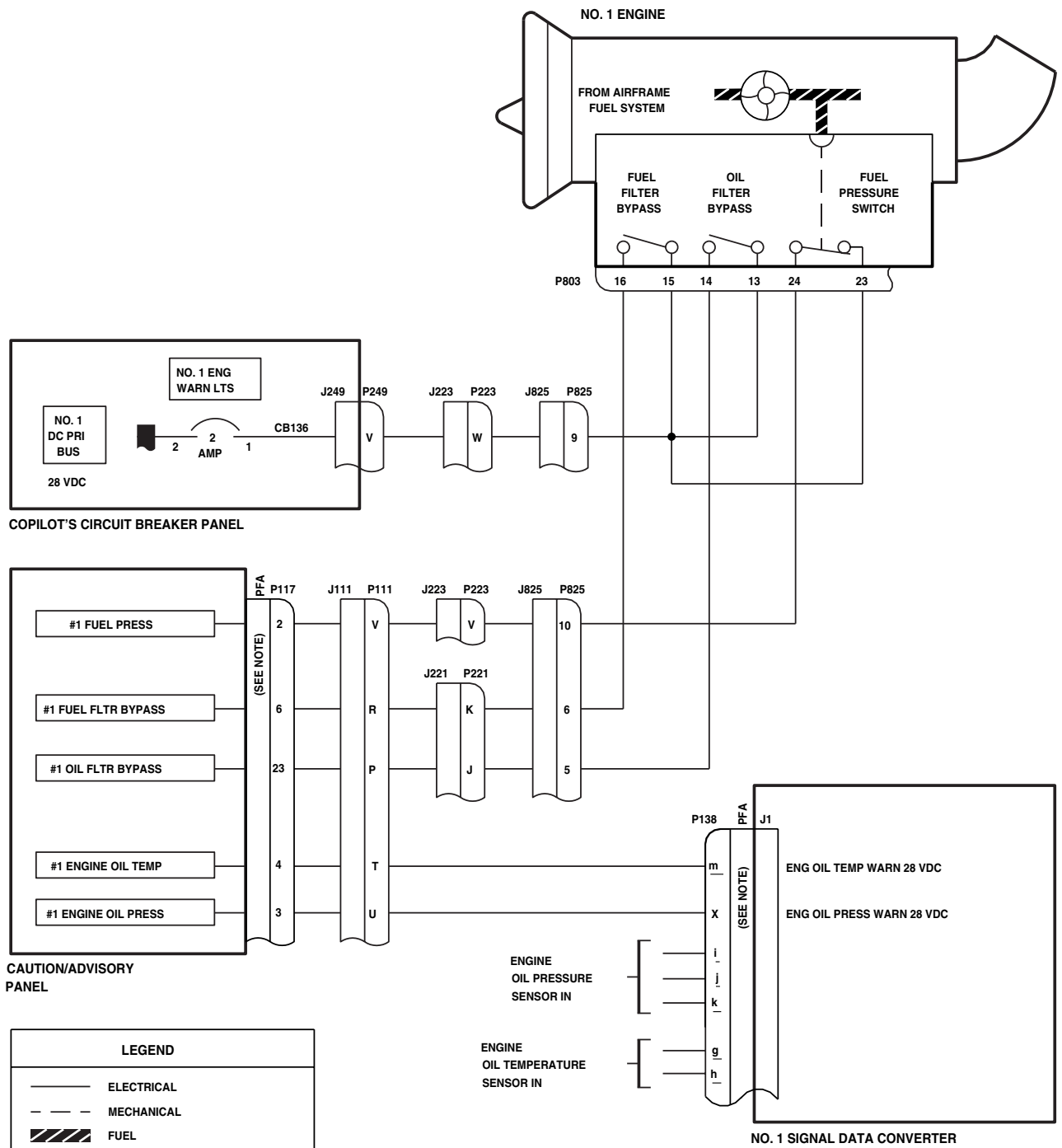


Figure 2. No. 1 Engine Instrument System Schematic Diagram. (Sheet 2 of 2)

SCHEMATICS - Continued



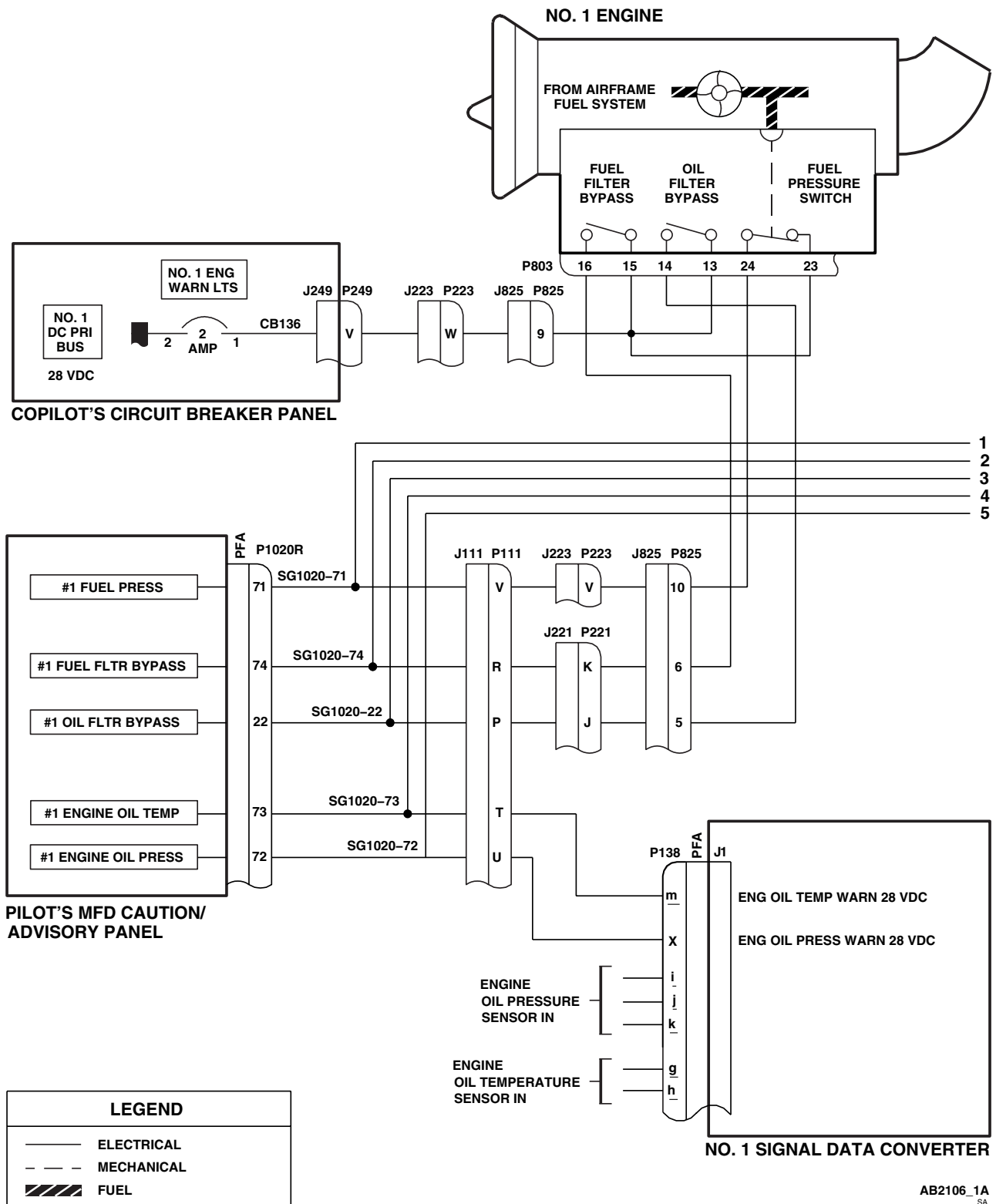
NOTE

W/O EMEP PIN FILTERED
ADAPTER (PFA) IS NOT INSTALLED.

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Figure 3. No. 1 Engine Warning Lights Schematic Diagram **UH-60A UH-60L EH-60A**

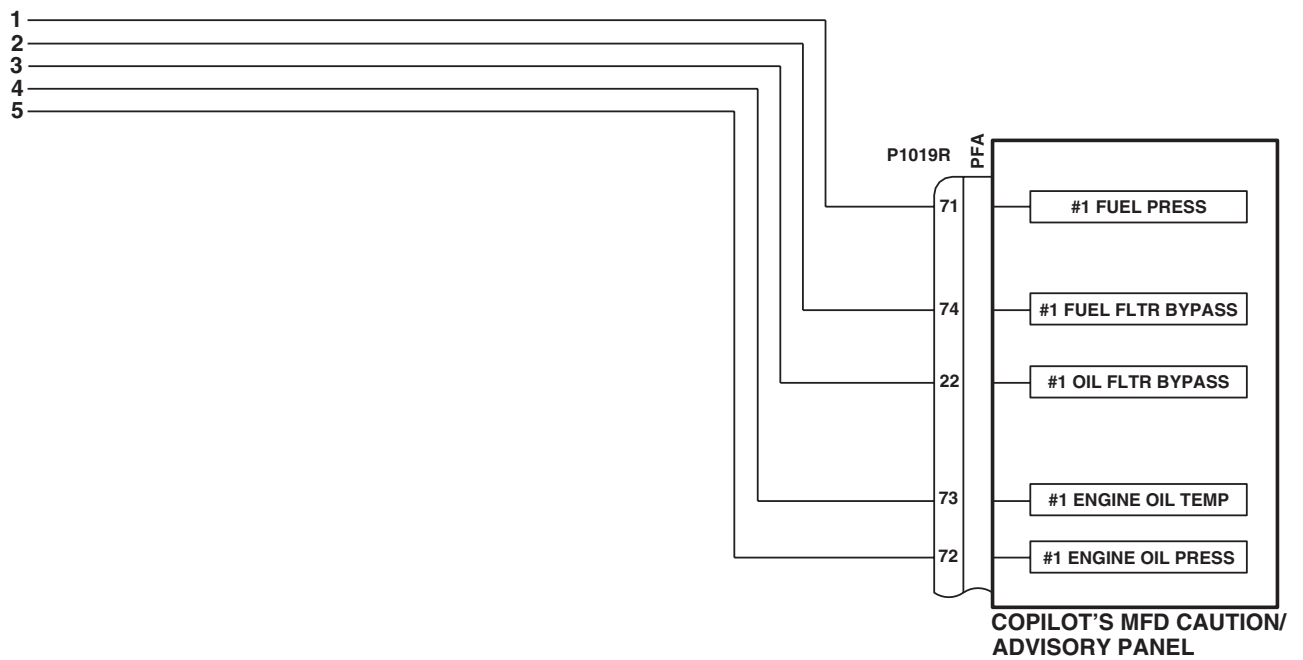
SCHEMATICS - Continued



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Figure 4. No. 1 Engine Warning Lights Schematic Diagram **HH-60A HH-60L** . (Sheet 1 of 2)

SCHEMATICS - Continued



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Figure 4. No. 1 Engine Warning Lights Schematic Diagram **HH-60A HH-60L** . (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL

ENGINE SYSTEM

NO. 2 ENGINE INSTRUMENTS AND WARNING LIGHTS SYSTEM

This WP supersedes WP 0074 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment

Multimeter, AN/PSM-45A

WP 0075 00

WP 0089 00

WP 0090 00

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 0100 00

Electrical Repairer Toolkit, SC 5180-99-B06

WP 0104 00

Engine Repairman's Toolkit, SC 5180-99-B07

WP 0481 00

Insulated Jumper Wire

WP 0482 00

WP 0824 00

Personnel Required

WP 0925 00

Aircraft Electrician MOS 15F (1)

WP 0975 00

Aircraft Powerplant Repairer MOS 15B (1)

WP 1685 00

UH-60 Helicopter Repairer MOS 15T (1)

WP 1747 00

References

TM 1-1520-237-MTF

TM 1-1520-237-10

TM 1-2840-248-23

Equipment Condition

External Electrical Power Available, WP 1685 00
or APU Operational, TM 1-1520-237-10

No. 2 Fuel Tank Serviced, TM 1-1520-237-10

NO. 2 ENGINE INSTRUMENTS AND WARNING LIGHTS SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
 - See Figure 1 for system location diagram, **UH-60A UH-60L EH-60A** Figure 2  and **HH-60A HH-60L** Figure 4  for warning system schematic diagrams, and Figure 3 for system schematic diagram as an aid in operational/troubleshooting.
1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
 2. Turn on electrical power. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed). 

Indication/Condition

- a. #2 FUEL PRESS capsule on caution/advisory panel or legend on MFD/caution/advisory panel shall be on.

NO. 2 ENGINE INSTRUMENTS AND WARNING LIGHTS SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- b. #2 ENGINE OIL PRESS capsule on caution/advisory panel or legend on MFD/caution/advisory panel shall be on.

Corrective Action

1. If Indication/Condition a. is not as specified, go to #2 FUEL PRESS CAPSULE OR LEGEND DOES NOT GO ON, in this work package.
2. If Indication/Condition b. is not as specified, go to #2 ENGINE OIL PRESS CAPSULE OR LEGEND DOES NOT GO ON, in this work package.

Step

3. Install an insulated jumper wire between P802-13 and P802-14.

Indication/Condition

#2 OIL FLTR BYPASS capsule on caution/advisory panel or legend on MFD/caution/advisory panel shall go on.

Corrective Action

If Indication/Condition is not as specified, go to #2 OIL FLTR BYPASS CAPSULE OR LEGEND DOES NOT GO ON, in this work package.

Step

4. Remove jumper wire from P802.
5. Install an insulated jumper wire between P802-15 and P802-16.

Indication/Condition

#2 FUEL FLTR BYPASS capsule on caution/advisory panel or legend on MFD/caution/advisory panel shall go on.

Corrective Action

If Indication/Condition is not as specified, go to #2 FUEL FLTR BYPASS CAPSULE OR LEGEND DOES NOT GO ON, in this work package.

Step

6. Remove jumper wire from P802.
7. Connect P802 to No. 2 engine (WP 0481 00).
8. Turn off electrical power.
9. With qualified personnel at controls, run No. 2 engine up to idle speed (TM 1-1520-237-10).

Indication/Condition

#2 FUEL PRESS capsule or legend shall go off.

Corrective Action

If Indication/Condition is not as specified, go to #2 FUEL PRESS CAPSULE OR LEGEND STAYS ON WHEN ENGINE IS AT IDLE SPEED, in this work package.

Step

10. With qualified personnel at controls, run No. 2 engine with main rotor turning at 100% Nr until engine instruments stabilize (TM 1-1520-237-10).

NO. 2 ENGINE INSTRUMENTS AND WARNING LIGHTS SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Engine status shall be as follows:

CAUTION/ADVISORY PANEL OR MFD/CAUTION/ADVISORY PANEL	CAPSULE OR LEGEND STATUS
#2 FUEL PRESS	OFF
#2 ENGINE OIL PRESS	OFF
#2 ENGINE OIL TEMP	OFF
#2 FUEL FLTR BYPASS	OFF
#2 OIL FLTR BYPASS	OFF

CENTRAL DISPLAY UNIT	INDICATION
Ng SPEED 2	(Refer to TM 1-1520-237-MTF for normal operating limits)
TGT TEMP 2	
ENG OIL TEMP 2	
ENG OIL PRESS 2	

PILOT'S AND COPILOT'S DISPLAY UNIT	INDICATION
% RPM 2	(Refer to TM 1-1520-237-MTF for normal operating limits)
% TRQ 2	

Corrective Action

1. If any of these caution/advisory panel capsules or legends are on:

CAPSULE OR LEGEND	GO TO
#2 ENGINE OIL PRESS	NO. 2 ENGINE OIL PRESSURE IS BELOW MINIMUM, in this work package.
#2 ENGINE OIL TEMP	NO. 2 ENGINE OIL TEMPERATURE (ENG OIL TEMP 2) IS TOO HIGH OR HAS NO INDICATION, in this work package.

NO. 2 ENGINE INSTRUMENTS AND WARNING LIGHTS SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. If #2 FUEL PRESS capsule or legend is on, troubleshoot engine for fuel low pressure caution light on during flight (TM 1-2840-248-23).
3. If #2 FUEL FLTR BYPASS or #2 OIL FLTR BYPASS capsule or legend is on, troubleshoot engine (TM 1-2840-248-23).
4. If No. 2 engine gas generator speed (% Ng 2) has no indication, go to NO. 2 ENGINE HAS NO GAS GENERATOR (%NG 2) INDICATION, in this work package.
5. If No. 2 engine turbine inlet temperature (TGT TEMP 2) has no indication, go to NO. 2 ENGINE TURBINE INLET TEMPERATURE (TGT TEMP 2) HAS NO INDICATION, in this work package.
6. If No. 2 engine oil temperature (ENG OIL TEMP 2) has no indication, go to NO. 2 ENGINE OIL TEMPERATURE (ENG OIL TEMP 2) IS TOO HIGH OR HAS NO INDICATION, in this work package.
7. If No. 2 engine oil pressure (ENG OIL PRESS 2):
 - Is too high (10 psi above normal), go to NO. 2 ENGINE OIL PRESSURE (ENG OIL PRESS 2) FLUCTUATES GREATER THAN 5 PSI, OR IS TOO HIGH, in this work package.
 - Has no indication, go to NO. 2 ENGINE HAS NO OIL PRESSURE INDICATION, in this work package.
 - Fluctuates greater than 5 psi, go to NO. 2 ENGINE OIL PRESSURE (ENG OIL PRESS 2) FLUCTUATES GREATER THAN 5 PSI, OR IS TOO HIGH, in this work package.
 - Is below minimum limits (45 psi), go to NO. 2 ENGINE OIL PRESSURE IS BELOW MINIMUM, in this work package.
8. If No. 2 engine power turbine speed (% RPM 2) has no indication, go to NO. 2 ENGINE HAS NO POWER TURBINE SPEED (%RPM ENG 2) INDICATION, in this work package.
9. If No. 2 engine has no torque indication (% TRQ 2), go to NO. 2 ENGINE HAS NO TORQUE (%TRQ 2) INDICATION, in this work package.

Step

11. Turn MFD switch to OFF.
12. Shut down No. 2 engine (TM 1-1520-237-10).

Indication/Condition

No. 2 engine shall shut down.

Corrective Action

None Required

#2 FUEL PRESS CAPSULE OR LEGEND DOES NOT GO ON**SYMPTOM**

#2 FUEL PRESS capsule or legend does not go on.

MALFUNCTION

#2 FUEL PRESS capsule or legend does not go on.

CORRECTIVE ACTION

1. Check helicopter configuration.

#2 FUEL PRESS CAPSULE OR LEGEND DOES NOT GO ON - Continued

- a. **EMEP** , go to **2**.
- b. **W/O EMEP** , go to **4**.
- 2. EMEP Remove and check pin filtered adapter P117 or HH-60A HH-60L P1019R or P1020R (WP 0925 00).**
 - a. If pin filtered adapter is good, go to **3**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **9**.
- 3. EMEP Install pin filtered adapter (WP 0925 00). Go to 4.**
- 4. Hold caution/advisory panel HH-60A HH-60L instrument panel INDICATOR LTS BRT/DIM-TEST switch to TEST.**
 - a. If #2 FUEL PRESS capsule or legend goes on, go to **5**.
 - b. If #2 FUEL PRESS capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **9**.
- 5. Check for 28 vdc between P117-50 or HH-60A HH-60L P1019R-55 or P1020R-55 and ground.**
 - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **9**.
 - b. If voltage is not as specified, go to **6**.
- 6. Check for 28 vdc between P802-23 and ground.**
 - a. If voltage is as specified, go to **7**.
 - b. If voltage is not as specified, go to **8**.
- 7. Check continuity between P117-50 or HH-60A HH-60L P1019R-55 or P1020R-55 and P802-24.**
 - a. If continuity is present, replace fuel pressure switch (WP 0975 00). Go to **9**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9**.
- 8. Check for 28 vdc between NO. 2 ENG WARN LTS circuit breaker terminal 1 and ground.**
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P802-23 (WP 1747 00). Go to **9**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **9**.
- 9. Procedure completed.**

#2 ENGINE OIL PRESS CAPSULE OR LEGEND DOES NOT GO ON**SYMPTOM**

#2 ENGINE OIL PRESS capsule or legend does not go on.

MALFUNCTION

#2 ENGINE OIL PRESS capsule or legend does not go on.

CORRECTIVE ACTION

- 1. Check helicopter configuration.**

#2 ENGINE OIL PRESS CAPSULE OR LEGEND DOES NOT GO ON - Continued

- a. **EMEP** , go to **2**.
- b. **W/O EMEP** , go to **4**.
2. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R (WP 0925 00).
 - a. If pin filtered adapter is good, go to **3**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **7**.
3. **EMEP** Install pin filtered adapter (WP 0925 00). Go to **4**.
4. Hold caution/advisory panel **HH-60A HH-60L** instrument panel INDICATOR LTS BRT/DIM-TEST switch to TEST.
 - a. If #2 ENGINE OIL PRESS capsule or legend goes on, go to **5**.
 - b. If #2 ENGINE OIL PRESS capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **7**.
5. Check for 28 vdc between P117-51 or **HH-60A HH-60L** P1019R-56 or P1020R-56 and ground.
 - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **7**.
 - b. If voltage is not as specified, go to **6**.
6. Check continuity between P117-51 or **HH-60A HH-60L** P1019R-56 or P1020R-56 and P137-X.
 - a. If continuity is present, troubleshoot instrument display system (WP 0089 00). Go to **7**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.
7. Procedure completed.

#2 OIL FLTR BYPASS CAPSULE OR LEGEND DOES NOT GO ON**SYMPTOM**

#2 OIL FLTR BYPASS capsule or legend does not go on.

MALFUNCTION

#2 OIL FLTR BYPASS capsule or legend does not go on.

CORRECTIVE ACTION

1. Hold caution/advisory panel **HH-60A HH-60L** instrument panel INDICATOR LTS BRT/DIM-TEST switch to TEST.
 - a. If #2 OIL FLTR BYPASS capsule or legend goes on, go to **2**.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **9**.
2. Remove jumper wire from P802. Go to **3**.
3. Check for 28 vdc between P802-13 and P802-14.
 - a. If voltage is as specified, troubleshoot engine oil filter bypass system (TM 1-2840-248-23). Go to **9**.

#2 OIL FLTR BYPASS CAPSULE OR LEGEND DOES NOT GO ON - Continued

- b. If voltage is not as specified, go to **4**.
- 4. Check for 28 vdc between P802-13 and ground.**
 - a. If voltage is as specified, go to **5**.
 - b. If voltage is not as specified, repair/replace wiring between NO. 2 ENG WARN LTS circuit breaker terminal 1 and P802-13 (WP 1747 00). Go to **9**.
- 5. Check helicopter configuration.**
 - a. **EMEP** , go to **6**.
 - b. **W/O EMEP** , go to **8**.
- 6. EMEP Remove and check pin filtered adapter P117 or HH-60A HH-60L P1019R or P1020R (WP 0925 00).**
 - a. If pin filtered adapter is good, go to **7**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **9**.
- 7. EMEP Install pin filtered adapter (WP 0925 00). Go to 8.**
- 8. Check for 28 vdc between P117-39 or HH-60A HH-60L P1019R-52 or P1020R-52 and ground.**
 - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **9**.
 - b. If voltage is not as specified, repair/replace wiring between P117-39 or HH-60A HH-60L P1019R-52 or P1020R-52 and P802-14 (WP 1747 00). Go to **9**.
- 9. Procedure completed.**

#2 FUEL FLTR BYPASS CAPSULE OR LEGEND DOES NOT GO ON**SYMPTOM**

#2 FUEL FLTR BYPASS capsule or legend does not go on.

MALFUNCTION

#2 FUEL FLTR BYPASS capsule or legend does not go on.

CORRECTIVE ACTION

- 1. Check helicopter configuration.**
 - a. **EMEP** , go to **2**.
 - b. **W/O EMEP** , go to **4**.
- 2. EMEP Remove and check pin filtered adapter P117 or HH-60A HH-60L P1019R or P1020R (WP 0925 00).**
 - a. If pin filtered adapter is good, go to **3**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **8**.
- 3. EMEP Install pin filtered adapter (WP 0925 00). Go to 4.**
- 4. Hold caution/advisory panel HH-60A HH-60L instrument panel INDICATOR LTS BRT/DIM-TEST switch to TEST.**
 - a. If #2 FUEL FLTR BYPASS capsule or legend goes on, go to **5**.

#2 FUEL FLTR BYPASS CAPSULE OR LEGEND DOES NOT GO ON - Continued

- b. If #2 FUEL FLTR BYPASS capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **8**.
- 5. Check for 28 vdc between P117-54 or HH-60A HH-60L P1019R-58 or P1020R-58 and ground.**
 - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **8**.
 - b. If voltage is not as specified, go to **6**.
- 6. Remove jumper wire from P802. Check for 28 vdc between P802-15 and ground.**
 - a. If voltage is as specified, go to **7**.
 - b. If voltage is not as specified, repair/replace wiring between NO. 2 ENG WARN LTS circuit breaker terminal 1 and P802-15 (WP 1747 00). Go to **8**.
- 7. Check continuity between P117-54 or HH-60A HH-60L P1019R-58 or P1020R-58 and P802-16.**
 - a. If continuity is present, troubleshoot engine fuel filter bypass system (TM 1-2840-248-23). Go to **8**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **8**.
- 8. Procedure completed.**

#2 FUEL PRESS CAPSULE OR LEGEND STAYS ON WHEN ENGINE IS AT IDLE SPEED**SYMPTOM**

#2 FUEL PRESS capsule or legend stays on.

MALFUNCTION

#2 FUEL PRESS capsule or legend stays on when engine is at idle speed.

CORRECTIVE ACTION

- 1. With No. 2 engine operating at ground idle speed, place upper console FUEL PUMP switch to FUEL PRIME. Go to 2.**
- 2. Check that #2 FUEL PRESS capsule or legend goes on.**
 - a. If capsule or legend goes off, go to **3**.
 - b. If capsule or legend does not go off, troubleshoot engine for FUEL LOW PRESSURE caution light at below flight idle speed (TM 1-2840-248-23). Go to **6**.
- 3. Shut down No. 2 engine (TM 1-1520-237-10). Go to 4.**
- 4. Disconnect electrical connector P802 from No. 2 engine fuel pressure switch. Go to 5.**
- 5. Check that #2 FUEL PRESS capsule or legend goes off.**
 - a. If capsule or legend goes off, troubleshoot airframe fuel system for possible leaks between engine and fuel tanks. Repair/replace components as required. Go to **6**.
 - b. If capsule or legend does not go off, repair/replace wiring between (WP 1747 00):
 - P117-50 and P802-24

#2 FUEL PRESS CAPSULE OR LEGEND STAYS ON WHEN ENGINE IS AT IDLE SPEED - Continued

- **HH-60A HH-60L** P1019R-71 or P1020R-71 and P803-24
- P236-V and P802-23
- c. If trouble remains, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **6**.

6. Procedure completed.**NO. 2 ENGINE OIL PRESSURE IS BELOW MINIMUM****SYMPTOM**

No. 2 engine oil pressure low.

MALFUNCTION

No. 2 engine oil pressure is below minimum.

CORRECTIVE ACTION**CAUTION**

If engine ran for more than 1 minute while oil pressure was below minimum limits, and indicating system was operating properly, replace engine.

1. Visually check engine for external oil leaks.

- a. If leaks are present, repair/replace item causing leak. Check oil tank level and service as necessary (TM 1-2840-248-23). Go to **2**.
- b. If leaks are not present, go to NO. 2 ENGINE OIL PRESSURE (ENG OIL PRESS 2) FLUCTUATES GREATER THAN 5 PSI, OR IS TOO HIGH, in this work package.

2. Procedure completed.**NO. 2 ENGINE OIL TEMPERATURE (ENG OIL TEMP 2) IS TOO HIGH OR HAS NO INDICATION****SYMPTOM**

No. 2 engine oil temperature too high or has no indication.

MALFUNCTION

No. 2 engine oil temperature (ENG OIL TEMP 2) is too high or has no indication.

CORRECTIVE ACTION**1. Check engine oil tank for proper level.**

- a. If oil level is at proper level, go to **2**.
- b. If oil level is not at proper level, troubleshoot engine oil system (TM 1-2840-248-23). Go to **16**.

2. Troubleshoot instrument display system, ENG OIL TEMP 2 scale (WP 0089 00).

- a. If instrument display system is good, go to **3**.

3. Install an insulated jumper wire between P802-9 and P802-10. Go to 4.**4. Check helicopter configuration.**

- a. **EMEP**, go to **5**.

NO. 2 ENGINE OIL TEMPERATURE (ENG OIL TEMP 2) IS TOO HIGH OR HAS NO INDICATION - Continued

- b. **W/O EMEP** , go to 7.
- 5. **EMEP** Remove and check pin filtered adapter P137 (WP 0925 00).
 - a. If pin filtered adapter is good, go to 6.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 16.
- 6. **EMEP** Install pin filtered adapter P137 (WP 0925 00). Go to 7.
- 7. Check continuity between P137-g and P137-h.
 - a. If continuity is present, go to 8.
 - b. If continuity is not present, go to 14.
- 8. Remove jumper wire from P802. Go to 9.
- 9. Install No. 1 engine oil temperature sensor into No. 2 engine. Go to 10.
- 10. Operate No. 2 engine (TM 1-1520-237-10). Go to 11.
- 11. Check that No. 2 engine temperature is normal.
 - a. If temperature is normal, go to 12.
 - b. If temperature is abnormal, troubleshoot engine oil system (TM 1-2840-248-23). Go to 16.
- 12. Install No. 1 engine oil temperature sensor (WP 0482 00). Go to 13.
- 13. Replace faulty oil temperature sensor in No. 2 engine (WP 0482 00). Go to 16.
- 14. Remove jumper wire from P802. Go to 15.
- 15. Repair/replace wiring, as required, between (WP 1747 00), then go to 16:
 - P137-g and P802-10
 - P137-h and P802-9
- 16. Procedure completed.

NO. 2 ENGINE HAS NO GAS GENERATOR (%NG 2) INDICATION**SYMPTOM**

No. 2 engine has no gas generator indication.

MALFUNCTION

No. 2 engine has no gas generator (%Ng 2) indication.

CORRECTIVE ACTION

- 1. Troubleshoot engine Ng indicating system (TM 1-2840-248-23).
 - a. If indicating system is good, go to 2.
- 2. Troubleshoot instrument display system, %Ng 2 scale (WP 0089 00).
 - a. If display system is good, go to 3.
 - b. Repair/replace wiring, as required, between (WP 1747 00), then go to 3:
 - P137-c and P802-20
 - P137-d and P802-21

NO. 2 ENGINE HAS NO GAS GENERATOR (%NG 2) INDICATION - Continued**3. Procedure completed.****NO. 2 ENGINE TURBINE INLET TEMPERATURE (TGT TEMP 2) HAS NO INDICATION****SYMPTOM**

No. 2 engine turbine inlet temperature has no indication.

MALFUNCTION

No. 2 engine turbine inlet temperature (TGT TEMP 2) has no indication.

CORRECTIVE ACTION

- 1. Troubleshoot instrument display system, TGT TEMP 2 scale (WP 0089 00).**
 - a. If display system is good, go to **2**.
- 2. Install an insulated jumper wire between P800-16 and P800-17. Go to 3.**
- 3. Check helicopter configuration.**
 - a. **EMEP** , go to **4**.
 - b. **W/O EMEP** , go to **6**.
- 4. EMEP Remove and check pin filtered adapter P137 (WP 0925 00).**
 - a. If pin filtered adapter is good, go to **5**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **11**.
- 5. EMEP Install pin filtered adapter P137 (WP 0925 00). Go to 6.**
- 6. Check continuity between P137-e and P137-f.**
 - a. If continuity is present, go to **7**.
 - b. If continuity is not present, go to **9**.
- 7. Remove jumper wire from P800. Go to 8.**
- 8. Troubleshoot engine indicating system for TGT (TM 1-2840-248-23). Go to 11.**
- 9. Remove jumper wire from P800. Go to 10.**
- 10. Repair/replace wiring, as required, between (WP 1747 00), then go to 11:**
 - P137-e and P800-17
 - P137-f and P800-16
- 11. Procedure completed.**

NO. 2 ENGINE OIL PRESSURE (ENG OIL PRESS 2) FLUCTUATES GREATER THAN 5 PSI, OR IS TOO HIGH**SYMPTOM**

No. 2 engine oil pressure fluctuates.

MALFUNCTION

No. 2 engine oil pressure (ENG OIL PRESS 2) fluctuates greater than 5 psi, or is too high.

CORRECTIVE ACTION

- 1. Check engine oil tank for proper level.**
 - a. If oil has proper level, go to **2**.
 - b. If oil does not have proper level, troubleshoot engine oil system (TM 1-2840-248-23). Go to **20**.
- 2. Troubleshoot instrument display system, ENG OIL PRESS 2 scale (WP 0089 00).**
 - a. If display system is good, go to **3**.
- 3. Install an insulated jumper wire between P802-6 and P802-7. Go to 4.**
- 4. Check helicopter configuration.**
 - a. **EMEP** , go to **5**.
 - b. **W/O EMEP** , go to **7**.
- 5. **EMEP** Remove and check pin filtered adapter P137 (WP 0925 00).**
 - a. If pin filtered adapter is good, go to **6**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **20**.
- 6. **EMEP** Install pin filtered adapter P137 (WP 0925 00). Go to 7.**
- 7. Check continuity between P137-i and P137-j.**
 - a. If continuity is present, go to **8**.
 - b. If continuity is not present, go to **18**.
- 8. Install an insulated jumper wire between P802-5 and P802-7. Go to 9.**
- 9. Check continuity between P137-j and P137-k.**
 - a. If continuity is present, go to **10**.
 - b. If continuity is not present, go to **16**.
- 10. Remove jumper wire from P802. Go to 11.**
- 11. Remove No. 1 engine oil pressure sensor and install in No. 2 engine (WP 0482 00). Go to 12.**
- 12. With a qualified personnel at the controls, motor No. 2 engine (TM 1-1520-237-10). Go to 13.**
- 13. If a positive oil pressure indication appears during motoring of engine, do a normal start and check for normal oil pressure. Go to 14.**
- 14. Check pressure indication during motoring of engine.**

NO. 2 ENGINE OIL PRESSURE (ENG OIL PRESS 2) FLUCTUATES GREATER THAN 5 PSI, OR IS TOO HIGH - Continued

- a. If a positive oil pressure indication appears during motoring of engine, do a normal start and go to **15**.

15. Check that oil indication is within limits.

- a. If indication is within limits, No. 2 engine oil system is operational. Install a new oil pressure sensor in No. 1 engine (WP 0482 00). Go to **20**.
- b. If indication is not within limits, troubleshoot engine oil system (TM 1-2840-248-23). Go to **20**.

16. Remove jumper wire from P802. Go to 17.**17. Repair/replace wiring between P137-k and P802-5 (WP 1747 00). Go to 20.****18. Remove jumper wire from connector P802. Go to 19.****19. Repair/replace wiring, as required, between (WP 1747 00), then go to 20:**

- P137-i and P802-6
- P137-j and P802-7

20. Procedure completed.**NO. 2 ENGINE HAS NO OIL PRESSURE INDICATION****SYMPTOM**

No. 2 engine has no oil pressure indication.

MALFUNCTION

No. 2 engine has no oil pressure indication.

CORRECTIVE ACTION**CAUTION**

If engine ran for more than 1 minute while oil pressure was below minimum limits, and indicating system was operating properly, replace engine.

1. Make sure engine green harness is connected to sensor.

- a. If harness is connected, go to NO. 2 ENGINE OIL PRESSURE (ENG OIL PRESS 2) FLUCTUATES GREATER THAN 5 PSI, OR IS TOO HIGH, in this work package.
- b. If harness is not connected, reconnect connector to sensor. Go to **2**.

2. Procedure completed.**NO. 2 ENGINE HAS NO POWER TURBINE SPEED (%RPM ENG 2) INDICATION****SYMPTOM**

No. 2 engine has no power turbine speed indication.

MALFUNCTION

No. 2 engine has no power turbine speed (%RPM ENG 2) indication.

CORRECTIVE ACTION**1. Troubleshoot instrument display system, %RPM ENGS 2 scale (WP 0089 00).**

NO. 2 ENGINE HAS NO POWER TURBINE SPEED (%RPM ENG 2) INDICATION - Continued

- a. If display system is good, go to **2**.
- 2. Install an insulated jumper wire between P800-8 and P800-9. Go to 3.**
- 3. Check helicopter configuration.**
 - a. **EMEP** , go to **4**.
 - b. **W/O EMEP** , go to **6**.
- 4. EMEP Remove and check pin filtered adapter P137 (WP 0925 00).**
 - a. If pin filtered adapter is good, go to **5**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **11**.
- 5. EMEP Install pin filtered adapter P137 (WP 0925 00). Go to 6.**
- 6. Check continuity between P137-Y and P137-Z.**
 - a. If continuity is present, go to **7**.
 - b. If continuity is not present, go to **9**.
- 7. Remove jumper wire from P800. Go to 8.**
- 8. Troubleshoot engine Np indicating system (TM 1-2840-248-23). Go to 11.**
- 9. Remove jumper wire from P800. Go to 10.**
- 10. Repair/replace wiring, as required, between (WP 1747 00), then go to 11:**
 - **P137-Y and P800-8**
 - **P137-Z and P800-9**
- 11. Procedure completed.**

NO. 2 ENGINE HAS NO TORQUE (%TRQ 2) INDICATION**SYMPTOM**

No. 2 engine has no torque indication.

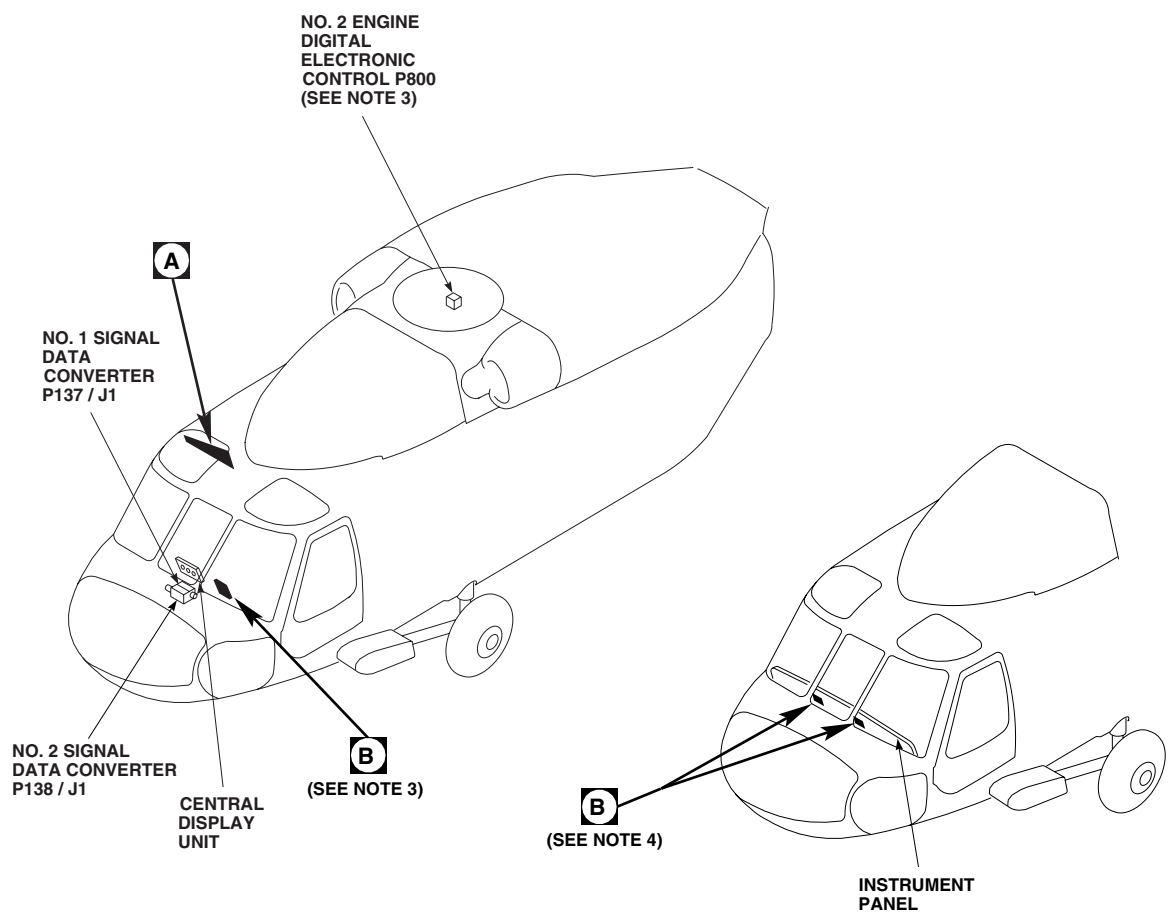
MALFUNCTION

No. 2 engine has no torque (%TRQ 2) indication.

CORRECTIVE ACTION

- 1. Troubleshoot engine overspeed protection system (WP 0075 00).**
 - a. If overspeed protection system is good, go to **2**.
- 2. Troubleshoot instrument display system, %TRQ 2 scale (WP 0089 00).**
 - a. If display system is good, go to **3**.
- 3. Repair/replace wiring, as required, between (WP 1747 00), then go to 4:**
 - **P137-a and P800-10**
 - **P137-b and P800-11**
- 4. Procedure completed.**

LOCATION



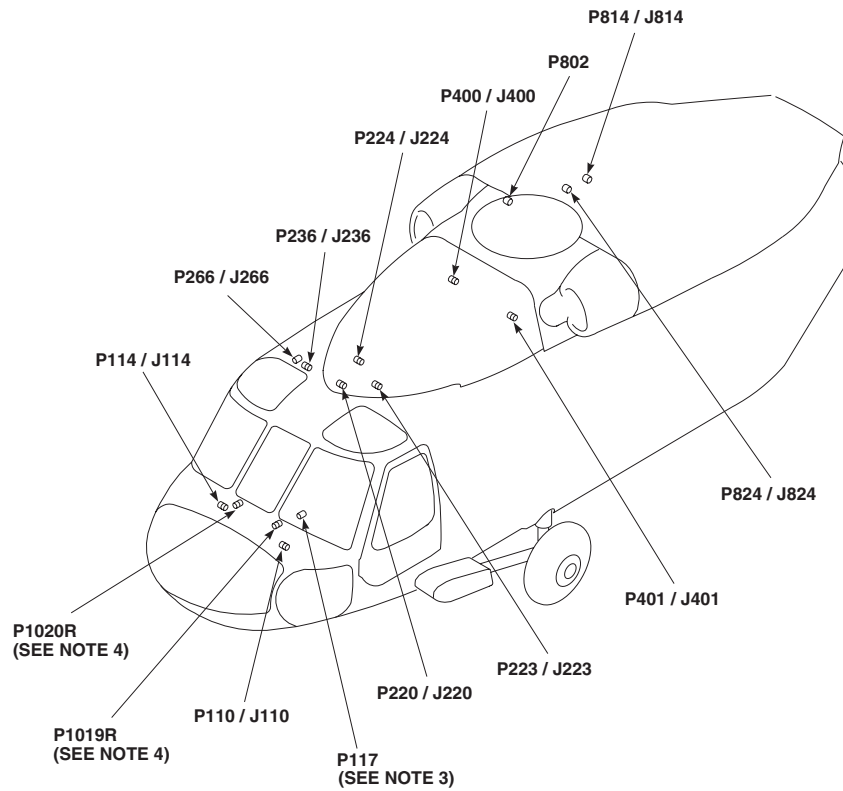
- NOTES 
- 1. **EH-60A**
 - 2. **UH-60A** **EH-60A**
HAVE AN ELECTRICAL CONTROL UNIT.
 - 3. **UH-60A** **UH-60L** **EH-60A**
 - 4. **HH-60L** **HH-60A**
 - 5. **UH-60A** **UH-60L** **HH-60A**
HH-60L

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P110 / J110	COCKPIT BL 7.5 RH, STA 197
P114 / J114	COCKPIT BL 8 RH, STA 200
P117 / J117	CAUTION / ADVISORY PANEL (SEE NOTE 3)
P137 / J1	NO. 2 SIGNAL DATA CONVERTER
P138 / J1	NO. 1 SIGNAL DATA CONVERTER
P220 / J220	JUNCTION BOX, MAIN ROTOR PYLON DECK
P223 / J223	JUNCTION BOX, MAIN ROTOR PYLON DECK
P224 / J224	JUNCTION BOX, MAIN ROTOR PYLON DECK
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 23.0 RH

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Figure 1. No. 2 Engine Instruments and Warning Lights System Location Diagram. (Sheet 1 of 4)

LOCATION - Continued

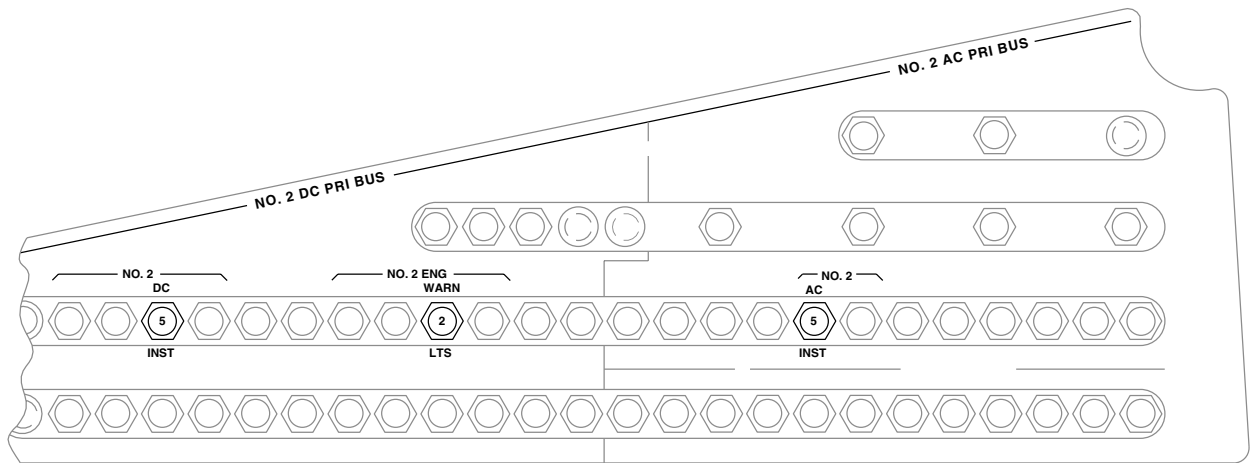


TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25.0 RH
P400 / J400	MAIN ROTOR PYLON DECK BL 24 RH, STA 318
P401 / J401	MAIN ROTOR PYLON DECK BL 24 RH, STA 318
P420 / J420	DISCONNECT (SEE NOTE 4)
P800	NO. 2 ENGINE DIGITAL ELECTRONIC CONTROL
P802	NO. 2 ENGINE INTERFACE DISCONNECT
P814 / J814	FIREWALL DISCONNECT BL 14 RH, STA 373
P824 / J824	FIREWALL DISCONNECT BL 14 RH, STA 356.25
P1019R / J3	COPILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 4)
P1020R / J3	PILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 4)

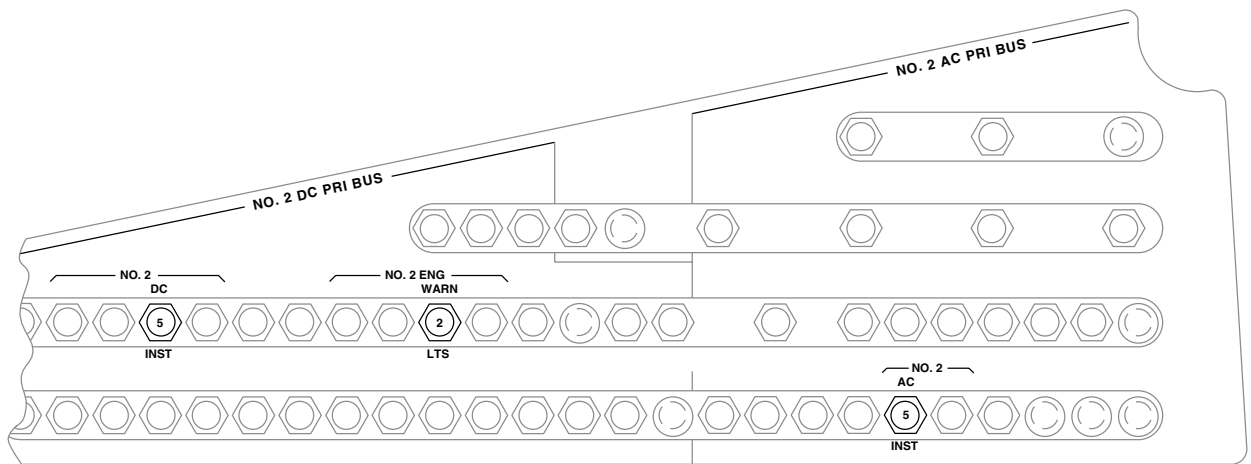
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Figure 1. No. 2 Engine Instruments and Warning Lights System Location Diagram. (Sheet 2 of 4)

LOCATION - Continued



(SEE NOTE 5)



(SEE NOTE 1)

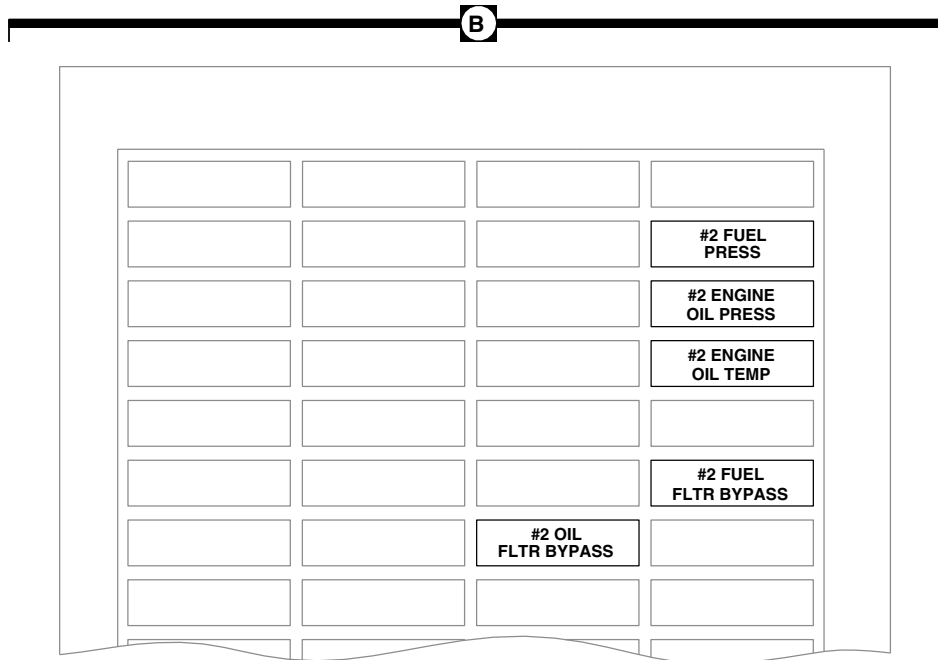
PILOT'S CIRCUIT BREAKER PANEL



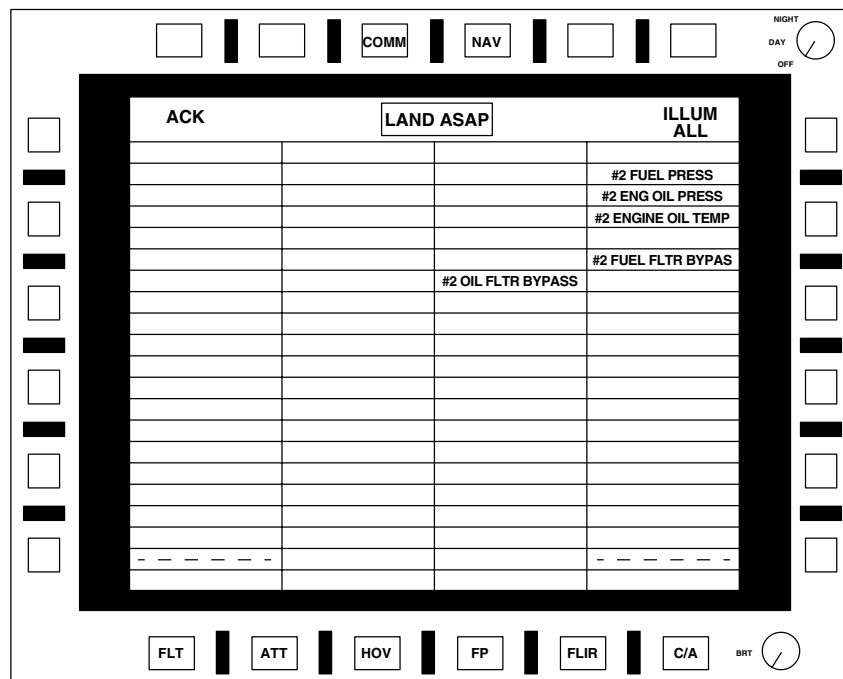
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Figure 1. No. 2 Engine Instruments and Warning Lights System Location Diagram. (Sheet 3 of 4)

LOCATION - Continued



CAUTION/ADVISORY PANEL
(SEE NOTE 3)

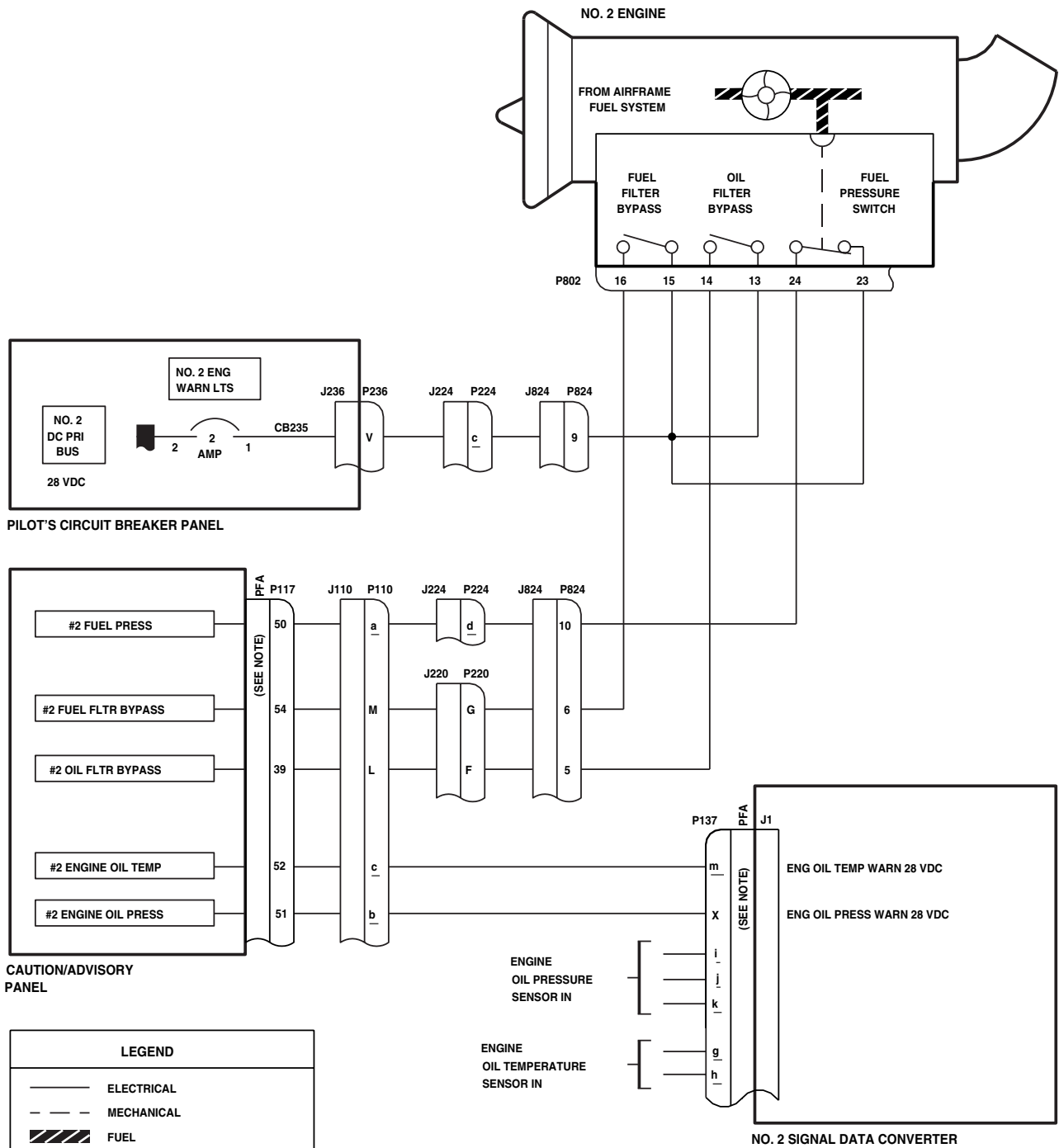


MULTIFUNCTION DISPLAY'S
CAUTION/ADVISORY GRID
(SEE NOTE 4)

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Figure 1. No. 2 Engine Instruments and Warning Lights System Location Diagram. (Sheet 4 of 4)

SCHEMATICS



NOTE

W/O EMEP PIN FILTERED
ADAPTER (PFA) IS NOT INSTALLED.

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Figure 2. No. 2 Engine Warning System Schematic Diagram **UH-60A UH-60L EH-60A** .

NO. 2 SIGNAL DATA CONVERTER

NOTES

1. **EH-60A** NO. 2 AC INST CIRCUIT BREAKER IS RELOCATED / IDENTIFIED AS CIRCUIT BREAKER CB220.
2. **UH-60A** **EH-60A** **HH-60A** HAVE AN ELECTRICAL CONTROL UNIT.
3. **W/O EMEP** PIN FILTERED ADAPTER (PFA) IS NOT INSTALLED.
4. **HUD** **HH-60L**

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Figure 3. No. 2 Engine Instrument System Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued

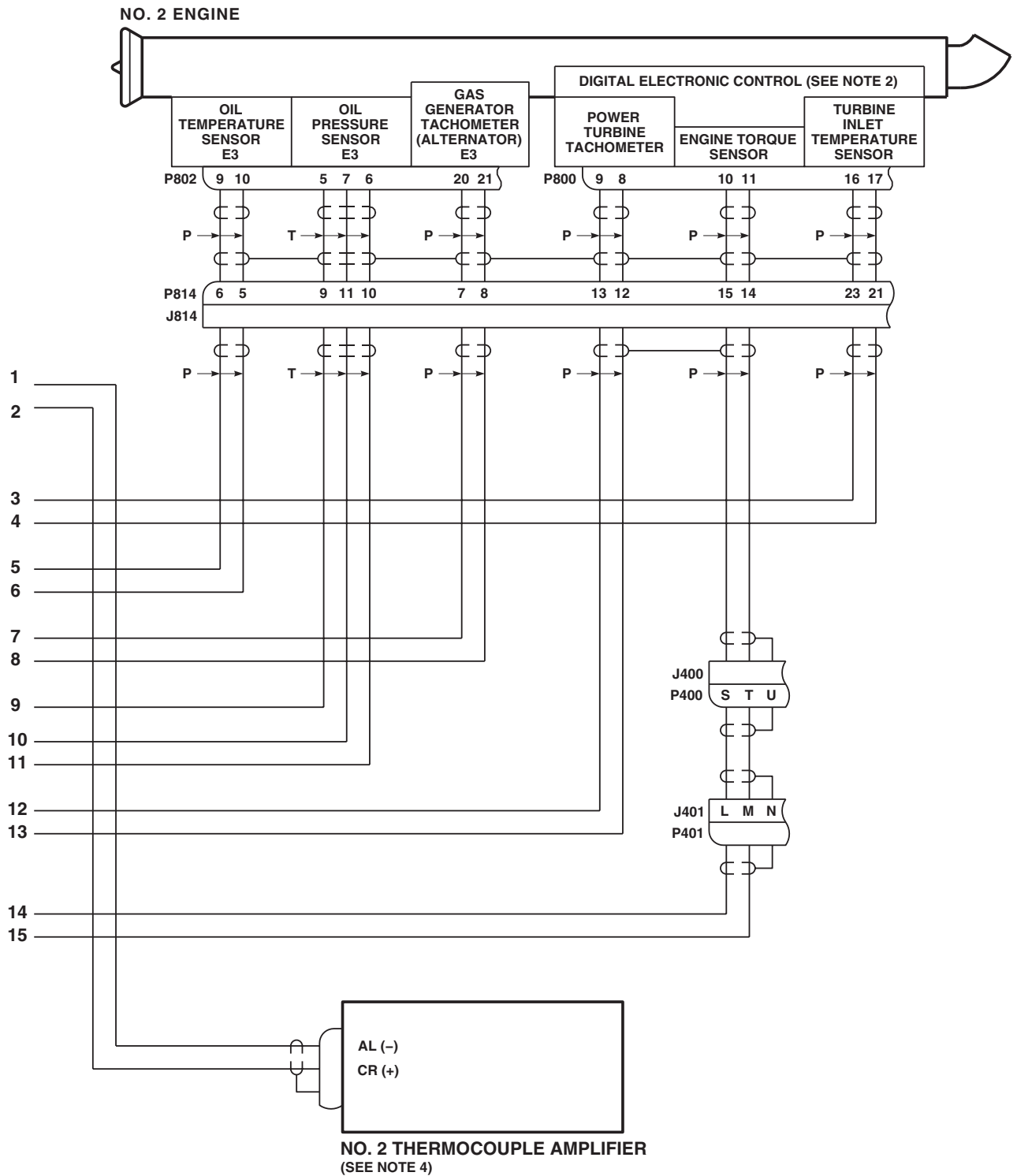
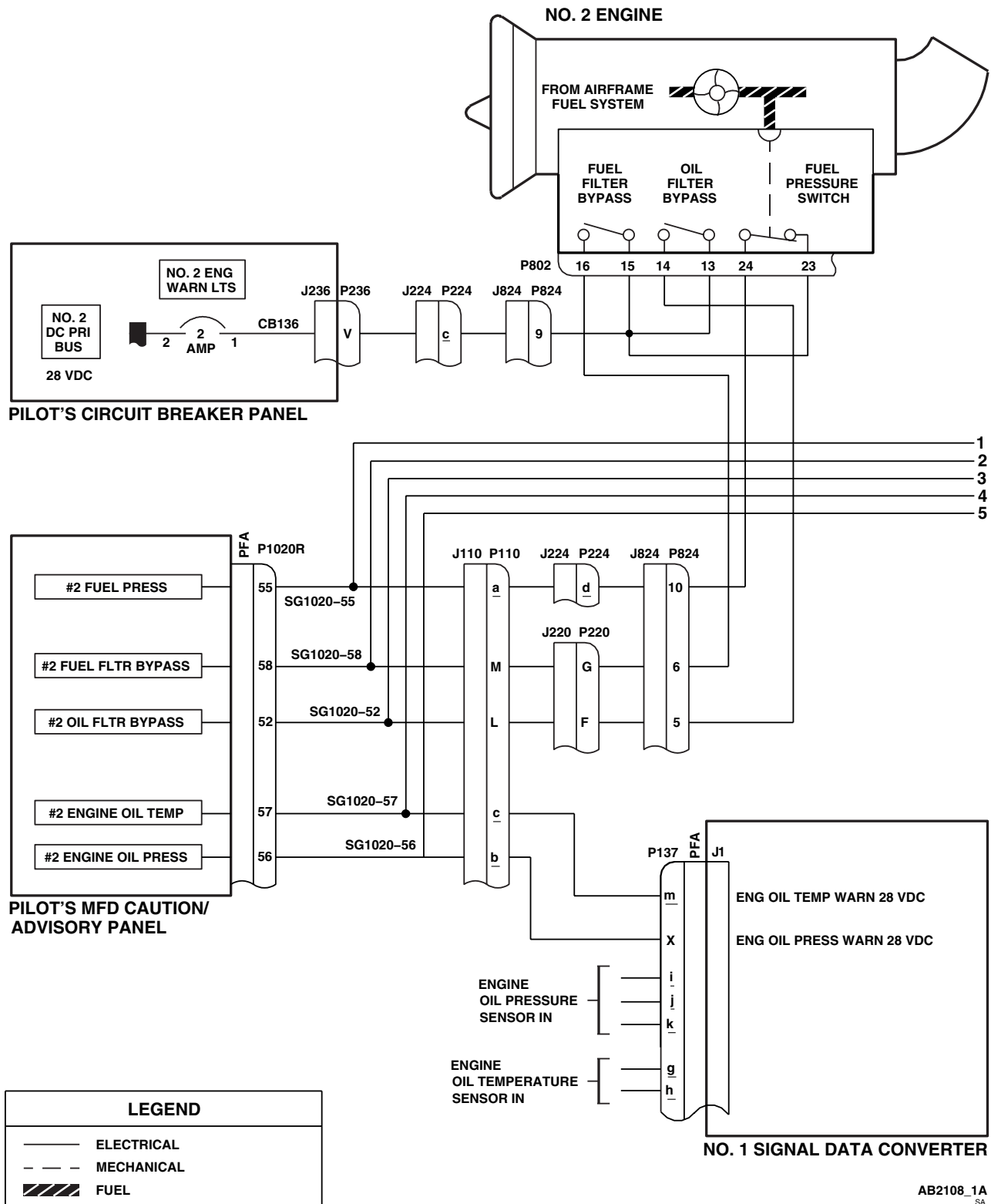
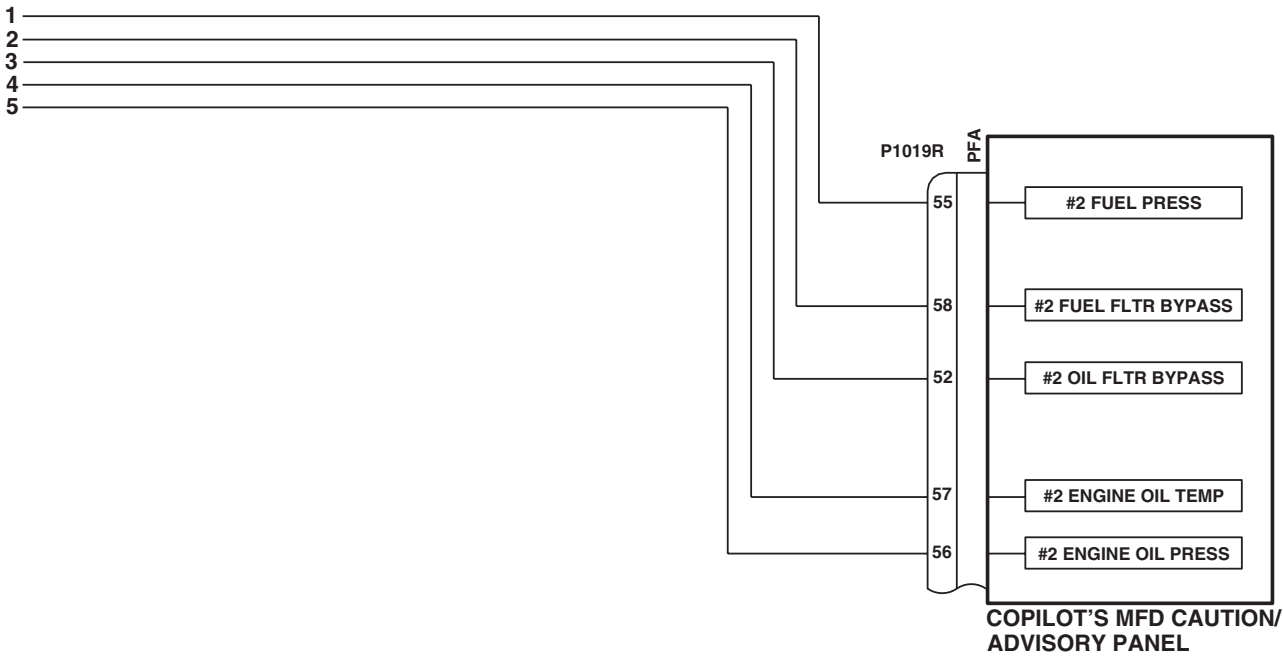
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Figure 3. No. 2 Engine Instrument System Schematic Diagram. (Sheet 2 of 2)

SCHEMATICS - Continued

Figure 4. No. 2 Engine Warning Lights System Schematic Diagram **HH-60A HH-60L** . (Sheet 1 of 2)

SCHEMATICS - Continued



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Figure 4. No. 2 Engine Warning Lights System Schematic Diagram **HH-60A HH-60L** . (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL

ENGINE SYSTEM

ENGINE OVERSPEED PROTECTION SYSTEM

This WP supersedes WP 0075 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment

Multimeter, AN/PSM-45A

WP 0079 00

WP 0104 00

WP 0824 00

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 0829 00

Electrical Repairer Toolkit, SC 5180-99-B06

WP 0924 00

Engine Repairman's Toolkit, SC 5180-99-B07

WP 1747 00

Personnel Required

Aircraft Electrician MOS 15F (1)

Aircraft Powerplant Repairer MOS 15B (1)

UH-60 Helicopter Repairer MOS 15T (1)

Equipment Condition

APU Operational, TM 1-1520-237-10

No. 1 and No. 2 Engines Operational,

TM 1-1520-237-10

References

TM 1-1520-237-10

TM 1-2840-248-23

ENGINE OVERSPEED PROTECTION SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step**NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for system location diagram, and **UH-60A EH-60A HH-60A** Figure 2  and **HH-60L UH-60L** Figure 3  for overspeed protection system schematic diagrams as an aid in operational/troubleshooting.

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER**CIRCUIT BREAKER PANEL**

NO. 1 ENG OVSP

Copilot's

NO. 2 ENG OVSP

Pilot's

NO. 1 ENG START CONTR

Upper Console

ENGINE OVERSPEED PROTECTION SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
NO. 2 ENG START CONTR 2. Start APU (TM 1-1520-237-10). 3. With qualified personnel at controls, run No. 1 and No. 2 engines at 100% Nr (TM 1-1520-237-10).	Pilot's
Indication/Condition Engines shall operate normally. Corrective Action If either engine does not start, troubleshoot engine start and ignition system (WP 0079 00).	
Step 4. Press upper console NO. 1 ENG OVSP TEST A button.	
Indication/Condition No. 1 engine gas generator speed (Ng SPEED 1) shall not decrease. Corrective Action If Indication/Condition is not as specified, go to NO. 1 ENGINE OVERSPEED CUTS IN WHEN TEST A BUTTON IS PRESSED, in this work package.	
Step 5. Press upper console NO. 1 ENG OVSP TEST B button.	
Indication/Condition No. 1 engine gas generator speed (Ng SPEED 1) shall not decrease. Corrective Action If Indication/Condition is not as specified, go to NO. 1 ENGINE OVERSPEED CUTS IN WHEN TEST B BUTTON IS PRESSED, in this work package.	
Step 6. Simultaneously press upper console NO. 1 ENG OVSP TEST A and NO. 1 ENG OVSP TEST B buttons.	
Indication/Condition a. UH-60L HH-60L The engine shall have a reduction in Np, Ng, and torque. ■ b. No. 1 gas generator speed (Ng) shall cycle. c. When test buttons are released, the engine shall reignite, and speed shall return to normal.	
Corrective Action 1. If Indication/Condition a. is not as specified, go to NO. 1 ENGINE DOES NOT SHUT DOWN WITH OVERSPEED CONDITION, in this work package. 2. If Indication/Conditions are not as specified, troubleshoot engine (TM 1-2840-248-23).	

ENGINE OVERSPEED PROTECTION SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE -
Continued**Step**

7. Press upper console NO. 2 ENG OVSP TEST A button.

Indication/Condition

No. 2 engine gas generator speed (Ng SPEED 2) shall not decrease.

Corrective Action

If Indication/Condition is not as specified, go to NO. 2 ENGINE OVERSPEED CUTS IN WHEN TEST A BUTTON IS PRESSED, in this work package.

Step

8. Press upper console NO. 2 ENG OVSP TEST B button.

Indication/Condition

No. 2 engine gas generator speed (Ng SPEED 2) shall not decrease.

Corrective Action

If Indication/Condition is not as specified, go to NO. 2 ENGINE OVERSPEED CUTS IN WHEN TEST B BUTTON IS PRESSED, in this work package.

Step

9. Simultaneously press upper console NO. 2 ENG OVSP TEST A and NO. 2 ENG OVSP TEST B buttons.

Indication/Condition

- a. **UH-60L HH-60L** The engine shall have a reduction in Np, Ng, and torque. ■
- b. No. 2 gas generator speed (Ng) and/or power turbine speed (Np) shall cycle.
- c. When test buttons are released, the engine shall reignite, and speed shall return to normal.

Corrective Action

1. If Indication/Condition a. is not as specified, go to NO. 2 ENGINE DOES NOT SHUT DOWN WITH OVERSPEED CONDITION, in this work package.
2. If Indication/Conditions are not as specified, troubleshoot engine (TM 1-2840-248-23).

Step

10. Shut down engines (TM 1-1520-237-10).

Indication/Condition

UH-60L HH-60L The engine shall have a reduction in Np, Ng, and torque. ■

Corrective Action

1. If No. 1 BITE shows 85%, go to NO. 1 ENGINE BITE READS 85%, in this work package.
2. If No. 2 BITE shows 85%, go to NO. 2 ENGINE BITE READS 85%, in this work package.

Step

11. Shut down APU (TM 1-1520-237-10).

ENGINE OVERSPEED PROTECTION SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE -
Continued**Indication/Condition**

APU shall shut down.

Corrective Action

None Required

NO. 1 ENGINE OVERSPEED CUTS IN WHEN TEST A BUTTON IS PRESSED**SYMPTOM**

No. 1 engine overspeed cuts in.

MALFUNCTION

No. 1 engine overspeed cuts in when TEST A button is pressed.

CORRECTIVE ACTION**1. Check continuity between P801-6 and P801-14.**

- a. If continuity is present, go to **2**.
- b. If continuity is not present, troubleshoot engine overspeed test system (TM 1-2840-248-23). Go to **4**.

2. Check continuity between TEST B switch terminal 1 and P801-14.

- a. If continuity is present, go to **3**.
- b. If continuity is not present, repair/replace wiring between terminal 1 and J815-26 (WP 1747 00). Go to **4**.
- c. If trouble remains, replace engine harness (TM 1-2840-248-23). Go to **4**.

3. Check continuity between TEST B switch terminal 2 and P801-6.

- a. If continuity is present, replace switch (WP 0829 00). Go to **4**.
- b. If continuity is not present, repair/replace wiring between terminal 2 and J815-27 (WP 1747 00). Go to **4**.
- c. If trouble remains, replace engine harness (TM 1-2840-248-23). Go to **4**.

4. Procedure completed.**NO. 1 ENGINE OVERSPEED CUTS IN WHEN TEST B BUTTON IS PRESSED****SYMPTOM**

No. 1 engine overspeed cuts in.

MALFUNCTION

No. 1 engine overspeed cuts in when TEST B button is pressed.

CORRECTIVE ACTION**1. Check continuity between P801-6 and P801-12.**

- a. If continuity is present, go to **2**.
- b. If continuity is not present troubleshoot engine overspeed test system (TM 1-2840-248-23). Go to **4**.

NO. 1 ENGINE OVERSPEED CUTS IN WHEN TEST B BUTTON IS PRESSED - Continued**2. Check continuity between TEST A switch terminal 1 and P801-12.**

- a. If continuity is present, go to **3**.
- b. If continuity is not present, repair/replace wiring between terminal 1 and J815-25 (WP 1747 00). Go to **4**.
- c. If trouble remains, replace engine harness (TM 1-2840-248-23). Go to **4**.

3. Check continuity between TEST A switch terminal 2 and P801-6.

- a. If continuity is present, replace switch (WP 0829 00). Go to **4**.
- b. If continuity is not present, repair/replace wiring between terminal 2 and J815-27 (WP 1747 00). Go to **4**.
- c. If trouble remains, replace engine harness (TM 1-2840-248-23). Go to **4**.

4. Procedure completed.**NO. 1 ENGINE DOES NOT SHUT DOWN WITH OVERSPEED CONDITION****SYMPTOM**

No. 1 engine does not shut down.

MALFUNCTION

No. 1 engine does not shut down with overspeed condition.

CORRECTIVE ACTION**1. Check for 28 vdc between P801-28 and ground.**

- a. If voltage is as specified, go to **2**.
- b. If voltage is not as specified, go to **8**.

2. Check for 28 vdc between P381-C and P381-B.

- a. If voltage is as specified, go to **3**.
- b. If voltage is not as specified, go to **5**.

3. Check for 28 vdc between P381-D and P381-B.

- a. If voltage is as specified, replace No. 1 overspeed relay assembly (WP 0924 00). Go to **9**.
- b. If voltage is not as specified, go to **4**.

4. Check for 28 vdc between NO. 1 ENG START circuit breaker terminal 1 and ground.

- a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P381-D (WP 1747 00). Go to **9**.
- b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **9**.

5. Check for 28 vdc between P381-C and ground.

- a. If voltage is as specified, repair/replace wiring between P381-B and GND381-1 (WP 1747 00). Go to **9**.
- b. If voltage is not as specified, go to **6**.

6. Check continuity between P381-C and J815-1.

- a. If continuity is present, go to **7**.

NO. 1 ENGINE DOES NOT SHUT DOWN WITH OVERSPEED CONDITION - Continued

- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9**.

7. Check continuity between P801-27 and P815-1.

- a. If continuity is present, troubleshoot engine (TM 1-2840-248-23). Go to **9**.
- b. If continuity is not present, replace engine harness (TM 1-2840-248-23). Go to **9**.

8. Check for 28 vdc between J815-2 and ground.

- a. If voltage is as specified, replace engine harness (TM 1-2840-248-23). Go to **9**.
- b. If voltage is not as specified, repair/replace wiring between J815-2 and NO. 1 ENG START circuit breaker terminal 1 (WP 1747 00). Go to **9**.

9. Procedure completed.**NO. 2 ENGINE OVERSPEED CUTS IN WHEN TEST A BUTTON IS PRESSED****SYMPTOM**

No. 2 engine overspeed cuts in.

MALFUNCTION

No. 2 engine overspeed cuts in when TEST A button is pressed.

CORRECTIVE ACTION**1. Check continuity between P800-6 and P800-14.**

- a. If continuity is present, go to **2**.
- b. If continuity is not present, troubleshoot engine overspeed test system (TM 1-2840-248-23). Go to **4**.

2. Check continuity between TEST B switch terminal 1 and P800-14.

- a. If continuity is present, go to **3**.
- b. If continuity is not present, repair/replace wiring between terminal 1 and J814-28 (WP 1747 00). Go to **4**.
- c. If trouble remains, replace engine harness (TM 1-2840-248-23). Go to **4**.

3. Check continuity between TEST B switch terminal 2 and P800-6.

- a. If continuity is present, replace switch (WP 0829 00). Go to **4**.
- b. If continuity is not present, repair/replace wiring between terminal 2 and J814-29 (WP 1747 00). Go to **4**.
- c. If trouble remains, replace engine harness (TM 1-2840-248-23). Go to **4**.

4. Procedure completed.

NO. 2 ENGINE OVERSPEED CUTS IN WHEN TEST B BUTTON IS PRESSED**SYMPTOM**

No. 2 engine overspeed cuts in.

MALFUNCTION

No. 2 engine overspeed cuts in when TEST B button is pressed.

CORRECTIVE ACTION**1. Check continuity between P800-6 and P800-12.**

- a. If continuity is present, go to **2**.
- b. If continuity is not present, troubleshoot engine overspeed test system (TM 1-2840-248-23). Go to **4**.

2. Check continuity between TEST A switch terminal 1 and P800-12.

- a. If continuity is present, go to **3**.
- b. If continuity is not present, repair/replace wiring between terminal 1 and J814-27 (WP 1747 00). Go to **4**.
- c. If trouble remains, replace engine harness (TM 1-2840-248-23). Go to **4**.

3. Check continuity between TEST A switch terminal 2 and P800-6.

- a. If continuity is present, replace switch (WP 0829 00). Go to **4**.
- b. If continuity is not present, repair/replace wiring between terminal 2 and J814-29 (WP 1747 00). Go to **4**.
- c. If trouble remains, replace engine harness (TM 1-2840-248-23). Go to **4**.

4. Procedure completed.**NO. 2 ENGINE DOES NOT SHUT DOWN WITH OVERSPEED CONDITION****SYMPTOM**

No. 2 engine does not shut down.

MALFUNCTION

No. 2 engine does not shut down with overspeed condition.

CORRECTIVE ACTION**1. Check for 28 vdc between P800-28 and ground.**

- a. If voltage is as specified, go to **2**.
- b. If voltage is not as specified, go to **8**.

2. Check for 28 vdc between P382-C and P382-B.

- a. If voltage is as specified, go to **3**.
- b. If voltage is not as specified, go to **5**.

3. Check for 28 vdc between P382-D and P382-B.

- a. If voltage is as specified, replace No. 2 overspeed relay assembly (WP 0924 00). Go to **9**.

NO. 2 ENGINE DOES NOT SHUT DOWN WITH OVERSPEED CONDITION - Continued

- b. If voltage is not as specified, go to **4**.
- 4. Check for 28 vdc between NO. 2 ENG START CONTR circuit breaker terminal 1 and ground.**
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P382-D (WP 1747 00). Go to **9**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **9**.
- 5. Check 28 vdc between P382-C and ground.**
 - a. If voltage is as specified, repair/replace wiring between P382-B and GND382-1 (WP 1747 00). Go to **9**.
 - b. If voltage is not as specified, go to **6**.
- 6. Check continuity between J814-1 and P382-C.**
 - a. If continuity is present, go to **7**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9**.
- 7. Check continuity between P800-27 and P814-1.**
 - a. If continuity is present, troubleshoot engine (TM 1-2840-248-23). Go to **9**.
 - b. If continuity is not present, replace engine harness (TM 1-2840-248-23). Go to **9**.
- 8. Check for 28 vdc between J814-2 and ground.**
 - a. If voltage is as specified, replace engine harness (TM 1-2840-248-23). Go to **9**.
 - b. If voltage is not as specified, repair/replace wiring between J814-2 and NO. 2 ENG START CONTR circuit breaker terminal 1 (WP 1747 00). Go to **9**.
- 9. Procedure completed.**

NO. 1 ENGINE BITE READS 85%**SYMPTOM**

No. 1 BITE reads 85%.

MALFUNCTION

No. 1 BITE reads 85%.

CORRECTIVE ACTION

- 1. Do NO. 1 ENGINE DOES NOT SHUT DOWN WITH OVERSPEED CONDITION, in this work package. Go to 2.**
- 2. Check continuity between J815-35 and P381-F.**
 - a. If continuity is present, go to **3**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 3. Check continuity between:**
 - **P803-17 and P815-35**
 - **P803-19 and P815-35**
 - a. If continuity is present, go to **4**.

NO. 1 ENGINE BITE READS 85% - Continued

- b. If continuity is not present, replace engine harness (TM 1-2840-248-23). Go to **5**.
- 4. Check continuity between P381-G and key ignition switch terminal 2.**
 - a. If continuity is present, replace No. 1 overspeed relay assembly (WP 0924 00). Go to **5**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 5. Procedure completed.**

NO. 2 ENGINE BITE READS 85%**SYMPTOM**

No. 2 engine BITE reads 85%.

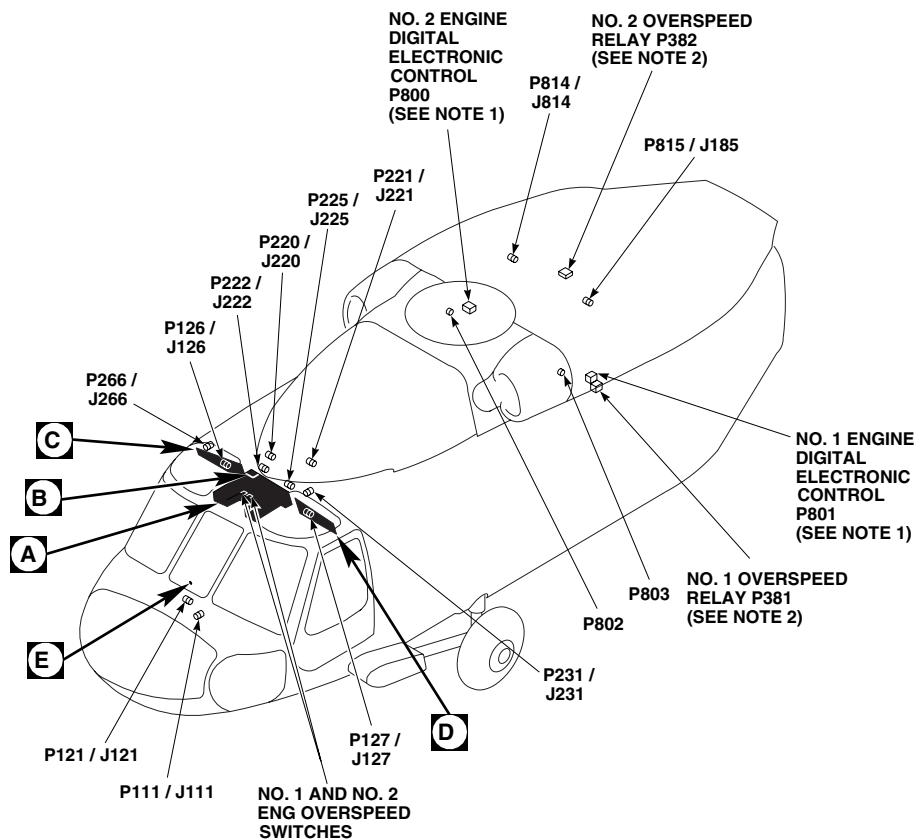
MALFUNCTION

No. 2 engine BITE reads 85%.

CORRECTIVE ACTION

- 1. Do NO. 2 ENGINE DOES NOT SHUT DOWN WITH OVERSPEED CONDITION, in this work package. Go to 2.**
- 2. Check continuity between J814-35 and P382-F.**
 - a. If continuity is present, go to **3**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 3. Check continuity between:**
 - **P802-17 and P814-35**
 - **P802-19 and P814-35**
 - a. If continuity is present, go to **4**.
 - b. If continuity is not present, replace engine harness (TM 1-2840-248-23). Go to **5**.
- 4. Check continuity between P382-G and key ignition switch terminal C2.**
 - a. If continuity is present, replace No. 2 overspeed relay assembly (WP 0924 00). Go to **5**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 5. Procedure completed.**

LOCATION



TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P111 / J111	COCKPIT, BL 7.5, STA 197
P121 / J121	COCKPIT, BL 10 RH, STA 197
P126 / J126	COCKPIT, BL 28 RH, STA 243
P127 / J127	COCKPIT, BL 28 LH, STA 243
P220 / J220	JUNCTION BOX, MAIN ROTOR PYLON DECK
P221 / J221	JUNCTION BOX, MAIN ROTOR PYLON DECK
P222 / J222	JUNCTION BOX, MAIN ROTOR PYLON DECK
P225 / J225	JUNCTION BOX, MAIN ROTOR PYLON DECK



NOTES

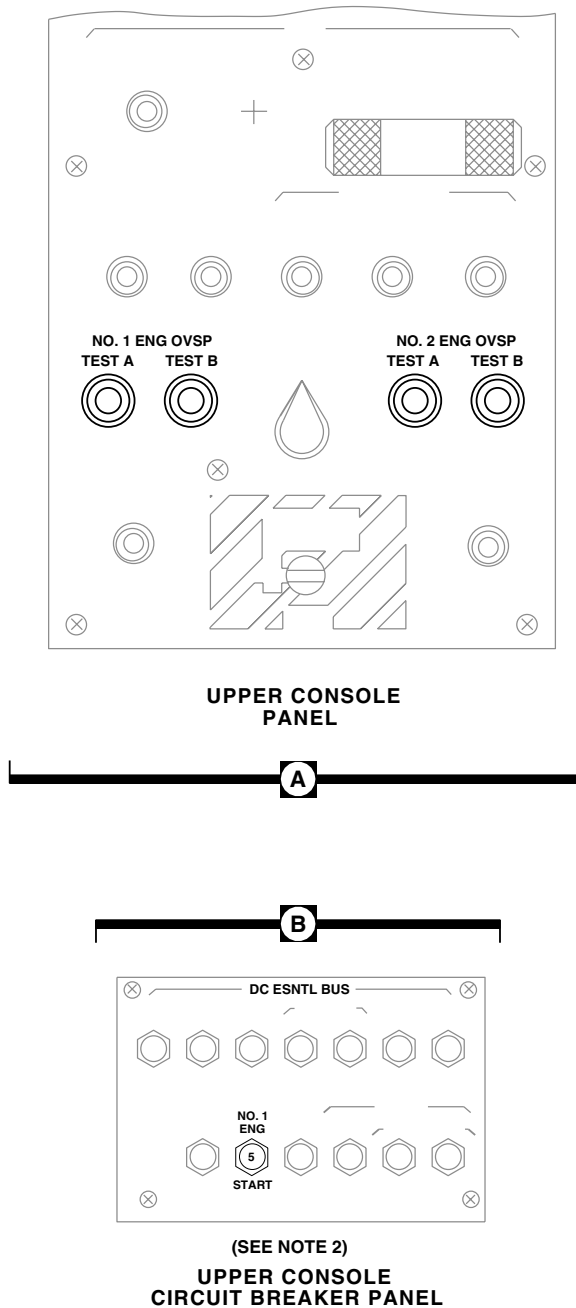
1. **UH-60A EH-60A HH-60A**
HAVE AN ELECTRICAL CONTROL UNIT.
2. **UH-60L HH-60L**
3. **HH-60A HH-60L**
ENGINE IGNITION ON LOWER CONSOLE.

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P231 / J231	CABIN CEILING, BL 8 RH, STA 247
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 25.0 RH
P381 / J381	NO. 1 OVERSPEED RELAY (SEE NOTE 2)
P382 / J382	NO. 2 OVERSPEED RELAY (SEE NOTE 2)
P800	NO. 2 ENGINE DIGITAL ELECTRONIC CONTROL
P801	NO. 1 ENGINE DIGITAL ELECTRONIC CONTROL
P802	NO. 2 ENGINE INTERFACE DISCONNECT
P803	NO. 1 ENGINE INTERFACE DISCONNECT
P814 / J814	FIREWALL DISCONNECT BL 14 RH, STA 373
P815 / J185	FIREWALL DISCONNECT BL 14 LH, STA 373

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Figure 1. Engine Overspeed Protection System Location Diagram. (Sheet 1 of 4)

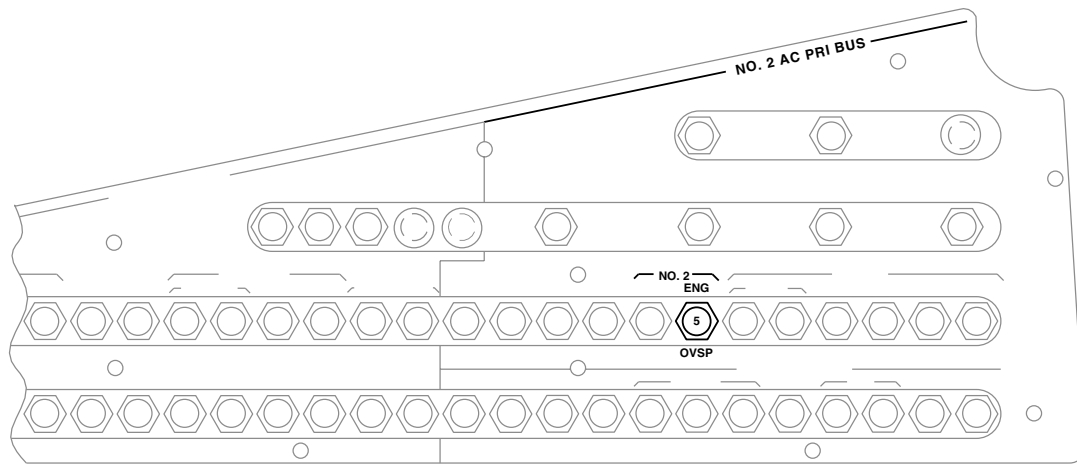
LOCATION - Continued



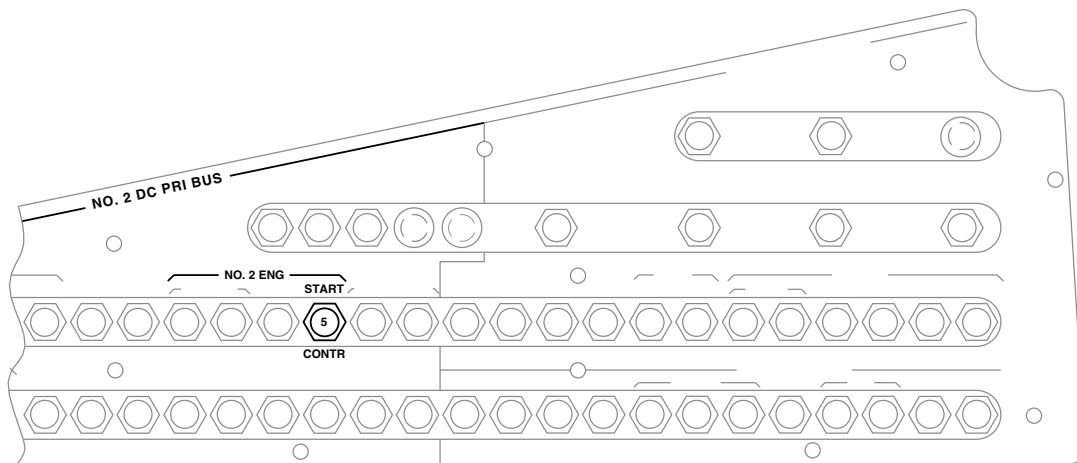
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Figure 1. Engine Overspeed Protection System Location Diagram. (Sheet 2 of 4)

LOCATION - Continued



(SEE NOTE 1)



(SEE NOTE 2)

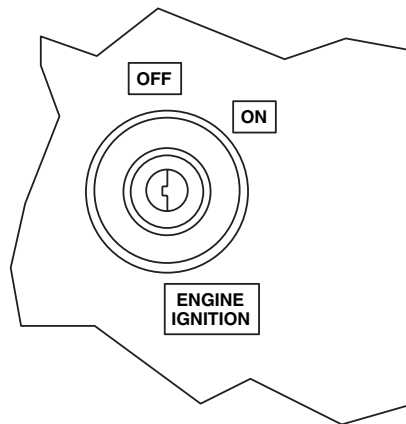
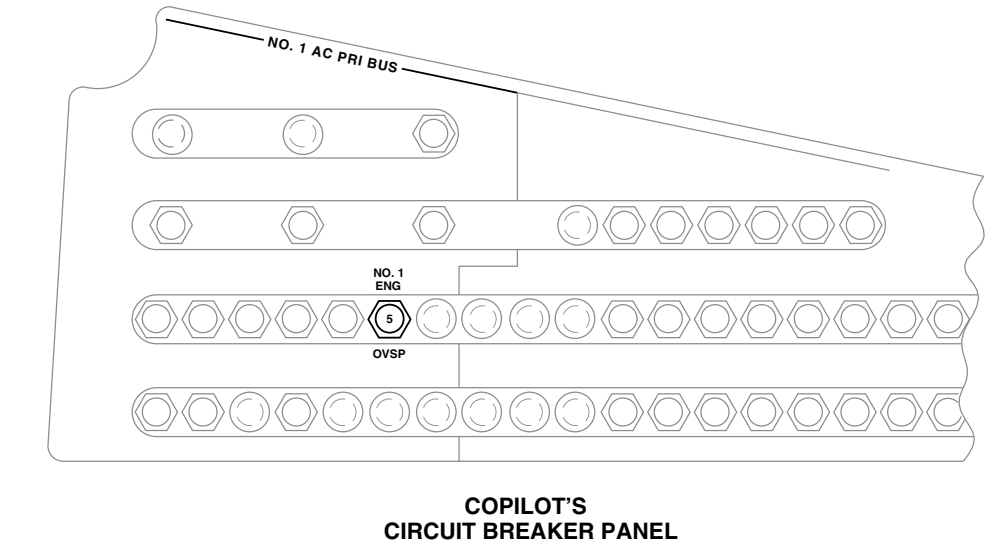
PILOT'S
CIRCUIT BREAKER PANEL

C

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Figure 1. Engine Overspeed Protection System Location Diagram. (Sheet 3 of 4)

LOCATION - Continued

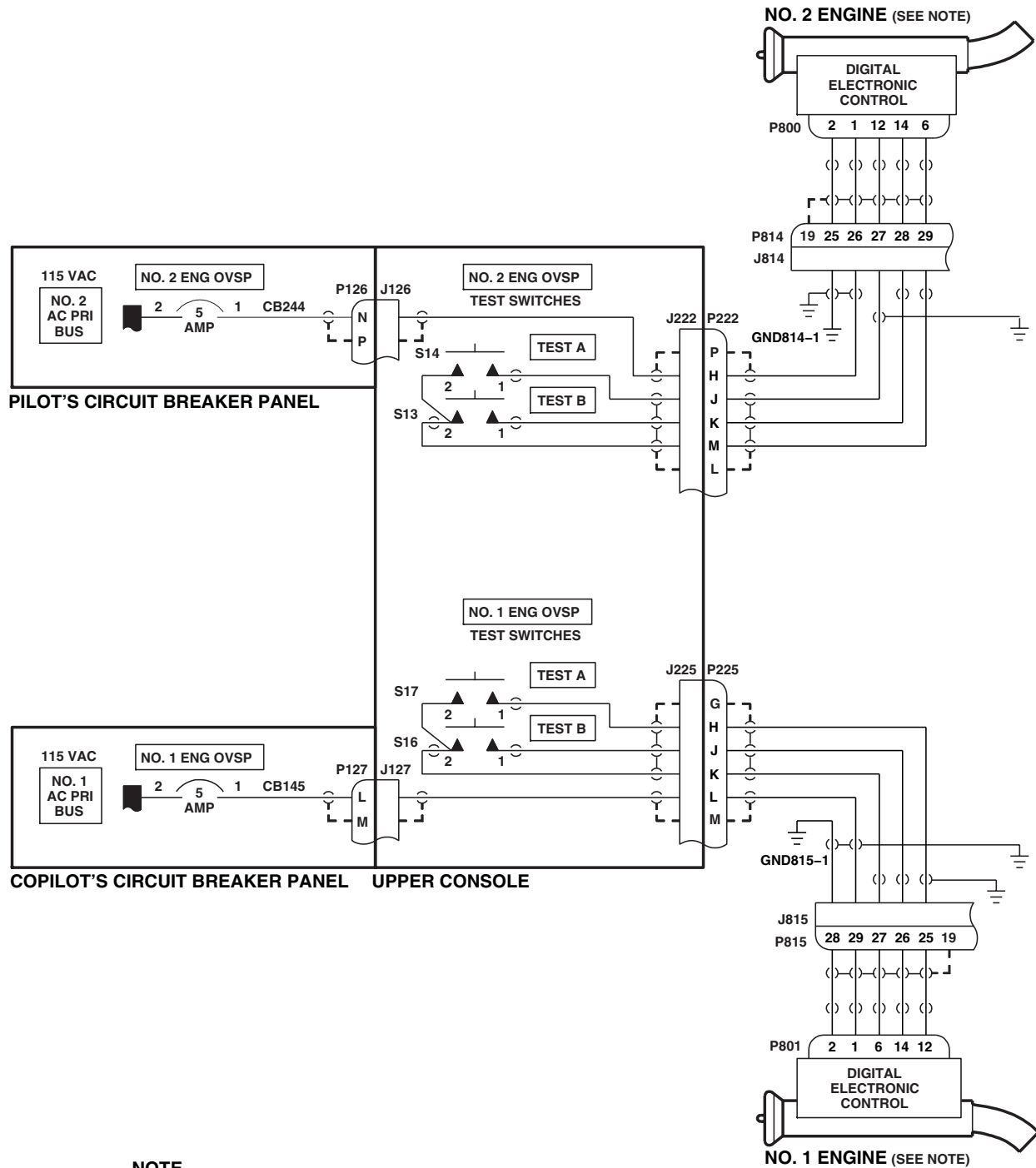


INSTRUMENT PANEL
(SEE NOTE 3)

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Figure 1. Engine Overspeed Protection System Location Diagram. (Sheet 4 of 4)

SCHEMATICS



NOTE

UH-60A EH-60A

HAVE AN ELECTRICAL CONTROL UNIT.

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Figure 2. Engine Overspeed Protection System Schematic Diagram UH-60A EH-60A HH-60A .

SCHEMATICS - Continued

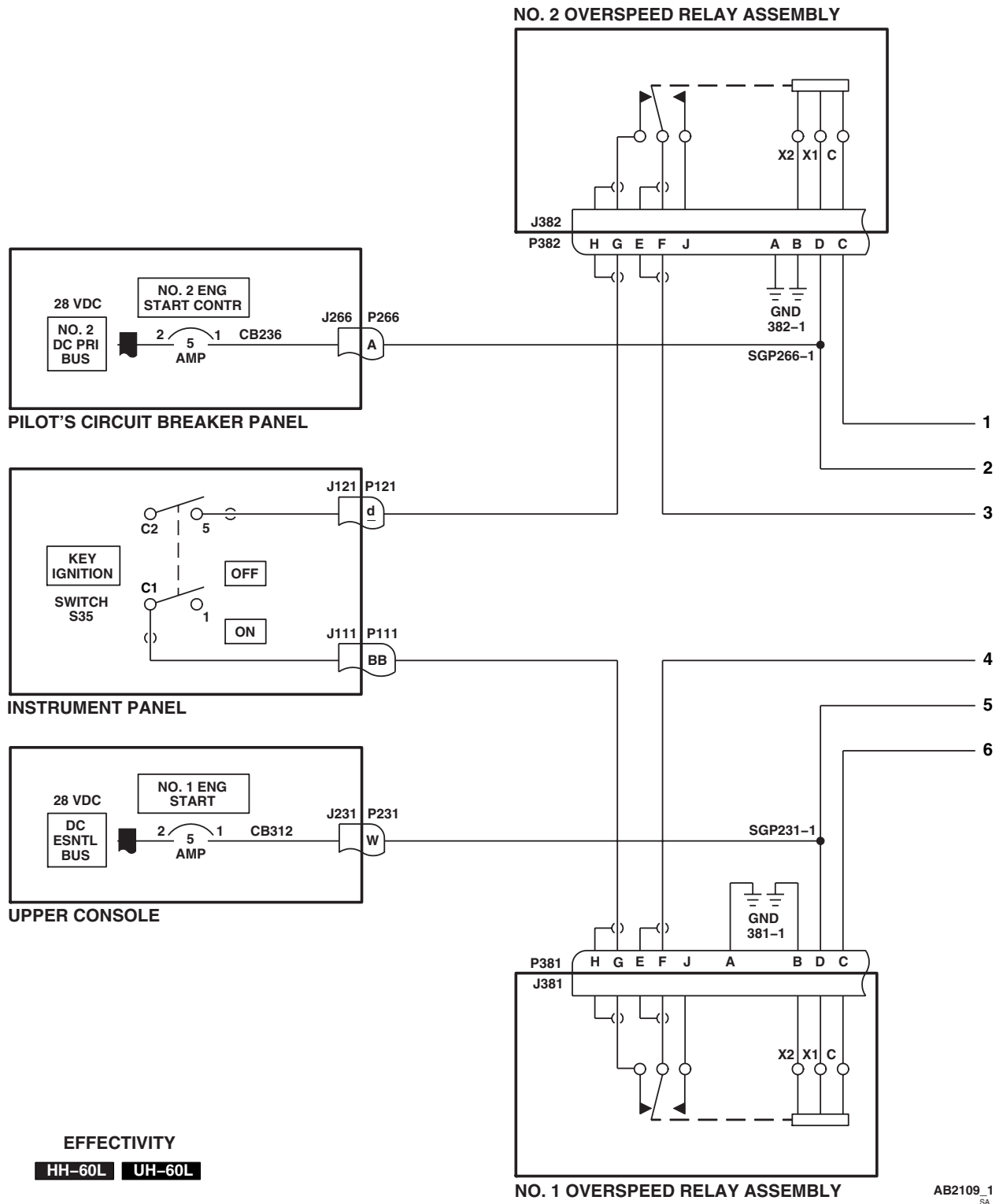
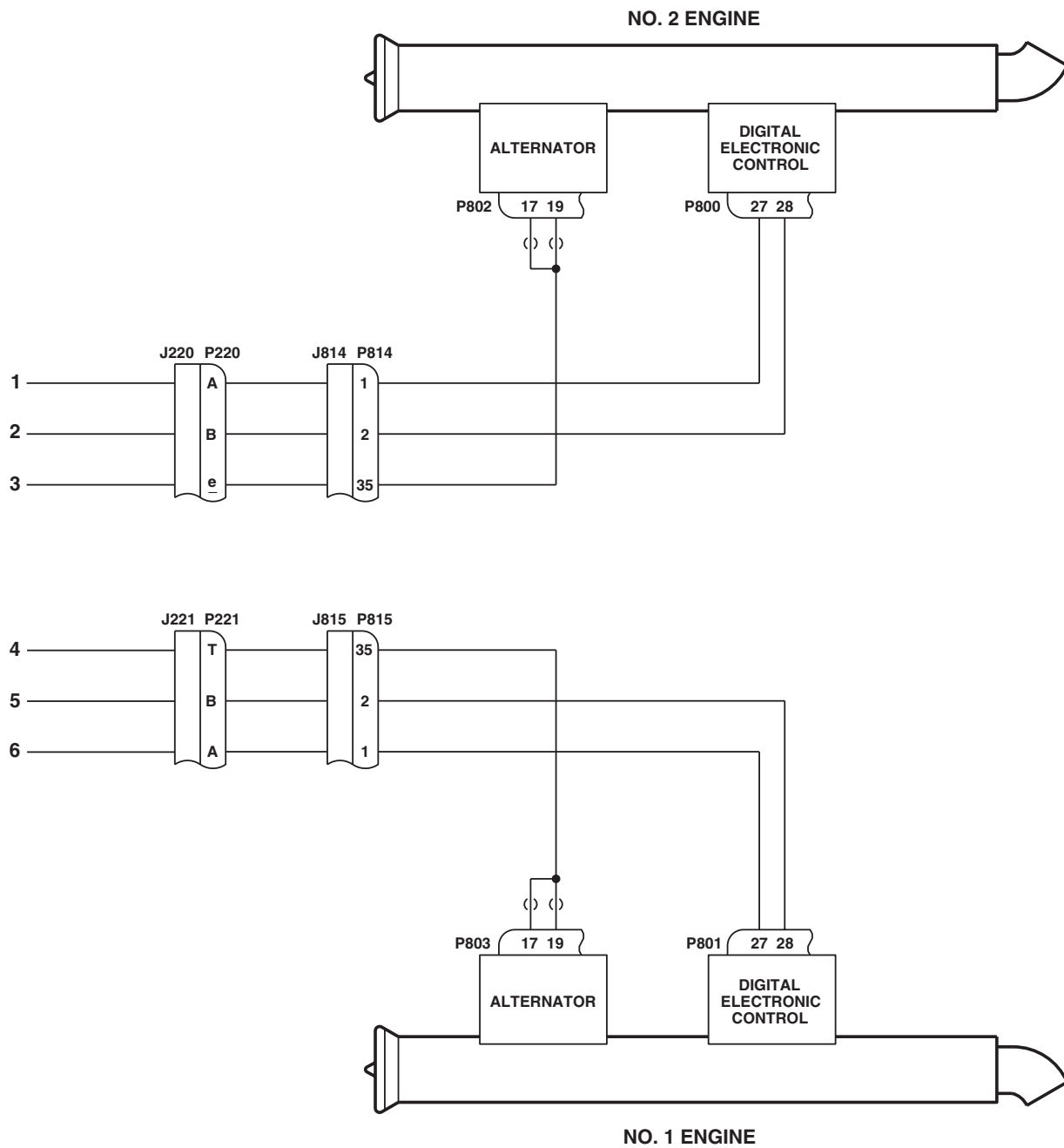


Figure 3. Engine Overspeed Protection System Schematic Diagram **HH-60L UH-60L** . (Sheet 1 of 2)

SCHEMATICS - Continued

AB2109 2
SAFigure 3. Engine Overspeed Protection System Schematic Diagram **HH-60L UH-60L** . (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL

ENGINE SYSTEM

ENGINE SPEED TRIM SYSTEM

This WP supersedes WP 0076 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment

Multimeter, AN/PSM-45A

WP 0105 00

WP 0500 00

WP 0824 00

WP 1066 00

WP 1747 00

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Electrical Repairer Toolkit, SC 5180-99-B06

Engine Repairman's Toolkit, SC 5180-99-B07

Equipment Condition

APU Operational, TM 1-1520-237-10

No. 1 and No. 2 Engines Operational,
TM 1-1520-237-10

No. 1 and No. 2 Fuel Tanks Serviced,
TM 1-1520-237-10

Personnel Required

Aircraft Electrician MOS 15F (1)

Aircraft Powerplant Repairer MOS 15B (1)

UH-60 Helicopter Repairer MOS 15T (1)

References

TM 1-1520-237-10

TM 1-2840-248-23

ENGINE SPEED TRIM SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0105 00).
- If one engine will not decrease below 100% Np, an associated torque split may be noticed.
- See Figure 1 for system location diagram, and **UH-60A EH-60A UH-60L** Figure 2 and **HH-60A HH-60L** Figure 3 for trim system schematic diagrams as an aid in operational/troubleshooting.

1. Make sure the SPEED TRIM circuit breaker on pilot's circuit breaker panel is pushed in.
2. Start APU (TM 1-1520-237-10).
3. With qualified personnel at controls, start No. 1 and No. 2 engines. Operate main rotor at 100% Nr until instruments stabilize (TM 1-1520-237-10).
4. Push and hold pilot's (or copilot's) collective stick ENG RPM or **HH-60A HH-60L** upper console ENGINE SPEED TRIM switch to DECR.

ENGINE SPEED TRIM SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

- a. %RPM 1 (Np) indication on pilot's and copilot's display unit shall decrease to 96% (bottom of green range).
- b. %RPM 2 (Np) indication on pilot's and copilot's display unit shall decrease to 96% (bottom of green range).
- c. No. 1 or No. 2 engine Np shall not go below 96% Np.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, go to NO. 1 AND NO. 2 ENGINE POWER TURBINE SPEED (%NP) CANNOT BE DECREASED BELOW 100%, in this work package.
2. If Indication/Condition a. is not as specified, go to NO. 1ENGINE POWER TURBINE SPEED (%NP) CANNOT BE INCREASED OR DECREASED, in this work package.
3. If Indication/Condition b. is not as specified, go to NO. 2 ENGINE POWER TURBINE SPEED (%NP) CANNOT BE INCREASED OR DECREASED, in this work package.
4. If Indication/Condition c. is not as specified, replace engine speed control box (WP 0500 00).

Step

5. Push and hold pilot's (or copilot's) collective stick ENG RPM or **HH-60A HH-60L** upper console ENGINE SPEED TRIM  switch to INCR.

Indication/Condition

- a. %RPM 1 (Np) indication on pilot's and copilot's display unit shall increase to 100% (top of green range).
- b. %RPM 2 (Np) indication on pilot's and copilot's display unit shall increase to 100% (top of green range).

Corrective Action

1. If Indication/Condition a. is not as specified, go to NO. 1ENGINE POWER TURBINE SPEED (%NP) CANNOT BE INCREASED OR DECREASED, in this work package.
2. If Indication/Condition b. is not as specified, go to NO. 2 ENGINE POWER TURBINE SPEED (%NP) CANNOT BE INCREASED OR DECREASED, in this work package.

Step

6. Shut down both engines (TM 1-1520-237-10).
7. Shut down APU (TM 1-1520-237-10).

Indication/Condition

APU shall shut down.

Corrective Action

None Required

NO. 1 AND NO. 2 ENGINE POWER TURBINE SPEED (%NP) CANNOT BE DECREASED BELOW 100%**SYMPTOM**

No. 1 and No. 2 engine power turbine speed (%Np) cannot be decreased.

MALFUNCTION

No. 1 and No. 2 engine power turbine speed (%Np) cannot be decreased below 100%.

CORRECTIVE ACTION

1. With copilot's collective stick ENG RPM or HH-60A HH-60L upper console ENGINE SPEED TRIM switch at DECR, check for 28 vdc between P901-J and ground.
 - a. If voltage is as specified, go to 2.
 - b. If voltage is not as specified, go to 3.
2. Check continuity between P901-L and ground.
 - a. If continuity is present, replace engine speed control assembly (WP 0500 00). Go to 10.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 10.
3. With pilot's collective stick ENG RPM or HH-60A HH-60L upper console ENGINE SPEED TRIM switch at DECR, check for 28 vdc between P901-J and ground.
 - a. If voltage is as specified, go to 4.
 - b. If voltage is not as specified, go to 7.
4. Check for 28 vdc between J103-k or HH-60A HH-60L P246-V and ground.
 - a. If voltage is as specified, go to 5 or HH-60A HH-60L 9.
 - b. If voltage is not as specified, go to 6.
5. Check continuity between:
 - J103-h and J103-j
 - J103-i and P901-J
 - HH-60A HH-60L P246-X and P901-J
 - HH-60A HH-60L P246-W and P901-K
 - a. If continuity is present, replace copilot's collective stick grip or HH-60A HH-60L upper console ENGINE SPEED TRIM switch (WP 1066 00). Go to 10.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to 10.
6. Check continuity between J104-g and J103-k.
 - a. If continuity is present, replace pilot's collective stick grip (WP 1066 00). Go to 10.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 10.
7. Check for 28 vdc between J104-k and ground.
 - a. If voltage is as specified, go to 8.
 - b. If voltage is not as specified, go to 9.
8. Check continuity between these points:

NO. 1 AND NO. 2 ENGINE POWER TURBINE SPEED (%NP) CANNOT BE DECREASED BELOW 100% - Continued

- J104-h and J104-j
- J104-i and P901-J
- a. If continuity is present, replace pilot's collective stick grip (WP 1066 00). Go to **10**.
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **10**.
- 9. Check continuity between and SPEED TRIM circuit breaker terminal 1 and J104-k HH-60A HH-60L or P246-V .**
 - a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **10**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10**.
- 10. Procedure completed.**

NO. 1 ENGINE POWER TURBINE SPEED (%NP) CANNOT BE INCREASED OR DECREASED

SYMPTOM

No. 1 engine power turbine speed (%Np) cannot adjusted.

MALFUNCTION

No. 1 engine power turbine speed (%Np) cannot be increased or decreased.

CORRECTIVE ACTION

- 1. Check that power turbine (%Np) can be increased or decreased.**
 - a. If power turbine speed cannot be increased, go to **2**.
 - b. If power turbine speed cannot be decreased, go to **5**.
- 2. Push pilot's or copilot's ENG RPM or HH-60A HH-60L upper console ENGINE SPEED TRIM switch to INCR until engine speed control motor stops. Go to 3.**
- 3. Check resistance between:**
 - P801-18 and P801-20 (2200-2400 ohms)
 - P801-19 and P801-20 (240-260 ohms)
 - P801-20 and P801-21 (120-130 ohms)
 - a. If resistance is as specified, troubleshoot engine (Np does not respond to Np demand trim) (TM 1-2840-248-23). Go to **8**.
 - b. If resistance is not as specified, go to **4**.
- 4. Check for resistance of 124 to 142 ohms between J901-A and J901-B.**
 - a. If resistance is as specified, repair/replace wiring, as required, between (WP 1747 00), then go to **8**:
 - P801-18 and P901-C
 - P801-19 and P901-A
 - P801-20 and P901-D
 - P801-21 and P901-B
 - b. If resistance is not as specified, replace engine speed control box (WP 0500 00). Go to **8**.

NO. 1 ENGINE POWER TURBINE SPEED (%NP) CANNOT BE INCREASED OR DECREASED - Continued

5. Push pilot's or copilot's ENG RPM or **HH-60A HH-60L** upper console ENGINE SPEED TRIM  switch to DECR until engine speed control motor stops. Go to 6.
6. Check resistance between:
 - P801-18 and P801-20 (2200-2400 ohms)
 - P801-19 and P801-20 (240-260 ohms)
 - P801-20 and P801-21 (860-900 ohms)
 - a. If resistance is as specified, troubleshoot engine (Np does not respond to Np demand trim) (TM 1-2840-248-23). Go to 8.
 - b. If resistance is not as specified, go to 7.
7. Check for resistance of 2200 to 2400 ohms between J901-C and J901-D.
 - a. If resistance is as specified, repair/replace wiring, as required, between (WP 1747 00), then go to 8:
 - P801-18 and P901-C
 - P801-19 and P901-A
 - P801-20 and P901-D
 - P801-21 and P901-B
 - b. If resistance is not as specified, replace engine speed control box (WP 0500 00). Go to 8.
8. Procedure completed.

NO. 2 ENGINE POWER TURBINE SPEED (%NP) CANNOT BE INCREASED OR DECREASED**SYMPTOM**

No. 2 engine power turbine speed (%Np) cannot be adjusted.

MALFUNCTION

No. 2 engine power turbine speed (%Np) cannot be increased or decreased.

CORRECTIVE ACTION

1. Check that power turbine speed (%Np) can be increased or decreased.
 - a. If power turbine speed cannot be increased, go to 2.
 - b. If power turbine speed cannot be decreased, go to 5.
2. Push pilot's or copilot's ENG RPM or **HH-60A HH-60L** upper console ENGINE SPEED TRIM  switch to INCR until engine speed control motor stops. Go to 3.
3. Check resistance between:
 - P800-18 and P800-20 (2200-2400 ohms)
 - P800-19 and P800-20 (240-260 ohms)
 - P800-20 and P800-21 (120-130 ohms)
 - a. If resistance is as specified, troubleshoot engine (Np does not respond to Np demand trim) (TM 1-2840-248-23). Go to 8.

NO. 2 ENGINE POWER TURBINE SPEED (%NP) CANNOT BE INCREASED OR DECREASED - Continued

b. If resistance is not as specified, go to **4**.

4. Check for resistance of 124 to 142 ohms between J901-E and J901-F.

a. If resistance is as specified, repair/replace wiring, as required, between (WP 1747 00), then go to **8**:

- P800-18 and P901-G
- P800-19 and P901-E
- P800-20 and P901-H
- P800-21 and P901-F

b. If resistance is not as specified, replace engine speed control box (WP 0500 00). Go to **8**.

5. Push pilot's or copilot's ENG RPM or HH-60A HH-60L upper console ENGINE SPEED TRIM switch to DECR until engine speed control motor stops. Go to 6.

6. Check resistance between:

- P800-18 and P800-20 (2200-2400 ohms)
- P800-19 and P800-20 (240-260 ohms)
- P800-20 and P800-21 (860-900 ohms)

a. If resistance is as specified, troubleshoot engine (Np does not respond to Np demand trim) (TM 1-2840-248-23). Go to **8**.

b. If resistance is as specified, go to **7**.

7. Check for resistance of 2200 to 2400 ohms between J901-G and J901-H.

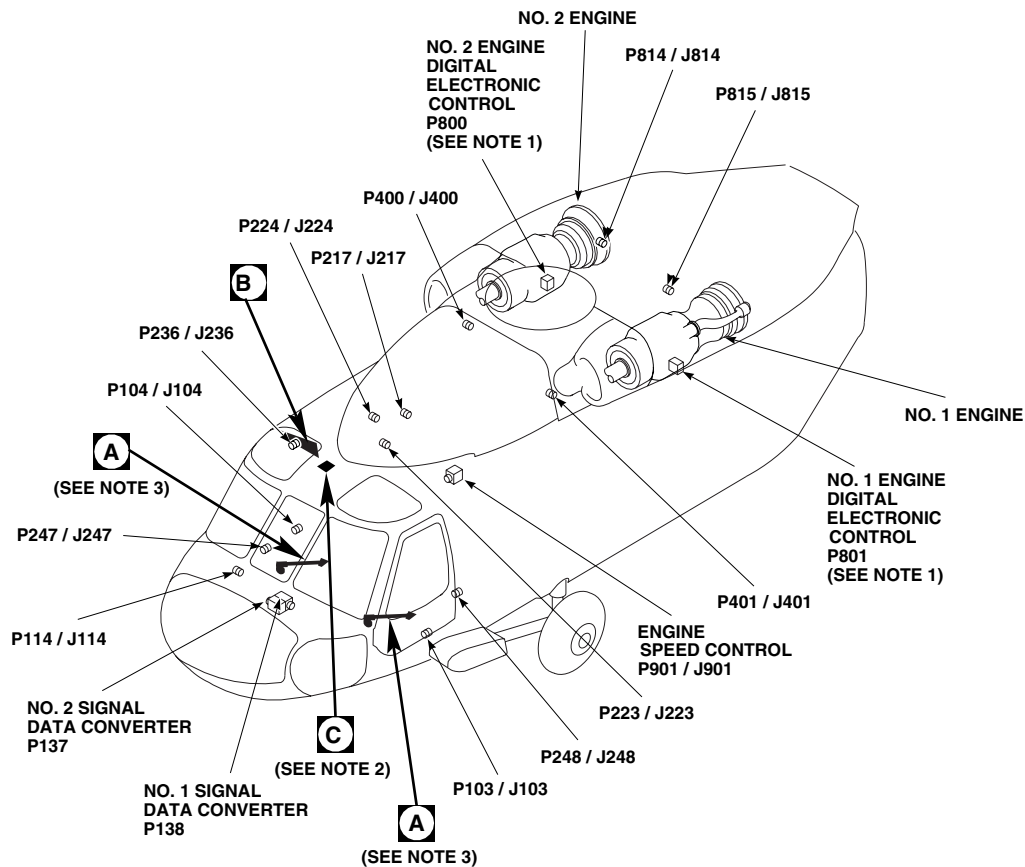
a. If resistance is as specified, repair/replace wiring, as required, between (WP 1747 00), then go to **8**:

- P800-18 and P901-G
- P800-19 and P901-E
- P800-20 and P901-H
- P800-21 and P901-F

b. If resistance is not as specified, replace engine speed control box (WP 0500 00). Go to **8**.

8. Procedure completed.

LOCATION



NOTES

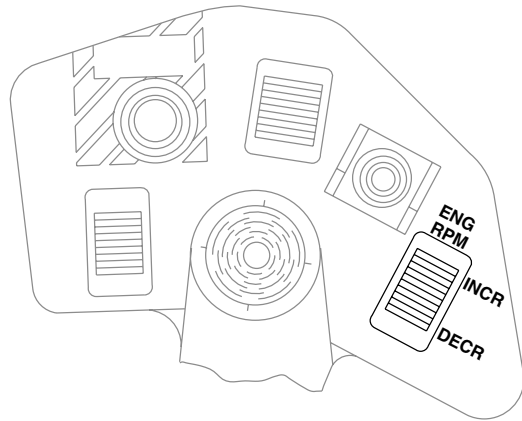
1. **UH-60A** **EH-60A**
HAVE AN ELECTRICAL CONTROL UNIT.
2. **HH-60L** **HH-60A**
3. **UH-60A** **EH-60A** **UH-60L**

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P103 / J103	COCKPIT TUB BL 32 LH, STA 247 (SEE NOTE 3)
P104 / J104	COCKPIT TUB BL 15 RH, STA 247 (SEE NOTE 3)
P114 / J114	COCKPIT BL 8 RH, STA 200
P137	NO. 2 SIGNAL DATA CONVERTER
P138	NO. 1 SIGNAL DATA CONVERTER
P217 / J217	CABIN CEILING BL 0, STA 284
P223 / J223	JUNCTION BOX, MAIN ROTOR PYLON DECK
P224 / J224	JUNCTION BOX, MAIN ROTOR PYLON DECK
P226 / J226	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 23 RH
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 23 RH
P246 / J246	ENGINE SPEED TRIM SWITCH (SEE NOTE 2)

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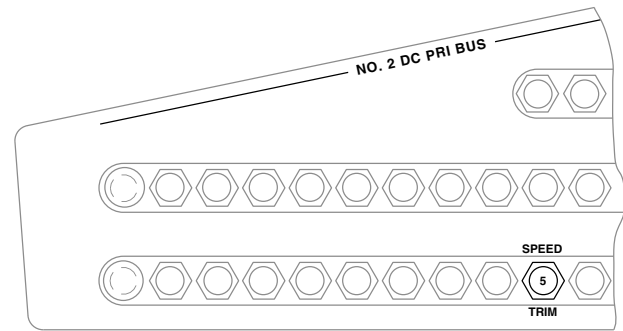
Figure 1. Engine Speed Trim System Location Diagram. (Sheet 1 of 2)

LOCATION - Continued



**PILOT'S AND COPILOT'S
COLLECTIVE STICK**
(SEE NOTE 3)

A

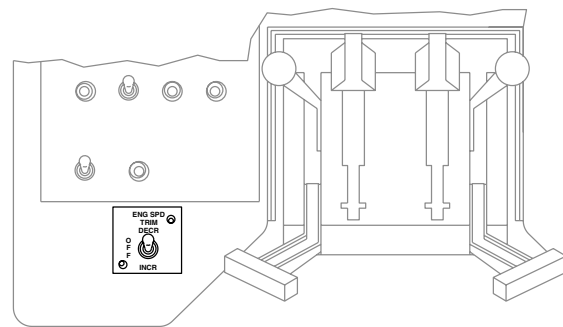


PILOT'S CIRCUIT BREAKER PANEL

B

C

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P247 / J247	CABIN BL 40 RH, STA 248
P248 / J248	CABIN BL 40 LH, STA 248
P400 / J400	MAIN ROTOR PYLON DECK BL 24 RH, STA 318
P401 / J401	MAIN ROTOR PYLON DECK BL 24 LH, STA 318
P800	NO. 2 ENGINE DIGITAL ELECTRONIC CONTROL (SEE NOTE 1)
P801	NO. 1 ENGINE DIGITAL ELECTRONIC CONTROL (SEE NOTE 1)
P814 / J814	FIREWALL DISCONNECT BL 14 LH, STA 373
P815 / J815	FIREWALL DISCONNECT BL 14 LH, STA 373
P901 / J901	ENGINE SPEED CONTROL
P922 / J922	DISCONNECT (SEE NOTE 2)

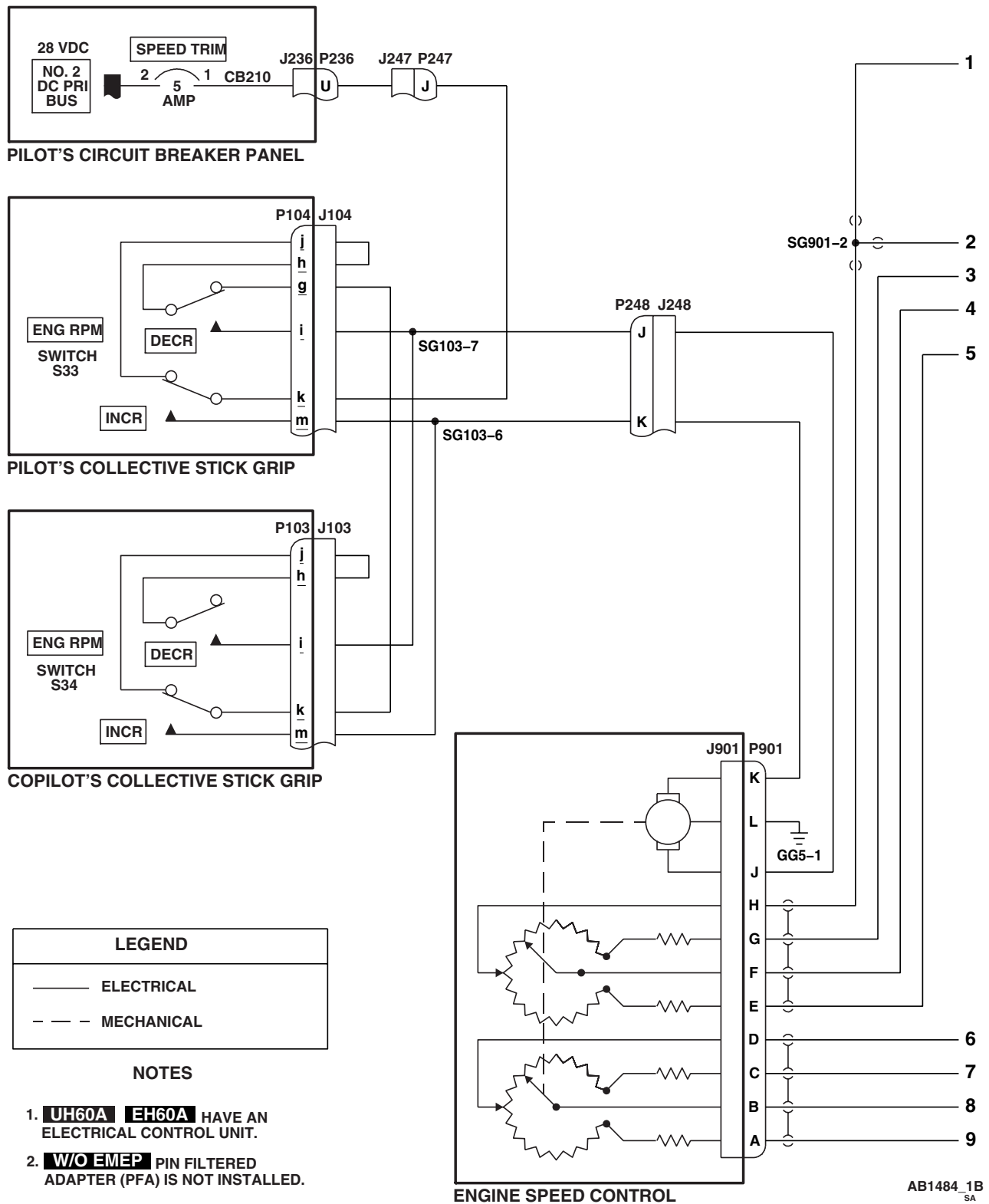


UPPER CONSOLE
(SEE NOTE 2)

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Figure 1. Engine Speed Trim System Location Diagram. (Sheet 2 of 2)

SCHEMATICS

Figure 2. Engine Speed Trim System Schematic Diagram **UH-60A EH-60A UH-60L** . (Sheet 1 of 2)

SCHEMATICS - Continued

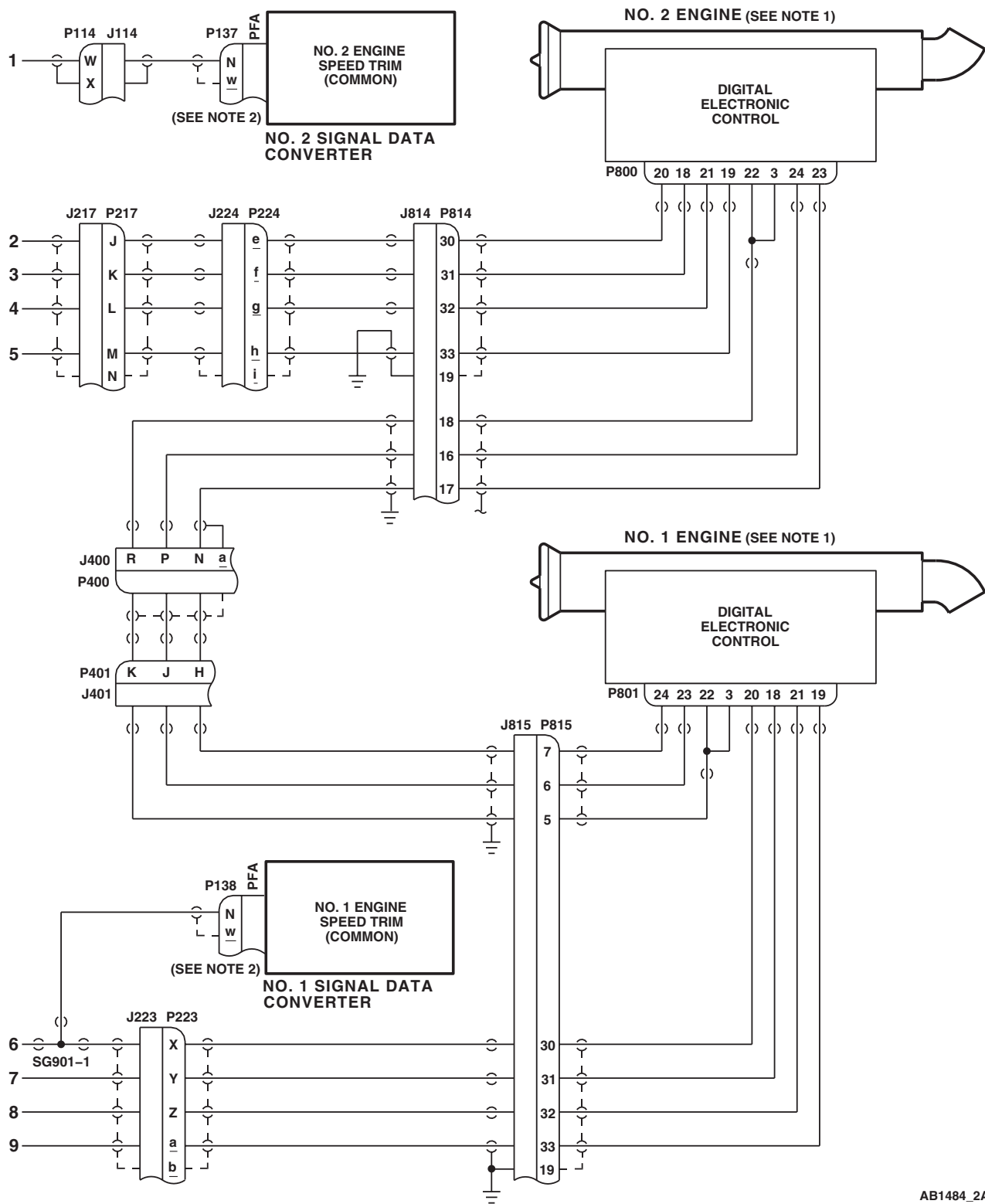
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Figure 2. Engine Speed Trim System Schematic Diagram UH-60A EH-60A UH-60L . (Sheet 2 of 2)

The diagram illustrates the electrical and mechanical connections for the engine speed control system. It is divided into three main sections: the Pilot's Circuit Breaker Panel, the Upper Console, and the Engine Speed Control unit.

Pilot's Circuit Breaker Panel: This panel contains a 28 VDC NO. 2 DC PRI BUS and a SPEED TRIM circuit breaker (CB210) rated at 5 AMP. The circuit breaker is connected to a terminal block (J236) with a terminal labeled U.

Upper Console: This console contains an ENGINE SPEED TRIM control with three positions: DECR (1), COM (2), and INCR (3). These positions are connected to a terminal block (J246) with terminals X, V, and W respectively. The terminal block is also connected to a terminal block (J922) with terminals U and T.

Engine Speed Control: This unit contains a motor and two potentiometers. The motor is connected to a terminal block (J901) with terminals K, L, and J. The potentiometers are connected to a terminal block (P901) with terminals H, G, F, E, D, C, B, and A. The terminal block is also connected to a terminal block (SG901-2) with terminals 1, 2, 3, 4, and 5.

Legend:

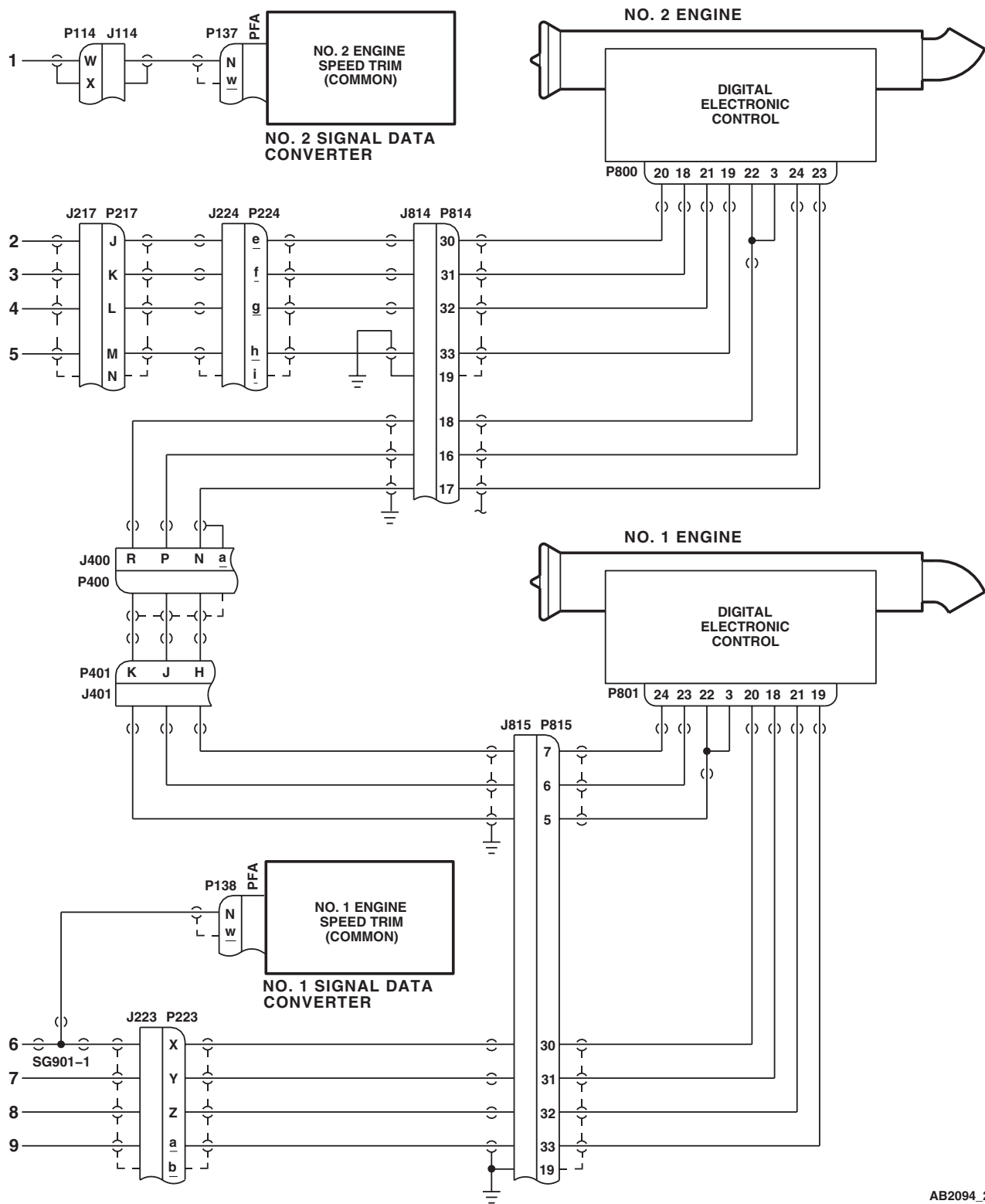
- ELECTRICAL
- - - MECHANICAL

Wiring Details:

- The 28 VDC NO. 2 DC PRI BUS is connected to the SPEED TRIM circuit breaker (CB210).
- The SPEED TRIM circuit breaker (CB210) is connected to terminal U of terminal block J236.
- Terminal U of J236 is connected to terminal X of terminal block J246.
- Terminal X of J246 is connected to terminal U of terminal block J922.
- Terminal V of J246 is connected to terminal V of terminal block J922.
- Terminal W of J246 is connected to terminal T of terminal block J922.
- Terminal U of J922 is connected to terminal K of terminal block J901.
- Terminal T of J922 is connected to terminal J of terminal block P901.
- Terminal K of J901 is connected to terminal H of P901.
- Terminal L of J901 is connected to terminal G of P901.
- Terminal J of P901 is connected to terminal F of P901.
- Terminal H of P901 is connected to terminal E of P901.
- Terminal G of P901 is connected to terminal D of P901.
- Terminal F of P901 is connected to terminal C of P901.
- Terminal E of P901 is connected to terminal B of P901.
- Terminal D of P901 is connected to terminal A of P901.
- Terminal C of P901 is connected to terminal 1 of SG901-2.
- Terminal B of P901 is connected to terminal 2 of SG901-2.
- Terminal A of P901 is connected to terminal 3 of SG901-2.
- Terminal 4 of SG901-2 is connected to terminal 4 of the engine speed control unit.
- Terminal 5 of SG901-2 is connected to terminal 5 of the engine speed control unit.

Figure 3. Engine Speed Trim System Schematic Diagram **HH-60A HH-60L**. (Sheet 1 of 2)

SCHEMATICS - Continued

AB2094_2
SAFigure 3. Engine Speed Trim System Schematic Diagram **HH-60A HH-60L** . (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL**ENGINE SYSTEM****ENGINE CHIP DETECTOR SYSTEM**

This WP supersedes WP 0077 00, dated 17 April 2006.

INITIAL SETUP:**Test Equipment**

Multimeter, AN/PSM-45A

WP 0090 00

WP 0100 00

WP 0104 00

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Electrical Repairer Toolkit, SC 5180-99-B06

WP 0824 00

WP 0925 00

WP 1685 00

WP 1747 00

Personnel Required

Aircraft Electrician MOS 15F (1)

UH-60 Helicopter Repairer MOS 15T (1)

Equipment Condition

External Electrical Power Available, WP 1685 00
or APU Operational, TM 1-1520-237-10

References

TM 1-1520-237-10

TM 1-2840-248-23

SPECIAL INSTRUCTIONS**NOTE**

See Figure 1 for system location diagram, and Figure 2 for system schematic diagram as an aid in operational/troubleshooting.

The engine chip detector system operational/troubleshooting procedure is subdivided as follows:

- Engine No. 1
- Engine No. 2

ENGINE NO. 1 OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step****NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
NO. 1 ENG WARN LTS	Copilot's
NO. 2 ENG WARN LTS	Pilot's
CAUT/ADVSY PNL	Upper Console
LIGHTS CAUT/ADVSY	Copilot's
HH-60A HH-60L MFD PILOT	Aux Circuit Breaker Panel
HH-60A HH-60L MFD COPILOT	Aux Circuit Breaker Panel

2. Remove No. 1 engine chip detector (TM 1-2840-248-23).
3. Turn on electrical power. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed).

Indication/Condition

CHIP #1 ENGINE capsule on caution/advisory panel or legend on MFD/caution/advisory panel shall be off.

Corrective Action

1. If Indication/Condition is not as specified, disconnect P803.
 - a. If CHIP #1 ENGINE capsule goes off, replace No. 1 engine chip detector (TM 1-2840-248-23).
 - b. If CHIP #1 ENGINE capsule does not go off, disconnect P117 or **HH-60A HH-60L** P1019R or P1020R.
 - (1) **EMEP** Remove and check pin filtered adapter. Replace if necessary (WP 0925 00).
 - (2) **EMEP** Install pin filtered adapter (WP 0925 00).
 - (a) If CHIP #1 ENGINE capsule goes off, repair/replace wiring between P117 or **HH-60A HH-60L** P1019R or P1020R and P803 (WP 1747 00).
 - (b) If CHIP #1 ENGINE capsule remains on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).

ENGINE NO. 1 OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

4. Position clean screwdriver with an insulated handle on chip detector.

Indication/Condition

CHIP #1 ENGINE capsule on caution/advisory panel or legend on MFD/caution/advisory panel shall go on.

Corrective Action

If Indication/Condition is not as specified, go to CHIP #1 ENGINE CAPSULE OR LEGEND DOES NOT GO ON, in this work package.

Step

5. Turn MFD switch to OFF.
6. Turn off electrical power.
7. Install No. 1 engine chip detector (TM 1-2840-248-23).

Indication/Condition

No. 1 engine chip detector shall be installed.

Corrective Action

None Required

ENGINE NO. 2 OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step****NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
NO. 1 ENG WARN LTS	Copilot's
NO. 2 ENG WARN LTS	Pilot's
CAUT/ADVSY PNL	Upper Console
LIGHTS CAUT/ADVSY	Copilot's

2. Remove No. 2 engine chip detector (TM 1-2840-248-23).
3. Turn on electrical power.

Indication/Condition

CHIP #2 ENGINE capsule on caution/advisory panel or legend on MFD/caution/advisory panel shall be off.

ENGINE NO. 2 OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition is not as specified, disconnect P802.
 - a. If CHIP #2 ENGINE capsule goes off, replace No. 2 engine chip detector (TM 1-2840-248-23).
 - b. If CHIP #2 ENGINE capsule does not go off, disconnect P117 or **HH-60A HH-60L** P1019R or P1020R **■**.
 - (1) **EMEP** Remove and check pin filtered adapter. Replace if necessary (WP 0925 00). **■**
 - (2) **EMEP** Install pin filtered adapter (WP 0925 00). **■**
 - (a) If CHIP #2 ENGINE capsule goes off, repair/replace wiring between P117 or **HH-60A HH-60L** P1019R or P1020R **■** and P802 (WP 1747 00).
 - (b) If CHIP #2 ENGINE capsule remains on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).

Step

4. Position clean screwdriver with an insulated handle on chip detector.

Indication/Condition

CHIP #2 ENGINE capsule on caution/advisory panel or legend on MFD/caution/advisory panel shall go on.

Corrective Action

If Indication/Condition is not as specified, go to CHIP #2 ENGINE CAPSULE OR LEGEND DOES NOT GO ON, in this work package.

Step

5. Turn MFD switch to OFF.
6. Turn off electrical power.
7. Install No. 2 engine chip detector (TM 1-2840-248-23).

Indication/Condition

No. 2 engine chip detector shall be installed.

Corrective Action

None Required

CHIP #1 ENGINE CAPSULE OR LEGEND DOES NOT GO ON**SYMPTOM**

CHIP #1 ENGINE capsule or legend does not go on.

MALFUNCTION

CHIP #1 ENGINE capsule or legend does not go on.

CORRECTIVE ACTION

1. Hold caution/advisory panel or **HH-60A HH-60L** instrument panel INDICATOR LTS **■** BRT/DIM-TEST switch to TEST. Go to 2.
2. Check that CHIP #1 ENGINE capsule or legend goes on.

CHIP #1 ENGINE CAPSULE OR LEGEND DOES NOT GO ON - Continued

- a. If capsule or legend goes on, go to **3**.
- b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **20**.
- 3. Install an insulated jumper wire between P803-11 and P803-12. Go to 4.**
- 4. Check that CHIP #1 ENGINE capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **5**.
 - b. If capsule or legend does not go on, go to **7**.
- 5. Remove jumper wire from P803. Go to 6.**
- 6. Troubleshoot engine chip detector (TM 1-2840-248-23). Go to 20.**
- 7. Remove jumper wire from P803. Go to 8.**
- 8. Check helicopter configuration.**
 - a. **EMEP** , go to **9**.
 - b. **W/O EMEP** , go to **11**.
- 9. EMEP Remove and check pin filtered adapters P117 and P118 or HH-60A HH-60L P1019R or P1020R (WP 0925 00).**
 - a. If pin filtered adapters are good, go to **10**.
 - b. If pin filtered adapters are not good, replace pin filtered adapters (WP 0925 00). Go to **20**.
- 10. EMEP Install pin filtered adapters (WP 0925 00). Go to 11 or HH-60A HH-60L 12.**
- 11. Check for 28 vdc between P118-g and ground.**
 - a. If voltage is as specified, go to **12**.
 - b. If voltage is not as specified, go to **19**.
- 12. Install an insulated jumper wire between J117-5 or HH-60A HH-60L J3-6 on MFD and ground. Go to 13.**
- 13. Check that CHIP #1 ENGINE caution capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **14**.
 - b. If capsule or legend does not go on, go to **16**.
- 14. Remove jumper wire from J117 or HH-60A HH-60L J3 on MFD . Go to 15.**
- 15. Repair/replace wiring between P117-5 or HH-60A HH-60L P1019R-6 or P1020R-6 and P803-11 (WP 1747 00). Go to 20.**
- 16. Check continuity between P803-12 and ground.**
 - a. If continuity is present, go to **17**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **20**.
- 17. Remove jumper wire from J117 or HH-60A HH-60L J3 on MFD . Go to 18.**
- 18. Troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 20.**

CHIP #1 ENGINE CAPSULE OR LEGEND DOES NOT GO ON - Continued

19. Check for 28 vdc between NO. 1 ENG WARN LTS circuit breaker terminal 1 and ground.

- a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P118-g (WP 1747 00). Go to **20**.
- b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **20**.

20. Procedure completed.

CHIP #2 ENGINE CAPSULE OR LEGEND DOES NOT GO ON**SYMPTOM**

CHIP #2 ENGINE capsule or legend does not go on.

MALFUNCTION

CHIP #2 ENGINE capsule or legend does not go on.

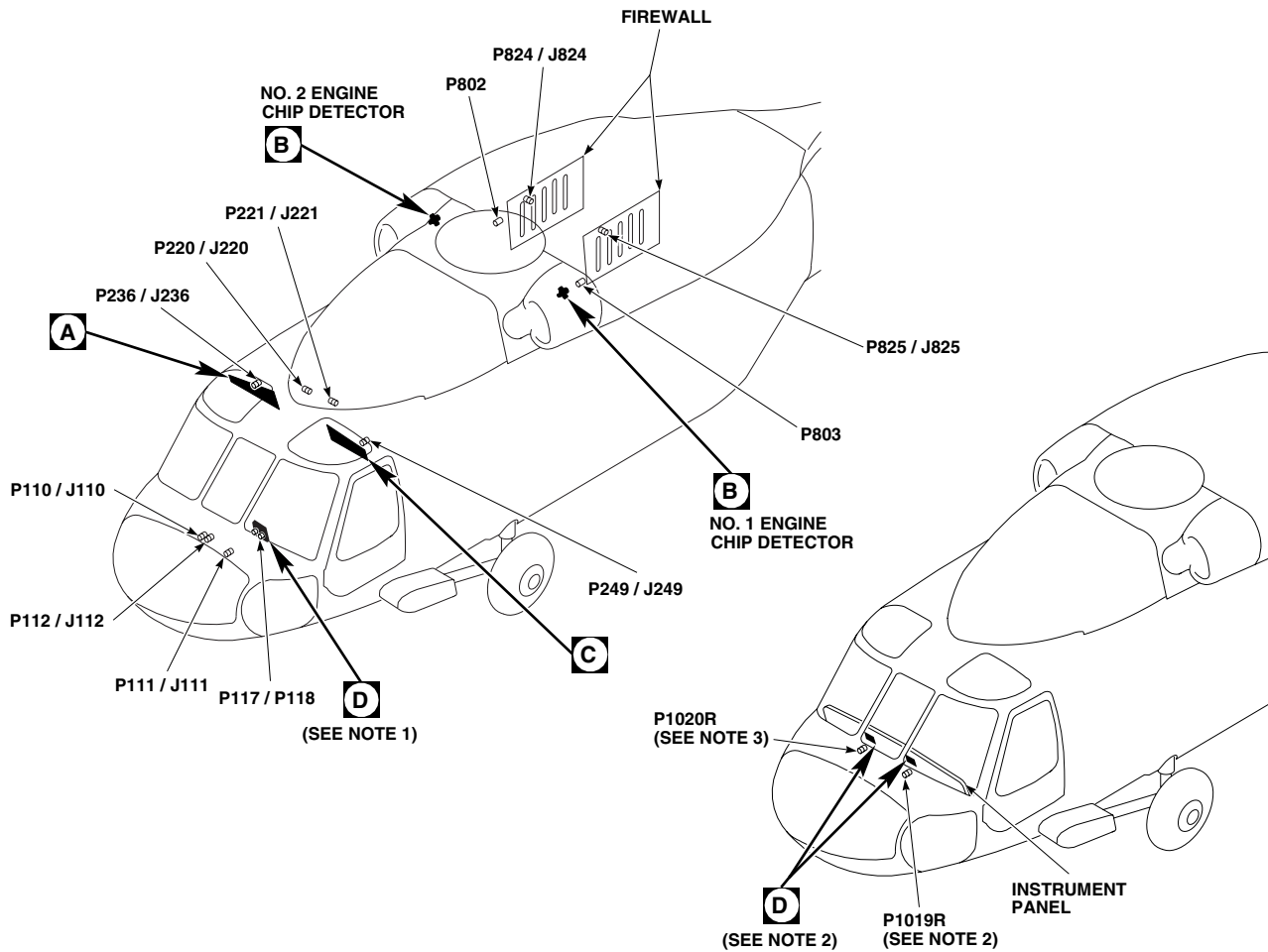
CORRECTIVE ACTION

- 1. Hold caution/advisory panel or HH-60A HH-60L instrument panel INDICATOR LTS BRT/DIM-TEST switch to TEST. Go to 2.**
- 2. Check that CHIP #2 ENGINE capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **3**.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **20**.
- 3. Install an insulated jumper wire between P802-11 and P802-12. Go to 4.**
- 4. Check that CHIP #2 ENGINE capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **5**.
 - b. If capsule or legend does not go on, go to **7**.
- 5. Remove jumper wire from P802. Go to 6.**
- 6. Troubleshoot engine chip detector (TM 1-2840-248-23). Go to 20.**
- 7. Remove jumper wire from P802. Go to 8.**
- 8. Check helicopter configuration.**
 - a. **EMEP**, go to **9**.
 - b. **W/O EMEP**, go to **11**.
- 9. EMEP Remove and check pin filtered adapters P117 and P118 or HH-60A HH-60L P1019R or P1020R (WP 0925 00).**
 - a. If pin filtered adapters are good, go to **10**.
 - b. If pin filtered adapters are not good, replace pin filtered adapters (WP 0925 00). Go to **20**.
- 10. EMEP Install pin filtered adapters (WP 0925 00). Go to 11 or HH-60A HH-60L 12.**
- 11. Check for 28 vdc between P118-h and ground.**
 - a. If voltage is as specified, go to **12**.

CHIP #2 ENGINE CAPSULE OR LEGEND DOES NOT GO ON - Continued

- b. If voltage is not as specified, go to **19**.
- 12. Install insulated jumper wire between J117-53 or HH-60A HH-60L J3-7 on MFD and ground. Go to 13.**
- 13. Check that CHIP #2 ENGINE capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **14**.
 - b. If capsule or legend does not go on, go to **16**.
- 14. Remove jumper wire from J117 or HH-60A HH-60L J3 on MFD . Go to 15.**
- 15. Repair/replace wiring between P117-53 or HH-60A HH-60L P1019R-7 or P1020R-7 and P802-11 (WP 1747 00). Go to 20.**
- 16. Check continuity between P802-12 and ground.**
 - a. If continuity is present, go to **17**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **20**.
- 17. Remove jumper wire from J117 or HH-60A HH-60L J3 on MFD . Go to 18.**
- 18. Troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 20.**
- 19. Check for 28 vdc between NO. 2 ENG WARN LTS circuit breaker terminal 1 and ground.**
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P118-h (WP 1747 00). Go to **20**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **20**.
- 20. Procedure completed.**

LOCATION



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P110 / J110	COCKPIT, BL 7.5 RH, STA 197
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P112 / J112	COCKPIT, BL 6 RH, STA 197
P117 / J117	CAUTION / ADVISORY PANEL (SEE NOTE 1)
P118 / J118	CAUTION / ADVISORY PANEL (SEE NOTE 1)
P220 / J220	JUNCTION BOX, MAIN ROTOR PYLON DECK
P221 / J221	JUNCTION BOX, MAIN ROTOR PYLON DECK

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 23 RH
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 11 LH
P802	NO. 2 ENGINE INTERFACE DISCONNECT
P803	NO. 1 ENGINE INTERFACE DISCONNECT
P824 / J824	FIREWALL DISCONNECT BL 14 RH, STA 338.25
P825 / J825	FIREWALL DISCONNECT BL 14 LH, STA 358.25
P1019R / J3	COPILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 2)
P1020R / J3	PILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 2)



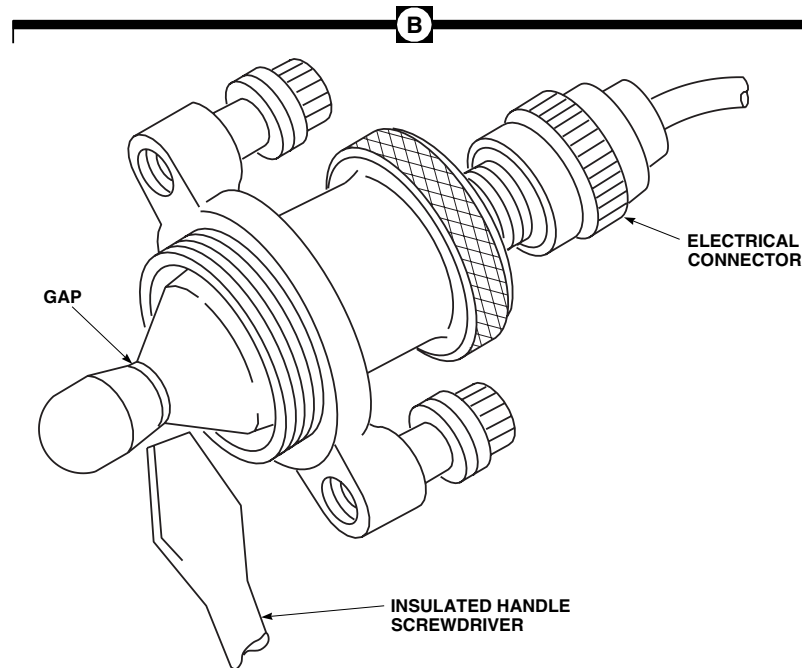
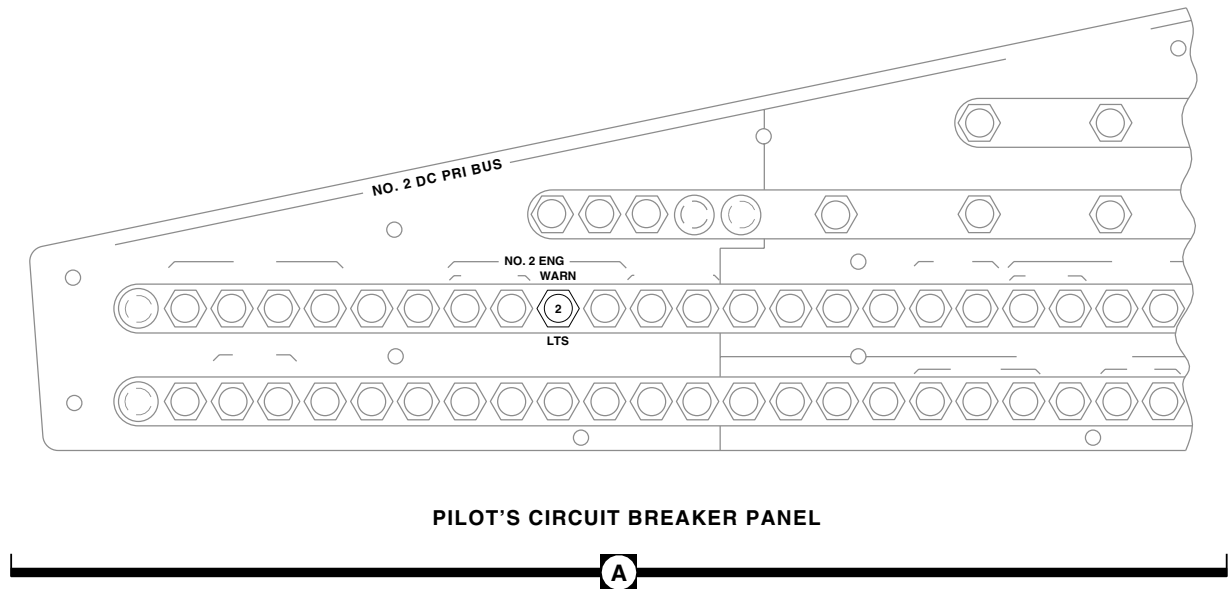
NOTES

1. UH-60A UH-60L EH-60A
2. HH-60A HH-60L

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Figure 1. Engine Chip Detector System Location Diagram. (Sheet 1 of 3)

LOCATION - Continued

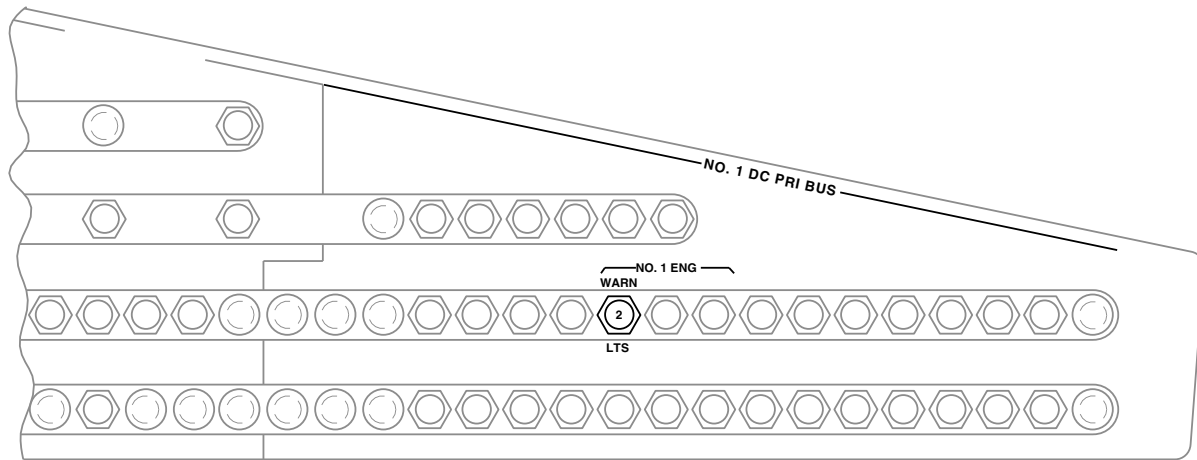


ENGINE CHIP DETECTOR

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Figure 1. Engine Chip Detector System Location Diagram. (Sheet 2 of 3)

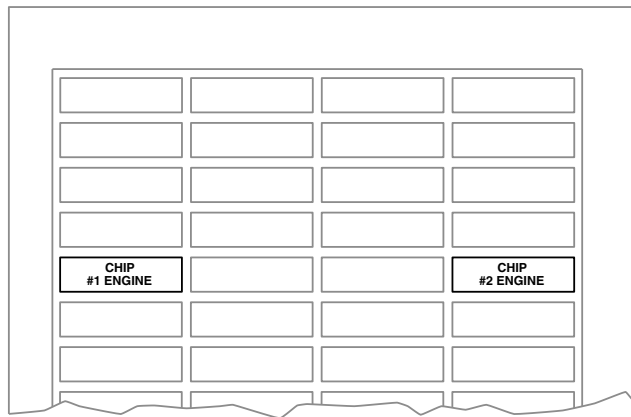
LOCATION - Continued



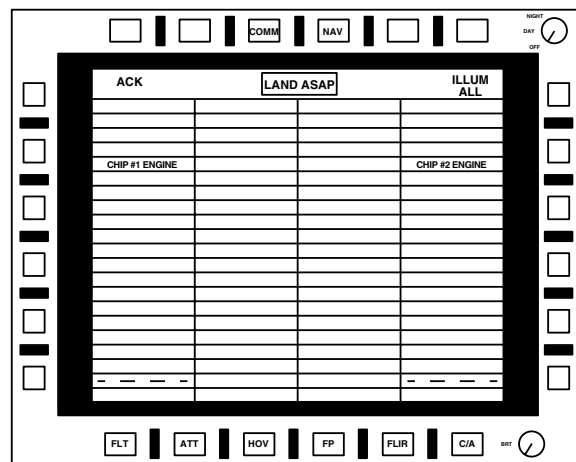
COPLOT'S CIRCUIT BREAKER PANEL

C

D



CAUTION/ADVISORY PANEL
(SEE NOTE 1)



MULTIFUNCTION DISPLAY'S
CAUTION/ADVISORY GRID
(SEE NOTE 2)

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Figure 1. Engine Chip Detector System Location Diagram. (Sheet 3 of 3)

SCHEMATICS

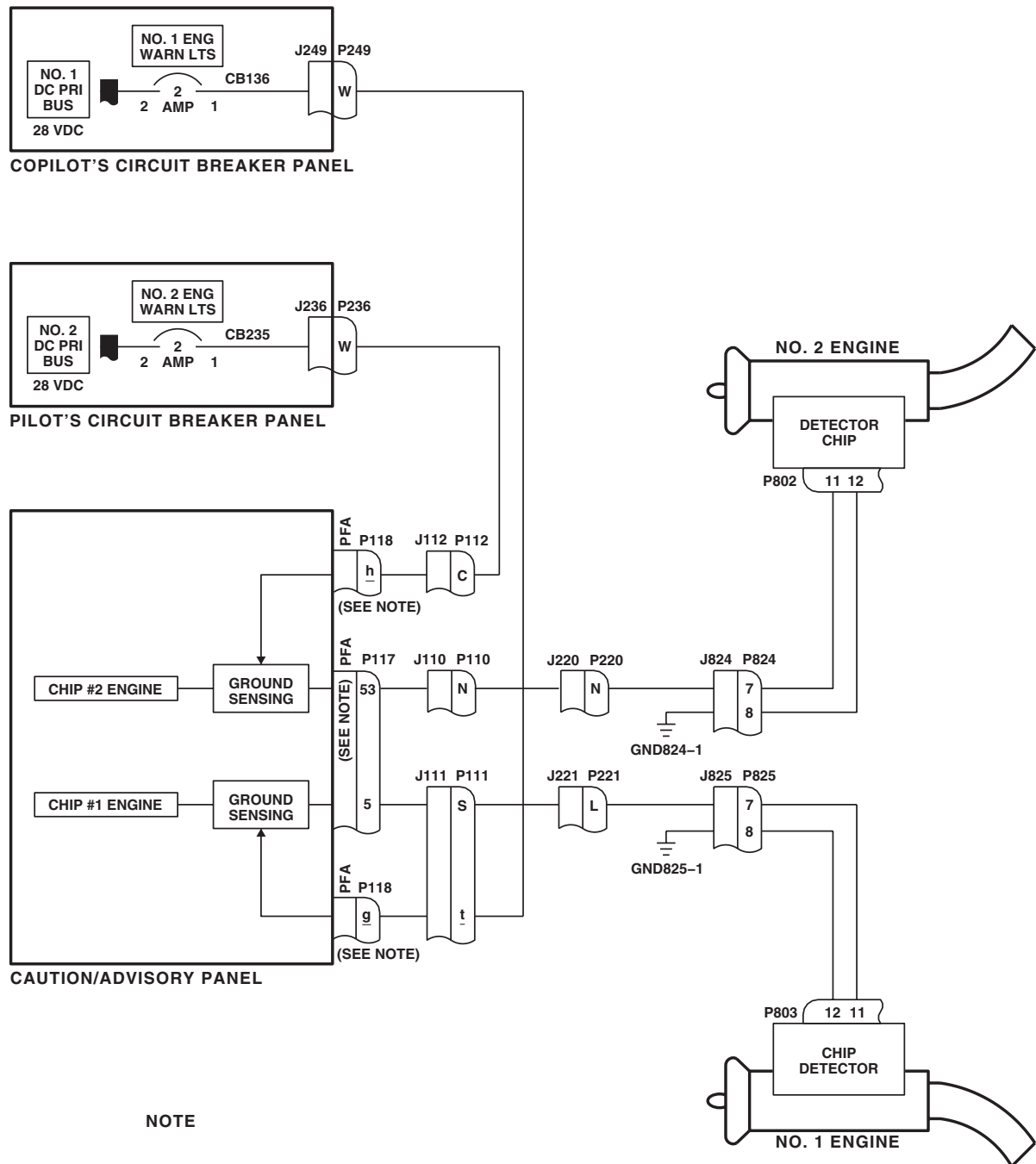
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Figure 2. Engine Chip Detector System Schematic Diagram.

UNIT LEVEL

ENGINE SYSTEM

ENGINE ANTI-ICE SYSTEM

This WP supersedes WP 0078 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment

Multimeter, AN/PSM-45A

WP 0104 00

WP 0121 00

WP 0484 00

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Electrical Repairer Toolkit, SC 5180-99-B06

Insulated Jumper Wire

WP 0824 00

WP 0825 00

WP 0828 00

WP 0926 00

Personnel Required

Aircraft Electrician MOS 15F (1)

UH-60 Helicopter Repairer MOS 15T (1)

WP 1216 00

WP 1217 00

WP 1747 00

References

TM 1-1520-237-10

TM 1-1520-253-10 HH-60A HH-60L

TM 1-2840-248-23

WP 0090 00

WP 0100 00

Equipment Condition

APU Operational, TM 1-1520-237-10

No. 1 and No. 2 Engines Operational,

TM 1-1520-237-10

No. 1 and No. 2 Fuel Tanks Serviced,

TM 1-1520-237-10

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for system location diagram and Figure 2 for system schematic diagram as an aid in operational/troubleshooting.

The engine anti-ice system operational/troubleshooting procedure is subdivided as follows:

- No. 1 Engine
- No. 2 Engine

NO. 1 ENGINE OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
CAUT/ADVSY PNL	Upper Console
LIGHTS CAUT/ADVSY	Copilot's
NO. 1 ENG ANTI-ICE	Copilot's
NO. 1 ENG ANTI-ICE WARN	Copilot's
NO. 1 ENG START	Upper Console
HH-60A HH-60L MFD PILOT	Aux Circuit Breaker Panel
HH-60A HH-60L MFD COPILOT	Aux Circuit Breaker Panel

2. Make sure upper console ENG ANTI-ICE NO. 1 switch is placed OFF.
3. Start APU (TM 1-1520-237-10 or **HH-60A HH-60L** TM 1-1520-253-10). **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed).
4. With qualified personnel at controls, start No. 1 engine and operate up to ground idle speed (TM 1-1520-237-10 or **HH-60A HH-60L** TM 1-1520-253-10).

Indication/Condition

- a. #1 ENG ANTI-ICE ON capsule or legend shall be on.
- b. #1 ENG INLET ANTI-ICE ON capsule or legend shall be off.

Corrective Action

1. If Indication/Condition a. is not as specified, go to #1 ENG ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO ON WHEN ENGINE IS AT GROUND IDLE SPEED, in this work package.
2. If Indication/Condition b. is not as specified, go to #1 ENG INLET ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO OFF WHEN ENGINE INLET IS COOLED DOWN BELOW 200°F (93.9°C), OR GOES ON WHEN ENGINE IS AT GROUND IDLE SPEED, in this work package.

Step

5. Advance No. 1 engine controls quadrant ENG POWER CONT lever to FLY.

NO. 1 ENGINE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

6. Pull collective stick up as required to get Ng SPEED 1 above 90%.

Indication/Condition

#1 ENG ANTI-ICE ON capsule or legend shall go off when Ng SPEED 1 is above 90%.

Corrective Action

If Indication/Condition is not as specified, replace No. 1 engine anti-ice start bleed valve (TM 1-2840-248-23).

Step

7. Place upper console ENG ANTI-ICE NO. 1 switch to ON.

Indication/Condition

- a. #1 ENG ANTI-ICE ON capsule or legend shall go on immediately.
- b. No. 1 engine TGT shall increase.
- c. No. 1 engine inlet anti-ice valve shall open, allowing engine bleed-air to heat up engine inlet.

NOTE

- When operating at outside air temperatures above 39.2° to 55.4°F (4° to 13°C), #1 ENG INLET ANTI-ICE ON capsule or legend may or may not come on when ENG ANTI-ICE NO. 1 switch is ON.
 - If ambient temperature is above 13°C (55.4°F), the #1 ENG INLET ANTI-ICE ON capsule or legend will not be on.
- d. When engine inlet temperature is above 200°F (93.3°C), temperature switch shall close and #1 ENG INLET ANTI-ICE ON capsule or legend shall go on.

Corrective Action

If Indication/Conditions are not as specified, go to #1 ENG INLET ANTI-ICE ON CAPSULE DOES NOT GO ON WHEN ANTI-ICE SYSTEM IS USED, in this work package.

Step

8. Place upper console ENG ANTI-ICE NO. 1 switch to OFF.

Indication/Condition

- a. #1 ENG ANTI-ICE ON capsule or legend shall go off immediately.
- b. No. 1 engine TGT shall decrease. #1 ENG INLET ANTI-ICE ON capsule or legend shall go off when engine inlet is cooled down below 200°F (93.3°C).

Corrective Action

1. If Indication/Condition a. is not as specified, go to #1 ENG ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO OFF IMMEDIATELY, in this work package.
2. If Indication/Condition b. is not as specified, go to #1 ENG INLET ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO OFF WHEN ENGINE INLET IS COOLED DOWN BELOW 200°F (93.9°C), OR GOES ON WHEN ENGINE IS AT GROUND IDLE SPEED, in this work package.

Step

9. Lower copilot's collective stick.

NO. 1 ENGINE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

#1 ENG ANTI-ICE ON capsule or legend shall go on.

Corrective Action

If Indication/Condition is not as specified, go to #1 ENG ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO ON WHEN ENGINE IS AT GROUND IDLE SPEED, in this work package.

Step

10. Shut down No. 1 engine (TM 1-1520-237-10 or **HH-60A HH-60L** TM 1-1520-253-10 **■**).
11. Shut down APU (TM 1-1520-237-10 or **HH-60A HH-60L** TM 1-1520-253-10 **■**).

Indication/Condition

APU shall shut down.

Corrective Action

None Required

**NO. 2 ENGINE OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**
Step**NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
CAUT/ADVSY PNL	Upper Console
LIGHTS CAUT/ADVSY	Copilot's
NO. 1 ENG START	Upper Console
NO. 2 ENG ANTI-ICE	Pilot's
NO. 2 ENG ANTI-ICE WARN	Pilot's
HH-60A HH-60L MFD PILOT	Aux Circuit Breaker Panel
HH-60A HH-60L MFD COPILOT	Aux Circuit Breaker Panel

2. Make sure upper console ENG ANTI-ICE NO. 2 switch is placed OFF.
3. Start APU (TM 1-1520-237-10 or **HH-60A HH-60L** TM 1-1520-253-10 **■**). **HH-60A HH-60L** Turn pilot's or

NO. 2 ENGINE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed). ■

4. With qualified personnel at controls, start No. 2 engine and operate up to ground idle speed (TM 1-1520-237-10 or **HH-60A HH-60L** TM 1-1520-253-10 ■).

Indication/Condition

- a. #2 ENG ANTI-ICE ON capsule or legend shall be on.
- b. #2 ENG INLET ANTI-ICE ON capsule or legend shall be off.

Corrective Action

1. If Indication/Condition a. is not as specified, go to #2 ENG ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO ON WHEN ENGINE IS AT GROUND IDLE SPEED, in this work package.
2. If Indication/Condition b. is not as specified, go to #2 ENG INLET ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO OFF WHEN ENGINE INLET IS COOLED DOWN BELOW 200°F (93.3°C), OR GOES ON WHEN ENGINE IS AT GROUND IDLE SPEED, in this work package.

Step

5. Advance No. 2 engine controls quadrant ENG POWER CONT lever to FLY.
6. Pull collective stick up as required to get Ng SPEED 2 above 90%.

Indication/Condition

#2 ENG ANTI-ICE ON capsule or legend shall go off when Ng SPEED 2 is above 90%.

Corrective Action

If Indication/Condition is not as specified, replace No. 2 engine anti-ice start bleed valve (TM 1-2840-248-23).

Step

7. Place upper console ENG ANTI-ICE NO. 2 switch to ON.

Indication/Condition

- a. #2 ENG ANTI-ICE ON capsule or legend shall go on immediately.
- b. No. 2 engine TGT shall increase.
- c. No. 2 engine inlet anti-ice valve shall open, allowing engine bleed-air to heat up engine inlet.

NOTE

- When operating at outside air temperatures above 39.2° to 55.4°F (4° to 13°C), #2 ENG INLET ANTI-ICE ON capsule or legend may or may not come on when ENG ANTI-ICE NO. 2 switch is ON.
 - If ambient temperature is above 13°C (55.4°F), the #2 ENG INLET ANTI-ICE ON capsule or legend will not go on.
- d. When engine inlet temperature is above 200°F (93.3°C), temperature switch shall close and #2 ENG INLET ANTI-ICE ON capsule or legend shall go on.

Corrective Action

If Indication/Conditions are not as specified, go to #2 ENG INLET ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO ON WHEN ANTI-ICE SYSTEM IS USED, in this work package.

NO. 2 ENGINE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

8. Place upper console ENG ANTI-ICE NO. 2 switch to OFF.

Indication/Condition

- a. #2 ENG ANTI-ICE ON capsule or legend shall go off immediately.
- b. No. 2 engine TGT shall decrease.
- c. #2 ENG INLET ANTI-ICE ON advisory capsule or legend shall go off when engine inlet is cooled down below 200°F (93.3°C).

Corrective Action

1. If Indication/Condition a. is not as specified, go to #2 ENG ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO OFF IMMEDIATELY, in this work package.
2. If Indication/Condition b. is not as specified, go to #2 ENG INLET ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO OFF WHEN ENGINE INLET IS COOLED DOWN BELOW 200°F (93.3°C), OR GOES ON WHEN ENGINE IS AT GROUND IDLE SPEED, in this work package.

Step

9. Lower pilot's collective stick.

Indication/Condition

#2 ENG ANTI-ICE ON capsule or legend shall go on.

Corrective Action

If Indication/Condition is not as specified, go to #2 ENG ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO ON WHEN ENGINE IS AT GROUND IDLE SPEED, in this work package.

Step

10. Turn MFD switch to OFF.
11. Shut down No. 2 engine (TM 1-1520-237-10 or **HH-60A HH-60L** TM 1-1520-253-10 **■**).
12. Shut down APU (TM 1-1520-237-10 or **HH-60A HH-60L** TM 1-1520-253-10 **■**).

Indication/Condition

APU shall shut down.

Corrective Action

None Required

#1 ENG ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO ON WHEN ENGINE IS AT GROUND IDLE SPEED**SYMPTOM**

#1 ENG ANTI-ICE ON capsule or legend does not go on.

MALFUNCTION

#1 ENG ANTI-ICE ON capsule or legend does not go on when engine is at ground idle speed.

CORRECTIVE ACTION

1. Hold caution/advisory panel **HH-60A HH-60L** instrument panel INDICATOR LTS **■** BRT/DIM-TEST switch to TEST. Go to 2.
2. Check if #1 ENGINE ANTI-ICE ON capsule or legend goes on.
 - a. If capsule or legend goes on, go to 3.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 16.
3. Install an insulated jumper wire between P803-3 and P803-4. Go to 4.
4. Check if #1 ENG ANTI-ICE ON capsule or legend goes on.
 - a. If capsule or legend goes on, go to 5.
 - b. If capsule or legend does not go on, go to 7.
5. Remove jumper wire from P803. Go to 6.
6. Troubleshoot No. 1 engine anti-ice start bleed valve (TM 1-2840-248-23). Go to 16.
7. Check helicopter configuration.
 - a. **EMEP** , go to 8.
 - b. **W/O EMEP** , go to 10.
8. **EMEP** Remove and check pin filtered adapter P118 or **HH-60A HH-60L** P1019R or **■** P1020R(WP 0926 00).
 - a. If pin filtered adapter is good, go to 9.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0926 00). Go to 16.
9. **EMEP** Install pin filtered adapter (WP 0926 00). Go to 10.
10. Check for 28 vdc between P118-A or **HH-60A HH-60L** P1019R-83 or P1020R-83 **■** and ground.
 - a. If voltage is as specified, go to 11.
 - b. If voltage is not as specified, go to 13.
11. Remove jumper wire from P803. Go to 12.
12. Troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 16.
13. Remove jumper wire from P803. Go to 14.
14. Check for 28 vdc between P803-3 and ground.

#1 ENG ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO ON WHEN ENGINE IS AT GROUND IDLE SPEED - Continued

- a. If voltage is as specified, repair/replace wiring between P803-4 and P118-A or **HH-60A HH-60L** P1019R-83 or P1020R-83 **■** (WP 1747 00). Go to **16**.
 - b. If voltage is not as specified, go to **15**.
- 15. Check for 28 vdc between NO. 1 ENG ANTI-ICE WARN circuit breaker terminal 1 and ground.**
- a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P803-3 (WP 1747 00). Go to **16**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **16**.
- 16. Procedure completed.**

#1 ENG INLET ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO ON WHEN ANTI-ICE SYSTEM IS USED**SYMPTOM**

#1 ENG INLET ANTI-ICE ON capsule or legend does not go on.

MALFUNCTION

#1 ENG INLET ANTI-ICE ON capsule or legend does not go on when anti-ice system is used.

CORRECTIVE ACTION

- 1. Visually check rubber hoses, clamps, and tubing for possible damage between engine inlet and engine anti-ice valve.**
 - a. If damage is present, repair/replace clamps, tubing, or rubber hoses as required (WP 0484 00). Go to **17**.
 - b. If damage is not present, go to **2**.
- 2. Hold caution/advisory panel **HH-60A HH-60L** instrument panel INDICATOR LTS **■** BRT/DIM-TEST switch to TEST. Go to 3.**
- 3. Check if #1 ENG INLET ANTI-ICE ON capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **4**.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **17**.
- 4. Install an insulated jumper wire between P821-A and P821-B. Go to 5.**
- 5. Check if #1 ENG INLET ANTI-ICE ON capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **6**.
 - b. If capsule or legend does not go on, go to **8**.
- 6. Remove jumper wire from P821. Go to 7.**
- 7. Replace No. 1 engine inlet temperature switch (WP 1217 00). Go to 17.**
- 8. Check helicopter configuration.**
 - a. **EMEP** **■**, go to **9**.
 - b. **W/O EMEP** **■**, go to **11**.

#1 ENG INLET ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO ON WHEN ANTI-ICE SYSTEM IS USED - Continued

9. **EMEP** Remove and check pin filtered adapter P118 or **HH-60A HH-60L** P1019R or P1020R (WP 0926 00).
 - a. If pin filtered adapter is good, go to 10.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0926 00). Go to 17.
10. **EMEP** Install pin filtered adapter (WP 0926 00). Go to 11.
11. Check for 28 vdc between P118-D or **HH-60A HH-60L** P1019R-29 or P1020R-29 and ground.
 - a. If voltage is as specified, go to 12.
 - b. If voltage is not as specified, go to 14.
12. Remove jumper wire from P821. Go to 13.
13. Troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 17.
14. Remove jumper wire from P821. Go to 15.
15. Check for 28 vdc between P821-A and ground.
 - a. If voltage is as specified, repair/replace wiring between P118-D or **HH-60A HH-60L** P1019R-29 or P1020R-29 and P821-B (WP 1747 00). Go to 17.
 - b. If voltage is not as specified, go to 16.
16. Check for 28 vdc between NO. 1 ENG ANTI-ICE WARN circuit breaker terminal 1 and ground.
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P821-A (WP 1747 00). Go to 17.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to 17.
17. Procedure completed.

#1 ENG ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO OFF IMMEDIATELY**SYMPTOM**

#1 ENG ANTI-ICE ON capsule or legend does not go off.

MALFUNCTION

#1 ENG ANTI-ICE ON capsule or legend does not go off immediately.

CORRECTIVE ACTION

1. With ENG ANTI-ICE ON NO. 1 switch placed to OFF, check for 28 vdc between P803-1 and P803-2.
 - a. If voltage is as specified, troubleshoot engine anti-ice/start bleed valve (TM 1-2840-248-23). Go to 3.
 - b. If voltage is not as specified, go to 2.
2. Check for 28 vdc between P803-1 and ground.
 - a. If voltage is as specified, repair/replace ground wire on P803-2 (WP 1747 00). Go to 3.

#1 ENG ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO OFF IMMEDIATELY - Continued

- b. If voltage is not as specified, repair/replace wiring between P225-N and P803-1 (WP 1747 00). Go to **3**.

3. Procedure completed.**#1 ENG INLET ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO OFF WHEN ENGINE INLET IS COOLED DOWN BELOW 200°F (93.3°C), OR GOES ON WHEN ENGINE IS AT GROUND IDLE SPEED****SYMPTOM**

#1 ENG INLET ANTI-ICE ON capsule or legend does not go off, or goes on.

MALFUNCTION

#1 ENG INLET ANTI-ICE ON capsule or legend does not go off when engine inlet is cooled down below 200°F (93.3°C), or goes on when engine is at ground idle speed.

CORRECTIVE ACTION

- 1. With ENG ANTI-ICE NO. 1 switch placed OFF, check for 28 vdc between P820-2 and P820-5.**
 - a. If voltage is as specified, go to **2**.
 - b. If voltage is not as specified, go to **3**.
- 2. With No. 1 engine inlet cooled down, check continuity between P821-A and P821-B of inlet temperature switch.**
 - a. If continuity is present, replace No. 1 engine inlet temperature switch (WP 1217 00). Go to **13**.
 - b. If continuity is not present, replace No. 1 engine inlet anti-ice valve (WP 1216 00). Go to **13**.
- 3. Check for 28 vdc between P820-2 and ground.**
 - a. If voltage is as specified, repair/replace wiring between P820-5 and ground (WP 1747 00). Go to **13**.
 - b. If voltage is not as specified, go to **4**.
- 4. Install an insulated jumper wire between P242-R and P242-S. Go to 5.**
- 5. Check for 28 vdc between P820-2 and ground.**
 - a. If voltage is as specified, go to **6**.
 - b. If voltage is not as specified, go to **8**.
- 6. Remove jumper wire from P242. Go to 7.**
- 7. Troubleshoot left relay panel (WP 0121 00). Go to 13.**
- 8. Remove jumper wire from P242. Go to 9.**
- 9. Check for 28 vdc between P242-R and ground.**
 - a. If voltage is as specified, repair/replace wiring between P242-S and P820-2 (WP 1747 00). Go to **13**.
 - b. If voltage is not as specified, go to **10**.
- 10. Check for 28 vdc between ENG ANTI-ICE NO. 1 switch terminal 1 and ground.**

#1 ENG INLET ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO OFF WHEN ENGINE INLET IS COOLED DOWN BELOW 200°F (93.3°C), OR GOES ON WHEN ENGINE IS AT GROUND IDLE SPEED -
Continued

- a. If voltage is as specified, repair/replace wiring between P242-R and switch terminal 1 (WP 1747 00). Go to **13**.
- b. If voltage is not as specified, go to **11**.
- 11. Check for 28 vdc between NO. 1 ENG ANTI-ICE circuit breaker terminal 1 and ground.**
 - a. If voltage is as specified, go to **12**.
 - b. If voltage is not as specified, repair/replace circuit breaker (WP 0824 00). Go to **13**.
- 12. Check continuity between NO. 1 ENG ANTI-ICE circuit breaker terminal 1 and ENG ANTI-ICE NO. 1 switch terminal 2.**
 - a. If continuity is present, replace switch (WP 0828 00). Go to **13**.
 - b. If continuity is not present, repair/replace wiring to switch (WP 1747 00). Go to **13**.
- 13. Procedure completed.**

#2 ENG ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO ON WHEN ENGINE IS AT GROUND IDLE SPEED

SYMPTOM

#2 ENG ANTI-ICE ON capsule or legend does not go on.

MALFUNCTION

#2 ENG ANTI-ICE ON capsule or legend does not go on when engine is at ground idle speed.

CORRECTIVE ACTION

- 1. Hold caution/advisory panel HH-60A HH-60L instrument panel INDICATOR LTS BRT/DIM-TEST switch to TEST. Go to 2.**
- 2. Check if #2 ENG ANTI-ICE ON capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **3**.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **16**.
- 3. Install an insulated jumper wire between P802-3 and P802-4. Go to 4.**
- 4. Check if #2 ENG ANTI-ICE ON capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **5**.
 - b. If capsule or legend does not go on, go to **7**.
- 5. Remove jumper wire from P802. Go to 6.**
- 6. Troubleshoot No. 2 engine anti-ice start bleed valve (TM 1-2840-248-23). Go to 16.**
- 7. Check helicopter configuration.**
 - a. **EMEP**, go to **8**.
 - b. **W/O EMEP**, go to **10**.
- 8. EMEP Remove and check pin filtered adapter P118 or HH-60A HH-60L P1019R or P1020R (WP 0926 00).**
 - a. If pin filtered adapter is good, go to **9**.

#2 ENG ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO ON WHEN ENGINE IS AT GROUND IDLE SPEED - Continued

- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0926 00). Go to **16**.
9. **EMEP** Install pin filtered adapter (WP 0926 00). Go to 10.
10. Check for 28 vdc between P118-P or **HH-60A HH-60L** P1019R-63 or P1020R-63 and ground.
 - a. If voltage is as specified, go to **11**.
 - b. If voltage is not as specified, go to **13**.
11. Remove jumper wire from P802. Go to 12.
12. Troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 16.
13. Remove jumper wire from P802. Go to 14.
14. Check for 28 vdc between P802-3 and ground.
 - a. If voltage is as specified, repair/replace wiring between P118-P or **HH-60A HH-60L** P1019R-63 or P1020R-63 and P802-4 (WP 1747 00). Go to **16**.
 - b. If voltage is not as specified, go to **15**.
15. Check for 28 vdc between NO. 2 ENG ANTI-ICE WARN circuit breaker terminal 1 and ground.
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P802-3 (WP 1747 00). Go to **16**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **16**.
16. Procedure completed.

#2 ENG INLET ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO ON WHEN ANTI-ICE SYSTEM IS USED

SYMPTOM

#2 ENG INLET ANTI-ICE ON capsule or legend does not go on.

MALFUNCTION

#2 ENG INLET ANTI-ICE ON capsule or legend does not go on when anti-ice system is used.

CORRECTIVE ACTION

1. Visually check rubber hoses, clamps, and tubing for possible damage between engine inlet and engine anti-ice valve.
 - a. If damage is present, repair/replace clamps, tubing, or rubber hoses as required (WP 0484 00). Go to **17**.
 - b. If damage is not present, go to **2**.
2. Hold caution/advisory panel **HH-60A HH-60L** instrument panel INDICATOR LTS BRT/DIM-TEST switch to TEST. Go to 3.
3. Check if #2 ENG INLET ANTI-ICE ON capsule or legend goes on.
 - a. If capsule or legend goes on, go to **4**.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **17**.

#2 ENG INLET ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO ON WHEN ANTI-ICE SYSTEM IS USED - Continued

4. Install an insulated jumper wire between P828-A and P828-B. Go to 5.
5. #2 ENG INLET ANTI-ICE ON capsule or legend should go on.
 - a. If capsule or legend goes on, go to 6.
 - b. If capsule or legend does not go on, go to 8.
6. Remove jumper wire from P828. Go to 7.
7. Replace No. 2 engine inlet temperature switch (WP 1217 00). Go to 17.
8. Check helicopter configuration.
 - a. **EMEP** , go to 9.
 - b. **W/O EMEP** , go to 11.
9. **EMEP** Remove and check pin filtered adapter P118 or **HH-60A HH-60L** P1019R or P1020R (WP 0926 00).
 - a. If pin filtered adapter is good, go to 10.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0926 00). Go to 17.
10. **EMEP** Install pin filtered adapter (WP 0926 00). Go to 11.
11. Check for 28 vdc between P118-J or **HH-60A HH-60L** P1019R-65 or P1020R-65 and ground.
 - a. If voltage is as specified, go to 12.
 - b. If voltage is not as specified, go to 14.
12. Remove jumper wire from P828. Go to 13.
13. Troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 17.
14. Remove jumper wire from P828. Go to 15.
15. Check for 28 vdc between P828-A and ground.
 - a. If voltage is as specified, repair/replace wiring between P118-J or **HH-60A HH-60L** P1019R-65 or P1020R-65 and P828-B (WP 1747 00). Go to 17.
 - b. If voltage is not as specified, go to 16.
16. Check for 28 vdc between NO. 2 ENG ANTI-ICE WARN circuit breaker terminal 1 and ground.
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P828-A (WP 1747 00). Go to 17.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to 17.
17. Procedure completed.

#2 ENG ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO OFF IMMEDIATELY**SYMPTOM**

#2 ENG ANTI-ICE ON capsule or legend does not go off.

MALFUNCTION

#2 ENG ANTI-ICE ON capsule or legend does not go off immediately.

CORRECTIVE ACTION

- 1. With ENG ANTI-ICE NO. 2 switch placed to OFF, check for 28 vdc between P802-1 and P802-2.**
 - a. If voltage is as specified, troubleshoot engine anti-ice start bleed valve (TM 1-2840-248-23). Go to **3**.
 - b. If voltage is not as specified, go to **2**.
- 2. Check for 28 vdc between P802-1 and ground.**
 - a. If voltage is as specified, repair/replace ground wire to P802-2 (WP 1747 00). Go to **3**.
 - b. If voltage is not as specified, repair/replace wiring between P222-R and P802-1 (WP 1747 00). Go to **3**.
- 3. Procedure completed.**

#2 ENG INLET ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO OFF WHEN ENGINE INLET IS COOLED DOWN BELOW 200°F (93.3°C), OR GOES ON WHEN ENGINE IS AT GROUND IDLE SPEED**SYMPTOM**

#2 ENG INLET ANTI-ICE ON capsule or legend does not go off when engine inlet is cooled down below 200°F (93.3°C), or goes on when engine is at ground idle speed.

MALFUNCTION

#2 ENG INLET ANTI-ICE ON capsule or legend does not go off or goes on.

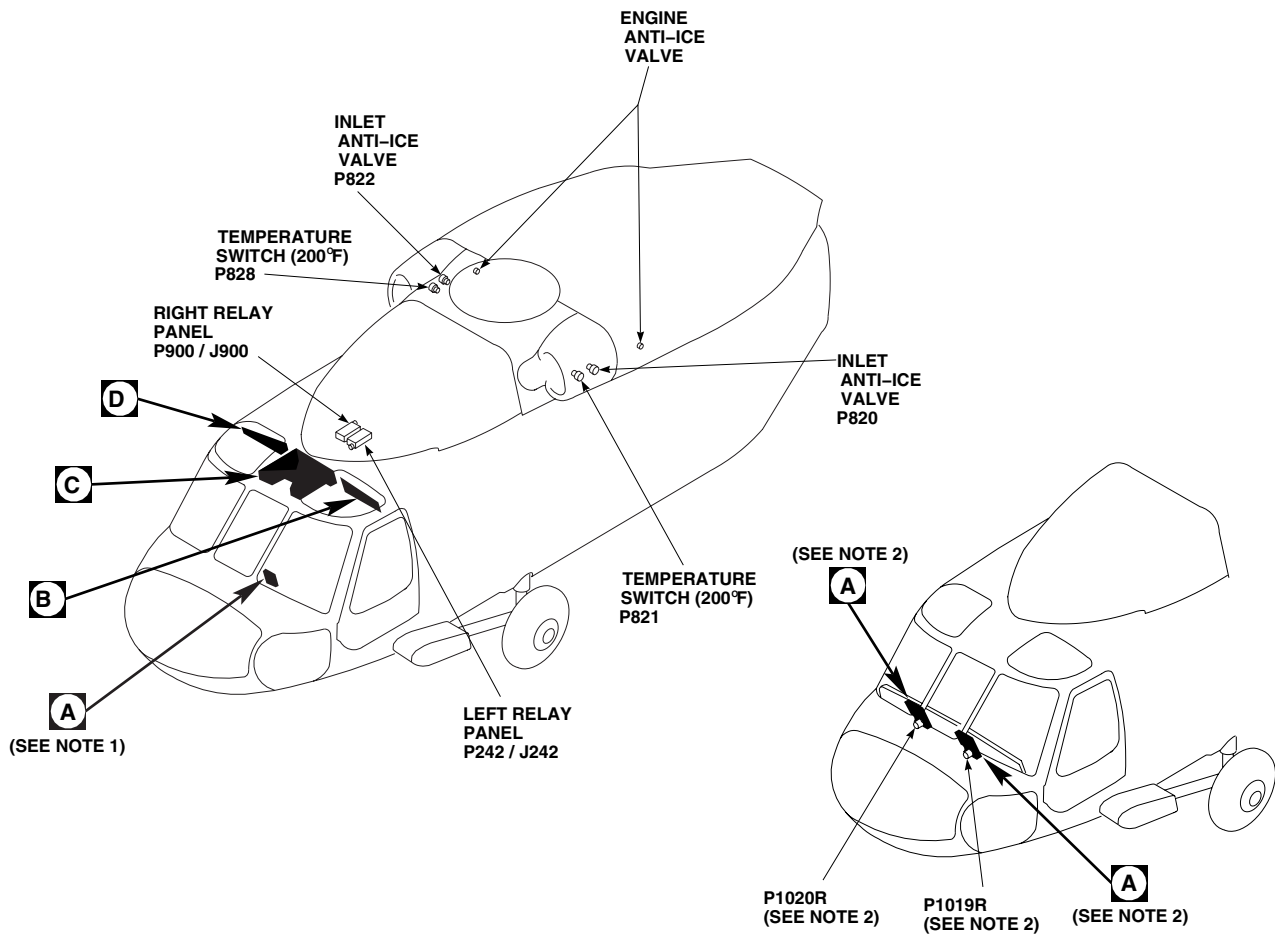
CORRECTIVE ACTION

- 1. With ENG ANTI-ICE NO. 2 switch placed OFF, check for 28 vdc between P822-2 and P822-5.**
 - a. If voltage is as specified, go to **2**.
 - b. If voltage is not as specified, go to **3**.
- 2. With No. 2 engine inlet cooled down, check continuity between P828-A and P828-B of inlet temperature switch.**
 - a. If continuity is present, replace No. 2 engine inlet temperature switch (WP 1217 00). Go to **13**.
 - b. If continuity is not present, replace No. 2 engine inlet anti-ice valve (WP 1216 00). Go to **13**.
- 3. Check for 28 vdc between P822-2 and ground.**
 - a. If voltage is as specified, repair/replace ground wire to P822-5 (WP 1747 00). Go to **13**.
 - b. If voltage is not as specified, go to **4**.
- 4. Install an insulated jumper wire between P900-K and P900-L. Go to 5.**

#2 ENG INLET ANTI-ICE ON CAPSULE OR LEGEND DOES NOT GO OFF WHEN ENGINE INLET IS COOLED DOWN BELOW 200°F (93.3°C), OR GOES ON WHEN ENGINE IS AT GROUND IDLE SPEED -
Continued

- 5. Check for 28 vdc between P822-2 and ground.**
 - a. If voltage is as specified, go to **6**.
 - b. If voltage is not as specified, go to **8**.
- 6. Remove jumper wire from P900. Go to 7.**
- 7. Replace right relay panel (WP 0825 00). Go to 13.**
- 8. Remove jumper wire from P900. Go to 9.**
- 9. Check for 28 vdc between P900-K and ground.**
 - a. If voltage is as specified, repair/replace wiring between P900-L and P822-2 (WP 1747 00). Go to **13**.
 - b. If voltage is not as specified, go to **10**.
- 10. Check for 28 vdc between ENG ANTI-ICE NO. 2 switch terminal 1 and ground.**
 - a. If voltage is as specified, repair/replace wiring between switch terminal 1 and P900-K (WP 1747 00). Go to **13**.
 - b. If voltage is not as specified, go to **11**.
- 11. Check for 28 vdc between NO. 2 ENG ANTI-ICE circuit breaker terminal 1 and ground.**
 - a. If voltage is as specified, go to **12**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **13**.
- 12. Check continuity between NO. 2 ENG ANTI-ICE circuit breaker terminal 1 and ENG ANTI-ICE NO. 2 switch terminal 2.**
 - a. If continuity is present, replace switch (WP 0828 00). Go to **13**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 13. Procedure completed.**

LOCATION



TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P110 / J110	COCKPIT, BL 7.5 RH, STA 197
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P118 / J118	CAUTION / ADVISORY PANEL (SEE NOTE 1)
P126 / J126	COCKPIT CEILING BL 28 RH, STA 243
P127 / J127	COCKPIT CEILING BL 28 LH, STA 243
P220 / J220	JUNCTION BOX, MAIN ROTOR PYLON DECK
P221 / J221	JUNCTION BOX, MAIN ROTOR PYLON DECK

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P222 / J222	JUNCTION BOX, MAIN ROTOR PYLON DECK
P225 / J225	JUNCTION BOX, MAIN ROTOR PYLON DECK
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P231 / J231	CABIN CEILING, BL 9 LH, STA 247
P242 / J242	LH RELAY PANEL
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23.0 LH
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25.0 RH



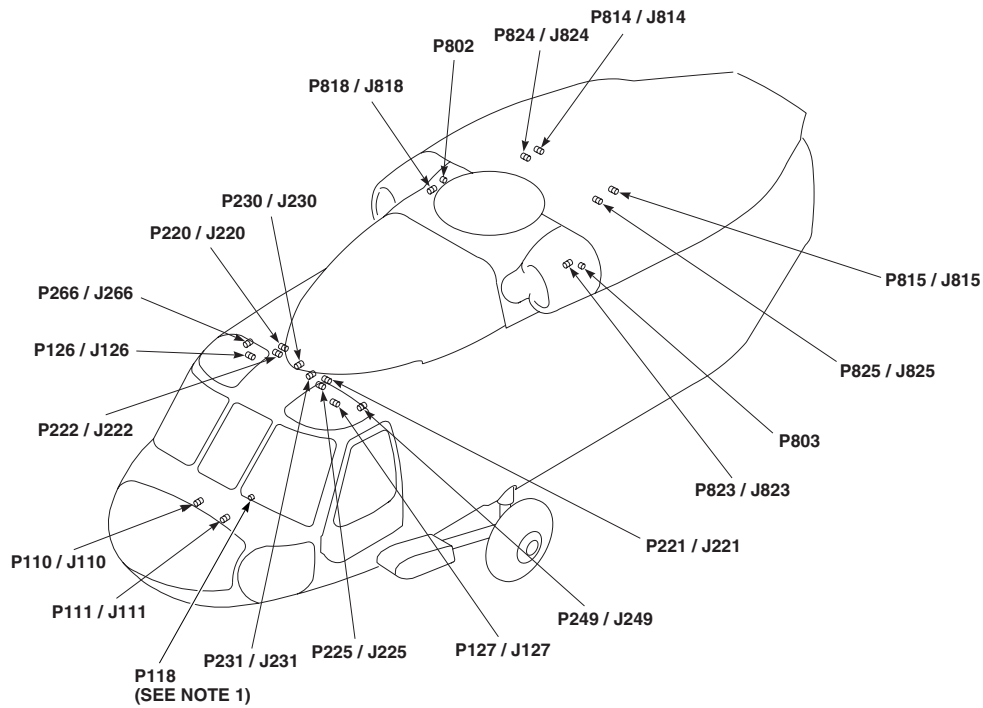
NOTES

1. UH-60A UH-60L EH-60A
2. HH-60A HH-60L

AA2666_1B
SA

Figure 1. Engine Anti-Ice System Location Diagram. (Sheet 1 of 5)

LOCATION - Continued



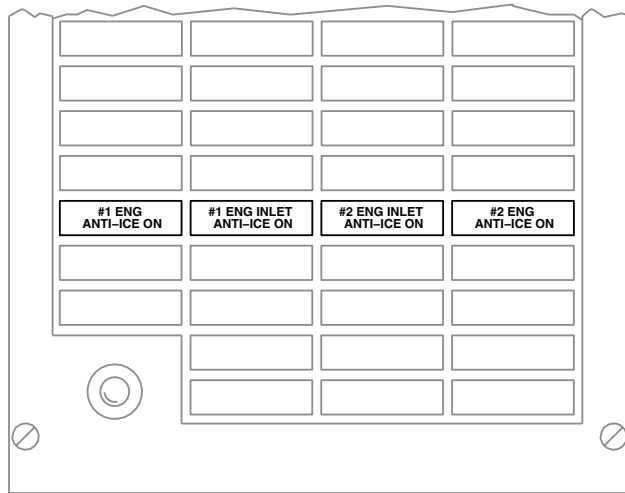
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P802	NO. 2 ENGINE INTERFACE DISCONNECT
P803	NO. 1 ENGINE INTERFACE DISCONNECT
P814 / J814	FIREWALL DISCONNECT BL 14 RH, STA 373
P815 / J815	FIREWALL DISCONNECT BL 14 LH, STA 373
P818 / J818	NO. 2 ENGINE INLET
P820	NO. 1 ENGINE INLET ANTI-ICE VALVE
P821	NO. 1 ENGINE INLET TEMPERATURE SWITCH

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P822	NO. 2 ENGINE INLET ANTI-ICE VALVE
P823 / J823	NO. 1 ENGINE INLET
P824 / J824	FIREWALL DISCONNECT BL 14 RH, STA 358.25
P825 / J825	FIREWALL DISCONNECT BL 14 LH, STA 358.25
P828	NO. 2 ENGINE INLET TEMPERATURE SWITCH
P900 / J900	RH RELAY PANEL
P1019R / J3	COPLOT'S MULTIFUNCTION DISPLAY (SEE NOTE 2)
P1020R / J3	PILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 2)

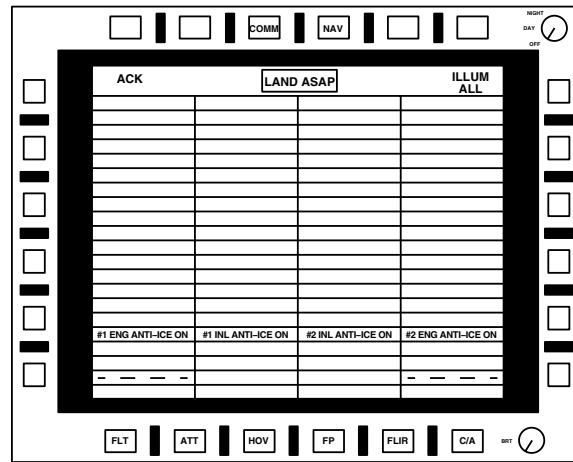
AA2666_2A
SA

Figure 1. Engine Anti-Ice System Location Diagram. (Sheet 2 of 5)

LOCATION - Continued



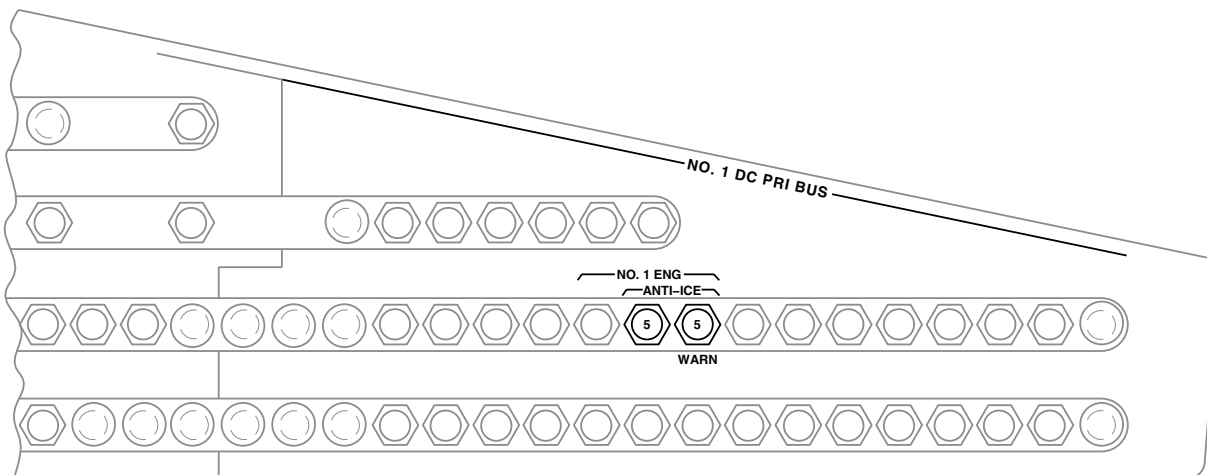
CAUTION/ADVISORY PANEL
(SEE NOTE 1)



**MULTIFUNCTION DISPLAY'S
CAUTION/ADVISORY GRID**
(SEE NOTE 2)

A

B

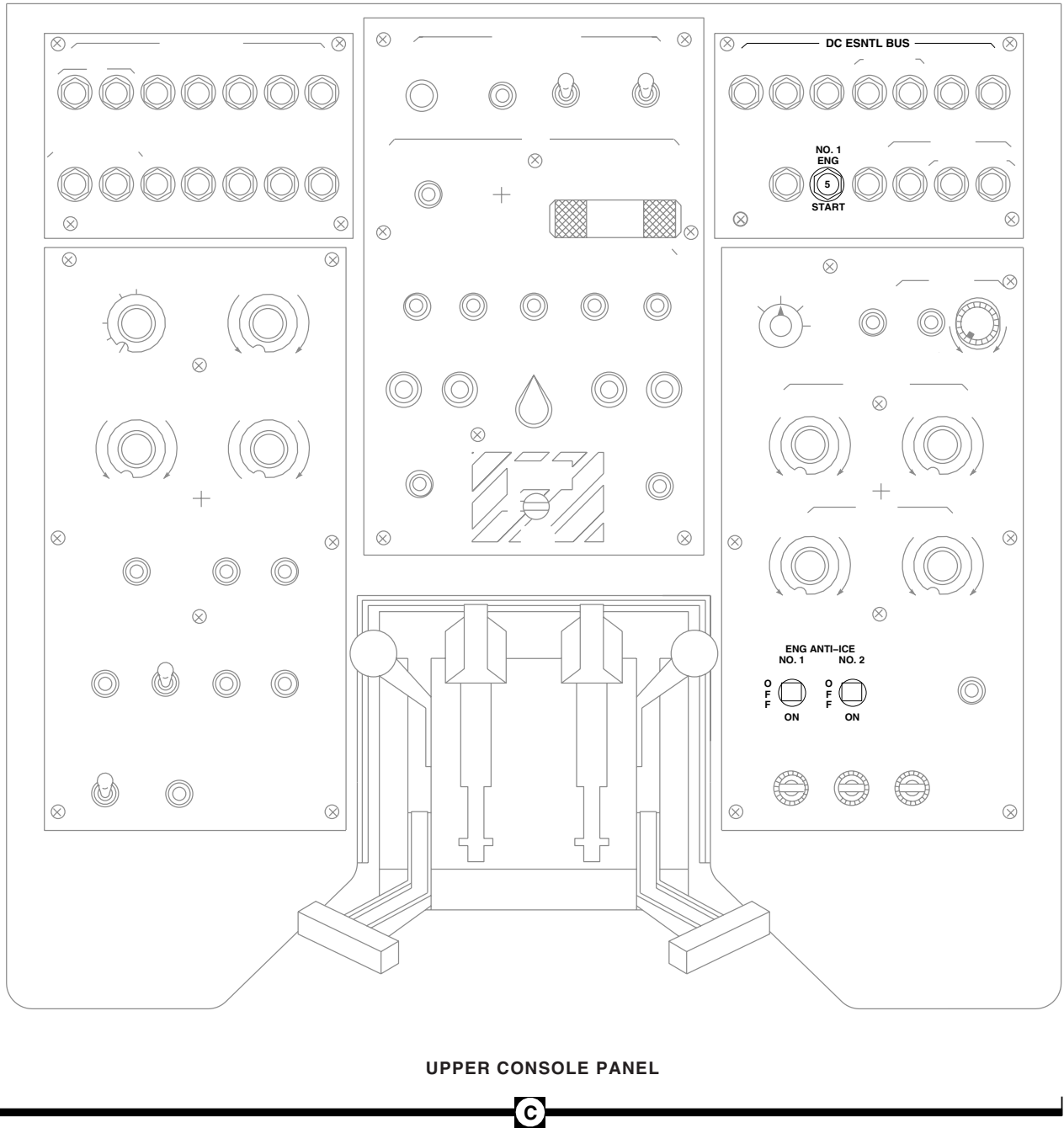


COPLOT'S CIRCUIT BREAKER PANEL

AA2666_3A
SA

Figure 1. Engine Anti-Ice System Location Diagram. (Sheet 3 of 5)

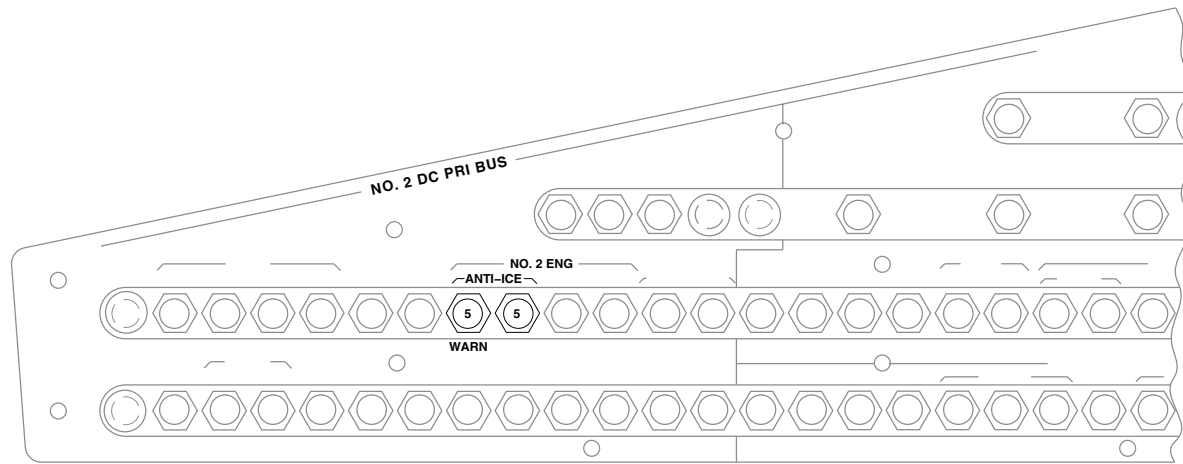
LOCATION - Continued



AA2666_4
SA

Figure 1. Engine Anti-Ice System Location Diagram. (Sheet 4 of 5)

LOCATION - Continued



PILOT'S CIRCUIT BREAKER PANEL

D

AA2666_5
SA

Figure 1. Engine Anti-Ice System Location Diagram. (Sheet 5 of 5)

SCHEMATICS - Continued

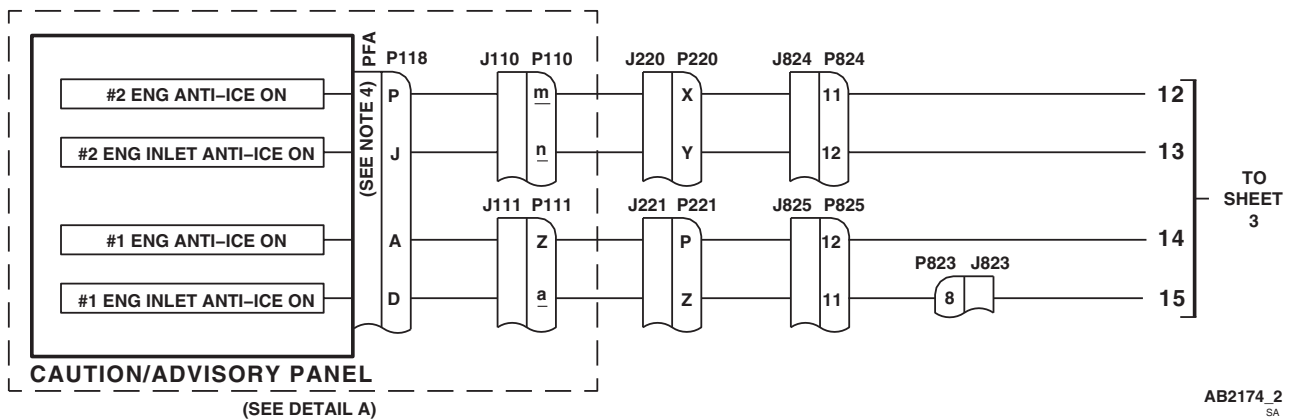
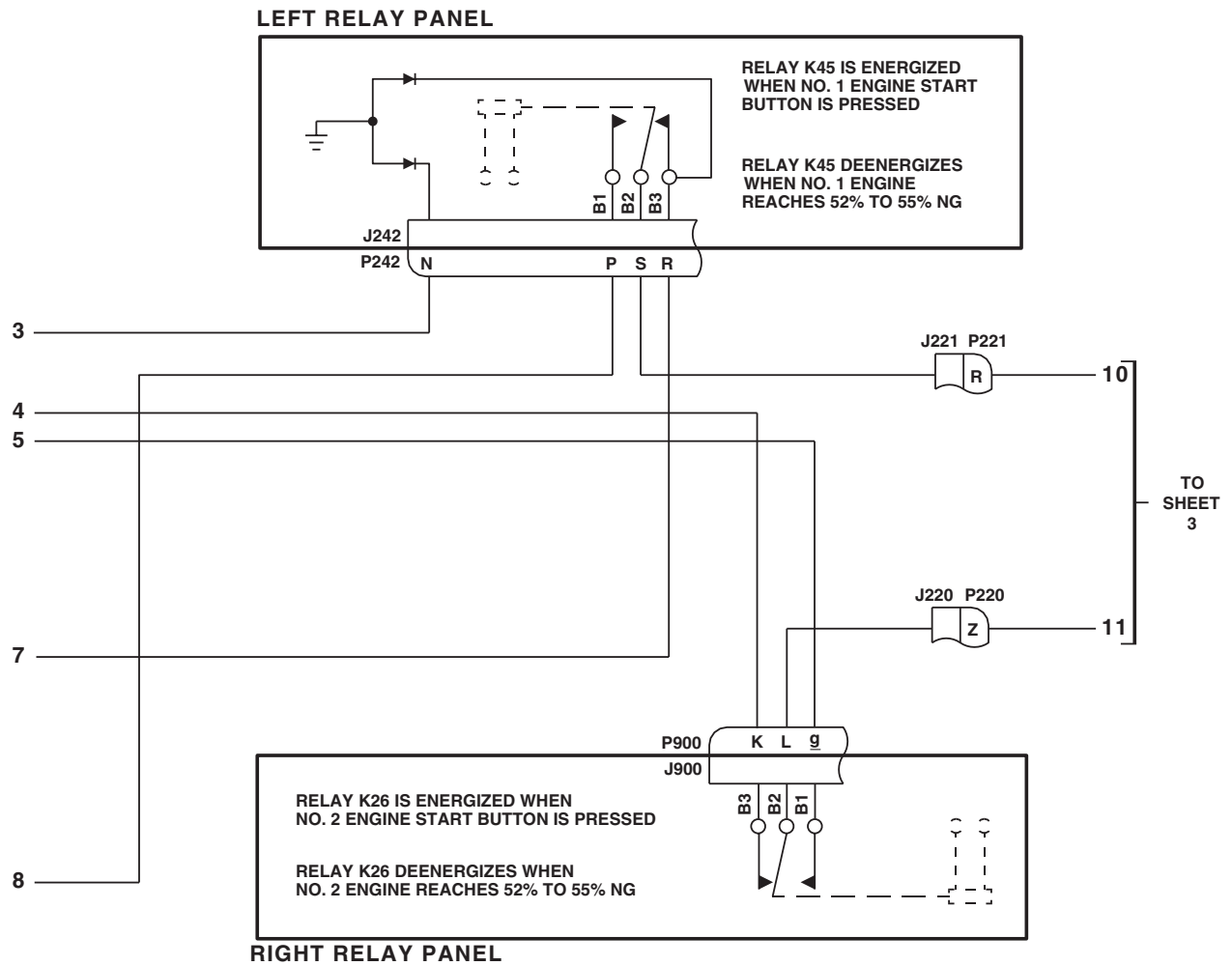


Figure 2. Engine Anti-Ice System Schematic Diagram. (Sheet 2 of 4)

SCHEMATICS - Continued

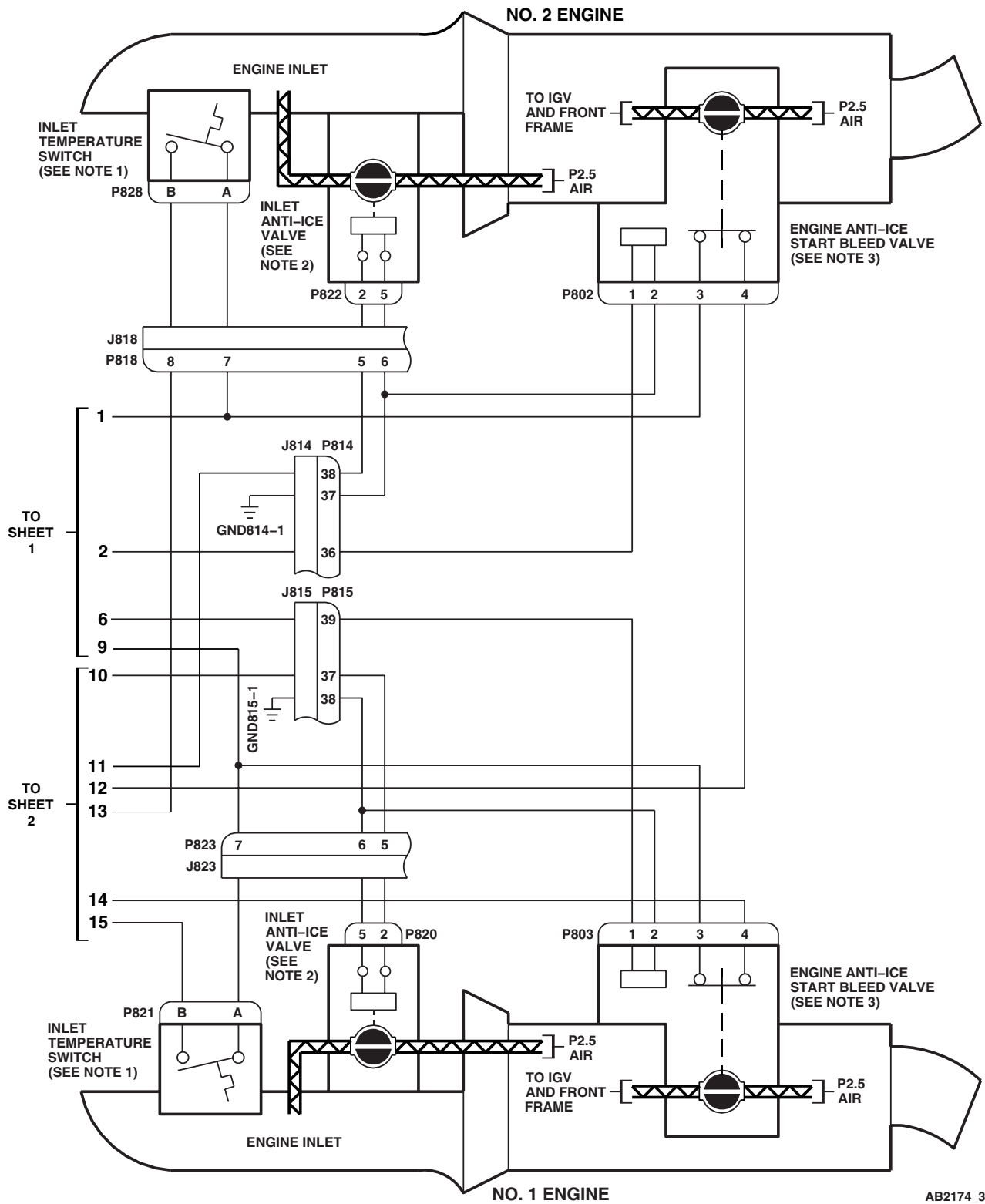


Figure 2. Engine Anti-Ice System Schematic Diagram. (Sheet 3 of 4)

SCHEMATICS - Continued

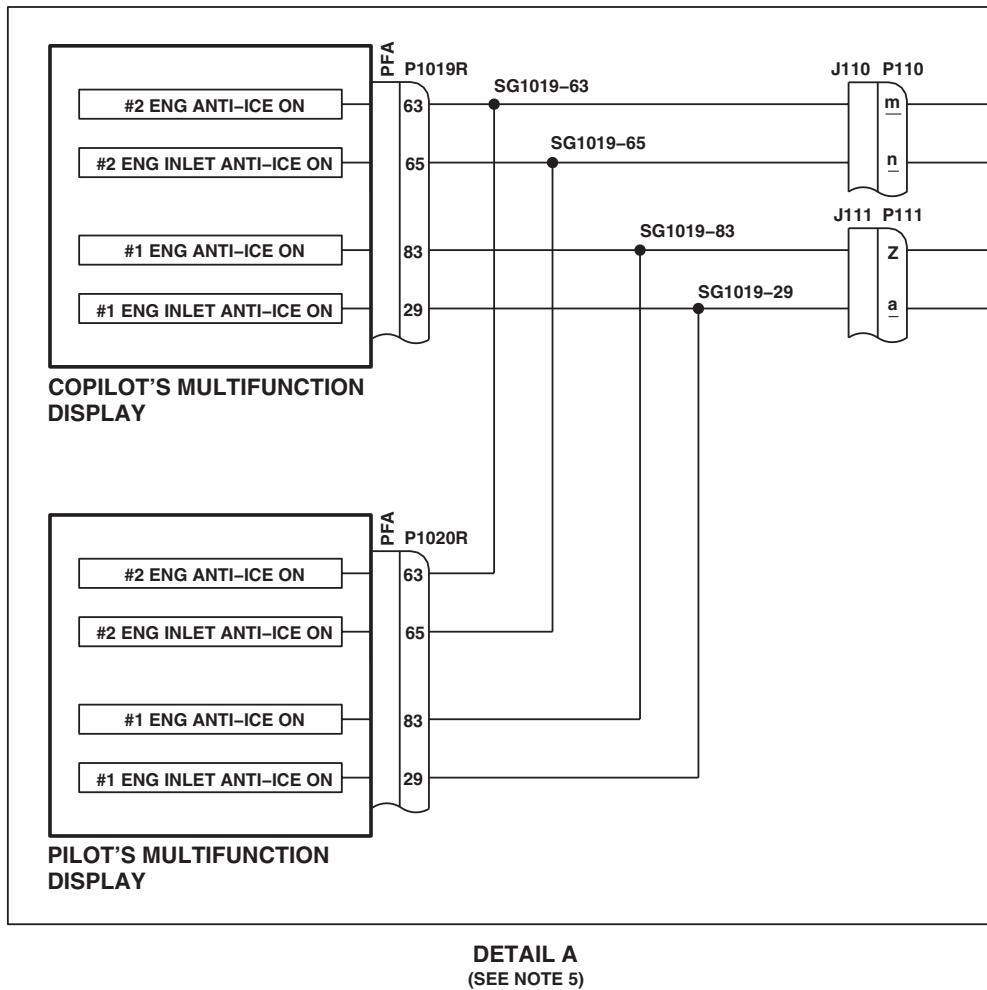
AB2174 4
SA

Figure 2. Engine Anti-Ice System Schematic Diagram. (Sheet 4 of 4)

END OF WORK PACKAGE

UNIT LEVEL

ENGINE SYSTEM

ENGINE START AND IGNITION SYSTEM

This WP supersedes WP 0079 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment

Multimeter, AN/PSM-45A

WP 0503 00

WP 0505 00

WP 0507 00

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Electrical Repairer Toolkit, SC 5180-99-B06

WP 0527 00

WP 0528 00

WP 0529 00

Personnel Required

Aircraft Electrician MOS 15F (1)

UH-60 Helicopter Repairer MOS 15T (1)

WP 0694 00

WP 0695 00

WP 0696 00

References

TM 1-1520-237-10

TM 1-2840-248-23

WP 0072 00

WP 0075 00

WP 0078 00

WP 0089 00

WP 0090 00

WP 0100 00

WP 0104 00

WP 0121 00

WP 0130 00

WP 0155 00

WP 0156 00

WP 0499 00

WP 0501 00

WP 0704 00

WP 0743 00

WP 0824 00

WP 0825 00

WP 0828 00

WP 0829 00

WP 0923 00

WP 0925 00

WP 0990 00

WP 1400 00

WP 1747 00

Equipment Condition

APU Operational, TM 1-1520-237-10

No. 1 and No. 2 Fuel Tanks Serviced,

TM 1-1520-237-10

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for system location diagram, and

UH-60L MOD HH-60A HH-60L Figure 2 and **EH-60A UH-60A W/O MOD** Figure 3 for system schematic diagrams as an aid in operational/troubleshooting.

The engine start and ignition system operational/troubleshooting procedure is subdivided as follows:

- No. 1 Engine
- No. 2 Engine

NO. 1 ENGINE OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
AIR SOURCE HEAT/START	Copilot's
FUEL PRIME BOOST	Lower Console
NO. 1 ENG START CONTR	Upper Console
NO. 2 ENG START CONTR	Lower Console
HH-60A HH-60L MFD PILOT	Aux Circuit Breaker Panel
HH-60A HH-60L MFD COPILOT	Aux Circuit Breaker Panel

2. Place upper console FUEL PUMP switch to OFF.
3. Place upper console BATT switch to ON.

Indication/Condition

PRIME BOOST PUMP ON capsule or legend shall not come on.

Corrective Action

If Indication/Condition is not as specified, go to PRIME BOOST PUMP ON CAPSULE OR LEGEND GOES ON WHEN POWER IS APPLIED TO HELICOPTER, in this work package.

Step

4. Start APU (TM 1-1520-237-10). **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed). ■

Indication/Condition

#1 ENG OUT capsule on pilot's and copilot's master caution panel shall be on.

Corrective Action

If Indication/Condition is not as specified, troubleshoot instrument display system (WP 0089 00).

NO. 1 ENGINE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

5. With qualified personnel at controls, start No. 1 engine and operate up to IDLE speed (TM 1-1520-237-10). Refer to NO. 1 ENGINE STARTING, in this work package for trouble analysis.

Table 1. No. 1 Engine Starting.

FAULT INDICATION	GO TO TROUBLESHOOTING TABLE
#1 ENGINE STARTER capsule or legend is on, but engine does not motor	NO. 1 ENGINE DOES NOT START (COMPRESSOR NOT ROTATING) (#1ENGINE STARTER CAPSULE OR LEGEND IS ON), in this work package
Engine motors, but does not accelerate to ground idle speed	ENGINE STARTER MOTORS ENGINE, BUT CANNOT REACH PROPER NG SPEED TO START ENGINE, in this work package
#1 ENGINE STARTER capsule or legend is not on and engine does not motor	NO. 1 ENGINE DOES NOT START (COMPRESSOR NOT ROTATING) (#1 ENGINE STARTER CAPSULE OR LEGEND IS OFF), in this work package
PRIME BOOST PUMP ON capsule or legend is not on	PRIME BOOST PUMP ON CAPSULE OR LEGEND IS NOT ON WHEN STARTER SWITCH IS PRESSED, in this work package
When No. 1 engine starter button is released, #1 ENGINE STARTER capsule or legend goes off and engine does not motor	NO. 1 ENGINE START MOTOR STOPS (#1 ENGINE STARTER CAPSULE OR LEGEND GOES OFF) WHEN START BUTTON IS RELEASED, in this work package
#1 ENGINE STARTER capsule or legend goes off, but engine continues to motor, and no noticeable change (or popping sounds) in APU is heard	NO. 1 ENGINE STARTS, BUT #1 ENGINE STARTER CAPSULE OR LEGEND IS OFF (APU IS NOT STALLING), in this work package
Engine motors, but engine does not ignite	NO. 1 ENGINE MOTORS, BUT DOES NOT IGNITE AND ACCELERATE TO IDLE SPEED, in this work package
#1 ENGINE STARTER capsule or legend remains on when No. 1 engine starter button is released, engine motors, but noticeable change (or popping sounds) in APU operation is heard	NO. 1 ENGINE STARTER MOTORS ENGINE (#1 ENGINE STARTER CAPSULE OR LEGEND IS ON), BUT APU STALLS (POPPING SOUND IS HEARD), in this work package
Binding is felt when engine POWER CONT level is advanced to IDLE	NO. 1 OR NO. 2 ENGINE POWER CONTROL LEVER IS MORE DIFFICULT TO MOVE THAN NORMAL, in this work package
Engine does not accelerate ground idle speed	ENGINE NO. 1 OR NO. 2 HAS A HUNG START (TGT INCREASES, BUT HANGS), in this work package
TGT is abnormally high	ENGINE NO. 1 OR NO. 2 TGT ADNORMALLY HIGH (TGT REACHES 1490F/810C BEFORE IDLE SPEED IS REACHED), in this work package

NO. 1 ENGINE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Table 1. No. 1 Engine Starting. - Continued**

FAULT INDICATION	GO TO TROUBLESHOOTING TABLE
If ground idle speed is too high or low	WP 0072 00
#1 ENG STARTER capsule or legend remains on after engine reaches 65% Ng	WP 0089 00/#1 ENGINE STARTER CAPSULE OR LEGEND DOES NOT GO OFF (ENGINE SPEED ABOVE 65% NG), in this work package
#1 ENGINE STARTER capsule or legend goes off before 52% Ng	NO. 1 OR NO. 2 ENGINE STARTER CAPSULE OR LEGEND GOES OFF BEFORE 52% NG ENGINE SPEED, in this work package.
PRIME BOOST PUMP ON capsule or legend does not go off	PRIME BOOST PUMP ON CAPSULE OR LEGEND GOES ON WHEN POWER IS APPLIED TO HELICOPTER, in this work package.

6. Shut down No. 1 engine (TM 1-1520-237-10).

Indication/Condition

#1 ENG OUT capsule shall go on before 55% Ng.

Corrective Action

If Indication/Condition is not as specified, troubleshoot instrument display system (WP 0089 00).

Step

7. With qualified personnel at controls and instrument panel IGNITION SWITCH at OFF, press and hold No. 1 engine starter button until a cockpit Ng indication is noted (above 10% Ng), and then, pull engine controls quadrant ENG POWER CONT lever down (TM 1-1520-237-10).

Indication/Condition

Engine starter shall motor engine. When engine POWER CONT lever is pulled down, engine shall stop motoring.

Corrective Action

If Indication/Condition is not as specified, go to NO. 1 OR NO. 2 ENGINE START MOTOR DOES NOT SHUTDOWN WHEN POWER CONTROL LEVER IS PULLED DOWN (SPEED BELOW 52% NG), in this work package.

Step

8. Remove P314 from starter speed switch located in cabin ceiling.
9. Place container under No. 1 engine overboard drain line.
10. With qualified personnel at controls and APU operating, have pilot press and hold No. 1 engine starter button. Advance engine controls quadrant ENG POWER CONT lever to between OFF and IDLE (TM 1-1520-237-10).

Indication/Condition

No. 1 engine starter shall continue to motor No. 1 engine when ENG POWER CONT lever is away from OFF.

NO. 1 ENGINE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If Indication/Condition is not as specified, go to NO. 1 ENGINE STARTER DOES NOT MOTOR ENGINE (COMPRESSOR IS NOT ROTATING) WHEN USING ENGINE STARTER OVERRIDE, in this work package.

Step

11. Move engine controls quadrant ENG POWER CONT lever to LOCKOUT.

Indication/Condition

Fuel from No. 1 engine overboard drain line shall drain into container.

Corrective Action

If Indication/Condition is not as specified, go to NO FUEL IS COMING OUT OF NO. 1 ENGINE FUEL DRAIN, in this work package.

Step

12. Connect P314 to starter speed switch.
13. Turn MFD switch to OFF.
14. Shut down APU (TM 1-1520-237-10).

Indication/Condition

APU shall shut down.

Corrective Action

None Required

NO. 2 ENGINE OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step****NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
AIR SOURCE HEAT/START	Copilot's
FUEL PRIME BOOST	Lower Console
NO. 1 ENG START CONTR	Upper Console
NO. 2 ENG START CONTR	Lower Console

NO. 2 ENGINE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**CIRCUIT BREAKER****CIRCUIT BREAKER PANEL****HH-60A HH-60L** MFD PILOT

Aux Circuit Breaker Panel

HH-60A HH-60L MFD COPILOT

Aux Circuit Breaker Panel

2. Place upper console FUEL PUMP switch to OFF, if not already done.
3. Place upper console BATT switch to ON, if not already done.

Indication/Condition

PRIME BOOST PUMP ON capsule or legend shall not come on.

Corrective Action

If Indication/Condition is not as specified, go to PRIME BOOST PUMP ON CAPSULE OR LEGEND GOES ON WHEN POWER IS APPLIED TO HELICOPTER, in this work package.

Step

4. Start APU (TM 1-1520-237-10). **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed). ■

Indication/Condition

#2 ENG OUT capsule on pilot's and copilot's master caution panel shall be on.

Corrective Action

If Indication/Condition is not as specified, troubleshoot instrument display system (WP 0089 00).

Step

5. With qualified personnel at controls, start No. 2 engine and operate engine up to IDLE speed (TM 1-1520-237-10). Refer to NO. 2 ENGINE STARTING, in this work package for trouble analysis.

Table 2. No. 2 Engine Starting.

FAULT INDICATION	GO TO TROUBLESHOOTING TABLE
#2 ENGINE STARTER capsule or legend is on, but engine does not motor	NO. 2 ENGINE DOES NOT START (COMPRESSOR NOT ROTATING) (#2 ENGINE STARTER CAPSULE OR LEGEND IS ON), in this work package
Engine motors, but does not accelerate to ground idle speed	ENGINE STARTER MOTORS ENGINE, BUT CANNOT REACH PROPER NG SPEED TO START ENGINE, in this work package
#2 ENGINE STARTER capsule or legend is not on and engine does not motor	NO. 2 ENGINE DOES NOT START (COMPRESSOR NOT ROTATING) (#2 ENGINE STARTER CAPSULE OR LEGEND IS OFF), in this work package

NO. 2 ENGINE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Table 2. No. 2 Engine Starting. - Continued**

FAULT INDICATION	GO TO TROUBLESHOOTING TABLE
PRIME BOOST PUMP ON capsule or legend is not on	PRIME BOOST PUMP ON CAPSULE OR LEGEND IS NOT ON WHEN STARTER SWITCH IS PRESSED, in this work package
When No. 2 engine starter button is released, #2 ENGINE STARTER capsule or legend goes off and engine does not motor	NO. 2 ENGINE START MOTOR STOPS (#2 ENGINE STARTER CAPSULE OR LEGEND GOES OFF) WHEN START BUTTON IS RELEASED, in this work package
#2 ENGINE STARTER capsule or legend goes off, but engine continues to motor, and no noticeable change (or popping sounds) in APU is heard	NO. 2 ENGINE STARTS, BUT #2 ENGINE STARTER CAPSULE OR LEGEND IS OFF (APU IS NOT STALLING), in this work package
Engine motors, but engine does not ignite	NO. 2 ENGINE STARTER MOTORS ENGINE (#2 ENGINE STARTER CAPSULE OR LEGEND IS ON), BUT APU STALLS (POPPING SOUND IS HEARD), in this work package
#2 ENGINE STARTER capsule or legend remains on when No. 2 engine starter button is released, engine motors, but noticeable change (or popping sounds) in APU operation is heard	NO. 2 ENGINE MOTORS, BUT DOES NOT IGNITE AND ACCELERATE TO IDLE SPEED, in this work package
Binding is felt when engine POWER CONT level is advanced to IDLE	NO. 1 OR NO. 2 ENGINE POWER CONTROL LEVER IS MORE DIFFICULT TO MOVE THAN NORMAL, in this work package
Engine does not accelerate ground idle speed	ENGINE NO. 1 OR NO. 2 HAS A HUNG START (TGT INCREASES, BUT HANGS), in this work package
TGT is abnormally high	ENGINE NO. 1 OR NO. 2 TGT ADNORMALLY HIGH (TGT REACHES 1490F/810C BEFORE IDLE SPEED IS REACHED), in this work package
If ground idle speed is too high or low	WP 0072 00
#2 ENG STARTER capsule or legend remains on after engine reaches 65% Ng	WP 0089 00/#2 ENGINE STARTER CAPSULE OR LEGEND DOES NOT GO OFF (ENGINE SPEED ABOVE 65% NG), in this work package
#2 ENGINE STARTER capsule or legend goes off before 52% Ng	NO. 1 OR NO. 2 ENGINE STARTER CAPSULE OR LEGEND GOES OFF BEFORE 52% NG ENGINE SPEED, in this work package.
PRIME BOOST PUMP ON capsule or legend does not go off	PRIME BOOST PUMP ON CAPSULE OR LEGEND GOES ON WHEN POWER IS APPLIED TO HELICOPTER, in this work package.

6. Shut down No. 2 engine (TM 1-1520-237-10).

NO. 2 ENGINE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

#2 ENG OUT capsule shall go on before 55% Ng.

Corrective Action

If Indication/Condition is not as specified, troubleshoot instrument display system (WP 0089 00).

Step

7. With qualified personnel at controls and instrument panel IGNITION SWITCH at OFF, press and hold No. 2 engine starter button until a cockpit Ng indication is noted (above 10% Ng), then pull engine controls quadrant ENG POWER CONT lever down.

Indication/Condition

Engine starter shall motor engine. When ENG POWER CONT lever is pulled down, engine shall stop motoring.

Corrective Action

If Indication/Condition is not as specified, go to NO. 1 OR NO. 2 ENGINE START MOTOR DOES NOT SHUTDOWN WHEN POWER CONTROL LEVER IS PULLED DOWN (SPEED BELOW 52% NG), in this work package.

Step

8. Remove P314 from starter speed switch located in cabin ceiling.
9. Place container under No. 2 engine overboard drain line.
10. With qualified personnel at controls and APU operating, have pilot press and hold No. 2 engine starter button. Advance engine control quadrant ENG POWER CONT lever to between OFF and IDLE (TM 1-1520-237-10).

Indication/Condition

No. 2 engine starter shall continue to motor No. 2 engine when ENG POWER CONT lever is away from OFF.

Corrective Action

If Indication/Condition is not as specified, go to NO. 2 ENGINE STARTER DOES NOT MOTOR ENGINE (COMPRESSOR IS NOT ROTATING) WHEN USING ENGINE STARTER OVERRIDE, in this work package.

Step

11. Move engine control quadrant ENG POWER CONT lever to LOCKOUT.

Indication/Condition

Fuel from No. 2 engine overboard drain line shall drain into container.

Corrective Action

If Indication/Condition is not as specified, go to NO FUEL COMING OUT OF NO. 2 FUEL DRAIN, in this work package.

Step

12. Connect P314 to starter speed switch.
13. Turn MFD switch to OFF.
14. Shut down APU (TM 1-1520-237-10).

NO. 2 ENGINE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

APU shall shut down.

Corrective Action

None Required

PRIME BOOST PUMP ON CAPSULE OR LEGEND GOES ON WHEN POWER IS APPLIED TO HELICOPTER**SYMPTOM**

PRIME BOOST PUMP ON capsule or legend goes on when power is applied to helicopter.

MALFUNCTION

PRIME BOOST PUMP ON capsule or legend goes on when power is applied to helicopter.

CORRECTIVE ACTION**1. Can prime boost pump be heard operating?**

- a. If pump can be heard, go to **2**.
- b. If pump cannot be heard, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **3**.

2. Check continuity between J241-B and J241-C.

- a. If continuity is present, replace right relay panel (WP 0825 00). Go to **3**.
- b. If continuity is not present, replace switch S19 (WP 0829 00). Go to **3**.

3. Procedure completed.**NO. 1 ENGINE DOES NOT START (COMPRESSOR NOT ROTATING) (#1 ENGINE STARTER CAPSULE OR LEGEND IS ON)****SYMPTOM**

No. 1 engine does not start (compressor not rotating) (#1 ENGINE STARTER capsule or legend is on).

MALFUNCTION

No. 1 engine does not start (compressor not rotating) (#1 ENGINE STARTER capsule or legend is on).

CORRECTIVE ACTION**1. Troubleshoot APU system for faulty start bypass valve (WP 0156 00).**

- a. If valve is malfunctioning, replace valve (WP 0743 00). Go to **7**.
- b. If valve is operating properly, go to **2**.

2. Check all clamps and rubber connectors on pneumatic lines between start motor and APU check valve for proper connection or damage.

- a. If connections are proper and no damage is present, go to **3**.
- b. If connections are not proper and/or damage is present, repair/replace clamps or connectors as required. Go to **7**.

3. Disconnect start motor from engine (WP 0694 00 or WP 0695 00). Go to 4.

NO. 1 ENGINE DOES NOT START (COMPRESSOR NOT ROTATING) (#1 ENGINE STARTER CAPSULE OR LEGEND IS ON) - Continued**NOTE**

Turning the gear train requires considerable effort to overcome resistance.

- 4. Check start motor for sheared/seized shaft by grasping shaft spline end and rotating shaft clockwise, making sure turbine wheel also rotates.**
 - a. If start motor has sheared/seized shaft, replace start motor (WP 0694 00 or WP 0695 00). Go to **7**.
 - b. If start motor does not have sheared/seized shaft, go to **5**.
- 5. Troubleshoot engine for seized condition (TM 1-2840-248-23). Go to 6.**
- 6. Install start motor (WP 0694 00 or WP 0695 00). Go to 7.**
- 7. Procedure completed.**

ENGINE STARTER MOTORS ENGINE, BUT CANNOT REACH PROPER NG SPEED TO START ENGINE**SYMPTOM**

Engine starter motors engine, but cannot reach proper Ng speed to start engine.

MALFUNCTION

Engine starter motors engine, but cannot reach proper Ng speed to start engine.

CORRECTIVE ACTION

- 1. With helicopter on level ground, verify starter has not been over serviced by removing oil fill plug and letting excess oil drain from starter. Go to 2.**
- 2. Check that oil is level with oil fill port.**
 - a. If oil is level with oil fill port, go to **3**.
 - b. If oil is not level with oil fill port, replace starter motor (WP 0694 00 or WP 0695 00). Go to **4**.
- 3. Check that engine starter can reach proper Ng.**
 - a. If engine starter can reach proper Ng, engine start system is operational. Go to **4**.
 - b. If engine starter can not reach proper Ng, replace starter motor (WP 0694 00 or WP 0695 00). Go to **4**.
- 4. Procedure completed.**

NO. 1 ENGINE DOES NOT START (COMPRESSOR NOT ROTATING) (#1 ENGINE STARTER CAPSULE OR LEGEND IS OFF)**SYMPTOM**

No. 1 engine does not start.

MALFUNCTION

No. 1 engine does not start (compressor not rotating) (#1 ENGINE STARTER capsule or legend is off).

CORRECTIVE ACTION

- 1. With APU shutdown and No. 1 engine START button pressed, check for 28 vdc between P440-1 and ground.**

NO. 1 ENGINE DOES NOT START (COMPRESSOR NOT ROTATING) (#1 ENGINE STARTER CAPSULE OR LEGEND IS OFF) - Continued

- a. If voltage is as specified, go to **2**.
 - b. If voltage is not as specified, go to **3**.
- 2. Check continuity between P440-2 and ground.**
 - a. If continuity is present, replace No. 1 engine start control valve (WP 0696 00). Go to **16**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **16**.
- 3. Check continuity between J105-J and J105-K.**
 - a. If continuity is present, go to **4**.
 - b. If continuity is not present, replace engine control quadrant assembly (WP 0501 00). Go to **16**.
- 4. With APU shut down, press No. 1 engine START button and check continuity between J105-G and J105-H.**
 - a. If continuity is present, go to **5**.
 - b. If continuity is not present, go to **15**.
- 5. Check for 28 vdc between P105-K and ground.**
 - a. If voltage is as specified, go to **6**.
 - b. If voltage is not as specified, go to **14**.
- 6. Check for 28 vdc between P105-G and ground.**
 - a. If voltage is as specified, go to **7**.
 - b. If voltage is not as specified, repair/replace wiring between NO. 1 ENG START circuit breaker terminal 1 and P105-G (WP 1747 00). Go to **16**.
- 7. With No. 1 engine START button pressed, check for 28 vdc between P242-T and ground.**
 - a. If voltage is as specified, go to **8**.
 - b. If voltage is not as specified, repair/replace wiring between P105-H and P242-T (WP 1747 00). Go to **16**.
- 8. Install an insulated jumper wire between P242-U and P242-Y. Go to 9.**
- 9. Check for 28 vdc between P440-1 and ground.**
 - a. If voltage is as specified, go to **10**.
 - b. If voltage is not as specified, go to **12**.
- 10. Remove jumper wire from P242. Go to 11.**
- 11. Troubleshoot left relay panel assembly (WP 0121 00). Go to 16.**
- 12. Remove jumper wire from P242. Go to 13.**
- 13. Repair/replace wiring, as required, between (WP 1747 00), then go to 16:**
 - P105-J and P242-U
 - P242-Y and P440-1
- 14. Check for 28 vdc between NO. 1 ENG START circuit breaker terminal 1 and ground.**

NO. 1 ENGINE DOES NOT START (COMPRESSOR NOT ROTATING) (#1 ENGINE STARTER CAPSULE OR LEGEND IS OFF) - Continued

- a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P105-K (WP 1747 00). Go to **16**.
- b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **16**.

15. Check No. 1 engine starter switch for proper adjustment (WP 0503 00).

- a. If switch has proper adjustment, replace switch (WP 0527 00). Go to **16**.
- b. If switch does not have proper adjustment, adjust switch (WP 0503 00). Go to **16**.

16. Procedure completed.**PRIME BOOST PUMP ON CAPSULE OR LEGEND IS NOT ON WHEN STARTER SWITCH IS PRESSED****SYMPTOM**

PRIME BOOST PUMP ON capsule or legend is not.

MALFUNCTION

PRIME BOOST PUMP ON capsule or legend is not on when STARTER switch is pressed.

CORRECTIVE ACTION**1. Check for 28 vdc between P241-B and ground.**

- a. If voltage is as specified, go to **2**.
- b. If voltage is not as specified, go to **4**.

2. Check for 28 vdc between P246-G and ground.

- a. If voltage is as specified, troubleshoot fuel prime boost system (WP 0130 00). Go to **5**.
- b. If voltage is not as specified, go to **3**.

3. Check continuity between P241-C and P246-G.

- a. If continuity is present, replace right relay panel (WP 0825 00). Go to **5**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.

4. Check for 28 vdc between P246-B and ground.

- a. If voltage is as specified, repair/replace wiring between P241-B and P246-B (WP 1747 00). Go to **5**.
- b. If voltage is not as specified, troubleshoot fuel prime boost system (WP 0130 00). Go to **5**.

5. Procedure completed.

NO. 1 ENGINE START MOTOR STOPS (#1 ENGINE STARTER CAPSULE OR LEGEND GOES OFF) WHEN START BUTTON IS RELEASED**SYMPTOM**

No. 1 engine start motor stops.

MALFUNCTION

No. 1 engine start motor stops (#1 ENGINE STARTER capsule or legend goes off) when START button is released.

CORRECTIVE ACTION

1. Disconnect P314 from starter speed switch. Go to 2.
2. Install an insulated jumper wire between P314-C and P314-D. Go to 3.
3. Make sure NO. 1 ENG POWER CONT lever is OFF. Go to 4.
4. With qualified personnel at controls and APU operating, press No. 1 engine START button, then release. Go to 5.
5. Check that #1 ENGINE STARTER capsule or legend goes on and engine continues to motor after START button is released.
 - a. If capsule or legend goes on and engine continues to motor, go to 6.
 - b. If capsule or legend does not go on and/or engine does not motor, go to 9.
6. Shut down APU (TM 1-1520-237-10). Go to 7.
7. Remove jumper wire from P314. Go to 8.
8. Replace starter speed switch (WP 0527 00). Go to 19.
9. Remove jumper wire from P314. Do not install connector to speed switch. Go to 10.
10. With qualified personnel at controls and APU operating, press No. 1 engine START button and hold. Go to 11.
11. Advance NO. 1 ENG POWER CONT lever to a position between OFF and IDLE. Go to 12.
12. Check that No. 1 engine continues to motor with ENG POWER CONT lever away from OFF.
 - a. If No. 1 engine continues to motor, go to 13.
 - b. If No. 1 engine does not continue to motor, go to 16.
13. Shut down APU (TM 1-1520-237-10). Go to 14.
14. Connect P314 to starter speed switch. Go to 15.
15. Repair/replace wiring, as required, between the following points (WP 1747 00), then go to 19:
 - P105-T and P314-D
 - P314-C and P440-5
16. Shut down APU (TM 1-1520-237-10). Go to 17.
17. Connect P314 to starter speed switch. Go to 18.

NO. 1 ENGINE START MOTOR STOPS (#1 ENGINE STARTER CAPSULE OR LEGEND GOES OFF) WHEN START BUTTON IS RELEASED - Continued

18. Go to NO. 1 ENGINE STARTER DOES NOT MOTOR ENGINE (COMPRESSOR IS NOT ROTATING) WHEN USING ENGINE STARTER OVERRIDE, in this work package.
19. Procedure completed.

NO. 1 ENGINE STARTS, BUT #1 ENGINE STARTER CAPSULE OR LEGEND IS OFF (APU IS NOT STALLING)

SYMPTOM

No. 1 engine starts, but #1 ENGINE STARTER capsule or legend is off.

MALFUNCTION

No. 1 engine starts, but #1 ENGINE STARTER capsule or legend is off (APU is not stalling).

CORRECTIVE ACTION

1. Hold caution/advisory panel **HH-60A HH-60L** instrument panel INDICATOR LTS **■** BRT/DIM-TEST switch to TEST. Go to 2.
2. Check that #1 ENGINE STARTER capsule or legend goes on.
 - a. If capsule or legend goes on, go to 3.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 12.
3. Install an insulated jumper wire between P440-3 and P440-5. Go to 4.
4. Check helicopter configuration.
 - a. **EMEP** , go to 5.
 - b. **W/O EMEP** , go to 7.
5. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R **■** (WP 0925 00).
 - a. If pin filtered adapter is good, go to 6.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 12.
6. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 7.
7. Check for 28 vdc between P117-7 or **HH-60A HH-60L** P1019R-75 or P1020R-75 **■** and ground.
 - a. If voltage is as specified, go to 8.
 - b. If voltage is not as specified, go to 10.
8. Remove jumper wire from P440. Go to 9.
9. Troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 12.
10. Remove jumper wire from P440. Go to 11.
11. Repair/replace wiring between P117-7 or **HH-60A HH-60L** P1019R-75 or P1020R-75 **■** and SG105-2 (WP 1747 00). Go to 12.
12. Procedure completed.

NO. 1 ENGINE MOTORS, BUT DOES NOT IGNITE AND ACCELERATE TO IDLE SPEED**SYMPTOM**

No. 1 engine motors, but does not ignite and accelerate.

MALFUNCTION

No. 1 engine motors, but does not ignite and accelerate to idle speed.

CORRECTIVE ACTION

1. Move ENG POWER CONT lever on control quadrant in cockpit to LOCKOUT and hold. Go to 2.
2. Move ENG FUEL SYS selector lever on control quadrant to DIR detent. Go to 3.
3. Place FUEL PUMP switch to FUEL PRIME. Go to 4.
4. Check that fuel is coming from overboard drain.
 - a. If fuel is coming from overboard drain, go to 5.
 - b. If fuel is not coming from overboard drain, troubleshoot fuel prime boost system (WP 0130 00). Go to 20.
5. Troubleshoot engine for no ignite (TM 1-2840-248-23).
 - a. If engine is good, go to 6.
6. Install an insulated jumper wire between P803-17 and P803-18. Go to 7.
7. Check continuity between P111-BB and P111-CC.
 - a. If continuity is present, go to 8.
 - b. If continuity is not present, go to 18.
8. Check continuity between P242-W and P242-X.
 - a. If continuity is present, go to 9.
 - b. If continuity is not present, go to 16.
9. Remove jumper wire from P803. Go to 10.
10. Check helicopter configuration.
 - a. HH-60L UH-60L , go to 11.
 - b. EH-60A HH-60A UH-60A , go to 14.
11. HH-60L UH-60L Check continuity between:
 - P803-17 and P815-35
 - P803-19 and P815-35
 - a. If continuity is present, go to 12.
 - b. If continuity is not present, repair/replace harness (TM 1-2840-248-23). Go to 20.
12. HH-60L UH-60L Check continuity between P381-F and J815-35.
 - a. If continuity is present, go to 13.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 20.

NO. 1 ENGINE MOTORS, BUT DOES NOT IGNITE AND ACCELERATE TO IDLE SPEED - Continued

13. **HH-60L UH-60L** Check continuity between P111-BB and P381-G.
 - a. If continuity is present, troubleshoot engine overspeed protection system (WP 0075 00). Go to **20**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **20**.
14. **EH-60A HH-60A UH-60A** On instrument panel, turn ENGINE IGNITION switch to ON. Go to **15**.
15. **EH-60A HH-60A UH-60A** Check continuity between J111-BB and J111-CC.
 - a. If continuity is present, replace keylock ignition switch (WP 0923 00). Go to **20**.
 - b. If continuity is not present, troubleshoot left relay panel (WP 0121 00). Go to **20**.
16. Remove jumper wire from P803. Go to **17**.
17. Repair/replace wiring, as required, between (WP 1747 00), then go to **20**:
 - P111-BB and P242-W
 - P111-CC and P242-X
18. Remove jumper wire from P803. Go to **19**.
19. Repair/replace wiring, as required, between (WP 1747 00), then go to **20**:
 - P111-BB and P803-18
 - P111-CC and P803-17
20. Procedure completed.

NO. 1 ENGINE STARTER MOTORS ENGINE (#1 ENGINE STARTER CAPSULE OR LEGEND IS ON), BUT APU STALLS (POPPING SOUND IS HEARD)**SYMPTOM**

No. 1 engine starter motors engine, but APU stalls.

MALFUNCTION

No. 1 engine starter motors engine (#1 ENGINE STARTER capsule or legend is on), but APU stalls (popping sound is heard).

CORRECTIVE ACTION

1. Start APU (TM 1-1520-237-10). Go to **2**.
2. Motor No. 2 engine (TM 1-1520-237-10). Go to **3**.
3. Check if APU stalls.
 - a. If APU stalls, go to **4**.
 - b. If APU does not stall, shut down APU (TM 1-1520-237-10). Replace right relay panel (WP 0825 00). Go to **17**.
4. Troubleshoot APU system, start bypass valve circuitry (WP 0156 00).
 - a. If APU system is good, go to **5**.
5. Install an insulated jumper wire between P440-3 and P440-5. Go to **6**.
6. Check helicopter configuration.

NO. 1 ENGINE STARTER MOTORS ENGINE (#1 ENGINE STARTER CAPSULE OR LEGEND IS ON), BUT APU STALLS (POPPING SOUND IS HEARD) - Continued

- a. **WINTER** , go to 7.
- b. **W/O WINTER** , go to 8.
7. **WINTER** Check for 28 vdc between P244B-M and ground.
 - a. If voltage is as specified, go to 9.
 - b. If voltage is not as specified, replace **WINTER** (WP 1400 00).
 - c. If trouble remains, go to 8.
8. Check for 28 vdc between P244-M and ground.
 - a. If voltage is as specified, go to 9.
 - b. If voltage is not as specified, go to 13.
9. Remove jumper wire from P440. Go to 10.
10. Check helicopter configuration.
 - a. **WINTER** , go to 11.
 - b. **W/O WINTER** , go to 12.
11. **WINTER** Replace **WINTER** (WP 1400 00).
 - a. If trouble remains, go to 12.
12. Repair/replace wiring between AIR SOURCE HEAT/START circuit breaker terminal 1 and P244-T (WP 1747 00). Go to 17.
13. Remove jumper wire from P440. Go to 14.
14. Check helicopter configuration.
 - a. **WINTER** , go to 15.
 - b. **W/O WINTER** , go to 16.
15. **WINTER** Replace **WINTER** (WP 1400 00).
 - a. If trouble remains, go to 16.
16. Repair/replace wiring between P117-7 or **HH-60A HH-60L** P1019R-75 or P1020R-75 and P244-M (WP 1747 00). Go to 17.
17. Procedure completed.

NO. 1 OR NO. 2 ENGINE POWER CONTROL LEVER IS MORE DIFFICULT TO MOVE THAN NORMAL

SYMPTOM

No. 1 or No. 2 engine power control lever difficult to move.

MALFUNCTION

No. 1 or No. 2 engine power control lever is more difficult to move than normal.

CORRECTIVE ACTION

1. Disconnect engine power available rotary input from HMU (WP 0507 00). Go to 2.
2. Cycle engine power control lever (on engine control quadrant), back and forth. Go to 3.

NO. 1 OR NO. 2 ENGINE POWER CONTROL LEVER IS MORE DIFFICULT TO MOVE THAN NORMAL -
Continued**3. Check that movement of lever is smooth.**

- a. If movement of lever is smooth, replace engine hydro-mechanical control unit (HMU) (TM 1-2840-248-23). Go to **7**.
- b. If movement of lever is not smooth, go to **4**.

4. Disconnect PAS push/pull cable from rotary input (WP 0507 00). Go to 5.**5. Cycle engine power control lever. Go to 6.****6. Check that movement of lever is smooth.**

- a. If movement is smooth, replace engine power available rotary input (WP 0499 00). Go to **7**.
- b. If movement is not smooth, replace fuselage power available push/pull cable (WP 0507 00). Go to **7**.

7. Procedure completed.**ENGINE NO. 1 OR NO. 2 HAS A HUNG START (TGT INCREASES, BUT HANGS)****SYMPTOM**

Engine No. 1 or No. 2 has a hung start.

MALFUNCTION

Engine No. 1 or No. 2 has a hung start (TGT increases, but hangs).

CORRECTIVE ACTION**1. Troubleshoot engine anti-ice system, engine inlet anti-ice valve (WP 0078 00).**

- a. If system is good, go to **2**.
- b. If system is not good, repair/replace components as required. Go to **5**.

2. Start engine (TM 1-1520-237-10). Go to 3.**3. Check that engine start is normal.**

- a. If engine start is normal, system is operational. Go to **4**.
- b. If engine start is abnormal, troubleshoot engine for hung start (TM 1-2840-248-23). Go to **4**.

4. Shut down engine (TM 1-1520-237-10). Go to 5.**5. Procedure completed.****ENGINE NO. 1 OR NO. 2 TGT ABNORMALLY HIGH (TGT REACHES 1490°F/810°C BEFORE IDLE SPEED IS REACHED)****SYMPTOM**

Engine No. 1 or No. 2 TGT abnormally high.

MALFUNCTION

Engine No. 1 or No. 2 TGT abnormally high (TGT reaches 1490°F/810°C before idle speed is reached).

CORRECTIVE ACTION**1. On upper console panel, place FUEL PUMP switch to FUEL PRIME. Go to 2.****2. Start engine (TM 1-1520-237-10). Go to 3.**

ENGINE NO. 1 OR NO. 2 TGT ABNORMALLY HIGH (TGT REACHES 1490°F/810°C BEFORE IDLE SPEED IS REACHED) - Continued

- 3. Check that TGT is normal during start.**
 - a. If TGT is normal, system is operational. Go to **7**.
 - b. If TGT is abnormal, go to **4**.
- 4. Replace engine start motor (WP 0694 00 or WP 0695 00). Go to 5.**
- 5. Start engine (TM 1-1520-237-10). Go to 6.**
- 6. Check that TGT is normal during start.**
 - a. If TGT is normal, system is operational. Go to **7**.
 - b. If TGT is abnormal, troubleshoot engine for abnormally high TGT during start (TM 1-2840-248-23). Go to **7**.
- 7. Shut down engine (TM 1-1520-237-10). Go to 8.**
- 8. Procedure completed.**

NO. 1 OR NO. 2 ENGINE STARTER CAPSULE OR LEGEND GOES OFF BEFORE 52% NG ENGINE SPEED**SYMPTOM**

No. 1 or No. 2 ENGINE STARTER capsule or legend goes off.

MALFUNCTION

No. 1 or No. 2 ENGINE STARTER capsule or legend goes off before 52% Ng engine speed.

CORRECTIVE ACTION

- 1. Replace starter speed switch (WP 0704 00). Go to 2.**
- 2. Start engine and operate up to idle speed (TM 1-1520-237-10). Go to 3.**
- 3. Check that # 1 ENGINE STARTER capsule or legend goes off between 52% and 65% Ng.**
 - a. If capsule or legend goes off when specified, engine start system is operational. Go to **4**.
 - b. If capsule or legend does not go off when specified, replace engine starter (WP 0694 00 or WP 0695 00). Go to **4**.
- 4. Shut down engine (TM 1-1520-237-10). Go to 5.**
- 5. Procedure completed.**

#1 ENGINE STARTER CAPSULE OR LEGEND DOES NOT GO OFF (ENGINE SPEED ABOVE 65% NG)**SYMPTOM**

#1 ENGINE STARTER capsule or legend does not go off.

MALFUNCTION

#1 ENGINE STARTER capsule or legend does not go off (engine speed above 65% Ng).

CORRECTIVE ACTION

- 1. Visually check No. 1 engine start control valve position indicator. Go to 2.**
- 2. Check No. 1 engine override switch for proper adjustment (WP 0505 00).**
 - a. If switch is adjusted properly, replace override switch (WP 0529 00). Go to **13**.

#1 ENGINE STARTER CAPSULE OR LEGEND DOES NOT GO OFF (ENGINE SPEED ABOVE 65% NG) - Continued

- b. If switch is not adjusted properly, adjust override switch (WP 0505 00). Go to **3**.
- 3. Check that indicator shows CLOSED.**
 - a. If indicator shows CLOSED, go to **4**.
 - b. If indicator does not show CLOSED, replace start control valve (WP 0696 00). Go to **13**.
- 4. Check continuity between J440-3 and J440-4.**
 - a. If continuity is present, go to **5**.
 - b. If continuity is not present, replace start control valve (WP 0696 00). Go to **13**.
- 5. Remove jumper wire from connector P807. Go to 12.**
- 6. Repair/replace wiring, as required, between the following points (WP 1747 00), then go to 13:**
 - P314-E and P807-1
 - P314-F and P807-3
- 7. Procedure completed.**

NO. 1 OR NO. 2 ENGINE START MOTOR DOES NOT SHUTDOWN WHEN POWER CONTROL LEVER IS PULLED DOWN (SPEED BELOW 52% NG)**SYMPTOM**

No. 1 or No. 2 engine start motor does not shutdown.

MALFUNCTION

No. 1 or No. 2 engine start motor does not shutdown when power control lever is pulled down (speed below 52% Ng).

CORRECTIVE ACTION

- 1. Adjust starter abort switch (WP 0529 00). Go to 2.**
- 2. Pull lever down, and check continuity between J105-J and J105-K.**
 - a. If continuity is present, replace engine control quadrant (WP 0501 00). Go to **3**.
 - b. If continuity is not present, engine starter system is operational. Go to **3**.
- 3. Procedure completed.**

NO. 1 ENGINE STARTER DOES NOT MOTOR ENGINE (COMPRESSOR IS NOT ROTATING) WHEN USING ENGINE STARTER OVERRIDE**SYMPTOM**

No. 1 engine starter does not motor engine.

MALFUNCTION

No. 1 engine starter does not motor engine (compressor is not rotating) when using engine starter override.

CORRECTIVE ACTION

- 1. With No. 1 engine START button pressed, check continuity between J105-S and J105-T.**
 - a. If continuity is present, go to **2**.

NO. 1 ENGINE STARTER DOES NOT MOTOR ENGINE (COMPRESSOR IS NOT ROTATING) WHEN USING ENGINE STARTER OVERRIDE - Continued

- b. If continuity is not present, replace engine control quadrant starter override switch (WP 0529 00). Go to **15**.
- 2. Install an insulated jumper wire between P105-K and P105-T. Go to 3.**
- 3. Check for 28 vdc between P242-V and ground.**
 - a. If voltage is as specified, go to **4**.
 - b. If voltage is not as specified, go to **13**.
- 4. Remove jumper wire from P105. Go to 5.**
- 5. Install insulated jumper wire between P105-K and P105-S. Go to 6.**
- 6. Check for 28 vdc between J221-Y and ground.**
 - a. If voltage is as specified, go to **7**.
 - b. If voltage is not as specified, go to **9**.
- 7. Remove jumper wire from P105. Go to 8.**
- 8. Troubleshoot left relay panel (WP 0121 00). Go to 15.**
- 9. Remove jumper wire from P105. Go to 10.**
- 10. Check helicopter configuration.**
 - a. **WINTER** , go to **11**.
 - b. **W/O WINTER** , go to **12**.
- 11. **WINTER** Replace **WINTER** (WP 1400 00).**
 - a. If trouble remains, go to **12**.
- 12. Repair/replace wiring between P105-S and P244-M (WP 1747 00). Go to 15.**
- 13. Remove jumper wire from P105. Go to 14.**
- 14. Repair/replace wiring between P105-T and P242-V (WP 1747 00). Go to 15.**
- 15. Procedure completed.**

NO FUEL IS COMING OUT OF NO. 1 ENGINE FUEL DRAIN**SYMPTOM**

No fuel.

MALFUNCTION

No fuel is coming out of No. 1 engine fuel drain.

CORRECTIVE ACTION

- 1. Visually Check Drain Tube For Obvious Obstructions.**
 - a. If obstructions are present, remove obstructions (WP 0990 00). Go to **3**.
 - b. If obstructions are not present, go to **2**.
- 2. Check for 28 vdc between TB1-7 and ground.**
 - a. If voltage is as specified, troubleshoot fuel prime boost system (WP 0130 00). Go to **3**.

NO FUEL IS COMING OUT OF NO. 1 ENGINE FUEL DRAIN - Continued

- b. If voltage is not as specified, repair/replace wiring between TB1 and ground (WP 1747 00). Go to **3**.

3. Procedure completed.**NO. 2 ENGINE DOES NOT START (COMPRESSOR NOT ROTATING) (#2 ENGINE STARTER CAPSULE OR LEGEND IS ON)****SYMPTOM**

No. 2 engine does not start.

MALFUNCTION

No. 2 engine does not start (compressor not rotating) (#2 ENGINE STARTER capsule or legend is on).

CORRECTIVE ACTION

- 1. Troubleshoot APU system for faulty start bypass valve (WP 0156 00).**
 - a. If APU system start bypass valve is faulty, replace valve (WP 0743 00). Go to **7**.
 - b. If APU system start bypass valve is good, go to **2**.
- 2. Check all clamps and rubber connections on pneumatic lines between start motor and APU check valve for proper connections or damage.**
 - a. If connections are good, go to **3**.
 - b. If connections are not good, repair/replace clamps or connections as required. Go to **7**.
- 3. Disconnect start motor from engine (WP 0694 00 or WP 0695 00). Go to 4.**

NOTE

Turning the gear train requires considerable effort to overcome resistance.

- 4. Check start motor for sheared/seized shaft by grasping shaft spline end and rotating clockwise, making sure turbine wheel also rotates.**
 - a. If start motor has sheared/seized shaft, replace start motor (WP 0694 00 or WP 0695 00). Go to **7**.
 - b. If start motor does not have sheared/seized shaft, go to **5**.
- 5. Check engine for seized condition (TM 1-2840-248-23). Go to 6.**
- 6. Install start motor (WP 0694 00 or WP 0695 00). Go to 7.**
- 7. Procedure completed.**

NO. 2 ENGINE DOES NOT START (COMPRESSOR NOT ROTATING) (#2 ENGINE STARTER CAPSULE OR LEGEND IS OFF)**SYMPTOM**

No. 2 engine does not start.

MALFUNCTION

No. 2 engine does not start (compressor not rotating) (#2 ENGINE STARTER capsule or legend is off).

CORRECTIVE ACTION

- 1. With APU shutdown and external power on, press No. 2 engine START button and check for 28 vdc between P450-1 and ground.**
 - a. If voltage is as specified, go to **2**.
 - b. If voltage is not as specified, go to **3**.
- 2. Check continuity between P450-2 and ground.**
 - a. If continuity is present, replace No. 2 engine start control valve (WP 0696 00). Go to **10**.
 - b. If continuity is not present, repair/replace wiring between P450-2 and ground (WP 1747 00). Go to **10**.
- 3. Check continuity between J106-J and J106-K.**
 - a. If continuity is present, go to **4**.
 - b. If continuity is not present, replace engine control quadrant assembly start abort switch (WP 0528 00). Go to **10**.
- 4. Press No. 2 engine start button and check continuity between J106-G and J106-H.**
 - a. If continuity is present, go to **5**.
 - b. If continuity is not present, go to **9**.
- 5. Check for 28 vdc between P106-K and ground.**
 - a. If voltage is as specified, go to **6**.
 - b. If voltage is not as specified, go to **7**.
- 6. Check for 28 vdc between P106-G and ground.**
 - a. If voltage is as specified, go to **8**.
 - b. If voltage is not as specified, repair/replace wiring between NO. 2 ENG START CONTR circuit breaker terminal 1 and P106-G (WP 1747 00). Go to **10**.
- 7. Check for 28 vdc between NO. 2 ENG START CONTR circuit breaker terminal 1 and ground.**
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P106-K (WP 1747 00). Go to **10**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **10**.
- 8. Check continuity between:**
 - **P241-a and P450-1**
 - **P241-b and P106-J**

NO. 2 ENGINE DOES NOT START (COMPRESSOR NOT ROTATING) (#2 ENGINE STARTER CAPSULE OR LEGEND IS OFF) - Continued

- **P241-R and P106-H**

- a. If continuity is as specified, replace right relay panel (WP 0825 00). Go to **10**.
- b. If continuity is not as specified, repair/replace wiring (WP 1747 00). Go to **10**.

9. Check No. 2 engine starter switch for proper adjustment (WP 0503 00).

- a. If switch is adjusted properly, replace starter switch (WP 0527 00). Go to **10**.
- b. If switch is not adjusted properly, adjust switch (WP 0503 00). Go to **10**.

10. Procedure completed.

NO. 2 ENGINE START MOTOR STOPS (#2 ENGINE STARTER CAPSULE OR LEGEND GOES OFF) WHEN START BUTTON IS RELEASED**SYMPTOM**

No. 2 engine start motor stops.

MALFUNCTION

No. 2 engine start motor stops (#2 ENGINE STARTER capsule or legend goes off) when START button is released.

CORRECTIVE ACTION

- 1. Disconnect P314 from starter speed switch. Go to 2.**
- 2. Install an insulated jumper wire between P314-J and P314-K. Go to 3.**
- 3. Make sure No. 2 engine POWER CONT LEVER is OFF. Go to 4.**
- 4. With qualified personnel at control and APU operating, press No. 2 engine START button, then release. Go to 5.**
- 5. Check that #2 ENGINE STARTER capsule or legend goes on and engine continues to motor after START button is released.**
 - a. If capsule or legend goes on and engine motors, go to **6**.
 - b. If capsule or legend does not go on and/or engine does not motor, go to **9**.
- 6. Shut down APU (TM 1-1520-237-10). Go to 7.**
- 7. Remove jumper wire from P314. Go to 8.**
- 8. Replace starter speed switch (WP 0704 00). Go to 19.**
- 9. Remove jumper wire from P314. Do not install connector to speed switch. Go to 10.**
- 10. With qualified personnel at controls and APU operating, press No. 2 engine START button and hold. Go to 11.**
- 11. Advance NO. 2 ENG POWER CONT LEVER to a position between OFF and IDLE. Go to 12.**
- 12. Check that No. 2 engine continues to motor with ENG POWER CONT lever away from OFF.**
 - a. If No. 2 engine continues to motor, go to **13**.
 - b. If No. 2 engine does not continue to motor, go to **16**.
- 13. Shut down APU (TM 1-1520-237-10). Go to 14.**

NO. 2 ENGINE START MOTOR STOPS (#2 ENGINE STARTER CAPSULE OR LEGEND GOES OFF) WHEN START BUTTON IS RELEASED - Continued

14. Connect P314 to starter speed switch. Go to 15.
15. Repair/replace wiring, as required, between (WP 1747 00), then go to 19:
 - P106-T and P314-K
 - P314-J and P450-5
16. Shut down APU (TM 1-1520-237-10). Go to 17.
17. Reconnect P314 to starter speed switch. Go to 18.
18. Go to NO. 1 OR NO. 2 ENGINE STARTER CAPSULE OR LEGEND GOES OFF BEFORE 52% NG ENGINE SPEED, in this work package.
19. Procedure completed.

NO. 2 ENGINE STARTS, BUT #2 ENGINE STARTER CAPSULE OR LEGEND IS OFF (APU IS NOT STALLING)

SYMPTOM

No. 2 engine starts, but #2 ENGINE STARTER capsule or legend is off.

MALFUNCTION

No. 2 engine starts, but #2 ENGINE STARTER capsule or legend is off (APU is not stalling).

CORRECTIVE ACTION

1. Hold caution/advisory panel **HH-60A HH-60L** instrument panel INDICATOR LTS **■** BRT/DIM-TEST switch to TEST. Go to 2.
2. Check that #2 ENGINE STARTER capsule or legend goes on.
 - a. If capsule or legend goes on, go to 3.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 12.
3. Install an insulated jumper wire between P450-3 and P450-5. Go to 4.
4. Check helicopter configuration.
 - a. **EMEP** , go to 5.
 - b. **W/O EMEP** , go to 7.
5. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R **■**
 - a. If pin filtered adapter is good, go to 6.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 12.
6. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 7.
7. Check for 28 vdc between P117-55 or **HH-60A HH-60L** P1019R-59 or P1020R-59 **■** and ground.
 - a. If voltage is as specified, go to 8.
 - b. If voltage is not as specified, go to 10.
8. Remove jumper wire from P450. Go to 9.

NO. 2 ENGINE STARTS, BUT #2 ENGINE STARTER CAPSULE OR LEGEND IS OFF (APU IS NOT STALLING) - Continued

9. Troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 12.
10. Remove jumper wire from P450. Go to 11.
11. Repair/replace wiring between P117-55 or **HH-60A HH-60L** P1019R-59 or P1020R-59 and P241-W (WP 1747 00). Go to 12.
12. Procedure completed.

NO. 2 ENGINE STARTER MOTORS ENGINE (#2 ENGINE STARTER CAPSULE OR LEGEND IS ON), BUT APU STALLS (POPPING SOUND IS HEARD)**SYMPTOM**

No. 2 engine starter motors engine, but APU stalls.

MALFUNCTION

No. 2 engine starter motors engine (#2 ENGINE STARTER capsule or legend is on), but APU stalls (popping sound is heard).

CORRECTIVE ACTION

1. Start APU (TM 1-1520-237-10). Go to 2.
2. Motor No. 1 engine (TM 1-1520-237-10). Go to 3.
3. Check if APU stalls.
 - a. If APU stalls, go to 4.
 - b. If APU does not stall, shut down APU (TM 1-1520-237-10). Replace right relay panel (WP 0825 00). Go to 16.
4. Troubleshoot APU system, start bypass valve circuitry (WP 0155 00 or WP 0156 00).
 - a. If the APU start bypass valve and circuitry operates properly, go to 5.
5. Install an insulated jumper wire between P450-3 and P450-5. Go to 6.
6. Check for 28 vdc between P241-W and ground.
 - a. If voltage is as specified, go to 7.
 - b. If voltage is not as specified, go to 14.
7. Remove jumper wire from P450. Go to 8.
8. Check helicopter configuration.
 - a. **WINTER** , go to 9.
 - b. **W/O WINTER** , go to 10.
9. **WINTER** With AIR SOURCE HEAT/START switch in APU position, check for 28 vdc between P244B-T and ground.
 - a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to 16.
 - b. If voltage is not as specified, replace **WINTER** (WP 1400 00).
 - c. If trouble remains, go to 10.

NO. 2 ENGINE STARTER MOTORS ENGINE (#2 ENGINE STARTER CAPSULE OR LEGEND IS ON), BUT APU STALLS (POPPING SOUND IS HEARD) - Continued

- 10. With AIR SOURCE HEAT/START switch in APU position, check for 28 vdc between P244-T and ground.**
 - a. If voltage is as specified, replace right relay panel (WP 0825 00). Go to **16**.
 - b. If voltage is not as specified, go to **11**.
- 11. Check continuity between AIR SOURCE HEAT/START switch terminal 1 and P244-T.**
 - a. If continuity is present, go to **12**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **16**.
- 12. Check for 28 vdc between AIR SOURCE HEAT/START switch terminal 2 and ground.**
 - a. If voltage is as specified, replace switch (WP 0828 00). Go to **16**.
 - b. If voltage is not as specified, go to **13**.
- 13. Check for 28 vdc between AIR SOURCE HEAT/START circuit breaker terminal 1 and ground.**
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and switch (WP 1747 00). Go to **16**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **16**.
- 14. Remove jumper wire from P450. Go to 15.**
- 15. Repair/replace wiring (WP 1747 00). Go to 16.**
- 16. Procedure completed.**

NO. 2 ENGINE MOTORS, BUT DOES NOT IGNITE AND ACCELERATE TO IDLE SPEED**SYMPTOM**

No. 2 engine motors, but does not ignite and accelerate.

MALFUNCTION

No. 2 engine motors, but does not ignite and accelerate to idle speed.

CORRECTIVE ACTION

- 1. Move ENG POWER CONT lever on control quadrant in cockpit to LOCKOUT and hold. Go to 2.**
- 2. Move ENG FUEL SYS selector lever on control quadrant to DIR detent. Go to 3.**
- 3. Place FUEL PUMP switch to FUEL PRIME. Go to 4.**
- 4. Check that fuel is coming from overboard drain.**
 - a. If fuel is coming from overboard drain, go to **5**.
 - b. If fuel is not coming from overboard drain, troubleshoot fuel prime boost system (WP 0130 00). Go to **20**.
- 5. Troubleshoot engine for no ignite (TM 1-2840-248-23).**
 - a. If engine is good, go to **6**.
- 6. Install an insulated jumper wire between P802-17 and P802-18. Go to 7.**

NO. 2 ENGINE MOTORS, BUT DOES NOT IGNITE AND ACCELERATE TO IDLE SPEED - Continued

7. Check continuity between P121-b and P121-d.
 - a. If continuity is present, go to 8.
 - b. If continuity is not present, go to 18.
8. Check continuity between P241-X and P241-Y.
 - a. If continuity is present, go to 9.
 - b. If continuity is not present, go to 16.
9. Remove jumper wire from P802. Go to 10.
10. Check helicopter configuration.
 - a. UH-60L HH-60L , go to 11.
 - b. EH-60A HH-60A UH-60A , go to 14.
11. HH-60L UH-60L Check continuity between:
 - P802-17 and P814-35
 - P802-19 and P814-35
 - a. If continuity is present, go to 12.
 - b. If continuity is not present, replace harness (TM 1-2840-248-23). Go to 20.
12. UH-60L HH-60L Check continuity between J814-35 and P382-F.
 - a. If continuity is present, go to 13.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 20.
13. UH-60L HH-60L Check continuity between P121-d and P382-G.
 - a. If continuity is present, troubleshoot engine overspeed protection system (WP 0075 00). Go to 20.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 20.
14. EH-60A UH-60A HH-60A On instrument panel, turn ENGINE IGNITION switch to ON. Go to 15.
15. EH-60A UH-60A HH-60A Check continuity between J121-b and J121-d.
 - a. If continuity is present, replace keylock ignition switch (WP 0923 00). Go to 20.
 - b. If continuity is not present, replace right relay panel (WP 0825 00). Go to 20.
16. Remove jumper wire from P802. Go to 17.
17. Repair/replace wiring, as required, between (WP 1747 00), then go to 20:
 - P121-b and P241-X
 - P121-d and P241-Y
18. Remove jumper wire from P802. Go to 19.
19. Repair/replace wiring, as required, between (WP 1747 00), then go to 20:
 - P121-b and P802-18
 - P121-d and P802-17

NO. 2 ENGINE MOTORS, BUT DOES NOT IGNITE AND ACCELERATE TO IDLE SPEED - Continued

20. Procedure completed.

#2 ENGINE STARTER CAPSULE OR LEGEND DOES NOT GO OFF (ENGINE SPEED ABOVE 65% NG)**SYMPTOM**

#2 ENGINE STARTER capsule or legend does not go off.

MALFUNCTION

#2 ENGINE STARTER capsule or legend does not go off (engine speed above 65% Ng).

CORRECTIVE ACTION

- 1. Visually check No. 2 engine start control valve position indicator. Go to 2.**
- 2. Check No. 2 engine override switch for proper adjustment (WP 0505 00).**
 - a. If switch is adjusted properly, replace switch (WP 0529 00). Go to **14**.
 - b. If switch is not adjusted properly, adjust switch (WP 0505 00). Go to **3**.
- 3. Check if indicator shows CLOSED.**
 - a. If indicator shows CLOSED, go to **4**.
 - b. If indicator does not show CLOSED, replace start control valve (WP 0696 00). Go to **14**.
- 4. Check continuity between J450-3 and J450-4.**
 - a. If continuity is present, go to **5**.
 - b. If continuity is not present, replace start control valve (WP 0696 00). Go to **14**.
- 5. Install an insulated jumper wire between P806-1 and P806-3. Go to 6.**
- 6. Check continuity between P314-L and P314-M.**
 - a. If continuity is present, go to **7**.
 - b. If continuity is not present, go to **11**.
- 7. Remove jumper wire from P806. Go to 8.**
- 8. Replace starter speed switch (WP 0704 00). Go to 9.**
- 9. Operate No. 2 engine up to idle speed (TM 1-1520-237-10). Go to 10.**
- 10. Check if #2 ENGINE STARTER capsule or legend goes off.**
 - a. If capsule or legend is off, engine start system is operational. Shut down engine (TM 1-1520-237-10). Go to **14**.
 - b. If capsule or legend is not off, shut down engine (TM 1-1520-237-10). Replace No. 2 engine start motor (WP 0694 00 or WP 0695 00). Go to **14**.
- 11. Shut down engine (TM 1-1520-237-10). Go to 12.**
- 12. Remove jumper wire from P806. Go to 13.**
- 13. Repair/replace wiring, as required, between (WP 1747 00), then go to 14:**
 - **P314-L and P806-1**
 - **P314-M and P806-3**
- 14. Procedure completed.**

NO. 2 ENGINE STARTER DOES NOT MOTOR ENGINE (COMPRESSOR IS NOT ROTATING) WHEN USING ENGINE STARTER OVERRIDE**SYMPTOM**

No. 2 engine starter does not motor engine.

MALFUNCTION

No. 2 engine starter does not motor engine (compressor is not rotating) when using engine starter override.

CORRECTIVE ACTION

- 1. With No. 2 engine START button pressed, check continuity between J106-S and J106-T.**
 - a. If continuity is present, go to **2**.
 - b. If continuity is not present, replace engine control quadrant starter override switch (WP 0529 00). Go to **13**.
- 2. Install an insulated jumper wire between P106-K and P106-T. Go to 3.**
- 3. Check for 28 vdc between P241-U and ground.**
 - a. If voltage is as specified, go to **4**.
 - b. If voltage is not as specified, go to **11**.
- 4. Remove jumper wire from P106. Go to 5.**
- 5. Install an insulated jumper wire between P106-K and P106-S. Go to 6.**
- 6. Check for 28 vdc between J220-x and ground.**
 - a. If voltage is as specified, go to **7**.
 - b. If voltage is not as specified, go to **9**.
- 7. Remove jumper wire from P106. Go to 8.**
- 8. Replace right relay panel (WP 0825 00). Go to 13.**
- 9. Remove jumper wire from P106. Go to 10.**
- 10. Repair/replace wiring between P106-S and P220-x (WP 1747 00). Go to 13.**
- 11. Remove jumper wire from P106. Go to 12.**
- 12. Repair/replace wiring between P106-T and P241-U (WP 1747 00). Go to 13.**
- 13. Procedure completed.**

NO FUEL COMING OUT OF NO. 2 FUEL DRAIN**SYMPTOM**

No fuel.

MALFUNCTION

No fuel coming out of No. 2 fuel drain.

CORRECTIVE ACTION

- 1. Visually check drain tube for obvious obstructions.**

NO FUEL COMING OUT OF NO. 2 FUEL DRAIN - Continued

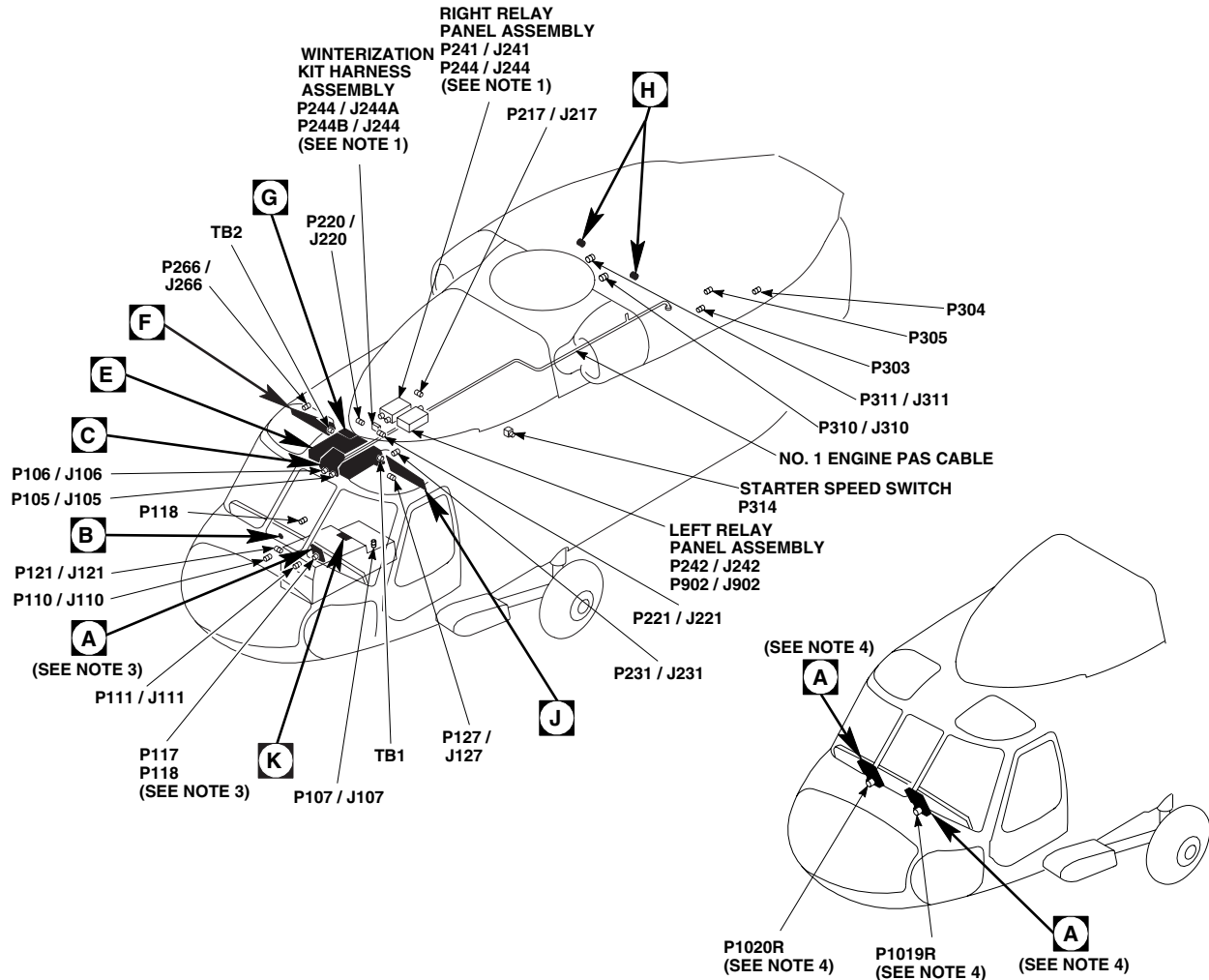
- a. If obstructions are present, remove obstructions (WP 0990 00). Go to **3**.
- b. If obstructions are not present, go to **2**.

2. Check for 28 vdc between TB2-7 and ground.

- a. If voltage is as specified, troubleshoot fuel prime boost system (WP 0130 00). Go to **3**.
- b. If voltage is not as specified, repair/replace wiring (WP 1747 00). Go to **3**.

3. Procedure completed.

LOCATION



NOTES

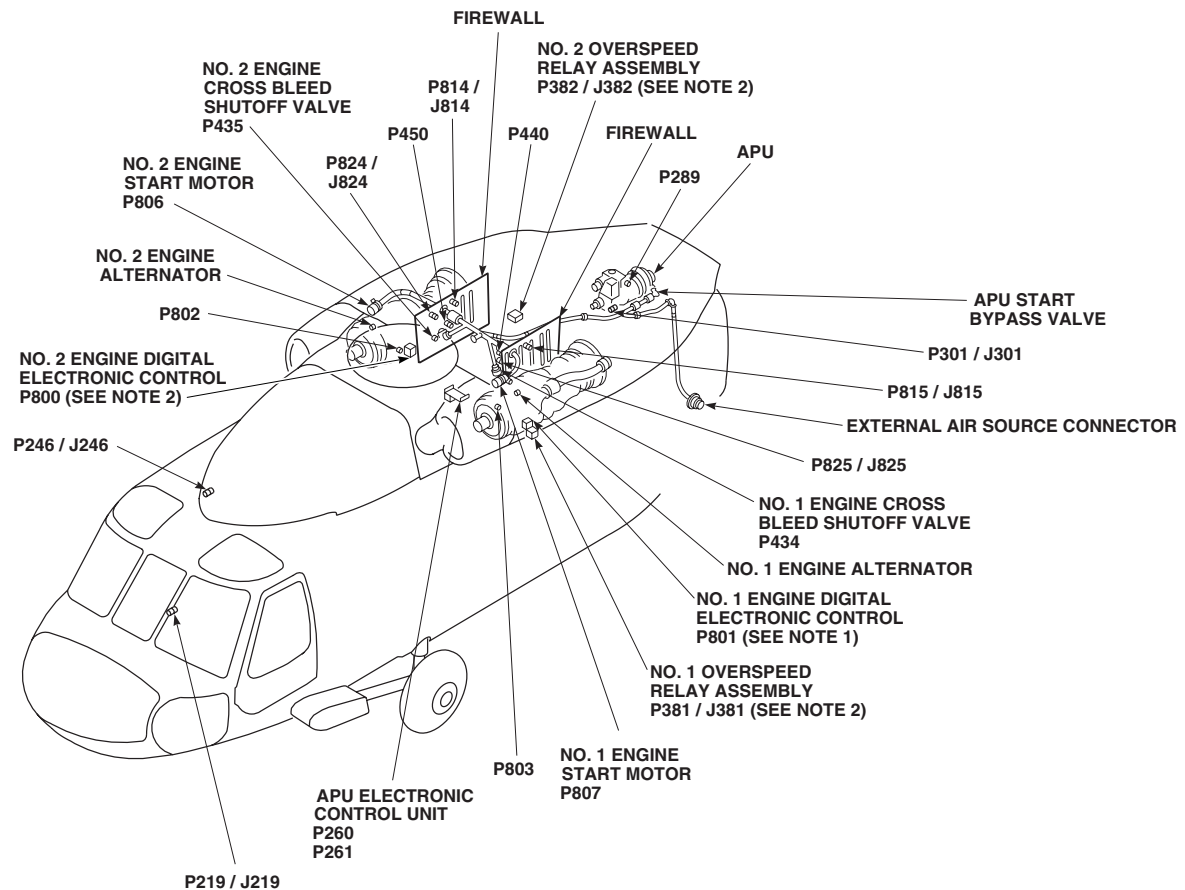
1. **WINTER**
2. **UH60L**
3. **UH60A UH60L EH60A**
4. **HH60A HH60L**
5. **HH60A HH60L** ENGINE IGNITION ON LOWER CONSOLE.

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P105 / J105	NO. 1 ENGINE CONTROL QUADRANT
P106 / J106	NO. 2 ENGINE CONTROL QUADRANT
P107 / J107	COCKPIT, BL 9 LH, STA 236
P110 / J110	COCKPIT, BL 7.5 RH, STA 197
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P117 / J117	CAUTION / ADVISORY PANEL (SEE NOTE 3)
P118 / J118	CAUTION / ADVISORY PANEL (SEE NOTE 3)
P121 / J121	COCKPIT, BL 10 RH, STA 197
P127 / J127	COCKPIT CEILING, BL 28 LH, STA 243

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Figure 1. Engine Start and Ignition System Location Diagram. (Sheet 1 of 6)

LOCATION - Continued



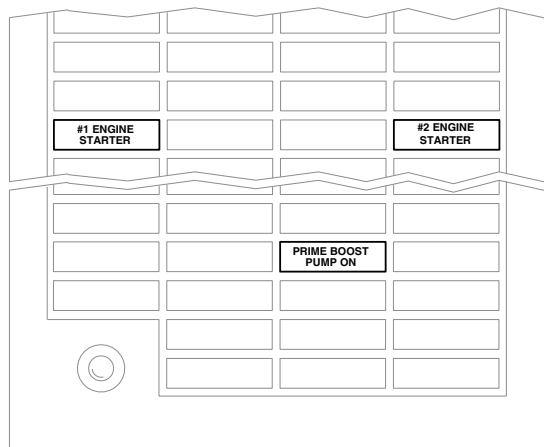
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P217 / J217	CABIN CEILING, BL 0, STA 284
P219 / J219	CABIN FLOOR, BL 25 RH, STA 247
P220 / J220	JUNCTION BOX, MAIN ROTOR PYLON DECK
P221 / J221	JUNCTION BOX, MAIN ROTOR PYLON DECK
P231 / J231	CABIN CEILING, BL 8 RH, STA 247
P241 / J241	RIGHT RELAY PANEL ASSEMBLY (SEE NOTE 1)
P242 / J242	LEFT RELAY PANEL ASSEMBLY
P244 / J244	RIGHT RELAY PANEL ASSEMBLY (SEE NOTE 1)

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P244 / J244A P244B / J244	WINTERIZATION KIT HARNESS ASSEMBLY (SEE NOTE 1)
P246 / J246	CABIN CEILING, BL 5 RH, STA 247
P260	APU ELECTRONIC CONTROL UNIT
P261	APU ELECTRONIC CONTROL UNIT
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 25 RH
P289	APU START BYPASS VALVE
P301 / J301	TRANSITION DECK, STA 415
P303	PRIME BOOST PUMP
P304	NO. 1 ENGINE PRIME SHUTOFF VALVE

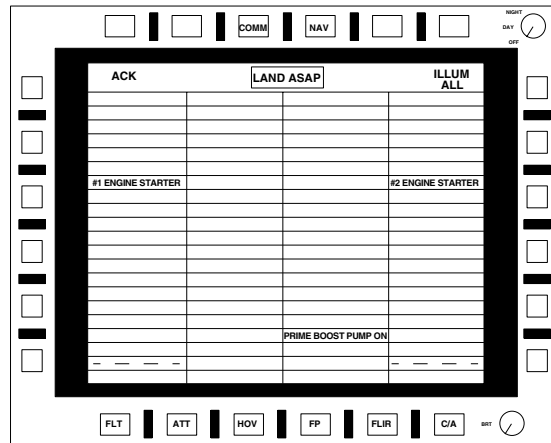
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Figure 1. Engine Start and Ignition System Location Diagram. (Sheet 2 of 6)

LOCATION - Continued



CAUTION/ADVISORY PANEL
(SEE NOTE 3)



**MULTIFUNCTION DISPLAY'S
CAUTION/ADVISORY GRID**
(SEE NOTE 4)

A

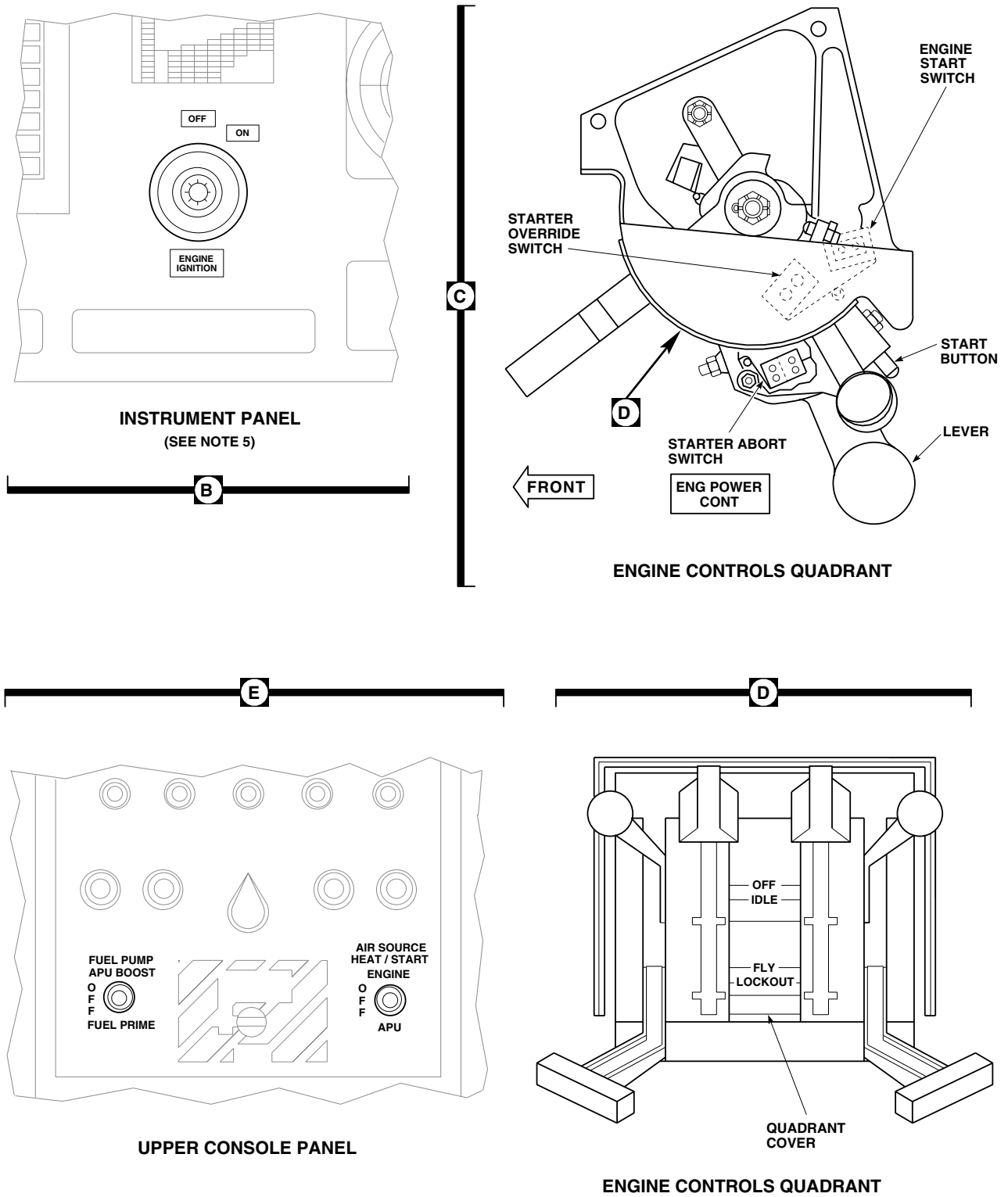
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P305	NO. 1 ENGINE PRIME SHUTOFF VALVE
P310 / J310	CABIN CEILING, BL 17 RH, STA 387
P311 / J311	CABIN CEILING, BL 16.5 LH, STA 383
P314	STARTER SPEED SWITCH
P381 / J381	NO. 1 OVERSPEED RELAY ASSEMBLY (SEE NOTE 2)
P382 / J382	NO. 2 OVERSPEED RELAY ASSEMBLY (SEE NOTE 2)
P434	NO. 1 ENGINE CROSSBLEED SHUTOFF VALVE
P435	NO. 2 ENGINE CROSSBLEED SHUTOFF VALVE
P440	NO. 1 ENGINE START CONTROL VALVE
P450	NO. 2 ENGINE START CONTROL VALVE
P800	NO. 2 ENGINE DIGITAL ELECTRONIC CONTROL (SEE NOTE 2)
P801	NO. 1 ENGINE DIGITAL ELECTRONIC CONTROL (SEE NOTE 2)

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P802	NO. 2 ENGINE INTERFACE DISCONNECT
P803	NO. 1 ENGINE INTERFACE DISCONNECT
P806	NO. 2 ENGINE START MOTOR
P807	NO. 1 ENGINE START MOTOR
P814 / J814	FIREWALL DISCONNECT BL 14 RH, STA 373
P815 / J815	FIREWALL DISCONNECT BL 14 LH, STA 373
P824 / J824	FIREWALL DISCONNECT BL 14 RH, STA 358.25
P825 / J825	FIREWALL DISCONNECT BL 14 LH, STA 358.25
P902 / J902	LEFT RELAY PANEL ASSEMBLY
TB1	COCKPIT CEILING, BL 9 LH, STA 236
TB2	COCKPIT CEILING, BL 9 RH, STA 236
P1019R / J3	COPILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 4)
P1020R / J3	PILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 4)

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Figure 1. Engine Start and Ignition System Location Diagram. (Sheet 3 of 6)

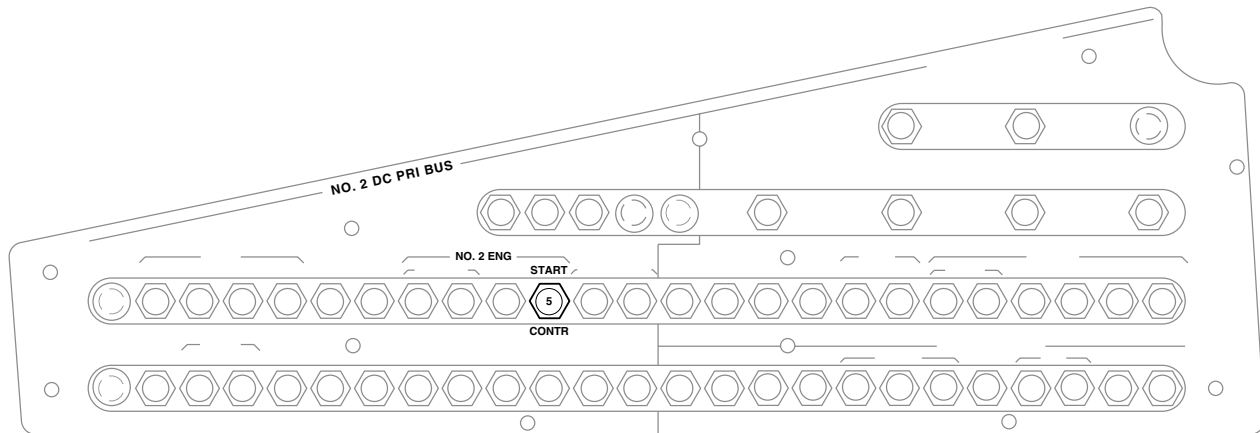
LOCATION - Continued



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Figure 1. Engine Start and Ignition System Location Diagram. (Sheet 4 of 6)

LOCATION - Continued

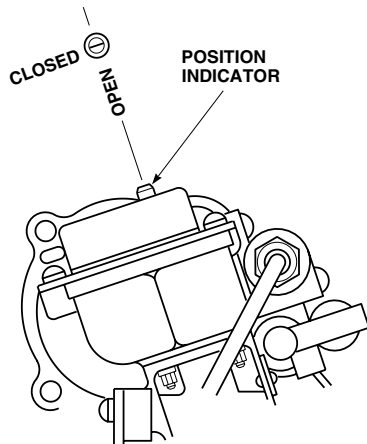


PILOT'S CIRCUIT BREAKER PANEL

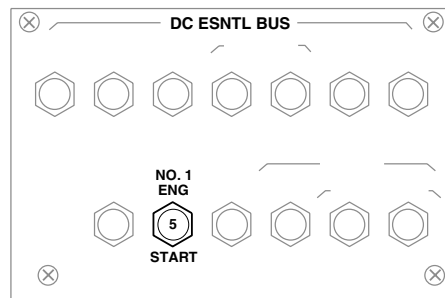
F

H

G



NO. 1 AND NO. 2 ENGINE
START CONTROL VALVE

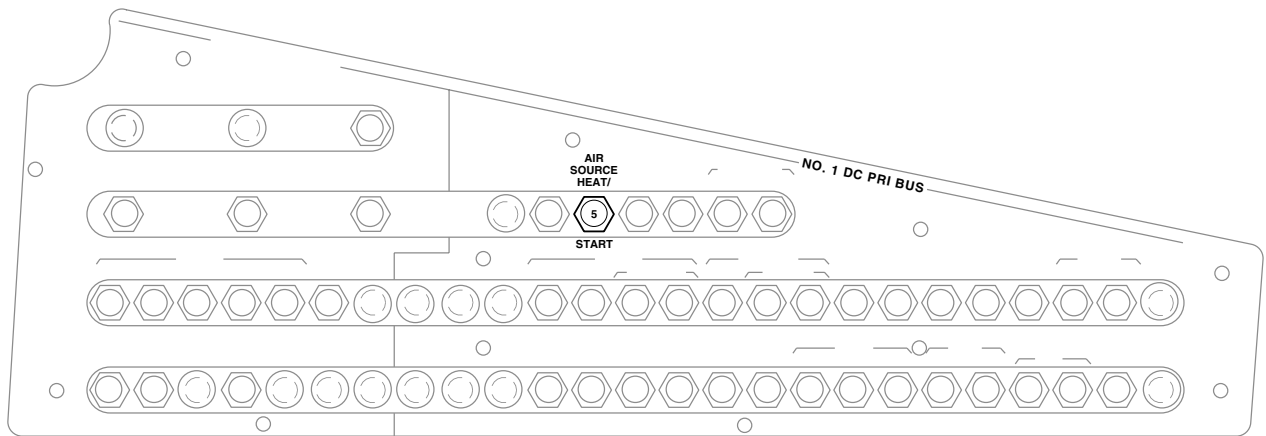


UPPER CONSOLE
CIRCUIT BREAKER PANEL

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Figure 1. Engine Start and Ignition System Location Diagram. (Sheet 5 of 6)

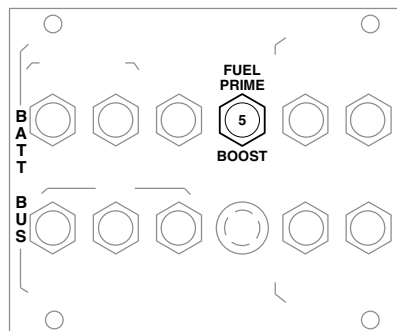
LOCATION - Continued



COPILOT'S CIRCUIT BREAKER PANEL

J

K

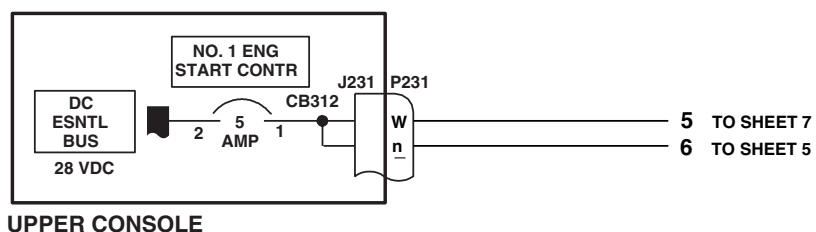
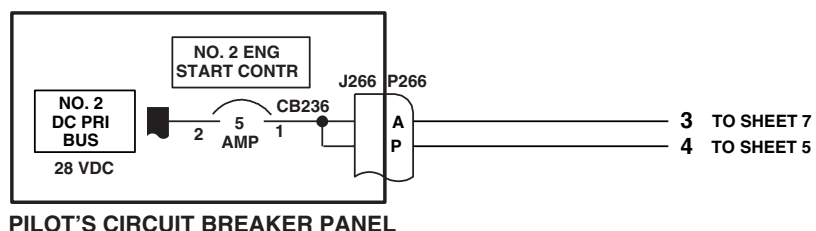
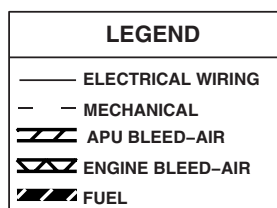
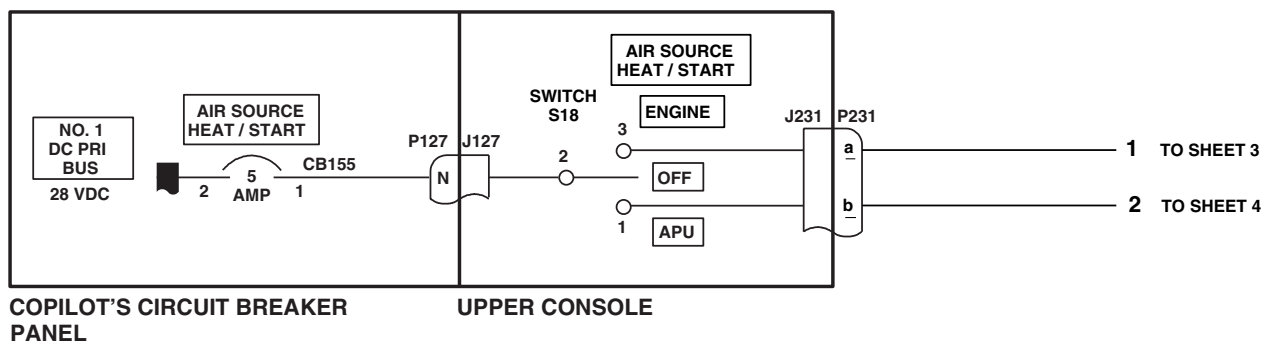


LOWER CONSOLE
CIRCUIT BREAKER PANEL

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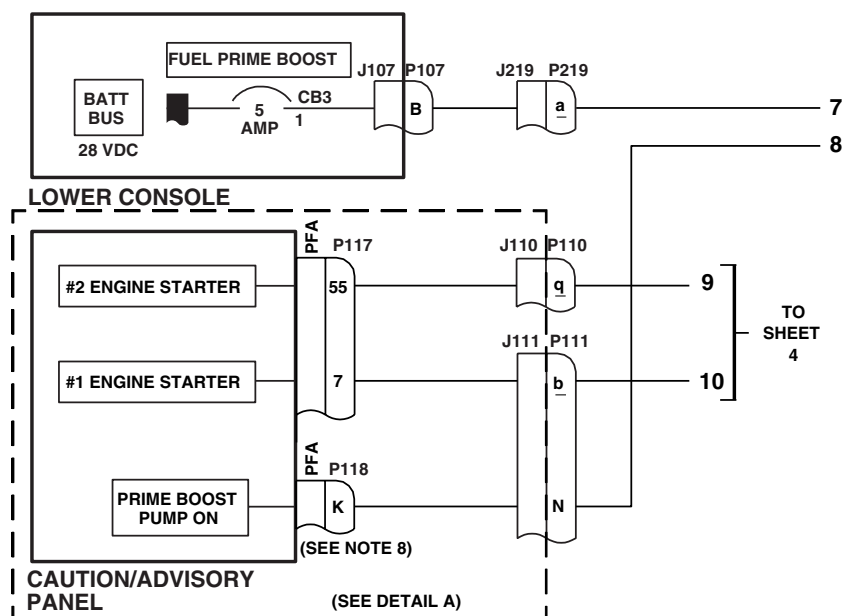
Figure 1. Engine Start and Ignition System Location Diagram. (Sheet 6 of 6)

SCHEMATICS



NOTES

1. POSITION OF SWITCHES ARE WITH **ENG POWER CONT** LEVER IN **OFF** POSITION.
2. ENGINE START SWITCH IS ACTUATED WHEN ENGINE START BUTTON IS PRESSED AND **ENG POWER CONT** LEVER IS **OFF** ONLY.
3. STARTER OVERRIDE SWITCH IS ACTUATED WHEN ENGINE STARTER BUTTON IS PRESSED, REGARDLESS OF **ENG POWER CONT** LEVER POSITION.
4. STARTER ABORT SWITCH IS ACTUATED WHEN **ENG POWER CONT** LEVER IS PULLED DOWN.
5. AIR PRESSURE OPENS VALVE WHEN SOLENOID IS ENERGIZED.
6. **WINTER**
7. **MOD**
8. **W/O EMEP** PIN FILTERED ADAPTER (PFA) IS NOT INSTALLED.
9. **HH-60A HH-60L**

AB2176_1A
SAFigure 2. Engine Start and Ignition System Schematic Diagram **UH-60L MOD HH-60A HH-60L** . (Sheet 1 of 10)

SCHEMATICS - Continued

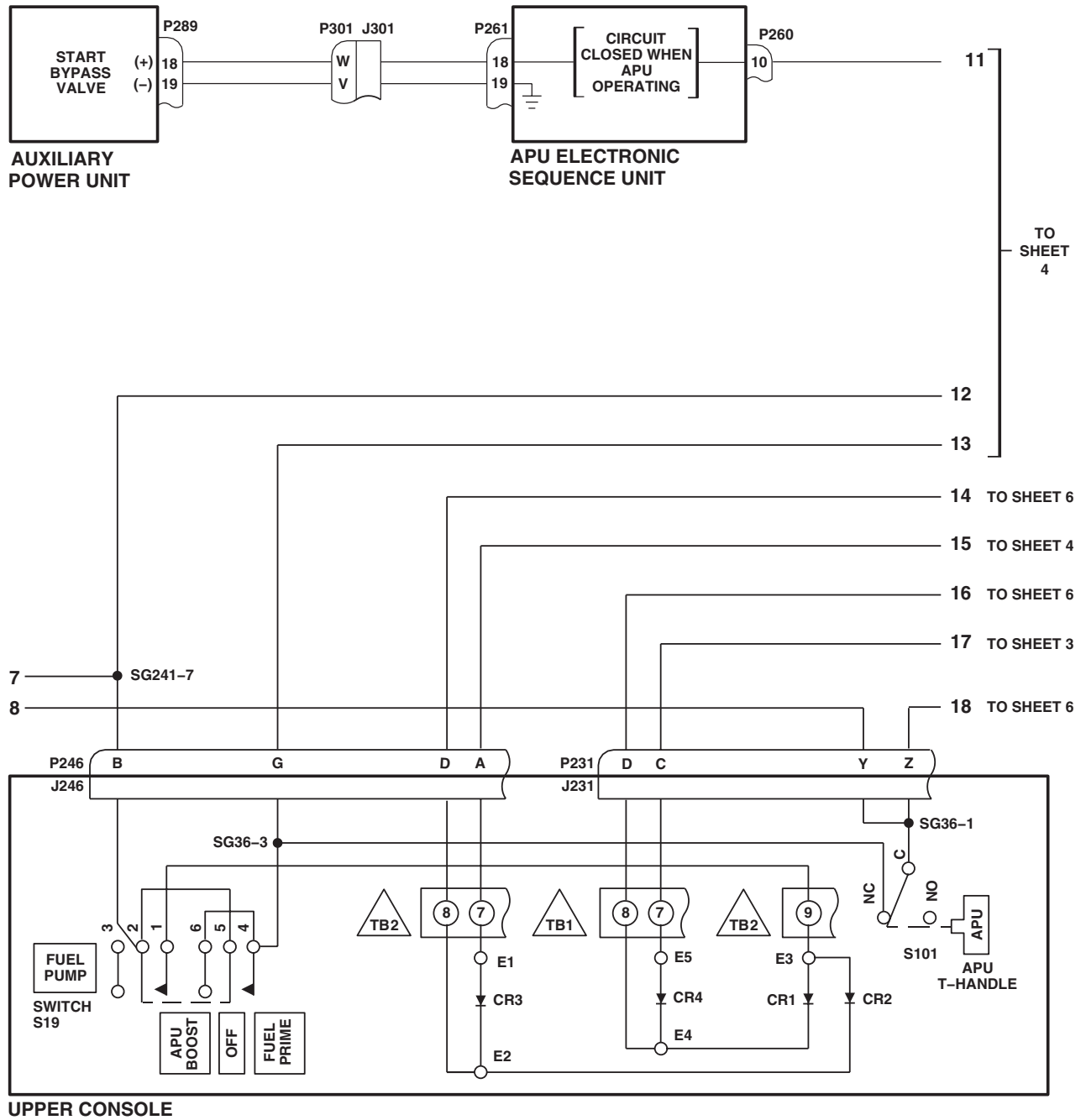


Figure 2. Engine Start and Ignition System Schematic Diagram UH-60L MOD HH-60A HH-60L . (Sheet 2 of 10)

SCHEMATICS - Continued

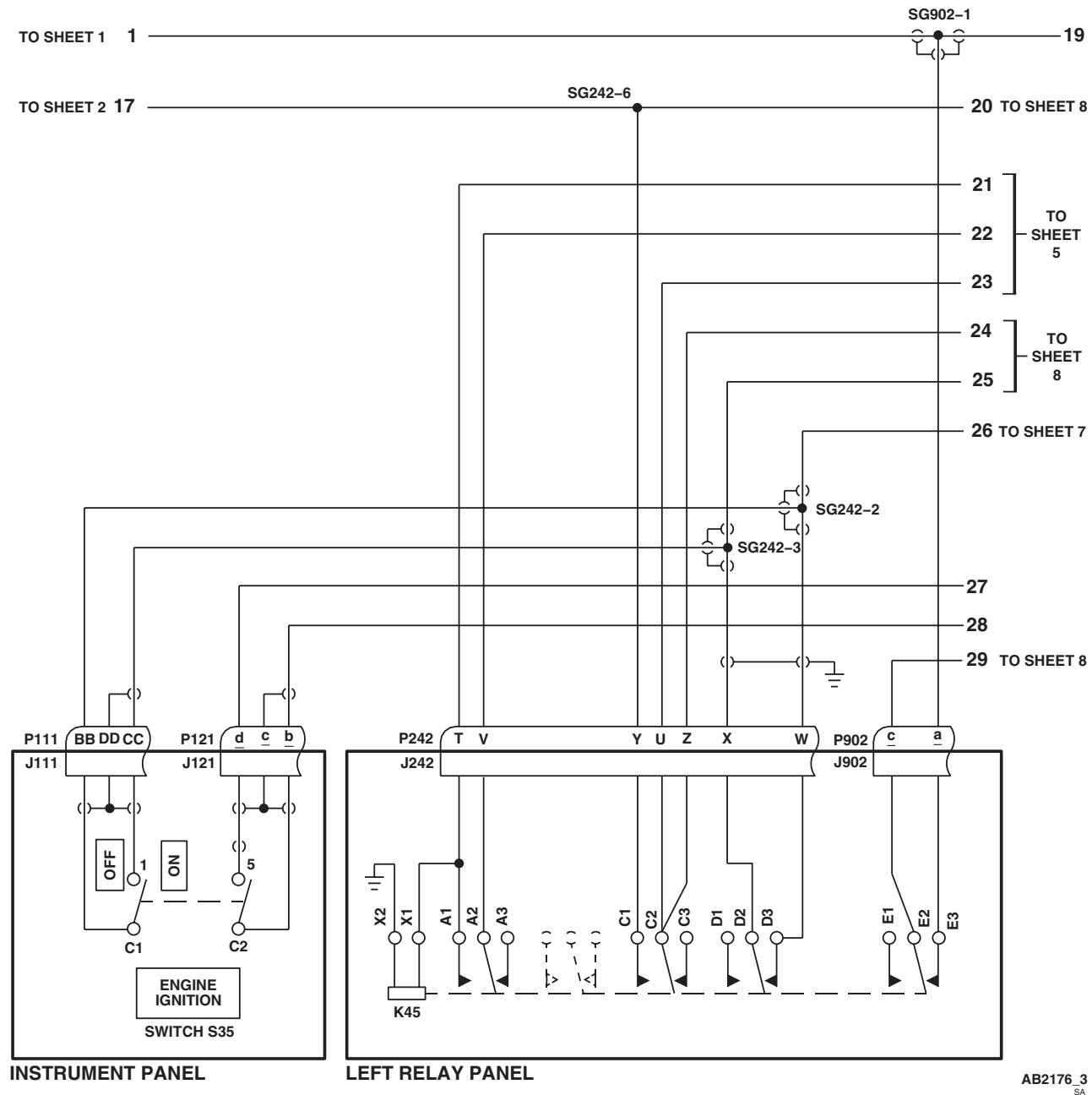
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Figure 2. Engine Start and Ignition System Schematic Diagram UH-60L MOD HH-60A HH-60L (Sheet 3 of 10)

SCHEMATICS - Continued

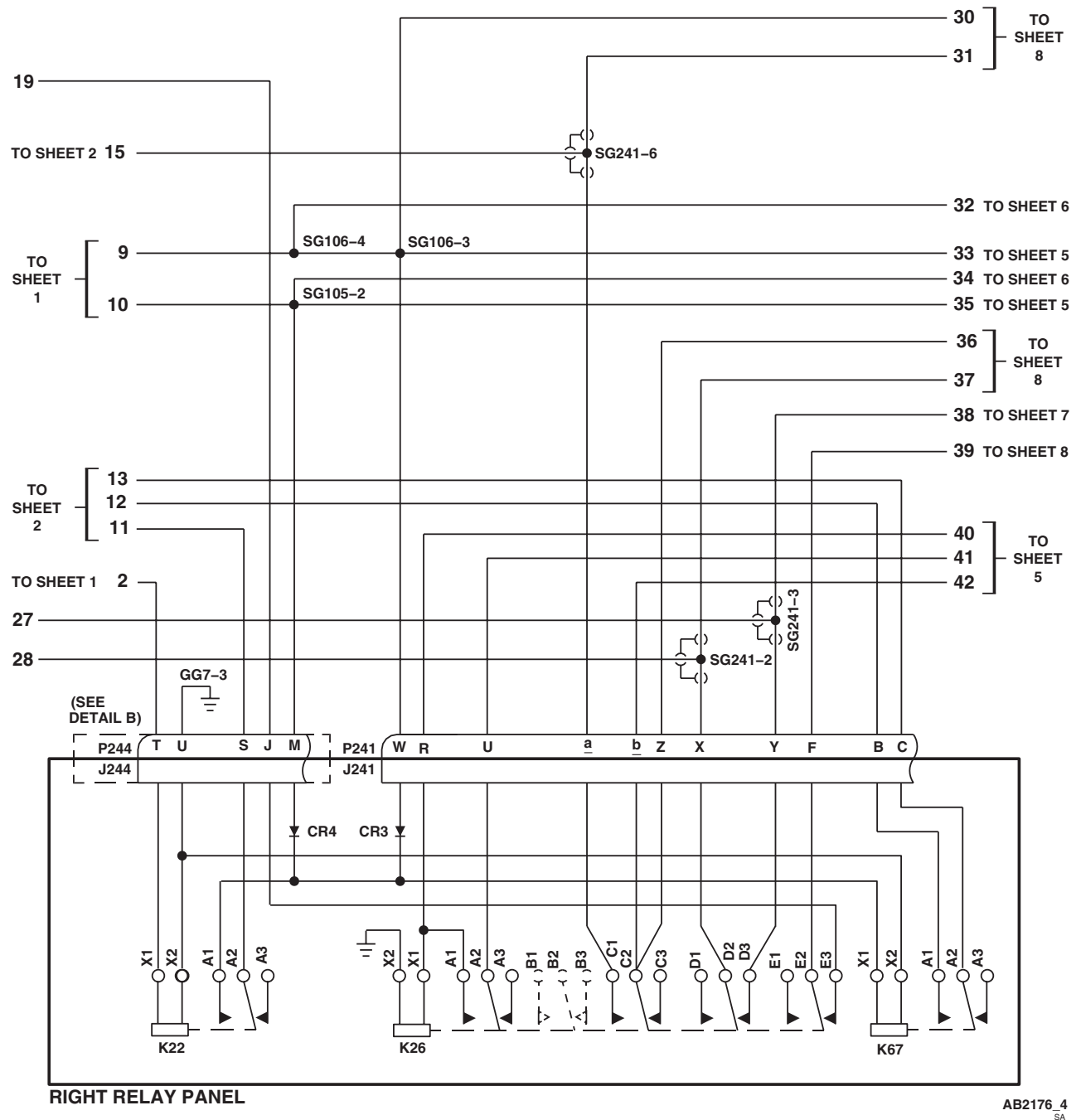


Figure 2. Engine Start and Ignition System Schematic Diagram **UH-60L MOD HH-60A HH-60L** . (Sheet 4 of 10)

SCHEMATICS - Continued

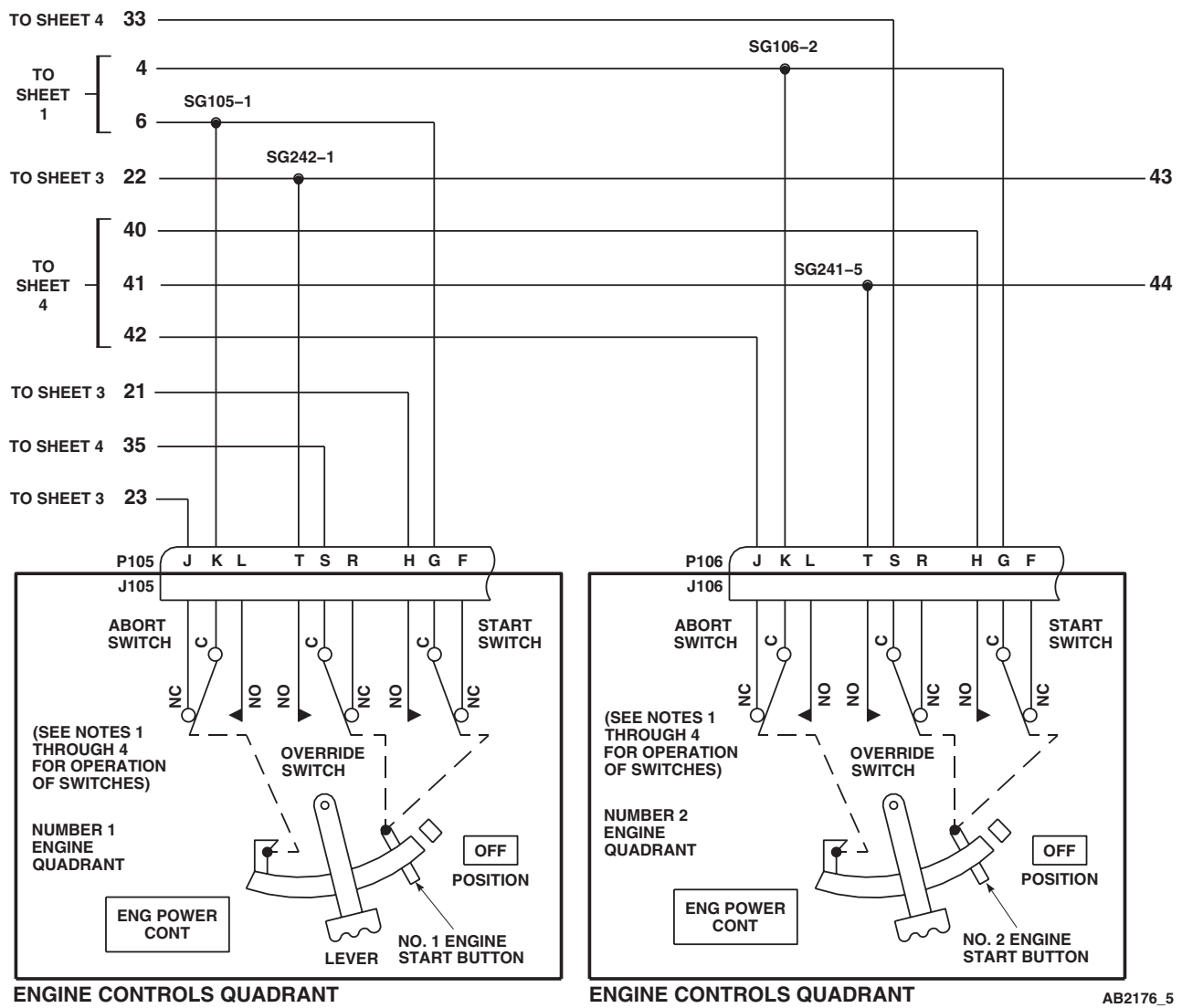


Figure 2. Engine Start and Ignition System Schematic Diagram UH-60L MOD HH-60A HH-60L . (Sheet 5 of 10)

TO SHEET 2

14

16

18

TO SHEET 4 34

43

44

TO SHEET 4 32

P217 Y A

J217

P314 J K C D B H

GG5-5

GG5-3

SG314-1

GND311-1

SG303-1

J311 P311 K J

GND310-3

SG304-2

P304 A C B

NO. 1 ENGINE PRIME SHUTOFF VALVE

ENGINE

FUEL TANK

PRIME BOOST PUMP

P303 A B C

SG305-1

P305 A B C

NO. 2 ENGINE PRIME SHUTOFF VALVE

ENGINE

45 TO SHEET 8

28 VDC IN WHEN START ABORT SWITCH CLOSED (UNACTUATED), AND NO. 2 ENGINE STARTER VALVE OPEN.

28 VDC OUT WHEN START BUTTON PRESSED AND NO. 2 ENGINE START VALVE OPEN. 28 VDC DISCONTINUED WHEN ENGINE SPEED BETWEEN 52% TO 65% Ng OR THE START ABORT SWITCH IS PRESSED.

28 VDC IN WHEN START ABORT SWITCH CLOSED (UNACTUATED), AND NO. 1 ENGINE STARTER VALVE OPEN.

28 VDC OUT WHEN START BUTTON PRESSED AND NO. 1 ENGINE START VALVE OPEN. 28 VDC DISCONTINUED WHEN ENGINE SPEED BETWEEN 52% TO 65% Ng OR THE START ABORT SWITCH IS PRESSED.

STARTER SPEED SWITCH

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Figure 2. Engine Start and Ignition System Schematic Diagram UH-60L MOD HH-60A HH-60L . (Sheet 6 of 10)

SCHEMATICS - Continued

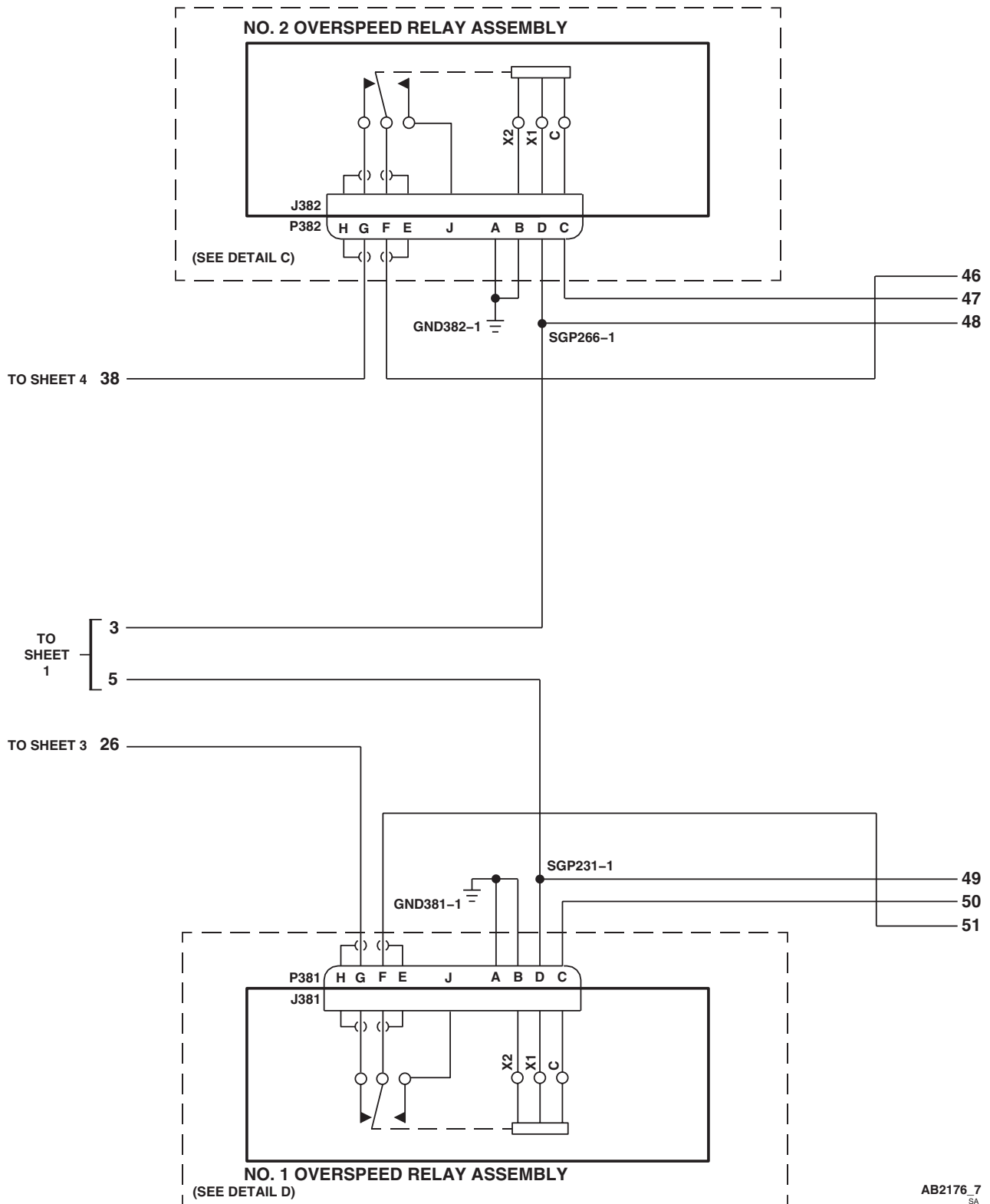


Figure 2. Engine Start and Ignition System Schematic Diagram UH-60L MOD HH-60A HH-60L . (Sheet 7 of 10)

SCHEMATICS - Continued

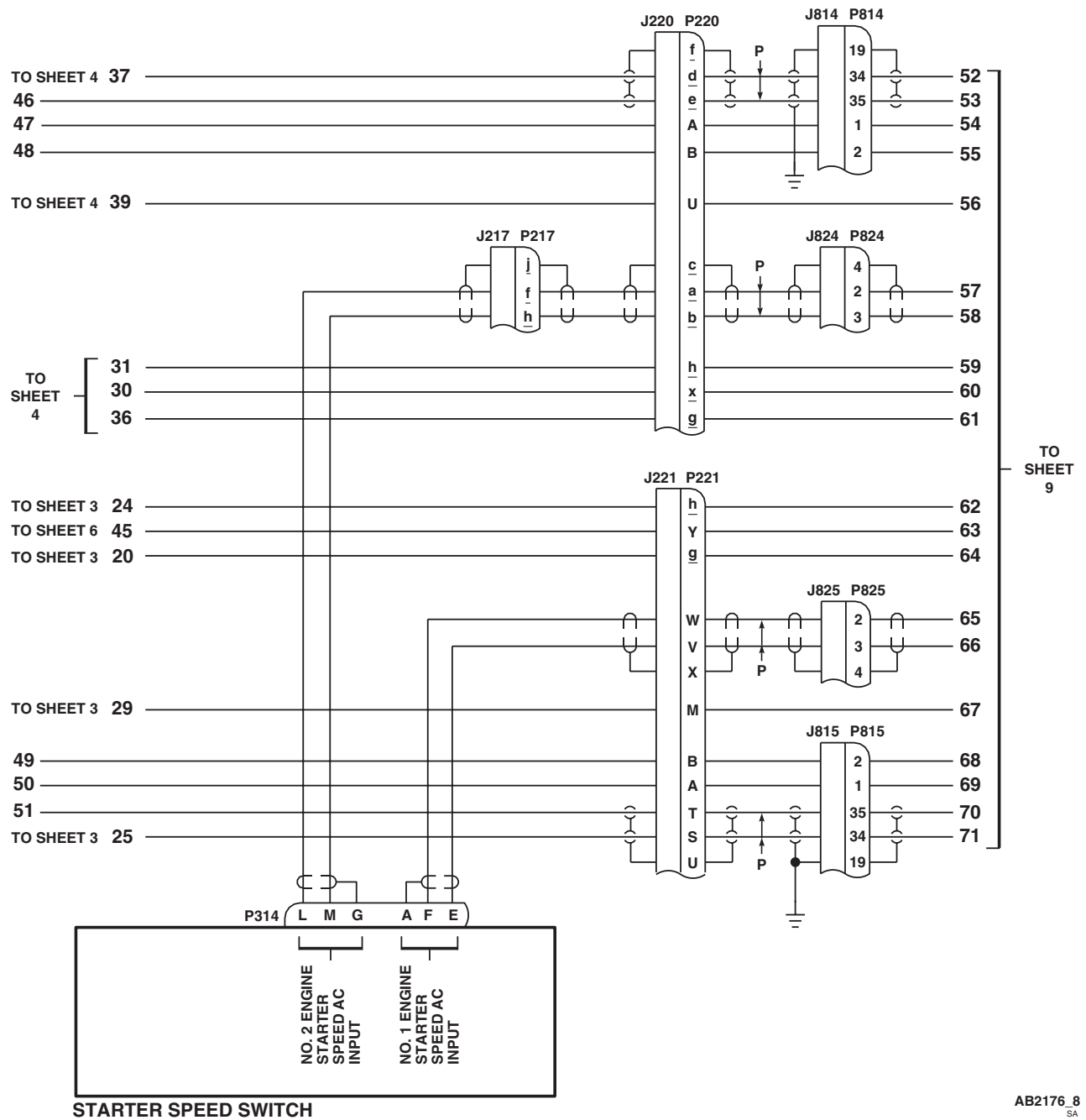
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Figure 2. Engine Start and Ignition System Schematic Diagram UH-60L MOD HH-60A HH-60L . (Sheet 8 of 10)

SCHEMATICS - Continued

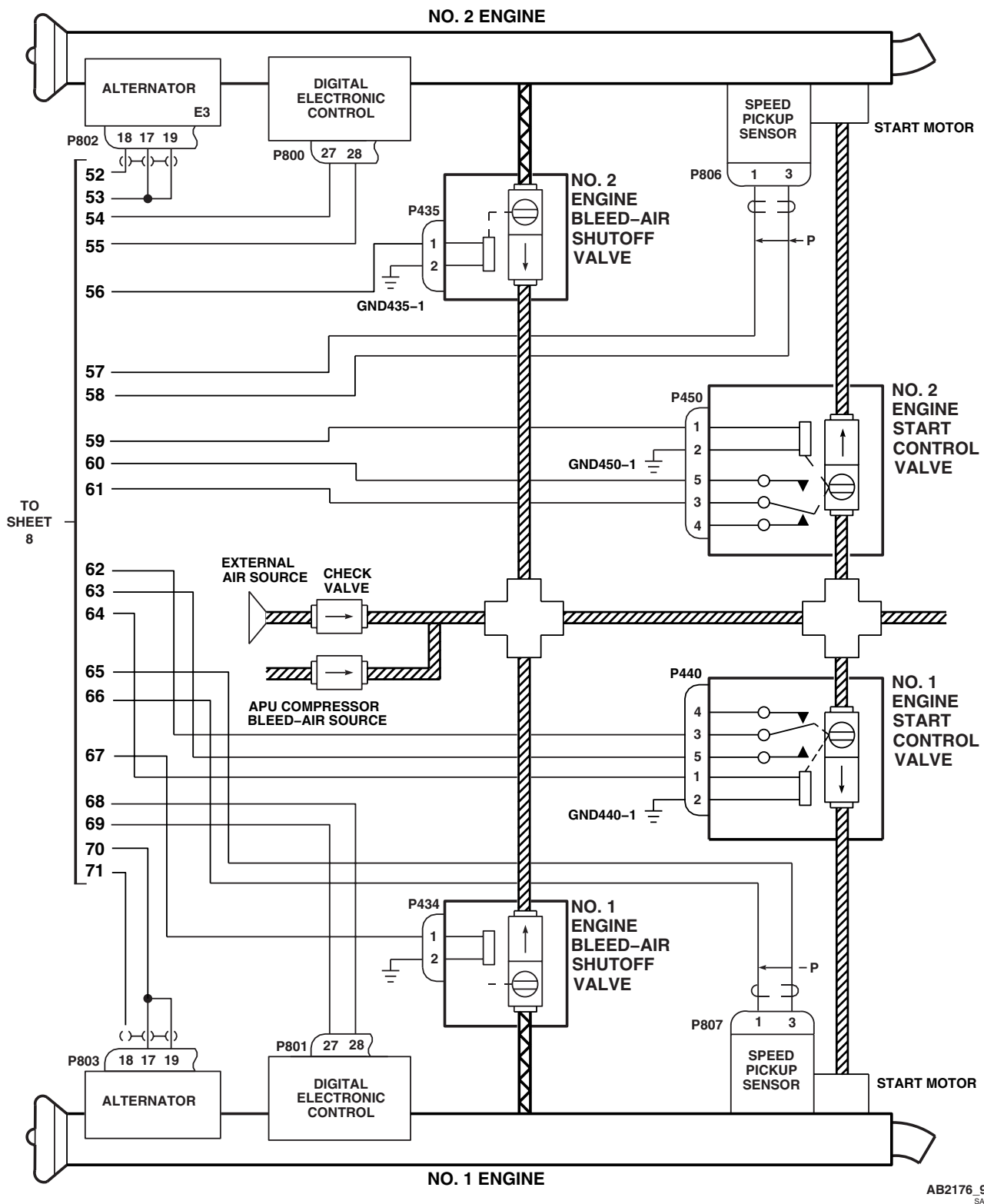
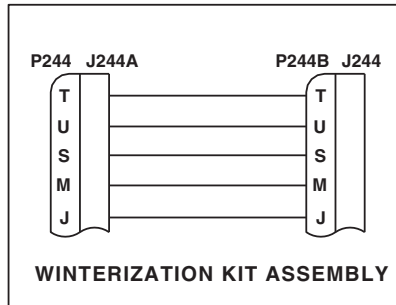
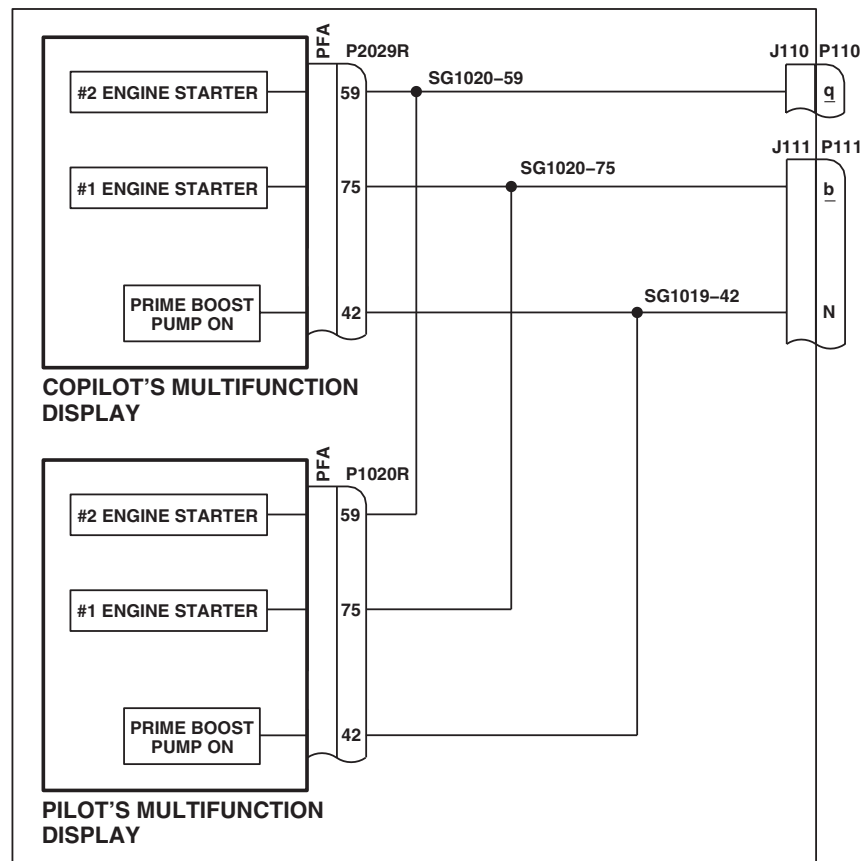
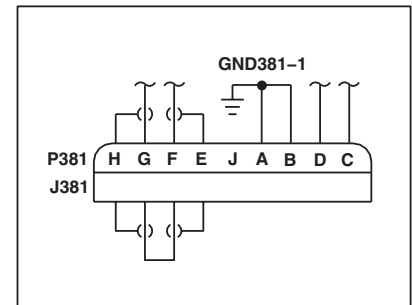
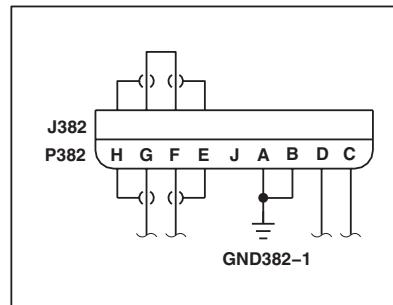


Figure 2. Engine Start and Ignition System Schematic Diagram UH-60L MOD HH-60A HH-60L . (Sheet 9 of 10)

SCHEMATICS - Continued



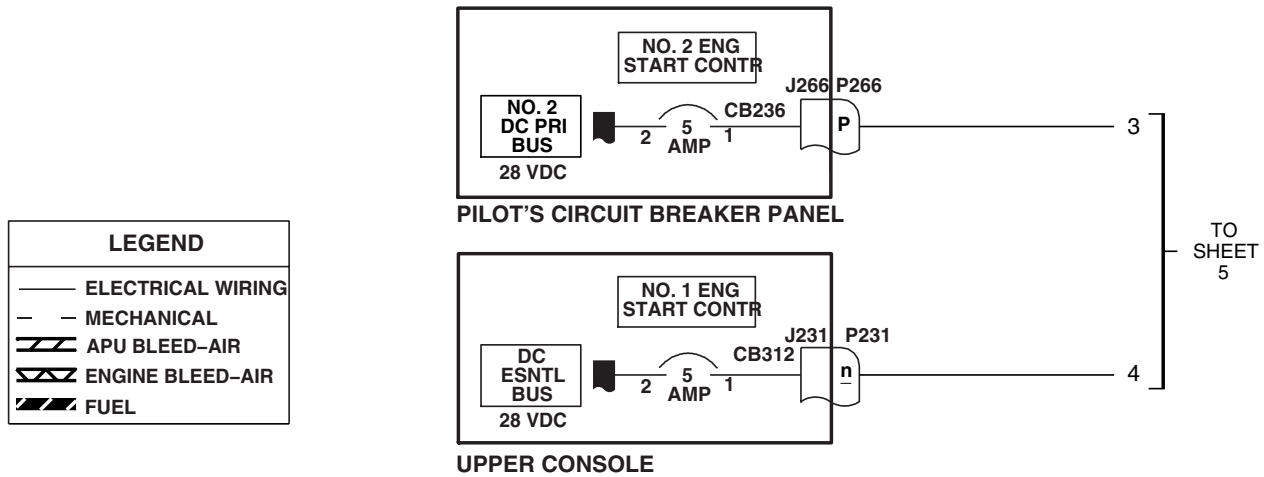
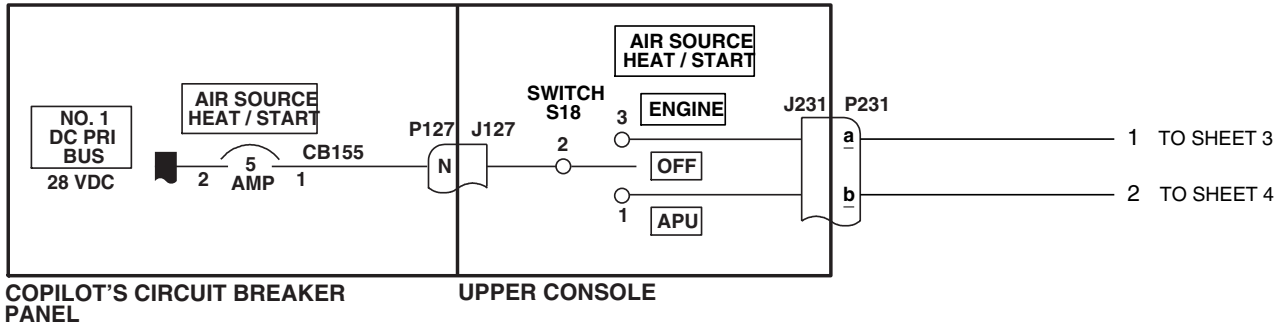
DETAIL B
(SEE NOTE 6)



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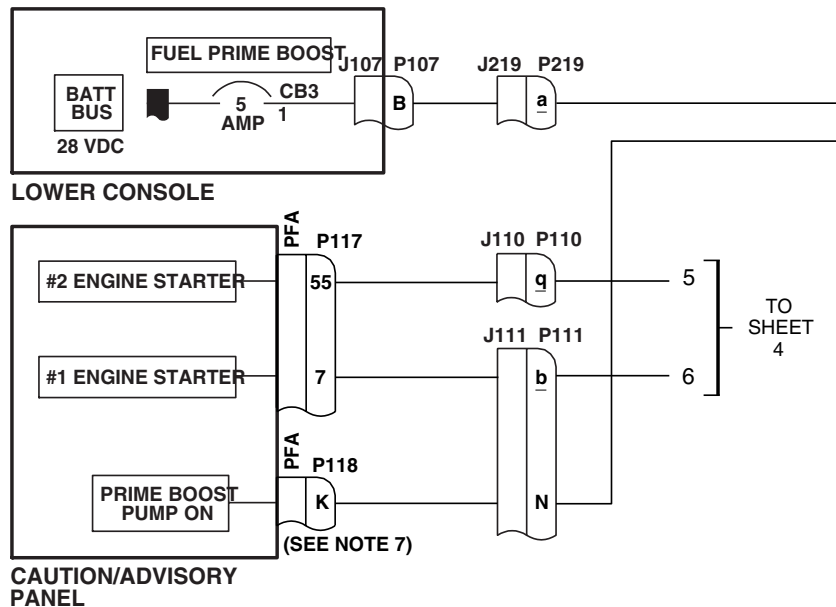
Figure 2. Engine Start and Ignition System Schematic Diagram **UH-60L MOD HH-60A HH-60L** . (Sheet 10 of 10)

SCHEMATICS - Continued



LEGEND	
—	ELECTRICAL WIRING
- -	MECHANICAL
///	APU BLEED-AIR
\\	ENGINE BLEED-AIR
///	FUEL

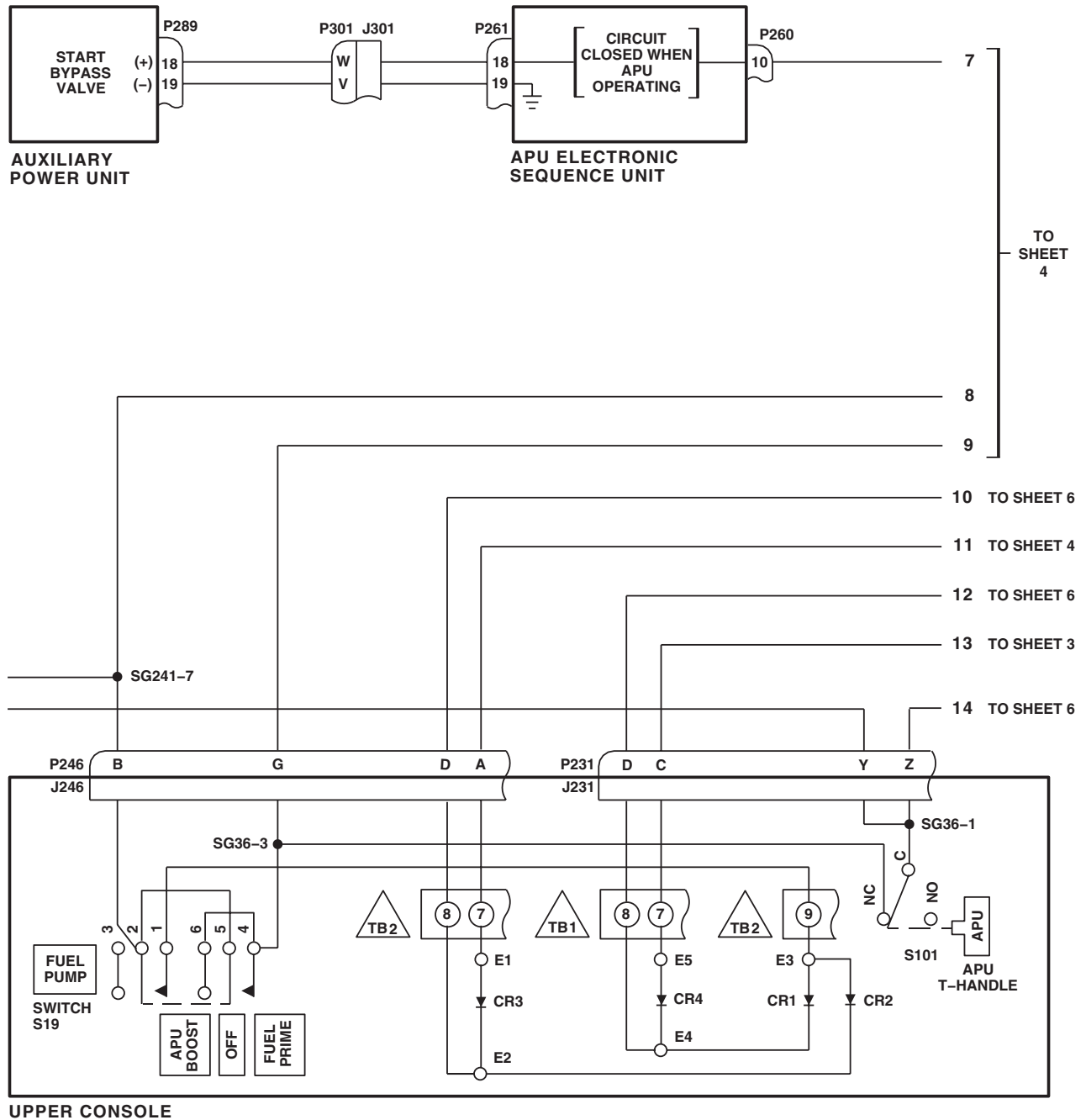
- NOTES**
1. POSITION OF SWITCHES ARE WITH **ENG POWER CONT** LEVER IN **OFF** POSITION.
 2. ENGINE START SWITCH IS ACTUATED WHEN ENGINE START BUTTON IS PRESSED AND **ENG POWER CONT** LEVER IS **OFF** ONLY.
 3. STARTER OVERRIDE SWITCH IS ACTUATED WHEN ENGINE STARTER BUTTON IS PRESSED, REGARDLESS OF **ENG POWER CONT** LEVER POSITION.
 4. STARTER ABORT SWITCH IS ACTUATED WHEN **ENG POWER CONT** LEVER IS PULLED DOWN.
 5. AIR PRESSURE OPENS VALVE WHEN SOLENOID IS ENERGIZED.
 6. **WINTER**
 7. **W/O EMEP** PIN FILTERED ADAPTER (PFA) IS NOT INSTALLED.



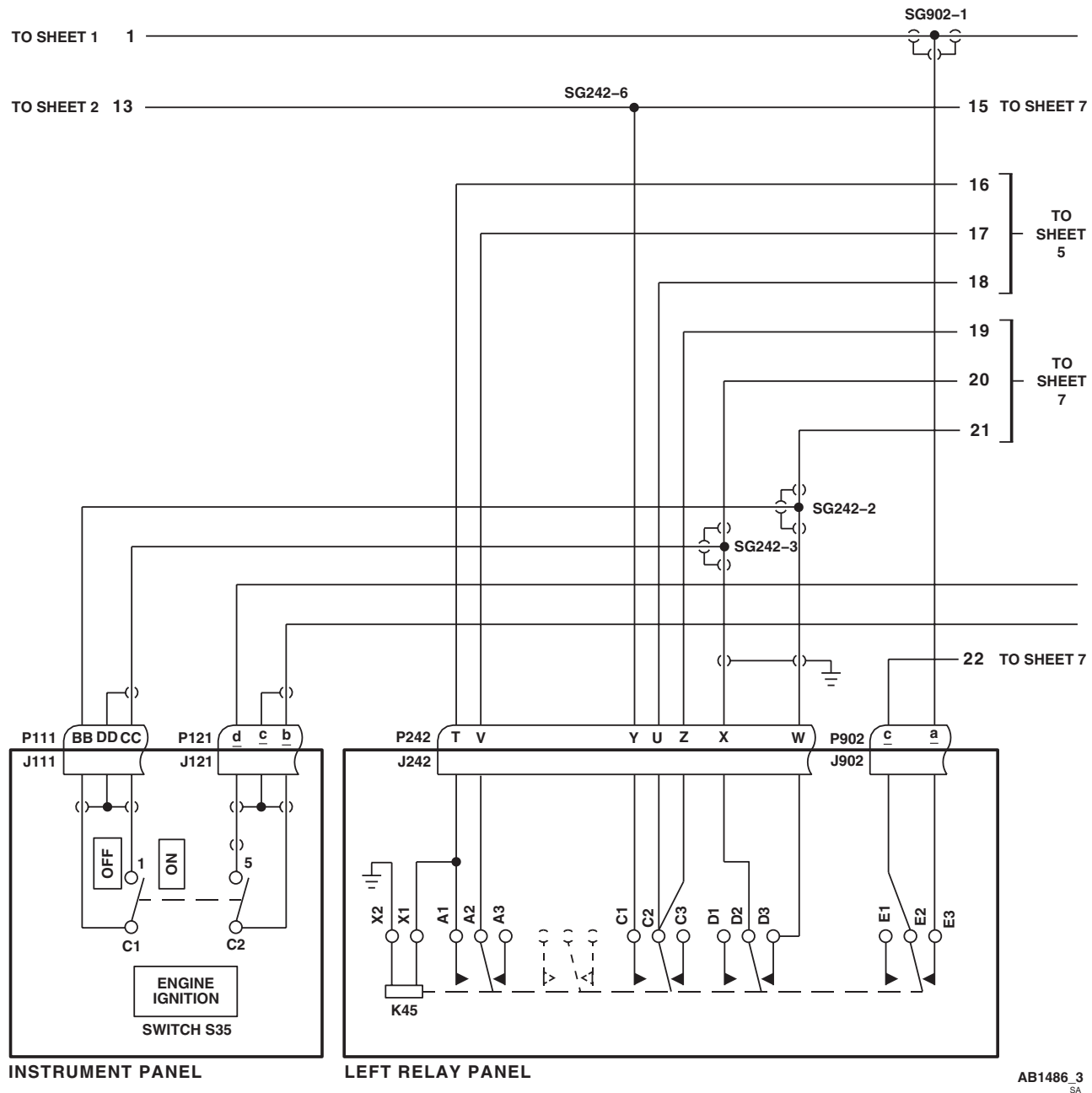
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Figure 3. Engine Start and Ignition System Schematic Diagram **EH-60A UH-60A W/O MOD** . (Sheet 1 of 8)

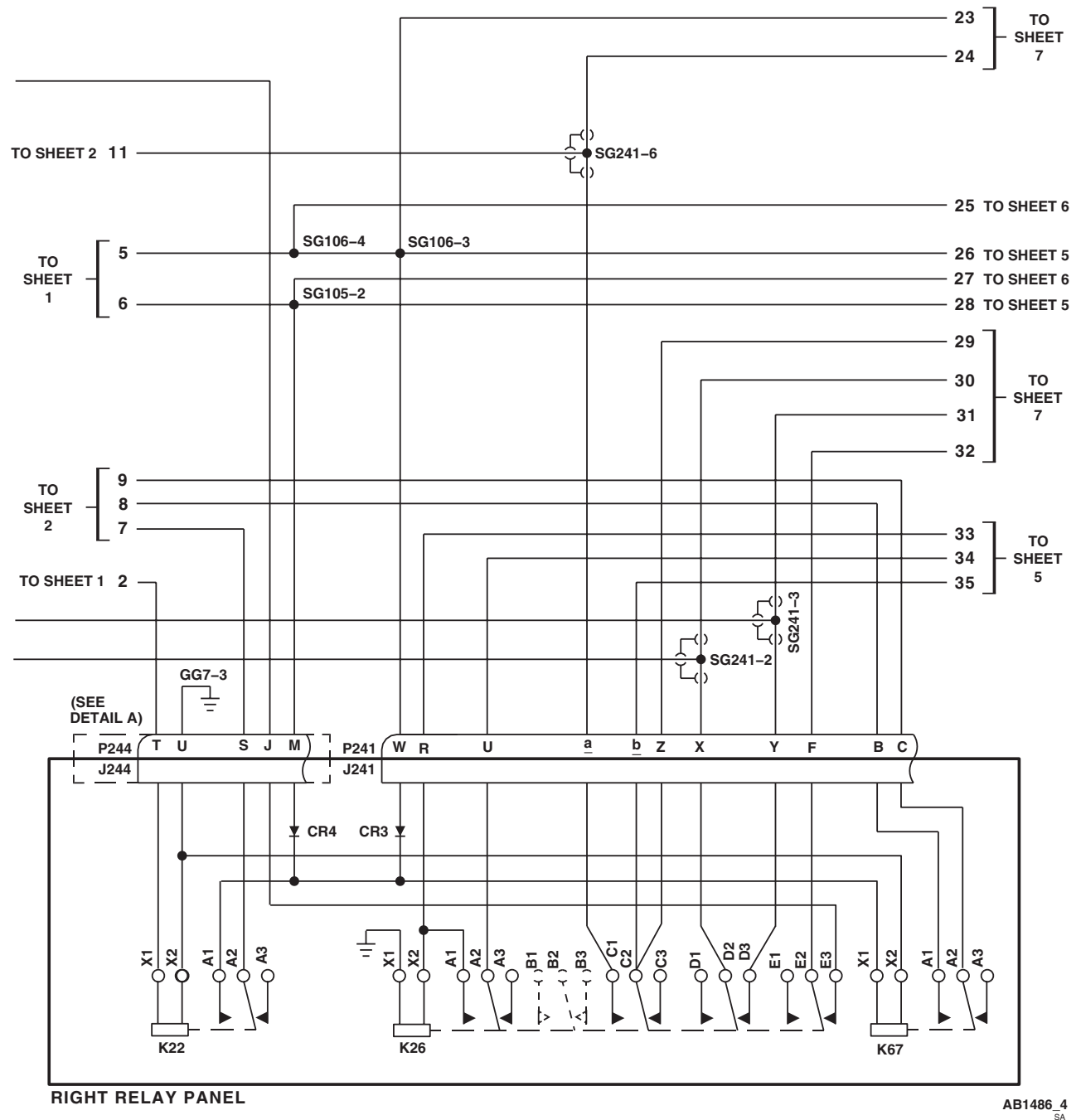
SCHEMATICS - Continued

Figure 3. Engine Start and Ignition System Schematic Diagram **EH-60A UH-60A W/O MOD** . (Sheet 2 of 8)

SCHEMATICS - Continued

Figure 3. Engine Start and Ignition System Schematic Diagram **EH-60A UH-60A W/O MOD** . (Sheet 3 of 8)

SCHEMATICS - Continued

Figure 3. Engine Start and Ignition System Schematic Diagram **EH-60A UH-60A W/O MOD** . (Sheet 4 of 8)

TO SHEET 4 26

TO SHEET 1 3

TO SHEET 1 4

TO SHEET 3 17

TO SHEET 4 33

TO SHEET 4 34

TO SHEET 4 35

TO SHEET 3 16

TO SHEET 4 28

TO SHEET 3 18

SG105-1

SG242-1

SG106-2

SG241-5

P105 J105

J K L T S R H G F

ABORT SWITCH

START SWITCH

NC

NO

NO

NC

NO

NC

(SEE NOTES 1 THROUGH 4 FOR OPERATION OF SWITCHES)

NUMBER 1 ENGINE QUADRANT

ENG POWER CONT

OVERVERRIDE SWITCH

LEVER

NO. 1 ENGINE STARTER BUTTON

OFF POSITION

ENGINE CONTROLS QUADRANT

P106 J106

J K L T S R H G F

ABORT SWITCH

START SWITCH

NC

NO

NO

NC

NO

NC

(SEE NOTES 1 THROUGH 4 FOR OPERATION OF SWITCHES)

NUMBER 2 ENGINE QUADRANT

ENG POWER CONT

OVERVERRIDE SWITCH

LEVER

NO. 2 ENGINE STARTER BUTTON

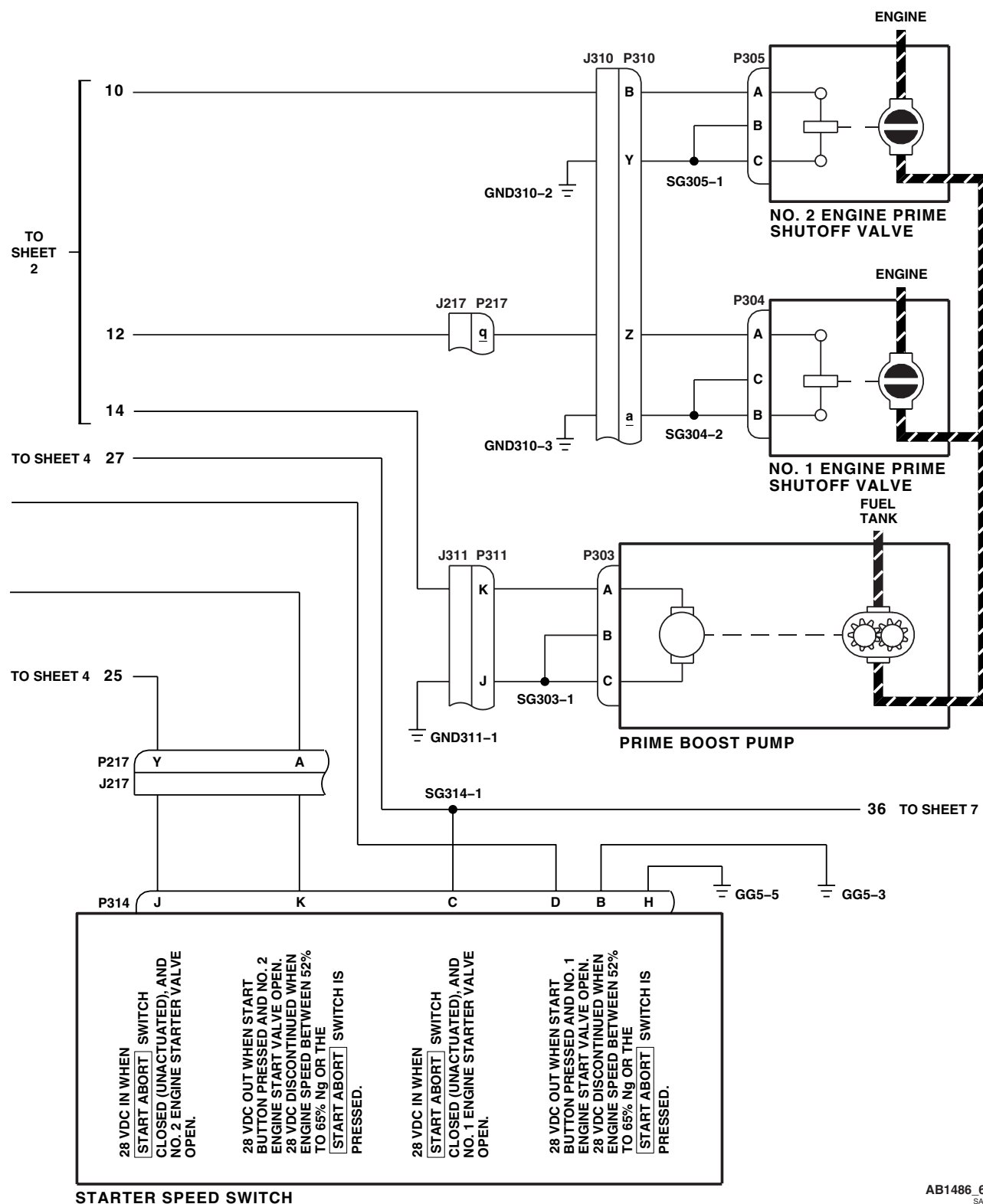
OFF POSITION

ENGINE CONTROLS QUADRANT

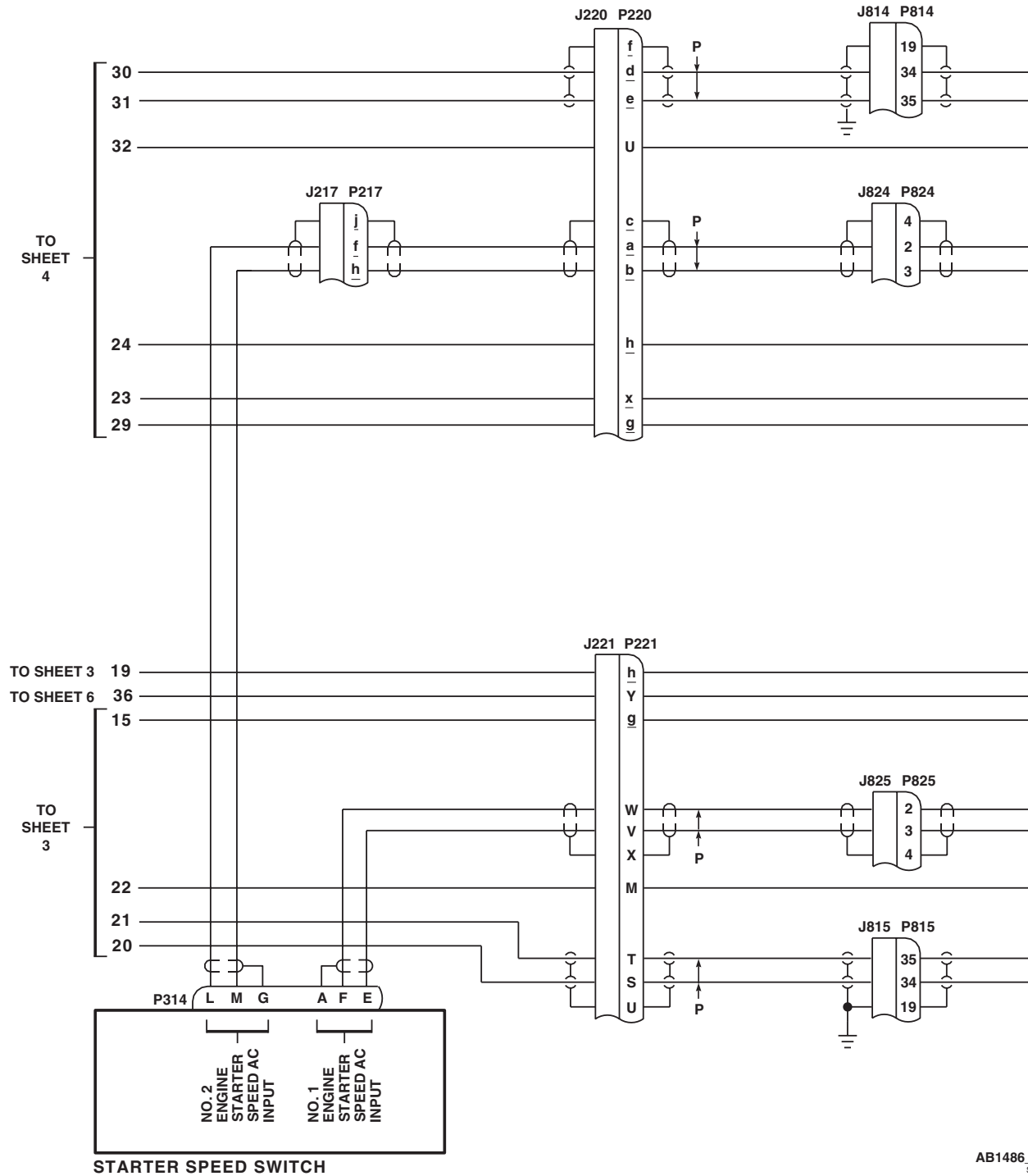
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Change 1

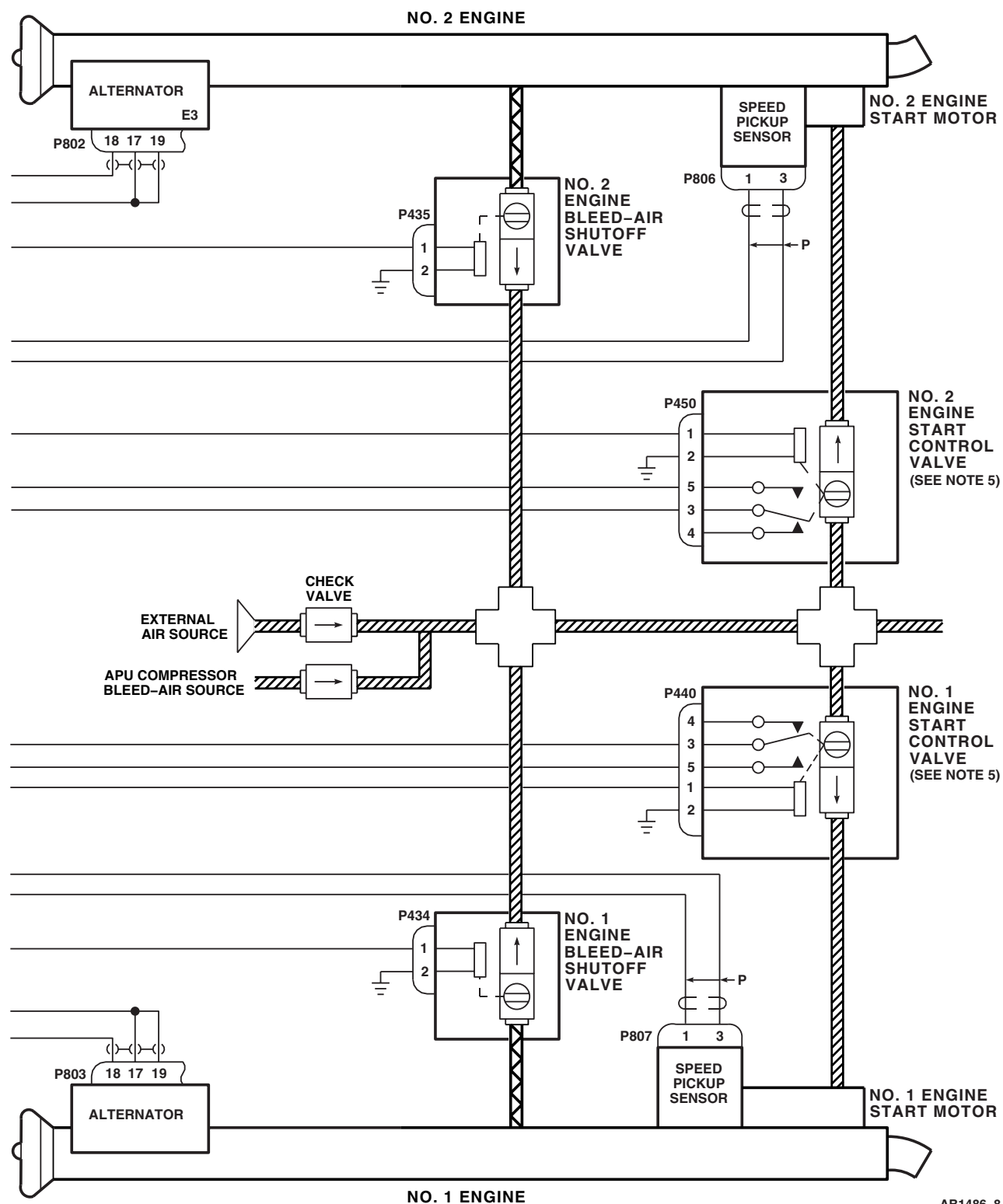
SCHEMATICS - Continued

Figure 3. Engine Start and Ignition System Schematic Diagram **EH-60A UH-60A W/O MOD** . (Sheet 6 of 8)

SCHEMATICS - Continued

AB1486.7
SAFigure 3. Engine Start and Ignition System Schematic Diagram **EH-60A UH-60A W/O MOD** . (Sheet 7 of 8)

SCHEMATICS - Continued

AB1486_8
SAFigure 3. Engine Start and Ignition System Schematic Diagram **EH-60A UH-60A W/O MOD** . (Sheet 8 of 8)

END OF WORK PACKAGE

UNIT LEVEL

ENGINE SYSTEM

ENGINE CONTROLS QUADRANT (AVIM)

INITIAL SETUP:

Test Equipment

Multimeter, AN/PSM-45A

WP 0527 00

WP 0528 00

Tools and Special Tools

Electrical Repairer Toolkit, SC 5180-99-B06

WP 0529 00

WP 0530 00

WP 0531 00

Personnel Required

Aircraft Electrician MOS 15F (1)

WP 1747 00

References

WP 0501 00

WP 0525 00

Equipment ConditionBetween 27 and 28 vdc, and 115 vac, 400 Hz Bench
Power Available

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for location diagram, and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

- For engine controls quadrant part Nos. 70301-01100-011 and 70301-01100-012, do steps 1. through 20.
- For engine controls quadrant part No. 70301-01100-013, do steps 1. through 23.

ENGINE CONTROLS QUADRANT (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

Engine Controls Quadrant (AVIM) Operational/Troubleshooting procedure of No. 1 or No. 2 engine controls quadrant is the same, except as noted.

1. Inspect engine controls quadrant for broken or burnt wires.

Indication/Condition

Wires shall not be broken or burnt.

Corrective Action

If Indication/Condition is not as specified, repair/replace wiring, as required (WP 1747 00).

Step

2. Check for general condition and security of components.

ENGINE CONTROLS QUADRANT (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Components shall be secure and in good condition.

Corrective Action

If Indication/Condition is not as specified, repair/replace components as required.

Step

3. Check quadrant levers for free movement.

Indication/Condition

All levers shall have free movement.

Corrective Action

If Indication/Condition is not as specified, replace engine controls quadrant (WP 0501 00).

Step

4. Check quadrant secondary idle stop latch for proper operation.

Indication/Condition

Secondary idle stop latch shall be in contact with secondary stop when engine POWER CONT lever is at IDLE position.

Corrective Action

If Indication/Condition is not as specified, replace engine controls quadrant (WP 0501 00).

Step

5. Connect No. 1 engine controls quadrant J105-A to 28 vdc bench power and J105-B to bench ground.

Indication/Condition

No. 1 engine fire extinguisher T-handle shall light evenly.

Corrective Action

1. If Indication/Condition is not as specified, check T-handle for burnt-out lamps.
 - a. If lamp is burnt-out, replace lamp (WP 0525 00).
 - b. If lamps are good, repair/replace wiring between J105-A and J105-B (WP 1747 00).

Step

6. Disconnect J105-A and J105-B from bench.
7. With No. 1 engine controls quadrant starter button not pressed, check continuity between J105-F and J105-G.

Indication/Condition

Continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between contacts C and NC of No. 1 engine start switch.
 - a. If continuity is not present, replace No. 1 engine starter switch (WP 0527 00).

ENGINE CONTROLS QUADRANT (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- b. If continuity is present, repair/replace wiring between J105-F and contact NC; and J105-G and contact C of No. 1 engine starter switch, as required (WP 1747 00).

Step

8. Press and hold No. 1 engine controls quadrant starter button. Check continuity between J105-G and J105-H.

Indication/Condition

Continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between contacts C and NO of No. 1 engine starter switch.
 - a. If continuity is not present, replace No. 1 engine starter switch (WP 0527 00).
 - b. If continuity is present, repair/replace wiring between J105-H and contact NO of No. 1 engine starter switch (WP 1747 00).

Step

9. Release No. 1 engine controls quadrant starter button.
10. Mechanically adjust starter override switch (WP 0529 00).
11. With No. 1 engine controls quadrant starter button not pressed, check continuity between J105-R and J105-S.

Indication/Condition

Continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between C and NC of No. 1 engine starter override switch.
 - a. If continuity is not present, replace No. 1 engine starter override switch (WP 0529 00).
 - b. If continuity is present, repair/replace wiring between J105-R and contact NC; and J105-S and contact C of No. 1 engine starter override switch, as required (WP 1747 00).

Step

12. Press and hold No. 1 engine controls quadrant starter button. Check continuity between J105-S and J105-T.

Indication/Condition

Continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between contacts C and NO of No. 1 engine starter override switch.
 - a. If continuity is not present, replace No. 1 engine starter override switch (WP 0529 00).
 - b. If continuity is present, repair/replace wiring between J105-T and contact NO of No. 1 engine starter override switch (WP 1747 00).

Step

13. Release No. 1 engine controls quadrant starter button.

ENGINE CONTROLS QUADRANT (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

14. Mechanically adjust starter abort switch (WP 0528 00).
15. With No. 1 engine controls quadrant engine POWER CONT lever pushed up, check continuity between J105-J and J105-K.

Indication/Condition

Continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between contacts C and NC of No. 1 engine starter abort switch.
 - a. If continuity is not present, replace No. 1 engine starter abort switch (WP 0528 00).
 - b. If continuity is present, repair/replace wiring between J105-J and contact NC; and J105-K and contact C of No. 1 engine starter abort switch, as required (WP 1747 00).

Step

16. With No. 1 engine controls quadrant engine POWER CONT lever pulled down, check continuity between J105-K and J105-L.

Indication/Condition

Continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between contacts C and NO of No. 1 engine starter abort switch.
 - a. If continuity is not present, replace No. 1 engine starter abort switch (WP 0528 00).
 - b. If continuity is present, repair/replace wiring between J105-L and contact NO of No. 1 engine starter abort switch (WP 1747 00).

Step

17. Mechanically adjust No. 1 engine fire extinguishing arming switch (WP 0530 00).
18. With No. 1 engine fire extinguishing T-handle in stowed position, check continuity between J105-N and J105-P.

Indication/Condition

Continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between contacts C and NO of No. 1 engine fire extinguishing arming switch.
 - a. If continuity is not present, replace No. 1 engine fire extinguishing arming switch (WP 0530 00).
 - b. If continuity is present, repair/replace wiring between J105-P and contact NO ; and J105-N and contact C of No. 1 engine fire extinguishing arming switch, as required (WP 1747 00).

Step

19. With No. 1 fire extinguishing T-handle in armed position, check continuity between J105-M and J105-N.

ENGINE CONTROLS QUADRANT (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Continuity shall be present.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between contacts C and NC of No. 1 engine fire extinguishing arming switch.
 - a. If continuity is not present, replace No. 1 engine fire extinguishing arming switch (WP 0530 00).
 - b. If continuity is present, repair/replace wiring between J105-M and NC of No. 1 engine fire extinguishing arming switch (WP 1747 00).

Step

20. Repeat steps 1. through 19. for No. 2 engine controls quadrant using J106.

NOTE

For engine controls quadrant part Nos. 70301-01100-011 and 70301-01100-012, operational/troubleshooting procedure is complete. For engine controls quadrant part No. 70301-01100-013, continue operational/troubleshooting procedure.

21. Connect left information panel P394-1 to 115 vac, 400 Hz bench power and P394-2 to bench ground. Observe information panel under reduced ambient light conditions.

Indication/Condition

Information panel shall light.

Corrective Action

If Indication/Condition is not as specified, replace panel (WP 0531 00).

Step

22. Connect right information panel P396-1 to 115 vac, 400 Hz bench power and P396-2 to bench ground. Observe information panel under reduced ambient light conditions.

Indication/Condition

Information panel shall light.

Corrective Action

If Indication/Condition is not as specified, replace panel (WP 0531 00).

Step

23. Connect center information panel P395-1 to 115 vac, 400 Hz bench power and P395-2 to bench ground. Observe panel under reduced ambient light conditions.

Indication/Condition

Center information panel shall light.

Corrective Action

1. If Indication/Condition is not as specified, do the following:
 - a. Disconnect 115 vac bench power.

ENGINE CONTROLS QUADRANT (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- b. Remove screws securing panel to center cover assembly.
- c. Unplug panel from cover assembly connector.
- d. Check continuity from P395 to disconnect plug on center cover assembly.
 - (1) If continuity is present, replace information panel (WP 0531 00).
 - (2) If continuity is not present, repair/replace wiring from P395 to center cover connector (WP 1747 00).

Step

- 24. Disconnect 115 vac bench power.

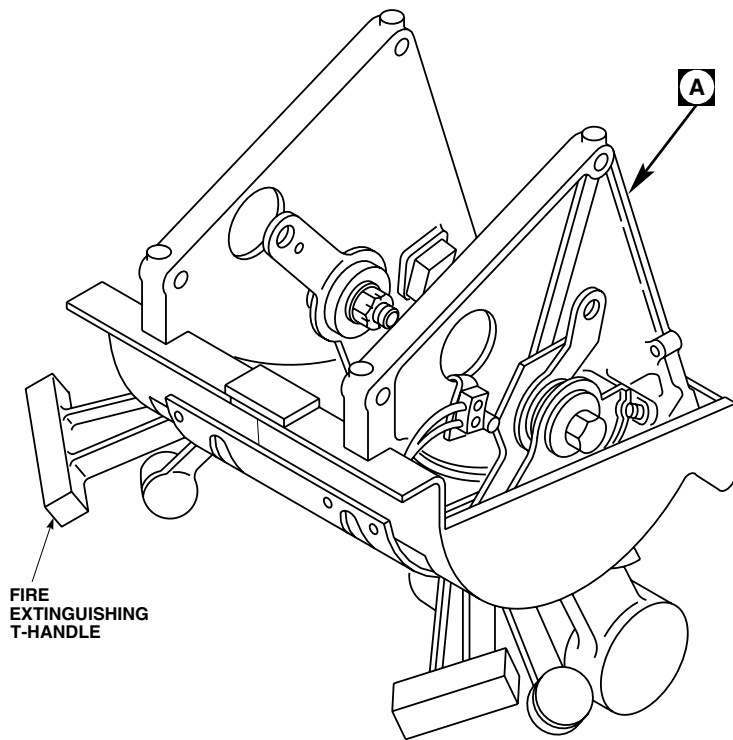
Indication/Condition

115 vac bench power shall be disconnected.

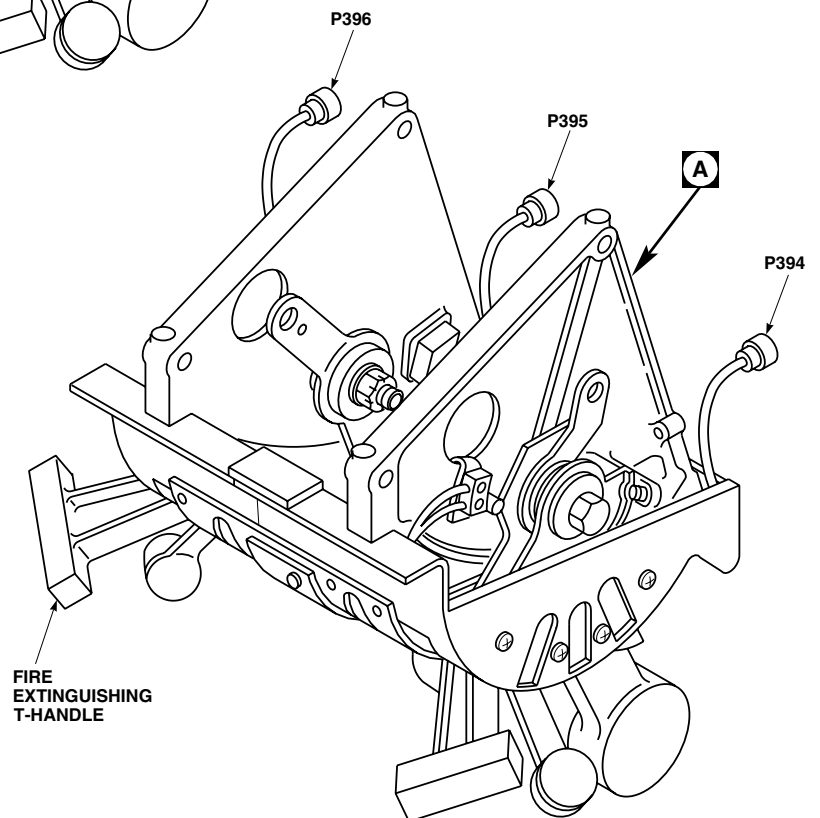
Corrective Action

None Required

LOCATION



(SEE NOTE 1)

FIRE
EXTINGUISHING
T-HANDLE

(SEE NOTE 2)

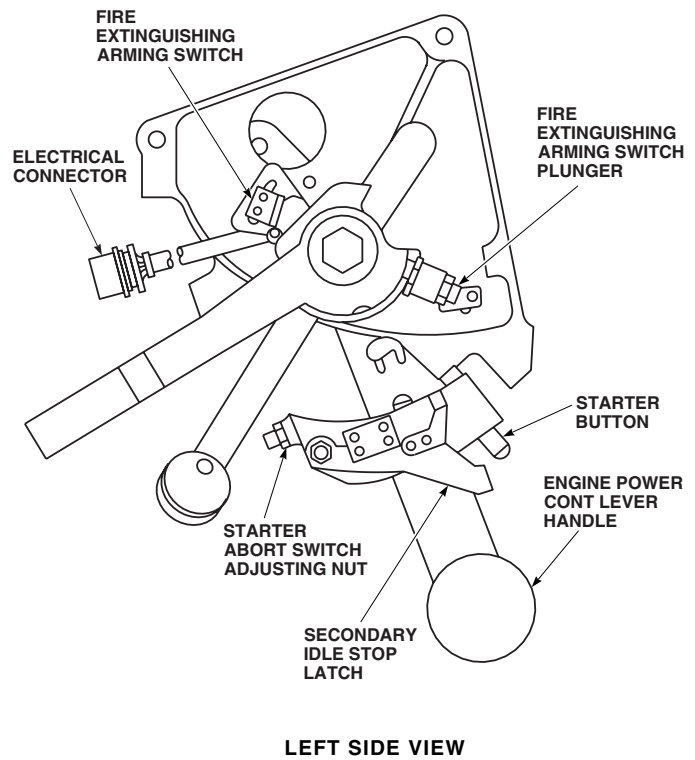
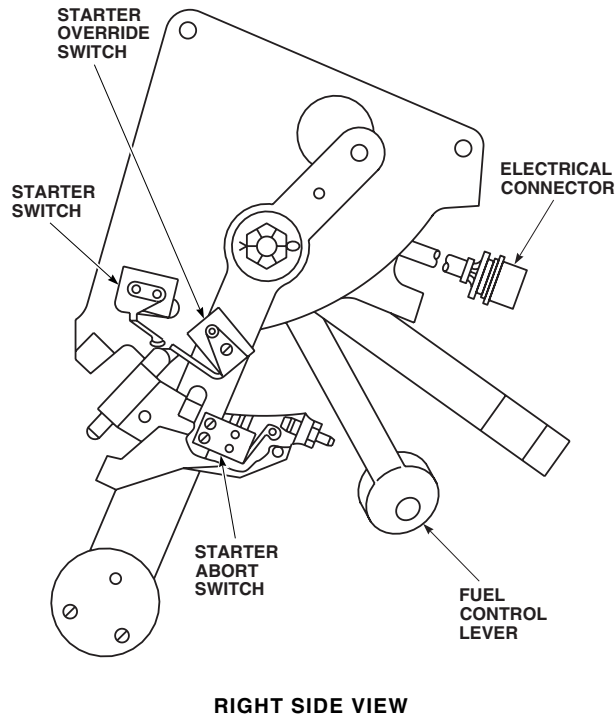
NOTES

1. ENGINE CONTROLS QUADRANT ASSEMBLIES PART NOS 70301-01100-011 AND 70301-01100-012.
2. ENGINE CONTROLS QUADRANT ASSEMBLY PART NO. 70301-01100-013.

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Figure 1. Engine Controls Quadrant Location Diagram. (Sheet 1 of 2)

LOCATION - Continued



LEFT (NO. 1 ENGINE) QUADRANT

A

AA2635 2
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Figure 1. Engine Controls Quadrant Location Diagram. (Sheet 2 of 2)

SCHEMATICS

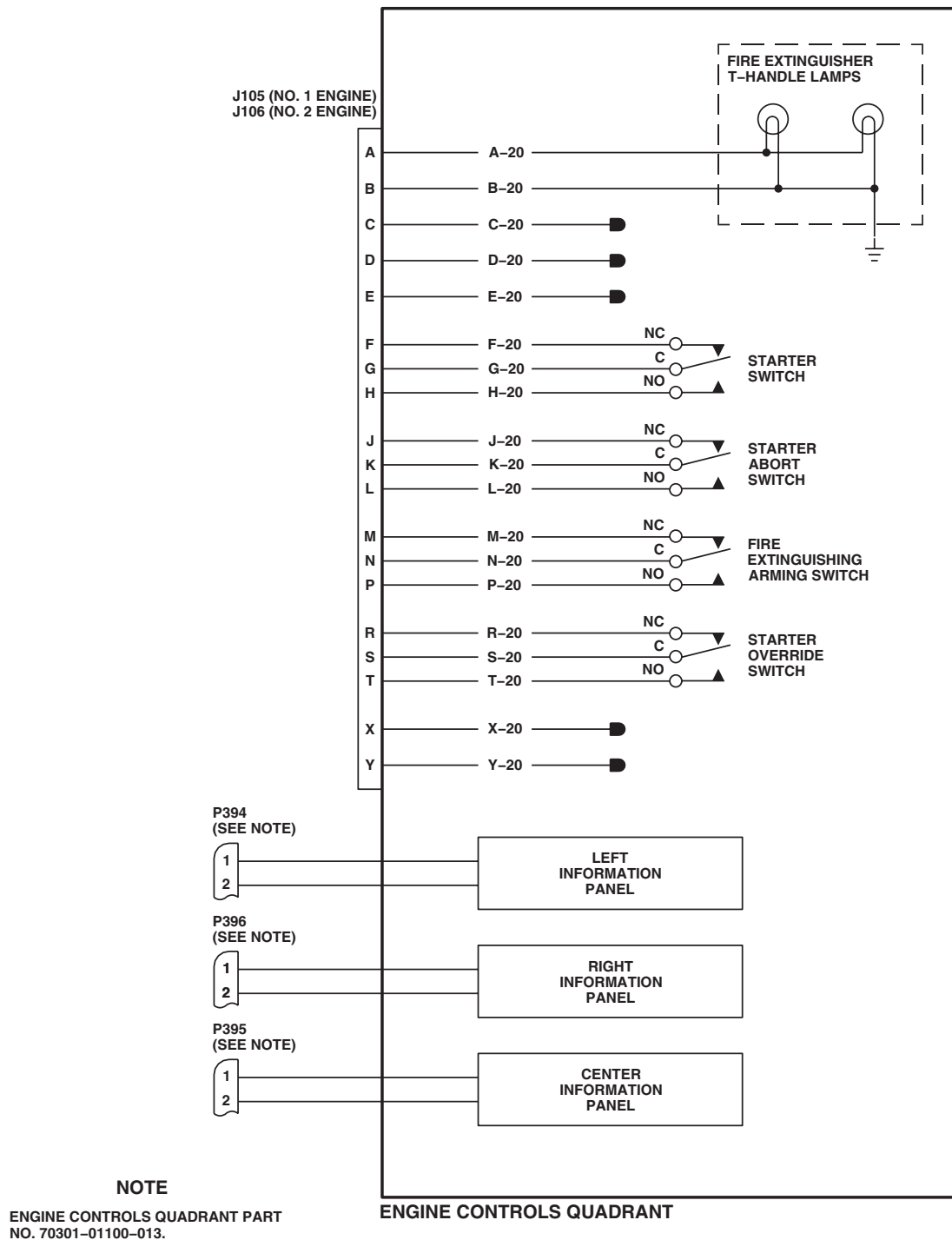
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Figure 2. Engine Control Quadrant Schematic Diagram.

UNIT LEVEL**ROTOR SYSTEM**

BIM® INDICATORS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

References

TM 1-1520-237-10

WP 0541 00

WP 0576 00

WP 0581 00

WP 1803 00

Material/Parts

Leak Detection Fluid, Item 175, WP 1803 00

Towel, Machinery Wiping, Item 344, WP 1803 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

SPECIAL INSTRUCTIONS**NOTE**

See Figure 1 for location diagram as an aid in operational/troubleshooting.

The BIM® Indicator operational/troubleshooting procedure is subdivided as follows:

- BIM® Indicator Packing Test (to determine if the indicator is leaking at the base).
- BIM® Indicator Test (to determine if the indicator is functioning properly).
- One-Hour Whirl Test (to determine the final integrity of blade spars returned to AVUM level from either AVIM or DEPOT level).

BIM® INDICATOR PACKING TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

WARNING

- Visual inspection of the indicator must be performed while viewing the indicator perpendicular to its axis.
- Upon inspection of each blade pressure indicator, if red is visible on all three inspection windows, maintenance must be performed prior to next flight. The cause of the unsafe condition must be positively identified and corrected. If the indicator is malfunctioning, it must be replaced, but only if the spar pressure is within permissible limits.

NOTE

- Small slivers of red visible through one or two of the inspection windows is not cause for maintenance action.
- When a blade is forwarded to AVIM or DEPOT, it must be accompanied by a statement that accurately describes what maintenance action or actions have been taken, and what was originally wrong with the blade to require these actions. If applicable, blade shall be marked to identify it as having had an unsafe BIM® indication.
- Do the following operational/troubleshooting procedure on all four main rotor blade BIM® indicators.
- When testing pressure indicator, press both thumbs on indicator's manual lever (grenade-type handle) to make sure that valve plunger under the lever is pushed all the way down to shut off all spar pressure to the indicator. Press on ridged part of handle, not on smooth tip. A partly depressed plunger may cause loss of spar pressure and a slow indication, or no indication at all.
- Do not hold indicator, as heat of hand may change internal reference pressure and result in erroneous indicator reading.
- Spar must be properly serviced before testing.

1. Apply leak detection fluid, Item 175, WP 1803 00, or mild liquid detergent solution over and around base of indicator.

Indication/Condition

No air bubbles shall form.

Corrective Action

If air bubbles form, replace BIM® indicator packing (WP 0581 00).

Step

2. Rinse test fluid using clean fresh water before fluid dries. Dry blade using machinery wiping towel, Item 344, WP 1803 00.

BIM® INDICATOR PACKING TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Blade shall be dry.

Corrective Action

None Required

**BIM® INDICATOR TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step****WARNING**

- Visual inspection of the indicator must be performed while viewing the indicator perpendicular to its axis.
- Upon inspection of each blade pressure indicator, if red is visible on all three inspection windows, maintenance must be performed prior to next flight. The cause of the unsafe condition must be positively identified and corrected. If the indicator is malfunctioning, it must be replaced, but only if the spar pressure is within permissible limits.

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 - Do not hold indicator, as heat of hand may change internal reference pressure and result in erroneous indicator reading.
 - Spar must be properly serviced before testing.
1. Press in and hold manual test lever. Do not place hand on glass bulb.

BIM® INDICATOR TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

BIM® indicator stripes shall turn from yellow to full red (unsafe) within 10 to 30 seconds at 20°F (-6.7°C) or above, or within the following time limits and extended upper limit temperatures:

TEMPERATURE	TIME
19°F to 0°F (-7.2°C to -17.8°C)	35 seconds
-1°F to - 20°F (-18.3°C to -28.9°C)	40 seconds
-21°F to - 40°F (-29.4°C to -40.0°C)	50 seconds
-41°F to - 60°F (-40.5°C to -51.1°C)	60 seconds

Corrective Action

1. If indicator does not change color or completely change color, check main rotor blade spar for proper pressure (WP 0541 00).
 - a. If spar pressure is good, replace BIM® indicator (WP 0581 00).
 - b. If spar pressure is not good, replace main rotor blade (WP 0576 00).
2. If indicator takes more than the maximum number of seconds given above or less than 10 seconds to change color completely, go to BIM® INDICATOR STRIPES ARE RED (DID NOT SNAP BACK TO YELLOW), in this work package.

Step

2. Release indicator lever.

Indication/Condition

Red indication must snap back to yellow immediately.

Corrective Action

If Indication/Condition is not as specified, go to BIM® INDICATOR STRIPES ARE RED (DID NOT SNAP BACK TO YELLOW), in this work package.

ONE-HOUR WHIRL TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step****WARNING**

- Visual inspection of the indicator must be performed while viewing the indicator perpendicular to its axis.
- Upon inspection of each blade pressure indicator, if red is visible on all three inspection windows, maintenance must be performed prior to next flight. The cause of the unsafe condition must be positively identified and corrected. If the indicator is malfunctioning, it must be replaced, but only if the spar pressure is within permissible limits.

NOTE

- Small slivers of red visible through one or two of the inspection windows is not cause for maintenance action.
 - When a blade is forwarded to AVIM or DEPOT, it must be accompanied by a statement that accurately describes what maintenance action or actions have been taken, and what was originally wrong with the blade to require these actions. If applicable, blade shall be marked to identify it as having had an unsafe BIM® indication.
 - Do the following operational/troubleshooting procedure on all four main rotor blade BIM® indicators.
 - When testing pressure indicator, press both thumbs on indicator's manual lever (grenade-type handle) to make sure that valve plunger under the lever is pushed all the way down to shut off all spar pressure to the indicator. Press on ridged part of handle, not on smooth tip. A partly depressed plunger may cause loss of spar pressure and a slow indication, or no indication at all.
 - Do not hold indicator, as heat of hand may change internal reference pressure and result in erroneous indicator reading.
 - Spar must be properly serviced before testing.
 - The whirl test is a final check on the integrity of a rotor blade spar that has had a loss of spar pressure. The test is conducted after all other corrective actions have been taken.
1. Measure and record spar temperature and pressure.
 2. With qualified personnel at controls, ground run helicopter for two half-hour periods at 100% RPM, low collective stick, and neutral cyclic stick and directional control (TM 1-1520-237-10).

Indication/Condition

At the end of first half-hour period, BIM® indicator stripes shall be yellow.

Corrective Action

If Indication/Condition is not as specified, send blade to AVIM level.

ONE-HOUR WHIRL TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

3. At the end of second half-hour period, measure and record spar pressure and temperature.
4. Correct for temperature variations as follows:
 - a. For temperature increases, subtract 0.07 psi per degree Fahrenheit (0.126 psi per degree Celsius) of increase.
 - b. For temperature decreases, add 0.07 psi per degree Fahrenheit (0.126 psi per degree Celsius) of decrease.

Indication/Condition

Corrected spar pressure shall be no greater than 0.25 psi from pressure measured in step 1.

EXAMPLE:**-INITIAL READING-**

PRESSURE	TEMPERATURE
18.0 (PSI)	73.0 °F (22.8 °C)

-READING AFTER 1 HOUR-

PRESSURE	TEMPERATURE
17.0 (PSI)	63.0 °F (17.2 °C)

Temperature change is 10.0°F (5.6°C) decrease. $10.0 \times 0.07 = 0.70$ PSI (for °F) or $5.6 \times 0.126 = 0.70$ PSI (for °C).
Correction to be added to reading. Therefore, the actual corrected pressure in the spar is $17.0 + 0.70 = 17.7$ PSI

Corrective Action

If Indication/Condition is not as specified, send blade to AVIM level.

BIM® INDICATOR STRIPES ARE RED (DID NOT SNAP BACK TO YELLOW)**SYMPTOM**

Indicator takes more than the maximum number of seconds given above or less than 10 seconds to change color completely. Red indication must snap back to yellow immediately.

MALFUNCTION

BIM® indicator stripes are red (did not snap back to yellow).

CORRECTIVE ACTION**NOTE**

If blade has had an unsafe (red) indication at any time within the last 30 flight hours, send blade to DEPOT.

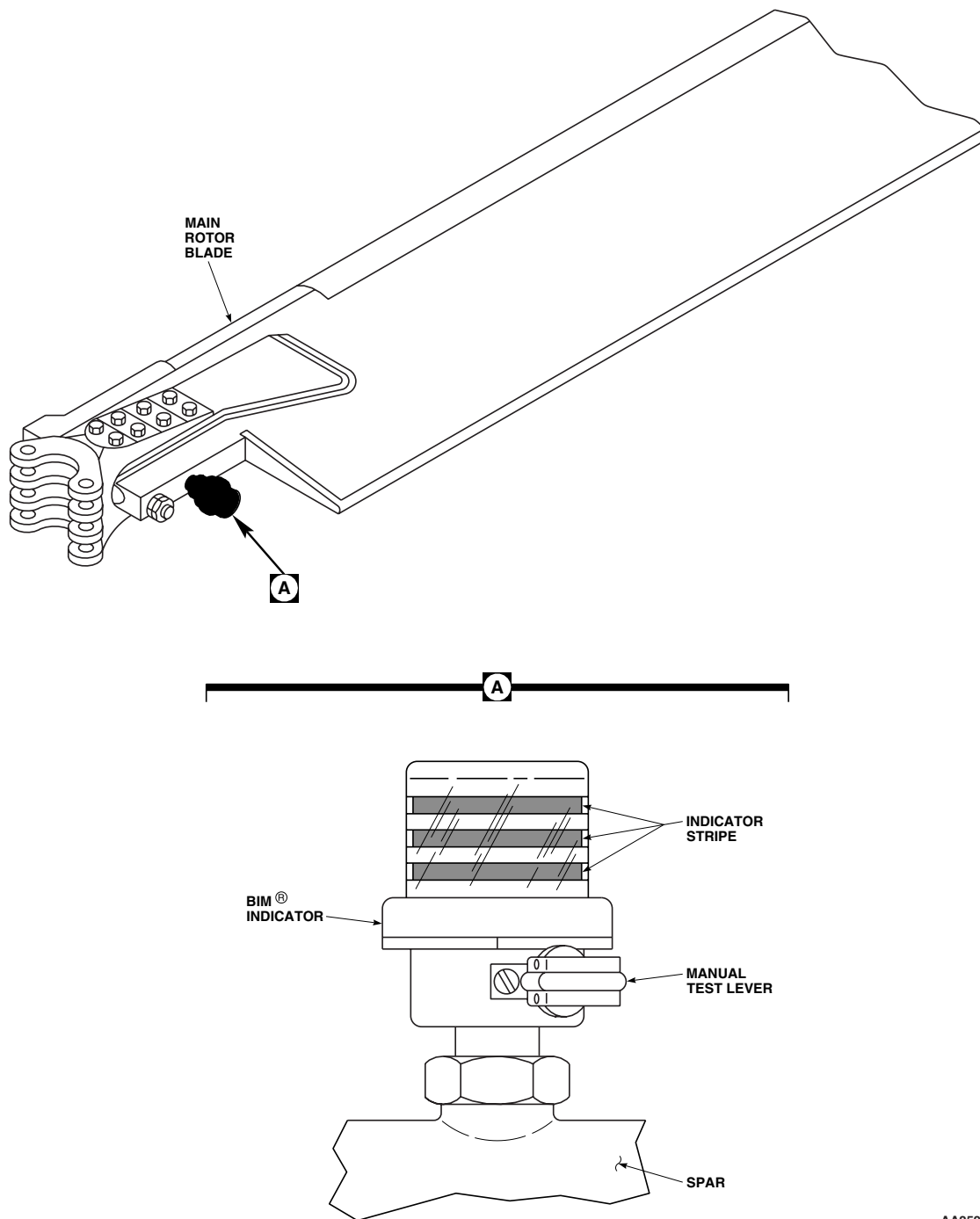
1. Check main rotor blade spar for proper pressure. Is pressure good?

- a. If pressure is good, go to **2**.
- b. If pressure is not good, replace main rotor blade (WP 0576 00). Go to **6**.

BIM® INDICATOR STRIPES ARE RED (DID NOT SNAP BACK TO YELLOW) - Continued

- 2. Wait for 6 hours. Go to 3.**
- 3. Check if BIM® indicator stripes return to yellow.**
 - a. If stripes return to yellow, go to **4**.
 - b. If stripes do not return to yellow, replace BIM® indicator (WP 0581 00). Go to **6**.
- 4. Perform ONE-HOUR WHIRL TEST, in this work package. Go to 5.**
- 5. Check if spar pressure is good.**
 - a. If spar pressure is good, continue blade in service. Go to **6**.
 - b. If spar pressure is not good, send blade to AVIM. Go to **6**.
- 6. Procedure completed.**

LOCATION



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Figure 1. BIM® Indicators Location Diagram.

END OF WORK PACKAGE

UNIT LEVEL
ROTOR SYSTEM
MAIN ROTOR BLADES (AVIM)

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 0544 00

WP 0581 00

WP 1624 00

Personnel Required

UH-60 Helicopter Repairer MOS 15T (1)

References

WP 0081 00

WP 0541 00

MAIN ROTOR BLADES (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step****CAUTION**

When AVIM has repaired a blade that had an unsafe indication on its BIM® pressure indicator, they must not repair that blade again if a second unsafe indication occurs within 30 flight-hours of the first. Should this happen, forward the blade to DEPOT.

NOTE

If a blade is sent to DEPOT level, it must have with it a statement that accurately describes the specific maintenance action or actions that were done at AVIM. The statement shall include spar pressure when blade was inducted into AVIM and when it left, and specific reason for forwarding blade to DEPOT, including identification of original problem, shall also be with blade. A blade component record card entry shall be made whenever a blade is removed from service stating the reason for removal.

1. Check main rotor blade spar pressure (WP 0541 00).

Indication/Condition

Spar pressure shall be correct, and BIM® indicator stripes shall be yellow.

Corrective Action

1. If spar pressure is correct but BIM® indicator stripes are red, perform a BIM® indicator test (WP 0081 00).
 - a. If BIM® indicator fails the test, replace indicator (WP 0581 00) and return blade to AVUM for one-hour whirl test.
2. If spar pressure is not correct, go to SPAR PRESSURE IS INCORRECT, in this work package.

SPAR PRESSURE IS INCORRECT**SYMPTOM**

Spar pressure shall be correct, and BIM® indicator stripes shall be yellow.

MALFUNCTION

Spar pressure is incorrect.

CORRECTIVE ACTION

- 1. Pressurize spar and test components for leakage (WP 1624 00).**
 - a. If spar leaks, go to **2**.
 - b. If spar does not leak, go to **4**.
- 2. Is cause of leak clearly identified and is leak reparable?**
 - a. If spar leak is clearly identified and reparable, go to **3**.
 - b. If spar leak is not clearly identified and reparable, send blade to DEPOT.
- 3. Repair blade (WP 0544 00).**
 - a. Confirm repair(s). Go to **4**.
 - b. Return blade to AVUM level for one hour whirl test (WP 0081 00). Go to **5**.
- 4. Has blade had previous BIM® indications within the last 30 hours?**
 - a. If blade had previous BIM® indications, send blade to DEPOT. Go to **5**.
 - b. If blade has not had previous BIM® indications, return blade to AVUM level for one hour whirl test (WP 0081 00). Go to **5**.
- 5. Procedure completed.**

END OF WORK PACKAGE

UNIT LEVEL
ROTOR SYSTEM
MEDIUM AND HIGH FREQUENCY VIBRATIONS

INITIAL SETUP:**Tools and Special Tools**

Aircraft Mechanic's Toolkit, SC5180-99-B01
 Powertrain Toolkit, SC5180-99-B13
 Torque Wrench, GGG-W-686

WP 0612 00

WP 0619 00

WP 0630 00

WP 0653 00

WP 0655 00

Personnel Required

Assistant (1)
 Powertrain Repairer MOS 15D (1)
 UH-60 Helicopter Repairer MOS 15T (1)

WP 0656 00

WP 0659 00

WP 0666 00

WP 0667 00

References

TM 1-1520-237-10

WP 0668 00

WP 0669 00

WP 0481 00

WP 0670 00

WP 0485 00

WP 0681 00

WP 0486 00

WP 1145 00

WP 0545 00

WP 1685 00

WP 0546 00

WP 1688 00

WP 0585 00

Equipment Condition

WP 0586 00

External Electrical Power Available, WP 1685 00
 and APU Operational, TM 1-1520-237-10

WP 0589 00

SPECIAL INSTRUCTIONS**NOTE**

See Figure 1 for location diagram as an aid in operational/troubleshooting.

The powertrain vibration operational/troubleshooting procedure is subdivided as follows:

- Medium-Frequency Vibrations
- High-Frequency Vibrations

MEDIUM-FREQUENCY VIBRATIONS OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step**

1. Medium-frequency vibrations are caused by the main and tail rotor drive shafts. Troubleshoot medium-frequency vibrations per MEDIUM-FREQUENCY VIBRATION FELT, in this work package.

Indication/Condition

Medium-frequency vibrations are felt.

MEDIUM-FREQUENCY VIBRATIONS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

Troubleshoot MEDIUM-FREQUENCY VIBRATION FELT, in this work package.

**HIGH-FREQUENCY VIBRATIONS OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step****NOTE**

High-frequency vibrations will normally come from the engine, main gear box input section, or tail rotor drive system, and are often as apparent in a ground run as in flight. The most important clue to high-frequency vibrations will be the associated sound. Tail rotor drive shaft vibrations are caused by an unbalanced drive shaft or a bad bearing. These vibrations can be identified during ground run by feeling the tail cone. The main gear box contains many possible sources of high-frequency vibrations such as the various gear box-mounted accessories, the accessories' gear train, oil cooler blower, the input bevel gears, and input free wheeling units. The engine gas generator or power turbine will normally produce a beat at various Ng and Np RPM combinations, or with Np RPM split from Nr. To the pilot, the only obvious evidence of excessive vibration will be greatly increased high-pitched noise levels. Combinations of these high-frequency vibrations in extreme cases could result in the pilot's sensing low or medium-frequency vibrations. These would be detected as vibrations which are affected only by variation in main rotor speed and may be just as apparent in ground run as in flight.

1. Troubleshoot high-frequency vibrations per HIGH-FREQUENCY FELT, in this work package.

Indication/Condition

High-frequency vibrations are felt.

Corrective Action

Troubleshoot high-frequency vibrations per HIGH-FREQUENCY FELT, in this work package.

MEDIUM-FREQUENCY VIBRATION FELT**SYMPTOM**

Medium-frequency vibrations are caused by the main and tail rotor drive shafts.

MALFUNCTION

Medium-frequency vibration felt.

CORRECTIVE ACTION

1. **Check tail rotor drive shaft for damage or missing weights, attaching hardware for looseness, or damaged couplings.**
 - a. If damage is found, replace damaged drive shafts or Thomas couplings, as required (WP 0666 00 through WP 0670 00). Go to **12**.
 - b. If no sign of damage is present, torque tail rotor drive shaft attaching bolts to required torque (WP 0666 00 through WP 0670 00). Go to **12**.
 - c. If no damage is found, go to **2**.

MEDIUM-FREQUENCY VIBRATION FELT - Continued

2. **Check tail rotor blades for damage (WP 0546 00).**
 - a. If blades are damaged, go to **3**.
 - b. If blades are not damaged, go to **5**.
3. **Replace tail rotor blades as required (WP 0585 00). Go to 4.**
4. **Torque bolts (WP 0585 00). Go to 12.**
5. **Check tail rotor blade retention plate(s) bolts for looseness.**
 - a. If bolts are loose, torque bolts (WP 0589 00). Go to **12**.
 - b. If bolts are not loose, go to **6**.
6. **Check tail rotor pitch change links for worn or binding bearings (WP 0545 00).**
 - a. If bearings are worn or binding, replace pitch change links rod ends, as required (WP 0586 00). Go to **12**.
 - b. If bearings are not worn or binding, go to **7**.
7. **Check tail rotor blades for out-of-track condition.**
 - a. If blades are out-of-track, check tail rotor for proper rigging. Rig as required. Go to **12**.
 - b. If blades are not out-of-track, go to **8**.
8. **Check tail rotor blades for out-of-balance condition.**
 - a. If blades are out-of-balance, balance tail rotor. Go to **12**.
 - b. If blades are not out-of-balance, go to **9**.
9. **Check tail rotor drive shaft for out-of-round condition.**
 - a. If drive shaft is out-of-round, replace drive shaft section as required (WP 0666 00) through (WP 0669 00). Go to **12**.
 - b. If drive shaft is not out-of-round, go to **10**.
10. **Check tail rotor control cables for proper tension.**
 - a. If cables do not have proper tension, adjust cable tension (WP 1145 00). Go to **12**.
 - b. If cables do have proper tension, go to **11**.
11. **Check tail gear box mounting bolts for looseness and improper torque.**
 - a. If bolts are loose or improperly torqued, torque bolts (WP 0681 00). Go to **12**.
 - b. If bolts are not loose or improperly torqued, check intermediate gear box mounting bolts for looseness and improper torque. Torque bolts (WP 0659 00). Go to **12**.
12. **Procedure completed.**

HIGH-FREQUENCY VIBRATION FELT**SYMPTOM**

High-frequency vibration felt.

MALFUNCTION

High-frequency vibration felt.

CORRECTIVE ACTION

1. **Check engine output shaft/input module flange hardware for proper torque (WP 0655 00).**
 - a. If torque is proper, go to **2**.
 - b. If torque is not proper, go to **6**.
2. **Inspect input module flange and coupling for wear and fretting (WP 0619 00).**
 - a. If flange and coupling is worn or fretting, replace engine output shaft and/or input module flange (WP 0486 00 or WP 0655 00). Go to **8**.
 - b. If flange and coupling is not worn or fretting, go to **3**.
3. **Check main transmission accessory module mounting bolts for proper torque (WP 0656 00).**
 - a. If torque is proper, go to **4**.
 - b. If torque is not proper, torque bolts (WP 0656 00). Go to **8**.
4. **Inspect main transmission mounting bolts for proper torque (WP 0612 00).**
 - a. If torque is proper, go to **5**.
 - b. If torque is not proper, go to **7**.
5. **Check engine mounts for looseness or too much play (WP 0485 00).**
 - a. If engine mounts are loose or have too much play, torque engine mounting bolts or replace mounts (WP 0485 00). Go to **8**.
 - b. If engine mounts are not loose or have too much play, then if engine has unusual noise during coast-down or during Np changes, replace engine (WP 0481 00). Go to **8**.
6. **Check flex coupling hardware for proper torque (WP 1688 00).**
 - a. If torque is proper, torque output shaft/input module hardware as required (WP 0486 00 or WP 0653 00). Go to **8**.
 - b. If trouble remains, replace engine output shaft (WP 0486 00). Go to **8**.
 - c. If torque is not proper, torque all hardware as required (WP 0486 00 or WP 0653 00). Go to **8**.
 - d. If trouble remains, replace engine output shaft and input module flange (WP 0486 00 and WP 0655 00). Go to **8**.
7. **Inspect main transmission barrel nuts for proper breakaway torque.**
 - a. If torque is proper, torque bolts (WP 0630 00). Go to **8**.
 - b. If torque is not proper, replace barrel nuts (WP 0630 00). Go to **8**.

HIGH-FREQUENCY VIBRATION FELT - Continued

8. **Procedure completed.**

LOCATION

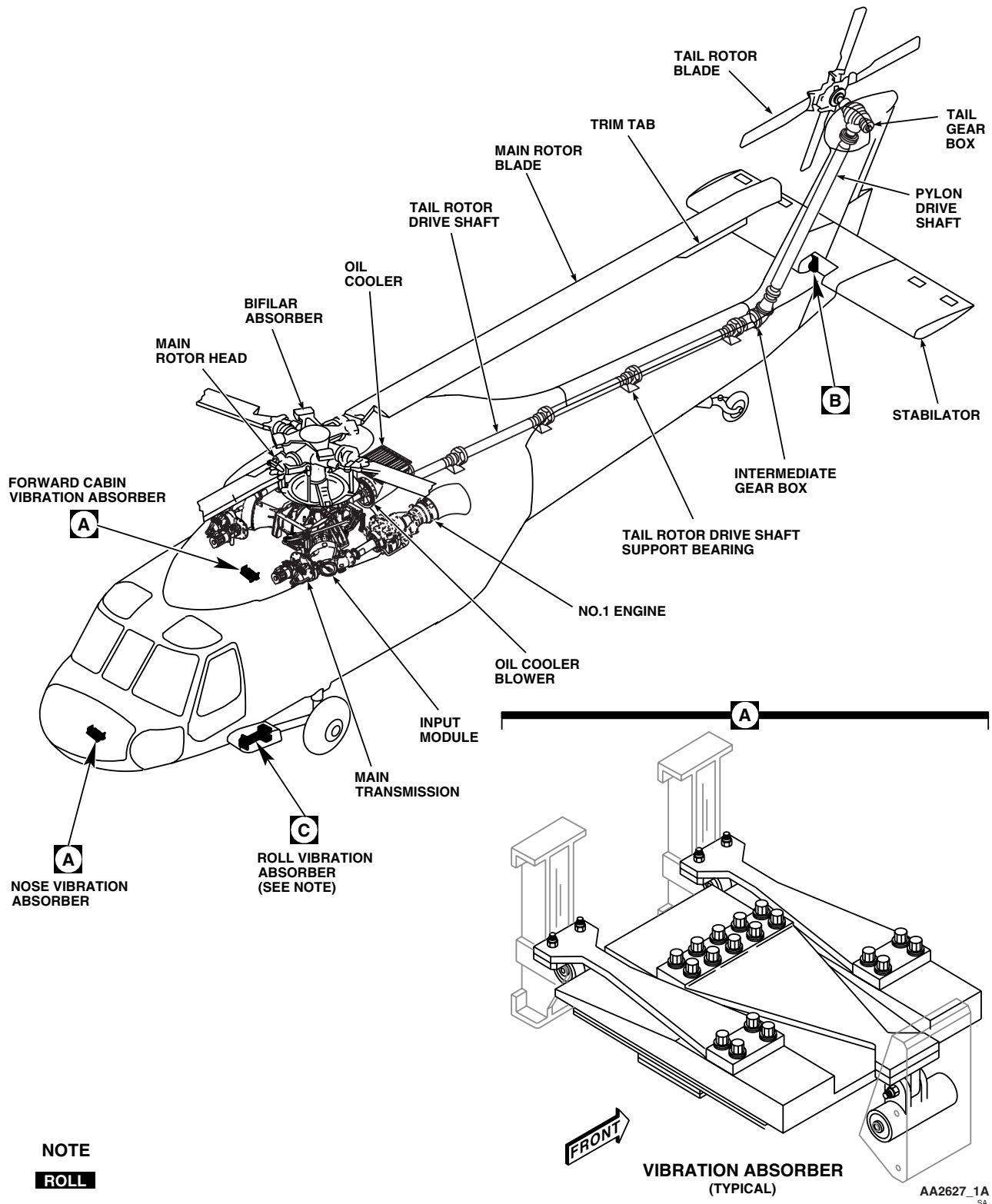


Figure 1. Medium and High Frequency Vibrations Location Diagram. (Sheet 1 of 2)

LOCATION - Continued

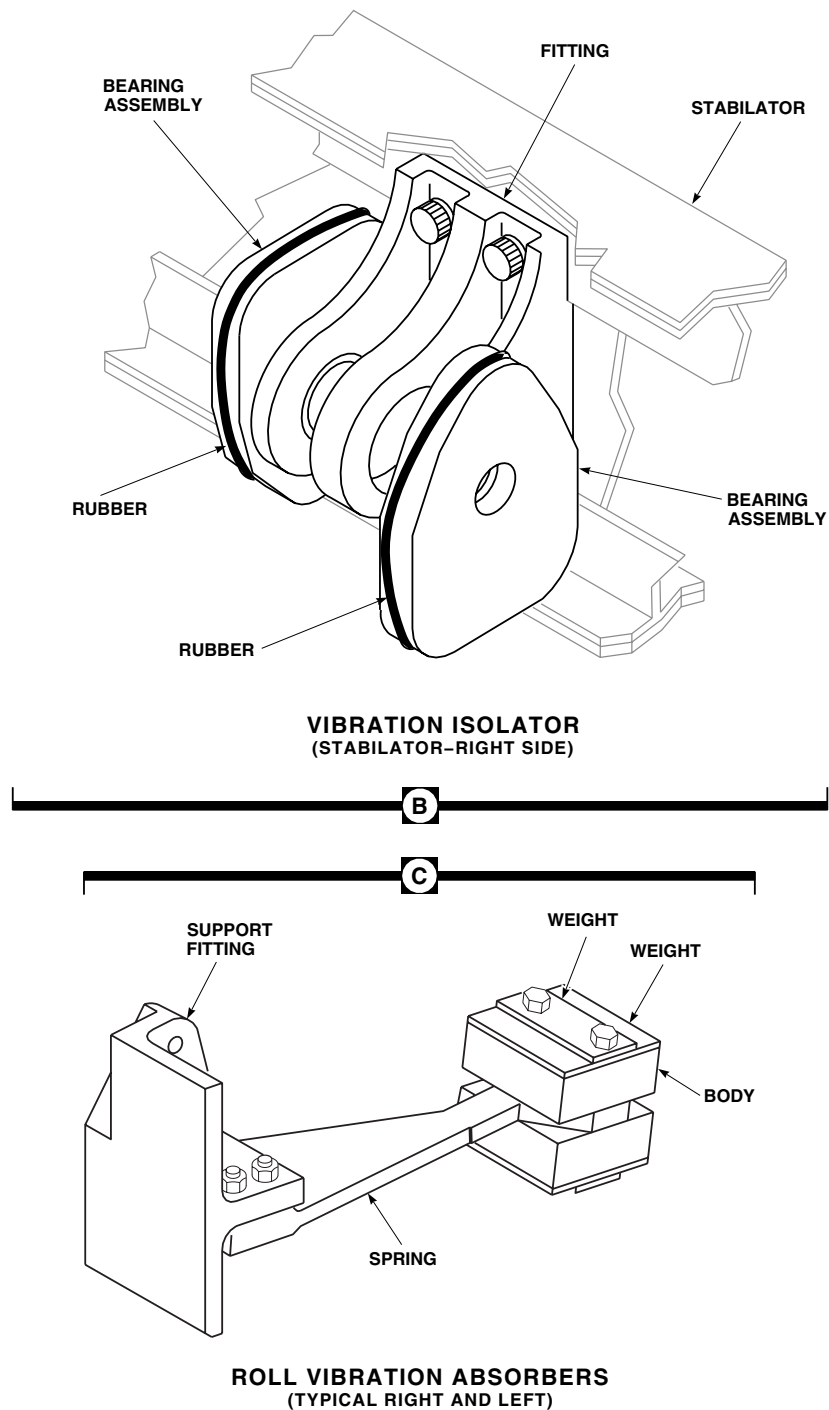
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Figure 1. Medium and High Frequency Vibrations Location Diagram. (Sheet 2 of 2)

END OF WORK PACKAGE

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UNIT LEVEL

DRIVE SYSTEM

GUST LOCK SYSTEM

This WP supersedes WP 0084 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment

Multimeter, AN/PSM-45A

WP 0100 00

WP 0105 00

WP 0638 00

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

Electrical Repairer Toolkit, SC 5180-99-B06

WP 0824 00

WP 0925 00

WP 1685 00

WP 1747 00

Personnel Required

Aircraft Electrician MOS 15F (1)

UH-60 Helicopter Repairer MOS 15T (1)

Equipment Condition

External Power Available, WP 1685 00

or APU Operational, TM 1-1520-237-10

References

TM 1-1520-237-10

WP 0090 00

GUST LOCK SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0105 00).
- See Figure 1 for location diagram, and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER

CIRCUIT BREAKER PANEL

LIGHTS ADVSY

Copilot's

HH-60A HH-60L MFD PILOT ■

Aux Circuit Breaker Panel

HH-60A HH-60L MFD COPILOT ■

Aux Circuit Breaker Panel

2. Turn on electrical power. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed). ■

GUST LOCK SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

3. Press button on gust lock handle and push handle up.

Indication/Condition

GUST LOCK capsule on the caution/advisory panel or legend on the MFD caution/advisory panel shall go on.

Corrective Action

If Indication/Condition is not as specified, go to GUST LOCK CAPSULE DOES NOT GO ON, in this work package.

Step

4. Press button on gust lock handle and pull handle down.

Indication/Condition

GUST LOCK capsule on the caution/advisory panel or legend on the MFD caution/advisory panel shall go off.

Corrective Action

If Indication/Condition is not as specified, replace gust lock switch (WP 0638 00).

Step

5. Press button on handle and push handle up.
6. **HH-60A HH-60L** Turn MFD switch to OFF. ■
7. Turn off electrical power.

Indication/Condition

Electrical power shall be off.

Corrective Action

None Required

GUST LOCK CAPSULE DOES NOT GO ON**SYMPTOM**

GUST LOCK capsule on the caution/advisory panel or legend on the MFD/caution/advisory panel shall go on.

MALFUNCTION

GUST LOCK capsule does not go on.

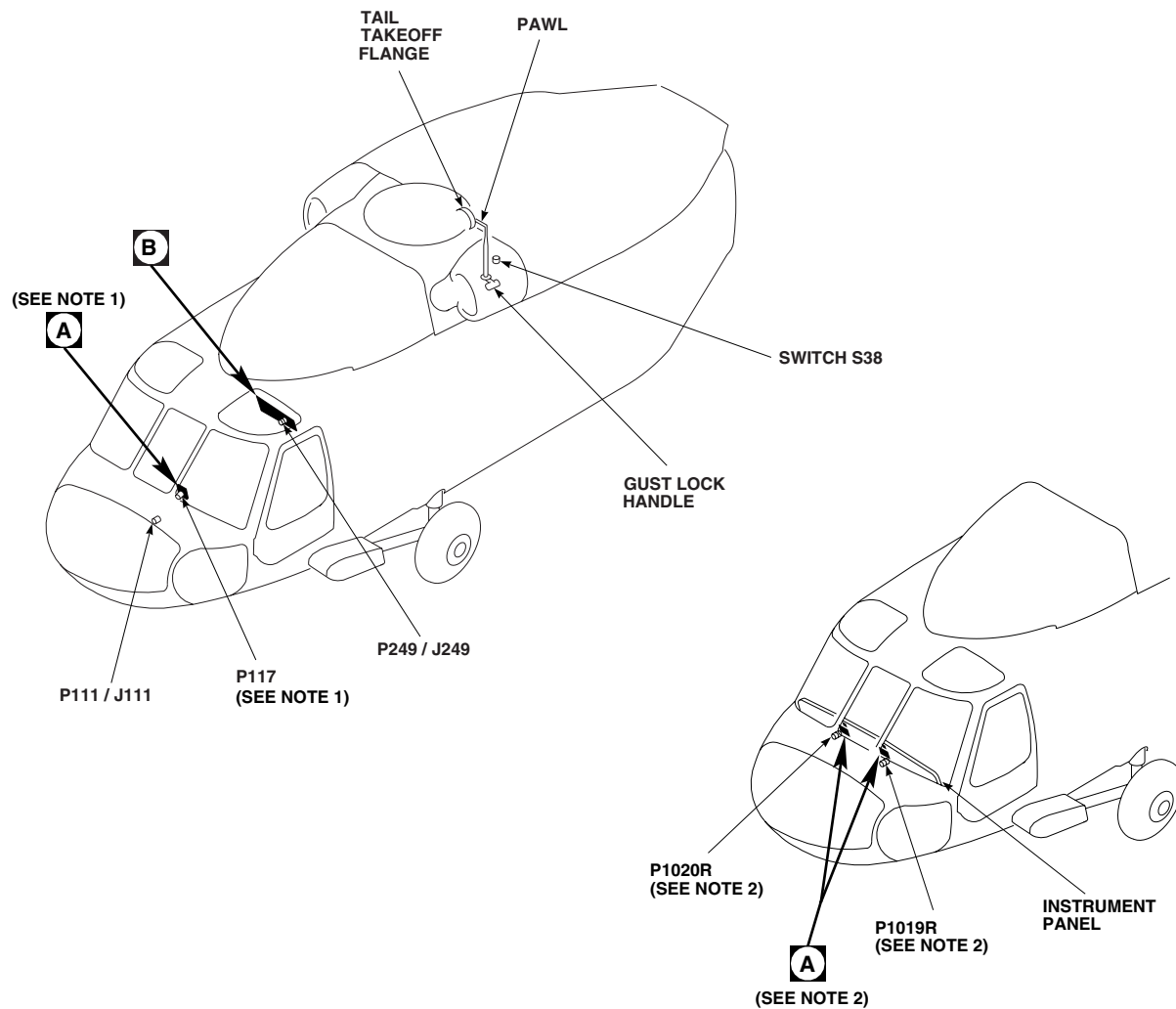
CORRECTIVE ACTION

1. Hold caution/advisory panel or MFD caution/advisory panel or **HH-60A HH-60L** instrument panel INDICATOR LTS ■ BRT/DIM-TEST switch to TEST. ■
Go to 2.
2. Check that GUST LOCK capsule or legend goes on.
 - a. If capsule or legend goes on, go to 3.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or **HH-60A HH-60L** MFD/caution/advisory warning system (WP 0100 00) ■
. Go to 14.
3. Check pawl for proper engagement.

GUST LOCK CAPSULE DOES NOT GO ON - Continued

- a. If pawl has proper engagement, go to **4**.
- b. If pawl does not have proper engagement, go to **12**.
- 4. Push button on gust lock switch in. Go to 5.**
- 5. Check that GUST LOCK capsule or legend goes on.**
 - a. If capsule or legend goes on, adjust switch (WP 0638 00). Go to **14**.
 - b. If capsule or legend does not go on, go to **6**.
- 6. Check helicopter configuration.**
 - a. **EMEP** , go to **7**.
 - b. **W/O EMEP** , go to **9**.
- 7. EMEP Remove and check pin filtered adapter P117 or HH-60A HH-60L P1019R or P1020R (WP 0925 00).**
 - a. If pin filtered adapter is good, go to **8**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **14**.
- 8. EMEP Install pin filtered adapter (WP 0925 00). Go to 9.**
- 9. Check for 28 vdc between P117-22 or HH-60A HH-60L P1019R-21 or P1020R-21 and ground.**
 - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or HH-60A HH-60L MFD/caution/advisory warning system (WP 0100 00) . Go to **14**.
 - b. If voltage is not as specified, go to **10**.
- 10. Check for 28 vdc between LIGHTS ADVSY circuit breaker terminal 1 and ground.**
 - a. If voltage is as specified, go to **11**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **14**.
- 11. Check continuity between:**
 - **LIGHTS ADVSY circuit breaker terminal 1 and GUST LOCK switch contact NC**
 - **P117-22 and GUST LOCK switch contact NO**
 - **HH-60A HH-60L P1019R-21 and GUST LOCK switch contact NO**
 - **HH-60A HH-60L P1020R-21 and GUST LOCK switch contact NO**
 - a. If continuity is present, replace GUST LOCK switch (WP 0638 00). Go to **14**.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **14**.
- 12. Turn rotor head slightly to engage pawl. Go to 13.**
- 13. Check pawl for proper engagement and that GUST LOCK capsule or legend goes on.**
 - a. If pawl has proper engagement and capsule or legend goes on, system is operational. Go to **14**.
 - b. If pawl does not have proper engagement, adjust rod end (WP 0638 00). Go to **14**.
- 14. Procedure completed.**

LOCATION



NOTES

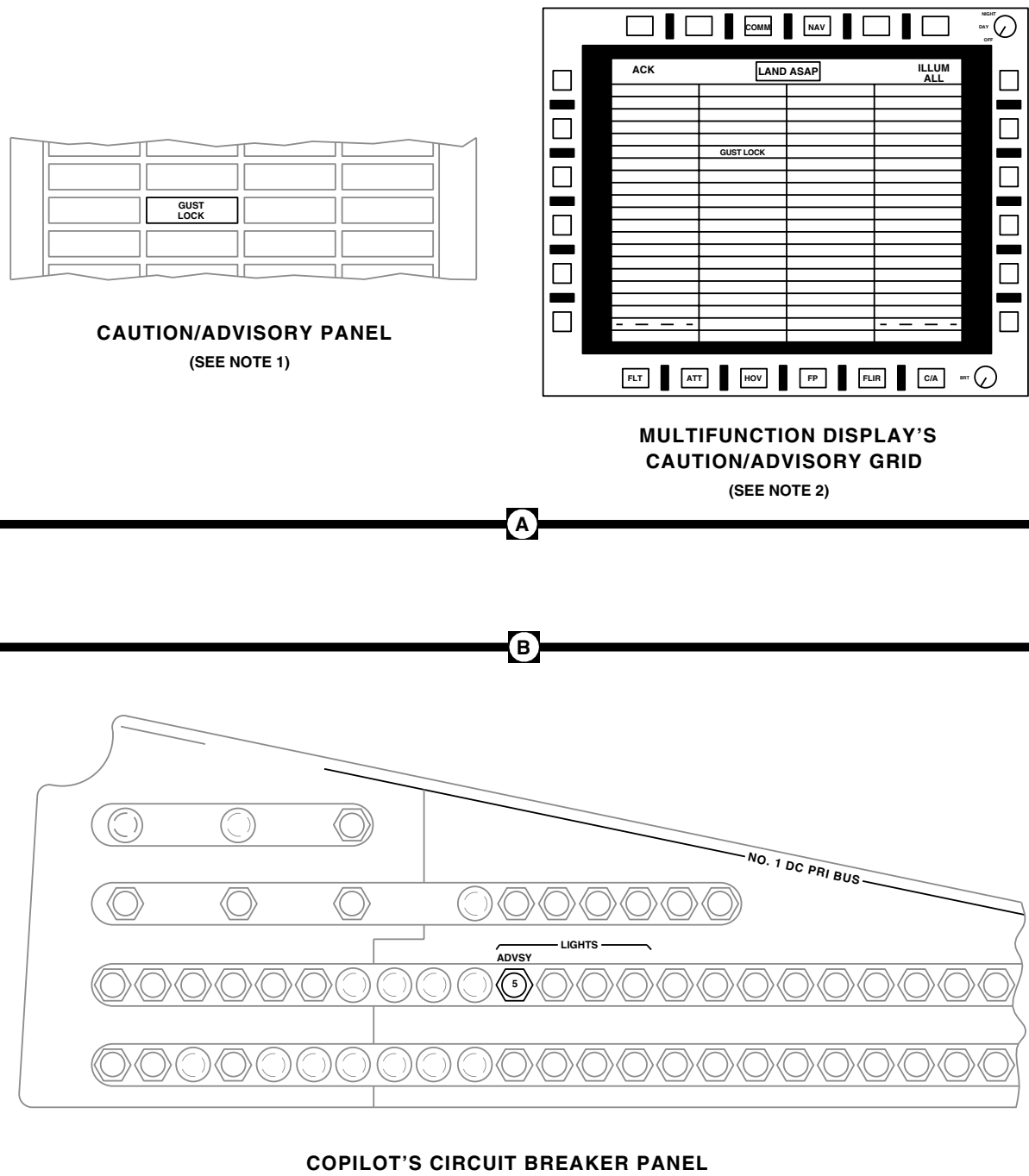
1. **UH-60A** **UH-60L** **EH-60A**
2. **HH-60A** **HH-60L**

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P111 / J111	COCKPIT, BL 7.5, STA 197
P117 / J117	CAUTION / ADVISORY PANEL (SEE NOTE 1)
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL, BL 23
P1019R / J3	COPLOT'S MULTIFUNCTION DISPLAY (SEE NOTE 2)
P1020R / J3	PILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 2)

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Figure 1. Gust Lock System Location Diagram. (Sheet 1 of 2)

LOCATION - Continued



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Figure 1. Gust Lock System Location Diagram. (Sheet 2 of 2)

SCHEMATICS

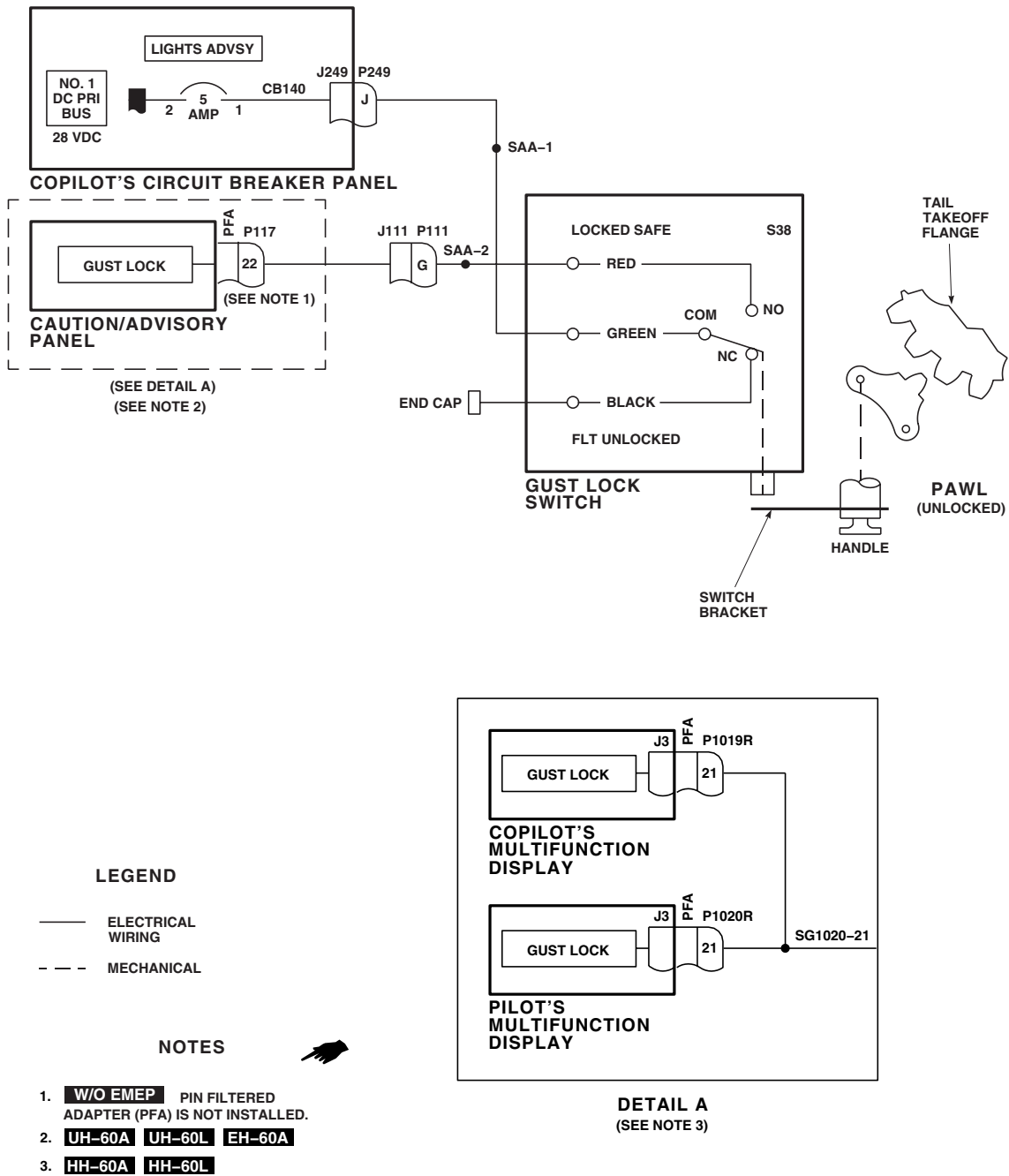
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Figure 2. Gust Lock System Schematic Diagram.

END OF WORK PACKAGE

UNIT LEVEL

DRIVE SYSTEM

TRANSMISSION CHIP DETECTOR, INSTRUMENTS, AND OIL WARNING SYSTEMS

This WP supersedes WP 0085 00, dated 15 December 2007.

INITIAL SETUP:

Test Equipment

Multimeter, AN/PSM-45A

WP 0647 00

WP 0648 00

WP 0649 00

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 0650 00

Electrical Repairer Toolkit, SC 5180-99-B06

WP 0653 00

Insulated Jumper Wire

WP 0665 00

WP 0676 00

Material/Parts

WP 0678 00

Towel, Machinery Wiping, Item 344, WP 1803 00

WP 0688 00

WP 0785 00

Personnel Required

Aircraft Electrician MOS 15F (1)

WP 0787 00

UH-60 Helicopter Repairer MOS 15T (1)

WP 0798 00

WP 0799 00

References

TM 1-1520-237-MTF

WP 0800 00

TM 1-1520-237-10

WP 0801 00

WP 0089 00

WP 0824 00

WP 0090 00

WP 0828 00

WP 0099 00

WP 0925 00

WP 0100 00

WP 0929 00

WP 0104 00

WP 1604 00

WP 0500 00

WP 1607 00

WP 0616 00

WP 1608 00

WP 0617 00

WP 1609 00

WP 0622 00

WP 1685 00

WP 0623 00

WP 1747 00

WP 0628 00

WP 1803 00

WP 0629 00

Equipment Condition

WP 0630 00

External Power Available, WP 1685 00

WP 0643 00

and APU Operational, TM 1-1520-237-10

WP 0644 00

Intermediate Gear Box Serviced, WP 1608 00

WP 0645 00

Main Gear Box Serviced, WP 1607 00

WP 0646 00

Tail Gear Box Serviced, WP 1609 00

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for system location diagram, **W/O MWO 50-26** Figure 2  and **MWO 50-26 UH-60A 83-23895-SUBQ UH-60L EH-60A HH-60A HH-60L** Figure 3  for chip detector system schematic diagrams, Figure 4 for instrument system schematic diagram, and Figure 5 for oil warning system schematic diagram as an aid in operational/troubleshooting.

The transmission chip detector, instruments, and oil warning operational/troubleshooting procedure is subdivided as follows:

- Improved Chip Detector Power Up
- Left Accessory Module Chip Detector
- Left Input Module Chip Detector
- Right Accessory Module Chip Detector
- Right Input Module Chip Detector
- Main Module Sump Chip Detector
- Intermediate Gear Box Chip Detector
- Tail Gear Box Chip Detector
- Instruments And Oil Warning
- Shutdown

IMPROVED CHIP DETECTOR POWER UP OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When a chip detector is installed after an operational/troubleshooting procedure, place upper console EXT PWR or GENERATORS APU switch to OFF, then ON, before removing next chip detector.
- Make sure upper console BATT switch is OFF.
- If the helicopter has any new, self-test chip detectors, the Improved Chip Detector section must be performed for those positions.

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.

IMPROVED CHIP DETECTOR POWER UP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. Turn on electrical power. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed). ■

Indication/Condition

- a. The following caution/advisory panel capsule or MFD legends shall be on for 45 to 75 seconds: CHIP ACCESS MDL-LH, CHIP INPUT MDL-LH, CHIP ACCESS MDL-RH, CHIP INPUT MDL-RH, CHIP TAIL XMSN, and CHIP INT XMSN.
- b. The following caution/advisory panel capsule or MFD legend shall be on after thirty (30) second delay and shall remain on for an additional thirty (30) seconds: CHIP MAIN MDL SUMP.

Corrective Action

1. If CHIP ACCESS MDL-LH capsule or legend is not as specified, go to CHIP ACCESS MDL-LH CAPSULE DOES NOT GO ON, in this work package.
2. If CHIP ACCESS MDL-LH capsule or legend goes on, but CHIP DET circuit breaker pops out, go to CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON, in this work package.
3. If CHIP INPUT MDL-LH capsule or legend is not as specified, go to CHIP INPUT MDL-LH CAPSULE DOES NOT GO ON, in this work package.
4. If CHIP INPUT MDL-LH capsule or legend goes on, but CHIP DET circuit breaker pops out, go to CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON, in this work package.
5. If CHIP ACCESS MDL-RH capsule or legend is not as specified, go to CHIP ACCESS MDL-RH CAPSULE DOES NOT GO ON, in this work package.
6. If CHIP ACCESS MDL-RH capsule or legend goes on, but CHIP DET circuit breaker pops out, go to CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON, in this work package.
7. If CHIP MAIN MDL SUMP capsule or legend is not as specified, go to CHIP INPUT MDL-LH CAPSULE DOES NOT GO ON, in this work package.

NOTE

On EFW (MFD) PN HH-60-00001A-01, no 30 second delay from initial power, delay active 120 seconds from initial power.

8. If CHIP INPUT MDL-RH capsule or legend goes on, but CHIP DET circuit breaker pops out, go to CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON, in this work package.
9. If CHIP INPUT MDL-RH capsule or legend is not as specified, go to CHIP INPUT MDL-RH CAPSULE DOES NOT GO ON, in this work package.
10. If CHIP MAIN MDL SUMP capsule or legend goes on before 30 seconds, troubleshoot caution/advisory warning system (WP 0090 00).
11. If CHIP MAIN MDL SUMP capsule or legend goes on, but CHIP DET circuit breaker pops out, go to CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON, in this work package.
12. If CHIP INT XMSN capsule or legend is not as specified, go to CHIP INT XMSN CAPSULE DOES NOT GO ON, in this work package.
13. If CHIP INT XMSN capsule or legend goes on, but CHIP DET circuit breaker pops out, go to CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON, in this work package.
14. If CHIP TAIL XMSN capsule or legend is not as specified, go to CHIP TAIL XMSN CAPSULE DOES NOT GO ON, in this work package.

IMPROVED CHIP DETECTOR POWER UP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

15. If CHIP TAIL XMSN capsule or legend goes on, but CHIP DET circuit breaker pops out, go to CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON, in this work package.

Step

3. Turn off electrical power.

Indication/Condition

CHIP ACCESS MDL-LH, CHIP INPUT MDL-LH, CHIP ACCESS MDL-RH, CHIP INPUT MDL-RH, CHIP MAIN MDL SUMP, and CHIP INT XMSN capsule or legends on caution/advisory panel shall go off after 5 seconds.

Corrective Action

1. If CHIP ACCESS MDL-LH is not as specified, disconnect P406.
 - a. If CHIP ACCESS MDL-LH capsule or legend goes off, replace chip detector (WP 0643 00).
 - b. If CHIP ACCESS MDL-LH capsule or legend remains on, disconnect P117 or **HH-60A HH-60L** P1019R or P1020R **■**.
 - (1) **EMEP** Remove and check pin filtered adapter (WP 0925 00). **■**
 - (2) **EMEP** Install pin filtered adapter (WP 0925 00). **■**
 - (a) If CHIP ACCESS MDL-LH capsule or legend goes off, repair/replace wiring between P117 or **HH-60A HH-60L** P1019R or P1020R **■** and P406 (WP 1747 00).
 - (b) If CHIP ACCESS MDL-LH capsule or legend remains on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).
2. If CHIP INPUT MDL-LH is not as specified, disconnect P403.
 - a. If CHIP INPUT MDL-LH capsule or legend goes off, replace chip detector (WP 0645 00).
 - b. If CHIP INPUT MDL-LH capsule or legend remains on, disconnect P117 or **HH-60A HH-60L** P1019R or P1020R **■**.
 - (1) **EMEP** Remove and check pin filtered adapter (WP 0925 00). **■**
 - (2) **EMEP** Install pin filtered adapter (WP 0925 00). **■**
 - (a) If CHIP INPUT MDL-LH capsule or legend goes off, repair/replace wiring between P117 or **HH-60A HH-60L** P1019R or P1020R **■** and P403 (WP 1747 00).
 - (b) If CHIP INPUT MDL-LH capsule or legend remains on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).
3. If CHIP ACCESS MDL-RH is not as specified, disconnect P405.
 - a. If CHIP ACCESS MDL-RH capsule or legend goes off, replace chip detector (WP 0643 00).
 - b. If CHIP ACCESS MDL-RH capsule or legend remains on, disconnect P117 or **HH-60A HH-60L** P1019R or P1020R **■**.
 - (1) **EMEP** Remove and check pin filtered adapter (WP 0925 00). **■**
 - (2) **EMEP** Install pin filtered adapter (WP 0925 00). **■**
 - (a) If CHIP ACCESS MDL-RH capsule or legend goes off, repair/replace wiring between P117 or **HH-60A HH-60L** P1019R or P1020R **■** and P405 (WP 1747 00).

IMPROVED CHIP DETECTOR POWER UP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- (b) If CHIP ACCESS MDL-RH capsule or legend remains on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).
4. If CHIP INPUT MDL-RH is not as specified, disconnect P404.
- If CHIP INPUT MDL-RH capsule or legend goes off, replace chip detector (WP 0645 00).
 - If CHIP INPUT MDL-RH capsule or legend remains on, disconnect P117 or **HH-60A HH-60L** P1019R or P1020R **■**.
 - EMEP** Remove and check pin filtered adapter (WP 0925 00). **■**
 - EMEP** Install pin filtered adapter (WP 0925 00). **■**
 - If CHIP INPUT MDL-RH capsule or legend goes off, repair/replace wiring between P117 or **HH-60A HH-60L** P1019R or P1020R **■** and P404 (WP 1747 00).
 - If CHIP INPUT MDL-RH capsule or legend remains on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).
5. If CHIP MAIN MDL SUMP is not as specified, disconnect P404.
- If CHIP MAIN MDL SUMP capsule or legend goes off, replace chip detector (WP 0646 00).
 - If CHIP MAIN MDL SUMP capsule or legend remains on, disconnect P117 or **HH-60A HH-60L** P1019R or P1020R **■**.
 - EMEP** Remove and check pin filtered adapter (WP 0925 00). **■**
 - EMEP** Install pin filtered adapter (WP 0925 00). **■**
 - If CHIP MAIN MDL SUMP capsule or legend goes off, repair/replace wiring between P117 or **HH-60A HH-60L** P1019R or P1020R **■** and P404 (WP 1747 00).
 - If CHIP MAIN MDL SUMP capsule or legend remains on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).
6. If CHIP INT XMSN is not as specified, visually check sensor for chips.
- If chips are found, do a serviceability check of intermediate gear box (WP 0623 00).
 - If chips are not found, disconnect P613.
 - If CHIP INT XMSN capsule or legend goes off, replace chip/temperature sensor (WP 0665 00).
 - If CHIP INT XMSN capsule or legend remains on, disconnect P117 or **HH-60A HH-60L** P1019R or P1020R **■**.
 - EMEP** Remove and check pin filtered adapter (WP 0925 00). **■**
 - EMEP** Install pin filtered adapter (WP 0925 00). **■**
 - 1 If CHIP INT XMSN capsule or legend goes off, repair/replace wiring between P117 or **HH-60A HH-60L** P1019R or P1020R **■** and P613 (WP 1747 00).
 - 2 If CHIP INT XMSN capsule or legend remains on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).
7. If CHIP TAIL XMSN is not as specified, check sensor for chips.
- If chips are found, do a serviceability check of tail gear box (WP 0629 00).
 - If chips are not found, disconnect P614.
 - If CHIP TAIL XMSN capsule or legend goes off, replace chip/temperature sensor (WP 0688 00).

IMPROVED CHIP DETECTOR POWER UP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- (2) If CHIP TAIL XMSN capsule or legend remains on, disconnect P117 or **HH-60A HH-60L** P1019R or P1020R **■**.
- (a) **EMEP** Remove and check pin filtered adapter (WP 0925 00). **■**
- (b) **EMEP** Install pin filtered adapter (WP 0925 00). **■**
- 1 If CHIP TAIL XMSN capsule or legend goes off, repair/replace wiring between P117 or **HH-60A HH-60L** P1019R or P1020R **■** and P614 (WP 1747 00).
- 2 If CHIP TAIL XMSN capsule or legend remains on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).

Step

4. Check main transmission for proper oil level and service as necessary (WP 1607 00).
5. Check intermediate gear box for proper oil level and service as necessary (WP 1608 00).
6. Check tail gear box for proper oil level and service as necessary (WP 1609 00).
7. **HH-60A HH-60L** Turn MFD switch to OFF **■**.
8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown must be performed.

Corrective Action

None Required

LEFT ACCESSORY MODULE CHIP DETECTOR OPERATIONAL/TROUBLESHOOTING PROCEDURE**Step****NOTE**

- If a circuit breaker pops out during operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When a chip detector is installed after an operational/troubleshooting procedure, place upper console EXT PWR or GENERATORS APU switch to OFF, then ON, before removing next chip detector.
- Make sure upper console BATT switch is OFF.
- For all other positions with the old-style chip detectors, the Chip Detector section must be performed.

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Remove chip detector from left accessory module and inspect for chips (WP 0643 00).
3. Clean all debris and oil off of chip detector using machinery towel wiping, Item 344, WP 1803 00.
4. Turn on electrical power.

LEFT ACCESSORY MODULE CHIP DETECTOR OPERATIONAL/TROUBLESHOOTING PROCEDURE -
Continued**Indication/Condition**

CHIP ACCESS MDL-LH capsule on the caution/advisory panel or legend on the MFD/caution/advisory panel shall be off.

Corrective Action

1. If Indication/Condition is not as specified, disconnect P406.
 - a. If CHIP ACCESS MDL-LH capsule or legend goes off, replace chip detector (WP 0643 00).
 - b. If CHIP ACCESS MDL-LH capsule or legend remains on, disconnect P117 or **HH-60A HH-60L** P1019R or P1020R **■**.
 - (1) **EMEP** Remove and check pin filtered adapter (WP 0925 00). **■**
 - (2) **EMEP** Install pin filtered adapter (WP 0925 00). **■**
 - (a) If CHIP ACCESS MDL-LH capsule or legend goes off, repair/replace wiring between P117 or **HH-60A HH-60L** P1019R or P1020R **■** and P406 (WP 1747 00).
 - (b) If CHIP ACCESS MDL-LH capsule or legend remains on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).

Step

5. Ground chip detector by placing a clean screwdriver with an insulated handle between each gap.

Indication/Condition

CHIP INPUT MDL-LH capsule or legend on caution/advisory panel or MFD/caution/advisory panel shall go on.

Corrective Action

1. If Indication/Condition is not as specified, go to CHIP ACCESS MDL-LH CAPSULE DOES NOT GO ON, in this work package.
2. If capsule or legend goes on, but CHIP DET circuit breaker pops out, go to CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON, in this work package.

Step

6. Turn off electrical power.
7. Install chip detector into left accessory module (WP 0643 00).
8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

LEFT INPUT MODULE CHIP DETECTOR OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When a chip detector is installed after an operational/troubleshooting procedure, place upper console EXT PWR or GENERATORS APU switch to OFF, then ON, before removing next chip detector.
- Make sure upper console BATT switch is OFF.
- For all other positions with the old-style chip detectors, the Chip Detector section must be performed.

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Remove left input module chip detector from main sump and inspect for chips (WP 0645 00).
3. Clean all debris and oil off of chip detector using machinery towel wiping, Item 344, WP 1803 00.
4. Turn on electrical power.

Indication/Condition

CHIP INPUT MDL-LH capsule on the caution/advisory panel or legend on the MFD/caution/advisory panel shall be off.

Corrective Action

1. If Indication/Condition is not as specified, disconnect P403.
 - a. If CHIP INPUT MDL-LH capsule or legend goes off, replace chip detector (WP 0645 00).
 - b. If CHIP INPUT MDL-LH capsule or legend remains on, disconnect P117 or **HH-60A HH-60L** P1019R or P1020R **■**.
 - (1) **EMEP** Remove and check pin filtered adapter (WP 0925 00). **■**
 - (2) **EMEP** Install pin filtered adapter (WP 0925 00). **■**
 - (a) If CHIP INPUT MDL-LH capsule or legend goes off, repair/replace wiring between P117 or **HH-60A HH-60L** P1019R or P1020R **■** and P403 (WP 1747 00).
 - (b) If CHIP INPUT MDL-LH capsule or legend remains on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).

Step

5. Ground chip detector by placing a clean screwdriver with an insulated handle between each gap.

Indication/Condition

CHIP INPUT MDL-LH capsule on the caution/advisory panel or legend on the MFD/caution/advisory panel shall go on.

LEFT INPUT MODULE CHIP DETECTOR OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition is not as specified, disconnect P403.
 - a. If CHIP INPUT MDL-LH capsule or legend goes off, replace chip detector (WP 0645 00).
 - b. If CHIP INPUT MDL-LH capsule or legend remains on, disconnect P117 or **HH-60A HH-60L** P1019R or P1020R **■**.
 - (1) **EMEP** Remove and check pin filtered adapter (WP 0925 00). **■**
 - (2) **EMEP** Install pin filtered adapter (WP 0925 00). **■**
 - (a) If CHIP INPUT MDL-LH capsule or legend goes off, repair/replace wiring between P117 or **HH-60A HH-60L** P1019R or P1020R **■** and P403 (WP 1747 00).
 - (b) If CHIP INPUT MDL-LH capsule or legend remains on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).
2. If Indication/Condition is not as specified, go to CHIP INPUT MDL-LH CAPSULE DOES NOT GO ON, in this work package.
3. If capsule or legend goes on, but CHIP DET circuit breaker pops out, go to CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON, in this work package.

Step

6. Turn off electrical power.
7. Install chip detector into main module sump (WP 0645 00).
8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

RIGHT ACCESSORY MODULE CHIP DETECTOR OPERATIONAL/TROUBLESHOOTING PROCEDURE**Step****NOTE**

- If a circuit breaker pops out during operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When a chip detector is installed after an operational/troubleshooting procedure, place upper console EXT PWR or GENERATORS APU switch to OFF, then ON, before removing next chip detector.
- Make sure upper console BATT switch is OFF.
- For all other positions with the old-style chip detectors, the Chip Detector section must be performed.

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.

RIGHT ACCESSORY MODULE CHIP DETECTOR OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. Remove chip detector from right accessory module and check for chips (WP 0643 00).
3. Clean all debris and oil off of chip detector using machinery towel wiping, Item 344, WP 1803 00.
4. Turn on electrical power.

Indication/Condition

CHIP ACCESS MDL-RH capsule on the caution/advisory panel or legend on the MFD/caution/advisory panel shall be off.

Corrective Action

1. If Indication/Condition is not as specified, disconnect P405.
 - a. If CHIP ACCESS MDL-RH capsule or legend goes off, replace chip detector (WP 0643 00).
 - b. If CHIP ACCESS MDL-RH capsule or legend remains on, disconnect P117 or **HH-60A HH-60L** P1019R or P1020R **■**.
 - (1) **EMEP** Remove and check pin filtered adapter (WP 0925 00). **■**
 - (2) **EMEP** Install pin filtered adapter (WP 0925 00). **■**
 - (a) If CHIP ACCESS MDL-RH capsule or legend goes off, repair/replace wiring between P117 or **HH-60A HH-60L** P1019R or P1020R **■** and P405 (WP 1747 00).
 - (b) If CHIP ACCESS MDL-RH capsule or legend remains on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).

Step

5. Ground chip detector by placing a clean screwdriver with an insulated handle between each gap.

Indication/Condition

CHIP ACCESS MDL-RH capsule on the caution/advisory panel or legend on the MFD/caution/advisory panel shall go on.

Corrective Action

1. If Indication/Condition is not as specified, go to CHIP ACCESS MDL-RH CAPSULE DOES NOT GO ON, in this work package.
2. If capsule or legend goes on, but CHIP DET circuit breaker pops out, go to CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON, in this work package.

Step

6. Turn off electrical power.
7. Install chip detector into right accessory module (WP 0643 00).
8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

RIGHT INPUT MODULE CHIP DETECTOR OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
 - When a chip detector is installed after an operational/troubleshooting procedure, place upper console EXT PWR or GENERATORS APU switch to OFF, then ON, before removing next chip detector.
 - Make sure upper console BATT switch is OFF.
 - For all other positions with the old-style chip detectors, the Chip Detector section must be performed.
1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
 2. Remove right input module chip detector from main sump and inspect for chips (WP 0645 00).
 3. Clean all debris and oil off of chip detector using machinery towel wiping, Item 344, WP 1803 00.
 4. Turn on electrical power.

Indication/Condition

CHIP INPUT MDL-RH capsule on the caution/advisory panel or legend on the MFD/caution/advisory panel shall be off.

Corrective Action

1. If Indication/Condition is not as specified, disconnect P404.
 - a. If CHIP INPUT MDL-RH capsule or legend goes off, replace chip detector (WP 0645 00).
 - b. If CHIP INPUT MDL-RH capsule or legend remains on, disconnect P117 or **HH-60A HH-60L** P1019R or P1020R **■**.
 - (1) **EMEP** Remove and check pin filtered adapter (WP 0925 00). **■**
 - (2) **EMEP** Install pin filtered adapter (WP 0925 00). **■**
 - (a) If CHIP INPUT MDL-RH capsule or legend goes off, repair/replace wiring between P117 or **HH-60A HH-60L** P1019R or P1020R **■** and P404 (WP 1747 00).
 - (b) If CHIP INPUT MDL-RH capsule or legend remains on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).

Step

5. Ground chip detector by placing a clean screwdriver with an insulated handle between each gap.

Indication/Condition

CHIP ACCESS MDL-RH capsule or legend on caution/advisory panel or MFD/caution/advisory panel shall go on.

RIGHT INPUT MODULE CHIP DETECTOR OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition is not as specified, go to CHIP INPUT MDL-RH CAPSULE DOES NOT GO ON, in this work package.
2. If capsule or legend goes on, but CHIP DET circuit breaker pops out, go to CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON, in this work package.

Step

6. Turn off electrical power.
7. Install chip detector into main module sump (WP 0645 00).
8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

MAIN MODULE SUMP CHIP DETECTOR OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step****NOTE**

- If a circuit breaker pops out during operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When a chip detector is installed after an operational/troubleshooting procedure, place upper console EXT PWR or GENERATORS APU switch to OFF, then ON, before removing next chip detector.
- Make sure upper console BATT switch is OFF.
- For all other positions with the old-style chip detectors, the Chip Detector section must be performed.

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Remove sump chip detector and remove strainer from detector and inspect for chips (WP 0646 00).
3. Clean all debris and oil off of chip detector using machinery towel wiping, Item 344, WP 1803 00.
4. Turn on electrical power.

Indication/Condition

CHIP MAIN MDL SUMP capsule on the caution/advisory panel or legend on the MFD/caution/advisory panel shall be off.

Corrective Action

1. If Indication/Condition is not as specified, disconnect P412.
 - a. If CHIP MAIN MDL SUMP capsule or legend goes off, replace chip detector (WP 0646 00).

MAIN MODULE SUMP CHIP DETECTOR OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- b. If CHIP MAIN MDL SUMP capsule or legend remains on, disconnect P117 or **HH-60A HH-60L** P1019R or P1020R **■**.
- (1) **EMEP** Remove and check pin filtered adapter (WP 0925 00). **■**
 - (2) **EMEP** Install pin filtered adapter (WP 0925 00). **■**
 - (a) If CHIP MAIN MDL SUMP capsule or legend goes off, repair/replace wiring between P117 or **HH-60A HH-60L** P1019R or P1020R **■** and P412 (WP 1747 00).
 - (b) If CHIP MAIN MDL SUMP capsule or legend remains on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).

Step

5. Ground chip detector by placing a clean screwdriver with an insulated handle between each gap for at least 30 seconds.

Indication/Condition

CHIP MAIN MDL SUMP capsule on the caution/advisory panel or legend on the MFD/caution/advisory panel shall go on 30 seconds after the detector was grounded.

Corrective Action

1. If Indication/Condition is not as specified, go to CHIP MAIN MDL SUMP CAPSULE DOES NOT GO ON, in this work package.
2. If capsule or legend goes on before 30 seconds, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).
3. If capsule or legend goes on, but CHIP DET circuit breaker pops out, go to CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON, in this work package.

Step

6. Turn off electrical power.
7. Install chip detector into main sump (WP 0646 00).
8. Check main transmission for proper oil level and service as necessary (WP 1607 00).
9. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

INTERMEDIATE GEAR BOX CHIP DETECTOR OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When a chip detector is installed after an operational/troubleshooting procedure, place upper console EXT PWR or GENERATORS APU switch to OFF, then ON, before removing next chip detector.
- Make sure upper console BATT switch is OFF.
- For all other positions with the old-style chip detectors, the Chip Detector section must be performed.

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Remove intermediate gear box chip/temperature sensor and inspect for chips (WP 0665 00).
3. Clean all debris and oil off of chip detector using machinery towel wiping, Item 344, WP 1803 00.
4. Turn on electrical power.

Indication/Condition

CHIP INT XMSN capsule on the caution/advisory panel or legend on the MFD/caution/advisory panel shall be off.

Corrective Action

1. If Indication/Condition is not as specified, visually check sensor for chips.
 - a. If chips are found, do a particle inspection of intermediate gear box (WP 0622 00).
 - b. If chips are not found, disconnect P613.
 - (1) If CHIP INT XMSN capsule or legend goes off, replace chip/temperature sensor (WP 0665 00).
 - (2) If CHIP INT XMSN capsule or legend remains on, disconnect P117 or **HH-60A HH-60L** P1019R or P1020R **■**.
 - (a) **EMEP** Remove and check pin filtered adapter (WP 0925 00). **■**
 - (b) **EMEP** Install pin filtered adapter (WP 0925 00). **■**
- 1 If CHIP INT XMSN capsule or legend goes off, repair/replace wiring between P117 or **HH-60A HH-60L** P1019R or P1020R **■** and P613 (WP 1747 00).
- 2 If CHIP INT XMSN capsule or legend remains on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).

Step

5. Ground chip/temperature sensor by placing a clean screwdriver with an insulated handle between each gap.

Indication/Condition

CHIP INT XMSN capsule or legend on caution/advisory panel or MFD/caution/advisory panel shall go on.

INTERMEDIATE GEAR BOX CHIP DETECTOR OPERATIONAL/TROUBLESHOOTING PROCEDURE -
Continued**Corrective Action**

1. If Indication/Condition is not as specified, go to CHIP INT XMSN CAPSULE DOES NOT GO ON, in this work package.
2. If capsule or legend goes on, but CHIP DET circuit breaker pops out, go to CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON, in this work package.

Step

6. Turn off electrical power.
7. Install chip/temperature sensor into intermediate gear box (WP 0665 00).
8. Check intermediate gear box for proper oil level and service as necessary (WP 1608 00).
9. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

TAIL GEAR BOX CHIP DETECTOR OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step****NOTE**

- If a circuit breaker pops out during operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
 - When a chip detector is installed after an operational/troubleshooting procedure, place upper console EXT PWR or GENERATORS APU switch to OFF, then ON, before removing next chip detector.
 - Make sure upper console BATT switch is OFF.
 - For all other positions with the old-style chip detectors, the Chip Detector section must be performed.
1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
 2. Remove chip/temperature sensor from tail gear box and inspect for chips (WP 0688 00).
 3. Clean all debris and oil off of chip detector using machinery towel wiping, Item 344, WP 1803 00.
 4. Turn on electrical power.

Indication/Condition

CHIP TAIL XMSN capsule on the caution/advisory panel or legend on the MFD/caution/advisory panel shall be off.

Corrective Action

1. If Indication/Condition is not as specified, check sensor for chips.
 - a. If chips are found, do a particle inspection of tail gear box (WP 0628 00).

TAIL GEAR BOX CHIP DETECTOR OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

b. If chips are not found, disconnect P614.

- (1) If CHIP TAIL XMSN capsule or legend goes off, replace chip/temperature sensor (WP 0688 00).
- (2) If CHIP TAIL XMSN capsule or legend remains on, disconnect P117 or **HH-60A HH-60L** P1019R or P1020R **■**.
 - (a) **EMEP** Remove and check pin filtered adapter (WP 0925 00). **■**
 - (b) **EMEP** Install pin filtered adapter (WP 0925 00). **■**
 - 1 If CHIP TAIL XMSN capsule or legend goes off, repair/replace wiring between P117 or **HH-60A HH-60L** P1019R or P1020R **■** and P614 (WP 1747 00).
 - 2 If CHIP TAIL XMSN capsule or legend remains on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).

Step

5. Ground chip/temperature sensor by placing a screwdriver with an insulated handle between each gap.

Indication/Condition

CHIP TAIL XMSN capsule on the caution/advisory panel or legend on the MFD/caution/advisory panel shall go on.

Corrective Action

1. If Indication/Condition is not as specified, go to CHIP TAIL XMSN CAPSULE DOES NOT GO ON, in this work package.
2. If capsule or legend goes on, but CHIP DET circuit breaker pops out, go to CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON, in this work package.

Step

6. Turn off electrical power.
7. Install chip/temperature sensor into tail gear box (WP 0688 00).
8. Check tail gear box for proper oil level and service as necessary (WP 1604 00).
9. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

INSTRUMENTS AND OIL WARNING OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When a chip detector is installed after an operational/troubleshooting procedure, place upper console EXT PWR or GENERATORS APU switch to OFF, then ON, before removing next chip detector.
- Make sure upper console BATT switch is OFF.

1. Check oil filter bypass button.

Indication/Condition

Oil filter bypass button shall not be extended.

Corrective Action

If Indication/Condition is not as specified, go to OIL FILTER BYPASS BUTTON EXTENDED, in this work package.

Step

2. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
3. Start APU (TM 1-1520-237-10).

Indication/Condition

MAIN XMSN OIL PRESS capsule or legend on caution/advisory panel shall be on.

Corrective Action

If Indication/Condition is not as specified, go to MAIN XMSN OIL PRESS CAPSULE DOES NOT GO ON, in this work package.

Step

4. With qualified personnel at controls, operate NO. 1 and NO. 2 engines with main rotor turning at 100% Nr until main transmission instruments have stabilized (TM 1-1520-237-10).

Indication/Condition

CAUTION/ADVISORY PANEL

CAPSULE STATUS

MAIN XMSN OIL TEMP

OFF

MAIN XMSN OIL PRESS

OFF

INSTRUMENTS AND OIL WARNING OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**CAUTION/ADVISORY PANEL****CAPSULE STATUS**

INT XMSN OIL TEMP

OFF

TAIL XMSN OIL TEMP

OFF

CENTRAL DISPLAY UNIT**INDICATION**

XMSN TEMP

(Refer to TM 1-1520-237-MTF for normal indications)

XMSN PRESS

(Refer to TM 1-1520-237-MTF for normal indications)

%RPM RTR

100%

Corrective Action

1. If these caution/advisory panel capsules or MFD/caution/advisory panel legends are on:

CAPSULE**GO TO**

MAIN XMSN OIL PRESS

XMSN PRESS INDICATION IS TOO LOW (MAIN XMSN OIL PRESS CAPSULE GOES ON), in this work package

MAIN XMSN OIL TEMP

XMSN TEMP INDICATION IS TOO HIGH OR IS NOT STEADY (MAIN XMSN OIL TEMP CAPSULE GOES ON), in this work package

INT XMSN OIL TEMP

INT XMSN OIL TEMP CAPSULE GOES ON, in this work package

TAIL XMSN OIL TEMP

TAIL XMSN OIL TEMP CAPSULE GOES ON, in this work package

2. If XMSN PRESS:

- is too low (MAIN XMSN OIL PRESS capsule or legend is on), go to XMSN PRESS INDICATION IS TOO LOW (MAIN XMSN OIL PRESS CAPSULE GOES ON), in this work package.
- is too high, go to XMSN PRESS HAS NO INDICATION, NOT STEADY, OR TOO HIGH, in this work package.
- is not steady, go to XMSN PRESS HAS NO INDICATION, NOT STEADY, OR TOO HIGH, in this work package.
- has no indication, go to XMSN PRESS HAS NO INDICATION, NOT STEADY, OR TOO HIGH, in this work package.
- suddenly drops (more than 10 psi) without fluctuations, go to PRESSURE SUDDENLY DROPS MORE THAN 10 PSI WITHOUT FLUCTUATIONS, in this work package.

INSTRUMENTS AND OIL WARNING OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

3. IF XMSN TEMP:

- is too high (MAIN XMSN OIL TEMP capsule or legend goes on) go to XMSN TEMP INDICATION IS TOO HIGH OR IS NOT STEADY (MAIN XMSN OIL TEMP CAPSULE GOES ON), in this work package.
- is not steady, go to XMSN TEMP INDICATION IS TOO HIGH OR IS NOT STEADY (MAIN XMSN OIL TEMP CAPSULE GOES ON), in this work package.
- has no indication, go to XMSN TEMP HAS NO INDICATION, in this work package.

4. If %RPM RTR:

- is not steady, go to PRESSURE SUDDENLY DROPS MORE THAN 10 PSI WITHOUT FLUCTUATIONS, in this work package.
- has no indication, make sure P443 is connected to main gear box.
 - a. If P443 is connected, go to %RPM RTR INDICATION IS NOT STEADY, in this work package.
 - b. If P443 is not connected, connect P443 to main gear box.

5. If any RTR OVERSPEED capsule or legend goes on, perform a special inspection (WP 1735 00).**Step**

5. Shut down NO. 1 and NO. 2 engines (TM 1-1520-237-10).
6. Turn on electrical power.
7. Install an insulated jumper wire between P408-A and P408-C.

Indication/Condition

MAIN XMSN OIL TEMP capsule or legend on caution/advisory panel or MFD/caution/advisory panel shall go on.

Corrective Action

If Indication/Condition is not as specified, go to MAIN XMSN OIL TEMP CAPSULE DOES NOT GO ON, in this work package.

Step

8. Remove jumper wire from P408 and install connector to main gear box.
9. Install an insulated jumper wire between P613-B and P613-C (intermediate gear box chip/temp detector connector).

Indication/Condition

INT XMSN OIL TEMP capsule or legend on caution/advisory panel or MFD/caution/advisory panel shall go on.

Corrective Action

If Indication/Condition is not as specified, go to INT XMSN OIL TEMP CAPSULE DOES NOT GO ON, in this work package.

Step

10. Remove jumper wire from P613.
11. Remove chip/temperature sensor (WP 0665 00).
12. Connect P613 to sensor.

INSTRUMENTS AND OIL WARNING OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

13. Connect an insulated jumper wire between spring and valve housing (Figure 1., Sheet 5, Detail F).

Indication/Condition

INT XMSN OIL TEMP capsule or legend on caution/advisory panel on MFD shall go on.

Corrective Action

If Indication/Condition is not as specified, replace chip/temperature sensor (WP 0665 00).

Step

14. Install chip/temperature sensor (WP 0665 00).
15. Install an insulated jumper wire between P614-B and P614-C (tail gear box chip/temperature detector connector).

Indication/Condition

TAIL XMSN OIL TEMP capsule on the caution/advisory panel or legend on the MFD/caution/advisory panel shall go on.

Corrective Action

If Indication/Condition is not as specified, go to TAIL XMSN OIL TEMP CAPSULE DOES NOT GO ON, in this work package.

Step

16. Remove jumper wire from P614.
17. Remove chip/temperature sensor (WP 0688 00).
18. Connect P614 to sensor.
19. Connect an insulated jumper wire between spring and valve housing (Figure 1., Sheet 5, Detail F).

Indication/Condition

TAIL XMSN OIL TEMP capsule or legend on caution/advisory panel or MFD/caution/advisory panel shall go on.

Corrective Action

If Indication/Condition is not as specified, replace chip/temperature sensor (WP 0688 00).

Step

20. Install chip/temperature sensor (WP 0688 00).
21. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Turn off electrical power.

Indication/Condition

Electrical power shall be off.

Corrective Action

None Required

CHIP ACCESS MDL-LH CAPSULE DOES NOT GO ON

SYMPTOM

CHIP ACCESS MDL-LH capsule does not go on.

MALFUNCTION

CHIP ACCESS MDL-LH capsule does not go on.

CORRECTIVE ACTION

1. Hold caution/advisory panel BRT/DIM-TEST **HH-60A HH-60L** instrument panel INDICATOR LTS **■** switch to TEST. Go to 2.
2. Check that CHIP ACCESS MDL-LH capsule or legend goes on.
 - a. If capsule or legend goes on, go to 3.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0089 00) or MFD/caution/advisory warning system (WP 0099 00). Go to 17.
3. Install an insulated jumper wire between J117-14 or **HH-60A HH-60L** P1019R-2 or P1020R-2 **■** and ground. Go to 4.
4. Check that CHIP ACCESS MDL-LH capsule or legend goes on.
 - a. If capsule or legend goes on, go to 5.
 - b. If capsule or legend does not go on, go to 10.
5. Remove jumper wire. Go to 6.
6. Check configuration of helicopter.
 - a. **EMEP** , go to 7.
 - b. **W/O EMEP** , go to 9.
7. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R **■** (WP 0929 00).
 - a. If pin filtered adapter is good, go to 8.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0929 00). Go to 17.
8. **EMEP** Install pin filtered adapter (WP 0929 00). Go to 9.

CHIP ACCESS MDL-LH CAPSULE DOES NOT GO ON - Continued

9. Repair/replace wiring between P117-14 or **HH-60A HH-60L** P1019R-2 or P1020R-2 and left accessory module chip detector (WP 1725 00). Go to 17.
10. Remove jumper wire. Go to 11.
11. Check configuration of helicopter.
 - a. **UH-60L UH-60A** , go to 12.
 - b. **HH-60A HH-60L** , go to 16.
12. Check configuration of helicopter.
 - a. **EMEP** , go to 13.
 - b. **W/O EMEP** , go to 15.
13. **EMEP** Remove and check pin filtered adapter P118 (WP 0929 00).
 - a. If pin filtered adapter is good, go to 14.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0929 00). Go to 17.
14. **EMEP** Install pin filtered adapter (WP 0929 00). Go to 15.
15. Check for 28 vdc between P118-f and ground.
 - a. If voltage is as specified, replace caution/advisory panel (WP 0787 00). Go to 17.
 - b. If voltage is not as specified, go to 16.
16. Check for 28 vdc between CHIP DET circuit breaker terminal 1 and ground.
 - a. If voltage is as specified, repair/replace wiring between P118-f and circuit breaker terminal 1 (WP 1725 00). Go to 17.
 - b. If voltage is not as specified, replace circuit breaker (WP 0828 00). Go to 17.
17. Procedure completed.

CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON**SYMPTOM**

CHIP DET circuit breaker pops when a chip detector light goes on.

MALFUNCTION

CHIP DET circuit breaker pops when a chip detector light goes on.

CORRECTIVE ACTION

1. Check configuration of helicopter.
 - a. **W/O MWO 50-26** , go to 2
 - b. **MWO 50-26** , go to 12.
2. **W/O MWO 50-26** Check that CHIP ACCESS MDL-RH capsule or legend is on.
 - a. If capsule or legend is on, go to 3.
 - b. If capsule or legend is not on, go to 4.
3. **W/O MWO 50-26** Check for 19,000 to 21,000 ohms between J125-E and J125-J.

CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON - Continued

- a. If resistance is as specified, troubleshoot for CHIP DET circuit breaker popping (WP 0104 00).
 - b. If trouble remains, go to CHIP ACCESS MDL-RH CAPSULE GOES ON, in this work package.
 - c. If resistance is not as specified, replace chip detector resistor unit (WP 0800 00). Go to **18**.
- 4. W/O MWO 50-26 Check that CHIP ACCESS MDL-LH capsule or legend is on.**
- a. If capsule or legend is on, go to **5**.
 - b. If capsule or legend is not on, go to **6**.
- 5. W/O MWO 50-26 Check for 19,000 to 21,000 ohms between J125-D and J125-J on resistor unit.**
- a. If resistance is as specified, troubleshoot for CHIP DET circuit breaker popping (WP 0104 00). Go to **18**.
 - b. If trouble remains, go to CHIP ACCESS MDL-LH CAPSULE GOES ON, in this work package.
 - c. If resistance is not as specified, replace chip detector resistor unit (WP 0800 00). Go to **18**.
- 6. W/O MWO 50-26 Check that CHIP INPUT MDL-LH capsule or legend is on.**
- a. If capsule or legend is on, go to **7**.
 - b. If capsule or legend is not on, go to **8**.
- 7. W/O MWO 50-26 Check for 19,000 to 21,000 ohms between J125-F and J125-J on resistor unit.**
- a. If resistance is as specified, troubleshoot for CHIP DET circuit breaker popping (WP 0104 00). Go to **18**.
 - b. If trouble remains, go to CHIP INPUT MDL-LH CAPSULE GOE ON, in this work package.
 - c. If resistance is not as specified, replace chip detector resistor unit (WP 0800 00). Go to **18**.
- 8. W/O MWO 50-26 Check that CHIP INPUT MDL-RH capsule or legend is on.**
- a. If capsule or legend is on, go to **9**.
 - b. If capsule or legend is not on, go to **10**.
- 9. W/O MWO 50-26 Check for 19,000 to 21,000 ohms between J125-G and J125-J.**
- a. If resistance is as specified, troubleshoot for CHIP DET circuit breaker popping (WP 0104 00). Go to **18**.
 - b. If trouble remains, go to CHIP INPUT MDL-RH CAPSULE GOES ON, in this work package.
 - c. If resistance is not as specified, replace chip detector resistor unit (WP 0800 00). Go to **18**.
- 10. W/O MWO 50-26 CHIP MAIN MDL SUMP capsule or legend was on. Go to 11.**
- 11. W/O MWO 50-26 Check for 19,000 to 21,000 ohms between J125-H and J125-J.**
- a. If resistance is as specified, troubleshoot for CHIP DET circuit breaker popping (WP 0104 00).
 - b. If trouble remains, go to CHIP MAIN MDL SUMP CAPSULE IS ON, in this work package.
 - c. If resistance is not as specified, replace chip detector resistor unit (WP 0800 00). Go to **18**.

CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON - Continued

12. **MWO 50-26** Check that **CHIP ACCESS MDL-RH capsule or legend is on.**
 - a. If capsule or legend is on, troubleshoot for CHIP DET circuit breaker popping (WP 0104 00). Go to **18.**
 - b. If trouble remains, go to **CHIP ACCESS MDL-RH CAPSULE GOES ON**, in this work package.
 - c. If capsule or legend is not on, go to **13.**
13. **MWO 50-26** Check that **CHIP ACCESS MDL-LH capsule or legend is on.**
 - a. If capsule or legend is on, troubleshoot for CHIP DET circuit breaker popping (WP 0104 00). Go to **18.**
 - b. If trouble remains, go to **CHIP ACCESS MDL-LH CAPSULE GOES ON**, in this work package.
 - c. If capsule or legend is not on, go to **14.**
14. **MWO 50-26** Check that **CHIP INPUT MDL-LH capsule or legend is on.**
 - a. If capsule or legend is on, troubleshoot CHIP DET circuit breaker popping (WP 0104 00). Go to **18.**
 - b. If trouble remains, go to **CHIP INPUT MDL-LH CAPSULE GOES ON**, in this work package.
 - c. If capsule or legend is not on, go to **15.**
15. **MWO 50-26** Check that **CHIP INPUT MDL-RH capsule or legend is on.**
 - a. If capsule or legend is on, troubleshoot for CHIP DET circuit breaker popping (WP 0104 00). Go to **18.**
 - b. If trouble remains, go to **CHIP INPUT MDL-RH CAPSULE GOES ON**, in this work package.
 - c. If capsule or legend is not on, go to **16.**
16. **MWO 50-26** **CHIP MAIN MDL SUMP capsule or legend was on. Go to 17.**
17. **MWO 50-26** **Troubleshoot for CHIP DET circuit breaker popping (WP 0104 00).**
 - a. If trouble remains, go to **CHIP MAIN MDL SUMP CAPSULE IS ON**, in this work package.
18. **Procedure completed.**

CHIP INPUT MDL-LH CAPSULE DOES NOT GO ON**SYMPTOM**

CHIP INPUT MDL-LH capsule does not go on.

MALFUNCTION

CHIP INPUT MDL-LH capsule does not go on.

CORRECTIVE ACTION

1. Hold caution/advisory panel or **HH-60A HH-60L** instrument panel **INDICATOR LTS** **BRT/DIM-TEST** switch to **TEST**. Go to **2.**
2. Check that **CHIP INPUT MDL-LH capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **3.**

CHIP INPUT MDL-LH CAPSULE DOES NOT GO ON - Continued

- b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0089 00) or MFD/caution/advisory warning system (WP 0099 00). Go to **12**.
- 3. Install an insulated jumper wire between J117-13 or HH-60A HH-60L P1019R-1 or P1020R-1 and ground. Go to 4.**
- 4. Check that CHIP INPUT MDL-LH capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **5**.
 - b. If capsule or legend does not go on, go to **10**.
- 5. Remove jumper wire. Go to 6.**
- 6. Check configuration of helicopter.**
 - a. **EMEP** , go to **7**.
 - b. **W/O EMEP** , go to **9**.
- 7. EMEP Remove and check pin filtered adapter P117 or HH-60A HH-60L P1019R or P1020R (WP 0929 00).**
 - a. If pin filtered adapter is good, go to **8**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0929 00). Go to **12**.
- 8. EMEP Install pin filtered adapter (WP 0929 00). Go to 9.**
- 9. Repair/replace wiring between P117-13 or HH-60A HH-60L P1019R-1 or P1020R-1 and left input module chip detector (WP 1725 00). Go to 12.**
- 10. Remove jumper wire. Go to 11.**
- 11. Replace caution/advisory panel (WP 0787 00) or MFD (WP 0801 00). Go to 12.**
- 12. Procedure completed.**

CHIP ACCESS MDL-RH CAPSULE DOES NOT GO ON**SYMPTOM**

CHIP ACCESS MDL-RH capsule does not go on.

MALFUNCTION

CHIP ACCESS MDL-RH capsule does not go on.

CORRECTIVE ACTION

- 1. Hold caution/advisory panel or HH-60A HH-60L instrument panel INDICATOR LTS BRT/DIM-TEST switch to TEST. Go to 2.**
- 2. Check that CHIP ACCESS MDL-RH capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **3**.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0089 00) or MFD/caution/advisory warning system (WP 0099 00). Go to **12**.
- 3. Install an insulated jumper wire between J117-62 or HH-60A HH-60L P1019R-5 or P1020R-5 and ground. Go to 4.**
- 4. Check that CHIP ACCESS MDL-RH capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **5**.

CHIP ACCESS MDL-RH CAPSULE DOES NOT GO ON - Continued

- b. If capsule or legend does not go on, go to **10**.
- 5. Remove jumper wire. Go to 6.**
- 6. Check configuration of helicopter.**
 - a. **EMEP** , go to **7**.
 - b. **W/O EMEP** , go to **9**.
- 7. EMEP Remove and check pin filtered adapter P117 or HH-60A HH-60L P1019R or P1020R (WP 0929 00).**
 - a. If pin filtered adapter is good, go to **8**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0929 00). Go to **12**.
- 8. EMEP Install pin filtered adapter (WP 0929 00). Go to 9.**
- 9. Repair/replace wiring between P117-62 or HH-60A HH-60L P1019R-5 or P1020R-5 and right accessory module chip detector (WP 1725 00). Go to 12.**
- 10. Remove jumper wire. Go to 11.**
- 11. Replace caution/advisory panel (WP 0787 00) or MFD (WP 0801 00). Go to 12.**
- 12. Procedure completed.**

CHIP INPUT MDL-RH CAPSULE DOES NOT GO ON**SYMPTOM**

CHIP INPUT MDL-RH capsule does not go on.

MALFUNCTION

CHIP INPUT MDL-RH capsule does not go on.

CORRECTIVE ACTION

- 1. Hold caution/advisory panel or HH-60A HH-60L instrument panel INDICATOR LTS BRT/DIM-TEST switch to TEST. Go to 2.**
- 2. Check that CHIP INPUT MDL-RH capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **3**.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0089 00) or MFD/caution/advisory warning system (WP 0099 00). Go to **12**.
- 3. Install an insulated jumper wire between J117-61 or HH-60A HH-60L P1019R-4 or P1020R-4 and ground. Go to 4.**
- 4. Check that CHIP INPUT MDL-RH capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **5**.
 - b. If capsule or legend does not go on, go to **10**.
- 5. Remove jumper wire. Go to 6.**
- 6. Check configuration of helicopter.**
 - a. **EMEP** , go to **7**.
 - b. **W/O EMEP** , go to **9**.

CHIP INPUT MDL-RH CAPSULE DOES NOT GO ON - Continued

7. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R (WP 0929 00).
 - a. If pin filtered adapter is good, go to 8.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0929 00). Go to 12.
8. **EMEP** Install pin filtered adapter (WP 0929 00). Go to 9.
9. Repair/replace wiring between P117-61 or **HH-60A HH-60L** P1019R-4 or P1020R-4 and right input module chip detector (WP 1725 00). Go to 12.
10. Remove jumper wire. Go to 11.
11. Replace caution/advisory panel (WP 0787 00) or MFD/caution/advisory panel (WP 0801 00). Go to 12.
12. Procedure completed.

CHIP MAIN MDL SUMP CAPSULE DOES NOT GO ON**SYMPTOM**

CHIP MAIN MDL SUMP capsule does not go on.

MALFUNCTION

CHIP MAIN MDL SUMP capsule does not go on.

CORRECTIVE ACTION

1. Hold caution/advisory panel or **HH-60A HH-60L** instrument panel INDICATOR LTS BRT/DIM-TEST switch to TEST. Go to 2.
2. Check that CHIP MAIN MDL SUMP capsule or legend goes on.
 - a. If capsule or legend goes on, go to 3.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0089 00) or MFD/caution/advisory warning system (WP 0099 00). Go to 13.
3. Install an insulated jumper wire between J117-30 or **HH-60A HH-60L** J3-11 on MFD and ground. Go to 4.
4. Wait 30 seconds. Go to 5.
5. Check that CHIP MAIN MDL SUMP capsule or legend goes on.
 - a. If capsule or legend goes on, go to 6.
 - b. If capsule or legend does not go on, go to 11.
6. Remove jumper wire. Go to 7.
7. Check configuration of helicopter.
 - a. **EMEP** , go to 8.
 - b. **W/O EMEP** , go to 10.
8. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R (WP 0929 00).
 - a. If pin filtered adapter is good, go to 9.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0929 00). Go to 13.

CHIP MAIN MDL SUMP CAPSULE DOES NOT GO ON - Continued

9. **EMEP** Install pin filtered adapter (WP 0929 00). Go to 10.
10. Repair/replace wiring between P117-30 or **HH-60A HH-60L** P1019R-11 or P1020R-11 and main module chip detector (WP 1725 00). Go to 13.
11. Remove jumper wire. Go to 12.
12. Replace caution/advisory panel (WP 0787 00) or MFD/caution/advisory panel (WP 0801 00). Go to 13.
13. Procedure completed.

CHIP INT XMSN CAPSULE DOES NOT GO ON**SYMPTOM**

CHIP INT XMSN capsule does not go on.

MALFUNCTION

CHIP INT XMSN capsule does not go on.

CORRECTIVE ACTION

1. Hold caution/advisory panel or **HH-60A HH-60L** instrument panel INDICATOR LTS and BRT/DIM-TEST switch to TEST. Go to 2.
2. Check that CHIP INT XMSN capsule or legend goes on.
 - a. If capsule or legend goes on, go to 3.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0089 00) or MFD/caution/advisory warning system (WP 0099 00). Go to 12.
3. Install an insulated jumper wire between J117-29 or **HH-60A HH-60L** P1019R-10 or P1020R-10 and ground. Go to 4.
4. Check that CHIP INT XMSN capsule or legend goes on.
 - a. If capsule or legend goes on, go to 5.
 - b. If capsule or legend does not go on, go to 10.
5. Remove jumper wire. Go to 6.
6. Check configuration of helicopter.
 - a. **EMEP** , go to 7.
 - b. **W/O EMEP** , go to 9.
7. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R (WP 0929 00).
 - a. If pin filtered adapter is good, go to 8.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0929 00). Go to 12.
8. **EMEP** Install pin filtered adapter (WP 0929 00). Go to 9.
9. Repair/replace wiring between P117-29 or **HH-60A HH-60L** P1019R-10 or P1020R-10 and intermediate gear box chip detector (WP 1747 00). Go to 12.
10. Remove jumper wire. Go to 11.

CHIP INT XMSN CAPSULE DOES NOT GO ON - Continued

11. Replace caution/advisory panel (WP 0787 00) or MFD/caution/advisory panel (WP 0801 00). Go to 12.

12. Procedure completed.

CHIP TAIL XMSN CAPSULE DOES NOT GO ON**SYMPTOM**

CHIP TAIL XMSN capsule does not go on.

MALFUNCTION

CHIP TAIL XMSN capsule does not go on.

CORRECTIVE ACTION

1. Hold caution/advisory panel or **HH-60A HH-60L** instrument panel INDICATOR LTS **■** BRT/DIM-TEST switch to TEST. Go to 2.
2. Check that CHIP TAIL XMSN capsule or legend goes on.
 - a. If capsule or legend goes on, go to 3.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0089 00) or MFD/caution/advisory warning system (WP 0099 00). Go to 12.
3. Install an insulated jumper wire between J117-45 or **HH-60A HH-60L** P1019R-3 or P1020R-3 **■** and ground. Go to 4.
4. Check that CHIP TAIL XMSN capsule or legend goes on.
 - a. If capsule or legend goes on, go to 5.
 - b. If capsule or legend does not go on, go to 10.
5. Remove jumper wire. Go to 6.
6. Check configuration of helicopter.
 - a. **EMEP** , go to 7.
 - b. **W/O EMEP** , go to 9.
7. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R **■** (WP 0929 00).
 - a. If pin filtered adapter is good, go to 8.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0929 00). Go to 12.
8. **EMEP** Install pin filtered adapter (WP 0929 00). Go to 9.
9. Repair/replace wiring between P117-45 or **HH-60A HH-60L** P1019R-3 or P1020R-3 **■** and tail gear box chip detector (WP 1747 00). Go to 12.
10. Remove jumper wire. Go to 11.
11. Replace caution/advisory panel (WP 0787 00) or MFD/caution/advisory panel (WP 0801 00). Go to 12.
12. Procedure completed.

OIL FILTER BYPASS BUTTON EXTENDED**SYMPTOM**

Oil filter bypass button extended.

MALFUNCTION

Oil filter bypass button extended.

CORRECTIVE ACTION**1. Refer to maintenance log book.**

- a. If first stage filter element has been replaced more than three times within 200 hours with no oil changes, replace filter bowl assembly (WP 0647 00). Go to **2**.
- b. If maintenance has been performed to the main transmission that caused oil to be drained, exposed to a non-purified container, or exposed to dust, perform serviceability inspection (WP 0616 00). Go to **2**.
- c. If the above steps have not occurred, perform a serviceability check (WP 0616 00). Go to **2**.

2. Procedure completed.**MAIN XMSN OIL PRESS CAPSULE DOES NOT GO ON****SYMPTOM**

MAIN XMSN OIL PRESS capsule does not go on.

MALFUNCTION

MAIN XMSN OIL PRESS capsule does not go on.

CORRECTIVE ACTION

- 1. Hold caution/advisory panel or HH-60A HH-60L instrument panel INDICATOR LTS BRT/DIM-TEST switch to TEST. Go to 2.**
- 2. Check that MAIN XMSN OIL PRESS capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **3**.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0089 00) or MFD/caution/advisory warning system (WP 0099 00). Go to **16**.
- 3. Install an insulated jumper wire between P409-3 and P409-4. Go to 4.**
- 4. Check that MAIN XMSN OIL PRESS capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **5**.
 - b. If capsule or legend does not go on, go to **7**.
- 5. Remove jumper wire. Go to 6.**
- 6. Replace low oil pressure sensor (WP 0649 00). Go to 16.**
- 7. Check configuration of helicopter.**
 - a. **EMEP** , go to **8**.
 - b. **W/O EMEP** , go to **10**.

MAIN XMSN OIL PRESS CAPSULE DOES NOT GO ON - Continued

8. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R **■** (WP 0929 00).
 - a. If pin filtered adapter is good, go to 9.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0929 00). Go to 16.
9. **EMEP** Install pin filtered adapter (WP 0929 00). Go to 10.
10. Check for 28 vdc between P117-16 or **HH-60A HH-60L** P1019R-78 or P1020R-78 **■** and ground.
 - a. If voltage is as specified, go to 11.
 - b. If voltage is not as specified, go to 13.
11. Remove jumper wire. Go to 12.
12. Troubleshoot caution/advisory warning system (WP 0089 00) or MFD/caution/advisory warning system (WP 0099 00). Go to 16.
13. Remove jumper wire. Go to 14.
14. Check for 28 vdc between P409-2 and ground.
 - a. If voltage is as specified, repair/replace wiring between P409-3 and P117-16 or **HH-60A HH-60L** P1019R-78 or P1020R-78 **■** (WP 1725 00). Go to 16.
 - b. If voltage is not as specified, go to 15.
15. Check for 28 vdc between MAIN XMSN circuit breaker terminal 1 and ground.
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P409-3 (WP 1725 00). Go to 16.
 - b. If voltage is not as specified, replace circuit breaker (WP 0828 00). Go to 16.
16. Procedure completed.

XMSN PRESS INDICATION IS TOO LOW (MAIN XMSN OIL PRESS CAPSULE GOES ON)**SYMPTOM**

XMSN PRESS indication is too low (MAIN XMSN OIL PRESS capsule goes on).

MALFUNCTION

XMSN PRESS indication is too low (MAIN XMSN OIL PRESS capsule goes on).

CORRECTIVE ACTION**NOTE**

A normally functioning transmission will occasionally experience a low pressure fluctuation when aircraft is in a nose up attitude. The following procedure should be implemented when aircraft is experiencing low pressure anomalies that cannot be attributed to nose up attitude.

1. Check main transmission oil for proper oil level using dipstick.
 - a. If oil level is good within full range, go to 2.

XMSN PRESS INDICATION IS TOO LOW (MAIN XMSN OIL PRESS CAPSULE GOES ON) - Continued

- b. If oil level is not good within full range, properly service main transmission (WP 1607 00). Monitor oil pressure during next flight (or ground run). If pressure anomalies continue, go to **2**. If oil pressure is normal, go to **8**.
- 2. Remove and inspect chip detector screen for aluminum debris (WP 0646 00).**
 - a. If debris is found, do not run helicopter, go to **3**.
 - b. If debris is not found, install chip detector screen (WP 0646 00). Go to **6**.
- 3. Drain transmission oil, removing one of the oil pumps (WP 0648 00). Go to 4.**
- 4. Inspect underside of carrier plate for cracks by walking the rotor system to rotate the carrier.**
 - a. If crack is found, replace transmission (WP 0630 00). Go to **7**.
 - b. If no crack is found, install oil pump (WP 0648 00). Go to **5**.
- 5. Replace oil filter (WP 0647 00). Go to 6.**
- 6. Make sure main transmission is properly serviced (WP 1607 00). Go to 7.**
- 7. If pressure is still not good, replace both pressure regulating valves (WP 0650 00). Go to 9.**
- 8. If pressure anomalies continue, contact AMCOM or Sikorsky Field Service Representative. Go to 9.**
- 9. Procedure completed.**

XMSN TEMP INDICATION IS TOO HIGH OR IS NOT STEADY (MAIN XMSN OIL TEMP CAPSULE GOES ON)**SYMPTOM**

XMSN TEMP indication is too high or is not steady (MAIN XMSN OIL TEMP capsule goes on).

MALFUNCTION

XMSN TEMP indication is too high or is not steady (MAIN XMSN OIL TEMP capsule goes on).

CORRECTIVE ACTION

- 1. Check main module and modules for any discoloration.**
 - a. If discoloration is present, replace affected module (WP 0653 00). Go to **5**.
 - b. If discoloration is not present, go to **2**.
- 2. Check main gear box for proper oil level.**
 - a. If oil level is good, go to **3**.
 - b. If oil level is not good, service main gear box (WP 1607 00). Go to **5**.
- 3. Troubleshoot XMSN TEMP scale on central display unit (WP 0089 00).**
 - a. If instrument display system is good, go to **4**.
- 4. Remove thermostatic valve from transmission oil cooler and check for proper operation.**
 - a. If operation is proper, replace transmission oil cooler (WP 0678 00). Go to **5**.
 - b. If operation is improper, replace thermostatic valve (WP 0676 00). Go to **5**.

XMSN TEMP INDICATION IS TOO HIGH OR IS NOT STEADY (MAIN XMSN OIL TEMP CAPSULE GOES ON) - Continued**5. Procedure completed.****INT XMSN OIL TEMP CAPSULE GOES ON****SYMPTOM**

INT XMSN OIL TEMP capsule goes on.

MALFUNCTION

INT XMSN OIL TEMP capsule goes on.

CORRECTIVE ACTION

- 1. Visually check intermediate gear box for signs of paint blistering or discoloration.**
 - a. If blistering or discoloration is present, do a serviceability check of intermediate gear box (WP 0622 00). Go to **7**.
 - b. If blistering or discoloration is not present, go to **2**.
- 2. Visually check intermediate gear box for proper oil level.**
 - a. If oil level is good, go to **3**.
 - b. If oil level is not good, do a serviceability check of intermediate gear box (WP 1608 00). Go to **7**.
- 3. Disconnect P613 from intermediate gear box. Go to 4.**
- 4. Check that INT XMSN OIL TEMP capsule or legend goes off.**
 - a. If capsule or legend goes off, go to **5**.
 - b. If capsule or legend does not go off, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **8**.
- 5. Replace intermediate gear box oil temperature sensor (WP 0665 00). Go to 6.**
- 6. Do a serviceability check of intermediate gear box (WP 0622 00). Go to 7.**
- 7. Procedure completed.**

TAIL XMSN OIL TEMP CAPSULE GOES ON**SYMPTOM**

TAIL XMSN OIL TEMP capsule goes on.

MALFUNCTION

TAIL XMSN OIL TEMP capsule goes on.

CORRECTIVE ACTION

- 1. Visually check tail rotor gear box for signs of paint blistering or discoloration.**
 - a. If blistering or discoloration is present, do a serviceability check of tail rotor gear box (WP 0628 00). Go to **7**.
 - b. If blistering or discoloration is not present, go to **2**.
- 2. Visually check tail rotor gear box for proper oil level.**
 - a. If oil level is good, go to **3**.

TAIL XMSN OIL TEMP CAPSULE GOES ON - Continued

- b. If oil level is not good, do a serviceability check of tail rotor gear box (WP 1604 00). Go to **7**.
- 3. Disconnect P614 from tail gear box. Go to 4.**
- 4. Check that TAIL XMSN OIL TEMP capsule or legend goes off.**
 - a. If capsule or legend goes off, go to **5**.
 - b. If capsule or legend is on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **7**.
- 5. Replace tail gear box chip detector (WP 0688 00). Go to 6.**
- 6. Do a serviceability check of intermediate gear box (WP 0622 00). Go to 7.**
- 7. Procedure completed.**

XMSN PRESS HAS NO INDICATION, NOT STEADY, OR TOO HIGH**SYMPTOM**

XMSN PRESS has no indication, not steady, or too high.

MALFUNCTION

XMSN PRESS has no indication, not steady, or too high.

CORRECTIVE ACTION

- 1. Check main module oil filter and screens between input and accessory modules for possible contamination.**
 - a. If contamination is present, do a serviceability check of affected modules or inputs (WP 0616 00). Go to **8**.
 - b. If contamination is not present, go to **2**.
- 2. Check main transmission oil for proper level using dipstick.**
 - a. If level is good, go to **3**.
 - b. If level is not good, properly service main transmission (WP 1607 00). Go to **8**.
 - c. If oil level is high, drain oil until oil is at proper level (WP 1607 00). Go to **8**.
- 3. Troubleshoot XMSN PRESS scale on central display unit (WP 0089 00).**
 - a. If instrument display system is good, go to **4**.
- 4. Check configuration of helicopter.**
 - a. **EMEP** , go to **5**.
 - b. **W/O EMEP** , go to **7**.
- 5. **EMEP** Remove and check pin filtered adapter P137 (WP 0925 00).**
 - a. If pin filtered adapter is good, go to **6**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **8**.
- 6. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 7.**
- 7. Check continuity between:**
 - **P137-T and P444-3**

XMSN PRESS HAS NO INDICATION, NOT STEADY, OR TOO HIGH - Continued

- **P137-W and P444-2**

- **P137-j and P444-1**

- a. If continuity is present, replace transmission oil pressure sensor (WP 0649 00). Go to **8**.
- b. If trouble remains, replace pressure regulating valves (WP 0650 00). Go to **8**.
- c. If trouble still remains, replace packings on both oil pumps (WP 0648 00). Go to **8**.
- d. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **8**.

8. Procedure completed.**XMSN EXPERIENCES SUSTAINED AND/OR SUDDEN PRESSURE DROP MORE THAN 10 PSI WITHOUT FLUCTUATIONS****SYMPTOM**

Pressure suddenly drops more than 10 psi without fluctuations.

MALFUNCTION

Pressure suddenly drops more than 10 psi without fluctuations.

CORRECTIVE ACTION**NOTE**

If situation permits, verify that pressure drop is not an instrumentation problem using a direct reading pressure gage. The gage should be calibrated and capable of accurately measuring pressures from zero to 100 psi. It shall be connected into the location of the pressure transmitter.

1. Inspect sump housing for oil leakage and/or cracks in the core wall (circular depression in the sump housing between pumps and filter bowl).

- a. If oil leakage and/or cracks in the core wall are visible, replace main module (WP 0630 00). Go to **2**.
- b. If oil leakage and/or cracks in the core wall are not visible, go to MAIN XMSN OIL PRESS Capsule Goes on, in this work package.

2. Procedure completed.**XMSN TEMP HAS NO INDICATION****SYMPTOM**

XMSN TEMP has no indication.

MALFUNCTION

XMSN TEMP has no indication.

CORRECTIVE ACTION**1. Make sure receptacle J412 is connected to main gear box.**

- a. If receptacle is connected, go to **2**.
- b. If receptacle is not connected, connect receptacle to main gear box. Go to **7**.

2. Troubleshoot XMSN TEMP scale on central display unit (WP 0089 00).

XMSN TEMP HAS NO INDICATION - Continued

- a. If instrument display system is good, go to **3**.
- 3. Check configuration of helicopter.**
 - a. **EMEP** , go to **4**.
 - b. **W/O EMEP** , go to **6**.
- 4. EMEP Remove and check pin filtered adapter P137 (WP 0925 00).**
 - a. If pin filtered adapter is good, go to **5**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **7**.
- 5. EMEP Install pin filtered adapter (WP 0925 00). Go to 6.**
- 6. Check continuity between these points:**
 - **P317-J and P412-C**
 - **P137-K and P412-D**
 - a. If continuity is present, replace transmission oil strainer (WP 0644 00). Go to **7**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.
- 7. Procedure completed.**

%RPM RTR INDICATION IS NOT STEADY**SYMPTOM**

%RPM RTR indication is not steady.

MALFUNCTION

%RPM RTR indication is not steady.

CORRECTIVE ACTION

- 1. Troubleshoot %RPM RTR scale on central display unit (WP 0089 00).**
 - a. If instrument display system is good, go to **2**.
- 2. Check configuration of helicopter.**
 - a. **EMEP** , go to **3**.
 - b. **W/O EMEP** , go to **5**.
- 3. EMEP Remove and check pin filtered adapters P137 and P138 (WP 0925 00).**
 - a. If pin filtered adapters are good, go to **4**.
 - b. If pin filtered adapters are not good, replace pin filtered adapter (WP 0925 00). Go to **6**.
- 4. EMEP Install pin filtered adapters (WP 0925 00). Go to 5.**
- 5. Check continuity between:**
 - **P443-1 and P137-B**
 - **P443-1 and P138-B**
 - **P443-2 and P137-A**
 - **P443-2 and P138-A**

%RPM RTR INDICATION IS NOT STEADY - Continued

- a. If continuity is present, replace Nr speed sensor (WP 0500 00). Go to **6**.
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **6**.

6. Procedure completed.**MAIN XMSN OIL TEMP CAPSULE DOES NOT GO ON****SYMPTOM**

MAIN XMSN OIL TEMP capsule does not go on.

MALFUNCTION

MAIN XMSN OIL TEMP capsule does not go on.

CORRECTIVE ACTION

- 1. Hold caution/advisory panel or HH-60A HH-60L instrument panel INDICATOR LTS BRT/DIM-TEST switch to TEST. Go to 2.**
- 2. Check that MAIN XMSN OIL TEMP capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **3**.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **9**.
- 3. Install an insulated jumper wire between P408-A and P408-C. Go to 4.**
- 4. Check that MAIN XMSN OIL TEMP capsule or legend goes on.**
 - a. If capsule or legend goes on, go to **5**.
 - b. If capsule or legend does not go on, go to **7**.
- 5. Remove jumper wire. Go to 6.**
- 6. Troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 9.**
- 7. Remove jumper wire. Go to 8.**
- 8. Check for 28 vdc between P408-A and ground.**
 - a. If voltage is as specified, repair/replace wiring between P408-C and P117-10 or HH-60A HH-60L P1019R-79 or P1020R-79 (WP 1747 00). Go to **9**.
 - b. If voltage is not as specified, repair/replace wiring between MAIN XMSN circuit breaker terminal 1 and P408-A (WP 1747 00). Go to **9**.
- 9. Procedure completed.**

INT XMSN OIL TEMP CAPSULE DOES NOT GO ON**SYMPTOM**

INT XMSN OIL TEMP capsule does not go on.

MALFUNCTION

INT XMSN OIL TEMP capsule does not go on.

CORRECTIVE ACTION

1. Hold caution/advisory panel or **HH-60A HH-60L** instrument panel INDICATOR LTS **■** BRT/DIM-TEST switch to TEST. Go to 2.
2. Check that INT XMSN OIL TEMP capsule or legend goes on.
 - a. If capsule or legend goes on, go to 3.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 18.
3. Install an insulated jumper wire between P613-B and P613-C. Go to 4.
4. Check that INT XMSN OIL TEMP capsule or legend goes on.
 - a. If capsule or legend goes on, go to 5.
 - b. If capsule or legend does not go on, go to 7.
5. Remove jumper wire. Go to 6.
6. Replace intermediate gear box chip/temperature sensor (WP 0665 00). Go to 18.
7. Remove jumper wire. Go to 8.
8. Check configuration of helicopter.
 - a. **UH-60L UH-60A** , go to 9.
 - b. **HH-60A HH-60L** , go to 16.
9. Check configuration of helicopter.
 - a. **EMEP** , go to 10.
 - b. **W/O EMEP** , go to 12.
10. **EMEP** Remove and check pin filtered adapter P118 (WP 0925 00).
 - a. If pin filtered adapter is good, go to 11.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 18.
11. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 12.
12. Check for 28 vdc between P118-f and ground.
 - a. If voltage is as specified, go to 13.
 - b. If voltage is not as specified, go to 17.
13. Check configuration of helicopter.
 - a. **EMEP** , go to 14.
 - b. **W/O EMEP** , go to 16.

INT XMSN OIL TEMP CAPSULE DOES NOT GO ON - Continued

14. **EMEP** Remove and check pin filtered adapter P117 (WP 0925 00).
 - a. If pin filtered adapter is good, go to 15.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 18.
15. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 16.
16. Check continuity between:
 - P613-B and ground
 - P613-C and P117-26
 - **HH-60A HH-60L** P613-C and P1019R-68 ■
 - **HH-60A HH-60L** P613-C and P1020R-68 ■
 - a. If continuity is present, replace caution/advisory panel (WP 0785 00) or MFD/caution/advisory warning system (WP 0799 00). Go to 18.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 18.
17. Check for 28 vdc between CHIP DET circuit breaker terminal 1 and ground.
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P118-f (WP 1747 00). Go to 18.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to 18.
18. Procedure completed.

TAIL XMSN OIL TEMP CAPSULE DOES NOT GO ON

SYMPTOM

TAIL XMSN OIL TEMP capsule does not go on.

MALFUNCTION

TAIL XMSN OIL TEMP capsule does not go on.

CORRECTIVE ACTION

1. Hold caution/advisory panel or **HH-60A HH-60L** instrument panel INDICATOR LTS ■ BRT/DIM-TEST switch to TEST. Go to 2.
2. Check that TAIL XMSN OIL TEMP capsule or legend goes on.
 - a. If capsule or legend goes on, go to 3.
 - b. If capsule or legend does not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 12.
3. Install an insulated jumper wire between P614-B and P614-C. Go to 4.
4. Check that TAIL XMSN OIL TEMP capsule or legend goes on.
 - a. If capsule or legend goes on, go to 5.
 - b. If capsule or legend does not go on, go to 7.
5. Remove jumper wire. Go to 6.
6. Replace tail gear box chip/temperature sensor (WP 0688 00). Go to 12.

TAIL XMSN OIL TEMP CAPSULE DOES NOT GO ON - Continued

7. Remove jumper wire. Go to 8.
8. Check configuration of helicopter.
 - a. **EMEP** , go to 9.
 - b. **W/O EMEP** , go to 11.
9. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R **■** (WP 0925 00).
 - a. If pin filtered adapter is good, go to 10.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 12.
10. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 11.
11. Check continuity between:
 - P614-B and ground
 - P614-C and P117-42
 - **HH-60A HH-60L** P614-C and P1019R-69 **■**
 - **HH-60A HH-60L** P614-C and P1020R-69 **■**
 - a. If continuity is present, replace caution/advisory panel (WP 0785 00) or MFD/caution/advisory warning system (WP 0799 00). Go to 12.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to 12.
12. Procedure completed.

CHIP ACCESS MDL-RH CAPSULE GOES ON

SYMPTOM

CHIP ACCESS MDL-RH capsule goes on.

MALFUNCTION

CHIP ACCESS MDL-RH capsule goes on.

CORRECTIVE ACTION

NOTE

If CHIP DET circuit breaker has popped, go to CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON, in this work package.

1. Remove chip detector from right accessory module (WP 0643 00). Go to 2.
2. Check chip detector.
 - a. If small amounts of fuzz are found, go to CHIP DETECTOR DOES NOT BURN OFF FUZZ, in this work package.
 - b. If chips are found, do a particle inspection of right accessory module (WP 0617 00). Go to 9.
 - c. If no chips or fuzz are found, go to 3.
3. Disconnect P117 from caution/advisory panel or **HH-60A HH-60L** P1019R or P1020R from MFD **■** . Go to 4.

CHIP ACCESS MDL-RH CAPSULE GOES ON - Continued**4. Check that CHIP ACCESS MDL-RH capsule or legend goes off.**

- a. If capsule or legend is off, go to **5**.
- b. If capsule or legend is on, replace caution/advisory panel (WP 0785 00) or MFD/caution/advisory warning system (WP 0799 00). Go to **9**.

5. Check configuration of helicopter.

- a. **EMEP** , go to **6**.
- b. **W/O EMEP** , go to **8**.

6. EMEP Remove and check pin filtered adapter P117 or HH-60A HH-60L P1019R or P1020R (WP 0925 00).

- a. If pin filtered adapter is good, go to **7**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **9**.

7. EMEP Install pin filtered adapter (WP 0925 00). Go to 8.**8. Repair/replace wiring between P117-62 or HH-60A HH-60L P1019R-5 or P1020R-5 and right accessory chip detector (WP 1747 00). Go to 9.****9. Procedure completed.****CHIP ACCESS MDL-LH CAPSULE GOES ON****SYMPTOM**

CHIP ACCESS MDL-LH capsule goes on.

MALFUNCTION

CHIP ACCESS MDL-LH capsule goes on.

CORRECTIVE ACTION**NOTE**

If CHIP DET circuit breaker has popped, go to CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON, in this work package.

1. Remove chip detector from left accessory module (WP 0643 00). Go to 2.**2. Check chip detector.**

- a. If small amounts of fuzz are found, go to CHIP DETECTOR DOES NOT BURN OFF FUZZ, in this work package.
- b. If chips are found, do a particle inspection of left accessory module (WP 0617 00). Go to **9**.
- c. If no fuzz or chips are found, go to **3**.

3. Disconnect P117 from caution/advisory panel or HH-60A HH-60L P1019R or P1020R from MFD . Go to 4.**4. Check that CHIP ACCESS MDL-LH capsule or legend goes off.**

- a. If capsule or legend is off, go to **5**.
- b. If capsule or legend is on, replace caution/advisory panel (WP 0785 00) or MFD/caution/advisory warning system (WP 0799 00). Go to **9**.

CHIP ACCESS MDL-LH CAPSULE GOES ON - Continued

5. Check configuration of helicopter.
 - a. **EMEP** , go to 6.
 - b. **W/O EMEP** , go to 8.
6. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R (WP 0925 00).
 - a. If pin filtered adapter is good, go to 7.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 9.
7. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 8.
8. Repair/replace wiring between P117-14 or **HH-60A HH-60L** P1019R-2 or P1020R-2 and left accessory module chip detector (WP 1747 00). Go to 9.
9. Procedure completed.

CHIP INPUT MDL-LH CAPSULE GOES ON**SYMPTOM**

CHIP INPUT MDL-LH capsule goes on.

MALFUNCTION

CHIP INPUT MDL-LH capsule goes on.

CORRECTIVE ACTION

1. If CHIP DET circuit breaker has popped, go to CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON, in this work package.
2. Remove left input module chip detector from main sump (WP 0645 00). Go to 2.
3. Check chip detector.
 - a. If small amounts of fuzz are found, go to CHIP DETECTOR DOES NOT BURN OFF FUZZ, in this work package.
 - b. If chips are found, do a particle inspection of left input module (WP 0617 00). Go to 10.
 - c. If no chips or fuzz are found, go to 3.
4. Visually check oil filter screen for chip(s).
 - a. If chips are present, do a serviceability check of left input module (WP 0616 00). Go to 10.
 - b. If chips are not present, go to 4.
5. Disconnect P117 from caution/advisory panel or **HH-60A HH-60L** P1019R or P1020R from MFD . Go to 5.
6. Check that CHIP INPUT MDL-LH capsule or legend goes off.
 - a. If capsule or legend is off, go to 6.
 - b. If capsule or legend is on, replace caution/advisory panel (WP 0785 00) or MFD/caution/advisory warning system (WP 0799 00). Go to 10.
7. Check configuration of helicopter.
 - a. **EMEP** , go to 7.

CHIP INPUT MDL-LH CAPSULE GOES ON - Continued

- b. **W/O EMEP** , go to **9**.
- 8. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R (WP 0925 00).
 - a. If pin filtered adapter is good, go to **8**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **10**.
- 9. **EMEP** Install pin filtered adapter (WP 0925 00). Go to **9**.
- 10. Repair/replace wiring between P117-13 or **HH-60A HH-60L** P1019R-1 or P1020R-1 and left input module chip detector (WP 1747 00). Go to **10**.
- 11. Procedure completed.

CHIP INPUT MDL-RH CAPSULE GOES ON**SYMPTOM**

CHIP INPUT MDL-RH capsule goes on.

MALFUNCTION

CHIP INPUT MDL-RH capsule goes on.

CORRECTIVE ACTION

- 1. If **CHIP DET** circuit breaker has popped, go to **CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON**, in this work package.
- 2. Remove right input module chip detector from main sump (WP 0645 00). Go to **2**.
- 3. Check chip detector.
 - a. If small amounts of fuzz are found, go to **CHIP DETECTOR DOES NOT BURN OFF FUZZ**, in this work package.
 - b. If chips are found, do a particle inspection of right input module (WP 0617 00). Go to **10**.
 - c. If no chips or fuzz are found, go to **3**.
- 4. Visually check oil filter screen for chip(s).
 - a. If chips are present, do a serviceability check of right input module (WP 0616 00). Go to **10**.
 - b. If chips are not present, go to **4**.
- 5. Disconnect P117 from caution/advisory panel or **HH-60A HH-60L** P1019R or P1020R from MFD . Go to **5**.
- 6. Check that **CHIP INPUT MDL-RH** capsule or legend goes off.
 - a. If capsule or legend is off, go to **6**.
 - b. If capsule or legend is on, replace caution/advisory panel (WP 0785 00) or MFD/caution/advisory warning system (WP 0799 00). Go to **10**.
- 7. Check configuration of helicopter.
 - a. **EMEP** , go to **7**.
 - b. **W/O EMEP** , go to **9**.
- 8. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R (WP 0925 00).

CHIP INPUT MDL-RH CAPSULE GOES ON - Continued

- a. If pin filtered adapter is good, go to **8**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **10**.
9. **EMEP** Install pin filtered adapter (WP 0925 00). Go to **9**.
10. Repair/replace wiring between P117-61 or **HH-60A HH-60L** P1019R-4 or P1020R-4 and right input module chip detector (WP 1747 00). Go to **10**.
11. Procedure completed.

CHIP MAIN MDL SUMP CAPSULE IS ON**SYMPTOM**

CHIP MAIN MDL SUMP capsule is on.

MALFUNCTION

CHIP MAIN MDL SUMP capsule is on.

CORRECTIVE ACTION

1. If **CHIP DET** circuit breaker has popped, go to **CHIP DET CIRCUIT BREAKER POPS WHEN A CHIP DETECTOR LIGHT GOES ON**, in this work package.
2. Remove chip detector from main module sump (WP 0646 00). Go to **2**.
3. Check chip detector.
 - a. If small amounts of fuzz are found, go to **CHIP DETECTOR DOES NOT BURN OFF FUZZ**, in this work package.
 - b. If chips are found, do a particle inspection of main module (WP 0617 00). Go to **9**.
 - c. If no chips or fuzz are found, go to **3**.
4. Disconnect P117 from caution/advisory panel or **HH-60A HH-60L** P1019R or P1020R from MFD . Go to **4**.
5. Check that **CHIP MAIN MDL SUMP** capsule or legend goes off.
 - a. If capsule or legend is off, go to **5**.
 - b. If capsule or legend is on, replace caution/advisory panel (WP 0785 00) or MFD/caution/advisory warning system (WP 0799 00). Go to **9**.
6. Check configuration of helicopter.
 - a. **EMEP** , go to **6**.
 - b. **W/O EMEP** , go to **8**.
7. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R (WP 0925 00).
 - a. If pin filtered adapter is good, go to **7**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **9**.
8. **EMEP** Install pin filtered adapter (WP 0925 00). Go to **8**.
9. Repair/replace wiring between P117-30 and right accessory chip detector (WP 1747 00). Go to **9**.
10. Procedure completed.

CHIP DETECTOR DOES NOT BURN OFF FUZZ**SYMPTOM**

Chip detector does not burn off fuzz.

MALFUNCTION

Chip detector does not burn off fuzz.

CORRECTIVE ACTION

1. **Check helicopter configuration.**
 - a. **W/O MWO 50-26** , go to 2.
 - b. **MWO 50-26** , go to 11.
2. **W/O MWO 50-26 Is right accessory module chip detector good?**
 - a. If chip detector is good, go to 3.
 - b. If chip detector is not good, go to 10.
3. **W/O MWO 50-26 Is left accessory module chip detector good?**
 - a. If chip detector is good, go to 4.
 - b. If chip detector is not good, go to 9.
4. **W/O MWO 50-26 Is left input module chip detector good?**
 - a. If chip detector is good, go to 5.
 - b. If chip detector is not good, go to 8.
5. **W/O MWO 50-26 Is right input module chip detector good?**
 - a. If chip detector is good, go to 6.
 - b. If chip detector is not good, go to 7.
6. **W/O MWO 50-26 Check for minimum of 18 vdc between P125-H and ground.**
 - a. If voltage is as specified, replace main module chip detector (WP 0646 00). Go to 20.
 - b. If voltage is not as specified, replace chip detector resistor unit (WP 0798 00). Go to 20.
7. **W/O MWO 50-26 Check for minimum of 18 vdc between P125-G and ground.**
 - a. If voltage is as specified, replace right input module chip detector (WP 0645 00). Go to 20.
 - b. If voltage is not as specified, replace chip detector resistor unit (WP 0798 00). Go to 20.
8. **W/O MWO 50-26 Check for minimum of 18 vdc between P125-F and ground.**
 - a. If voltage is as specified, replace left input module chip detector (WP 0645 00). Go to 20.
 - b. If voltage is not as specified, replace chip detector resistor unit (WP 0798 00). Go to 20.
9. **W/O MWO 50-26 Check for minimum of 18 vdc between P125-D and ground.**
 - a. If voltage is as specified, replace left accessory module chip detector (WP 0643 00). Go to 20.
 - b. If voltage is not as specified, replace chip detector resistor unit (WP 0798 00). Go to 20.
10. **W/O MWO 50-26 Check for minimum of 18 vdc between P125-E and ground.**

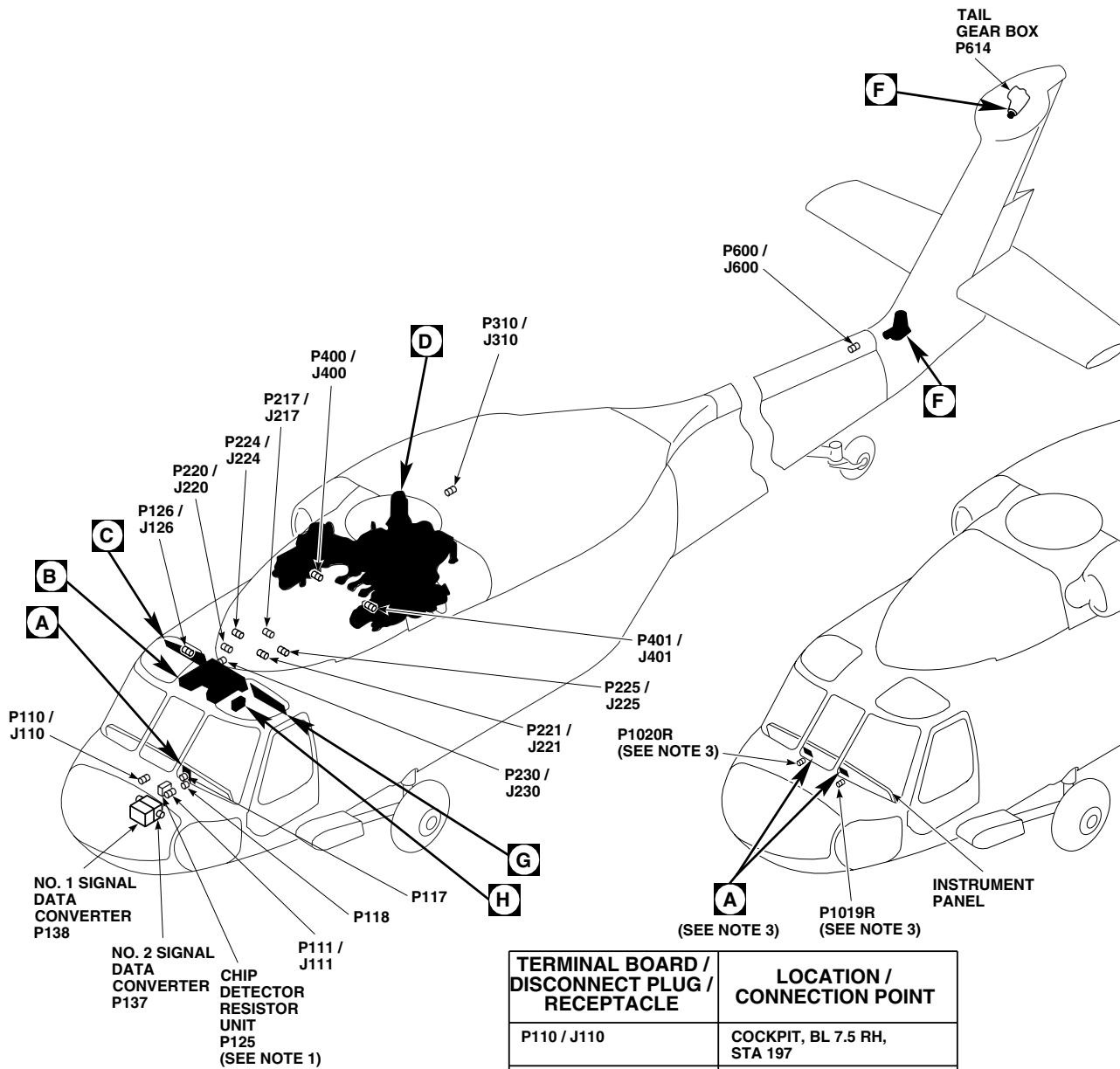
CHIP DETECTOR DOES NOT BURN OFF FUZZ - Continued

- a. If voltage is as specified, replace right accessory module chip detector (WP 0643 00). Go to **20**.
- b. If voltage is not as specified, replace chip detector resistor unit (WP 0798 00). Go to **20**.
- 11. **MWO 50-26 Is right accessory module chip detector good?**
 - a. If chip detector is good, go to **12**.
 - b. If chip detector is not good, go to **19**.
- 12. **MWO 50-26 Is left accessory module chip detector good?**
 - a. If chip detector is good, go to **13**.
 - b. If chip detector is not good, go to **18**.
- 13. **MWO 50-26 Is left input module chip detector good?**
 - a. If chip detector is good, go to **14**.
 - b. If chip detector is not good, go to **17**.
- 14. **MWO 50-26 Is right input module chip detector good?**
 - a. If chip detector is good, go to **15**.
 - b. If chip detector is not good, go to **16**.
- 15. **MWO 50-26 Check for 28 vdc between P412-A and ground.**
 - a. If voltage is as specified, replace main module chip detector (WP 0646 00). Go to **20**.
 - b. If voltage is not as specified, replace caution/advisory panel (WP 0785 00) or MFD/caution/advisory warning system (WP 0799 00). Go to **20**.
- 16. **MWO 50-26 Check for 28 vdc between P404-A and ground.**
 - a. If voltage is as specified, replace right input module chip detector (WP 0645 00). Go to **20**.
 - b. If voltage is not as specified, replace caution/advisory panel (WP 0785 00) or MFD/caution/advisory warning system (WP 0799 00). Go to **20**.
- 17. **MWO 50-26 Check for 28 vdc between P403-A and ground.**
 - a. If voltage is as specified, replace left input module chip detector (WP 0645 00). Go to **20**.
 - b. If voltage is not as specified, replace caution/advisory panel (WP 0785 00) or MFD/caution/advisory warning system (WP 0799 00). Go to **20**.
- 18. **MWO 50-26 Check for 28 vdc between P406-A and ground.**
 - a. If voltage is as specified, replace left accessory module chip detector (WP 0643 00). Go to **20**.
 - b. If voltage is not as specified, replace caution/advisory panel (WP 0785 00) or MFD/caution/advisory warning system (WP 0799 00). Go to **20**.
- 19. **MWO 50-26 Check for 28 vdc between P405-A and ground.**
 - a. If voltage is as specified, replace right accessory module chip detector (WP 0643 00). Go to **20**.
 - b. If voltage is not as specified, replace caution/advisory panel (WP 0785 00) or MFD/caution/advisory warning system (WP 0799 00). Go to **20**.

CHIP DETECTOR DOES NOT BURN OFF FUZZ - Continued

20. Procedure completed.

LOCATION



TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P110 / J110	COCKPIT, BL 7.5 RH, STA 197
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P117 / J117	CAUTION / ADVISORY PANEL (SEE NOTE 2)
P118 / J118	CAUTION / ADVISORY PANEL (SEE NOTE 2)
P125	CHIP DETECTOR RESISTOR UNIT (SEE NOTE 1)
P126 / J126	COCKPIT CEILING BL 28 RH, STA 243
P137	NO. 2 SIGNAL DATA CONVERTER
P138	NO. 1 SIGNAL DATA CONVERTER

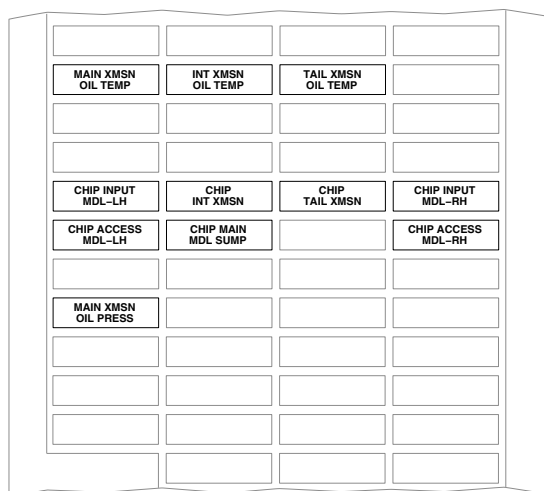
NOTES

1. **W/O MWO 50-26**
2. **UH60A** **UH60L** **EH60A**
3. **HH60A** **HH60L**

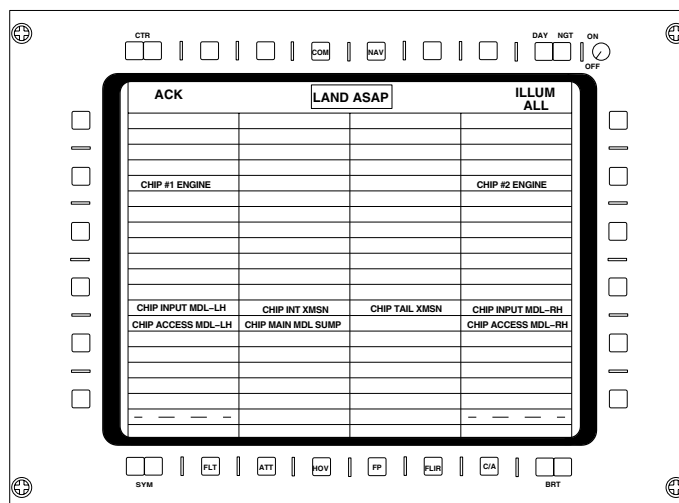
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Figure 1. Transmission Chip Detector, Instruments, and Oil Warning Systems Location Diagram. (Sheet 1 of 5)

LOCATION - Continued



CAUTION/ADVISORY PANEL
(SEE NOTE 2)



**MULTIFUNCTION DISPLAY'S
CAUTION/ADVISORY GRID**
(SEE NOTE 3)

A

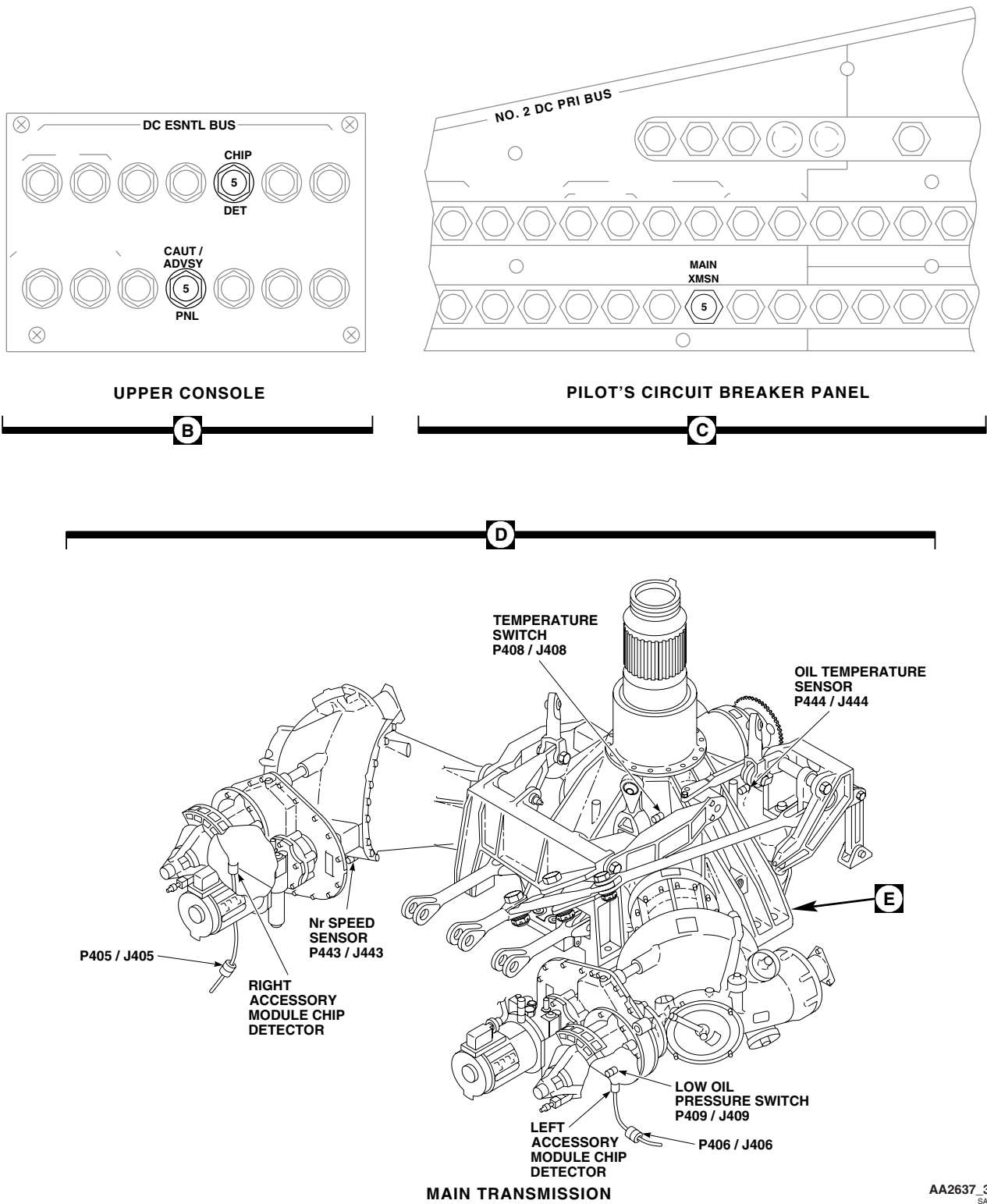
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P217 / J217	CABIN CEILING, BL 0, STA 284
P220 / J220	MAIN ROTOR PYLON DECK BL 3 RH, STA 243
P221 / J221	JUNCTION BOX MAIN ROTOR PYLON DECK BL 3 LH, STA 245.5
P224 / J224	JUNCTION BOX MAIN ROTOR PYLON DECK
P225 / J225	JUNCTION BOX MAIN ROTOR PYLON DECK BL 3 LH, STA 244
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P310 / J310	CABIN CEILING, BL 17 RH, STA 387
P400 / J400	MAIN ROTOR PYLON DECK BL 24 RH, STA 318
P401 / J401	MAIN ROTOR PYLON DECK BL 24 LH, STA 318
P403 / J403	CHIP DETECTOR LEFT INPUT MODULE
P404 / J404	CHIP DETECTOR RIGHT INPUT MODULE
P405 / J405	CHIP DETECTOR RIGHT ACCESSORY MODULE
P406 / J406	CHIP DETECTOR LEFT ACCESSORY MODULE

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P408 / J408	MAIN TRANSMISSION TEMPERATURE SWITCH
P409 / J409	MAIN TRANSMISSION LOW OIL PRESSURE SWITCH
P412 / J412	MAIN TRANSMISSION SUMP TEMPERATURE SENSOR
P443 / J443	Nr SPEED SENSOR
P444 / J444	MAIN TRANSMISSION OIL PRESSURE SENSOR
P600 / J600	TAIL CONE, BL 0, STA 600
P613	CHIP DETECTOR INTERMEDIATE GEAR BOX
P614	CHIP DETECTOR TAIL GEAR BOX
P802	NO. 2 ENGINE INTERFACE DISCONNECT (SEE NOTE 3)
P803	NO. 1 ENGINE INTERFACE DISCONNECT (SEE NOTE 3)
P824 / J824	FIREWALL DISCONNECT BL 14RH, STA 338.25 (SEE NOTE 3)
P825 / J825	FIREWALL DISCONNECT BL 14LH, STA 358.25 (SEE NOTE 3)
P1019R	COPILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 3)
P1020R	PILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 3)

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Figure 1. Transmission Chip Detector, Instruments, and Oil Warning Systems Location Diagram. (Sheet 2 of 5)

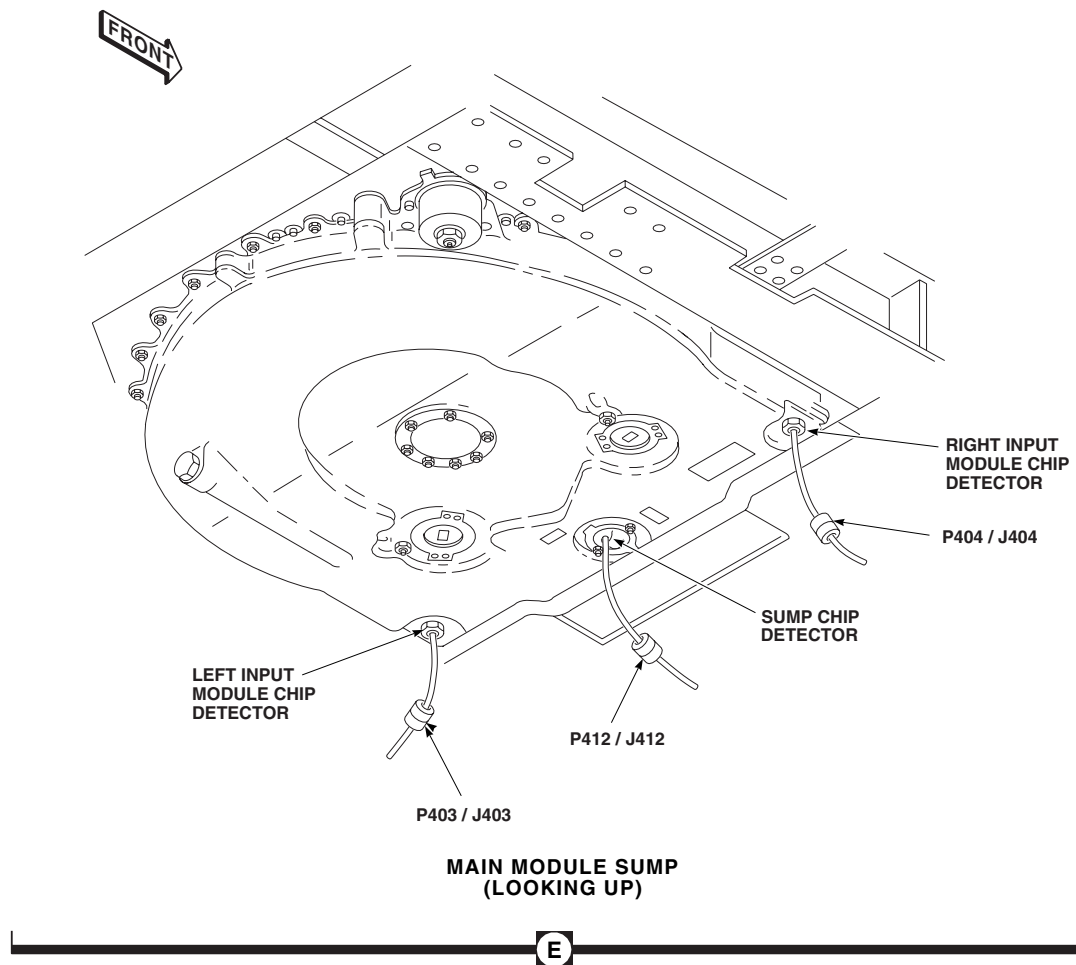
LOCATION - Continued



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Figure 1. Transmission Chip Detector, Instruments, and Oil Warning Systems Location Diagram. (Sheet 3 of 5)

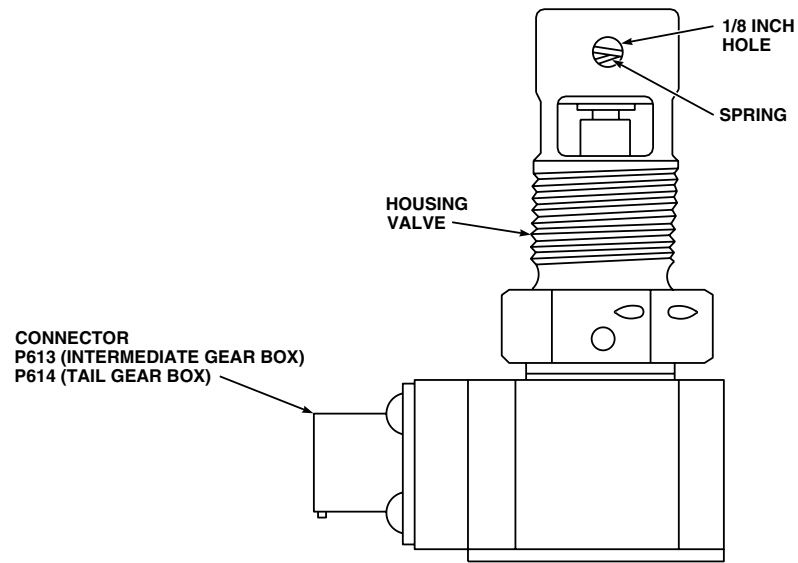
LOCATION - Continued



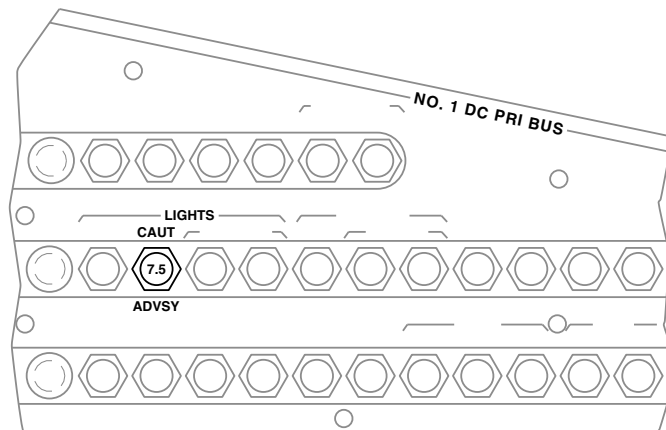
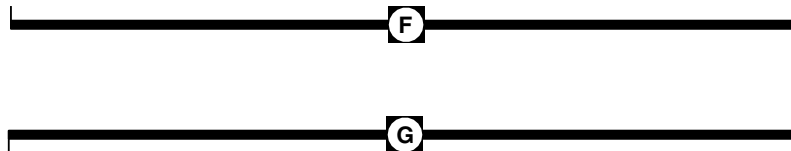
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Figure 1. Transmission Chip Detector, Instruments, and Oil Warning Systems Location Diagram. (Sheet 4 of 5)

LOCATION - Continued



CHIP DETECTOR/TEMPERATURE SENSOR

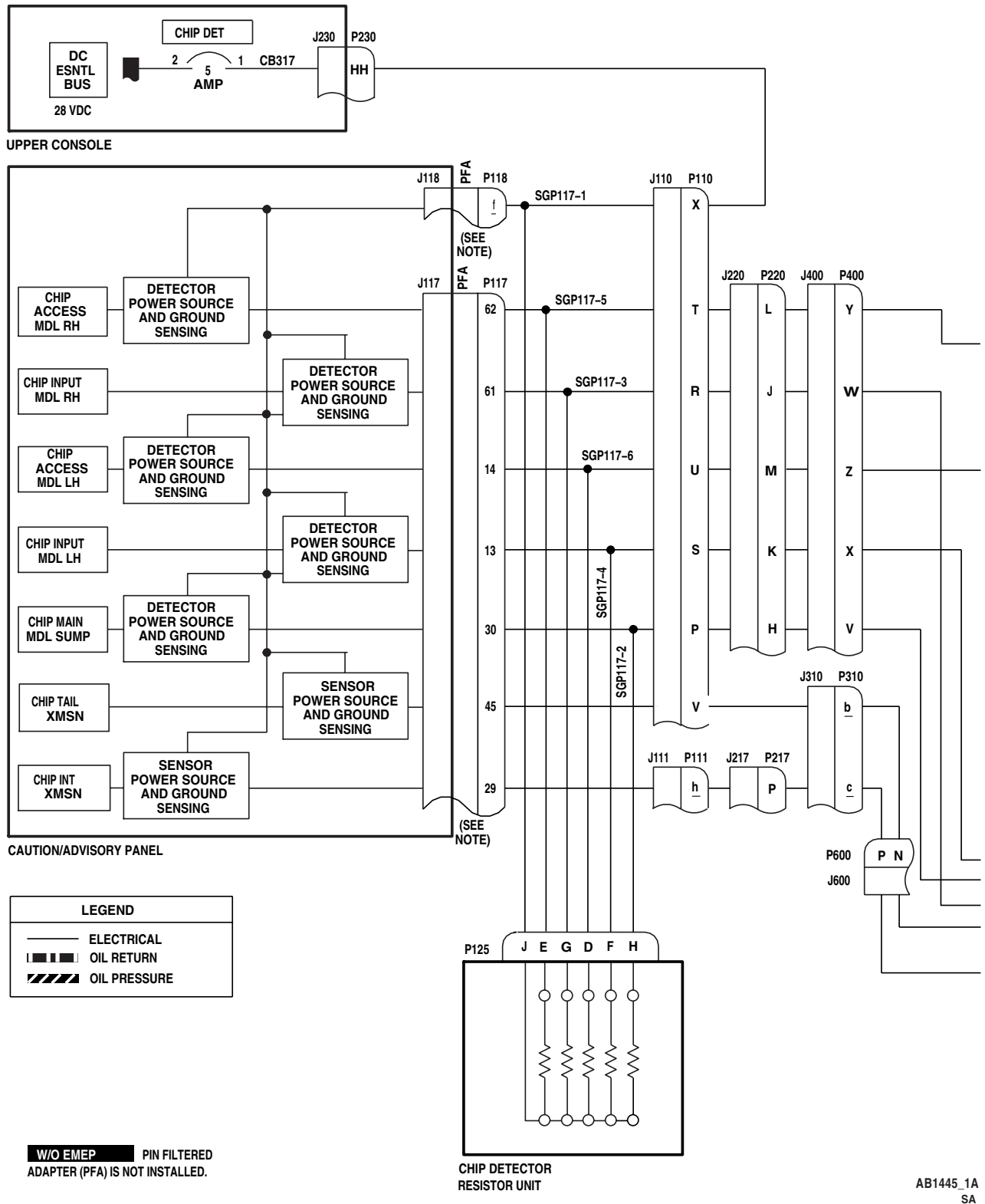


COPLOT'S CIRCUIT BREAKER PANEL

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Figure 1. Transmission Chip Detector, Instruments, and Oil Warning Systems Location Diagram. (Sheet 5 of 5)

SCHEMATICS

Figure 2. Transmission Chip Detector System Schematic Diagram **W/O MWO 50-26** . (Sheet 1 of 2)

SCHEMATICS - Continued

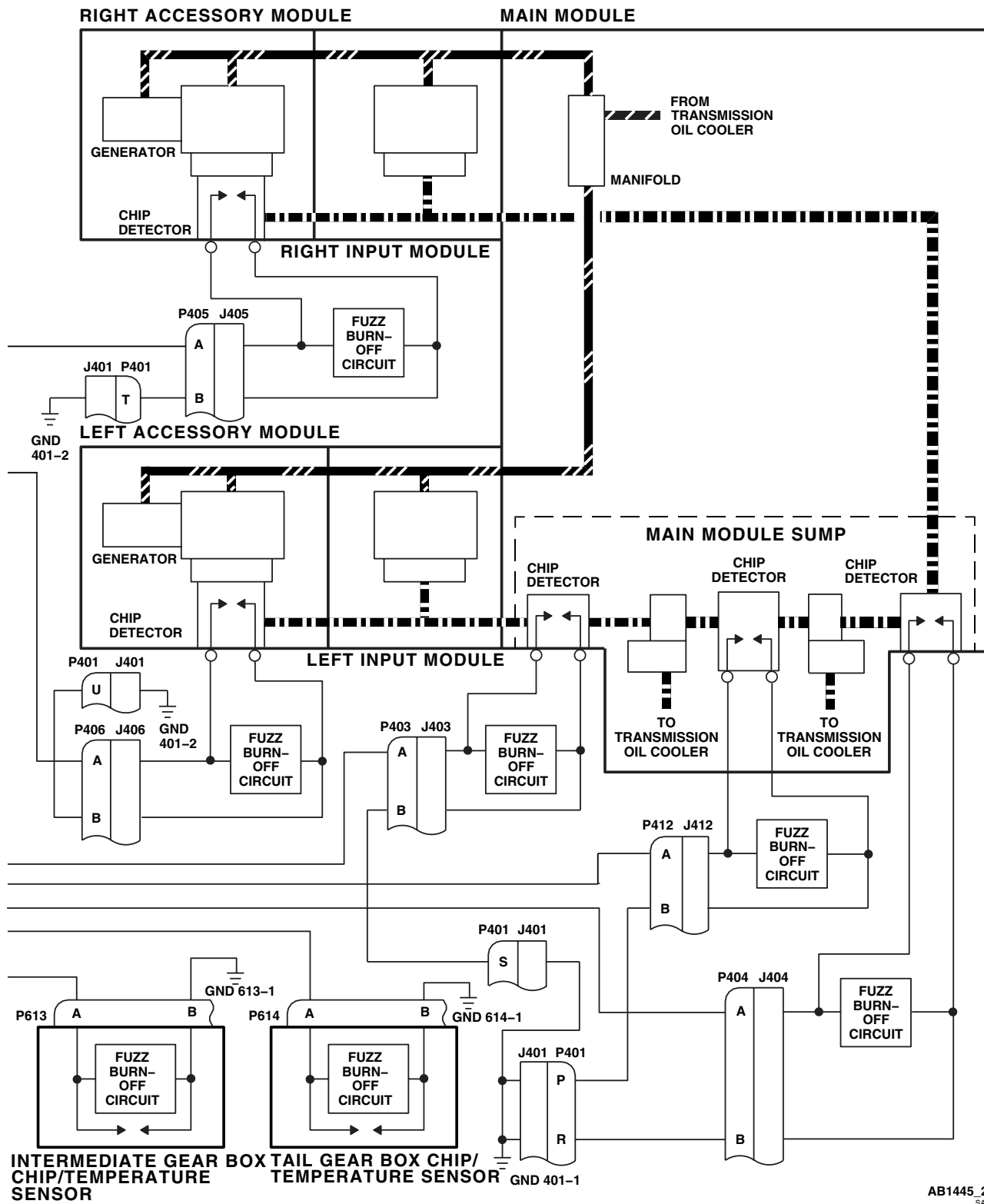


Figure 2. Transmission Chip Detector System Schematic Diagram W/O MWO 50-26 . (Sheet 2 of 2)

SCHEMATICS - Continued

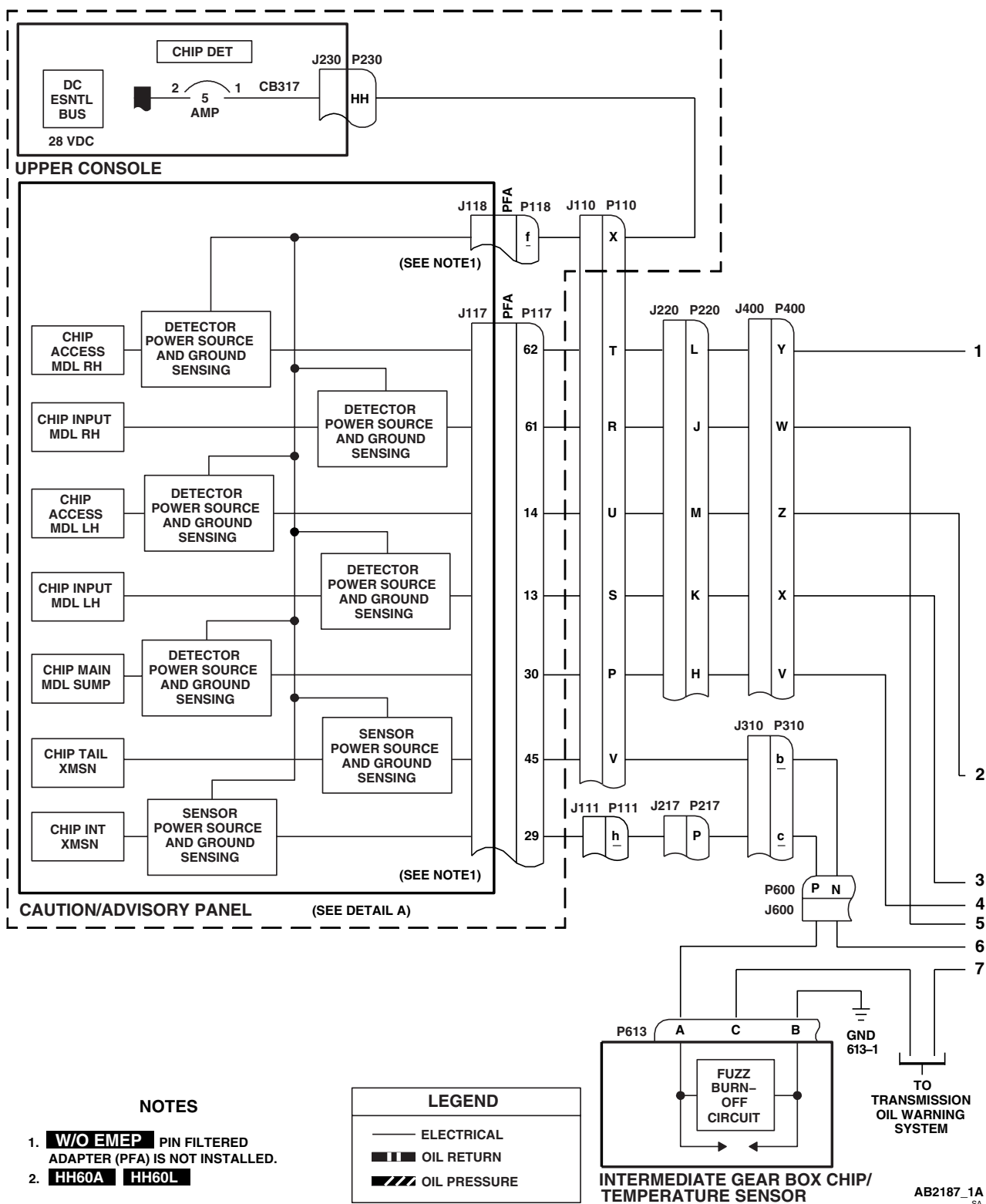


Figure 3. Transmission Chip Detector System Schematic Diagram

MWO 50-26 UH-60A 83-23895-SUBQ UH-60L EH-60A HH-60A HH-60L . (Sheet 1 of 3)

SCHEMATICS - Continued

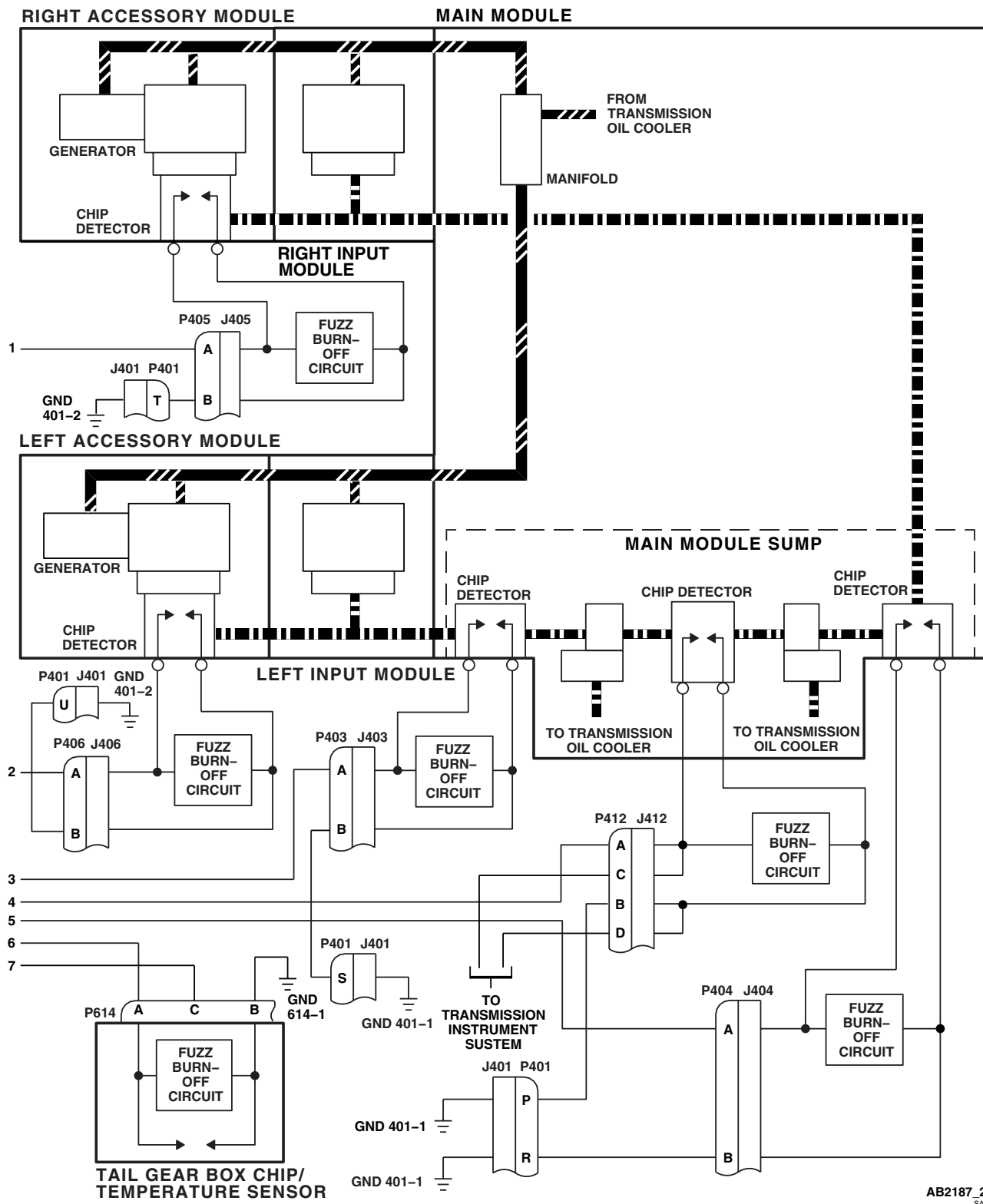


Figure 3. Transmission Chip Detector System Schematic Diagram

MWO 50-26 UH-60A 83-23895-SUBQ UH-60L EH-60A HH-60A HH-60L . (Sheet 2 of 3)

SCHEMATICS - Continued

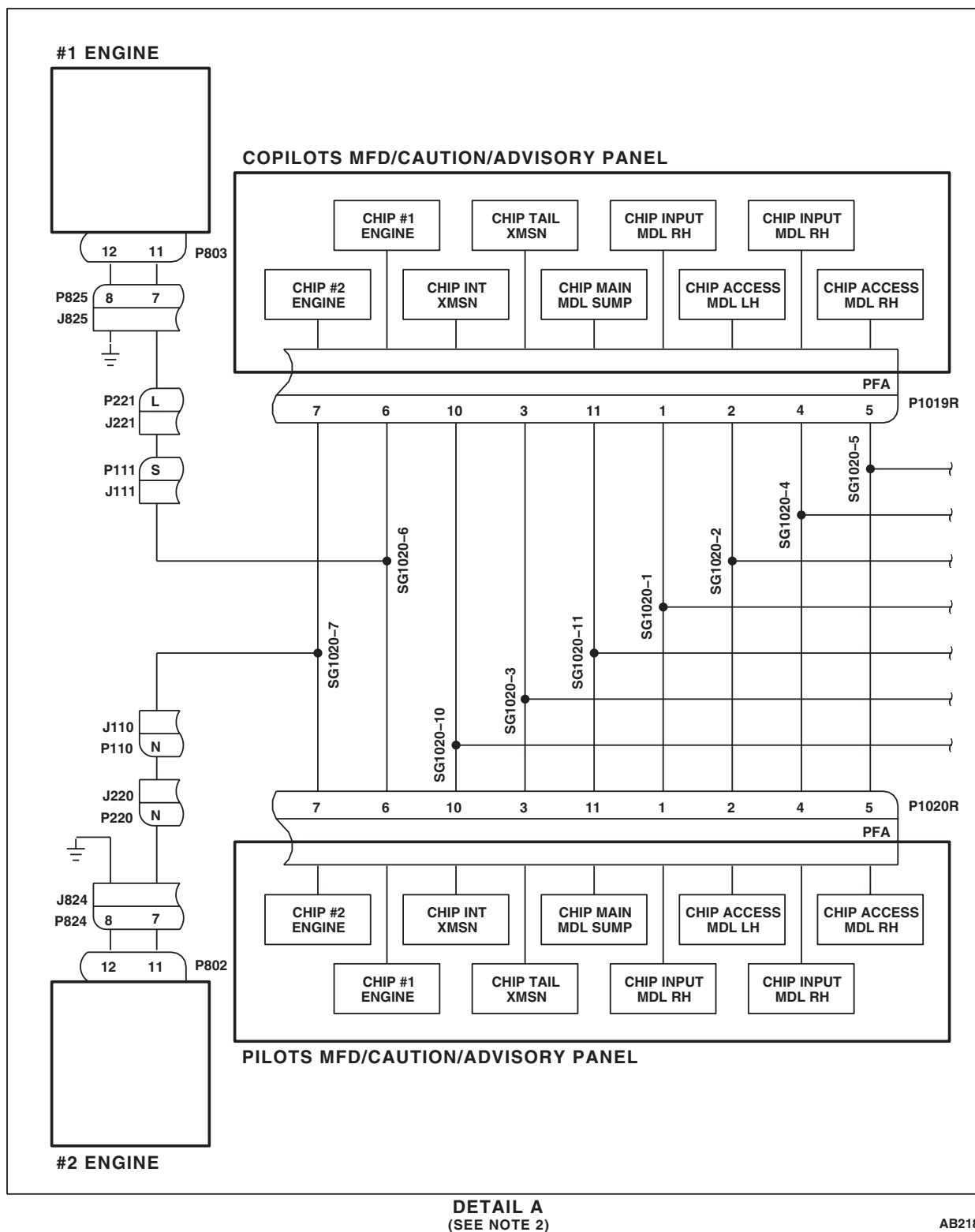


Figure 3. Transmission Chip Detector System Schematic Diagram

MWO 50-26 UH-60A 83-23895-SUBQ UH-60L EH-60A HH-60A HH-60L . (Sheet 3 of 3)

SCHEMATICS - Continued

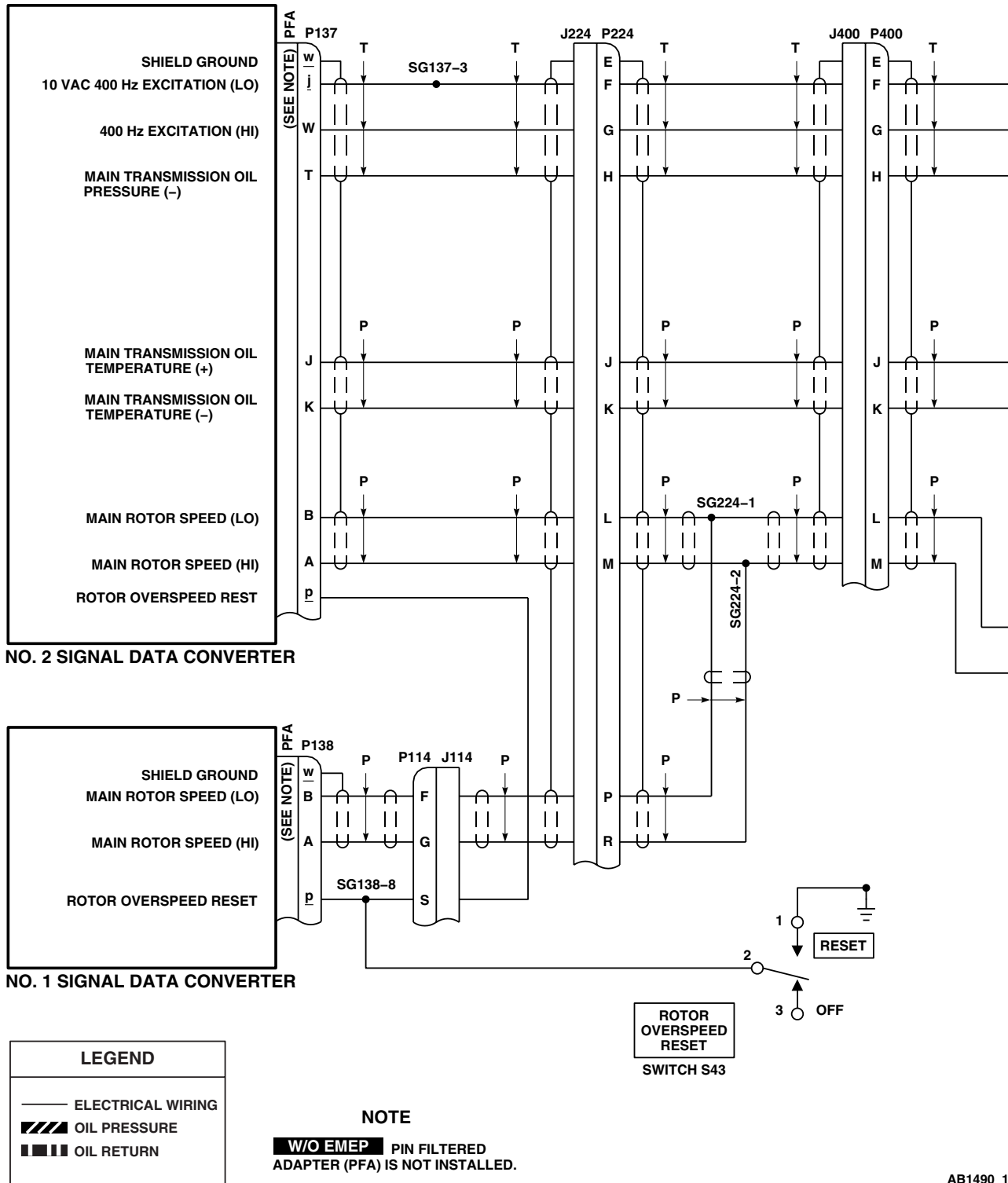


Figure 4. Transmission Instrument System Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued

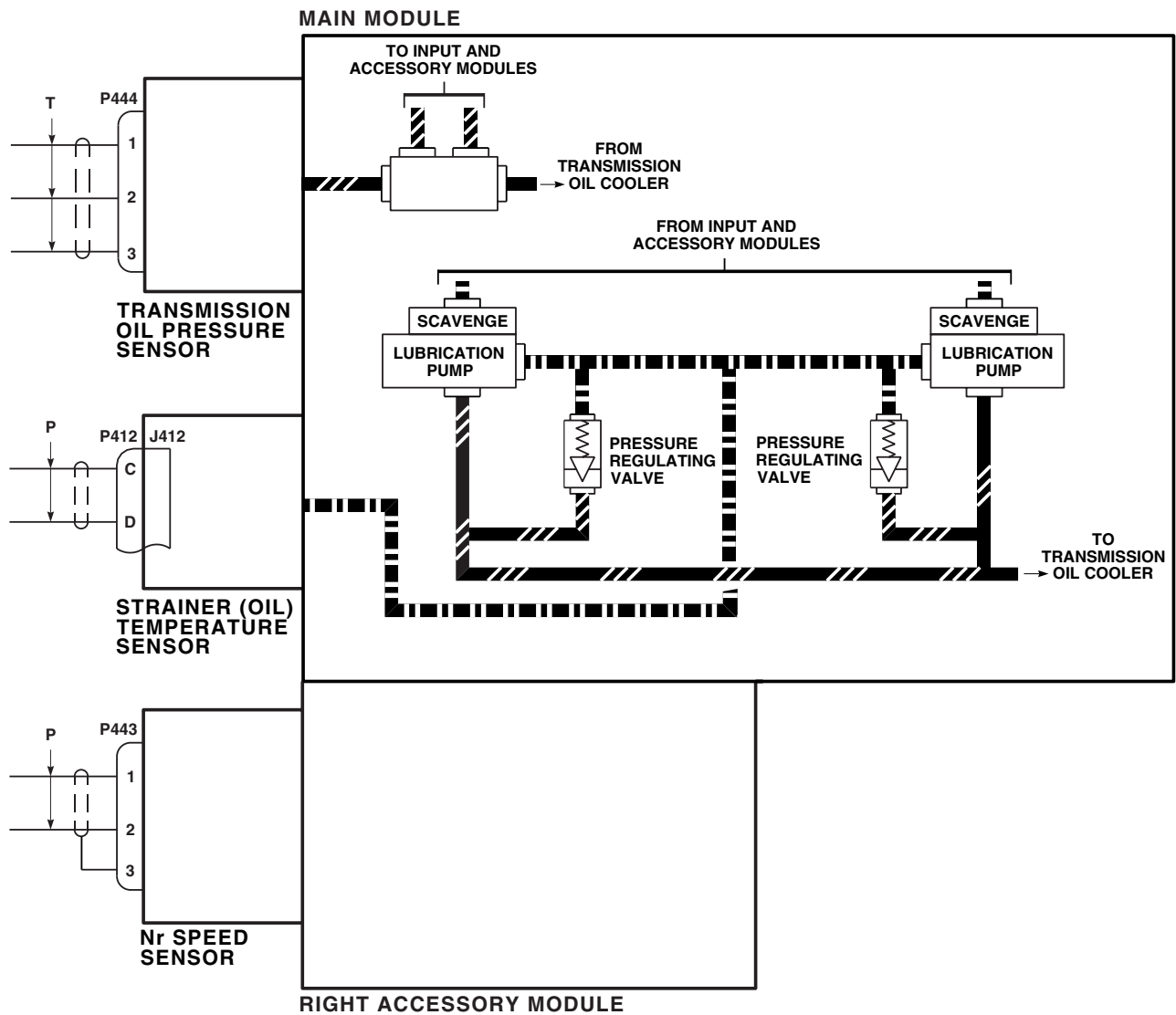
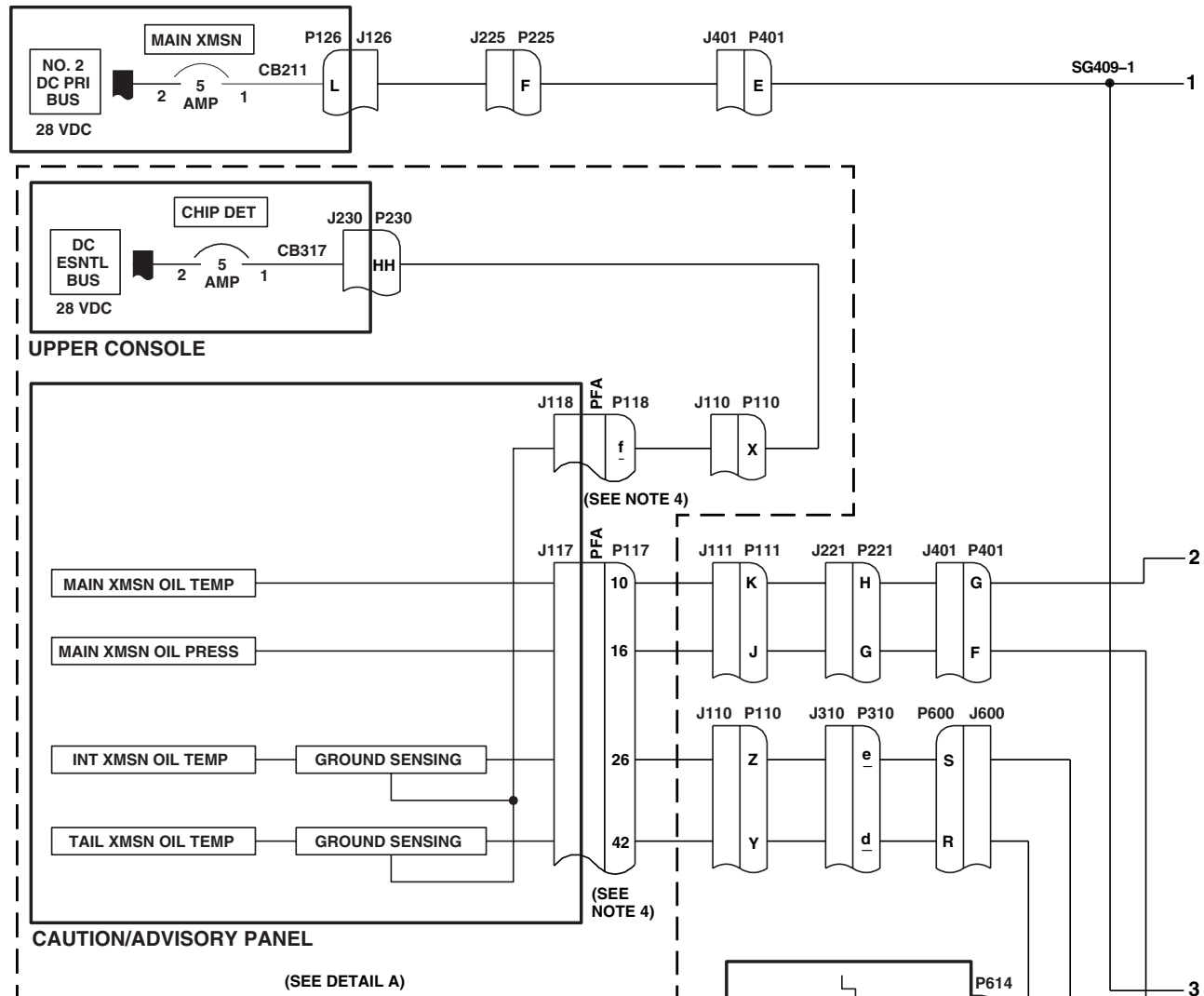
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Figure 4. Transmission Instrument System Schematic Diagram. (Sheet 2 of 2)

SCHEMATICS - Continued

PILOT'S CIRCUIT BREAKER PANEL



NOTES

1. SWITCH CLOSSES WHEN TEMPERATURE IS ABOVE 134 °C TO 146 °C.
2. SWITCH CLOSSES WHEN OIL PRESSURE IS BELOW 12 TO 16 PSI.
3. SWITCH CLOSSES WHEN TEMPERATURE IS ABOVE 112 °C TO 121 °C.
4. **W/O EMEP** PIN FILTERED ADAPTER (PFA) IS NOT INSTALLED.
5. **HH60A** **HH60L**

LEGEND

- ELECTRICAL WIRING
- /// OIL PRESSURE
- ||| OIL RETURN

TAIL GEAR BOX CHIP TEMPERATURE DETECTOR

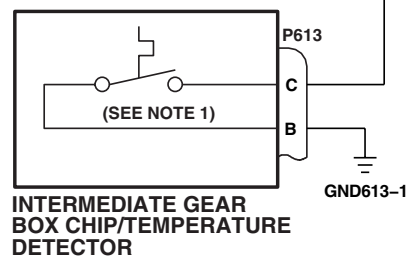


Figure 5. Transmission Oil Warning System Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued

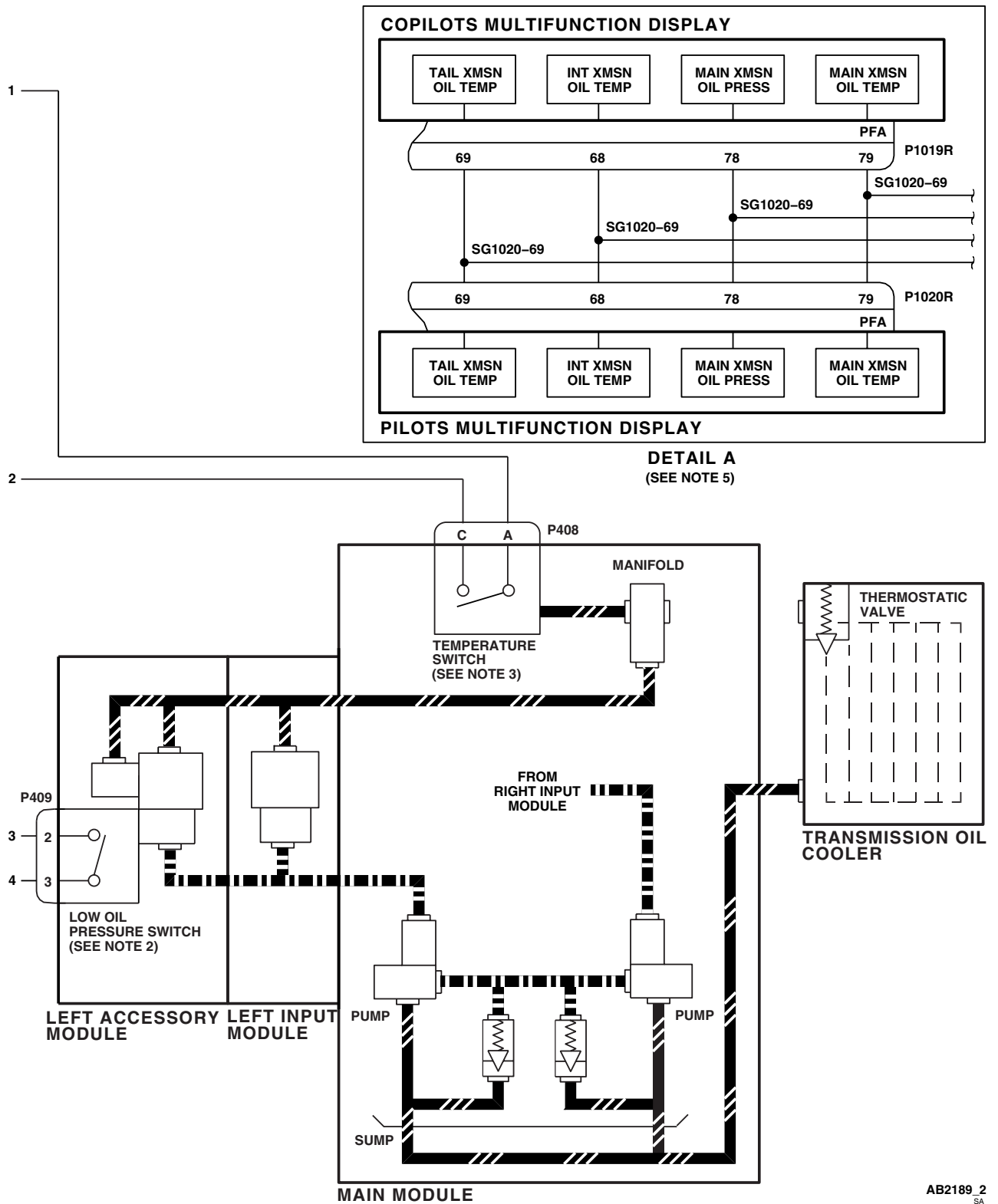


Figure 5. Transmission Oil Warning System Schematic Diagram. (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL
PNEUDRAULIC SYSTEM
HYDRAULIC SYSTEMS

This WP supersedes WP 0086 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment

Multimeter, AN/PSM-45A

WP 0733 00

WP 0734 00

WP 0742 00

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 0748 00

Aviation Ground Power Unit (AGPU), 83-36A

WP 0824 00

Electrical Repairer Toolkit, SC 5180-99-B06

WP 0825 00

Extractor Tool, M81939/14-02

WP 0828 00

Insulated Jumper Wire (2-inch long, 2 required)

WP 0830 00

Locally-Made Drag Beam Wedge, Figure 167,

WP 0870 00

WP 1805 00

WP 0873 00

WP 0925 00

Personnel Required

Aircraft Electrician MOS 15F (1)

WP 1061 00

UH-60 Helicopter Repairer MOS 15T (1)

WP 1092 00

WP 1148 00

WP 1149 00

References

TM 1-1520-237-10

WP 1399 00

TM 11-1520-237-23

WP 1400 00

TM 11-1520-249-23-2

WP 1603 00

WP 0090 00

WP 1605 00

WP 0100 00

WP 1606 00

WP 0101 00

WP 1684 00

WP 0102 00

WP 1685 00

WP 0104 00

WP 1747 00

WP 0120 00

WP 1805 00

WP 0136 00

WP 0443 00

WP 0706 00

WP 0707 00

WP 0708 00

WP 0718 00

WP 0719 00

WP 0720 00

WP 0721 00

WP 0724 00

WP 0725 00

WP 0726 00

WP 0728 00

Equipment Condition

APU Accumulator, Properly Serviced, WP 1605 00

Backup Pump Module Reservoir, Properly Serviced,

WP 1603 00

External Electrical Power Available, WP 1685 00

and APU Operational, TM 1-1520-237-10

No. 1 Pump Module Reservoir, Properly Serviced,

WP 1603 00

No. 2 Pump Module Reservoir, Properly Serviced,

WP 1603 00

Winterization Kit Accumulator, if installed, Properly

Serviced, WP 1606 00

SPECIAL INSTRUCTIONS**NOTE**

See Figure 1 for component location diagram and Figure 2, Figure 3, and Figure 4 for schematic diagrams as an aid in operational/troubleshooting.

An individual procedure is to be done after a particular component change. All procedures are to be done if a complete checkout of the hydraulic system is required.

The hydraulic system operational/troubleshooting procedure is subdivided as follows:

- Backup Pump Operation-General Information
- Switch And Circuit Breaker Setup
- APU Accumulator-Recharge Using APU
- APU Accumulator-Recharge Using External Power
- Hydraulic Leak Test
- No. 1 Hydraulic System (first stage of primary and tail rotor servos)
- No. 2 Hydraulic System (second stage of primary servos and pilot assist module and servos)
- Backup Hydraulics System (No. 1 & No. 2 systems and second stage of tail rotor servo)
- Hydraulic Logic Module Test
- Shutdown

CAUTION

To prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.

Table 1. Temperature Requirements.

FAT F° (C°)	OPERATING TIME (MINUTES)	COOLDOWN TIME, (PUMP OFF) (MINUTES)
90° (32°) max	Unlimited	-
91° (33°) to 100° (38°)	24	72
101° (39°) to 125° (52°)	16	48

**BACKUP PUMP OPERATION-GENERAL INFORMATION OPERATIONAL/TROUBLESHOOTING
PROCEDURE****PROCEDURE****Step****CAUTION**

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
 - If external hydraulic pressure is applied to backup pump, pull BACKUP PUMP PWR circuit breaker (upper console circuit breaker panel).
1. Whenever the backup pump is turned on, either by placing the BACKUP HYD PUMP switch to ON or by simulating a failure, the depressurization valve in the backup pump module is energized, allowing the pump to go into bypass (low pressure output) while the pump motor is accelerating to rated speed. If APU GEN is supplying helicopter power, there is a 4 second delay after the pump is turned on. If main GEN is supplying helicopter power, there is a 0.5 second delay after the pump is turned on, the depressurization valve is deenergized and the backup pump strokes up to rated pressure (3000 psig), causing the BACK-UP PUMP ON capsule on the caution/advisory panel or **HH-60A HH-60L** legend on the MFD/caution/advisory panel to go on.
 2. Whenever the backup pump is turned on by a simulated failure (when BACKUP HYD PUMP switch is placed to AUTO), the backup pump will continue to operate for approximately 90 seconds (180 seconds when winterization kit is installed), even if the failure input is removed.

Indication/Condition

Backup pump shall operate for approximately 90 seconds (180 seconds when winterization kit is installed) when BACKUP HYD PUMP switch is placed to AUTO.

Corrective Action

None Required

SWITCH AND CIRCUIT BREAKER SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- If external hydraulic pressure is applied to backup pump, pull BACKUP PUMP PWR circuit breaker (upper console circuit breaker panel).

NOTE

- Before starting operational/troubleshooting procedure, make sure all tubes, hoses, and components are correctly installed and connected. Check quick-disconnects for security.
- To prevent air in hydraulic cart from going into helicopter hydraulic system, loop pressure and return hoses over a high point on the helicopter, such as the engine cowling or main gear box fairing.
- Whenever ground return lines from aviation ground power unit are connected to helicopter, the reservoir of the pump module will deplete, causing its respective RSVR LOW capsule on the caution/advisory panel or **HH-60A HH-60L** legend on the MFD/caution/advisory panel  to go on.
- With none of the hydraulic systems pressurized and external hydraulic power connected to No. 1 and No. 2 pump modules, these capsules on the caution/advisory panel **HH-60A HH-60L** legends on the MFD/caution/advisory panel  shall be on:

#1 HYD PUMP

#2 HYD PUMP

#1 PRI SERVO PRESS

#2 PRI SERVO PRESS

#1 TAIL RTR SERVO

SAS OFF

BOOST SERVO OFF

#1 RSVR LOW

#2 RSVR LOW

- Ignore TRIM FAIL and FLT PATH STAB capsules or legends.
- These capsules on the caution/advisory panel or **HH-60A HH-60L** legends on the MFD/caution/advisory panel  shall be off:

BACK-UP PUMP ON

SWITCH AND CIRCUIT BREAKER SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

BACK-UP RSVR LOW

#2 TAIL RTR SERVO ON

APU ACCUM LOW

1. Make sure switches are placed as indicated:

SWITCH	LOCATION	POSITION
TAIL SERVO	Lower Console	NORMAL
BACKUP HYD PUMP	Upper Console	OFF
HYD LEAK TEST	Upper Console	NORM
SVO OFF	Pilot's and Copilot's Collective Stick Grip	Centered

2. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
AC ESNTL BUS WARN	Pilot's
BACKUP HYD CONTR	Upper Console
BACKUP PUMP PWR	Copilot's
BATT & ESNTL DC WARN EXT PWR CONTR	Lower Console
BATT BUS CONTR	Lower Console
BATT BUS SPLY	Upper Console
BATT CHGR	Pilot's AC
BATT CHGR	Pilot's DC
CAUT/ADVSY PNL	Upper Console
CMPTR TRIM	Copilot's
CMPTR TRIM	Pilot's
DC ESNTL BUS SPLY	Copilot's
ESNTL BUS AC & CONV WARN	Lower Console
ESNTL BUS SENSE	Upper Console

SWITCH AND CIRCUIT BREAKER SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
NO. 1 AC INST	Copilot's
NO. 1 CONVERTER	Copilot's
NO. 1 DC INST	Copilot's
NO. 1 SERVO CONTR	Copilot's
NO. 1 SERVO WARN	Copilot's
NO. 2 AC INST	Pilot's
NO. 2 CONVERTER CMPTR	Pilot's
NO. 2 DC INST	Pilot's
NO. 2 SERVO CONTR	Pilot's
NO. 2 SERVO WARN	Pilot's
SAS AMPL	Pilot's
SAS BOOST	Upper Console
T RTR SERVO WARN	Copilot's
HH-60A HH-60L MFD PILOT	Aux Circuit Breaker Panel
HH-60A HH-60L MFD COPILOT	Aux Circuit Breaker Panel
HH-60A HH-60L PILOT CDU	Aux Circuit Breaker Panel
HH-60A HH-60L COPILOT CDU	Aux Circuit Breaker Panel
HH-60A HH-60L CDU BACKUP	Aux Circuit Breaker Panel

Indication/Condition

All circuit breakers listed above shall be pushed in.

Corrective Action

None Required

APU ACCUMULATOR-RECHARGE USING APU OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- If external hydraulic pressure is applied to backup pump, pull BACKUP PUMP PWR circuit breaker (upper console circuit breaker panel).

NOTE

- Before starting operational/troubleshooting procedure, make sure all tubes, hoses, and components are correctly installed and connected. Check quick-disconnects for security.
- To prevent air in hydraulic cart from going into helicopter hydraulic system, loop pressure and return hoses over a high point on the helicopter, such as the engine cowling or main gear box fairing.
- Whenever ground return lines from aviation ground power unit are connected to helicopter, the reservoir of the pump module will deplete, causing its respective RSVR LOW capsule on the caution/advisory panel or **HH-60A HH-60L** legend on the MFD/caution/advisory panel  to go on.
- With none of the hydraulic systems pressurized and external hydraulic power connected to No. 1 and No. 2 pump modules, these capsules on the caution/advisory panel or **HH-60A HH-60L** legends on the MFD/caution/advisory panel  shall be on:

#1 HYD PUMP

#2 HYD PUMP

#1 PRI SERVO PRESS

#2 PRI SERVO PRESS

#1 TAIL RTR SERVO

SAS OFF

BOOST SERVO OFF

#1 RSVR LOW

#2 RSVR LOW

- Ignore TRIM FAIL and FLT PATH STAB capsules or legends.
- These capsules on the caution/advisory panel or **HH-60A HH-60L** legends on the MFD/caution/advisory panel  shall be off:

BACK-UP PUMP ON

APU ACCUMULATOR-RECHARGE USING APU OPERATIONAL/TROUBLESHOOTING PROCEDURE -
Continued

BACK-UP RSVR LOW

#2 TAIL RTR SERVO ON

APU ACCUM LOW

1. Do SETUP, in this work package, if not already done.
2. Place upper console BATT switch to ON and start APU (TM 1-1520-237-10).
3. Place upper console GENERATORS APU switch to ON. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed). ■

Indication/Condition

- a. APU ACCUM LOW capsule shall go on.
- b. After 4 seconds, backup pump shall operate.
- c. BACK-UP PUMP ON capsule or legend goes on.
- d. After 90 seconds (180 seconds if winterization kit is installed), backup pump shall shut down.
- e. BACK-UP PUMP ON capsule or legend shall go off.
- f. APU ACCUM LOW capsule or legend shall go off.
- g. APU piston position indicator (tape) shall indicate from 63% to 85% (from 126% to 170%, the sum of the two accumulator tapes when the winterization kit is installed).
- h. APU pressure gage shall indicate from 2700 to 3100 psig.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, replace accumulator pressure switch (WP 0748 00).
2. If Indication/Condition a. is not as specified, go to APU ACCUM LOW CAPSULE OR LEGEND DOES NOT GO ON, in this work package.
3. If Indication/Conditions b. and c. are not as specified, go to BACKUP PUMP DOES NOT OPERATE, in this work package.
4. If Indication/Condition b. occurs, but Indication/Condition c. is not as specified, go to BACKUP PUMP OPERATES BUT BACK-UP PUMP ON CAPSULE DOES NOT GO ON, in this work package.
5. If Indication/Conditions d., e., and f. are not as specified, go to BACKUP PUMP DOES NOT SHUT OFF AND APU ACCUM LOW CAPSULE OR LEGEND IS ON, in this work package.
6. If Indication/Condition d. is not as specified, but Indication/Condition f. occurs, replace right relay panel WP 0825 00).
7. If Indication/Condition d. is not as specified, check if copilot's BACKUP PUMP PWR circuit breaker or upper console BACKUP HYD CONTR circuit breaker has popped.
 - a. If either one has popped, troubleshoot system circuit breakers (WP 0104 00).
 - b. If circuit breakers are engaged, go to BACKUP PUMP OPERATES THEN SHUTS DOWN, in this work package.
8. If APU tape indicates 0 and APU pressure gage reads 1450 psig, check backup pump module lines for leakage.
 - a. If leaks are found, repair/replace components as required (WP 0734 00).

APU ACCUMULATOR-RECHARGE USING APU OPERATIONAL/TROUBLESHOOTING PROCEDURE -
Continued

9. If Indication/Condition g. is not as specified, service accumulator(s) (WP 1605 00).
 - a. If accumulator(s) does not hold pressure, replace accumulator(s) (WP 0742 00).
 - (1) If trouble remains, disconnect P423 from backup pump and check for 115 vac between pins A, B, or C, and ground.
 - (a) If 115 vac is present at any pin, replace relay K19, NO. 3 HYD PWR PUMP relay (WP 0873 00).

Step

4. Place upper console GENERATORS APU switch to OFF/RESET.
5. Shut down APU (TM 1-1520-237-10).
6. Turn MFD switch to OFF.
7. Place BATT switch to OFF.
8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

APU ACCUMULATOR-RECHARGE USING EXTERNAL POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- If external hydraulic pressure is applied to backup pump, pull BACKUP PUMP PWR circuit breaker (upper console circuit breaker panel).

NOTE

- Before starting operational/troubleshooting procedure, make sure all tubes, hoses, and components are correctly installed and connected. Check quick-disconnects for security.
- To prevent air in hydraulic cart from going into helicopter hydraulic system, loop pressure and return hoses over a high point on the helicopter, such as the engine cowling or main gear box fairing.
- Whenever ground return lines from aviation ground power unit are connected to helicopter, the reservoir of the pump module will deplete, causing its respective RSVR LOW capsule on the caution/advisory panel or **HH-60A HH-60L** legend on the MFD/caution/advisory panel  to go on.
- With none of the hydraulic systems pressurized and external hydraulic power connected to No. 1 and No. 2 pump modules, these capsules on the caution/advisory panel or **HH-60A HH-60L** legends on the MFD/caution/advisory panel  shall be on:

#1 HYD PUMP

#2 HYD PUMP

#1 PRI SERVO PRESS

#2 PRI SERVO PRESS

#1 TAIL RTR SERVO

SAS OFF

BOOST SERVO OFF

#1 RSVR LOW

#2 RSVR LOW

- Ignore TRIM FAIL and FLT PATH STAB capsules or legends.
- These capsules on the caution/advisory panel or **HH-60A HH-60L** legends on the MFD/caution/advisory panel  shall be off:

BACK-UP PUMP ON

APU ACCUMULATOR-RECHARGE USING EXTERNAL POWER OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

BACK-UP RSVR LOW

#2 TAIL RTR SERVO ON

APU ACCUM LOW

1. Do SETUP, in this work package, if not already done.
2. Connect external electrical power to helicopter.
3. Turn on electrical power. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed). ■
4. Pull up manual lever of APU start valve solenoid for 30 seconds to discharge APU accumulator, and then push lever down.

Indication/Condition**NOTE**

APU start motor will spin and APU compressor will rotate until hydraulic pressure is discharged.

- a. APU ACCUM LOW capsule or legend shall go on.
- b. After 4 seconds, backup pump shall operate.
- c. BACK-UP PUMP ON capsule or legend goes on.
- d. After 90 seconds (180 seconds, if winterization kit is installed), backup pump shall shut down.
- e. BACK-UP PUMP ON capsule or legend shall go off.
- f. APU ACCUM LOW capsule or legend shall go off.
- g. APU position indicator (tape) shall indicate from 63% to 85% (from 126% to 170%, the sum of the two accumulator tapes when the winterization kit is installed).
- h. APU pressure gage shall indicate from 2700 to 3100 psig.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, replace accumulator pressure switch (WP 0748 00).
2. If Indication/Condition a. is not as specified, go to APU ACCUM LOW CAPSULE OR LEGEND DOES NOT GO ON, in this work package.
3. If Indication/Conditions b. and c. are not as specified, go to BACKUP PUMP DOES NOT OPERATE, in this work package.
4. If Indication/Condition b. occurs, but Indication/Condition c. is not as specified, go to BACKUP PUMP OPERATES BUT BACK-UP PUMP ON CAPSULE DOES NOT GO ON, in this work package.
5. If Indication/Conditions d., e., and f. are not as specified, go to BACKUP PUMP DOES NOT SHUT OFF AND APU ACCUM LOW CAPSULE OR LEGEND IS ON, in this work package.
6. If Indication/Condition d. is not as specified, but Indication/Condition f. occurs, replace right relay panel WP 0825 00).
7. If Indication/Condition d. is not as specified, check if copilot's BACKUP PUMP PWR circuit breaker or upper console BACKUP HYD CONTR circuit breaker has popped.
 - a. If either one has popped, troubleshoot system circuit breakers (WP 0104 00).

**APU ACCUMULATOR-RECHARGE USING EXTERNAL POWER OPERATIONAL/TROUBLESHOOTING
PROCEDURE** - Continued

- b. If circuit breakers are engaged, go to BACKUP PUMP OPERATES THEN SHUTS DOWN, in this work package.
- 8. If APU tape indicates 0 and APU pressure gage reads 1450 psig, check backup pump module lines for leakage.
 - a. If leaks are found, repair/replace components as required (WP 0734 00).
- 9. If Indication/Condition g. is not as specified, service accumulator(s) (WP 1605 00).
 - a. If accumulator(s) does not hold pressure, replace accumulator(s) (WP 0742 00).
 - (1) If trouble remains, disconnect P423 from backup pump and check for 115 vac between pins A, B, or C and ground.
 - (a) If 115 vac is present at any pin, replace relay K19, NO. 3 HYD PWR PUMP relay (WP 0873 00).

Step

- 5. Turn MFD switch to OFF.
- 6. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

HYDRAULIC LEAK TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- If external hydraulic pressure is applied to backup pump, pull BACKUP PUMP PWR circuit breaker (upper console circuit breaker panel).

NOTE

- Before starting operational/troubleshooting procedure, make sure all tubes, hoses, and components are correctly installed and connected. Check quick-disconnects for security.
- To prevent air in hydraulic cart from going into helicopter hydraulic system, loop pressure and return hoses over a high point on the helicopter, such as the engine cowling or main gear box fairing.
- Whenever ground return lines from aviation ground power unit are connected to helicopter, the reservoir of the pump module will deplete, causing its respective RSVR LOW capsule on the caution/advisory panel or **HH-60A HH-60L** legend on the MFD/caution/advisory panel  to go on.
- With none of the hydraulic systems pressurized and external hydraulic power connected to No. 1 and No. 2 pump modules, these capsules on the caution/advisory panel or **HH-60A HH-60L** legend on the MFD/caution/advisory panel  shall be on:

#1 HYD PUMP

#2 HYD PUMP

#1 PRI SERVO PRESS

#2 PRI SERVO PRESS

#1 TAIL RTR SERVO

SAS OFF

BOOST SERVO OFF

#1 RSVR LOW

#2 RSVR LOW

- Ignore TRIM FAIL and FLT PATH STAB capsules or legends.
- These capsules on the caution/advisory panel or **HH-60A HH-60L** legend on the MFD/caution/advisory panel  shall be off:

BACK-UP PUMP ON

HYDRAULIC LEAK TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

BACK-UP RSVR LOW

#2 TAIL RTR SERVO ON

APU ACCUM LOW

1. Do APU ACCUMULATOR RECHARGE USING APU, in this work package, if not already done.
2. If necessary, service No. 1, No. 2, and backup pump modules until reservoirs are full (indicator in green area of decal) (WP 1603 00).
3. Pull out BACKUP HYD CONTR circuit breaker (upper console circuit breaker panel).
4. Turn on electrical power. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed). ■
5. Place upper console HYD LEAK TEST switch to TEST.

Indication/Condition

#1 RSVR LOW, #2 RSVR LOW, and BACK-UP RSVR LOW capsules or legends shall go on.

Corrective Action

1. If Indication/Condition is not as specified, place and hold caution/advisory panel **HH-60A HH-60L** INDICATOR LTS ■ BRT/DIM-TEST switch to TEST.
 - a. If capsules or legends do not go on, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00).
 - b. If any capsule or legend does not go on, go to CAUTION/ADVISORY CAPSULES OR MFD/CAUTION ADVISORY LEGENDS DO NOT GO ON, in this work package.

Step**NOTE**

The #1 HYD PUMP, #2 HYD PUMP, #1 PRI SERVO PRESS, or #2 PRI SERVO PRESS capsules or legends may flicker when the HYD LEAK TEST switch is moved from TEST to RESET, and then to NORM. This is considered normal.

6. Place upper console HYD LEAK TEST switch to RESET and then to NORM.

Indication/Condition

- a. #1 RSVR LOW capsule or legend shall go off.
- b. #2 RSVR LOW and BACK-UP RSVR LOW capsules or legends shall go off.

Corrective Action

1. If Indication/Condition a. is not as specified, replace left relay panel (WP 0825 00).
2. If Indication/Condition b. is not as specified, replace right relay panel (WP 0825 00).

Step

7. Push in upper console BACKUP HYD CONTR circuit breaker and place upper console BACKUP HYD PUMP switch to AUTO.
8. Place upper console HYD LEAK TEST switch to TEST.

HYDRAULIC LEAK TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

- a. #1 RSVR LOW, #2 RSVR LOW, and BACK-UP RSVR LOW capsules or legends shall go on.
- b. Backup hydraulic pump shall run.
- c. BACK-UP PUMP ON capsule or legend shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, go to CAUTION/ADVISORY CAPSULES OR MFD/ CAUTION ADVISORY LEGENDS DO NOT GO ON, in this work package.
2. If Indication/Condition b. is not as specified, check for 28 vdc between P244-H and ground.
 - a. If voltage is as specified, replace right relay panel (WP 0825 00).
 - b. If voltage is not as specified, repair/replace wiring (WP 1747 00).
3. If Indication/Condition c. is not as specified, go to BACKUP PUMP OPERATES BUT BACK-UP PUMP ON CAPSULE DOES NOT GO ON, in this work package.

Step

9. Place MISC SW panel TAIL SERVO switch to BACKUP.

Indication/Condition

- a. #1 TAIL RTR SERVO capsule or legend shall go on.
- b. #2 TAIL RTR SERVO ON capsule or legend shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, go to #1 TAIL RTR SERVO CAPSULE OR LEGEND DOES NOT GO ON WHEN TAIL SERVO SWITCH IS PLACED TO BACKUP, in this work package.
2. If Indication/Condition b. is not as specified, go to #2 TAIL RTR SERVO ON CAPSULE OR LEGEND DOES NOT GO ON, in this work package.

Step

10. Place MISC SW panel TAIL SERVO switch to normal.

Indication/Condition

- a. #1 TAIL RTR SERVO capsule or legend shall go off.
- b. #2 TAIL RTR SERVO capsule or legend shall go off.

Corrective Action

1. If Indication/Condition a. is not as specified, go to #1 TAIL RTR SERVO CAPSULE OR LEGEND REMAINS ON WHEN NO. 1 PUMP MODULE IS PRESSURIZED, in this work package.
2. If Indication/Condition b. is not as specified, go to #2 TAIL RTR SERVO ON CAPSULE OR LEGEND REMAINS ON WHEN NO. 1 PUMP MODULE IS PRESSURIZED, in this work package.

Step

11. Place upper console HYD LEAK TEST switch to RESET and then back to NORM, and place BACKUP HYD PUMP switch to OFF.

HYDRAULIC LEAK TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

- a. Backup pump shall shut down.
- b. BACK-UP PUMP ON capsule or legend shall go off.
- c. #1 RSVR LOW, #2 RSVR LOW, and BACK-UP RSVR LOW capsules or legend shall go off.

Corrective Action

1. If Indication/Condition a. is not as specified, check for 28 vdc between P244-D and ground with HYD LEAK TEST switch in RESET position.
 - a. If voltage is not as specified, repair/replace wiring between P244-D and HYD LEAK TEST switch (WP 1747 00).
2. If Indication/Condition b. is not as specified, check continuity between J445-1 and J445-3.
 - a. If continuity is present, replace backup pump pressure switch (WP 0748 00).

Step

12. Insert locally-made drag beam wedge (WP 1805 00) in left drag beam switch to simulate flight condition.
13. Place upper console HYD LEAK TEST switch to TEST.

Indication/Condition

- a. #1 RSVR LOW capsule or legend shall not go on.
- b. #2 RSVR LOW capsule or legend shall not go on.
- c. BACK-UP RSVR LOW capsule or legend shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, check No. 1 pump module sight glass window.
 - a. If indicator is in red area of decal, service the pump reservoir (WP 1603 00).
 - b. If indicator is in green area of decal, replace left relay panel WP 0825 00).
2. If Indication/Condition b. is not as specified, check No. 2 pump module sight glass window.
 - a. If indicator is in red area of decal, service the respective pump reservoir (WP 1603 00).
 - b. If indicator is in green area, replace right relay panel (WP 0825 00).

Step

14. Place upper console HYD LEAK TEST switch to RESET and then to NORM, and remove locally-made wedge.
15. Pull out upper console BACKUP HYD CONTR circuit breaker, place upper console BACKUP HYD PUMP switch to OFF.
16. Connect external hydraulic power to No. 1, No. 2, and backup pump modules, but do not pressurize (WP 1684 00).
17. Increase hydraulic power to No. 1, No. 2, and backup systems to between 3000 and 3100 psig.

Indication/Condition

Hydraulic fluid shall not leak from No. 1, No. 2, or backup pump module reservoir relief valve.

Corrective Action

If Indication/Condition is not as specified, replace pump module (WP 0706 00).

HYDRAULIC LEAK TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

18. Decrease hydraulic pressure to No. 1, No. 2, and backup pump to 0 psig.
19. Disconnect external hydraulic power from backup pump.
20. Turn MFDs switches to OFF.
21. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

NO. 1 HYDRAULIC SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- If external hydraulic pressure is applied to backup pump, pull BACKUP PUMP PWR circuit breaker (upper console circuit breaker panel).

NOTE

- Before starting operational/troubleshooting procedure, make sure all tubes, hoses, and components are correctly installed and connected. Check quick-disconnects for security.
- To prevent air in hydraulic cart from going into helicopter hydraulic system, loop pressure and return hoses over a high point on the helicopter, such as the engine cowling or main gear box fairing.
- Whenever ground return lines from aviation ground power unit are connected to helicopter, the reservoir of the pump module will deplete, causing its respective RSVR LOW capsule on the caution/advisory panel or **HH-60A HH-60L** legend on the MFD/caution/advisory panel  to go on.
- With none of the hydraulic systems pressurized and external hydraulic power connected to No. 1 and No. 2 pump modules, these capsules on the caution/advisory panel or **HH-60A HH-60L** legends on the MFD/caution/advisory panel  shall be on:

#1 HYD PUMP

#2 HYD PUMP

#1 PRI SERVO PRESS

#2 PRI SERVO PRESS

#1 TAIL RTR SERVO

SAS OFF

BOOST SERVO OFF

#1 RSVR LOW

#2 RSVR LOW

- Ignore TRIM FAIL and FLT PATH STAB capsules or legends.
- These capsules on the caution/advisory panel or **HH-60A HH-60L** legends on the MFD/caution/advisory panel  shall be off:

BACK-UP PUMP ON

NO. 1 HYDRAULIC SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

BACK-UP RSVR LOW

#2 TAIL RTR SERVO ON

APU ACCUM LOW

1. Do HYDRAULIC LEAK TEST, in this work package, if not already done.
2. Slowly increase No. 2 hydraulic pressure to between 3000 and 3100 psig, make sure MISC SW panel TAIL SERVO switch is in NORMAL, and place upper console EXT PWR switch first to RESET and then to ON.

Indication/Condition

- a. #1 HYD PUMP capsule or legend on caution/advisory panel **HH-60A HH-60L** MFD/caution/advisory panel  shall go on.
- b. #1 PRI SERVO PRESS capsule or legend on caution/advisory panel **HH-60A HH-60L** MFD/caution/advisory panel  shall go on.
- c. #1 TAIL RTR SERVO capsule or legend on caution/advisory panel **HH-60A HH-60L** MFD/caution/advisory panel  shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, go to # 1 HYD PUMP CAPSULE OR LEGEND DOES NOT GO ON, in this work package.
2. If Indication/Condition b. is not as specified, go to #1 PRI SERVO PRESS CAPSULE OR LEGEND DOES NOT GO ON, in this work package.
3. If Indication/Condition c. is not as specified, go to #1 TAIL RTR SERVO CAPSULE OR LEGEND DOES NOT GO ON, in this work package.

Step

3. Slowly increase No. 1 hydraulic pressure to between 3000 and 3100 psig.

Indication/Condition**NOTE**

The #1 HYD PUMP and #2 PRI SERVO PRESS capsules or legends may flicker when #1 TAIL RTR SERVO capsule goes off (No. 1 tail rotor servo pressurized). This is considered normal.

- a. #1 HYD PUMP capsule or legend shall go off.
- b. #1 PRI SERVO PRESS capsule or legend shall go off.
- c. #1 TAIL RTR SERVO capsule or legend shall go off.

Corrective Action

1. If Indication/Condition a. is not as specified, check for popped pressure filter.
 - a. If filter is popped, replace filter (WP 0708 00).
 - b. If filter is not popped, replace pump pressure switch (WP 0721 00).
2. If Indication/Condition b. is not as specified, go to #1 PRI SERVO PRESS CAPSULE OR LEGEND REMAINS ON WHEN NO. 1 SYSTEM IS PRESSURIZED, in this work package.
3. If Indication/Condition c. is not as specified, go to #1 TAIL RTR SERVO CAPSULE OR LEGEND REMAINS ON WHEN NO. 1 PUMP MODULE IS PRESSURIZED, in this work package.

NO. 1 HYDRAULIC SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

4. Place either pilot's or copilot's collective stick SVO OFF switch to 1ST STG.

Indication/Condition

#1 PRI SERVO PRESS capsule or legend shall go on.

Corrective Action

If Indication/Condition is not as specified, go to #1 PRI SERVO PRESS CAPSULE OR LEGEND DOES NOT GO ON WHEN 1ST STAGE IS SHUT OFF, in this work package.

Step

5. Place pilot's or copilot's collective stick SVO OFF switch to center position.

Indication/Condition

#1 PRI SERVO PRESS capsule or legend shall go off.

Corrective Action

1. If capsule or legend does not go off, perform the following steps:
 - a. Interchange first and second stage servo pressure switches (WP 1092 00).
 - b. Retest No. 1 hydraulic system.
 - (1) If trouble remains, replace malfunctioning pressure switch (WP 1092 00).
 - (2) If trouble still remains, replace malfunctioning servo cylinder (WP 0719 00).

Step

6. Place MISC SW panel TAIL SERVO switch to BACKUP.

Indication/Condition

#1 TAIL RTR SERVO capsule or legend shall go on.

Corrective Action

If Indication/Condition is not as specified, go to #1 TAIL RTR SERVO CAPSULE OR LEGEND DOES NOT GO ON WHEN TAIL SERVO SWITCH IS PLACED TO BACKUP, in this work package.

Step

7. Decrease No. 1 and No. 2 hydraulic pressure to zero.
8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

NO. 2 HYDRAULIC SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- If external hydraulic pressure is applied to backup pump, pull BACKUP PUMP PWR circuit breaker (upper console circuit breaker panel).

NOTE

- Before starting operational/troubleshooting procedure, make sure all tubes, hoses, and components are correctly installed and connected. Check quick-disconnects for security.
- To prevent air in hydraulic cart from going into helicopter hydraulic system, loop pressure and return hoses over a high point on the helicopter, such as the engine cowling or main gear box fairing.
- Whenever ground return lines from aviation ground power unit are connected to helicopter, the reservoir of the pump module will deplete, causing its respective RSVR LOW capsule on the caution/advisory panel or **HH-60A HH-60L** legend on the MFD/caution/advisory panel ■ shall to go on.
- With none of the hydraulic systems pressurized and external hydraulic power connected to No. 1 and No. 2 pump modules, these capsules on the caution/advisory panel or **HH-60A HH-60L** legends on the MFD/caution/advisory panel ■ shall be on:

#1 HYD PUMP

#2 HYD PUMP

#1 PRI SERVO PRESS

#2 PRI SERVO PRESS

#1 TAIL RTR SERVO

SAS OFF

BOOST SERVO OFF

#1 RSVR LOW

#2 RSVR LOW

- Ignore TRIM FAIL and FLT PATH STAB capsules or legends.
- These capsules on the caution/advisory panel or **HH-60A HH-60L** legends on the MFD/caution/advisory panel ■ shall be off:

BACK-UP PUMP ON

NO. 2 HYDRAULIC SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

BACK-UP RSVR LOW

#2 TAIL RTR SERVO ON

APU ACCUM LOW

1. Do HYDRAULIC LEAK TEST, in this work package, if not already done.
2. Slowly increase No. 1 hydraulic pressure to between 3000 and 3100 psig and place upper console EXT PWR switch first to RESET, and then to ON.

Indication/Condition

- a. #2 HYD PUMP capsule or legend on caution/advisory panel **HH-60A HH-60L** MFD/caution/advisory panel  shall go on.
- b. #2 PRI SERVO PRESS capsule or legend on caution/advisory panel **HH-60A HH-60L** MFD/caution/advisory panel  shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, go to #2 HYD PUMP CAPSULE DOES NOT GO ON, in this work package.
2. If Indication/Condition b. is not as specified, go to #2 PRI SERVO PRESS CAPSULE DOES NOT GO ON, in this work package.

Step

3. Check pilot assist module pressure reducer for popped indicator button.

Indication/Condition

Pressure reducer indicator button shall not be popped.

Corrective Action

If Indication/Condition is not as specified, go to PILOT ASSIST MODULE PRESSURE REDUCER INDICATOR BUTTON IS POPPED, in this work package.

Step

4. Slowly increase No. 2 hydraulic pressure to between 3000 and 3100 psig.

Indication/Condition

- a. #2 HYD PUMP capsule or legend shall go off.
- b. #2 PRI SERVO PRESS capsule or legend shall go off.

Corrective Action

1. If Indication/Condition a. is not as specified, check for popped pressure filter.
 - a. If filter is popped, replace filter (WP 0708 00).
 - b. If filter is not popped, replace pump pressure switch (WP 0726 00).
2. If Indication/Condition b. is not as specified, go to #2 PRI SERVO PRESS CAPSULE REMAINS ON WHEN NO. 2 SYSTEM IS PRESSURIZED, in this work package.

Step

5. Place either pilot's or copilot's collective stick SVO OFF switch to 2ND STG.

NO. 2 HYDRAULIC SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

#2 PRI SERVO PRESS capsule or legend shall go on.

Corrective Action

If Indication/Condition is not as specified, go to #2 PRI SERVO PRESS CAPSULE DOES NOT GO ON WHEN 2ND STAGE IS SHUT OFF, in this work package.

Step

6. Place either pilot's or copilot's collective stick SVO OFF switch to center position.

Indication/Condition

#2 PRI SERVO PRESS capsule or legend shall go off.

Corrective Action

1. If capsule or legend does not go off, perform the following steps:
 - a. Interchange first and second stage servo pressure switches (WP 0726 00).
 - b. Retest No. 2 hydraulic system.
 - (1) If trouble remains, replace malfunctioning pressure switch (WP 0726 00).
 - (a) If trouble still remains, replace malfunctioning servo cylinder (WP 0724 00).

Step**NOTE**

- The following steps check out the pilot assist servos.
 - When both SAS 1 and SAS 2 are disengaged, the MASTER CAUTION capsule or legend may go on when the TRIM switch on the stabilator control/auto flight control panel or either of the TRIM REL switches on the cyclic sticks is pressed. If this occurs, press to reset the MASTER CAUTION capsule.
7. Make sure all pilot-assist servos BOOST, SAS 1, SAS 2, TRIM, and FPS switches on the stabilator control/auto flight control panel are disengaged.

Indication/Condition

- a. BOOST, SAS 1, SAS 2, TRIM, and FPS legends on stabilator control/auto flight control panel shall be off.
- b. SAS OFF and BOOST SERVO OFF capsules or legends on caution/advisory panel **HH-60A HH-60L** MFD/caution/advisory panel shall be on.
- c. TRIM FAIL and FLT PATH STAB caution advisories shall be off.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, troubleshoot digital AFCS (TM 11-1520-237-23 or TM 11-1520-249-23-2).
2. If Indication/Conditions c. and d. are not as specified, simultaneously press and release both fail advisory reset pushbuttons on the stabilator control/auto flight control panel to reset.

Step

8. Press face of stabilator control/auto flight control panel TRIM switch on.

NO. 2 HYDRAULIC SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

- a. TRIM button light shall go on.
- b. TRIM FAIL and FLT PATH STAB caution advisories shall be off.

Corrective Action

1. If Indication/Condition a. is not as specified, troubleshoot digital AFCS (TM 11-1520-237-23 or TM 11-1520-249-23-2).
2. If Indication/Condition b. is not as specified, simultaneously press and release both fail advisory reset push-buttons on the stabilator control/auto flight control panel to reset.

Step

9. Move cyclic stick back and forth.

Indication/Condition

Stick shall be hard to move.

Corrective Action

If Indication/Condition is not as specified, troubleshoot digital AFCS (TM 11-1520-237-23 or TM 11-1520-249-23-2).

Step**NOTE**

Do not try to oppose stick movement. If hand is held on stick, follow stick movement.

10. Move cyclic stick STICK TRIM switch FWD and AFT.

Indication/Condition

- a. TRIM FAIL capsule or legend shall not go on.
- b. Stick shall move slowly by itself.

Corrective Action

If Indication/Condition a. or b. is not as specified, troubleshoot digital AFCS (TM 11-1520-237-23 or TM 11-1520-249-23-2).

Step

11. Turn boost servo on and then off, by pressing stabilator control/auto flight control panel BOOST switch.

Indication/Condition

BOOST SERVO OFF capsule or legend shall go off and then on.

Corrective Action

If Indication/Condition is not as specified, troubleshoot digital AFCS (TM 11-1520-237-23 or TM 11-1520-249-23-2).

NO. 2 HYDRAULIC SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step****NOTE**

The #2 HYD PUMP, #2 PRI SERVO OFF, and BOOST SERVO OFF capsules or legends may flicker when either SAS 1 or SAS 2 is turned on after both have been off. This is considered normal.

12. Turn SAS actuators on, and then off, by pressing stabilator control/auto flight control panel SAS 1 and SAS 2 switches, one at a time. Observe SAS OFF capsule or legend.

Indication/Condition

SAS OFF capsule or legend shall go off when either switch is engaged (lit), and shall go on only if both switches are disengaged (not lit).

Corrective Action

If Indication/Condition is not as specified, troubleshoot digital AFCS (TM 11-1520-237-23 or TM 11-1520-249-23-2).

Step

13. Slowly decrease hydraulic pressure to zero.
14. Place upper console EXT PWR switch to OFF.
15. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

BACKUP HYDRAULICS SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- If external hydraulic pressure is applied to backup pump, pull BACKUP PUMP PWR circuit breaker (upper console circuit breaker panel).

NOTE

- Before starting operational/troubleshooting procedure, make sure all tubes, hoses, and components are correctly installed and connected. Check quick-disconnects for security.
- To prevent air in hydraulic cart from going into helicopter hydraulic system, loop pressure and return hoses over a high point on the helicopter, such as the engine cowling or main gear box fairing.
- Whenever ground return lines from aviation ground power unit are connected to helicopter, the reservoir of the pump module will deplete, causing its respective RSVR LOW capsule on the caution/advisory panel or **HH-60A HH-60L** legend on the MFD/caution/advisory panel  to go on.
- With none of the hydraulic systems pressurized and external hydraulic power connected to No. 1 and No. 2 pump modules, these capsules on the caution/advisory panel or **HH-60A HH-60L** legends on the MFD/caution/advisory panel  shall be on:

#1 HYD PUMP

#2 HYD PUMP

#1 PRI SERVO PRESS

#2 PRI SERVO PRESS

#1 TAIL RTR SERVO

SAS OFF

BOOST SERVO OFF

#1 RSVR LOW

#2 RSVR LOW

- Ignore TRIM FAIL and FLT PATH STAB capsules or legends.
- These capsules on the caution/advisory panel or **HH-60A HH-60L** legends on the MFD/caution/advisory panel  shall be off:

BACK-UP PUMP ON

BACKUP HYDRAULICS SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

BACK-UP RSVR LOW

#2 TAIL RTR SERVO ON

APU ACCUM LOW

1. Do HYDRAULIC LEAK TEST, in this work package, if not already done.
2. On hydraulic cart, turn off hydraulic pressure to first and second stage (No. 1 and No. 2) hydraulics.
3. Place upper console BACKUP HYD PUMP switch to OFF and MISC SW panel TAIL SERVO switch to BACKUP.
4. Make sure BACKUP PUMP PWR circuit breaker (copilot's circuit breaker panel) is engaged and BACKUP HYD CONTR circuit breaker (upper console circuit breaker panel) is pulled out.
5. Place upper console EXT PWR switch first to RESET and then to ON.

Indication/Condition

#1 HYD PUMP AND #2 HYD PUMP capsules or legends shall go on.

Corrective Action

If Indication/Condition is not as specified, do NO. 1 HYDRAULIC SYSTEM or the NO. 2 HYDRAULIC SYSTEM in this work package, as required.

Step

6. Check pilot assist module pressure reducer for popped indicator button.

Indication/Condition

Pressure reducer indicator shall not be popped.

Corrective Action

If Indication/Condition is not as specified, go to PILOT ASSIST MODULE PRESSURE REDUCER INDICATOR BUTTON IS POPPED, in this work package.

Step

7. Pressurize hydraulic cart backup hydraulic system to between 3000 and 3100 psig.

Indication/Condition

- a. #1 PRI SERVO PRESS capsule or legend shall go off.
- b. #1 TAIL RTR SERVO capsule or legend shall go on.
- c. #2 PRI SERVO PRESS capsule or legend shall go off.
- d. #2 TAIL RTR SERVO ON capsule or legend shall go on.
- e. BACK-UP PUMP ON capsule or legend shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, go to #1 PRI SERVO PRESS CAPSULE OR LEGEND REMAINS ON WHEN NO. 1 SYSTEM IS PRESSURIZED, in this work package.
2. If Indication/Condition b. is not as specified, go to #1 TAIL RTR SERVO CAPSULE OR LEGEND DOES NOT GO ON WHEN TAIL SERVO SWITCH IS PLACED TO BACKUP, in this work package.
3. If Indication/Condition c. is not as specified, go to #2 PRI SERVO PRESS CAPSULE DOES NOT GO ON WHEN 2ND STAGE IS SHUT OFF, in this work package.

BACKUP HYDRAULICS SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

4. If Indication/Condition d. is not as specified, go to #2 TAIL RTR SERVO ON CAPSULE OR LEGEND DOES NOT GO ON, in this work package.
5. If Indication/Condition e. is not as specified, go to BACKUP PUMP OPERATES BUT BACK-UP PUMP ON CAPSULE DOES NOT GO ON, in this work package.

Step

8. Decrease hydraulic pressure to approximately 1000 psig, and push in upper console BACKUP HYD CONTR circuit breaker.
9. Place upper console BACKUP HYD PUMP switch to ON.

Indication/Condition

After 4 seconds, backup pump shall operate.

Corrective Action

If Indication/Condition is not as specified, go to BACKUP PUMP FAILS TO RUN WITH BACKUP HYD PUMP SWITCH PLACED ON, in this work package.

Step

10. Place upper console BACKUP HYD PUMP switch to AUTO.

Indication/Condition

Backup pump continues to operate.

Corrective Action

If Indication/Condition is not as specified, replace BACKUP HYD PUMP switch (WP 0828 00).

Step

11. Place upper console BACKUP HYD PUMP switch to OFF.

Indication/Condition

Backup pump shall shut down.

Corrective Action

If Indication/Condition is not as specified, go to BACKUP PUMP RUNS CONTINUOUSLY WITH NO FAULT DISPLAYED ON THE CAUTION/ADVISORY PANEL OR MFD/CAUTION/ADVISORY PANEL, in this work package.

Step

12. Insert locally-made wedge in left drag beam switch to simulate flight condition.

Indication/Condition

- a. Backup pump shall operate.
- b. BACK-UP PUMP ON capsule or legend shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, go to BACKUP PUMP DOES NOT OPERATE, in this work package.

BACKUP HYDRAULICS SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. If Indication/Condition b. is not as specified, go to BACKUP PUMP OPERATES BUT BACK-UP PUMP ON CAPSULE DOES NOT GO ON, in this work package.

Step

13. Remove locally-made wedge.
14. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

HYDRAULIC LOGIC MODULE TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

- In the following operational/troubleshooting procedures, to prevent overheating backup pump hydraulic system, use Table 1 for limits when running backup pump.
- If external hydraulic pressure is applied to backup pump, pull BACKUP PUMP PWR circuit breaker (upper console circuit breaker panel).

NOTE

- Before starting operational/troubleshooting procedure, make sure all tubes, hoses, and components are correctly installed and connected. Check quick-disconnects for security.
- To prevent air in hydraulic cart from going into helicopter hydraulic system, loop pressure and return hoses over a high point on the helicopter, such as the engine cowling or main gear box fairing.
- Whenever ground return lines from aviation ground power unit are connected to helicopter, the reservoir of the pump module will deplete, causing its respective RSVR LOW capsule on the caution/advisory panel or **HH-60A HH-60L** legend on the MFD/caution/advisory panel  to go on.
- With none of the hydraulic systems pressurized and external hydraulic power connected to No. 1 and No. 2 pump modules, these capsules on the caution/advisory panel or **HH-60A HH-60L** legends on the MFD/caution/advisory panel  shall be on:

#1 HYD PUMP

#2 HYD PUMP

#1 PRI SERVO PRESS

#2 PRI SERVO PRESS

#1 TAIL RTR SERVO

SAS OFF

BOOST SERVO OFF

#1 RSVR LOW

#2 RSVR LOW

- Ignore TRIM FAIL and FLT PATH STAB capsules or legends.
- These capsules on the caution/advisory panel or **HH-60A HH-60L** legends on the MFD/caution/advisory panel  shall be off:

BACK-UP PUMP ON

HYDRAULIC LOGIC MODULE TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

BACK-UP RSVR LOW

#2 TAIL RTR SERVO ON

APU ACCUM LOW

1. Do SWITCH AND CIRCUIT BREAKER SETUP, in this work package, if not already done.
2. Connect external electrical power to helicopter.
3. Place upper console BATT switch ON.

Indication/Condition

EXT PWR CONNECTED, #1 CONV, #2 CONV, and AC ESS BUS OFF capsules on the caution/advisory panel shall go on.

Corrective Action

None Required

Step

4. Place upper console EXT PWR switch first to RESET, and then to ON.

Indication/Condition

#1 CONV, #2 CONV, and AC ESS BUS OFF capsules on the caution/advisory panel shall go off.

Corrective Action

None Required

Step

5. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed).
6. **HH-60A HH-60L** Press C/A (caution/advisory) button on the right lower corner of the MFD display to view C/A page.
7. Ensure the switches listed below are as indicated and the capsules on the caution/advisory panel or **HH-60A HH-60L** the legends on the pilot's or copilot's inboard MFD are on.

SWITCH	LOCATION	POSITION
SERVO OFF	Collective grip	CENTERED
TAIL ROTOR SERVO	Miscellaneous switch panel	NORMAL
BACKUP HYD PUMP	Upper Console	OFF
BOOST	AFCS control panel	ON
TRIM	AFCS control panel	OFF
SAS 1	AFCS control panel	ON

HYDRAULIC LOGIC MODULE TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

SWITCH	LOCATION	POSITION
SAS 2	AFCS control panel	ON
ROTARY	SAS/FPS computer	GND

8. Connect external hydraulic power to No. 1 and No. 2 pump modules, but do not pressurize (WP 1684 00).

9. Slowly increase hydraulic power to No. 1 and No. 2 pump systems to between 3000 and 3100 psig.

Indication/Condition

The following capsules on the caution/advisory panel or **HH-60A HH-60L** legends on the pilot's or copilot's inboard MFD  shall not go on:

- #1 PRI SERVO PRESS
- #2 PRI SERVO PRESS
- BOOST SERVO OFF
- SAS or SAS OFF
- TRIM FAIL
- #1 TAIL RTR SERVO

Corrective Action

If any of the capsules on the caution/advisory panel or **HH-60A HH-60L** legends on the pilot's or copilot's inboard MFD  go on, check both pilot's and copilot's collective grip SERVO OFF switches are in the center position, the BOOST (or SAS/BOOST), TRIM, SAS 1, SAS 2, or SAS switches on the AFCS control panel are in the ON position.

Step**NOTE**

The aircraft must be powered on and the NO. 1 SERVO CONTR and NO. 2 SERVO CONTR circuit breakers pushed in for at least one hour prior to doing the following steps. If an output transistor in the hydraulic logic module in the left or right relay panel has an excessive voltage leak, it may cause failure of the backup hydraulic system and loss of the second stage tail rotor. The procedure in the following steps is intended to verify operation of the logic modules and backup hydraulic system. If transistors in the logic modules are leaking output voltage, the leak may only appear after an extended period of operation.

10. Pull out NO. 2 DC INST circuit breaker (pilot's circuit breaker panel).

11. Place upper console BACKUP HYD PUMP switch to AUTO.

HYDRAULIC LOGIC MODULE TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

NOTE

The stopwatch is used to measure the operating time of the backup pump from when it goes on in step 12. until it goes off in step 14.

12. Simultaneously start stopwatch and place upper console HYD LEAK TEST switch to TEST.

Indication/Condition

- a. The following capsules on the caution/advisory panel or **HH-60A HH-60L** legends on the pilot's or copilot's inboard MFD  shall go on:
- SAS OFF
 - BOOST SERVO OFF
 - #1 RSVR LOW
 - #2 RSVR LOW
 - BACKUP RSVR LOW
 - #1 TAIL RTR SERVO
 - #2 TAIL RTR SERVO ON
 - BACKUP PUMP ON
- b. Electric motor of backup pump shall start.

Corrective Action

If BACKUP PUMP ON capsule on the caution/advisory panel or **HH-60A HH-60L** legend on the pilot's or copilot's inboard MFD  does not go on or #2 TAIL RTR SERVO ON capsule on the caution/advisory panel or **HH-60A HH-60L** legend on the pilot's or copilot's inboard MFD  flickers or does not go on, replace right relay panel (WP 0825 00).

Step

13. Place upper console HYD LEAK TEST switch to NORM.

Indication/Condition

There shall be no change in the caution/advisory panel capsules or **HH-60A HH-60L** the pilot's or copilot's inboard MFD legends  .

Corrective Action

None Required

Step

14. Momentarily place upper console HYD LEAK TEST switch to RESET and then back to NORM.

Indication/Condition

- a. The following capsules on the caution/advisory panel or **HH-60A HH-60L** legends on the pilot's or copilot's inboard MFD  shall go off immediately:
- SAS OFF

HYDRAULIC LOGIC MODULE TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- BOOST SERVO OFF
 - #1 RSVR LOW
 - #2 RSVR LOW
 - BACKUP RSVR LOW
 - #1 TAIL RTR SERVO
 - #2 TAIL RTR SERVO ON
 - BACKUP PUMP ON
- b. BACKUP PUMP ON capsule on the caution/advisory panel or **HH-60A HH-60L** legend on the pilot's or copilot's inboard MFD **■** shall go off and the pump motor shall stop after approximately 80 to 100 seconds (170 to 190 seconds when winterization kit is installed).

Corrective Action

None Required

Step

15. Place upper console BACKUP HYD PUMP switch to OFF.
16. Push in NO. 2 DC INST circuit breaker (pilot's circuit breaker panel).
17. Decrease hydraulic pressure to No. 1 and No. 2, to 0 psig.
18. Disconnect external hydraulic power from No.1 and No. 2 hydraulic systems.
19. **HH-60A HH-60L** On pilot's or copilot's inboard MFD, turn OFF switch to OFF. **■**
20. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step**

1. Turn off hydraulic power (WP 1684 00).
2. Turn off electrical power.

Indication/Condition

Electrical power shall be off.

Corrective Action

None Required

APU ACCUM LOW CAPSULE OR LEGEND DOES NOT GO ON**SYMPTOM**

APU ACCUM LOW capsule or legend does not go on.

MALFUNCTION

APU ACCUM LOW capsule or legend does not go on.

CORRECTIVE ACTION**1. Check helicopter configuration.**

- a. **EMEP** , go to 2.
- b. **W/O EMEP** , go to 4.

2. EMEP Remove and check pin filtered adapter P118 or HH-60A HH-60L P1019R or P1020R (WP 0925 00).

- a. If pin filtered adapter is good, go to 3.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 8.

3. EMEP Install pin filtered adapter (WP 0925 00). Go to 4.**4. Check for 28 vdc between P118-C or HH-60A HH-60L P1019R-16 or P1020R-16 and ground.**

- a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 8.
- b. If voltage is not as specified, go to 5.

5. Check for 28 vdc between P302-2 and ground.

- a. If voltage is as specified, go to 6.
- b. If voltage is not as specified, go to 7.

6. Check continuity between P118-C or HH-60A HH-60L P1019R-16 or P1020R-16 and P302-1.

- a. If continuity is present, replace APU accumulator pressure switch (WP 0748 00). Go to 8.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 8.

7. Check for 28 vdc between BACKUP HYD CONTR circuit breaker terminal 1 and ground.

- a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P302-2 (WP 1747 00). Go to 8.
- b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to 8.

8. Procedure completed.**BACKUP PUMP DOES NOT OPERATE****SYMPTOM**

Backup pump does not operate.

MALFUNCTION

Backup pump does not operate.

BACKUP PUMP DOES NOT OPERATE - Continued**CORRECTIVE ACTION**

1. Place **BACKUP HYD PUMP** switch to **ON**, and check that pump operates.
 - a. If pump operates, go to **2**.
 - b. If pump does not operate, go to **BACKUP PUMP FAILS TO RUN WITH BACKUP HYD PUMP SWITCH PLACED ON**, in this work package.
2. Place **BACKUP HYD PUMP** switch to **OFF**. Go to **3**.
3. Check helicopter configuration.
 - a. **W/O WINTER** , go to **4**.
 - b. **WINTER** , go to **5**.
4. **W/O WINTER** Check for 28 vdc between P244-B and ground.
 - a. If voltage is as specified, go to **7**.
 - b. If voltage is not as specified, repair/replace wiring between P244-B and P302-1 (WP 1747 00). Go to **9**.
5. **WINTER** Check for 28 vdc between J244B-B and ground.
 - a. If voltage is as specified, go to **7**.
 - b. If voltage is not as specified, go to **6**.
6. **WINTER** Check continuity between J244A-B and J244B-B.
 - a. If continuity is present, go to **7**.
 - b. If continuity is not present, replace **WINTER** (WP 1400 00). Go to **9**.
7. Check for 28 vdc between **BACKUP HYD PUMP** switch terminal 3 and ground.
 - a. If voltage is as specified, replace switch (WP 0828 00). Go to **9**.
 - b. If voltage is not as specified, go to **8**.
8. Check continuity between **BACKUP HYD PUMP** switch terminal 3 and P241-L.
 - a. If continuity is present, replace right relay panel (WP 0825 00). Go to **9**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9**.
9. Procedure completed.

BACKUP PUMP OPERATES BUT BACK-UP PUMP ON CAPSULE DOES NOT GO ON**SYMPTOM**

Backup pump operates. BACK-UP PUMP ON capsule does not go on.

MALFUNCTION

Backup pump operates. BACK-UP PUMP ON capsule does not go on.

CORRECTIVE ACTION

1. Check helicopter configuration.
 - a. **EMEP** , go to **2**.
 - b. **W/O EMEP** , go to **4**.

BACKUP PUMP OPERATES BUT BACK-UP PUMP ON CAPSULE DOES NOT GO ON - Continued

2. **EMEP** Remove and check pin filtered adapter P118-R or **HH-60A HH-60L** P1019R or P1020R (WP 0925 00).
 - a. If pin filtered adapter is good, go to 3.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 8.
3. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 4.
4. Check for 28 vdc between P118-R or **HH-60A HH-60L** P1019R-51 or P1020R-51 and ground.
 - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 8.
 - b. If voltage is not as specified, go to 5.
5. Check for 28 vdc between P445-3 and ground.
 - a. If voltage is as specified, go to 6.
 - b. If voltage is not as specified, go to 7.
6. Check continuity between P118-R or **HH-60A HH-60L** P1019R-51 or P1020R-51 and P445-1.
 - a. If continuity is present, replace backup pump pressure switch (WP 0733 00). Go to 8.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 8.
7. Check for 28 vdc between NO. 2 SERVO WARN circuit breaker terminal 1 and ground.
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P445-3 (WP 1747 00). Go to 8.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to 8.
8. Procedure completed.

BACKUP PUMP DOES NOT SHUT OFF AND APU ACCUM LOW CAPSULE OR LEGEND IS ON**SYMPTOM**

Backup pump does not shut off. APU ACCUM LOW capsule or legend is on.

MALFUNCTION

Backup pump does not shut off. APU ACCUM LOW capsule or legend is on.

CORRECTIVE ACTION

1. Check APU accumulator pressure gage for proper charge.
 - a. If charge is correct, go to 2.
 - b. If charge is not correct, service APU accumulator (WP 1605 00). Go to 4.
2. Check helicopter configuration.
 - a. **WINTER** , go to 3.
 - b. **W/O WINTER** , replace APU accumulator pressure switch (WP 0748 00). Go to 4.
3. **WINTER** Check continuity between J302-1 and J302-3.

BACKUP PUMP DOES NOT SHUT OFF AND APU ACCUM LOW CAPSULE OR LEGEND IS ON - Continued

- a. If continuity is present, replace APU accumulator pressure switch (WP 0748 00). Go to **4**.
- b. If continuity is not present, replace winterization kit relay (WP 1400 00). Go to **4**.

4. Procedure completed.**BACKUP PUMP OPERATES THEN SHUTS DOWN****SYMPTOM**

Backup pump operates then shuts down.

MALFUNCTION

Backup pump operates then shuts down.

CORRECTIVE ACTION

- 1. Allow pump to cool down. See equipment conditions. Go to 2.**
- 2. Place locally-made wedge in left drag beam switch to simulate weight off wheels. Go to 3.**
- 3. Check that backup pump starts and continues to run.**
 - a. If backup pump starts and continues to run, remove locally-made wedge and replace thermal switch (WP 0870 00). Go to **4**.
 - b. If backup pump does not start, remove locally-made wedge and replace K19 NO. 3 HYD PWR PUMP relay (WP 0873 00). Go to **4**.
- 4. Procedure completed.**

CAUTION/ADVISORY CAPSULES OR MFD/CAUTION/ADVISORY LEGENDS DO NOT GO ON**SYMPTOM**

Caution/advisory capsules or MFD/caution/advisory legends do not go on.

MALFUNCTION

Caution/advisory capsules or MFD/caution/advisory legends do not go on.

CORRECTIVE ACTION

- 1. Does #1 RSVR LOW capsule or legend remain off?**
 - a. If capsule or legend remains off, go to **2**.
 - b. If capsule or legend goes on, go to **16**.
- 2. Check sight glass window on No. 1 hydraulic pump for full indication.**
 - a. If sight glass indication is full, go to **3**.
 - b. If sight glass does not indicate full, service pump reservoir (WP 1603 00). Go to **41**.
- 3. Check helicopter configuration.**
 - a. **EMEP** , go to **4**.
 - b. **W/O EMEP** , go to **6**.
- 4. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R (WP 0925 00).**

CAUTION/ADVISORY CAPSULES OR MFD/CAUTION/ADVISORY LEGENDS DO NOT GO ON - Continued

- a. If pin filtered adapter is good, go to **5**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **41**.
- 5. EMEP Install pin filtered adapter (WP 0925 00). Go to 6.**
- 6. Check for 28 vdc between P117-32 or HH-60A HH-60L P1019R-28 or P1020R-28 and ground.**
 - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **41**.
 - b. If voltage is not as specified, go to **7**.
- 7. Check for 28 vdc between P902-t and ground.**
 - a. If voltage is as specified, go to **8**.
 - b. If voltage is not as specified, go to **9**.
- 8. Check continuity between P117-32 or HH-60A HH-60L P1019R-28 or P1020R-28 and P902-y.**
 - a. If continuity is present, replace left relay panel (WP 0825 00). Go to **41**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
- 9. Check for 28 vdc between HYD LEAK TEST switch terminal 6 and ground.**
 - a. If voltage is as specified, repair/replace wiring between switch terminal 6 and P902-t (WP 1747 00). Go to **41**.
 - b. If voltage is not as specified, go to **10**.
- 10. Check for 28 vdc between HYD LEAK TEST switch terminal 5 and ground.**
 - a. If voltage is as specified, replace switch (WP 0828 00). Go to **41**.
 - b. If voltage is not as specified, go to **11**.
- 11. Check for 28 vdc between P447-3 and ground.**
 - a. If voltage is as specified, go to **12**.
 - b. If voltage is not as specified, go to **13**.
- 12. Check continuity between HYD LEAK TEST switch terminal 5 and P447-2.**
 - a. If continuity is present, remove pump and send to AVIM (WP 0706 00). Go to **41**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
- 13. Check helicopter configuration.**
 - a. **WINTER** , go to **14**.
 - b. **W/O WINTER** , go to **15**.
- 14. WINTER Check continuity between J244B-X and P447-3.**
 - a. If continuity is present, replace right relay panel (WP 0825 00). Go to **41**.
 - b. If continuity is not present, replace **WINTER** (WP 1400 00). Go to **41**.
 - c. If trouble remains, go to **15**.
- 15. Check continuity between P244-X and P447-3.**

CAUTION/ADVISORY CAPSULES OR MFD/CAUTION/ADVISORY LEGENDS DO NOT GO ON - Continued

- a. If continuity is present, replace right relay panel (WP 0825 00). Go to **41**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
- 16. Check that #2 RSVR LOW capsule or legend remains off.**
 - a. If capsule or legend remains off, go to **17**.
 - b. If capsule or legend does not remain off, go to **28**.
- 17. Check sight glass window on No. 2 hydraulic pump for full indication.**
 - a. If pump indication is full, go to **18**.
 - b. If pump indication is not full, service pump reservoir (WP 1603 00). Go to **41**.
- 18. Check for 28 vdc between P117-48 or HH-60A HH-60L P1019R-64 or P1020R-64 and ground.**
 - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **41**.
 - b. If voltage is not as specified, go to **19**.
- 19. Check for 28 vdc between P900-a and ground.**
 - a. If voltage is as specified, go to **20**.
 - b. If voltage is not as specified, go to **23**.
- 20. Check helicopter configuration.**
 - a. **WINTER** , go to **21**.
 - b. **W/O WINTER** , go to **22**.
- 21. WINTER Check continuity between J244B-g and P117-48 or HH-60A HH-60L P1019R-64 or P1020R-64 .**
 - a. If continuity is present, replace right relay panel (WP 0825 00). Go to **41**.
 - b. If continuity is not present, replace **WINTER** (WP 1400 00). Go to **41**.
 - c. If trouble remains, go to **22**.
- 22. Check continuity between P117-48 or HH-60A HH-60L P1019R-64 or P1020R-64 and P244-g.**
 - a. If continuity is present, replace right relay panel (WP 0825 00). Go to **41**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
- 23. Check for 28 vdc between HYD LEAK TEST switch terminal 12 and ground.**
 - a. If voltage is as specified, repair/replace wiring between switch terminal 12 and P900-a (WP 1747 00). Go to **41**.
 - b. If voltage is not as specified, go to **24**.
- 24. Check for 28 vdc between HYD LEAK TEST switch terminal 11 and ground.**
 - a. If voltage is as specified, replace switch (WP 0828 00). Go to **41**.
 - b. If voltage is not as specified, go to **25**.
- 25. Check for 28 vdc between P432-3 and ground.**

CAUTION/ADVISORY CAPSULES OR MFD/CAUTION/ADVISORY LEGENDS DO NOT GO ON - Continued

- a. If voltage is as specified, go to **26**.
- b. If voltage is not as specified, go to **27**.
- 26. Check continuity between HYD LEAK TEST switch terminal 11 and P432-2.**
 - a. If continuity is present, remove pump and send to AVIM (WP 0706 00). Go to **41**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
- 27. Check for 28 vdc between NO. 2 SERVO CONTR circuit breaker terminal 1 and ground.**
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P432-3 (WP 1747 00). Go to **41**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **41**.
- 28. BACK-UP PUMP RSVR LOW capsule or legend does not go on. Check sight glass window on back up hydraulic pump for full indication.**
 - a. If pump shows full indications, go to **29**.
 - b. If pump does not show full indication, service pump reservoir (WP 1603 00). Go to **41**.
- 29. Check for 28 vdc between P117-64 or HH-60A HH-60L P1019R-23 or P1020R-23 and ground.**
 - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **41**.
 - b. If voltage is not as specified, go to **30**.
- 30. Check helicopter configuration.**
 - a. **WINTER** , go to **31**.
 - b. **W/O WINTER** , go to **33**.
- 31. WINTER Check for 28 vdc between J244B-f and ground.**
 - a. If voltage is as specified, go to **32**.
 - b. If voltage is not as specified, replace **WINTER** (WP 1400 00). Go to **41**.
 - c. If trouble remains, go to **33**.
- 32. WINTER Check continuity between J244B-f and P117-64 or HH-60A HH-60L P1019R-50 or P1020R-50 .**
 - a. If continuity is present, go to **35**.
 - b. If continuity is not present, replace **WINTER** (WP 1399 00). Go to **41**.
 - c. If trouble remains, go to **34**.
- 33. Check for 28 vdc between P244-f and ground.**
 - a. If voltage is as specified, go to **34**.
 - b. If voltage is not as specified, go to **35**.
- 34. Check continuity between P117-64 or HH-60A HH-60L P1019R-50 or P1020R-50 and P244-f.**
 - a. If continuity is present, replace right relay panel (WP 0825 00). Go to **41**.

CAUTION/ADVISORY CAPSULES OR MFD/CAUTION/ADVISORY LEGENDS DO NOT GO ON - Continued

- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
- 35. Check for 28 vdc between P442-2 and ground.**
 - a. If voltage is as specified, go to **36**.
 - b. If voltage is not as specified, go to **39**.
- 36. Check helicopter configuration.**
 - a. **WINTER** , go to **37**.
 - b. **W/O WINTER** , go to **38**.
- 37. WINTER Check continuity between J244B-e and P442-3.**
 - a. If continuity is present, remove pump and send to AVIM (WP 0706 00). Go to **41**.
 - b. If continuity is not present, replace **WINTER** (WP 1400 00). Go to **41**.
 - c. If trouble remains, go to **38**.
- 38. Check continuity between P244-e and P442-3.**
 - a. If continuity is present, remove pump and send to AVIM (WP 0706 00). Go to **41**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **41**.
- 39. Check for 28 vdc between HYD LEAK TEST switch terminal 9 and ground.**
 - a. If voltage is as specified, repair/replace wiring between switch terminal 9 and P442-2 (WP 1747 00). Go to **41**.
 - b. If voltage is not as specified, go to **40**.
- 40. Check for 28 vdc between HYD LEAK TEST switch terminal 8 and ground.**
 - a. If voltage is as specified, replace switch (WP 0828 00). Go to **41**.
 - b. If voltage is not as specified, repair/replace wiring between switch terminal 8 and P432-2 (WP 1747 00). Go to **41**.
- 41. Procedure completed.**

#1 TAIL RTR SERVO CAPSULE OR LEGEND DOES NOT GO ON WHEN TAIL SERVO SWITCH IS PLACED TO BACKUP**SYMPTOM**

#1 TAIL RTR SERVO capsule or legend does not go on.

MALFUNCTION

#1 TAIL RTR SERVO capsule or legend does not go on when TAIL SERVO switch is placed to BACKUP.

CORRECTIVE ACTION

- 1. Check for 28 vdc between P902-y and ground.**
 - a. If voltage is as specified, replace left relay panel (WP 0825 00). Go to **4**.
 - b. If voltage is not as specified, go to **2**.
- 2. Check for 28 vdc between P135-S and ground.**
 - a. If voltage is as specified, go to **3**.

#1 TAIL RTR SERVO CAPSULE OR LEGEND DOES NOT GO ON WHEN TAIL SERVO SWITCH IS PLACED TO BACKUP - Continued

- b. If voltage is not as specified, repair/replace wiring between P135-S and NO. 1 SERVO CONTR circuit breaker terminal 1 (WP 1747 00). Go to **4**.

3. Check continuity between P135-T and P902-v.

- a. If continuity is present, replace miscellaneous switch panel (WP 0830 00). Go to **4**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.

4. Procedure completed.

#2 TAIL RTR SERVO ON CAPSULE OR LEGEND DOES NOT GO ON

SYMPTOM

#2 TAIL RTR SERVO ON capsule or legend does not go on.

MALFUNCTION

#2 TAIL RTR SERVO ON capsule or legend does not go on.

CORRECTIVE ACTION

1. Check helicopter configuration.

- a. **EMEP** , go to **2**.
- b. **W/O EMEP** , go to **4**.

2. **EMEP** Remove and check pin filtered adapter P118 or **HH-60A HH-60L** P1019R or P1020R (WP 0925 00).

- a. If pin filtered adapter is good, go to **3**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **8**.

3. **EMEP** Install pin filtered adapter (WP 0925 00). Go to **4**.

4. Check for 28 vdc between P118-S or **HH-60A HH-60L** P1019R-67 or P1020R-67 and ground.

- a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **8**.
- b. If voltage is not as specified, go to **5**.

5. Check for 28 vdc between P609-1 and ground.

- a. If voltage is as specified, go to **6**.
- b. If voltage is not as specified, go to **7**.

6. Check continuity between P118-S or **HH-60A HH-60L** P1019R-67 or P1020R-67 and P609-3.

- a. If continuity is present, replace 2nd stage pressure switch (WP 0726 00). Go to **8**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.

7. Check for 28 vdc between T RTR SERVO WARN circuit breaker terminal 1 and ground.

- a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P609-1 (WP 1747 00). Go to **8**.
- b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **8**.

#2 TAIL RTR SERVO ON CAPSULE OR LEGEND DOES NOT GO ON - Continued**8. Procedure completed.****#1 TAIL RTR SERVO CAPSULE OR LEGEND REMAINS ON WHEN NO. 1 PUMP MODULE IS PRESSURIZED****SYMPTOM**

#1 TAIL RTR SERVO capsule or legend remains on.

MALFUNCTION

#1 TAIL RTR SERVO capsule or legend remains on when No. 1 pump module is pressurized.

CORRECTIVE ACTION

- 1. Disconnect P610. Go to 2.**
- 2. Move pilot's or copilot's pedals.**
 - a. If pedals move normally, replace 1st stage pressure switch (WP 1149 00). Go to **5**.
 - b. If pedals do not move normally, go to **3**.
- 3. Grasp 1st stage pressure line at tail rotor servo. Go to 4.**
- 4. Cycle TAIL SERVO switch from NORMAL to BACKUP several times and check that a jump is felt in pressure line.**
 - a. If jump is felt, replace tail rotor servo (WP 1148 00). Go to **5**.
 - b. If jump is not felt, replace No. 1 transfer module (WP 0718 00). Go to **5**.
- 5. Procedure completed.**

#2 TAIL RTR SERVO ON CAPSULE OR LEGEND REMAINS ON WHEN NO. 1 PUMP MODULE IS PRESSURIZED**SYMPTOM**

#2 TAIL RTR SERVO ON capsule or legend remains on.

MALFUNCTION

#2 TAIL RTR SERVO ON capsule or legend remains on when No. 1 pump module is pressurized.

CORRECTIVE ACTION

- 1. Disconnect P609. Go to 2.**
- 2. Move pilot's or copilot's pedals. Check that pedals move normally.**
 - a. If pedals move normally, replace 2nd stage pressure switch (WP 1149 00). Go to **5**.
 - b. If pedals do not move normally, go to **3**.
- 3. Grasp 2nd stage pressure line at tail rotor servo. Go to 4.**
- 4. Cycle TAIL SERVO switch from NORMAL to BACKUP several times and check that a jump is felt in pressure line.**
 - a. If jump is felt, replace tail rotor servo (WP 1148 00). Go to **5**.
 - b. If jump is not felt, replace backup pump module (WP 0725 00). Go to **5**.
- 5. Procedure completed.**

#1 HYD PUMP CAPSULE OR LEGEND DOES NOT GO ON**SYMPTOM**

#1 HYD PUMP capsule or legend does not go on.

MALFUNCTION

#1 HYD PUMP capsule or legend does not go on.

CORRECTIVE ACTION

1. Check for 28 vdc between P117-24 or HH-60A HH-60L P1019R-23 or P1020R-23 and ground.
 - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 8.
 - b. If voltage is not as specified, go to 2.
2. Check for 28 vdc between P421-6 and ground.
 - a. If voltage is as specified, go to 3.
 - b. If voltage is not as specified, go to 7.
3. Check helicopter configuration.
 - a. EMEP , go to 4.
 - b. W/O EMEP , go to 6.
4. EMEP Remove and check pin filtered adapter P117 or HH-60A HH-60L P1019R or P1020R (WP 0925 00).
 - a. If pin filtered adapter is good, go to 5.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 8.
5. EMEP Install pin filtered adapter (WP 0925 00). Go to 6.
6. Check continuity between P117-24 or HH-60A HH-60L P1019R-23 or P1020R-23 and P421-1.
 - a. If continuity is present, replace pressure switch (WP 0721 00). Go to 8.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 8.
7. Check for 28 vdc between NO. 1 SERVO WARN circuit breaker terminal 1 and ground.
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P421-6 (WP 1747 00). Go to 8.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to 8.
8. Procedure completed.

#1 PRI SERVO PRESS CAPSULE OR LEGEND DOES NOT GO ON**SYMPTOM**

#1 PRI SERVO PRESS capsule or legend does not go on.

MALFUNCTION

#1 PRI SERVO PRESS capsule or legend does not go on.

#1 PRI SERVO PRESS CAPSULE OR LEGEND DOES NOT GO ON - Continued

CORRECTIVE ACTION**1. Check helicopter configuration.**

- a. **EMEP** , go to **2**.
- b. **EMEP** , go to **2**.
- c. **W/O EMEP** , go to **4**.

2. EMEP Remove and check pin filtered adapter P117 or HH-60A HH-60L P1019R or P1020R (WP 0925 00).

- a. If pin filtered adapter is good, go to **3**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **7**.

3. EMEP Install pin filtered adapter (WP 0925 00). Go to 4.**4. Check for 28 vdc between P117-8 or HH-60A HH-60L P1019R-60 or P1020R-60 and ground.**

- a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **7**.
- b. If voltage is not as specified, go to **5**.

5. Check continuity between:

- P117-8 and P424-1
- P117-8 and P425-1
- P117-8 and P426-1
- HH-60A HH-60L P1019R-76 and P424-1
- HH-60A HH-60L P1019R-76 and P425-1
- HH-60A HH-60L P1019R-76 and P426-1
- HH-60A HH-60L P1020R-76 and P424-1
- HH-60A HH-60L P1020R-76 and P425-1
- HH-60A HH-60L P1020R-76 and P426-1

- a. If continuity is present, go to **6**.
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **7**.

6. Check continuity between:

- J424-1 and J424-6
- J425-1 and J425-6
- J426-1 and J426-6

- a. If continuity is present, repair/replace wiring between NO. 1 SERVO WARN circuit breaker as required (WP 1747 00), go to **7**:
 - Terminal 1 and P424-6

#1 PRI SERVO PRESS CAPSULE OR LEGEND DOES NOT GO ON - Continued

- Terminal 1 and P425-6
 - Terminal 1 and P426-6
- b. If continuity is not present, replace 1st stage pressure switch as required (WP 0721 00). Go to **7**.

7. Procedure completed.**#1 TAIL RTR SERVO CAPSULE OR LEGEND DOES NOT GO ON****SYMPTOM**

#1 TAIL RTR SERVO capsule or legend does not go on.

MALFUNCTION

#1 TAIL RTR SERVO capsule or legend does not go on.

CORRECTIVE ACTION**1. Check helicopter configuration.**

- a. **EMEP** , go to **2**.
- b. **W/O EMEP** , go to **4**.

2. EMEP Remove and check pin filtered adapter P117 or HH-60A HH-60L P1019R or P1020R (WP 0925 00).

- a. If pin filtered adapter is good, go to **3**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **9**.

3. EMEP Install pin filtered adapter (WP 0925 00). Go to 4.**4. Check for 28 vdc between P117-57 or HH-60A HH-60L P1019R-66 or P1020R-66 and ground.**

- a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **9**.
- b. If voltage is not as specified, go to **5**.

5. Check for 28 vdc between P243-M and ground.

- a. If voltage is as specified, go to **6**.
- b. If voltage is not as specified, go to **7**.

6. Check continuity between P117-57 or HH-60A HH-60L P1019R-66 or P1020R-66 and P902-z.

- a. If continuity is present, replace left relay panel (WP 0825 00). Go to **9**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9**.

7. Check for 28 vdc between P610-7 and ground.

- a. If voltage is as specified, go to **8**.
- b. If voltage is not as specified, repair/replace wiring between P610-7 and NO. 1 SERVO CONTR circuit breaker terminal 1 (WP 1747 00). Go to **9**.

8. Check continuity between P243-M and P610-4.

#1 TAIL RTR SERVO CAPSULE OR LEGEND DOES NOT GO ON - Continued

- a. If continuity is present, replace 1st stage pressure switch (WP 0721 00). Go to **9**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9**.

9. Procedure completed.**#1 PRI SERVO PRESS CAPSULE OR LEGEND REMAINS ON WHEN NO. 1 SYSTEM IS PRESSURIZED****SYMPTOM**

#1 PRI SERVO PRESS capsule or legend remains on.

MALFUNCTION

1 PRI SERVO PRESS capsule or legend remains on when No. 1 system is pressurized.

CORRECTIVE ACTION**1. Remove P424.**

- a. If capsule or legend goes off, replace lateral primary servo 1st stage pressure switch (WP 0721 00). Go to **3**.
- b. If capsule or legend does not go off, go to **2**.

2. Remove P425.

- a. If capsule or legend goes off, replace aft primary servo 1st stage pressure switch (WP 0721 00). Go to **3**.
- b. If capsule or legend does not go off, replace forward primary servo 1st stage pressure switch (WP 0721 00). Go to **3**.

3. Procedure completed.**#1 PRI SERVO PRESS CAPSULE OR LEGEND DOES NOT GO ON WHEN 1ST STAGE IS SHUT OFF****SYMPTOM**

#1 PRI SERVO PRESS capsule or legend does not go on.

MALFUNCTION

#1 PRI SERVO PRESS capsule or legend does not go on when 1st stage is shut off.

CORRECTIVE ACTION**1. If pilot's SVO OFF switch was used, place copilot's SVO OFF switch to 1ST STG. If copilot's SVO OFF switch was used, place pilot's SVO OFF switch to 1ST STG. Check that capsule or legend goes on.**

- a. If capsule or legend goes on, replace copilot's or pilot's collective stick as required (WP 1061 00). Go to **14**.
- b. If capsule or legend does not go on, go to **2**.

2. Check for 28 vdc between P417-1 and P417-2.

- a. If voltage is as specified, replace No. 1 transfer module (WP 0720 00). Go to **14**.
- b. If voltage is not as specified, go to **3**.

3. Check for 28 vdc between P417-1 and ground.

#1 PRI SERVO PRESS CAPSULE OR LEGEND DOES NOT GO ON WHEN 1ST STAGE IS SHUT OFF -

Continued

- a. If voltage is as specified, repair/replace wiring between P417-2 and ground (WP 1747 00). Go to **14**.
- b. If voltage is not as specified, go to **4**.
- 4. Check for 28 vdc between P429-9 and ground.**
 - a. If voltage is as specified, go to **5**.
 - b. If voltage is not as specified, go to **6**.
- 5. Check continuity between P417-1 and P429-4.**
 - a. If continuity is present, replace forward primary servo 2nd stage pressure switch (WP 0726 00). Go to **14**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 6. Check for 28 vdc between P428-9 and ground.**
 - a. If voltage is as specified, go to **7**.
 - b. If voltage is not as specified, go to **8**.
- 7. Check continuity between P428-4 and P429-9.**
 - a. If continuity is present, replace aft primary servo 2nd stage pressure switch (WP 0726 00). Go to **14**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 8. Check for 28 vdc between P427-9 and ground.**
 - a. If voltage is as specified, go to **9**.
 - b. If voltage is not as specified, go to **10**.
- 9. Check continuity between P427-4 and P428-9.**
 - a. If continuity is present, replace lateral primary servo 2nd stage pressure switch (WP 0726 00). Go to **14**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.
- 10. With pilot's SVO OFF switch at 1ST STG, check for 28 vdc between:**
 - **J103-f and ground**
 - **J104-f and ground**
 - a. If voltage is as specified, go to **11**.
 - b. If voltage is not as specified, repair/replace wiring between NO. 1 SERVO CONTR circuit breaker as required (WP 1747 00). Go to **14**.
 - terminal 1 and J103-f
 - terminal 1 and J104-f
- 11. Check helicopter configuration.**
 - a. **UH-60A 85-24415 - SUBQ MOD UH-60L EH-60A HH-60A HH-60L** , go to **12**.
 - b. **UH-60A 77-22714 - 85-24414 W/O MOD** , go to **13**.

#1 PRI SERVO PRESS CAPSULE OR LEGEND DOES NOT GO ON WHEN 1ST STAGE IS SHUT OFF - Continued

12. **UH-60A 85-24415 SUBQ, MOD, UH-60L, EH-60A, HH-60A, AND HH-60L** Check continuity between:
 - J103-J and P427-9
 - J104-J and P427-9
 - a. If continuity is present, troubleshoot pilot's or copilot's collective stick as required (WP 0136 00). Go to **14**.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **14**.
13. **UH-60A 77-22714 - 77-22723 W/O MOD** Check continuity between:
 - J103-a and P427-9
 - J104-a and P427-9
 - a. If continuity is present, troubleshoot pilot's or copilot's collective stick as required (WP 0136 00). Go to **14**.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **14**.
14. **Procedure completed.**

#2 HYD PUMP CAPSULE DOES NOT GO ON

SYMPTOM

#2 HYD PUMP capsule does not go on.

MALFUNCTION

#2 HYD PUMP capsule does not go on.

CORRECTIVE ACTION

1. **Check helicopter configuration.**
 - a. **EMEP** , go to **2**.
 - b. **W/O EMEP** , go to **4**.
2. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R (WP 0925 00).
 - a. If pin filtered adapter is good, go to **3**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **8**.
3. **EMEP** Install pin filtered adapter (WP 0925 00). Go to **4**.
4. **Check for 28 vdc between P117-40 or HH-60A HH-60L P1019R-53 or P1020R-53 and ground.**
 - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to **8**.
 - b. If voltage is not as specified, go to **5**.
5. **Check for 28 vdc between P420-6 and ground.**
 - a. If voltage is as specified, go to **6**.
 - b. If voltage is not as specified, go to **7**.

#2 HYD PUMP CAPSULE DOES NOT GO ON - Continued

6. Check continuity between P117-40 or **HH-60A HH-60L** P1019R-53 or P1020R-53 and P420-1.
 - a. If continuity is present, replace No. 2 hydraulic pump pressure switch (WP 0726 00). Go to 8.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 8.
7. Check for 28 vdc between NO. 2 SERVO WARN circuit breaker terminal 1 and ground.
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P420-6 (WP 1747 00). Go to 8.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to 8.
8. Procedure completed.

#2 PRI SERVO PRESS CAPSULE DOES NOT GO ON**SYMPTOM**

#2 PRI SERVO PRESS capsule does not go on.

MALFUNCTION

#2 PRI SERVO PRESS capsule does not go on.

CORRECTIVE ACTION

1. Check helicopter configuration.
 - a. **EMEP** , go to 2.
 - b. **W/O EMEP** , go to 4.
2. **EMEP** Remove and check pin filtered adapter P117 or **HH-60A HH-60L** P1019R or P1020R (WP 0925 00).
 - a. If pin filtered adapter is good, go to 3.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 7.
3. **EMEP** Install pin filtered adapter (WP 0925 00). Go to 4.
4. Check for 28 vdc between P117-56 or **HH-60A HH-60L** P1019R-60 or P1020R-60 and ground.
 - a. If voltage is as specified, troubleshoot caution/advisory warning system (WP 0090 00) or MFD/caution/advisory warning system (WP 0100 00). Go to 7.
 - b. If voltage is not as specified, go to 5.
5. Check continuity between:
 - P117-56 and P427-1
 - P117-56 and P428-1
 - P117-56 and P429-1
 - **HH-60A HH-60L** P1019R-60 and P427-1
 - **HH-60A HH-60L** P1019R-60 and P428-1

#2 PRI SERVO PRESS CAPSULE DOES NOT GO ON - Continued

- **HH-60A HH-60L P1019R-60 and P429-1** ■
- **HH-60A HH-60L P1020R-60 and P427-1** ■
- **HH-60A HH-60L P1020R-60 and P428-1** ■
- **HH-60A HH-60L P1020R-60 and P429-1** ■

- a. If continuity is present, go to **6**.
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **7**.

6. Check continuity between:

- **J427-1 and J427-6**
- **J428-1 and J428-6**
- **J429-1 and J429-6**

- a. If continuity is present, repair/replace wiring from NO. 2 SERVO WARN circuit breaker as required (WP 1747 00), go to **7**:

- Terminal 1 and P427-6
- Terminal 1 and P428-6
- Terminal 1 and P429-6

- b. If continuity is not present, replace 2nd stage pressure switch as required (WP 0726 00). Go to **7**.

7. Procedure completed.**PILOT ASSIST MODULE PRESSURE REDUCER INDICATOR BUTTON IS POPPED****SYMPTOM**

Pilot assist module pressure reducer indicator button is popped.

MALFUNCTION

Pilot assist module pressure reducer indicator button is popped.

CORRECTIVE ACTION**1. Turn on backup hydraulic system to reset button. Check that button resets.**

- a. If button resets, go to **2**.
- b. If button does not reset, replace pilot assist module (WP 0728 00). Go to **4**.

2. Do not engage/disengage solenoids or run No. 1 or No. 2 hydraulic systems. Go to 3.**3. Run backup hydraulic system for two (2) minutes.**

- a. If button is popped, replace pilot assist module (WP 0728 00). Go to **4**.

4. Procedure completed.

#2 PRI SERVO PRESS CAPSULE REMAINS ON WHEN NO. 2 SYSTEM IS PRESSURIZED**SYMPTOM**

#2 PRI SERVO PRESS capsule remains on.

MALFUNCTION

#2 PRI SERVO PRESS capsule remains on when No. 2 system is pressurized.

CORRECTIVE ACTION**1. Remove P427.**

- a. If capsule or legend goes off, replace lateral primary servo 2nd stage pressure switch (WP 0726 00). Go to **3**.
- b. If capsule or legend does not go off, go to **2**.

2. Remove P428.

- a. If capsule or legend goes off, replace forward primary servo 2nd stage pressure switch (WP 0726 00). Go to **3**.
- b. If capsule or legend does not go off, replace aft primary servo 2nd stage pressure switch (WP 0726 00). Go to **3**.

3. Procedure completed.**#2 PRI SERVO PRESS CAPSULE DOES NOT GO ON WHEN 2ND STAGE IS SHUT OFF****SYMPTOM**

#2 PRI SERVO PRESS capsule does not go on.

MALFUNCTION

#2 PRI SERVO PRESS capsule does not go on when 2nd stage is shut off.

CORRECTIVE ACTION**1. If pilot's SVO OFF switch was used, place copilot's SVO OFF switch to 2ND STG. If copilot's SVO OFF switch was used, place pilot's SVO OFF switch to 2ND STG. Go to 2.****2. Check that #2 PRI SERVO PRESS capsule or legend goes on.**

- a. If capsule or legend goes on, replace pilot's or copilot's collective stick as required (WP 1061 00). Go to **13**.
- b. If capsule or legend does not go on, go to **3**.

3. Check for 28 vdc between P416-1 and P416-2.

- a. If voltage is as specified, replace No. 2 transfer module (WP 0725 00). Go to **13**.
- b. If voltage is not as specified, go to **4**.

4. Check for 28 vdc between P416-1 and ground.

- a. If voltage is as specified, repair/replace wiring between P416-2 and ground (WP 1747 00). Go to **13**.
- b. If voltage is not as specified, go to **5**.

5. Check for 28 vdc between P426-9 and ground.

#2 PRI SERVO PRESS CAPSULE DOES NOT GO ON WHEN 2ND STAGE IS SHUT OFF - Continued

- a. If voltage is as specified, go to **6**.
- b. If voltage is not as specified, go to **7**.
- 6. Check continuity between P416-1 and P426-4.**
 - a. If continuity is present, replace forward primary servo 1st stage pressure switch (WP 0721 00). Go to **13**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 7. Check for 28 vdc between P425-9 and ground.**
 - a. If voltage is as specified, go to **8**.
 - b. If voltage is not as specified, go to **9**.
- 8. Check continuity between P425-4 and P426-9.**
 - a. If continuity is present, replace aft primary servo 1st stage pressure switch (WP 0721 00). Go to **13**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 9. Check for 28 vdc between P424-9 and ground.**
 - a. If voltage is as specified, go to **10**.
 - b. If voltage is not as specified, go to **11**.
- 10. Check continuity between P424-4 and P425-9.**
 - a. If continuity is present, replace lateral primary servo 1st stage pressure switch (WP 0721 00). Go to **13**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 11. Check for 28 vdc between:**
 - **J103-c and ground**
 - **J104-c and ground**
 - a. If voltage is as specified, go to **12**.
 - b. If voltage is not as specified, repair/replace wiring from NO. 2 SERVO CONTR circuit breaker as required (WP 1747 00), go to **13**:
 - terminal 1 and J103-c
 - terminal 1 and J104-c
- 12. Check continuity between:**
 - **J103-d and P424-9**
 - **J104-d and P424-9**
 - a. If continuity is present, troubleshoot pilot's or copilot's collective stick as required (WP 0136 00). Go to **13**.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **13**.
- 13. Procedure completed.**

BACKUP PUMP FAILS TO RUN WITH BACKUP HYD PUMP SWITCH PLACED ON**SYMPTOM**

Backup pump fails to run.

MALFUNCTION

Backup pump fails to run with BACKUP HYD PUMP switch placed on.

CORRECTIVE ACTION

- 1. Place locally-made wedge (Figure 167, WP 1805 00) in left drag beam switch to simulate weight off wheels. Go to 2.**
- 2. Check that backup pump runs.**
 - a. If backup pump runs, go to **3**.
 - b. If backup pump does not run, go to **6**.
- 3. Remove locally-made wedge. Go to 4.**
- 4. Check continuity between NO. 3 HYD PUMP RELAY terminal X2 and P422-2.**
 - a. If continuity is present, go to **5**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **32**.
- 5. Check continuity between P422-1 and ground.**
 - a. If continuity is present, replace backup pump module (WP 0707 00). Go to **32**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **32**.
- 6. Remove locally-made wedge. Go to 7.**
- 7. Check for 115 vac between:**
 - **P423-A and P423-D**
 - **P423-B and P423-D**
 - **P423-C and P423-D**
 - a. If voltage is as specified, replace backup pump (WP 0707 00). Go to **32**.
 - b. If voltage is not as specified, go to **8**.
- 8. Check continuity between P423-D and ground.**
 - a. If continuity is present, go to **9**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **32**.
- 9. Check for 115 vac between K19 NO. 3 HYD PUMP RELAY:**
 - **Terminal A2 and ground**
 - **Terminal B2 and ground**
 - **Terminal C2 and ground**
 - a. If voltage is as specified, repair/replace wiring between relay and P423 as required (WP 1747 00). Go to **32**.
 - b. If voltage is not as specified, go to **10**.

BACKUP PUMP FAILS TO RUN WITH BACKUP HYD PUMP SWITCH PLACED ON - Continued

10. Check for 115 vac between:
 - K19-A1 and ground
 - K19-B1 and ground
 - K19-C1 and ground
 - a. If voltage is as specified, go to **11**.
 - b. If voltage is not as specified, go to **30**.
11. Check for 28 vdc between K19-X1 and K19-X2.
 - a. If voltage is as specified, replace relay K19 (WP 0873 00). Go to **32**.
 - b. If voltage is not as specified, go to **12**.
12. Check for 28 vdc between K19-X1 and ground.
 - a. If voltage is as specified, go to **13**.
 - b. If voltage is not as specified, go to **17**.
13. Check for 28 vdc between P902-CC and ground.
 - a. If voltage is as specified, go to **14**.
 - b. If voltage is not as specified, repair/replace wiring between P902-CC and BACKUP HYD CONTR circuit breaker terminal 1 (WP 1747 00). Go to **32**.
14. Check continuity between K19-X2 and P902-BB.
 - a. If continuity is present, go to **15**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **32**.
15. Check continuity between:
 - J163-A and ground
 - J163-C and P242-L
 - a. If continuity is present, go to **16**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **32**.
16. Check continuity between P163-A and P163-C.
 - a. If continuity is present, replace left relay panel (WP 0825 00). Go to **32**.
 - b. If continuity is not present, replace left drag beam switch (WP 0443 00). Go to **32**.
17. Check helicopter configuration.
 - a. **WINTER** , go to **18**.
 - b. **W/O WINTER** , go to **20**.
18. **WINTER** Check for 28 vdc between J244B-C and ground.
 - a. If voltage is as specified, go to **19**.
 - b. If voltage is not as specified, replace **WINTER** (WP 1400 00). Go to **32**.
 - c. If trouble remains, go to **20**.
19. **WINTER** Check continuity between J244B-A and K19-X1.

BACKUP PUMP FAILS TO RUN WITH BACKUP HYD PUMP SWITCH PLACED ON - Continued

- a. If continuity is present, go to **22**.
- b. If continuity is not present, replace **WINTER** (WP 1400 00). Go to **32**.
- c. If trouble remains, go to **21**.

20. Check for 28 vdc between P244-C and ground.

- a. If voltage is as specified, go to **21**.
- b. If voltage is not as specified, go to **27**.

21. Check continuity between K19-X1 and P244-A.

- a. If continuity is present, go to **22**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **32**.

22. Check for 28 vdc between P241-p and ground.

- a. If voltage is as specified, troubleshoot right relay panel (WP 0120 00). Go to **32**.
- b. If voltage is not as specified, go to **23**.

23. Check for 28 vdc between:

- **P204-E and ground**
- **P206-E and ground**
- **P212-E and ground**

- a. If voltage is as specified, go to **24**.
- b. If voltage is not as specified, go to **25**.

24. Check continuity between:

- **P204-F and P241-p**
- **P206-F and P241-p**
- **P212-F and P241-p**

- a. If continuity is present, troubleshoot DC electrical system (WP 0102 00). Go to **32**.
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **32**.

25. Check for 28 vdc between BACKUP HYD PUMP switch terminal 3 and ground.

- a. If voltage is as specified, repair/replace wiring from BACKUP HYD PUMP switch as required (WP 1747 00). Go to **32**.
 - **Terminal 3 and P204-E**
 - **Terminal 3 and P206-E**
 - **Terminal 3 and P212-E**
- b. If voltage is not as specified, go to **26**.

26. Check for 28 vdc between BACKUP HYD PUMP switch terminal 2 and ground.

- a. If voltage is as specified, replace switch (WP 0828 00). Go to **32**.

BACKUP PUMP FAILS TO RUN WITH BACKUP HYD PUMP SWITCH PLACED ON - Continued

- b. If voltage is not as specified, repair/replace wiring between BACKUP HYD CONTR circuit breaker terminal 1 and BACKUP HYD PUMP switch terminal 2 (WP 1747 00). Go to **32**.
- 27. Check helicopter configuration.**
- a. **WINTER** , go to **28**.
- b. **W/O WINTER** , go to **29**.
- 28. WINTER Check continuity between J244B-C and BACKUP HYD CONTR circuit breaker terminal 1.**
- a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **32**.
- b. If continuity is not present, replace **WINTER** (WP 1400 00). Go to **32**.
- c. If trouble remains, go to **29**.
- 29. Check continuity between P244-C and BACKUP HYD CONTR circuit breaker terminal 1.**
- a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **32**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **32**.
- 30. Check for 115 vac between HYD PWR PUMP circuit breaker:**
- **Terminal A2 and ground**
 - **Terminal B2 and ground**
 - **Terminal C2 and ground**
- a. If voltage is as specified, repair/replace wiring between circuit breaker and relay as required (WP 1747 00). Go to **32**.
- b. If voltage is not as specified, go to **31**.
- 31. Check for 115 vac between HYD PWR PUMP circuit breaker:**
- **Terminal A1 and ground**
 - **Terminal B1 and ground**
 - **Terminal C1 and ground**
- a. If voltage is as specified, replace circuit breaker (WP 0824 00). Go to **32**.
- b. If voltage is not as specified, troubleshoot AC electrical system (WP 0101 00). Go to **32**.
- 32. Procedure completed.**

**BACKUP PUMP RUNS CONTINUOUSLY WITH NO FAULT LIGHT DISPLAYED ON THE CAUTION/
ADVISORY PANEL HH-60A HH-60L MFD/Caution/Advisory Panel ■****SYMPTOM**

Backup pump runs continuously.

MALFUNCTION

Backup pump runs continuously with no fault light displayed on the caution/advisory panel

HH-60A HH-60L MFD/caution/advisory panel ■ .

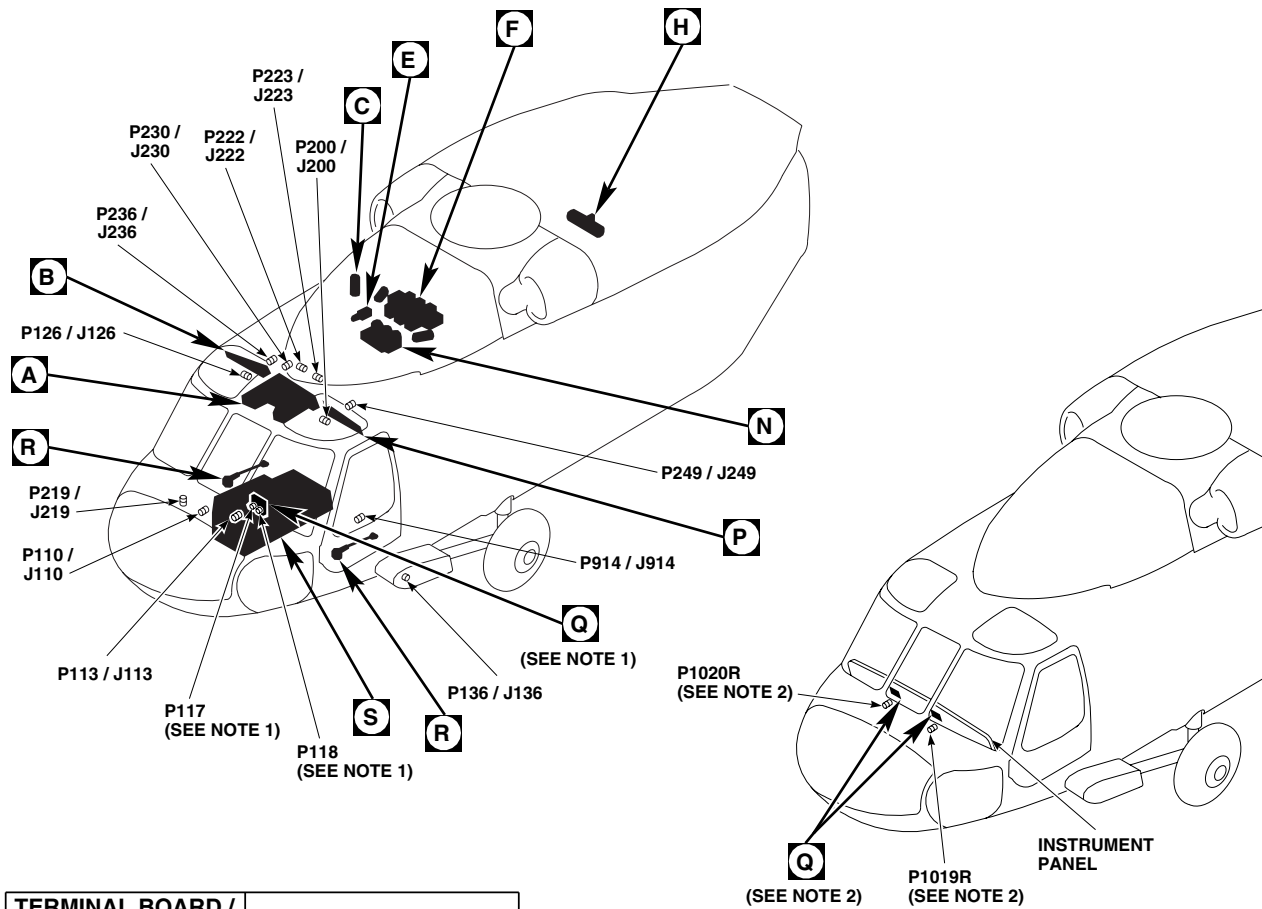
CORRECTIVE ACTION**1. Check continuity as indicated between:**

- P420-9 and P420-4 (open)
- P420-7 and P420-4 (continuity)
- P420-5 and P420-6 (open)
- P420-6 and P420-1 (continuity)
- P421-9 and P421-4 (open)
- P421-7 and P421-4 (continuity)
- P421-5 and P421-6 (open)
- P421-6 and P421-1 (continuity)

- a. If continuity is present, replace BACKUP HYD PUMP switch S31 (WP 0828 00). Go to **2**.
- b. If continuity is not present, replace applicable hydraulic pump pressure switch (WP 0721 00 or WP 0726 00). Go to **2**.

2. Procedure completed.

LOCATION



TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P3R / J3R	STABILATOR CONTROL / AUTO FLIGHT CONTROL PANEL
P4R / J4R	STABILATOR CONTROL / AUTO FLIGHT CONTROL PANEL
P103 / J103	COCKPIT TUB, BL 32 LH, STA 247
P104 / J104	COCKPIT TUB, BL 15 RH, STA 247
P110 / J110	COCKPIT, BL 7.5 RH, STA 197
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P113 / J113	COCKPIT, BL 6 LH, STA 197
P114 / J114	COCKPIT, BL 8 RH, STA 200
P117 / J117	CAUTION / ADVISORY PANEL (SEE NOTE 1)

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P118 / J118	CAUTION / ADVISORY PANEL (SEE NOTE 1)
P126 / J126	COCKPIT CEILING, BL 28 RH, STA 243
P127 / J127	COCKPIT CEILING, BL 28 LH, STA 243
P135 / J135	MISCELLANEOUS SWITCH PANEL
P137 / J1	NO. 2 SIGNAL DATA CONVERTER
P138 / J1	NO. 1 SIGNAL DATA CONVERTER
P142 / J142	COCKPIT, BL 3 RH, STA 210
P163 / J163	LEFT DRAG BEAM SWITCH
P200 / J200	COCKPIT CEILING, BL 13 LH, STA 246



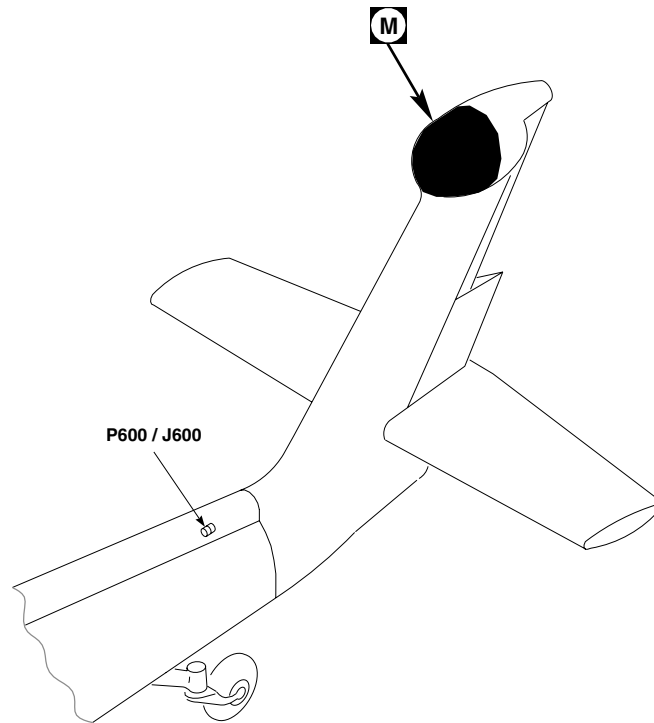
NOTES

1. EH-60A UH-60A UH-60L
2. HH-60L HH-60A

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Figure 1. Hydraulic Systems Location Diagram. (Sheet 1 of 13)

LOCATION - Continued



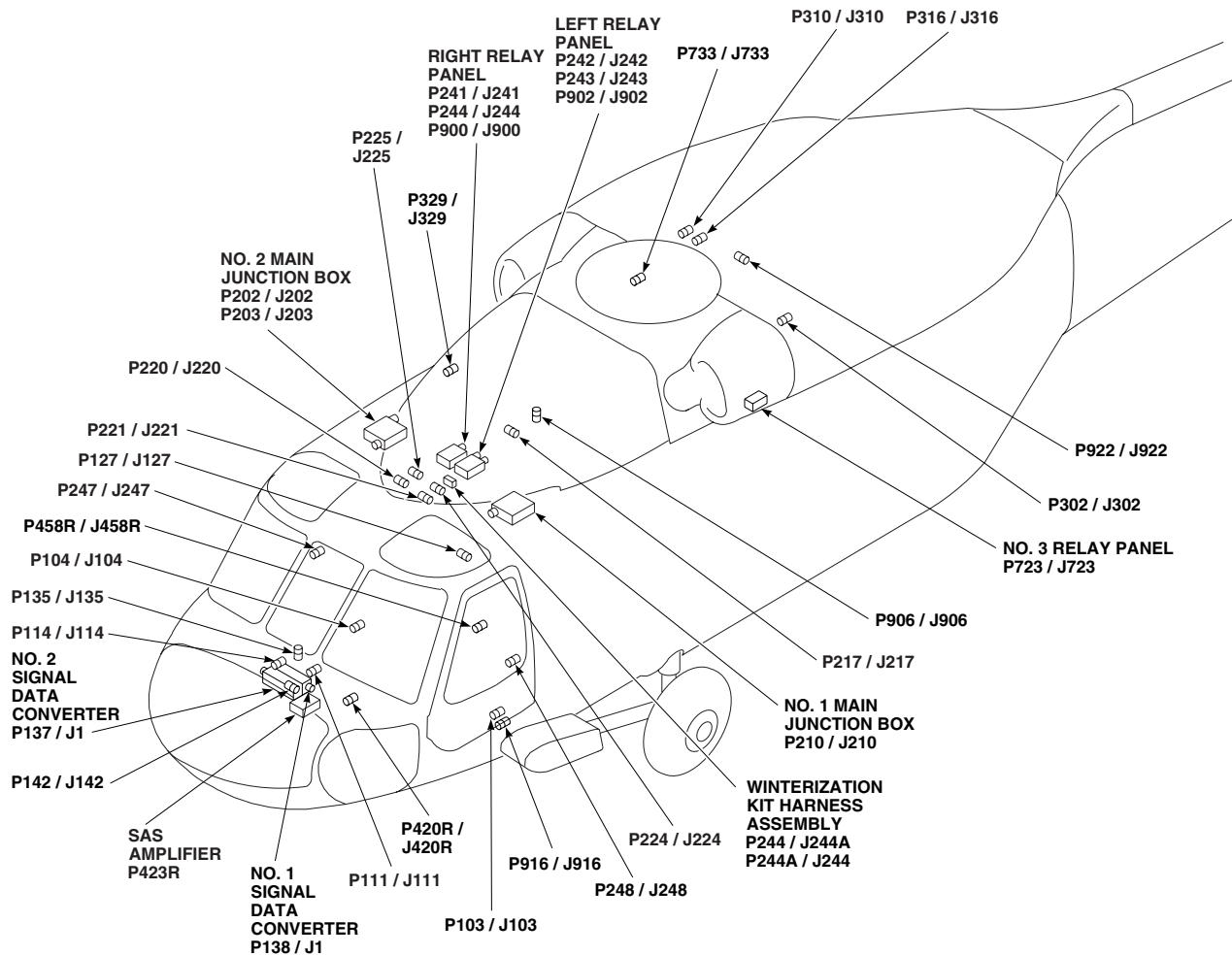
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P202 / J202	NO. 2 MAIN ELECTRICAL JUNCTION BOX
P203 / J203	NO. 2 MAIN ELECTRICAL JUNCTION BOX
P210 / J210	NO. 1 MAIN ELECTRICAL JUNCTION BOX
P217 / J217	CABIN CEILING, BL 0, STA 284
P219 / J219	CABIN FLOOR, BL 25 RH, STA 242
P220 / J220	JUNCTION BOX, MAIN ROTOR PYLON DECK
P221 / J221	JUNCTION BOX, MAIN ROTOR PYLON DECK
P222 / J222	JUNCTION BOX, MAIN ROTOR PYLON DECK
P223 / J223	JUNCTION BOX, MAIN ROTOR PYLON DECK

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P224 / J224	JUNCTION BOX, MAIN ROTOR PYLON DECK
P225 / J225	JUNCTION BOX, MAIN ROTOR PYLON DECK
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P236 / J236	BEHIND PILOT'S CIRCUIT BREAKER PANEL, BL 23.0 RH
P241 / J241	RIGHT RELAY PANEL ASSEMBLY
P242 / J242	LEFT RELAY PANEL ASSEMBLY
P243 / J243	LEFT RELAY PANEL ASSEMBLY
P244 / J244	RIGHT RELAY PANEL ASSEMBLY
P244 / J244A J244B / J244	WINTERIZATION KIT HARNESS ASSEMBLY

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Figure 1. Hydraulic Systems Location Diagram. (Sheet 2 of 13)

LOCATION - Continued



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P247 / J247	CABIN, BL 40 RH, STA 248
P248 / J248	CABIN, BL 40 LH, STA 248
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL 23.0 LH
P302 / J302	APU ACCUMULATOR PRESSURE SWITCH
P310 / J310	CABIN CEILING, BL 17 RH, STA 387
P316 / J316	CABIN CEILING, BL 17 RH, STA 387
P329 / J329	MAIN ROTOR BLADE DE-ICE JUNCTION BOX
P416 / J416	2ND STAGE SERVO SHUTOFF VALVE
P417 / J417	1ST STAGE SERVO SHUTOFF VALVE

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P420 / J420	NO. 2 HYDRAULIC PUMP PRESSURE SWITCH
P420R / J420R	COCKPIT, BL 0, STA 191.5
P421 / J421	NO. 1 HYDRAULIC PUMP PRESSURE SWITCH
P422 / J422	BACKUP PUMP MOTOR THERMAL SWITCH
P422R	SAS / FPS COMPUTER
P423 / J423	BACKUP PUMP MOTOR
P423R	SAS AMPLIFIER
P424 / J424	1ST STAGE PRESSURE SWITCH, LATERAL PRIMARY SERVO
P425 / J425	1ST STAGE PRESSURE SWITCH, AFT PRIMARY SERVO
P426 / J426	1ST STAGE PRESSURE SWITCH, FORWARD PRIMARY SERVO

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Figure 1. Hydraulic Systems Location Diagram. (Sheet 3 of 13)

LOCATION - Continued

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT	TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P427 / J427	2ND STAGE PRESSURE SWITCH, LATERAL PRIMARY SERVO	P464R / J464R	PITCH TRIM ACTUATOR ASSEMBLY
P428 / J428	2ND STAGE PRESSURE SWITCH, AFT PRIMARY SERVO	P466R / J466R	PITCH TRIM TURN-ON VALVE
P428R / J11	SAS / FPS COMPUTER	P467R / J467R	SAS PRESSURE SWITCH
P429 / J429	2ND STAGE PRESSURE SWITCH, FORWARD PRIMARY SERVO	P468R / J468R	SAS SHUTOFF VALVE
P432 / J432	NO. 2 HYDRAULIC PUMP MODULE, FLUID LEVEL SWITCH	P469R / J469R	BOOST SHUTOFF VALVE
P433	2ND STAGE TAIL ROTOR SHUTOFF VALVE	P470R / J470R	YAW BOOST SERVO PRESSURE SWITCH
P441 / J441	1ST STAGE TAIL ROTOR SHUTOFF VALVE	P600 / J600	TAIL CONE, BL 0, STA 650
P442 / J442	BACKUP PUMP MODULE FLUID LEVEL SWITCH	P609 / J609	TAIL ROTOR SERVO, 2ND STAGE PRESSURE SWITCH
P445 / J445	BACKUP PUMP PRESSURE SWITCH	P610 / J610	TAIL ROTOR SERVO, 1ST STAGE PRESSURE SWITCH
P446 / J446	DEPRESSURIZATION VALVE	P723 / J723	NO. 3 RELAY PANEL
P447 / J447	NO.1 HYDRAULIC PUMP MODULE, FLUID LEVEL SWITCH	P733 / J733	BL 24.71, STA 360, WL 265.30
P448	PILOT ASSIST SHUTOFF VALVE	P900 / J900	RH RELAY PANEL ASSEMBLY
P458R / J458R	TOP DECK, BL 4 LH, STA 262	P902 / J902	LH RELAY PANEL ASSEMBLY
P460R / J460R	PITCH TRIM SAS MODULE	P906 / J906	CABIN CEILING, BL 0, STA 284
P461R / J461R	ROLL SAS MODULE	P914 / J914	COCKPIT, BL 24 LH, STA 247
P462R / J462R	YAW SAS MODULE	P916 / J916	CABIN TUB, BL 25 RH, STA 247
P463R / J463R	COLLECTIVE BOOST SERVO PRESSURE SWITCH	P922 / J922	CABIN CEILING, BL 10.7 LH, STA 380
		P1019R / J3	COPILOT'S MFD'S / CAUTION ADVISORY PANEL (SEE NOTE 2)
		P1020R / J3	PILOT'S MFD'S / CAUTION ADVISORY PANEL (SEE NOTE 2)

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Figure 1. Hydraulic Systems Location Diagram. (Sheet 4 of 13)

LOCATION - Continued

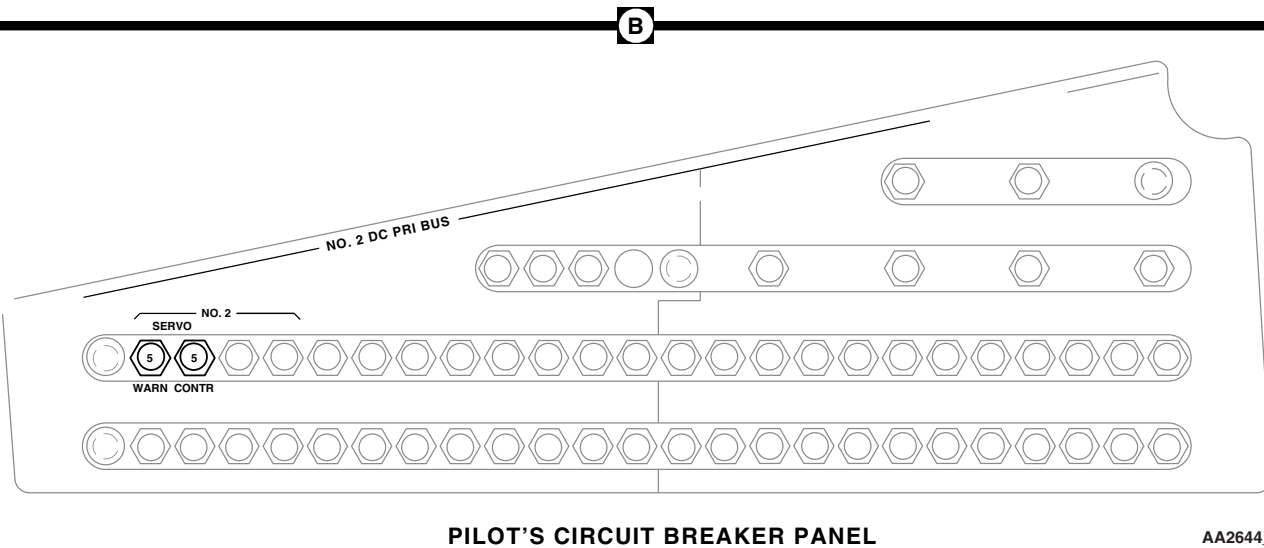
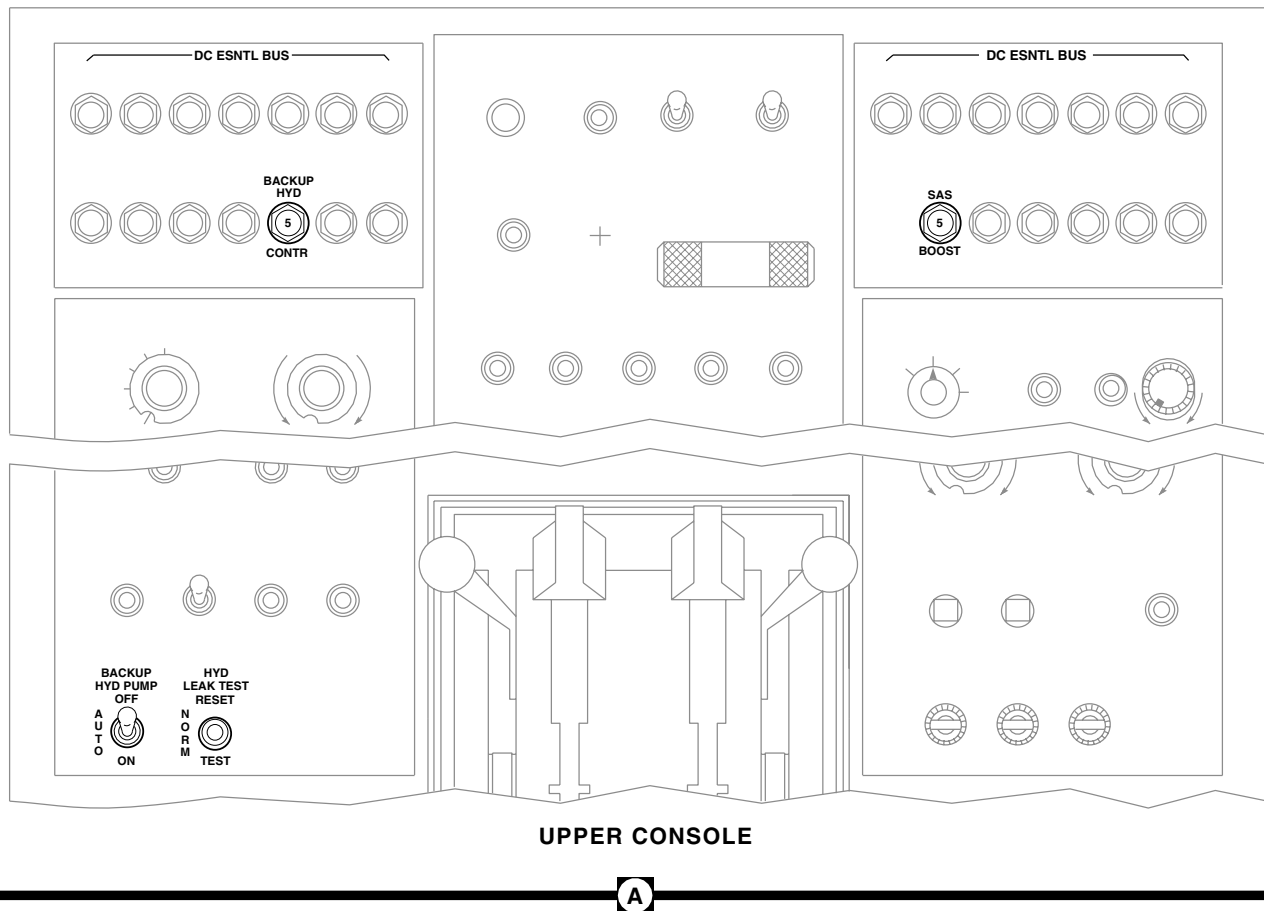


Figure 1. Hydraulic Systems Location Diagram. (Sheet 5 of 13)

LOCATION - Continued

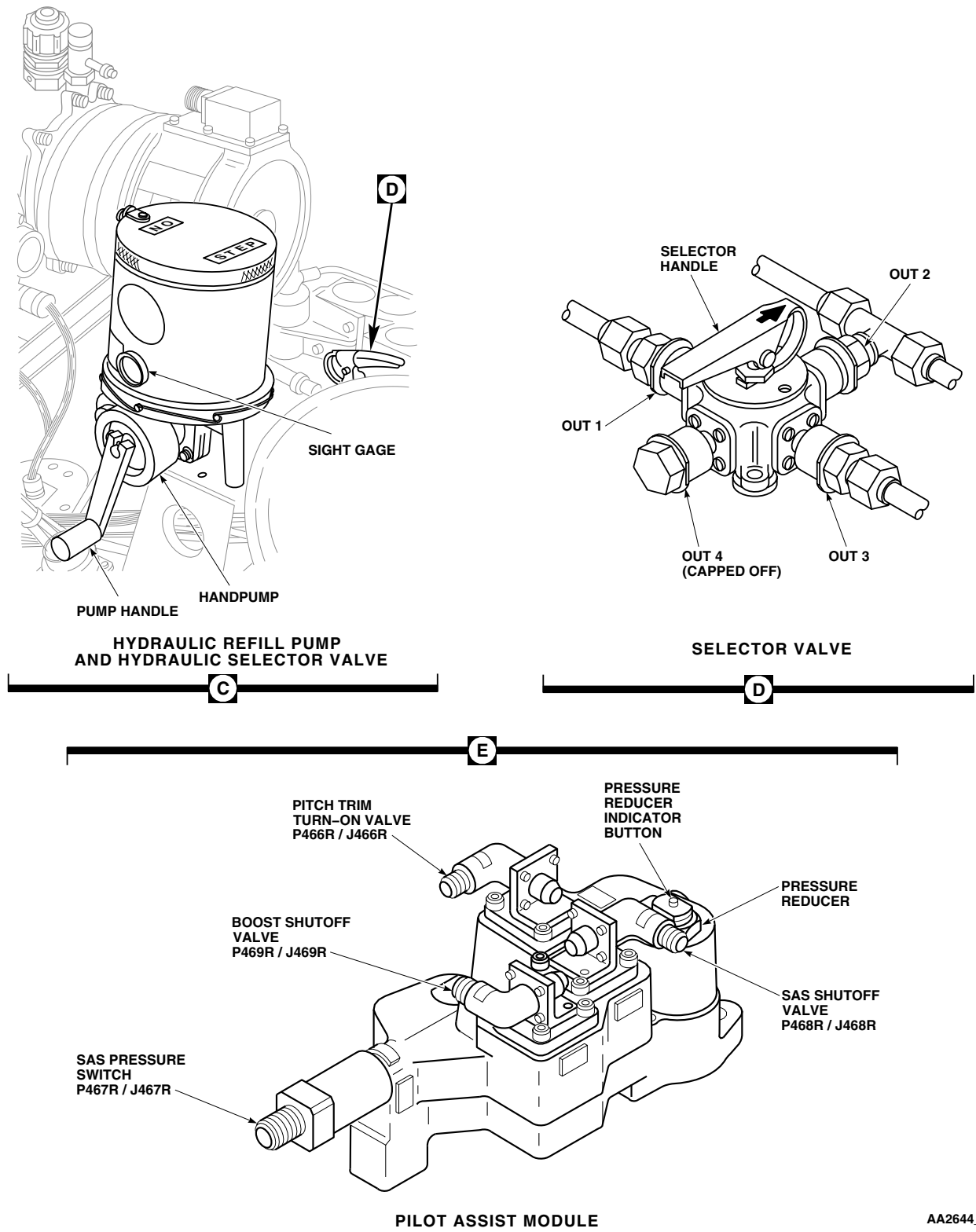
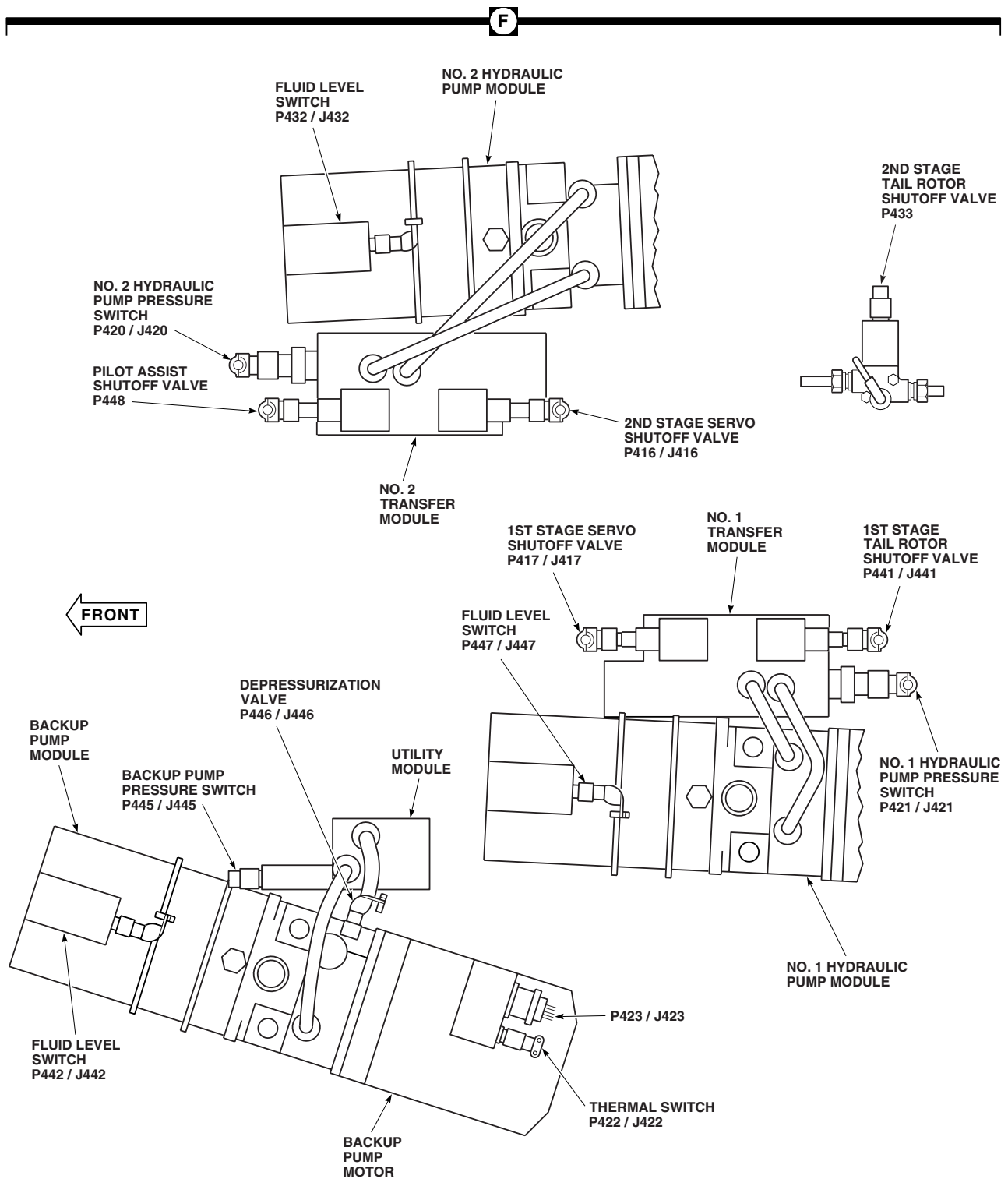


Figure 1. Hydraulic Systems Location Diagram. (Sheet 6 of 13)

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LOCATION - Continued

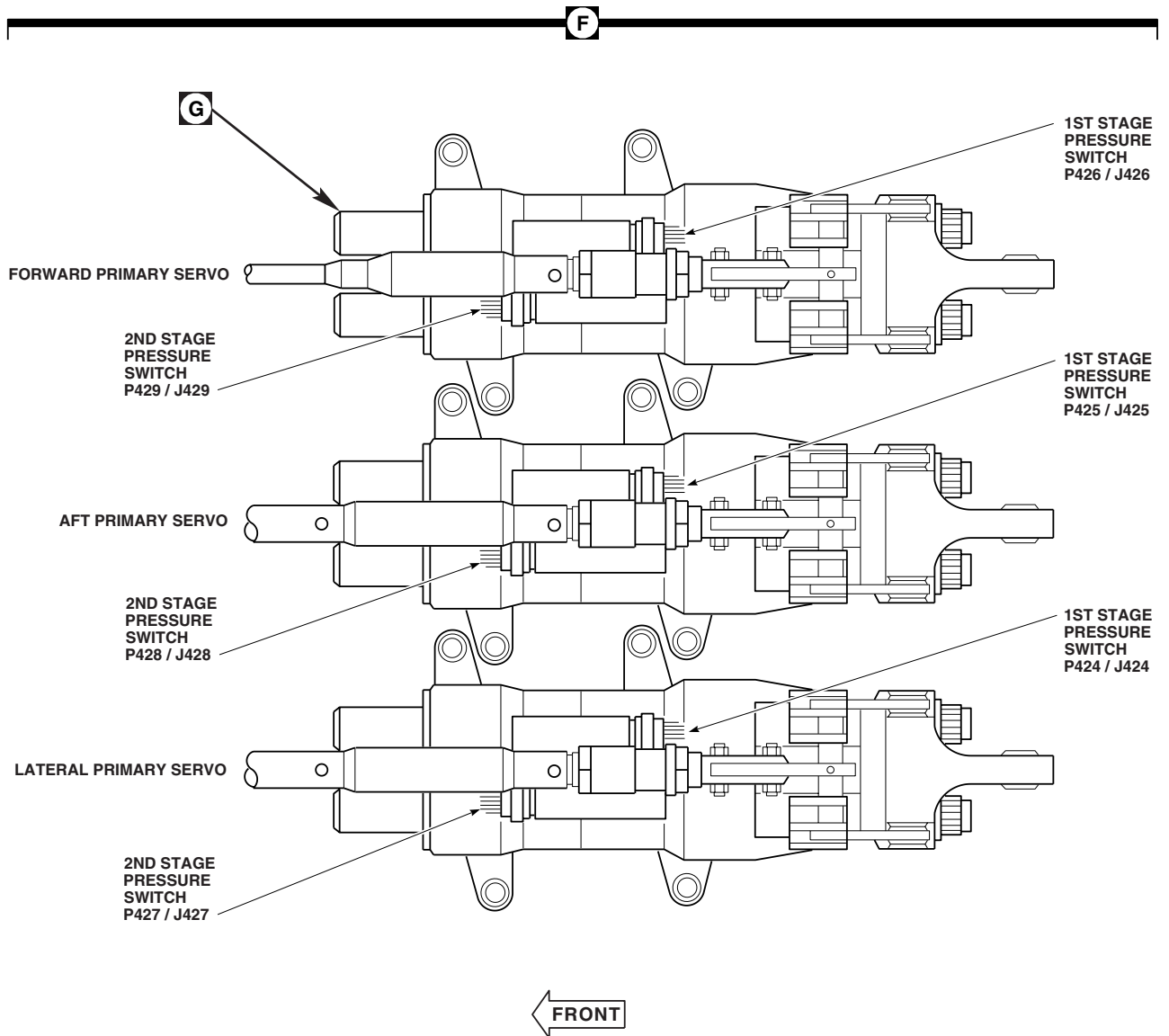


MAIN ROTOR PYLON HYDRAULIC COMPONENTS

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Figure 1. Hydraulic Systems Location Diagram. (Sheet 7 of 13)

LOCATION - Continued

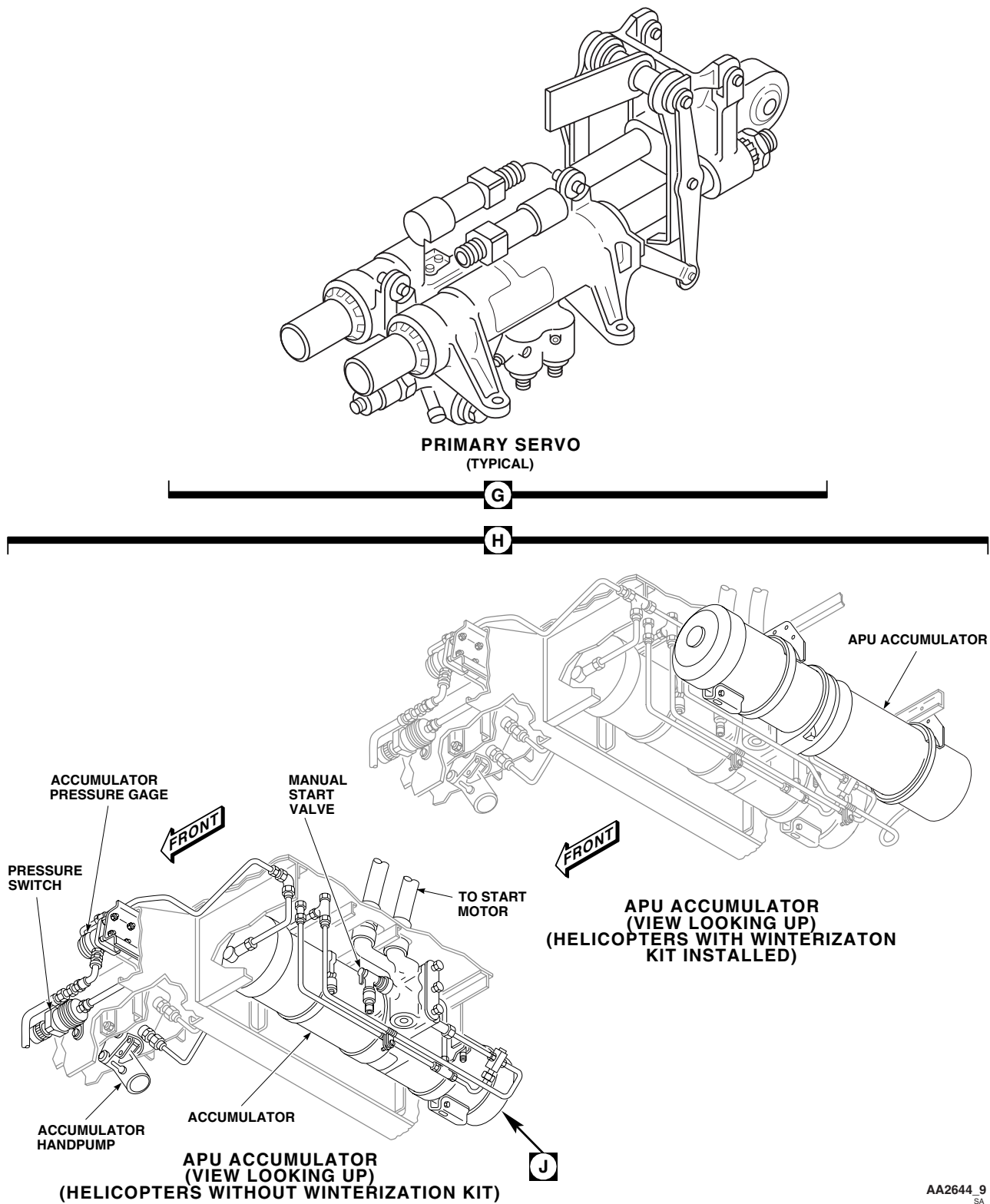


MAIN ROTOR PYLON HYDRAULIC COMPONENTS

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Figure 1. Hydraulic Systems Location Diagram. (Sheet 8 of 13)

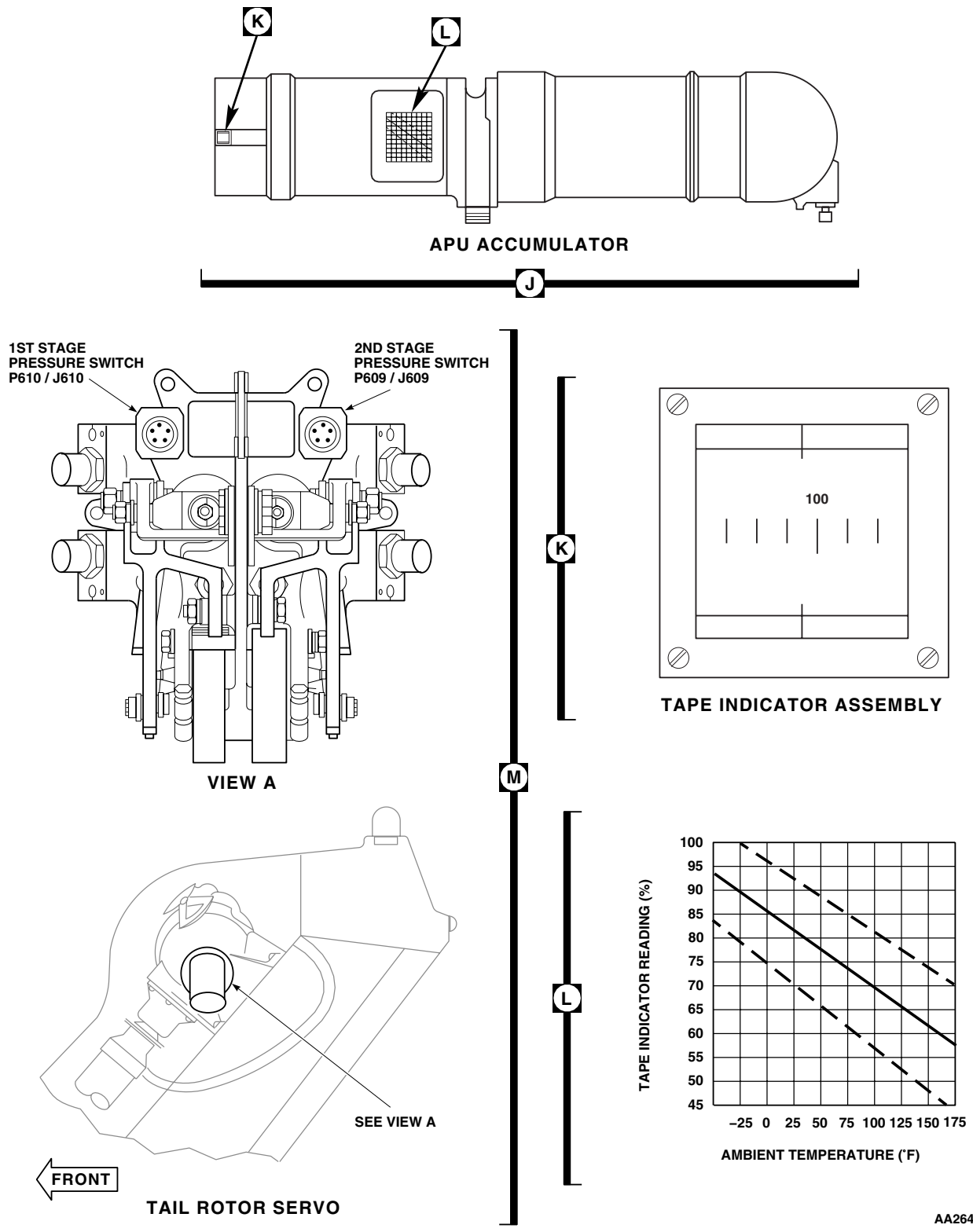
LOCATION - Continued



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Figure 1. Hydraulic Systems Location Diagram. (Sheet 9 of 13)

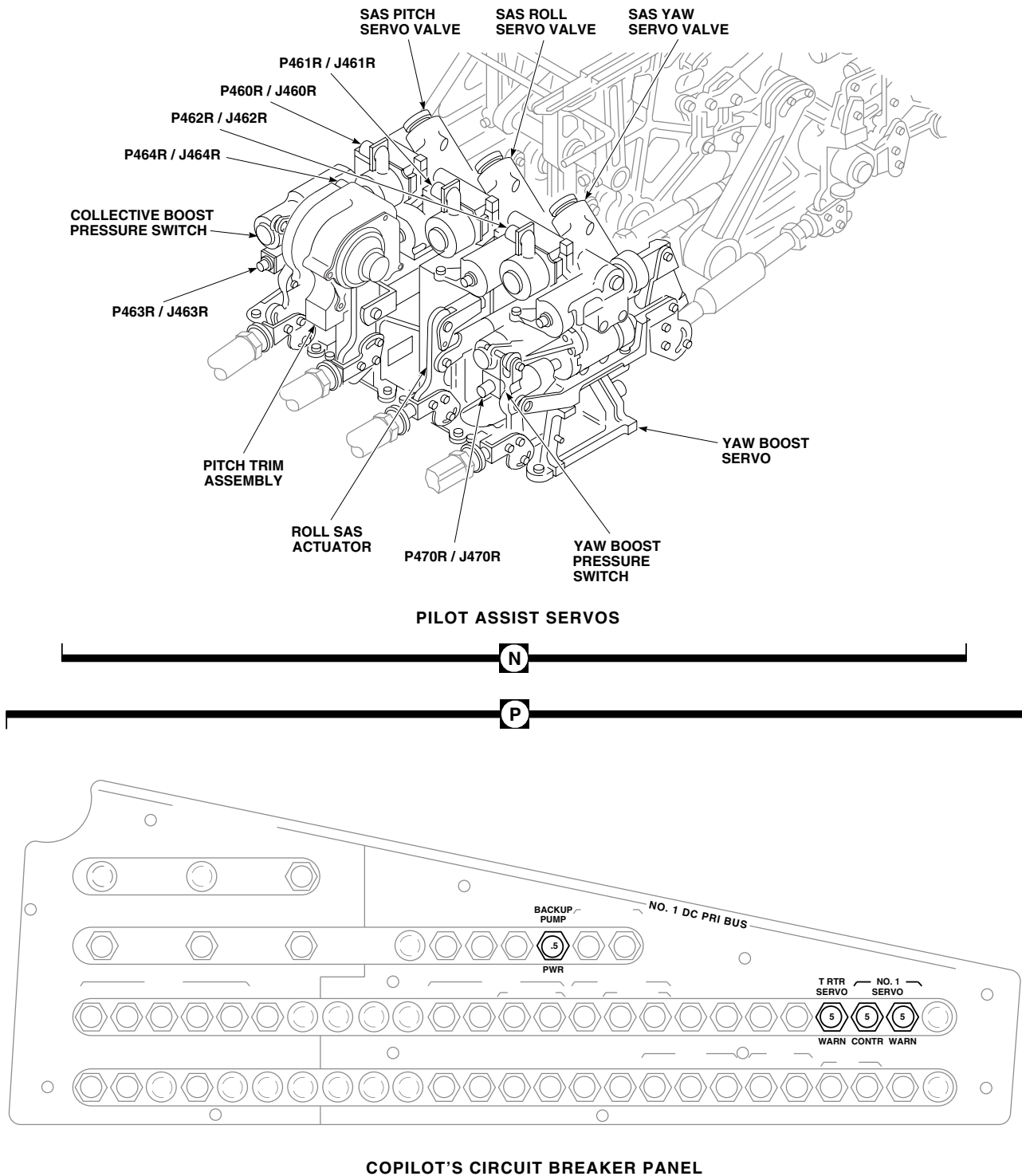
LOCATION - Continued



AA2644_10A
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Figure 1. Hydraulic Systems Location Diagram. (Sheet 10 of 13)

LOCATION - Continued



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SA

Figure 1. Hydraulic Systems Location Diagram. (Sheet 11 of 13)

Q



LOCATION - Continued

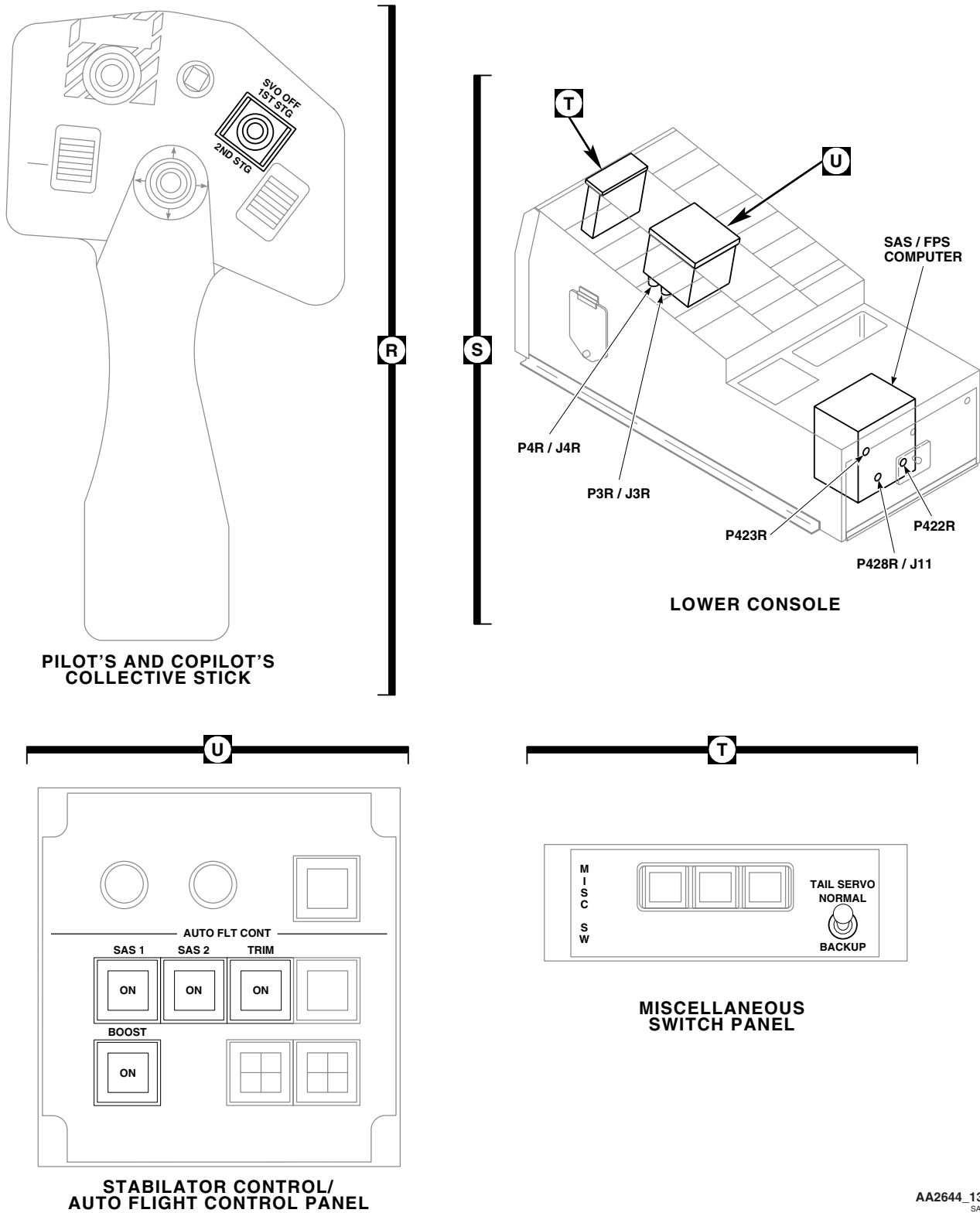
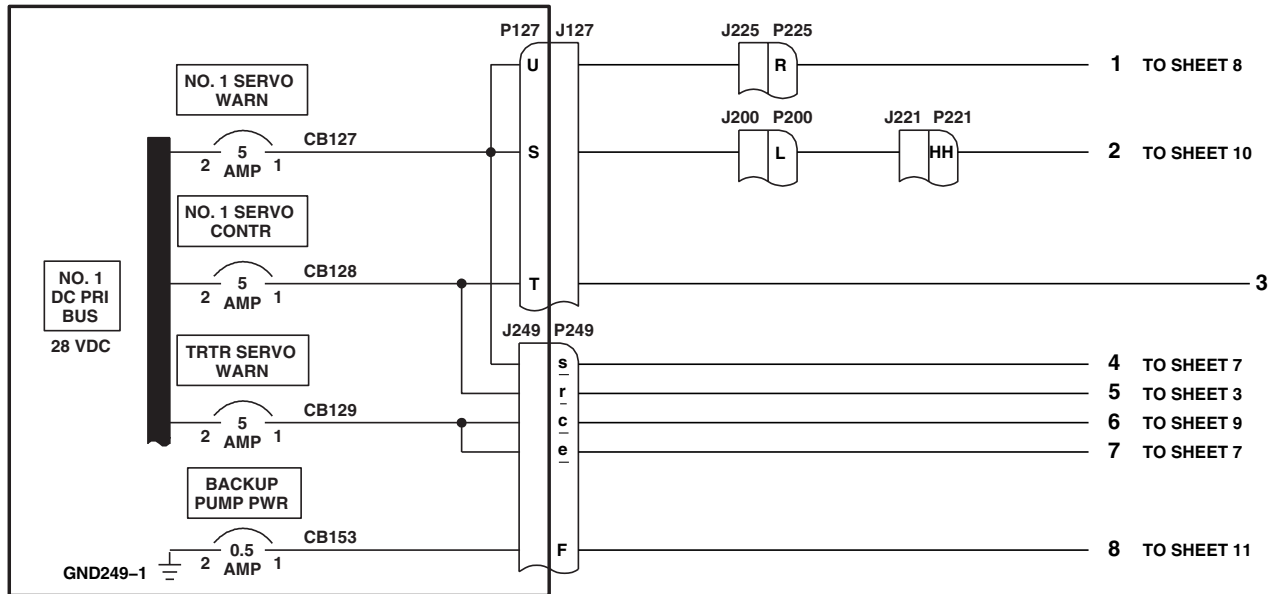


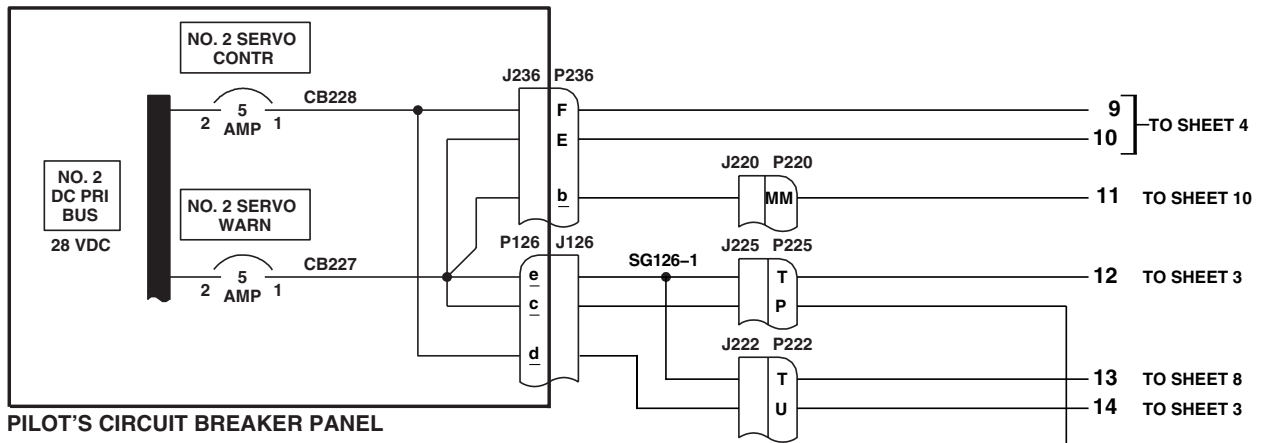
Figure 1. Hydraulic Systems Location Diagram. (Sheet 13 of 13)

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SCHEMATICS



COPILOT'S CIRCUIT BREAKER PANEL



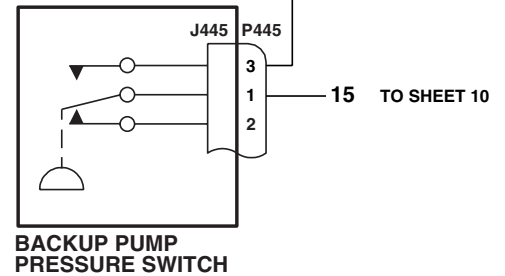
PILOT'S CIRCUIT BREAKER PANEL

NOTES

1. SWITCH S31 SHOWN IN THE OFF POSITION. CONNECTIONS BETWEEN CONTACTS 1 AND 2, AND 5 AND 6 ARE MADE IN THE AUTO POSITION. CONNECTIONS BETWEEN CONTACTS 2 AND 3, AND 5 AND 6 ARE MADE WITH S31 ON.
2. **WINTER**
3. SWITCH SHOWN WITH WEIGHT ON WHEELS.
4. THIS VALVE (WHEN ENERGIZED) ALLOWS THE BACKUP PUMP TO BE IN BYPASS UNTIL THE MOTOR IS AT RATED SPEED.
5. HELICOPTERS EQUIPPED WITH AUX CABIN HEATER.
6. **W/O EMEP** PIN FILTERED ADAPTER (PFA) IS NOT INSTALLED.
7. **UH-60A 77-22714 - 85-24414**

NOTES

8. **EH-60A** **UH-60A** **UH-60L**
9. **HH-60L** **HH-60A**



BACKUP PUMP PRESSURE SWITCH

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Figure 2. Hydraulic Systems Schematic Diagram. (Sheet 1 of 15)

SCHEMATICS - Continued

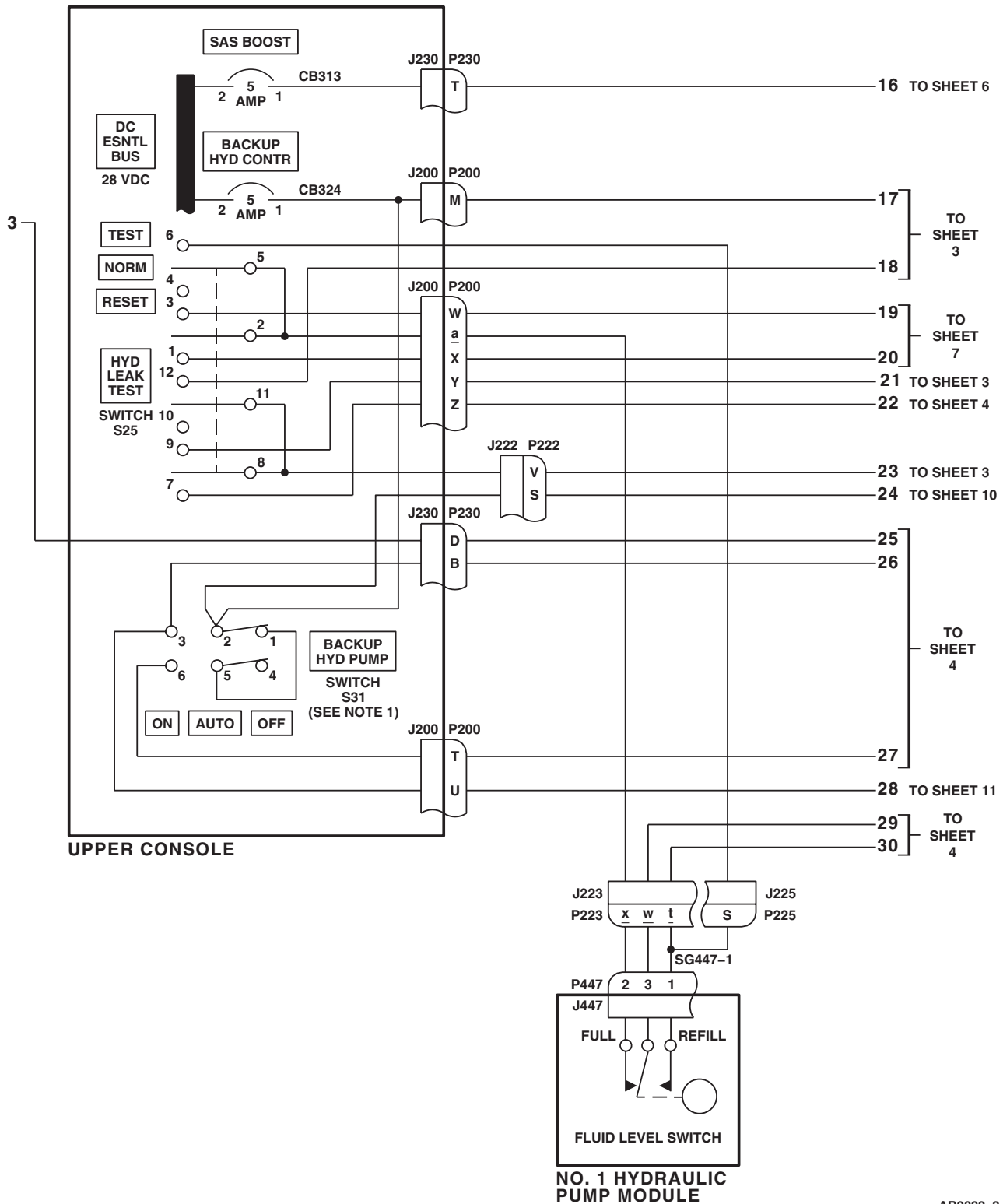
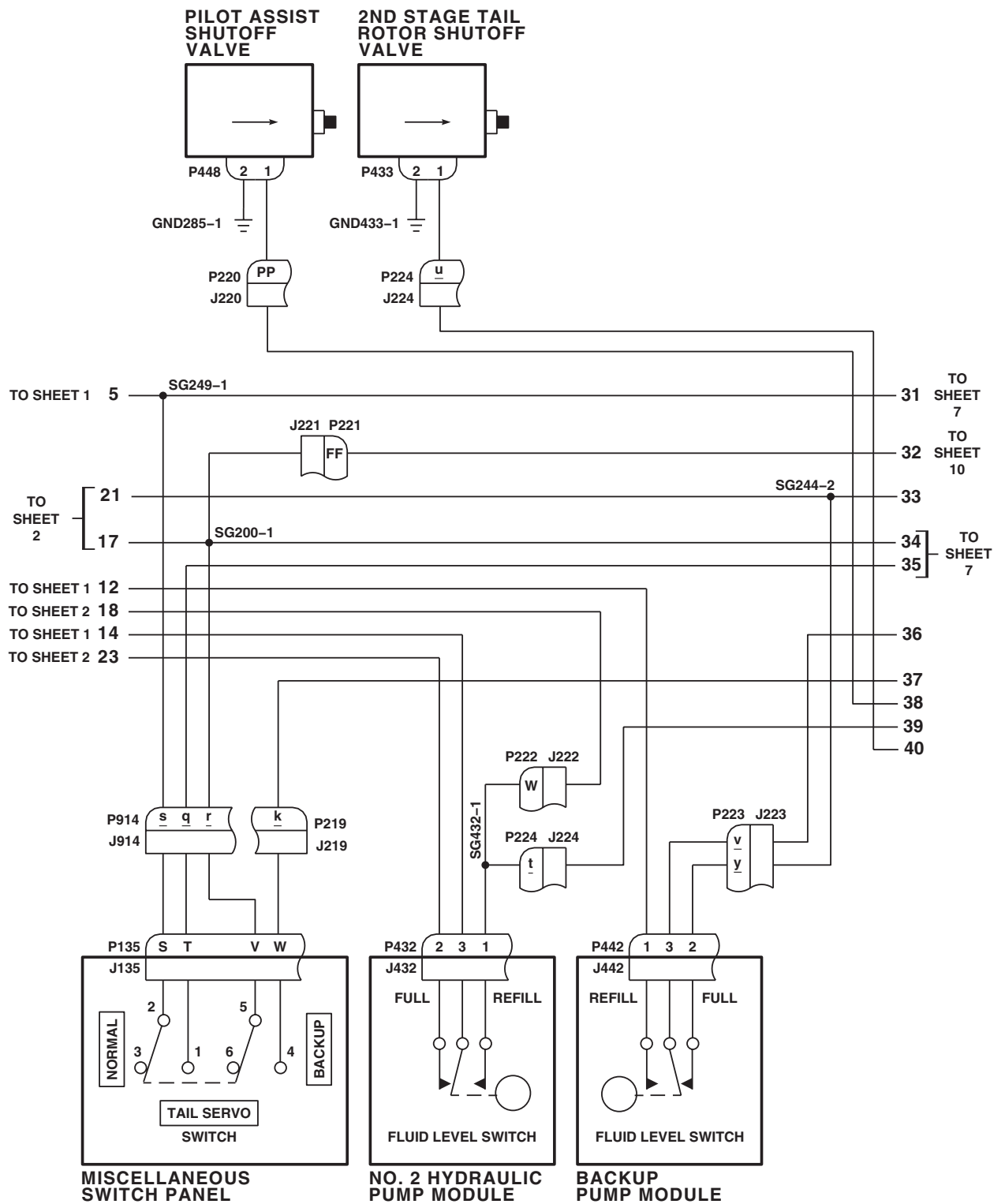


Figure 2. Hydraulic Systems Schematic Diagram. (Sheet 2 of 15)

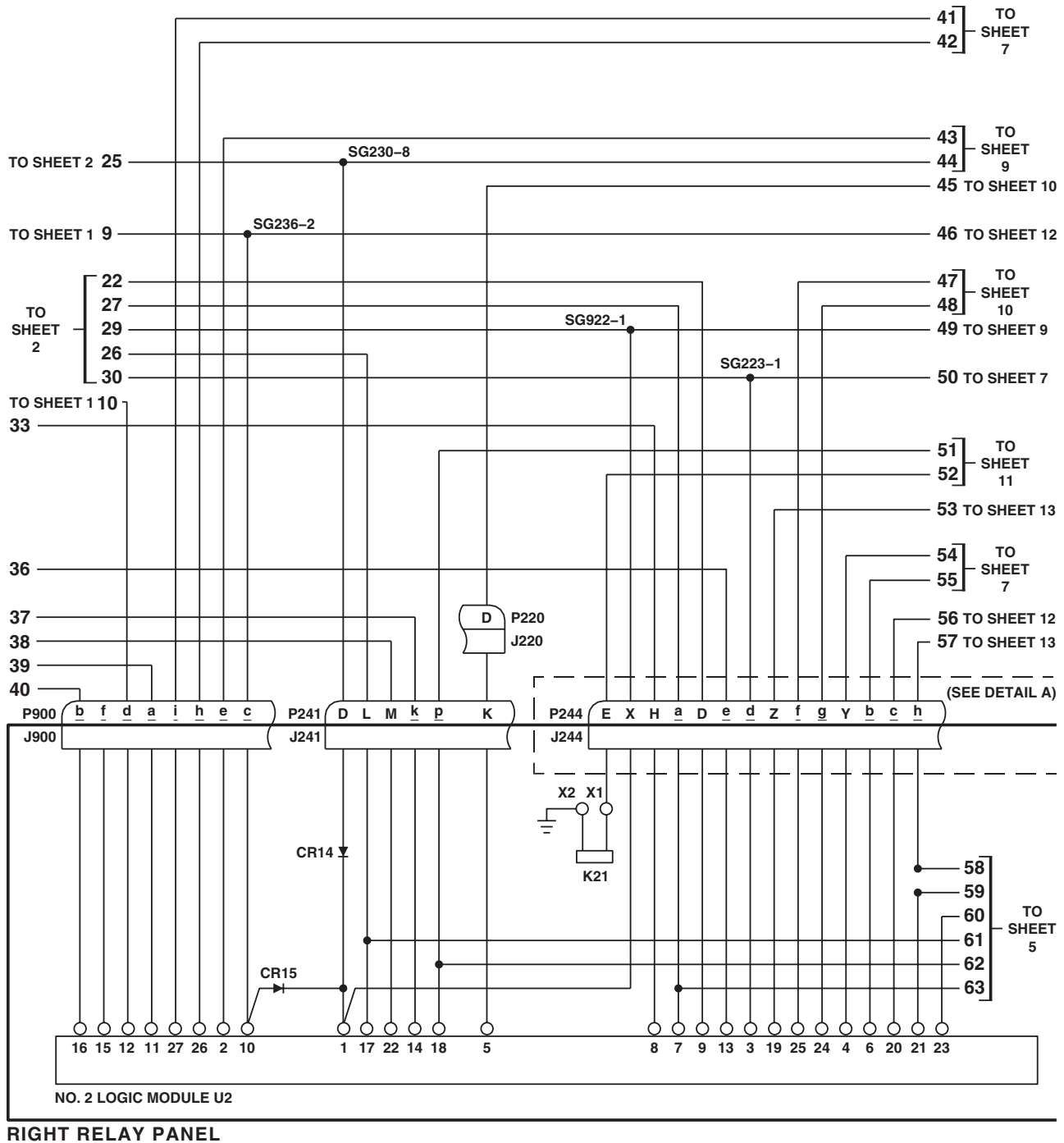
SCHEMATICS - Continued



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Figure 2. Hydraulic Systems Schematic Diagram. (Sheet 3 of 15)

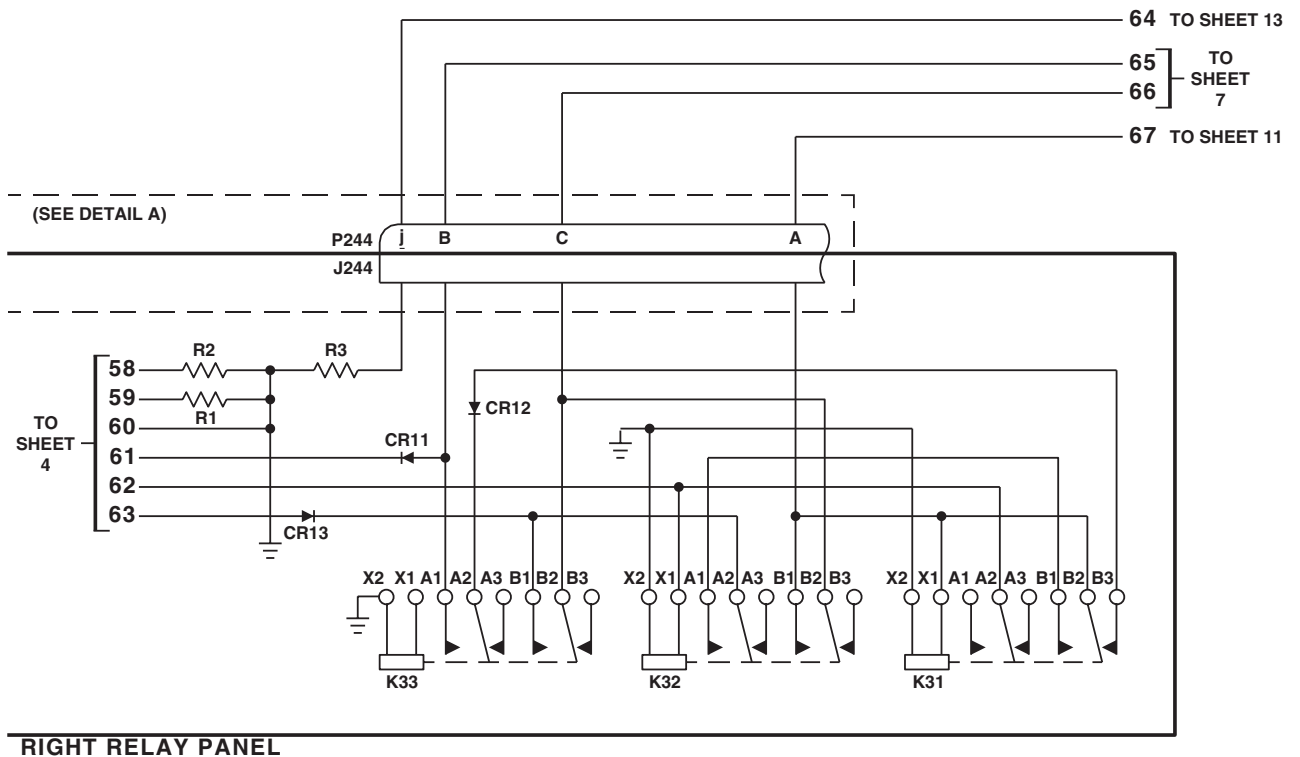
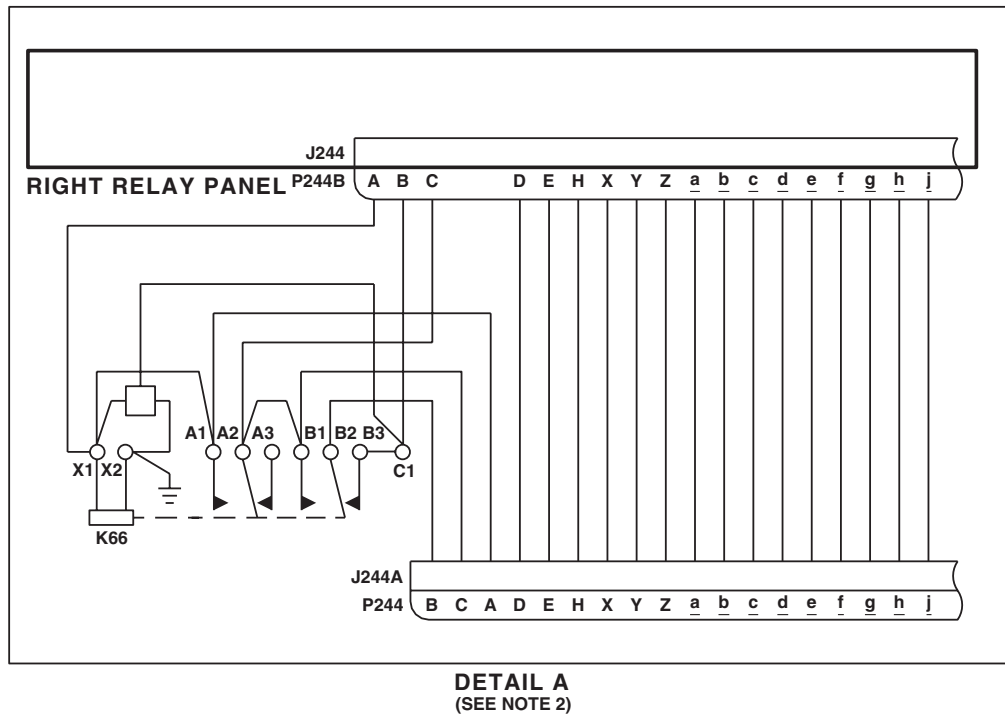
SCHEMATICS - Continued



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Figure 2. Hydraulic Systems Schematic Diagram. (Sheet 4 of 15)

SCHEMATICS - Continued



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SA

Figure 2. Hydraulic Systems Schematic Diagram. (Sheet 5 of 15)

SCHEMATICS - Continued

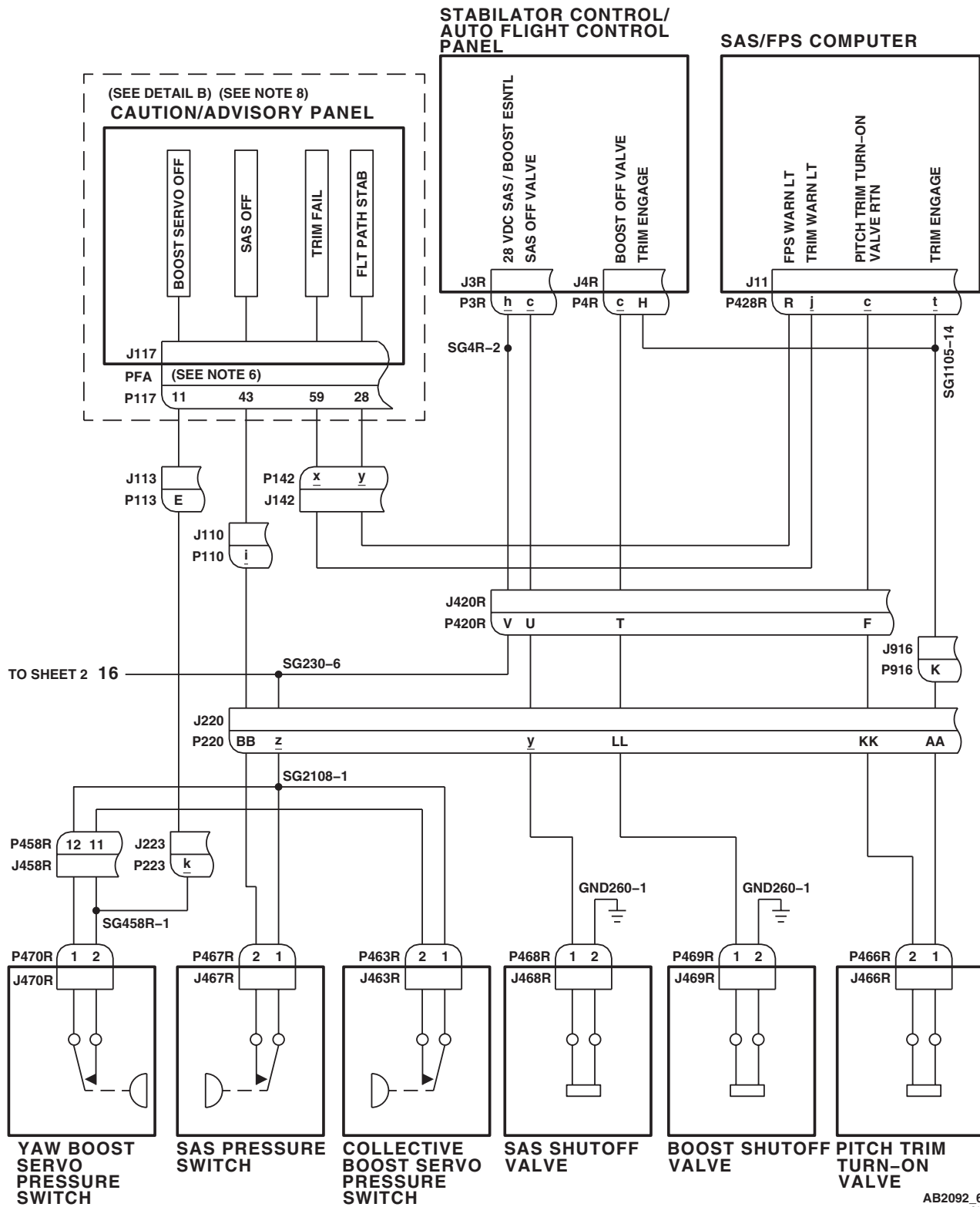


Figure 2. Hydraulic Systems Schematic Diagram. (Sheet 6 of 15)

SCHEMATICS - Continued

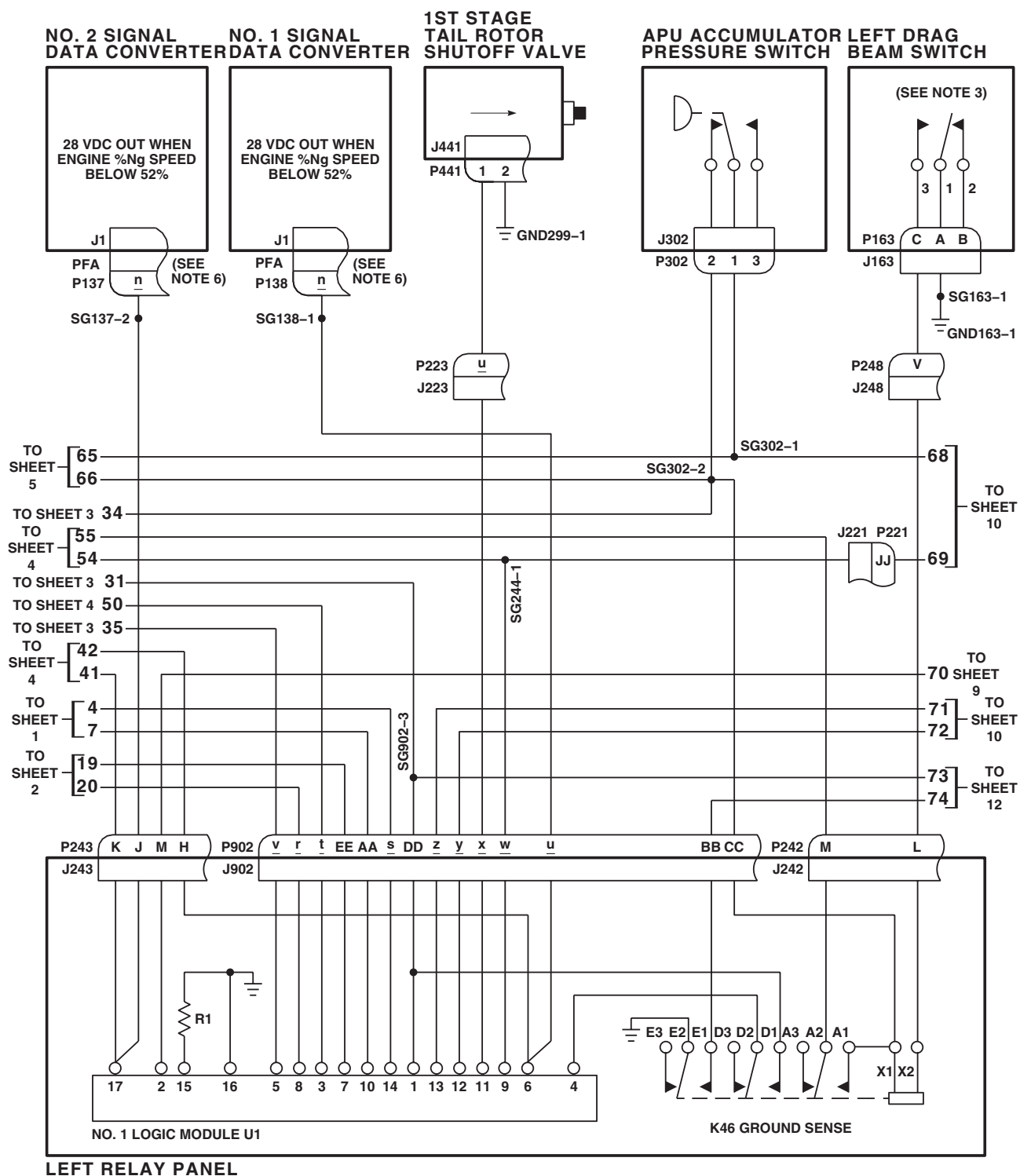
AB2092.7
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Figure 2. Hydraulic Systems Schematic Diagram. (Sheet 7 of 15)

SCHEMATICS - Continued

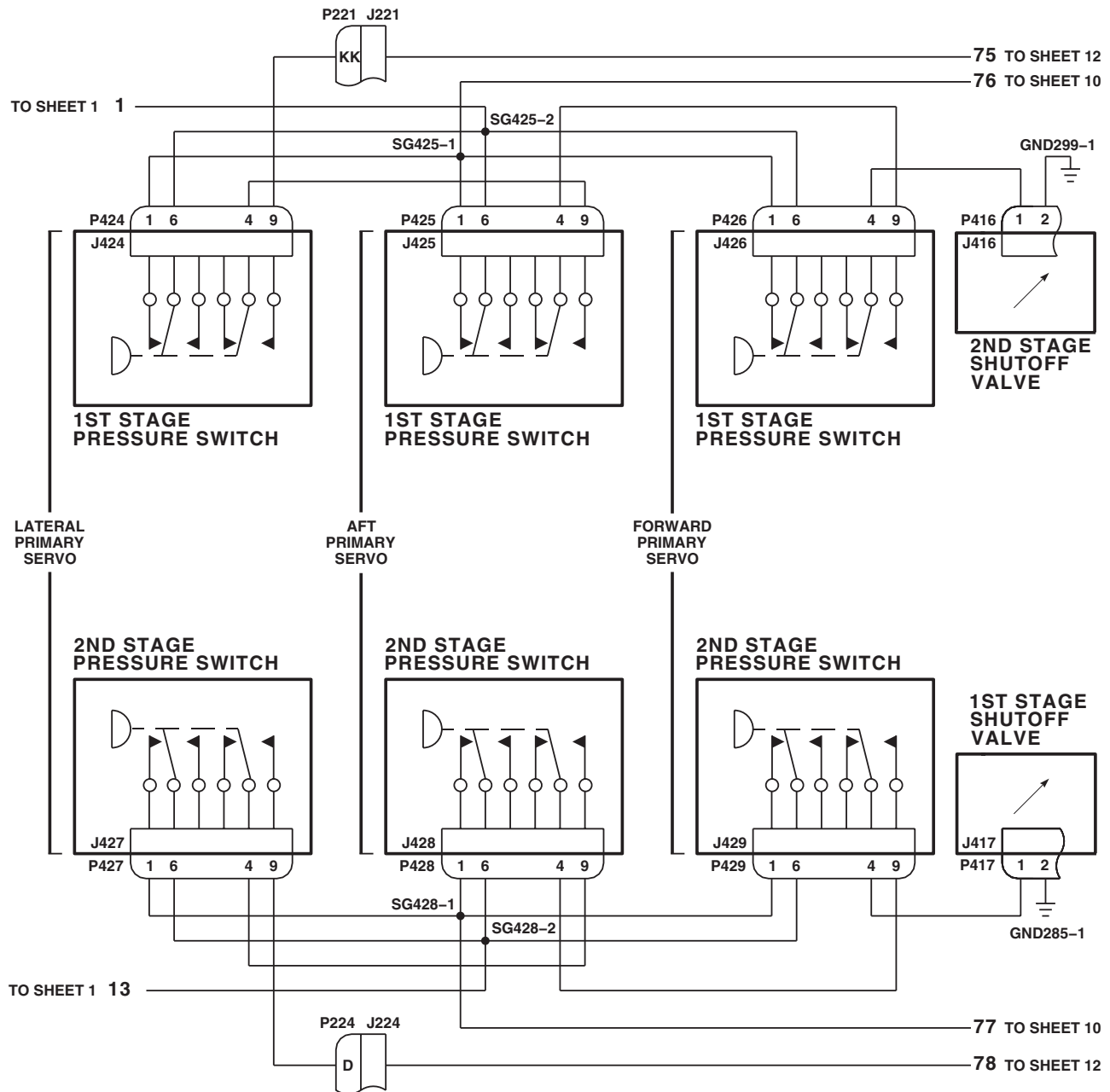
AB2092 8
SA

Figure 2. Hydraulic Systems Schematic Diagram. (Sheet 8 of 15)

SCHEMATICS - Continued

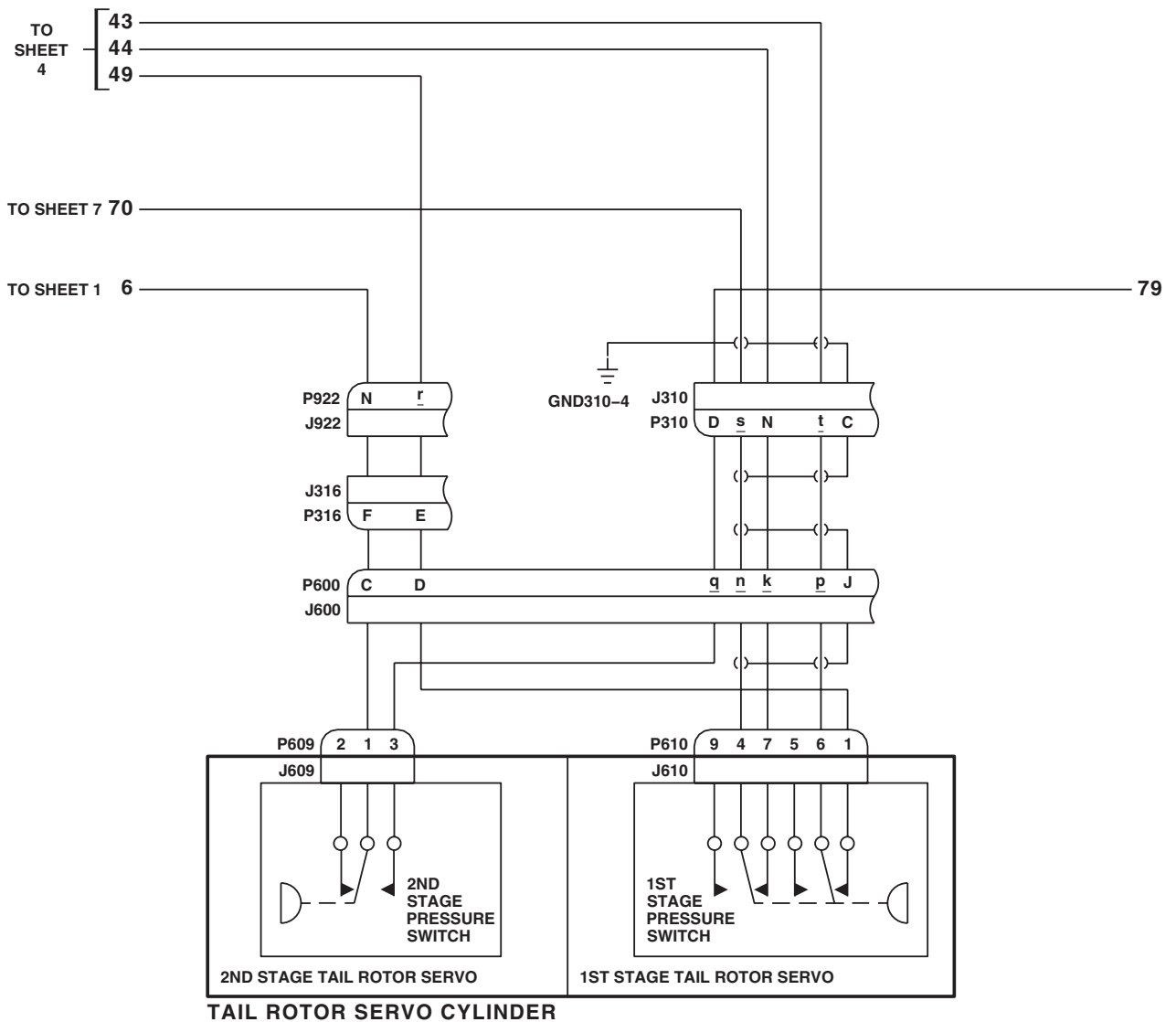
AB2092_9
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Figure 2. Hydraulic Systems Schematic Diagram. (Sheet 9 of 15)

SCHEMATICS - Continued

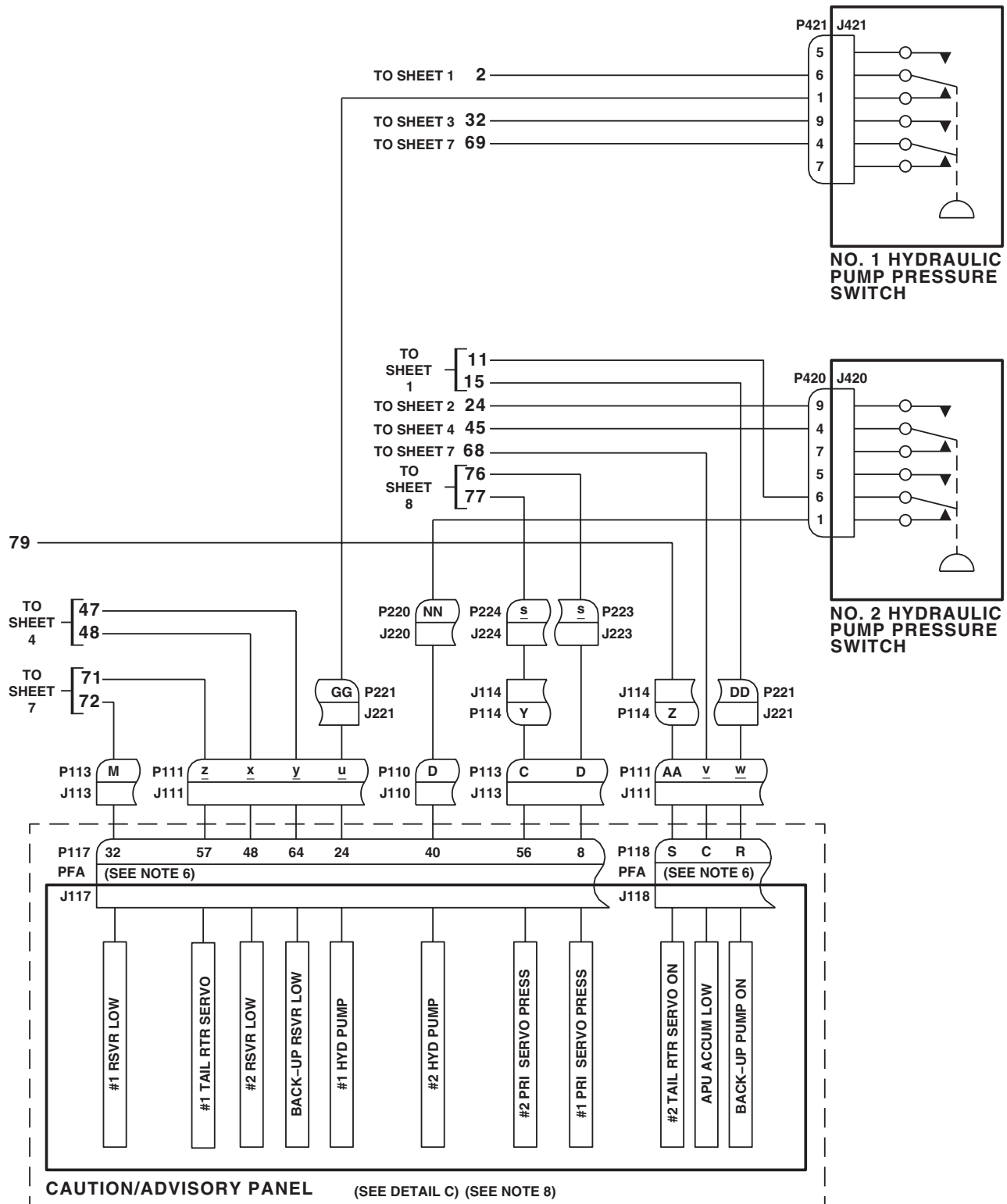
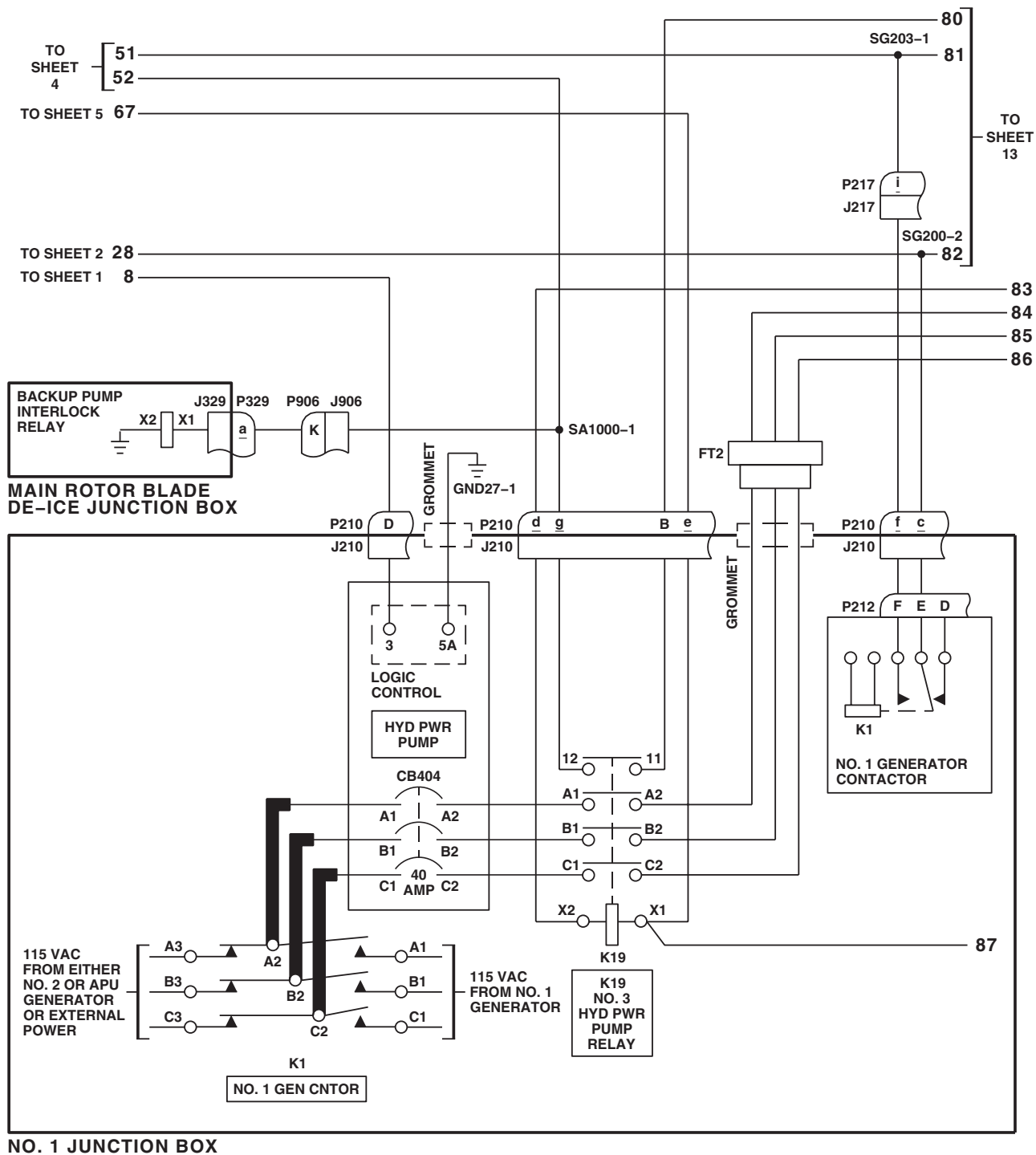


Figure 2. Hydraulic Systems Schematic Diagram. (Sheet 10 of 15)

SCHEMATICS - Continued



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SA

Figure 2. Hydraulic Systems Schematic Diagram. (Sheet 11 of 15)

SCHEMATICS - Continued

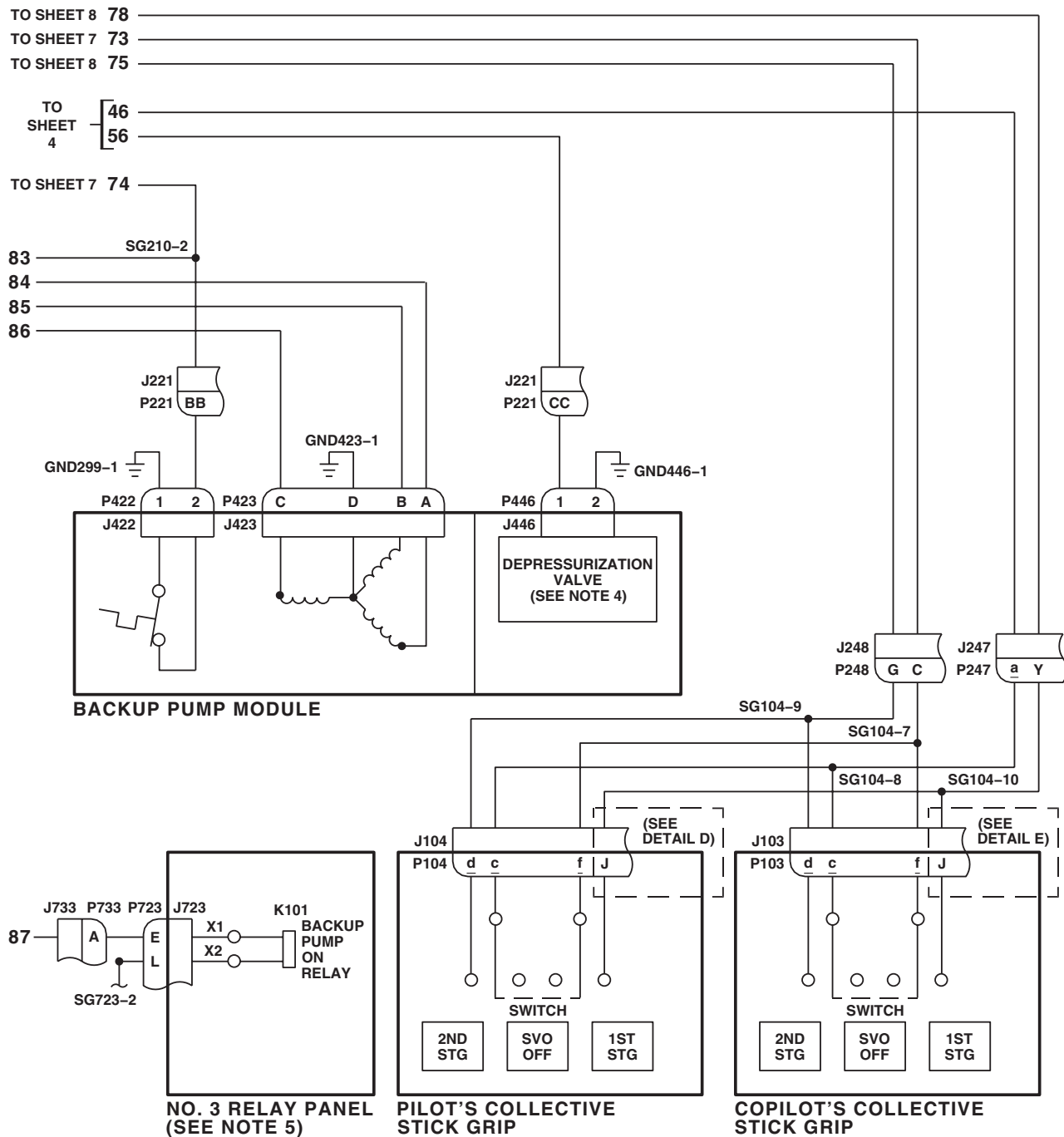
AB2092_12A
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Figure 2. Hydraulic Systems Schematic Diagram. (Sheet 12 of 15)

SCHEMATICS - Continued

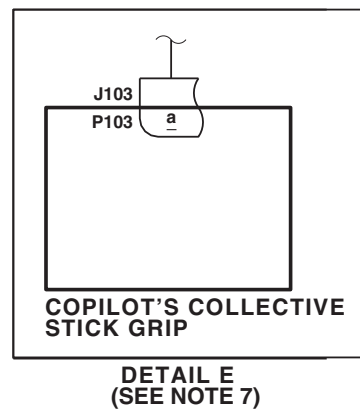
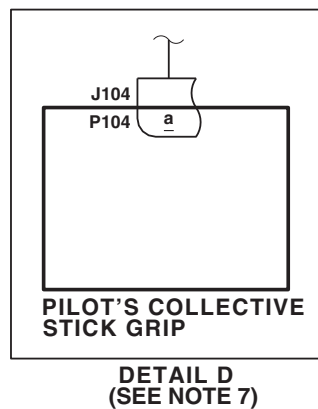
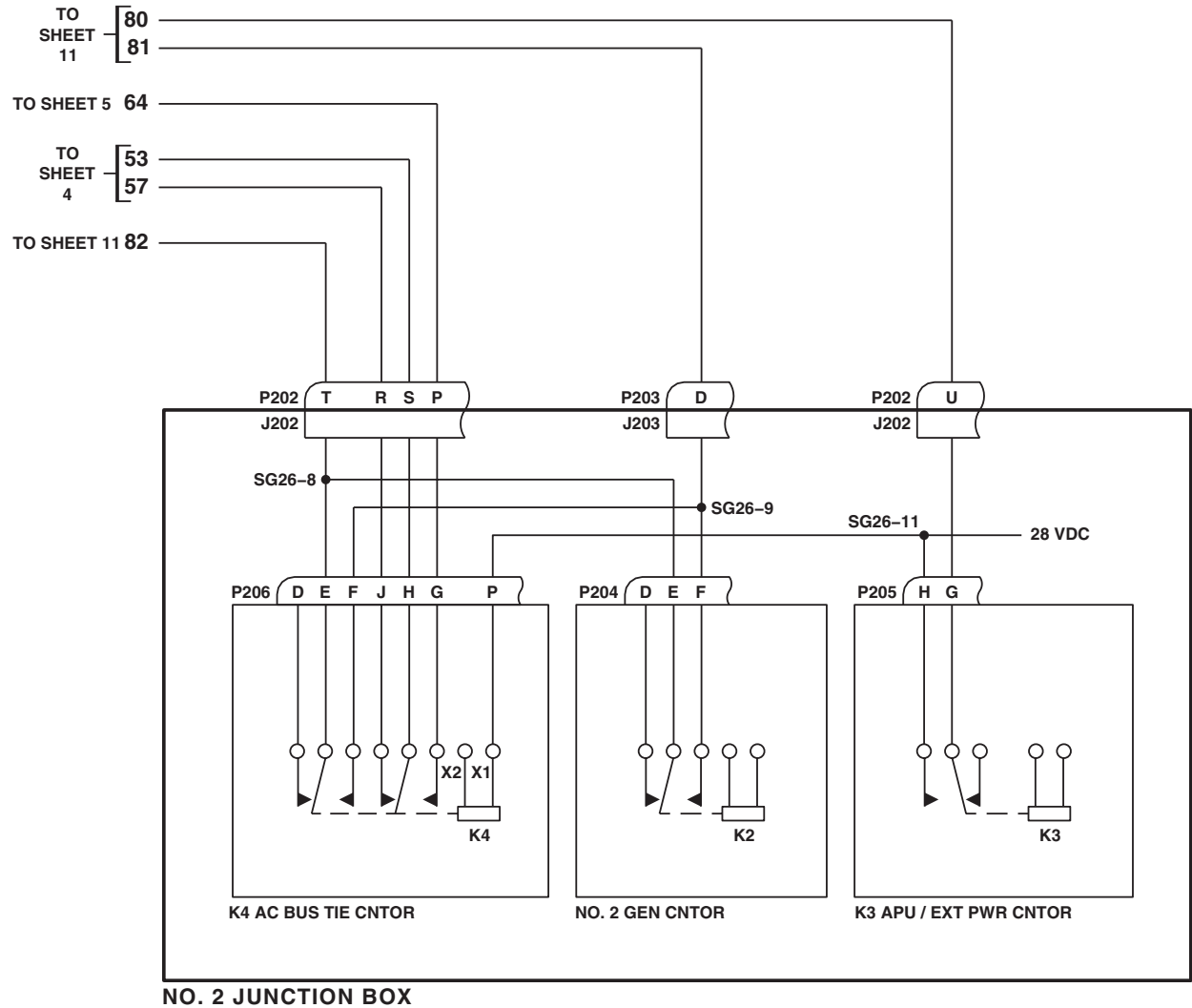
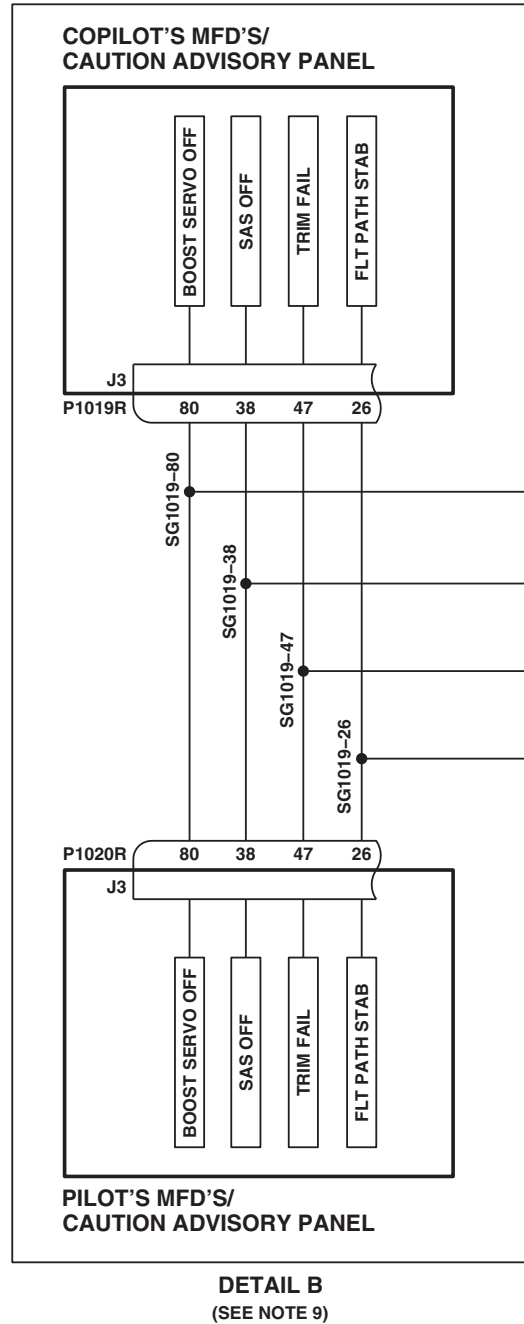
AB2092_13
SA

Figure 2. Hydraulic Systems Schematic Diagram. (Sheet 13 of 15)

SCHEMATICS - Continued



AB2092_14
SA

Figure 2. Hydraulic Systems Schematic Diagram. (Sheet 14 of 15)

SCHEMATICS - Continued

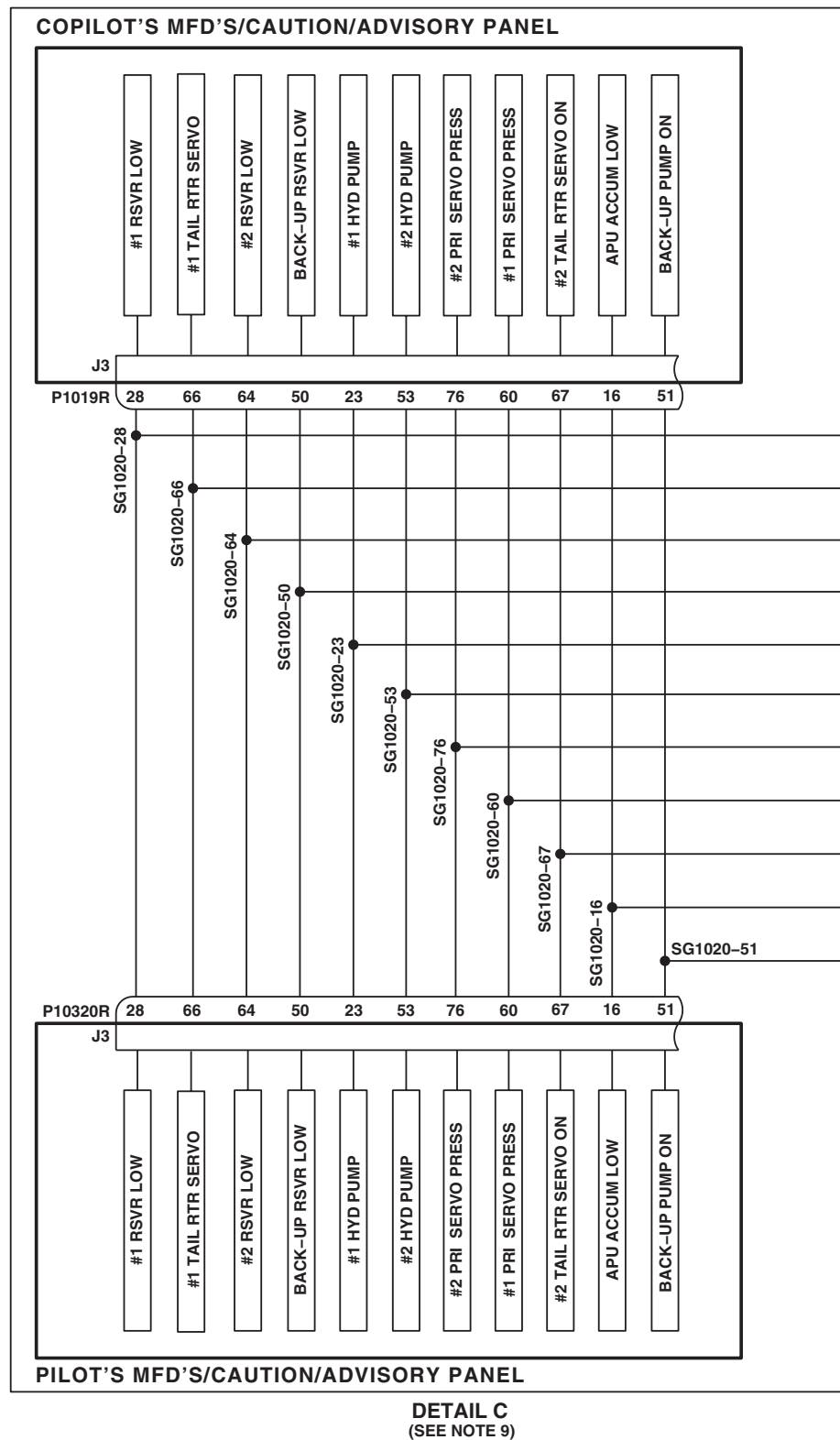


Figure 2. Hydraulic Systems Schematic Diagram. (Sheet 15 of 15)

SCHEMATICS - Continued

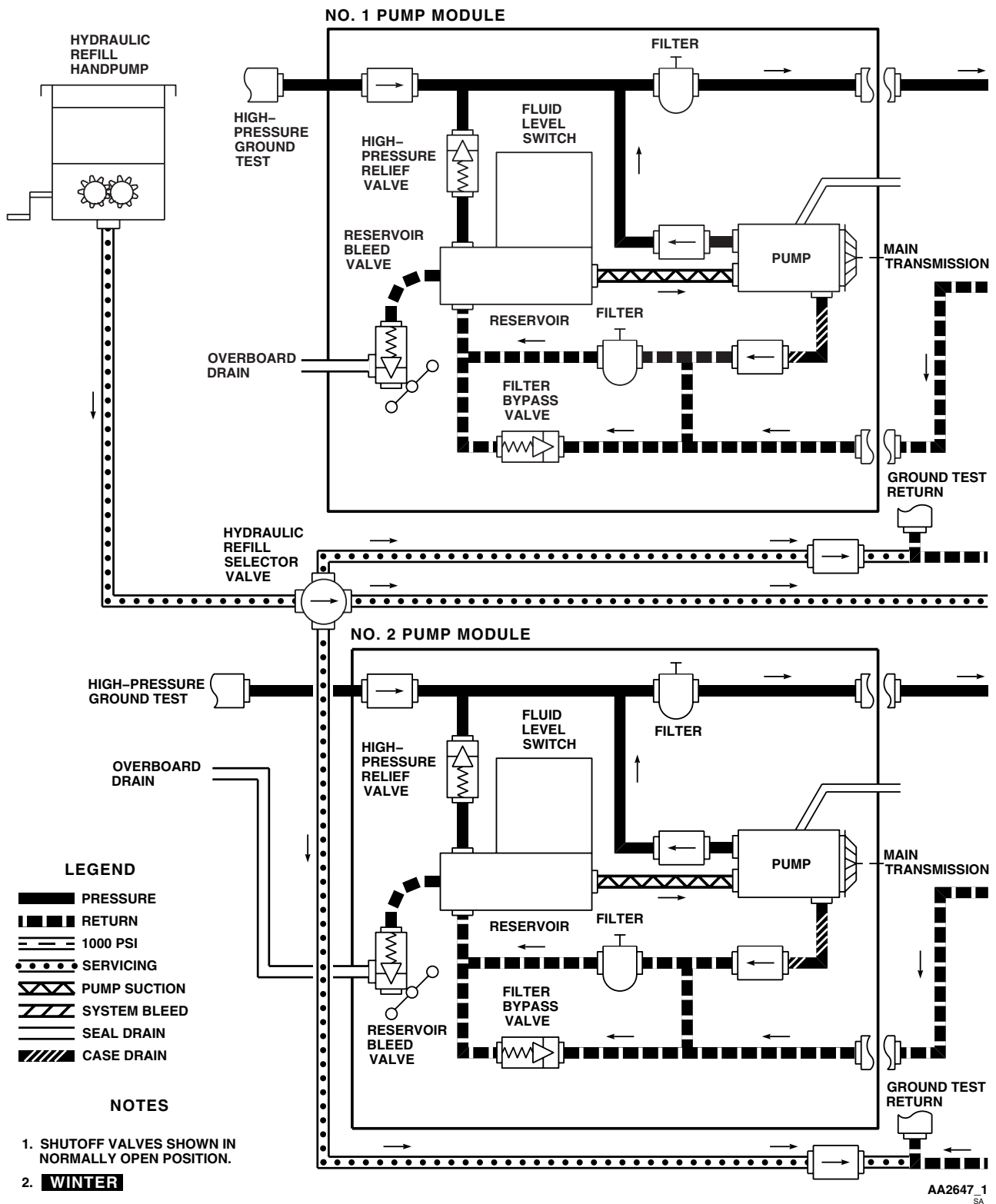


Figure 3. Hydraulic Systems Power Schematic. (Sheet 1 of 8)

SCHEMATICS - Continued

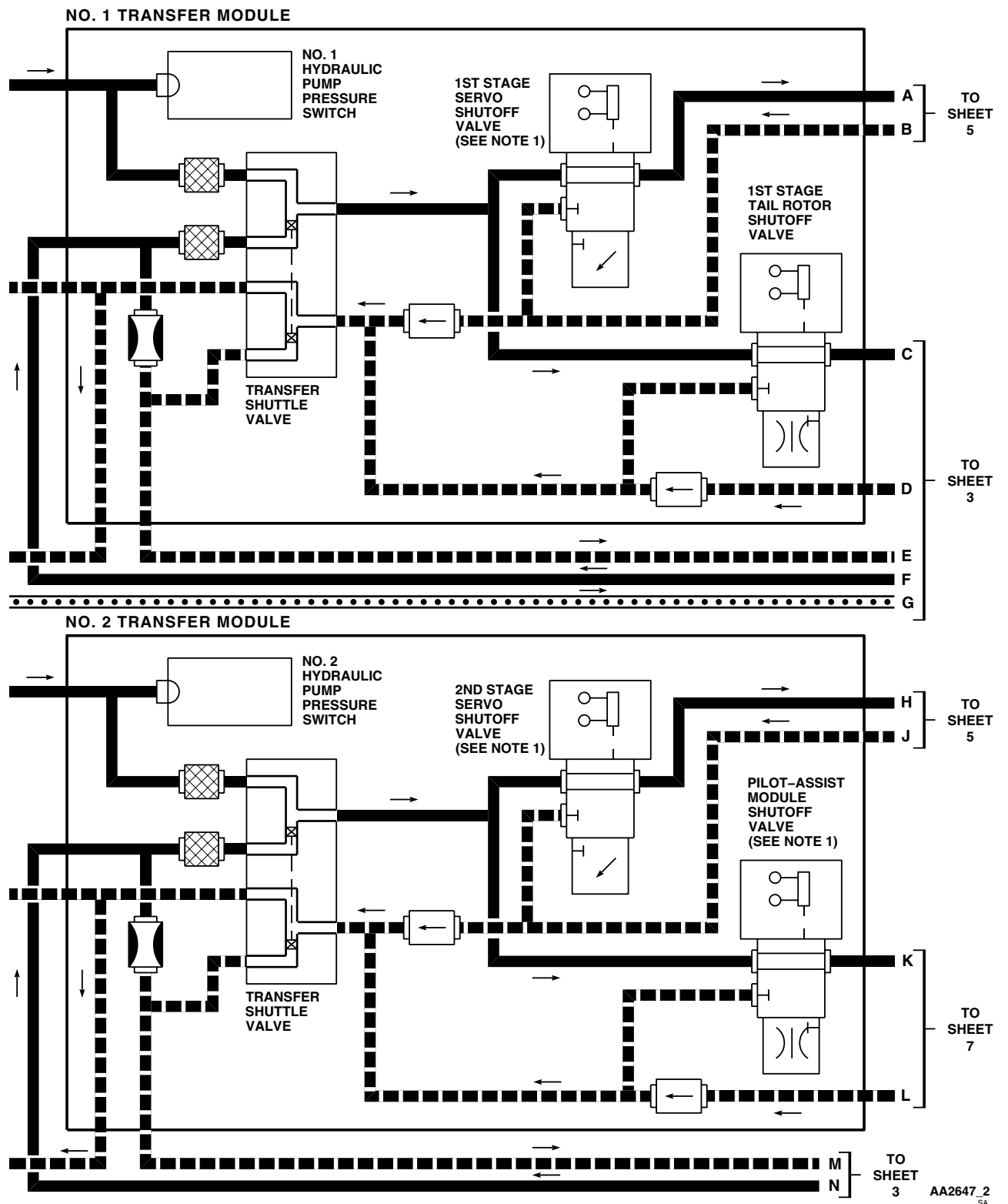


Figure 3. Hydraulic Systems Power Schematic. (Sheet 2 of 8)

SCHEMATICS - Continued

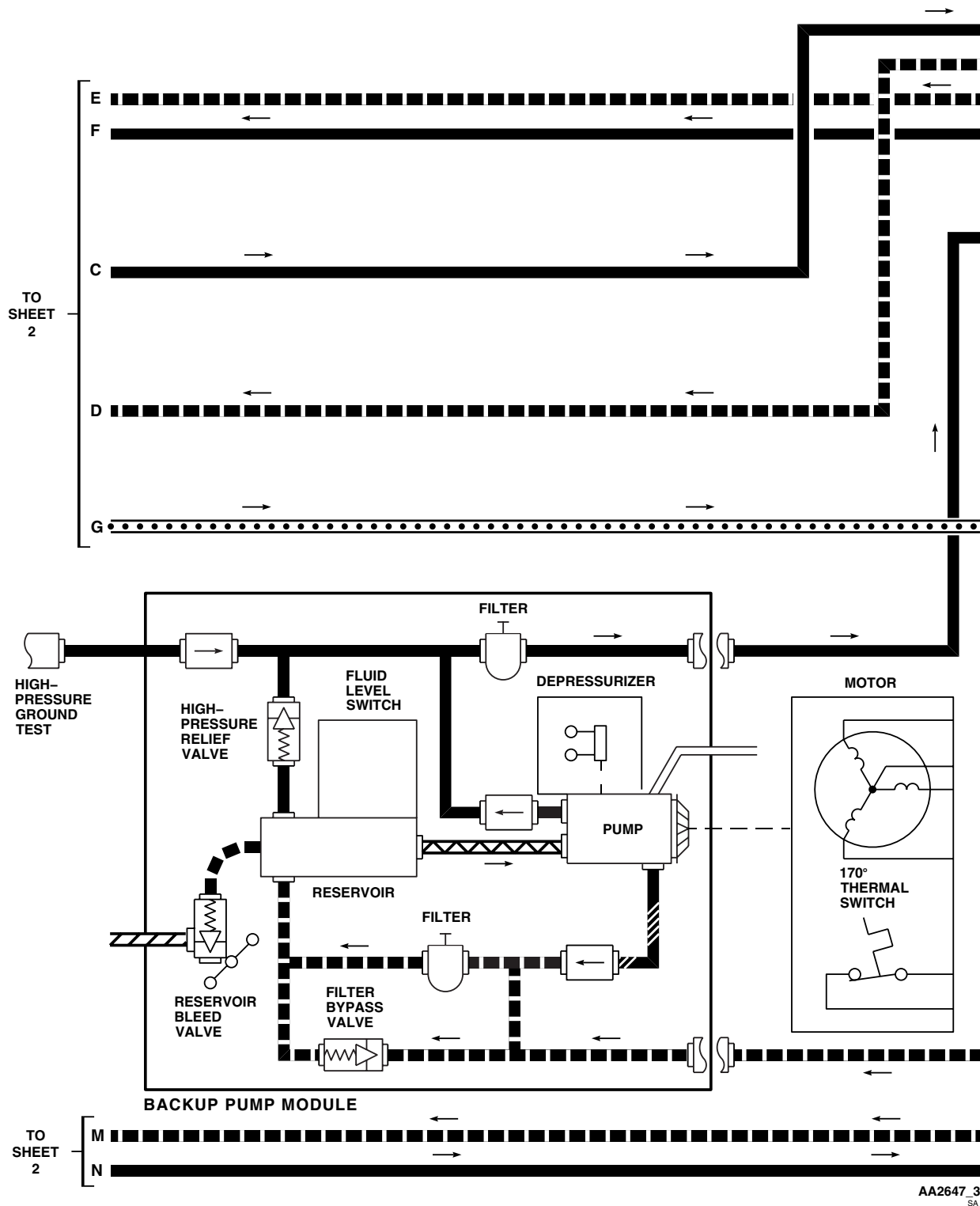
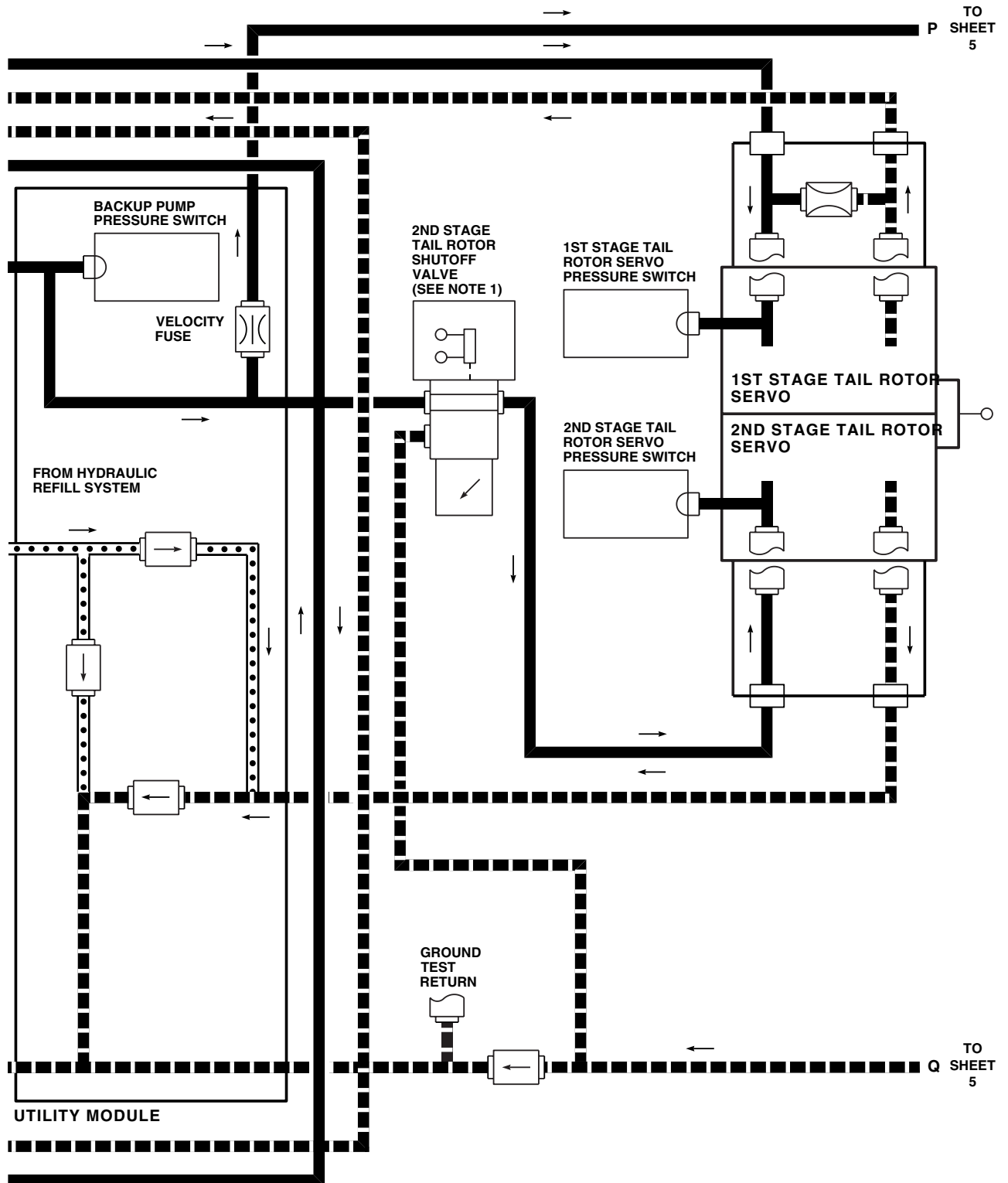


Figure 3. Hydraulic Systems Power Schematic. (Sheet 3 of 8)

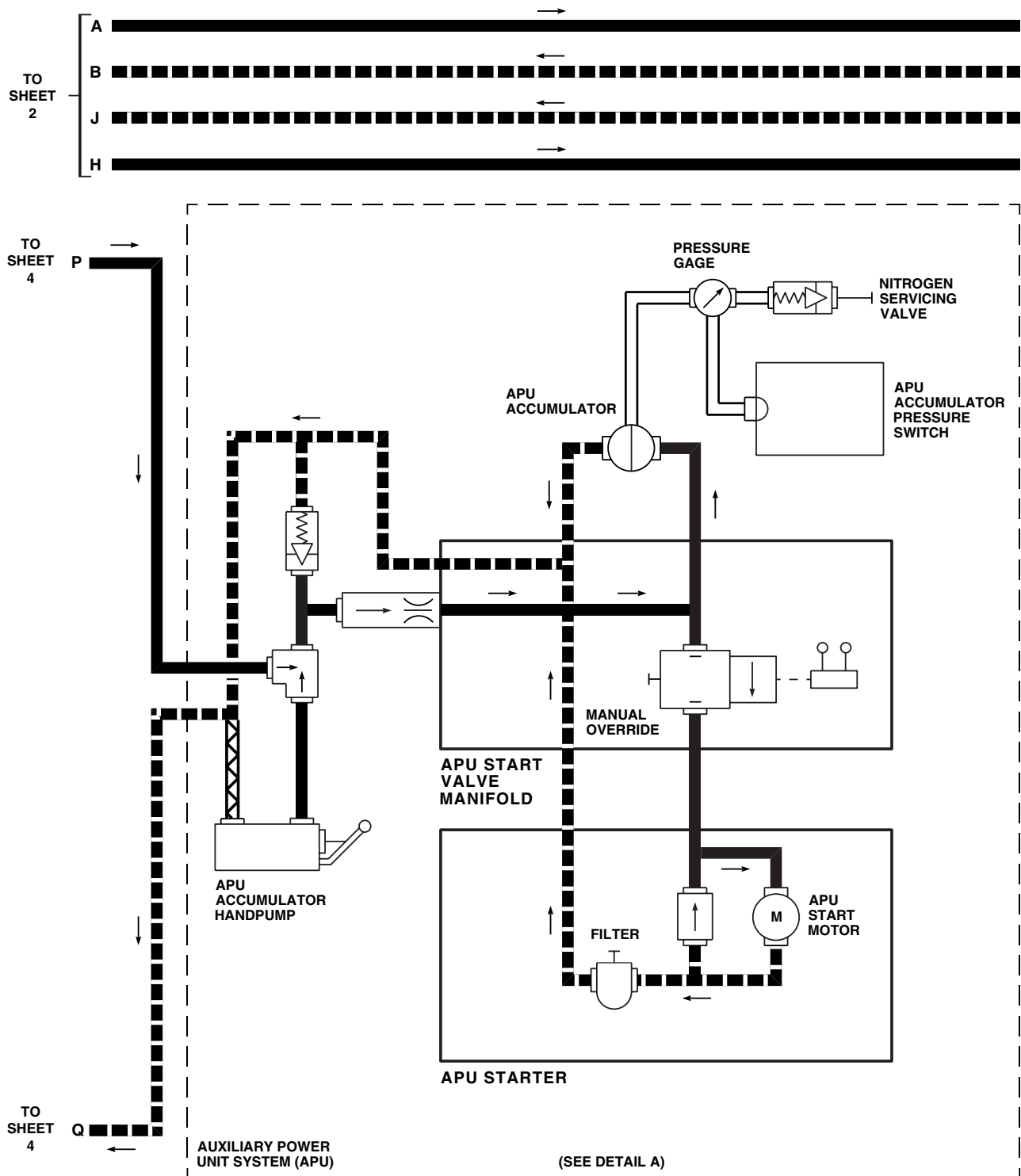
SCHEMATICS - Continued



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Figure 3. Hydraulic Systems Power Schematic. (Sheet 4 of 8)

SCHEMATICS - Continued



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SA

Figure 3. Hydraulic Systems Power Schematic. (Sheet 5 of 8)

SCHEMATICS - Continued

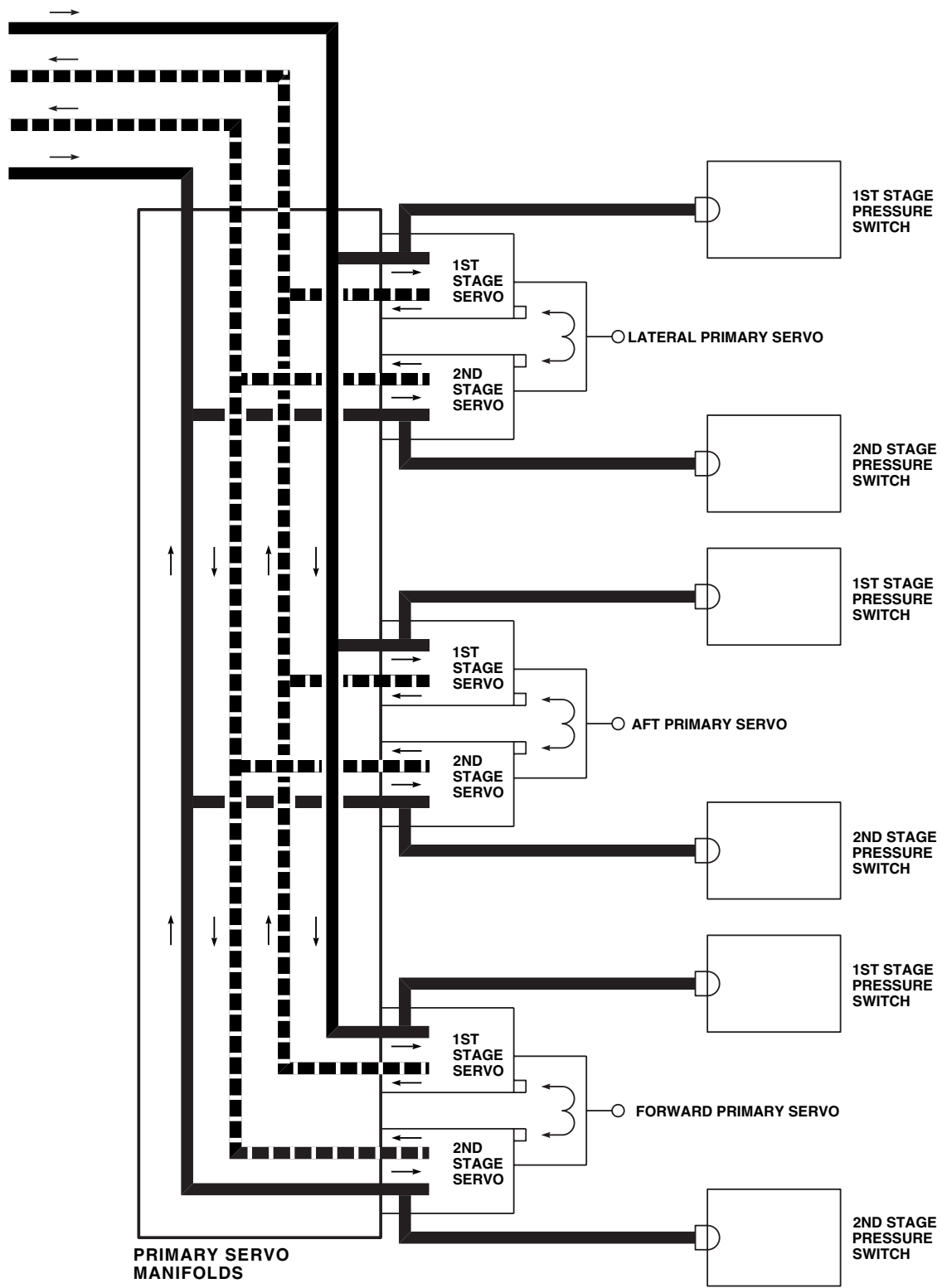
AA2647_6
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Figure 3. Hydraulic Systems Power Schematic. (Sheet 6 of 8)

SCHEMATICS - Continued

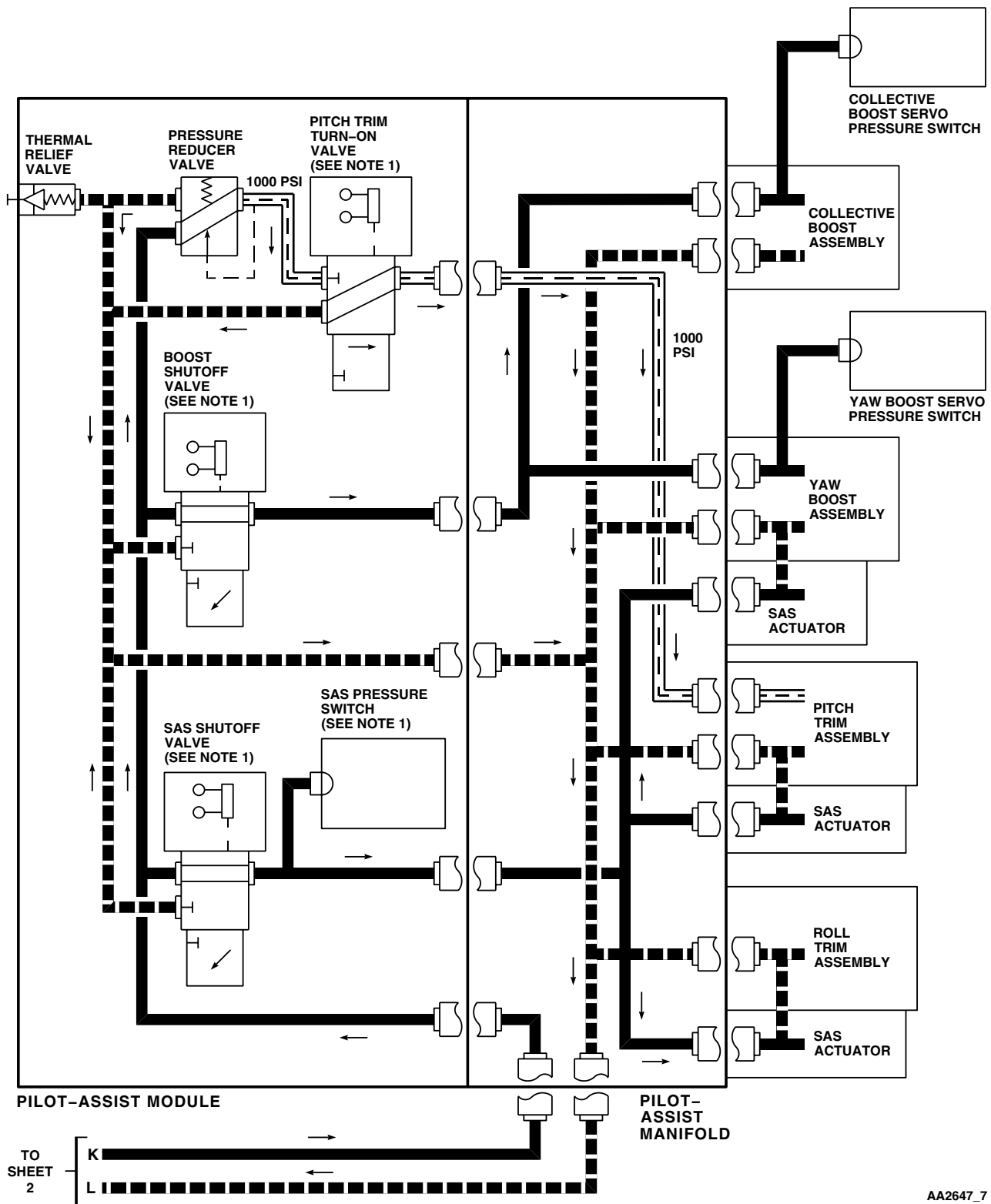


Figure 3. Hydraulic Systems Power Schematic. (Sheet 7 of 8)

SCHEMATICS - Continued

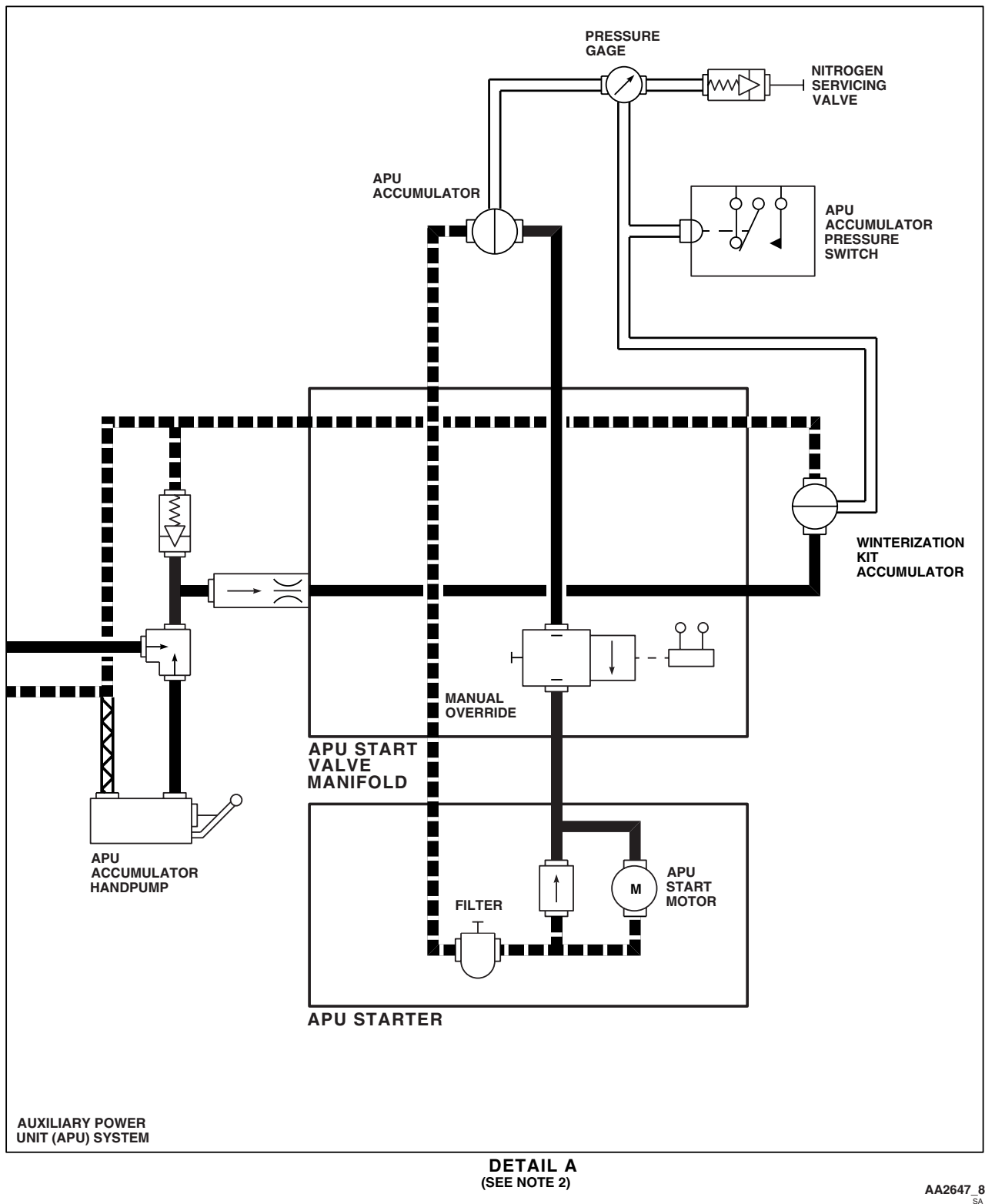
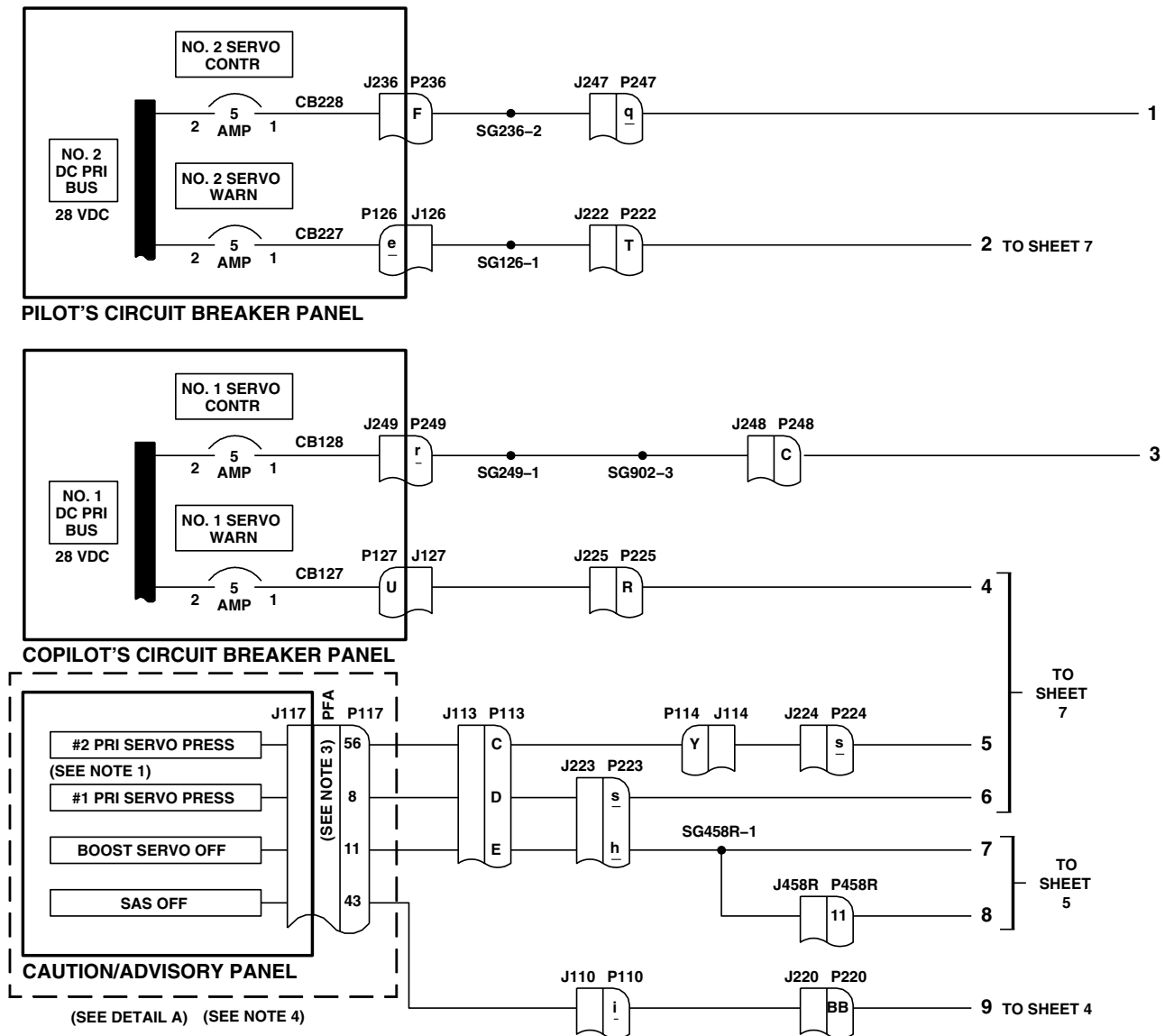


Figure 3. Hydraulic Systems Power Schematic. (Sheet 8 of 8)

SCHEMATICS - Continued



NOTES

- #1 AND #2 PRI SERVO CAUTION CAPSULES GO ON WHEN RESPECTIVE HYDRAULIC STAGE IS OFF.
- W/O MOD** PIN J IS IDENTIFIED AS PIN a ON PILOT'S AND COPILOT'S COLLECTIVE STICK CONNECTORS.
- W/O EMEP** PIN FILTERED ADAPTER (PFA) IS NOT INSTALLED.
- UH60L UH60A EH60A**
- HH60A HH60L**

LEGEND	
	PRESSURE (1000 PSI)
	PRESSURE (3050 PSI)
	RETURN
	ELECTRICAL
	MECHANICAL

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Figure 4. Hydraulic Systems Primary and Pilot Assist Servos Schematic Diagram. (Sheet 1 of 9)

SCHEMATICS - Continued

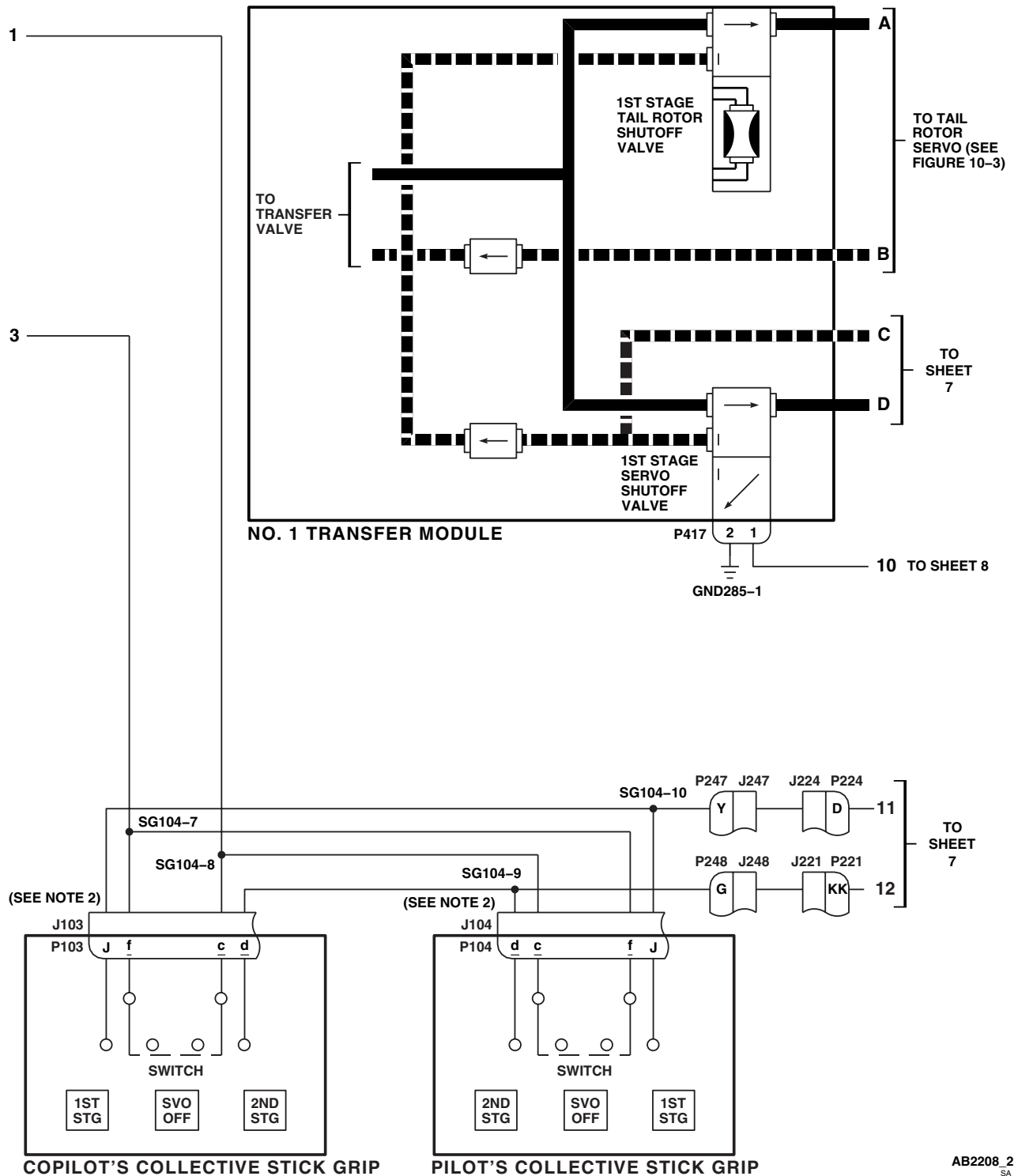


Figure 4. Hydraulic Systems Primary and Pilot Assist Servos Schematic Diagram. (Sheet 2 of 9)

SCHEMATICS - Continued

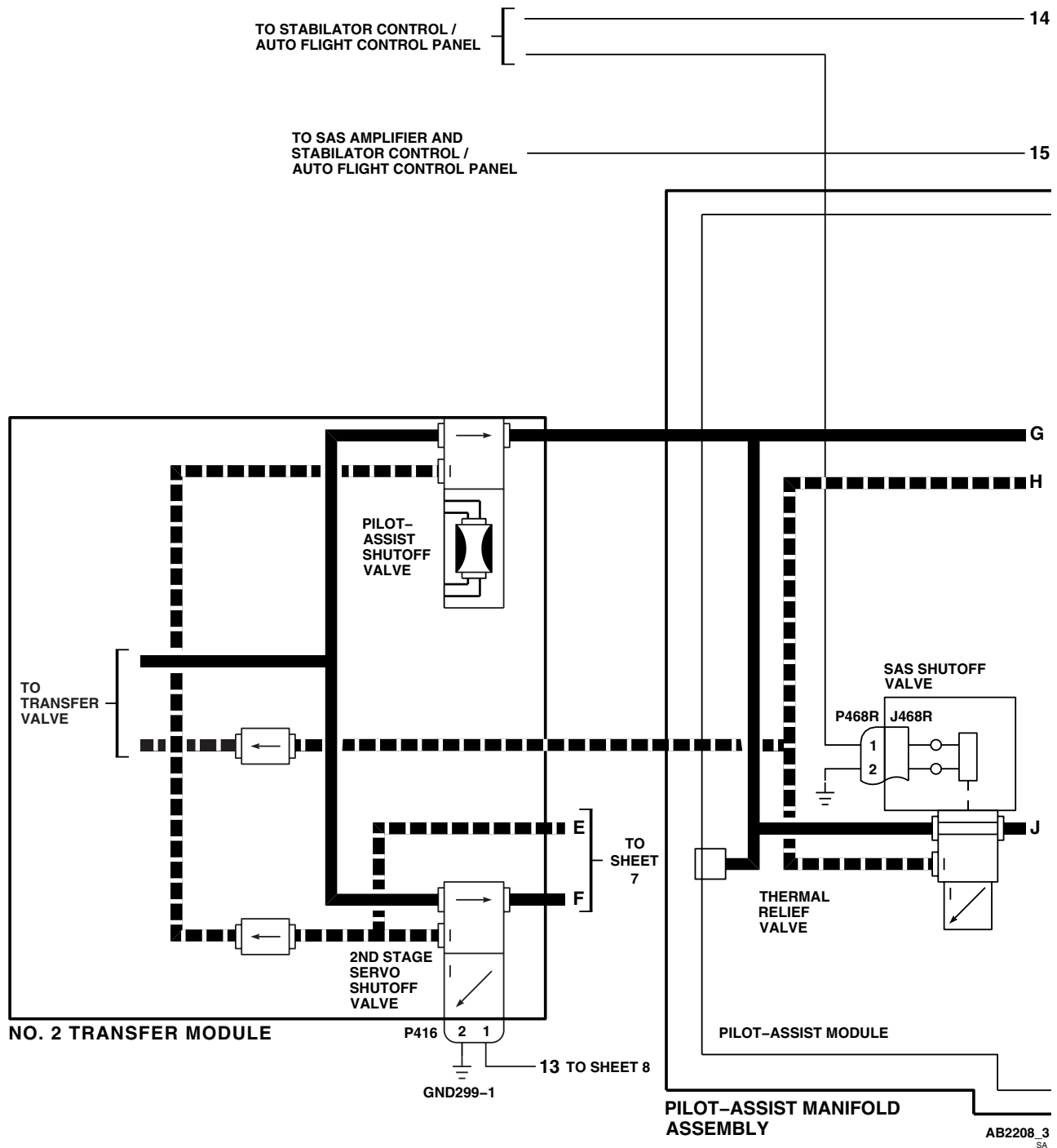


Figure 4. Hydraulic Systems Primary and Pilot Assist Servos Schematic Diagram. (Sheet 3 of 9)

SCHEMATICS - Continued

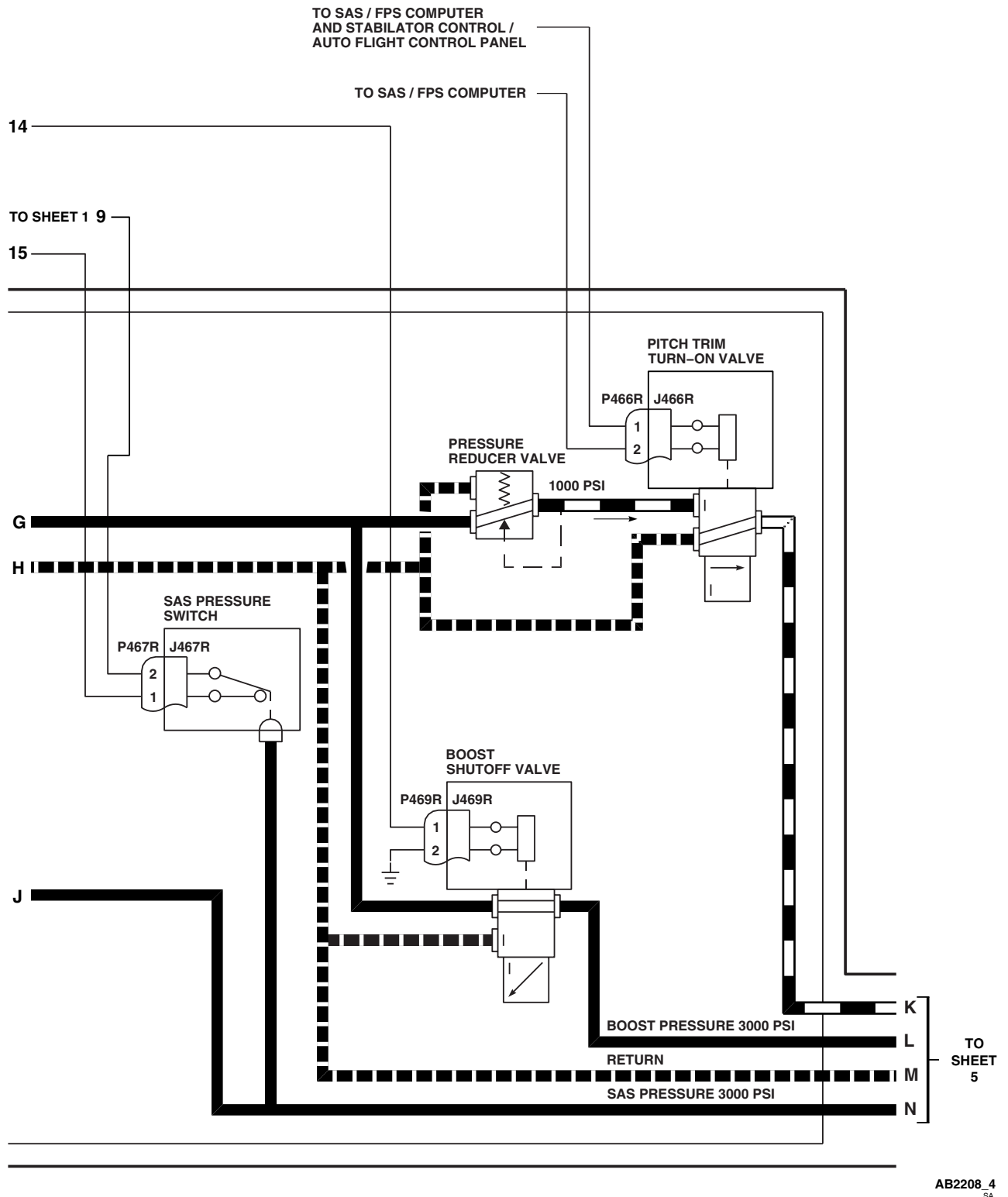


Figure 4. Hydraulic Systems Primary and Pilot Assist Servos Schematic Diagram. (Sheet 4 of 9)

SCHEMATICS - Continued

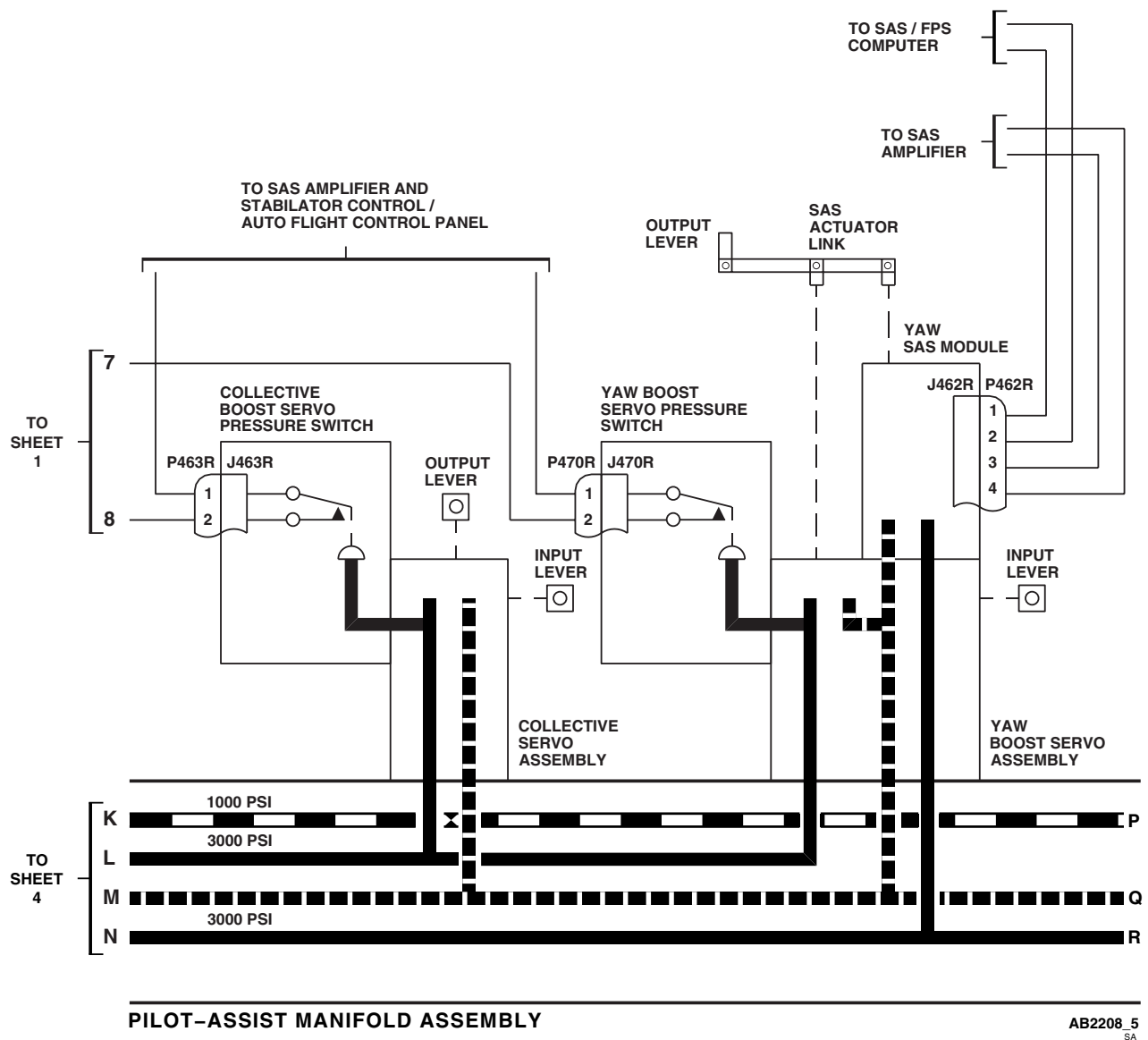


Figure 4. Hydraulic Systems Primary and Pilot Assist Servos Schematic Diagram. (Sheet 5 of 9)

SCHEMATICS - Continued

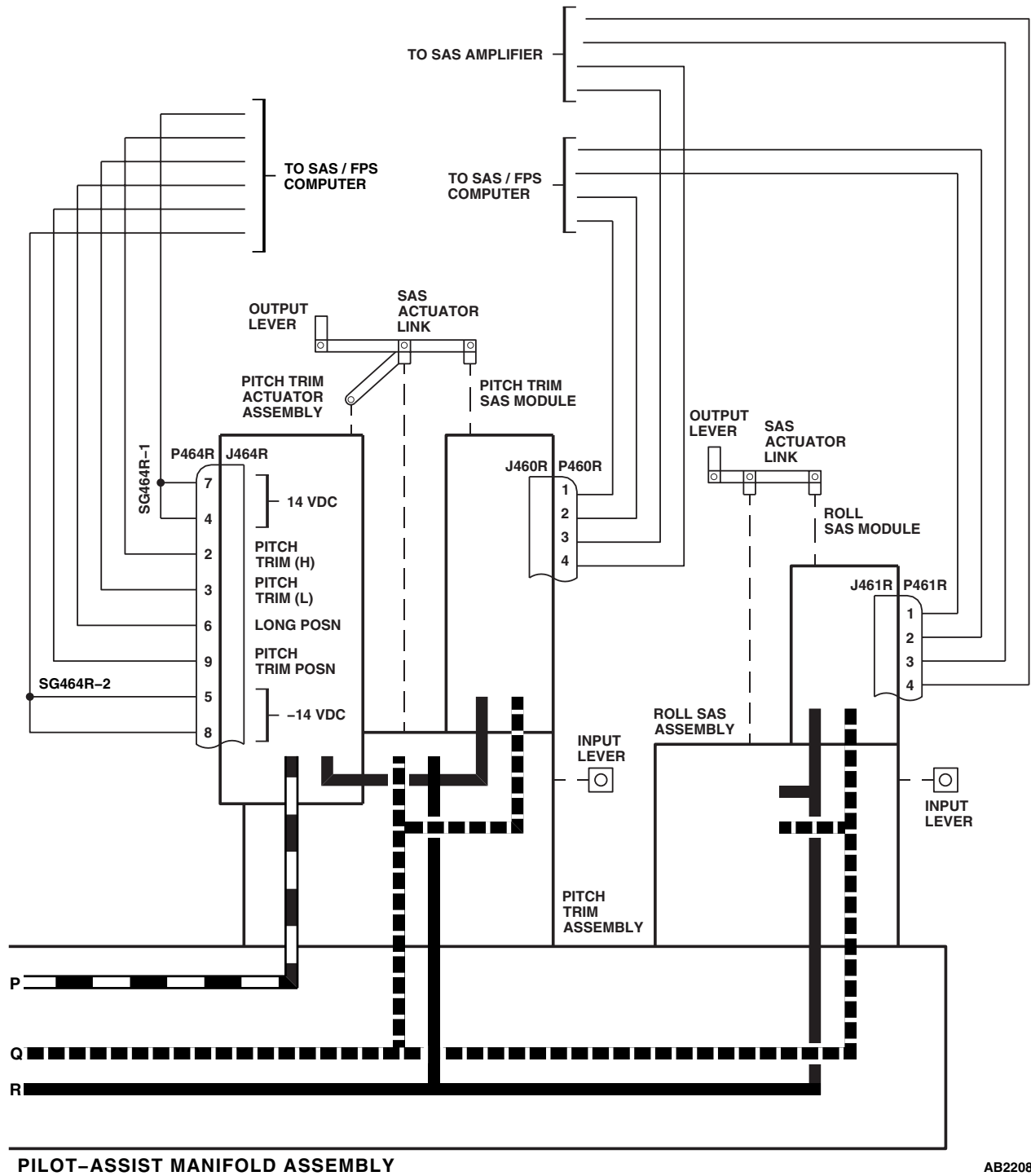
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Figure 4. Hydraulic Systems Primary and Pilot Assist Servos Schematic Diagram. (Sheet 6 of 9)

SCHEMATICS - Continued

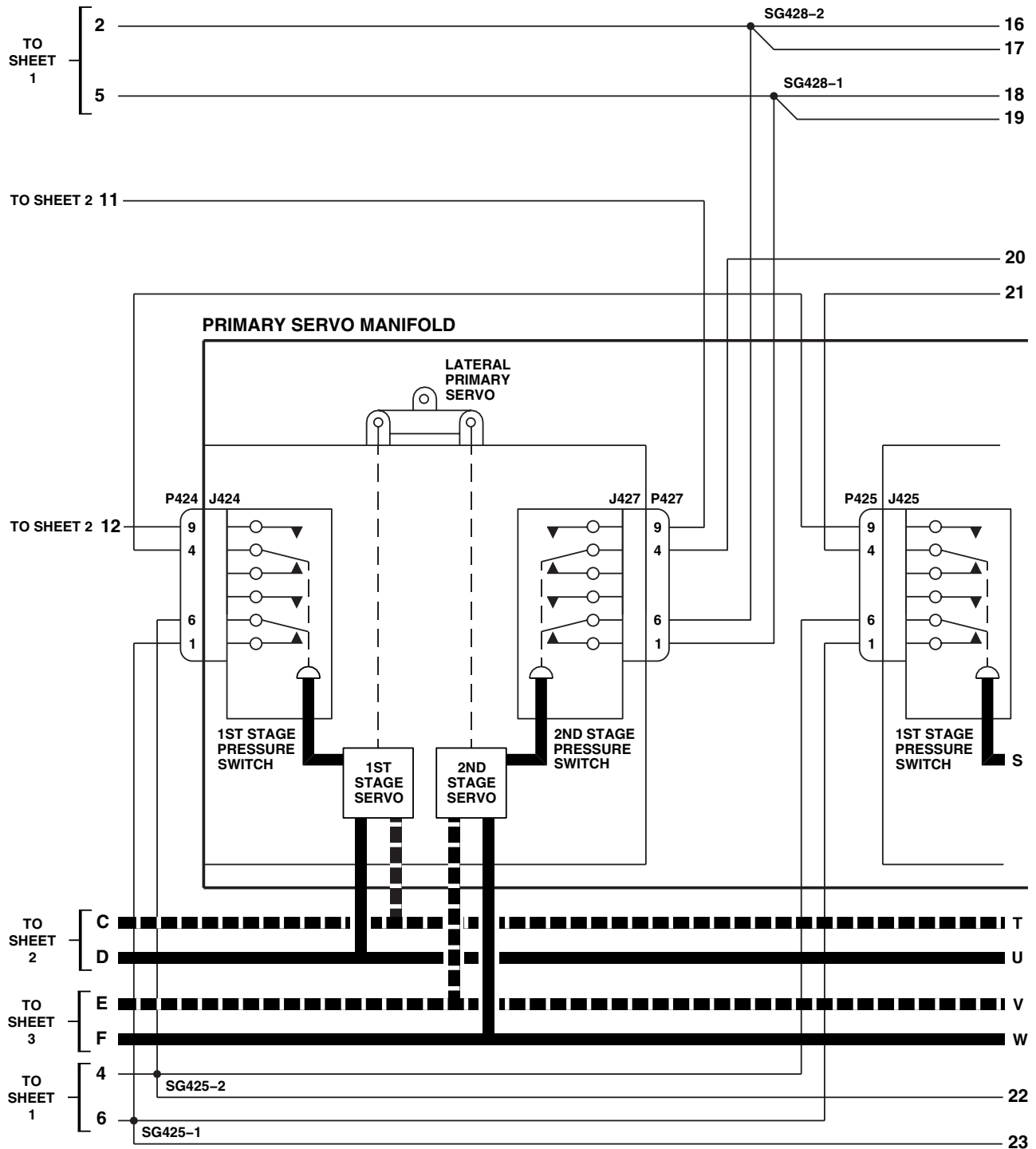


Figure 4. Hydraulic Systems Primary and Pilot Assist Servos Schematic Diagram. (Sheet 7 of 9)

SCHEMATICS - Continued

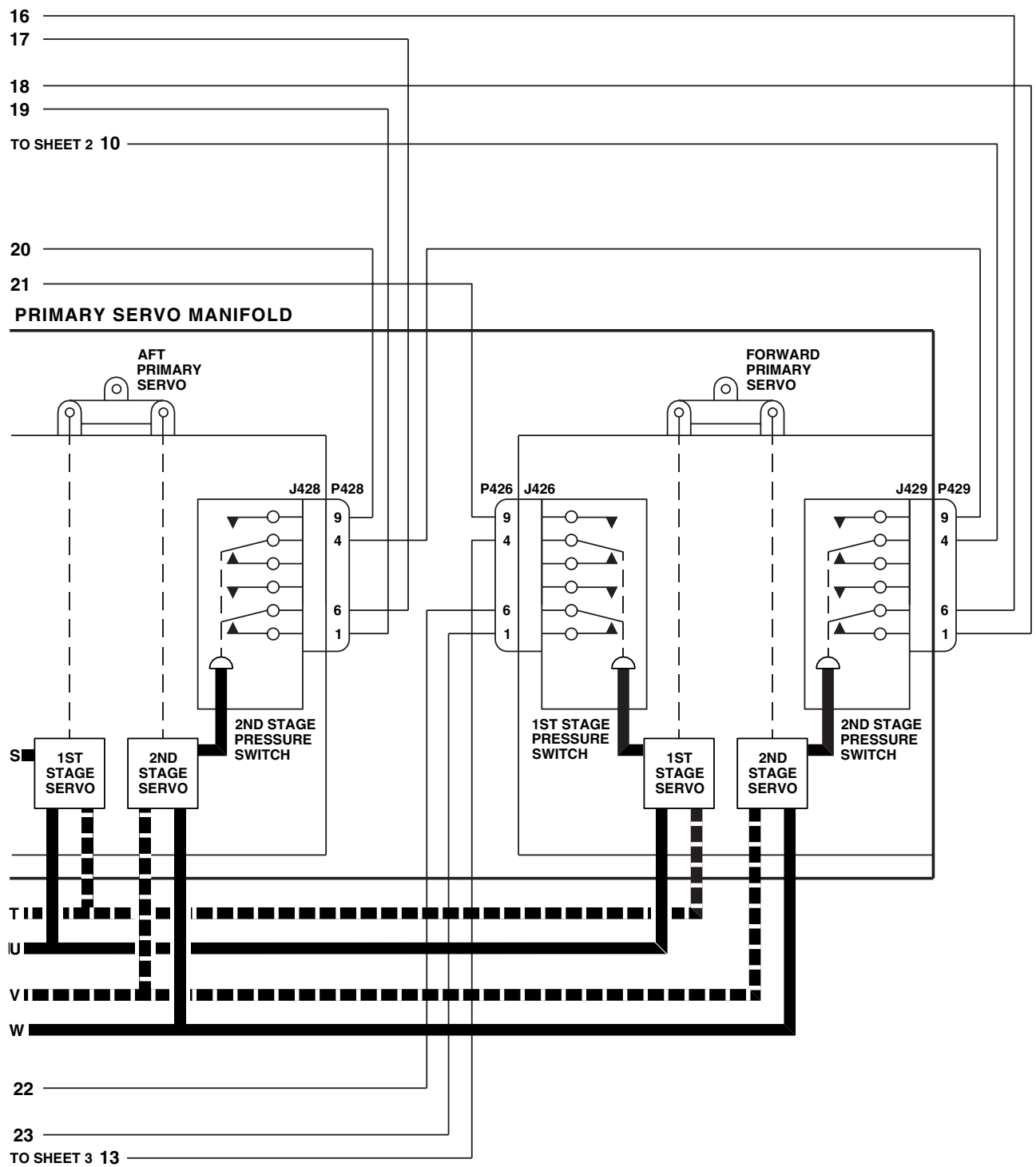
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Figure 4. Hydraulic Systems Primary and Pilot Assist Servos Schematic Diagram. (Sheet 8 of 9)

SCHEMATICS - Continued

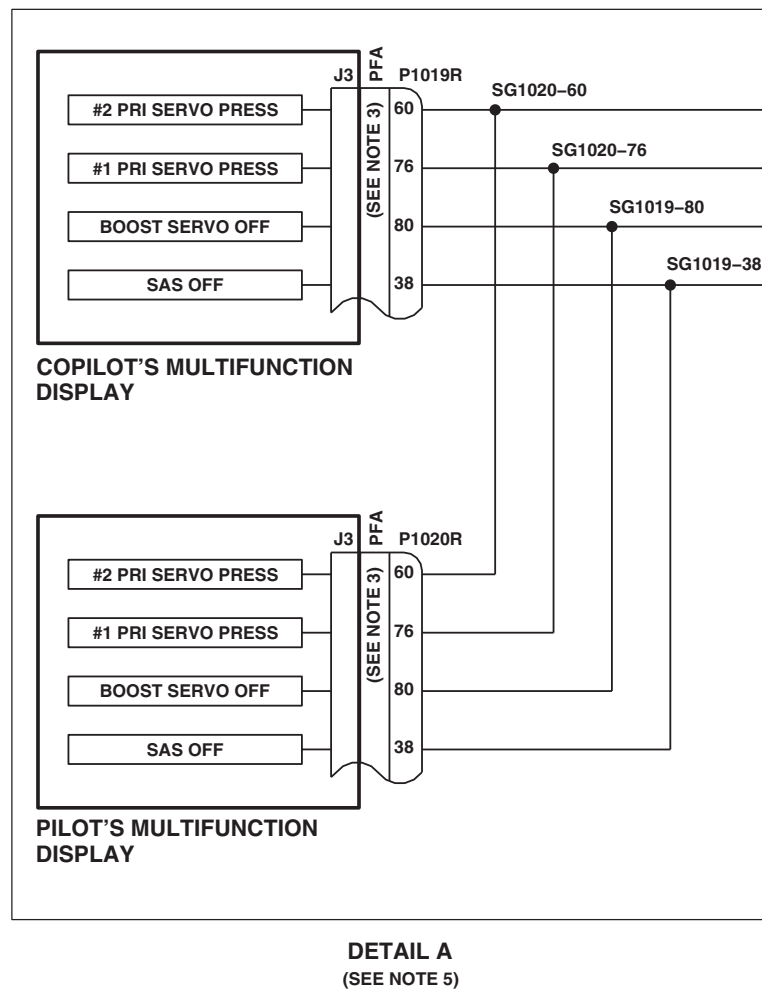
AB2208 9
SA

Figure 4. Hydraulic Systems Primary and Pilot Assist Servos Schematic Diagram. (Sheet 9 of 9)

END OF WORK PACKAGE

UNIT LEVEL
INSTRUMENT SYSTEMS
PITOT-STATIC SYSTEM

This WP supersedes WP 0087 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment

Multimeter, AN/PSM-45A
 Pitot and Static System Test Set, 7365

WP 0110 00

WP 0767 00

WP 0769 00

WP 0770 00

Tools and Special Tools

Electrical Repairer Toolkit, SC 5180-99-B06
 Locally-Made Pitot-Static Test Set Power Cable Assembly, Figure 49, WP 1805 00, Sheet 1
 Locally-Made Pitot-Static Test Set Tee Air Hose Assembly, Figure 49, WP 1805 00, Sheet 3
 Pitot-Static Head Adapters, (2 Required)

WP 0771 00

WP 0772 00

WP 0824 00

WP 1685 00

WP 1747 00

WP 1803 00

WP 1805 00

Material/Parts

Ethylene Glycol, Technical, Item 123, WP 1803 00

Equipment Condition

External Electrical Power Available, WP 1685 00
 or APU Operational, TM 1-1520-237-10

Personnel Required

Aircraft Electrician MOS 15F (1)

References

TM 1-1520-237-10
 WP 0105 00

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for component location diagram,

UH-60A UH-60L HH-60A HH-60L Figure 2  and **EH-60A** Figure 3  for altimeter schematic diagrams, and **MWO 50-42 UH-60A 82-23748-SUBQ UH-60L EH-60A HH-60A HH-60L** Figure 4  and **UH-60A 77-22714 - 82-23747** Figure 5  for Pitot-Static system test setup diagrams as an aid in operational/troubleshooting.

The Pitot-Static operational/troubleshooting procedure is subdivided as follows:

- Setup
- Altimeter/Vertical Speed Indicator
- Airspeed Indicator
- Barometric Altimeter Lighted Bezel Assembly
- Shutdown

SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

CAUTION

Damage could occur to Pitot-Static components if reverse pressure is applied to Pitot-Static heads during operational/troubleshooting procedure. Make sure test set and connections are properly set up before applying power to the test set. Use extreme caution when removing the Pitot-Static test set and adapters. Slowly return Pitot and static pressure to 0. Do not pull Pitot-Static adapters off Pitot heads until all readings are at 0, since this could cause damage to components.

NOTE

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0105 00).

1. Connect electrical power.
2. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER

CIRCUIT BREAKER PANEL

PILOT ALTM

Pilot's

CPLT ALTM

Copilot's

3. Make sure upper console PITOT HEAT switch is placed OFF.
4. Remove covers from Pitot tubes.
5. Bleed Pitot-Static lines (WP 0769 00).
6. Attach locally-made tee air hoses to test set (WP 1805 00).
7. Attach P2 of locally-made power cable to power connector at rear of test set.
8. Attach P1 of power cable to 28 vdc cabin utility receptacle.

NOTE

Qualified personnel may use external power or power test set using vac or vdc power.

9. Cap all tee air hoses.
10. Do a leak check of test set and tee air hoses.
11. Attach tee air hoses to Pitot-Static head adapters, making sure Pitot and static lines are not crossed.
12. **UH-60A 82-23748 - SUBQ UH-60L EH-60A MWO 50-42 HH-60A HH-60L** Attach static tee air hoses to pilot's and copilot's vertical speed indicators. ■

SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**CAUTION**

To prevent damage to Pitot-Static heads, use only pure technical ethylene glycol Item 117, WP 1803 00, (ethylene glycol) and not antifreeze.

NOTE

Make sure that vent holes on Pitot-Static head adapters are not plugged.

13. Wet inside surface of Pitot-Static head adapters with water to permit easier installation and prevent wear of sealing rings. In freezing temperatures, use mixture of one-half water and one-half ethylene glycol.

WARNING

To prevent injury to personnel, make sure Pitot heaters are off (PITOT HEAT switch on upper console panel is OFF), and Pitot heads are cool.

14. Using twisting motion, slide Pitot-Static head adapter connected to test set over pilot's (right) Pitot head until fully seated.
15. Using twisting motion, slide other Pitot-Static head adapter over copilot's (left) Pitot head until fully seated.
16. Turn on electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

**ALTIMETER/VERTICAL SPEED INDICATOR OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step****NOTE**

On the following steps, altimeter/vertical speed indicator readings shall not exceed tolerance ratings specified on test set calibration sticker.

1. Do SETUP, in this work package, if not already done.
2. Set pilot's and copilot's vertical speed indicator pointers to indicate 0 fpm with zero adjust screw (lower left side of instrument).
3. Set test set and pilot's and copilot's altimeters to scale reading of 29.92 in. hg.
4. Listen to pilot's and copilot's altimeters to make sure internal vibrator is working.

Indication/Condition

- a. Pilot's altimeter vibrator shall be working.
- b. Copilot's altimeter vibrator shall be working.

Corrective Action

1. If Indication/Condition a. is not as specified, go to PILOT'S ALIMETER VIBRATOR DOES NOT WORK, in this work package.

ALTIMETER/VERTICAL SPEED INDICATOR OPERATIONAL/TROUBLESHOOTING PROCEDURE -
Continued

2. If Indication/Condition b. is not as specified, go to COPILOT'S ALTIMETER VIBRATOR DOES NOT WORK, in this work package.

Step

5. Set test set altitude indicator to 0 feet.
6. While monitoring pilot's altimeter footage needle, turn pilot's altimeter barometric pressure knob from fully clockwise to fully counterclockwise.

Indication/Condition

- a. Pilot's altimeter footage indication shall change about 2500 feet.
- b. Pilot's altimeter footage needle shall not bind or skip.

Corrective Action

If Indication/Conditions are not as specified, replace malfunctioning altimeter (WP 0772 00).

Step

7. Set pilot's altimeter barometric scale to 29.92.

Indication/Condition

Pilot's altimeter shall indicate between -50 and 50 feet.

Corrective Action

If Indication/Condition is not as specified, replace pilot's altimeter (WP 0772 00).

Step

8. Repeat steps 6. and 7. for copilot's altimeter.
9. Set test set altimeter indicator to 500 feet.

Indication/Condition

Pilot's and copilot's altimeters shall indicate 470 to 530 feet.

Corrective Action

If Indication/Condition is not as specified, replace malfunctioning altimeter (WP 0772 00).

Step**CAUTION**

To prevent damage to equipment, do not exceed 5,000 fpm climb or descent on Pitot-Static test set.

10. Set test set altitude indicator to 1,000 feet, and allow readings to stabilize.

Indication/Condition

- a. Pilot's and copilot's altimeters shall indicate from 965 to 1,035 feet.
- b. **UH-60A 82-23748 - SUBQ UH-60L EH-60A MWO 50-42 HH-60A HH-60L** Pilot's and copilot's vertical speed indicators shall indicate UP within 250 fpm of test set vertical speed indicator. ■

ALTIMETER/VERTICAL SPEED INDICATOR OPERATIONAL/TROUBLESHOOTING PROCEDURE -
Continued**Corrective Action**

1. If Indication/Conditions a. and b. are not as specified, check for leaky adapter sealing rings and loose tee air hose connections to pilot's and copilot's vertical speed indicators and altimeters.
 - a. If leaky seals are present, replace seals.
 - b. If loose fittings are present, tighten fittings.
 - (1) If trouble remains, replace pilot's or copilot's altimeter or vertical speed indicator, as required (WP 0772 00) or (WP 0770 00).
2. If Indication/Condition a. is not as specified, replace pilot's or copilot's altimeter, as required (WP 0772 00).
3. If Indication/Condition b. is not as specified, replace pilot's or copilot's vertical speed indicator, as required (WP 0770 00).

Step

11. Set test set altimeter indicator to 2,500 feet.

Indication/Condition

Pilot's and copilot's altimeters indicate 2,430 to 2,570 feet.

Corrective Action

If Indication/Condition is not as specified, replace malfunctioning altimeter (WP 0772 00).

Step

12. Set test set altimeter indicator to 5,000 feet.

Indication/Condition

Pilot's and copilot's altimeters shall indicate 4,920 to 5,080 feet.

Corrective Action

If Indication/Condition is not as specified, replace malfunctioning altimeter (WP 0772 00).

Step

13. Set test set altimeter indicator to 10,000 feet.

Indication/Condition

Pilot's and copilot's altimeters shall indicate 9,900 to 10,100 feet.

Corrective Action

If Indication/Condition is not as specified, replace malfunctioning altimeter (WP 0772 00).

Step

14. Set test set altitude indicator to 15,500 feet.
15. Close off vacuum source from test set.
16. Using helicopter's clock or a watch, check pilot's and copilot's altimeter indications for 2 minutes, at 1-minute periods.

ALTIMETER/VERTICAL SPEED INDICATOR OPERATIONAL/TROUBLESHOOTING PROCEDURE -
Continued**Indication/Condition**

Pilot's and copilot's altimeters indications shall not change by more than 150 fpm.

Corrective Action

If Indication/Condition is not as specified, go to PILOT'S AND COPILOT'S ALTIMETERS CHANGE MORE THAN 150 FPM, in this work package.

Step**CAUTION**

Use extreme caution when removing the Pitot-Static test set and adapters. Do not exceed the maximum allowable feet per minute descent on Pitot-Static test set. Slowly return Pitot and static pressure to 0. Do not pull Pitot-Static adapters off Pitot heads until all readings are at 0, since this could cause damage to components.

17. Open Pitot-Static test set pressure release valves and return pressure to 0 psi.
18. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

AIRSPPEED INDICATOR OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step**

1. Do Setup, in this work package, if not already done.
2. Set test set airspeed indicator to 200 knots and allow readings to stabilize.

Indication/Condition

Pilot's and copilot's airspeed indicators shall indicate from 196.5 to 203.5 knots.

Corrective Action

1. If Indication/Condition is not as specified, check for leaky adapter sealing rings, and loose tee air hose connections to pilot's and copilot's airspeed indicators.
 - a. If leaky seals are present, replace seals.
 - b. If loose fittings are present, tighten fittings.
 - (1) If trouble remains, replace pilot's or copilot's airspeed indicator, as required (WP 0771 00).

Step

3. Using helicopter's clock or a watch, check pilot's airspeed indicator indication for 2 minutes.

Indication/Condition

Pilot's airspeed indicator shall indicate 196.5 to 203.5 knots.

AIRSPEED INDICATOR OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition is not as specified, visually check for damaged Pitot-Static head.
 - a. If Pitot-Static head is damaged, replace Pitot-Static head (WP 0767 00).
 - b. If Pitot-Static is not damaged, replace airspeed indicator (WP 0771 00).

Step

4. Slowly decrease test set pressure until test set airspeed indicator indicates 100 knots.

Indication/Condition

Pilot's airspeed indicator shall indicate 98 to 102 knots.

Corrective Action

If Indication/Condition is not as specified, replace airspeed indicator (WP 0771 00).

Step

5. Set test set airspeed indicator to 0 knots.
6. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

BAROMETRIC ALTIMETER LIGHTED BEZEL ASSEMBLY OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step**

1. Turn INST LT PILOT FLT control from OFF to BRT.

Indication/Condition

Pilot's barometric altimeter lighted bezel assembly shall increase in brilliance.

Corrective Action

If Indication/Condition is not as specified, go to PILOT'S BAROMETRIC ALTIMETER LIGHTED BEZEL DOES NOT WORK, in this work package.

Step

2. Turn INST LT PILOT FLT control from BRT to OFF.

Indication/Condition

Pilot's barometric altimeter lighted bezel assembly shall decrease in brilliance and go off.

Corrective Action

If Indication/Condition is not as specified, troubleshoot instrument panel and consoles indicator lights dimming (WP 0110 00).

**BAROMETRIC ALTIMETER LIGHTED BEZEL ASSEMBLY OPERATIONAL/TROUBLESHOOTING
PROCEDURE - Continued****Step**

3. Turn CPLT FLT INST LTS control from OFF to BRT.

Indication/Condition

Copilot's barometric altimeter lighted bezel assembly shall increase in brilliance.

Corrective Action

If Indication/Condition is not as specified, go to COPILOT'S BAROMETRIC ALTIMETER LIGHTED BEZEL DOES NOT WORK, in this work package.

Step

4. Turn CPLT FLT INST LTS control from BRT to OFF.

Indication/Condition

Copilot's barometric altimeter lighted bezel assembly shall decrease in brilliance and go off.

Corrective Action

If Indication/Condition is not as specified, troubleshoot instrument panel and console lights dimming (WP 0110 00).

Step

5. If no other tests are to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step****CAUTION**

Damage could occur to Pitot-Static components if reverse pressure is applied to Pitot-Static heads during operational/troubleshooting procedure. Make sure test set and connections are properly set up before applying power to the test set. Use extreme caution when removing the Pitot-Static test set and adapters. Slowly return Pitot and static pressure to 0. Do not pull Pitot-Static adapters off Pitot heads until all readings are at 0, since this could cause damage to components.

1. Turn off and disconnect external electrical power.
2. Disconnect test setup.
3. If ethylene glycol was used to install Pitot-Static head adapters, wipe off Pitot heads with soft rag after removing adapters.
4. Install covers on Pitot tubes.

SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Covers shall be installed on Pitot tubes.

Corrective Action

None Required

PILOT'S ALTIMETER VIBRATOR DOES NOT WORK**SYMPTOM**

Pilot's altimeter vibrator shall be working.

MALFUNCTION

Pilot's altimeter vibrator does not work.

CORRECTIVE ACTION**1. Check for 28 vdc between P306R-M and P306R-P.**

- a. If voltage is as specified, replace altimeter (WP 0772 00). Go to **4**.
- b. If voltage is not as specified, go to **2**.

2. Check continuity between P306R-M and ground.

- a. If continuity is present, go to **3**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.

3. Check continuity between P306R-P and PILOT ALTM circuit breaker terminal 1.

- a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **4**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.

4. Procedure completed.**COPILOT'S ALTIMETER VIBRATOR DOES NOT WORK****SYMPTOM**

Copilot's altimeter vibrator shall be working.

MALFUNCTION

Copilot's altimeter vibrator does not work.

CORRECTIVE ACTION**1. Check for 28 vdc between P1-M and P1-P.**

- a. If voltage is as specified, replace altimeter (WP 0772 00). Go to **4**.
- b. If voltage is not as specified, go to **2**.

2. Check continuity between P1-M and ground.

- a. If continuity is present, go to **3**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.

3. Check continuity between P1-P and COPILLOT ALTM circuit breaker terminal 1.

- a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **4**.

COPILOT'S ALTIMETER VIBRATOR DOES NOT WORK - Continued

- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.

4. Procedure completed.**PILOT'S AND COPILOT'S ALTIMETERS CHANGE MORE THAN 150 FPM****SYMPTOM**

Pilot's and copilot's altimeters indications shall not change by more than 150 FPM.

MALFUNCTION

Pilot's and copilot's altimeters change more than 150 FPM.

CORRECTIVE ACTION**1. Check that static line connections are tight.**

- a. If connections are tight, go to **2**.
- b. If connections are not tight, tighten static line connections (WP 0769 00). Go to **4**.

2. Visually check for damaged static lines.

- a. If damaged lines are present, replace static tubing (WP 0769 00). Go to **4**.
- b. If damaged lines are not present, go to **3**.

3. Visually check for damaged Pitot-Static heads.

- a. If damaged heads are present, replace Pitot-Static head assembly, as required (WP 0767 00). Go to **4**.
- b. If damaged heads are not present, replace pilot's or copilot's altimeter, as required (WP 0772 00). Go to **4**.

4. Procedure completed.**PILOT'S BAROMETRIC ALTIMETER LIGHTED BEZEL DOES NOT WORK****SYMPTOM**

Pilot's barometric altimeter lighted bezel shall increase in brilliance.

MALFUNCTION

Pilot's barometric altimeter lighted bezel does not work.

CORRECTIVE ACTION**1. Turn INST LT PILOT FLT control to BRT. Go to 2.****2. Check for 28 vdc between P183-A and ground.**

- a. If voltage is not as specified, troubleshoot instrument panel and consoles indicator lights dimming (WP 0110 00). Go to **4**.
- b. If voltage is as specified, go to **3**.

3. Check continuity between P183-B and ground.

- a. If continuity is present, replace lighted bezel assembly (WP 0772 00). Go to **4**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.

4. Procedure completed.

COPILOT'S BAROMETRIC ALTIMETER LIGHTED BEZEL DOES NOT WORK**SYMPTOM**

Copilot's barometric altimeter lighted bezel shall increase in brilliance.

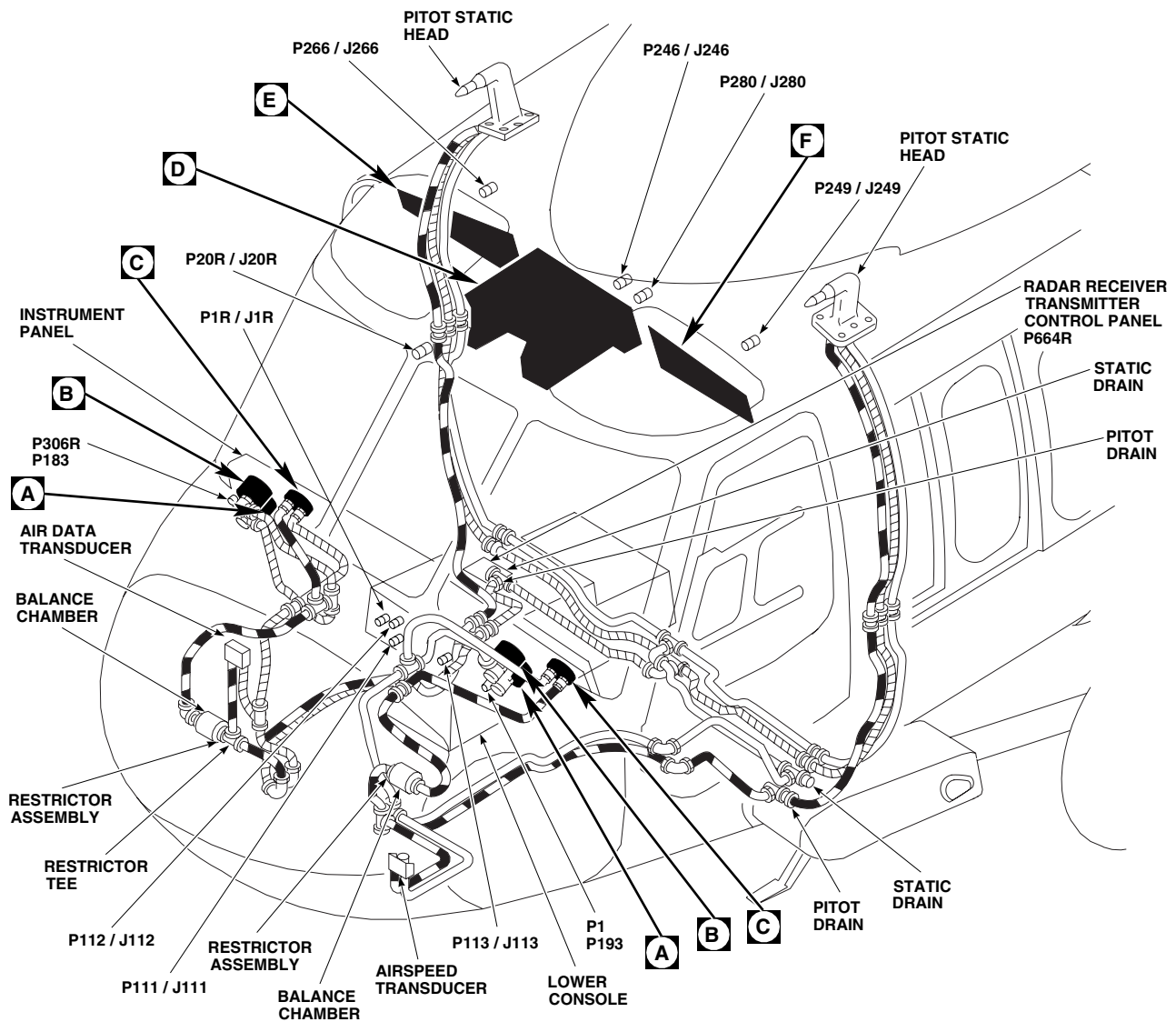
MALFUNCTION

Copilot's barometric altimeter lighted bezel does not work.

CORRECTIVE ACTION

- 1. Turn CPLT FLT INST LTS control to BRT. Go to 2.**
- 2. Check for 28 vdc between P193-A and ground.**
 - a. If voltage is not as specified, troubleshoot instrument panel and consoles indicator lights dimming (WP 0110 00). Go to **4**.
 - b. If voltage is as specified, go to **3**.
- 3. Check continuity between P193-B and ground.**
 - a. If continuity is present, replace lighted bezel assembly (WP 0772 00). Go to **4**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.
- 4. Procedure completed.**

LOCATION



NOTES

1. **UH-60A UH-60L HH-60A HH-60L**
PILOT'S BAROMETRIC ALTIMETER ENCODER.
2. **EH-60A** PILOT'S AND COPILOT'S BAROMETRIC ALTIMETER ENCODER.
3. **MWO 50-42**
UH-60A 82-23748 - SUBQ
UH-60L EH-60A HH-60A HH-60L
PILOT'S AND COPILOT'S VERTICAL SPEED INDICATORS ARE VENTED TO CABIN PRESSURE, NOT STATIC SYSTEM PRESSURE.
4. **EH-60A**

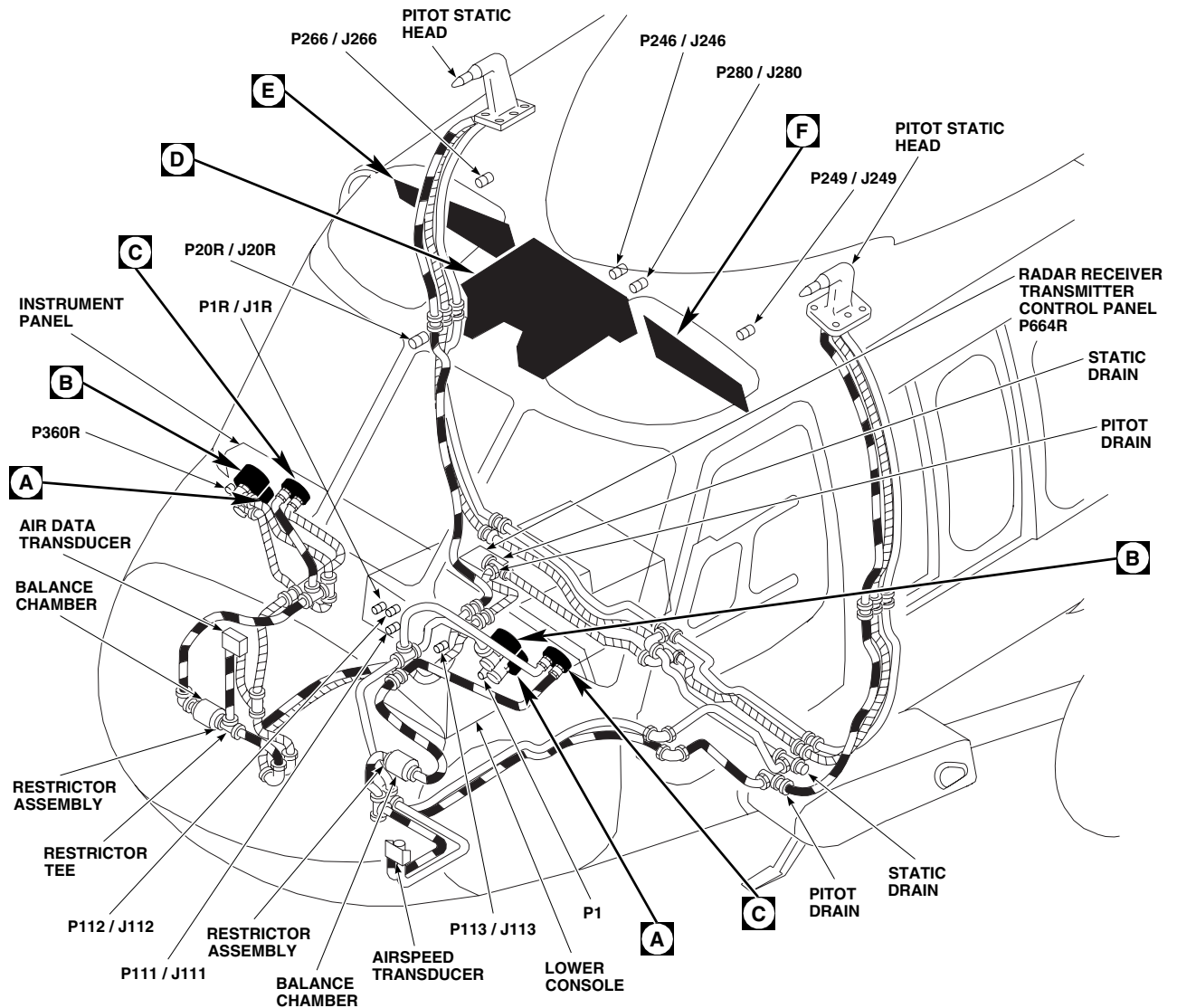
LEGEND

	PITOT
	STATIC (1)
	STATIC (2)

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SA

Figure 1. Pitot-Static System Location Diagram. (Sheet 1 of 5)

LOCATION - Continued



TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P1	COPILOT'S BAROMETRIC ALTIMETER ENCODER
P1R / J1R	COCKPIT BL 6 LH, STA 210
P20R / J20R	CABIN BL 40 RH, STA 248
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P112 / J112	COCKPIT BL 6 RH, STA 197
P113 / J113	COCKPIT BL 6 RH, STA 197
P183	PILOT'S BAROMETRIC ALTIMETER LIGHTED, BEZEL ASSEMBLY

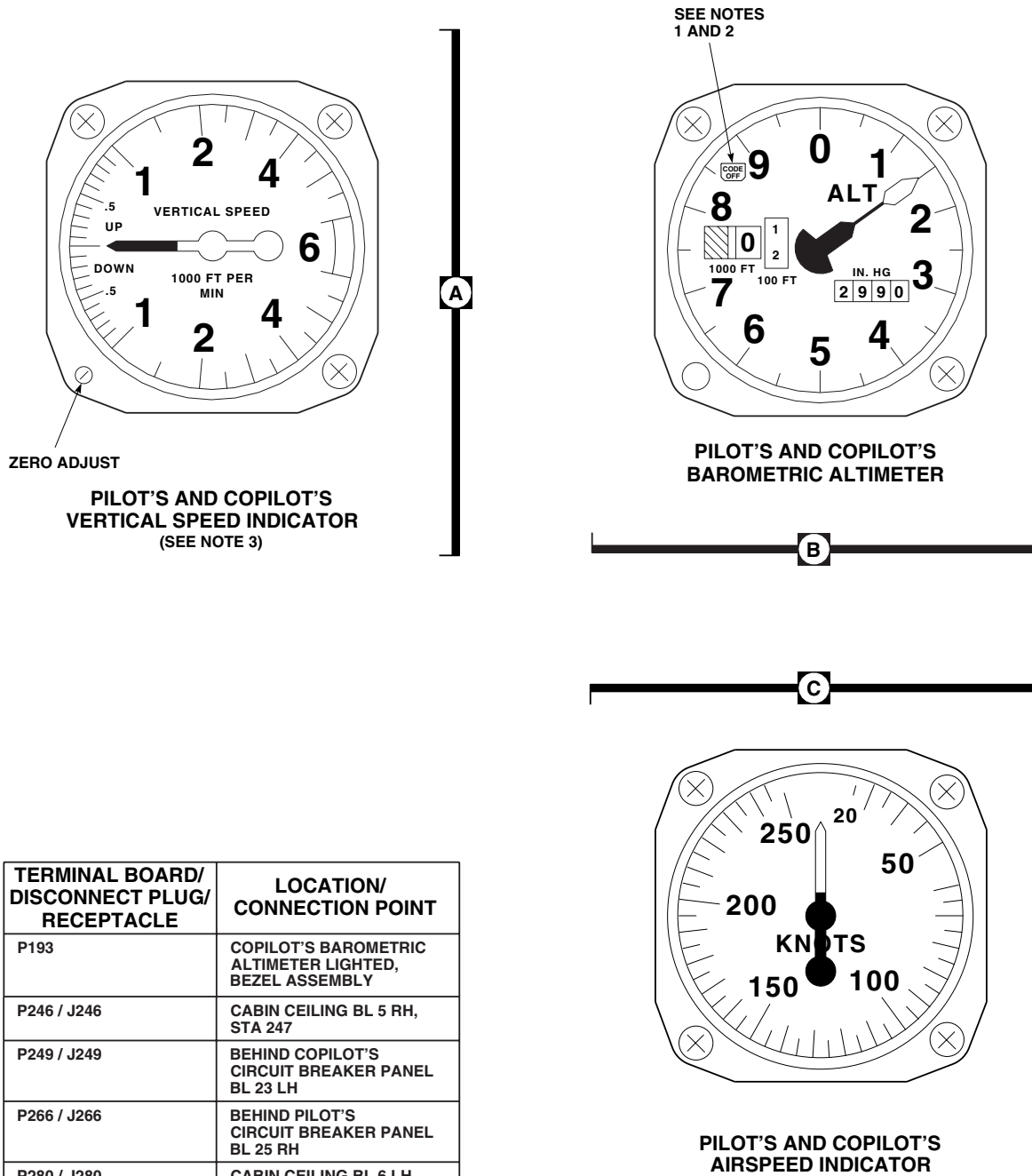
EFFECTIVITY

UH-60A 77-22714-82-23747

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Figure 1. Pitot-Static System Location Diagram. (Sheet 2 of 5)

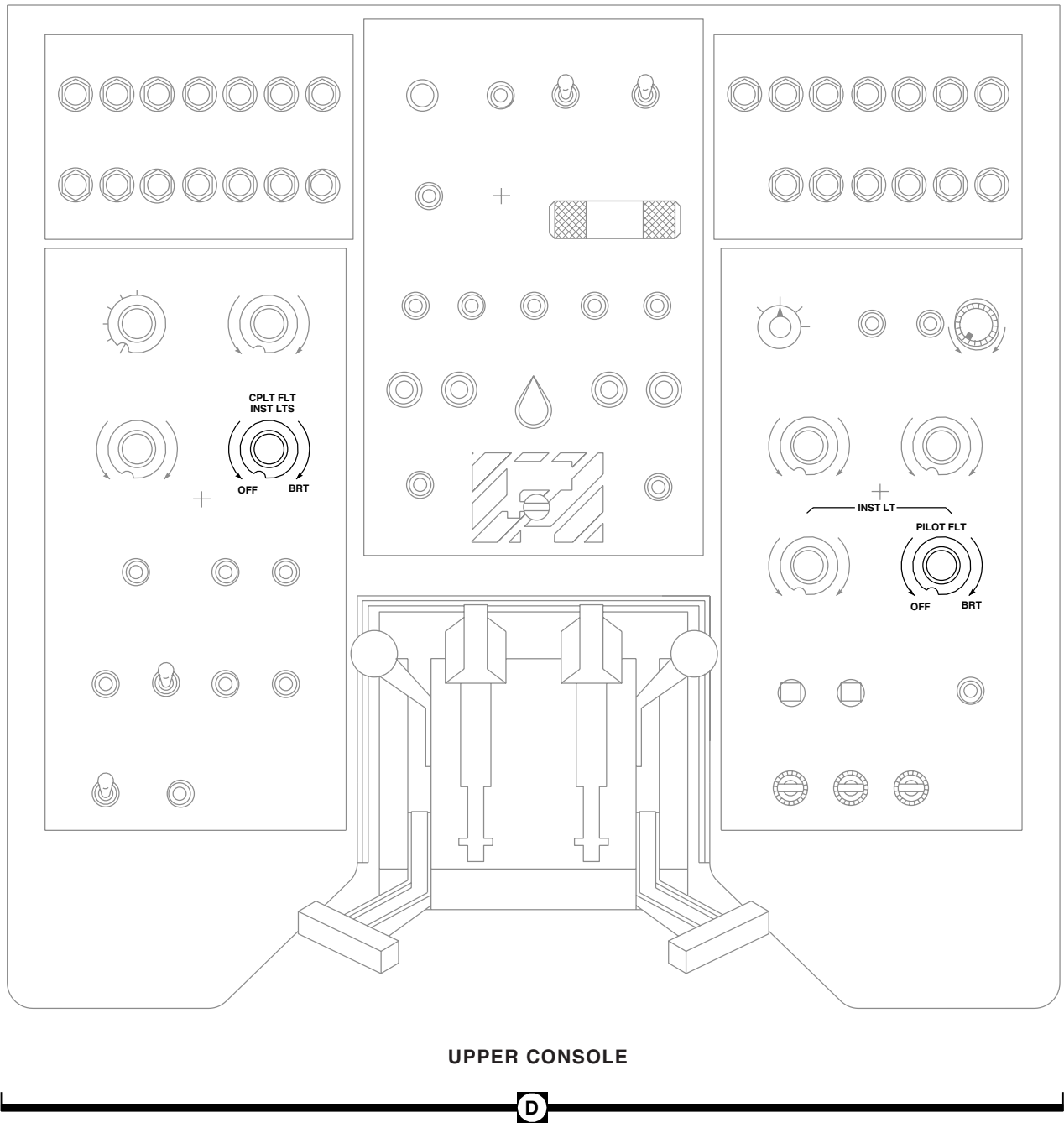
LOCATION - Continued



AA2606_3A
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Figure 1. Pitot-Static System Location Diagram. (Sheet 3 of 5)

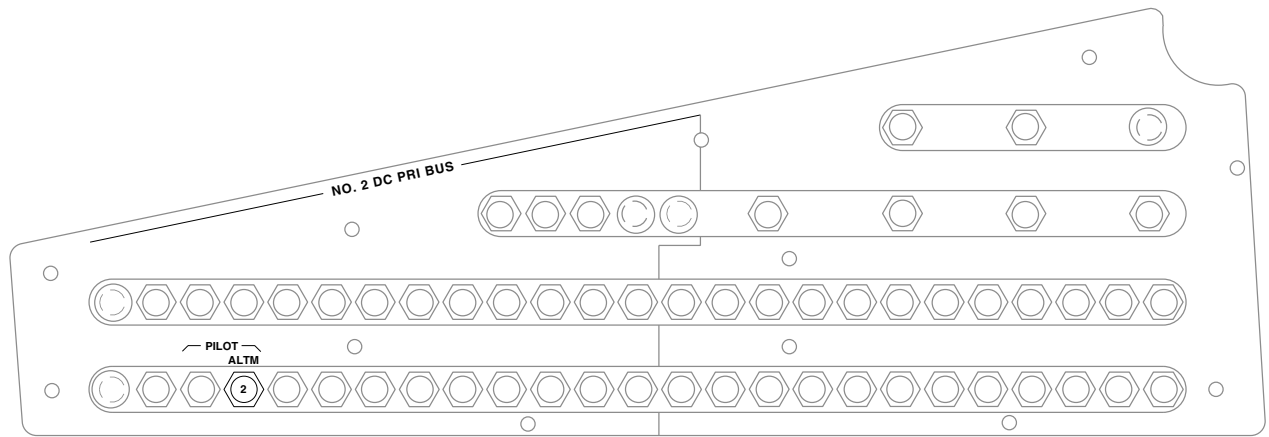
LOCATION - Continued



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Figure 1. Pitot-Static System Location Diagram. (Sheet 4 of 5)

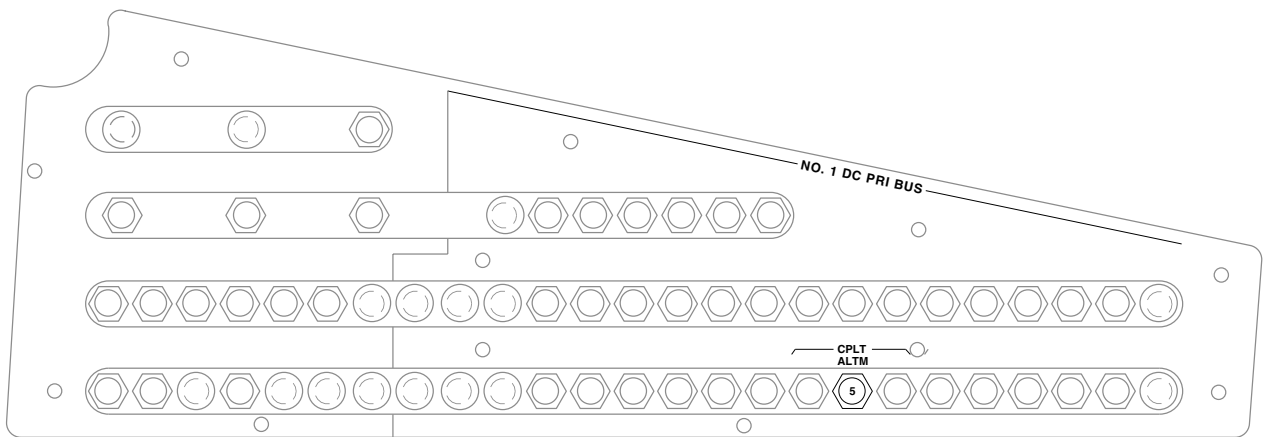
LOCATION - Continued



PILOT'S CIRCUIT BREAKER PANEL

E

F

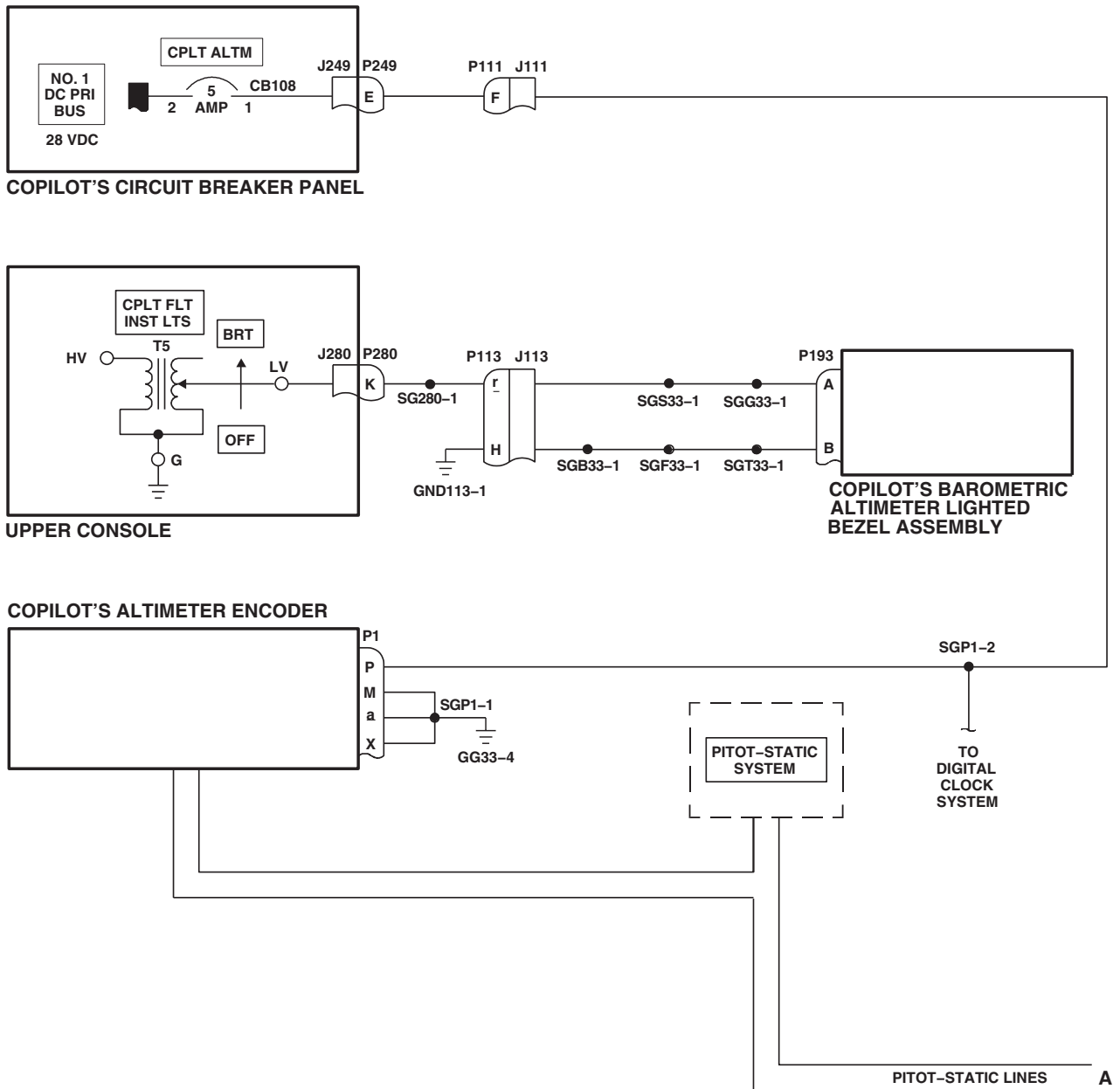


COPILLOT'S CIRCUIT BREAKER PANEL

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SA

Figure 1. Pitot-Static System Location Diagram. (Sheet 5 of 5)

SCHEMATICS



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Figure 2. Altimeter Schematic Diagram **UH-60A UH-60L HH-60A HH-60L** . (Sheet 1 of 2)

SCHEMATICS - Continued

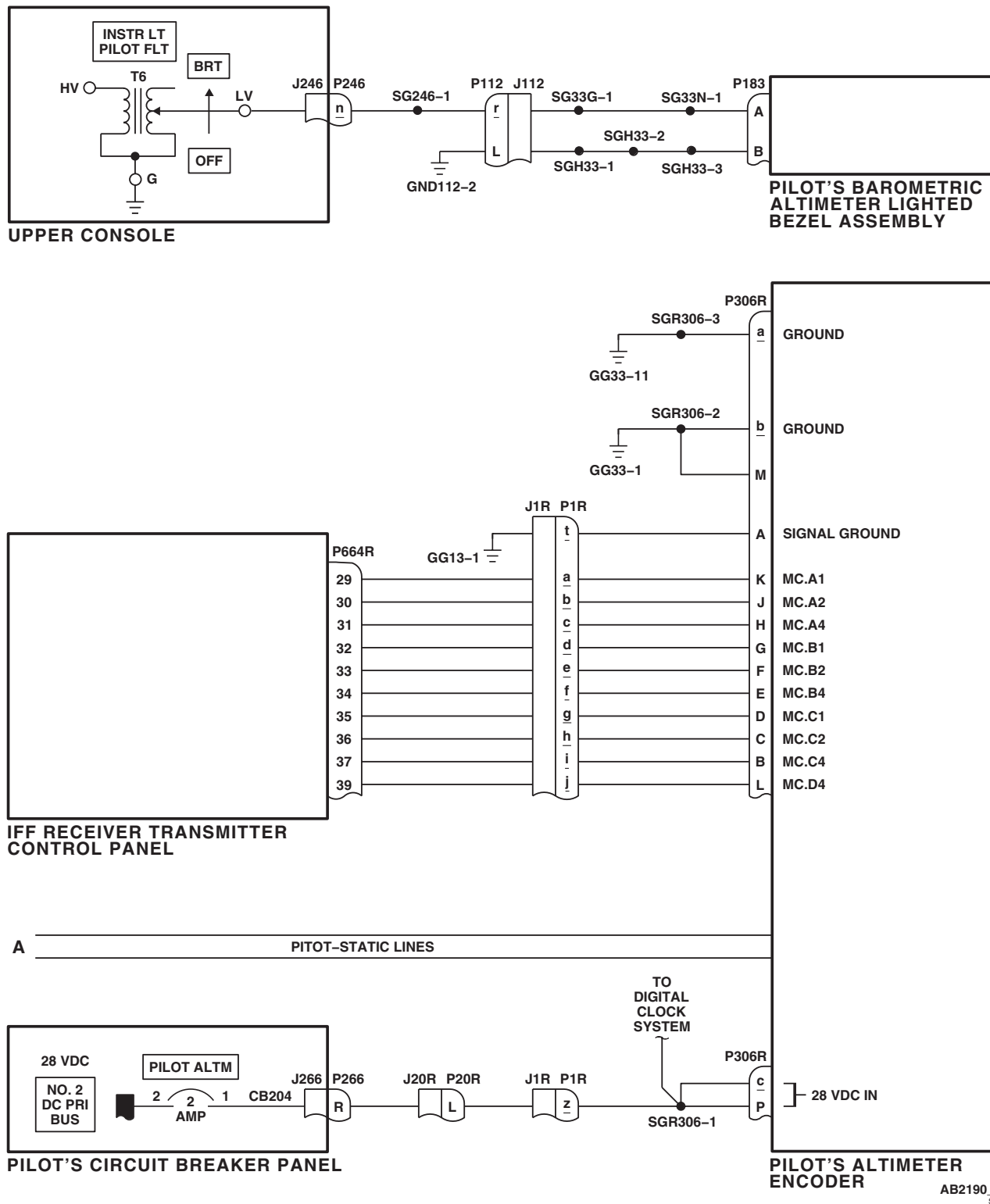


Figure 2. Altimeter Schematic Diagram **UH-60A UH-60L HH-60A HH-60L** . (Sheet 2 of 2)

COPILOT'S CIRCUIT BREAKER PANEL

28 VDC NO. 1 DC PRI BUS

CPTLT ALTM

2 5 1 CB108

AMP

J249 P249 P111 J111

UPPER CONSOLE COPILOT'S ALTIMETER ENCODER

HV T5 BRT LV

CPTLT FLT INST LTS

OFF

G

J280 P280 P113 J113

SG280-1 GND113-1 SGB33-1 SGF33-1 SGT33-1

COPILOT'S BAROMETRIC ALTIMETER LIGHTED BEZEL ASSEMBLY

P193 A B

COPILOT'S CIRCUIT BREAKER PANEL

28 VDC NO. 2 DC PRI BUS

PILOT ALTM

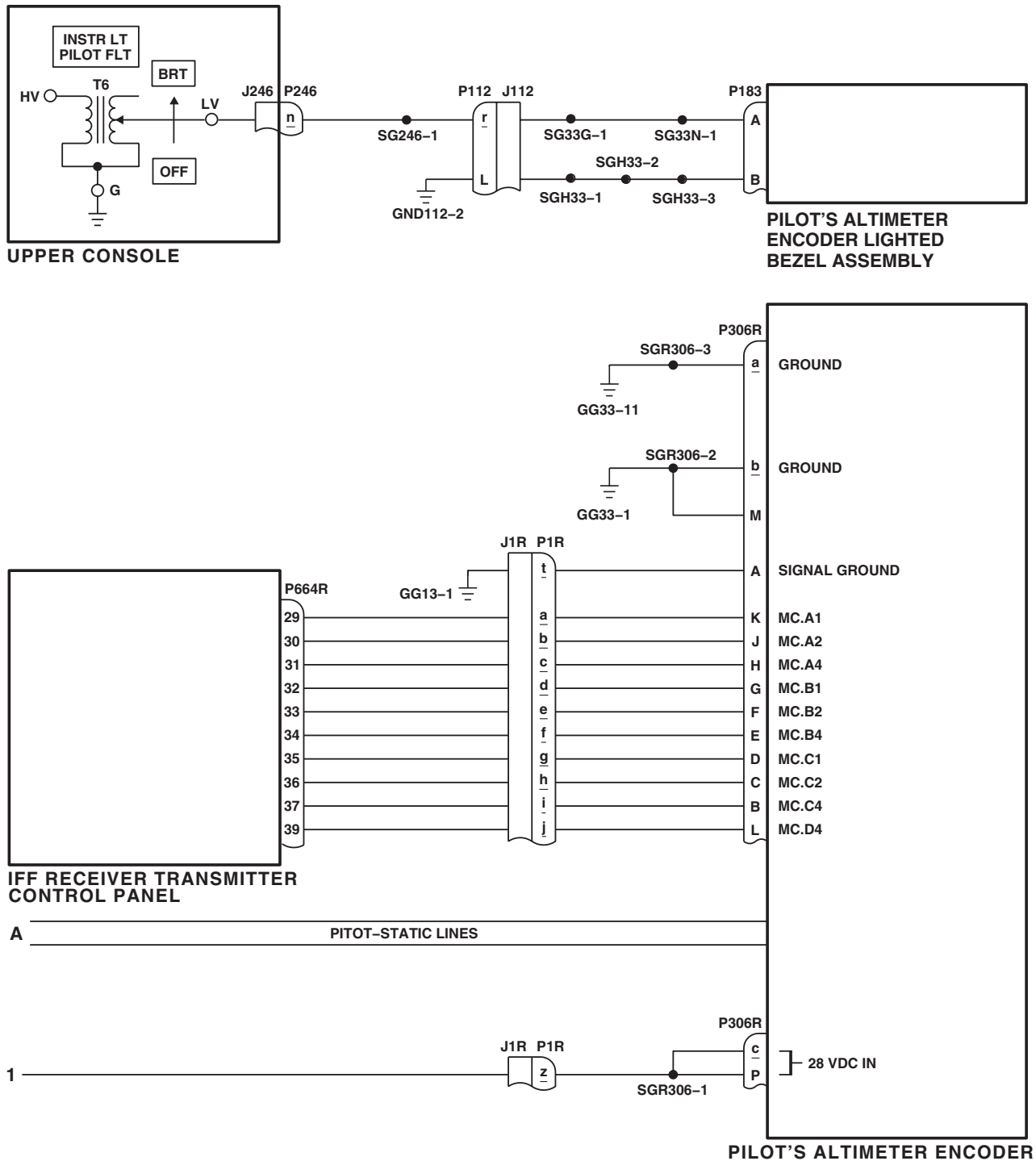
2 2 1 CB204

AMP

J266 P266 J20R P20R

Change 1

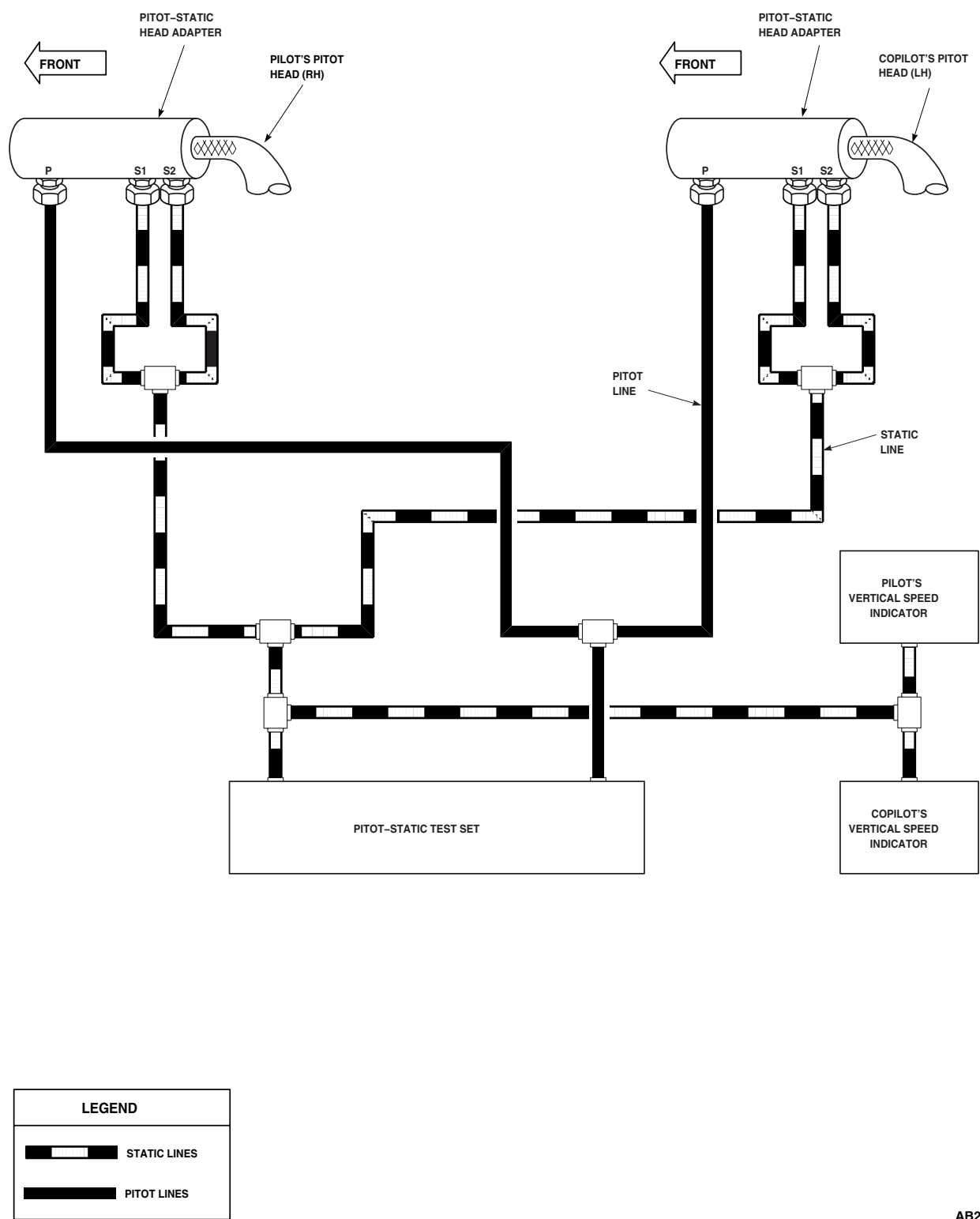
SCHEMATICS - Continued



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SA

Figure 3. Altimeter Schematic Diagram **EH-60A** . (Sheet 2 of 2)

TEST SETUP



AB2298B
SA

Figure 4. Pitot-Static System Test Setup Diagram
MWO 50-42 UH-60A 82-23748-SUBQ UH-60L EH-60A HH-60A HH-60L .

TEST SETUP - Continued

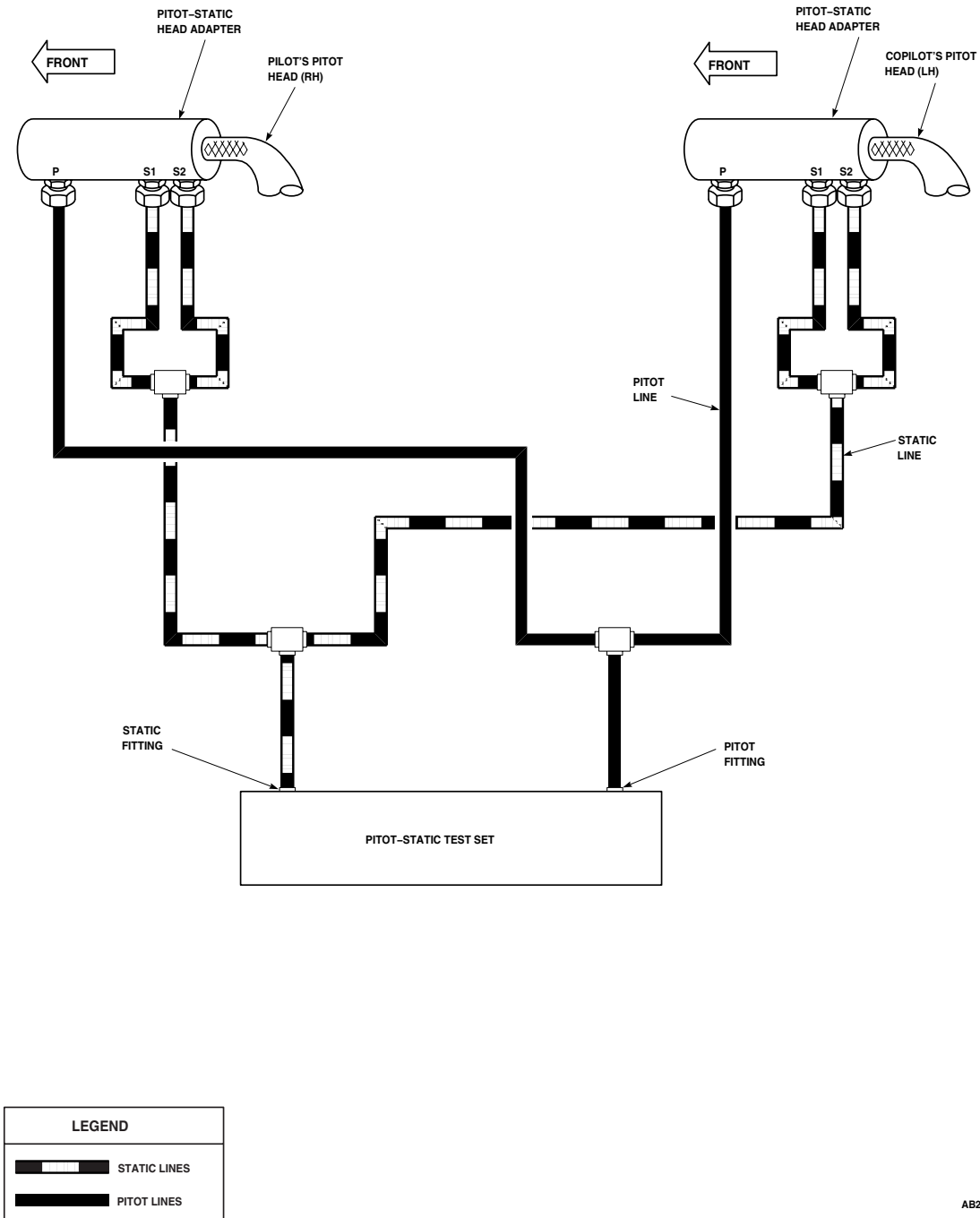


Figure 5. Pitot-Static System Test Setup Diagram **UH-60A 77-22714 - 82-23747** .

END OF WORK PACKAGE

UNIT LEVEL
INSTRUMENT SYSTEMS
PITOT HEATER

This WP supersedes WP 0088 00, dated 17 April 2006.

INITIAL SETUP:**Test Equipment**

Multimeter, AN/PSM-45A

WP 0120 00

WP 0121 00

WP 0769 00

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01

WP 0824 00

Electrical Repairer Toolkit, SC 5180-99-B06

WP 0828 00

WP 1685 00

Personnel Required

Aircraft Electrician MOS 15F (1)

WP 1747 00

Tactical Transport Helicopter Repairer MOS 15T (1)

Equipment Condition

External Electrical Power Connected, WP 1685 00

or APU Operational, TM 1-1520-237-10

References

TM 1-1520-237-10

WP 0104 00

PITOT HEATER OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- See Figure 1 for component location diagram, and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER**CIRCUIT BREAKER PANEL****UH-60L**

LEFT PITOT HEAT

Copilot's

NO. 1 ENG ANTI ICE

Copilot's

RT PITOT HEAT

Pilot's

EH-60A

PITOT HEATER OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
LEFT PITOT HEAT	Copilot's
NO. 1 ENG ANTI ICE	Copilot's
RIGHT PITOT HEAT	Pilot's
HH-60A HH-60L MFD PILOT	Aux Circuit Breaker Panel
HH-60A HH-60L MFD COPILOT	Aux Circuit Breaker Panel
2. Make sure upper console PITOT HEAT switch is placed OFF. 3. Remove covers from Pitot tubes. 4. Turn on electrical power. HH-60A HH-60L Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed).	

WARNING

To avoid skin burn, carefully feel for heat at Pitot tubes.

5. Carefully feel for heat at left and right Pitot tubes.

Indication/Condition

No heat shall be felt.

Corrective Action

If Indication/Condition is not as specified, replace PITOT HEAT switch S30 (WP 0828 00).

PROCEDURE**Step**

6. Place upper console PITOT HEAT switch to ON and carefully feel for heat at Pitot tubes.

Indication/Condition

- a. Heat shall be felt at left Pitot tube.
- b. LFT PITOT HEAT capsule or legend shall be off.
- c. Heat shall be felt at right Pitot tube.
- d. RT PITOT HEAT capsule or legend shall be off.

Corrective Action

1. If all Indication/Conditions are not as specified, replace PITOT HEAT switch S30 (WP 0828 00).
2. If Indication/Conditions a. and b. are not as specified, go to NO HEAT AT LEFT PITOT TUBE AND LFT PITOT HEAT CAPSULE OR LEGEND IS ON, in this work package.
3. If Indication/Conditions c. and d. are not as specified, go to NO HEAT AT RIGHT PITOT TUBE AND RT PITOT HEAT CAPSULE OR LEGEND IS ON, in this work package.

PITOT HEATER OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

7. Place upper console PITOT HEAT switch to OFF.
8. Turn off electrical power.
9. Turn pilot's or copilot's MFD switch OFF.
10. After Pitot tubes have cooled, install covers.

Indication/Condition

Covers shall be installed.

Corrective Action

None Required

NO HEAT AT LEFT PITOT TUBE AND LFT PITOT HEAT CAPSULE OR LEGEND IS ON**SYMPTOM**

Heat shall be felt at right pitot tube and LFT PITOT HEAT capsule or legend shall be off.

MALFUNCTION

No heat at left pitot tube and LFT PITOT HEAT capsule or legend is on.

CORRECTIVE ACTION**1. Check resistance between:**

- J269-A and J269-D (9 to 29 ohms)
- J269-B and J269-C (45 to 55 ohms)

a. If resistance is as specified, go to **2**.

b. If resistance is not as specified, replace left Pitot-static tube assembly (WP 0769 00). Go to **9**.

2. Check continuity between:

- P269-C and ground
- P269-D and ground

a. If continuity is present, go to **3**.

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **9**.

3. Check for 9 to 29 ohms between P242-K and ground.

a. If resistance is as specified, go to **4**.

b. If resistance is not as specified, repair/replace wiring between P242-K and P269-A (WP 1747 00). Go to **9**.

4. Check for 45 to 55 ohms between P242-m and ground.

a. If resistance is as specified, go to **5**.

b. If resistance is not as specified, repair/replace wiring between P242-m and P269-B (WP 1747 00). Go to **9**.

5. Check for 115 vac between P242-r and ground.

NO HEAT AT LEFT PITOT TUBE AND LFT PITOT HEAT CAPSULE OR LEGEND IS ON - Continued

- a. If voltage is as specified, troubleshoot left relay panel (WP 0121 00). Go to **9**.
- b. If voltage is not as specified, go to **6**.
- 6. Check for 115 vac between PITOT HEAT switch terminal 6 and ground.**
 - a. If voltage is as specified, repair/replace wiring between switch terminal 6 and P242-r (WP 1747 00). Go to **9**.
 - b. If voltage is not as specified, go to **7**.
- 7. Check for 115 vac between PITOT HEAT switch terminal 5 and ground.**
 - a. If voltage is as specified, replace switch S30 (WP 0828 00). Go to **9**.
 - b. If voltage is not as specified, go to **8**.
- 8. Check for 115 vac between LEFT PITOT HEAT circuit breaker terminal 1 and ground.**
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and switch terminal 5 (WP 1747 00). Go to **9**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **9**.
- 9. Procedure completed.**

NO HEAT AT RIGHT PITOT TUBE AND RT PITOT HEAT CAPSULE OR LEGEND IS ON**SYMPTOM**

Heat shall be felt at right pitot tube and RT PITOT HEAT capsule or legend shall be off.

MALFUNCTION

No heat at right pitot tube and RT PITOT HEAT capsule or legend is on.

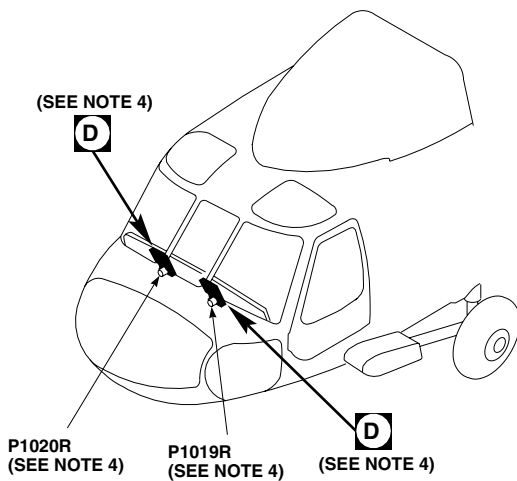
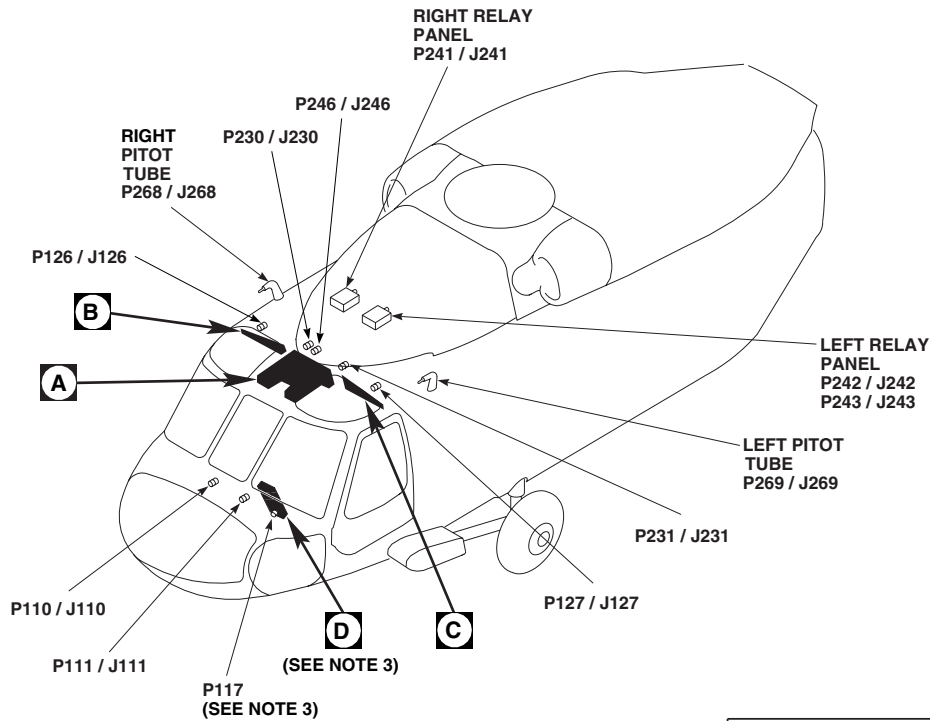
CORRECTIVE ACTION

- 1. Check resistance between:**
 - **J268-A and J268-D (9 to 29 ohms)**
 - **J268-B and J268-C (45 to 55 ohms)**
 - a. If resistance is as specified, go to **2**.
 - b. If resistance is not as specified, replace right Pitot-static tube assembly (WP 0769 00). Go to **9**.
- 2. Check continuity between:**
 - **P268-C and ground**
 - **P268-D and ground**
 - a. If continuity is present, go to **3**.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **9**.
- 3. Check for 9 to 29 ohms between P241-c and ground.**
 - a. If resistance is as specified, go to **4**.
 - b. If resistance is not as specified, repair/replace wiring between P241-c and P268-A (WP 1747 00). Go to **9**.

NO HEAT AT RIGHT PITOT TUBE AND RT PITOT HEAT CAPSULE OR LEGEND IS ON - Continued

- 4. Check for 45 to 55 ohms between P241-r and ground.**
 - a. If resistance is as specified, go to **5**.
 - b. If resistance is not as specified, repair/replace wiring between P241-r and P268-B (WP 1747 00). Go to **9**.
- 5. Check for 115 vac between PITOT HEAT switch terminal 3 and ground.**
 - a. If voltage is as specified, go to **6**.
 - b. If voltage is not as specified, go to **7**.
- 6. Check continuity between PITOT HEAT switch terminal 3 and P241-m.**
 - a. If continuity is present, troubleshoot right relay panel (WP 0120 00). Go to **9**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9**.
- 7. Check for 115 vac between PITOT HEAT switch terminal 2 and ground.**
 - a. If voltage is as specified, replace switch (WP 0828 00). Go to **9**.
 - b. If voltage is not as specified, go to **8**.
- 8. Check continuity between RT PITOT HEAT circuit breaker terminal 1 and PITOT HEAT switch terminal 2.**
 - a. If continuity is present, replace circuit breaker (WP 0824 00). Go to **9**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9**.
- 9. Procedure completed.**

LOCATION



NOTES

1. UH-60A UH-60L HH-60A
HH-60L
2. EH-60A
3. UH-60A UH-60L EH-60A
4. HH-60A HH-60L

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P110 / J110	COCKPIT, BL 7.5 RH, STA 197
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P117 / J117	CAUTION / ADVISORY PANEL (SEE NOTE 3)
P126 / J126	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 28 RH, STA 243
P127 / J127	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 28 LH, STA 243
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P231 / J231	CABIN CEILING, BL 9 LH, STA 247
P241 / J241	RIGHT RELAY PANEL
P242 / J242	LEFT RELAY PANEL
P243 / J243	LEFT RELAY PANEL
P246 / J246	CABIN CEILING, BL 5 RH, STA 247
P268 / J268	RIGHT PITOT TUBE
P269 / J269	LEFT PITOT TUBE
P1019R / J3	COPILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 4)
P1020R / J3	PILOT'S MULTIFUNCTION DISPLAY (SEE NOTE 4)

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Figure 1. Pitot Heater System Location Diagram. (Sheet 1 of 3)

LOCATION - Continued

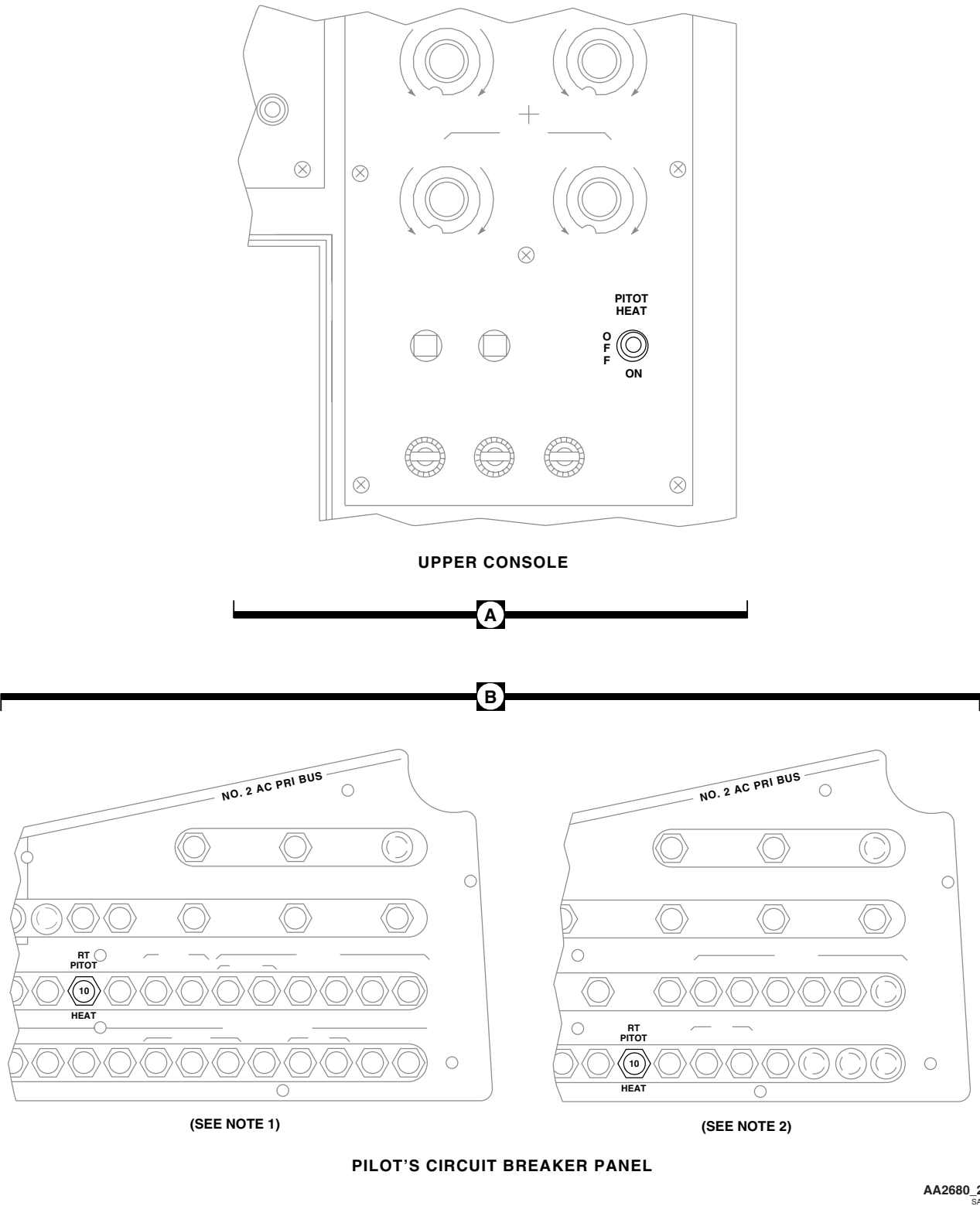
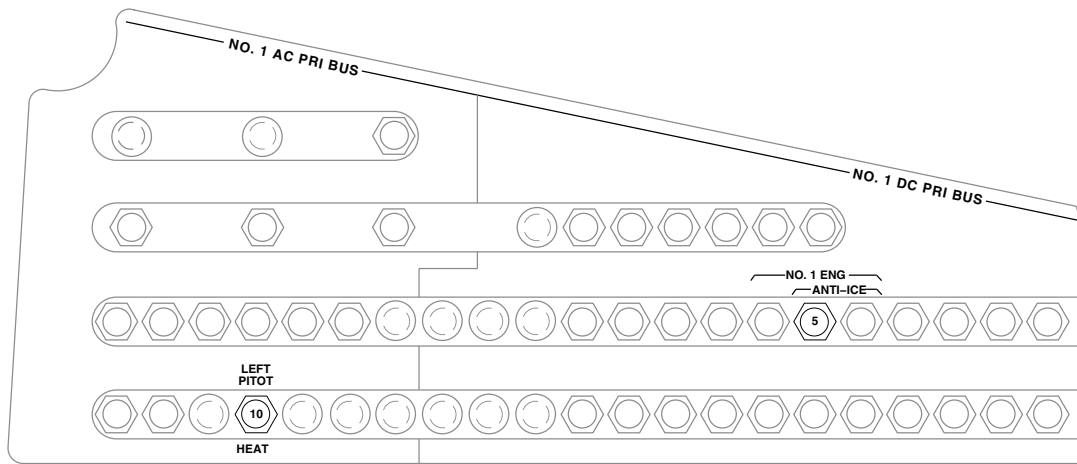


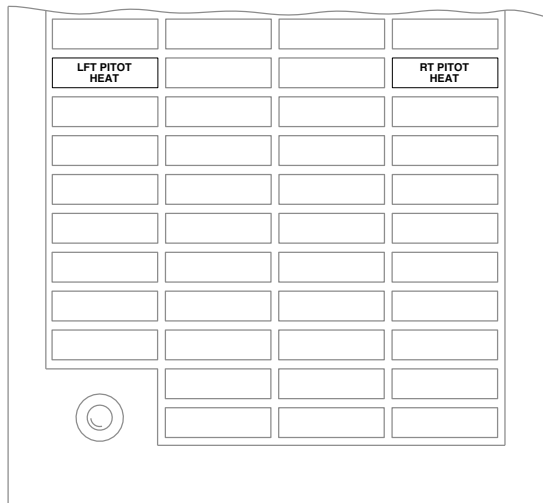
Figure 1. Pitot Heater System Location Diagram. (Sheet 2 of 3)

LOCATION - Continued

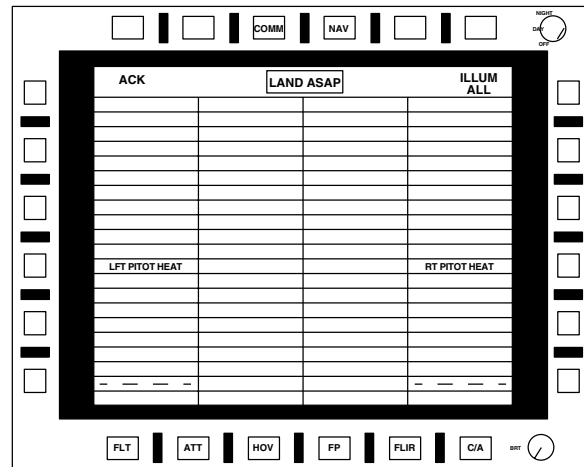


C
COPILOT'S CIRCUIT BREAKER PANEL

D



CAUTION/ADVISORY PANEL
(SEE NOTE 3)



MULTIFUNCTION DISPLAY'S CAUTION/ADVISORY GRID
(SEE NOTE 4)

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Figure 1. Pitot Heater System Location Diagram. (Sheet 3 of 3)

SCHEMATICS

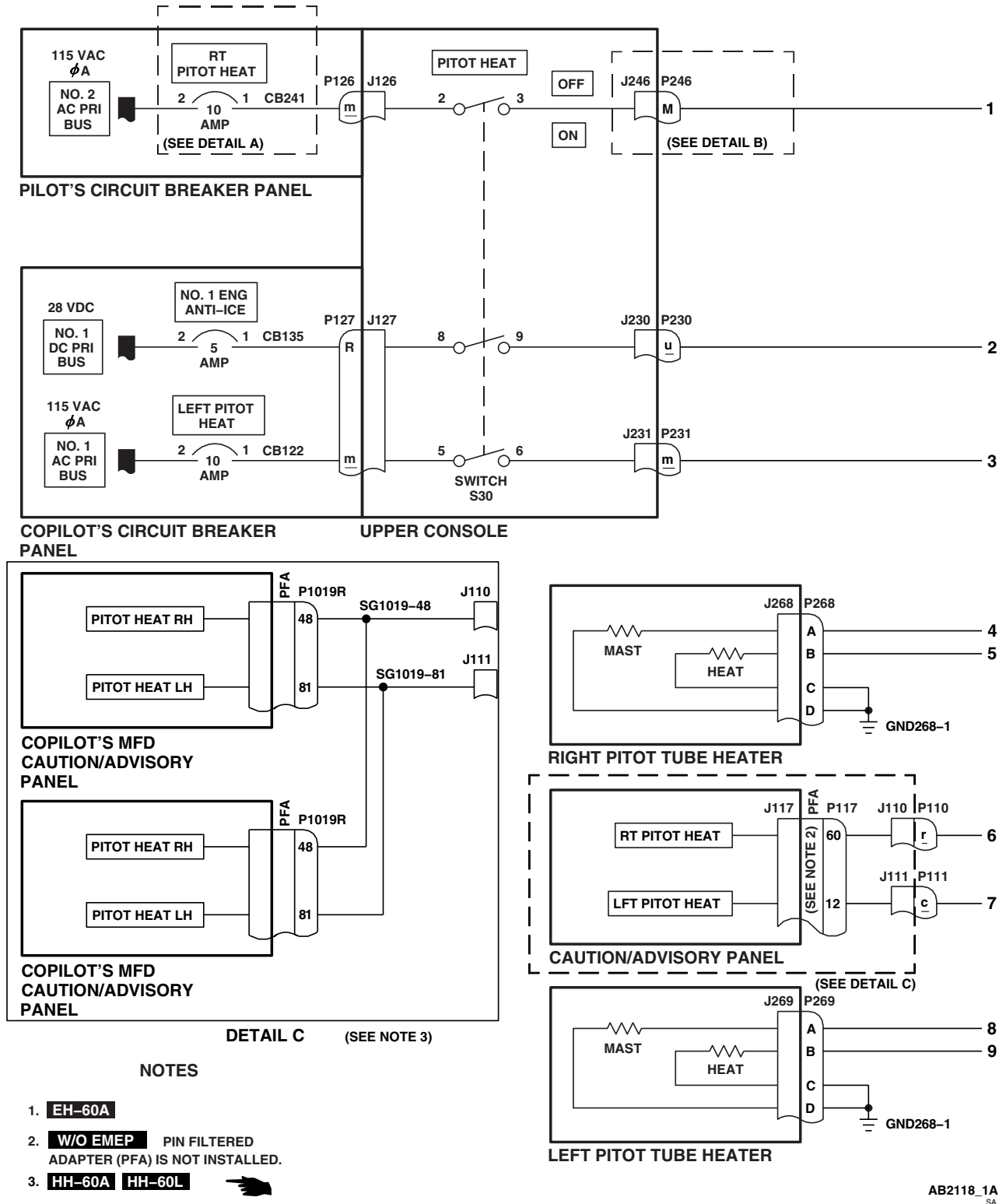
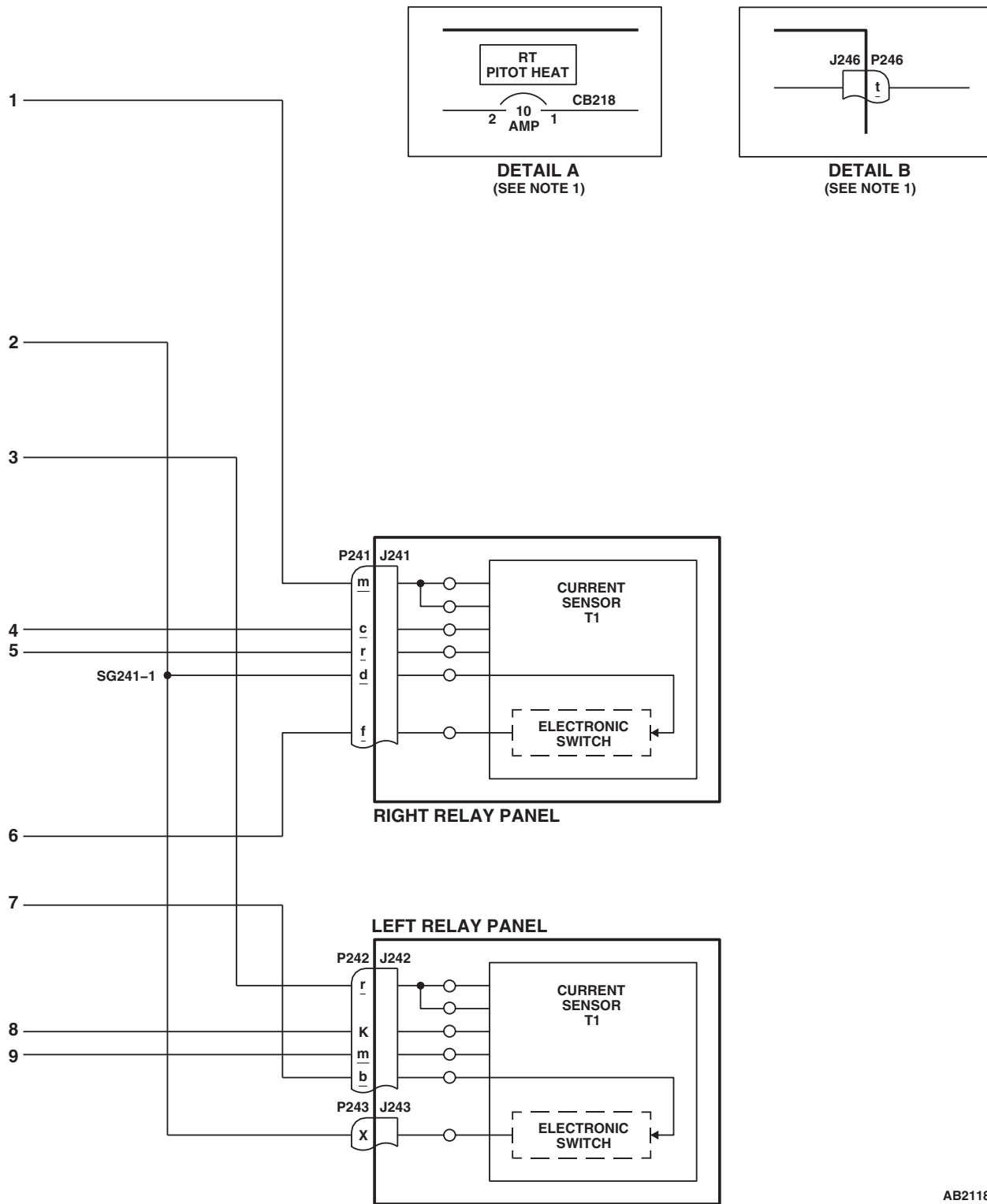


Figure 2. Pitot Heater System Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued



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Figure 2. Pitot Heater System Schematic Diagram. (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL

INSTRUMENT SYSTEMS

INSTRUMENT DISPLAY SYSTEM

This WP supersedes WP 0089 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment

Multimeter, AN/PSM-45A

WP 0780 00

WP 0781 00

WP 0782 00

Tools and Special Tools

Electrical Repairer Toolkit, SC 5180-99-B06

WP 0783 00

Line Test Set, Instrument Display, 245-476853-000

WP 0784 00

WP 0785 00

Personnel Required

Aircraft Electrician MOS 15F (One)

WP 0786 00

WP 0790 00

WP 0791 00

References

TM 1-1520-237-10

WP 0797 00

WP 0073 00

WP 0824 00

WP 0074 00

WP 0881 00

WP 0085 00

WP 0925 00

WP 0104 00

WP 1685 00

WP 0105 00

WP 1747 00

WP 0132 00

Equipment Condition

WP 0776 00

External Electrical Power Available, WP 1685 00
or APU Operational, TM 1-1520-237-10

WP 0778 00

WP 0779 00

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

The instrument display system (IDS) operational/troubleshooting procedure is subdivided as follows:

- Setup
- % RPM R (main rotor speed)
- FUEL QTY
- ENG OIL PRESS
- ENG OIL TEMP
- Ng SPEED (engine gas generator tachometer)
- % TRQ (engine torque)

SPECIAL INSTRUCTIONS - Continued

- XMSN PRESS (main transmission oil pressure)
- XMSN TEMP (main transmission oil temperature)
- % RPM 1-2 (engine power turbine tachometer)
- TGT TEMP (turbine gas temperature)
- Lamp Power Supply
- Faceplate Lighting
- Shutdown

SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step****NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Place central display unit (CDU) DIGITS switch to ON and DIM control to about midposition.
3. Connect line test set (LTS) 28 vdc cable between utility receptacle J258 in upper cabin area behind pilot's seat and LTS.
4. Connect LTS 115 vac cable between utility receptacle J257 in upper cabin area behind pilot's seat and LTS.
5. Disconnect cable assemblies from J1 on No. 1 signal data converter (No. 1 SDC) and J1 on No. 2 signal data converter (No. 2 SDC).
6. Position LTS switches as follows:

INPUT POWER	OFF
All CAL-NOR switches	NOR
All HI-LO switches	HI
All NOR controls	Fully counterclockwise
7. Connect LTS cable assembly 419-741 between NO. 1 SIGNAL DATA CONVERTER on LTS and J1 on No. 1 SDC.
8. Connect second LTS cable assembly 419-741 between NO. 2 SIGNAL DATA CONVERTER on LTS and J1 on No. 2 SDC.
9. Turn on electrical power. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed). ■
10. Place LTS - INPUT POWER switch to SIM ONLY.

SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

- a. All CDU and PDU (pilot's and copilot's) vertical displays shall be at zero, digital readouts shall indicate about zero, and the following LTS lights shall go on:
 - LOW ROTOR SPD
 - NO. 1 ENGINE OUT
 - NO. 2 ENGINE OUT
 - NO. 1 LOW OIL PRESS
 - NO. 2 LOW OIL PRESS
- b. Ng SPEED 1 or 2 vertical display or digital readout shall indicate between 0 and 0.5%.

Corrective Action

1. If LOW ROTOR SPD light on LTS is off, check RTR SPD fuse.
 - a. If fuse is good, replace CDU (WP 0776 00).
 - b. If fuse is not good, replace fuse (WP 0779 00).
2. If NO. 1 or NO. 2 ENGINE OUT light on LTS is off, go to NO. 1 ENGINE OUT AND/OR NO. 2 ENGINE OUT LIGHTS ARE OFF, in this work package.
3. If NO. 1 or NO. 2 LOW OIL PRESS light on LTS is off, go to NO. 1 LOW OIL PRESS AND/OR NO. 2 LOW OIL PRESS LIGHTS ARE OFF, in this work package.
4. If Ng SPEED readout is not as specified, do an Ng SPEED check, in this work package.
 - a. If trouble remains, troubleshoot No. 1 or No. 2 engine instruments and warning lights system, as applicable (WP 0073 00 or WP 0074 00).
5. If ENG OIL PRESS 1 or 2 display does not indicate zero, do ENG OIL PRESS, in this work package.
 - a. If trouble remains, troubleshoot No. 1 or No. 2 engine instruments and warning lights system, as applicable (WP 0073 00 or WP 0074 00).
6. If XMSN PRESS display does not indicate zero, do XMSN PRESS, in this work package.
 - a. If trouble remains, troubleshoot transmission chip detector, instruments, and oil warning system (WP 0085 00).
7. If % RPM 1 or 2 display does not indicate zero, do % RPM 1-2, in this work package.
 - a. If trouble remains, troubleshoot No. 1 or No. 2 engine instruments and warning lights system, as applicable (WP 0073 00 or WP 0074 00).
8. If % RPM R display does not indicate zero, do % RPM R, in this work package.
 - a. If trouble remains, troubleshoot transmission chip detector, instruments, and oil warning system (WP 0085 00).
9. If % TRQ vertical displays or digital readouts do not indicate zero, do % TRQ, in this work package.
 - a. If trouble remains, troubleshoot No. 1 or No. 2 engine instruments and warning lights system, as applicable (WP 0073 00 or WP 0074 00).

Step

11. Momentarily press LTS-LOW ROTOR SPD reset button.

SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

- a. RTR OVERSPEED lights on PDUs (pilot's and copilot's) shall be off.
- b. CHAN 1 warning light on CDU shall be off.
- c. CHAN 2 warning light on CDU shall be off.
- d. ENG OIL PRESS, XMSN PRESS, and Ng SPEED vertical displays, plus the associated Ng digital displays on CDU shall all indicate zero.
- e. The % RPM and % TRQ vertical displays, and the associated digital readouts on PDUs (pilot's and copilot's) shall indicate zero.

Corrective Action

- 1. If Indication/Condition a. is not as specified, go to RTR OVERSPEED LIGHT IS ON, in this work package.
- 2. If Indication/Condition b. is not as specified, go to CHAN 1 LIGHT IS ON, in this work package.
- 3. If Indication/Condition c. is not as specified, go to CHAN 2 LIGHT IS ON, in this work package.
- 4. If Indication/Condition d. or e. is not as specified, note trouble and continue operational/troubleshooting procedure.
- 5. If Indication/Conditions d. and e. are not as specified, replace the CDU (WP 0776 00).

Step

- 12. Press and hold CDU PUSH TO TEST switch and turn DIM control fully counterclockwise, then fully clockwise. Next, adjust DIM control for normal display.

Indication/Condition

- a. All CDU vertical scale lamps shall go on.
- b. All CDU digital readouts shall indicate 888, except for TOTAL FUEL, which shall indicate 8888.
- c. Three RTR OVERSPEED lights on PDUs (pilot's and copilot's) shall go on.
- d. CHAN 1 and CHAN 2 warning lights on CDU shall go on.
- e. CDU display lighting shall vary with CDU's DIM control setting, except CHAN 1 and CHAN 2 warning lights.

Corrective Action

- 1. If any two or more adjacent lamps on one vertical scale are off, or if one lamp between the green and amber, amber and red, or green and red ranges on a scale is off, go to WITH CDU PUSH TO TEST SWITCH PRESSED, SCALES OR RANGES ARE OFF , in this work package.
- 2. If alternate vertical scale lamps are off, note trouble and continue with steps 13. and 14.
- 3. **EMEP** If Indication/Condition b. is not as specified, replace direct view filament (DVF) module (WP 0778 00).
 - a. If trouble remains, replace CDU (WP 0776 00).
- 4. **W/O EMEP** If Indication/Condition b. is not as specified, replace CDU (WP 0776 00).
- 5. If Indication/Condition c. is not as specified, go to WITH CDU PUSH TO TEST SWITCH PRESSED, ONE OR MORE RTR OVERSPEED LIGHTS ARE OFF, in this work package.
- 6. If Indication/Condition d. or e. is not as specified, replace CDU (WP 0776 00).

SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

7. If DIM control varies alternate display lamps only, go to CDU DIM CONTROL VARIES PDU DISPLAY LIGHTING, in this work package.

Step

13. Release CDU PUSH TO TEST switch.
14. Press and hold pilot's PDU LT switch.

Indication/Condition

- a. All PDU (pilot's) vertical scale lamps, except the top segments (130% RPM), shall go on.
- b. All PDU (pilot's) digital readouts shall indicate 188.

Corrective Action

1. If Indication/Condition a. is not as specified and alternate CDU scale lamps were off in step 12., replace No. 2 SDC (WP 0781 00).
2. **EMEP** If any two or more adjacent lamps on one vertical scale are off, or if one lamp between the green and amber, amber and red, or green and red ranges on a scale is off, replace bar graph filaments (BGF) lamp (WP 0784 00). ■
 - a. If trouble remains, replace pilot's PDU (WP 0782 00).
3. **W/O EMEP** If any two or more adjacent lamps on one vertical scale are off, or if one lamp between the green and amber, amber and red, or green and red ranges on a scale is off, replace pilot's PDU (WP 0782 00). ■
4. **EMEP** If Indication/Condition b. is not as specified, replace direct view filament (DVF) module (WP 0778 00). ■
 - a. If trouble remains, replace pilot's PDU (WP 0782 00).
5. **W/O EMEP** If Indication/Condition b. is not as specified, replace pilot's PDU (WP 0782 00). ■

Step

15. Release pilot's PDU LT switch.
16. Press and hold copilot's PDU LT switch.

Indication/Condition

- a. All PDU (copilot's) vertical scale lamps, except the top segments (130% RPM), shall go on.
- b. All PDU (copilot's) digital readouts shall indicate 188.

Corrective Action

1. If all vertical scale lamps are off and alternate CDU scale lamps were off in step 12., replace No. 1 SDC (WP 0781 00).
2. **EMEP** If any two or more adjacent lamps on one vertical scale are off, or if one lamp between the green and amber, amber and red, or green and red ranges on a scale is off, replace BGF lamp. ■
 - a. If trouble remains, replace copilot's PDU (WP 0782 00).
3. **W/O EMEP** If any two or more adjacent lamps on one vertical scale are off, or if one lamp between the green and amber, amber and red, or green and red ranges on a scale is off, replace copilot's PDU (WP 0782 00). ■
4. **EMEP** If Indication/Condition b. is not as specified, replace DVF module. ■

SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- a. If trouble remains, replace copilot's PDU (WP 0782 00).
5. **W/O EMEP** If Indication/Condition b. is not as specified, replace copilot's PDU (WP 0782 00). ■

Step

17. Release copilot's PDU LT switch.

NOTE

Make sure the CDU DIM control is at midtravel.

18. Using a flashlight, position light beam on PDUs (pilot's and copilot's) and CDU faceplates separately.

Indication/Condition

All displays shall increase in brightness as light beam is positioned on each unit.

Corrective Action

1. If Indication/Condition is not as specified for PDUs (pilot's and copilot's), interchange PDU's.
 - a. If trouble remains on same PDU, replace display unit (WP 0782 00).
 - b. If trouble does not remain on same PDU, repair/replace wiring between P153-4 and P157-30, or between P156-4 and P158-30 as required (WP 1747 00).
2. If Indication/Condition is not as specified, replace CDU (WP 0776 00).

Step

19. Press CDU PUSH TO TEST switch and adjust DIM control to get a normal viewing brightness.

Indication/Condition

Switch shall be pressed and adjusted to get normal viewing brightness.

Corrective Action

None Required

% RPM R OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step**

1. Do SETUP, in this work package, if not already done.

NOTE

With line test set connected, RTR OVERSPEED lamps on PDUs (pilot's and copilot's) are reset by ROTOR OVERSPD RESET switch on line test set.

2. Slowly turn LTS - ROTOR SPEED variable control fully clockwise.

Indication/Condition

- a. % RPM R vertical scale indication on PDUs (pilot's and copilot's) shall advance smoothly to full scale.
- b. At 96% indication, lower red and amber scale segments shall go off.
- c. LOW ROTOR SPD light on LTS shall go off.

% RPM R OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- d. Left RTR OVERSPEED light (indicating about 127%) on PDUs (pilot's and copilot's) shall go on.
- e. Center RTR OVERSPEED light (indicating about 137%) shall go on.
- f. Right RTR OVERSPEED light (indicating about 142%) shall go on.

Corrective Action

- 1. If Indication/Condition a. is not as specified, go to PILOT'S OR COPILOT'S PDU %RPM DISPLAY IS INOPERATIVE, in this work package.
- 2. If Indication/Condition b. is not as specified, replace malfunctioning PDU (WP 0782 00).
- 3. If Indication/Condition c. is not as specified, replace CDU (WP 0776 00).
- 4. If Indication/Conditions d., e., and f. are not as specified, go to ONE OR MORE RTR OVERSPEED LIGHTS ARE OFF, in this work package.

Step

- 3. Slowly turn LTS - ROTOR SPEED variable control fully counterclockwise.

Indication/Condition

- a. % RPM R vertical scale indications on PDUs (pilot's and copilot's) shall decrease smoothly.
- b. At 95% indication, lower red and amber scale segments shall go on.
- c. LOW ROTOR SPD light on LTS shall go on.
- d. All three RTR OVERSPEED lights on PDUs (pilot's and copilot's) shall remain ON.

Corrective Action

- 1. If Indication/Condition a. or b. is not as specified, replace malfunctioning display unit (WP 0782 00).
- 2. If Indication/Condition c. is not as specified, check RTR SPD fuse.
 - a. If fuse is good, replace CDU (WP 0776 00).
 - b. If fuse is not good, replace fuse (WP 0782 00).
- 3. If Indication/Condition d. is not as specified, replace CDU (WP 0776 00).

Step

- 4. Press and release LTS - ROTOR OVER SPD RESET switch.

Indication/Condition

Three RTR OVERSPEED lights on PDUs (pilot's and copilot's) shall go off.

Corrective Action

If Indication/Condition is not as specified, replace CDU (WP 0776 00).

Step

- 5. Place LTS - ROTOR SPEED CAL-NOR switch to CAL.

Indication/Condition

- a. % RPM R (pilot's PDU) scale shall indicate 101% or 102%.
- b. % RPM R (copilot's PDU) scale shall indicate 101% or 102%.

% RPM R OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition a. is not as specified, replace No. 2 SDC (WP 0781 00).
2. If Indication/Condition b. is not as specified, replace No. 1 SDC (WP 0781 00).

Step

6. Place LTS - ROTOR SPEED HI-LO switch to LO. LTS LOW ROTOR SPD lamp shall go on.

Indication/Condition

- a. % RPM R (pilot's PDU) scales shall be completely blank (0%).
- b. % RPM R (copilot's PDU) scales shall be completely blank (0%).

Corrective Action

1. If Indication/Condition a. is not as specified, replace No. 2 SDC (WP 0781 00).
2. If Indication/Condition b. is not as specified, replace No. 1 SDC (WP 0781 00).

Step

7. Place LTS - ROTOR SPEED CAL-NOR switch to NOR.
8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

FUEL QTY OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step**

1. Do SETUP, in this work package, if not already done.
2. Slowly turn LTS - FUEL QUANTITY NO. 1 variable control fully clockwise.

Indication/Condition

- a. FUEL QTY 1 vertical scale indication on CDU shall advance smoothly to full scale.
- b. TOTAL FUEL digital readout on CDU shall advance to about 1500.
- c. At about 250 pounds indication, amber scale segment shall go off.

Corrective Action

1. If Indication/Condition a. or b. is not as specified, go to FUEL QTY 1, FUEL QTY 2, AND/OR TOTAL FUEL DISPLAY INOPERATIVE, in this work package.
2. If Indication/Condition c. is not as specified, replace CDU (WP 0776 00).

Step

3. Slowly turn LTS - FUEL QUANTITY NO. 1 variable control fully counterclockwise.

FUEL QTY OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

- a. FUEL QTY 1 vertical scale indication on CDU shall decrease smoothly to zero.
- b. TOTAL FUEL digital readout on CDU shall decrease to zero.
- c. At about 200 pounds indication, amber scale segment shall go on.

Corrective Action

If Indication/Conditions are not as specified, replace CDU (WP 0776 00).

Step

4. Place LTS - FUEL QUANTITY NO. 1 CAL-NOR switch to CAL.

Indication/Condition

- a. FUEL QTY 1 vertical scale on CDU shall indicate 1200 pounds.
- b. TOTAL FUEL digital readout on CDU shall indicate between 1210 and 1230 pounds.

Corrective Action

If Indication/Conditions are not as specified, replace CDU (WP 0776 00).

Step

5. Place LTS - FUEL QUANTITY NO. 1 CAL-NOR switch to NOR.
6. Slowly turn LTS - FUEL QUANTITY NO. 2 variable control fully clockwise.

Indication/Condition

- a. FUEL QTY 2 vertical scale indication on CDU shall advance smoothly to full scale.
- b. TOTAL FUEL digital readout on CDU shall advance to about 1500.
- c. At about 250 pounds indication, amber scale segment shall go off.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, go to FUEL QTY 1, FUEL QTY 2, AND/OR TOTAL FUEL DISPLAY INOPERATIVE, in this work package.
2. If Indication/Condition c. is not as specified, replace CDU (WP 0776 00).

Step

7. Slowly turn LTS - FUEL QUANTITY NO. 2 variable control fully counterclockwise.

Indication/Condition

- a. FUEL QTY 2 scale on CDU shall decrease smoothly to zero.
- b. TOTAL FUEL digital readout on CDU shall decrease to zero.
- c. At about 200 pounds indication, amber scale segment shall go on.

Corrective Action

If Indication/Conditions are not as specified, replace CDU (WP 0776 00).

FUEL QTY OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

8. Place LTS - FUEL QUANTITY NO. 1 CAL-NOR switch to CAL, FUEL QUANTITY NO. 2 CAL-NOR switch to CAL, and NO. 2 HI-LO switch to LO.

Indication/Condition

- a. FUEL QTY 2 vertical scale on CDU shall be completely blank.
- b. TOTAL FUEL digital readout on CDU shall indicate between 1210 and 1230 pounds.

Corrective Action

If Indication/Conditions are not as specified, replace CDU (WP 0776 00).

Step

9. Place LTS - FUEL QUANTITY NO. 1 HI-LO switch to LO.

Indication/Condition

- a. FUEL QTY 1 vertical scale on CDU shall be completely blank.
- b. TOTAL FUEL digital readout on CDU shall indicate between 0 and 10 pounds.

Corrective Action

If Indication/Conditions are not as specified, replace CDU (WP 0776 00).

Step

10. Place LTS - FUEL QUANTITY NO. 2 HI-LO switch to HI.

Indication/Condition

- a. FUEL QTY 2 vertical scale on CDU shall indicate 1200 pounds.
- b. TOTAL FUEL digital readout on CDU shall indicate between 1210 and 1230 pounds.

Corrective Action

If Indication/Conditions are not as specified, replace CDU (WP 0776 00).

Step

11. Place LTS - FUEL QUANTITY NO. 1 HI-LO switch to HI.

Indication/Condition

TOTAL FUEL digital readout on CDU shall indicate between 2420 and 2460 pounds.

Corrective Action

If Indication/Condition is not as specified, replace CDU (WP 0776 00).

Step

12. Place LTS - FUEL QUANTITY NO. 1 and NO. 2 CAL-NOR switches to NORM.
13. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

FUEL QTY OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

None Required

ENG OIL PRESS OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE
Step

1. Do SETUP, in this work package, if not already done.
2. Slowly turn LTS - ENGINE OIL PRESS NO. 1 variable control fully clockwise.

Indication/Condition

- a. ENG OIL PRESS 1 vertical scale indication on CDU shall advance smoothly to full scale.
- b. **UH-60L HH-60L** At 22 psi indication, NO. 1 LOW OIL PRESS light on LTS shall go off. ■
- c. **EH-60A UH-60A HH-60A** At 25 psi indication, NO. 1 LOW OIL PRESS light on LTS shall go off. ■
- d. **UH-60L HH-60L** At 26 psi indication, lower red and amber scale segments shall go off. ■
- e. **EH-60A UH-60A HH-60A** At 47.5 psi indication, lower red and amber scale segments shall go off. ■

Corrective Action

1. If Indication/Condition a. is not as specified, go to ENG OIL PRESS 1 AND/OR ENG OIL PRESS 2 DISPLAY INOPERATIVE, in this work package.
2. If Indication/Condition b., c., d., or e. is not as specified, replace CDU (WP 0776 00).

Step

3. Slowly turn LTS - ENGINE OIL PRESS NO. 1 variable control fully counterclockwise.

Indication/Condition

- a. ENG OIL PRESS 1 vertical scale indication shall decrease smoothly.
- b. **UH-60L HH-60L** At 26 psi indication, lower red and amber scale segments shall go on. ■
- c. **EH-60A UH-60A HH-60A** At 45 psi indication, lower red and amber scale segments shall go on. ■
- d. **UH-60L HH-60L** At 22 psi indication, NO. 1 LOW OIL PRESS light on LTS shall go on. ■
- e. **EH-60A UH-60A HH-60A** At 20 psi indication, NO. 1 LOW OIL PRESS light on LTS shall go on. ■

Corrective Action

1. If Indication/Condition a., b., or c. is not as specified, replace CDU (WP 0776 00).
2. If Indication/Condition d. or e. is not as specified, go to WITH CDU PUSH TO TEST SWITCH PRESSED, ONE OR MORE RTR OVERSPEED LIGHTS ARE OFF, in this work package.

Step

4. Place LTS - ENGINE OIL PRESS NO. 1 CAL-NOR switch to CAL.

Indication/Condition

ENG OIL PRESS 1 scale on CDU shall indicate 110 psi or 120 psi.

ENG OIL PRESS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If Indication/Condition is not as specified, replace No. 1 SDC (WP 0781 00).

Step

5. Place LTS - ENGINE OIL PRESSURE NO. 1 HI-LO switch to LO.

Indication/Condition

ENG OIL PRESS 1 scale on CDU shall be completely blank.

Corrective Action

If Indication/Condition is not as specified, replace No. 1 SDC (WP 0781 00).

Step

6. Place LTS - ENGINE OIL PRESS NO. 1 CAL-NOR switch to NOR.
7. Slowly turn LTS - ENGINE OIL PRESS NO. 2 variable control fully clockwise.

Indication/Condition

- a. ENG OIL PRESS 2 vertical scale indications on CDU shall advance smoothly to full scale.
- b. **UH-60L HH-60L** At 22 psi indication, NO. 2 LOW OIL PRESS light on LTS shall go off. ■ |
- c. **EH-60A UH-60A HH-60A** At 25 psi indication, NO. 2 LOW OIL PRESS light on LTS shall go off. ■ |
- d. **UH-60L HH-60L** At 26 psi indication, lower red and amber scale segments shall go off. ■ |
- e. **EH-60A UH-60A HH-60A** At 47.5 psi indication, lower red and amber scale segments shall go off. ■ |

Corrective Action

1. If Indication/Condition a. is not as specified, go to ENG OIL PRESS 1 AND/OR ENG OIL PRESS 2 DISPLAY INOPERATIVE, in this work package.
2. If Indication/Condition b., c., d., or e. is not as specified, replace CDU (WP 0776 00).

Step

8. Slowly turn LTS - ENGINE OIL PRESS NO. 2 variable control fully counterclockwise.

Indication/Condition

- a. ENG OIL PRESS 2 vertical scale indication on CDU shall decrease smoothly.
- b. **UH-60L HH-60L** At 26 psi indication, lower red and amber scale segments shall go on. ■ |
- c. **EH-60A UH-60A HH-60A** At 45 psi indication, lower red and amber scale segments shall go on. ■ |
- d. **UH-60L HH-60L** At 22 psi indication, NO. 2 LOW OIL PRESS light on LTS shall go on. ■ |
- e. **EH-60A UH-60A HH-60A** At 20 psi indication, NO. 2 LOW OIL PRESS light on LTS shall go on. ■ |

Corrective Action

1. If Indication/Condition a., b., or c. is not as specified, replace CDU (WP 0776 00).
2. If Indication/Condition d. or e. is not as specified, go to WITH CDU PUSH TO TEST SWITCH PRESSED, ONE OR MORE RTR OVERSPEED LIGHTS ARE OFF, in this work package.

ENG OIL PRESS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

9. Place LTS - ENGINE OIL PRESS NO. 2 CAL-NOR switch to CAL.

Indication/Condition

ENG OIL PRESS 2 scale on CDU shall indicate 110 or 120 psi.

Corrective Action

If Indication/Condition is not as specified, replace No. 2 SDC (WP 0781 00).

Step

10. Place LTS - ENG OIL PRESS NO. 2 HI-LO switch to LO.

Indication/Condition

ENG OIL PRESS 2 scale on CDU shall be completely blank.

Corrective Action

If Indication/Condition is not as specified, replace No. 2 SDC (WP 0781 00).

Step

11. Place LTS - ENGINE OIL PRESS NO. 2 CAL-NOR switch to NOR.
12. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

ENG OIL TEMP OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step**

1. Do SETUP, in this work package, if not already done.
2. Slowly turn LTS - ENGINE OIL TEMP NO. 1 variable control fully clockwise.

Indication/Condition

- a. ENG OIL TEMP 1 vertical scale indication on CDU shall advance smoothly to full scale.
- b. At 302°F (150°C) indication, NO. 1 OIL TEMP light on LTS shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, go to ENG OIL TEMP 1 AND/OR ENG OIL TEMP 2 DISPLAY INOPERATIVE, in this work package.
2. If Indication/Condition b. is not as specified, go to NO. 1 HIGH OIL TEMP AND/OR NO. 2 HIGH OIL TEMP LIGHT IS OFF, in this work package.

ENG OIL TEMP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

3. Slowly turn LTS - ENGINE OIL TEMP NO. 1 variable control full counterclockwise.

Indication/Condition

- a. ENG OIL TEMP 1 vertical scale indication on CDU shall decrease smoothly.
- b. At 284°F (140°C) indication, NO. 1 HIGH OIL TEMP light on LTS shall go off.

Corrective Action

If Indication/Conditions are not as specified, replace CDU (WP 0776 00).

Step

4. Place LTS - ENGINE OIL TEMP NO. 1 CAL-NOR switch to CAL.

Indication/Condition

ENG OIL TEMP 1 scale on CDU shall indicate 284°F (140°C).

Corrective Action

If Indication/Condition is not as specified, replace No. 1 SDC (WP 0781 00).

Step

5. Place LTS - NO. 1 ENGINE OIL TEMP HI-LO switch to LO.

Indication/Condition

- a. **UH-60L HH-60L** ENG OIL TEMP 1 scale on CDU shall be completely blank. ■
- b. **EH-60A UH-60A HH-60A** ENG OIL TEMP 1 scale on CDU shall be blank. ■

Corrective Action

If Indication/Condition a. or b. is not as specified, replace No. 1 SDC (WP 0781 00).

Step

6. Place LTS - NO. 1 ENGINE OIL TEMP CAL-NOR switch to NOR.
7. Slowly turn LTS - ENGINE OIL TEMP NO. 2 variable control fully clockwise.

Indication/Condition

- a. ENG OIL TEMP 2 vertical scale indication on CDU shall advance smoothly to full scale.
- b. At 302°F (150°C) indication, NO. 2 HIGH OIL TEMP light on LTS shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, go to ENG OIL TEMP 1 AND/OR ENG OIL TEMP 2 DISPLAY INOPERATIVE, in this work package.
2. If Indication/Condition b. is not as specified, go to NO. 1 HIGH OIL TEMP AND/OR NO. 2 HIGH OIL TEMP LIGHT IS OFF, in this work package.

Step

8. Slowly turn LTS - ENGINE OIL TEMP NO. 2 variable control fully counterclockwise.

ENG OIL TEMP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

- a. ENG OIL TEMP 2 vertical scale indication on CDU shall decrease smoothly.
- b. At 284°F (140°C) indication, NO. 2 HIGH OIL TEMP light on LTS shall go off.

Corrective Action

If Indication/Conditions are not as specified, replace CDU (WP 0776 00).

Step

9. Place LTS - ENGINE OIL TEMP NO. 2 CAL-NOR switch to CAL.

Indication/Condition

ENG OIL TEMP 2 scale on CDU shall indicate 284°F (140°C).

Corrective Action

If Indication/Condition is not as specified, replace No. 2 SDC (WP 0781 00).

Step

10. Place LTS - NO. 2 ENGINE OIL TEMP HI-LO switch to LO.

Indication/Condition

- a. **UH-60L HH-60L** ENG OIL TEMP 2 scale on CDU shall be completely blank. ■
- b. **EH-60A UH-60A HH-60A** ENG OIL TEMP 2 scale on CDU shall be blank. ■

Corrective Action

If Indication/Condition a. or b. is not as specified, replace No. 2 SDC (WP 0781 00).

Step

11. Place LTS - ENGINE OIL TEMP NO. 2 CAL-NOR switch to NOR.
12. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

NG SPEED OPERATIONAL/TROUBLESHOOTING PROCEDURE**Step**

1. Do SETUP, in this work package, if not already done.
2. Slowly turn LTS - GAS GEN SPEED NO. 1 variable control fully clockwise.

Indication/Condition

- a. Ng SPEED 1 vertical scale on CDU shall advance smoothly to full scale.
- b. Ng SPEED digital readout indications on CDU shall advance smoothly to full scale.

NG SPEED OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- c. At about 55% indication, NO. 1 ENGINE OUT light on LTS shall go off.

Corrective Action

1. If Indication/Condition a. is not as specified, go to NG SPEED 1 AND/OR NG SPEED 2 VERTICAL DISPLAY INOPERATIVE, in this work package.
2. If Indication/Condition b. is not as specified, go to NG SPEED 1 AND/OR NG SPEED 2 DIGITAL DISPLAY INOPERATIVE, in this work package.
3. If Indication/Condition c. is not as specified, replace CDU (WP 0776 00).

Step

3. Slowly turn LTS - GAS GEN SPEED NO. 1 variable control fully counterclockwise.

Indication/Condition

- a. Ng SPEED 1 vertical scale and digital readout indications on CDU shall decrease smoothly.
- b. At about 55% indication, NO. 1 ENGINE OUT light on LTS shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, replace CDU (WP 0776 00).
2. If Indication/Condition b. is not as specified, go to WITH CDU PUSH TO TEST SWITCH PRESSED, SCALES OR RANGES ARE OFF , in this work package.

Step

4. Place LTS - GAS GEN SPEED NO. 1 CAL-NOR switch to CAL.

Indication/Condition

- a. Ng SPEED 1 vertical scale shall indicate 98% or 100%.
- b. Ng SPEED 1 digital readout shall indicate between 99.5% and 100.5%.

Corrective Action

If Indication/Conditions are not as specified, replace No. 1 SDC (WP 0781 00).

Step

5. Place LTS - GAS GEN SPEED NO. 1 HI-LO switch to LO.

Indication/Condition

- a. Ng SPEED 1 vertical scale on CDU shall be completely blank.
- b. Ng SPEED 1 digital readout on CDU shall indicate between 0 and 0.5%.

Corrective Action

If Indication/Conditions are not as specified, replace No. 1 SDC (WP 0781 00).

Step

6. Place LTS - GAS GEN SPEED NO. 1 CAL-NOR switch to NOR.
7. Slowly turn LTS - GAS GEN SPEED NO. 2 variable control fully clockwise.

Indication/Condition

- a. Ng SPEED 2 vertical scale indication on CDU shall advance smoothly to full scale.

NG SPEED OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- b. Ng SPEED 2 digital readout indication on CDU shall advance smoothly to full scale.
- c. At about 55% indication, NO. 2 ENGINE OUT light on LTS shall go off.

Corrective Action

- 1. If Indication/Condition a. is not as specified, go to NG SPEED 1 AND/OR NG SPEED 2 VERTICAL DISPLAY INOPERATIVE, in this work package.
- 2. If Indication/Condition b. is not as specified, go to NG SPEED 1 AND/OR NG SPEED 2 DIGITAL DISPLAY INOPERATIVE, in this work package.
- 3. If Indication/Condition c. is not as specified, replace CDU (WP 0776 00).

Step

- 8. Slowly turn LTS - GAS GEN SPEED NO. 2 variable control fully counterclockwise.

Indication/Condition

- a. Ng SPEED 2 vertical scale and digital readout indications on CDU shall decrease smoothly.
- b. At about 55% indication, NO. 2 ENGINE OUT light on LTS shall go on.

Corrective Action

- 1. If Indication/Condition a. is not as specified, replace CDU (WP 0776 00).
- 2. If Indication/Condition b. is not as specified, go to WITH CDU PUSH TO TEST SWITCH PRESSED, SCALES OR RANGES ARE OFF , in this work package.

Step

- 9. Place LTS - GAS GEN SPEED NO. 2 CAL-NOR switch to CAL.

Indication/Condition

- a. Ng SPEED 2 vertical scale on CDU shall indicate 98% or 100%.
- b. Ng SPEED 2 digital readout on CDU shall indicate between 99.5% and 100.5%.

Corrective Action

If Indication/Conditions are not as specified, replace No. 2 SDC (WP 0781 00).

Step

- 10. Place LTS - GAS GEN SPEED NO. 2 HI/LO switch to LO.

Indication/Condition

- a. Ng SPEED 2 vertical scale on CDU shall be completely blank.
- b. Ng SPEED 2 digital readout on CDU shall indicate between 0 and 0.5%.

Corrective Action

If Indication/Conditions are not as specified, replace No. 2 SDC (WP 0781 00).

Step

- 11. Place LTS - GAS GEN SPEED NO. 2 CAL-NOR switch to NOR.
- 12. If no other check is to be done, go to SHUTDOWN, in this work package.

NG SPEED OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Shutdown shall be performed.

Corrective Action

None Required

**% TRQ OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**
Step

1. Do SETUP, in this work package, if not already done.
2. Slowly turn LTS - TORQUE NO. 1 variable control fully clockwise.

Indication/Condition

% TRQ 1 (pilot's and copilot's PDU) vertical scale and digital readout indications shall advance smoothly to full scale.

Corrective Action

If Indication/Condition is not as specified, go to % TRQ 1 AND/OR % TRQ 2 VERTICAL DISPLAY OR % TRQ DIGITAL READOUT INOPERATIVE, in this work package.

Step

3. Slowly turn LTS - TORQUE NO. 1 variable control fully counterclockwise.

Indication/Condition

- a. % TRQ 1 (pilot's PDU) vertical scale and digital readout indications shall decrease smoothly.
- b. % TRQ 1 (copilot's PDU) vertical scale and digital readout indications shall decrease smoothly.

Corrective Action

1. If Indication/Condition a. is not as specified, replace pilot's PDU (WP 0782 00).
2. If Indication/Condition b. is not as specified, replace copilot's PDU (WP 0782 00).

Step

4. Place LTS - TORQUE NO. 1 CAL-NOR switch to CAL.

Indication/Condition

- a. % TRQ 1 (pilot's PDU) vertical scales shall indicate 100% or 105%.
- b. % TRQ 1 (pilot's PDU) UH-60A HH-60A digital readouts (T-700 engine) shall indicate between 104% and 106%, UH-60L HH-60L digital readouts (T-701C engine) shall indicate between 103% and 105% .
- c. % TRQ 1 (copilot's PDU) vertical scales shall indicate 100% or 105%.
- d. % TRQ 1 (copilot's PDU) UH-60A HH-60A digital readouts (T-700 engine) shall indicate between 104% and 106%, UH-60L HH-60L digital readouts (T-701C engine) shall indicate between 103% and 105% .

Corrective Action

1. If Indication/Condition a. or b. is not as specified, replace No. 2 SDC (WP 0781 00).

% TRQ OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. If Indication/Condition c. or d. is not as specified, replace No. 1 SDC (WP 0781 00).

Step

5. Place LTS - TORQUE NO. 1 HI-LO switch to LO.

Indication/Condition

- a. % TRQ 1 (pilot's PDU) digital readouts shall be 0 or 1%.
- b. % TRQ 1 (pilot's PDU) vertical scales shall be completely blank.
- c. % TRQ 1 (copilot's PDU) digital readouts shall be 0 or 1%.
- d. % TRQ 1 (copilot's PDU) vertical scales shall be completely blank.

Corrective Action

1. If Indication/Condition a. or b. is not as specified, replace No. 2 SDC (WP 0781 00).
2. If Indication/Condition c. or d. is not as specified, replace No. 1 SDC (WP 0781 00).

Step

6. Place LTS - TORQUE NO. 1 CAL-NOR switch to NOR.
7. Slowly turn LTS - TORQUE NO. 2 variable control fully clockwise.

Indication/Condition

% TRQ 2 (pilot's and copilot's PDU) vertical scale and digital readout indications shall advance smoothly to full scale.

Corrective Action

If Indication/Condition is not as specified, go to % TRQ 1 AND/OR % TRQ 2 VERTICAL DISPLAY OR % TRQ DIGITAL READOUT INOPERATIVE, in this work package.

Step

8. Slowly turn LTS - TORQUE NO. 2 variable control fully counterclockwise.

Indication/Condition

- a. % TRQ 2 (pilot's PDU) vertical scale and digital readout indication shall decrease smoothly.
- b. % TRQ 2 (copilot's PDU) vertical scale and digital readout indication shall decrease smoothly.

Corrective Action

1. If Indication/Condition a. is not as specified, replace pilot's PDU (WP 0782 00).
2. If Indication/Condition b. is not as specified, replace copilot's PDU (WP 0782 00).

Step

9. Place LTS - TORQUE NO. 2 CAL-NOR switch to CAL.

Indication/Condition

- a. % TRQ 2 (pilot's PDU) vertical scales shall indicate 100% or 105%.
- b. % TRQ 2 (pilot's PDU) UH-60A HH-60A digital readouts (T-700 engine) shall indicate between 104% and 106%, UH-60L HH-60L digital readouts (T-701C engine) shall indicate between 103% and 105% .
- c. % TRQ 2 (copilot's PDU) vertical scales shall indicate 100% or 105%.

% TRQ OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- d. % TRQ 2 (copilot's PDU) **UH-60A HH-60A** digital readouts (T-700 engine) shall indicate between 104% and 106%, **UH-60L HH-60L** digital readouts (T-701C engine) shall indicate between 103% and 105% .

Corrective Action

1. If Indication/Condition a. or b. is not as specified, replace No. 2 SDC (WP 0781 00).
2. If Indication/Condition c. or d. is not as specified, replace No. 1 SDC (WP 0781 00).

Step

10. Place LTS - TORQUE NO. 2 HI-LO switch to LO.

Indication/Condition

- a. % TRQ 2 (pilot's PDU) digital readout shall be 0 or 1%.
- b. % TRQ 2 (pilot's PDU) vertical scales shall be completely blank.
- c. % TRQ 2 (copilot's PDU) digital readout shall be 0 or 1%.
- d. % TRQ 2 (copilot's PDU) vertical scales shall be completely blank.

Corrective Action

1. If Indication/Condition a. or b. is not as specified, replace No. 2 SDC (WP 0781 00).
2. If Indication/Condition c. or d. is not as specified, replace No. 1 SDC (WP 0781 00).

Step

11. Place LTS - TORQUE NO. 2 CAL-NOR switch to NOR.
12. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

XMSN PRESS OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step**

1. Do SETUP, in this work package, if not already done.
2. Slowly turn LTS - XMSN OIL PRESS variable control fully clockwise.

Indication/Condition

- a. XMSN PRESS vertical scale indication on CDU shall advance smoothly to full scale.
- b. At 32.5 psi indication, lower red and amber scale segments shall go off.

Corrective Action

1. If Indication/Condition a. is not as specified, go to XMSN PRESS DISPLAY INOPERATIVE, in this work package.
2. If Indication/Condition b. is not as specified, replace CDU (WP 0776 00).

XMSN PRESS OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

3. Slowly turn LTS - XMSN OIL PRESS variable control fully counterclockwise.

Indication/Condition

- a. XMSN PRESS vertical scale indication on CDU shall decrease smoothly.
- b. At 30 psi indication, lower red and amber scale segments shall go on.

Corrective Action

If Indication/Conditions are not as specified, replace CDU (WP 0776 00).

Step

4. Place LTS - XMSN OIL PRESS CAL-NOR switch to CAL.

Indication/Condition

XMSN PRESS vertical scale shall indicate 130 psi.

Corrective Action

If Indication/Condition is not as specified, replace CDU (WP 0776 00).

Step

5. Place LTS - XMSN OIL PRESS HI-LO switch to LO.

Indication/Condition

XMSN PRESS vertical scale shall be completely blank.

Corrective Action

If Indication/Condition is not as specified, replace CDU (WP 0776 00).

Step

6. Place LTS - XMSN OIL PRESS CAL-NOR switch to NOR.
7. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

XMSN TEMP OPERATIONAL/TROUBLESHOOTING PROCEDURE**Step**

1. Do SETUP, in this work package, if not already done.
2. Slowly turn LTS - XMSN OIL TEMP variable control fully clockwise.

Indication/Condition

XMSN TEMP vertical scale indication on CDU shall advance smoothly to full scale.

XMSN TEMP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If Indication/Condition is not as specified, go to XMSN TEMP DISPLAY INOPERATIVE, in this work package.

Step

3. Slowly turn LTS - XMSN OIL TEMP variable control fully counterclockwise.

Indication/Condition

XMSN TEMP vertical scale indication on CDU shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace CDU (WP 0776 00).

Step

4. Place LTS - XMSN OIL TEMP CAL-NOR switch to CAL.

Indication/Condition

XMSN TEMP vertical scale on CDU shall indicate 266°F (130°C).

Corrective Action

If Indication/Condition is not as specified, replace CDU (WP 0776 00).

Step

5. Place LTS - XMSN OIL TEMP HI-LO switch to LO.

Indication/Condition

- a. **UH-60L HH-60L** XMSN TEMP vertical scale on CDU shall be completely blank. ■
- b. **EH-60A UH-60A HH-60A** XMSN TEMP vertical scale on CDU shall be blank. ■

Corrective Action

If Indication/Condition a. or b. is not as specified, replace CDU (WP 0776 00).

Step

6. Place LTS - XMSN OIL TEMP CAL-NOR switch to NOR.
7. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

% RPM 1-2 OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step**

1. Do SETUP, in this work package, if not already done.

% RPM 1-2 OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. Slowly turn LTS - ENGINE % RPM NO. 1 variable control fully clockwise.

Indication/Condition

- a. % RPM 1 (pilot's and copilot's PDU) vertical scale indications shall advance smoothly to full scale.
- b. **UH-60L HH-60L** At 96% indication, pilot's PDU lower red and amber scale segments shall go off. ■
- c. **EH-60A UH-60A HH-60A** At 96% indication, pilot's PDU lower red and amber scale segments shall go off. ■
- d. **UH-60L HH-60L** At 96% indication, copilot's PDU lower red and amber scale segments shall go off. ■
- e. **EH-60A UH-60A HH-60A** At 96% indication, copilot's PDU lower red and amber scale segments shall go off. ■

Corrective Action

1. If Indication/Condition a. is not as specified, go to PDU'S % RPM 1 OR % RPM 2 DISPLAY INOPERATIVE, in this work package.
2. If Indication/Condition b. or c. is not as specified, replace pilot's PDU (WP 0782 00).
3. If Indication/Condition d. or e. is not as specified, replace copilot's PDU (WP 0782 00).

Step

3. Slowly turn LTS - ENGINE % RPM NO. 1 variable control fully counterclockwise.

Indication/Condition

- a. % RPM 1 (pilot's PDU) vertical scale indications shall decrease smoothly.
- b. **UH-60L HH-60L** At 95% indication, pilot's PDU lower red and amber scale segments shall go on. ■
- c. **EH-60A UH-60A HH-60A** At 95% indication, pilot's PDU lower red and amber scale segments shall go on. ■
- d. % RPM 1 (copilot's PDU) vertical scale indications shall decrease smoothly.
- e. **UH-60L HH-60L** At 95% indication, copilot's PDU lower red and amber scale segments shall go on. ■
- f. **EH-60A UH-60A HH-60A** At 95% indication, copilot's PDU lower red and amber scale segments shall go on. ■

Corrective Action

1. If Indication/Condition a., b., or c. is not as specified, replace pilot's PDU (WP 0782 00).
2. If Indication/Condition d., e., or f. is not as specified, replace copilot's PDU (WP 0782 00).

Step

4. Place LTS - ENGINE % RPM NO. 1 CAL-NOR switch to CAL.

Indication/Condition

- a. % RPM 1 (pilot's PDU) vertical scales shall indicate 95% or 96%.
- b. % RPM 1 (copilot's PDU) vertical scales shall indicate 95% or 96%.

Corrective Action

1. If Indication/Condition a. is not as specified, replace No. 2 SDC (WP 0781 00).
2. If Indication/Condition b. is not as specified, replace No. 1 SDC (WP 0781 00).

% RPM 1-2 OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

5. Place LTS - ENGINE % RPM NO. 1 HI-LO switch to LO.

Indication/Condition

- a. % RPM 1 (pilot's PDU) vertical scales shall be completely blank.
- b. % RPM 1 (copilot's PDU) vertical scales shall be completely blank.

Corrective Action

1. If Indication/Condition a. is not as specified, replace No. 2 SDC (WP 0781 00).
2. If Indication/Condition b. is not as specified, replace No. 1 SDC (WP 0781 00).

Step

6. Place LTS - ENGINE % RPM NO. 1 CAL-NOR switch to NOR.
7. Slowly turn LTS - ENGINE % RPM NO. 2 variable control fully clockwise.

Indication/Condition

- a. PDUs (pilot's and copilot's) % RPM 2 vertical scale indications shall advance smoothly to full scale.
- b. **UH-60L HH-60L** At 96% indication, pilot's PDU lower red and amber scale segments shall go off. ■
- c. **EH-60A UH-60A HH-60A** At 96% indication, pilot's PDU lower red and amber scale segments shall go off. ■
- d. **UH-60L HH-60L** At 96% indication, copilot's PDU lower red and amber scale segments shall go off. ■
- e. **EH-60A UH-60A HH-60A** At 96% indication, copilot's PDU lower red and amber scale segments shall go off. ■

Corrective Action

1. If Indication/Condition a. is not as specified, go to PDU'S % RPM 1 OR % RPM 2 DISPLAY INOPERATIVE, in this work package.
2. If Indication/Condition b. or c. is not as specified, replace pilot's PDU (WP 0782 00).
3. If Indication/Condition d. or e. is not as specified, replace copilot's PDU (WP 0782 00).

Step

8. Slowly turn LTS - ENGINE % RPM NO. 2 variable control fully counterclockwise.

Indication/Condition

- a. % RPM 2 (pilot's PDU) vertical scale indications shall decrease smoothly.
- b. **UH-60L** At 95% indication, pilot's PDU lower red and amber scale segments shall go on. ■
- c. **EH-60A UH-60A HH-60A** At 95% indication, pilot's PDU lower red and amber scale segments shall go on. ■
- d. % RPM 2 (copilot's PDU) vertical scale indications shall decrease smoothly.
- e. **UH-60L HH-60L** At 95% indication, copilot's PDU lower red and amber scale segments shall go on. ■
- f. **EH-60A UH-60A HH-60A** At 95% indication, copilot's PDU lower red and amber scale segments shall go on. ■

% RPM 1-2 OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition a., b., or c. is not as specified, replace pilot's PDU (WP 0782 00).
2. If Indication/Condition d., e., or f. is not as specified, replace copilot's PDU (WP 0782 00).

Step

9. Place LTS - ENGINE % RPM NO. 2 CAL-NOR switch to CAL, and HI-LO switch to HI.

Indication/Condition

- a. % RPM 2 (pilot's PDU) vertical scales shall indicate 95% or 96%.
- b. % RPM 2 (copilot's PDU) vertical scales shall indicate 95% or 96%.

Corrective Action

1. If Indication/Condition a. is not as specified, replace No. 2 SDC (WP 0781 00).
2. If Indication/Condition b. is not as specified, replace No. 1 SDC (WP 0781 00).

Step

10. Place LTS - ENGINE % RPM NO. 2 HI-LO switch to LO.

Indication/Condition

- a. % RPM 2 (pilot's PDU) vertical scales shall be completely blank.
- b. % RPM 2 (copilot's PDU) vertical scales shall be completely blank.

Corrective Action

1. If Indication/Condition a. is not as specified, replace No. 2 SDC (WP 0781 00).
2. If Indication/Condition b. is not as specified, replace No. 1 SDC (WP 0781 00).

Step

11. Place LTS - ENGINE % RPM NO. 2 CAL-NOR switch to NOR.
12. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

TGT TEMP OPERATIONAL/TROUBLESHOOTING PROCEDURE**Step**

1. Do SETUP, in this work package, if not already done.
2. Slowly turn LTS - TURBINE GAS TEMP NO. 1 variable control fully clockwise.

Indication/Condition

- a. TGT TEMP 1 vertical scale indication on CDU shall advance smoothly to full scale.

TGT TEMP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- b. TGT TEMP 1 digital readout indication on CDU shall advance smoothly.

Corrective Action

1. If Indication/Condition a. is not as specified, go to TGT TEMP 1 OR TGT TEMP 2 VERTICAL DISPLAY INOPERATIVE, in this work package.
2. If Indication/Condition b. is not as specified, go to TGT TEMP 1 AND/OR TGT TEMP 2 DIGITAL DISPLAY INOPERATIVE, in this work package.

Step

3. Slowly turn LTS - TURBINE GAS TEMP NO. 1 variable control fully counterclockwise.

Indication/Condition

TGT TEMP 1 vertical scale and digital readout indications on CDU shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace CDU (WP 0776 00).

Step

4. Place LTS - TURBINE GAS TEMP NO. 1 CAL-NOR switch to CAL, and TURBINE GAS TEMP NO. 1 HI-LO switch to HI.

Indication/Condition

- a. TGT TEMP 1 vertical scale on CDU shall indicate 1472°F (800°C).
- b. TGT TEMP 1 digital readout on CDU shall indicate between 1482° and 1504°F (806° and 818°C).

Corrective Action

If Indication/Conditions are not as specified, replace No. 1 SDC (WP 0781 00).

Step

5. Place LTS - TURBINE GAS TEMP NO. 1 HI-LO switch to LO.

Indication/Condition

- a. TGT TEMP 1 vertical scale on CDU shall be completely blank.
- b. TGT TEMP 1 digital readout on CDU shall indicate 32° to 46°F (0° to 8°C).

Corrective Action

If Indication/Conditions are not as specified, replace CDU (WP 0776 00).

Step

6. Place LTS - TURBINE GAS TEMP NO. 1 CAL-NOR switch to NOR.
7. Slowly turn LTS - TURBINE GAS TEMP NO. 2 variable control fully clockwise.

Indication/Condition

- a. TGT TEMP 2 vertical scale indication on CDU shall advance smoothly to full scale.
- b. TGT TEMP 2 digital readout indication on CDU shall advance smoothly.

TGT TEMP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition a. is not as specified, go to TGT TEMP 1 OR TGT TEMP 2 VERTICAL DISPLAY INOPERATIVE, in this work package.
2. If Indication/Condition b. is not as specified, go to TGT TEMP 1 AND/OR TGT TEMP 2 DIGITAL DISPLAY INOPERATIVE, in this work package.

Step

8. Slowly turn LTS - TURBINE GAS TEMP NO. 2 variable control fully counterclockwise.

Indication/Condition

TGT TEMP 2 vertical scale and digital readout indications on CDU shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace CDU (WP 0776 00).

Step

9. Place LTS - TURBINE GAS TEMP NO. 2 CAL-NOR switch to CAL and HI-LO switch to HI.

Indication/Condition

- a. TGT TEMP 2 vertical scale on CDU shall indicate 1472°F (800°C).
- b. TGT TEMP 2 digital readout on CDU shall indicate between 1482° to 1504°F (806° and 818°C).

Corrective Action

If Indication/Conditions are not as specified, replace No. 2 SDC (WP 0781 00).

Step

10. Place LTS - TURBINE GAS TEMP NO. 2 HI-LO switch to LO.

Indication/Condition

- a. TGT TEMP 2 vertical scale on CDU shall be completely blank.
- b. TGT TEMP 2 digital readout on CDU shall indicate 32° to 46°F (0° to 8°C).

Corrective Action

If Indication/Conditions are not as specified, replace CDU (WP 0776 00).

Step

11. Place LTS - TURBINE GAS TEMP NO. 2 CAL-NOR switch to NOR.
12. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

LAMP POWER SUPPLY OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP, in this work package, if not already done.
2. Turn all LTS variable controls fully clockwise.
3. Place and hold LTS - LAMP SUPPLY OVERLOAD switch to NO. 1.

Indication/Condition

- a. All lamps on copilot's PDU shall be off, except RTR OVERSPEED lamps.
- b. Alternate lamps on CDU shall be off.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, replace No. 1 SDC (WP 0781 00).
2. If Indication/Condition a. is not as specified, replace copilot's PDU (WP 0782 00).
3. If Indication/Condition b. is not as specified, replace CDU (WP 0776 00).

Step

4. Release LTS - LAMP SUPPLY OVERLOAD switch.

Indication/Condition

All PDU (pilot's and copilot's) and CDU lamps shall go on.

Corrective Action

If Indication/Condition is not as specified, replace No. 1 SDC (WP 0781 00).

Step

5. Place and hold LTS - LAMP SUPPLY OVERLOAD switch to NO. 2.

Indication/Condition

- a. All lamps on pilot's PDU shall be off, except RTR OVERSPEED lamps.
- b. Alternate lamps on CDU shall be off.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, replace No. 2 SDC (WP 0781 00).
2. If Indication/Condition a. is not as specified, replace pilot's PDU (WP 0782 00).
3. If Indication/Condition b. is not as specified, replace CDU (WP 0776 00).

Step

6. Release LTS - LAMP SUPPLY OVERLOAD switch.

Indication/Condition

All PDU (pilot's and copilot's) and CDU lamps shall go on.

LAMP POWER SUPPLY OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If Indication/Condition is not as specified, replace No. 2 SDC (WP 0781 00).

Step

7. Press LTS ROTOR OVERSPD RESET switch (located in helicopter's nose) to reset RTR OVERSPEED lights if lights remain lit.
8. Position LTS switches as follows:

INPUT POWER	OFF
All CAL-NOR switches	NOR
All HI-LO switches	HI
All NOR controls	Fully counterclockwise
9. Turn off electrical power.
10. Disconnect LTS setup from helicopter.

Indication/Condition

LTS setup shall be disconnected from helicopter.

Corrective Action

None Required

FACEPLATE LIGHTING OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE
Step**NOTE**

Ensure Shutdown has been performed before proceeding to step 1.

1. Turn on electrical power.
2. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch ON and press T6 switch (illuminates all MFD legends for 10 seconds, then only the active caution and advisory legends will be displayed).
3. Turn upper console INST LT NON FLT control to BRT.

Indication/Condition

CDU faceplate lights shall go on bright.

Corrective Action

If Indication/Condition is not as specified, go to CDU FACEPLATE LIGHTING INOPERATIVE, in this work package.

Step

4. Turn upper console INST LT PLT FLT control to BRT.

FACEPLATE LIGHTING OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

PDU (pilot's) faceplate lights shall go on bright.

Corrective Action

If Indication/Condition is not as specified, go to PDU FACEPLATE LIGHTING INOPERATIVE, in this work package.

Step

5. Turn upper console CPLT FLT INST LTS control to BRT.

Indication/Condition

PDU (copilot's) faceplate lights shall go on bright.

Corrective Action

If Indication/Condition is not as specified, go to COPILOT'S PDU FACEPLATE LIGHTING INOPERATIVE, in this work package.

Step

6. Turn upper console INST LT NON FLT, INST LT PLT FLT, and CPLT FLT INST LTS controls to comfortable viewing level.

Indication/Condition

- a. TGT TEMP vertical displays and associated TGT digital readouts on CDU shall indicate approximate ambient temperature of helicopter (assuming cooldown has occurred).
- b. ENG OIL TEMP vertical displays on CDU shall indicate approximate ambient temperature of helicopter (assuming cooldown has occurred).
- c. XMSN TEMP vertical displays on CDU shall indicate approximate ambient temperature of helicopter (assuming cooldown has occurred).
- d. FUEL QTY vertical displays on CDU shall indicate actual fuel on helicopter and associated TOTAL FUEL digital readout shall indicate sum of two vertical scale readings.
- e. #1 ENG OUT, #2 ENG OUT, and LOW ROTOR RPM capsules on pilot's and copilot's master warning panels shall go on (LOW ROTOR RPM capsule flashes, others are on steady).
- f. #1 ENGINE OIL PRESS and #2 ENGINE OIL PRESS capsules or legends shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, do TGT TEMP using line test set.
 - a. If trouble remains, troubleshoot No. 1 or No. 2 engine instruments and warning lights system, as required (WP 0073 00 or WP 0074 00).
2. If Indication/Condition b. is not as specified, do ENG OIL TEMP using line test set.
 - a. If trouble remains, troubleshoot No. 1 or No. 2 engine instruments and warning lights system, as required (WP 0073 00 or WP 0074 00).
3. If Indication/Condition c. is not as specified, do XMSN TEMP using line test set.
4. If Indication/Condition d. is not as specified, do FUEL QTY using line test set.
 - a. If trouble remains, troubleshoot fuel quantity system (WP 0132 00).

FACEPLATE LIGHTING OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

5. If Indication/Condition e. is not as specified, go to MASTER WARNING PANEL LOW ROTOR RPM, #1 ENG OUT, OR #2 ENG OUT CAPSULE IS OFF, in this work package.
6. If Indication/Condition f. is not as specified, go to #1 ENGINE OIL PRESS OR #2 ENGINE OIL PRESS CAPSULE OR LEGENDS DO NOT GO ON, in this work package.

Step

7. **HH-60A HH-60L** Turn pilot's or copilot's multifunction display (MFD) switch OFF. **■**
8. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

**SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**
Step

1. Turn off electrical power.
2. Disconnect LTS setup from helicopter.

Indication/Condition

LTS setup shall be disconnected from helicopter.

Corrective Action

None Required

MASTER WARNING PANEL LOW ROTOR RPM, #1 ENG OUT, OR #2 ENG OUT CAPSULE IS OFF**SYMPTOM**

Master warning panel LOW ROTOR RPM, #1 ENG OUT, or #2 ENG OUT capsule is off.

MALFUNCTION

Master warning panel LOW ROTOR RPM, #1 ENG OUT, or #2 ENG OUT capsule is off.

CORRECTIVE ACTION

1. Check helicopter configuration.
 - a. **EMEP** , go to 2.
 - b. **W/O EMEP** , go to 10.
2. **EMEP** Remove and check pin filtered adapters P117 or **HH-60A HH-60L** P1019R or P1020R **■** , P137, and P138 (WP 0925 00).
 - a. If pin filtered adapters are good, go to 3.
 - b. If pin filtered adapters are not good, replace pin filtered adapters as required (WP 0925 00). Go to 13.

MASTER WARNING PANEL LOW ROTOR RPM, #1 ENG OUT, OR #2 ENG OUT CAPSULE IS OFF -
Continued

3. **EMEP** Install pin filtered adapters (WP 0925 00). Go to 4.
4. **EMEP** Disconnect P130, P131, P137, P138, P242, and P902. Go to 5.
5. **EMEP** Check continuity between:
 - P130-3 and P138-n
 - P130-3 and P242-a
 - P130-3 and P902-u
 - P130-9 and P137-u
 - P130-9 and P138-u
 - P130-9 and P243-h
 - P130-10 and P137-n
 - P130-10 and P243-a
 - P130-10 and P243-J
 - a. If continuity is present, go to 6.
 - b. If continuity is not present, replace pin filtered connector P130 (WP 0925 00). Go to 13.
6. **EMEP** Check continuity between each pin of P130 and shell.
 - a. If continuity is present, replace pin filtered connector P130 (WP 0925 00). Go to 13.
 - b. If continuity is not present, go to 7.
7. **EMEP** Check continuity between:
 - P130-3 and P131-3
 - P130-9 and P131-9
 - P130-10 and P131-10
 - a. If continuity is present, go to 8.
 - b. If continuity is not present, replace pin filtered connector P131 (WP 0925 00). Go to 13.
8. **EMEP** Check continuity between each pin of P131 and shell.
 - a. If continuity is present, replace pin filtered connector P131 (WP 0925 00). Go to 13.
 - b. If continuity is not present, go to 9.
9. **EMEP** Connect P130, P131, P137, P138, P242, and P902 to their respective units. Go to 10.
10. Hold caution/advisory panel BRT/DIM-TEST **HH-60A HH-60L** instrument panel INDICATOR LTS **■** switch to TEST. Go to 11.
11. Check that all master warning panel capsules go on.
 - a. If capsules go on, repair/replace wiring, as required, between (WP 1747 00), then go to 13:
 - P130-3 and P138-n
 - P130-9 and P137-u

MASTER WARNING PANEL LOW ROTOR RPM, #1 ENG OUT, OR #2 ENG OUT CAPSULE IS OFF -
Continued

- P130-9 and P138-u
 - P130-10 and P137-n
 - P131-3 and P138-n
 - P131-9 and P137-u
 - P131-9 and P138-u
 - P131-10 and P137-n
- b. If capsules do not go on, go to **12**.
- 12. Check capsule lamps.**
- a. If lamps are good, replace master warning panel (WP 0790 00). Go to **13**.
- b. If lamps are not good, replace lamps as required (WP 0791 00). Go to **13**.
- 13. Procedure completed.**

NO. 1 ENGINE OUT AND/OR NO. 2 ENGINE OUT LIGHTS ARE OFF**SYMPTOM**

NO. 1 ENGINE OUT and/or NO. 2 ENGINE OUT lights are off.

MALFUNCTION

NO. 1 ENGINE OUT and/or NO. 2 ENGINE OUT lights are off.

CORRECTIVE ACTION

- 1. Are both lights off?**
- a. If both lights are off, replace CDU (WP 0776 00). Go to **7**.
- b. If both lights are not off, go to **2**.
- 2. Is NO. 1 ENGINE OUT light off?**
- a. If light is off, go to **3**.
- b. If light is on, go to **5**.
- 3. Check if Ng 1 fuse is good.**
- a. If fuse is good, go to **4**.
- b. If fuse is not good, replace fuse (WP 0779 00). Go to **7**.
- 4. Check continuity between P151-31 and P154-31.**
- a. If continuity is present, replace No. 1 SDC (WP 0781 00). Go to **7**.
- b. If trouble remains, replace CDU (WP 0776 00). Go to **7**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.
- 5. Check if Ng 2 fuse is good.**
- a. If fuse is good, go to **6**.
- b. If fuse is not good, replace fuse (WP 0779 00). Go to **7**.
- 6. Check continuity between P152-31 and P155-31.**

NO. 1 ENGINE OUT AND/OR NO. 2 ENGINE OUT LIGHTS ARE OFF - Continued

- a. If continuity is present, replace No. 2 SDC (WP 0781 00). Go to **7**.
- b. If trouble remains, replace CDU (WP 0776 00). Go to **7**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.

7. Procedure completed.**NO. 1 LOW OIL PRESS AND/OR NO. 2 LOW OIL PRESS LIGHTS ARE OFF****SYMPTOM**

NO. 1 LOW OIL PRESS and/or NO. 2 LOW OIL PRESS lights are off.

MALFUNCTION

NO. 1 LOW OIL PRESS and/or NO. 2 LOW OIL PRESS lights are off.

CORRECTIVE ACTION**1. Are both lights off?**

- a. If both lights are off, replace CDU (WP 0776 00). Go to **7**.
- b. If both lights are not off, go to **2**.

2. Check if NO. 1 LOW OIL PRESS light is off.

- a. If light is off, go to **3**.
- b. If light is on, go to **5**.

3. Check if NO. 1 OIL PRESS fuse is good.

- a. If fuse is good, go to **4**.
- b. If fuse is not good, replace fuse (WP 0779 00). Go to **7**.

4. Check continuity between P151-30 and P154-30.

- a. If continuity is present, replace No. 1 SDC (WP 0781 00). Go to **7**.
- b. If trouble remains, replace CDU (WP 0776 00). Go to **7**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.

5. Check if No. 2 OIL PRESS fuse is good.

- a. If fuse is good, go to **6**.
- b. If fuse is not good, replace fuse (WP 0779 00). Go to **7**.

6. Check continuity between P152-30 and P155-30.

- a. If continuity is present, replace No. 2 SDC (WP 0781 00). Go to **7**.
- b. If trouble remains, replace CDU (WP 0776 00). Go to **7**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.

7. Procedure completed.

#1 ENGINE OIL PRESS OR #2 ENGINE OIL PRESS CAPSULE OR LEGENDS DO NOT GO ON**SYMPTOM**

#1 ENGINE OIL PRESS or #2 ENGINE OIL PRESS capsule or legends do not go on.

MALFUNCTION

#1 ENGINE OIL PRESS or #2 ENGINE OIL PRESS capsule or legends do not go on.

CORRECTIVE ACTION

1. Hold caution/advisory panel BRT/DIM-TEST or HH-60A HH-60L instrument panel INDICATOR LTS switch to TEST. Go to 2.
2. Check that capsules or legends go on.
 - a. If capsules or legends go on, go to 3.
 - b. If capsules or legends do not go on, go to 7.
3. Check helicopter configuration.
 - a. EMEP, go to 4.
 - b. W/O EMEP, go to 6.
4. EMEP Remove and check pin filtered adapters P117 or HH-60A HH-60L P1019R or P1020R, P137, and P138 (WP 0925 00).
 - a. If pin filtered adapters are good, go to 5.
 - b. If pin filtered adapters are not good, replace pin filtered adapter (WP 0925 00). Go to 8.
5. EMEP Install pin filtered adapters (WP 0925 00). Go to 6.
6. Check continuity between:
 - P117-3 and P138-X
 - P117-51 and P137-X
 - HH-60A HH-60L P1019R-56 and P138-X
 - HH-60A HH-60L P1020R-56 and P138-X
 - HH-60A HH-60L P1019R-73 and P137-X
 - HH-60A HH-60L P1020R-73 and P137-X
 - a. If continuity is present, system is operational. Go to 8.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to 8.
7. Check capsule for burnt out lamps.
 - a. If lamps are burnt out, replace lamps (WP 0786 00). Go to 8.
 - b. If lamps are not burnt out, replace caution/advisory panel (WP 0785 00). Go to 8.
8. Procedure completed.

RTR OVERSPEED LIGHT IS ON**SYMPTOM**

RTR OVERSPEED light is on.

MALFUNCTION

RTR OVERSPEED light is on.

CORRECTIVE ACTION

- 1. Place and hold rotor overspeed reset switch to RESET. Go to 2.**
- 2. Check that lights go off.**
 - a. If lights go off, system is operational. Go to **9**.
 - b. If lights do not go off, go to **3**.
- 3. With switch held in RESET position, check continuity between rotor overspeed reset switch terminals 1 and 2.**
 - a. If continuity is present, go to **4**.
 - b. If continuity is not present, replace switch (WP 0797 00). Go to **9**.
- 4. Check continuity between rotor overspeed reset switch terminal 1 and ground.**
 - a. If continuity is present, go to **5**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9**.
- 5. Check helicopter configuration.**
 - a. **EMEP** , go to **6**.
 - b. **W/O EMEP** , go to **8**.
- 6. EMEP Remove and check pin filtered adapters P137 and P138 (WP 0925 00).**
 - a. If pin filtered adapters are good, go to **7**.
 - b. If pin filtered adapters are not good, replace as required (WP 0925 00). Go to **9**.
- 7. EMEP Install pin filtered adapters (WP 0925 00). Go to 8.**
- 8. Check continuity between rotor overspeed reset switch terminal 2 and P137-p and P138-p.**
 - a. If continuity is present, replace CDU (WP 0776 00). Go to **9**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9**.
- 9. Procedure completed.**

CHAN 1 LIGHT IS ON**SYMPTOM**

CHAN 1 light is on.

MALFUNCTION

CHAN 1 light is on.

CORRECTIVE ACTION**1. Check helicopter configuration.**

- a. **EMEP** , go to **2**.
- b. **W/O EMEP** , go to **4**.

2. EMEP Remove and check pin filtered adapter P138 (WP 0925 00).

- a. If pin filtered adapter is good, go to **3**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **9**.

3. EMEP Install pin filtered adapter (WP 0925 00). Go to 4.**4. Check for 115 vac between P138-CC and P138-DD.**

- a. If voltage is as specified, go to **5**.
- b. If voltage is not as specified, go to **7**.

5. Check for 28 vdc between P138-AA and P138-DD.

- a. If voltage is as specified, replace No. 1 SDC (WP 0781 00). Go to **9**.
- b. If trouble remains, replace CDU (WP 0776 00). Go to **9**.
- c. If trouble still remains, replace PDU (WP 0782 00). Go to **9**.
- d. If voltage is not as specified, go to **6**.

6. Check for 28 vdc between NO. 1 DC INST circuit breaker terminal 1 and ground.

- a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P138-AA (WP 1747 00). Go to **9**.
- b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **9**.

7. Check continuity between P138-DD and ground.

- a. If continuity is present, go to **8**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9**.

8. Check for 115 vac between NO. 1 AC INST circuit breaker terminal 1 and ground.

- a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P138-AA (WP 1747 00). Go to **9**.
- b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **9**.

9. Procedure completed.

CHAN 2 LIGHT IS ON**SYMPTOM**

CHAN 2 light is on.

MALFUNCTION

CHAN 2 light is on.

CORRECTIVE ACTION

- 1. Check helicopter configuration.**
 - a. **EMEP** , go to **2**.
 - b. **W/O EMEP** , go to **4**.
- 2. EMEP Remove and check pin filtered adapter P137 (WP 0925 00).**
 - a. If pin filtered adapter is good, go to **3**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **9**.
- 3. EMEP Install pin filtered adapter (WP 0925 00). Go to 4.**
- 4. Check for 115 vac between P137-CC and P137-DD.**
 - a. If voltage is as specified, go to **5**.
 - b. If voltage is not as specified, go to **7**.
- 5. Check for 28 vdc between P137-AA and P137-DD.**
 - a. If voltage is as specified, replace No. 2 SDC (WP 0781 00). Go to **9**.
 - b. If trouble remains, replace CDU (WP 0776 00). Go to **9**.
 - c. If trouble still remains, replace PDU (WP 0782 00). Go to **9**.
 - d. If voltage is not as specified, go to **6**.
- 6. Check for 28 vdc between NO. 2 DC INST circuit breaker terminal 1 and ground.**
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P137-AA (WP 1747 00). Go to **9**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **9**.
- 7. Check continuity between P137-DD and ground.**
 - a. If continuity is present, go to **8**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9**.
- 8. Check for 115 vac between NO. 2 AC INST circuit breaker terminal 1 and ground.**
 - a. If voltage is as specified, repair/replace wiring between circuit breaker terminal 1 and P137-CC (WP 1747 00). Go to **9**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **9**.
- 9. Procedure completed.**

WITH CDU PUSH TO TEST SWITCH PRESSED, SCALES OR RANGES ARE OFF**SYMPTOM**

With CDU PUSH TO TEST switch pressed, two or more adjacent lamps on one scale are off, or one lamp between green and amber, amber and red, or green and red ranges on a scale is off.

MALFUNCTION

With CDU PUSH TO TEST switch pressed, two or more adjacent lamps on one scale are off, or one lamp between green and amber, amber and red, or green and red ranges on a scale is off.

CORRECTIVE ACTION**NOTE**

These modules are interchangeable:

Transmission Oil Temp (A5)

No. 1 Engine Oil Temp (A7)

No. 2 Engine Oil Temp (A8)

No. 1 TGT (A11)

No. 2 TGT (A12)

No. 1 Ng Speed (A13)

No. 2 Ng Speed (A14)

No. 1 Fuel Quantity (A3)

No. 2 Fuel Quantity (A4)

No. 1 Engine Oil Press (A9)

No. 2 Engine Oil Press (A10)

1. Interchange modules in accordance with above. Does problem follow modules?

- a. If problem follows module, go to **2**.
- b. If problem does not follow module, replace CDU (WP 0776 00). Go to **5**.

2. Check helicopter configuration.

- a. **EMEP** , go to **3**.
- b. **W/O EMEP** , go to **4**.

3. **EMEP Replace malfunctioning BGF lamp (WP 0778 00).**

- a. If trouble remains, replace CDU (WP 0776 00). Go to **5**.

4. **W/O EMEP Replace display lamp assembly (WP 0780 00).**

- a. If trouble remains, replace CDU (WP 0776 00). Go to **5**.

5. Procedure completed.

WITH CDU PUSH TO TEST SWITCH PRESSED, ONE OR MORE RTR OVERSPEED LIGHTS ARE OFF**SYMPTOM**

With CDU PUSH TO TEST switch pressed, one or more RTR OVERSPEED lights are off.

MALFUNCTION

With CDU PUSH TO TEST switch pressed, one or more RTR OVERSPEED lights are off.

CORRECTIVE ACTION**1. Is same light off on both PDU's?**

- a. If the same light is off, replace CDU (WP 0776 00). Go to **4**.
- b. If different lights are off, go to **2**.

2. Interchange PDU's. Go to 3.**3. Is trouble on same PDU?**

- a. If trouble is on same display unit, replace malfunctioning PDU or CDU (WP 0776 00). Go to **4**.
- b. If trouble is on different display unit, repair/replace wiring, as required, between (WP 1747 00), then go to **4**:
 - P153-5 and P157-31
 - P153-6 and P157-32
 - P153-7 and P157-33
 - P156-5 and P158-31
 - P156-6 and P158-32
 - P156-7 and P158-33

4. Procedure completed.**CDU DIM CONTROL VARIES PDU DISPLAY LIGHTING****SYMPTOM**

CDU DIM control varies PDU display lighting.

MALFUNCTION

CDU DIM control varies PDU display lighting.

CORRECTIVE ACTION**1. Check that CDU DIM control varies PDU display lighting.**

- a. If PDU display lighting varies, go to **2**.
- b. If PDU display lighting does not vary, go to **3**.

2. Check continuity between P151-19 and P154-19.

- a. If continuity is present, replace No 1. SDC (WP 0781 00). Go to **4**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.

CDU DIM CONTROL VARIES PDU DISPLAY LIGHTING - Continued**3. Check continuity between P152-19 and P155-19.**

- a. If continuity is present, replace No. 2 SDC (WP 0781 00). Go to **4.**
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**

4. Procedure completed.**PILOT'S OR COPILOT'S PDU % RPM DISPLAY IS INOPERATIVE****SYMPTOM**

Pilot's or copilot's PDU % RPM display is inoperative.

MALFUNCTION

Pilot's or copilot's PDU % RPM display is inoperative.

CORRECTIVE ACTION**1. Interchange PDU's. Go to 2.****2. Is trouble on same PDU?**

- a. If trouble is on same PDU, replace malfunctioning PDU (WP 0782 00). Go to **4.**
- b. If trouble is on different PDU, go to **3.**

3. Check continuity between P149-18 and P157-18 (copilot's PDU), or between P150-18 and P158-18 (pilot's PDU).

- a. If continuity is present, replace No. 1 SDC or No. 2 SDC as required (WP 0781 00). Go to **4.**
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4.**

4. Procedure completed.**ONE OR MORE RTR OVERSPEED LIGHTS ARE OFF****SYMPTOM**

One or more RTR OVERSPEED lights are off.

MALFUNCTION

One or more RTR OVERSPEED lights are off.

CORRECTIVE ACTION**1. Is same light off on pilot's and copilot's PDU?**

- a. If the same light is off, replace CDU (WP 0776 00). Go to **4.**
- b. If different light is off, go to **2.**

2. Interchange PDU's. Go to 3.**3. Is trouble on same PDU?**

- a. If trouble is still on same display unit, replace malfunctioning PDU (WP 0782 00). Go to **4.**
- b. If trouble is on different display unit, repair/replace wiring, as required, between (WP 1747 00), then go to **4:**
 - P153-5 and P157-31

ONE OR MORE RTR OVERSPEED LIGHTS ARE OFF - Continued

- P153-6 and P157-32
- P153-7 and P157-33
- P156-5 and P158-31
- P156-6 and P158-32
- P156-7 and P158-33

4. Procedure completed.**FUEL QTY 1, FUEL QTY 2, AND/OR TOTAL FUEL DISPLAY INOPERATIVE****SYMPTOM**

FUEL QTY 1, FUEL QTY 2, and/or TOTAL FUEL display inoperative.

MALFUNCTION

FUEL QTY 1, FUEL QTY 2, and/or TOTAL FUEL display inoperative.

CORRECTIVE ACTION**1. Are all three displays inoperative?**

- a. If all three displays are inoperative, replace CDU (WP 0776 00). Go to **7**.
- b. If only 1 or 2 displays are inoperative, go to **2**.

2. Is FUEL QTY 1 inoperative?

- a. If FUEL QTY 1 is inoperative, go to **3**.
- b. If FUEL QTY 1 is operating, go to **4**.

3. Check continuity between P151-41 and P154-41.

- a. If continuity is present, replace No. 1 SDC (WP 0781 00). Go to **7**.
- b. If trouble remains, replace CDU (WP 0776 00). Go to **7**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.

4. Is FUEL QTY 2 inoperative?

- a. If FUEL QTY 2 is inoperative, go to **5**.
- b. If FUEL QTY 2 is operative, go to **6**.

5. Check continuity between P152-41 and P155-41.

- a. If continuity is present, replace No. 2 SDC (WP 0781 00). Go to **7**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.
- c. If trouble remains, replace CDU (WP 0776 00). Go to **7**.

6. Check continuity between P151-43 and P154-43.

- a. If continuity is present, replace No. 1 SDC (WP 0781 00). Go to **7**.
- b. If trouble remains, replace CDU (WP 0776 00). Go to **7**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.

7. Procedure completed.

ENG OIL PRESS 1 AND/OR ENG OIL PRESS 2 DISPLAY INOPERATIVE**SYMPTOM**

ENG OIL PRESS 1 and/or ENG OIL PRESS 2 display inoperative.

MALFUNCTION

ENG OIL PRESS 1 and/or ENG OIL PRESS 2 display inoperative.

CORRECTIVE ACTION**1. Are both displays inoperative?**

- a. If both displays are inoperative, replace CDU (WP 0776 00). Go to **5**.
- b. If only one display is inoperative, go to **2**.

2. Is ENG OIL PRESS 1 inoperative?

- a. If ENG OIL PRESS 1 is inoperative, go to **3**.
- b. If ENG OIL PRESS 1 is operative, go to **4**.

3. Check continuity between P151-45 and P154-45.

- a. If continuity is present, replace No. 1 SDC (WP 0781 00). Go to **5**.
- b. If trouble remains, replace CDU (WP 0776 00). Go to **5**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.

4. Check continuity between P152-45 and P155-45.

- a. If continuity is present, replace No. 2 SDC (WP 0781 00). Go to **5**.
- b. If trouble remains, replace CDU (WP 0776 00). Go to **5**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.

5. Procedure completed.**ENG OIL TEMP 1 AND/OR ENG OIL TEMP 2 DISPLAY INOPERATIVE****SYMPTOM**

ENG OIL TEMP 1 and/or ENG OIL TEMP 2 display inoperative.

MALFUNCTION

ENG OIL TEMP 1 and/or ENG OIL TEMP 2 display inoperative.

CORRECTIVE ACTION**1. Are both displays inoperative?**

- a. If both displays are inoperative, replace CDU (WP 0776 00). Go to **5**.
- b. If only one display is inoperative, go to **2**.

2. Is ENG OIL PRESS 1 inoperative?

- a. If ENG OIL PRESS 1 is inoperative, go to **3**.
- b. If ENG OIL PRESS 1 is operative, go to **4**.

3. Check continuity between P151-44 and P154-44.

ENG OIL TEMP 1 AND/OR ENG OIL TEMP 2 DISPLAY INOPERATIVE - Continued

- a. If continuity is present, replace No. 1 SDC (WP 0781 00). Go to **5**.
 - b. If trouble remains, replace CDU (WP 0776 00). Go to **5**.
 - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 4. Check continuity between P152-44 and P155-44.**
- a. If continuity is present, replace No. 2 SDC (WP 0781 00). Go to **5**.
 - b. If trouble remains, replace CDU (WP 0776 00). Go to **5**.
 - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.
- 5. Procedure completed.**

NO. 1 HIGH OIL TEMP AND/OR NO. 2 HIGH OIL TEMP LIGHT IS OFF**SYMPTOM**

NO. 1 HIGH OIL TEMP and/or NO. 2 HIGH OIL TEMP light is off.

MALFUNCTION

NO. 1 HIGH OIL TEMP and/or NO. 2 HIGH OIL TEMP light is off.

CORRECTIVE ACTION

- 1. Are both lights off?**
 - a. If both lights are off, replace CDU (WP 0776 00). Go to **7**.
 - b. If only one light is off, go to **2**.
- 2. Is NO. 1 HIGH OIL TEMP light off?**
 - a. If NO. 1 HIGH OIL TEMP light is off, go to **3**.
 - b. If NO. 1 HIGH OIL TEMP light is on, go to **5**.
- 3. Check if OIL TEMP 1 fuse is good.**
 - a. If fuse is good, go to **4**.
 - b. If fuse is not good, replace fuse (WP 0779 00). Go to **7**.
- 4. Check continuity between P151-32 and P154-32.**
 - a. If continuity is present, replace No. 1 SDC (WP 0781 00). Go to **7**.
 - b. If trouble remains, replace CDU (WP 0776 00). Go to **7**.
 - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.
- 5. Check if OIL TEMP 2 fuse is good.**
 - a. If fuse is good, go to **6**.
 - b. If fuse is not good, replace fuse (WP 0779 00). Go to **7**.
- 6. Check continuity between P152-32 and P155-32.**
 - a. If continuity is present, replace No. 2 SDC (WP 0781 00). Go to **7**.
 - b. If trouble remains, replace CDU (WP 0776 00). Go to **7**.
 - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.

NO. 1 HIGH OIL TEMP AND/OR NO. 2 HIGH OIL TEMP LIGHT IS OFF - Continued**7. Procedure completed.****NG SPEED 1 AND/OR NG SPEED 2 VERTICAL DISPLAY INOPERATIVE****SYMPTOM**

Ng SPEED 1 and/or Ng SPEED 2 vertical display inoperative.

MALFUNCTION

Ng SPEED 1 and/or Ng SPEED 2 vertical display inoperative.

CORRECTIVE ACTION**1. Are both displays inoperative?**

- a. If both displays are inoperative, replace CDU (WP 0776 00). Go to **5**.
- b. If only 1 display is inoperative, go to **2**.

2. Is Ng SPEED 1 vertical display inoperative?

- a. If Ng SPEED 1 vertical display is inoperative, go to **3**.
- b. If Ng SPEED 1 display is operative, go to **4**.

3. Check continuity between P151-47 and P154-47.

- a. If continuity is present, replace No. 1 SDC (WP 0781 00). Go to **5**.
- b. If trouble remains, replace CDU (WP 0776 00). Go to **5**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.

4. Check continuity between P152-47 and P155-47.

- a. If continuity is present, replace No. 2 SDC (WP 0781 00). Go to **5**.
- b. If trouble remains, replace CDU (WP 0776 00). Go to **5**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.

5. Procedure completed.**NG SPEED 1 AND/OR NG SPEED 2 DIGITAL DISPLAY INOPERATIVE****SYMPTOM**

Ng SPEED 1 and/or Ng SPEED 2 digital display inoperative.

MALFUNCTION

Ng SPEED 1 and/or Ng SPEED 2 digital display inoperative.

CORRECTIVE ACTION**1. Are both digital displays inoperative?**

- a. If both displays are inoperative, replace CDU (WP 0776 00). Go to **5**.
- b. If only one display is inoperative, go to **2**.

2. Is Ng SPEED 1 digital display inoperative?

- a. If Ng SPEED 1 display is inoperative, go to **3**.
- b. If Ng SPEED 1 display is operating, go to **4**.

NG SPEED 1 AND/OR NG SPEED 2 DIGITAL DISPLAY INOPERATIVE - Continued**3. Check continuity between P151-49 and P154-49.**

- a. If continuity is present, replace No. 1 SDC (WP 0781 00). Go to **5**.
- b. If trouble remains, replace CDU (WP 0776 00). Go to **5**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.

4. Check continuity between P152-49 and P155-49.

- a. If continuity is present, replace No. 2 SDC (WP 0781 00). Go to **5**.
- b. If trouble remains, replace CDU (WP 0776 00). Go to **5**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **5**.

5. Procedure completed.**% TRQ 1 AND/OR % TRQ 2 VERTICAL DISPLAY OR % TRQ DIGITAL READOUT INOPERATIVE****SYMPTOM**

% TRQ 1 and/or % TRQ 2 vertical display or % TRQ digital readout inoperative.

MALFUNCTION

% TRQ 1 and/or % TRQ 2 vertical display or % TRQ digital readout inoperative.

CORRECTIVE ACTION**1. Are both vertical displays inoperative?**

- a. If both displays are inoperative, replace PDU (WP 0782 00). Go to **6**.
- b. If only one display is inoperative, go to **2**.

2. Is CHAN 1 light on?

- a. If CHAN 1 light is on, replace No. 2 SDC (WP 0781 00). Go to **6**.
- b. If CHAN 1 light is off, go to **3**.

3. Is CHAN 2 light on?

- a. If CHAN 2 light is on, replace No. 2 SDC (WP 0781 00). Go to **6**.
- b. If CHAN 2 light is off, go to **4**.

4. Interchange PDU's. Go to 5.**5. Is trouble on same PDU?**

- a. If trouble is on same PDU, replace PDU (WP 0782 00). Go to **6**.
- b. If trouble is on different PDU, repair/replace wiring, as required, between (WP 1747 00), then go to **6**:
 - P149-20 and P157-20
 - P149-21 and P157-21
 - P149-22 and P157-22
 - P149-23 and P157-23
 - P150-20 and P158-20

% TRQ 1 AND/OR % TRQ 2 VERTICAL DISPLAY OR % TRQ DIGITAL READOUT INOPERATIVE - Continued

- P150-21 and P158-21
- P150-22 and P158-22
- P150-23 and P158-23

6. Procedure completed.**XMSN PRESS DISPLAY INOPERATIVE****SYMPTOM**

XMSN PRESS display inoperative.

MALFUNCTION

XMSN PRESS display inoperative.

CORRECTIVE ACTION**1. Check continuity between P152-28 and P155-28.**

- a. If continuity is present, replace No. 2 SDC (WP 0781 00). Go to **2**.
- b. If trouble remains, replace CDU (WP 0776 00). Go to **2**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **2**.

2. Procedure completed.**XMSN TEMP DISPLAY INOPERATIVE****SYMPTOM**

XMSN TEMP display inoperative.

MALFUNCTION

XMSN TEMP display inoperative.

CORRECTIVE ACTION**1. Check continuity between:**

- **P152-22 and P155-22**
- **P152-23 and P155-23**

- a. If continuity is present, replace No. 2 SDC (WP 0781 00). Go to **2**.
- b. If trouble remains, replace CDU (WP 0776 00). Go to **2**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **2**.

2. Procedure completed.

PDU'S % RPM 1 OR % RPM 2 DISPLAY INOPERATIVE**SYMPTOM**

PDU's % RPM 1 or % RPM 2 display inoperative.

MALFUNCTION

PDU's % RPM 1 or % RPM 2 display inoperative.

CORRECTIVE ACTION**1. Is CHAN 1 light on?**

- a. If CHAN 1 light is on, replace No. 1 SDC (WP 0781 00). Go to **5**.
- b. If CHAN 1 light is not on, go to **2**.

2. Is CHAN 2 light on?

- a. If CHAN 2 light is on, replace No. 2 SDC (WP 0781 00). Go to **5**.
- b. If CHAN 2 light is off, go to **3**.

3. Interchange PDU's. Go to 4.**4. Is trouble on same PDU?**

- a. If trouble is on same display unit, replace malfunctioning display unit (WP 0782 00). Go to **5**.
- b. If trouble is on different display unit, repair/replace wiring, as required, between (WP 1747 00), then go to **5**:
 - P149-17 and P157-17
 - P149-19 and P157-19
 - P150-17 and P158-17
 - P150-19 and P158-19

5. Procedure completed.**TGT TEMP 1 OR TGT TEMP 2 VERTICAL DISPLAY INOPERATIVE****SYMPTOM**

TGT TEMP 1 or TGT TEMP 2 vertical display inoperative.

MALFUNCTION

TGT TEMP 1 or TGT TEMP 2 vertical display inoperative.

CORRECTIVE ACTION**1. Is CHAN 1 light on?**

- a. If CHAN 1 light is on, replace No. 1 SDC (WP 0781 00). Go to **4**.
- b. If trouble remains, replace CDU (WP 0776 00). Go to **4**.
- c. If CHAN 1 light is not on, go to **2**.

2. Is CHAN 2 light on?

- a. If CHAN 2 light is on, replace No. 2 SDC (WP 0781 00). Go to **4**.

TGT TEMP 1 OR TGT TEMP 2 VERTICAL DISPLAY INOPERATIVE - Continued

b. If CHAN 2 light is off, go to **3**.

3. Check continuity between:

- **P151-46 and P154-46**
- **P152-46 and P155-46**

a. If continuity is present, replace CDU (WP 0776 00). Go to **4**.

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **4**.

4. Procedure completed.

TGT TEMP 1 AND/OR TGT TEMP 2 DIGITAL DISPLAY INOPERATIVE**SYMPTOM**

TGT TEMP 1 and/or TGT TEMP 2 digital display inoperative.

MALFUNCTION

TGT TEMP 1 and/or TGT TEMP 2 digital display inoperative.

CORRECTIVE ACTION

1. Are both digital displays inoperative?

a. If both displays are inoperative, replace CDU (WP 0776 00). Go to **7**.

b. If only one display is inoperative, go to **2**.

2. Is CHAN 1 light on?

a. If CHAN 1 light is on, replace No. 1 or No. 2 SDC as applicable (WP 0781 00). Go to **7**.

b. If trouble remains, replace CDU (WP 0776 00). Go to **7**.

c. If CHAN 1 light is off, go to **3**.

3. Is CHAN 2 light on?

a. If CHAN 2 light is on, replace No. 2 SDC (WP 0781 00). Go to **7**.

b. If trouble remains, replace CDU (WP 0776 00). Go to **7**.

c. If CHAN 2 light is off, go to **4**.

4. Is TGT TEMP 1 digital display inoperative?

a. If display is inoperative, go to **5**.

b. If display is operating, go to **6**.

5. Check continuity between P151-48 and P154-48.

a. If continuity is present, replace CDU (WP 0776 00). Go to **7**.

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.

6. Check continuity between P152-48 and P155-48.

a. If continuity is present, replace CDU (WP 0776 00). Go to **7**.

b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **7**.

7. Procedure completed.

CDU FACEPLATE LIGHTING INOPERATIVE**SYMPTOM**

CDU faceplate lighting inoperative.

MALFUNCTION

CDU faceplate lighting inoperative.

CORRECTIVE ACTION

- 1. Turn upper console INST LTS NON FLT control to BRT. Go to 2.**
- 2. Remove CDU faceplate and check for 115 vac at faceplate lighting connector.**
 - a. If voltage is as specified, replace CDU faceplate (WP 0776 00). Go to **9**.
 - b. If voltage is not as specified, go to **3**.
- 3. Check for 115 vac between P154-128 and ground.**
 - a. If voltage is as specified, replace CDU (WP 0776 00). Go to **9**.
 - b. If voltage is not as specified, go to **4**.
- 4. Check continuity between P151-128 and P154-128.**
 - a. If continuity is present, go to **5**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9**.
- 5. Check helicopter configuration.**
 - a. **EMEP** , go to **6**.
 - b. **W/O EMEP** , go to **8**.
- 6. EMEP Remove and check pin filtered adapter P138 (WP 0925 00).**
 - a. If pin filtered adapter is good, go to **7**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **9**.
- 7. EMEP Install pin filtered adapter (WP 0925 00). Go to 8.**
- 8. Check for 5 vac between P138-GG and ground.**
 - a. If voltage is as specified, replace No. 1 SDC (WP 0781 00). Go to **9**.
 - b. If voltage is not as specified, repair/replace wiring between P280-L and P138-GG (WP 1747 00).
 - c. If trouble remains, troubleshoot instrument panel lights (WP 0105 00). Go to **9**.
- 9. Procedure completed.**

PDU FACEPLATE LIGHTING INOPERATIVE**SYMPTOM**

PDU faceplate lighting inoperative.

MALFUNCTION

PDU faceplate lighting inoperative.

CORRECTIVE ACTION

- 1. Interchange pilot's PDU and copilot's PDU, and check that faceplate on PDU lights.**
 - a. If faceplate lights, replace malfunctioning PDU faceplate (WP 0783 00). Go to **10**.
 - b. If faceplate does not light, go to **2**.
- 2. Check for 115 vac between P158-99 and P158-100.**
 - a. If voltage is as specified, replace PDU (WP 0782 00). Go to **10**.
 - b. If voltage is not as specified, go to **3**.
- 3. Check continuity between P150-99 and P158-99.**
 - a. If continuity is present, go to **4**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **10**.
- 4. Check for 115 vac between P386-C and ground.**
 - a. If voltage is as specified, replace No. 2 SDC (WP 0781 00). Go to **10**.
 - b. If voltage is not as specified, go to **5**.
- 5. Check for 5 vac between P386-A and ground.**
 - a. If voltage is as specified, go to **6**.
 - b. If voltage is not as specified, repair/replace wiring between P246-a and P386-A (WP 1747 00). Go to **10**.
 - c. If trouble remains, troubleshoot instrument panel lights (WP 0105 00). Go to **10**.
- 6. Check helicopter configuration.**
 - a. **EMEP** , go to **7**.
 - b. **W/O EMEP** , go to **9**.
- 7. EMEP Remove and check pin filtered adapter P137 (WP 0925 00).**
 - a. If pin filtered adapter is good, go to **8**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **10**.
- 8. EMEP Install pin filtered adapter (WP 0925 00). Go to 9.**
- 9. Repair/replace wiring between P137-FF and P386-C (WP 1747 00).**
 - a. If trouble remains, replace pilots flight instruments 5v/115v transformer (WP 0881 00). Go to **10**.
- 10. Procedure completed.**

COPILOT'S PDU FACEPLATE LIGHTING INOPERATIVE**SYMPTOM**

Copilot's PDU faceplate lighting inoperative.

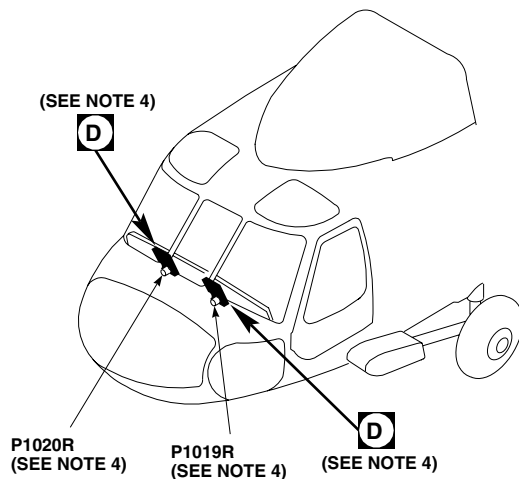
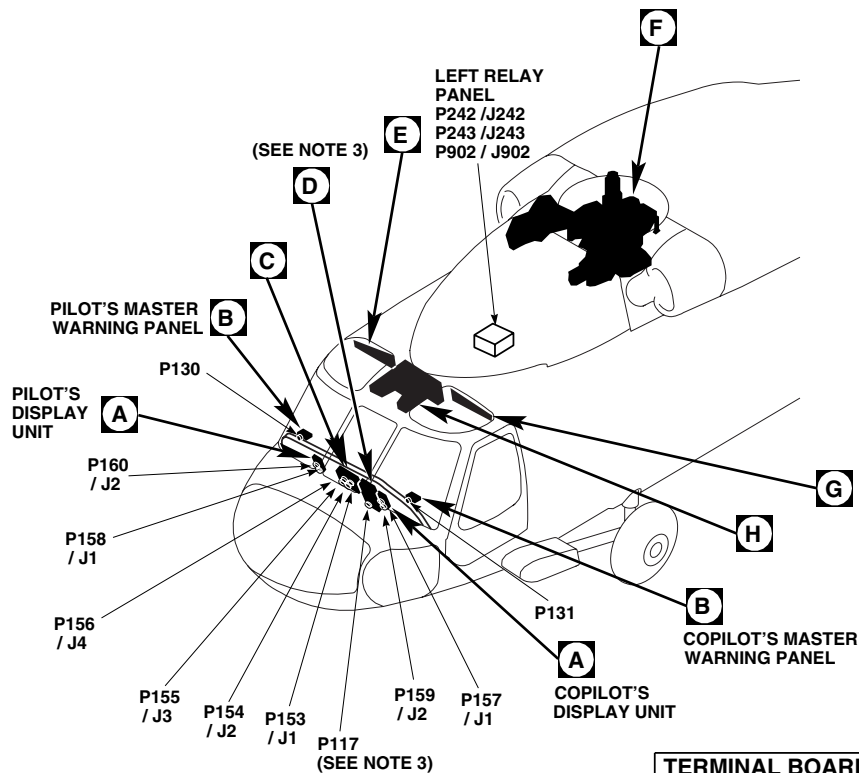
MALFUNCTION

Copilot's PDU faceplate lighting inoperative.

CORRECTIVE ACTION

- 1. Interchange copilot's and pilot's PDU, and check that faceplate on copilot's PDU lights.**
 - a. If faceplate lights, replace malfunctioning PDU faceplate (WP 0783 00). Go to **9**.
 - b. If faceplate does not light, go to **2**.
- 2. Check for 115 vac between P157-99 and P157-100.**
 - a. If voltage is as specified, replace PDU (WP 0782 00). Go to **9**.
 - b. If voltage is not as specified, go to **3**.
- 3. Check continuity between:**
 - **P149-99 and P157-99**
 - **P149-100 and P157-100**
 - a. If continuity is present, go to **4**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9**.
- 4. Check helicopter configuration.**
 - a. **EMEP** , go to **5**.
 - b. **W/O EMEP** , go to **7**.
- 5. EMEP Remove and check pin filtered adapter P138 (WP 0925 00).**
 - a. If pin filtered adapter is good, go to **6**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **9**.
- 6. EMEP Install pin filtered adapter (WP 0925 00). Go to 7.**
- 7. Check for 115 vac between P138-FF and ground.**
 - a. If voltage is as specified, replace No. 1 SDC (WP 0781 00). Go to **9**.
 - b. If voltage is not as specified, go to **8**.
- 8. Check for 5 vac between P387-A and ground.**
 - a. If voltage is as specified, repair/replace wiring between P138-FF and P387-C (WP 1747 00). Go to **9**.
 - b. If trouble remains, replace pilots flight instrument 5v/115v transformer (WP 0881 00). Go to **9**.
 - c. If voltage is not as specified, repair/replace wiring between P246-g and P387-A (WP 1747 00). Go to **9**.
- 9. Procedure completed.**

LOCATION



NOTES

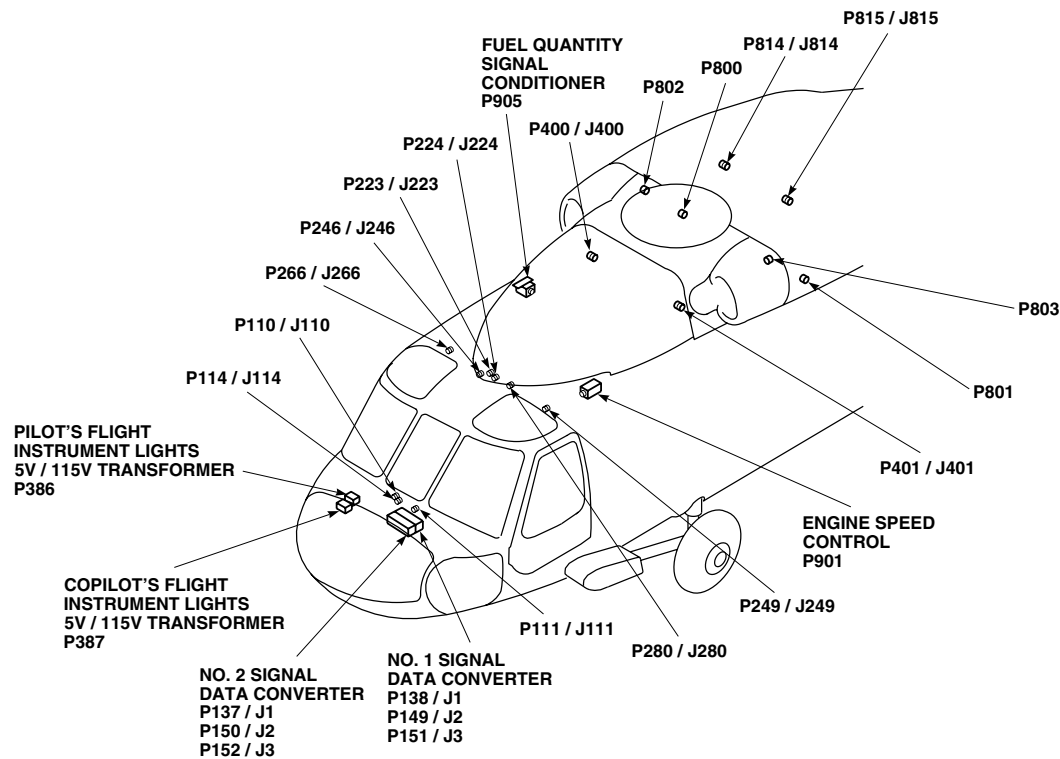
1. EH-60A
2. UH-60A UH-60L HH-60A
HH-60L
3. UH-60A UH-60L EH-60A
4. HH-60L HH-60A

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P110 / J110	COCKPIT BL 7.5 RH, STA 197
P111 / J111	COCKPIT BL 7.5 LH, STA 197
P114 / J114	COCKPIT BL 8 RH, STA 200
P117 / J117	CAUTION / ADVISORY PANEL (SEE NOTE 3)
P130	PILOT'S MASTER WARNING PANEL
P131	COPILOT'S MASTER WARNING PANEL
P137 / J1	NO. 2 SIGNAL DATA CONVERTER
P138 / J1	NO. 1 SIGNAL DATA CONVERTER
P149 / J2	NO. 1 SIGNAL DATA CONVERTER
P150 / J2	NO. 2 SIGNAL DATA CONVERTER
P151 / J3	NO. 1 SIGNAL DATA CONVERTER
P152 / J3	NO. 2 SIGNAL DATA CONVERTER
P153 / J1	CENTRAL DISPLAY UNIT
P154 / J2	CENTRAL DISPLAY UNIT
P1019R / J3	COPILOT MULTIFUNCTION DISPLAY (SEE NOTE 4)
P1020R / J3	PILOT MULTIFUNCTION DISPLAY (SEE NOTE 4)

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Figure 1. Instrument Display System Location Diagram. (Sheet 1 of 6)

LOCATION - Continued



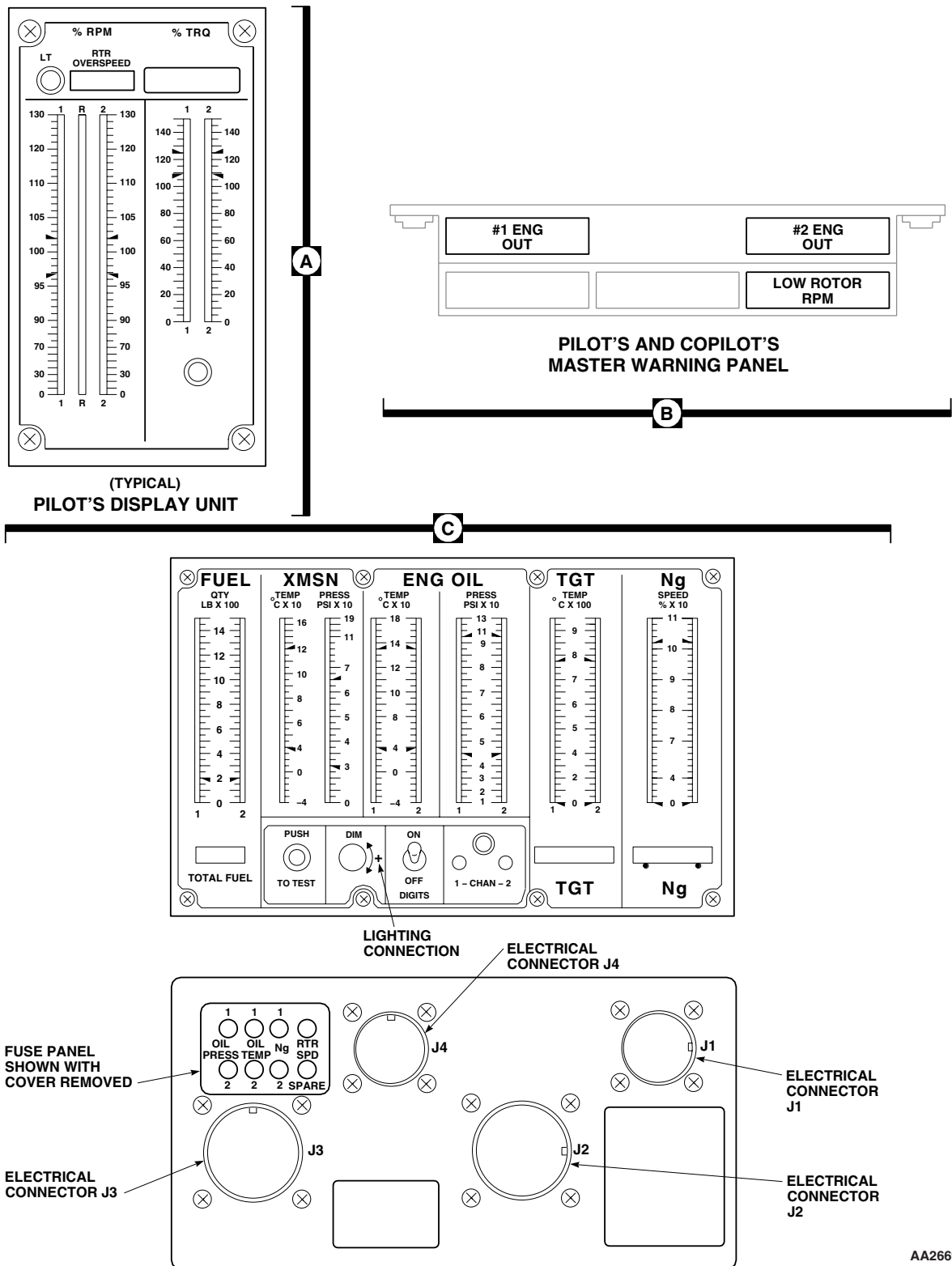
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P155 / J3	CENTRAL DISPLAY UNIT
P156 / J4	CENTRAL DISPLAY UNIT
P157 / J1	COPILOT'S DISPLAY UNIT
P158 / J1	PILOT'S DISPLAY UNIT
P159 / J2	COPILOT'S DISPLAY UNIT
P160 / J2	PILOT'S DISPLAY UNIT
P223 / J223	COCKPIT CEILING BL 3 LH, STA 243
P224 / J224	COCKPIT CEILING BL 3 RH, STA 245
P242 / J242	LEFT RELAY PANEL
P243 / J243	LEFT RELAY PANEL
P246 / J246	CABIN CEILING BL 5 RH, STA 247
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH, STA 247
P266 / J266	BEHIND PILOT'S CIRCUIT BREAKER PANEL BL 25 RH, STA 247
P280 / J280	CABIN CEILING BL 6 LH, STA 247
P386	PILOT'S FLIGHT INSTRUMENT LIGHTS 5V / 115V TRANSFORMER

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P387	COPILOT'S FLIGHT INSTRUMENT LIGHTS 5V / 115V TRANSFORMER
P400 / J400	CABIN CEILING BL 24 RH, STA 318
P401 / J401	CABIN CEILING BL 24 LH, STA 318
P412	MAIN XMSN OIL TEMPERATURE SENSOR
P443	MAIN ROTOR SPEED SENSOR
P444	MAIN XMSN OIL PRESSURE SENSOR
P800	NO. 2 ENGINE
P801	NO. 1 ENGINE
P802	NO. 2 ENGINE
P803	NO. 1 ENGINE
P814 / J814	ENGINE FIREWALL BL 14 RH, STA 373
P815 / J815	ENGINE FIREWALL BL 14 LH, STA 373
P901	ENGINE SPEED CONTROL
P902 / J902	LEFT RELAY PANEL
P905	FUEL QUANTITY SIGNAL CONDITIONER

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Figure 1. Instrument Display System Location Diagram. (Sheet 2 of 6)

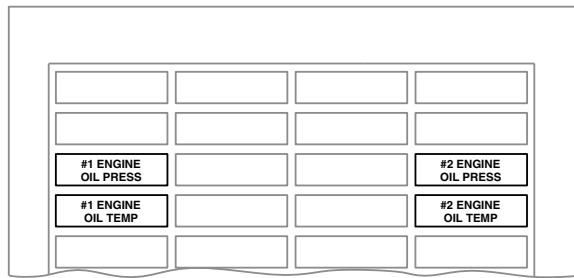
LOCATION - Continued



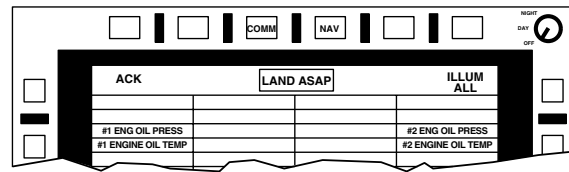
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Figure 1. Instrument Display System Location Diagram. (Sheet 3 of 6)

LOCATION - Continued



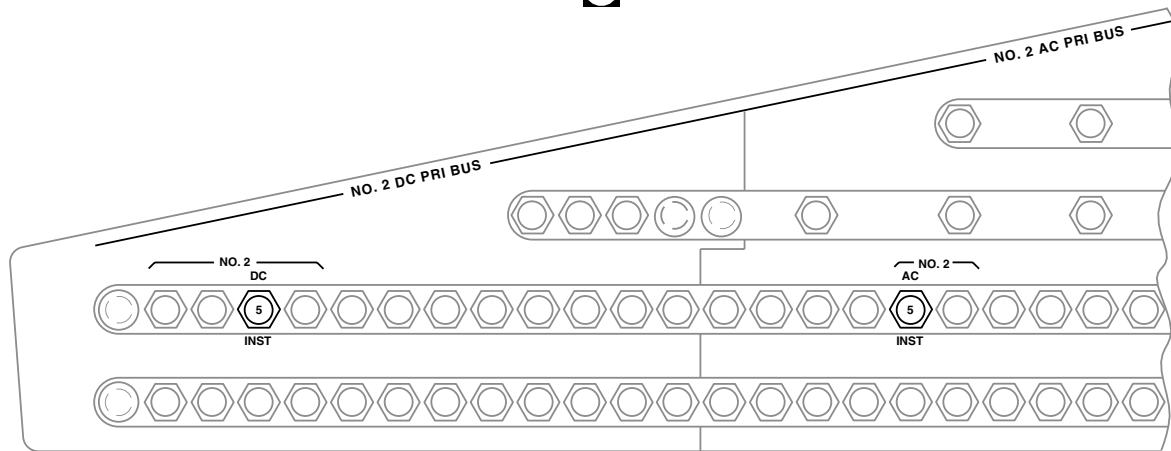
CAUTION/ADVISORY PANEL
(SEE NOTE 3)



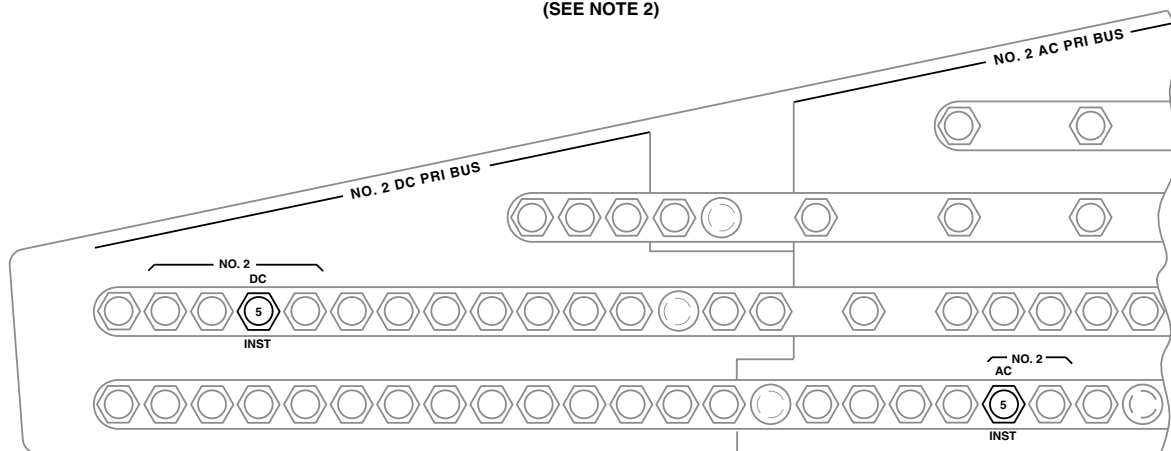
**MULTIFUNCTION DISPLAY'S
CAUTION/ADVISORY GRID**
(SEE NOTE 4)

D

E



(SEE NOTE 2)



(SEE NOTE 1)

PILOT'S CIRCUIT BREAKER PANEL

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Figure 1. Instrument Display System Location Diagram. (Sheet 4 of 6)

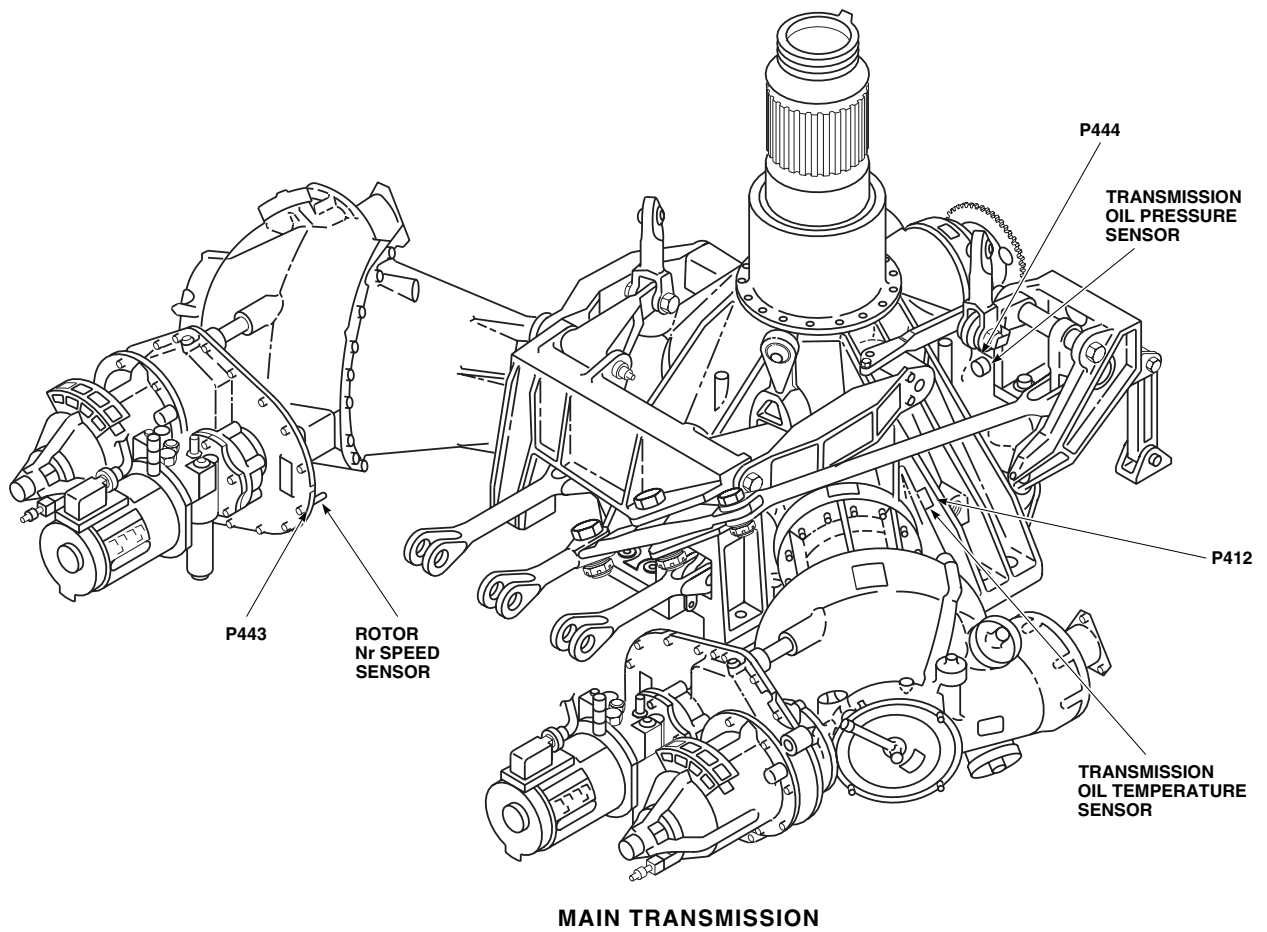
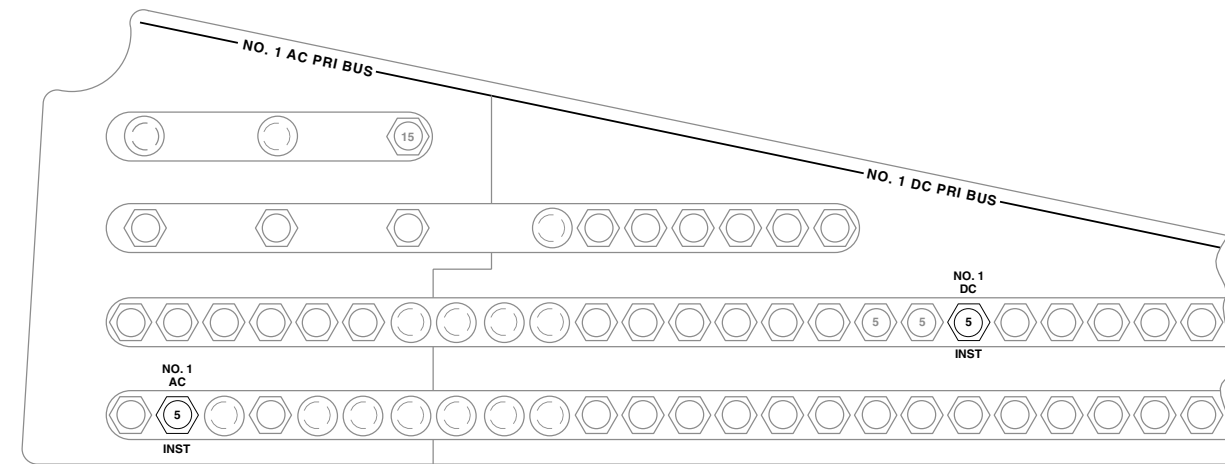
LOCATION - Continued**F**AA2669_5A
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Figure 1. Instrument Display System Location Diagram. (Sheet 5 of 6)

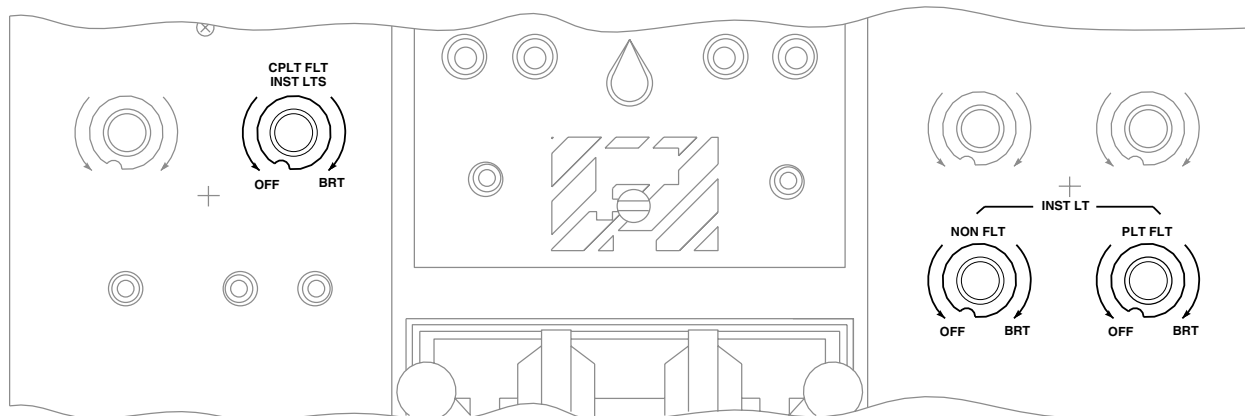
LOCATION - Continued



COPILOT'S CIRCUIT BREAKER PANEL

G

H



UPPER CONSOLE

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Figure 1. Instrument Display System Location Diagram. (Sheet 6 of 6)

SCHEMATICS

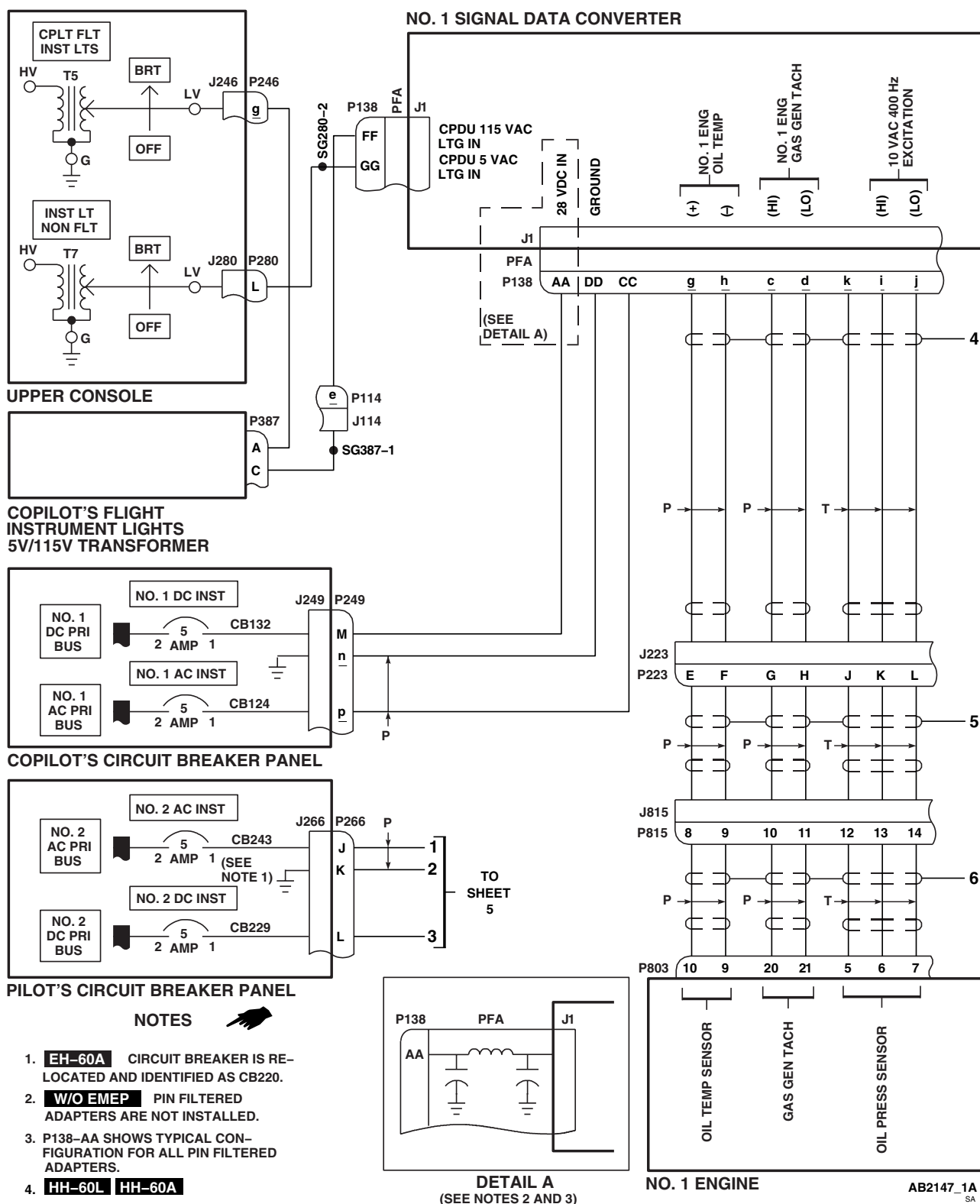


Figure 2. Instrument Display System Schematic Diagram. (Sheet 1 of 22)

SCHEMATICS - Continued

NO. 1 SIGNAL DATA CONVERTER

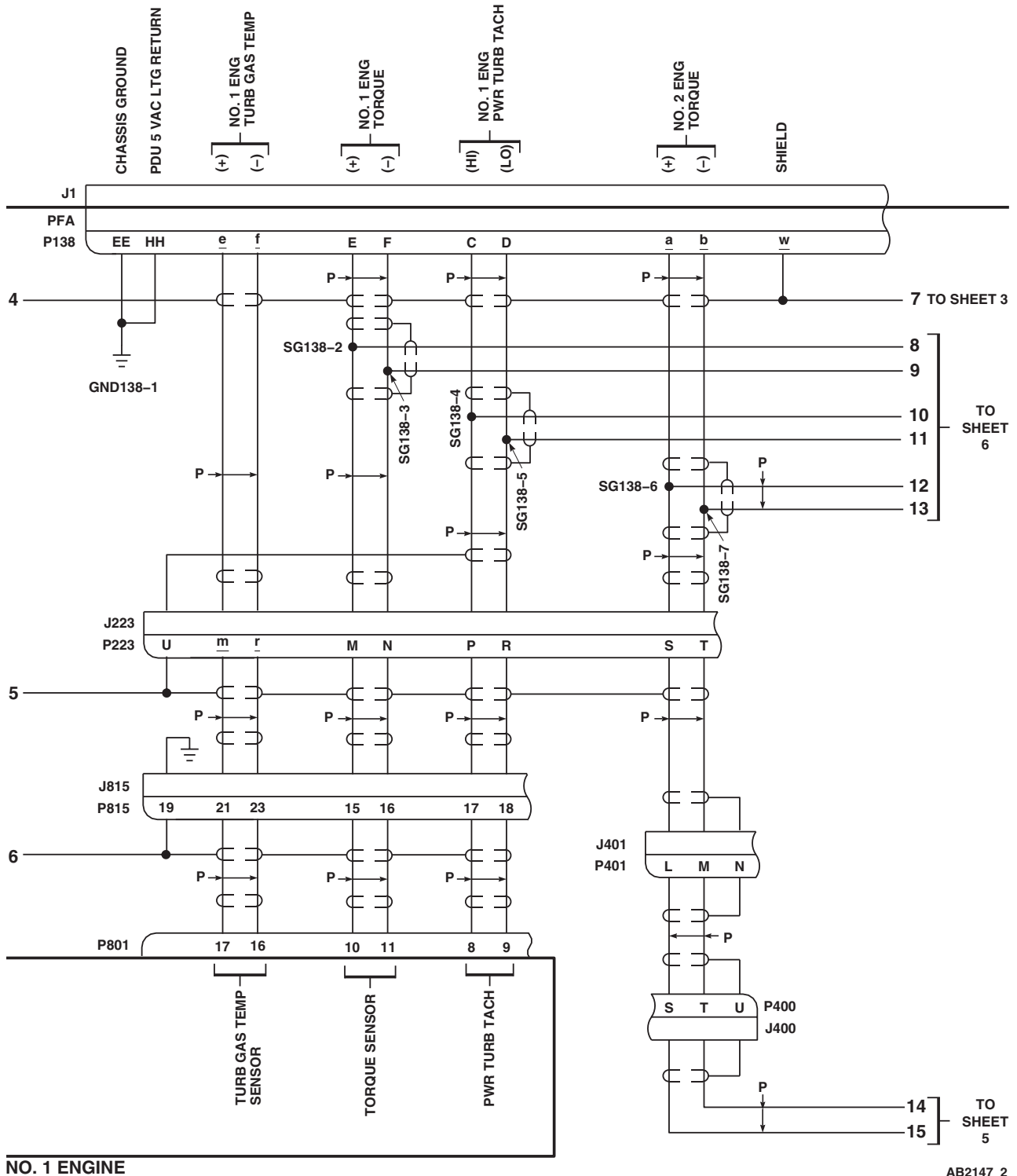
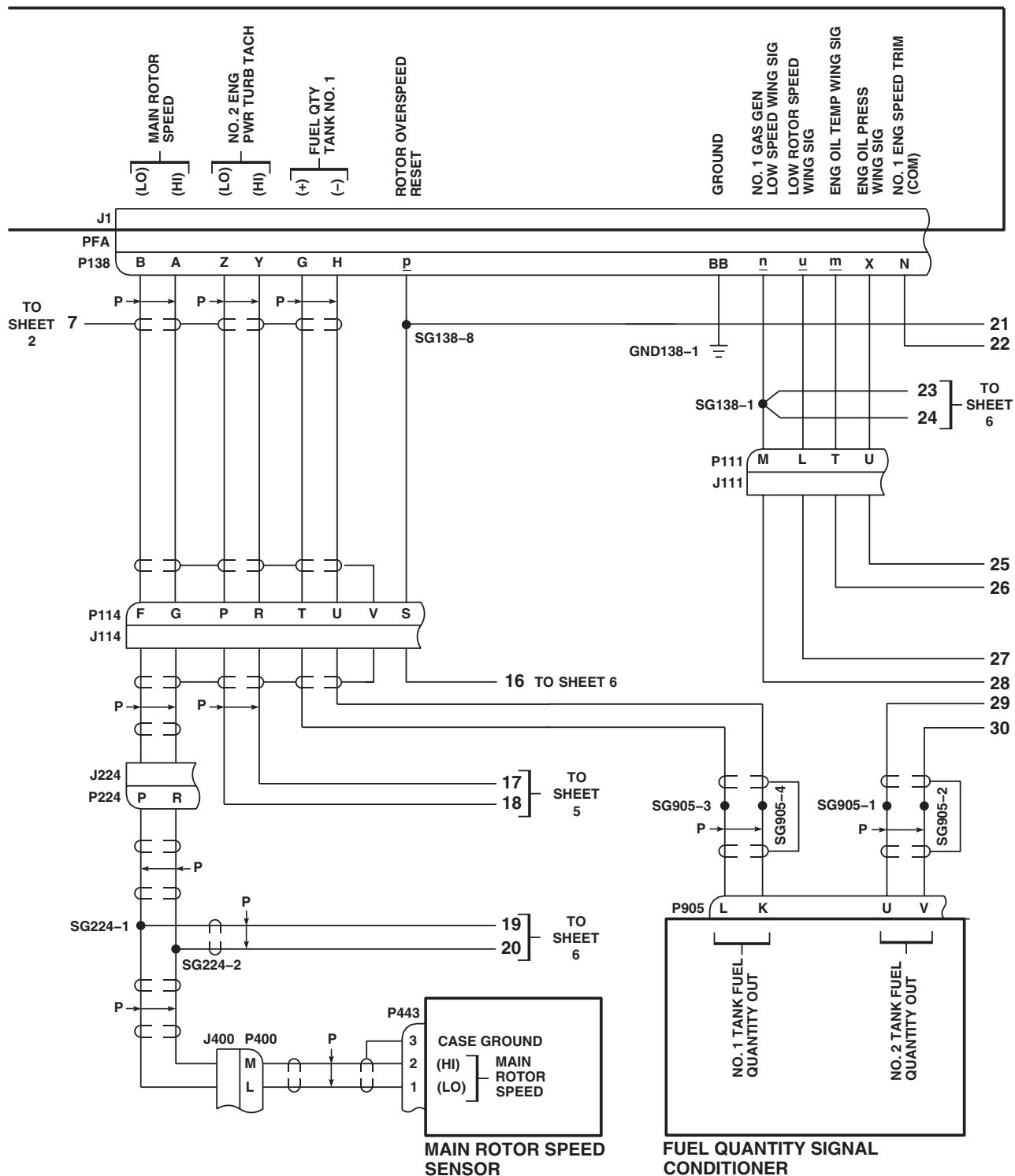


Figure 2. Instrument Display System Schematic Diagram. (Sheet 2 of 22)

SCHEMATICS - Continued

NO. 1 SIGNAL DATA CONVERTER



AB2147_3
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Figure 2. Instrument Display System Schematic Diagram. (Sheet 3 of 22)

SCHEMATICS - Continued

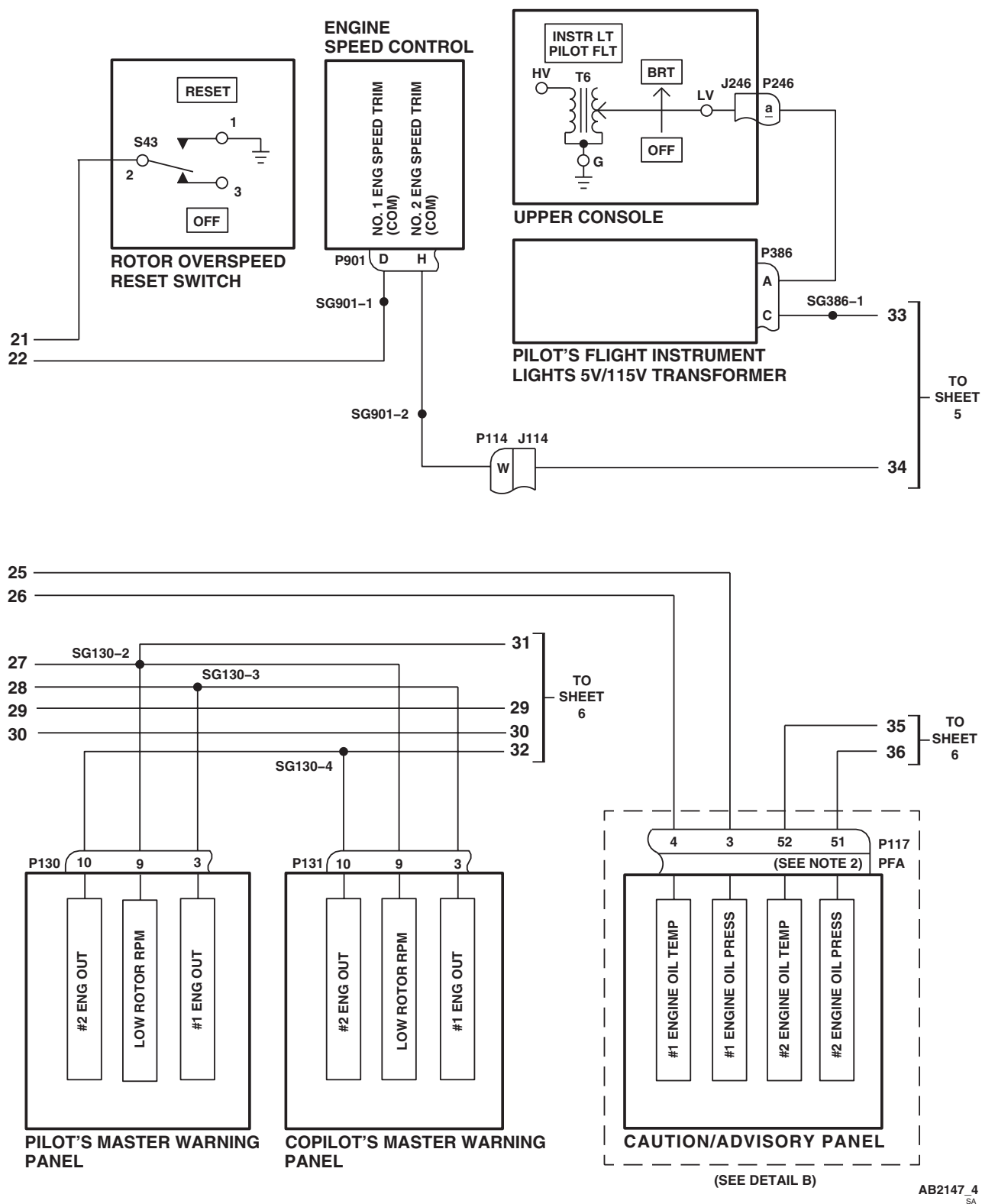


Figure 2. Instrument Display System Schematic Diagram. (Sheet 4 of 22)

SCHEMATICS - Continued

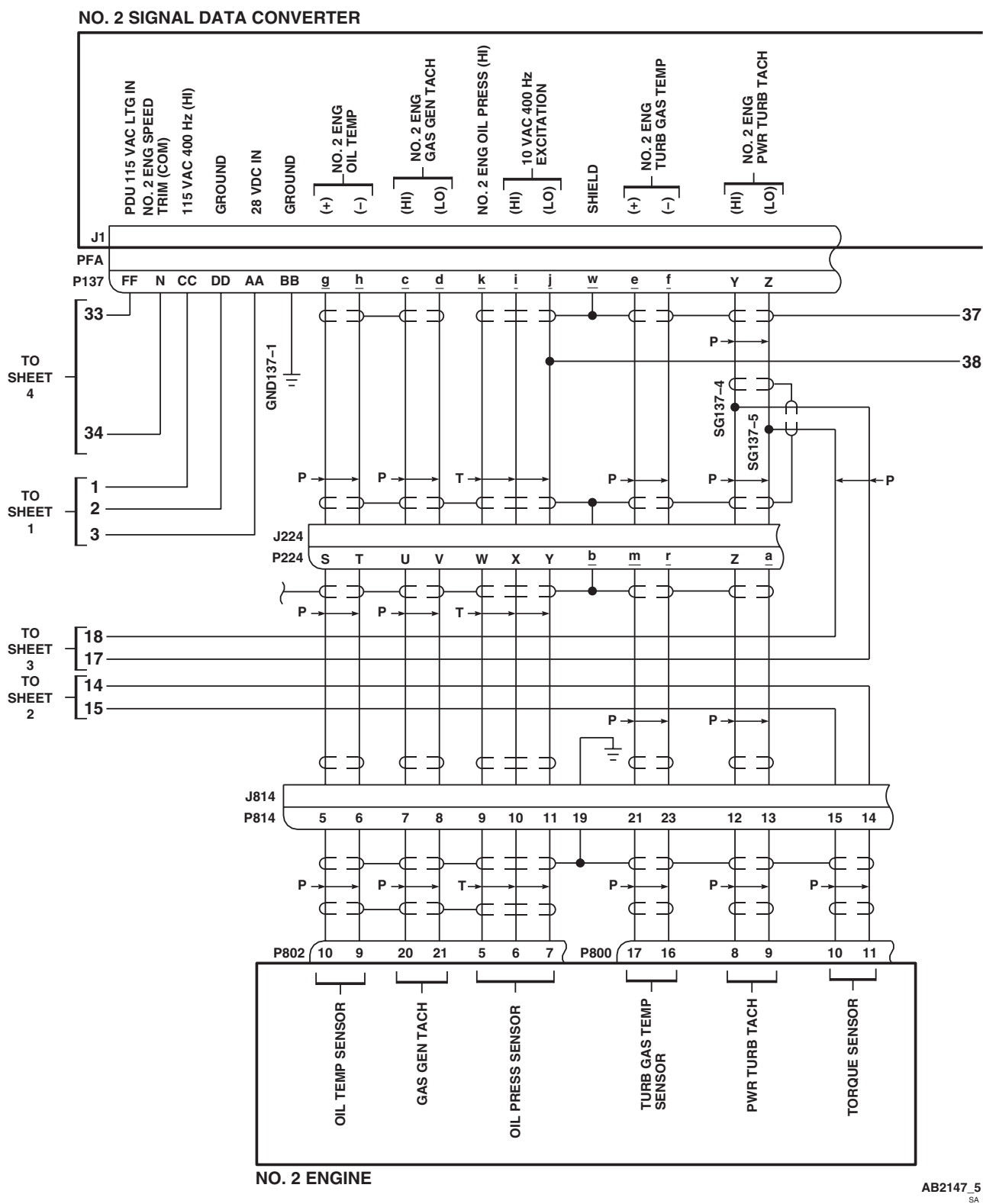


Figure 2. Instrument Display System Schematic Diagram. (Sheet 5 of 22)

SCHEMATICS - Continued

NO. 2 SIGNAL DATA CONVERTER

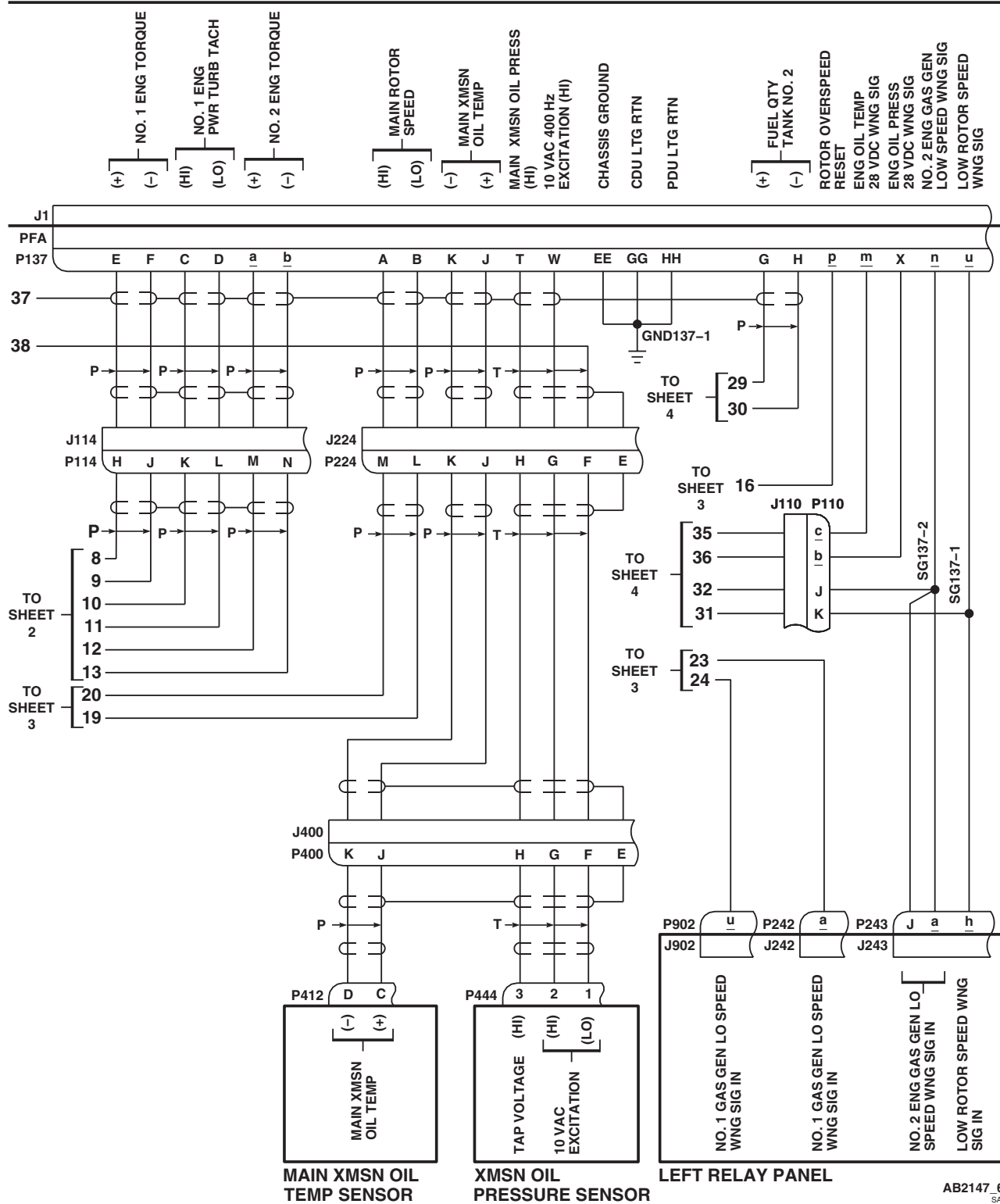


Figure 2. Instrument Display System Schematic Diagram. (Sheet 6 of 22)

SCHEMATICS - Continued

NO. 1 SIGNAL DATA CONVERTER

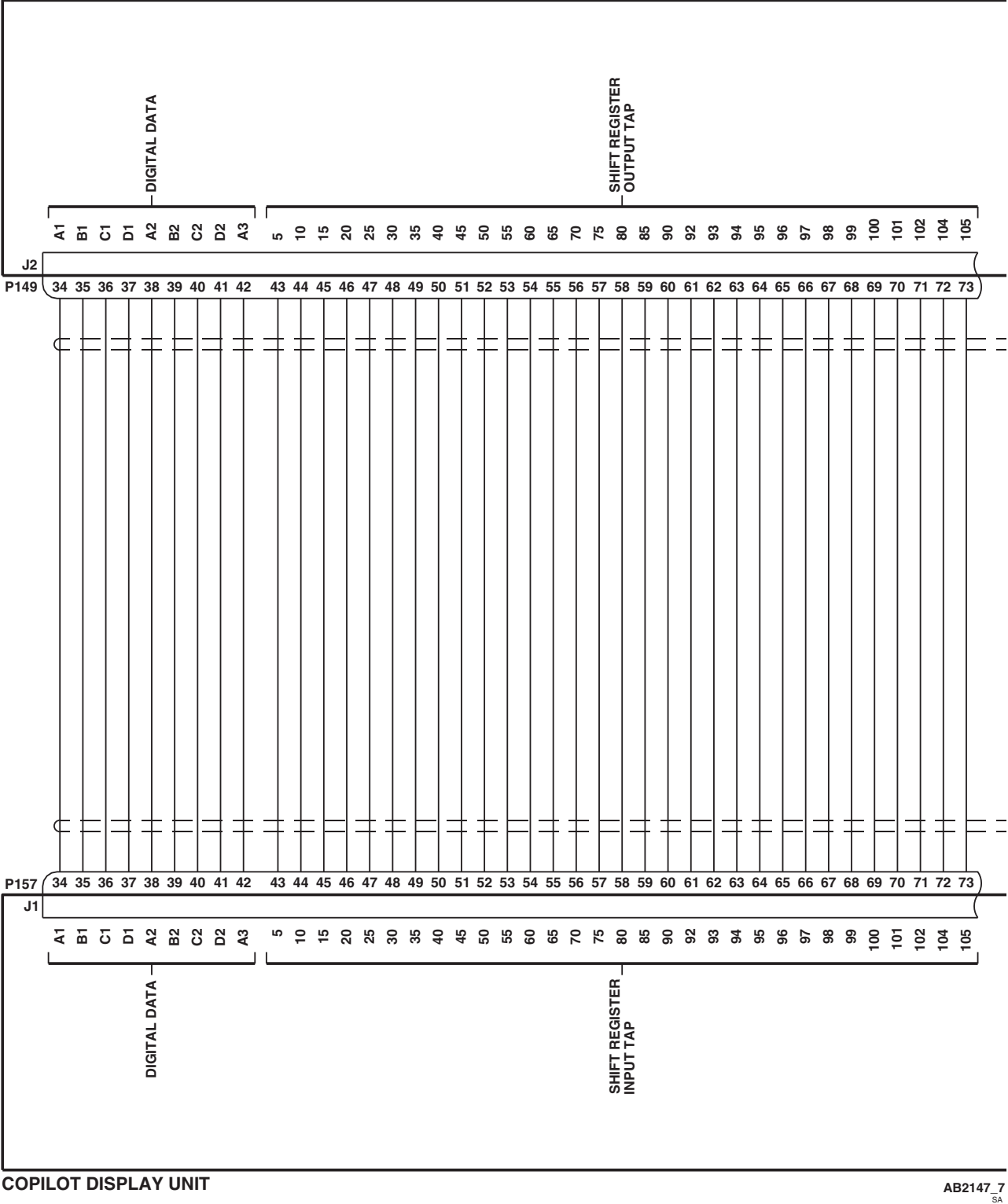
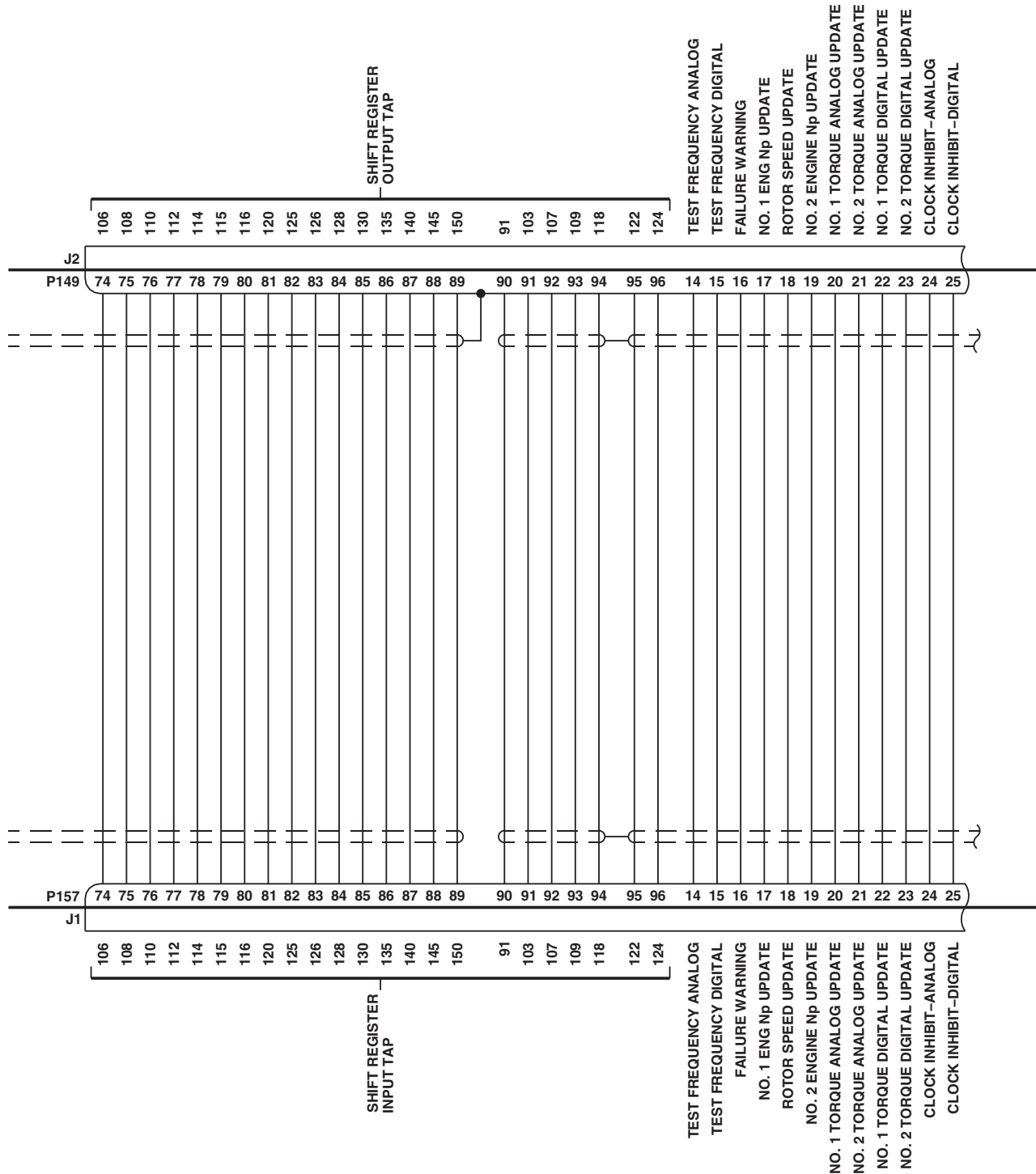


Figure 2. Instrument Display System Schematic Diagram. (Sheet 7 of 22)

SCHEMATICS - Continued

NO. 1 SIGNAL DATA CONVERTER



COPILLOT DISPLAY UNIT

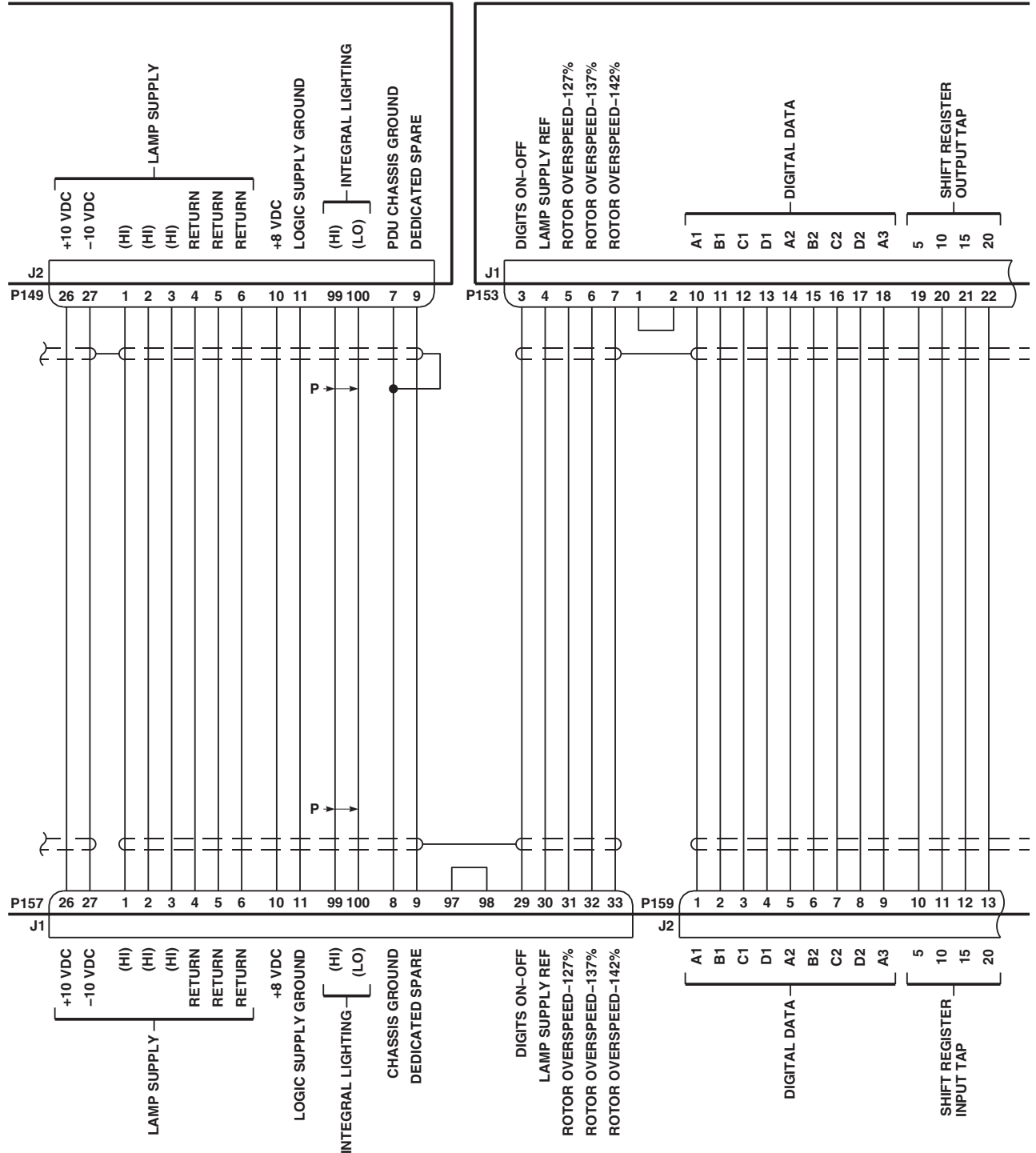
AB2147 8
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Figure 2. Instrument Display System Schematic Diagram. (Sheet 8 of 22)

SCHEMATICS - Continued

NO. 1 SIGNAL DATA CONVERTER

CENTRAL DISPLAY UNIT



COPILOT DISPLAY UNIT

AB2147_9
SA

Figure 2. Instrument Display System Schematic Diagram. (Sheet 9 of 22)

SCHEMATICS - Continued

CENTRAL DISPLAY UNIT

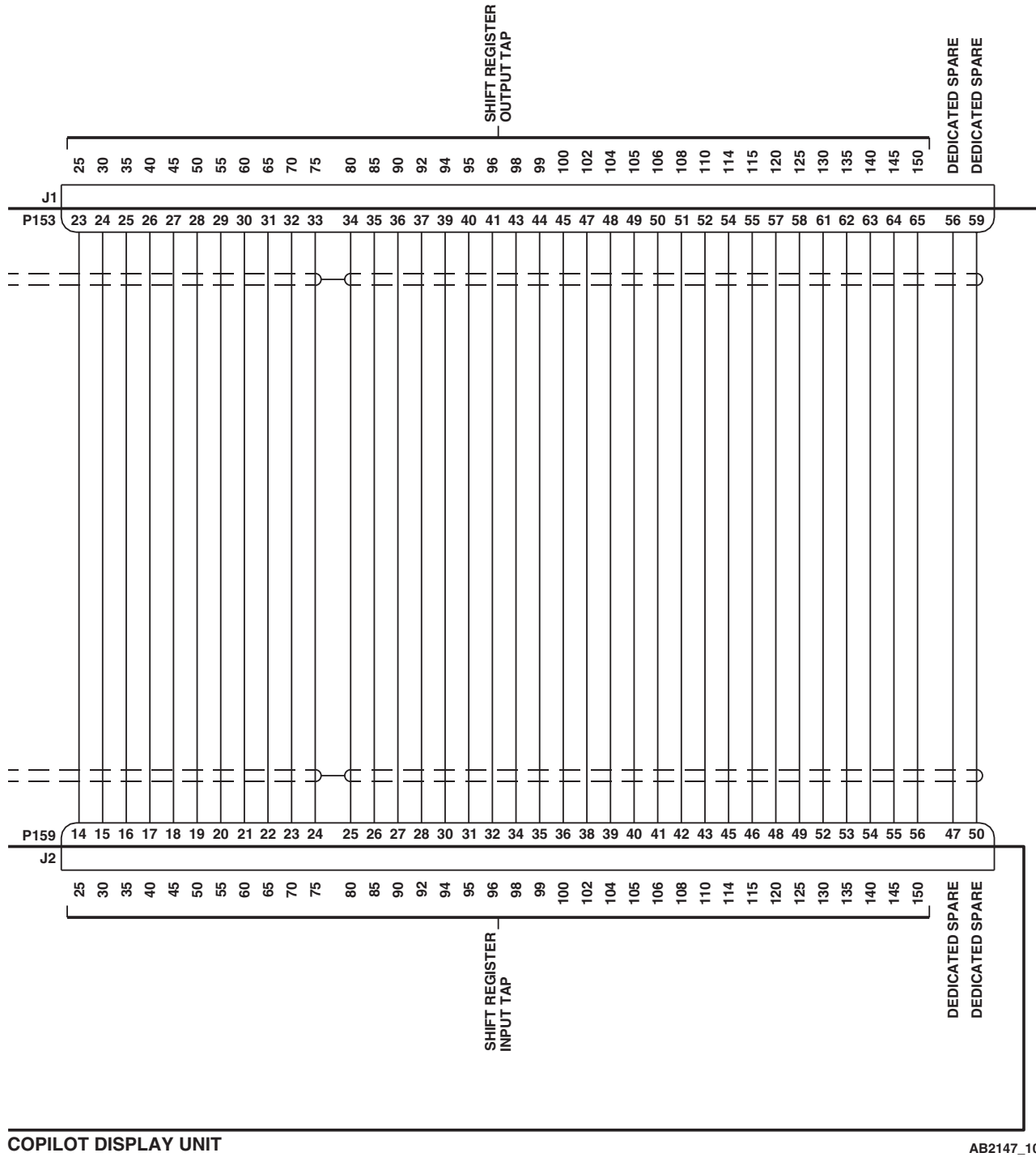


Figure 2. Instrument Display System Schematic Diagram. (Sheet 10 of 22)

SCHEMATICS - Continued

CENTRAL DISPLAY UNIT

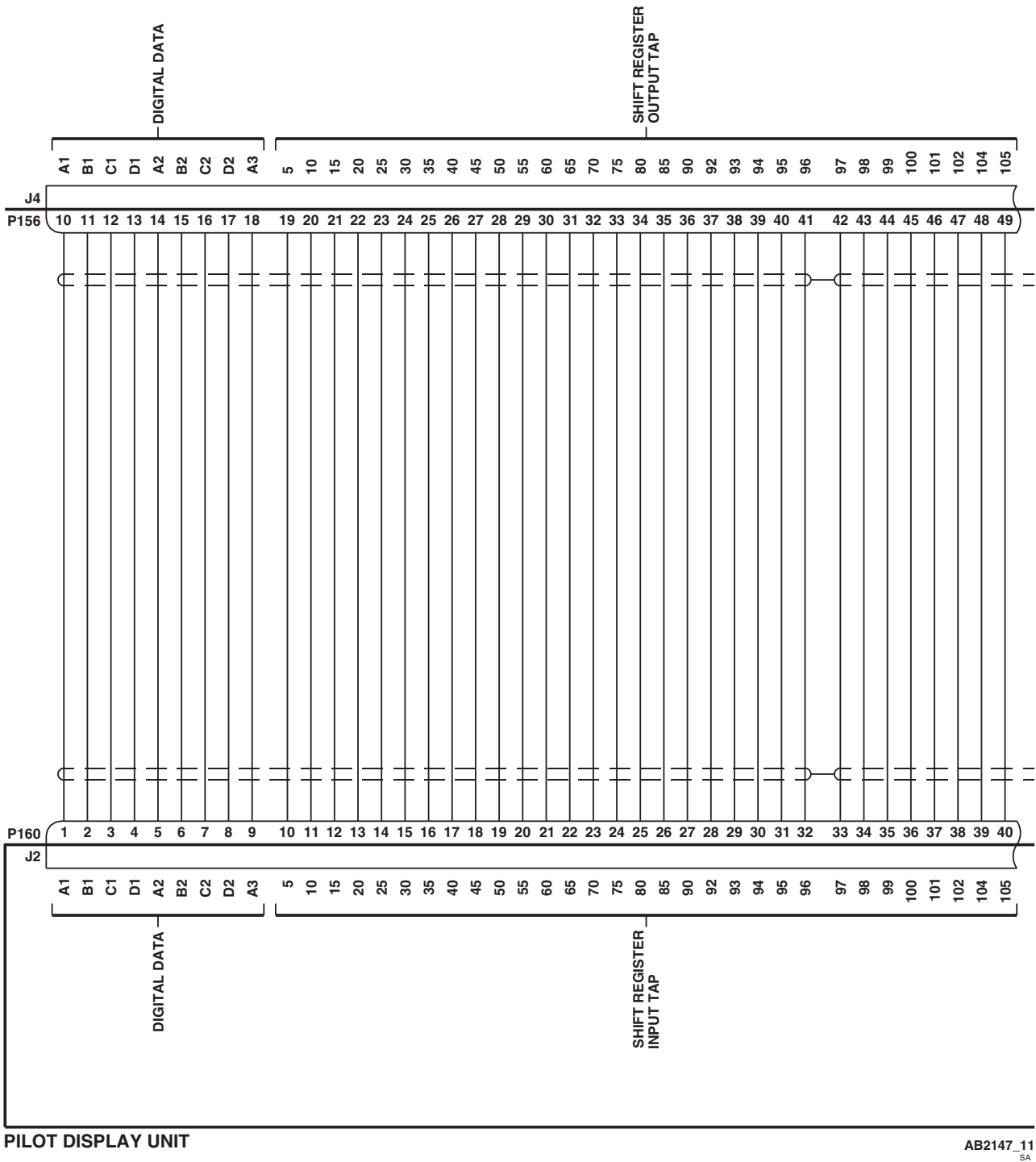
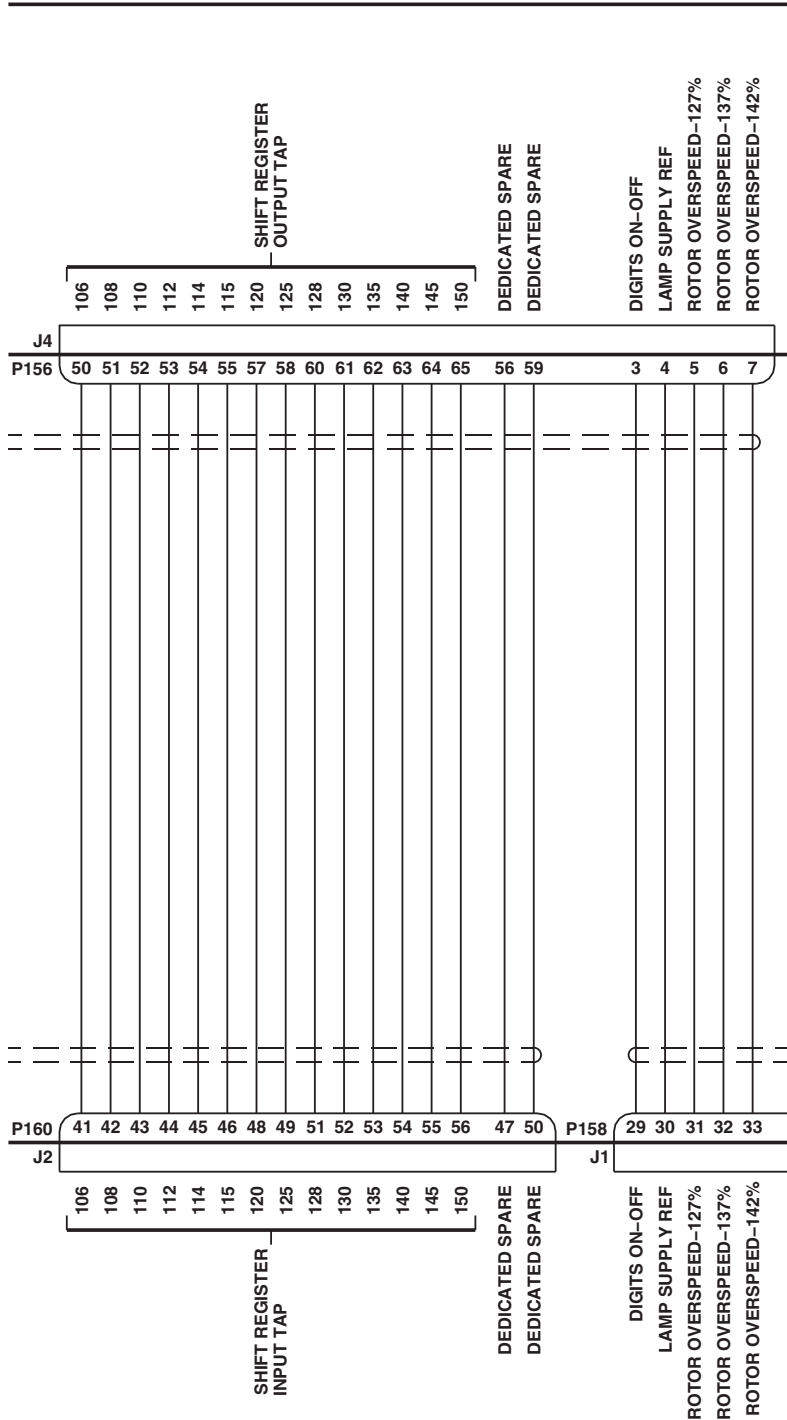


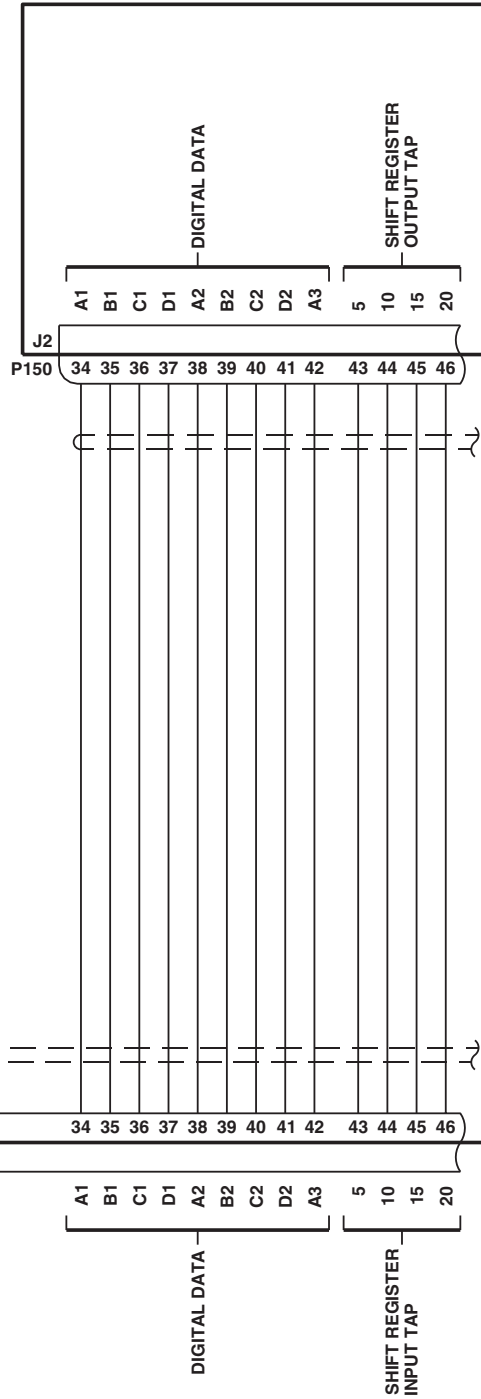
Figure 2. Instrument Display System Schematic Diagram. (Sheet 11 of 22)

SCHEMATICS - Continued

CENTRAL DISPLAY UNIT



NO. 2 SIGNAL DATA CONVERTER



PILOT DISPLAY UNIT

AB2147_12
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Figure 2. Instrument Display System Schematic Diagram. (Sheet 12 of 22)

SCHEMATICS - Continued

NO. 2 SIGNAL DATA CONVERTER

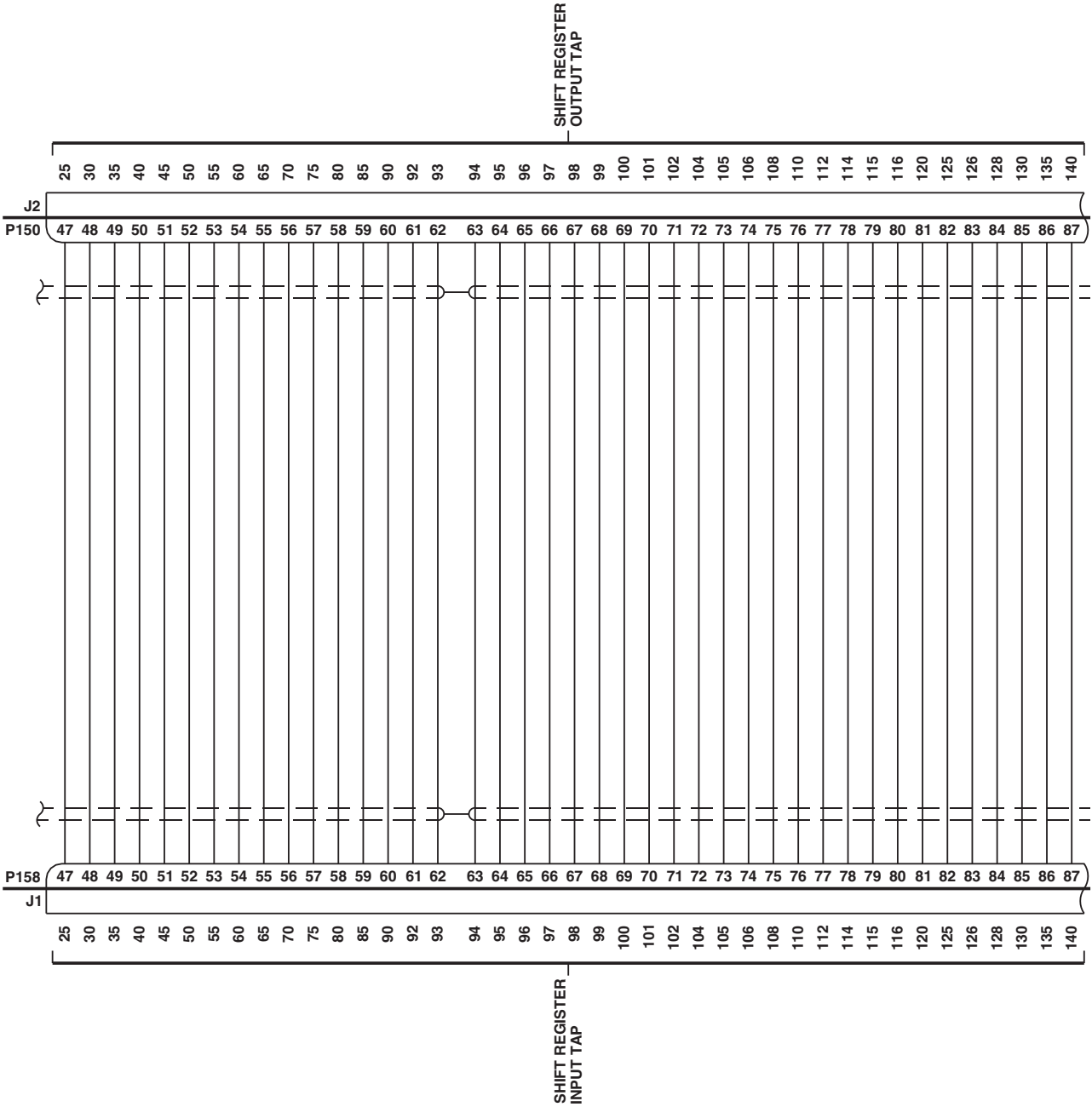


Figure 2. Instrument Display System Schematic Diagram. (Sheet 13 of 22)

SCHEMATICS - Continued

NO. 2 SIGNAL DATA CONVERTER

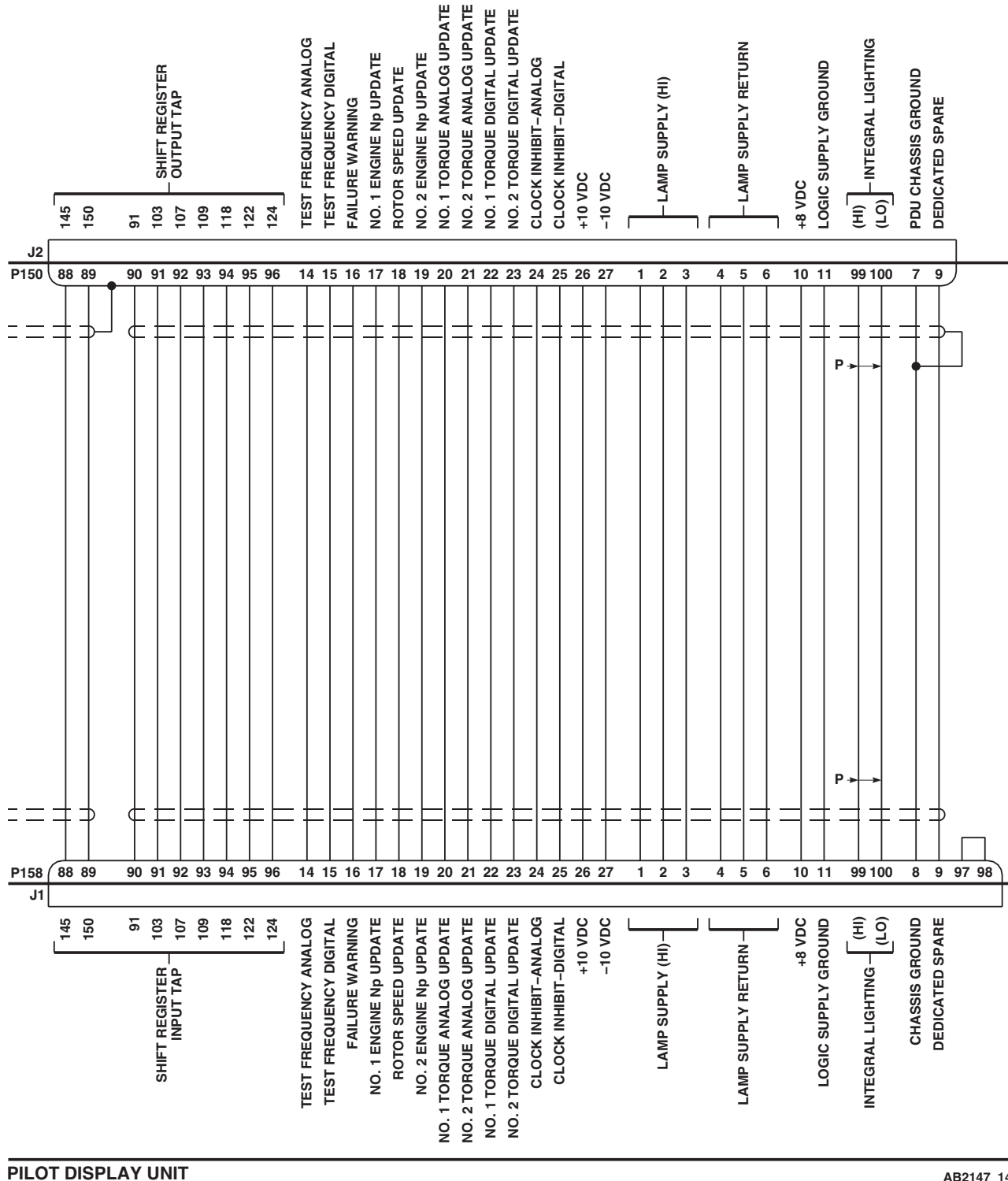


Figure 2. Instrument Display System Schematic Diagram. (Sheet 14 of 22)

SCHEMATICS - Continued

NO. 2 SIGNAL DATA CONVERTER

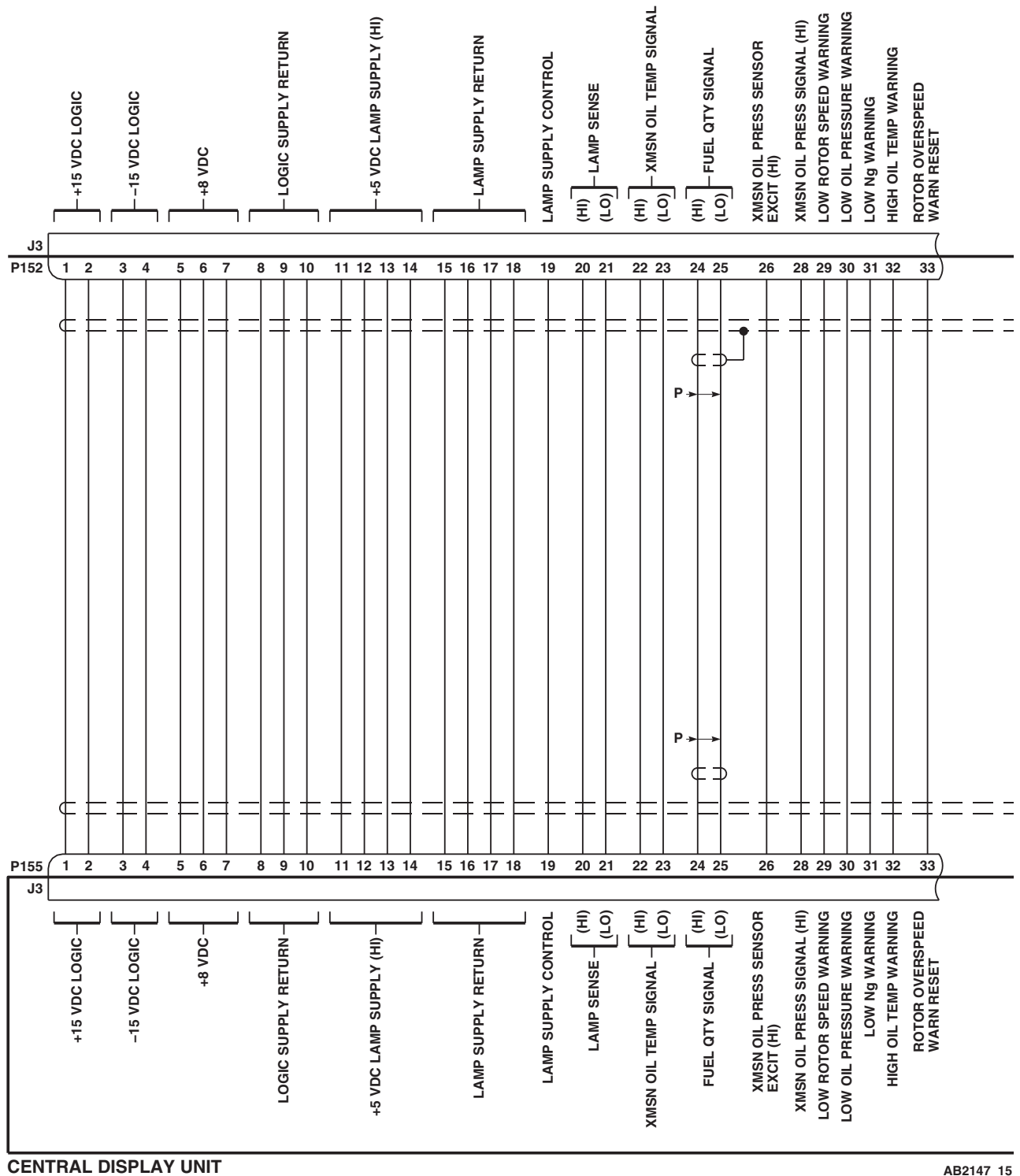
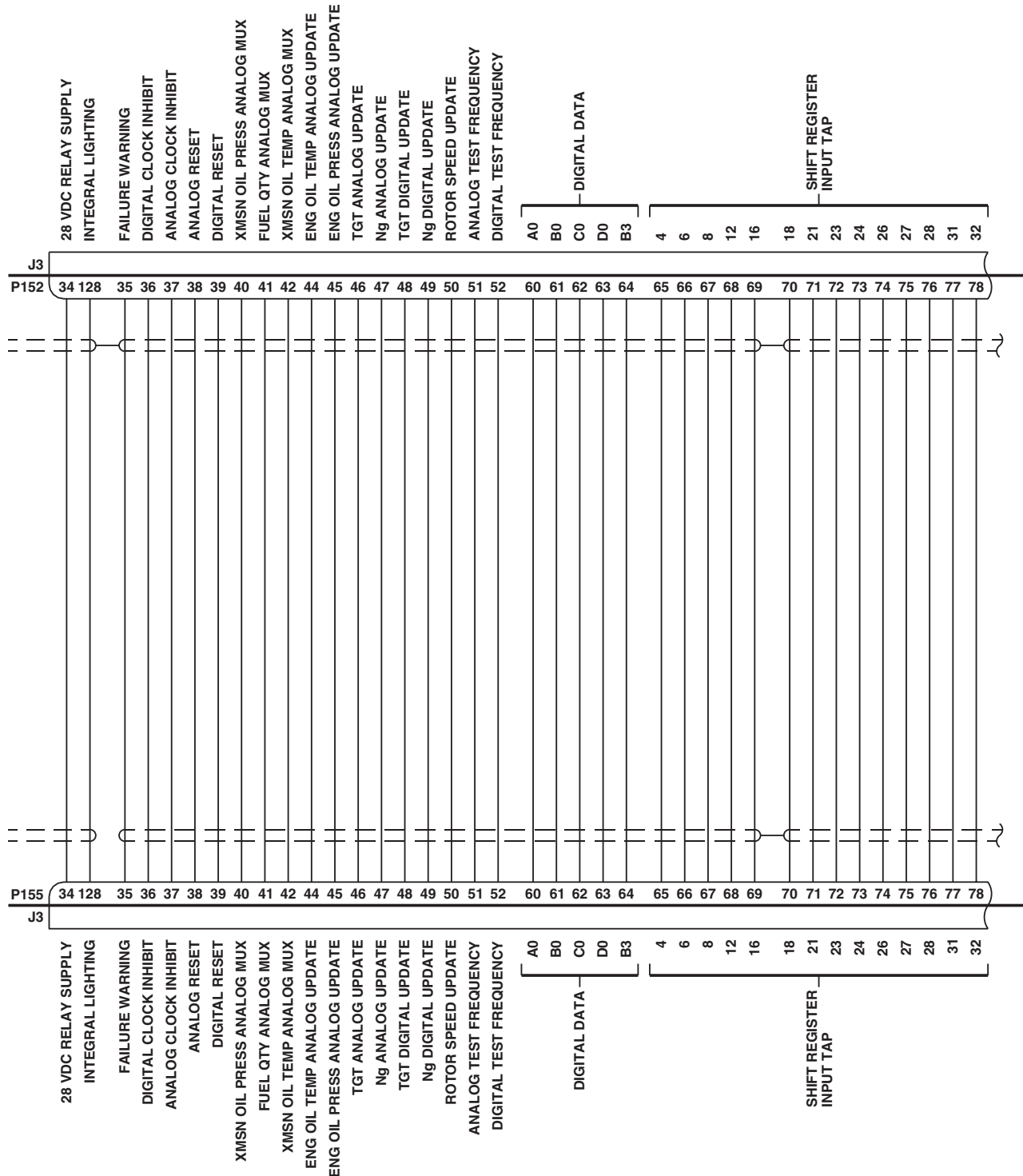


Figure 2. Instrument Display System Schematic Diagram. (Sheet 15 of 22)

SCHEMATICS - Continued

NO. 2 SIGNAL DATA CONVERTER



CENTRAL DISPLAY UNIT

AB2147_16
SA

Figure 2. Instrument Display System Schematic Diagram. (Sheet 16 of 22)

SCHEMATICS - Continued

NO. 2 SIGNAL DATA CONVERTER

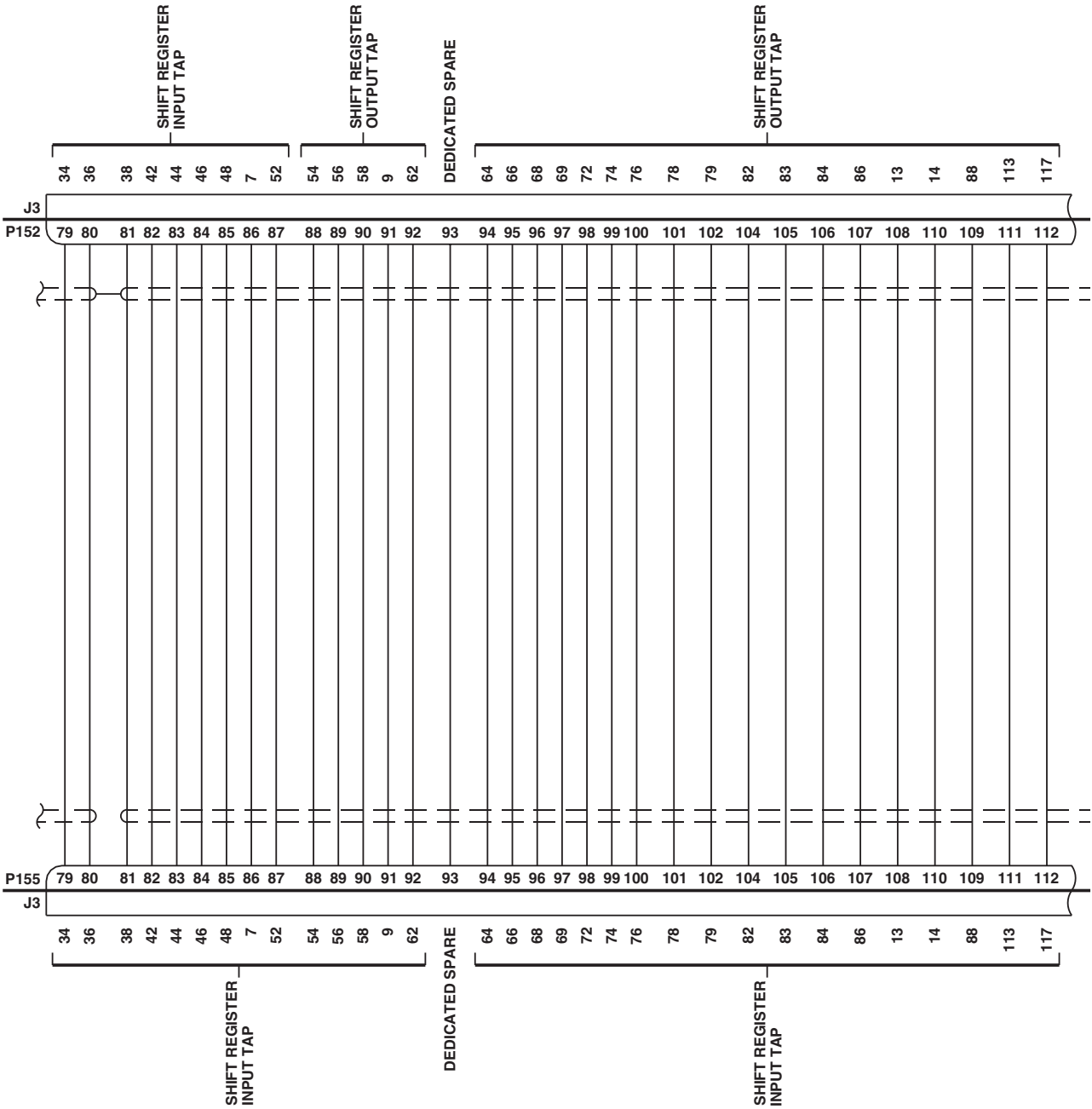
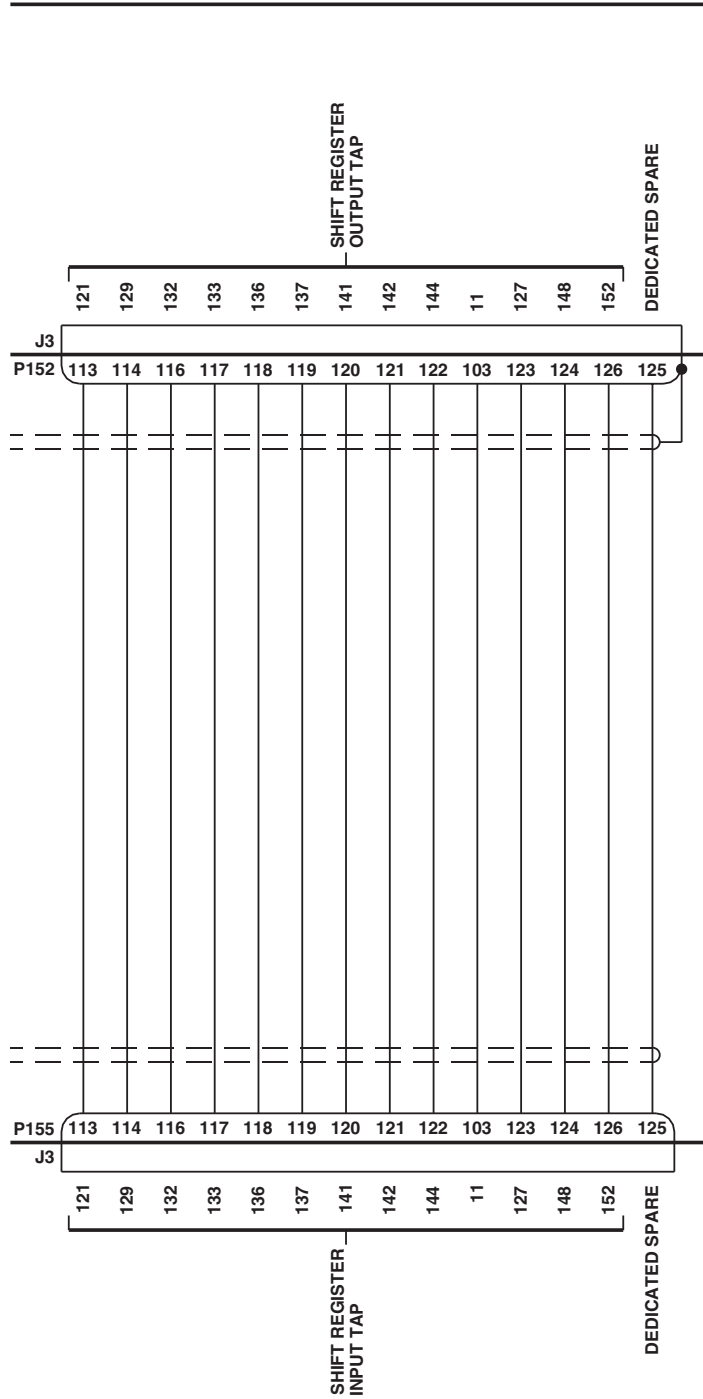


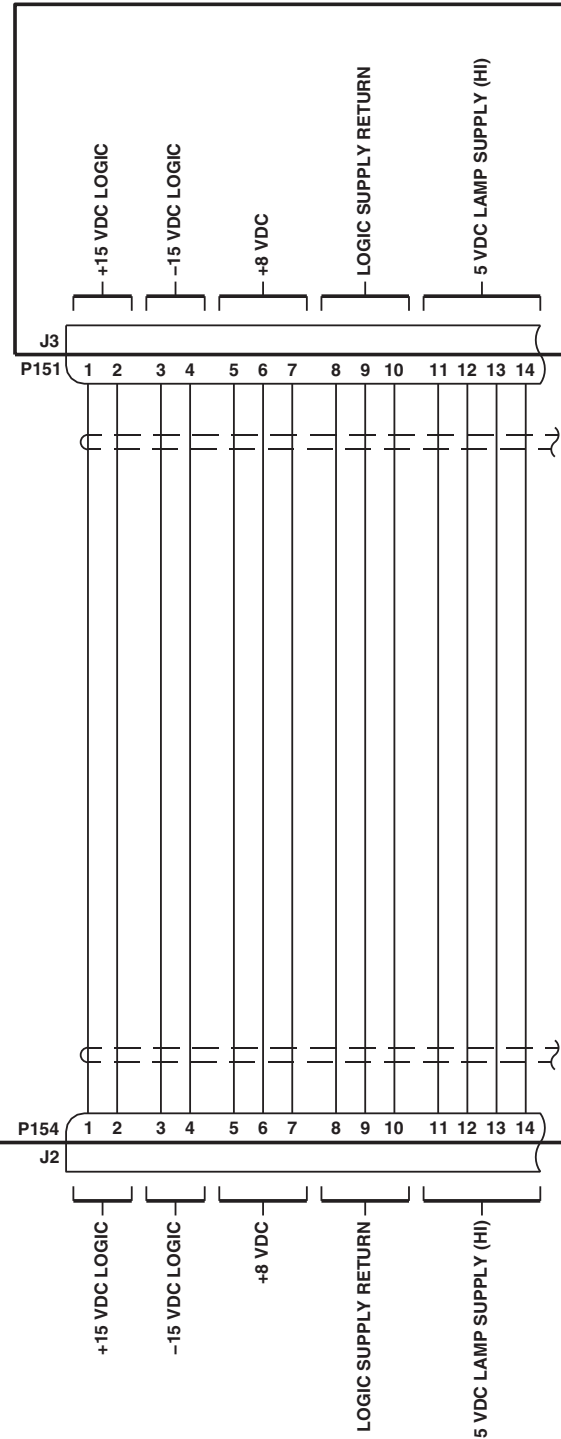
Figure 2. Instrument Display System Schematic Diagram. (Sheet 17 of 22)

SCHEMATICS - Continued

NO. 2 SIGNAL DATA CONVERTER



NO. 1 SIGNAL DATA CONVERTER



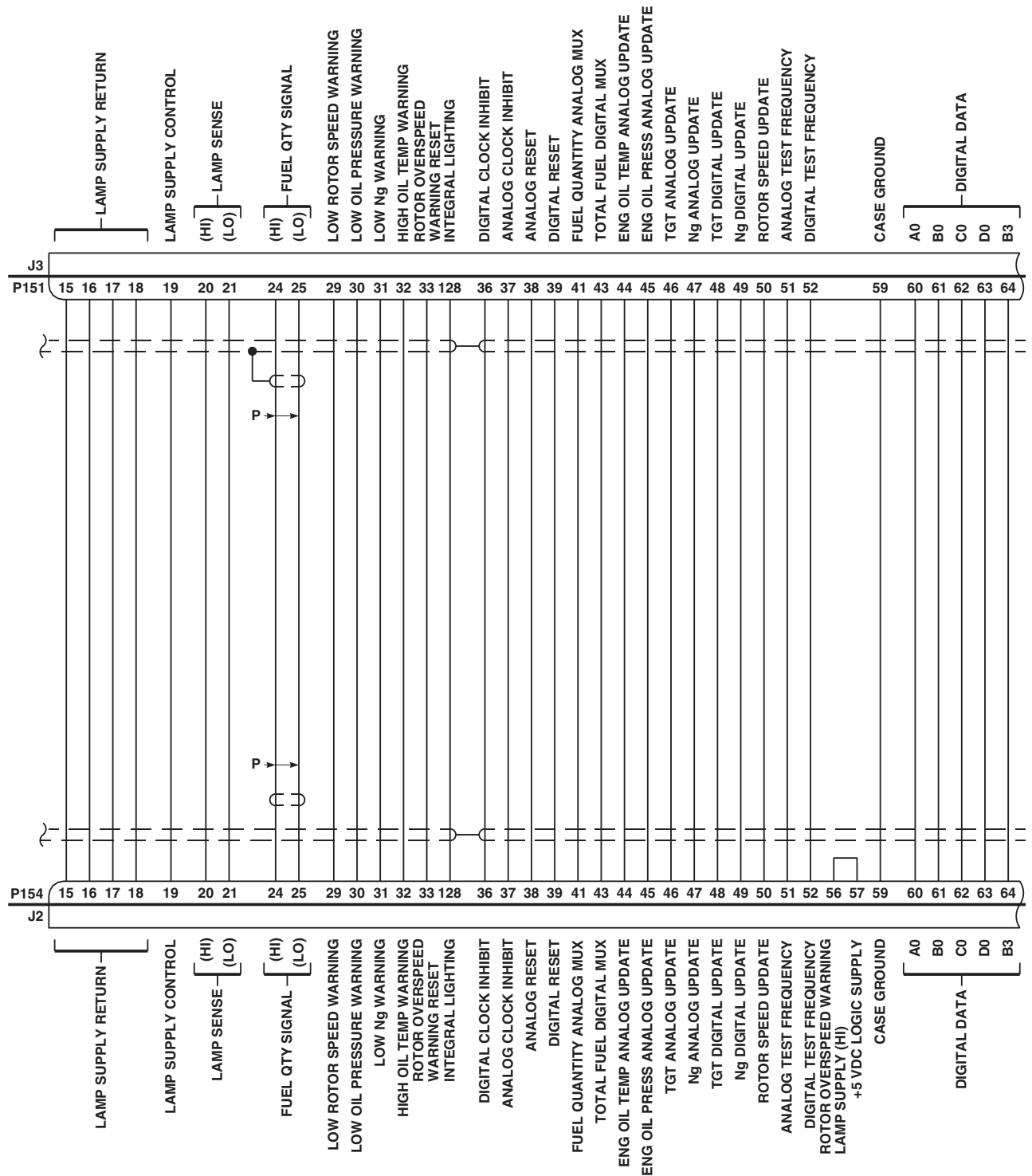
CENTRAL DISPLAY UNIT

AB2147_18
SA

Figure 2. Instrument Display System Schematic Diagram. (Sheet 18 of 22)

SCHEMATICS - Continued

NO. 1 SIGNAL DATA CONVERTER



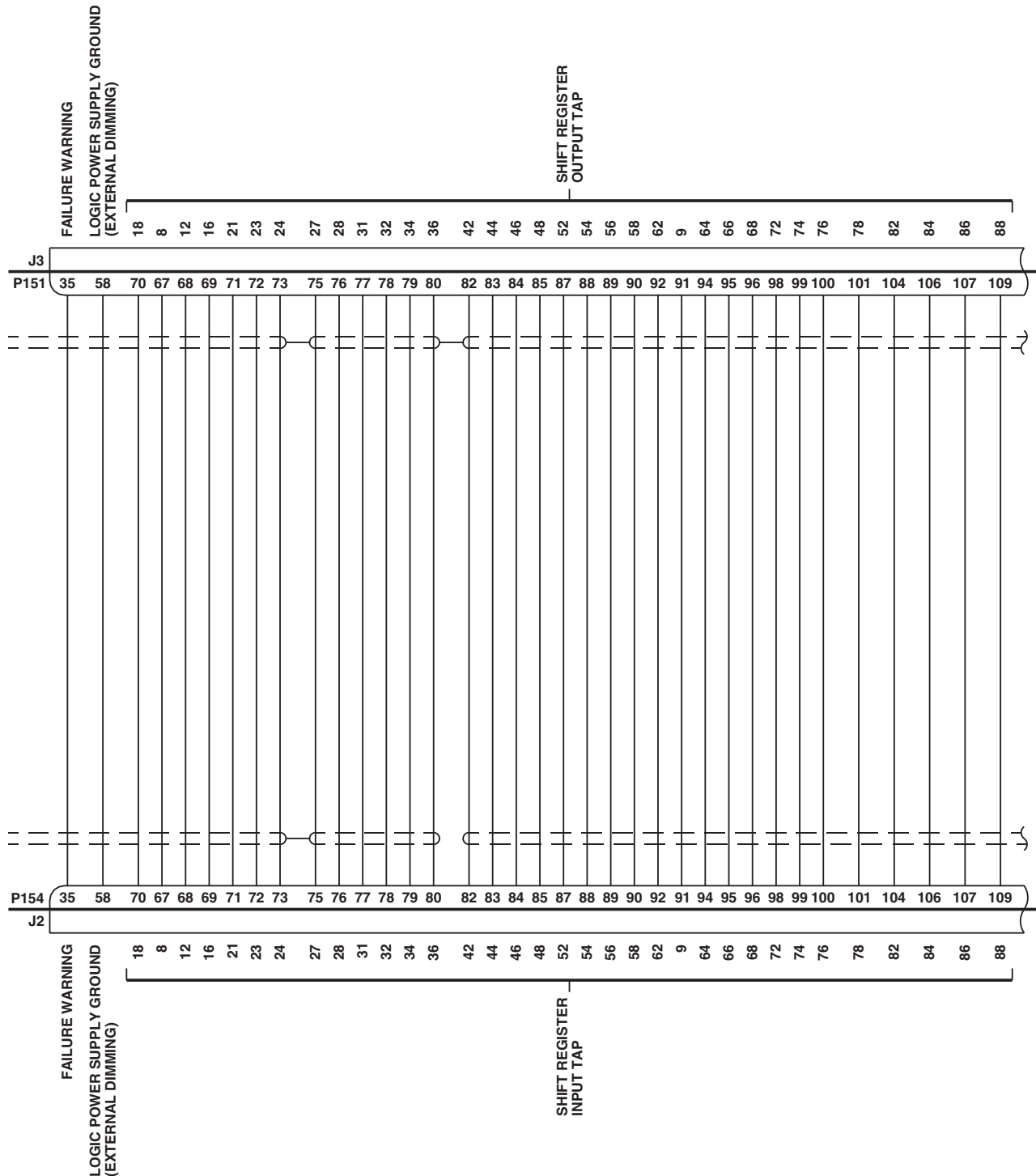
CENTRAL DISPLAY UNIT

AB2147_19
SA

Figure 2. Instrument Display System Schematic Diagram. (Sheet 19 of 22)

SCHEMATICS - Continued

NO. 1 SIGNAL DATA CONVERTER



CENTRAL DISPLAY UNIT

AB2147_20
SA

Figure 2. Instrument Display System Schematic Diagram. (Sheet 20 of 22)

SCHEMATICS - Continued

NO. 1 SIGNAL DATA CONVERTER

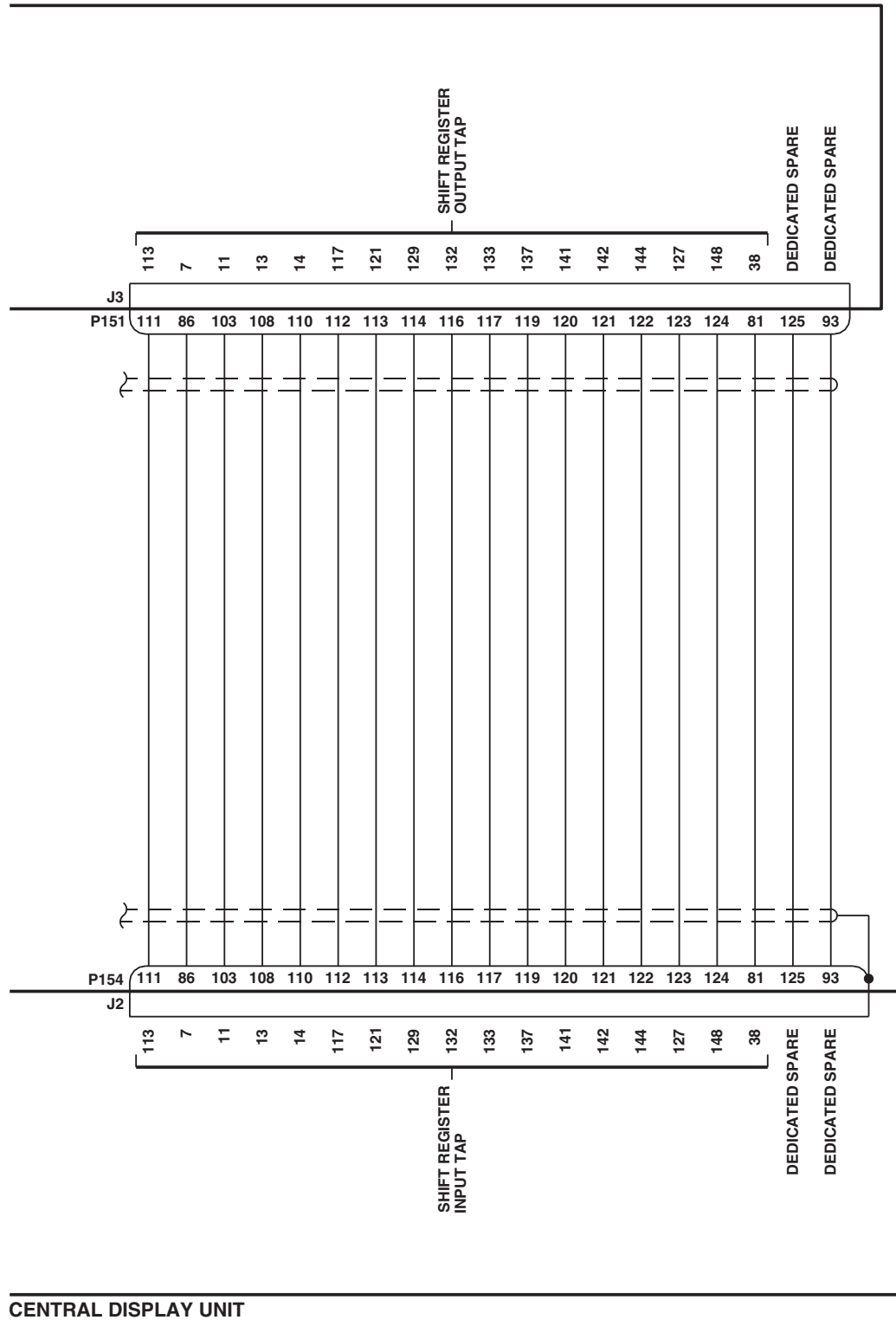
AB2147_21
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Figure 2. Instrument Display System Schematic Diagram. (Sheet 21 of 22)

SCHEMATICS - Continued

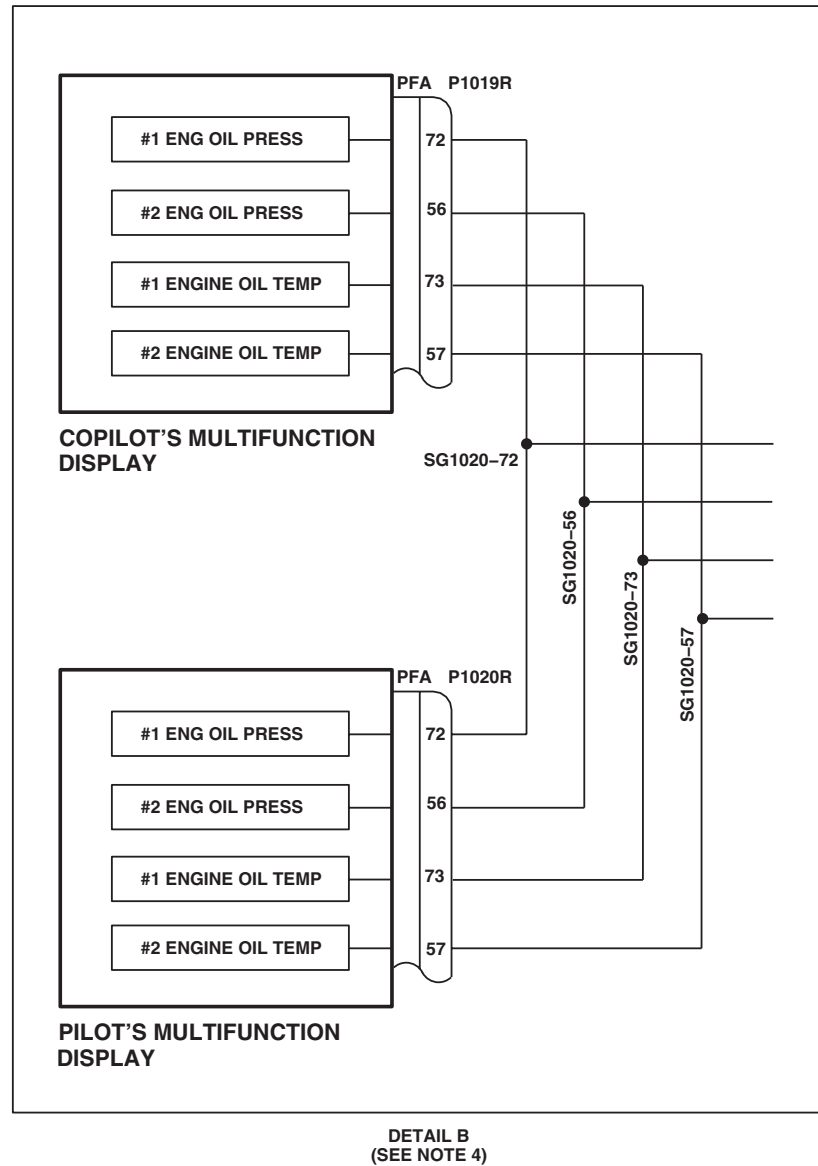
AB2147_22
SA

Figure 2. Instrument Display System Schematic Diagram. (Sheet 22 of 22)

END OF WORK PACKAGE

UNIT LEVEL**INSTRUMENT SYSTEMS****CAUTION/ADVISORY WARNING SYSTEM**

This WP supersedes WP 0090 00, dated 21 March 2007.

INITIAL SETUP:**Test Equipment**

Headset - Microphone, H-101A-U
Multimeter, AN/PSM-45A

WP 0101 00

WP 0104 00

WP 0785 00

WP 0786 00

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer Toolkit, SC 5180-99-B06
Locally-Made Drag Beam Wedge, Figure 167,
WP 1805 00
Locally-Made Infrared Light, Test Set, Figure 149,
WP 1805 00

WP 0788 00

WP 0789 00

WP 0790 00

WP 0798 00

WP 0824 00

WP 0825 00

WP 0925 00

Personnel Required

Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

WP 1050 00

WP 1685 00

WP 1747 00

WP 1805 00

References

TM 1-1520-237-10
TM 11-1520-237-23
TM 11-1520-249-23-2
WP 0089 00

Equipment Condition

External Electrical Power Available, WP 1685 00
or APU Operational, TM 1-1520-237-10

CAUTION/ADVISORY WARNING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- Step 1. must be performed when a system malfunction has been reported; otherwise, continue to step 2.
 - **W/O NVG II** , represents UH-60A helicopters and UH-60L helicopters prior to serial No. 96-26723, not modified by MWO 55-1520-237-50-20.
 - A momentary flicker of some caution/advisory panel capsules is normal and requires no corrective action.
 - If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
 - See Figure 1 for component location diagram, and Figure 2 for schematic diagram as an aid in operational/troubleshooting.
 - Two variations of the caution/advisory panel exist, however their system operations are identical. For this procedure the variations have no impact, both caution/advisory panels utilize light capsules to illuminate the panel: the legacy caution/advisory panels use incandescent lights and the caution/advisory panels (LED) use light emitting diodes.
1. **EMEP** Perform pin filtered connector check of P130 and P131. Refer to PIN FILTERED CONNECTOR CHECK, in this work package. ■
 2. Make sure the following circuit breakers are pushed in:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
LIGHTS CAUT/ADVSY	Copilot's
LIGHTS ADVSY	Copilot's
LIGHTS LWR CSL	Copilot's
CPLT MODE SELECT	Copilot's
PILOT'S MODE SELECT	Pilot's
CHIP DET	Upper console
FIRE DET NO. 1 ENG	Upper console

CAUTION/ADVISORY WARNING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

3. Pull out the following circuit breakers:

CIRCUIT BREAKER	CIRCUIT BREAKER PANEL
BACKUP HYD CONTR	Upper console
ICS PILOT	Upper console
NO. 1 GEN WARN	Copilot's
NO. 2 GEN WARN	Pilot's
NO. 2 AC INST	Pilot's
NO. 2 DC INST	Pilot's

4. Turn on electrical power.
5. Turn upper console INST LTS PILOT FLT control to OFF, then turn slightly clockwise to get out of OFF detent.
6. Momentarily press copilot's master warning panel MASTER CAUTION PRESS TO RESET capsule.

Indication/Condition

If pilot's and copilot's MASTER CAUTION PRESS TO RESET capsules were on, they shall go off.

Corrective Action

If Indication/Condition is not as specified, go to MASTER CAUTION PRESS TO RESET CAPSULES DO NOT GO OFF WHEN RESET, in this work package.

PROCEDURE**Step****NOTE**

If caution/advisory capsules do not go on, press and release copilot's cyclic stick PNL LTS switch. The dimming mode of the caution/advisory panel may get out of synchronization with the other panel lights; that is, when the PNL LTS button is pressed, the caution/advisory panel capsules may go on bright when the other lights go off. To correct this condition, place the caution/advisory panel capsules at high intensity, using the PNL LTS switch. Remove power from the helicopter and then reapply power.

7. Place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM.

NOTE

In step 8, start with CAUT/ADVSY NVG DIMMING control in fully counterclockwise position.

8. Turn instrument panel CAUT/ADVSY NVG DIMMING control fully clockwise, then counterclockwise.

Indication/Condition

Caution/advisory capsules shall increase and then decrease in brightness.

CAUTION/ADVISORY WARNING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If Indication/Condition is not as specified, replace CAUT/ADVSY NVG DIMMING control (WP 0788 00).
2. If no caution/advisory capsules are on, go to NO CAPSULES ARE ON, in this work package.

Step

9. Place power switch of locally-made infrared light test set (WP 1805 00) to ON.

Indication/Condition

POWER ON lamp (green) on locally-made infrared light test set shall go on.

Corrective Action

If Indication/Condition is not as specified, replace locally-made infrared light test set, locally-made infrared light test set lamp, or locally-made infrared light test set battery, as required.

Step

10. Place locally-made infrared light test set IR detector lead over test set's POWER ON lamp (green).

Indication/Condition

Locally-made infrared light test set IR LAMP ON lamp (amber) shall go on.

Corrective Action

If Indication/Condition is not as specified, replace locally-made infrared light test set.

Step

11. Turn CAUT/ADVSY NVG DIMMING and MA WRN NVG DIMMING controls fully clockwise.
12. **W/O NVG II** While holding down the caution/advisory panel BRT/DIM-TEST switch, place locally-made infrared light test set IR detector lead over each caution/advisory panel and master warning capsules that are not flashing. ■

Indication/Condition

- a. Locally-made infrared light test set IR LIGHT ON lamp shall go on each time detector lead is placed over capsules that are not flashing.
- b. Master warning capsules will not illuminate if NVG is installed.

Corrective Action

If Indication/Condition is not as specified, go to CAUTION/ADVISORY CAPSULES DO NOT DIM FOR NIGHT VISION GOGGLE VIEWING, in this work package.

Step**NOTE**

Make sure instrument panel MA WRN NVG DIMMING and CAUT/ADVSY NVG DIMMING controls are rotated fully clockwise.

13. Push in NO. 1 GEN WARN circuit breaker (copilot's circuit breaker panel).

Indication/Condition

Pilot's and copilot's MASTER CAUTION PRESS TO RESET capsules shall go on and #1 GEN caution capsule shall go on.

CAUTION/ADVISORY WARNING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If Indication/Condition is not as specified, troubleshoot AC electrical system (WP 0101 00).

Step

14. Momentarily press copilot's master warning panel MASTER CAUTION PRESS TO RESET capsule.

Indication/Condition

Both MASTER CAUTION PRESS TO RESET capsules shall go off.

Corrective Action

If Indication/Condition is not as specified, go to MASTER CAUTION PRESS TO RESET CAPSULES DO NOT GO OFF WHEN RESET, in this work package.

Step

15. Push in NO. 2 GEN WARN circuit breaker (pilot's circuit breaker panel).

Indication/Condition

- a. Both MASTER CAUTION PRESS TO RESET capsules shall go on.
- b. #2 GEN caution capsule shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, check if lamps are good.
 - a. If lamps are good, replace caution advisory or master caution panel (WP 0785 00).
 - b. If lamps are not good, replace lamps (WP 0786 00).
2. If Indication/Condition b. is not as specified, troubleshoot AC electrical system (WP 0101 00).

Step

16. Turn instrument panel MA WRN NVG DIMMING control fully counterclockwise then fully clockwise.

Indication/Condition

All master warning panel capsules which are on shall decrease in brilliance then increase in brilliance.

Corrective Action

If Indication/Condition is not as specified, go to WARNING CAPSULES DO NOT DIM FOR NIGHT VISION GOGGLE VIEWING, in this work package.

Step

17. Momentarily press pilot's master warning panel MASTER CAUTION PRESS TO RESET capsule.

Indication/Condition

Both MASTER CAUTION PRESS TO RESET capsules shall go off.

Corrective Action

If Indication/Condition is not as specified, go to MASTER CAUTION PRESS TO RESET CAPSULES DO NOT GO OFF WHEN RESET, in this work package.

CAUTION/ADVISORY WARNING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

18. Connect headset to pilot's ICS.
19. Push in ICS PILOT circuit breaker (upper console circuit breaker panel).

WARNING

To prevent injury to personnel, make sure area around the helicopter's stabilator is clear of personnel and equipment.

NOTE

Make sure landing gear drag beam switch is in ground position and stabilator control/auto flight control panel AUTO CONTROL ON legend is on.

20. Place stabilator control/auto flight control panel MAN SLEW switch DOWN or UP depending on the stabilator's position.

Indication/Condition

Beeping tone shall be heard in headset.

Corrective Action

If Indication/Condition is not as specified, go to NO STABILATOR WARNING TONE HEARD IN HEADSET, in this work package.

Step

21. Press MASTER CAUTION PRESS TO RESET capsule located on the pilot's master warning panel.
22. Push in NO. 2 DC INST and NO. 2 AC INST circuit breakers (pilot's circuit breaker panel).

Indication/Condition

- a. NO. 1 ENG OUT and NO. 2 ENG OUT master warning capsules shall go on.
- b. LOW ROTOR RPM capsules shall flash (master warning panels).
- c. MASTER CAUTION capsules shall go on.
- d. A steady tone shall be heard in headset.

Corrective Action

If Indication/Conditions are not as specified, troubleshoot instrument display system (IDS) (WP 0089 00).

Step

23. Press pilot's master warning panel MASTER CAUTION PRESS TO RESET capsule and check for tone in headset.

Indication/Condition

No tone shall be heard.

Corrective Action

If Indication/Condition is not as specified, replace left relay panel (WP 0825 00).

Step

24. Push in BACKUP HYD CONTR circuit breaker (upper console circuit breaker panel).

CAUTION/ADVISORY WARNING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

25. Insert locally-made wedge in left drag beam switch to simulate inflight (weight-off-wheels) condition.

Indication/Condition

Steady tone shall be heard in headset.

Corrective Action

If Indication/Condition is not as specified, go to NO STABILATOR WARNING TONE HEARD IN HEADSET, in this work package.

Step

26. Press pilot's master warning panel MASTER CAUTION PRESS TO RESET capsule and check for tone in headset.

Indication/Condition

Steady tone shall be heard in headset.

Corrective Action

If Indication/Condition is not as specified, go to NO STABILATOR WARNING TONE HEARD IN HEADSET, in this work package.

Step

27. Remove locally-made wedge (WP 1805 00) from left drag beam switch.

Indication/Condition

No tone shall be heard.

Corrective Action

If Indication/Condition is not as specified, troubleshoot digital AFCS (TM 11-1520-237-23 or TM 11-1520-249-23-2).

Step**WARNING**

To prevent injury to personnel, make sure area around the helicopter's stabilator is clear of personnel and equipment.

NOTE

In the following step, disregard vertical instrument display indications.

28. Reset aircraft power by using the EXT PWR RESET switch (center of the upper console), and then turn on.

Indication/Condition

- a. NO. 1 ENG OUT and NO. 2 ENG OUT caution capsules shall go on.
- b. LOW ROTOR RPM capsules shall flash and MASTER CAUTION PRESS TO RESET capsules shall go on.
- c. A steady tone shall be heard in headset.

Corrective Action

If Indication/Conditions are not as specified, troubleshoot instrument display system (IDS) (WP 0089 00).

CAUTION/ADVISORY WARNING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

29. Make sure stabilator control/auto flight control panel AUTO CONTROL ON legend is on.
30. Place stabilator control/auto flight control panel MAN SLEW switch to the DOWN position.

Indication/Condition

- a. Caution/advisory panel STABILATOR capsule shall go on.
- b. Tone shall change from steady to beeping.

Corrective Action

1. If Indication/Condition a. is not as specified, troubleshoot stabilator unit (TM 11-1520-237-23 or TM 11-1520-249-23-2).
2. If Indication/Condition b. is not as specified, replace audible warning unit (WP 0789 00).
 - a. If trouble remains, replace left relay panel (WP 0825 00).

Step

31. Momentarily press either master warning panel MASTER CAUTION PRESS TO RESET capsule.

Indication/Condition

- a. Tone shall change from beeping to steady.
- b. MASTER CAUTION PRESS TO RESET capsules shall go off.

Corrective Action

1. If Indication/Condition a. is not as specified, troubleshoot digital AFCS (TM 11-1520-237-23 or TM 11-1520-249-23-2).
2. If Indication/Condition a. is not as specified, go to MASTER CAUTION PRESS TO RESET CAPSULES DO NOT GO OFF WHEN RESET, in this work package.

Step

32. Momentarily press either master warning panel MASTER CAUTION PRESS TO RESET capsule.

Indication/Condition

Tone shall not be heard in headset.

Corrective Action

If Indication/Condition is not as specified, troubleshoot digital AFCS (TM 11-1520-237-23 or TM 11-1520-249-23-2).

Step

33. Insert locally-made wedge in left drag beam switch to simulate inflight (weight-off-wheels) condition.

Indication/Condition

Steady tone shall be heard in headset.

Corrective Action

If Indication/Condition is not as specified, go to NO STABILATOR WARNING TONE HEARD IN HEADSET, in this work package.

CAUTION/ADVISORY WARNING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

34. Momentarily press stabilator control/auto flight control panel AUTO CONTROL RESET switch.

Indication/Condition

- a. Caution/advisory panel STABILATOR capsule shall go off.
- b. Stabilator control/auto flight control panel AUTO CONTROL switch ON legend shall go on.

Corrective Action

If Indication/Conditions are not as specified, troubleshoot stabilator unit (TM 11-1520-237-23 or TM 11-1520-249-23-2).

Step

35. Press and hold stabilator control/auto flight control panel TEST switch.

Indication/Condition

- a. Caution/advisory panel STABILATOR capsule shall go on.
- b. MASTER CAUTION PRESS TO RESET capsules shall go on.
- c. Stabilator control/auto flight control panel AUTO CONTROL switch ON legend shall go off.
- d. Beeping tone shall be heard in headset.

Corrective Action

1. If Indication/Condition a. is not as specified, troubleshoot stabilator unit (TM 11-1520-237-23 or TM 11-1520-249-23-2).
2. If Indication/Conditions b., c., or d. are not as specified, troubleshoot digital AFCS (TM 11-1520-237-23 or TM 11-1520-249-23-2).

Step

36. Momentarily press either master warning panel MASTER CAUTION PRESS TO RESET capsule.

Indication/Condition

- a. Tone shall change from beeping to steady.
- b. MASTER CAUTION PRESS TO RESET capsules shall go off.

Corrective Action

1. If Indication/Condition a. is not as specified, troubleshoot digital AFCS (TM 11-1520-237-23 or TM 11-1520-249-23-2).
2. If Indication/Condition b. is not as specified, go to MASTER CAUTION PRESS TO RESET CAPSULES DO NOT GO OFF WHEN RESET, in this work package.

Step

37. Remove locally-made wedge from left drag beam switch.

Indication/Condition

No tone shall be heard in headset.

CAUTION/ADVISORY WARNING SYSTEM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If Indication/Condition is not as specified, troubleshoot digital AFCS (TM 11-1520-237-23 or TM 11-1520-249-23-2).

Step

38. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

PIN FILTERED CONNECTOR CHECK**SYMPTOM**

Perform pin filtered connector check.

MALFUNCTION

Pin filtered connector check.

CORRECTIVE ACTION

1. Disconnect P118, P130, P131, P137, P243, and P902. Go to 2.
2. Make sure INST LT PILOT FLT switch is OFF. Go to 3.
3. Remove and check pin filtered adapter from P118 (WP 0925 00).
 - a. If pin filtered adapter is good, go to 4.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to 9.
4. Install pin filtered adapter to P118 (WP 0925 00). Go to 5.
5. Check continuity between each pin of P130 and connector shell.
 - a. If continuity is present, replace pin filtered connector (WP 0925 00). Go to 9.
 - b. If continuity is not present, go to 6.
6. Check continuity between:
 - P118-a and P130-6
 - P118-e and P130-2
 - P118-i and P130-8
 - P118-j and P130-1
 - P118-W and P130-7
 - P118-X and P130-5
 - P130-3 and P138-n
 - P130-3 and P242-a

PIN FILTERED CONNECTOR CHECK - Continued

- P130-4 and P130-5
- P130-4 and P243-G
- P130-6 and P243-U
- P130-7 and INST LT PILOT FLT switch terminal A
- P130-9 and P138-u
- P130-9 and P243-h
- P130-10 and P137-n
- P130-10 and P243-a
- P130-11 and GG33-5
- P131-4 and P902-H

a. If continuity is present, go to **7**.

b. If continuity is not present, replace pin filtered connector P130 (WP 0925 00). Go to **9**.

7. Check continuity between each pin of P131 and connector shell.

a. If continuity is present, replace pin filtered connector (WP 0925 00). Go to **9**.

b. If continuity is not present, go to **8**.

8. Check continuity between:

- P130-1 and P131-1
- P130-2 and P131-2
- P130-3 and P131-3
- P130-5 and P131-5
- P130-6 and P131-6
- P130-7 and P131-7
- P130-8 and P131-8
- P130-9 and P131-9
- P130-10 and P131-10

a. If continuity is present, go to **9**.

b. If continuity is not present, replace pin filtered connector P131 (WP 0925 00). Go to **9**.

9. Connect P118, P130, P131, P137, P243, and P902. Go to 10.

10. Procedure completed. Go to CAUTION/ADVISORY WARNING SYSTEM, step 2 of Operational/Troubleshooting Procedure, in this work package.

MASTER CAUTION PRESS TO RESET CAPSULES DO NOT GO OFF WHEN RESET**SYMPTOM**

MASTER CAUTION PRESS TO RESET capsules shall either go on or off.

MALFUNCTION

MASTER CAUTION PRESS TO RESET capsules do not go off when reset.

CORRECTIVE ACTION

- 1. Check helicopter configuration.**
 - a. **EMEP** , go to **2**.
 - b. **W/O EMEP** , go to **4**.
- 2. EMEP Remove and check pin filtered adapter from P118 (WP 0925 00).**
 - a. If pin filtered adapter is good, go to **3**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **16**.
- 3. EMEP Install pin filtered adapter (WP 0925 00). Go to 4.**
- 4. Press pilot's capsule and check that both capsules go off.**
 - a. If both capsules go off, go to **5**.
 - b. If both capsules do not off, go to **6**.
- 5. Check continuity between:**
 - **P118-a and P130-6**
 - **P118-W and P130-7**
 - a. If continuity is present, replace copilot's master warning panel (WP 0790 00). Go to **16**.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **16**.
- 6. Press copilot's capsule and check that both capsules go off.**
 - a. If both capsules go off, go to **7**.
 - b. If both capsules do not go off, go to **8**.
- 7. Check continuity between:**
 - **P118-a and P131-6**
 - **P118-W and P131-7**
 - a. If continuity is present, replace pilot's master warning panel (WP 0790 00). Go to **16**.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **16**.
- 8. Check for 28 vdc between:**
 - **P130-7 and P130-11**
 - **P131-7 and P131-11**
 - a. If voltage is as specified, go to **9**.
 - b. If voltage is not as specified, go to **14**.

MASTER CAUTION PRESS TO RESET CAPSULES DO NOT GO OFF WHEN RESET - Continued

9. Check helicopter configuration.

- a. **W/O MWO 50-26** , go to **10**.
- b. **MWO 50-26** , go to **13**.

10. W/O MWO 50-26 Check for 28 vdc between P125-B and ground while pressing pilot's MASTER CAUTION PRESS TO RESET capsule.

- a. If voltage is as specified, go to **11**.
- b. If voltage is not as specified, repair/replace wiring between P125-B and P130-6 (WP 1747 00). Go to **16**.

11. W/O MWO 50-26 Turn off electrical power, check for diode front to back resistance ratio of more than 10 to 1 between pins B and C of the chip detector resistor unit connector.

- a. If resistance is more than 10 to 1, go to **12**.
- b. If resistance is not more than 10 to 1, replace chip detector resistor unit (WP 0798 00). Go to **16**.

12. W/O MWO 50-26 Check continuity between P118-a and P125-C.

- a. If continuity is present, replace caution/advisory panel (WP 0785 00). Go to **16**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **16**.

13. MWO 50-26 Check for 28 vdc between P118-a and ground while pressing pilot's MASTER CAUTION PRESS TO RESET capsule.

- a. If voltage is as specified, replace caution/advisory panel (WP 0785 00). Go to **16**.
- b. If voltage is not as specified, repair/replace wiring, as required, between (WP 1747 00), then go to **16**:
 - P118-a and P130-6
 - P118-a and P131-6

14. Check continuity between:

- **GG33-2 and P131-11**
- **GG33-5 and P130-11**

- a. If continuity is present, go to **15**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **16**.

15. W/O EMEP Check for 28 vdc between CAUT/ADVSY PNL circuit breaker terminal 1 and ground.

- a. If voltage is as specified, repair/replace wiring, as required, between (WP 1747 00), then go to **16**:
 - Terminal 1 of circuit breaker and P130-7
 - Terminal 1 of circuit breaker and P131-7
- b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **16**.

16. Procedure completed.

NO CAPSULES ARE ON**SYMPTOM**

Caution/advisory capsules shall increase and then decrease in brightness.

MALFUNCTION

No capsules are on.

CORRECTIVE ACTION**1. Check helicopter configuration.**

- a. **EMEP** , go to **2**.
- b. **W/O EMEP** , go to **4**.

2. EMEP Remove and check pin filtered adapter from P118.

- a. If pin filtered adapter is good, go to **3**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **11**.

3. EMEP Install pin filtered adapter (WP 0925 00). Go to 4.**4. Check that master warning panel capsules, stabilator control/auto flight control panel capsules, and HSI/VSI mode select panel switch legends are off.**

- a. If all legends are off, go to **5**.
- b. If all legends are not off, go to **6**.

5. Check continuity between:

- **P118-e** and **P130-2**
- **P118-e** and **P131-2**

- a. If continuity is present, replace caution/advisory panel (WP 0785 00). Go to **11**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **11**.

6. Check that HSI/VSI mode select panel switch legends are off.

- a. If select panel switch legends are off, repair/replace wiring, as required, between (WP 1747 00), then go to **11**:
 - **P118-e** and **P300R-27**
 - **P118-e** and **P317R-27**
- b. If select panel switch legends are not off, go to **7**.

7. Check that stabilator control/auto flight control panel capsules are off.

- a. If capsules are off, repair/replace wiring between P4R-D and P118-e (WP 1747 00). Go to **11**.
- b. If capsules are not off, go to **8**.

8. Check that capsules on both master warning panels are off.

- a. If master warning panels are off, go to **9**.
- b. If master warning panels are not off, go to **10**.

NO CAPSULES ARE ON - Continued**9. Check continuity between:**

- **P118-e and P130-2**
- **P118-e and P131-2**

- a. If continuity is present, replace caution/advisory panel (WP 0785 00). Go to **11**.
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **11**.

10. Check continuity between:

- **P118-e and P130-2**
- **P118-e and P131-2**

- a. If continuity is present, replace malfunctioning master warning panel (WP 0790 00). Go to **11**.
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **11**.

11. Procedure completed.**CAUTION/ADVISORY CAPSULES DO NOT DIM FOR NIGHT VISION GOGGLE VIEWING****SYMPTOM**

Locally-made infrared light test set IR LIGHT ON lamp shall go on.

MALFUNCTION

Caution/advisory capsules do not dim for night vision goggle viewing.

CORRECTIVE ACTION**1. Check helicopter configuration.**

- a. **EMEP** , go to **2**.
- b. **W/O EMEP** , go to **4**.

2. EMEP Remove and check pin filtered adapter from P118 (WP 0925 00).

- a. If pin filtered adapter is good, go to **3**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **14**.

3. EMEP Install pin filtered adapter (WP 0925 00). Go to 4.**4. Check that lower console panel lights can be turned off by pressing cyclic stick PNL LTS switch.**

- a. If lower console panel lights can be turned off, go to **5**.
- b. If lower console panel lights can not be turned off, go to **7**.

5. Check for resistance of 0 to 1.5 K ohms between P118-n and P118-r.

- a. If resistance is as specified, replace caution/advisory panel (WP 0785 00). Go to **14**.
- b. If resistance is not as specified, go to **6**.

6. Check continuity between:

- **P118-n and R3-2**
- **P118-r and R3-1**

CAUTION/ADVISORY CAPSULES DO NOT DIM FOR NIGHT VISION GOGGLE VIEWING - Continued

- a. If continuity is present, replace CAUT/ADVSY NVG DIMMING control R3 (WP 0788 00). Go to **14**.
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **14**.
- 7. Check for 28 vdc between:**
 - **J120-a and ground**
 - **J120-N and ground**
 - a. If voltage is as specified, go to **8**.
 - b. If voltage is not as specified, repair/replace wiring, as required, between (WP 1747 00), then go to **14**:
 - J120-a and P902-j
 - J120-N and P902-j
- 8. Check for 28 vdc between J119-N and ground.**
 - a. If voltage is as specified, go to **9**.
 - b. If voltage is not as specified, go to **13**.
- 9. While pressing pilot's cyclic stick PNL LTS switch, check for 28 vdc between J119-a and ground.**
 - a. If voltage is as specified, go to **10**.
 - b. If voltage is not as specified, go to **12**.
- 10. While pressing pilot's cyclic stick PNL LTS switch, check for 28 vdc between P118-p and ground.**
 - a. If voltage is as specified, replace caution/advisory panel (WP 0785 00). Go to **14**.
 - b. If voltage is not as specified, go to **11**.
- 11. Check continuity between:**
 - **J119-b and P118-p**
 - **J119-P and P118-p**
 - a. If continuity is present, replace copilot's cyclic stick (WP 1050 00). Go to **14**.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **14**.
- 12. Check continuity between:**
 - **J119-a and J120-b**
 - **J119-N and J120-P**
 - a. If continuity is present, replace pilot's cyclic stick (WP 1050 00). Go to **14**.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **14**.
- 13. Check continuity between J119-N and J120-P.**
 - a. If continuity is present, replace pilot's cyclic stick (WP 1050 00). Go to **14**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **14**.

CAUTION/ADVISORY CAPSULES DO NOT DIM FOR NIGHT VISION GOGGLE VIEWING - Continued**14. Procedure completed.****WARNING CAPSULES DO NOT DIM FOR NIGHT VISION GOGGLE VIEWING****SYMPTOM**

All master warning panel capsules which are on shall decrease in brilliance then increase in brilliance.

MALFUNCTION

Warning capsules do not dim for night vision goggle viewing.

CORRECTIVE ACTION**1. Check helicopter configuration.**

- a. **EMEP** , go to **2**.
- b. **W/O EMEP** , go to **4**.

2. EMEP Remove and check pin filtered adapter from P118 (WP 0925 00).

- a. If pin filtered adapter is good, go to **3**.
- b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **6**.

3. EMEP Install pin filtered adapter (WP 0925 00). Go to 4.**4. Check for resistance of 0 to 1.5 K ohms between P118-k and P118-m.**

- a. If resistance is as specified, replace caution/advisory panel (WP 0785 00). Go to **6**.
- b. If resistance is not as specified, go to **5**.

5. Check continuity between:

- **P118-k and R2-1**
- **P118-m and R2-2**

- a. If continuity is present, replace MA WRN NVG DIMMING control R2 (WP 0788 00). Go to **6**.
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **6**.

6. Procedure completed.**NO STABILATOR WARNING TONE HEARD IN HEADSET****SYMPTOM**

Beeping tone shall be heard in headset.

MALFUNCTION

No stabilator warning tone heard in headset.

CORRECTIVE ACTION**1. Is steady tone heard in headset?**

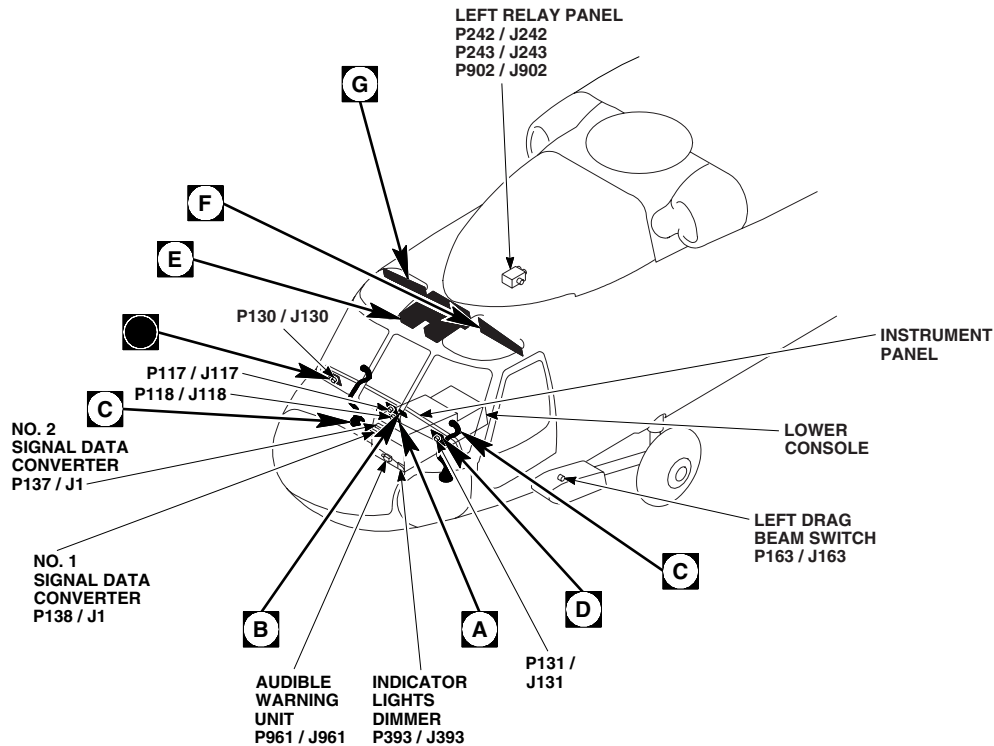
- a. If tone is steady in headset, go to **2**.
- b. If tone is not steady in headset, go to **7**.

2. Check for 28 vdc between P243-S and ground.

NO STABILATOR WARNING TONE HEARD IN HEADSET - Continued

- a. If voltage is as specified, replace left relay panel (WP 0825 00). Go to **12**.
- b. If voltage is not as specified, go to **3**.
- 3. Check helicopter configuration.**
 - a. **EMEP** , go to **4**.
 - b. **W/O EMEP** , go to **6**.
- 4. EMEP Remove and check pin filtered adapter from P117 (WP 0925 00).**
 - a. If pin filtered adapter is good, go to **5**.
 - b. If pin filtered adapter is not good, replace pin filtered adapter (WP 0925 00). Go to **12**.
- 5. EMEP Install pin filtered adapter (WP 0925 00). Go to 6.**
- 6. Repair/replace wiring between P117-27 and P243-S (WP 1747 00). Go to 12.**
- 7. Press pilot's master warning panel MASTER CAUTION PRESS TO RESET capsule, and listen for a steady tone in headset.**
 - a. If steady tone is heard in headset, go to **8**.
 - b. If steady tone is not heard in headset, go to **9**.
- 8. Check continuity between P242-G and P961-D.**
 - a. If continuity is present, replace left relay panel (WP 0825 00). Go to **12**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **12**.
- 9. Check for 28 vdc between P961-A and P961-B.**
 - a. If voltage is as specified, go to **10**.
 - b. If voltage is not as specified, go to **11**.
- 10. Check continuity between:**
 - **GND961-1 and P961-B**
 - **P242-E and P656R-M**
 - a. If continuity is present, replace audible warning unit (WP 0789 00). Go to **12**.
 - b. If trouble remains, troubleshoot intercommunication system (TM 11-1520-237-23). Go to **12**.
 - c. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **12**.
- 11. Check continuity between:**
 - **P242-H and P961-A**
 - **P961-B and ground**
 - a. If continuity is present, replace left relay panel (WP 0825 00). Go to **12**.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **12**.
- 12. Procedure completed.**

LOCATION



NOTES

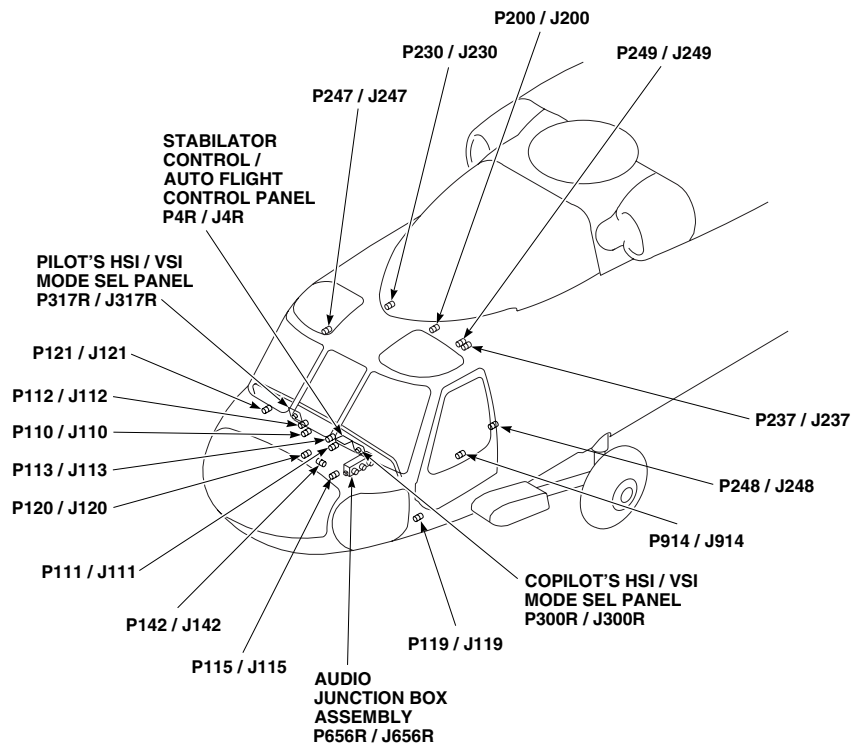
1. **EH-60A**
CARGO HOOK OPEN CAPSULE IS AIR COND ON CAPSULE.
2. **EH-60A**
CAPSULE IS ANTENNA EXTENDED.
3. **ESSS**
4. **EH-60A**
CAPSULE IS ANTENNA RETRACTED.
5. **EH-60A**
CAPSULE IS CABIN HEAT ON.
6. **UH-60A UH-60L**
7. **77-27714 - 96-26723**

TERMINAL BOARD / DISCONNECT PLUG / RECEPTACLE	LOCATION / CONNECTION POINT
P4R / J4R	STABILATOR CONTROL / AUTO FLIGHT CONTROL PANEL
P110 / J110	COCKPIT, BL 7.5 RH, STA 197
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P112 / J112	COCKPIT, BL 6 RH, STA 197
P113 / J113	COCKPIT, BL 6 LH, STA 197
P115 / J115	COCKPIT, BL 6 RH, STA 210
P117 / J117	CAUTION / ADVISORY PANEL
P118 / J118	CAUTION / ADVISORY PANEL
P119 / J119	COCKPIT TUB, BL 30 LH, STA 220
P120 / J120	COCKPIT TUB, BL 30 RH, STA 220
P121 / J121	COCKPIT, BL 10 RH, STA 195
P130 / J130	PILOT'S MASTER WARNING PANEL

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Figure 1. Caution/Advisory Warning System Location Diagram. (Sheet 1 of 7)

LOCATION - Continued



TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P131 / J131	COPILOT'S MASTER WARNING PANEL
P137 / J1	NO. 2 SIGNAL DATA CONVERTER
P138 / J1	NO. 1 SIGNAL DATA CONVERTER
P142 / J142	COCKPIT, BL 3 RH, STA 210
P163 / J163	LEFT DRAG BEAM SWITCH
P200 / J200	CABIN CEILING, BL 13 LH, STA 246
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P237 / J237	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 25 LH
P242 / J242	LEFT RELAY PANEL
P243 / J243	LEFT RELAY PANEL

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P247 / J247	CABIN, BL 40 RH, STA 248
P248 / J248	CABIN, BL 40 LH, STA 248
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P300R / J300R	COPILOT'S HSI / VSI MODE SELECT PANEL
P317R / J317R	PILOT'S HSI / VSI MODE SELECT PANEL
P393 / J393	INDICATOR LIGHTS DIMMER
P656R / J656R	JUNCTION BOX ASSEMBLY
P902 / J902	LEFT RELAY PANEL
P914 / J914	COCKPIT, BL 24 LH, STA 247
P961 / J961	AUDIBLE WARNING UNIT

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Figure 1. Caution/Advisory Warning System Location Diagram. (Sheet 2 of 7)

LOCATION - Continued

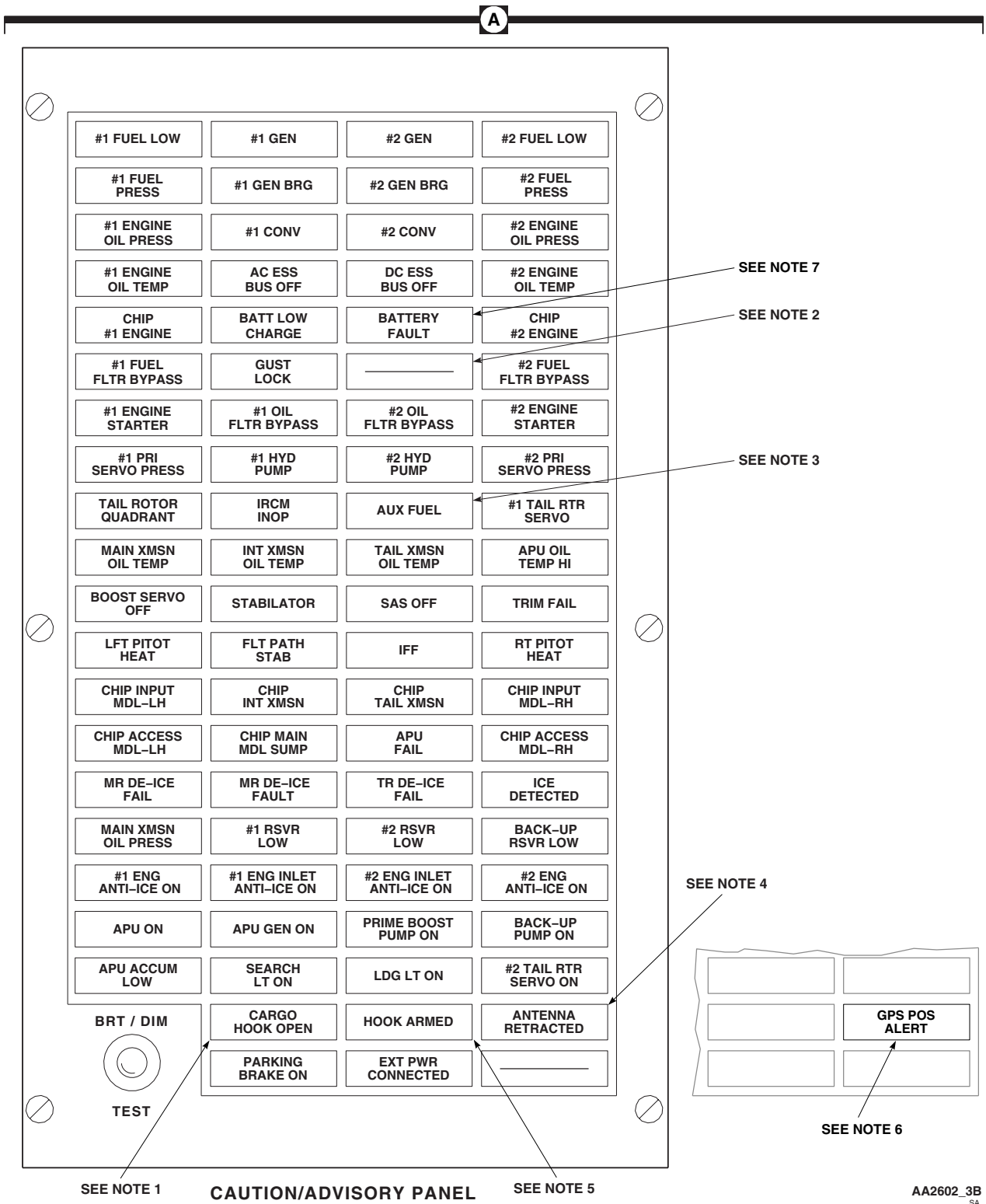
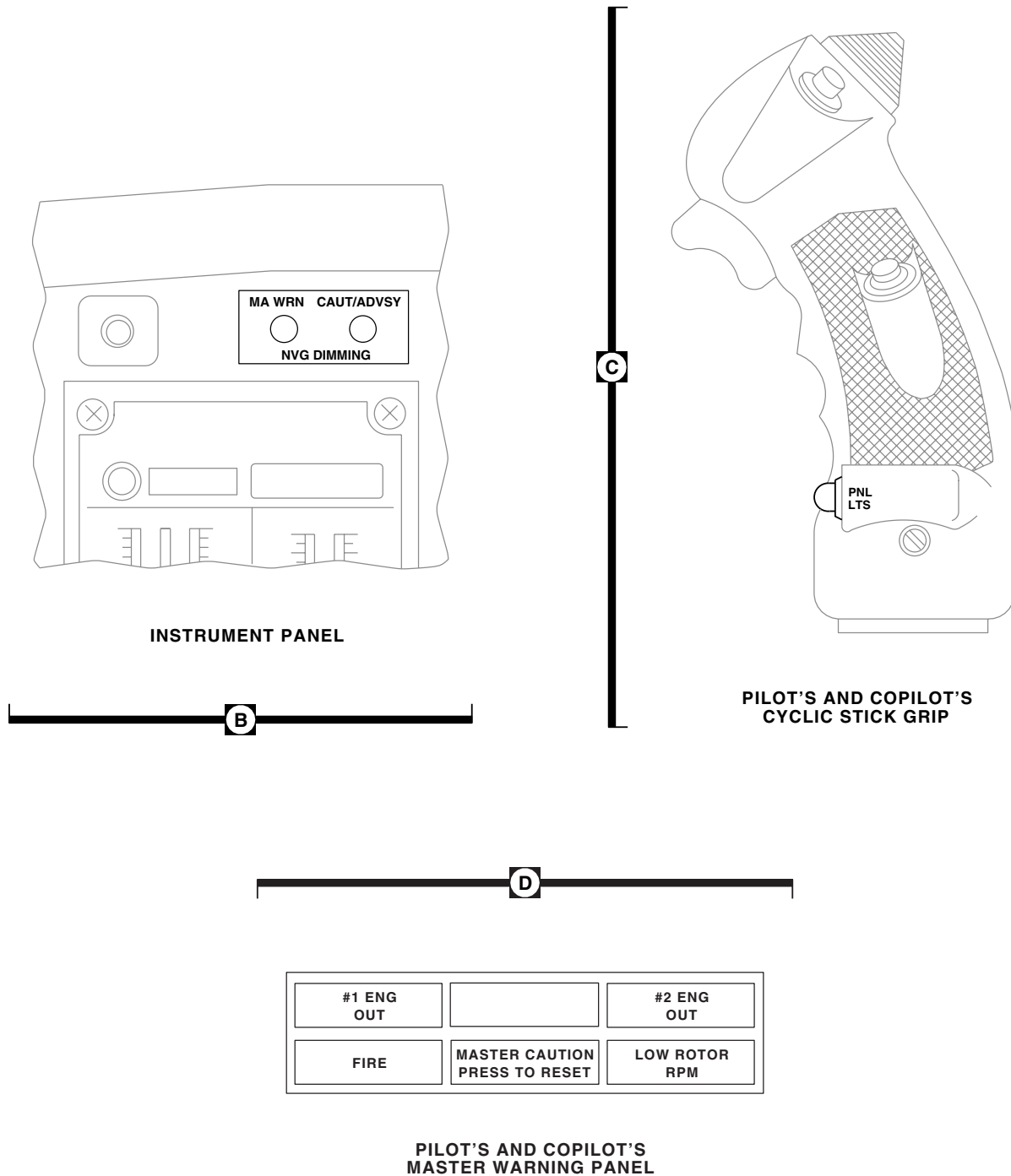


Figure 1. Caution/Advisory Warning System Location Diagram. (Sheet 3 of 7)

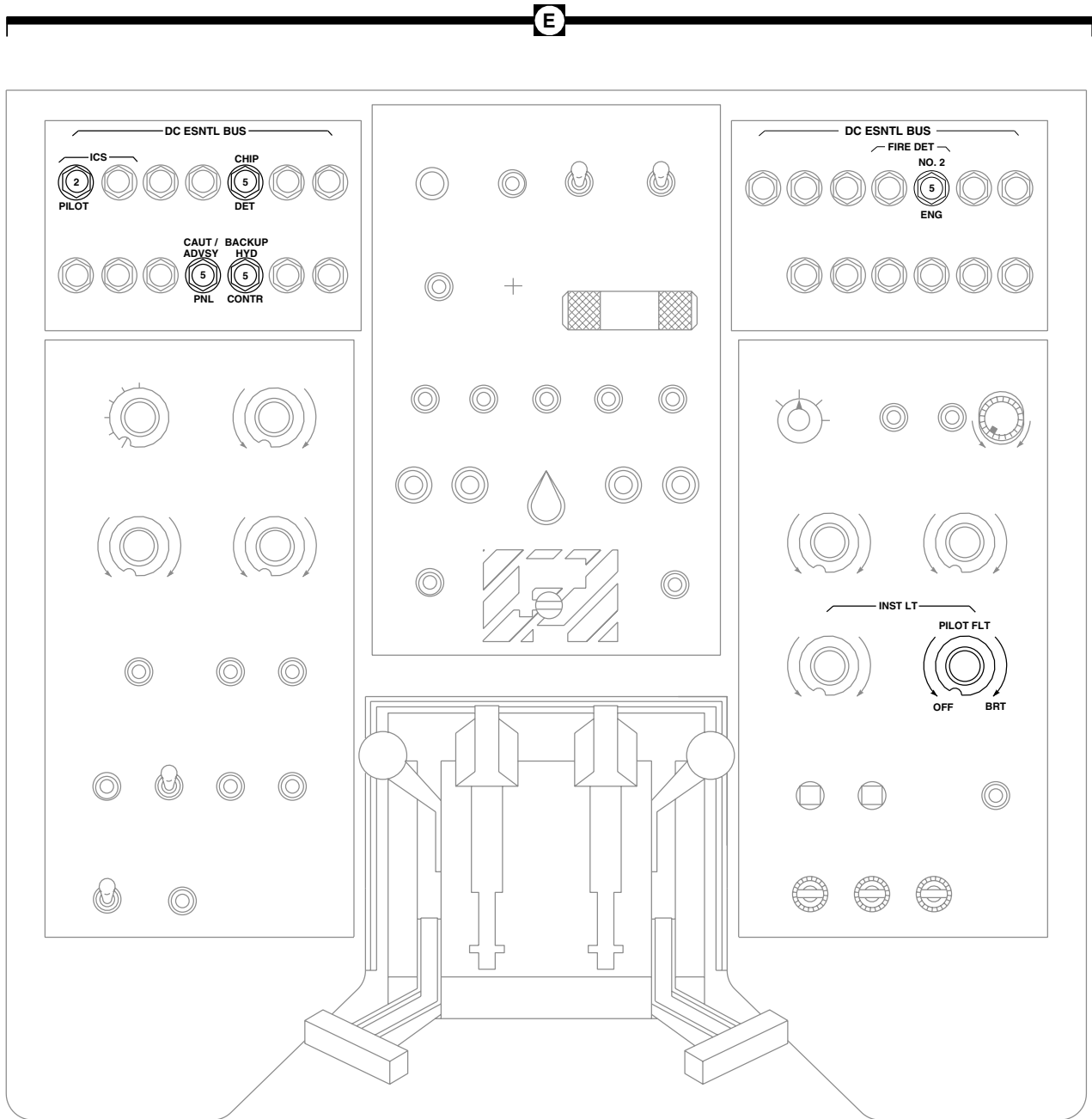
LOCATION - Continued



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Figure 1. Caution/Advisory Warning System Location Diagram. (Sheet 4 of 7)

LOCATION - Continued

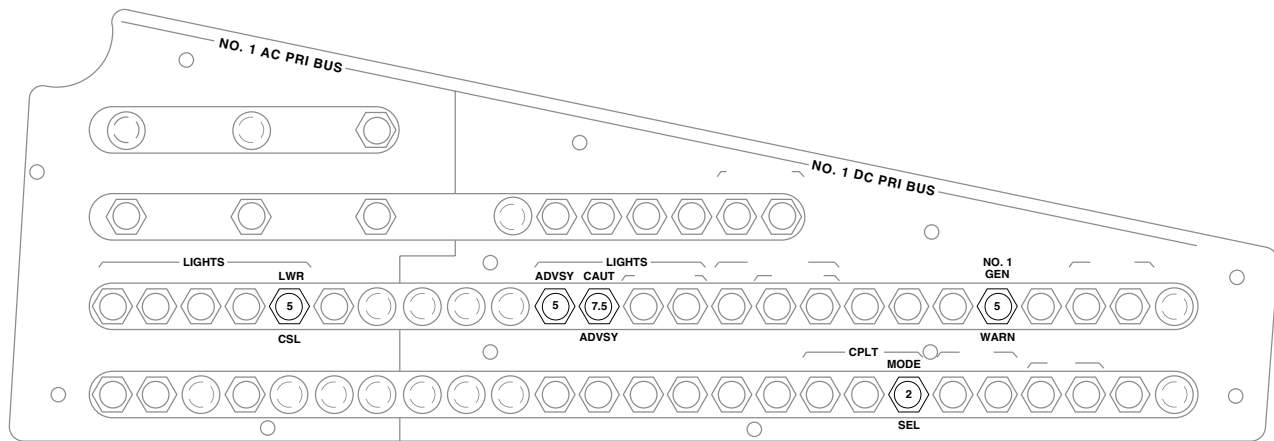


UPPER CONSOLE

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Figure 1. Caution/Advisory Warning System Location Diagram. (Sheet 5 of 7)

LOCATION - Continued



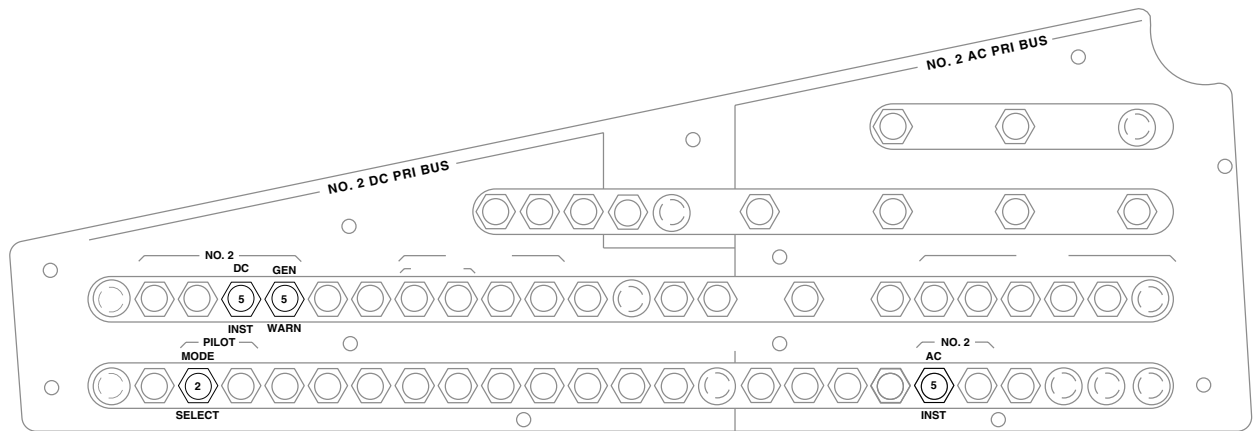
COPLOT'S CIRCUIT BREAKER PANEL

F

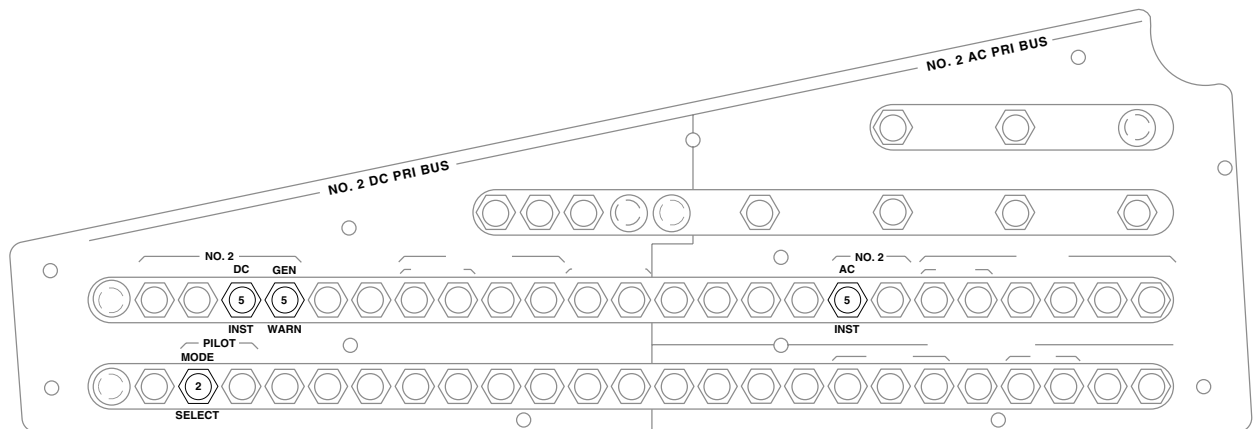
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Figure 1. Caution/Advisory Warning System Location Diagram. (Sheet 6 of 7)

LOCATION - Continued



(SEE NOTE 4)



PILOT'S CIRCUIT BREAKER PANEL

G

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Figure 1. Caution/Advisory Warning System Location Diagram. (Sheet 7 of 7)

SCHEMATICS

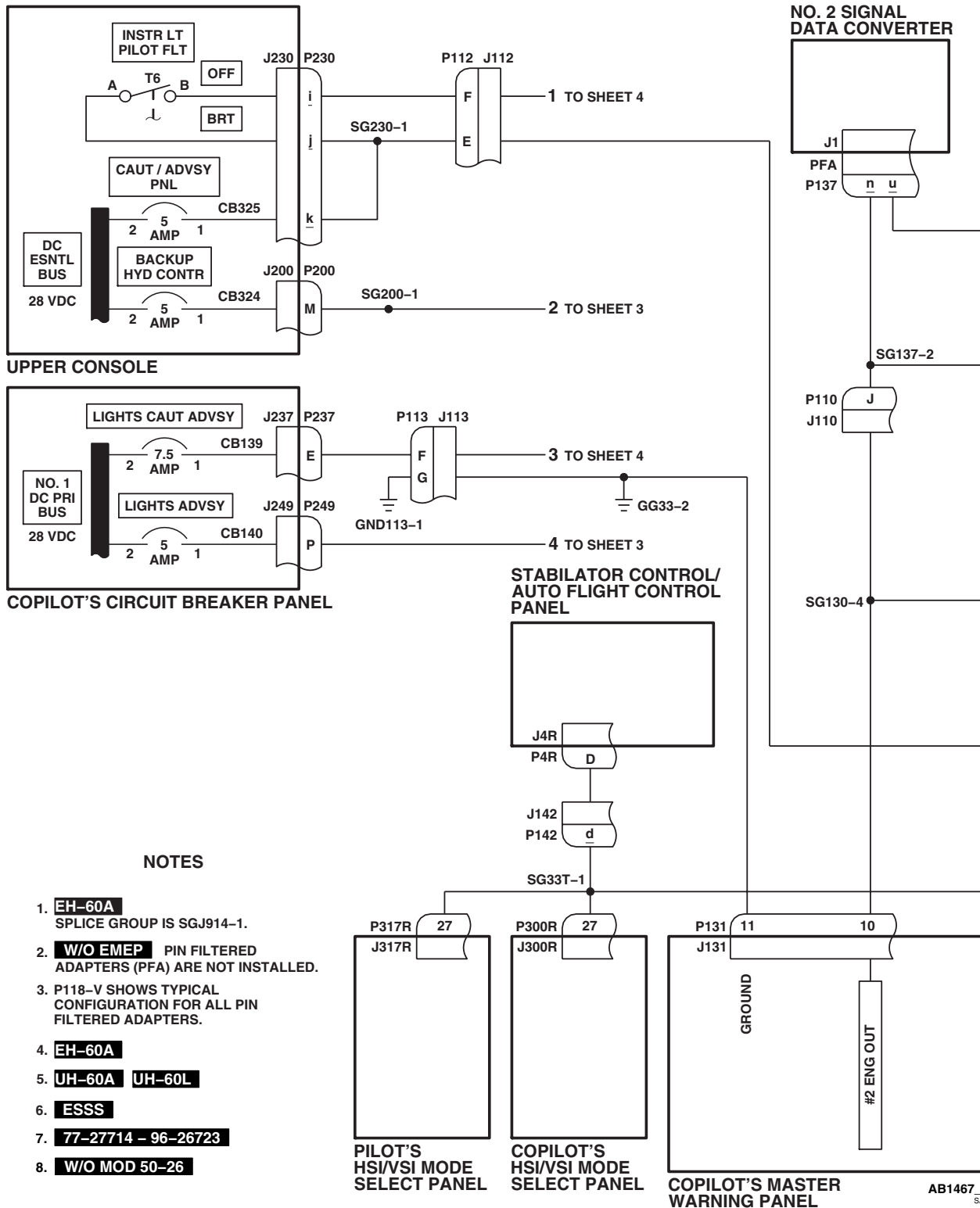


Figure 2. Caution/Advisory Warning System Schematic Diagram. (Sheet 1 of 10)

SCHEMATICS - Continued

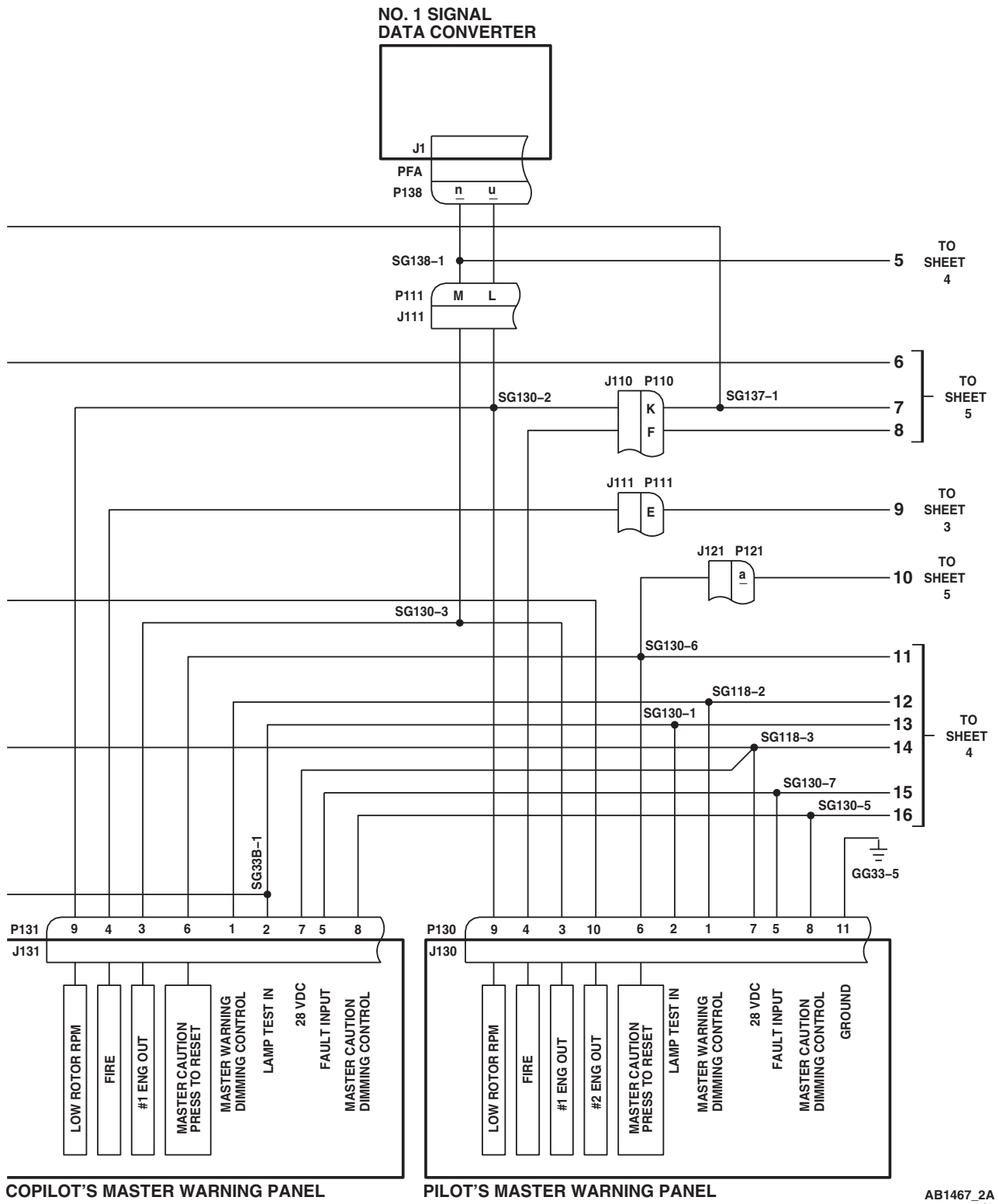


Figure 2. Caution/Advisory Warning System Schematic Diagram. (Sheet 2 of 10)

SCHEMATICS - Continued

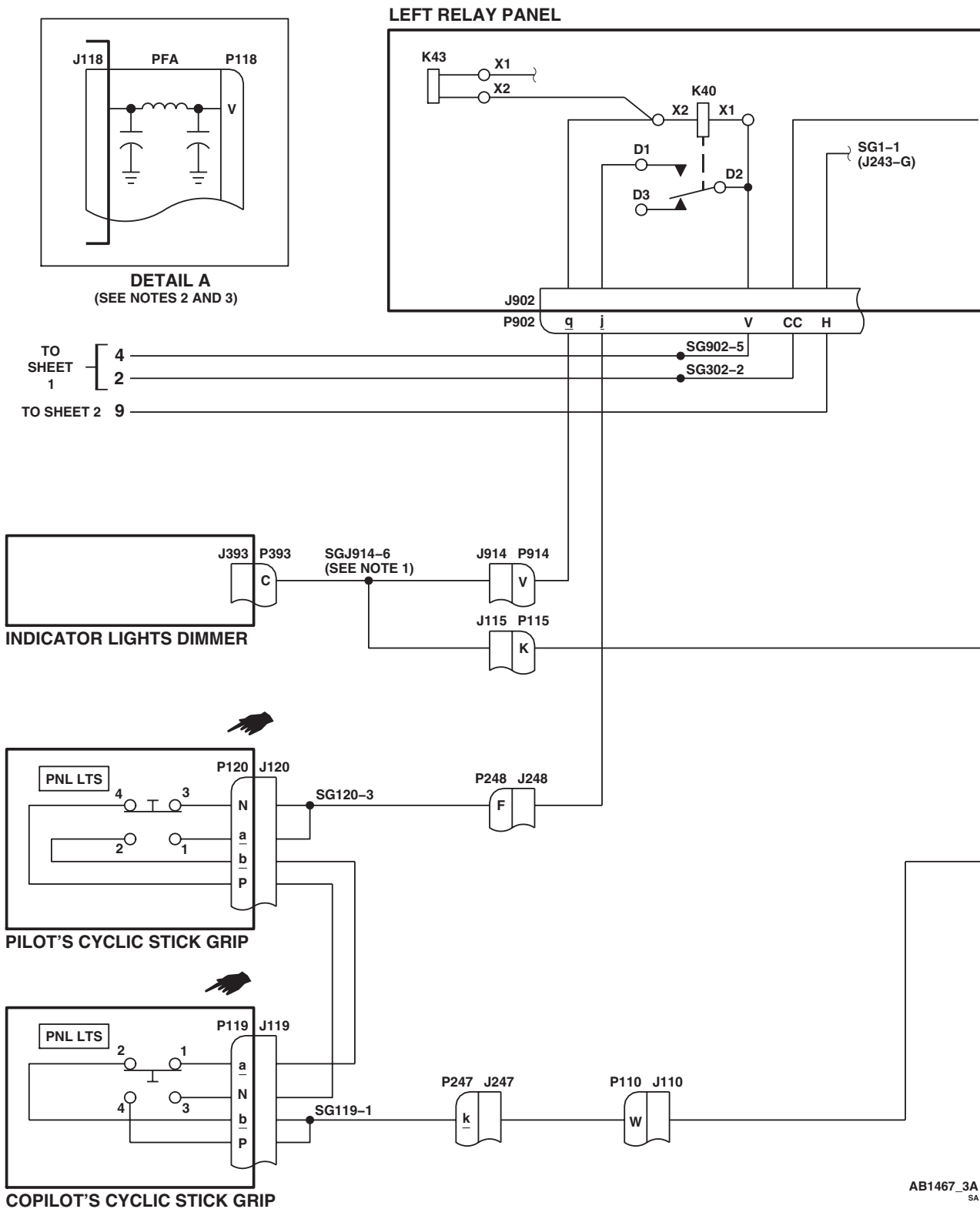
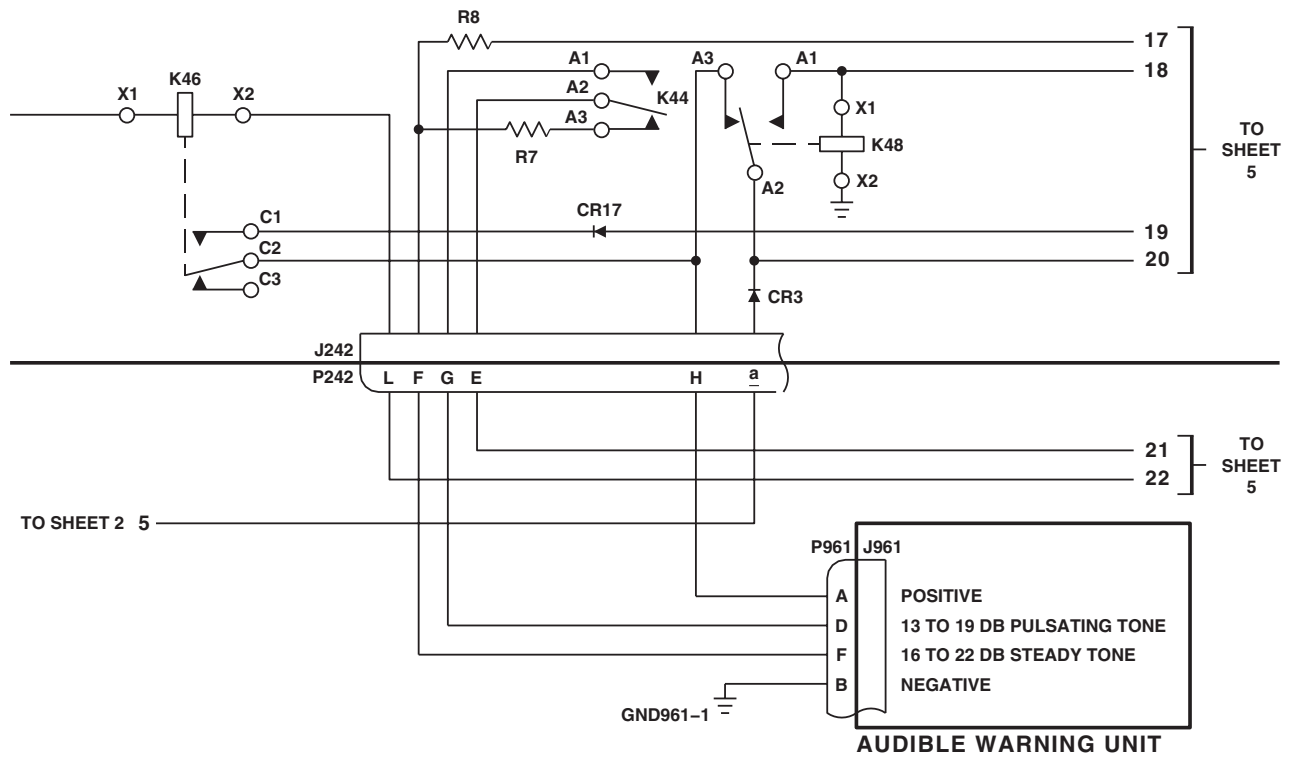


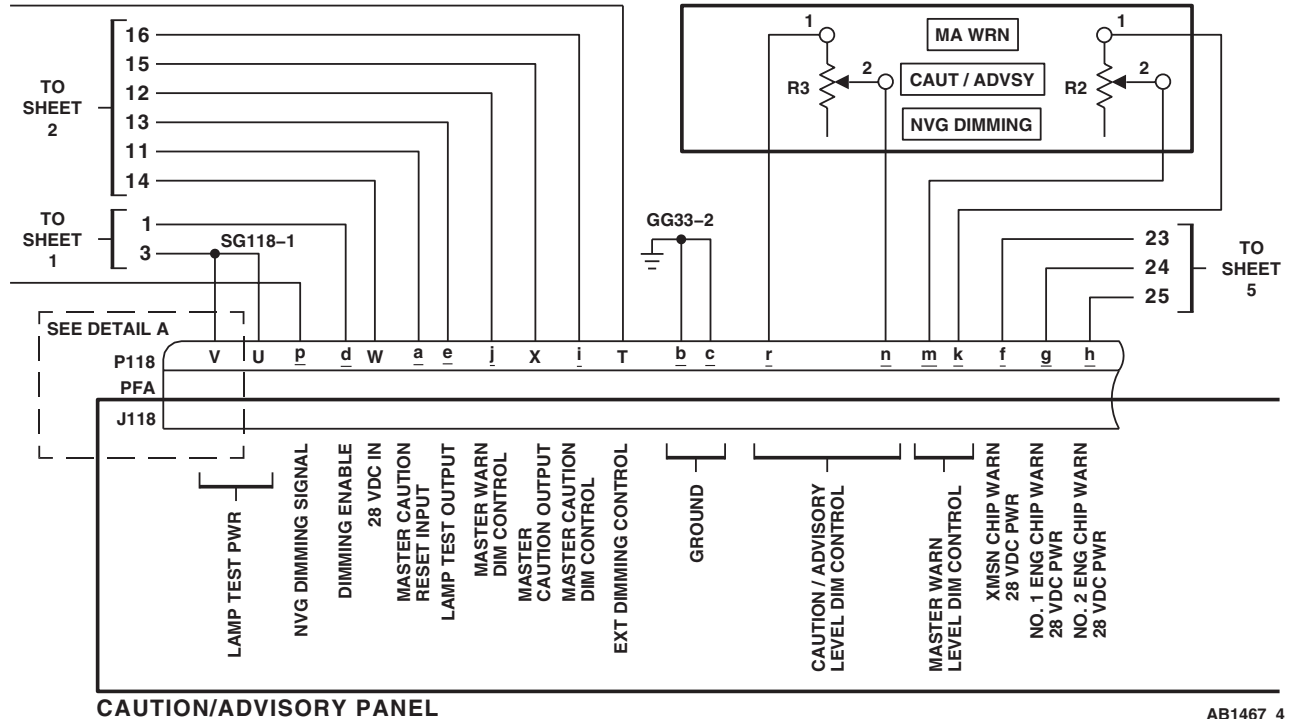
Figure 2. Caution/Advisory Warning System Schematic Diagram. (Sheet 3 of 10)

SCHEMATICS - Continued

LEFT RELAY PANEL



INSTRUMENT PANEL



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Figure 2. Caution/Advisory Warning System Schematic Diagram. (Sheet 4 of 10)

SCHEMATICS - Continued

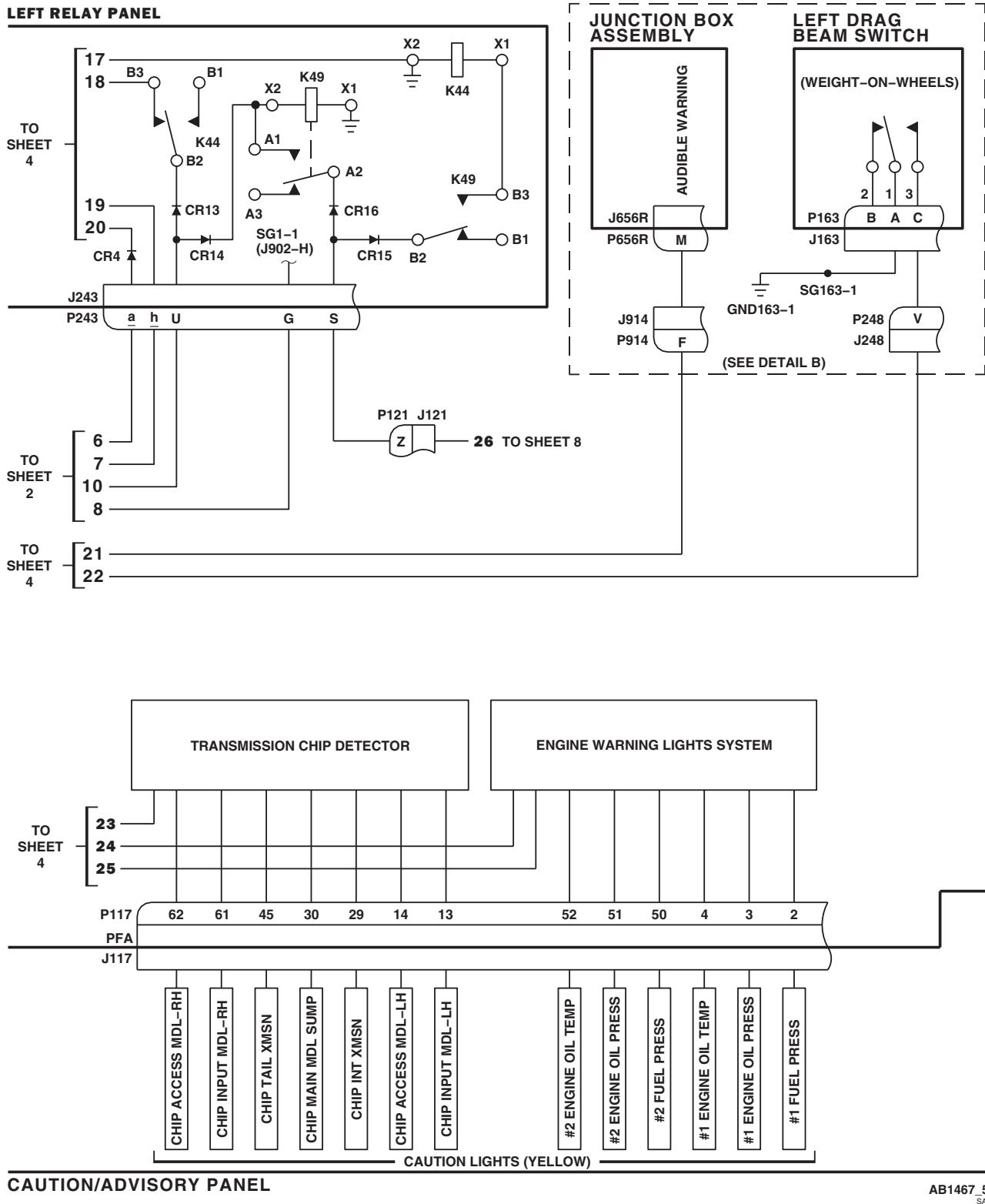
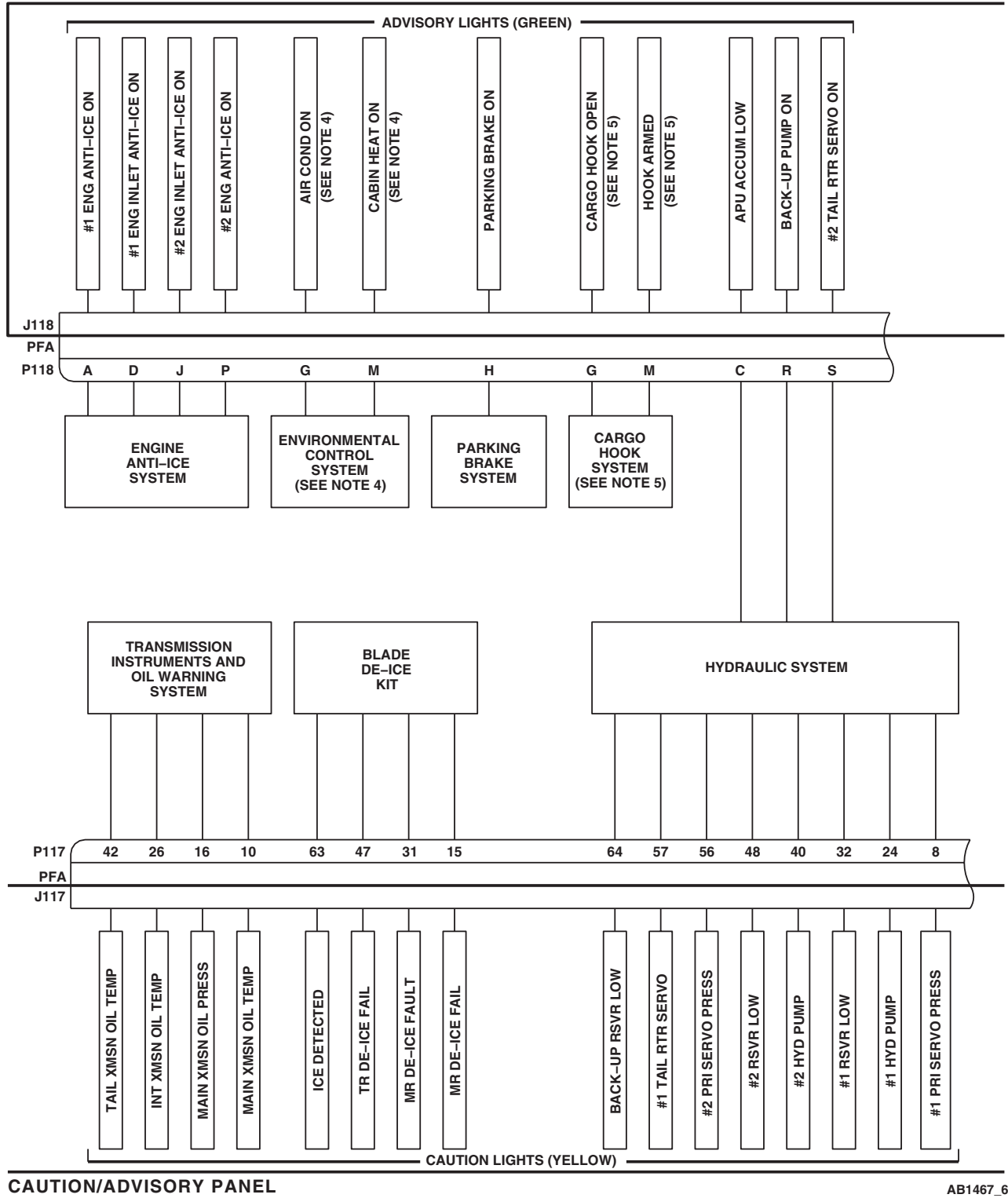


Figure 2. Caution/Advisory Warning System Schematic Diagram. (Sheet 5 of 10)

SCHEMATICS - Continued

CAUTION/ADVISORY PANEL

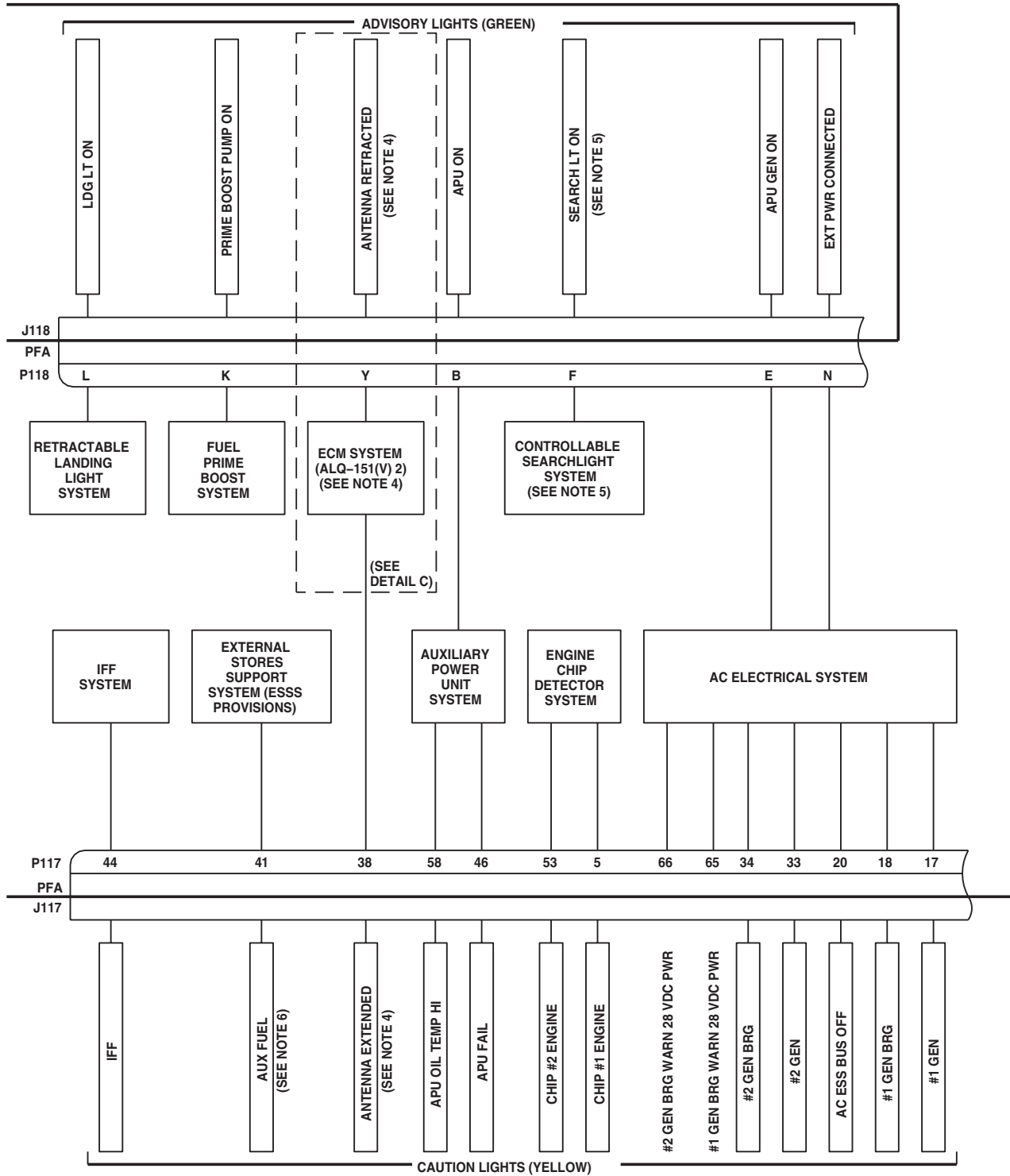


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Figure 2. Caution/Advisory Warning System Schematic Diagram. (Sheet 6 of 10)

SCHEMATICS - Continued

CAUTION/ADVISORY PANEL



CAUTION/ADVISORY PANEL

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Figure 2. Caution/Advisory Warning System Schematic Diagram. (Sheet 7 of 10)

SCHEMATICS - Continued

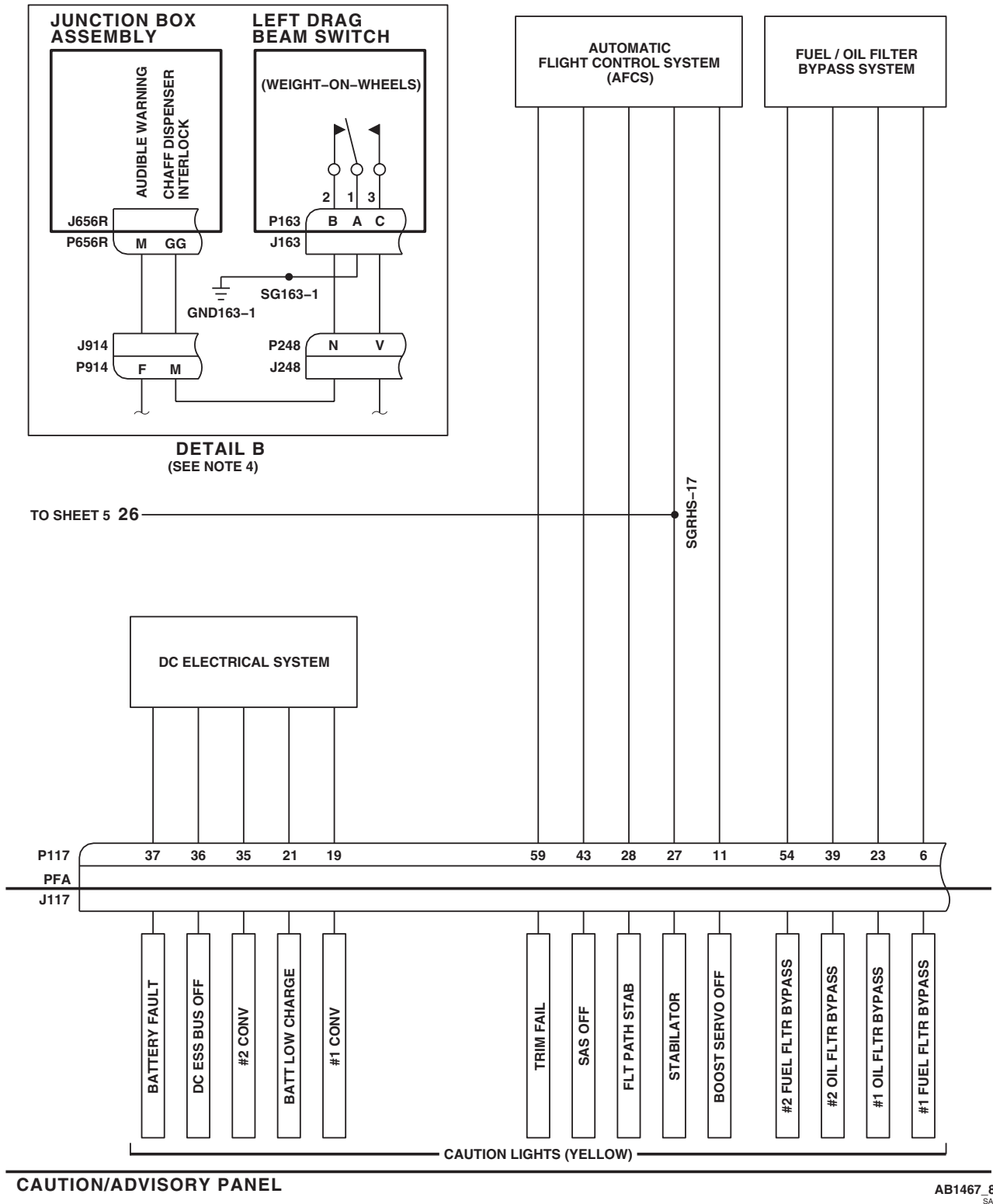


Figure 2. Caution/Advisory Warning System Schematic Diagram. (Sheet 8 of 10)

SCHEMATICS - Continued

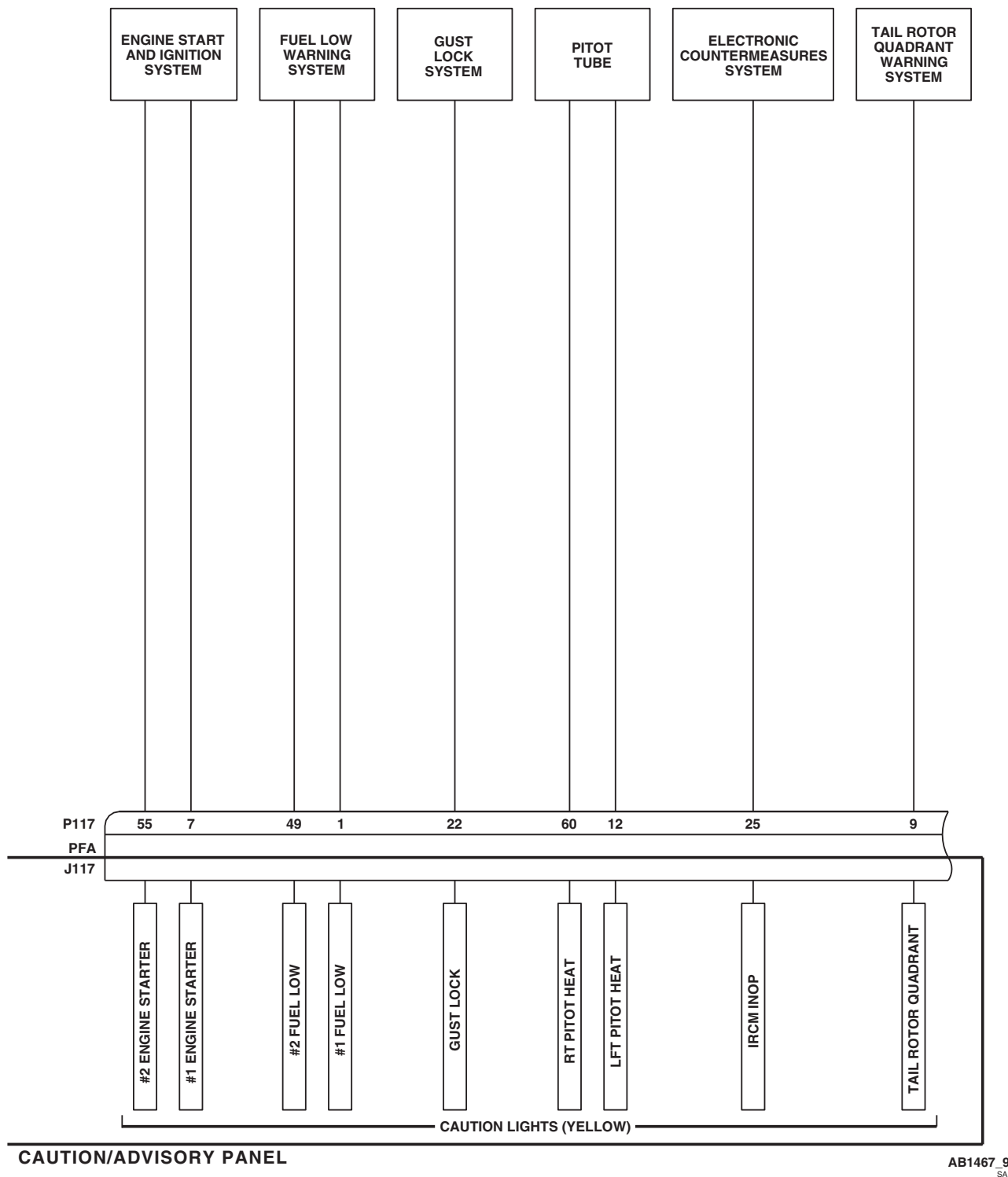
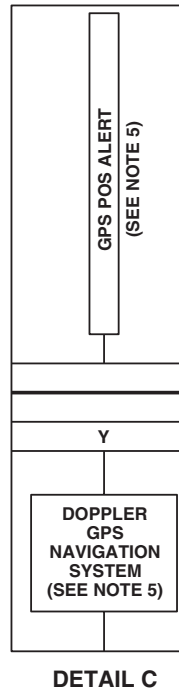


Figure 2. Caution/Advisory Warning System Schematic Diagram. (Sheet 9 of 10)

SCHEMATICS - Continued



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Figure 2. Caution/Advisory Warning System Schematic Diagram. (Sheet 10 of 10)

END OF WORK PACKAGE

UNIT LEVEL
INSTRUMENT SYSTEMS
DIGITAL CLOCK

This WP supersedes WP 0091 00, dated 17 April 2006.

INITIAL SETUP:**Tools and Special Tools**

Electrical Repairer Toolkit, SC 5180-99-B06
Multimeter, AN/PSM-45A

WP 0795 00

WP 0824 00

WP 0875 00

WP 1685 00

WP 1747 00

Personnel Required

Avionics Mechanic MOS 15N (1)

References

TM 1-1520-237-10
WP 0104 00
WP 0794 00

Equipment Condition

External Electrical Power Available, WP 1685 00
or APU Operational, TM 1-1520-237-10

SPECIAL INSTRUCTIONS**NOTE**

See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

The digital clock operational/troubleshooting procedure is subdivided as follows:

- Digital Clock
- LC-6 Digital Clock

DIGITAL CLOCK OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step****NOTE**

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).
- When power is initially applied, the LC-6 clock must be initialized by pushing all three buttons simultaneously.
- The LC-6 digital clock has provisions for an internal battery to maintain time when the aircraft power is off. If the clock does not keep time when aircraft electrical power is not applied, the battery must be replaced (WP 0795 00).

1. Turn on electrical power.

DIGITAL CLOCK OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. Make sure all circuit breakers are pushed in except those noted as being pulled for maintenance or safety.
3. Digital clock shall go on and automatically execute a segment test during the first 5 seconds of operation.

Indication/Condition

Verify segments illuminate.

Corrective Action

1. If only some segments illuminate, replace digital clock (WP 0794 00).
2. If no segments illuminate, go to SEGMENTS DO NOT ILLUMINATE, in this work package.

Step

4. Turn INST LT PILOT FLT and/or CPLT FLT INST LTS control to BRT.

Indication/Condition

Clock integral lighting shall go bright.

Corrective Action

If Indication/Condition is not as specified, go to INTEGRAL LIGHTING DOES NOT GO ON, in this work package.

Step

5. Turn INST LT PILOT FLT and/or CPLT FLT INST LTS control to OFF.

Indication/Condition

Clock integral lighting shall go off.

Corrective Action

If Indication/Condition is not as specified, replace INST LT PILOT FLT and/or CPLT FLT INST LTS dimmer control (WP 0875 00).

Step

6. Depress SEL and CTRL buttons simultaneously. When both hour digits flash, hold down CTRL button until desired hour is reached.
7. Press SEL button. When both minute digits are flashing, hold down CTRL button until desired minutes are reached.
8. Press SEL button. When both second digits are flashing, hold down CTRL button until desired seconds are reached.
9. Press SEL button.

Indication/Condition

Selected time shall be displayed and clock timer will begin.

Corrective Action

If Indication/Condition is not as specified, replace digital clock (WP 0794 00).

Step

10. With digital clock in clock mode, press SEL to initiate elapsed time mode.
11. Press CTRL button.

DIGITAL CLOCK OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Elapsed timer shall begin.

Corrective Action

If Indication/Condition is not as specified, replace digital clock (WP 0794 00).

Step

12. Press CTRL button.

Indication/Condition

Elapsed timer shall stop.

Corrective Action

If Indication/Condition is not as specified, replace digital clock (WP 0794 00).

Step

13. Press CTRL button again.

Indication/Condition

Elapsed timer shall reset.

Corrective Action

If Indication/Condition is not as specified, replace digital clock (WP 0794 00).

Step

14. Press CTRL button again.

Indication/Condition

Elapsed timer shall begin at 0.0 seconds.

Corrective Action

If Indication/Condition is not as specified, replace digital clock (WP 0794 00).

Step

15. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

LC-6 DIGITAL CLOCK OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

NOTE

- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Turn on electrical power.
2. Make sure all circuit breakers are pushed in except those noted as being pulled for maintenance or safety.
3. Digital clock shall turn on and display either the current local time, if previously set, or 12:00:00, if not.

Indication/Condition

Verify LCD display illuminates.

Corrective Action

1. If only some segments illuminate, replace digital clock (WP 0794 00).
2. If no segments illuminate, go to SEGMENTS DO NOT ILLUMINATE, in this work package.

Step

4. Turn INST LT PILOT FLT and/or CPLT FLT INST LTS control to BRT.

Indication/Condition

Clock integral lighting shall go bright.

Corrective Action

If Indication/Condition is not as specified, go to INTEGRAL LIGHTING DOES NOT GO ON, in this work package.

Step

5. Turn INST LT PILOT FLT and/or CPLT FLT INST LTS control to OFF.

Indication/Condition

Clock integral lighting shall go off.

Corrective Action

If Indication/Condition is not as specified, replace INST LT PILOT FLT and/or CPLT FLT INST LTS dimmer control (WP 0875 00).

Step

6. Press the MODE button until LT is displayed under the time.
7. To set the hour, press the SET button once, then press the ADV button and hold until correct hour is displayed.
8. To set the minutes, press the SET button once, so that minutes and seconds are displayed. Press the ADV button, until the desired minute is displayed, then press the SET button to hold the time.
9. When the time standard shows the exact time displayed on the LC-6, press the ST/SP button to activate time start.

LC-6 DIGITAL CLOCK OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Correct local time is displayed, and seconds count up.

Corrective Action

If Indication/Condition is not as specified, replace digital clock (WP 0794 00).

Step

10. Press the MODE button until UTC is displayed under the time.
11. To set the hour, press the SET button once, then press the ADV button and hold until correct hour is displayed.

Indication/Condition**Corrective Action****Step**

12. Minutes and seconds do not need to be set if local time is correctly set.
13. Press SET button to display minutes and seconds, then press SET again to activate the complete display.

Indication/Condition

Correct UTC time is displayed, and seconds count up.

Corrective Action

If Indication/Condition is not as specified, replace digital clock (WP 0794 00).

Step**NOTE**

Flight mode is not functional.

14. Press the MODE button until SW is displayed under the time.

Indication/Condition**Corrective Action****Step**

15. Press RST button to set time at zero (0). Press ST/SP button to start the timer. To stop, press the ST/SP button again. Press the RST button to clear the timer.

Indication/Condition**NOTE**

If the RST button is pressed when the timer is counting, timer will continue counting to zero.

RST sets timer to zero, and ST/SP starts and stops the timer.

Corrective Action

If Indication/Condition is not as specified, replace digital clock (WP 0794 00).

LC-6 DIGITAL CLOCK OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

16. Press the MODE button until DC is displayed under the time.
17. Press SET button twice. Hour display will show zero. Press ADV button and hold until the desired hour is displayed.
18. Press SET button again, and press the ADV button until correct minutes are displayed.
19. Press the SET button again, so that zeros show in the seconds display. Press SET button.
20. Press the ST/SP button to start counting down.
21. When the Down Counter reaches zero, the display will flash. Press the ST/SP button to stop flashing.

Indication/Condition

Counter counts down, and ST/SP starts and stops the timer.

Corrective Action

If Indication/Condition is not as specified, replace digital clock (WP 0794 00).

Step

22. Turn off electrical power.

Indication/Condition

Power shall be off.

Corrective Action

None Required

SEGMENTS DO NOT ILLUMINATE**SYMPTOM**

Verify segments illuminate.

MALFUNCTION

Segments do not illuminate.

CORRECTIVE ACTION

- 1. Check that segments on pilot's clock illuminate.**
 - a. If segments illuminate, go to **5**.
 - b. If segments do not illuminate, go to **2**.
- 2. Check for 28 vdc between P186-1 and ground.**
 - a. If voltage is as specified, go to **3**.
 - b. If voltage is not as specified, go to **4**.
- 3. Check continuity between:**
 - **GG33-8 and P186-2**
 - **GG33-8 and P186-5**
 - a. If continuity is present, replace digital clock (WP 0794 00). Go to **9**.

SEGMENTS DO NOT ILLUMINATE - Continued

- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9**.
- 4. Check for 28 vdc between PILOT ALTM circuit breaker terminal 1 and ground.**
 - a. If voltage is as specified, repair/replace wiring between P186-1 and circuit breaker (WP 1747 00). Go to **9**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **9**.
- 5. Check that segments on copilot's clock illuminate.**
 - a. If segments illuminate, continue operational/troubleshooting procedure.
 - b. If segments do not illuminate, go to **6**.
- 6. Check for 28 vdc between P195-1 and ground.**
 - a. If voltage is as specified, go to **7**.
 - b. If voltage is not as specified, go to **8**.
- 7. Check continuity between:**
 - **GG33-1 and P195-2**
 - **GG33-1 and P195-5**
 - a. If continuity is present, replace digital clock (WP 0794 00). Go to **9**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **9**.
- 8. Check for 28 vdc between CPLT ALTM circuit breaker terminal 1 and ground.**
 - a. If voltage is as specified, repair/replace wiring between circuit breaker and P195-1 (WP 1747 00). Go to **9**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **9**.
- 9. Procedure completed.**

INTEGRAL LIGHTING DOES NOT GO ON**SYMPTOM**

Clock integral lighting shall go bright.

MALFUNCTION

Integral lighting does not go on.

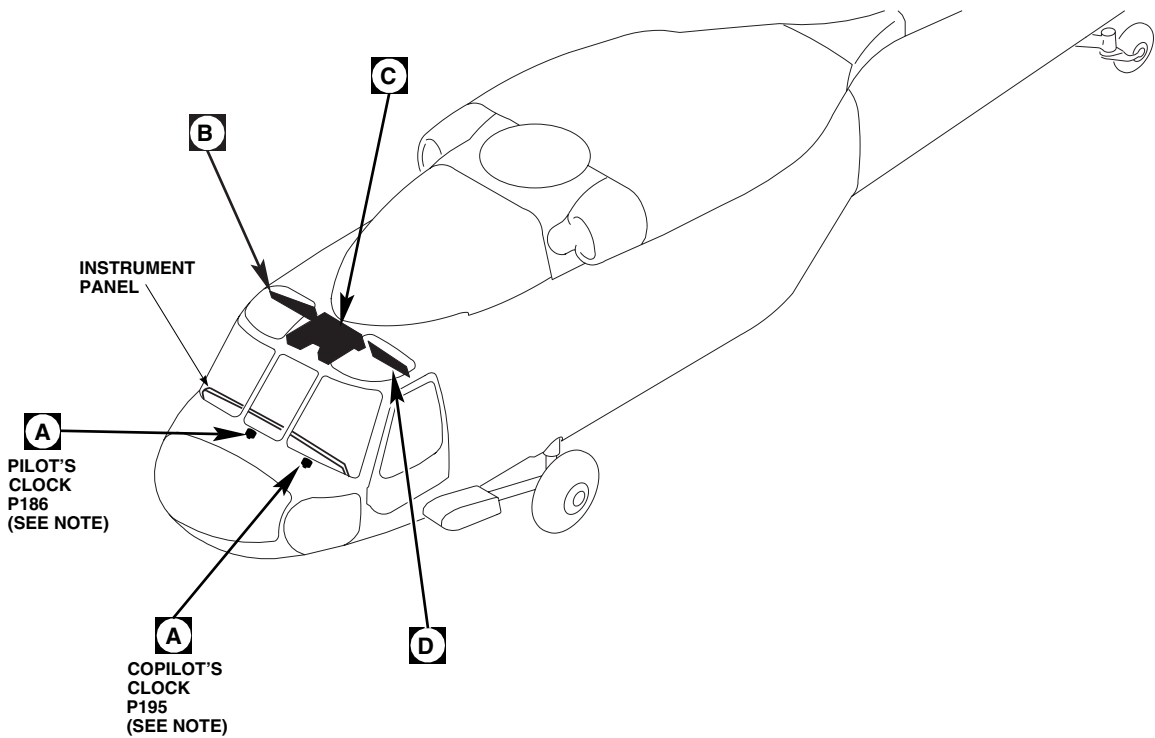
CORRECTIVE ACTION

- 1. Check that pilot's digital clock integral lighting goes on.**
 - a. If pilot's digital clock integral lighting goes on, go to **7**.
 - b. If pilot's digital clock integral lighting does not go on, go to **2**.
- 2. Check for 5 vac between P186-3 and ground.**
 - a. If voltage is as specified, go to **3**.
 - b. If voltage is not as specified, go to **4**.
- 3. Check continuity between P186-4 and ground.**
 - a. If continuity is present, replace digital clock (WP 0794 00). Go to **13**.

INTEGRAL LIGHTING DOES NOT GO ON - Continued

- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 4. Check for 5 vac between INST LT PILOT FLT control terminal LV and ground.**
 - a. If voltage is as specified, repair/replace wiring between control and P186-3 (WP 1747 00). Go to **13**.
 - b. If voltage is not as specified, go to **5**.
- 5. Check for 115 vac between HV terminal of INST LT PILOT FLT control and ground.**
 - a. If voltage is as specified, replace control (WP 0875 00). Go to **13**.
 - b. If voltage is not as specified, go to **6**.
- 6. Check for 115 vac between LIGHTS PLT FLT circuit breaker terminal 1 and ground.**
 - a. If voltage is as specified, repair/replace wiring between circuit breaker and HV terminal of INST LT PILOT FLT control (WP 1747 00). Go to **13**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **13**.
- 7. Check that copilot's digital clock integral lighting goes on.**
 - a. If copilot's digital clock integral lighting goes on, continue operational/troubleshooting procedure.
 - b. If copilot's digital clock integral lighting does not go on, go to **8**.
- 8. Check for 5 vac between P195-3 and ground.**
 - a. If voltage is as specified, go to **9**.
 - b. If voltage is not as specified, go to **10**.
- 9. Check continuity between P195-4 and ground.**
 - a. If continuity is present, replace digital clock (WP 0794 00). Go to **13**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **13**.
- 10. Check for 5 vac between CPLT FLT INST LTS control terminal LV and ground.**
 - a. If voltage is as specified, repair/replace wiring between control and P195-3 (WP 1747 00). Go to **13**.
 - b. If voltage is not as specified, go to **11**.
- 11. Check for 115 vac between HV terminal of CPLT FLT INST LTS control and ground.**
 - a. If voltage is as specified, replace CPLT FLT INST LTS control (WP 0875 00). Go to **13**.
 - b. If voltage is not as specified, go to **12**.
- 12. Check for 115 vac between LIGHTS CPLT FLT circuit breaker terminal 1 and ground.**
 - a. If voltage is as specified, repair/replace wiring between circuit breaker and control (WP 1747 00). Go to **13**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **13**.
- 13. Procedure completed.**

LOCATION



NOTE

DIGITAL OR LC-6 DIGITAL MAY BE INSTALLED.

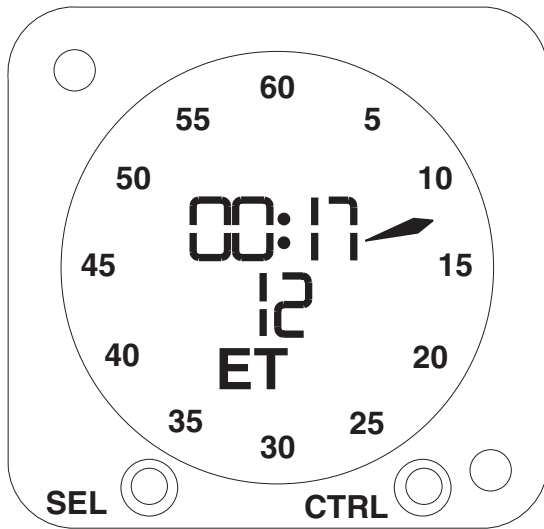
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P1R / J1R	INSTRUMENT PANEL / AVIONIC NOSE DISCONNECT
P20R / J20R	AVIONICS NOSE DISCONNECT
P111 / J111	INSTRUMENT PANEL DISCONNECT
P112 / J112	INSTRUMENT PANEL DISCONNECT
P113 / J113	INSTRUMENT PANEL DISCONNECT
P126 / J126	PILOT'S CIRCUIT BREAKER PANEL DISCONNECT
P127 / J127	COPILOT'S CIRCUIT BREAKER PANEL DISCONNECT

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P186 / J186	PILOT'S DIGITAL CLOCK DISCONNECT (SEE NOTE)
P195 / J195	COPILOT'S DIGITAL CLOCK DISCONNECT (SEE NOTE)
P246 / J246	UPPER CONSOLE DISCONNECT
P249 / J249	COPILOT'S CIRCUIT BREAKER PANEL DISCONNECT
P266 / J266	PILOT'S CIRCUIT BREAKER PANEL DISCONNECT
P280 / J280	UPPER CONSOLE DISCONNECT

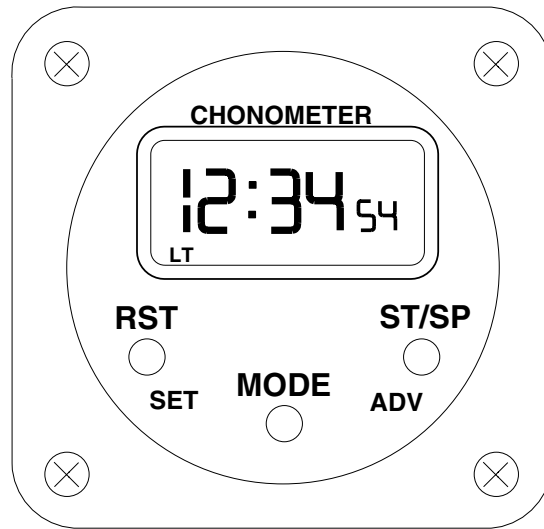
AA8920_1B
SA

Figure 1. Digital Clock Location Diagram. (Sheet 1 of 3)

LOCATION - Continued



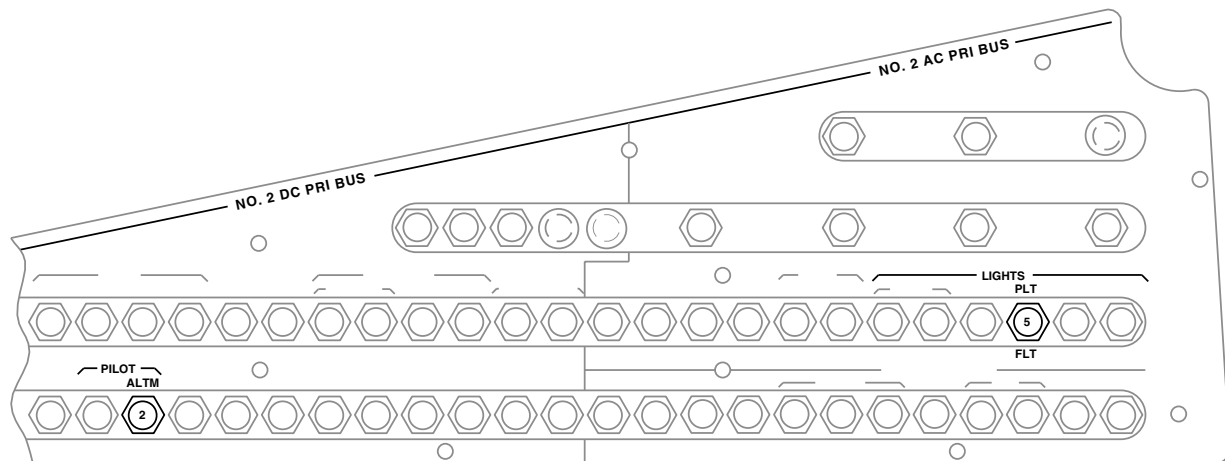
DIGITAL CLOCK
(SEE NOTE)



LC-6 DIGITAL CLOCK
(SEE NOTE)

A

B

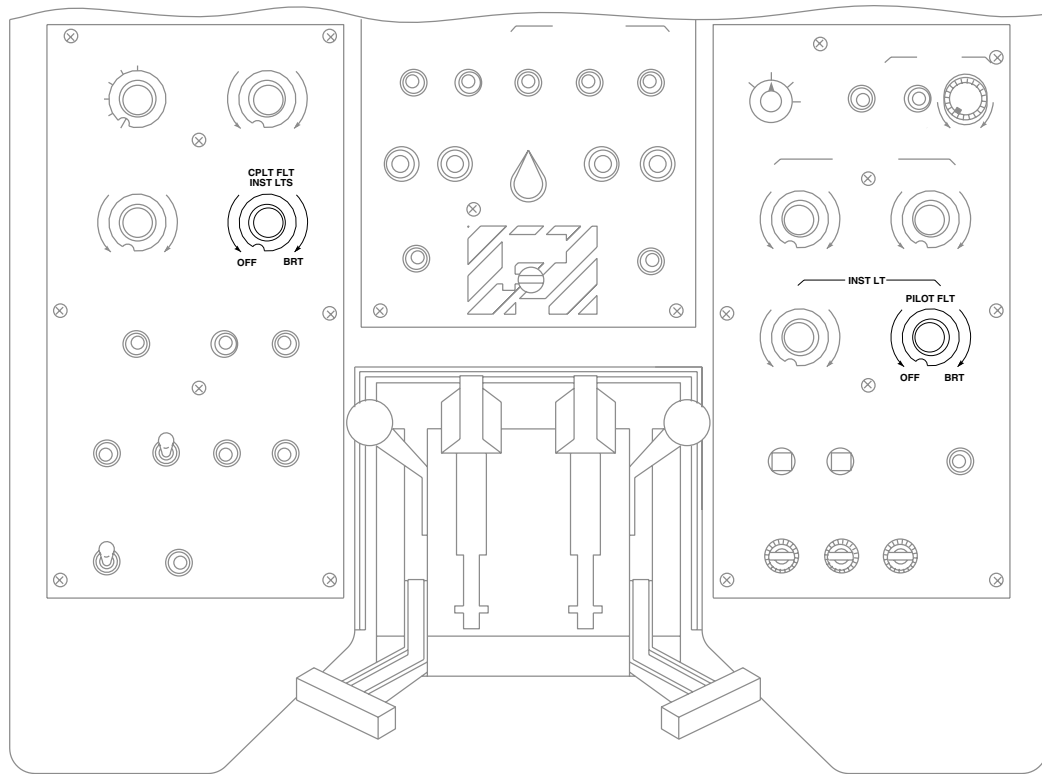


PILOT'S CIRCUIT BREAKER PANEL

AA8920_2B
SA

Figure 1. Digital Clock Location Diagram. (Sheet 2 of 3)

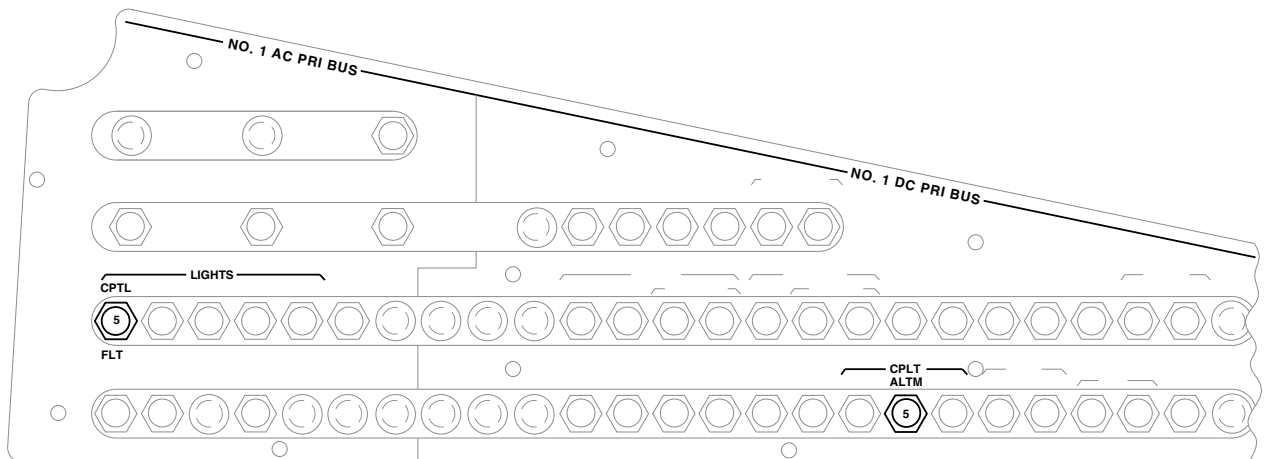
LOCATION - Continued



UPPER CONSOLE

C

D



COPILLOT CIRCUIT BREAKER PANEL

AA8920_3
SA

Figure 1. Digital Clock Location Diagram. (Sheet 3 of 3)

SCHEMATICS

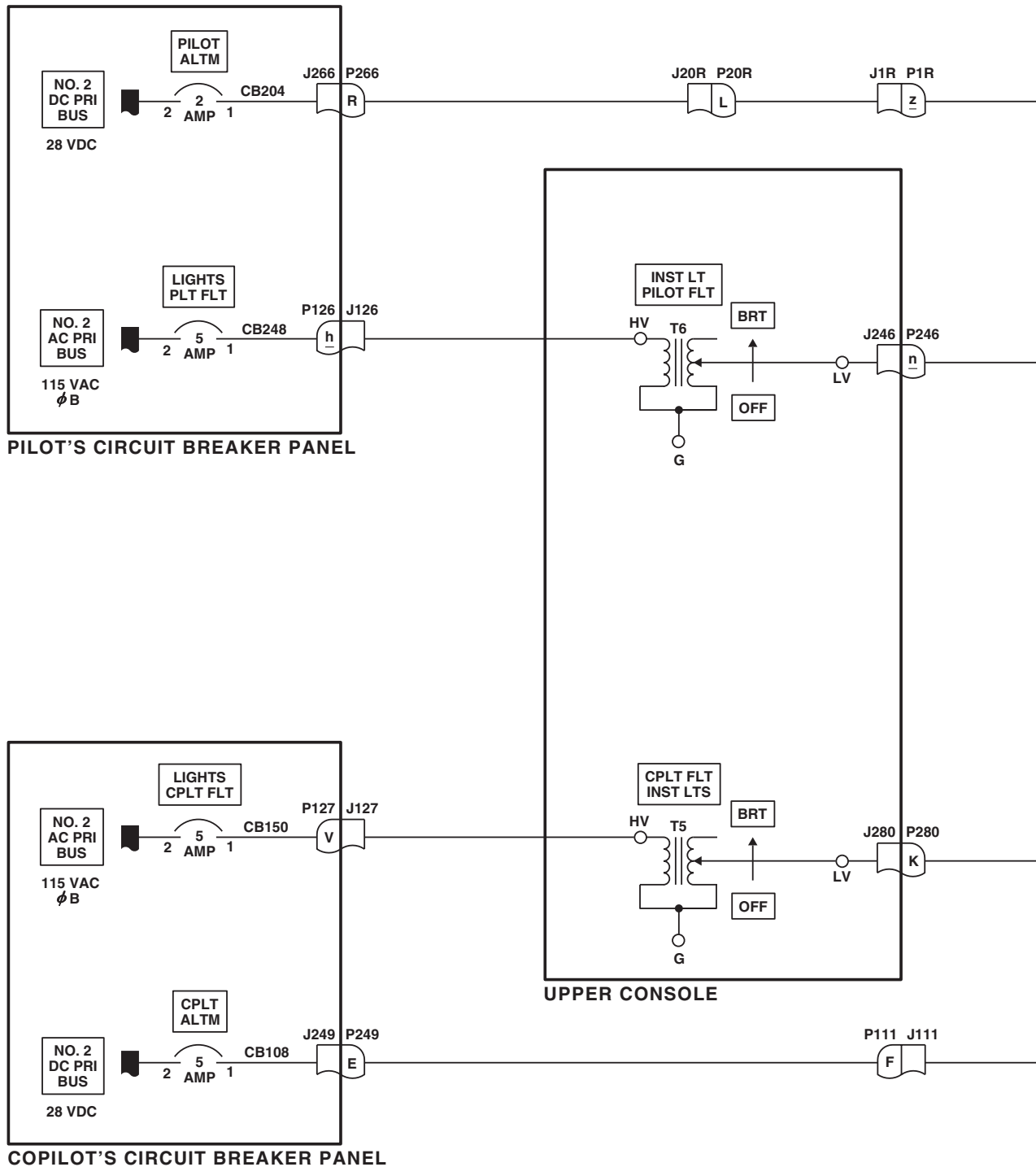
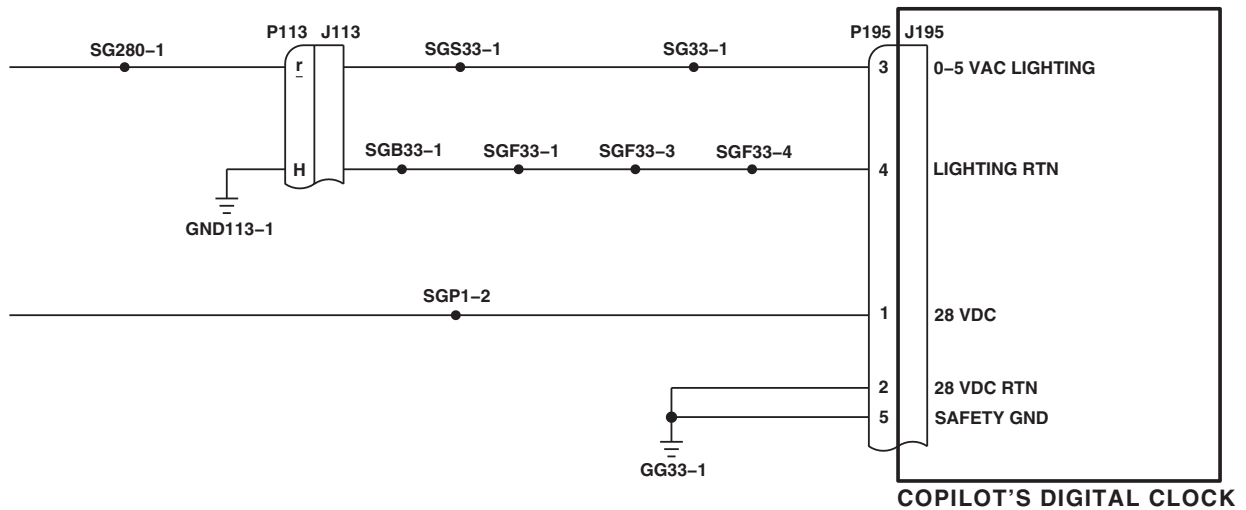
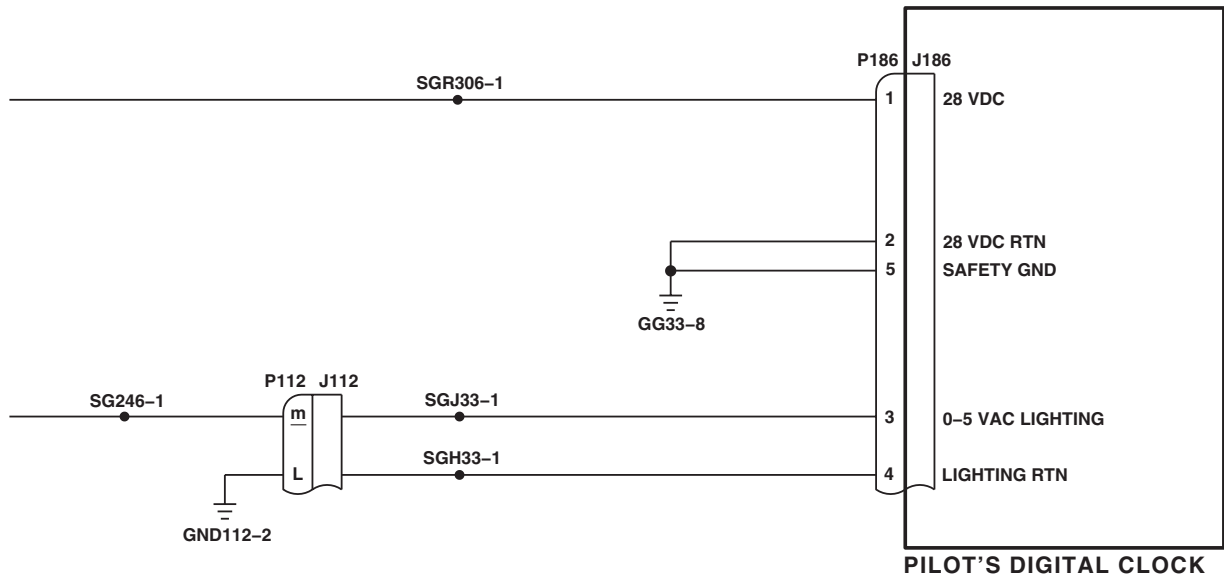
AB1540_1
SA

Figure 2. Digital Clock Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued



AB1540_2
SA

Figure 2. Digital Clock Schematic Diagram. (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL**INSTRUMENT SYSTEMS**

CENTRAL DISPLAY UNIT (70450-01043-122) (AVIM)

INITIAL SETUP:**Test Equipment**

Bench Test Set, 245-476854-000
Multimeter, AN/PSM-45A

WP 0779 00

WP 0780 00

WP 0802 00

WP 0803 00

WP 0804 00

Tools and Special Tools

Electrical Repairer Toolkit, SC 5180-99-B06

References

TM 11-1520-253-13&P
WP 0777 00

Equipment Condition

Bench Power Available, 28 vdc, 115 vac, 400 Hz,
Single-Phase

SPECIAL INSTRUCTIONS**NOTE**

See Figure 1 for component location diagram and Figure 2 for central display unit test setup.

The central display unit (CDU) operational/troubleshooting procedure is subdivided as follows:

- Setup/Power On
- Digital and Analog Update
- CDU Push To Test
- Display Auto-Dim
- Logic Power Supply Redundancy
- Regulator Voltage And Single SDC
- Analog Parameter Indication
- Digital Parameter Indication
- No. 1 Fuel Quantity (analog)
- No. 2 Fuel Quantity (analog)
- Main Transmission Oil Temperature (analog)
- Main Transmission Oil Pressure (analog)
- Parameter Measurement (digital)
- Failure Mode

SPECIAL INSTRUCTIONS - Continued

- Shutdown

SETUP/POWER ON OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step**

1. Make sure simulator tester INPUT POWER switch is OFF.
2. Connect cable assembly, P/N 217-419742-000, between simulator tester INTERCONNECT connector and unit tester J5.
3. Connect simulator tester dc power connector to 28 vdc bench power.
4. Connect simulator tester ac power connector to 115 vac, 400 Hz, single-phase bench power.
5. Place simulator tester INPUT POWER switch to ON.

Indication/Condition

Simulator tester INPUT POWER 115 VAC and 28 VDC lamps shall go on.

Corrective Action

If Indication/Conditions are not as specified, troubleshoot bench test set (TM 55-4920-412-13&P).

Step

6. Momentarily press simulator tester LAMP TEST switch.

Indication/Condition

- a. All simulator tester lamps shall go on when LAMP TEST switch is pressed.
- b. All UNIT tester lamps shall go on when LAMP TEST switch is pressed.

Corrective Action

If Indication/Conditions are not as specified, troubleshoot bench test set (TM 55-4920-412-13&P).

Step

7. Place simulator tester INPUT POWER switch to OFF.
8. Connect cable assembly, P/N 217-419744-000, between unit tester J1 and CDU J1.
9. Connect cable assembly, P/N 217-419748-000, between unit tester J2 and CDU J2.
10. Connect cable assembly, P/N 217-419748-000, between unit tester J3 and CDU J3.
11. Connect cable assembly, P/N 217-419744-000, between unit tester J4 and CDU J4.
12. Place simulator tester switches as follows:

SWITCH/CONTROL	POSITION
All CAL-NOR switches	NOR
All HI-LO switches	LO

SETUP/POWER ON OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

SWITCH/CONTROL	POSITION
All NOR controls	Fully counterclockwise
INTEGRAL LIGHTING	INT

13. Place unit tester DISPLAY CONTROLS section switches and controls as follows:

SWITCH/CONTROL	POSITION
DIGITAL SET	0
DIGITS	ON
DISPLAY UPDATE	ON
ANALOG SET	STOP
ROTOR OVER-SPEED 127%	Off (down)
ROTOR OVER-SPEED 137%	Off (down)
ROTOR OVER-SPEED 142%	Off (down)

14. Place unit tester SIGNAL DATA CONVERTER CONVERSION switch to 1.

15. Place unit tester CENTRAL DISPLAY UNIT section switches and controls as follows:

SWITCH/CONTROL	POSITION
PILOTS	Fully counterclockwise
COPILOTS	Fully counterclockwise
P/S CONT	NORMAL
SDC FAIL	Off (center)

16. Position unit tester INTER CAL SELECT switch to OFF.

17. Place CDU - DIGITS switch to ON and turn DIM control fully clockwise.

18. Place simulator tester INPUT POWER switch to ON.

Indication/Condition

- a. Simulator tester INPUT POWER 115 VAC and 28 VDC lamps shall go on.
- b. After about 3 seconds, CENTRAL DISPLAY UNIT SENSING A and B lamps on unit tester shall be on.
- c. CDU integral panel lighting shall go on without any dark spots.

SETUP/POWER ON OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition b. is not as specified, CDU chassis assembly is malfunctioning.
2. If Indication/Condition c. is not as specified, replace CDU faceplate (WP 0777 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

19. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**DIGITAL AND ANALOG UPDATE OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place unit tester switches as follows:

SWITCH	POSITION
INTER CAL SELECT	21
DISPLAY UPDATE	ON
DIGITAL SET	5
DIGITS	ON
ANALOG SET	FAST

Indication/Condition

- a. All CDU vertical scales shall continually cycle from minimum to maximum.
- b. TGT 1, TGT 2, Ng SPEED 1, and Ng SPEED 2 digital readouts shall indicate 555.
- c. TOTAL FUEL digital readout shall indicate 1550.

Corrective Action

1. If Indication/Conditions are not as specified, record result obtained and continue with operational/troubleshooting procedure.

Step

3. If no other check is to be done, go to SHUTDOWN, in this work package.

DIGITAL AND ANALOG UPDATE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Shutdown shall be performed.

Corrective Action

None Required

CDU PUSH TO TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Press, then release CDU - PUSH TO TEST switch.

Indication/Condition

With PUSH TO TEST switch pressed:

- All CDU vertical scale lamps shall go on.
- TGT 1, TGT 2, Ng SPEED 1, and Ng SPEED 2 digital readouts shall indicate 888.
- TOTAL FUEL digital readout shall indicate 8888.
- CHAN 1 and CHAN 2 lamps shall go on.
- Unit tester 127%, 137%, and 142% ROTOR OVERSPEED lamps shall be on.

Corrective Action

1. If none of the Indication/Conditions are as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
2. If any vertical scale lamps do not go on, replace display lamp assembly (WP 0780 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
3. If CHAN 1 and CHAN 2 lamps do not go on, replace digital readout module A18 (WP 0804 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
4. If any digital readout is not as specified, replace digital readout module A18 (WP 0804 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
5. If ROTOR OVERSPEED lamps do not go on, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

3. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

DISPLAY AUTO-DIM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.

NOTE

Make sure unit tester INTER CAL SELECT switch is placed to 21.

2. Turn CDU - DIM control fully counterclockwise.

Indication/Condition

CDU display lighting shall go off.

Corrective Action

1. If Indication/Condition is not as specified, replace interface 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

NOTE

Opaque object must remain in place for steps 3. through 13.

3. Cover light sensor, located between CHAN lamps on CDU, with an opaque object.
4. Set multimeter to indicate dc voltage and connect leads between PDU AD/O and SDC COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate less than 0.5 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace interface 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

5. Turn CDU - DIM control clockwise, but not enough to activate the auto-dim override switch.

Indication/Condition

Multimeter shall indicate less than 1.75 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace interface 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

6. Connect voltmeter between CDU AD/P and SDC COM test jacks on unit tester.
7. Adjust unit tester PILOTS control to obtain between 1.95 and 2.05 vdc on voltmeter.

DISPLAY AUTO-DIM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

CDU display lighting shall go on.

Corrective Action

1. If Indication/Condition is not as specified, replace interface 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

8. Connect multimeter between PDU AD/O and SDC COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 3.3 and 4.1 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace interface 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

9. Turn unit tester COPILOTS control fully clockwise.

Indication/Condition

Multimeter shall indicate between 4.9 and 5.9 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace interface 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

10. Slowly turn CDU - DIM control fully counterclockwise.

Indication/Condition

CDU display lighting intensity shall gradually dim.

Corrective Action

1. If Indication/Condition is not as specified, replace interface 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

11. Turn unit tester PILOTS and COPILOTS controls fully counterclockwise.
12. Position CDU - DIM control to mid-position.
13. Remove opaque object from light sensor.
14. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

DISPLAY AUTO-DIM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

None Required

LOGIC POWER SUPPLY REDUNDANCY OPERATIONAL/TROUBLESHOOTING PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place unit tester LOGIC P/S switch to 1.
3. Connect multimeter between SDC +15.5V and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 14 and 21 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

4. Connect multimeter between SDC -15.5V and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between -21 and -14 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

5. Connect multimeter between SDC 8V and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 6.5 and 10.5 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

6. Place unit tester LOGIC P/S switch to 2.
7. Repeat steps 3. through 5.
8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

LOGIC POWER SUPPLY REDUNDANCY OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

None Required

REGULATOR VOLTAGE AND SINGLE SDC OPERATIONAL/TROUBLESHOOTING PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place unit tester LOGIC P/S switch to 1.
3. Connect multimeter between CDU 5V and SDC COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 4.75 and 5.25 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

4. Connect multimeter between CDU +10 and SDC COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 9.5 and 10.5 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

5. Connect multimeter between CDU -10 and SDC COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between -10.5 and -9.5 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

6. Disconnect multimeter.
7. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

REGULATOR VOLTAGE AND SINGLE SDC OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

None Required

ANALOG PARAMETER INDICATION OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester INPUT POWER switch to ON.

NOTE

Cycling rate of vertical scales is controlled by unit tester ANALOG SET switch.

3. Place unit tester ANALOG SET switch to SLOW.

Indication/Condition

- a. FUEL QTY 1 vertical scale on CDU shall continually cycle from 0 to full scale.
- b. At about 250 lbs indication, FUEL QTY 1 lower amber scale segment on CDU shall go off.
- c. FUEL QTY 2 vertical scale on CDU shall continually cycle from 0 to full scale.
- d. At about 250 lbs indication, FUEL QTY 2 lower amber scale segment on CDU shall go off.
- e. XMSN TEMP vertical scale on CDU shall continually cycle from -58°F (-50°C) to full scale.
- f. XMSN PRESS vertical scale on CDU shall continually cycle from 0 to full scale.
- g. At about 37.5 psi, XMSN PRESS lower red and amber scale segments on CDU shall go off.
- h. ENG OIL TEMP 1 vertical scale on CDU shall continually cycle from -58°F (-50°C) to full scale.
- i. Unit tester HIGH OIL 1 TEMP lamp shall go on whenever scale indication is greater than 302°F (150°C).
- j. ENG OIL TEMP 2 vertical scale on CDU shall continually cycle from -58°F (-50°C) to full scale.
- k. Unit tester HIGH OIL 2 TEMP lamp shall go on when scale indication is greater than 302°F (150°C).
- l. ENG OIL PRESS 1 vertical scale on CDU shall continually cycle from 10 psi to full scale.
- m. At about 47.5 psi indication, ENG OIL PRESS 1 lower red and amber scale segments on CDU shall go off.
- n. Unit tester LOW OIL 1 PRESS lamp shall be on when scale indication is less than 25 psi.
- o. ENG OIL PRESS 2 vertical scale on CDU shall continually cycle from 10 psi to full scale.
- p. At about 47.5 psi indication, ENG OIL PRESS 2 lower red and amber scale segments on CDU shall go off.
- q. Unit tester LOW OIL 2 PRESS lamp shall be on when scale indication is less than 25 psi.
- r. TGT TEMP 1 vertical scale on CDU shall continually cycle from 0 to full scale.
- s. TGT TEMP 2 vertical scale on CDU shall continually cycle from 0 to full scale.
- t. Ng SPEED 1 vertical scale on CDU shall continually cycle from 0 to full scale.
- u. Unit tester LOW ENG 1 OUT lamp shall be on when scale indication is less than 55%.
- v. Ng SPEED 2 vertical scale on CDU shall continually cycle from 0 to full scale.
- w. Unit tester LOW ENG 2 OUT lamp shall be on when scale indication is less than 55%.

ANALOG PARAMETER INDICATION OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition a. is not as specified, replace display driver module A3 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
2. If Indication/Condition b. is not as specified, replace display driver module A3 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
3. If Indication/Condition c. is not as specified, replace display driver module A4 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
4. If Indication/Condition d. is not as specified, replace display driver module A4 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
5. If Indication/Condition e. is not as specified, replace display driver module A5 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
6. If Indication/Condition f. is not as specified, replace display driver module A6 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
7. If Indication/Condition g. is not as specified, replace display driver module A6 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
8. If Indication/Condition h. is not as specified, replace display driver module A7 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
9. If Indication/Condition i. is not as specified, check CDU OIL TEMP 1 fuse.
 - a. If fuse is not good, replace fuse (WP 0779 00).
 - b. If fuse is good, replace voltage regulator module A17 (WP 0803 00).
 - c. If trouble remains, the CDU chassis assembly is malfunctioning.
10. If HIGH OIL 1 TEMP lamp does not go off when scale indication is less than 302°F (150°C), replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
11. If Indication/Condition j. is not as specified, replace display driver module A8 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
12. If Indication/Condition k. is not as specified, check CDU OIL TEMP 2 fuse.
 - a. If fuse is not good, replace fuse (WP 0779 00).
 - b. If fuse is good, replace voltage regulator module A17 (WP 0803 00).
 - (1) If trouble remains, the CDU chassis assembly is malfunctioning.
13. If HIGH OIL 2 TEMP lamp does not go off when scale indication is less than 302°F (150°C), replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
14. If Indication/Condition l. is not as specified, replace display driver module A9 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

ANALOG PARAMETER INDICATION OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

15. If Indication/Condition m. is not as specified, replace display driver module A9 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
16. If Indication/Condition n. is not as specified, check CDU OIL PRESS 1 fuse.
 - a. If fuse is not good, replace fuse (WP 0779 00).
 - b. If fuse is good, replace voltage regulator module A17 (WP 0803 00).
 - (1) If trouble remains, the CDU chassis assembly is malfunctioning.
17. If LOW OIL 1 PRESS lamp does not go off when scale indication is more than 25 psi, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
18. If Indication/Condition o. is not as specified, replace display driver module A10 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
19. If Indication/Condition p. is not as specified, replace display driver module A10 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
20. If Indication/Condition q. is not as specified, check CDU OIL PRESS 2 fuse.
 - a. If fuse is not good, replace fuse (WP 0779 00).
 - b. If fuse is good, replace voltage regulator module A17 (WP 0803 00).
 - (1) If trouble remains, the CDU chassis assembly is malfunctioning.
21. If LOW OIL 2 PRESS lamp does not go off when scale indication is more than 25 psi, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
22. If Indication/Condition r. is not as specified, replace display driver module A11 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
23. If Indication/Condition s. is not as specified, replace display driver module A12 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
24. If Indication/Condition t. is not as specified, replace display driver module A13 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
25. If Indication/Condition u. is not as specified, check CDU Ng 1 fuse.
 - a. If fuse is not good, replace fuse (WP 0779 00).
 - b. If fuse is good, replace voltage regulator module A17 (WP 0803 00).
 - (1) If trouble remains, the CDU chassis assembly is malfunctioning.
26. If LOW ENG 1 OUT lamp does not go off when scale indication is more than 55%, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
27. If Indication/Condition v. is not as specified, replace display driver module A14 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
28. If Indication/Condition w. is not as specified, check CDU Ng 2 fuse.

ANALOG PARAMETER INDICATION OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- a. If fuse is not good, replace fuse (WP 0779 00).
- b. If fuse is good, replace voltage regulator module A17 (WP 0803 00).
 - (1) If trouble remains, the CDU chassis assembly is malfunctioning.
29. If LOW ENG 2 OUT lamp does not go off when scale indication is more than 55%, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

4. Place unit tester ANALOG SET switch to STOP when unit tester INTERFACE MONITOR indicates less than 700.
5. Momentarily press unit tester RESET switch.

Indication/Condition

- a. Unit tester LOW ROTOR SPEED lamp shall be on.
- b. Unit tester 127%, 137%, and 142% ROTOR OVERSPEED lamps shall go off.

Corrective Action

1. If Indication/Condition a. is not as specified, check CDU RTR SP fuse.
 - a. If fuse is not good, replace fuse (WP 0779 00).
 - b. If fuse is good, replace voltage regulator module A17 (WP 0803 00).
 - (1) If trouble remains, the CDU chassis assembly is malfunctioning.
2. If Indication/Condition b. is not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

6. Place unit tester ANALOG SET switch to SLOW.

Indication/Condition

- a. Unit tester LOW ROTOR SPEED lamp shall go off when unit tester INTERFACE MONITOR indicates more than 950.
- b. Unit tester 127% ROTOR OVERSPEED lamp shall go on when INTERFACE MONITOR initially indicates between 1269 and 1271, and shall remain on during subsequent INTERFACE MONITOR cycles.
- c. Unit tester 137% ROTOR OVERSPEED lamp shall go on when INTERFACE MONITOR initially indicates between 1369 and 1371, and shall remain on during subsequent INTERFACE MONITOR cycles.
- d. Unit tester 142% ROTOR OVERSPEED lamp shall go on when INTERFACE MONITOR initially indicates between 1419 and 1421, and shall remain on during subsequent INTERFACE MONITOR cycles.

Corrective Action

1. If Indication/Condition a. is not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
2. If Indication/Conditions b. through d. are not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

ANALOG PARAMETER INDICATION OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

7. Place unit tester ANALOG SET switch to STOP when unit tester INTERFACE MONITOR indicates less than 1200.
8. Momentarily press unit tester RESET switch.

Indication/Condition

Unit tester 127%, 137%, and 142%, ROTOR OVERSPEED lamps shall go off.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

9. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**DIGITAL PARAMETER INDICATION OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**
Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Set unit tester DIGITAL SET thumbwheel to following positions:

DIGITAL SET	TOTAL FUEL	1 TGT 21 Ng SPEED 2
0	0	0
1	1110	111
2	2220	222
3	3330	333
4	440	444
5	1550	555
6	2660	666
7	3770	777
8	880	888

DIGITAL PARAMETER INDICATION OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

DIGITAL SET	TOTAL FUEL	1 TGT 21 Ng SPEED 2
9	1990	999

Indication/Condition

CDU TOTAL FUEL, TGT 1, TGT 2, Ng SPEED 1, and Ng SPEED 2 digital readouts shall indicate corresponding numbers.

Corrective Action

1. If Indication/Condition is not as specified, replace digital readout module A18 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

3. Place CDU - DIGITS switch to OFF.

Indication/Condition

CDU digital readouts shall go off.

Corrective Action

1. If Indication/Condition is not as specified, replace digital readout module A18 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

4. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**NO. 1 FUEL QUANTITY (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**
Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place CDU - DIGITS switch to ON.
3. Place simulator tester NO. 1 FUEL QUANTITY CAL-NOR switch to CAL and HI-LO switch to HI.
4. Position unit tester INTER CAL SELECT switch to 15 and DISPLAY UPDATE SWITCH to OFF.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1216 and 1234.

NO. 1 FUEL QUANTITY (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

5. Place simulator tester NO. 1 FUEL QUANTITY HI-LO SWITCH to LO.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 0 and 9.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

6. Place simulator tester NO. 1 FUEL QUANTITY CAL-NOR switch to NOR.
7. Slowly turn simulator tester NO. 1 FUEL QUANTITY variable control fully clockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall increase smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

8. Slowly turn simulator tester NO. 1 FUEL QUANTITY variable control fully counterclockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall decrease smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

9. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

NO. 2 FUEL QUANTITY (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester NO. 2 FUEL QUANTITY CAL-NOR switch to CAL and HI-LO switch to HI.
3. Position unit tester INTER CAL SELECT switch to 16.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1216 and 1234.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

4. Place simulator tester NO. 2 FUEL QUANTITY HI-LO switch to LO.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 0 and 9.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

5. Place simulator tester NO. 2 FUEL QUANTITY CAL-NOR switch to NOR.
6. Slowly turn simulator tester NO. 2 FUEL QUANTITY variable control fully clockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall increase smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

7. Slowly turn simulator tester No. 2 FUEL QUANTITY variable control fully counterclockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall decrease smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

NO. 2 FUEL QUANTITY (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

MAIN TRANSMISSION OIL TEMPERATURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester MAIN XMSN OIL TEMP CAL-NOR switch to CAL and HI-LO switch to HI.
3. Position unit tester INTER CAL SELECT switch to 17.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1216 and 1264.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

4. Place simulator tester MAIN XMSN OIL TEMP HI-LO switch to LO.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 0 and 18.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

5. Place simulator tester MAIN XMSN OIL TEMP CAL-NOR switch to NOR.
6. Slowly turn simulator tester MAIN XMSN OIL TEMP variable control fully clockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall increase smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

MAIN TRANSMISSION OIL TEMPERATURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

7. Slowly turn simulator tester MAIN XMSN OIL TEMP variable control fully counterclockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall decrease smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

MAIN TRANSMISSION OIL PRESSURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester XMSN OIL PRESS CAL-NOR switch to CAL and HI-LO switch to HI.
3. Position unit tester INTER CAL SELECT switch to 18.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1108 and 1132.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

4. Place simulator tester XMSN OIL PRESS HI-LO switch to LO.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 0 and 12.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

5. Place simulator tester XMSN OIL PRESS CAL-NOR switch to NOR.

MAIN TRANSMISSION OIL PRESSURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

6. Slowly turn simulator tester XMSN OIL PRESS variable control fully clockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall increase smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

7. Slowly turn XMSN OIL PRESS variable control fully counterclockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall decrease smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**PARAMETER MEASUREMENT (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester NO. 1 FUEL QUANTITY and NO. 2 FUEL QUANTITY CAL-NOR switches to CAL.
3. Position unit tester INTER CAL SELECT switch to 19.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 0 and 7.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

4. Place simulator tester NO. 1 FUEL QUANTITY HI-LO switch to HI.

PARAMETER MEASUREMENT (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Unit tester INTERFACE MONITOR shall indicate between 1218 and 1232.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

5. Place simulator tester NO. 2 FUEL QUANTITY HI-LO switch to HI.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 2443 and 2457.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

6. Place simulator tester NO. 1 FUEL QUANTITY CAL-NOR switch to NOR.
7. Slowly turn simulator tester NO. 1 FUEL QUANTITY variable control fully clockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall increase smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

8. Slowly turn simulator tester NO. 1 FUEL QUANTITY variable control full counterclockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall decrease smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

9. Position unit tester INTER CAL SELECT switch to OFF.
10. Place unit tester DISPLAY UPDATE switch to ON.
11. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

PARAMETER MEASUREMENT (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

None Required

**FAILURE MODE OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Position unit tester INTER CAL SELECT switch to 21.
3. Place unit tester SDC FAILURE switch to 1 and release.

Indication/Condition

With SDC FAILURE switch to 1, CDU CHAN 1 lamp shall go on.

Corrective Action

If Indication/Condition is not as specified, the display unit chassis assembly is malfunctioning.

Step

4. Place unit tester SDC FAILURE switch to 2 and release.

Indication/Condition

With SDC FAILURE switch to 2, CDU CHAN 2 lamp shall go on.

Corrective Action

If Indication/Condition is not as specified, the display unit chassis assembly is malfunctioning.

Step

5. Place unit tester DISPLAY UPDATE switch to OFF.

Indication/Condition

- a. All CDU vertical displays shall blank out.
- b. All CDU digital displays shall blank out.
- c. CDU CHAN 1 and CHAN 2 lamps shall go on.

Corrective Action

If Indication/Condition is not as specified, display unit chassis assembly is malfunctioning.

Step

6. Place unit tester ANALOG SET switch to STOP when INTERFACE MONITOR indicates less than 930.
7. Press, then release unit tester RESET switch.

Indication/Condition

Unit tester 127%, 137%, and 142%, lamps shall go off.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A17 (WP 0803 00).

FAILURE MODE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Place simulator tester INPUT POWER switch to OFF.
2. Disconnect cable assemblies.

Indication/Condition

Cable assemblies shall be disconnected.

Corrective Action

None Required

LOCATION

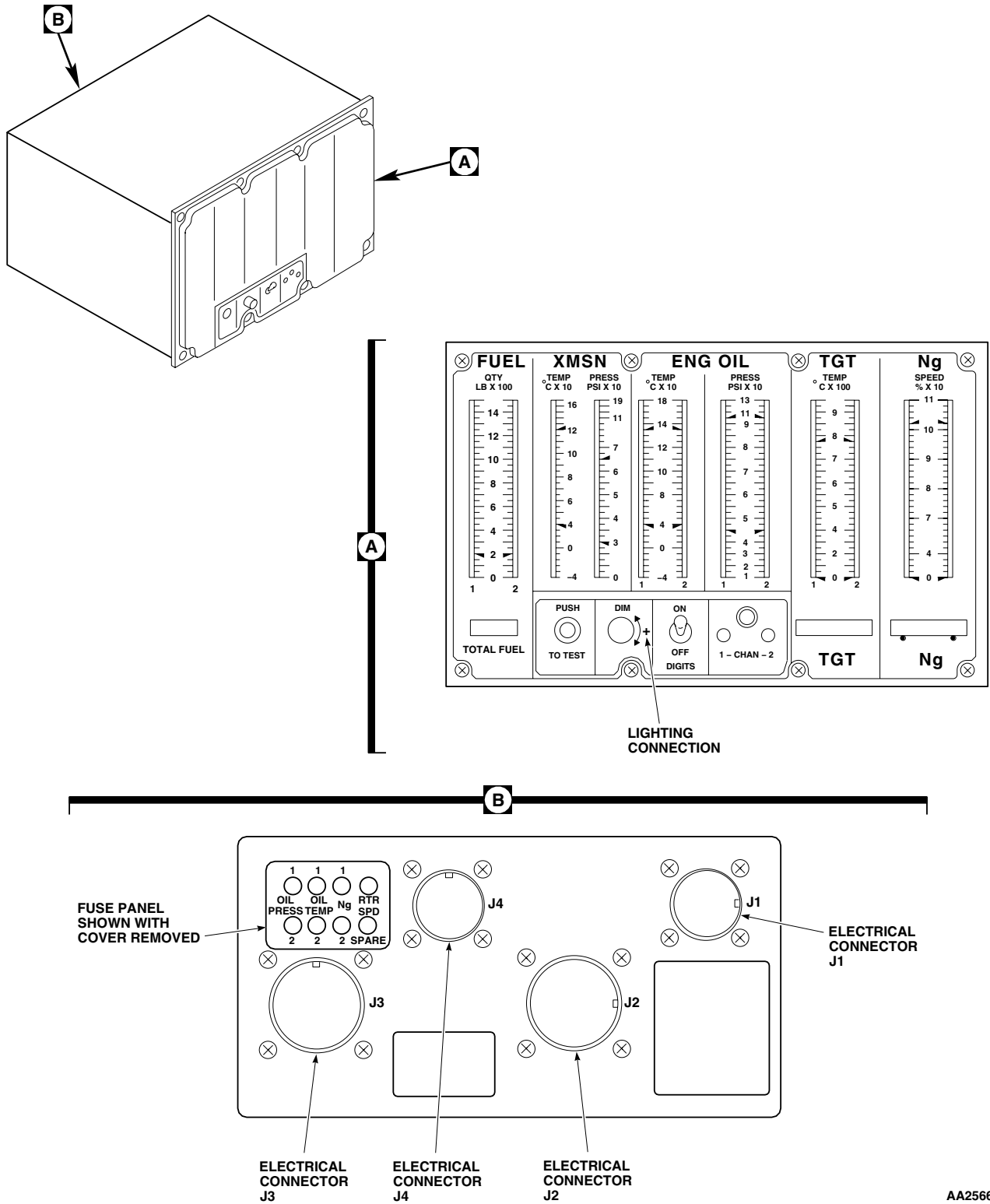
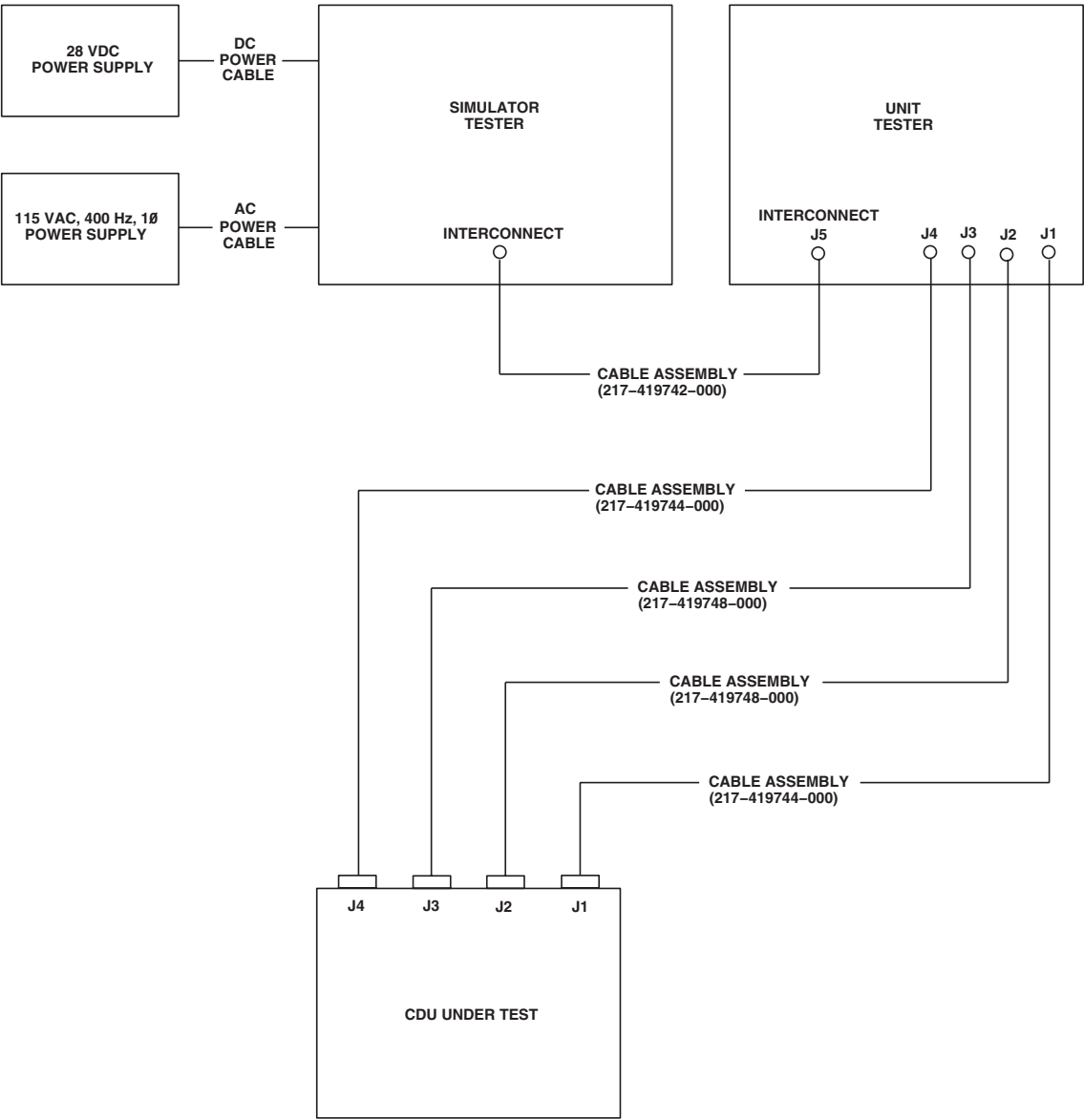


Figure 1. Central Display Unit (CDU).

AA2566
SA

TEST SETUP



AA6578
SA

Figure 2. Central Display Unit Test Setup.

END OF WORK PACKAGE

UNIT LEVEL
INSTRUMENT SYSTEMS

CENTRAL DISPLAY UNIT (70450-01043-125, 70450-21943-118, OR 70450-01916-105) (AVIM)

INITIAL SETUP:**Test Equipment**

Bench Test Set, 245-476854-001
Multimeter, AN/PSM-45A

WP 0780 00

WP 0802 00

WP 0803 00

WP 0804 00

Tools and Special Tools

Electrical Repairer Toolkit, SC 5180-99-B06

Equipment Condition

Bench Power Available, 28 vdc, 115 vac, 400 Hz,
Single-Phase

References

TM 55-4920-412-13&P
WP 0779 00

SPECIAL INSTRUCTIONS**NOTE**

See Figure 1 for component location diagram.

The central display unit (CDU) operational/troubleshooting procedure is subdivided as follows:

- Setup/Power On
- Digital and Analog Update
- CDU Push To Test
- Display Auto-Dim
- Regulator Voltage And Single SDC
- Analog Parameter Indication
- Digital Parameter Indication
- No. 1 Fuel Quantity (analog)
- No. 2 Fuel Quantity (analog)
- Main Transmission Oil Temperature (analog)
- Main Transmission Oil Pressure (analog)
- Parameter Measurement (digital)
- Failure Mode
- Shutdown

SETUP/POWER ON OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Make sure simulator tester INPUT POWER switch is OFF.
2. Connect cable assembly, 217-419742-000, between simulator tester INTERCONNECT connector and unit tester J5.
3. Connect simulator tester dc power connector to 28 vdc bench power.
4. Connect simulator tester ac power connector to 115 vac, 400 Hz, single-phase bench power.
5. Place simulator tester INPUT POWER switch to ON.

Indication/Condition

Simulator tester INPUT POWER 115 VAC and 28 VDC lamps shall go on.

Corrective Action

If Indication/Condition is not as specified, troubleshoot bench test set (TM 55-4920-412-13&P).

Step

6. Press simulator tester LAMP TEST switch, then release.

Indication/Condition

- a. All simulator tester lamps shall go on when LAMP TEST switch is pressed.
- b. All UNIT tester lamps shall go on when LAMP TEST switch is pressed.

Corrective Action

If Indication/Conditions are not as specified, troubleshoot bench test set (TM 55-4920-412-13&P).

Step

7. Place simulator tester INPUT POWER switch to OFF.
8. Connect cable assembly, P/N 217-419744-000, between unit tester J1 and central display unit (CDU) J1.
9. Connect cable assembly, P/N 217-419748-000, between unit tester J2 and CDU J2.
10. Connect cable assembly, P/N 217-419748-000, between unit tester J3 and CDU J3.
11. Connect cable assembly, P/N 217-419744-000, between unit tester J4 and CDU J4.
12. Place simulator tester switches as follows:

SWITCH/CONTROL	POSITION
All CAL-NOR switches	NOR
All HI-LO switches	LO
All NOR controls	Fully counterclockwise

SETUP/POWER ON OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

SWITCH/CONTROL	POSITION
INTEGRAL	INT
LIGHTING	

13. Place unit tester DISPLAY CONTROLS section switches and controls as follows:

SWITCH/CONTROL	POSITION
DIGITAL SET	0
DIGITS	ON
DISPLAY UPDATE	ON
ANALOG SET	STOP
ROTOR OVER-SPEED 127%	Off (down)
ROTOR OVER-SPEED 137%	Off (down)
ROTOR OVER-SPEED 142%	Off (down)

14. Place unit tester SIGNAL DATA CONVERTER CONVERSION switch to 2.

15. Place unit tester CENTRAL DISPLAY UNIT section switches and controls as follows:

SWITCH/CONTROL	POSITION
PILOTS	Fully counterclockwise
COPILOTS	Fully counterclockwise
P/S CONT	NORMAL
SDC FAIL	Center position

16. Position unit tester INTER CAL SELECT switch OFF.

17. Place CDU - DIGITS switch to ON and turn DIM control fully clockwise.

18. Place simulator tester INPUT POWER switch to ON.

Indication/Condition

- a. Simulator tester INPUT POWER 115 VAC and 28 VDC lamps shall go on.
- b. After approximately 3 seconds, CENTRAL DISPLAY UNIT SENSING A and B lamps on unit tester shall go on.

SETUP/POWER ON OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- c. CDU integral panel lighting shall go on without any dark spots.

Corrective Action

1. If Indication/Condition b. is not as specified, CDU chassis assembly is malfunctioning.
2. If Indication/Condition c. is not as specified, replace CDU faceplate (WP 0779 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

DIGITAL AND ANALOG UPDATE OPERATIONAL/TROUBLESHOOTING PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place unit tester switches as follows:

SWITCH	POSITION
INTER CAL SELECT	21
DISPLAY UPDATE	ON
DIGITAL SET	5
DIGITS	ON
ANALOG SET	FAST

Indication/Condition

- a. All CDU vertical scales shall continually cycle from minimum to maximum.
- b. TGT 1, TGT 2, Ng SPEED 1, and Ng SPEED 2 digital readouts shall indicate 555.
- c. TOTAL FUEL digital readout shall indicate 1550.

Corrective Action

If Indication/Conditions are not as specified, record result obtained and continue with operational/troubleshooting procedure.

Step

3. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

CDU PUSH TO TEST OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Momentarily press CDU - LAMP TEST switch.

Indication/Condition

With LAMP TEST switch pressed:

- All vertical scale lamps shall go on.
- TGT 1, TGT 2, Ng SPEED 1, and Ng SPEED 2 digital readouts shall indicate 888.
- TOTAL FUEL digital readout shall indicate 8888.
- CHAN 1 and CHAN 2 lamps shall go on.
- Unit tester 127%, 137%, and 142% ROTOR OVERSPEED lamps shall be on.

Corrective Action

1. If all Indication/Conditions are not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
2. If Indication/Condition a. is not as specified, replace display unit lamp assembly (WP 0780 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
3. If Indication/Condition b. or c. is not as specified, replace digital readout module A18 (WP 0804 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
4. If Indication/Condition d. is not as specified, replace digital readout module A18 (WP 0804 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
5. If Indication/Condition e. is not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

3. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

DISPLAY AUTO-DIM OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.

NOTE

Make sure unit tester INTER CAL SELECT switch is placed to 21.

2. Turn CDU - DIM control fully counterclockwise.

Indication/Condition

CDU display lighting shall go off.

Corrective Action

1. If Indication/Condition is not as specified, replace interface 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

NOTE

Opaque object must remain in place for steps 3. through 13.

3. Cover light sensor, located between CHAN lamps on CDU, with an opaque object.
4. Set multimeter to indicate dc voltage and connect leads between PDU AD/O and SDC COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate less than 0.5 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace interface 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

5. Turn CDU - DIM control clockwise, but not enough to activate the auto-dim override switch.

Indication/Condition

Multimeter shall indicate less than 1.75 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace interface 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

6. Connect voltmeter between CDU AD/P and SDC COM test jacks on unit tester.
7. Adjust unit tester PILOTS control to obtain between 1.95 and 2.05 vdc on voltmeter.

DISPLAY AUTO-DIM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

CDU display lighting shall go on.

Corrective Action

1. If Indication/Condition is not as specified, replace interface 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

8. Connect multimeter between PDU AD/O and SDC COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 3.3 and 4.1 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace interface 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

9. Turn unit tester COPILOTS control fully clockwise.

Indication/Condition

Multimeter shall indicate between 4.9 and 5.9 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace interface 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

10. Slowly turn CDU - DIM control fully counterclockwise.

Indication/Condition

CDU display lighting intensity shall gradually dim.

Corrective Action

If Indication/Condition is not as specified, replace interface 1 module A15 (WP 0803 00).

Step

11. Turn unit tester PILOTS and COPILOTS controls fully counterclockwise.
12. Position CDU - DIM control to mid-position.
13. Remove opaque object from light sensor.
14. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

DISPLAY AUTO-DIM OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

None Required

**REGULATOR VOLTAGE AND SINGLE SDC OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place unit tester LOGIC P/S switch to 1.
3. Connect multimeter between CDU 5V and SDC COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 4.75 and 5.25 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

4. Connect multimeter between CDU +10 and SDC COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 9.5 and 10.5 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

5. Connect multimeter between CDU -10 and SDC COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between -10.5 and -9.5 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

6. Disconnect multimeter.
7. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

REGULATOR VOLTAGE AND SINGLE SDC OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

None Required

ANALOG PARAMETER INDICATION OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester INPUT POWER switch to ON.
3. Place unit tester ANALOG SET switch to SLOW.

Indication/Condition**NOTE**

Cycling rate of vertical scales is controlled by unit tester ANALOG SET switch.

- a. FUEL QTY 1 vertical scale shall continually cycle from 0 to full scale.
- b. At approximately 250 lbs indication, FUEL QTY 1 lower amber scale segment shall go off.
- c. FUEL QTY 2 vertical scale shall continually cycle from 0 to full scale.
- d. At approximately 250 lbs indication, FUEL QTY 2 lower amber scale segment shall go off.
- e. XMSN TEMP vertical scale shall continually cycle from -58°F (-50°C) to full scale.
- f. XMSN PRESS vertical scale shall continually cycle from 0 to full scale.
- g. At approximately 37.5 psi, XMSN PRESS lower red and amber scale segments shall go off.
- h. ENG OIL TEMP 1 vertical scale shall continually cycle from -58°F (-50°C) to full scale.
- i. Unit tester HIGH OIL 1 TEMP lamp shall go on whenever scale indication is greater than 302°F (150°C).
- j. ENG OIL TEMP 2 vertical scale shall continually cycle from -58°F (-50°C) to full scale.
- k. Unit tester HIGH OIL 2 TEMP lamp shall go on when scale indication is greater than 302°F (150°C).
- l. ENG OIL PRESS 1 vertical scale shall continually cycle from 10 psi to full scale.
- m. At approximately 26 psi indication, ENG OIL PRESS 1 lower red and amber scale segments shall go off.
- n. Unit tester LOW OIL 1 PRESS lamp shall be on when scale indication is less than 24 psi.
- o. ENG OIL PRESS 2 vertical scale shall continually cycle from 10 psi to full scale.
- p. At approximately 26 psi indication, ENG OIL PRESS 2 lower red and amber scale segments shall go off.
- q. Unit tester LOW OIL 2 PRESS lamp shall be on when scale indication is less than 24 psi.
- r. TGT TEMP 1 vertical scale shall continually cycle from 0 to full scale.
- s. TGT TEMP 2 vertical scale shall continually cycle from 0 to full scale.
- t. Ng SPEED 1 vertical scale shall continually cycle from 0 to full scale.
- u. Unit tester LOW ENG 1 OUT lamp shall be on when scale indication is less than 55%.
- v. Ng SPEED 2 vertical scale shall continually cycle from 0 to full scale.
- w. Unit tester LOW ENG 2 OUT lamp shall be on when scale indication is less than 55%.

ANALOG PARAMETER INDICATION OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition a. is not as specified, replace display driver module A3 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
2. If Indication/Condition b. is not as specified, replace display driver module A3 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
3. If Indication/Condition c. is not as specified, replace display driver module A4 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
4. If Indication/Condition d. is not as specified, replace display driver module A4 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
5. If Indication/Condition e. is not as specified, replace display driver module A5 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
6. If Indication/Condition f. is not as specified, replace display driver module A6 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
7. If Indication/Condition g. is not as specified, replace display driver module A6 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
8. If Indication/Condition h. is not as specified, replace display driver module A7 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
9. If Indication/Condition i. is not as specified, check CDU OIL TEMP 1 fuse.
 - a. If fuse is good, replace voltage regulator module A17 (WP 0802 00).
 - b. If fuse is not good, replace fuse (WP 0779 00).
 - (1) If trouble remains, the CDU chassis assembly is malfunctioning.
10. If HIGH OIL 1 TEMP lamp does not go off when scale indication is less than 302°F (150°C), replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
11. If Indication/Condition j. is not as specified, replace display driver module A8 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
12. If Indication/Condition k. is not as specified, check CDU OIL TEMP 2 fuse.
 - a. If fuse is good, replace voltage regulator module A17 (WP 0803 00).
 - b. If fuse is not good, replace fuse (WP 0779 00).
 - (1) If trouble remains, the CDU chassis assembly is malfunctioning.
13. If HIGH OIL 2 TEMP lamp does not go off when scale indication is less than 302°F (150°C), replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
14. If Indication/Condition l. is not as specified, replace display driver module A9 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

ANALOG PARAMETER INDICATION OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

15. If Indication/Condition m. is not as specified, replace display driver module A9 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
16. If Indication/Condition n. is not as specified, check CDU OIL PRESS 1 fuse.
 - a. If fuse is good, replace voltage regulator module A17 (WP 0803 00).
 - b. If fuse is not good, replace fuse (WP 0779 00).
 - (1) If trouble remains, the CDU chassis assembly is malfunctioning.
17. If LOW OIL 1 PRESS lamp does not go off when scale indication is more than 25 psi, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
18. If Indication/Condition o. is not as specified, replace display driver module A10 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
19. If Indication/Condition p. is not as specified, replace display driver module A10 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
20. If Indication/Condition q. is not as specified, check CDU OIL PRESS 2 fuse.
 - a. If fuse is good, replace voltage regulator module A17 (WP 0803 00).
 - b. If fuse is not good, replace fuse (WP 0779 00).
 - (1) If trouble remains, the CDU chassis assembly is malfunctioning.
21. If LOW OIL 2 PRESS lamp does not go off when scale indication is more than 25 psi, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
22. If Indication/Condition r. is not as specified, replace display driver module A11 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
23. If Indication/Condition s. is not as specified, replace display driver module A12 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
24. If Indication/Condition t. is not as specified, replace display driver module A13 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
25. If Indication/Condition u. is not as specified, check CDU Ng 1 fuse.
 - a. If fuse is good, replace voltage regulator module A17 (WP 0803 00).
 - b. If fuse is not good, replace fuse (WP 0779 00).
 - (1) If trouble remains, the CDU chassis assembly is malfunctioning.
26. If LOW ENG 1 OUT lamp does not go off when scale indication is more than 55%, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
27. If Indication/Condition v. is not as specified, replace display driver module A14 (WP 0802 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
28. If Indication/Condition w. is not as specified, check CDU Ng 2 fuse.

ANALOG PARAMETER INDICATION OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- a. If fuse is good, replace voltage regulator module A17 (WP 0803 00).
- b. If fuse is not good, replace fuse (WP 0779 00).
 - (1) If trouble remains, the CDU chassis assembly is malfunctioning.
- 29. If LOW ENG 2 OUT lamp does not go off when scale indication is more than 55%, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

- 4. Place unit tester ANALOG SET switch to STOP when unit tester INTERFACE MONITOR indicates less than 700.
- 5. Momentarily press unit tester RESET switch.

Indication/Condition

- a. Unit tester LOW ROTOR SPEED lamp shall be on.
- b. Unit tester 127%, 137%, and 142% ROTOR OVERSPEED lamps shall go off.

Corrective Action

- 1. If Indication/Condition a. is not as specified, check CDU RTR SP fuse.
 - a. If fuse is good, replace voltage regulator module A17 (WP 0803 00).
 - b. If fuse is not good, replace fuse (WP 0779 00).
 - (1) If trouble remains, the CDU chassis assembly is malfunctioning.
- 2. If Indication/Condition b. is not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

- 6. Place unit tester ANALOG SET switch to SLOW.

Indication/Condition

- a. Unit tester LOW ROTOR SPEED lamp shall go off when unit tester INTERFACE MONITOR indicates more than 950.
- b. Unit tester 127% ROTOR OVERSPEED lamp shall go on when INTERFACE MONITOR initially indicates between 1269 and 1271 and shall remain on during subsequent INTERFACE MONITOR cycles.
- c. Unit tester 137% ROTOR OVERSPEED lamp shall go on when INTERFACE MONITOR initially indicates between 1369 and 1371 and shall remain on during subsequent INTERFACE MONITOR cycles.
- d. Unit tester 142% ROTOR OVERSPEED lamp shall go on when INTERFACE MONITOR initially indicates between 1419 and 1421 and shall remain on during subsequent INTERFACE MONITOR cycles.

Corrective Action

- 1. If Indication/Condition a. is not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.
- 2. If Indication/Conditions b. through d. are not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

ANALOG PARAMETER INDICATION OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

7. Place unit tester ANALOG SET switch to STOP when unit tester INTERFACE MONITOR indicates less than 1200.
8. Momentarily press unit tester RESET switch.

Indication/Condition

Unit tester 127%, 137%, and 142% ROTOR OVERSPEED lamps shall go off.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

9. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**DIGITAL PARAMETER INDICATION OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Set unit tester DIGITAL SET thumbwheel to following positions:

DIGITAL SET	TOTAL FUEL	1 TGT 21 Ng SPEED 2
0	0	0
1	1110	111
2	2220	222
3	3330	333
4	440	444
5	1550	555
6	2660	666
7	3770	777
8	880	888

DIGITAL PARAMETER INDICATION OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

DIGITAL SET	TOTAL FUEL	1 TGT 21 Ng SPEED 2
9	1990	999

Indication/Condition

CDU TOTAL FUEL, TGT 1, TGT 2, Ng SPEED 1, and Ng SPEED 2 digital readouts shall indicate corresponding numbers.

Corrective Action

1. If Indication/Condition is not as specified, replace digital readout module A18 (WP 0804 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

3. Place CDU - DIGITS switch to OFF.

Indication/Condition

CDU digital readouts shall go off.

Corrective Action

1. If Indication/Condition is not as specified, replace digital readout module A18 (WP 0804 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

4. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

NO. 1 FUEL QUANTITY (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place CDU - DIGITS switch to ON.
3. Place simulator tester NO. 1 FUEL QUANTITY CAL-NOR switch to CAL and HI-LO switch to HI.
4. Position unit tester INTER CAL SELECT switch to 15 and DISPLAY UPDATE SWITCH to OFF.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1216 and 1234.

NO. 1 FUEL QUANTITY (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

5. Place simulator tester NO. 1 FUEL QUANTITY HI-LO SWITCH to LO.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 0 and 9.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

6. Place simulator tester NO. 1 FUEL QUANTITY CAL-NOR switch to NOR.
7. Slowly turn simulator tester NO. 1 FUEL QUANTITY variable control fully clockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall increase smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

8. Slowly turn simulator tester NO. 1 FUEL QUANTITY variable control fully counterclockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall decrease smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

9. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

NO. 2 FUEL QUANTITY (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester NO. 2 FUEL QUANTITY CAL-NOR switch to CAL and HI-LO switch to HI.
3. Position unit tester INTER CAL SELECT switch to 16.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1216 and 1234.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

4. Place simulator tester NO. 2 FUEL QUANTITY HI-LO switch to LO.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 0 and 9.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

5. Place simulator tester NO. 2 FUEL QUANTITY CAL-NOR switch to NOR.
6. Slowly turn simulator tester NO. 2 FUEL QUANTITY variable control fully clockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall increase smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

7. Slowly turn simulator tester NO. 2 FUEL QUANTITY variable control fully counterclockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall decrease smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

NO. 2 FUEL QUANTITY (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

MAIN TRANSMISSION OIL TEMPERATURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester MAIN XMSN OIL TEMP CAL-NOR switch to CAL and HI-LO switch to HI.
3. Position unit tester INTER CAL SELECT switch to 17.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1216 and 1264.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

4. Place simulator tester MAIN XMSN OIL TEMP HI-LO switch to LO.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 0 and 18.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

5. Place simulator tester MAIN XMSN OIL TEMP CAL-NOR switch to NOR.
6. Slowly turn simulator tester MAIN XMSN OIL TEMP variable control fully clockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall increase smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

**MAIN TRANSMISSION OIL TEMPERATURE (ANALOG) OPERATIONAL/TROUBLESHOOTING
PROCEDURE - Continued****Step**

7. Slowly turn simulator tester MAIN XMSN OIL TEMP variable control fully counterclockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall decrease smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**MAIN TRANSMISSION OIL PRESSURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester XMSN OIL PRESS CAL-NOR switch to CAL and HI-LO switch to HI.
3. Position unit tester INTER CAL SELECT switch to 18.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1108 and 1132.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

4. Place simulator tester XMSN OIL PRESS HI-LO switch to LO.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 0 and 12.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

5. Place simulator tester XMSN OIL PRESS CAL-NOR switch to NOR.

MAIN TRANSMISSION OIL PRESSURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE -
Continued

6. Slowly turn simulator tester XMSN OIL PRESS variable control fully clockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall increase smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

7. Slowly turn XMSN OIL PRESS variable control fully counterclockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall decrease smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 1 module A15 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

PARAMETER MEASUREMENT (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester NO. 1 FUEL QUANTITY and NO. 2 FUEL QUANTITY CAL-NOR switches to CAL.
3. Position unit tester INTER CAL SELECT switch to 19.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 0 and 7.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

4. Place simulator tester NO. 1 FUEL QUANTITY HI-LO switch to HI.

PARAMETER MEASUREMENT (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Unit tester INTERFACE MONITOR shall indicate between 1218 and 1232.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

5. Place simulator tester NO. 2 FUEL QUANTITY HI-LO switch to HI.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 2443 and 2457.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

6. Place simulator tester NO. 1 FUEL QUANTITY CAL-NOR switch to NOR.
7. Slowly turn simulator tester NO. 1 FUEL QUANTITY variable control fully clockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall increase smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

8. Slowly turn simulator tester NO. 1 FUEL QUANTITY variable control full counterclockwise.

Indication/Condition

Unit tester INTERFACE MONITOR count shall decrease smoothly.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2 module A16 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

Step

9. Place unit tester INTER CAL SELECT switch to OFF.
10. Place unit tester DISPLAY UPDATE switch to ON.
11. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

PARAMETER MEASUREMENT (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

None Required

**FAILURE MODE OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place unit tester SDC FAILURE switch to 1 and release.

Indication/Condition

With SDC FAILURE switch to 1, CDU CHAN 1 lamp shall go on.

Corrective Action

If Indication/Condition is not as specified, the CDU chassis assembly is malfunctioning.

Step

3. Place unit tester SDC FAILURE switch to 2 and release.

Indication/Condition

With SDC FAILURE switch to 2, CDU CHAN 2 lamp shall go on.

Corrective Action

If Indication/Condition is not as specified, the CDU chassis assembly is malfunctioning.

Step

4. Place unit tester DISPLAY UPDATE switch to OFF.

Indication/Condition

- a. All CDU vertical displays shall blank out.
- b. All CDU digital displays shall blank out.
- c. CHAN 1 and CHAN 2 lamps shall go on.

Corrective Action

If Indication/Conditions are not as specified, CDU chassis assembly is malfunctioning.

Step

5. Momentarily press unit tester RESET switch.

Indication/Condition

Unit tester 127%, 137%, and 142% lamps shall go off.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A17 (WP 0803 00).
 - a. If trouble remains, the CDU chassis assembly is malfunctioning.

FAILURE MODE OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

6. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Place simulator tester INPUT POWER switch to OFF.
2. Disconnect cable assemblies.

Indication/Condition

Cable assemblies shall be disconnected.

Corrective Action

None Required

LOCATION

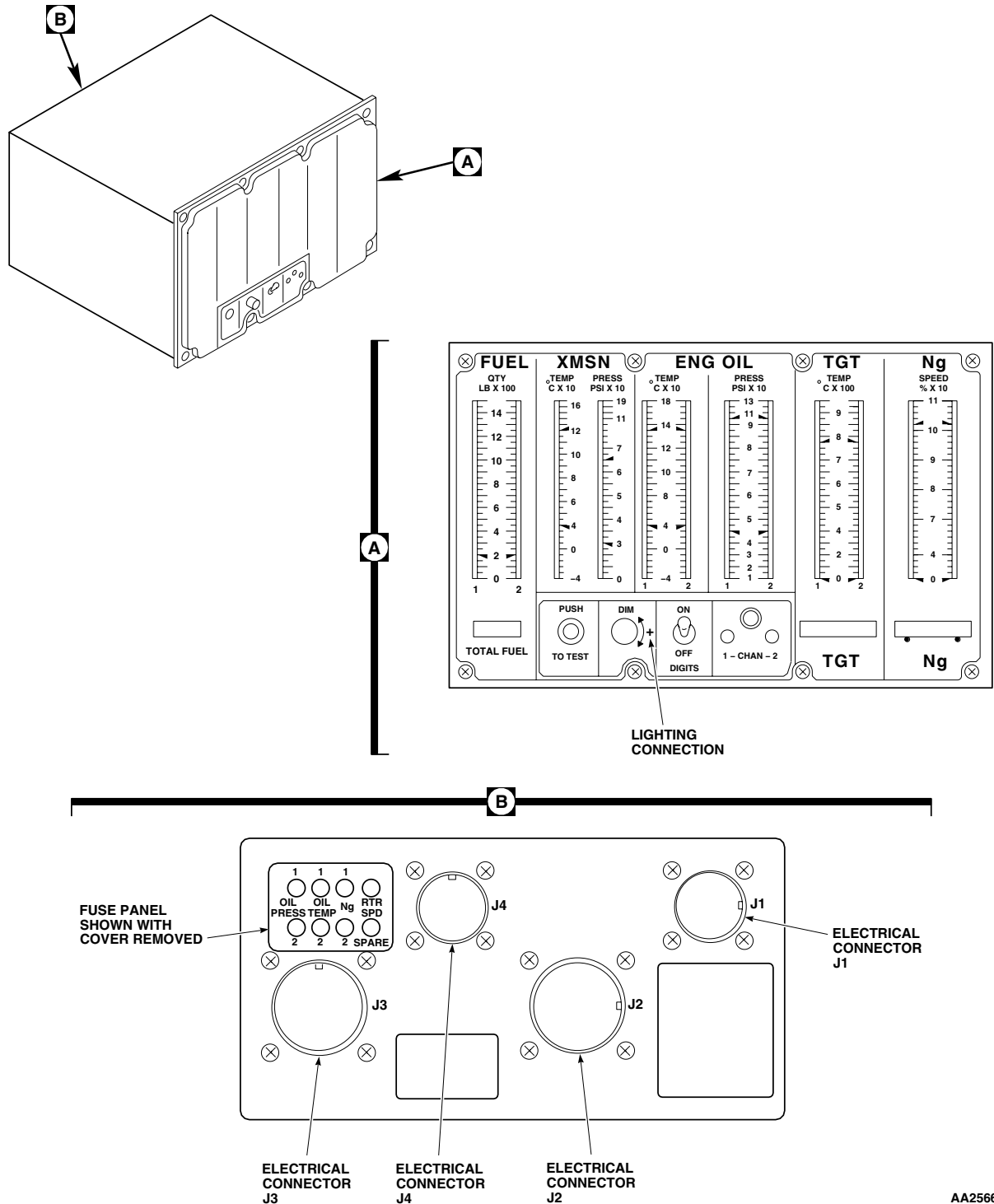


Figure 1. Central Display Unit (CDU).

UNIT LEVEL
INSTRUMENT SYSTEMS
PILOTS DISPLAY UNIT (AVIM)

INITIAL SETUP:**Test Equipment**

Bench Test Set, 245-476854-000
Bench Test Set, 245-476854-001
Multimeter, AN/PSM-45A
Oscilloscope, 7603

Tools and Special Tools

Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required

Aircraft Electrician MOS 15F (1)

References

TM 55-4920-412-13&P
WP 0808 00
WP 0809 00
WP 0812 00
WP 0813 00

Equipment Condition

Bench Power Available, 28 vdc, 115 vac, 400 Hz,
Single-Phase

SPECIAL INSTRUCTIONS**NOTE**

See Figure 1 for component location diagram, Figure 2 for pilots display unit test setup, and Figure 3 for simulator front panel.

The pilots display unit (PDU) operational/troubleshooting procedure is subdivided as follows:

- Setup/Power On
- 5 VDC Regulator
- Clock Inhibit
- Lamp/Digit
- Parameter Indication (Analog)
- Parameter Indication (Digital)
- Failure Mode
- Rotor Overspeed Lamp
- Autodim
- Shutdown

SETUP/POWER ON OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step**

1. Make sure simulator tester INPUT POWER switch is OFF.

SETUP/POWER ON OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. Connect cable assembly, 217-419742-000, between simulator tester INTERCONNECT connector and unit tester J5 connector.
3. Connect simulator tester dc power connector to 28 vdc bench power.
4. Connect simulator tester ac power connector to 115 vac, 400 Hz, single-phase bench power.
5. Place simulator tester INPUT POWER switch to ON.

Indication/Condition

Simulator tester INPUT POWER 115 VAC and 28 VDC lamps shall go on.

Corrective Action

If Indication/Condition is not as specified, troubleshoot bench test set (TM 55-4920-412-13&P).

Step

6. Press simulator tester LAMP TEST switch, then release.

Indication/Condition

All simulator tester lamps shall go on when LAMP TEST switch is pressed.

Corrective Action

If Indication/Condition is not as specified, troubleshoot bench test set (TM 55-4920-412-13&P).

Step

7. Place simulator tester INPUT POWER switch to OFF.
8. Connect cable assembly, 217-419745-000, between display unit connector J1 and unit tester connectors J6 and J7.

NOTE

For 100-601137-000, 100-601184-000, 100-601581-000 (70450-01916-104), and 100-601561-000 (70450-01043-124) PDUs, apply 115 vac 400 hz to simulator front panel integral lighting section HI and LO test points.

9. Place simulator tester INTEGRAL LIGHTING switch to EXT.
10. Place unit tester DISPLAY CONTROLS section switches as follows:

SWITCH	POSITION
DIGITAL SET	1
DIGITS	ON
ROTOR OVERSPEED 127%, 137%, and 142%	Off (Down)
ANALOG SET	STOP
DISPLAY UPDATE	ON

11. Position unit tester INTER CAL SELECT switch to 21.

SETUP/POWER ON OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

12. Place unit tester CENTRAL DISPLAY UNIT P/S CONT switch to REMOTE.
13. Place simulator tester INPUT POWER switch to ON.

Indication/Condition

Simulator tester INPUT POWER 115 VAC and 28 VDC lamps shall go on.

Corrective Action

If Indication/Condition is not as specified, troubleshoot bench test set (TM 55-4920-412-13&P).

5 VDC REGULATOR OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Set multimeter to indicate dc voltage and connect leads between PDU 5V and SDC COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 4.75 and 5.25 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace pilot display autodim module A9 (WP 0813 00).
 - a. If trouble remains, the display unit chassis assembly is malfunctioning.

Step

3. Disconnect multimeter.
4. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

CLOCK INHIBIT OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Connect oscilloscope between PDU left INHIB and SDC COM test jacks on unit tester.

Indication/Condition

Oscilloscope shall display 4 to 6 volt peak-to-peak square waves.

Corrective Action

1. If Indication/Condition is not as specified, replace pilot display autodim module A9 (WP 0813 00).

CLOCK INHIBIT OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- a. If trouble remains, the display unit chassis assembly is malfunctioning.

Step

3. Connect oscilloscope between PDU right INHIB and SDC COM test jacks on unit tester.

Indication/Condition

Oscilloscope shall display 4 to 6 volt peak-to-peak square waves.

Corrective Action

1. If Indication/Condition is not as specified, replace pilot display autodim module A9 (WP 0813 00).
 - a. If trouble remains, the display unit chassis assembly is malfunctioning.

Step

4. Disconnect oscilloscope.
5. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

LAMP/DIGIT OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Press display unit TEST switch and turn unit tester P/S CONTROL fully counterclockwise, then fully clockwise. Next, adjust so you can see display normally. Release TEST switch.

Indication/Condition

With TEST switch pressed:

- a. All display unit vertical scale lamps shall go on.
- b. Both display unit digital readouts shall indicate 188.
- c. Display unit lighting shall vary with P/S CONTROL setting.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, display unit chassis assembly is malfunctioning.
2. If Indication/Condition a. is not as specified, replace pilot's display driver module (A3, A4, A5, A6, and/or A7) as required (WP 0808 00).
 - a. If trouble remains, the display unit chassis assembly is malfunctioning.
3. If Indication/Condition b. is not as specified, replace torque digital readout module A8 (WP 0812 00).
 - a. If trouble remains, the display unit chassis assembly is malfunctioning.
4. If Indication/Condition c. is not a specified, replace pilot display autodim module A9 (WP 0813 00).

LAMP/DIGIT OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- a. If trouble remains, the display unit chassis assembly is malfunctioning.

Step

3. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**PARAMETER INDICATION (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place unit tester ANALOG SET switch to FAST.

Indication/Condition**NOTE**

Cycling rate of vertical scales is controlled by unit tester ANALOG SET switch.

- a. Display unit % RPM 1 vertical scale shall continually cycle from 0 to full scale.
- b. At 96%, % RPM 1 low red and yellow scale segments shall go off.
- c. Display unit % RPM 2 vertical scale shall continually cycle from 0 to full scale.
- d. At 96%, % RPM 2 low red and yellow scale segments shall go off.
- e. Display unit % RPM RTR vertical scale shall continually cycle from 0 to full scale.
- f. At 96%, % RPM RTR low red and yellow scale segments shall go off.
- g. Display unit % TRQ 1 vertical scale shall continually cycle from 0 to full scale.
- h. Display unit % TRQ 2 vertical scale shall continually cycle from 0 to full scale.

Corrective Action

1. If Indication/Conditions a., c., e., g., and h. are not as specified, replace pilot's display driver module (A3, A4, A5, A6 and/or A7) as required (WP 0809 00).
 - a. If trouble remains, the display unit chassis assembly is malfunctioning.
2. If Indication/Conditions b., d., and f. are not as specified, replace pilot's display driver module (A3, A4 and/or A5) as required (WP 0809 00).
 - a. If trouble remains, replace pilot display autodim module A9 (WP 0813 00).
 - (1) If trouble still remains, the display unit chassis assembly is malfunctioning.

Step

3. If no other check is to be done, go to SHUTDOWN, in this work package.

PARAMETER INDICATION (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Shutdown shall be performed.

Corrective Action

None Required

**PARAMETER INDICATION (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Set unit tester to the following positions:

DIGITAL SET	% TORQUE DIGITAL
0	0
1	111
2	22
3	133
4	44
5	155
6	66
7	177
8	88
9	199

Indication/Condition

Display unit % TRQ digital readouts shall indicate corresponding number.

Corrective Action

1. If digital readouts are not as specified, replace torque digital readout module A8 (WP 0812 00).
 - a. If trouble remains, the display unit chassis assembly is malfunctioning.

Step

3. If no other check is to be done, go to SHUTDOWN, in this work package.

PARAMETER INDICATION (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Shutdown shall be performed.

Corrective Action

None Required

**FAILURE MODE OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place unit tester DISPLAY UPDATE switch to OFF.

Indication/Condition

- a. All display unit vertical displays shall blank out.
- b. Display unit digital displays shall blank out.
- c. Unit tester FAIL WARN lamp shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, replace pilot's display driver (A3, A4, A5, A6, and/or A7) as required (WP 0809 00).
2. If Indication/Condition b. is not as specified, replace torque digital readout module A8 (WP 0812 00).
3. If Indication/Condition c. is not as specified, display unit chassis assembly is malfunctioning.

Step

3. Place unit tester DISPLAY UPDATE switch to ON.

Indication/Condition

- a. All display unit vertical displays shall continually cycle from 0 to full scale.
- b. Display unit digital displays shall indicate 199.
- c. Unit tester FAIL WARN lamp shall go off.

Corrective Action

If Indication/Conditions are not as specified, the display unit chassis assembly is malfunctioning.

Step

4. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

ROTOR OVERSPEED LAMP OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place unit tester 127% ROTOR OVERSPEED switch up.

Indication/Condition

Left RTR OVERSPEED lamp on display unit shall go on.

Corrective Action

If Indication/Condition is not as specified, display unit chassis assembly is malfunctioning.

Step

3. Place unit tester 137% ROTOR OVERSPEED switch up.

Indication/Condition

Center RTR OVERSPEED lamp on display unit shall go on.

Corrective Action

If Indication/Condition is not as specified, display unit chassis assembly is malfunctioning.

Step

4. Place unit tester 142% ROTOR OVERSPEED switch up.

Indication/Condition

Right RTR OVERSPEED lamp on display unit shall go on.

Corrective Action

If Indication/Condition is not as specified, display unit chassis assembly is malfunctioning.

Step

5. Place unit tester 127%, 137%, and 142% ROTOR OVERSPEED switches down.

Indication/Condition

Three RTR OVERSPEED lamps on display unit shall go off.

Corrective Action

If Indication/Condition is not as specified, display unit chassis assembly is malfunctioning.

Step

6. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

ROTOR OVERSPEED LAMP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

None Required

**AUTODIM OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place unit tester P/S CONT switch to NORMAL.
3. Cover light sensor, in lower right corner of display unit, with opaque object.

Indication/Condition

All display unit vertical scale and digital readout lighting shall go off or decrease in brightness.

Corrective Action

1. If Indication/Condition is not as specified, replace pilot display autodim module A9 (WP 0813 00).
 - a. If trouble remains, the display unit chassis assembly is malfunctioning.

Step

4. Remove opaque object from light sensor.

Indication/Condition

All display unit vertical scale and digital readout lighting shall increase in brightness.

Corrective Action

1. If Indication/Condition is not as specified, replace pilot display autodim module A9 (WP 0813 00).
 - a. If trouble remains, the display unit chassis assembly is malfunctioning.

Step

5. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Place simulator tester INPUT POWER switch to OFF.
2. Disconnect cable assemblies.

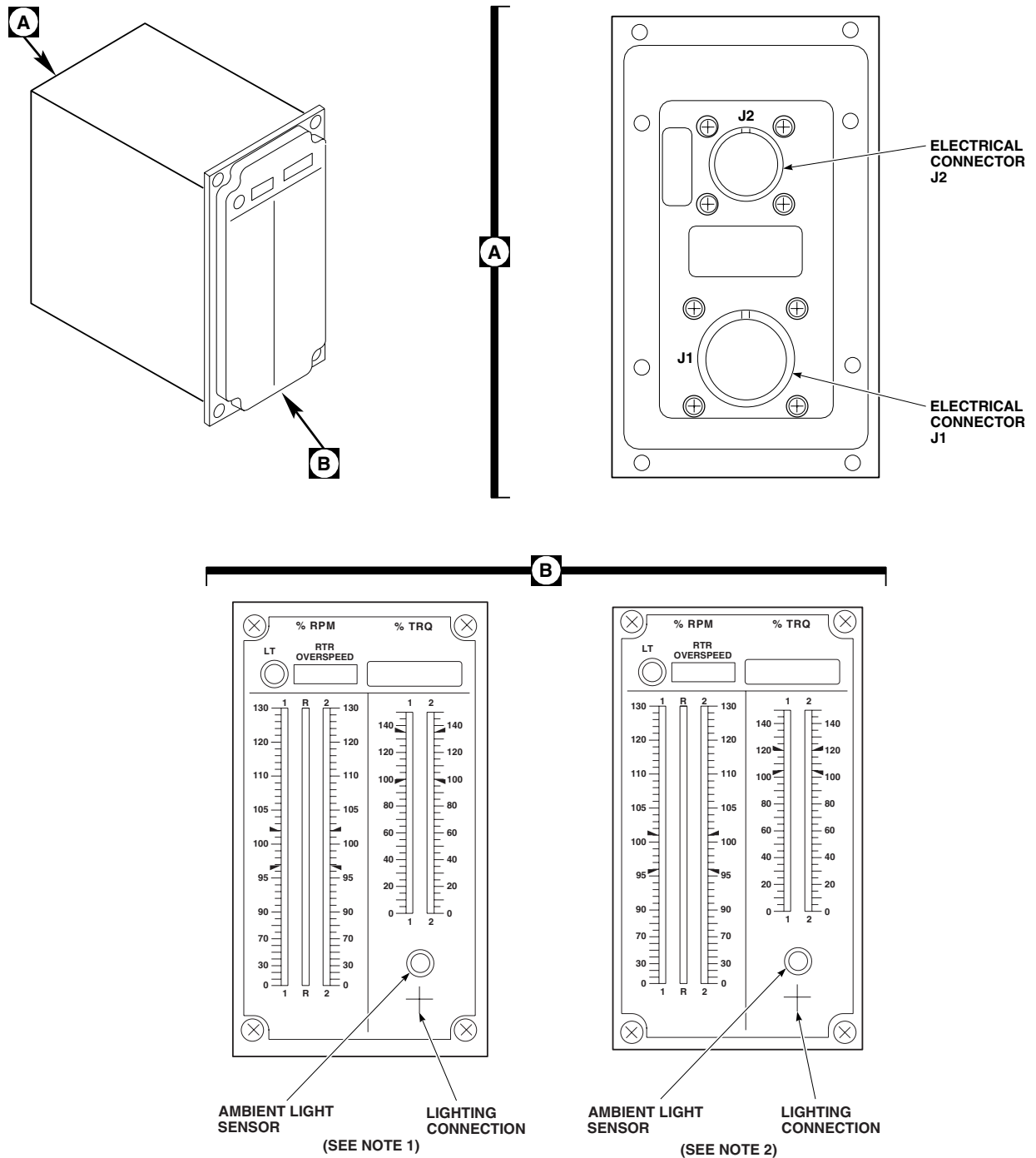
SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Cable assemblies shall be disconnected.

Corrective Action

None Required

LOCATION



NOTES

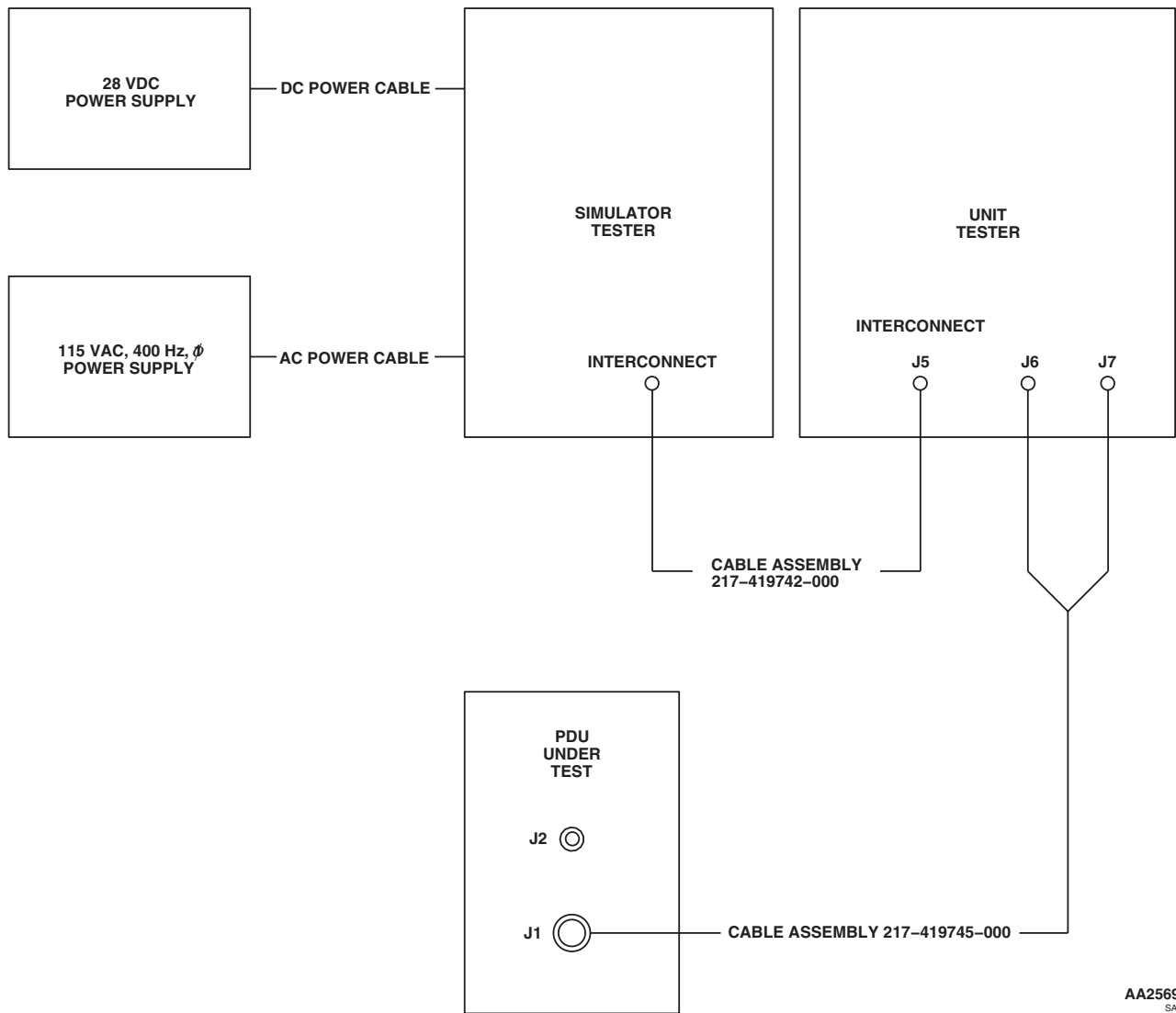
1. PDU 70450-01043-123 OR 70450-01916-104.
2. PDU 70450-01043-121.

PILOT'S AND COPILOT'S DISPLAY UNIT

AA2568
SA

Figure 1. Pilot's/Copilot's Display Unit.

TEST SETUP



AA2569
SA

Figure 2. Pilots Display Unit Test Setup.

TEST SETUP - Continued

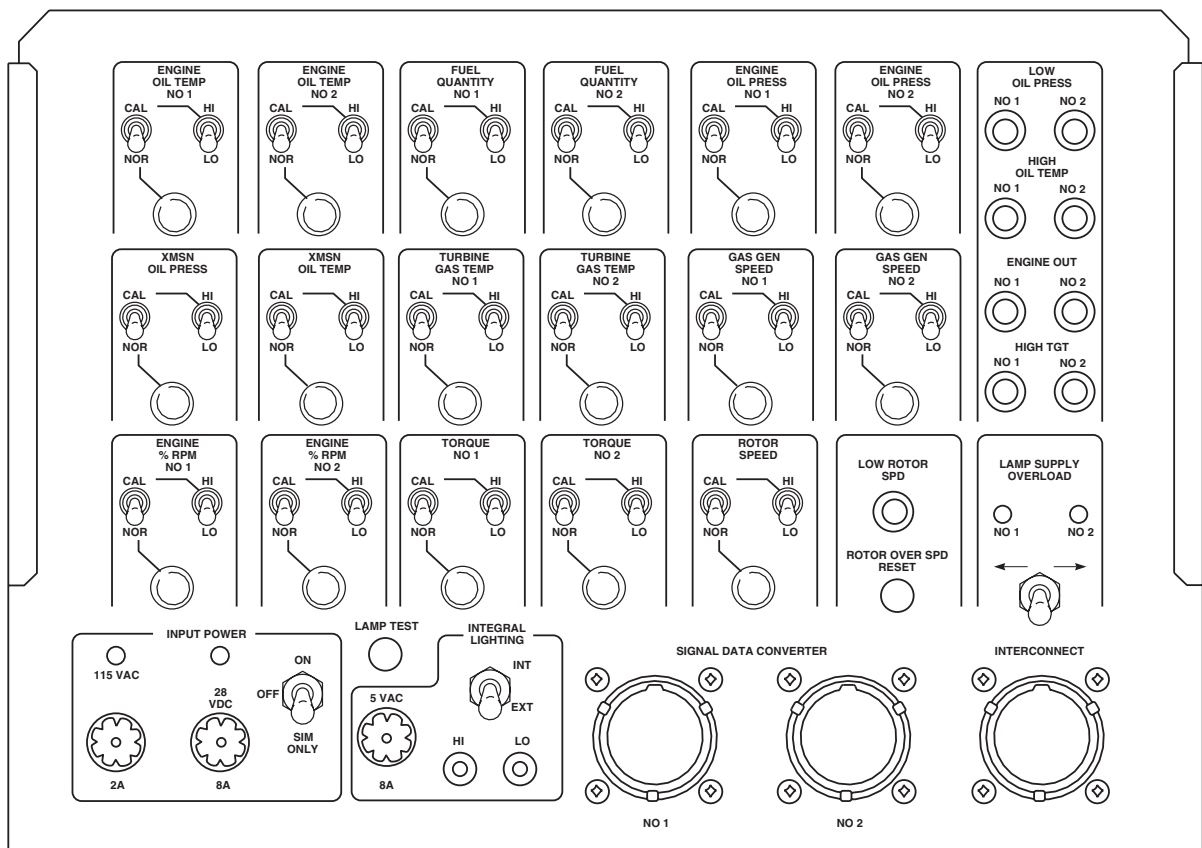
AA1737
SA

Figure 3. Simulator, Front Panel.

END OF WORK PACKAGE

0094 00-13/14 Blank

UNIT LEVEL
INSTRUMENT SYSTEMS

SIGNAL DATA CONVERTER (SDC) (70450-01043-112 OR 70450-01043-126) (AVIM)

INITIAL SETUP:**Test Equipment**

Bench Test Set, 245-476854-000
Bench Test Set, 245-476854-001
Multimeter, AN/PSM-45A
Oscilloscope, 7603

WP 0806 00

WP 0807 00

Equipment Condition

Bench Power Available, 28 vdc, 115 vac, 400 Hz,
Single-Phase

Tools and Special Tools

Electrical Repairer Toolkit, SC 5180-99-B06

References

TM 55-4920-412-13&P
WP 0805 00

SPECIAL INSTRUCTIONS**NOTE**

See Figure 1 for component location diagram and Figure 2 for SDC test setup.

The signal data converter operational/troubleshooting procedure is subdivided as follows:

- Setup/Power On
- Logic Power Supply/Voltage Regulator
- Lamp Power Supply
- Processor Failure Switch-Over
- Lamp Power Supply Failure
- Sensor Excitation Voltage
- Clock Monitor
- Clock Inhibit (Digital)
- Clock Inhibit (Analog)
- Sensor Interwiring/Warning Output
- No. 1 Engine % RPM (Analog)
- No. 2 Engine % RPM (Analog)
- Rotor Speed (Analog)
- No. 1 Torque (Analog)

SPECIAL INSTRUCTIONS - Continued

- No. 2 Torque (Analog)
- Engine Oil Temperature (Analog)
- Engine Oil Pressure (Analog)
- Turbine Gas Temperature (Analog)
- Gas Generator Speed (Analog)
- No. 1 Torque (Digital)
- No. 2 Torque (Digital)
- Turbine Gas Temperature (Digital)
- Gas Generator Speed (Digital)
- Rotor Speed (Analog Relay)
- Integral Lighting
- Shutdown

SETUP/POWER ON OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step**

1. Make sure simulator tester INPUT POWER switch is OFF.
2. Connect cable assembly, 217-419742-000, between simulator tester INTERCONNECT connector and unit tester J5 connector.
3. Connect simulator tester dc power connector to 28 vdc bench power.
4. Connect simulator tester ac power connector to 115 vac, 400 Hz, single-phase bench power.
5. Place simulator tester INPUT POWER switch to ON.

Indication/Condition

Simulator tester INPUT POWER 115 VAC and 28 VDC lamps shall go on.

Corrective Action

If Indication/Condition is not as specified, troubleshoot bench test set (TM 55-4920-412-13&P).

Step

6. Press simulator tester LAMP TEST switch, then release.

Indication/Condition

- a. All simulator tester lamps shall go on when LAMP TEST switch is pressed.
- b. All unit tester lamps shall go on when LAMP TEST switch is pressed.

Corrective Action

If Indication/Conditions are not as specified, troubleshoot bench test set (TM 55-4920-412-13&P).

SETUP/POWER ON OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

7. Place simulator tester INPUT POWER switch to OFF.
8. On test set 245-476854-000, connect cable assembly 217-419741-000, or on test set 245-476854-001, connect cable assembly 217-419740-000, between simulator tester No. 1 SIGNAL DATA CONVERTER connector and signal data converter (SDC) J1.
9. Connect cable assembly, 217-419747-000, between unit tester J8 and SDC J2.
10. Connect cable assembly, 217-419748-000, between unit tester J9 and SDC J3.
11. Place simulator tester switches and controls as follows:

SWITCH/CONTROL	POSITION
All CAL-NOR switches	NOR
All HI-LO switches	LO
All NOR controls	Fully counterclockwise
INTEGRAL LIGHTING	EXT

12. Place unit tester SIGNAL DATA CONVERTER section switches and controls as follows:

SWITCH/CONTROL	POSITION
CLOCK INHIBIT	OFF
FAILURES PROC	OFF
FAILURES TEST FREQ	OFF
P/S CONTROL	Full
CONVERSION	1

13. Place unit tester CENTRAL DISPLAY UNIT section P/S CONT switch to REMOTE.
14. Place unit tester DISPLAY CONTROLS section DISPLAY UPDATE switch to ON.
15. Place unit tester INTER CAL SELECT switch to OFF.
16. Place simulator tester INPUT POWER switch to ON.

Indication/Condition

- a. Simulator tester INPUT POWER 115 VAC and 28 VDC, and No. 1 LAMP SUPPLY OVERLOAD lamps shall go on.
- b. The following unit tester lamps shall go on:
 - MONITORS SHIFT REG

SETUP/POWER ON OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- MONITORS MUX ANALOG
- MONITORS DIGITAL O/P
- MONITORS MUX DIGITAL
- TEST FREQ ANA (flash)
- TEST FREQ DIG (flash)
- PROCESSOR SELECTION ANALOG A or B
- PROCESSOR SELECTION DIGITAL A or B

Corrective Action

1. If simulator tester No. 1 LAMP SUPPLY OVERLOAD lamp does not go on, replace lamp power supply (WP 0806 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.
2. If none of the unit tester lamps go on, do LOGIC POWER SUPPLY/VOLTAGE REGULATOR, in this work package.
3. If MONITORS SHIFT REG lamp does not go on, replace analog processor No. 1, module A9 (WP 0805 00).
 - a. If trouble remains, replace analog processor No. 2 module A8 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.
4. If MONITORS MUX ANALOG lamp does not go on, replace analog processor No. 2, module A8 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.
5. If MONITORS DIGITAL O/P lamp does not go on, replace digital processor module A10 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.
6. If MONITORS MUX DIGITAL lamp does not go on, replace digital processor module A10 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.
7. If TEST FREQ ANA lamp does not flash, replace analog processor No. 2, module A8 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.
8. If TEST FREQ DIG lamp does not flash, replace digital processor module A10 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

17. Place unit tester SIGNAL DATA CONVERTER FAILURES TEST FREQ switch to ON.

Indication/Condition

- a. Unit tester SIGNAL DATA CONVERTER TEST FREQ ANA lamp shall flash.
- b. Unit tester SIGNAL DATA CONVERTER TEST FREQ DIG lamp shall flash.

Corrective Action

1. If Indication/Condition a. is not as specified, replace analog processor No. 2, module A8 (WP 0805 00).

SETUP/POWER ON OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- a. If trouble remains, replace analog processor No. 1, module A9 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.
2. If Indication/Condition b. is not as specified, replace digital processor module A10 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

**LOGIC POWER SUPPLY/VOLTAGE REGULATOR OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Set multimeter to indicate dc voltage and connect leads between SDC +15.5 and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 14 and 21 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace logic power supply (WP 0807 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

3. Connect multimeter between SDC -15.5V and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between -21 and -14 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace logic power supply (WP 0807 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Connect multimeter between SDC 8V and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 6.5 and 10.5 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace logic power supply (WP 0807 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

5. Place unit tester SIGNAL DATA CONVERTER FAILURES TEST FREQ switch to OFF.

Indication/Condition

Multimeter shall indicate between 6.5 and 10.5 vdc.

LOGIC POWER SUPPLY/VOLTAGE REGULATOR OPERATIONAL/TROUBLESHOOTING PROCEDURE -
Continued**Corrective Action**

1. If Indication/Condition is not as specified, replace logic power supply (WP 0807 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

6. Place unit tester SIGNAL DATA CONVERTER FAILURES TEST FREQ switch to ON.
7. Connect multimeter between SDC +10V and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 9.5 and 10.5 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A11 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

8. Connect multimeter between SDC -10V and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between -10.5 and -9.5 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A11 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

9. Connect multimeter between SDC 5V and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 4.75 and 5.25 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A11 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

10. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

LAMP POWER SUPPLY OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Connect digital multimeter between SDC L/S and COM test jacks on unit tester.
3. Adjust unit tester SIGNAL DATA CONVERTER P/S CONTROL to obtain 4.9 to 5.1 vdc indication on multimeter.

Indication/Condition

Simulator tester No. 1 LAMP SUPPLY OVERLOAD lamp shall be on.

Corrective Action

1. If Indication/Condition is not as specified, replace lamp power supply (WP 0806 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Connect digital multimeter between SDC CONT and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 4.5 and 4.9 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A11 (WP 0806 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

5. Adjust unit tester SIGNAL DATA CONVERTER P/S CONTROL to obtain 0.99 to 1.01 vdc indication on multimeter.

Indication/Condition

No. 1 LAMP SUPPLY OVERLOAD lamp shall be on dimly.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A11 (WP 0807 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

6. Connect digital multimeter between SDC L/S and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 1.1 and 1.5 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace lamp power supply (WP 0806 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

LAMP POWER SUPPLY OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

7. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**PROCESSOR FAILURE SWITCH-OVER OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Momentarily place unit tester SIGNAL DATA CONVERTER PROCESSOR SELECTION switch to up position.

Indication/Condition

- a. If not already on, SIGNAL DATA CONVERTER PROCESSOR SELECTION ANALOG A lamp on the unit tester shall go on and remain on.
- b. If not already on, SIGNAL DATA CONVERTER DIGITAL A lamp on the unit tester shall go on and remain on.

Corrective Action

1. If Indication/Condition a. is not as specified, replace analog processor No. 1, module A9 (WP 0805 00).
 - a. If trouble remains, replace analog processor No. 2, module A8 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.
2. If Indication/Condition b. is not as specified, replace digital processor module A10 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

3. Place and hold unit tester SIGNAL DATA CONVERTER PROC FAILURES switch to A.

Indication/Condition

After approximately 5 seconds, the following shall occur:

- PROCESSOR SELECTION ANALOG A lamp shall go off.
- PROCESSOR SELECTION ANALOG B lamp shall go on.
- Unit tester SIGNAL DATA CONVERTER FAIL WARN lamp shall go on.

Corrective Action

1. If Indication/Conditions are not as specified, replace analog processor No. 1, module A9 (WP 0805 00).
 - a. If trouble remains, replace analog processor No. 2, module A8 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

PROCESSOR FAILURE SWITCH-OVER OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

4. Release unit tester PROC FAILURES switch.

Indication/Condition

FAIL WARN lamp shall go off.

Corrective Action

1. If Indication/Conditions are not as specified, replace analog processor No. 1, module A9 (WP 0805 00).
 - a. If trouble remains, replace analog processor No. 2, module A8 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

5. Place and hold unit tester PROC FAILURES switch to B.

Indication/Condition

After approximately 5 seconds, the following shall occur:

- PROCESSOR SELECTION ANALOG B lamp shall go off.
- PROCESSOR SELECTION ANALOG A lamp shall go on.
- FAIL WARN lamp shall go on.

Corrective Action

1. If Indication/Conditions are not as specified, replace analog processor No. 1, module A9 (WP 0805 00).
 - a. If trouble remains, replace analog processor No. 2, module A8 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

6. Release unit tester PROC FAILURES switch.

Indication/Condition

FAIL WARN lamp shall go off.

Corrective Action

1. If Indication/Conditions are not as specified, replace analog processor No. 1, module A9 (WP 0805 00).
 - a. If trouble remains, replace analog processor No. 2, module A8 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

7. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

LAMP POWER SUPPLY FAILURE OPERATIONAL/TROUBLESHOOTING PROCEDURE PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. If not already connected, connect multimeter between SDC L/S and COM test jacks on unit tester.
3. Turn unit tester SIGNAL DATA CONVERTER P/S CONTROL fully clockwise.

Indication/Condition

Simulator tester No. 1 LAMP SUPPLY OVERLOAD lamp shall be on.

Corrective Action

1. If Indication/Condition is not as specified, replace lamp power supply (WP 0806 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place and hold simulator tester LAMP SUPPLY OVERLOAD switch to No. 1.

Indication/Condition

- a. Multimeter shall indicate less than 1.0 vdc.
- b. Simulator tester No. 1 LAMP SUPPLY OVERLOAD lamp shall go off.

Corrective Action

1. If Indication/Condition is not as specified, replace lamp power supply (WP 0806 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

5. Release simulator tester LAMP SUPPLY OVERLOAD switch.

Indication/Condition

- a. Multimeter shall indicate between 4.8 and 5.8 vdc.
- b. Simulator tester No. 1 LAMP SUPPLY OVERLOAD lamp shall go on.

Corrective Action

If Indication/Conditions are not as specified, replace lamp power supply (WP 0806 00).

Step

6. Disconnect digital multimeter.
7. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

SENSOR EXCITATION VOLTAGE OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Connect multimeter between SDC AC1 and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 7.5 and 11.0 vac.

Corrective Action

1. If Indication/Condition is not as specified, replace logic power supply (WP 0807 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

3. Connect multimeter between SDC AC2 and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 7.5 and 11.0 vac.

Corrective Action

1. If Indication/Condition is not as specified, replace logic power supply (WP 0807 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Connect multimeter between SDC AC3 and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 7.5 and 11.0 vac.

Corrective Action

1. If Indication/Condition is not as specified, replace logic power supply (WP 0807 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

5. Disconnect multimeter.
6. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

CLOCK MONITOR OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Connect oscilloscope vertical input leads between SDC CLOCK and COM test jacks on unit tester.

Indication/Condition

Oscilloscope shall display square waves with these requirements:

- AMPLITUDE - 4 to 6 Volts
- PERIOD - 6.8 to 8.8 Microseconds

Corrective Action

1. If Indication/Condition is not as specified, replace digital processor module A10 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

3. Disconnect oscilloscope.
4. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

CLOCK INHIBIT (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place and hold unit tester SIGNAL DATA CONVERTER CLOCK INHIBIT switch to DIG.

Indication/Condition

- a. Both SIGNAL DATA CONVERTER PROCESSOR SELECTION DIGITAL A and B lamps on unit tester shall be off.
- b. The following unit tester SIGNAL DATA CONVERTER lamps shall go off:
 - TEST FREQ DIG
 - MONITORS DIGITAL O/P
 - MONITORS MUX DIGITAL
- c. Unit tester SIGNAL DATA CONVERTER FAIL WARN lamp shall go on.

CLOCK INHIBIT (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If Indication/Conditions are not as specified, replace digital processor module A10 (WP 0805 00).

Step

3. Release unit tester CLOCK INHIBIT switch.

Indication/Condition

- a. The following lamps shall go on:
 - TEST FREQ DIG (flash)
 - MONITORS DIGITAL O/P)
 - MONITORS MUX DIGITAL
 - Either PROCESSOR SELECTION DIGITAL A or B
- b. FAIL WARN lamp shall go off.

Corrective Action

If Indication/Conditions are not as specified, replace digital processor module A10 (WP 0805 00).

Step

4. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

CLOCK INHIBIT (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place and hold unit tester SIGNAL DATA CONVERTER CLOCK INHIBIT switch to ANA.

Indication/Condition

- a. Both SIGNAL DATA CONVERTER PROCESSOR SELECTION ANALOG A and B lamps on unit tester shall be off.
- b. The following unit tester SIGNAL DATA CONVERTER lamps shall go off:
 - TEST FREQ ANA
 - MONITORS SHIFT REG
 - MONITORS MUX ANALOG
- c. Unit tester SIGNAL DATA CONVERTER FAIL WARN lamp shall go on.

CLOCK INHIBIT (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Conditions are not as specified, replace analog processor No. 1, module A9 (WP 0805 00).
 - a. If trouble remains, replace analog processor No. 2, module A8 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

3. Release unit tester CLOCK INHIBIT switch.

Indication/Condition

- a. The following lamps shall go on:
 - TEST FREQ ANA (flash)
 - MONITORS SHIFT REG
 - MONITORS MUX ANALOG
 - Either PROCESSOR SELECTION ANALOG A or B
- b. FAIL WARN lamp shall go off.

Corrective Action

1. If Indication/Condition is not as specified, replace analog processor No. 1, module A9 (WP 0805 00).
 - a. If trouble remains, replace analog processor No. 2, module A8 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

4. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**SENSOR INTERWIRING/WARNING OUTPUT OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Make sure simulator tester No. 1 FUEL QUANTITY CAL-NOR switch is placed to NOR.
3. Connect digital multimeter between SDC FUEL H and FUEL L test jacks on unit tester.
4. Turn simulator tester No. 1 FUEL QUANTITY variable control fully clockwise, then fully counterclockwise.

Indication/Condition

Multimeter shall indicate between -0.5 and 0.5 vdc when the No. 1 FUEL QUANTITY control is fully counterclockwise, and shall indicate between 6.5 and 8.5 vdc when the No. 1 FUEL QUANTITY control is fully clockwise.

SENSOR INTERWIRING/WARNING OUTPUT OPERATIONAL/TROUBLESHOOTING PROCEDURE -
Continued**Corrective Action**

If Indication/Condition is not as specified, the signal data converter chassis assembly is malfunctioning.

Step

5. Connect digital multimeter between SDC WIRE and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 4.5 and 5.5 vdc.

Corrective Action

If Indication/Condition is not as specified, the signal data converter chassis assembly is malfunctioning.

Step

6. Momentarily press unit tester SIGNAL DATA CONVERTER WARN OUTPUTS button.

Indication/Condition

With WARNING OUTPUT button pressed, the following lamps on simulator tester shall go on:

- LOW ROTOR SPD
- NO. 1 LOW OIL PRESS
- NO. 1 HIGH OIL TEMP
- NO. 1 ENGINE OUT

Corrective Action

If Indication/Condition is not as specified, the signal data converter chassis assembly is malfunctioning.

Step

7. Connect digital multimeter between SDC W/C and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 4.5 and 5.5 vdc.

Corrective Action

If Indication/Condition is not as specified, the signal data converter chassis assembly is malfunctioning.

Step

8. Momentarily press simulator tester ROTOR OVER SPD RESET button.

Indication/Condition

With ROTOR OVER SPD RESET button pressed, multimeter shall indicate between -0.5 and 0.5 vdc.

Corrective Action

If Indication/Condition is not as specified, the signal data converter chassis assembly is malfunctioning.

Step

9. Disconnect digital multimeter.

SENSOR INTERWIRING/WARNING OUTPUT OPERATIONAL/TROUBLESHOOTING PROCEDURE -
Continued

10. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

NO. 1 ENGINE % RPM (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 1 ENGINE % RPM CAL-NOR switch to CAL and No. 1 ENGINE % RPM HI-LO switch to HI.
3. Position unit tester INTER CAL SELECT switch to 1.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 952 and 962.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 ENGINE % RPM HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 5.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

Step

5. Place simulator tester No. 1 ENGINE % RPM CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 ENGINE % RPM variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 1 ENGINE % RPM variable control fully counterclockwise.

NO. 1 ENGINE % RPM (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

NO. 2 ENGINE % RPM (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 2 ENGINE % RPM CAL-NOR switch to CAL and HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 2.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 952 and 962.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 2 ENGINE % RPM HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 5.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

Step

5. Place simulator tester No. 2 ENGINE % RPM CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 2 ENGINE % RPM variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

NO. 2 ENGINE % RPM (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 2 ENGINE % RPM variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**ROTOR SPEED (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester ROTOR SPEED CAL-NOR switch to CAL and HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 3.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1016 and 1026.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester ROTOR SPEED HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 5.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

ROTOR SPEED (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

5. Place simulator tester ROTOR CAL-NOR switch to NOR.
6. Slowly turn simulator tester ROTOR SPEED variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

7. Slowly turn simulator tester ROTOR SPEED variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

NO. 1 TORQUE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step**

1. Do SETUP/POWER ON, if not already done.
2. Place simulator tester No. 1 TORQUE CAL-NOR switch to CAL and No. 1 TORQUE HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 4.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1042 and 1062.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 TORQUE HI-LO switch to LO.

NO. 1 TORQUE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

INTERFACE MONITOR shall indicate between 0 and 10.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

5. Place simulator tester No. 1 TORQUE CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 TORQUE variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 1 TORQUE variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**NO. 2 TORQUE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 2 TORQUE CAL-NOR switch to CAL and No. 2 TORQUE HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 5.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1042 and 1062.

NO. 2 TORQUE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 2 TORQUE HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 10.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

5. Place simulator tester No. 2 TORQUE CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 2 TORQUE variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 2 TORQUE variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

ENGINE OIL TEMPERATURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.

ENGINE OIL TEMPERATURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. Place simulator tester No. 1 ENGINE OIL TEMP CAL-NOR switch to CAL and No. 1 ENGINE OIL TEMP HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 6.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1177 and 1223.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 ENGINE OIL TEMP HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 16.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

Step

5. Place simulator tester No. 1 ENGINE OIL TEMP CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 ENGINE OIL TEMP variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 1 ENGINE OIL TEMP variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

ENGINE OIL PRESSURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 1 ENGINE OIL PRESS CAL-NOR switch to CAL and No. 1 ENGINE OIL PRESS HI-LO switch to HI.
3. Position unit tester INTER CAL SELECT switch to 7.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 900 and 940.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 ENGINE OIL PRESS HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 20.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

Step

5. Place simulator tester No. 1 ENGINE OIL PRESS CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 ENGINE OIL PRESS variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 1 ENGINE OIL PRESS variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

ENGINE OIL PRESSURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Shutdown shall be performed.

Corrective Action

None Required

TURBINE GAS TEMPERATURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 1 TURBINE GAS TEMP CAL-NOR switch to CAL and No. 1 TURBINE GAS TEMP HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 8.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate 1267 and 1283.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 TURBINE GAS TEMP HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 8.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

Step

5. Place simulator tester No. 1 TURBINE GAS TEMP CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 TURBINE GAS TEMP variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 1 TURBINE GAS TEMP variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

TURBINE GAS TEMPERATURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**GAS GENERATOR SPEED (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 1 GAS GEN SPEED CAL-NOR switch to CAL and No. 1 GAS GEN SPEED HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 9.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 995 and 1005.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 GAS GEN SPEED HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 5.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

5. Place simulator tester No. 1 GAS GEN SPEED CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 GAS GEN SPEED variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

GAS GENERATOR SPEED (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

7. Slowly turn simulator tester No. 1 GAS GEN SPEED variable control counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**NO. 1 TORQUE (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 1 TORQUE CAL-NOR switch to CAL and No. 1 TORQUE HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 10.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1042 and 1062.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).
 - a. If trouble remains, replace digital processor, module A10 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 TORQUE HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 10.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

5. Place simulator tester No. 1 TORQUE CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 TORQUE variable control fully clockwise.

NO. 1 TORQUE (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 1 TORQUE variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

NO. 2 TORQUE (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 2 TORQUE CAL-NOR switch to CAL and No. 2 TORQUE HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 11.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1042 and 1062.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).
 - a. If trouble remains, replace digital processor, module A10 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 2 TORQUE HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 10.

NO. 2 TORQUE (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

5. Place simulator tester No. 2 TORQUE CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 2 TORQUE variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 2 TORQUE variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

TURBINE GAS TEMPERATURE (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 1 TURBINE GAS TEMP CAL-NOR switch to CAL and No. 1 TURBINE GAS TEMP HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 12.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 807 and 817.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).
 - a. If trouble remains, replace digital processor module A10 (WP 0805 00).

TURBINE GAS TEMPERATURE (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 TURBINE GAS TEMP HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 8.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

Step

5. Place simulator tester No. 1 TURBINE GAS TEMP CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 TURBINE GAS TEMP variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 1 TURBINE GAS TEMP variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**GAS GENERATOR SPEED (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 1 GAS GEN SPEED CAL-NOR switch to CAL and No. 1 GAS GEN SPEED HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 13.

GAS GENERATOR SPEED (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Unit tester INTERFACE MONITOR shall indicate between 995 and 1005.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).
 - a. If trouble remains, replace digital processor module A10 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 GAS GEN SPEED HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 5.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

5. Place simulator tester No. 1 GAS GEN SPEED CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 GAS GEN SPEED variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 1 GAS GEN SPEED variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

ROTOR SPEED (ANALOG RELAY) OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester ROTOR SPEED CAL-NOR switch to CAL and ROTOR SPEED HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 14.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1016 and 1026.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester ROTOR SPEED HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 5.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

5. Place simulator tester ROTOR CAL-NOR switch to NOR.
6. Slowly turn simulator tester ROTOR SPEED variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

7. Slowly turn simulator tester ROTOR SPEED variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

ROTOR SPEED (ANALOG RELAY) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Shutdown shall be performed.

Corrective Action

None Required

**INTEGRAL LIGHTING OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester INTEGRAL LIGHTING switch to INT.

Indication/Condition

Both SIGNAL DATA CONVERTER INT LIGHT lamps on unit tester shall go on.

Corrective Action

If Indication/Condition is not as specified, the signal data converter chassis assembly is malfunctioning.

Step

3. Place simulator tester INTEGRAL LIGHTING switch to EXT.
4. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Place simulator tester INPUT POWER switch to OFF.
2. Disconnect cable assemblies.

Indication/Condition

Cable assemblies shall be disconnected.

Corrective Action

None Required

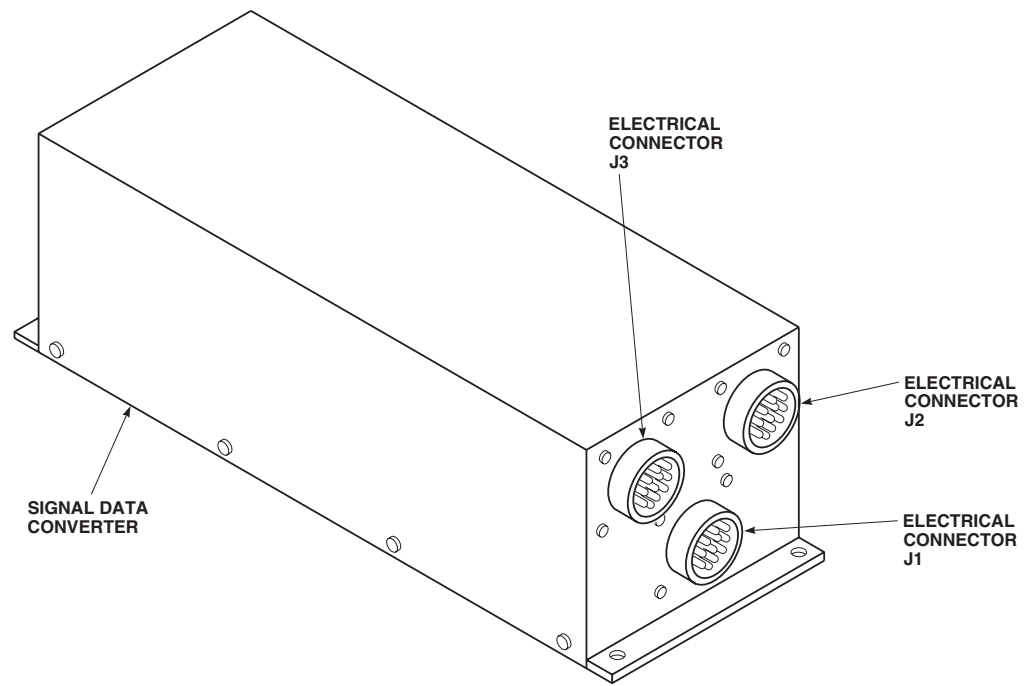
LOCATIONAA2570
SA

Figure 1. Signal Data Converter (SDC).

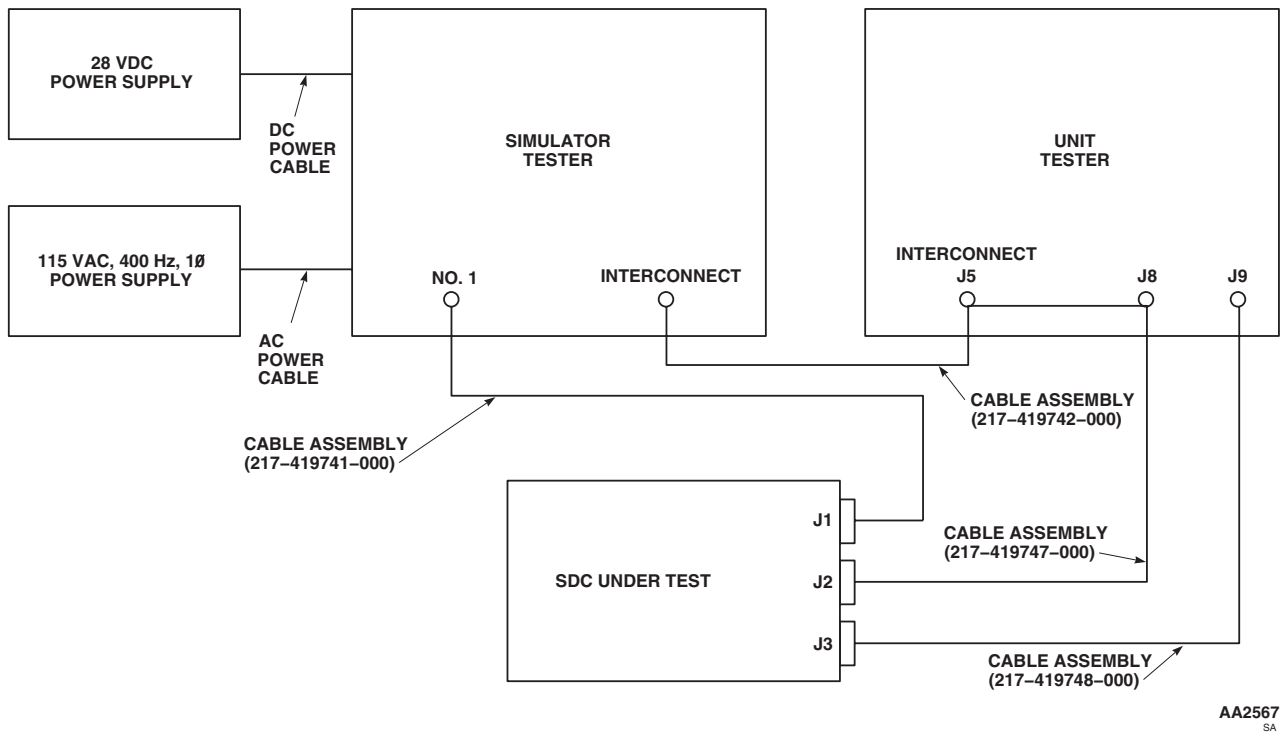
TEST SETUP

Figure 2. Signal Data Converter Test Setup.

END OF WORK PACKAGE

UNIT LEVEL**INSTRUMENT SYSTEMS**

SIGNAL DATA CONVERTER (SDC) (70450-21943-110 OR 70450-01916-103) (AVIM)

INITIAL SETUP:**Test Equipment**

Bench Test Set, 245-476854-001
Multimeter, AN/PSM-45A
Oscilloscope, 7603

WP 0807 00

Equipment Condition

Bench Power Available, 28 vdc, 115 vac, 400 Hz,
Single-Phase

Tools and Special Tools

Electrical Repairer Toolkit, SC 5180-99-B06

References

WP 0805 00
WP 0806 00

SPECIAL INSTRUCTIONS**NOTE**

See Figure 1 for component location diagram and Figure 2 for SDC test setup.

The signal data converter operational/troubleshooting procedure is subdivided as follows:

- Setup/Power On
- Logic Power Supply/Voltage Regulator
- Lamp Power Supply
- Processor Failure Switch-Over
- Lamp Power Supply Failure
- Sensor Excitation Voltage
- Clock Monitor
- Clock Inhibit (Digital)
- Clock Inhibit (Analog)
- Sensor Interwiring/Warning Output
- No. 1 Engine % RPM (Analog)
- No. 2 Engine % RPM (Analog)
- Rotor Speed (Analog)
- No. 1 Torque (Analog)

SPECIAL INSTRUCTIONS - Continued

- No. 2 Torque (Analog)
- Engine Oil Temperature (Analog)
- Engine Oil Pressure (Analog)
- Turbine Gas Temperature (Analog)
- Gas Generator Speed (Analog)
- No. 1 Torque (Digital)
- No. 2 Torque (Digital)
- Turbine Gas Temperature (Digital)
- Gas Generator Speed (Digital)
- Rotor Speed (Analog Relay)
- Integral Lighting
- Shutdown

SETUP/POWER ON OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step**

1. Make sure simulator tester INPUT POWER switch is OFF.
2. Connect cable assembly, 217-419742-000, between simulator tester INTERCONNECT connector and unit tester J5 connector.
3. Connect simulator tester dc power connector to 28 vdc bench power.
4. Connect simulator tester ac power connector to 115 vac, 400 Hz, single-phase bench power.
5. Place simulator tester INPUT POWER switch to ON.

Indication/Condition

Simulator tester INPUT POWER 115 VAC and 28 VDC lamps shall go on.

Corrective Action

If Indication/Condition is not as specified, troubleshoot bench test set.

Step

6. Press simulator tester LAMP TEST switch, then release.

Indication/Condition

- a. All simulator tester lamps shall go on when LAMP TEST switch is pressed.
- b. All unit tester lamps shall go on when LAMP TEST switch is pressed.

Corrective Action

If Indication/Conditions are not as specified, troubleshoot bench test set.

SETUP/POWER ON OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

7. Place simulator tester INPUT POWER switch to OFF.
8. Connect cable assembly, 217-419741-000, between simulator tester No. 1 SIGNAL DATA CONVERTER connector and signal data converter (SDC) J1.
9. Connect cable assembly, 217-419747-000, between unit tester J8 and SDC J2.
10. Connect cable assembly, 217-419748-000, between unit tester J9 and SDC J3.
11. Place simulator tester switches and controls as follows:

SWITCH/CONTROL	POSITION
All CAL-NOR switches	NOR
All HI-LO switches	LO
All NOR controls	Fully counterclockwise
INTEGRAL	EXT
LIGHTING	

12. Place unit tester SIGNAL DATA CONVERTER section switches and controls as follows:

SWITCH/CONTROL	POSITION
CLOCK INHIBIT	OFF
FAILURES PROC	OFF
P/S CONTROL	Midposition
CONVERSION	1
FAILURES TEST	OFF
FREQ	

13. Place unit tester CENTRAL DISPLAY UNIT section P/S CONT switch to REMOTE.
14. Place unit tester DISPLAY CONTROLS section DISPLAY UPDATE switch to ON.
15. Place unit tester INTER CAL SELECT switch to OFF.
16. Place simulator tester INPUT POWER switch to ON.

Indication/Condition

- a. Simulator tester INPUT POWER 115 VAC and 28 VDC and No. 1 LAMP SUPPLY OVERLOAD lamps shall go on.

SETUP/POWER ON OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

b. The following unit tester lamps shall go on:

- MONITORS SHIFT REG
- MONITORS MUX ANALOG
- MONITORS DIGITAL O/P
- MONITORS MUX DIGITAL
- TEST FREQ ANA (flash)
- TEST FREQ DIG (flash)
- PROCESSOR SELECTION ANALOG A or B
- PROCESSOR SELECTION DIGITAL A or B

Corrective Action

1. If simulator tester No. 1 LAMP SUPPLY OVERLOAD lamp does not go on, replace lamp power supply (WP 0806 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.
2. If none of the unit tester lamps go on, do LOGIC POWER SUPPLY/VOLTAGE REGULATOR, in this package.
3. If MONITORS SHIFT REG lamp does not go on, replace analog processor No. 1, module A9 (WP 0805 00).
 - a. If trouble remains, replace analog processor No. 2 module A8 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.
4. If MONITORS MUX ANALOG lamp does not go on, replace analog processor No. 2, module A8 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.
5. If MONITORS DIGITAL O/P lamp does not go on, replace digital processor module A10 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.
6. If MONITORS MUX DIGITAL lamp does not go on, replace digital processor module A10 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.
7. If TEST FREQ ANA lamp does not flash, replace analog processor No. 2, module A8 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.
8. If TEST FREQ DIG lamp does not flash, replace digital processor module A10 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

17. Place unit tester SIGNAL DATA CONVERTER FAILURES TEST FREQ switch to ON.

Indication/Condition

- a. Unit tester SIGNAL DATA CONVERTER TEST FREQ ANA lamp shall flash.
- b. Unit tester SIGNAL DATA CONVERTER TEST FREQ DIG lamp shall flash.

SETUP/POWER ON OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition a. is not as specified, replace analog processor No. 2, module A8 (WP 0805 00).
 - a. If trouble remains, replace analog processor No. 1, module A9 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.
2. If Indication/Condition b. is not as specified, replace digital processor module A10 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

LOGIC POWER SUPPLY/VOLTAGE REGULATOR OPERATIONAL/TROUBLESHOOTING PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Set multimeter to indicate dc voltage and connect leads between SDC +15.5 and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 14 and 21 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace logic power supply (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

3. Connect multimeter between SDC -15.5V and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between -21 and -14 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace logic power supply (WP 0807 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Connect multimeter between SDC 8V and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 6.5 and 10.5 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace logic power supply (WP 0807 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

5. Place unit tester SIGNAL DATA CONVERTER FAILURES TEST FREQ switch to OFF.

LOGIC POWER SUPPLY/VOLTAGE REGULATOR OPERATIONAL/TROUBLESHOOTING PROCEDURE -
Continued**Indication/Condition**

Multimeter shall indicate between 6.5 and 10.5 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace logic power supply (WP 0807 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

6. Place unit tester SIGNAL DATA CONVERTER FAILURES TEST FREQ switch to ON.
7. Connect multimeter between SDC +10V and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 9.5 and 10.5 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A11 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

8. Connect multimeter between SDC -10V and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between -10.5 and -9.5 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A11 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

9. Connect multimeter between SDC 5V and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 4.75 and 5.25 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A11 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

10. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be perform.

Corrective Action

None Required

LAMP POWER SUPPLY OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Connect multimeter between SDC L/S and COM test jacks on unit tester.
3. Adjust unit tester SIGNAL DATA CONVERTER P/S CONTROL to obtain 4.9 to 5.1 vdc indication on multimeter.

Indication/Condition

Simulator tester No. 1 LAMP SUPPLY OVERLOAD lamp shall be on.

Corrective Action

1. If Indication/Condition is not as specified, replace lamp power supply (WP 0806 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Connect multimeter between SDC CONT and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 4.5 and 4.9 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A11 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

5. Adjust unit tester SIGNAL DATA CONVERTER P/S CONTROL to obtain 1.70 to 1.90 vdc indication on multimeter.

Indication/Condition

No. 1 LAMP SUPPLY OVERLOAD lamp shall be dim.

Corrective Action

1. If Indication/Condition is not as specified, replace voltage regulator module A11 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

6. Connect multimeter between SDC L/S and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 2.0 and 2.2 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace lamp power supply (WP 0806 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

LAMP POWER SUPPLY OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

7. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**PROCESSOR FAILURE SWITCH-OVER OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Momentarily place unit tester SIGNAL DATA CONVERTER PROCESSOR SELECTION switch to up position.

Indication/Condition

- a. If not already on, SIGNAL DATA CONVERTER PROCESSOR SELECTION ANALOG A lamp on the unit tester shall go on and remain on.
- b. If not already on, SIGNAL DATA CONVERTER PROCESSOR DIGITAL A lamp on the unit tester shall go on and remain on.

Corrective Action

1. If Indication/Condition a. is not as specified, replace analog processor No. 1, module A9 (WP 0805 00).
 - a. If trouble remains, replace analog processor No. 2, module A8 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.
2. If Indication/Condition b. is not as specified, replace digital processor module A10 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

3. Place and hold unit tester SIGNAL DATA CONVERTER PROC FAILURES switch to A.

Indication/Condition

After approximately 5 seconds, the following shall occur:

- PROCESSOR SELECTION ANALOG A lamp shall go off.
- PROCESSOR SELECTION ANALOG B lamp shall go on.
- Unit tester SIGNAL DATA CONVERTER FAIL WARN lamp shall go on.

Corrective Action

1. If Indication/Conditions are not as specified, replace analog processor No. 1, module A9 (WP 0805 00).
 - a. If trouble remains, replace analog processor No. 2, module A8 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

PROCESSOR FAILURE SWITCH-OVER OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

4. Release unit tester PROC FAILURES switch.

Indication/Condition

FAIL WARN lamp shall go off.

Corrective Action

1. If Indication/Condition is not as specified, replace analog processor No. 1, module A9 (WP 0805 00).
 - a. If trouble remains, replace analog processor No. 2, module A8 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

5. Place and hold unit tester PROC FAILURES switch to B.

Indication/Condition

After approximately 5 seconds, the following shall occur:

- PROCESSOR SELECTION ANALOG B lamp shall go off.
- PROCESSOR SELECTION ANALOG A lamp shall go on.
- FAIL WARN lamp shall go on.

Corrective Action

1. If Indication/Conditions are not as specified, replace analog processor No. 1, module A9 (WP 0805 00).
 - a. If trouble remains, replace analog processor No. 2, module A8 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

6. Release unit tester PROC FAILURES switch.

Indication/Condition

FAIL WARN lamp shall go off.

Corrective Action

1. If Indication/Condition is not as specified, replace analog processor No. 1, module A9 (WP 0805 00).
 - a. If trouble remains, replace analog processor No. 2, module A8 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

7. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

LAMP POWER SUPPLY FAILURE OPERATIONAL/TROUBLESHOOTING PROCEDURE PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. If not already connected, connect multimeter between SDC L/S and COM test jacks on unit tester.
3. Turn unit tester SIGNAL DATA CONVERTER P/S CONTROL fully clockwise.

Indication/Condition

Simulator tester No. 1 LAMP SUPPLY OVERLOAD lamp shall be on.

Corrective Action

1. If Indication/Condition is not as specified, replace lamp power supply (WP 0806 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place and hold simulator tester LAMP SUPPLY OVERLOAD switch to No. 1.

Indication/Condition

- a. Multimeter shall indicate less than 1.0 vdc.
- b. Simulator tester No. 1 LAMP SUPPLY OVERLOAD lamp shall go off.

Corrective Action

1. If Indication/Condition is not as specified, replace lamp power supply (WP 0806 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

5. Release simulator tester LAMP SUPPLY OVERLOAD switch.

Indication/Condition

- a. Multimeter shall indicate between 4.8 and 5.8 vdc.
- b. Simulator tester No. 1 LAMP SUPPLY OVERLOAD lamp shall go on.

Corrective Action

If Indication/Conditions are not as specified, replace lamp power supply (WP 0806 00).

Step

6. Disconnect digital multimeter.
7. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

SENSOR EXCITATION VOLTAGE OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Set multimeter to indicate ac voltage and connect leads between SDC AC1 and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 7.5 and 11.0 vac.

Corrective Action

1. If Indication/Condition is not as specified, replace logic power supply (WP 0807 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

3. Connect multimeter between SDC AC2 and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 7.5 and 11.0 vac.

Corrective Action

1. If Indication/Condition is not as specified, replace logic power supply (WP 0807 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Connect multimeter between SDC AC3 and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 7.5 and 11.0 vac.

Corrective Action

1. If Indication/Condition is not as specified, replace logic power supply (WP 0807 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

5. Disconnect multimeter.
6. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

CLOCK MONITOR OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Connect oscilloscope vertical input leads between SDC CLOCK and COM test jacks on unit tester.

Indication/Condition

Oscilloscope shall display square wave with these requirements:

- AMPLITUDE - 4 to 6 Volts
- PERIOD - 6.8 to 8.8 Microseconds

Corrective Action

1. If Indication/Condition is not as specified, replace digital processor module A10 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

3. Disconnect oscilloscope.
4. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed

Corrective Action

None Required

CLOCK INHIBIT (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place and hold unit tester SIGNAL DATA CONVERTER CLOCK INHIBIT switch to DIG.

Indication/Condition

- a. Both SIGNAL DATA CONVERTER PROCESSOR SELECTION DIGITAL A and B lamps on unit tester shall be off.
- b. The following unit tester SIGNAL DATA CONVERTER lamps shall go off:
 - TEST FREQ DIG
 - MONITORS DIGITAL O/P
 - MONITORS MUX DIGITAL
- c. Unit tester SIGNAL DATA CONVERTER FAIL WARN lamp shall go on.

CLOCK INHIBIT (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If Indication/Conditions are not as specified, replace digital processor module A10 (WP 0805 00).

Step

3. Release unit tester CLOCK INHIBIT switch.

Indication/Condition

- a. The following lamps shall go on:
 - TEST FREQ DIG (flash)
 - MONITORS DIGITAL O/P
 - MONITORS MUX DIGITAL
 - Either PROCESSOR SELECTION DIGITAL A or B
- b. FAIL WARN lamp shall go off.

Corrective Action

If Indication/Conditions are not as specified, replace digital processor module A10 (WP 0805 00).

Step

4. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

CLOCK INHIBIT (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place and hold unit tester SIGNAL DATA CONVERTER CLOCK INHIBIT switch to ANA.

Indication/Condition

- a. Both SIGNAL DATA CONVERTER PROCESSOR SELECTION ANALOG A and B lamps on unit tester shall be off.
- b. The following unit tester SIGNAL DATA CONVERTER lamps shall go off:
 - TEST FREQ ANA
 - MONITORS SHIFT REG
 - MONITORS MUX ANALOG
- c. Unit tester SIGNAL DATA CONVERTER FAIL WARN lamp shall go on.

CLOCK INHIBIT (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Conditions are not as specified, replace analog processor No. 1, module A9 (WP 0805 00).
 - a. If trouble remains, replace analog processor No. 2, module A8 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

3. Release unit tester CLOCK INHIBIT switch.

Indication/Condition

- a. The following lamps shall go on:
 - TEST FREQ ANA (flash)
 - MONITORS SHIFT REG
 - MONITORS MUX ANALOG
 - Either PROCESSOR SELECTION ANALOG A or B
- b. FAIL WARN lamp shall go off.

Corrective Action

1. If Indication/Conditions are not as specified, replace analog processor No. 1, module A9 (WP 0805 00).
 - a. If trouble remains, replace analog processor No. 2, module A8 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

4. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**SENSOR INTERWIRING/WARNING OUTPUT OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Make sure No. 1 FUEL QUANTITY CAL-NOR switch on simulator tester is placed to NOR.
3. Set multimeter to indicate dc voltage and connect multimeter between SDC FUEL H and FUEL L test jacks on unit tester.
4. Turn simulator tester No. 1 FUEL QUANTITY variable control fully clockwise, then fully counterclockwise.

SENSOR INTERWIRING/WARNING OUTPUT OPERATIONAL/TROUBLESHOOTING PROCEDURE -
Continued**Indication/Condition**

Multimeter shall indicate between -0.5 and 0.5 vdc when the No. 1 FUEL QUANTITY control is fully counterclockwise, and shall indicate between 6.5 and 8.5 vdc when the No. 1 FUEL QUANTITY control is fully clockwise.

Corrective Action

If Indication/Condition is not as specified, the signal data converter chassis assembly is malfunctioning.

Step

5. Connect multimeter between SDC WIRE and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 4.5 and 5.5 vdc.

Corrective Action

If Indication/Condition is not as specified, the signal data converter chassis assembly is malfunctioning.

Step

6. Momentarily press unit tester SIGNAL DATA CONVERTER WARN OUTPUTS button.

Indication/Condition

- a. With WARNING OUTPUT button pressed, the following simulator tester lamps shall go on:

- LOW ROTOR SPD
- NO. 1 LOW OIL PRESS
- NO. 1 HIGH OIL TEMP
- NO. 1 ENGINE OUT

- b. If Indication/Condition is not as specified, the signal data converter chassis assembly is malfunctioning.

Corrective Action

None Required

Step

7. Connect multimeter between SDC W/C and COM test jacks on unit tester.

Indication/Condition

Multimeter shall indicate between 4.5 and 5.5 vdc.

Corrective Action

If Indication/Condition is not as specified, the signal data converter chassis assembly is malfunctioning.

Step

8. Momentarily press simulator tester ROTOR OVER SPD RESET button.

Indication/Condition

With ROTOR OVER SPD RESET button pressed, multimeter shall indicate between -0.5 and 0.5 vdc.

SENSOR INTERWIRING/WARNING OUTPUT OPERATIONAL/TROUBLESHOOTING PROCEDURE -
Continued**Corrective Action**

If Indication/Condition is not as specified, the signal data converter chassis assembly is malfunctioning.

Step

9. Disconnect multimeter.
10. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

NO. 1 ENGINE % RPM (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 1 ENGINE % RPM CAL-NOR switch to CAL and No. 1 ENGINE % RPM HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 1.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 952 and 962.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 ENGINE % RPM HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 5.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

Step

5. Place simulator tester No. 1 ENGINE % RPM CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 ENGINE % RPM variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

NO. 1 ENGINE % RPM (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 1 ENGINE % RPM variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

NO. 2 ENGINE % RPM (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE**Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 2 ENGINE % RPM CAL-NOR switch to CAL and No. 2 ENGINE % RPM HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 2.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 952 and 962.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 2 ENGINE % RPM HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 5.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

NO. 2 ENGINE % RPM (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

5. Place simulator tester No. 2 ENGINE % RPM CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 2 ENGINE % RPM variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 2 ENGINE % RPM variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**ROTOR SPEED (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester ROTOR SPEED CAL-NOR switch to CAL and ROTOR SPEED HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 3.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1016 and 1026.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester ROTOR SPEED HI-LO switch to LO.

ROTOR SPEED (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

INTERFACE MONITOR shall indicate between 0 and 5.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

5. Place simulator tester ROTOR CAL-NOR switch to NOR.
6. Slowly turn simulator tester ROTOR SPEED variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

7. Slowly turn simulator tester ROTOR SPEED variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**NO. 1 TORQUE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 1 TORQUE CAL-NOR switch to CAL and No. 1 TORQUE HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 4.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1042 and 1062.

NO. 1 TORQUE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 TORQUE HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 10.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

5. Place simulator tester No. 1 TORQUE CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 TORQUE variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 1 TORQUE variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**NO. 2 TORQUE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, if not already done.

NO. 2 TORQUE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

2. Place simulator tester No. 2 TORQUE CAL-NOR switch to CAL and No. 2 TORQUE HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 5.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1042 and 1062.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 2 TORQUE HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 10.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

5. Place simulator tester No. 2 TORQUE CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 2 TORQUE variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 2 TORQUE variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

ENGINE OIL TEMPERATURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 1 ENGINE OIL TEMP CAL-NOR switch to CAL and No. 1 ENGINE OIL TEMP HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 6.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1177 and 122A3.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 ENGINE OIL TEMP HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 16.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

Step

5. Place simulator tester No. 1 ENGINE OIL TEMP CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 ENGINE OIL TEMP variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 1 ENGINE OIL TEMP variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 2, module A6 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

ENGINE OIL TEMPERATURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Shutdown shall be performed.

Corrective Action

None Required

**ENGINE OIL PRESSURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 1 ENGINE OIL PRESS CAL-NOR switch to CAL and No. 1 ENGINE OIL PRESS HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 7.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 555 and 595.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 ENGINE OIL PRESS HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 20.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

Step

5. Place simulator tester No. 1 ENGINE OIL PRESS CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 ENGINE OIL PRESS variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 1 ENGINE OIL PRESS variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

ENGINE OIL PRESSURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**TURBINE GAS TEMPERATURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 1 TURBINE GAS TEMP CAL-NOR switch to CAL and No. 1 TURBINE GAS TEMP HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 8.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate 1267 and 1283.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 TURBINE GAS TEMP HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 8.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

Step

5. Place simulator tester No. 1 TURBINE GAS TEMP CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 TURBINE GAS TEMP variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

TURBINE GAS TEMPERATURE (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

7. Slowly turn simulator tester No. 1 TURBINE GAS TEMP variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**GAS GENERATOR SPEED (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 1 GAS GEN SPEED CAL-NOR switch to CAL and No. 1 GAS GEN SPEED HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 9.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 995 and 1005.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 GAS GEN SPEED HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 5.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

5. Place simulator tester No. 1 GAS GEN SPEED CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 GAS GEN SPEED variable control fully clockwise.

GAS GENERATOR SPEED (ANALOG) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 1 GAS GEN SPEED variable control counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**NO. 1 TORQUE (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 1 TORQUE CAL-NOR switch to CAL and No. 1 TORQUE HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 10.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1031 and 1051.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).
 - a. If trouble remains, replace digital processor, module A10 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 TORQUE HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 10.

NO. 1 TORQUE (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

5. Place simulator tester No. 1 TORQUE CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 TORQUE variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 1 TORQUE variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

NO. 2 TORQUE (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 2 TORQUE CAL-NOR switch to CAL and No. 2 TORQUE HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 11.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1031 and 1051.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).
 - a. If trouble remains, replace digital processor, module A10 (WP 0805 00).

NO. 2 TORQUE (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

(1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 2 TORQUE HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 10.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

5. Place simulator tester No. 2 TORQUE CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 2 TORQUE variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 2 TORQUE variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 3, module A5 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**TURBINE GAS TEMPERATURE (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 1 TURBINE GAS TEMP CAL-NOR switch to CAL and No. 1 TURBINE GAS TEMP HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 12.

TURBINE GAS TEMPERATURE (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Unit tester INTERFACE MONITOR shall indicate between 807 and 817.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).
 - a. If trouble remains, replace digital processor module A10 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 TURBINE GAS TEMP HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 8.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

Step

5. Place simulator tester No. 1 TURBINE GAS TEMP CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 TURBINE GAS TEMP variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 1 TURBINE GAS TEMP variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 4, module A4 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

GAS GENERATOR SPEED (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE

PROCEDURE

Step

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester No. 1 GAS GEN SPEED CAL-NOR switch to CAL and No. 1 GAS GEN SPEED HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 13.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 995 and 1005.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).
 - a. If trouble remains, replace digital processor module A10 (WP 0805 00).
 - (1) If trouble still remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester No. 1 GAS GEN SPEED HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 5.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

5. Place simulator tester No. 1 GAS GEN SPEED CAL-NOR switch to NOR.
6. Slowly turn simulator tester No. 1 GAS GEN SPEED variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

7. Slowly turn simulator tester No. 1 GAS GEN SPEED variable control fully counterclockwise.

Indication/Condition

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

GAS GENERATOR SPEED (DIGITAL) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**ROTOR SPEED (ANALOG RELAY) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester ROTOR SPEED CAL-NOR switch to CAL and ROTOR SPEED HI-LO switch to HI.
3. Place unit tester INTER CAL SELECT switch to 14.

Indication/Condition

Unit tester INTERFACE MONITOR shall indicate between 1016 and 1026.

Corrective Action

1. If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).
 - a. If trouble remains, the signal data converter chassis assembly is malfunctioning.

Step

4. Place simulator tester ROTOR SPEED HI-LO switch to LO.

Indication/Condition

INTERFACE MONITOR shall indicate between 0 and 5.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

5. Place simulator tester ROTOR CAL-NOR switch to NOR.
6. Slowly turn simulator tester ROTOR SPEED variable control fully clockwise.

Indication/Condition

INTERFACE MONITOR indication shall increase smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

7. Slowly turn simulator tester ROTOR SPEED variable control fully counterclockwise.

ROTOR SPEED (ANALOG RELAY) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

INTERFACE MONITOR indication shall decrease smoothly.

Corrective Action

If Indication/Condition is not as specified, replace interface No. 1, module A7 (WP 0805 00).

Step

8. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**INTEGRAL LIGHTING OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Do SETUP/POWER ON, in this work package, if not already done.
2. Place simulator tester INTEGRAL LIGHTING switch to INT.

Indication/Condition

Both SIGNAL DATA CONVERTER INT LIGHT lamps on unit tester shall go on.

Corrective Action

If Indication/Condition is not as specified, the signal data converter chassis assembly is malfunctioning.

Step

3. Place simulator tester INTEGRAL LIGHTING switch to EXT.
4. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

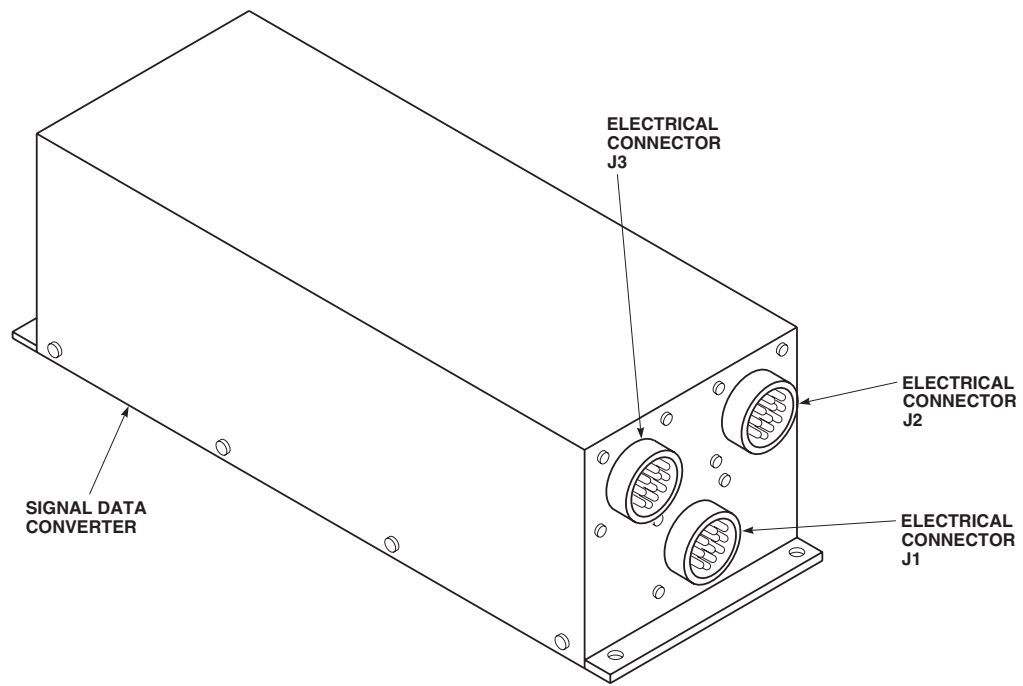
1. Place simulator tester INPUT POWER switch to OFF.
2. Disconnect cable assemblies.

SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Cable assemblies shall be disconnected.

Corrective Action

None Required

LOCATION

AA2570
SA

Figure 1. Signal Data Converter (SDC).

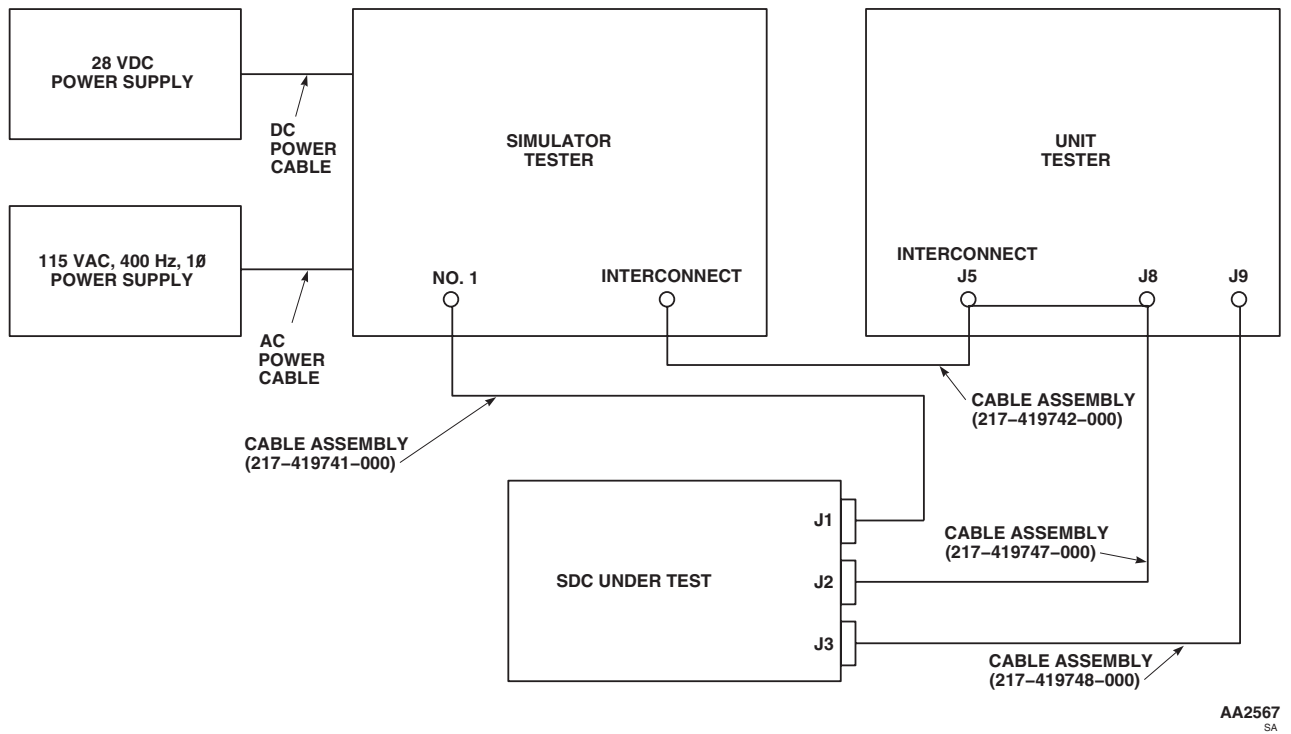
TEST SETUP

Figure 2. Signal Data Converter Test Setup.

END OF WORK PACKAGE

UNIT LEVEL**INSTRUMENT SYSTEMS****CAUTION/ADVISORY PANEL (AVIM)**

This WP supersedes WP 0097 00, dated 17 April 2006.

INITIAL SETUP:**Test Equipment**

Decade Resistor, ZM57U
Multimeter, AN/PSM-45A

Tools and Special Tools

Electrical Repairer Toolkit, SC 5180-99-B06
Locally-Made Caution/Advisory Test Adapter, Figure
53, WP 1805 00

Personnel Required

Aircraft Electrician MOS 15F (1)

References

WP 0815 00
WP 0816 00
WP 0817 00
WP 0818 00
WP 0821 00
WP 1747 00
WP 1805 00

Equipment Condition

27 to 28 vdc Bench Power Available

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step****NOTE**

- In the following operational/troubleshooting procedure, all voltage indications are dependent upon the input voltage being between 27 and 28 vdc.
- In the following operational/troubleshooting procedure, reference to caution/advisory panel is abbreviated as panel.
- Two variations of the caution/advisory panel exist. While system operations are identical, the internal operations are different. The internal operation differences are due to the insertion of light emitting diodes to illuminate the caution/advisory panel (LED) panel's light capsules where the legacy caution/advisory panels use incandescence lights. Before proceeding, insure that you are using the correct section of this manual for your application.
- See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Connect test adapter (WP 1805 00) CB1 and CB2 input terminals to bench power.
2. Make sure test adapter circuit breakers (circuit breakers) CB1 and CB2 are pulled out.
3. Make sure test adapter switches S2 and S3 are off.
4. Adjust test adapter resistors R1 and R2 fully counterclockwise.
5. Push in circuit breakers CB1 and CB2.
6. Connect No. 1 multimeter between test points TP8 and TP9.

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Indication/Condition

No. 1 multimeter shall indicated between 27 and 28 vdc.

Corrective Action

If Indication/Condition is not as specified, verify bench power is between 27 and 28 vdc.

Step

7. Connect test adapter test point (test point) TP9 to power ground.
8. Pull out circuit breakers CB1 and CB2.
9. Install an insulated jumper wire between test points TP8 and J117-65.
10. Install an insulated jumper wire between test points TP8 and J117-66.
11. Connect test adapter P118 to J118 at rear of panel.
12. Connect test adapter P117 to J117 at rear of panel.
13. Push in circuit breakers CB1 and CB2.
14. Make sure bench power is between 27 and 28 vdc.
15. Connect No. 1 multimeter between test points TP4 and TP5.
16. Connect No. 2 multimeter between test points TP1 and TP2.
17. Place and hold caution/advisory panel BRT/DIM-TEST switch at TEST.

Indication/Condition

- a. All caution/advisory panel capsules (capsules) on front of panel shall go on brightly.
- b. Multimeter No. 1 shall indicate between 22 and 26 vdc.
- c. Multimeter No. 2 shall indicate between 22 and 26 vdc.

Corrective Action

1. If Indication/Condition a. is not as specified, go to NO CAUTION/ADVISORY PANEL CAPSULES GO ON, in this work package.
2. If any single capsule or group of capsules listed by channel card in CAPSULES AND ASSOCIATED CHANNEL CARDS, in this work package, do not go on, replace associated channel card (WP 0815 00).
3. If Indication/Conditions b. and c. are not as specified, replace control card A5 (WP 0815 00).
 - a. If trouble remains, replace chassis-mounted Q1 (WP 0817 00).
4. If Indication/Condition b. is not as specified, repair/replace wiring, as required, between (WP 1747 00):
 - J118-V and J118-e
 - J118-U and J118-e
5. If Indication/Condition c. is not as specified, replace channel card A4 (WP 0815 00).
6. If only caution capsules do not go on, replace control card A5 (WP 0815 00).
 - a. If trouble remains, replace chassis-mounted Q3 (WP 0817 00).
7. If only advisory capsules do not go on, replace control card A5 (WP 0815 00).
 - a. If trouble remains, replace chassis-mounted Q2 (WP 0817 00).

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

18. Release caution/advisory panel BRT/DIM-TEST switch.

Indication/Condition

Switch handle shall return to center, spring-loaded, neutral position.

Corrective Action

If Indication/Condition is not as specified, replace switch (WP 0816 00).

Step

19. Install an insulated jumper wire between test points TP8 and TB1-A.
20. Connect No. 1 multimeter between test points TP8 and TP6.
21. Connect No. 2 multimeter between test points TP1 and TP2.

Indication/Condition

- a. No. 1 multimeter shall indicate greater than 25 vdc.
- b. No. 2 multimeter shall indicate 0 vdc.
- c. Capsule position 17 shall go on.
- d. Test adapter lamp DS1 shall remain off.

Corrective Action

1. If Indication/Condition a. is not as specified, replace control card A5 (WP 0815 00).
 - a. If trouble remains, replace chassis-mounted Q2 (WP 0817 00).
2. If Indication/Condition b. is not as specified, replace channel card A4 (WP 0815 00).
3. If Indication/Condition c. is not as specified, replace channel card A2 (WP 0815 00).
4. If Indication/Condition d. is not as specified, replace channel card A4 (WP 0815 00).

Step

22. Place test adapter switch S2 on.
23. Place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM and release.

Indication/Condition

No. 1 multimeter shall indicate between 3.0 and 4.5 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace control card A5 (WP 0815 00).
 - a. If trouble remains, replace chassis-mounted Q2 (WP 0817 00).

Step

24. Slowly adjust test adapter resistor R1 from fully counterclockwise to fully clockwise.

Indication/Condition

No. 1 multimeter shall indicate between 3.0 and 4.5 vdc (fully counterclockwise) to between 12 and 13 vdc (fully clockwise).

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition is not as specified, replace control card A5 (WP 0815 00).
 - a. If trouble remains, replace chassis-mounted Q2 (WP 0817 00).

Step

25. Adjust test adapter resistor R1 fully counterclockwise.
26. Place test adapter switch S3 on.
27. Adjust test adapter resistor R1 from fully counterclockwise to fully clockwise.

Indication/Condition

No. 1 multimeter shall indicate between 3.0 and 4.5 vdc and not change.

Corrective Action

1. If Indication/Condition is not as specified, replace control card A5 (WP 0815 00).
 - a. If trouble remains, replace chassis-mounted Q2 (WP 0817 00).

Step

28. Adjust test adapter resistor R1 fully counterclockwise.
29. Momentarily pull out then reset circuit breaker CB1.
30. Place test adapter switches S2 and S3 off.

Indication/Condition

Multimeter shall indicate between 3.0 and 4.5 vdc.

Corrective Action

If Indication/Condition is not as specified, replace control card A5 (WP 0815 00).

Step

31. Remove insulated jumper wire from between test points TP8 and TB1-A.
32. Momentarily pull out then reset circuit breaker CB1.
33. Install an insulated jumper wire between test points TP9 and J117-34.
34. Connect No. 1 multimeter between test points TP8 and TP7.

Indication/Condition

- a. Multimeter shall indicate greater than 25 vdc.
- b. Capsule position 42 shall go on.
- c. Test adapter lamp DS1 shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, replace control card A5 (WP 0815 00).
 - a. If trouble remains, replace chassis-mounted Q3 (WP 0817 00).
2. If Indication/Condition b. or c. is not as specified, replace channel card A3 (WP 0815 00).

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

35. Connect No. 1 multimeter between test points TP1 and TP2.
36. Momentarily press test adapter switch S1.

Indication/Condition

- a. Multimeter shall indicate 0 vdc.
- b. Test adapter lamp DS1 shall go off.

Corrective Action

If Indication/Conditions are not as specified, replace channel card A4 (WP 0815 00).

Step

37. Make sure test adapter switches S2 and S3 are off.
38. Momentarily pull out then reset circuit breaker CB1.
39. Momentarily install an insulated jumper wire between TP9 and J117-34.
40. Set decade resistor to 85 ohms.
41. Momentarily connect decade resistor, set to 85 ohms, between test points TP8 and TP3.
42. Connect No. 1 multimeter between test points TP8 and TP3.

Indication/Condition

- a. Multimeter shall indicate less than 0.3 vdc.
- b. Capsule position 42 shall go on.

Corrective Action

If Indication/Conditions are not as specified, replace channel card A3 (WP 0815 00).

Step

43. Place test adapter switch S2 on.
44. Place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM and release.

Indication/Condition

Multimeter shall indicate greater than 25 vdc.

Corrective Action

If Indication/Condition is not as specified, replace channel card A4 (WP 0815 00).

Step

45. Disconnect decade resistor connections from TP8 and TP3.
46. Connect No. 1 multimeter between TP8 and TP7.

Indication/Condition

Multimeter shall indicate between 3.0 and 4.5 vdc.

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition is not as specified, replace control card A5 (WP 0815 00).
 - a. If trouble remains, replace chassis-mounted Q3 (WP 0817 00).

Step

47. Slowly adjust test adapter resistor R1 from fully counterclockwise to fully clockwise.

Indication/Condition

No. 1 multimeter shall indicate between 3.0 and 4.5 vdc (fully counterclockwise) to between 12 and 13 vdc (fully clockwise).

Corrective Action

1. If Indication/Condition is not as specified, replace control card A5 (WP 0815 00).
 - a. If trouble remains, replace chassis-mounted Q3 (WP 0817 00).

Step

48. Adjust test adapter resistor R1 fully counterclockwise.
49. Place test adapter switch S3 on.
50. Adjust test adapter resistor R1 from fully counterclockwise to fully clockwise.

Indication/Condition

No. 1 multimeter shall indicate between 3.0 and 4.5 vdc and not change.

Corrective Action

1. If Indication/Condition is not as specified, replace control card A5 (WP 0815 00).
 - a. If trouble remains, replace chassis-mounted Q3 (WP 0817 00).

Step

51. Adjust test adapter resistor R1 fully counterclockwise.
52. Place test adapter switches S2 and S3 off.
53. Momentarily pull out then reset circuit breaker CB1.
54. Connect No. 2 multimeter between test points TP1 and TP2.

Indication/Condition

No. 2 multimeter shall indicate greater than 25 vdc.

Corrective Action

If Indication/Condition is not as specified, replace channel card A4 (WP 0815 00).

Step

55. Place test adapter switch S2 on.
56. Place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM and release.

Indication/Condition

No. 2 multimeter shall indicate between 3.0 and 4.5 vdc.

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If Indication/Condition is not as specified, replace channel card A4 (WP 0815 00).

Step

57. Slowly adjust test adapter resistor R2 from fully counterclockwise to fully clockwise.

Indication/Condition

No. 2 multimeter shall indicate between 3.0 and 4.5 vdc (fully counterclockwise) to between 14 and 15 vdc (fully clockwise).

Corrective Action

If Indication/Condition is not as specified, replace channel card A4 (WP 0815 00).

Step

58. Adjust test adapter resistor R2 fully counterclockwise.

59. Connect No. 1 multimeter between test points TP8 and TP7.

60. Place test adapter switch S3 on.

61. Slowly adjust test adapter resistor R2 from fully counterclockwise to fully clockwise.

Indication/Condition

No. 1 multimeter shall indicate between 3.0 and 4.5 vdc and not change.

Corrective Action

If Indication/Condition is not as specified, replace channel card A4 (WP 0815 00).

Step

62. Adjust test adapter resistor R2 fully counterclockwise.

63. Place test adapter S2 and S3 off.

64. Momentarily pull out then reset circuit breaker CB1.

65. Connect No. 1 multimeter between test points TP8 and TP5.

Indication/Condition

Multimeter shall indicate greater than 25 vdc.

Corrective Action

1. If Indication/Condition is not as specified, replace control card A5 (WP 0815 00).
 - a. If trouble remains, replace chassis-mounted Q1 (WP 0817 00).

Step

66. Place test adapter switch S2 on.

67. Place caution/advisory panel BRT/DIM-TEST switch to BRT/DIM and release.

Indication/Condition

Multimeter shall indicate between 3.0 and 4.5 vdc.

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Condition is not as specified, replace channel card A4 (WP 0815 00).
 - a. If trouble remains, replace chassis-mounted Q1 (WP 0817 00).

Step

68. Slowly adjust test adapter resistor R2 from fully counterclockwise to fully clockwise.

Indication/Condition

No. 1 multimeter shall indicate between 3.0 and 4.5 vdc (fully counterclockwise) to between 14.5 and 15.5 vdc (fully clockwise).

Corrective Action

1. If Indication/Condition is not as specified, replace channel card A4 (WP 0815 00).
 - a. If trouble remains, replace chassis-mounted Q1 (WP 0817 00).

Step

69. Adjust test adapter resistor R2 fully counterclockwise.
70. Place test adapter switch S3 on.
71. Adjust test adapter resistor R2 from fully counterclockwise to fully clockwise.

Indication/Condition

No. 1 multimeter shall indicate between 3.0 and 4.5 vdc (fully counterclockwise) to between 14.5 and 15.5 vdc (fully clockwise).

Corrective Action

1. If Indication/Condition is not as specified, replace control card A5 (WP 0815 00).
 - a. If trouble remains, replace chassis-mounted Q1 (WP 0817 00).

Step

72. Adjust test adapter resistor R2 fully counterclockwise.
73. Place test adapter S2 and S3 off.
74. Momentarily pull out then reset circuit breaker CB1.

Indication/Condition

Test adapter lamp DS1 shall go on.

Corrective Action

If Indication/Condition is not as specified, replace channel card A5 (WP 0815 00).

Step

75. Momentarily press test adapter switch S1.

Indication/Condition

Test adapter lamp DS1 shall go off and remain off.

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If Indication/Condition is not as specified, replace channel card A4 (WP 0815 00).

Step

76. Momentarily install an insulated jumper between test point TP8 (28 vdc) and test points listed in ADVISORY CAPSULES AND ASSOICATED CHANNEL CARDS, in this work package.

Indication/Condition

Each advisory capsule shall go on brightly when 28 vdc is applied to indicated test point and shall go off when 28 vdc is removed.

Corrective Action

1. If Indication/Condition is not as specified, replace indicated channel card (WP 0815 00).
2. If none of the capsules go on when 28 vdc is applied, replace overvoltage protector CR3 (WP 0817 00).

Step

77. Momentarily install an insulated jumper wire between test point TP8 (28 vdc) and test points listed in POSITIVE INPUT CAUTION CAPSULES, in this work package.

Indication/Condition

- a. Each caution capsule and test adapter lamp DS1 shall go on brightly when 28 vdc is applied to indicated test point and shall go off when 28 vdc is removed.
- b. Capsule positions 1 and 62 shall flash.

Corrective Action

1. If Indication/Conditions are not as specified, replace indicated channel card (WP 0815 00).
2. If none of the capsules go on when 28 vdc is applied, replace overvoltage protector CR3 (WP 0821 00).

Step

78. Connect one side of decade resistor to test point TP9 (28 vdc).
79. Set decade resistor to 50 ohms.
80. Connect other side of decade resistor to test point J117-5.

Indication/Condition

- a. Capsule position 5 shall go on when signal is applied.
- b. Test adapter lamp DS1 shall go on.
- c. Capsule position 5 shall go off when signal is removed.
- d. Test adapter lamp DS1 shall go off when signal is removed.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, adjust decade resistor between 45 and 55 ohms and repeat step 80.
2. If Indication/Conditions c. and d. are not as specified, replace channel card A3 (WP 0815 00).

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

81. Connect other side of decade resistor to test point J117-53.

Indication/Condition

- a. Capsule position 66 shall go on when signal is applied.
- b. Test adapter lamp DS1 shall go on.
- c. Capsule position 66 shall go off when signal is removed.
- d. Test adapter lamp DS1 shall go off when signal is removed.

Corrective Action

- 1. If Indication/Conditions a. and b. are not as specified, adjust decade resistor between 45 and 55 ohms and repeat step 81.
- 2. If Indication/Conditions c. and d. are not as specified, replace channel card A3 (WP 0815 00).

Step

82. Momentarily install an insulated jumper wire between test points TP9 (28 vdc) and J117-18.

Indication/Condition

- a. Capsule position 21 shall go on when signal is applied and go off when signal is removed.
- b. Test adapter lamp DS1 shall go on.

Corrective Action

If Indication/Conditions are not as specified, replace channel card A3 (WP 0815 00).

Step

83. Momentarily install an insulated jumper wire between test points TP9 (28 vdc) and J117-34.

Indication/Condition

- a. Capsule position 42 shall go on when signal is applied and go off when signal is removed.
- b. Test adapter lamp DS1 shall go on.

Corrective Action

If Indication/Condition is not as specified, replace channel card A3 (WP 0815 00).

Step

84. Momentarily connect decade resistor, set to 50 ohms, between test points TP9 (28 vdc) and J117-30.

Indication/Condition

- a. Capsule position 33 shall go on between 30 and 33 seconds after signal is applied and shall go off when signal is removed.
- b. Test adapter lamp DS1 shall go on.

Corrective Action

If Indication/Conditions are not as specified, replace channel card A3 (WP 0815 00).

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

85. Momentarily connect decade resistor, set to 50 ohms, between test points TP9 (28 vdc) and J117-61.

Indication/Condition

- a. Capsule position 74 shall go on when signal is applied and shall go off when signal is removed.
- b. Test adapter lamp DS1 shall go on.

Corrective Action

If Indication/Conditions are not as specified, replace channel card A3 (WP 0815 00).

Step

86. Connect decade resistor, set to 50 ohms, between test points TP9 (28 vdc) and J117-62.

Indication/Condition

- a. Capsule position 75 shall go on when signal is applied and shall go off when signal is removed.
- b. Test adapter lamp DS1 shall go on.

Corrective Action

If Indication/Conditions are not as specified, replace channel card A3 (WP 0815 00).

Step

87. Momentarily install an insulated jumper wire between test points TP9 (28 vdc) and J117-26.

Indication/Condition

- a. Capsule position 29 shall go on when signal is applied and shall go off when signal is removed.
- b. Test adapter lamp DS1 shall go on.

Corrective Action

If Indication/Conditions are not as specified, replace channel card A4 (WP 0815 00).

Step

88. Momentarily install an insulated jumper wire between test points TP9 (28 vdc) and J117-42.

Indication/Condition

- a. Capsule position 50 shall go on when signal is applied and shall go off when signal is removed.
- b. Test adapter lamp DS1 shall go on.

Corrective Action

If Indication/Conditions are not as specified, replace channel card A4 (WP 0815 00).

Step

89. Momentarily connect decade resistor, set to 50 ohms, between test points TP9 and J117-29.

Indication/Condition

- a. Capsule position 32 shall go on when signal is applied and shall go off when signal is removed.
- b. Test adapter lamp DS1 shall go on.

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

1. If Indication/Conditions are not as specified, adjust decade resistor between 45 and 55 ohms and repeat step 90.
 - a. If trouble remains, replace channel card A4 (WP 0815 00).

Step

90. Momentarily connect decade resistor, set to 50 ohms, between test points TP9 and J117-45.

Indication/Condition

- a. Capsule position 53 shall go on when signal is applied and shall go off when signal is removed.
- b. Test adapter lamp DS1 shall go on.

Corrective Action

1. If Indication/Conditions are not as specified, adjust decade resistor between 45 and 55 ohms and repeat step.
 - a. If trouble remains, replace channel card A4 (WP 0815 00).

Step

91. Momentarily connect decade resistor, set to 50 ohms, between test points TP9 and J117-13.

Indication/Condition

- a. Capsule position 13 shall go on when signal is applied and shall go off when signal is removed.
- b. Test adapter lamp DS1 shall go on.

Corrective Action

If Indication/Conditions are not as specified, replace channel card A4 (WP 0815 00).

Step

92. Momentarily connect decade resistor, set to 50 ohms, between test points TP9 and J117-14.

Indication/Condition

- a. Capsule position 14 shall go on when signal is applied and shall go off when signal is removed.
- b. Test adapter lamp DS1 shall go on.

Corrective Action

If Indication/Conditions are not as specified, replace channel card A4 (WP 0815 00).

Step

93. Remove bench power from test adapter.
94. Disconnect decade resistor.
95. Disconnect test adapter from caution/advisory panel.

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Table 1. Capsules And Associated Channel Cards.**

CARDS TO BE REPLACED	CAPSULE POSITION NO.
A1	2, 3, 4, 6, 7, 8, 9, 10, 11, 12, 15, 16, 20, 22, 23, 24, 25, 26, 27, 28, 30, 31, 34, 35, 41, 43, 44
A2	17, 18, 19, 45, 46, 47, 48, 49, 51, 52, 54, 55, 56, 63, 64, 65, 67, 68, 69, 70, 71, 72, 73, 76, 77
A3	5, 21, 33, 36, 37, 38, 39, 40, 42, 57, 58, 59, 60, 61, 66, 74, 75, 78, 79, 80
A4	1, 13, 14, 29, 32, 50, 53, 62, 81, 82

Table 2. Advisory Capsules And Associated Channel Cards.

TEST ADAPTER TEST POINT	CAPSULE POSITION NO.	CHANNEL CARD
TB1-A	17	A2
TB1-B	18	A2
TB1-C	19	A2
TB1-D	36	A3
TB1-E	37	A3
TB1-F	38	A3
TB1-G	39	A3
TB1-H	40	A3
TB1-J	57	A3
TB1-K	58	A3
TB1-L	59	A3
TB1-M	60	A3
TB1-N	61	A3
TB1-P	78	A3
TB1-R	79	A3
TB1-S	80	A3

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Table 2. Advisory Capsules And Associated Channel Cards. - Continued**

TEST ADAPTER TEST POINT	CAPSULE POSITION NO.	CHANNEL CARD
TB1-Y	81	A4
TB1-Z	82	A4

Table 3. Positive Input Caution Capsules.

TEST ADAPTER TEST POINT	CAPSULE POSITION NO.	CHANNEL CARD
J117-1	*1	A4
J117-2	2	A1
J117-3	3	A1
J117-4	4	A1
J117-6	6	A1
J117-7	7	A1
J117-8	8	A1
J117-9	9	A1
J117-10	10	A1
J117-11	11	A1
J117-12	12	A1
J117-15	15	A1
J117-16	16	A1
J117-17	20	A1
J117-19	22	A1
J117-20	23	A1
J117-21	24	A1
J117-22	25	A1
J117-23	26	A1

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Table 3. Positive Input Caution Capsules. - Continued**

TEST ADAPTER TEST POINT	CAPSULE POSITION NO.	CHANNEL CARD
J117-24	27	A1
J117-25	28	A1
J117-27	30	A1
J117-28	31	A1
J117-31	34	A1
J117-32	35	A1
J117-33	41	A1
J117-35	43	A1
J117-36	44	A1
J117-37	45	A2
J117-38	46	A2
J117-39	47	A2
J117-40	48	A2
J117-41	49	A2
J117-43	51	A2
J117-44	52	A2
J117-46	54	A2
J117-47	55	A2
J117-48	56	A2
J117-49	*62	A4
J117-50	63	A2
J117-51	64	A2
J117-52	65	A2
J117-54	67	A2

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Table 3. Positive Input Caution Capsules. - Continued**

TEST ADAPTER TEST POINT	CAPSULE POSITION NO.	CHANNEL CARD
J117-55	68	A2
J117-56	69	A2
J117-57	70	A2
J117-58	71	A2
J117-59	72	A2
J117-60	73	A2
J117-63	76	A2
J117-64	77	A2
*Indicates capsule flashes		

Indication/Condition

None Required

Corrective Action

None Required

NO CAUTION/ADVISORY PANEL CAPSULES GO ON**SYMPTOM**

All caution/advisory panel capsules shall go brightly.

MALFUNCTION

No caution/advisory panel capsules go on.

CORRECTIVE ACTION**1. With switch S1 held at TEST, check for 28 vdc between K1-X1 and K1-X2.**

- a. If voltage is as specified, go to **2**.
- b. If voltage is not as specified, go to **3**.

2. Check for 28 vdc between J118-e and test point TP-9.

- a. If voltage is as specified repair/replace wiring, as required between (WP 1747 00), then go to **5**:
 - J118-e and XA1-27
 - J118-e and XA1-72
 - J118-e and XA2-27

NO CAUTION/ADVISORY PANEL CAPSULES GO ON - Continued

- J118-e and XA2-72
 - J118-e and XA3-25
 - J118-e and XA3-70
 - J118-e and XA4-15
- b. If voltage is not as specified, replace relay K1 (WP 0818 00). Go to **5**.
- 3. Check for 28 vdc between S1-1 and test point TP9.**
- a. If voltage is as specified, go to **4**.
- b. If voltage is not as specified, repair/replace wiring as required between (WP 1747 00), then go to **5**:
- J118-U and S1-1
 - J118-V and S1-1
- 4. With switch S1 held at TEST, check for 28 vdc between S1-2 and test point TP9.**
- a. If voltage is as specified, repair/replace wiring, as required, between (WP 1747 00), then go to **5**:
- K1-X1 and S1-2
 - K1-X2 and ground
- b. If voltage is not as specified, replace switch S1 (WP 0816 00). Go to **5**.
- 5. Procedure completed.**

LOCATION

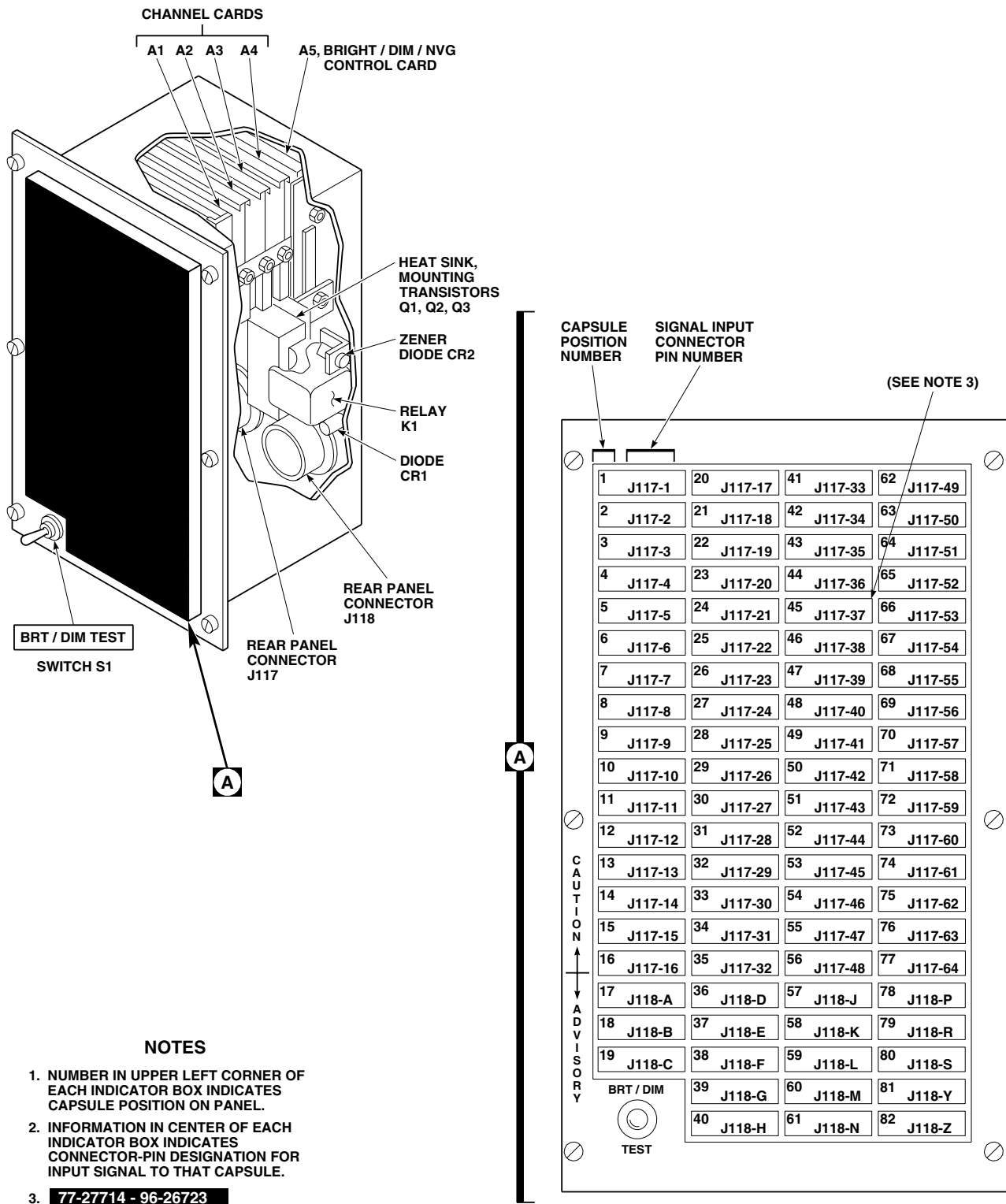
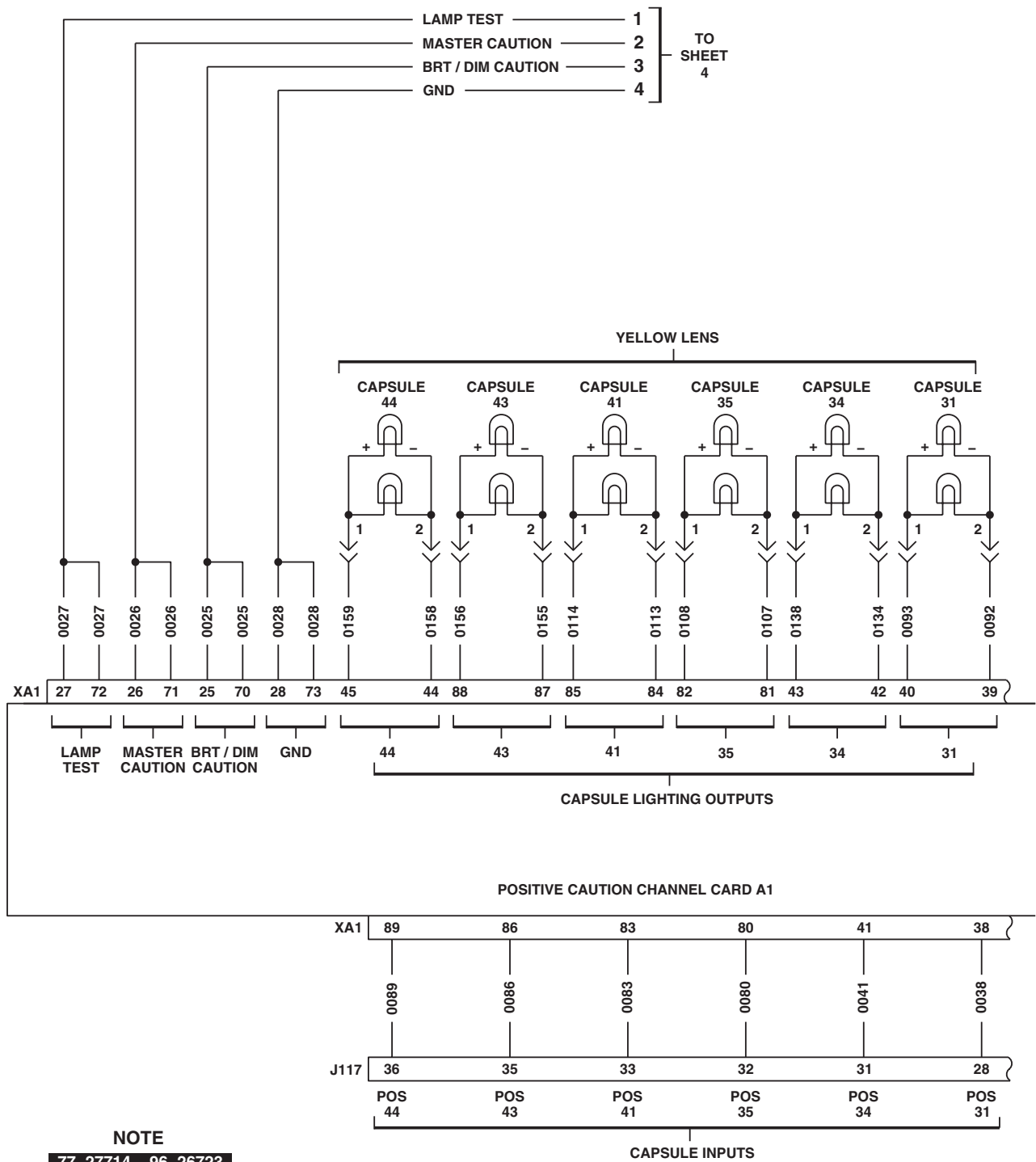
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Figure 1. Caution/Advisory Panel Components Location Diagram.

SCHEMATICS



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Figure 2. Caution/Advisory Panel Schematic Diagram. (Sheet 1 of 14)

SCHEMATICS - Continued

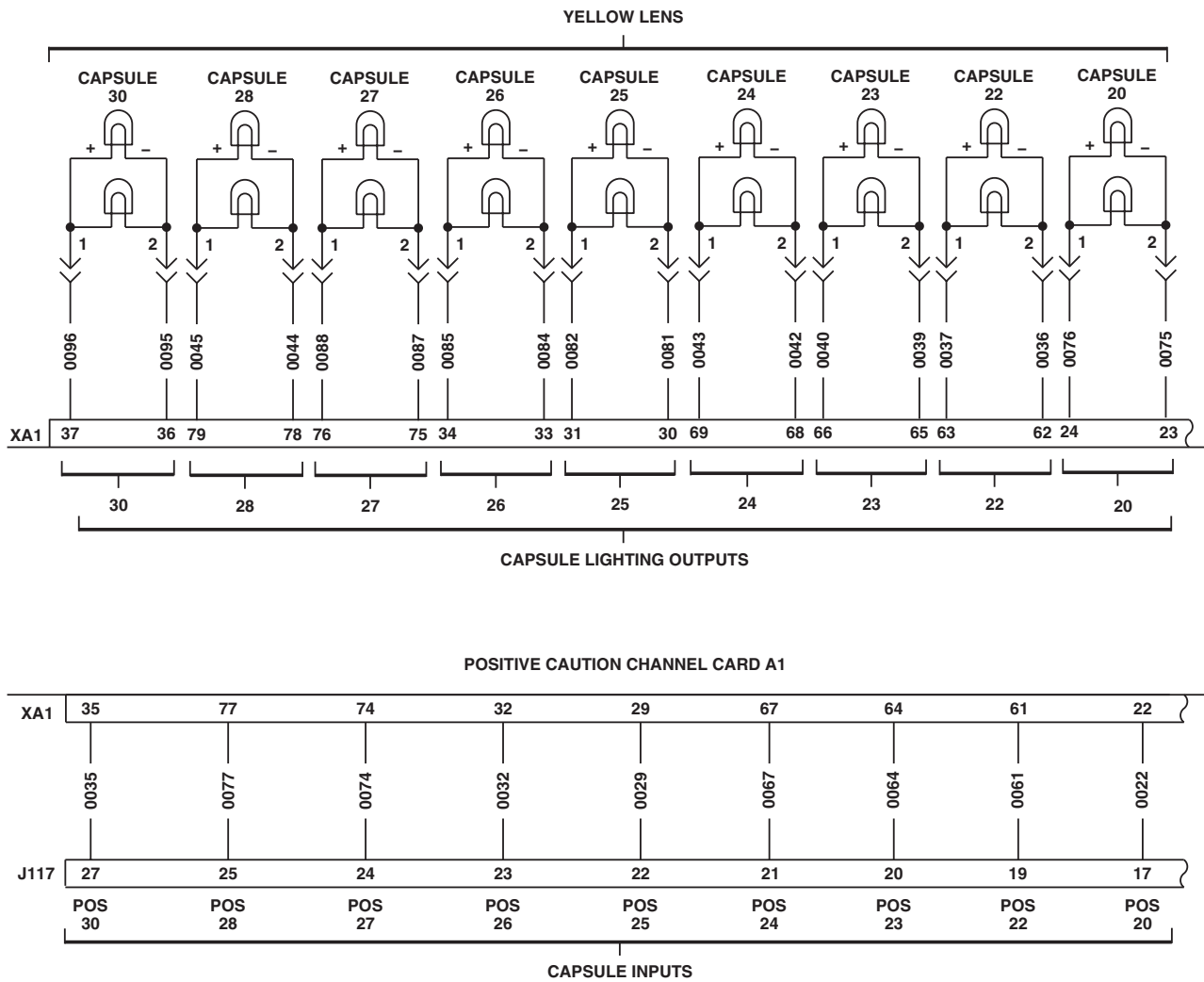


Figure 2. Caution/Advisory Panel Schematic Diagram. (Sheet 2 of 14)

SCHEMATICS - Continued

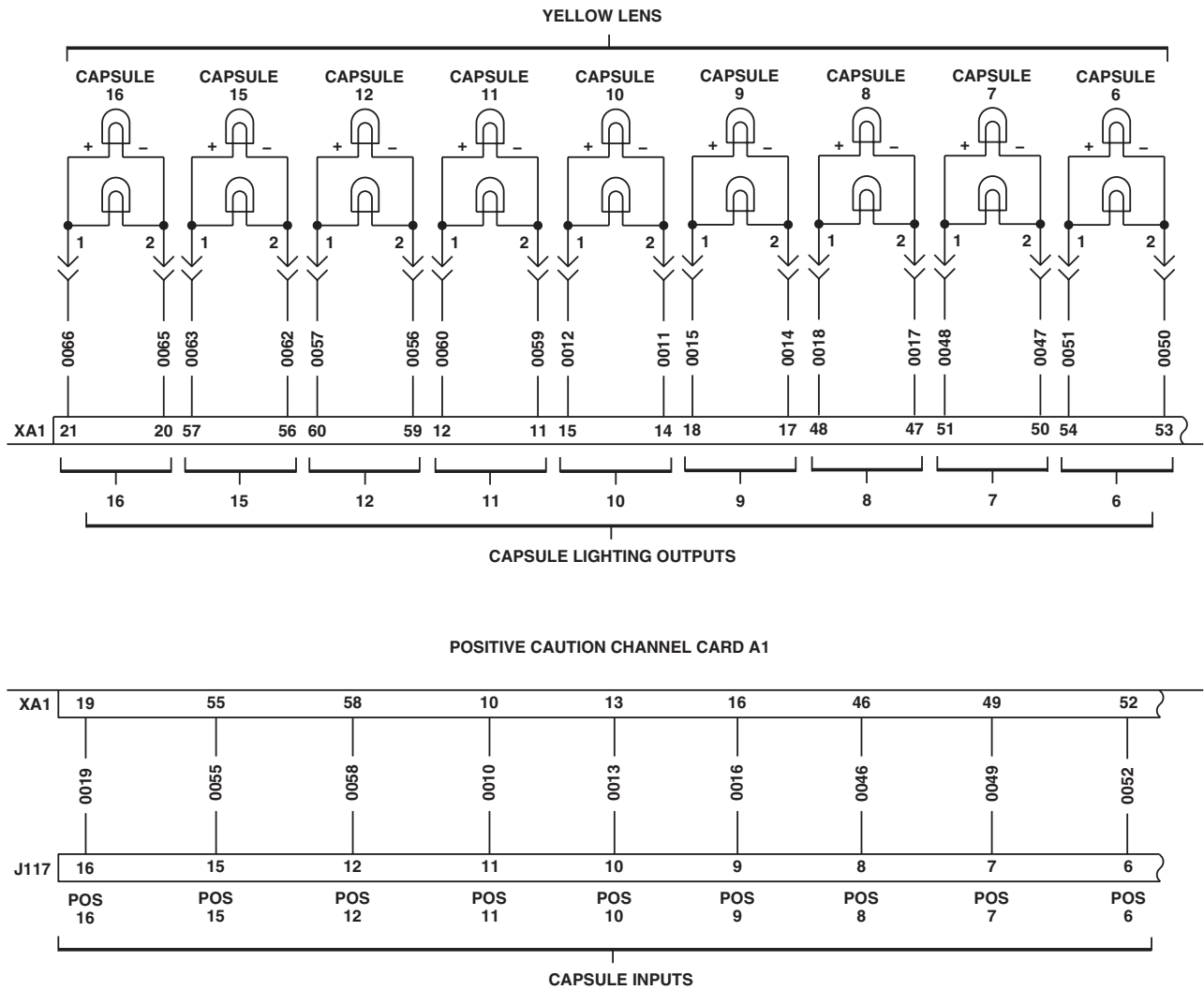


Figure 2. Caution/Advisory Panel Schematic Diagram. (Sheet 3 of 14)

SCHEMATICS - Continued

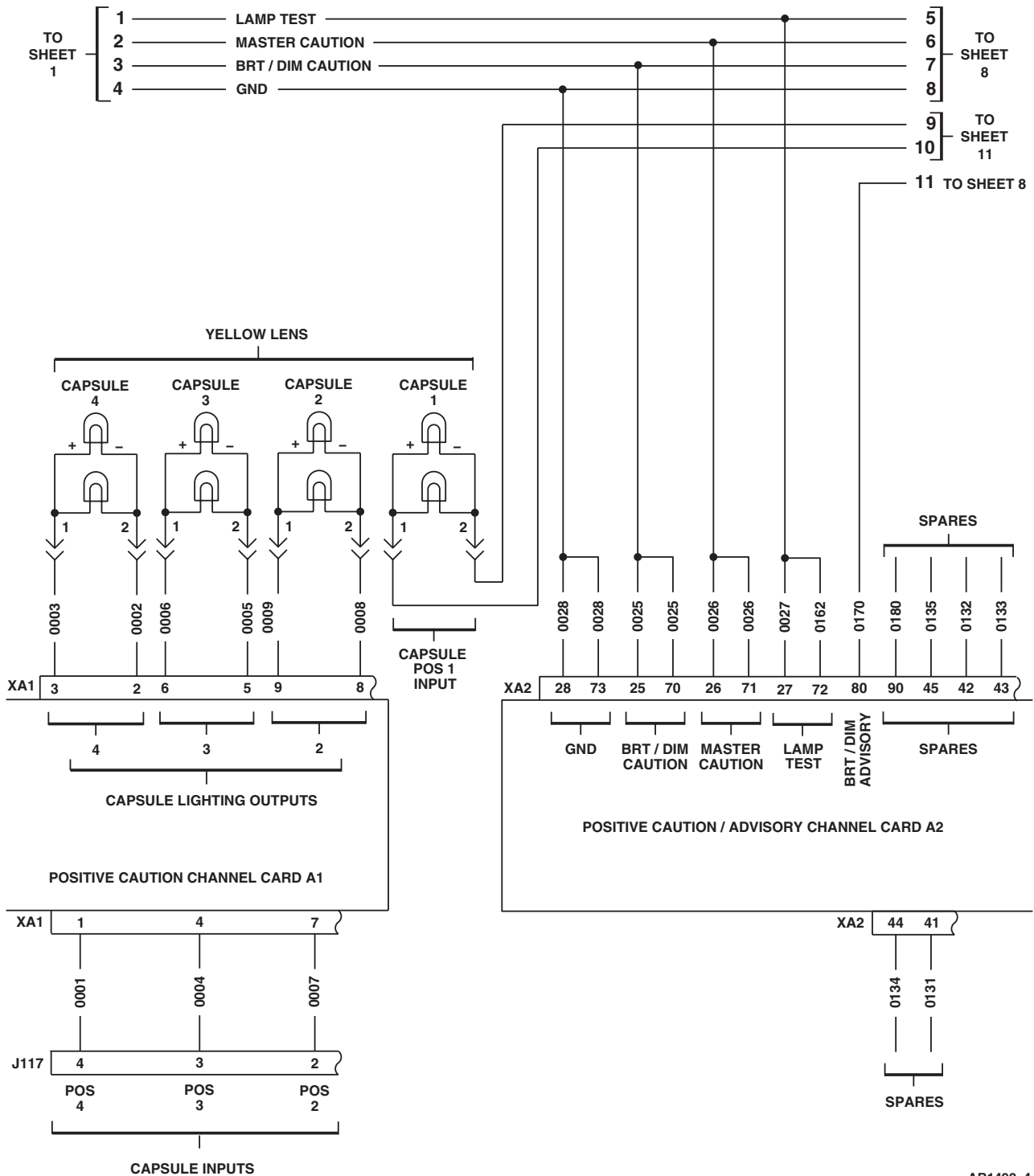
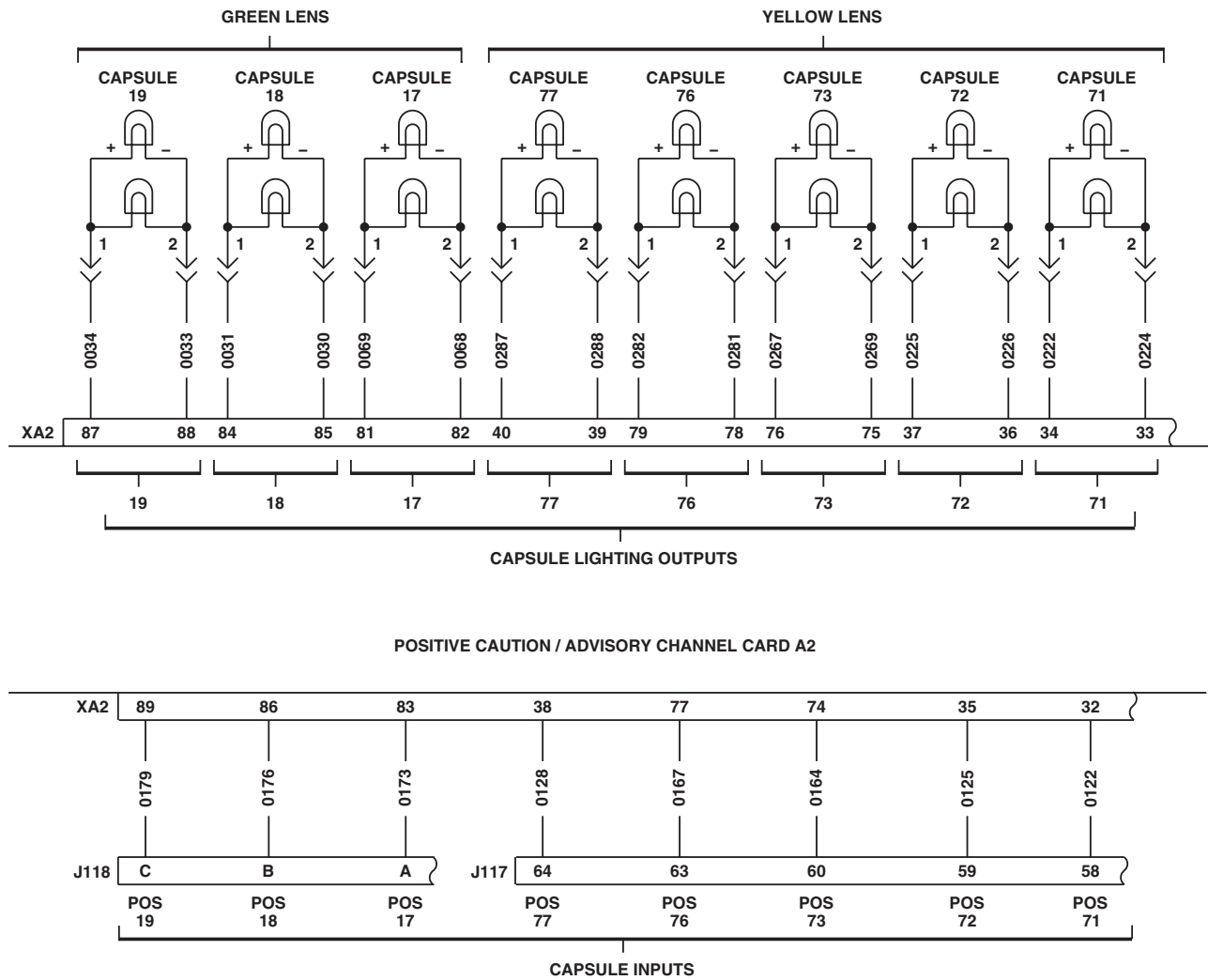


Figure 2. Caution/Advisory Panel Schematic Diagram. (Sheet 4 of 14)

SCHEMATICS - Continued



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Figure 2. Caution/Advisory Panel Schematic Diagram. (Sheet 5 of 14)

SCHEMATICS - Continued

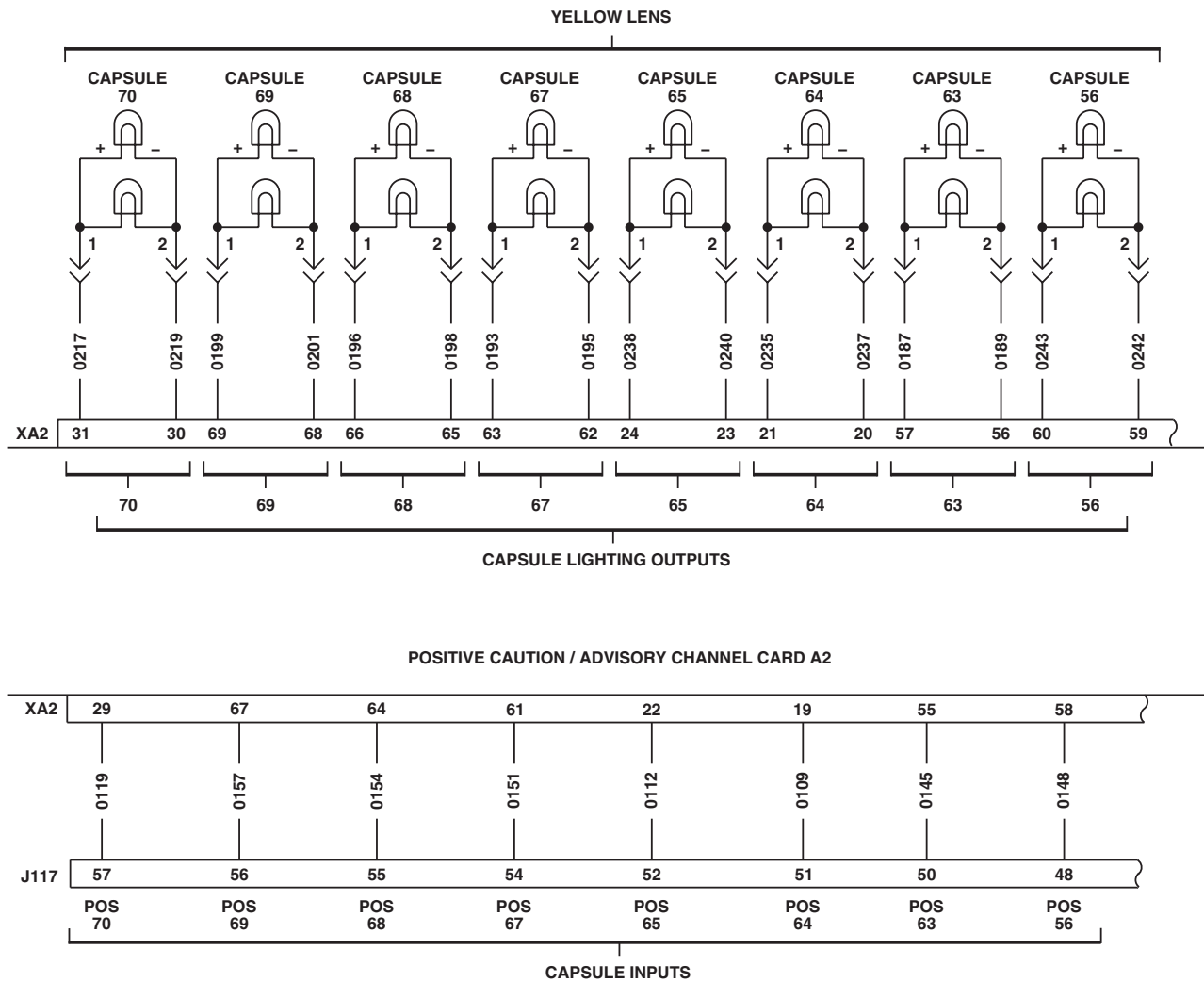


Figure 2. Caution/Advisory Panel Schematic Diagram. (Sheet 6 of 14)

SCHEMATICS - Continued

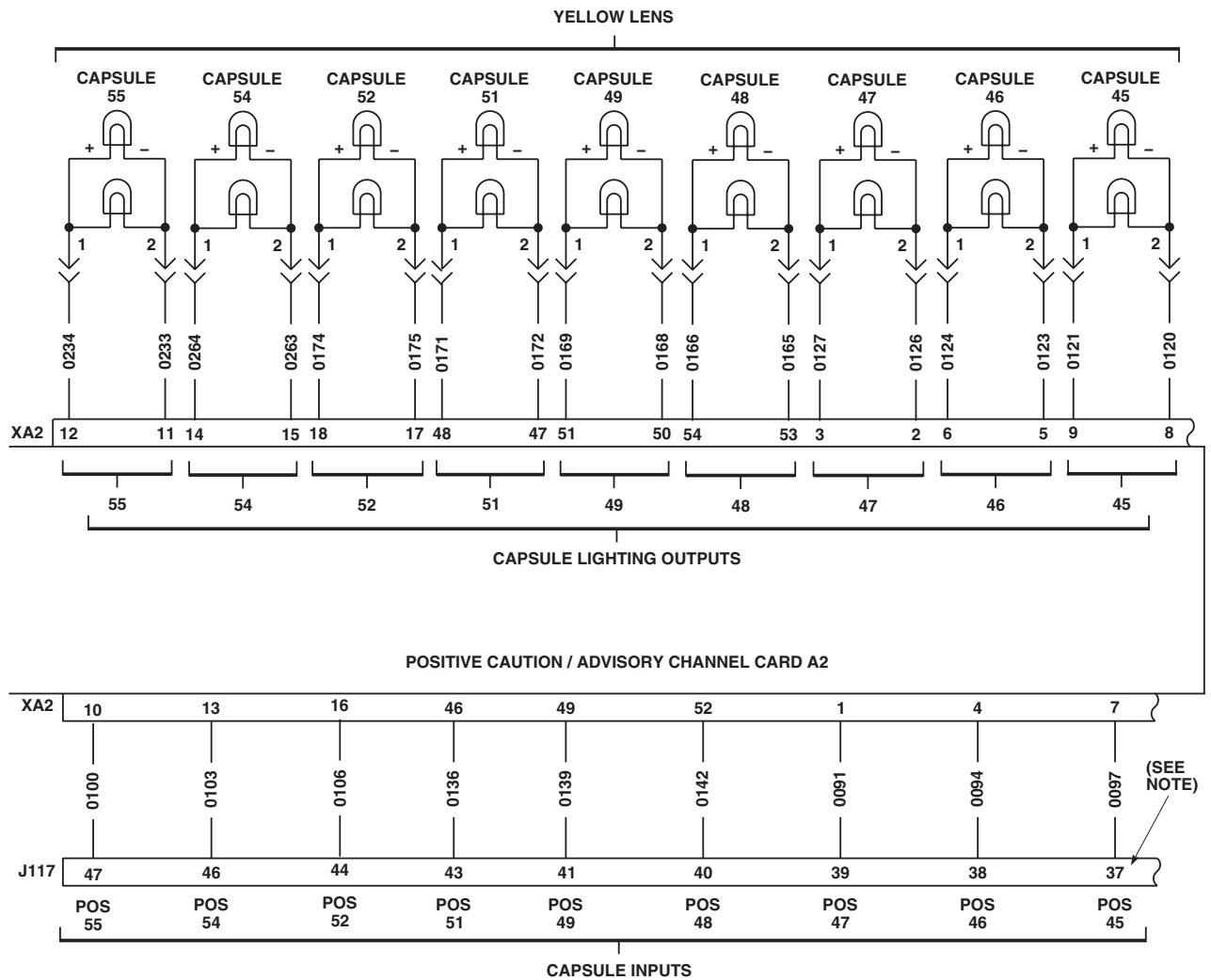
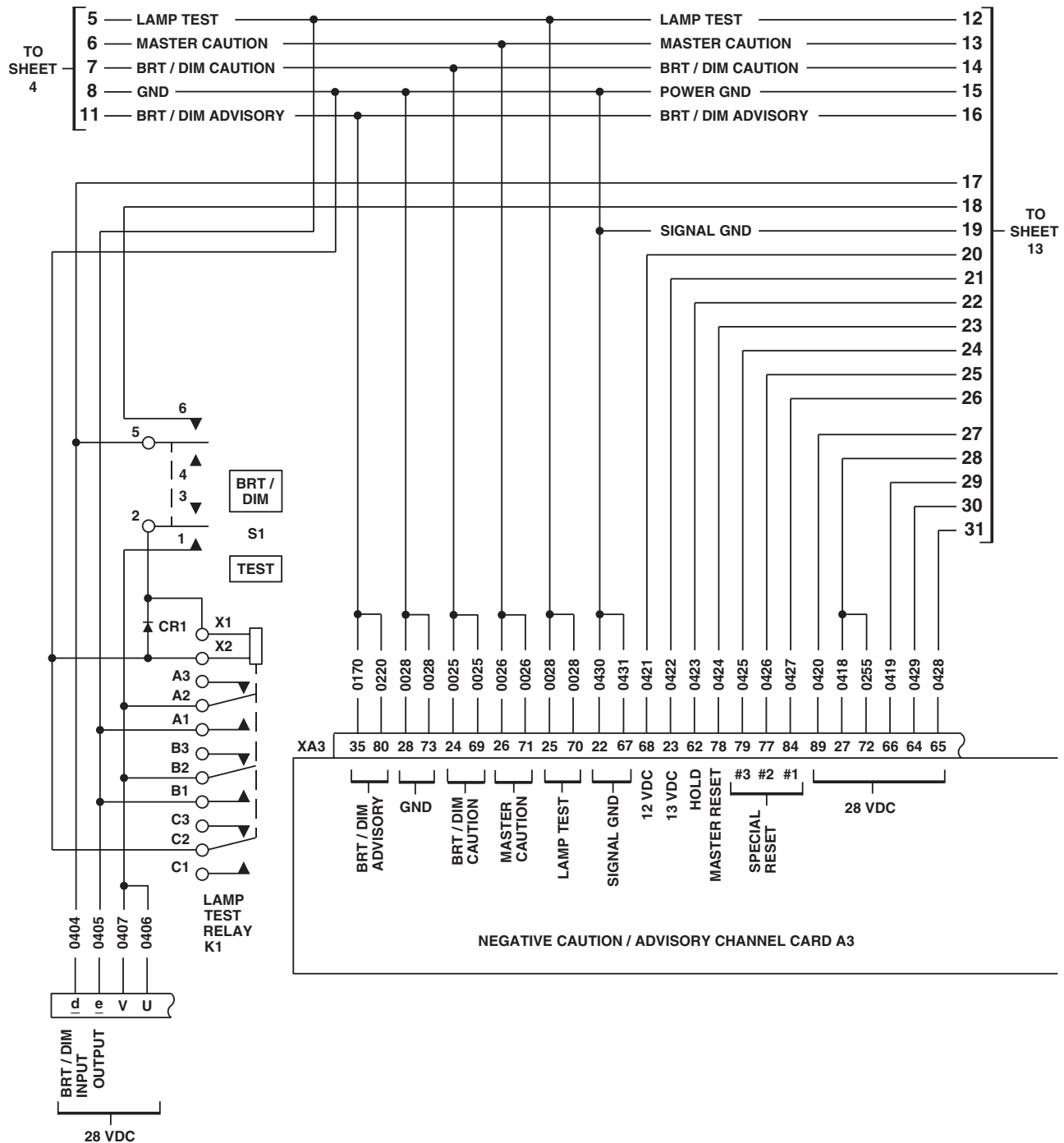
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Figure 2. Caution/Advisory Panel Schematic Diagram. (Sheet 7 of 14)

SCHEMATICS - Continued



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Figure 2. Caution/Advisory Panel Schematic Diagram. (Sheet 8 of 14)

SCHEMATICS - Continued

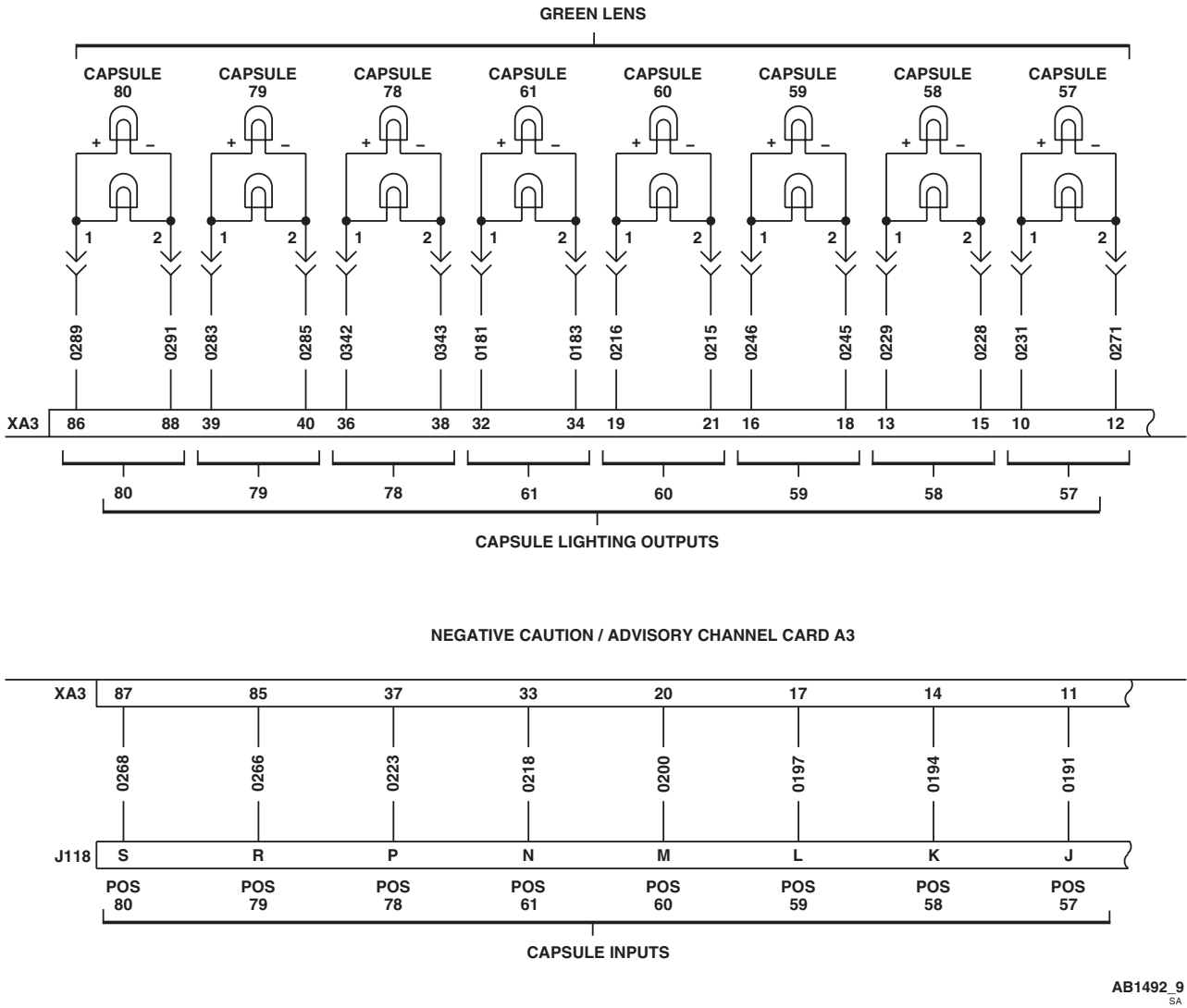


Figure 2. Caution/Advisory Panel Schematic Diagram. (Sheet 9 of 14)

SCHEMATICS - Continued

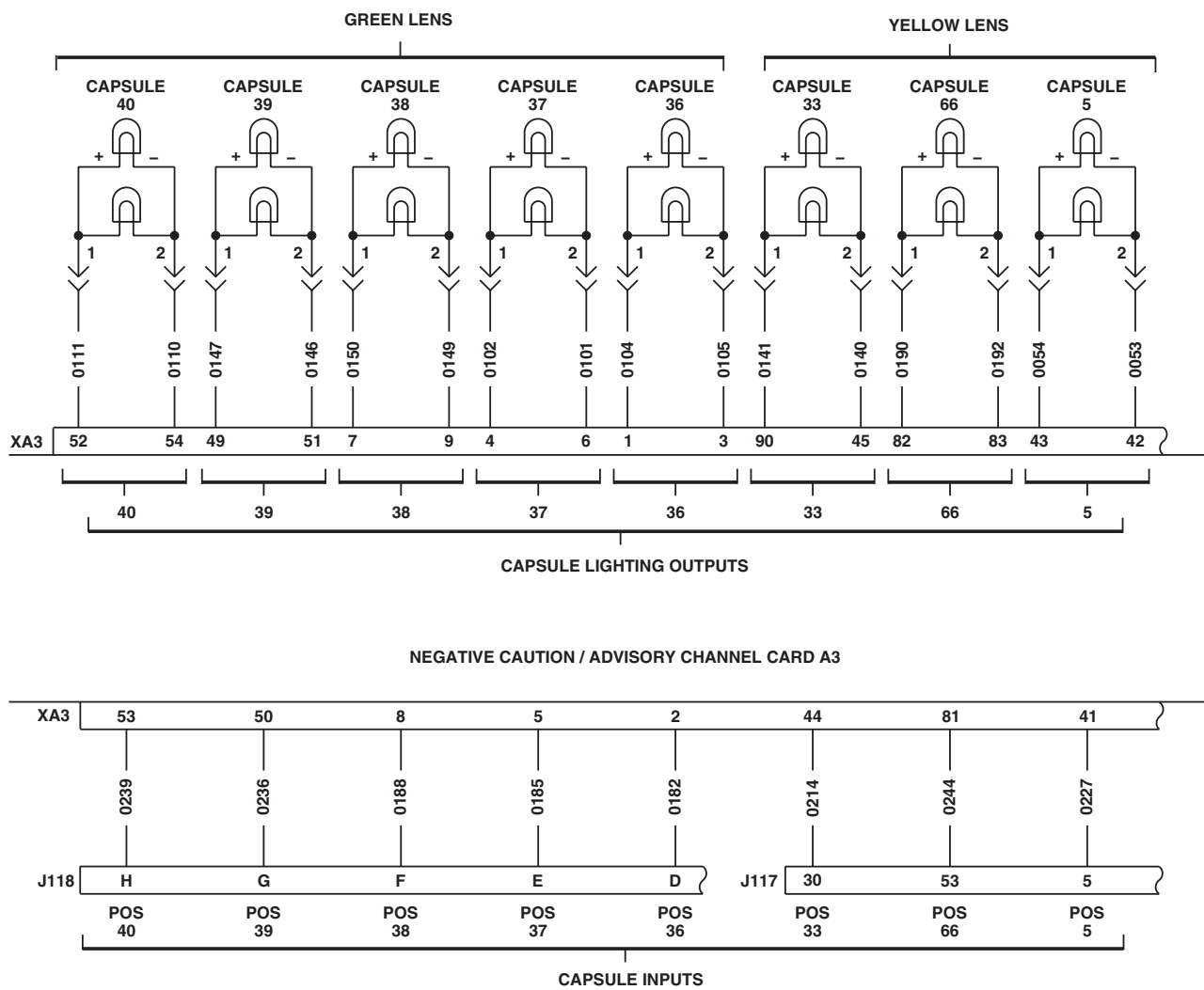
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Figure 2. Caution/Advisory Panel Schematic Diagram. (Sheet 10 of 14)

SCHEMATICS - Continued

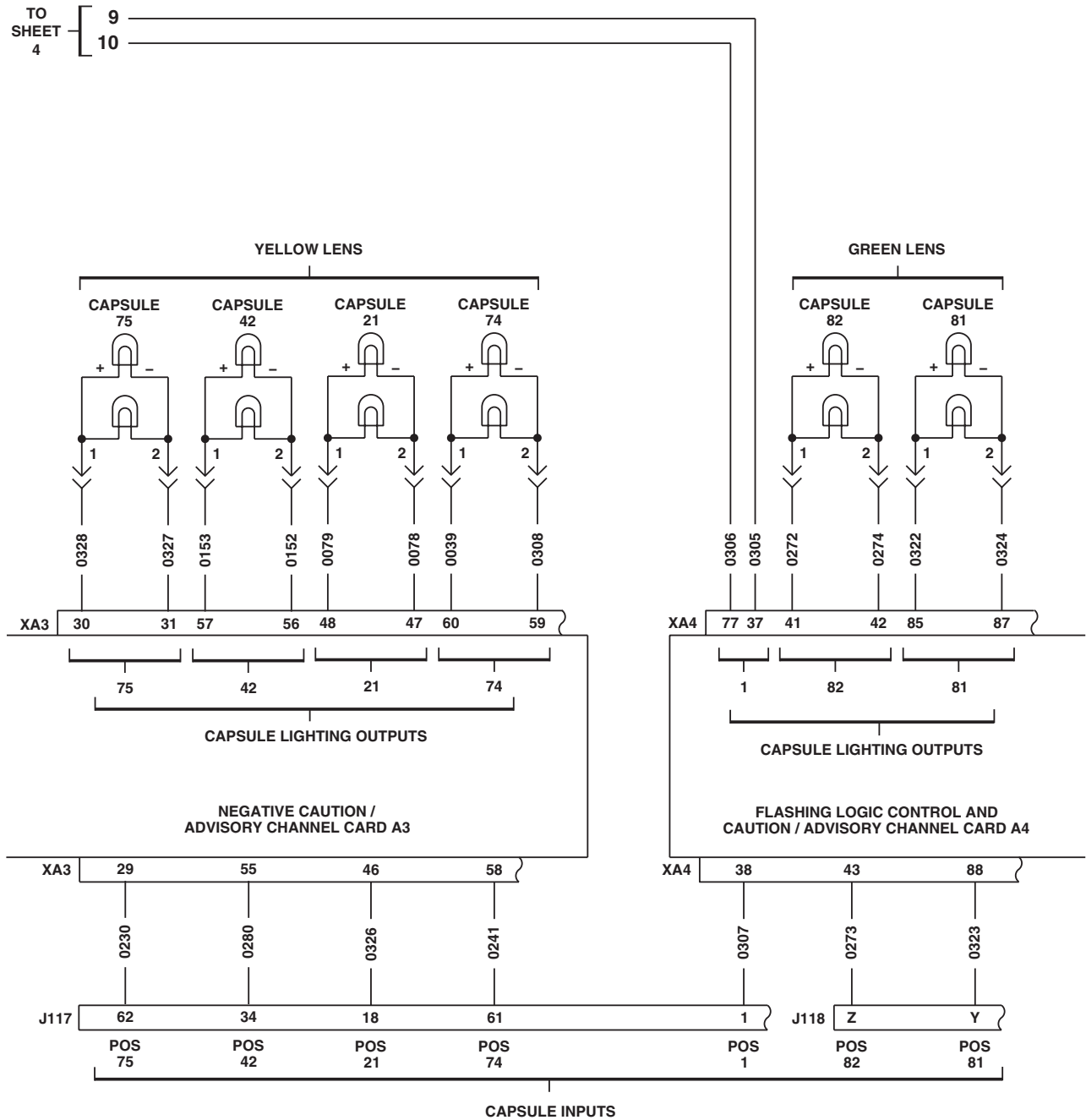
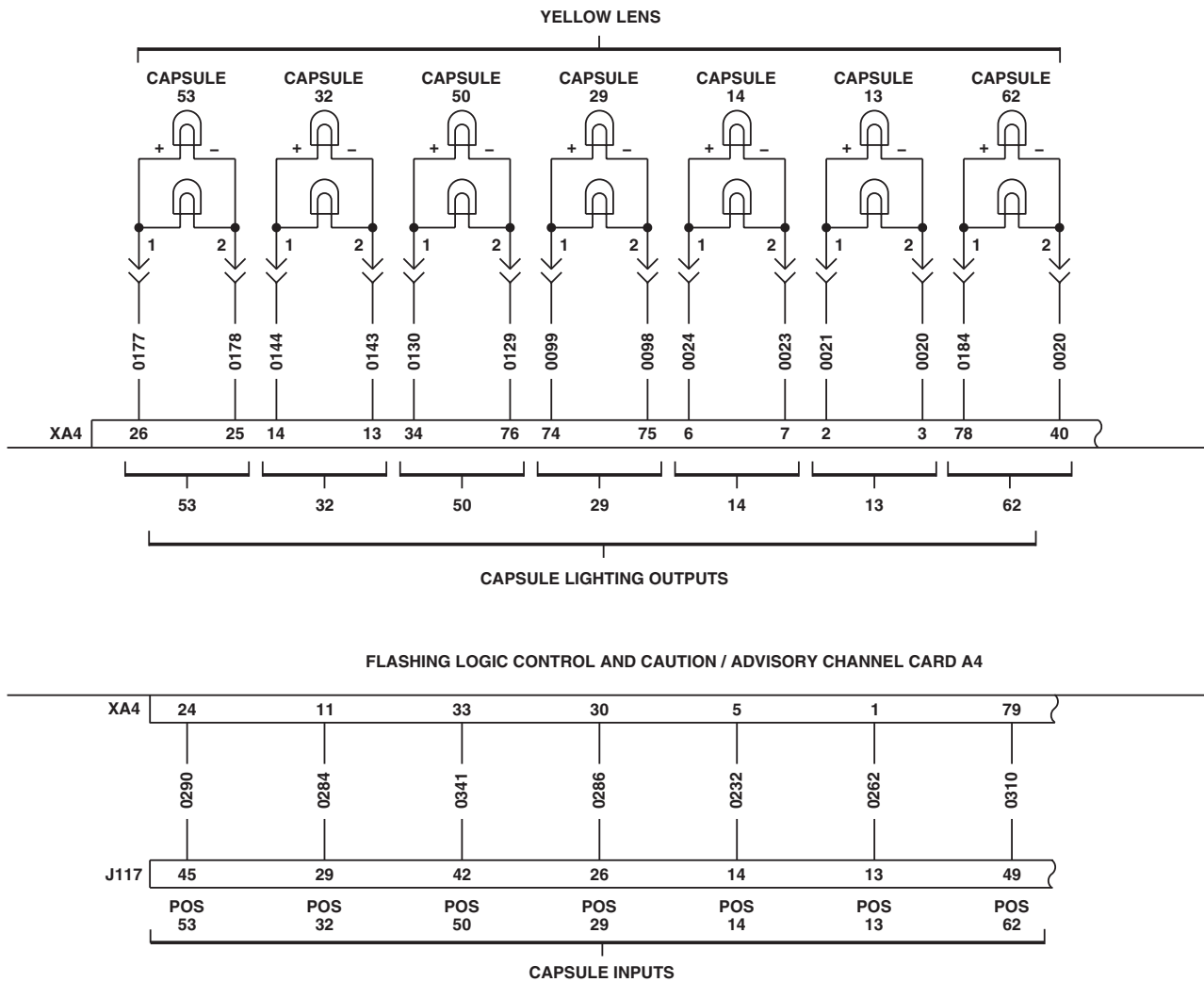
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Figure 2. Caution/Advisory Panel Schematic Diagram. (Sheet 11 of 14)

SCHEMATICS - Continued



AB1492_12
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Figure 2. Caution/Advisory Panel Schematic Diagram. (Sheet 12 of 14)

SCHEMATICS - Continued

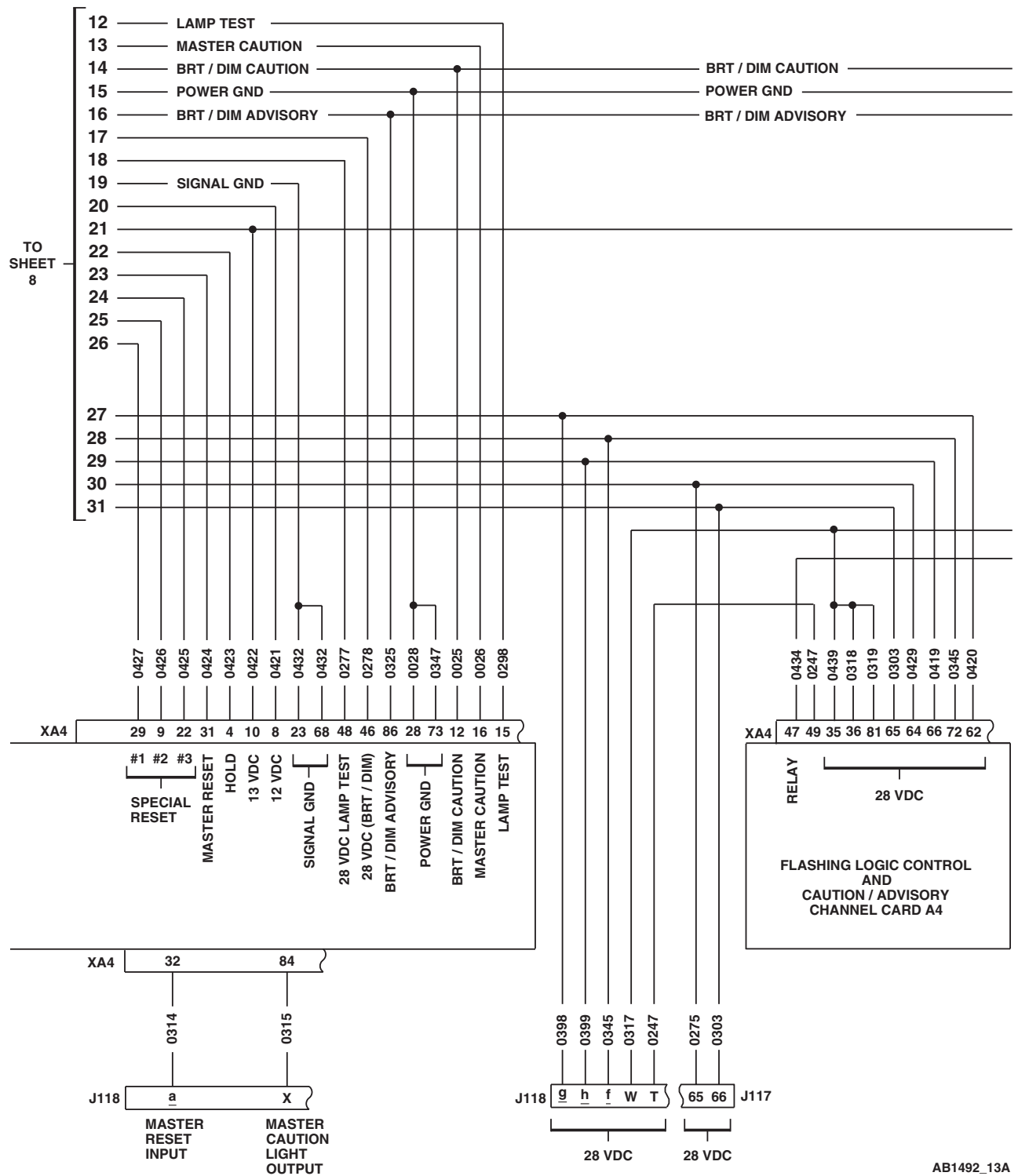


Figure 2. Caution/Advisory Panel Schematic Diagram. (Sheet 13 of 14)

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Figure 2. Caution/Advisory Panel Schematic Diagram. (Sheet 14 of 14)

UNIT LEVEL
INSTRUMENT SYSTEMS
CAUTION/ADVISORY PANEL (LED) (AVIM)

INITIAL SETUP:**Test Equipment**

Multimeter, AN/PSM-45A (2 required)

Tools and Special Tools

Electrical Repairer Toolkit, SC 5180-99-B06

Locally-Made Caution/Advisory Test Adapter, Figure
53, WP 1805 00

Personnel Required

Aircraft Electrician MOS 15F (1)

References

WP 0816 00

WP 0822 01

WP 0822 02

WP 0842 00

WP 1805 00

Equipment Condition

27 to 28 vdc Bench Power Available

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step****NOTE**

- In the following operational/troubleshooting procedure, all voltage indications are dependent upon the input voltage being between 27 and 28 vdc.
- In the following operational/troubleshooting procedure, reference to caution/advisory panel is abbreviated as panel.

1. Connect test adapter (WP 1805 00) CB1 and CB2 input terminals to bench power.
2. Make sure test adapter circuit breakers CB1 and CB2 are pulled out.
3. Make sure test adapter switches S2 and S3 are off.
4. Adjust test adapter resistors R1 and R2 fully counterclockwise.
5. Push in circuit breakers CB1 and CB2.
6. Connect No. 1 multimeter between test points TP8 and TP9.

Indication/Condition

No. 1 multimeter shall indicate between 27 and 28 vdc.

Corrective Action

If Indication/Condition is not as specified, verify bench power is between 27 and 28 vdc.

Step

7. Connect test adapter test point TP9 to power ground.
8. Pull out circuit breakers CB1 and CB2.
9. Install insulated jumper wire between test points TP8 and J117-65.

CAUTION/ADVISORY PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

10. Install insulated jumper wire between test points TP8 and J117-66.
11. Connect test adapter P118 to J118 at rear of panel.
12. Connect test adapter P117 to J117 at rear of panel.
13. Push in circuit breakers CB1 and CB2.
14. Make sure bench power is between 27 and 28 vdc.
15. Connect No. 1 multimeter between test points TP4 and TP5.
16. Connect No. 2 multimeter between test points TP1 and TP2.
17. Press and release caution/advisory panel S1 TEST switch.

Indication/Condition

Switch handle shall return to center, spring-loaded, neutral position.

Corrective Action

If Indication/Condition is not as specified, replace switch (WP 0816 00).

Step

18. Place and hold caution/advisory panel S1 Test switch to TEST.

Indication/Condition

- a. All caution/advisory panel capsules (capsules) on front of panel shall go on brightly.
- b. Multimeter No. 1 shall indicate between 22 and 28 vdc.
- c. Multimeter No. 2 shall indicate between 22 and 28 vdc.

Corrective Action**NOTE**

Each light capsule has 3 light emitting diodes (LED) inside. With 1 LED out, the caution/advisory panel operations will not be impacted: with 2 LEDs out, the light capsule will be dim and operation will be minimally impacted, replace with new caution/advisory panel at first opportunity.

1. If any single light capsule or group of light capsules do not light, replace caution advisory panel (WP 0842 00).
2. If results indicated on multimeters No. 1 and No. 2 are not as specified, remove all power and check all connectors between test adapter and caution/advisory panel. Turn power back on and repeat **18**. If results are not as specified, go to WP 0097 00, Table 1.
3. Remove bench power from test adapter.
4. Disconnect test adapter from caution/advisory panel.
5. Procedure completed.

NO CAUTION/ADVISORY PANEL CAPSULES GO ON**SYMPTOM**

All caution/advisory panel capsules shall go brightly.

MALFUNCTION

No caution/advisory panel capsules go on.

CORRECTIVE ACTION

1. Turn off bench power.
2. Disconnect test adapter connectors P117 from J117 and P118 from J118.

CAUTION

Damage to lightbox will result if Electro-Static Discharge (ESD) procedures are not followed during removal/replacement of lightbox assembly.

3. Separate caution/advisory panel lightbox assembly from input module assembly (WP 0842 00).
4. Connect multimeter leads (+) to Pin 1 and (-) to Pin 4 of CCA interconnect wiring harness.
5. Connect test adapter P117 to J117 and P118 to J118.
6. Turn on bench power.
7. Multimeter shall indicate between 5.6 and 5.8 vdc.
 - a. If result is not as specified, go to 8.
 - b. If result is as specified, go to 10.
8. Turn off bench power.
9. Disconnect CCA interconnect wiring harness from input module assembly by grasping connector, depressing connector locking tab and pulling connector away from chassis assembly.

NOTE

Wiring harness is a straight-through 6 pin harness. Pin 1 connects to pin 1, pin 2 connects to pin 2, pin 3 connects to pin 3, etc.

10. Check interconnect wiring harness wiring for continuity.
 - a. If continuity is not present, replace wiring harness (WP 0822 01) and go to 11.
 - b. If continuity is present, go to 11.

NOTE

Switch is a single-pole, double throw type with a center-off position.

11. Place lightbox assembly S1 test switch to center position and check for continuity.
 - a. If continuity is present, replace S1 test switch (WP 0822 02) and repeat 11.
 - b. If continuity is not present, go to 12.

NO CAUTION/ADVISORY PANEL CAPSULES GO ON - Continued

- 12. Place lightbox assembly S1 test switch to TEST position and check for continuity between pins 2 and 1.**
 - a. If continuity is not present, replace S1 test switch (WP 0822 02) and go to **11**.
 - b. If continuity is present, go to **13**.
- 13. Place lightbox assembly S1 test switch to TEST position and check for continuity between pins 2 and 3.**
 - a. If continuity is present, replace S1 test switch (WP 0822 02) and go to **11**.
 - b. If continuity is not present, go to **14**.
- 14. Place lightbox assembly S1 test switch to BRT/DIM position and check for continuity between pins 2 and 3.**
 - a. If continuity is not present, replace S1 test switch (WP 0822 02) and go to **11**.
 - b. If continuity is present, go to **15**.
- 15. Place lightbox assembly S1 test switch to BRT/DIM position and check for continuity between pins 2 and 1.**
 - a. If continuity is present, replace S1 test switch (WP 0822 02) and go to **11**.
 - b. If continuity is not present, present, go to **16**.
- 16. Assemble caution/advisory panel.**
- 17. Procedure completed.**

END OF WORK PACKAGE

UNIT LEVEL
INSTRUMENT SYSTEMS
CHIP DETECTOR RESISTOR UNIT (AVIM)

INITIAL SETUP:**Test Equipment**

Multimeter, AN/PSM-45A

References

WP 0823 00

WP 1747 00

Tools and Special Tools

Electrical Repairer Toolkit, SC 5180-99-B06

Personnel Required

Aircraft Electrician MOS 15F (1)

CHIP DETECTOR RESISTOR UNIT (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE

Step**NOTE**

See Figure 1 for component location diagram and Figure 2 for schematic diagram as an aid in operational/troubleshooting.

1. Remove cover assembly from unit and inspect for broken or burnt wires, security of components, and general conditions.

Indication/Condition

Wires shall not be broken or burnt, and components shall be secure and in good condition.

Corrective Action

If Indication/Condition is not as specified, repair/replace wiring as required (WP 1747 00).

Step

2. Check resistance between J3-H and J3-J.

Indication/Condition

Multimeter shall indicate between 19,000 and 21,000 ohms.

Corrective Action

1. If Indication/Condition is not as specified, check for between 19,000 and 21,000 ohms between terminals 1 and 7 on cover assembly.
 - a. If resistance is not as specified, replace R1 (WP 0823 00).
 - b. If resistance is as specified, repair/replace wiring or connector as required (WP 1747 00).

Step

3. Check resistance between J3-G and J3-J.

CHIP DETECTOR RESISTOR UNIT (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Multimeter shall indicate between 19,000 and 21,000 ohms.

Corrective Action

1. If Indication/Condition is not as specified, check for between 19,000 and 21,000 ohms between terminals 2 and 8 on cover assembly.
 - a. If resistance is not as specified, replace R2 (WP 0823 00).
 - b. If resistance is as specified, repair/replace wiring or connector as required (WP 1747 00).

Step

4. Check resistance between J3-F and J3-J.

Indication/Condition

Multimeter shall indicate between 19,000 and 21,000 ohms.

Corrective Action

1. If Indication/Condition is not as specified, check for between 19,000 and 21,000 ohms between terminals 3 and 9 on cover assembly.
 - a. If resistance is not as specified, replace R3 (WP 0823 00).
 - b. If resistance is as specified, repair/replace wiring or connector as required (WP 1747 00).

Step

5. Check resistance between J3-E and J3-J.

Indication/Condition

Multimeter shall indicate between 19,000 and 21,000 ohms.

Corrective Action

1. If Indication/Condition is not as specified, check for between 19,000 and 21,000 ohms between terminals 4 and 10 on cover assembly.
 - a. If resistance is not as specified, replace R4 (WP 0823 00).
 - b. If resistance is as specified, repair/replace wiring or connector as required (WP 1747 00).

Step

6. Check resistance between J3-D and J3-J.

Indication/Condition

Multimeter shall indicate between 19,000 and 21,000 ohms.

Corrective Action

1. If Indication/Condition is not as specified, check for between 19,000 and 21,000 ohms between terminals 5 and 11 on cover assembly.
 - a. If resistance is not as specified, replace R5 (WP 0823 00).
 - b. If resistance is as specified, repair/replace wiring or connector as required (WP 1747 00).

CHIP DETECTOR RESISTOR UNIT (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

7. Check resistance between J3-B and J3-C. Record indication. Reverse leads and record indication again.

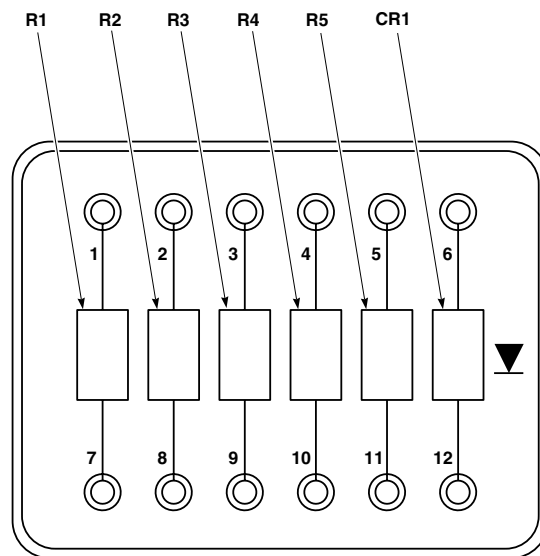
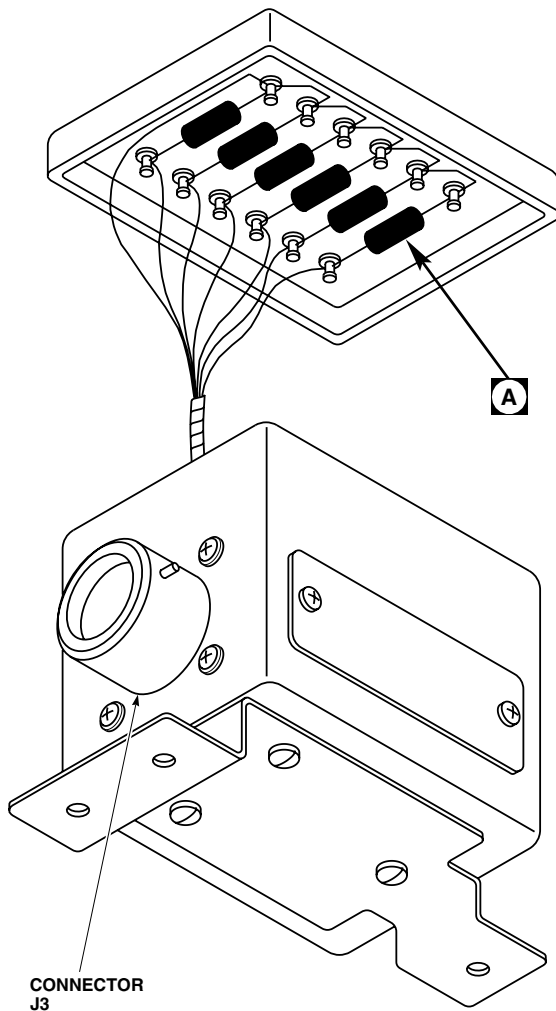
Indication/Condition

One indication shall be at least 10 times larger than the other.

Corrective Action

1. If Indication/Condition is not as specified, repeat step g. with multimeter connected to terminals 6 and 12 of cover assembly. One indication shall be at least 10 times larger than the other.
 - a. If resistance is not as specified, replace CR1 (WP 0823 00).
 - b. If resistance is as specified, repair/replace wiring or connector as required (WP 1747 00).

LOCATION



COVER ASSEMBLY

AA2440
SA

Figure 1. Chip Detector Resistor Unit Components Location Diagram.

SCHEMATICS

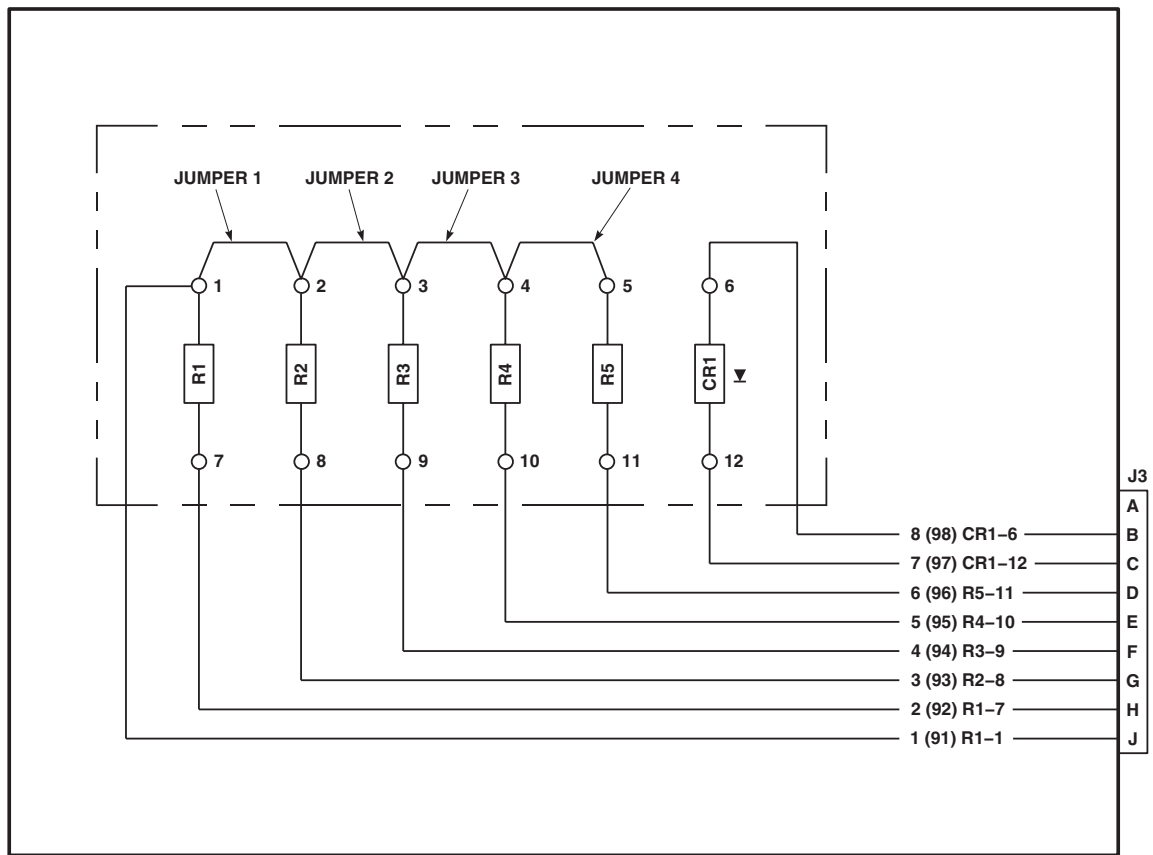
AB1448
SA

Figure 2. Chip Detector Resistor Unit Schematic Diagram.

END OF WORK PACKAGE

UNIT LEVEL
INSTRUMENT SYSTEMS
MISCELLANEOUS SWITCH PANEL (AVIM)
This WP supersedes WP 0099 00, dated 17 April 2006.

INITIAL SETUP:**Test Equipment**

Multimeter, AN/PSM-45A

Tools and Special Tools

Electrical Repairer Toolkit, SC 5180-99-B06

Locally-Made Switch Panel Test Harness, Figure 52,
WP 1805 00**Personnel Required**

Aircraft Electrician MOS 15F (1)

References

WP 0832 00

WP 0950 00

WP 0951 00

WP 1805 00

Equipment ConditionBench Power Available, Between 27 and 28 vdc, and
Between 114 and 116 vac, 395 to 405 Hz, 1ø

MISCELLANEOUS SWITCH PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE

Step**NOTE**

See Figure 1 for component location diagram, Figure 2 for schematic diagram, and Figure 3 for miscellaneous switch panel schematic diagram as an aid in operational/troubleshooting.

1. Remove cover from panel and inspect for broken or burnt wires, security of components, and general condition.

Indication/Condition

Wires shall not be broken or burnt, and components shall be secure and in good condition.

Corrective Action

If Indication/Condition is not as specified, repair/replace as required.

Step

2. Check switch panel test harness for continuity between:

CONNECTOR-PIN	TERMINAL BOARD-POST
P1-D	1
P1-E	2
P1-F	3

MISCELLANEOUS SWITCH PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

CONNECTOR-PIN	TERMINAL BOARD-POST
P1-G	4
P1-H	5
P2-A	6
P2-C	7
P2-D	8
P2-F	9
P2-G	10
P2-H	11
P2-J	12
P2-K	13
P2-L	14
P2-M	15
P2-N	16
P2-P	17
P2-R	18
P2-T	19
P2-S	20
P2-U	21
P2-W	22
P2-V	23
P2-X	24

Indication/Condition

Continuity shall be present.

Corrective Action

If Indication/Condition is not as specified, replace switch panel test harness.

MISCELLANEOUS SWITCH PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

3. Connect locally-made test harness (WP 1805 00) to miscellaneous switch panel as follows: P1 to J659R, and P2 to J135.
4. Check continuity between the following test harness terminal block posts:
 - 1 and 2
 - 2 and 3

Indication/Condition

Meter shall not indicate continuity.

Corrective Action

If Indication/Condition is not as specified, replace GYRO ERECT switch (WP 0950 00).

Step

5. Press and hold MISC SW panel GYRO ERECT switch while checking continuity between the following terminal block posts:
 - 1 and 2
 - 2 and 3

Indication/Condition

- a. Meter shall indicate continuity between terminal block posts 1 and 2.
- b. Meter shall indicate continuity between terminal block posts 2 and 3.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, check continuity between J659R-E and GYRO ERECT switch, section A terminal C.
 - a. If continuity is present, replace GYRO ERECT switch (WP 0950 00).
 - b. If continuity is not present, repair/replace wiring.
2. If Indication/Condition a. is not as specified, check continuity between J659R-D and GYRO ERECT switch, section A terminal NO.
 - a. If continuity is present, replace GYRO ERECT switch (WP 0950 00).
 - b. If continuity is not present, repair/replace wiring.
3. If Indication/Condition b. is not as specified, check continuity between J659R-F and GYRO ERECT switch, section C terminal NO, and between GYRO ERECT switch, section C terminal C, and section A terminal C.
 - a. If continuity is present, replace GYRO ERECT switch (WP 0950 00).
 - b. If continuity is not present, repair/replace wiring.

Step

6. Press and hold MISC SW panel FUEL IND TEST switch while checking continuity between the following terminal block posts:
 - 11 and 12
 - 13 and 14

MISCELLANEOUS SWITCH PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

- 15 and 16
- 17 and 18

Indication/Condition

Meter shall not indicate continuity.

Corrective Action

If Indication/Condition is not as specified, replace FUEL IND TEST switch (WP 0950 00).

Step

7. Release MISC SW panel FUEL IND TEST switch and check continuity between the following terminal block posts:

- 11 and 12
- 13 and 14
- 15 and 16
- 17 and 18

Indication/Condition

- a. Meter shall indicate continuity between terminal block posts 11 and 12.
- b. Meter shall indicate continuity between terminal block posts 13 and 14.
- c. Meter shall indicate continuity between terminal block posts 15 and 16.
- d. Meter shall indicate continuity between terminal block posts 17 and 18.

Corrective Action

1. If more than one Indication/Condition is not as specified, replace FUEL IND TEST switch (WP 0950 00).
2. If Indication/Condition a. is not as specified, check continuity between J135-H and FUEL IND TEST switch, section A terminal C, and between J135-J and FUEL IND TEST switch, section A terminal NC.
 - a. If continuity is present, replace FUEL IND TEST switch (WP 0950 00).
 - b. If continuity is not present, repair/replace wiring as required.
3. If Indication/Condition b. is not as specified, check continuity between J135-K and FUEL IND TEST switch, section B terminal C, and between J135-L and FUEL IND TEST switch, section B terminal NC.
 - a. If continuity is present, replace FUEL IND TEST switch (WP 0950 00).
 - b. If continuity is not present, repair/replace wiring as required.
4. If Indication/Condition c. is not as specified, check continuity between J135-M and FUEL IND TEST switch, section C terminal C, and between J135-N and FUEL IND TEST switch, section C terminal NC.
 - a. If continuity is present, replace FUEL IND TEST switch (WP 0950 00).
 - b. If continuity is not present, repair/replace wiring as required.
5. If Indication/Condition d. is not as specified, check continuity between J135-P and FUEL IND TEST switch, section D terminal C, and between J135-R and FUEL IND TEST switch, section D terminal NC.
 - a. If continuity is present, replace FUEL IND TEST switch (WP 0950 00).
 - b. If continuity is not present, repair/replace wiring as required.

MISCELLANEOUS SWITCH PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

8. Place MISC SW panel TAIL SERVO switch to NORMAL.
9. Check continuity between the following terminal block posts:
 - 19 and 20
 - 22 and 23

Indication/Condition

Meter shall not indicate continuity.

Corrective Action

If Indication/Condition is not as specified, replace TAIL SERVO switch (WP 0951 00).

Step

10. Check continuity between the following terminal block posts:
 - 20 and 21
 - 23 and 24

Indication/Condition

- a. Meter shall indicate continuity between terminal block posts 20 and 21.
- b. Meter shall indicate continuity between terminal block posts 23 and 24.

Corrective Action

1. If Indication/Condition a. is not as specified, check continuity between J135-S and TAIL SERVO switch terminal 2, and between J135-U and TAIL SERVO switch terminal 3.
 - a. If continuity is present, replace TAIL SERVO switch (WP 0951 00).
 - b. If continuity is not present, repair/replace wiring as required.
2. If Indication/Condition b. is not as specified, check continuity between J135-V and TAIL SERVO switch terminal 5, and between J135-X and TAIL SERVO switch terminal 6.
 - a. If continuity is present, replace TAIL SERVO switch (WP 0951 00).
 - b. If continuity is not present, repair/replace wiring as required.

Step

11. Place MISC SW panel TAIL SERVO switch to BACKUP.
12. Check continuity between the following terminal block posts:
 - 19 and 20
 - 22 and 23

Indication/Condition

- a. Meter shall indicate continuity between terminal block posts 19 and 20.
- b. Meter shall indicate continuity between terminal block posts 22 and 23.

MISCELLANEOUS SWITCH PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

Corrective Action

1. If Indication/Condition a. is not as specified, check continuity between J135-T and TAIL SERVO switch terminal 1.
 - a. If continuity is present, replace TAIL SERVO switch (WP 0951 00).
 - b. If continuity is not present, repair/replace wiring as required.
2. If Indication/Condition b. is not as specified, check continuity between J135-W and TAIL SERVO switch terminal 4.
 - a. If continuity is present, replace TAIL SERVO switch (WP 0951 00).
 - b. If continuity is not present, repair/replace wiring as required.

Step

13. Apply 115 vac to terminal block post 4 and ground to 5. Apply 28 vdc to terminal block post 7, 10 and ground to 5.

Indication/Condition

- a. LOCK legend shall light evenly on TAIL WHEEL pushbutton switch.
- b. UNLK legend shall light evenly on TAIL WHEEL pushbutton switch.
- c. Information plate legends shall light evenly.

Corrective Action

1. If Indication/Conditions a., b., and c. are not as specified, repair/replace wiring between J659R-H and J1-GND.
2. If Indication/Conditions a. and b. are not as specified, check continuity between J1-GND and TAIL WHEEL switch, lamp terminal C.
 - a. If continuity is present, replace TAIL WHEEL switch (WP 0950 00).
 - b. If continuity is not present, repair/replace wiring as required.
3. If Indication/Condition a. is not as specified, replace lamp that is not lit (WP 0832 00).
 - a. If trouble remains, check continuity between J135-G and TAIL WHEEL switch lamp terminal 1, and between TAIL WHEEL switch lamp terminal 1 and terminal 4.
 - (1) If continuity is present, replace TAIL WHEEL switch (WP 0950 00).
 - (2) If continuity is not present, repair/replace wiring.
4. If Indication/Condition b. is not as specified, replace lamp that is not lit (WP 0832 00).
 - a. If trouble remains, check continuity between J135-C and TAIL WHEEL switch lamp terminal 3, and between TAIL WHEEL switch lamp terminal 2 and terminal 3.
 - (1) If continuity is present, replace TAIL WHEEL switch (WP 0950 00).
 - (2) If continuity is not present, repair/replace wiring.
5. If Indication/Condition c. is not as specified, replace information plate lamps as required (WP 0832 00).
 - a. If trouble remains, check continuity between J659R-G and J1-1.
 - (1) If continuity is present, replace information panel (WP 0832 00).
 - (2) If continuity is not present, repair/replace wiring.

MISCELLANEOUS SWITCH PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

14. Remove 28 vdc and 115 vac from test harness.
15. Press MISC SW panel TAIL WHEEL switch once.
16. Check continuity between terminal block posts 6 and 8.

Indication/Condition

Meter shall indicate no continuity.

Corrective Action

1. If Indication/Condition is not as specified, press TAIL WHEEL switch, then check continuity between posts 6 and 8 in the UNLK position.
 - a. If continuity is present in UNLK position, continue with operational/troubleshooting procedure.
 - b. If continuity is not present in UNLK position, check continuity between J135-A and TAIL WHEEL switch, section A terminal NC, and between J135-D and TAIL WHEEL switch, section A terminal C.
 - (1) If continuity is present, replace TAIL WHEEL switch (WP 0950 00).
 - (2) If continuity is not present, repair/replace wiring.

Step

17. Check continuity between the following terminal block posts 9 and 8.

Indication/Condition

Meter shall not indicate continuity.

Corrective Action

If Indication/Condition is not as specified, replace TAIL WHEEL switch (WP 0950 00).

Step

18. Press miscellaneous switch panel TAIL WHEEL switch once.
19. Check continuity between terminal block posts 9 and 5.

Indication/Condition

Meter shall indicate continuity.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between J135-F and TAIL WHEEL switch, section C terminal C.
 - a. If continuity is present, replace TAIL WHEEL switch (WP 0950 00).
 - b. If continuity is not present, repair/replace wiring.

Step

20. Press miscellaneous switch panel TAIL WHEEL switch once.
21. Check continuity between terminal block posts:
 - 6 and 9
 - 8 and 5

MISCELLANEOUS SWITCH PANEL (AVIM) OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Meter shall indicate continuity.

Corrective Action

1. If Indication/Condition is not as specified, check continuity between TAIL WHEEL switch, section A terminal NC and TAIL WHEEL switch, section C terminal NO.
 - a. If continuity is present, replace TAIL WHEEL switch (WP 0950 00).
 - b. If continuity is not present, repair/replace wiring.

Step

22. Remove test harness from panel connectors J659R and J135.

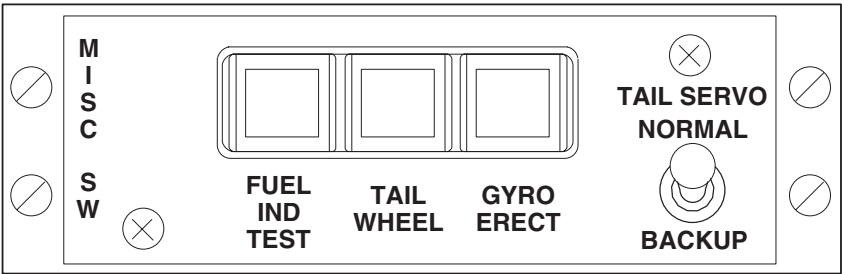
Indication/Condition

Panels shall be removed.

Corrective Action

None Required

LOCATION



AA2444
SA

Figure 1. Typical Miscellaneous Switch Panel.

SCHEMATICS

EFFECTIVITY
70450-01090-049

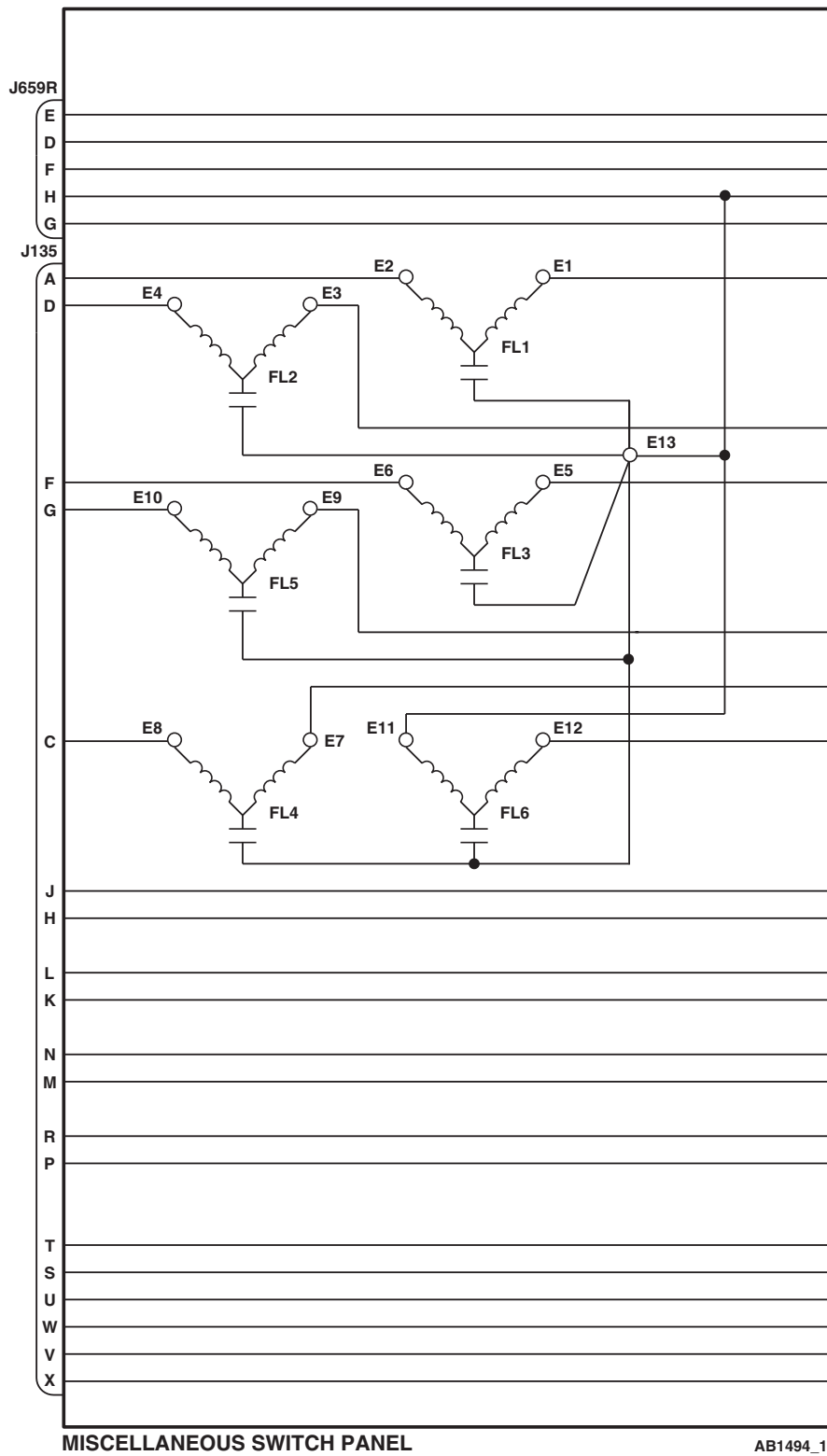


Figure 2. Miscellaneous Switch Panel Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued

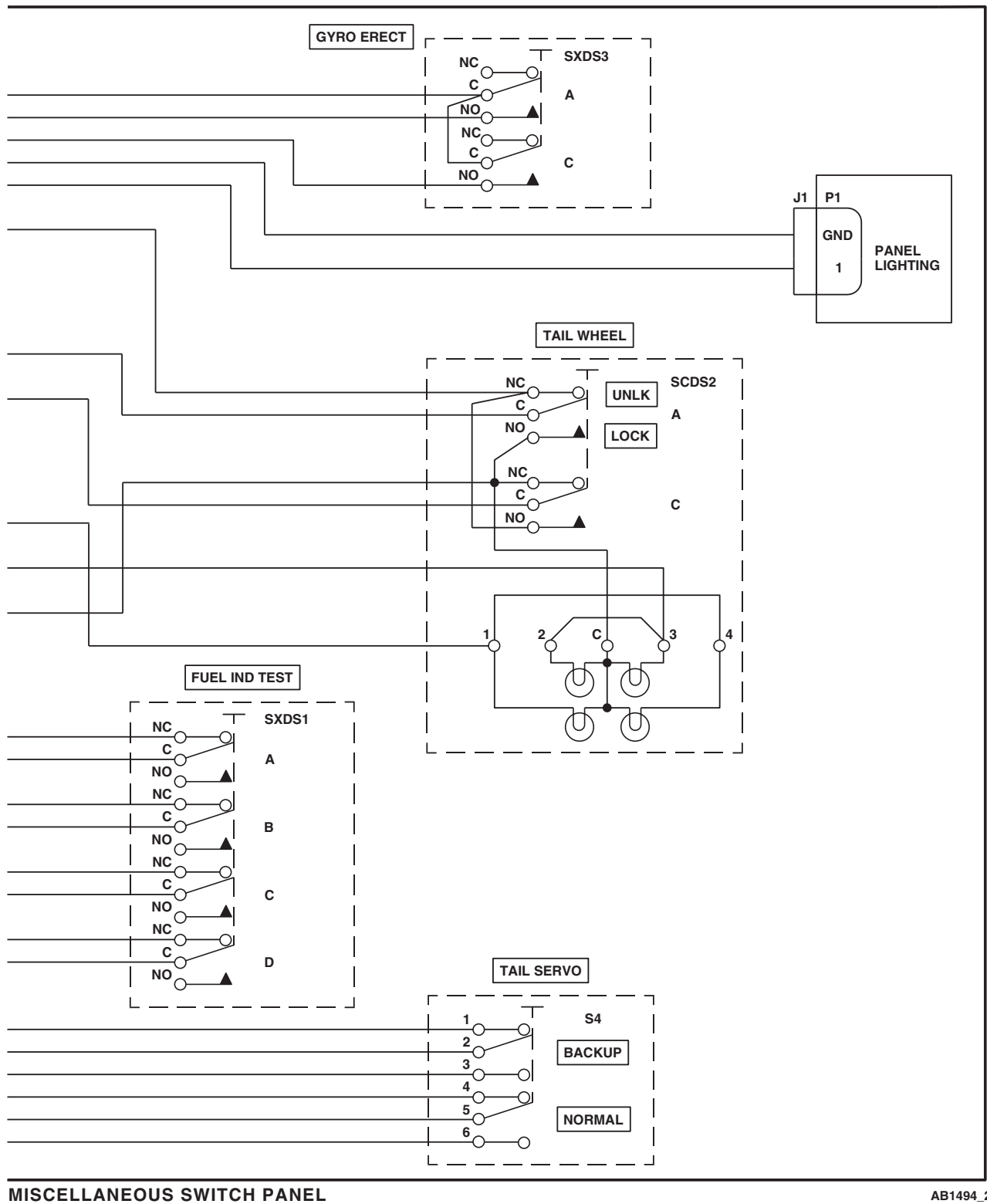
AB1494.2
SA

Figure 2. Miscellaneous Switch Panel Schematic Diagram. (Sheet 2 of 2)

SCHEMATICS - Continued

EFFECTIVITY
70450-01090-044

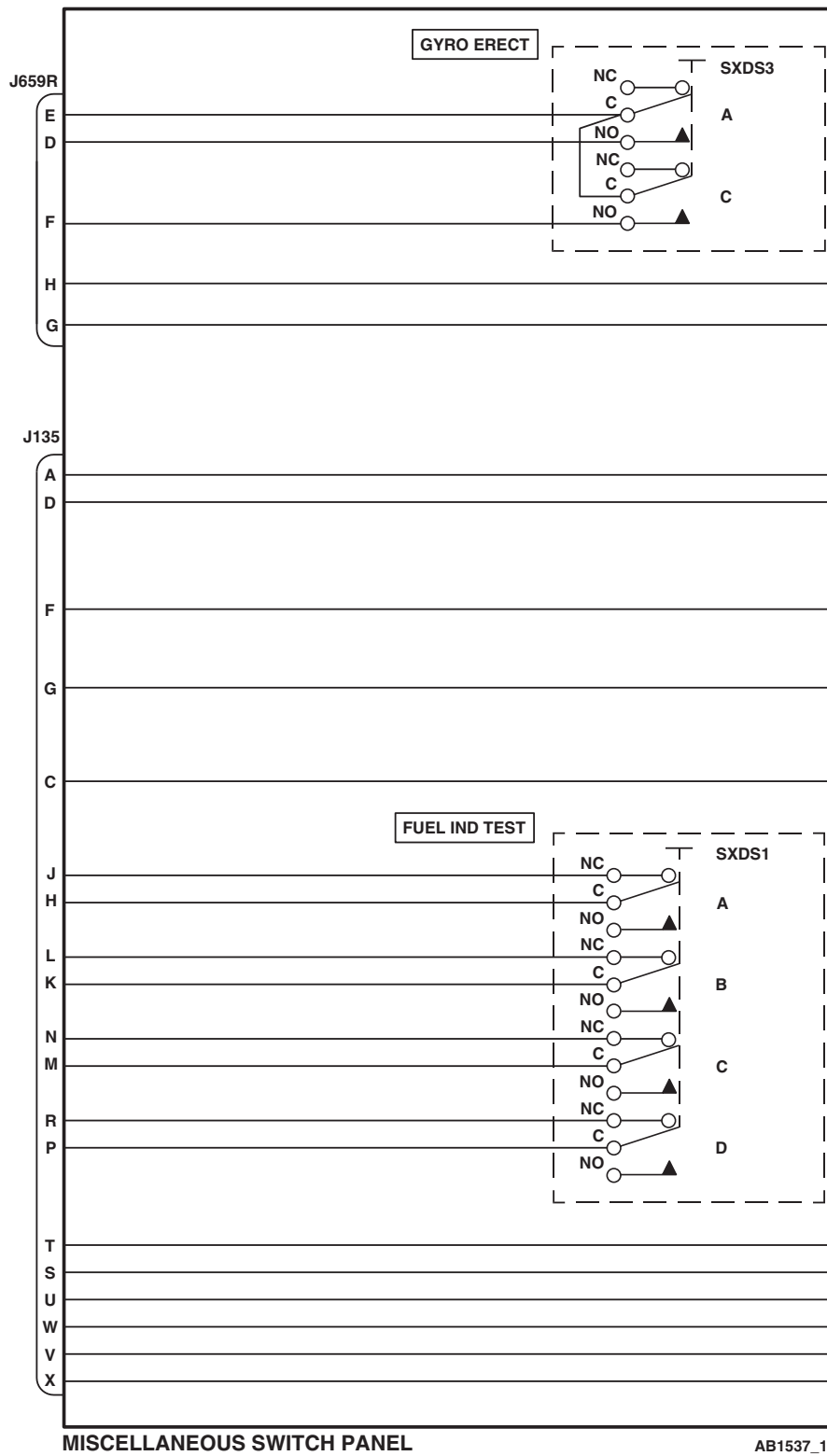


Figure 3. Miscellaneous Switch Panel Schematic Diagram. (Sheet 1 of 2)

SCHEMATICS - Continued

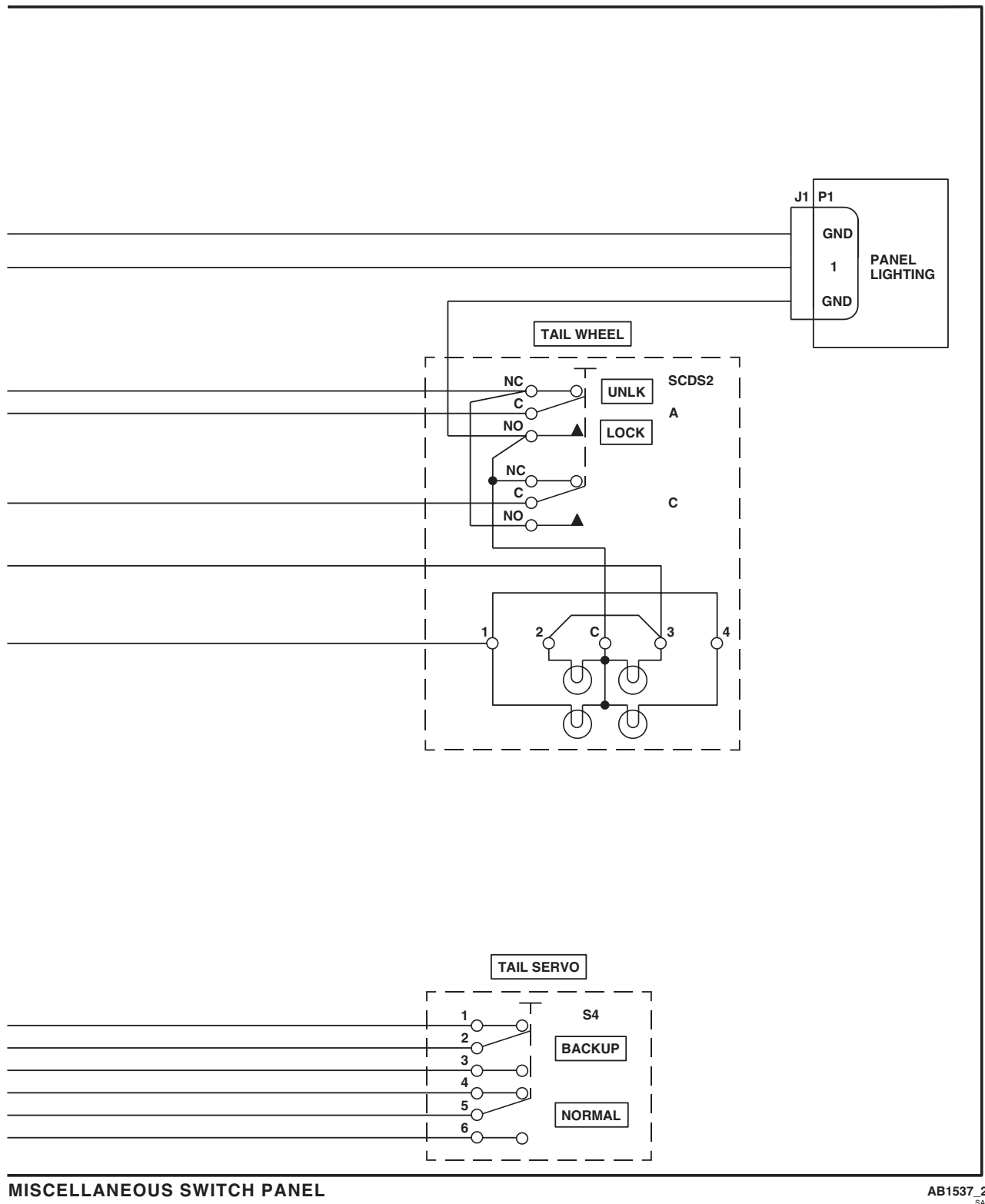


Figure 3. Miscellaneous Switch Panel Schematic Diagram. (Sheet 2 of 2)

END OF WORK PACKAGE

UNIT LEVEL

INSTRUMENT SYSTEMS

MFD/CAUTION/ADVISORY WARNING SYSTEM **HH-60A HH-60L**

This WP supersedes WP 0100 00, dated 17 April 2006.

INITIAL SETUP:

Test Equipment

Headset, Microphone, H-101A/U
Multimeter, AN/PSM-45A

WP 0104 00

WP 0126 00

WP 0141 00

WP 0788 00

Tools and Special Tools

Aircraft Mechanic's Toolkit, SC 5180-99-B01
Electrical Repairer Toolkit, SC 5180-99-B06
Locally-Made Drag Beam Wedge, Item 167,
WP 1805 00

WP 0790 00

WP 0791 00

WP 0793 00

WP 0799 00

WP 0800 00

Personnel Required

Aircraft Electrician MOS 15F (1)
UH-60 Helicopter Repairer MOS 15T (1)

WP 0824 00

WP 0825 00

WP 1685 00

WP 1747 00

References

TM 1-1520-237-10
TM 11-1520-237-23
WP 0089 00
WP 0090 00

Equipment Condition

External Electrical Power Available, WP 1685 00
or APU Operational, TM 1-1520-237-10

SPECIAL INSTRUCTIONS

NOTE

See Figure 1 for component location diagram, **HH-60A** Figure 2 **■** and **HH-60L** Figure 3 **■** for schematic diagram as an aid in operational/troubleshooting.

The MFD/Caution/Advisory Warning System operational/troubleshooting procedure is subdivided as follows:

- Setup
- Pilot Multifunction Display (MFD) Built-In Test (BIT) Check **PN 893369-03**
- Copilot MFD BIT Check **PN 893369-03**
- Pilot MFD BIT Check **HH-60-00001A-01**
- Copilot MFD BIT Check **HH-60-00001A-01**
- Operational Check
- Shutdown

SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step****NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure all circuit breakers are pushed in, except those noted being pulled for maintenance or safety.
2. Apply electrical power to helicopter.
3. Turn pilot's and copilot's MFD switches ON.

Indication/Condition

Caution and advisory legend grid shall appear.

Corrective Action

1. If Indication/Condition is not as specified, do the following:
 - a Set NIGHT/DAY switch as required.
 - b Turn BRT control clockwise and/or counterclockwise as required to adjust brightness of MFD screen as desired.
 - a. If trouble remains on pilot's MFD, go to CAUTION AND ADVISORY LEGEND GRID DOES NOT APPEAR ON PILOT'S MFD, in this work package.
 - b. If trouble remains on copilot's MFD, go to CAUTION AND ADVISORY LEGEND GRID DOES NOT APPEAR ON COPILOT'S MFD, in this work package.

Step

4. Press STS key on Control Display Unit (CDU) to get to SYSTEM STATUS.

Indication/Condition

The letter G shall appear next to MFD-1 and MFD-2.

Corrective Action

1. If Indication/Condition is not as specified, repair/replace wiring between the following points (WP 1747 00):
 - P1019R-90 and P1020R-88
 - P1019R-91 and P1020R-89
 - P1019R-88 and P1020R-90
 - P1019R-89 and P1020R-91
2. If a letter F appears next to MFD-1, go to LETTER F APPEARS NEXT TO MFD-1, in this work package.
3. If a letter F appears next to MFD-2, go to LETTER F APPEARS NEXT TO MFD-2, in this work package.

SETUP OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

5. Press TST key on CDU to go to TEST MENU.
6. Press soft key to go to BUS TEST.
7. Press soft key to start bus IBIT.

Indication/Condition

The letter G shall appear next to MFD-1 and MFD-2 in BUS A and BUS B columns.

Corrective Action

If a letter F appears next to either MFD-1 and/or MFD-2 in BUS A and/or BUS B column, troubleshoot 1553 Data Bus (TM 11-1520-237-23).

PILOT MULTIFUNCTION DISPLAY (MFD) BUILT-IN TEST (BIT) CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE PN 8933639-03**PROCEDURE****Step****NOTE**

- While performing MFD BIT check if any results FAIL, continue with check.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Do SETUP, in this work package, if not already done.
2. Press TST key on CDU to go to TEST MENU.
3. Press soft key 6 (MFD) to go to MFD TESTS.
4. Press soft key 6 (MFD-2) on CDU to start MFD-2 IBIT.
5. Press soft key R6 (CONTINUE) on MFD to go to IBIT status menu.

Indication/Condition

IBIT status menu shall be displayed.

Corrective Action

None Required

Step

6. Press soft key R3 (BEZEL) to go to BEZEL SWITCH TEST menu.

Indication/Condition

BEZEL SWITCH TEST menu shall be displayed.

Corrective Action

None Required

**PILOT MULTIFUNCTION DISPLAY (MFD) BUILT-IN TEST (BIT) CHECK OPERATIONAL/
TROUBLESHOOTING PROCEDURE PN 8933639-03 - Continued****Step**

7. Press soft key R6 (CONTINUE) to go to BEZEL SWITCH TEST page.

Indication/Condition

BEZEL SWITCH TEST page shall be displayed.

Corrective Action

None Required

Step**NOTE**

Bezel switch L6 (RETURN) shall be the last switch pressed during bezel switch test.

8. Press each bezel switch to test.

Indication/Condition

Good switches will have their display shown in inverse video.

Corrective Action

None Required

Step

9. Press soft key L6 (RETURN) to go to BEZEL SWITCH TEST menu.
10. If each bezel switch checked in step 8. passed, press soft key L5 (PASS), if any bezel switch checked in step 8. failed, press soft key R5 (FAIL).
11. Press soft key L6 (RETURN) to go to IBIT status menu.
12. Press soft key R4 (GLASS) to go to GLASS SURFACE TEST menu.

Indication/Condition

GLASS SURFACE TEST menu shall be displayed.

Corrective Action

None Required

Step

13. Press soft key L2 (RED).

Indication/Condition

Center box shall display color of selected soft key, when pressed.

Corrective Action

None Required

Step

14. Press R6 (CONTINUE) to go to glass test page.

**PILOT MULTIFUNCTION DISPLAY (MFD) BUILT-IN TEST (BIT) CHECK OPERATIONAL/
TROUBLESHOOTING PROCEDURE PN 8933639-03** - Continued**Indication/Condition**

Display shall change to test pattern, displaying color of soft key selected.

Corrective Action

None Required

Step

15. Press soft key L1.

Indication/Condition

Horizontal line shall be displayed next to soft key selected, displaying color selected from glass surface test.

Corrective Action

None Required

Step

16. Repeat step 15. for soft keys L2 through L5 and R1 through R6.

Indication/Condition

Horizontal line shall be displayed next to soft key selected, displaying color selected from glass surface test.

Corrective Action

None Required

Step

17. Press soft key T1.

Indication/Condition

Vertical line shall be displayed next to soft key selected, displaying color selected from glass surface test.

Corrective Action

None Required

Step

18. Repeat step 17. for soft keys T2 through T5 and B1 through B6.

Indication/Condition

Vertical line shall be displayed next to soft key selected, displaying color selected from glass surface test.

Corrective Action

None Required

Step

19. Press soft key L6 (RETURN) to go to GLASS SURFACE TEST menu.
20. Repeat steps 13. through 19. for each color soft key.

PILOT MULTIFUNCTION DISPLAY (MFD) BUILT-IN TEST (BIT) CHECK OPERATIONAL/ TROUBLESHOOTING PROCEDURE PN 8933639-03 - Continued

21. If Indication/Conditions are as specified as each soft key is pressed in steps 13. through 19., press soft key L5 (PASS); if Indication/Conditions are not as specified as soft keys were pressed in steps 13. through 19., press soft key R5 (FAIL).
22. Press soft key L6 (RETURN) to go to IBIT status menu.
23. Press soft key R5 (VIDEO) to go to VIDEO TEST menu.

Indication/Condition

VIDEO TEST menu shall be displayed.

Corrective Action

None Required

Step

24. Press soft key R6 (CONTINUE) to go to video test page.

Indication/Condition

Display shall show eight bands of green varying from dark to light from left to right.

Corrective Action

None Required

Step

25. Press soft key L6 (RETURN) to go to VIDEO TEST menu.
26. If display indicated results as specified, press soft key L5 (PASS), if display indicated results not as specified, press soft key R5 (FAIL).
27. Press soft key L6 (RETURN) to go to IBIT status menu.
28. Press soft key R6 (BIT HISTORY) to go to BIT HISTORY display.

Indication/Condition

BIT HISTORY display shall be displayed.

Corrective Action

None Required

Step**NOTE**

Up and down arrow keys may or may not appear. This depends on if the number of entries exceed one page. Step 23. may not be able to be performed, continue with next step.

29. Press soft key L1 (ARROW UP) and soft key R1 (ARROW DOWN) to scroll up and down through the pages.
30. Press soft key L6 (RETURN) to go to IBIT status menu.

Indication/Condition

IBIT status menu shall be displayed.

PILOT MULTIFUNCTION DISPLAY (MFD) BUILT-IN TEST (BIT) CHECK OPERATIONAL/ TROUBLESHOOTING PROCEDURE PN 8933639-03 - Continued**Corrective Action**

If IBIT status menu indicates any FAIL status, replace MFD (WP 0799 00).

Step

31. Press switch L6 (MAIN/RETURN).
32. Press switch L6 (ABORT/EXIT) to exit IBIT.
33. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

COPILOT MFD BIT CHECK HH-60A OPERATIONAL/TROUBLESHOOTING PROCEDURE PROCEDURE**Step****NOTE**

- While performing MFD BIT check if any results FAIL, continue with check.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Do SETUP, in this work package, if not already done.
2. Press TST key on CDU to go to TEST MENU.
3. Press soft key 6 (MFD) to go to MFD TESTS.
4. Press soft key 1 (MFD-1) on CDU to start MFD-1 IBIT.
5. Press soft key R6 (CONTINUE) on MFD to go to IBIT status menu.

Indication/Condition

IBIT status menu shall be displayed.

Corrective Action

None Required

Step

6. Press soft key R3 (BEZEL) to go to BEZEL SWITCH TEST menu.

Indication/Condition

BEZEL SWITCH TEST menu shall be displayed.

Corrective Action

None Required

COPILOT MFD BIT CHECK HH-60A OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Step**

7. Press soft key R6 (CONTINUE) to go to BEZEL SWITCH TEST page.

Indication/Condition

BEZEL SWITCH TEST page shall be displayed.

Corrective Action

None Required

Step**NOTE**

Bezel switch L6 (RETURN) shall be the last switch pressed during bezel switch test.

8. Press each bezel switch to test.

Indication/Condition

Good switches will have their display shaded.

Corrective Action

None Required

Step

9. Press soft key L6 (RETURN) to go to BEZEL SWITCH TEST menu.
10. If each bezel switch checked in step 6. passed, press soft key L5 (PASS); if any bezel switch checked in step 6. failed, press soft key R5 (FAIL).
11. Press soft key L6 (RETURN) to go to IBIT status menu.
12. Press soft key R4 (GLASS) to go to GLASS SURFACE TEST menu.

Indication/Condition

GLASS SURFACE TEST menu shall be displayed.

Corrective Action

None Required

Step

13. Press soft key L2 (RED).

Indication/Condition

Center box shall display color of selected soft key, when pressed.

Corrective Action

None Required

Step

14. Press soft key R6 (CONTINUE) to go to glass test page.

COPILOT MFD BIT CHECK HH-60A OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

Display shall change to test pattern, displaying color of selected soft key.

Corrective Action

None Required

Step

15. Press soft key L1.

Indication/Condition

Horizontal line shall be displayed next to soft key selected, displaying color selected from glass surface test.

Corrective Action

None Required

Step

16. Repeat step 15. for soft keys L2 through L5 and R1 through R6.

Indication/Condition

Horizontal line shall be displayed next to soft key selected, displaying color selected from glass surface test.

Corrective Action

None Required

Step

17. Press soft key T1.

Indication/Condition

Vertical line shall be displayed next to soft key selected, displaying color selected from glass surface test.

Corrective Action

None Required

Step

18. Repeat step 17. for soft keys T2 through T5 and B1 through B6.

Indication/Condition

Vertical line shall be displayed next to soft key selected, displaying color selected from glass surface test.

Corrective Action

None Required

Step

19. Press soft key L6 (RETURN) to go to GLASS SURFACE TEST menu.
20. Repeat steps 13. through 19. for each color soft key.

COPILOT MFD BIT CHECK HH-60A OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

21. If Indication/Conditions are as specified as each soft key is pressed in steps 13. through 19., press soft key L5 (PASS); if Indication/Conditions are not as specified as soft keys were pressed in steps 13. through 19., press soft key R5 (FAIL).
22. Press soft key L6 (RETURN) to go to IBIT status menu.
23. Press soft key R5 (VIDEO) to go to VIDEO TEST menu.

Indication/Condition

VIDEO TEST menu shall be displayed.

Corrective Action

None Required

Step

24. Press soft key R6 (CONTINUE) to go to video test page.

Indication/Condition

Display shall show eight bands of green varying from dark to light from left to right.

Corrective Action

None Required

Step

25. Press soft key L6 (RETURN) to go to VIDEO TEST menu.
26. If display indicated results as specified, press soft key L5 (PASS), if display indicated results not as specified, press soft key R5 (FAIL).
27. Press soft key L6 (RETURN) to go to IBIT status menu.
28. Press soft key R2 (BIT HISTORY) to go to BIT HISTORY display.

Indication/Condition

BIT HISTORY display shall be displayed.

Corrective Action

None Required

Step**NOTE**

Up and down arrow keys may or may not appear. This depends on if the number of entries exceed one page. Step 23. may not be able to be performed, continue with next step.

29. Press soft key L1 (ARROW UP) and soft key R1 (ARROW DOWN) to scroll up and down through the pages.
30. Press soft key L6 (RETURN) to go to IBIT status menu.

Indication/Condition

IBIT status menu shall be displayed.

COPILOT MFD BIT CHECK HH-60A OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued**Corrective Action**

If IBIT status menu indicates any FAIL status, replace MFD (WP 0799 00).

Step

31. Press switch L6 (MAIN/RETURN).
32. Press switch L6 (ABORT/EXIT) to exit IBIT.
33. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

PILOT MULTIFUNCTION DISPLAY (MFD) BUILT-IN TEST (BIT) CHECK OPERATIONAL/TROUBLESHOOTING PROCEDURE **PN HH-60-00001A-01****PROCEDURE****Step****NOTE**

- While performing MFD BIT check if any results FAIL, continue with check.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Do SETUP, in this work package, if not already done.
2. Press TST key on CDU to go to TEST MENU.
3. Press soft key (MFD ON) to go to MFD TESTS.
4. Press soft key (MFD2) on CDU to start MFD2 IBIT.
5. Press soft key (CONTINUE) on MFD to go to IBIT status page.

Indication/Condition

IBIT status page shall be displayed.

Corrective Action

None Required

Step

6. Press soft key (BIT HISTORY) on MFD to check BIT history.
7. Press soft key (PAGE UP) and (PAGE DOWN) to scroll through pages, as required.

**PILOT MULTIFUNCTION DISPLAY (MFD) BUILT-IN TEST (BIT) CHECK OPERATIONAL/
TROUBLESHOOTING PROCEDURE PN HH-60-00001A-01 - Continued****NOTE**

The BIT history page contains four columns. The first column is the item test, for example VM; the second column is the test message, for example the HEATERS FAILED; the third column is the test type, for example P; and the fourth column is the time stamp from the CDU of when the error occurred, for example XX YY ZZ.

8. Record date, aircraft tail number, MFD serial number, and MFD location then record the information from each column on the BIT history screen.
9. After recording BIT history, press soft key (RETURN) to go to IBIT entry screen.
10. Press soft key (CLEAR BIT HISTORY) to clear history.
11. BEZEL SWITCH TEST menu shall be displayed.
12. Press soft key (BEZEL) to go BEZEL SWITCH TEST menu.

Indication/Condition

BEZEL SWITCH TEST menu shall be displayed.

Corrective Action

None Required

Step

13. Press soft key (Continue) to go to BEZEL SWITCH TEST page.

Indication/Condition

BEZEL SWITCH TEST page shall be displayed.

Corrective Action**NOTE**

Bezel switch (RETURN) shall be the last switch pressed during bezel switch test.

None Required

Step

14. Press each bezel and rocker switch to test (SYM is not used).

Indication/Condition

Good switches will have their display shaded and rocker switches will properly response to type selected.

Corrective Action

None Required

Step

15. Press soft key (RETURN) to go to BEZEL SWITCH TEST menu.
16. If each bezel and rocker switch checked in step 21. passed, press soft key (PASS). IF any bezel or rocker switch checked in step 13. failed, press soft key (FAIL).
17. Press soft key (RETURN) to go to IBIT status menu.
18. Press soft key (GLASS) to go to GLASS SURFACE TEST page.

PILOT MULTIFUNCTION DISPLAY (MFD) BUILT-IN TEST (BIT) CHECK OPERATIONAL/ TROUBLESHOOTING PROCEDURE PN HH-60-00001A-01 - Continued**Indication/Condition**

GLASS SURFACE TEST page shall be displayed.

Corrective Action

None Required

Step

19. Press soft key to select color.

Indication/Condition

Box shall display color of selected soft key, when pressed.

Corrective Action

None Required

Step

20. Press soft keys R1 to R6.

Indication/Condition

Display shall change to bands of color of soft key selected.

Corrective Action

None Required

Step

21. Press soft key (RETURN) to go GLASS SURFACE TEST menu.
22. Repeat step 24. through p for each color soft key.
23. If results are specified as each soft key is pressed in step 24. through to 27., press soft key (PASS); if results are not as specified as soft keys were pressed in steps 24. through 27., press soft key (FAIL).
24. Press soft key (RETURN) once to go to IBIT entry screen display.
25. Press soft key (RUN ALL) to initiate remaining test for PSA, CPU, MEM, GM, VM, and I/O.

Indication/Condition

Screen shall momentarily display TESTING (in place of N/A next to TEST) then screen shall display PASS for each TEST.

Corrective Action

1. If results are as specified, go to step 25.
2. If results are not specified, select soft key that indicates test FAIL to initiate individual IBIT.
 - a. If status display PASS, go to step 26.
 - b. If status continues to display FAIL, do step 5.
3. If BIT HISTORY display FAIL, turn pilot MFD OFF (for more than 5 seconds) and then turn back ON. After pilot MFD reboots the SBIT (start-up BIT) may clear the nuisance error MFD malfunction.
4. If FAIL is still displayed after SBIT, replace MFD (WP 0799 00).

PILOT MULTIFUNCTION DISPLAY (MFD) BUILT-IN TEST (BIT) CHECK OPERATIONAL/ TROUBLESHOOTING PROCEDURE PN HH-60-00001A-01 - Continued**Step**

26. Press switch (ABORT/RETURN) to IBIT.
27. If no other check is to be done, go to (SHUTDOWN,) in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

COPILOT MFD BIT CHECK HH-60L OPERATIONAL/TROUBLESHOOTING PROCEDURE PROCEDURE**Step****NOTE**

- While performing MFD BIT check if any results FAIL, continue with check.
- If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Do SETUP, in this work package, if not already done.
2. Press TST key on CDU to go to TEST MENU.
3. Press soft key (MFD) to go to MFD TESTS.
4. Press soft key (MFD1) on CDU to start MFD1 IBIT.
5. Press soft key (CONTINUE) on MFD to go to IBIT status menu.

Indication/Condition

IBIT status menu shall be displayed.

Corrective Action

None Required

Step

6. Press soft key (BIT HISTORY) on MFD to check BIT history.
7. Press soft key (PAGE UP) and (PAGE DOWN) to scroll through pages, as required.
8. Press soft key (PAGE UP) and (PAGE DOWN) to scroll through pages, as required.

NOTE

The BIT HISTORY page contains four columns. The first column is the item test, for example VM; the second column is the test message, for example HEATERS FAILED; the third column is the test type, for example P; and the fourth column is the time stamp from the CDU when the error occurred, for example XX YY ZZ.

9. Record date, aircraft tail number, MFD serial number, and MFD location, then record the information for each column on the BIT history screen.

COPILOT MFD BIT CHECK HH-60L OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

10. After recording BIT history, press soft key (RETURN) to go to IBIT entry screen.
11. Press soft key (CLEAR BIT HISTORY) to clear history.
12. Press soft key (BEZEL) to go to BEZEL SWITCH TEST menu.
13. Press soft key L6 (RETURN) to go to IBIT status menu.
14. Press soft key R4 (GLASS) to go to GLASS SURFACE TEST menu.

Indication/Condition

BEZEL SWITCH TEST menu shall be displayed.

Corrective Action

None Required

Step

15. Press soft key L2 (RED).

Indication/Condition

Center box shall display color of selected soft key, when pressed.

Corrective Action

None Required

Step

16. Press soft key (CONTINUE) to go to BEZEL SWITCH TEST page.

Indication/Condition

BEZEL SWITCH TEST page shall be displayed.

Corrective Action

None Required

Step**NOTE**

Bezel switch (RETURN) shall be the last switch pressed during bezel switch test.

17. Press each bezel and rocker switch to test (SYM is not used).

Indication/Condition

Good bezel switches will have their display shaded and rocker switches will properly response to type selected.

Corrective Action

None Required

Step

18. Press soft key (RETURN) to go to BEZEL SWITCH TEST menu.
19. If each bezel and rocker switch checked in step 16. passed, press soft key (PASS). If any bezel or rocker switch checked in step 16. failed, press soft key (FAIL).

COPILOT MFD BIT CHECK HH-60L OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

20. Press soft key (RETURN) to go to IBIT status menu.

21. Press soft key (GLASS) to go to GLASS SURFACE TEST page.

Indication/Condition

GLASS SURFACE TEST page shall be displayed.

Corrective Action

None Required

Step

22. Press soft key to select color.

Indication/Condition

Box shall display color of selected soft key, when pressed.

Corrective Action

None Required

Step

23. Press soft key (CONTINUE).

24. Press soft keys through R1 to R6.

Indication/Condition

Display shall change to bands of color of soft key selected.

Corrective Action

None Required

Step

25. Press soft key (RETURN) to go to GLASS SURFACE TEST menu.

26. Repeat steps 20. through 22. for each color soft key.

27. If result are as specified as each soft key is pressed in steps 20. through 22., press soft key (PASS); if results are not specified as soft keys were pressed in steps 20. through 22., press soft key (FAIL).

28. Press soft key (RETURN) once to go to IBIT entry screen display.

29. Press soft key (RUN ALL) to initiate remaining test for PSA, CPU, MEM, GM, VM, and I/O.

Indication/Condition

Screen shall momentarily display TESTING (in place of N/A next to TEST) then screen shall display PASS for each test.

Corrective Action

1. If result are as specified, go to step 27.

2. If results are not specified, select soft key that indicates test FAIL to initiate individual IBIT.

a. If status displays PASS press (RETURN), go to step 27.

b. If status continues to display FAIL, do steps 8. and 9.

COPILOT MFD BIT CHECK HH-60L OPERATIONAL/TROUBLESHOOTING PROCEDURE - Continued

3. If BIT HISTORY display FAIL, turn pilot MFD OFF (for more than 5 seconds) and then turn back ON. After MFD reboots, SBIT (start-up BIT) may clear the nuisance error and MFD malfunction.
4. If FAIL is still displayed after SBIT, replace MFD (WP 0799 00)

Step

30. Press switch (ABORT/RETURN) to exit IBIT.
31. If not other check is to be done, go to shutdown.

Indication/Condition

Shut down shall be performed .

Corrective Action

None required

OPERATIONAL CHECK/TROUBLESHOOTING PROCEDURE**PROCEDURE****Step****NOTE**

If a circuit breaker pops out during the operational/troubleshooting procedure, a short circuit is indicated (WP 0104 00).

1. Make sure all circuit breakers are pushed in, except those noted as being pulled for maintenance or safety.
2. On upper console, pull out BACKUP HYD CONTR circuit breaker.
3. On upper console, turn INST LT PILOT FLT-OFF/BRT control at least 1/4 turn clockwise from OFF.

NOTE

If the # 1 ENG OUT, # 2 ENG OUT, LOW ROTOR RPM, and MASTER CAUTION PRESS TO RESET capsules come on dim, set INDICATOR LTS/BRT/DIM/TEST switch to BRT/DIM to select bright lighting mode.

4. Turn on external electrical power.

Indication/Condition

These capsules on pilot's and copilot's master warning panels shall be on bright: # 1 ENG OUT, # 2 ENG OUT, LOW ROTOR RPM (FLASHES).

Corrective Action

If any capsule does not go on, go to step 78.

Step

5. On instrument panel, set INDICATOR LTS BRT/DIM-TEST switch to TEST long enough to observe results, and then release.

Indication/Condition

- a. On pilot's master warning panel, these capsules shall be on: # 1 ENG OUT, # 2 ENG OUT, FIRE, LOW ROTOR RPM (FLASHES).

OPERATIONAL CHECK/TROUBLESHOOTING PROCEDURE - Continued

- b. On copilot's master warning panel, these capsules shall be on: # 1 ENG OUT, # 2 ENG OUT, FIRE, LOW ROTOR RPM (FLASHES).
- c. All stabilator control/auto flight control panel failure advisory legends shall go on.
- d. All legends on pilot's HSI/VSI MODE SEL panels shall go on.
- e. All legends on copilot's HSI/VSI MODE SEL panels shall go on.
- f. Switch legends on CIS mode select panel shall go on.
- g. Mode lamps marked GA, DH, and MB on pilot's and copilot's VSI shall go on.

Corrective Action

- 1. If any capsule in Indication/Condition a. or b. does not go on, check and replace burnt out lamp (WP 0791 00).
- 2. If Indication/Condition a. is not as specified, go to PILOT'S #1 ENG OUT, #2 ENG OUT, FIRE, LOW ROTOR CAPSULES DO NOT GO IN TEST MODE, in this work package.
- 3. If Indication/Condition b. is not as specified, go to COPILOT'S #1 ENG OUT, #2 ENG OUT, FIRE, LOW ROTOR RPM CAPSULES DO NOT GO ON IN TEST MODE, in this work package.
- 4. If Indication/Conditions c., d., and e. are not as specified, go to STABILATOR CONTROL/FLT CONT PANEL AND/OR PILOT'S AND COPILOT'S MODE SELECT PANEL LEGENDS DO NOT GO ON, in this work package.
- 5. If Indication/Conditions f. and g. are not as specified, troubleshoot Instrument Panel and Consoles Indicator Lights Dimming (WP 0126 00).
- 6. If pilot's or copilot's FIRE capsules do not go on, troubleshoot fire detection system (WP 0141 00).
- 7. If pilot's or copilot's # 1 ENG OUT capsules do not go on, go to PILOT'S AND COPILOT'S #1 ENG OUT CAPSULE DOES NOT GO ON, in this work package.
- 8. If pilot's or copilot's # 2 ENG OUT capsules do not go on, go to PILOT'S AND COPILOT'S #2 ENG OUT CAPSULE DOES NOT GO ON, in this work package.
- 9. If pilot's or copilot's LOW ROTOR RPM capsules do not go on, go to PILOT'S OR COPILOT'S LOW ROTOR RPM CAPSULE DOES NOT GO ON, in this work package.

Step**NOTE**

After initial MFD turn on, pressing key T6 on the MFDs will display all legends for approximately 10 seconds; then only the active legends will be displayed. Also, pressing key B6 (C/A) will display all legends (key can be used to toggle MFD grid on and off).

- 6. On pilot's and copilot's MFDs, set ON/OFF switch to ON.

Indication/Condition

Caution and advisory legend grid shall appear.

Corrective Action

- 1. If Indication/Condition is not as specified, do the following:
 - a. Set NIGHT/DAY switch as required.

OPERATIONAL CHECK/TROUBLESHOOTING PROCEDURE - Continued

- b. Turn BRT control clockwise and/or counterclockwise as required to adjust brightness of MFD screen as desired.

- (1) If trouble remains, replace MFDs as required (WP 0799 00).

Step

7. On pilot's and copilot's MFDs, press soft key T6.

Indication/Condition

All caution and advisory legends shall be displayed for approximately 10 seconds; then only active legends shall be displayed.

Corrective Action

If Indication/Condition is not as specified, replace malfunctioning MFD (WP 0799 00).

Step

8. On pilot's and copilot's MFDs, press soft key B6 (C/A).

Indication/Condition

The MFD screen shall be cleared, and the C/A WNDW icon shall be displayed.

Corrective Action

If Indication/Condition is not as specified, replace malfunctioning MFD (WP 0799 00).

Step

9. On pilot's and copilot's MFDs, press soft key R3 (C/A WNDW).

Indication/Condition

C/A WNDW shall change to a popup window displaying existing cautions or advisories.

Corrective Action

If Indication/Condition is not as specified, replace malfunctioning MFD (WP 0799 00).

Step**NOTE**

Up and down arrow keys may or may not appear. This depends on if the number of caution and advisory entries exceed one page. If step 10. may not be able to be performed, continue with next step.

10. On pilot's and copilot's MFDs, press soft key R2 (UP ARROW) and press soft key R4 (DOWN ARROW) to scroll up and down through the listed entries.

Indication/Condition

The caution and advisory list shall scroll up and scroll down to list of cautions and advisories.

Corrective Action

If Indication/Condition is not as specified, replace malfunctioning MFD (WP 0799 00).

OPERATIONAL CHECK/TROUBLESHOOTING PROCEDURE - Continued**Step**

11. On pilot's and copilot's MFDs, press soft key B6 (C/A).

Indication/Condition

Caution and advisory legend grid shall appear.

Corrective Action

If Indication/Condition is not as specified, replace malfunctioning MFD (WP 0799 00).

Step

12. Connect headset to pilot's ICS.
13. Momentarily press stabilator control/auto flight control panel AUTO CONTROL RESET switch.

Indication/Condition

- a. Stabilator control/auto flight control panel AUTO CONTROL switch ON legend shall go on.
- b. STABILATOR caution capsule on MFDs shall go off.
- c. Steady tone shall be heard in headset.

Corrective Action

1. If Indication/Conditions a. and b. are not as specified, troubleshoot stabilator unit (TM 11-1520-237-23).
2. If Indication/Condition c. is not as specified, go to STEADY TONE NOT HEARD IN HEADSET, in this work package.

Step

14. Press and hold stabilator control/auto flight control panel TEST switch.

Indication/Condition

- a. Beeping tone shall be heard in headset.
- b. Stabilator control/auto flight control panel AUTO CONTROL switch ON legend shall go off.
- c. STABILATOR caution capsule on MFDs shall go on.
- d. MASTER CAUTION PRESS TO RESET capsules shall go on.

Corrective Action

1. If Indication/Condition a. is not as specified, go to BEEPING TONE IS NOT HEARD, in this work package.
2. If Indication/Condition b., c., or d. is not as specified, troubleshoot digital AFCS (TM 11-1520-237-23).

Step

15. Release stabilator control/auto flight control panel TEST switch.

NOTE

After the MASTER CAUTION PRESS TO RESET capsules are pressed to reset the caution / warning circuits, they will remain off until one or more faults occur.

16. Momentarily press either the pilot's or the copilot's master warning panel MASTER CAUTION PRESS TO RESET capsule.

OPERATIONAL CHECK/TROUBLESHOOTING PROCEDURE - Continued**Indication/Condition**

- a. MASTER CAUTION PRESS TO RESET capsules shall go off.
- b. Tone shall no longer be heard.

Corrective Action

If Indication/Conditions are not as specified, go to MASTER CAUTION PRESS TO RESET CAPSULES DO NOT GO OFF AND TONE STILL HEARD, in this work package.

Step

17. On upper console, push in BACKUP HYD CONTR circuit breaker.
18. Insert locally-made drag beam wedge in left drag beam switch.

Indication/Condition

Beeping tone shall be heard in headset.

Corrective Action

If Indication/Condition is not as specified, go to BEEPING TONE IS NOT HEARD, in this work package.

Step

19. Remove wedge from left drag beam switch, and pull out BACKUP HYD CONTR circuit breaker.

Indication/Condition

No tone shall be heard.

Corrective Action

If Indication/Condition is not as specified, troubleshoot instrument display system (WP 0089 00).

Step

20. Press instrument panel INDICATOR LTS BRT/DIM-TEST switch to BRT/DIM and release to select dim lighting mode.
21. Slowly rotate pilot's and copilot's MA WRN NVG DIMMING controls counterclockwise.

Indication/Condition

1 ENG OUT, # 2 ENG OUT, LOW ROTOR RPM, and MASTER CAUTION PRESS TO RESET capsules go from very dim to dim (NVG).

Corrective Action

1. If Indication/Condition is not as specified, check continuity between these points: P1000-A and pilot's MA WRN NVG DIMMING control terminal 2, P1000-B and pilot's MA WRN NVG DIMMING control terminal 1, P1000-H and copilot's MA WRN NVG DIMMING control terminal 2, P1000-G and copilot's MA WRN NVG DIMMING control terminal 1.
 - a. If continuity is present, replace pilot's or copilot's master warning dimming control (WP 0793 00).
 - (1) If trouble remains, replace master warning dimmer (WP 0788 00).
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00).

OPERATIONAL CHECK/TROUBLESHOOTING PROCEDURE - Continued**Step**

22. Return both controls to fully clockwise position.
23. If no other check is to be done, go to SHUTDOWN, in this work package.

Indication/Condition

Shutdown shall be performed.

Corrective Action

None Required

**SHUTDOWN OPERATIONAL/TROUBLESHOOTING PROCEDURE
PROCEDURE****Step**

1. Turn pilot's and copilot's MFD switches OFF.
2. Turn off electrical power.
3. On upper console, push in BACKUP HYD CONTR circuit breaker.

Indication/Condition

BACKUP HYD CONTR circuit breaker shall be pushed in.

Corrective Action

None Required

CAUTION AND ADVISORY LEGEND GRID DOES NOT APPEAR ON PILOT'S MFD**SYMPTOM**

Caution and advisory legend grid shall appear.

MALFUNCTION

Caution and advisory legend grid does not appear on pilot's MFD.

CORRECTIVE ACTION

1. Check for 28 vdc between:
 - P1018R-1 and ground
 - P1018R-3 and ground
 - P1018R-2 and ground
 - P1018R-4 and ground
 - P1018R-5 and ground
 - a. If voltage is as specified, go to **3**.
 - b. If voltage is not as specified, go to **2**.
2. Check for 28 vdc between PLT MFD circuit breaker terminal 1 and ground.
 - a. If voltage is as specified, repair/replace wiring as required (WP 1747 00). Go to **4**.

CAUTION AND ADVISORY LEGEND GRID DOES NOT APPEAR ON PILOT'S MFD - Continued

b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **4**.

3. Check continuity between:

- P1018R-6 and ground
- P1018R-7 and ground
- P1018R-8 and ground
- P1018R-9 and ground
- P1018R-10 and ground
- P1018R-20 and ground

a. If continuity is present, replace pilot's MFD (WP 0799 00). Go to **4**.

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **4**.

4. Procedure completed.

CAUTION AND ADVISORY LEGEND GRID DOES NOT APPEAR ON COPILOT'S MFD**SYMPTOM**

Caution and advisory legend grid shall appear.

MALFUNCTION

Caution and advisory legend grid does not appear on copilot's MFD.

CORRECTIVE ACTION

1. Check for 28 vdc between:

- P1017R-1 and ground
- P1017R-3 and ground
- P1017R-2 and ground
- P1017R-4 and ground

a. If voltage is as specified, go to **3**.

b. If voltage is not as specified, go to **2**.

2. Check for 28 vdc between COPLT MFD circuit breaker terminal 1 and ground.

a. If voltage is as specified, repair/replace wiring as required (WP 1747 00). Go to **4**.

b. If voltage is not as specified, replace circuit breaker as required (WP 0824 00). Go to **4**.

3. Check continuity between:

- P1017R-5 and ground
- P1017R-6 and ground
- P1017R-7 and ground
- P1017R-8 and ground
- P1017R-20 and ground

CAUTION AND ADVISORY LEGEND GRID DOES NOT APPEAR ON COPILOT'S MFD - Continued

- a. If continuity is present, replace copilot's MFD (WP 0799 00). Go to **4**.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **4**.
- 4. Procedure completed.**

LETTER F APPEARS NEXT TO MFD-1**SYMPTOM**

The letter G shall appear next to MFD-1 and MFD-2.

MALFUNCTION

Letter F appears next to MFD-1.

CORRECTIVE ACTION

- 1. Check continuity between pins 119 and 121 of connector P1020R.**
 - a. If continuity is present, replace pilot's MFD (WP 0799 00). Go to **2**.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **2**.
- 2. Procedure completed.**

LETTER F APPEARS NEXT TO MFD-2**SYMPTOM**

The letter G shall appear next to MFD-1 and MFD-2.

MALFUNCTION

Letter F appears next to MFD-2.

CORRECTIVE ACTION

- 1. Check continuity between:**
 - **P1019R-116 and P1019R-117**
 - **P1019R-117 and P1019R-118**
 - **P1019R-118 and P1019R-121**
 - a. If continuity is present, replace copilot's MFD (WP 0799 00). Go to **2**.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go too.**2**.
- 2. Procedure completed.**

PILOT'S # 1 ENG OUT, # 2 ENG OUT, FIRE, LOW ROTOR RPM CAPSULES DO NOT GO ON IN TEST MODE**SYMPTOM**

On pilot's master warning panel, these capsules shall be on: # 1 ENG OUT # 2 ENG OUT FIRE LOW ROTOR RPM (flashes).

MALFUNCTION

Pilot's # 1 ENG OUT, # 2 ENG OUT, FIRE, LOW ROTOR RPM capsules do not go on in TEST mode.

CORRECTIVE ACTION

- 1. With INDICATOR LTS switch in TEST, check for 28 vdc between P130-2 and ground.**
 - a. If voltage is as specified, go to **2**.
 - b. If voltage is not as specified, go to **3**.
- 2. Check continuity between P130-1 and P1000-L.**
 - a. If continuity is present, replace master warning dimmer (WP 0793 00). Go to **6**.
 - b. If trouble remains, replace master warning panel (WP 0790 00). Go to **6**.
 - c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.
- 3. With INDICATOR LTS switch in TEST, check for 28 vdc between switch HH-60A S20 and HH-60L S120 terminal 3 and ground.**
 - a. If voltage is as specified, repair/replace wiring between switch S20 terminal 3 and P130-2 (WP 1747 00). Go to **6**.
 - b. If voltage is not as specified, go to **4**.
- 4. Check for 28 vdc between switch terminal 2 and ground.**
 - a. If voltage is as specified, replace switch (WP 0800 00). Go to **6**.
 - b. If voltage is not as specified, go to **5**.
- 5. Check for 28 vdc between terminal 1 of CAUT/ADVS PNL circuit breaker and ground.**
 - a. If voltage is as specified, repair/replace wiring between switch terminal 2 and circuit breaker (WP 1747 00). Go to **6**.
 - b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **6**.
- 6. Procedure completed.**

COPILOT'S # 1 ENG OUT, # 2 ENG OUT, FIRE, LOW ROTOR RPM CAPSULES DO NOT GO ON IN TEST MODE**SYMPTOM**

On copilot's master warning panel, these capsules shall be on: # 1 ENG OUT # 2 ENG OUT FIRE LOW ROTOR RPM (flashes).

MALFUNCTION

Copilot's # 1 ENG OUT, # 2 ENG OUT, FIRE, LOW ROTOR RPM capsules do not go on in TEST mode.

CORRECTIVE ACTION

- 1. With INDICATOR LTS switch in TEST, check for 28 vdc between P131-2 and ground.**
 - a. If voltage is as specified, go to **2**.
 - b. If voltage is not as specified, repair/replace wiring between terminal 3 of INDICATOR LTS switch **HH-60A** S20 **HH-60L** S120 and P131-2 (WP 1747 00). Go to **3**.
- 2. Check continuity between P131-1 and P1000-N.**
 - a. If continuity is present, replace copilot's master warning panel (WP 0790 00). Go to **3**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3**.
- 3. Procedure completed.**

STABILATOR CONTROL/AUTO FLT CONT PANEL AND/OR PILOT'S AND COPILOT'S MODE SELECT PANEL LEGENDS DO NOT GO ON**SYMPTOM**

All stabilator control/auto flt control panel failure advisory legends shall go on.

MALFUNCTION

Stabilator control/auto flt cont panel and/or pilot's and copilot's mode select panel legends do not go on.

CORRECTIVE ACTION

- 1. Check continuity between:**
 - P131-2 and **HH-60A** P4R-D **HH-60L** P121-z
 - P131-2 and P300R-27
 - P131-2 and P317R-27
 - a. If continuity is present, troubleshoot Command Instrument Set and HSI/VSI Mode Select Unit (TM 11-1520-237-23). Go to **2**.
 - b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **2**.
- 2. Procedure completed.**

PILOT'S OR COPILOT'S # 1 ENG OUT CAPSULE DOES NOT GO ON**SYMPTOM**

Pilot's or copilot's # 1 ENG OUT capsule shall go on.

MALFUNCTION

Pilot's or copilot's # 1 ENG OUT capsule does not go on.

CORRECTIVE ACTION

1. If pilot's # 1 ENG OUT capsule does not go on, go to 2. If copilot's # 1 ENG OUT capsule does not go on, go to 4.
2. Check for 28 vdc between P130-3 and ground.
 - a. If voltage is as specified, replace pilot's master warning panel (WP 0790 00). Go to 6.
 - b. If voltage is not as specified, go to 3.
3. Check continuity between:
 - P138-n and P130-3
 - P138-n and P242-a
 - a. If continuity is present, troubleshoot instrument display system (WP 0089 00). Go to 6.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 6.
4. Check for 28 vdc between P131-3 and ground.
 - a. If voltage is as specified, replace copilot's master warning panel (WP 0790 00). Go to 6.
 - b. If voltage is not as specified, go to 5.
5. Check continuity between P138-n and P131-3.
 - a. If continuity is present, troubleshoot instrument display system (WP 0089 00). Go to 6.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to 6.
6. Procedure completed.

PILOT'S OR COPILOT'S # 2 ENG OUT CAPSULE DOES NOT GO ON**SYMPTOM**

Pilot's or copilot's # 2 ENG OUT capsule shall go on.

MALFUNCTION

Pilot's or copilot's # 2 ENG OUT capsule does not go on.

CORRECTIVE ACTION

1. If pilot's # 2 ENG OUT capsule does not go on, go to 2. If copilot's # 2 ENG OUT capsule does not go on, go to 4.
2. Check for 28 vdc between P130-10 and ground.
 - a. If voltage is as specified, replace pilot's master warning panel (WP 0790 00). Go to 6.
 - b. If voltage is not as specified, go to 3.
3. Check continuity between P137-n and P130-10.

PILOT'S OR COPILOT'S # 2 ENG OUT CAPSULE DOES NOT GO ON - Continued

- a. If continuity is present, troubleshoot instrument display system (WP 0089 00). Go to **6**.
- b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.
- 4. Check for 28 vdc between P131-10 and ground.**
 - a. If voltage is as specified, replace copilot's master warning panel (WP 0790 00). Go to **6**.
 - b. If voltage is not as specified, go to **5**.
- 5. Check continuity between P137-n and P131-10.**
 - a. If continuity is present, troubleshoot instrument display system (WP 0089 00). Go to **6**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.
- 6. Procedure completed.**

PILOT'S OR COPILOT'S LOW ROTOR RPM CAPSULE DOES NOT GO ON**SYMPTOM**

Pilot's or copilot's LOW ROTOR RPM capsule shall go on.

MALFUNCTION

Pilot's or copilot's LOW ROTOR RPM capsule does not go on.

CORRECTIVE ACTION

- 1. If pilot's LOW ROTOR RPM capsule does not go on, go to 2. If copilot's LOW ROTOR RPM capsule does not go on, go to 4.**
- 2. Check for 28 vdc between P130-9 and ground.**
 - a. If voltage is as specified, replace pilot's master warning panel (WP 0790 00). Go to **6**.
 - b. If voltage is not as specified, go to **3**.
- 3. Check continuity between P138-u and P130-9.**
 - a. If continuity is present, troubleshoot instrument display system (WP 0089 00). Go to **6**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.
- 4. Check for 28 vdc between P131-9 and ground.**
 - a. If voltage is as specified, replace copilot's master warning panel (WP 0790 00). Go to **6**.
 - b. If voltage is not as specified, go to **5**.
- 5. Check continuity between P138-u and P131-9.**
 - a. If continuity is present, troubleshoot instrument display system (WP 0089 00). Go to **6**.
 - b. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **6**.
- 6. Procedure completed.**

STEADY TONE NOT HEARD IN HEADSET**SYMPTOM**

Steady tone shall be heard in headset.

MALFUNCTION

Steady tone not heard in headset.

CORRECTIVE ACTION**1. Check continuity between:**

- P961-F and P242-F
- P242-E and HH-60A P656R-M HH-60L P139R-63
- P242-H and P961-A
- P961-B and ground

a. If continuity is present, go to **2**.

b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **3**.

2. Check for 28 vdc between P961-A and ground.

a. If voltage is as specified, replace audible warning unit (TM 11-1520-237-23). Go to **3**.

b. If voltage is not as specified, repair/replace wiring as required (WP 1747 00). Go to **3**.

c. If trouble remains, replace left relay panel (WP 0825 00). Go to **3**.

3. Procedure completed.**BEEPING TONE IS NOT HEARD****SYMPTOM**

Beeping tone shall be heard.

MALFUNCTION

Beeping tone is not heard.

CORRECTIVE ACTION**1. Check for 28 vdc between:**

- P243-S and ground
- P243-h and ground

a. If voltage is as specified, go to **2**.

b. If voltage is not as specified, repair/replace wiring between (WP 1747 00), then go to **3**:

- P243-S and HH-60A P4R-M HH-60L P121-z
- P243-h and P137-u
- P137-u and P138-u
- P961-D and P243-G

BEEPING TONE IS NOT HEARD - Continued**2. Check continuity between P138-u and P243-h.**

- a. If continuity is present, replace audible warning unit (TM 11-1520-237-23).
- b. If trouble remains, replace left relay panel (WP 0825 00). Go to **3**.
- c. If continuity is not present, repair/replace wiring (WP 1747 00). Go to **3**.
- d. If trouble remains, troubleshoot Caution/Advisory Warning System (WP 0090 00). Go to **3**.

3. Procedure completed.**MASTER CAUTION PRESS TO RESET CAPSULES DO NOT GO OFF AND TONE STILL HEARD****SYMPTOM**

MASTER CAUTION PRESS TO RESET capsules go off and tone is not heard.

MALFUNCTION

MASTER CAUTION PRESS TO RESET capsules do not go off and tone still heard.

CORRECTIVE ACTION**1. Check continuity between:**

- P130-7 and CAUT/ADVSY PNL circuit breaker terminal 1
- P131-7 and CAUT/ADVSY PNL circuit breaker terminal 1
- P130-6 and P1020R-12
- P130-6 and P1019R-12
- P130-6 and P1020R-13
- P130-6 and P1019R-13
- P131-6 and P1020R-12
- P131-6 and P1019R-12
- P131-6 and P1020R-13
- P131-6 and P1019R-13
- P130-5 and P1020R-14
- P130-5 and P1019R-14
- P131-5 and P1020R-15
- P131-5 and P1019R-15

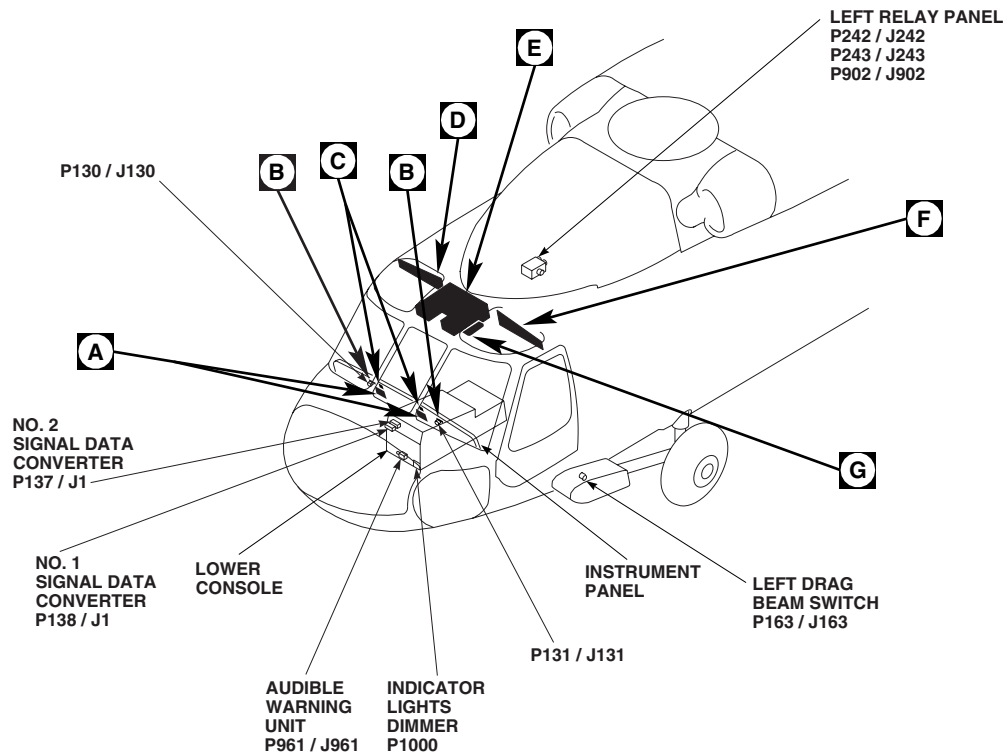
- a. If continuity is present, go to **2**.
- b. If continuity is not present, repair/replace wiring as required (WP 1747 00). Go to **3**.

2. Check for 28 vdc between CAUT/ADVSY PNL circuit breaker terminal 1 and ground.

- a. If voltage is as specified, repair/replace wiring as required (WP 1747 00). Go to **3**.
- b. If voltage is not as specified, replace circuit breaker (WP 0824 00). Go to **3**.

3. Procedure completed.

LOCATION

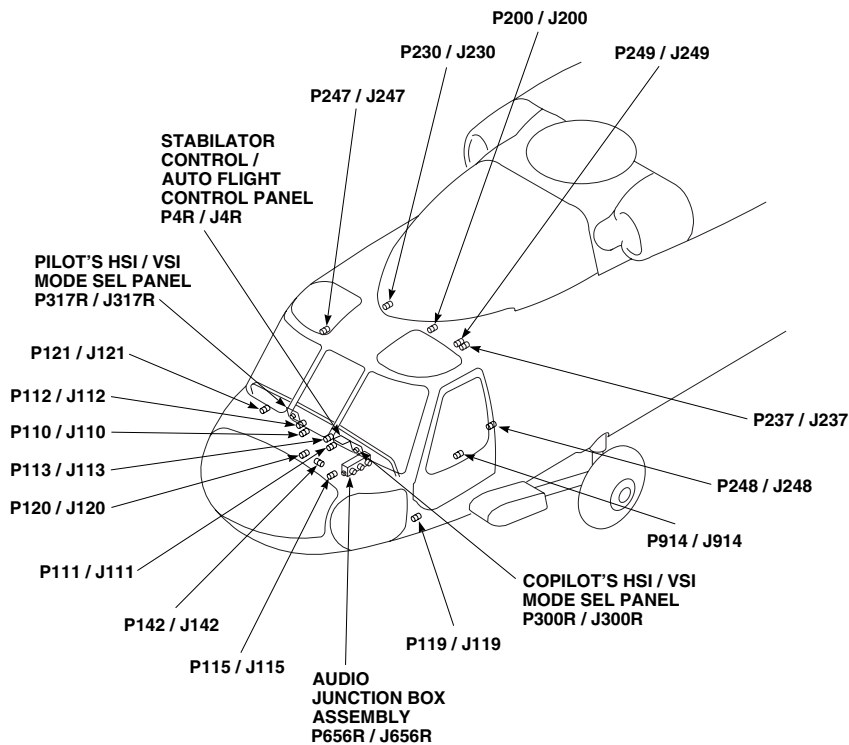


TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
J119	COPILOT CYCLIC STICK (SEE NOTE 1)
J120	PILOT CYCLIC STICK (SEE NOTE 1)
P1R / J1R	DISCONNECT
P4R / J4R	STABILATOR CONTROL / AUTO FLIGHT CONTROL PANEL
P11R / J11R	DISCONNECT (SEE NOTE 1)
P20R / J20R	DISCONNECT (SEE NOTE 1)
P102 / J2	NAVIGATION INTERFACE UNIT (SEE NOTE 1)
P110 / J110	COCKPIT, BL 7.5 RH, STA 197
P111 / J111	COCKPIT, BL 7.5 LH, STA 197
P112 / J112	COCKPIT, BL 6 RH, STA 197
P113 / J113	COCKPIT, BL 6 LH, STA 197
P115 / J115	COCKPIT, BL 6 RH, STA 210
P121 / J121	COCKPIT, BL 10 RH, STA 195
P129 / J129	COPILOT'S AUXILIARY CIRCUIT BREAKER PANEL
P130 / J130	PILOT'S MASTER WARNING PANEL

AB0338_1A
SA

Figure 1. MFD/Caution/Advisory Warning System Location Diagram. (Sheet 1 of 6)

LOCATION - Continued



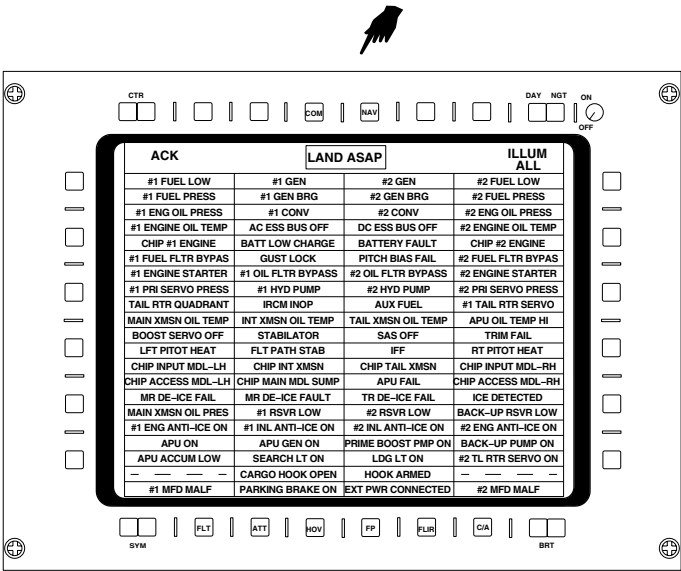
TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P131 / J131	COPILOT'S MASTER WARNING PANEL
P137 / J1	NO. 2 SIGNAL DATA CONVERTER
P138 / J1	NO. 1 SIGNAL DATA CONVERTER
P142 / J142	COCKPIT, BL 3 RH, STA 210
P163 / J163	LEFT DRAG BEAM SWITCH
P200 / J200	CABIN CEILING, BL 13 LH, STA 246
P230 / J230	CABIN CEILING, BL 8 RH, STA 247
P237 / J237	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 25 LH
P242 / J242	LEFT RELAY PANEL
P243 / J243	LEFT RELAY PANEL

TERMINAL BOARD/ DISCONNECT PLUG/ RECEPTACLE	LOCATION/ CONNECTION POINT
P247 / J247	CABIN, BL 40 RH, STA 248
P248 / J248	CABIN, BL 40 LH, STA 248
P249 / J249	BEHIND COPILOT'S CIRCUIT BREAKER PANEL BL 23 LH
P298 / J298	NO. 3 RELAY PANEL (SEE NOTE 1)
P300R / J300R	COPILOT'S HSI / VSI MODE SELECT PANEL
P317R / J317R	PILOT'S HSI / VSI MODE SELECT PANEL
P393 / J393	INDICATOR LIGHTS DIMMER
P656R / J656R	JUNCTION BOX ASSEMBLY
P902 / J902	LEFT RELAY PANEL
P914 / J914	COCKPIT, BL 24 LH, STA 247
P961 / J961	AUDIBLE WARNING UNIT
P2002R / J3	SIGNAL DATA CONVERTER (SEE NOTE 1)

AB0338 2
SA

Figure 1. MFD/Caution/Advisory Warning System Location Diagram. (Sheet 2 of 6)

LOCATION - Continued



MULTIFUNCTION DISPLAY'S
CAUTION/ADVISORY GRID
(SEE NOTE 1)

A

B

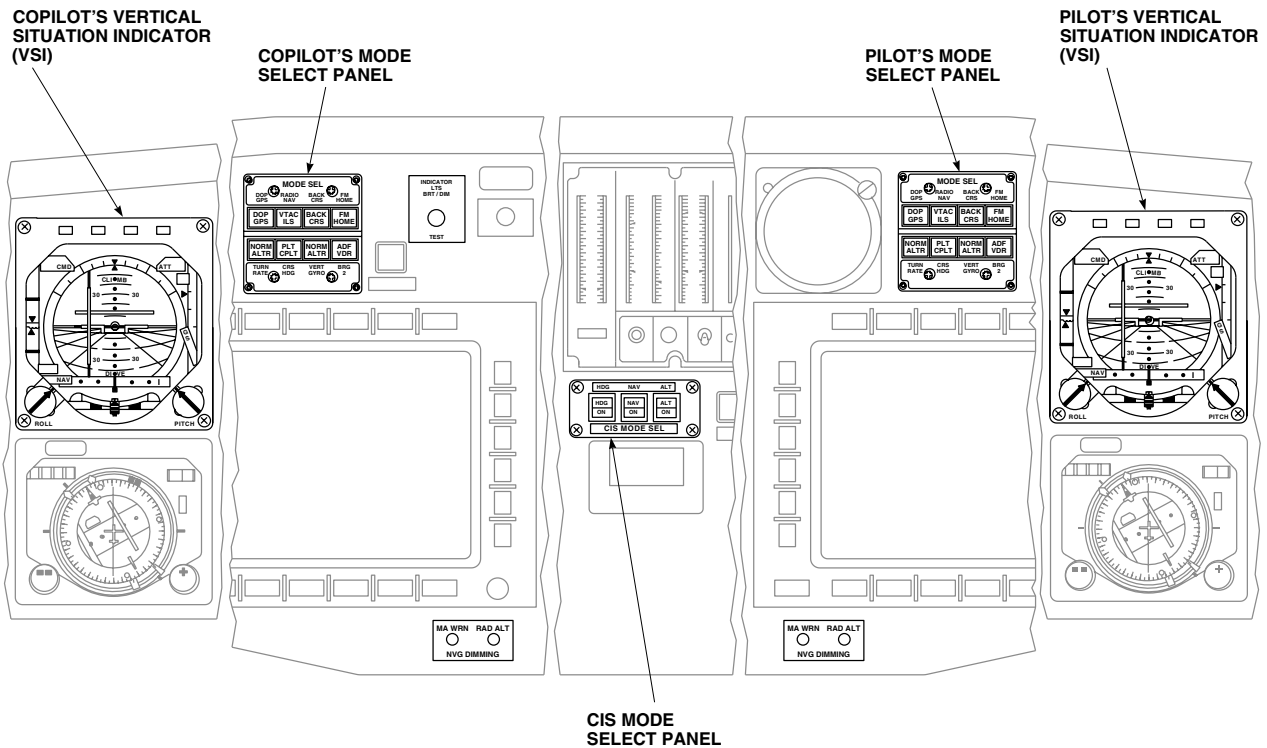


PILOT'S AND COPILOT'S
MASTER WARNING PANEL

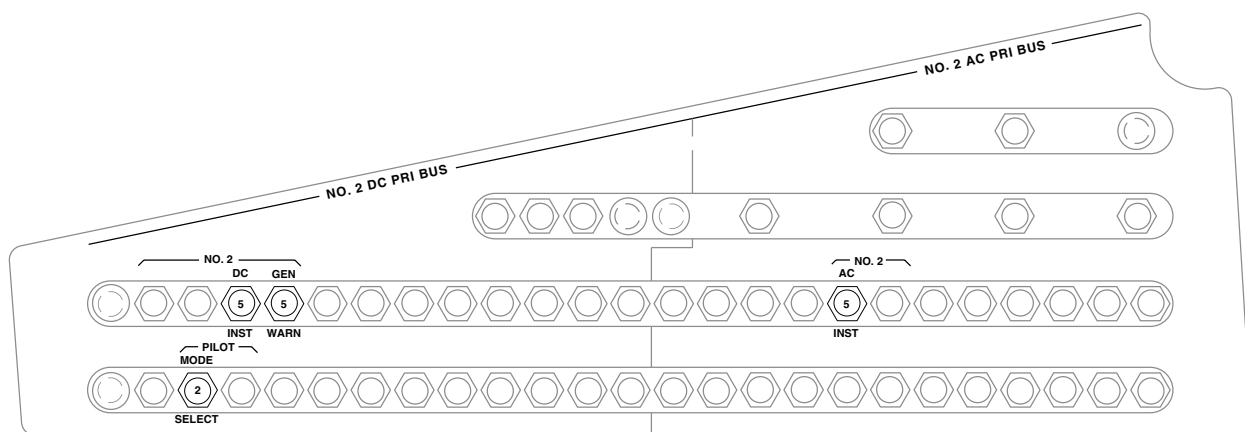
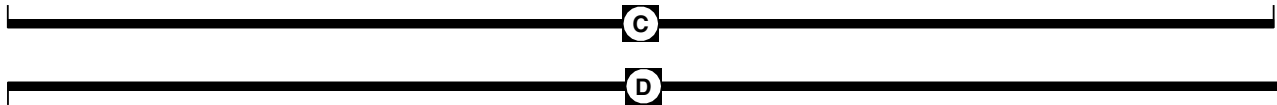
AB0338_3A
SA

Figure 1. MFD/Caution/Advisory Warning System Location Diagram. (Sheet 3 of 6)

LOCATION - Continued



INSTRUMENT PANEL

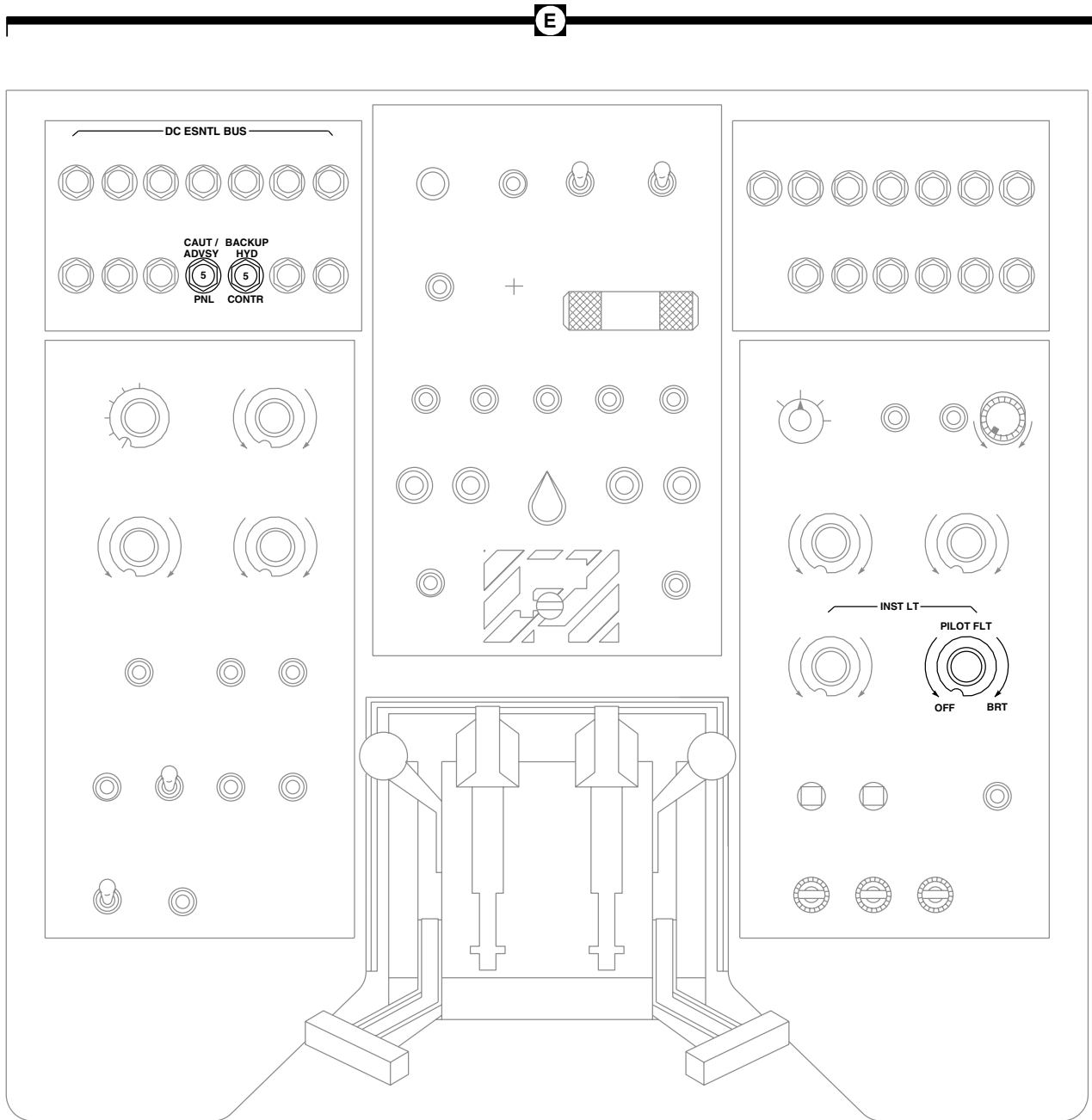


PILOT'S CIRCUIT BREAKER PANEL

AB0338 4
SA

Figure 1. MFD/Caution/Advisory Warning System Location Diagram. (Sheet 4 of 6)

LOCATION - Continued

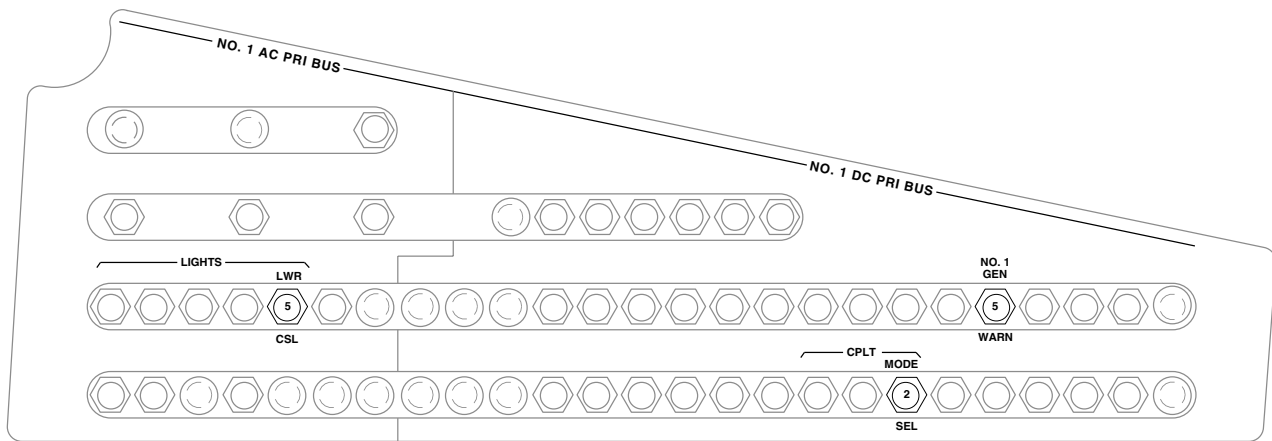


UPPER CONSOLE

AB0338_5
SA

Figure 1. MFD/Caution/Advisory Warning System Location Diagram. (Sheet 5 of 6)

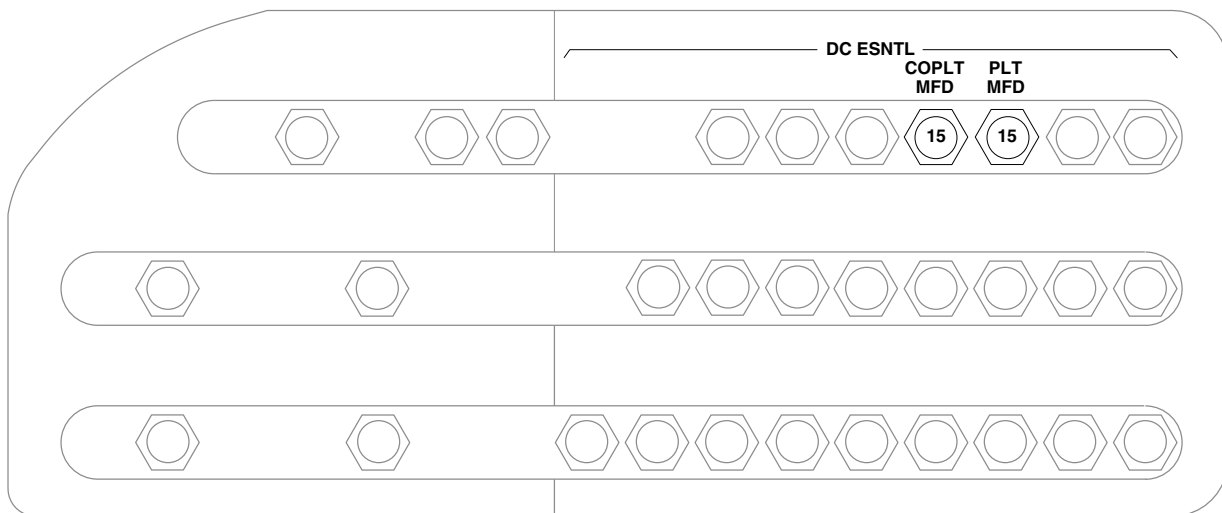
LOCATION - Continued



COPILOT'S CIRCUIT BREAKER PANEL

F

G

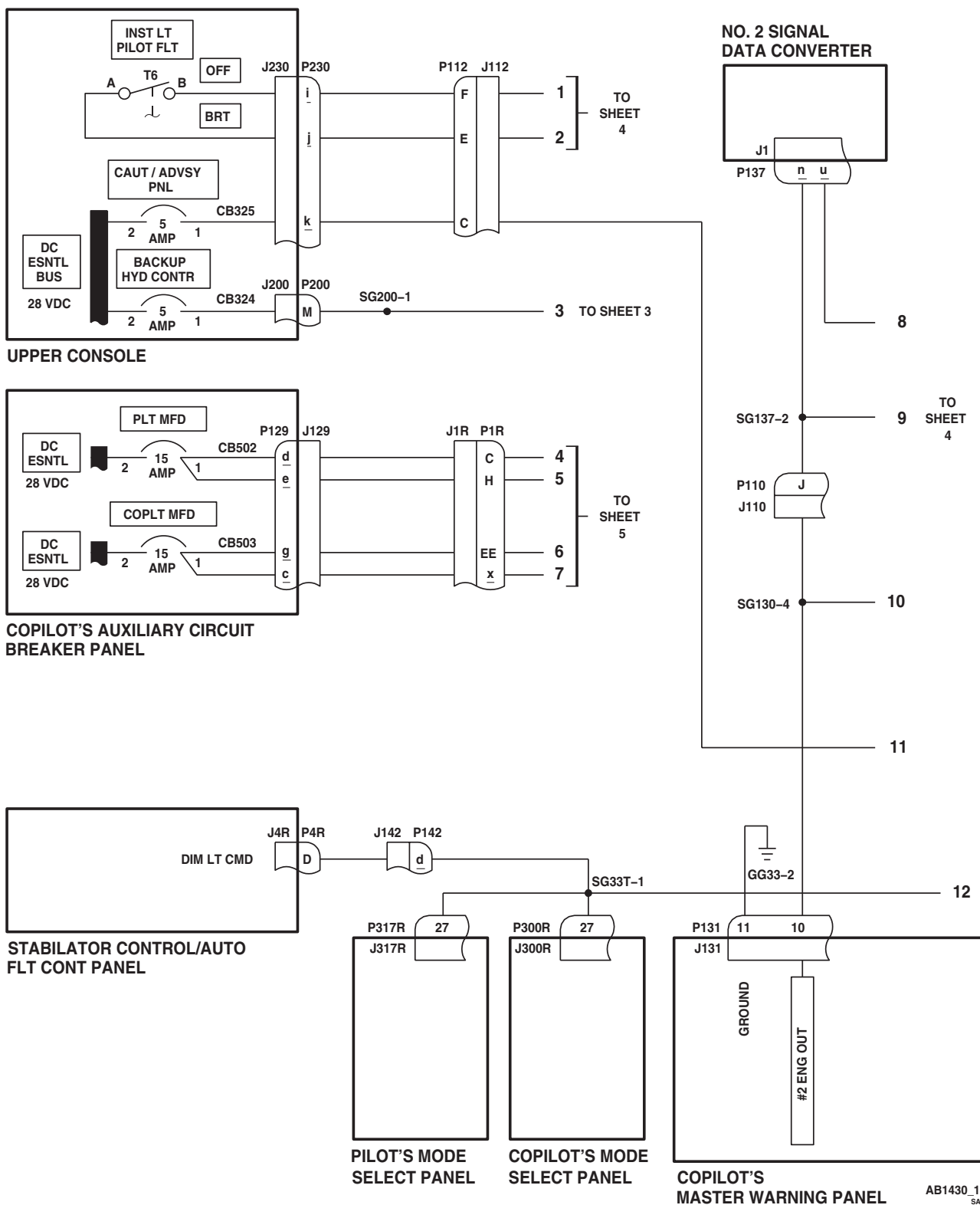


COPILOT'S AUXILIARY CIRCUIT BREAKER PANEL

AB0338 6
SA

Figure 1. MFD/Caution/Advisory Warning System Location Diagram. (Sheet 6 of 6)

SCHEMATICS

Figure 2. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60A** . (Sheet 1 of 14)

NO. 1 SIGNAL DATA CONVERTER

COPILOT'S MASTER WARNING PANEL

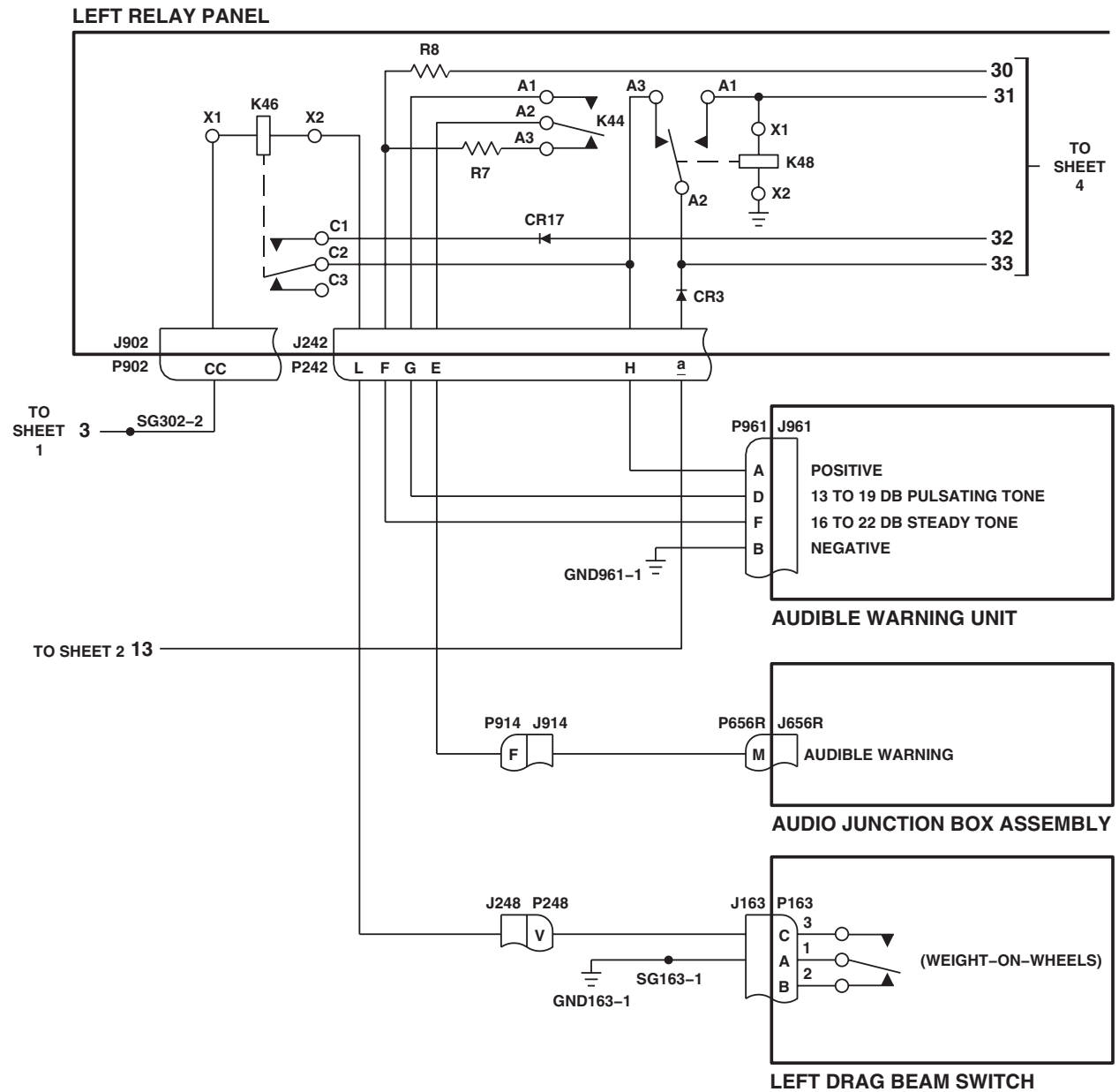
PILOT'S MASTER WARNING PANEL

Wiring Details:

- NO. 1 SIGNAL DATA CONVERTER:**
 - J1 P138 (n, u)
 - SG138-1
 - P111 J111 (M, L)
 - J110 P110 (K)
 - SG137-1
 - SG130-2
 - SG1019-12
 - SG1019-13
 - SG1020-13
 - J199 P199 (C)
 - J121 P121 (a)
 - SG1020-12
 - SG130-3
 - SG130-1
 - SGJ112-3
 - SG33B-1
 - GG33-5
- COPILOT'S MASTER WARNING PANEL:**
 - P131 J131 (9, 4, 3, 6, 1, 2, 7, 5, 8)
 - LOW ROTOR RPM
 - FIRE
 - #1 ENG OUT
 - MASTER CAUTION PRESS TO RESET
 - MASTER WARNING DIMMING CONTROL
 - LAMP TEST IN
 - 28 VDC
 - FAULT INPUT
 - MASTER CAUTION DIMMING CONTROL
- PILOT'S MASTER WARNING PANEL:**
 - P130 J130 (9, 4, 3, 10, 6, 2, 7, 1, 5, 8, 11)
 - LOW ROTOR RPM
 - FIRE
 - #1 ENG OUT
 - #2 ENG OUT
 - MASTER CAUTION PRESS TO RESET
 - LAMP TEST IN
 - 28 VDC
 - MASTER WARNING DIMMING CONTROL
 - FAULT INPUT
 - MASTER CAUTION DIMMING CONTROL
 - GROUND
- Connections to Other Sheets:**
 - 13 TO SHEET 3
 - 14 TO SHEET 4
 - 15 TO SHEET 4
 - 16 TO SHEET 6
 - 17 TO SHEET 6
 - 18 TO SHEET 6
 - 19 TO SHEET 6
 - 20 TO SHEET 4
 - 21 TO SHEET 4
 - 22 TO SHEET 4
 - 23 TO SHEET 4
 - 24 TO SHEET 4
 - 25 TO SHEET 6
 - 26 TO SHEET 6
 - 27 TO SHEET 6
 - 28 TO SHEET 6
 - 29 TO SHEET 6

Figure 2. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60A**. (Sheet 2 of 14)

SCHEMATICS - Continued

AB1430_3
SAFigure 2. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60A** . (Sheet 3 of 14)

SCHEMATICS - Continued

LEFT RELAY PANEL

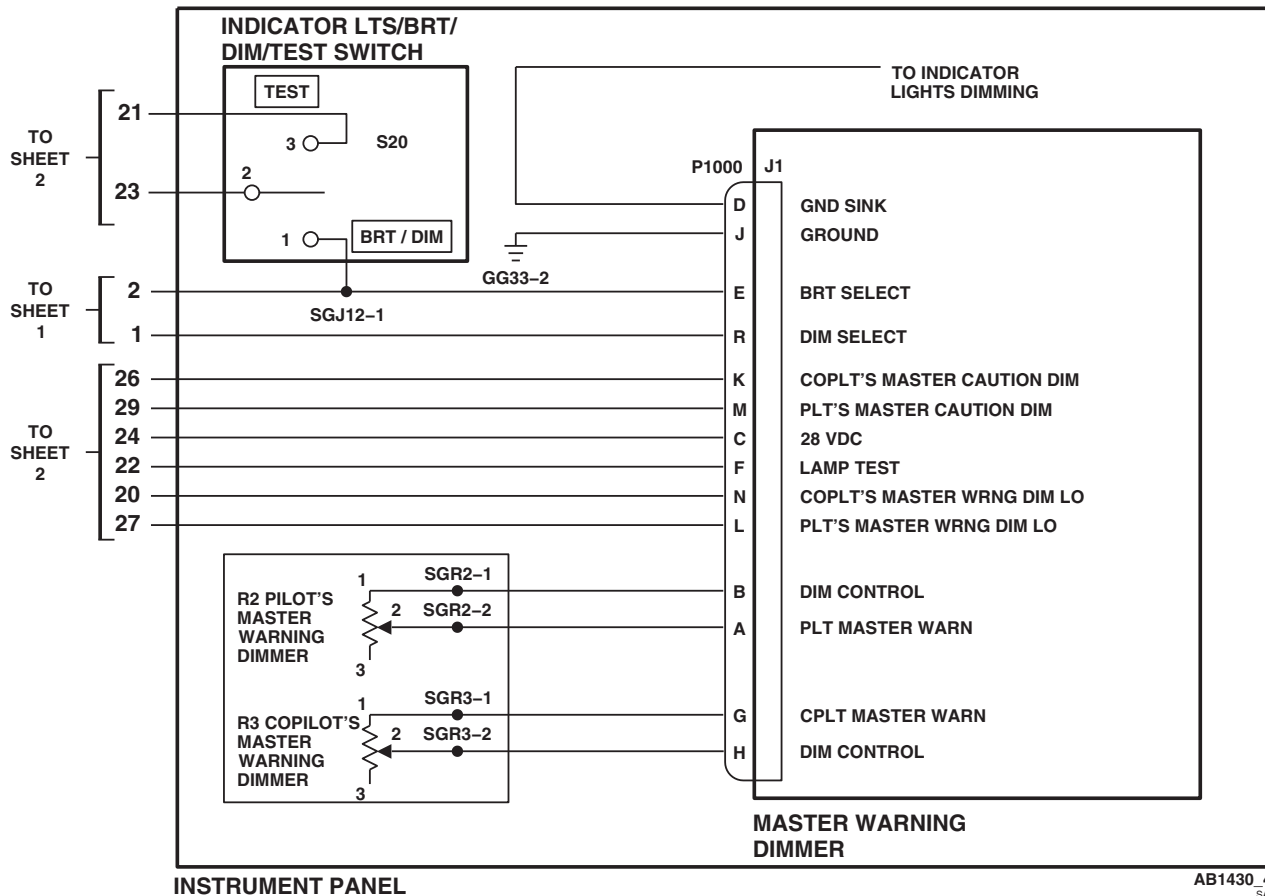
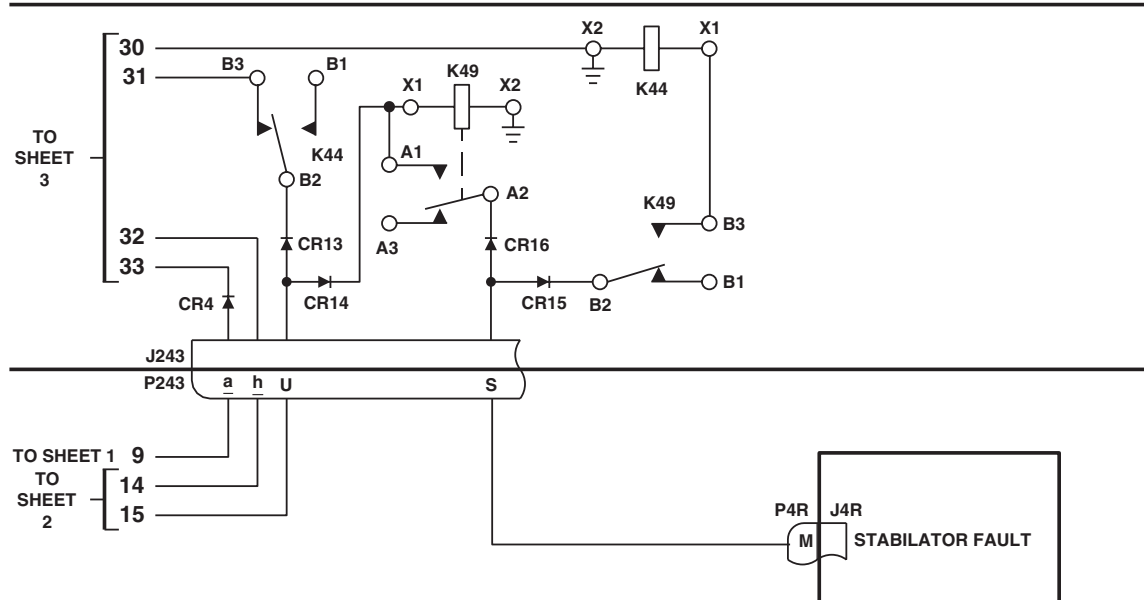


Figure 2. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60A** . (Sheet 4 of 14)

SCHEMATICS - Continued

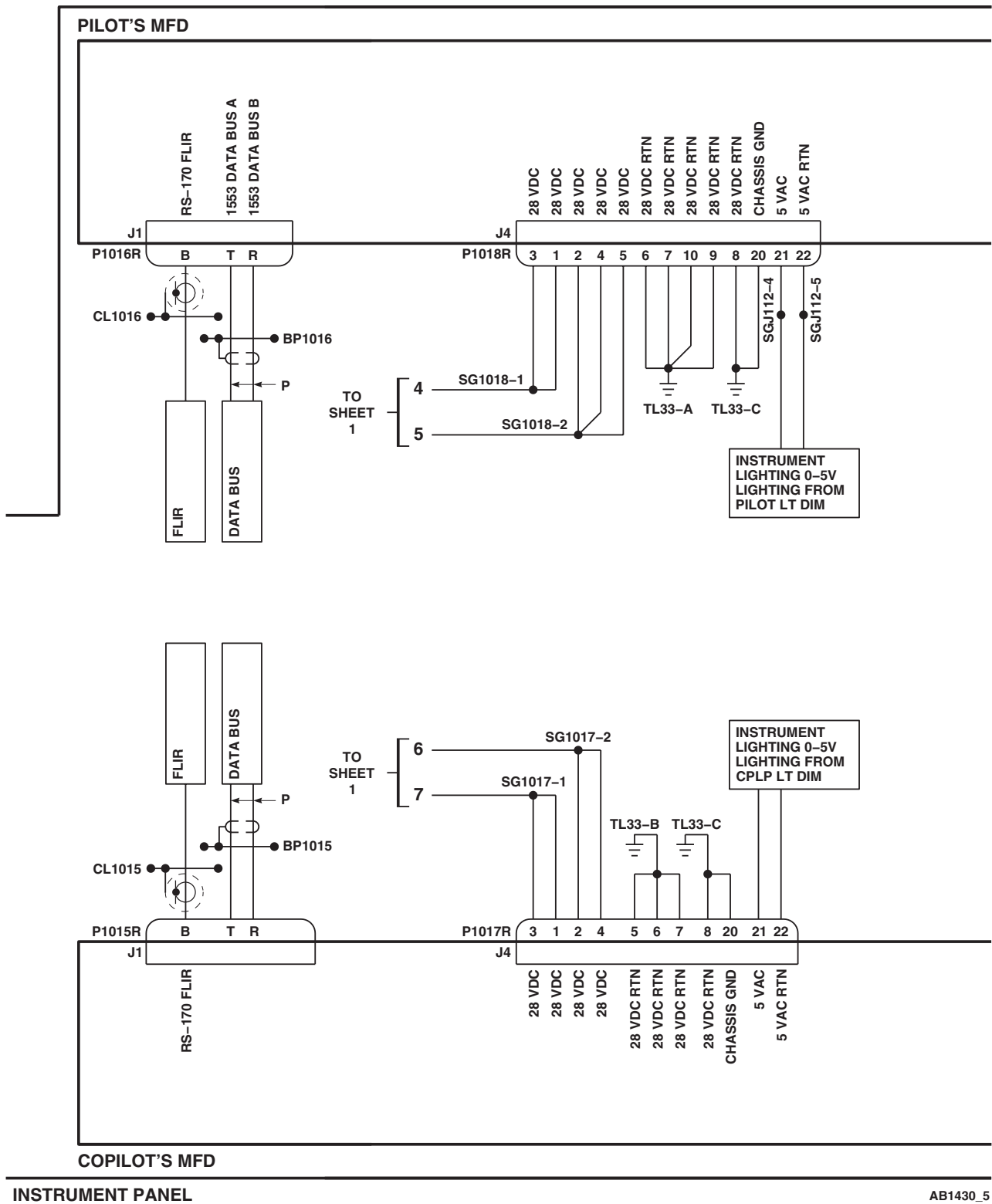
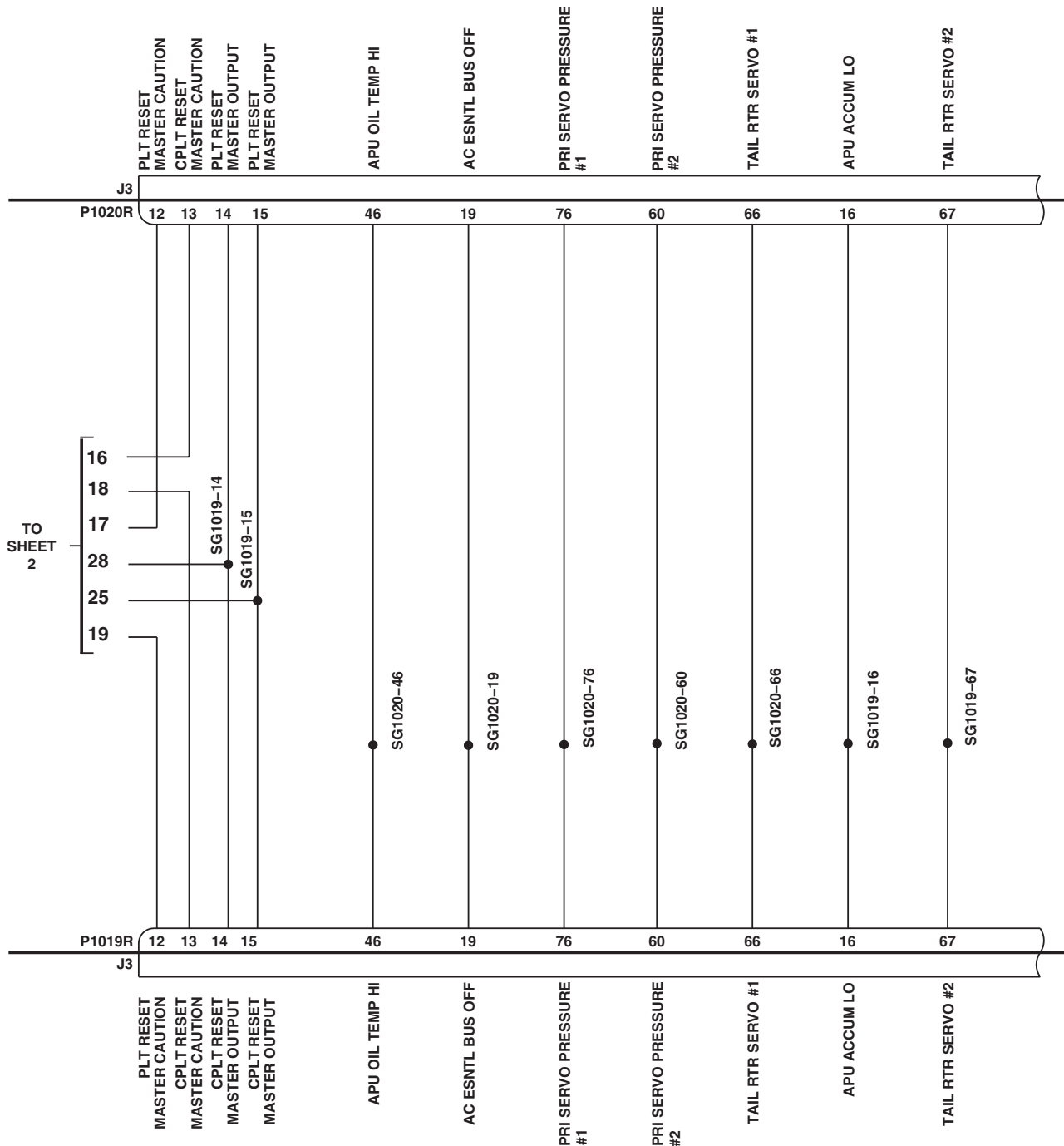


Figure 2. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60A** . (Sheet 5 of 14)

SCHEMATICS - Continued

PILOT'S MFD



COPILLOT'S MFD

INSTRUMENT PANEL

AB1430.6
SA

Figure 2. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60A** . (Sheet 6 of 14)

SCHEMATICS - Continued

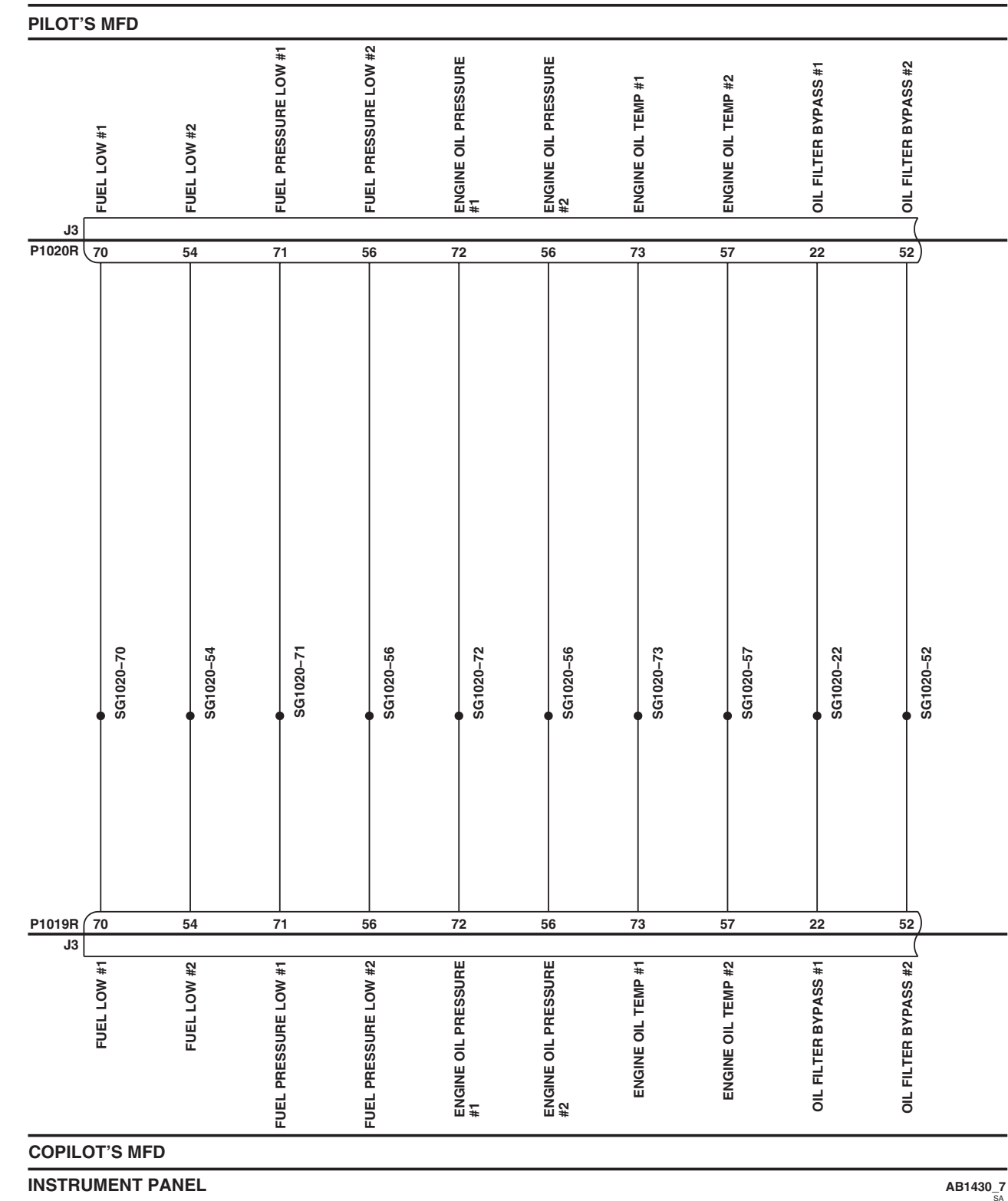


Figure 2. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60A** . (Sheet 7 of 14)

SCHEMATICS - Continued

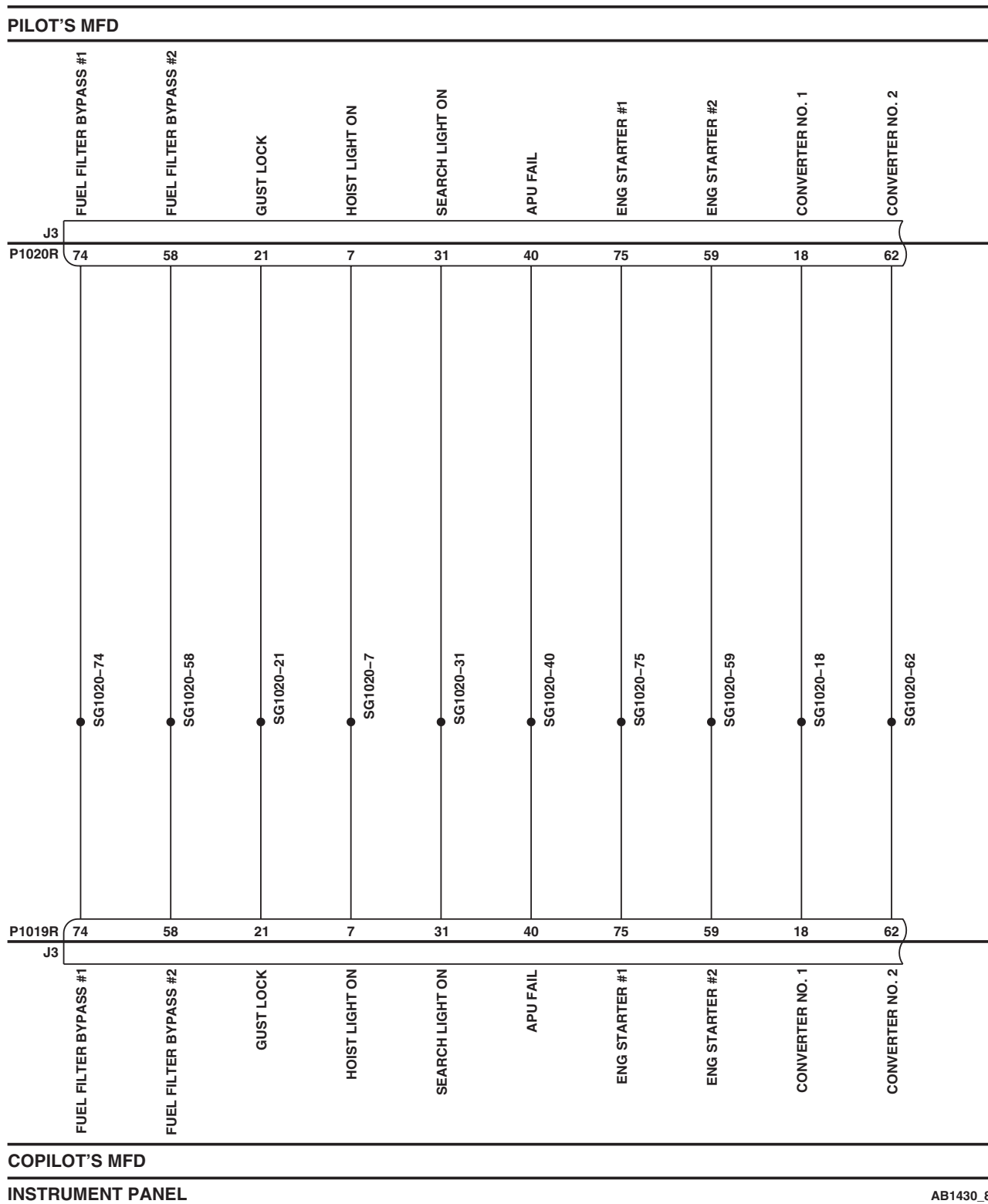


Figure 2. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60A** . (Sheet 8 of 14)

SCHEMATICS - Continued

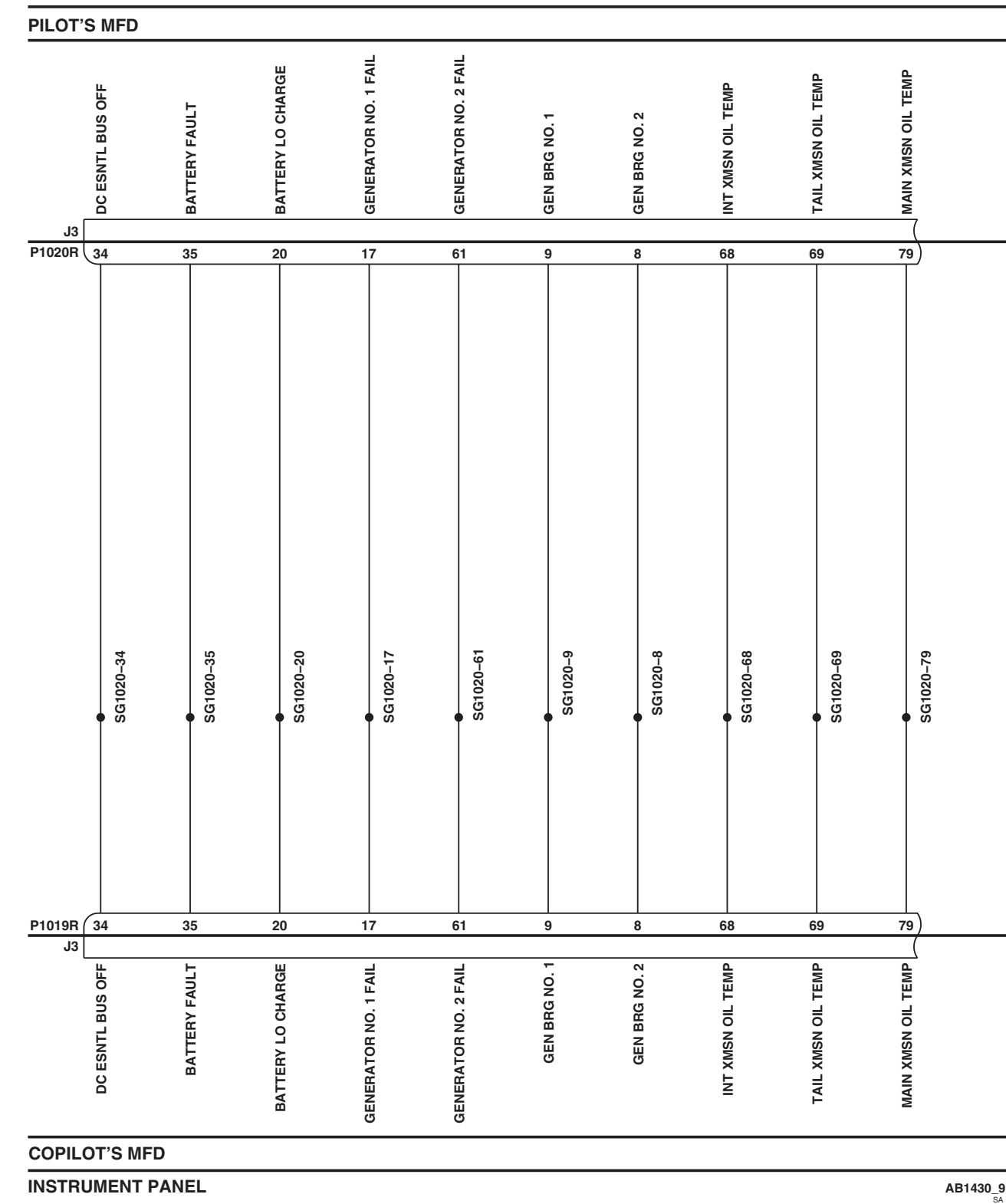


Figure 2. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60A** . (Sheet 9 of 14)

SCHEMATICS - Continued

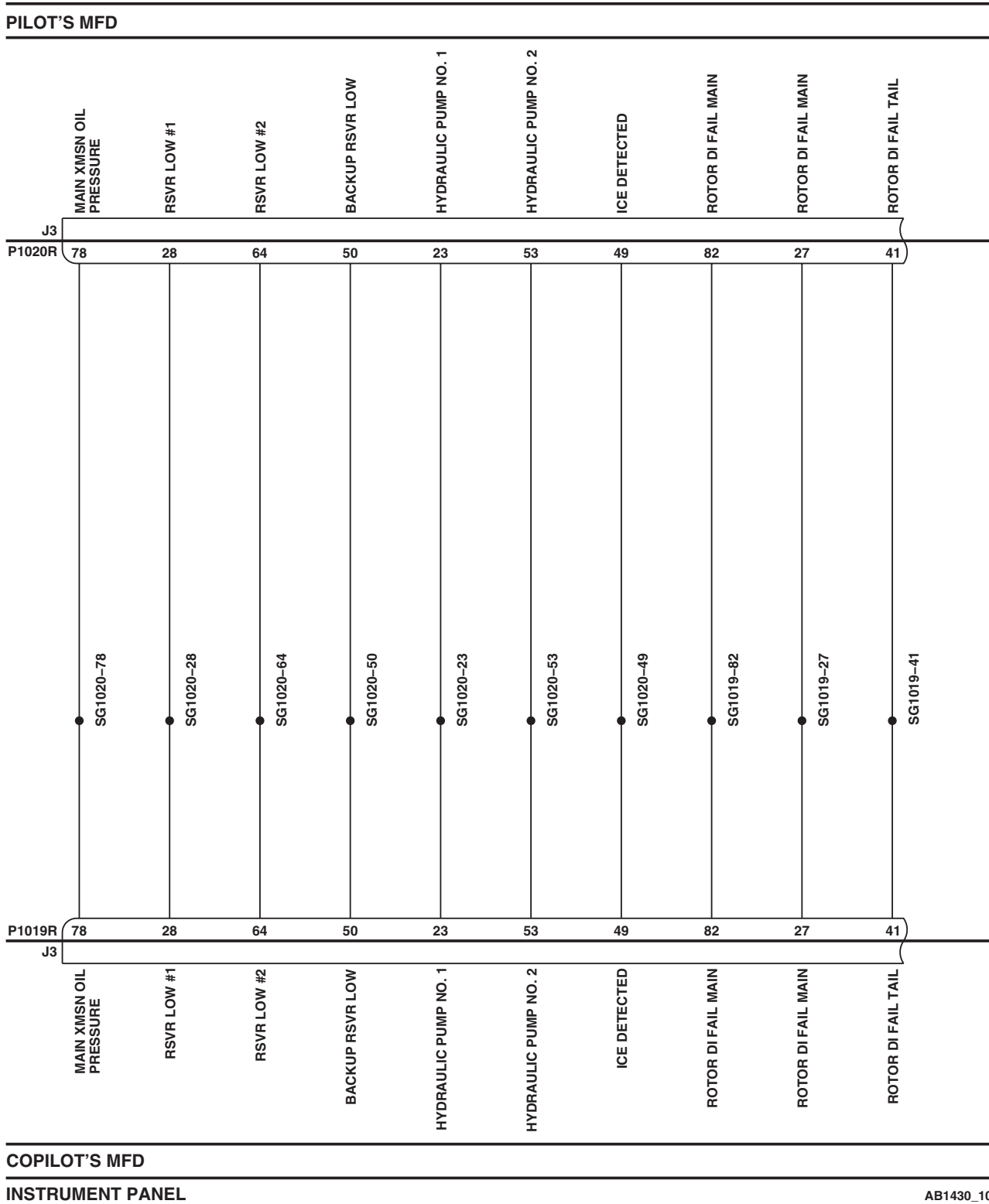
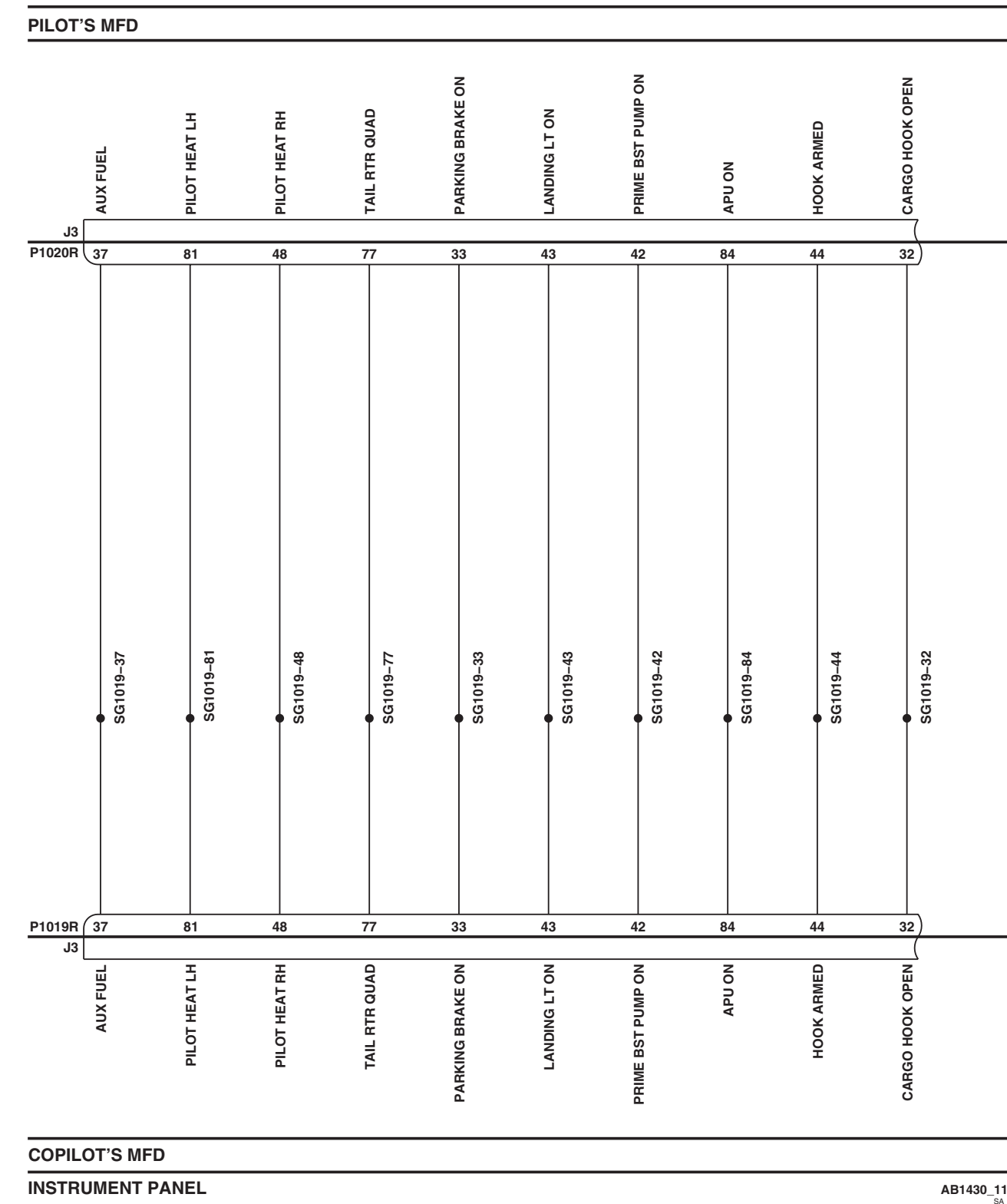


Figure 2. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60A** . (Sheet 10 of 14)

SCHEMATICS - Continued



SCHEMATICS - Continued

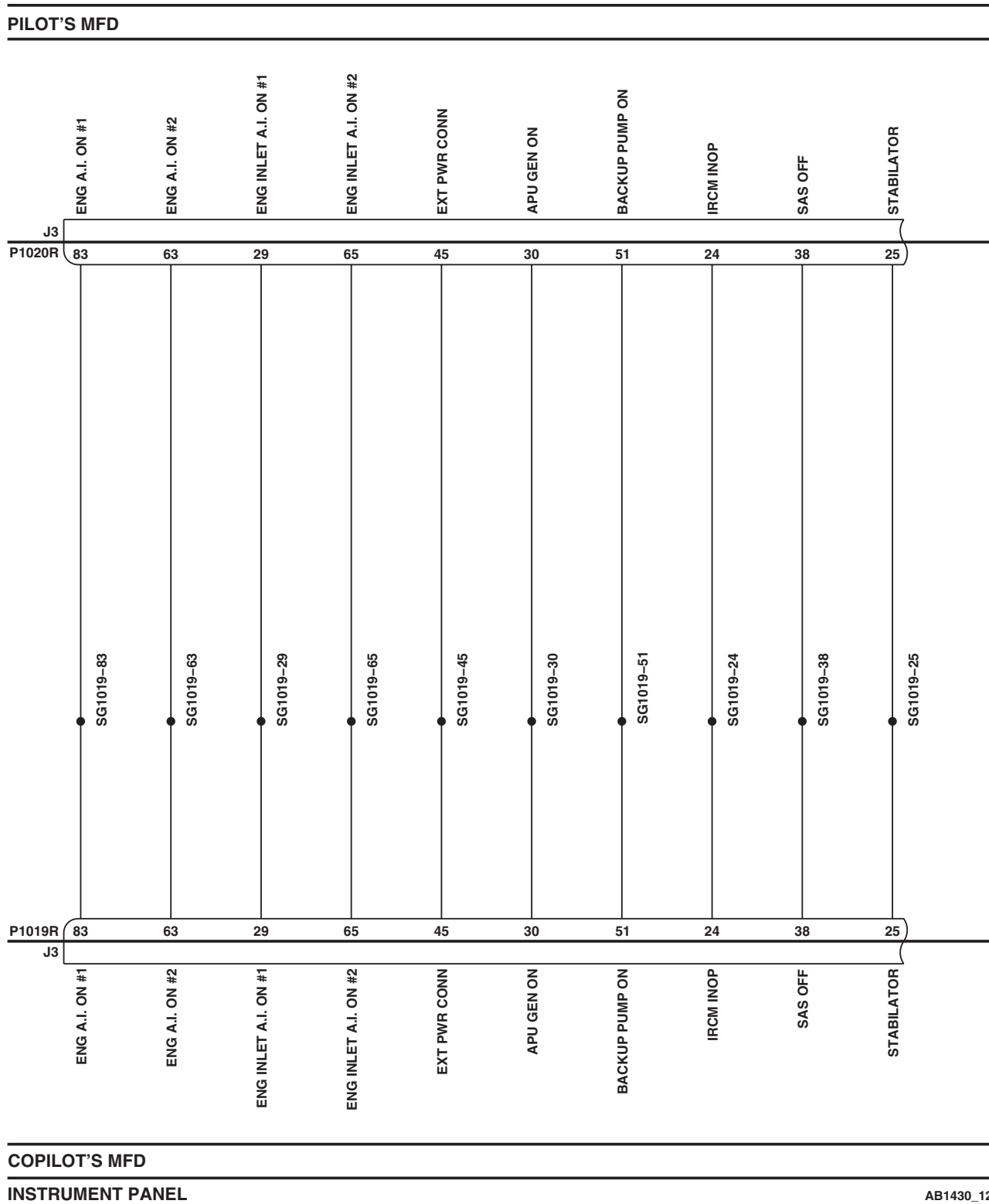


Figure 2. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60A** . (Sheet 12 of 14)

SCHEMATICS - Continued

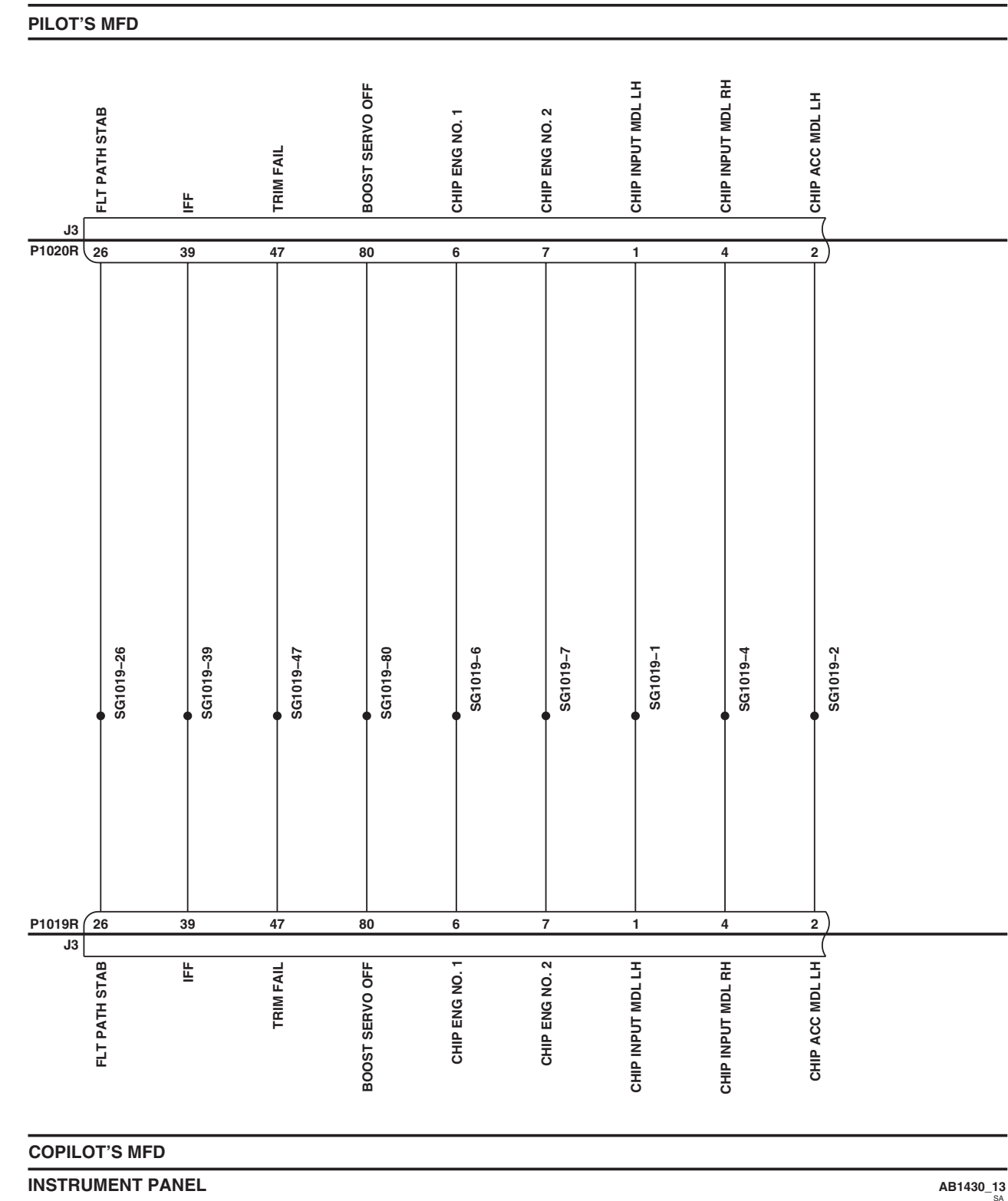


Figure 2. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60A** . (Sheet 13 of 14)

SCHEMATICS - Continued

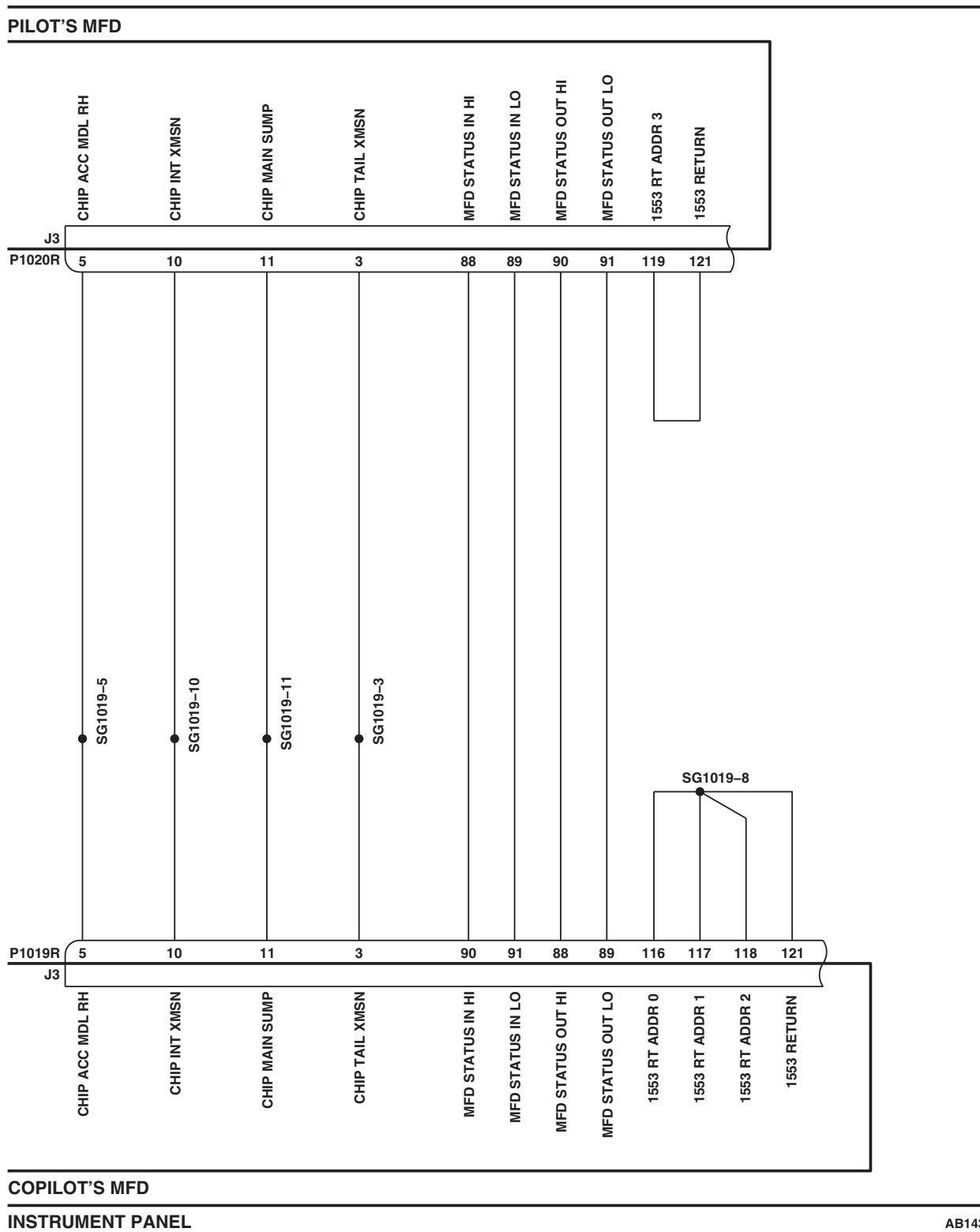
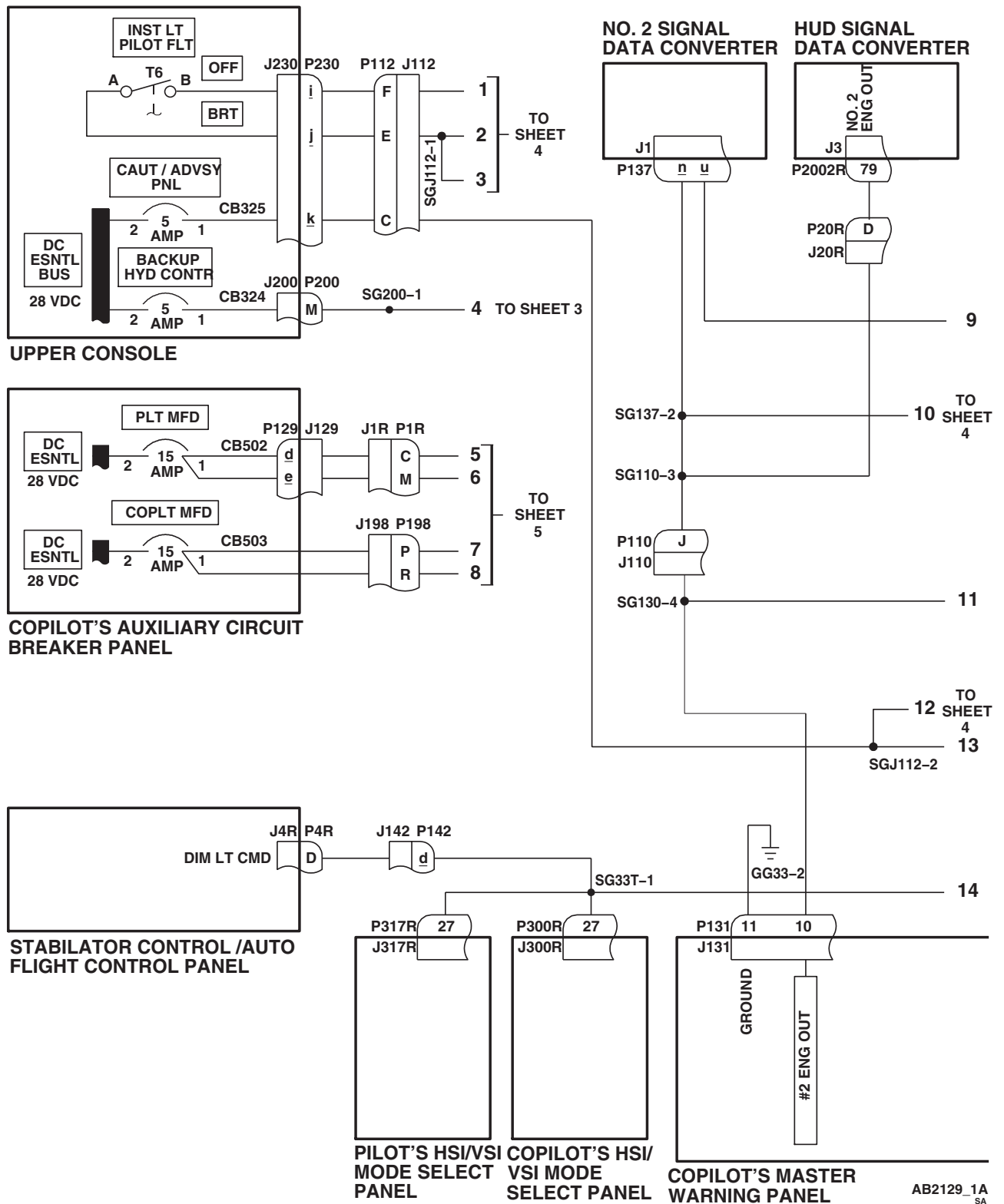
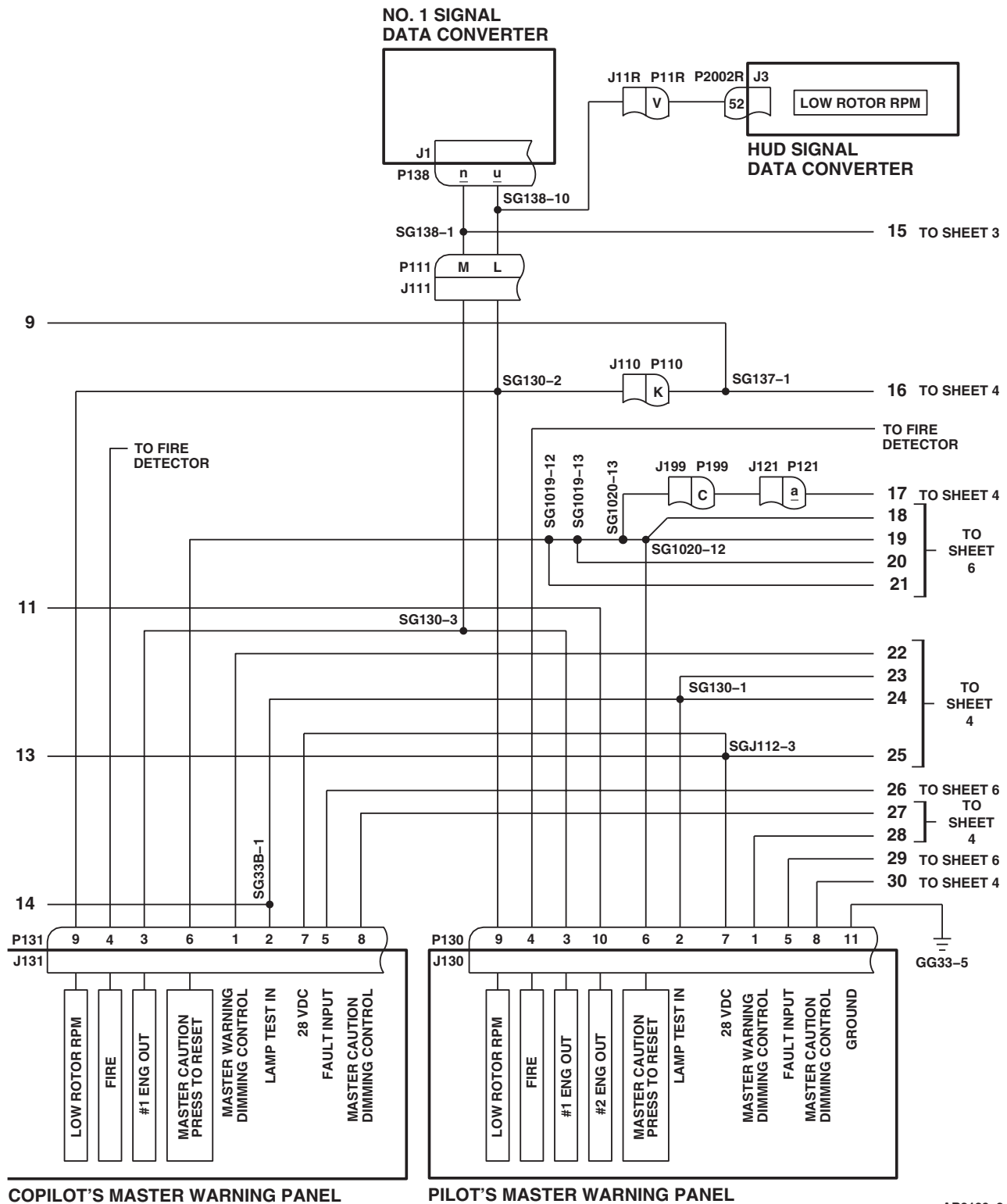


Figure 2. MFD/CAUTION/ADVISORY WARNING System Schematic Diagram **HH-60A** . (Sheet 14 of 14)

SCHEMATICS - Continued

Figure 3. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60L** . (Sheet 1 of 14)

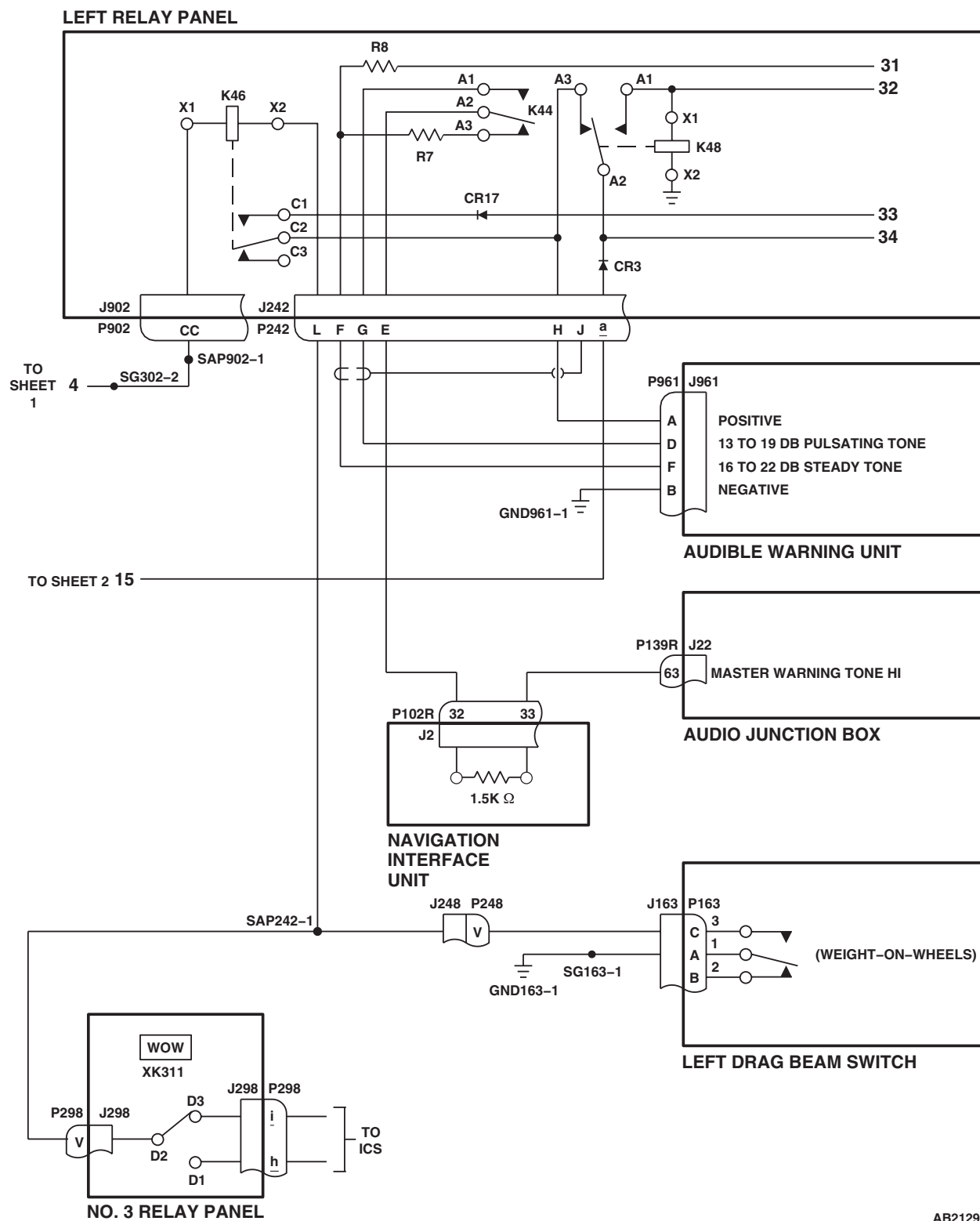
SCHEMATICS - Continued



AB2129 2
SA

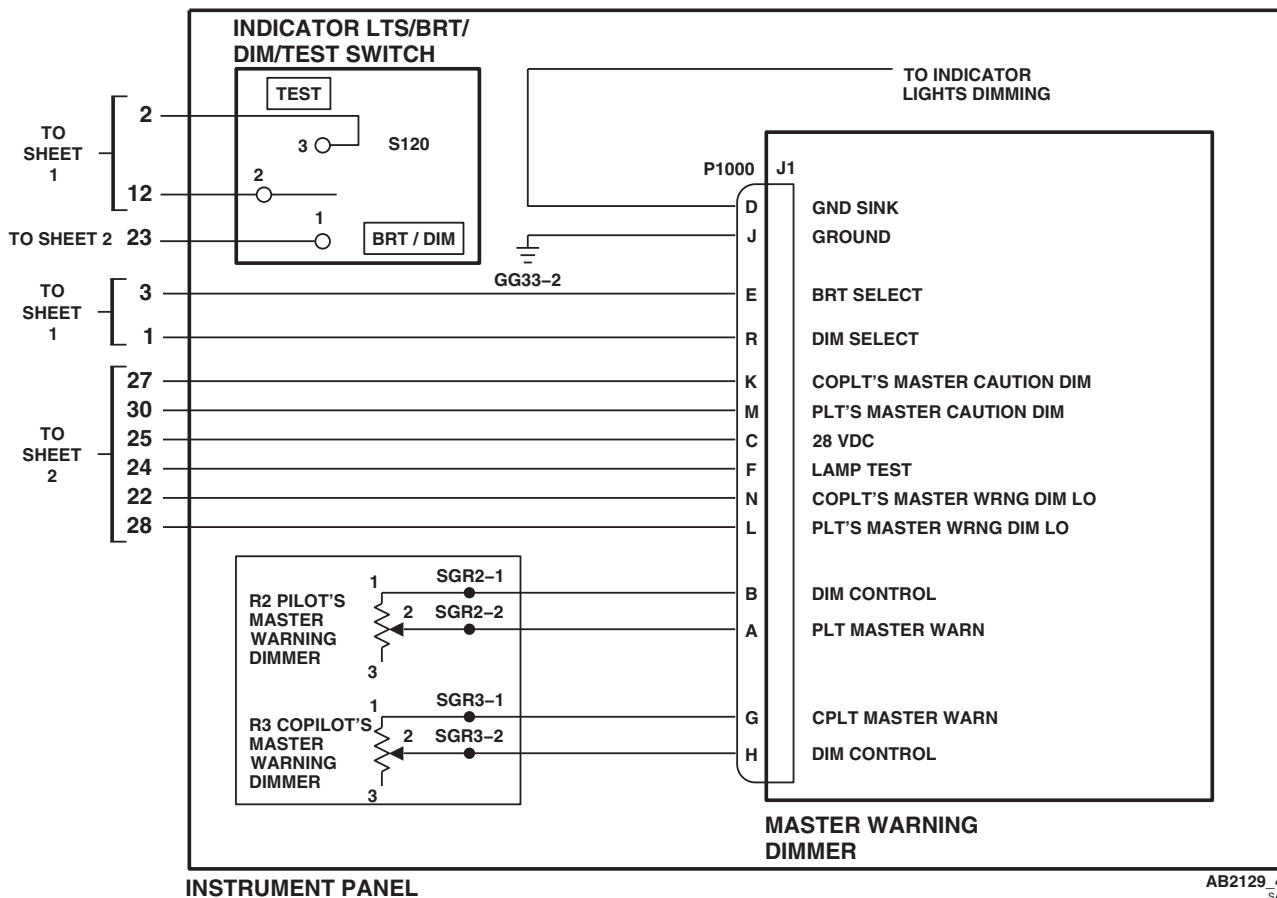
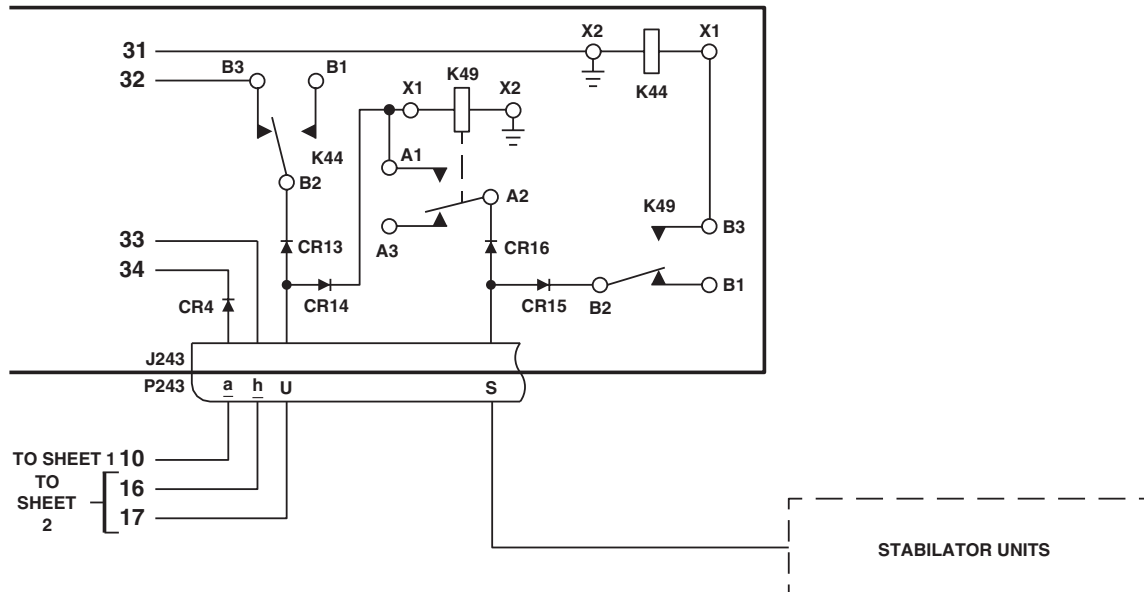
Figure 3. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60L** . (Sheet 2 of 14)

SCHEMATICS - Continued

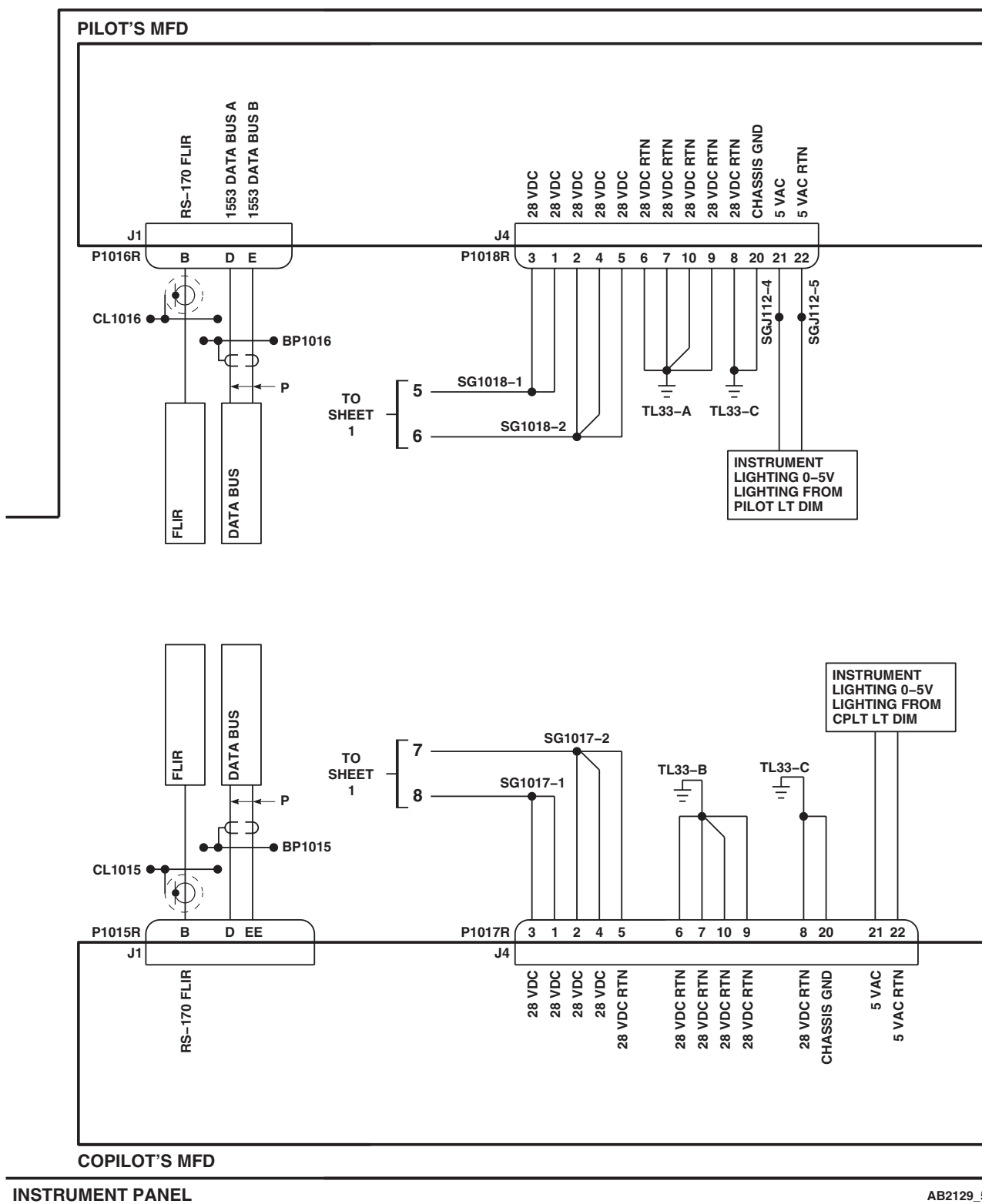
AB2129_3
SAFigure 3. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60L** . (Sheet 3 of 14)

SCHEMATICS - Continued

LEFT RELAY PANEL

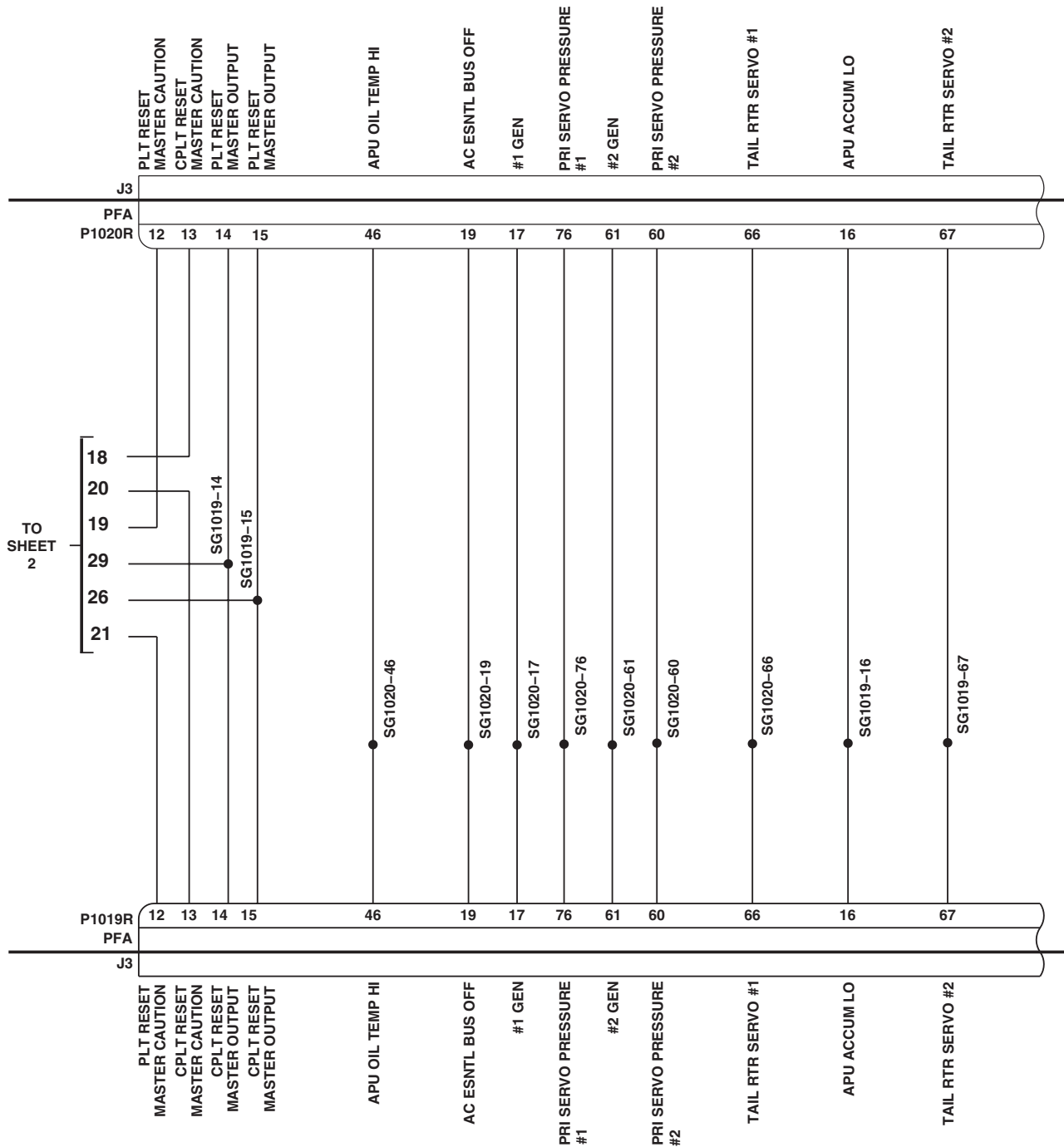
AB2129 4
SAFigure 3. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60L** . (Sheet 4 of 14)

SCHEMATICS - Continued

Figure 3. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60L** . (Sheet 5 of 14)

SCHEMATICS - Continued

PILOT'S MFD



COPILLOT'S MFD

INSTRUMENT PANEL

AB2129 6
SA

Figure 3. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60L** . (Sheet 6 of 14)

SCHEMATICS - Continued

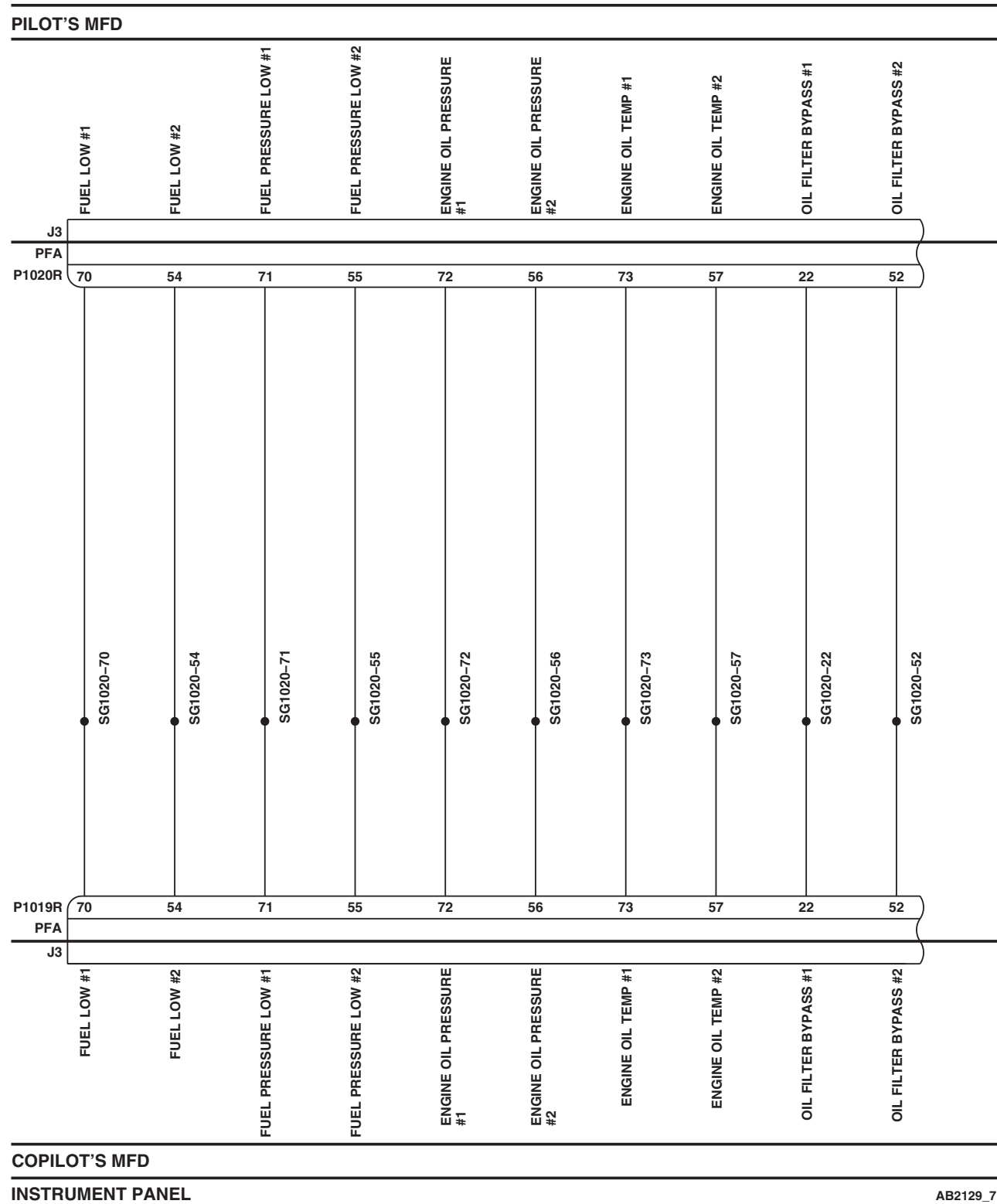


Figure 3. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60L** . (Sheet 7 of 14)

SCHEMATICS - Continued

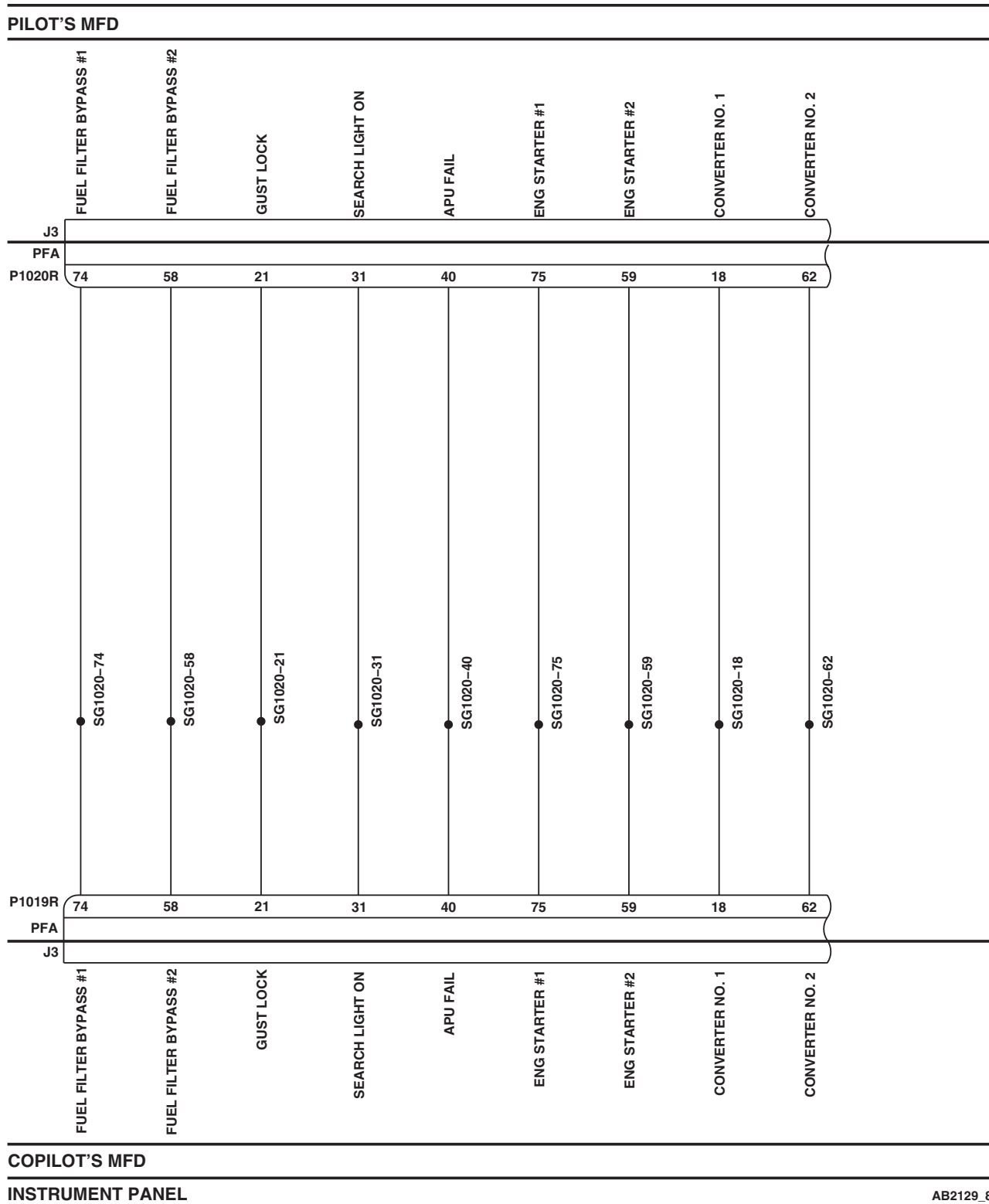


Figure 3. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60L** . (Sheet 8 of 14)

PILOT'S MFD

AB2129_9
SA

HH-60L

SCHEMATICS - Continued

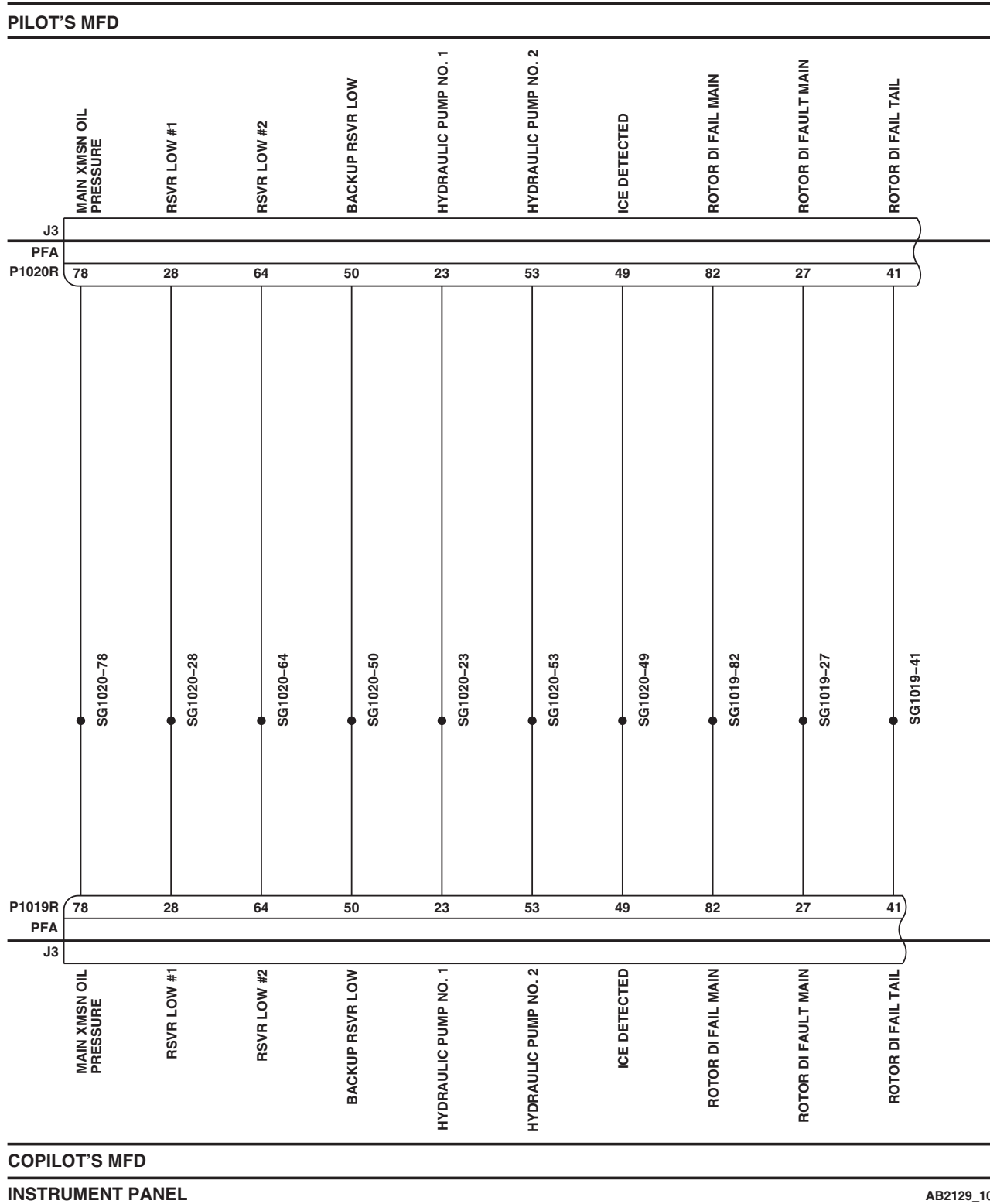
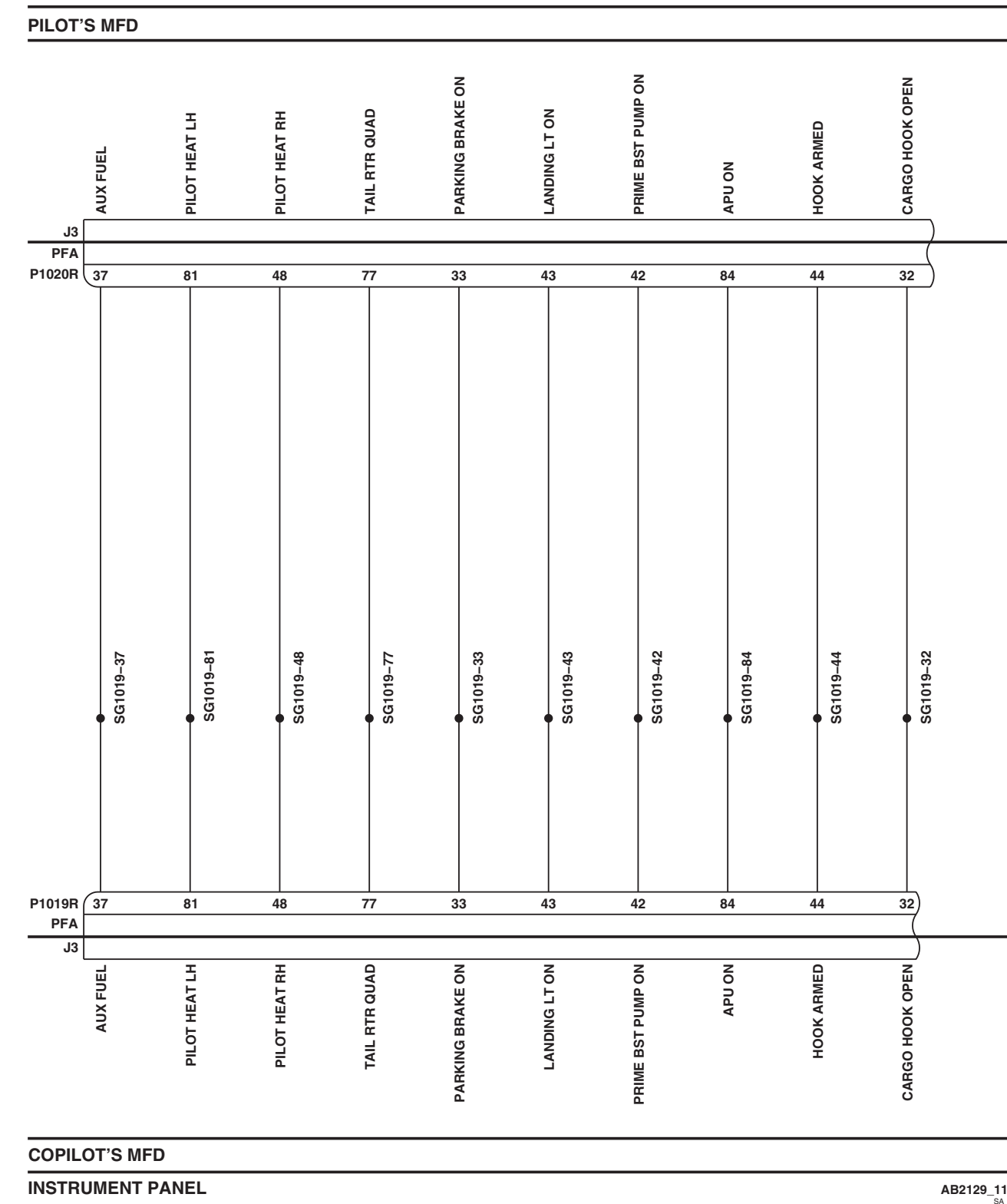


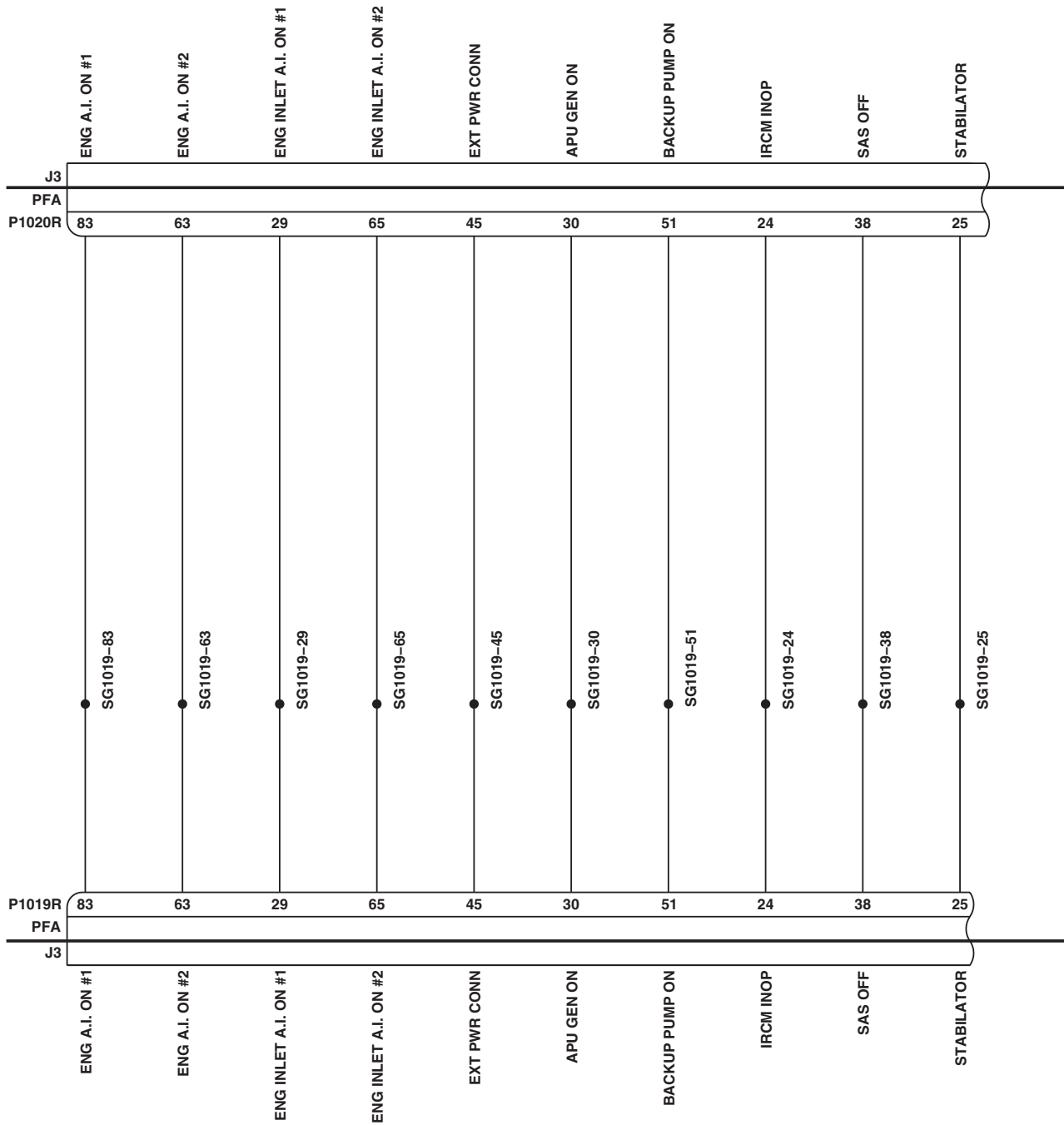
Figure 3. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60L** . (Sheet 10 of 14)

SCHEMATICS - Continued



SCHEMATICS - Continued

PILOT'S MFD



COPLOT'S MFD

INSTRUMENT PANEL

AB2129_12
SA

Figure 3. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60L** . (Sheet 12 of 14)

SCHEMATICS - Continued

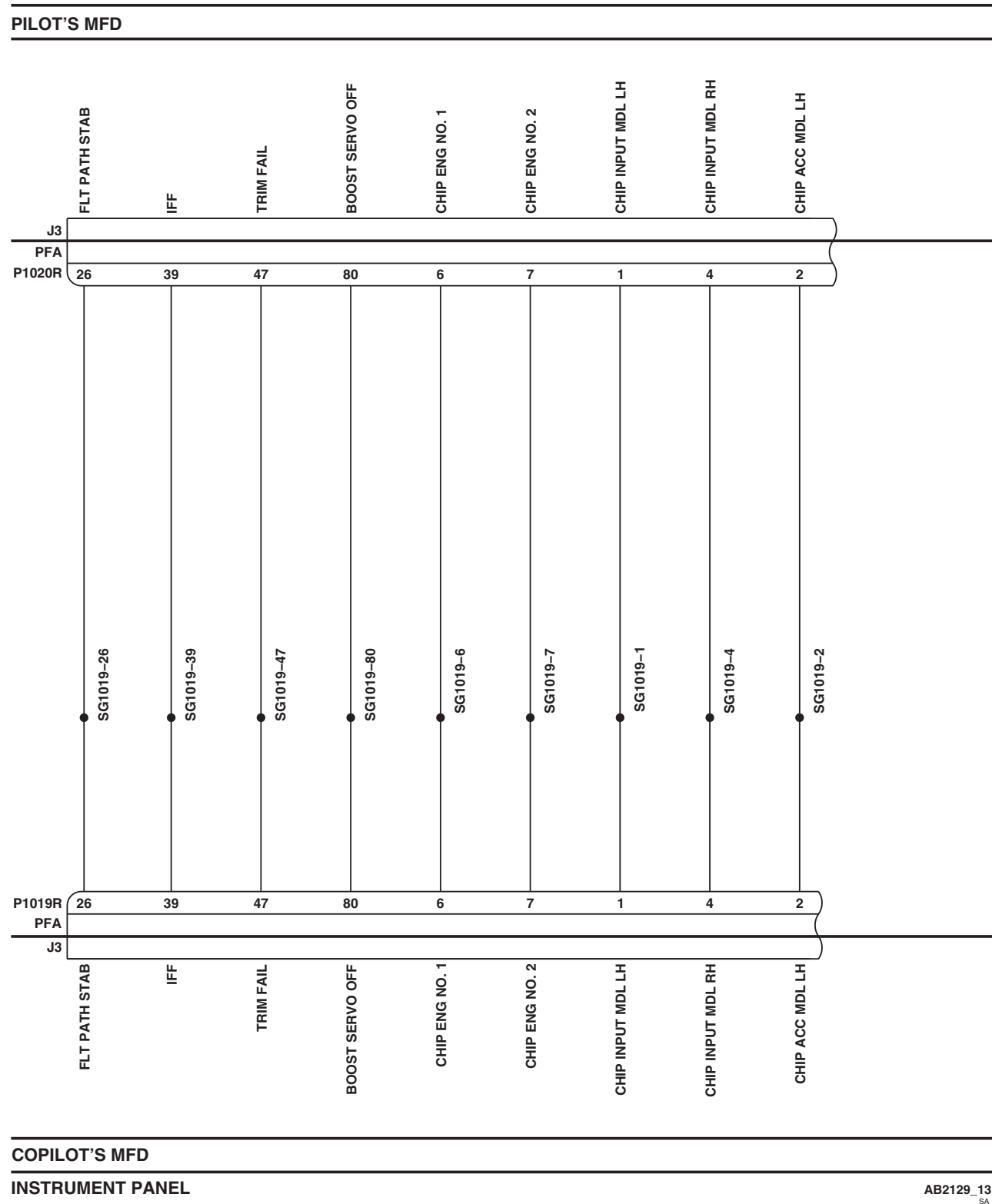


Figure 3. MFD/Caution/Advisory Warning System Schematic Diagram **HH-60L** . (Sheet 13 of 14)

SCHEMATICS - Continued

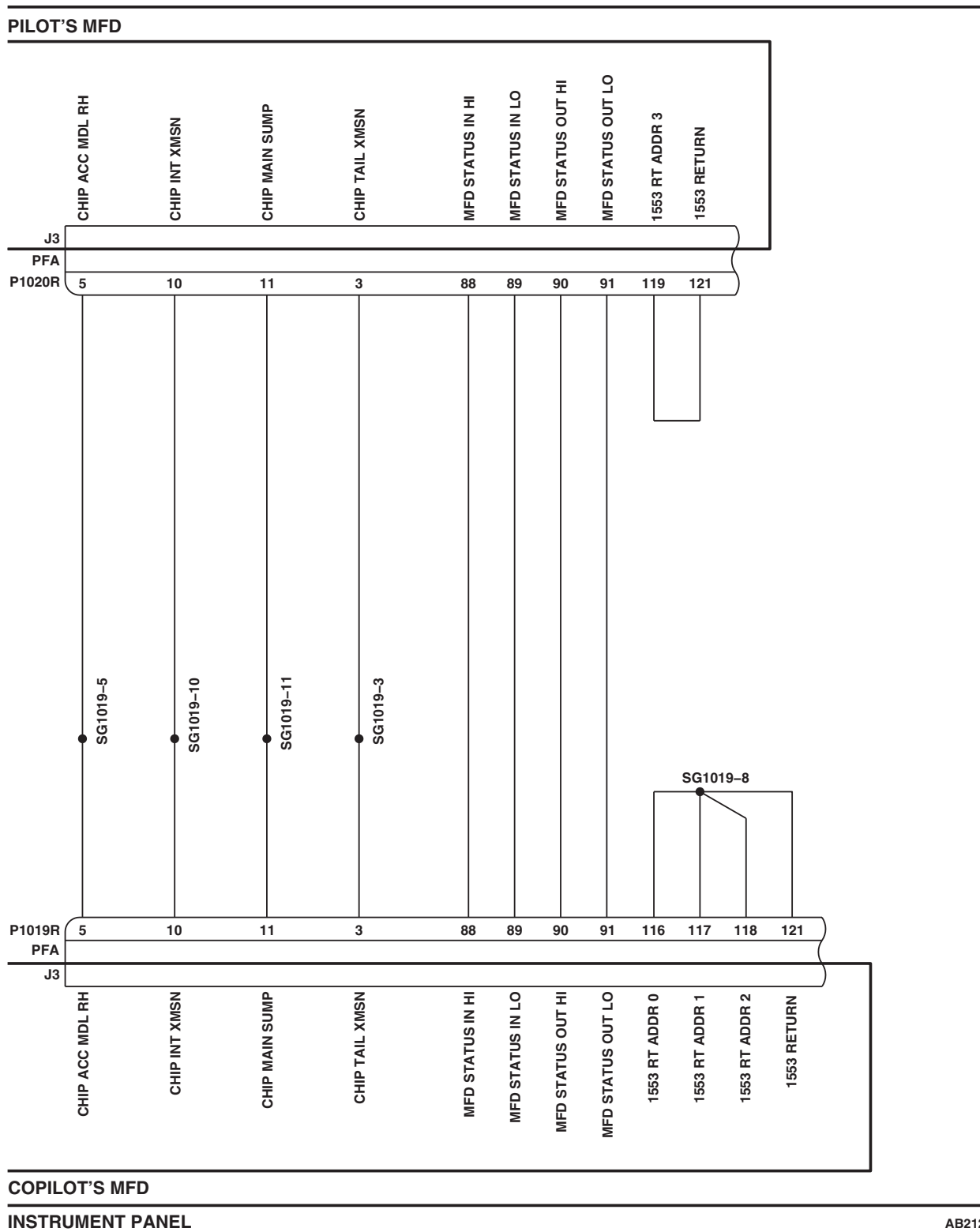


Figure 3. MFD/CAUTION/ADVISORY WARNING System Schematic Diagram **HH-60L** . (Sheet 14 of 14)

END OF WORK PACKAGE

By Order of the Secretary of the Army:

Official:

A handwritten signature in black ink, appearing to read "Joyce E. Morrow". The signature is fluid and cursive, with the first letters of each name being capitalized and prominent.

JOYCE E. MORROW

*Administrative Assistant to the
Secretary of the Army*

0608612

PETER J. SCHOOMAKER
*General, United States Army
Chief of Staff*

DISTRIBUTION:

To be distributed in accordance with Initial Distribution Number (IDN) 311285, requirements for TM 1-1520-237-23-2.

These are the instructions for sending an electronic 2028

The following format must be used if submitting an electronic 2028. The subject line must be exactly the same and all fields must be included; however only the following fields are mandatory: 1, 3, 4, 5, 6, 7, 8, 9, 10, 13, 15, 16, 17, and 27.

From: "Whomever" <whomever@wherever.army.mil>

To: 2028@redstone.army.mil

Subject: DA Form 2028

1. From: Joe Smith
2. Unit: home
3. Address: 4300 Park
4. City: Hometown
5. St: MO
6. Zip: 77777
7. Date Sent: 19--OCT--93
8. Pub no: 55--2840--229--23
9. Pub Title: TM
10. Publication Date: 04--JUL--85
11. Change Number: 7
12. Submitter Rank: MSG
13. Submitter FName: Joe
14. Submitter MName: T
15. Submitter LName: Smith
16. Submitter Phone: 123-123-1234
17. Problem: 1
18. Page: 2
19. Paragraph: 3
20. Line: 4
21. NSN: 5
22. Reference: 6
23. Figure: 7
24. Table: 8
25. Item: 9
26. Total: 123
27. Text:

This is the text for the problem below line 27.

TO: (Forward direct to addressee listed in publication) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, 35898				FROM: (Activity and location) (Include ZIP Code) MSG, Jane Q. Doe 1234 Any Street Nowhere Town, AL 34565			DATE 8/30/02	
PART II – REPAIR PARTS AND SPECIAL TOOL LISTS AND SUPPLY CATALOGS/SUPPLY MANUALS								
PUBLICATION NUMBER				DATE		TITLE		
PAGE NO.	COLM NO.	LINE NO.	NATIONAL STOCK NUMBER	REFERENCE NO.	FIGURE NO.	ITEM NO.	TOTAL NO. OF MAJOR ITEMS SUPPORTED	RECOMMENDED ACTION
PART III – REMARKS (Any general remarks or recommendations, or suggestions for improvement of publications and blank forms. Additional blank sheets may be used if more space is needed.)								
EXAMPLE								
TYPED NAME, GRADE OR TITLE MSG, Jane Q. Doe, SFC				TELEPHONE EXCHANGE/AUTOVON, PLUS EXTENSION 788-1234			SIGNATURE	

RECOMMENDED CHANGES TO PUBLICATIONS AND BLANK FORMS For use of this form, see AR 25-30; the proponent agency is ODISC4.						Use Par reverse) for Repair Parts and Special Tool Lists (RPSTL) and Supply Catalogs/Supply Manuals (SC/SM)	DATE
TO: (Forward to proponent of publication or form)(Include ZIP Code) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, AL 35898 - 5000						FROM: (Activity and location)(Include ZIP Code)	
PART 1 – ALL PUBLICATIONS (EXCEPT RPSTL AND SC/SM) AND BLANK FORMS							
PUBLICATION/FORM NUMBER						DATE	TITLE
ITEM NO.	PAGE NO.	PARA- GRAPH	LINE NO. *	FIGURE NO.	TABLE NO.	RECOMMENDED CHANGES AND REASON	
* Reference to line numbers within the paragraph or subparagraph.							
TYPED NAME, GRADE OR TITLE					TELEPHONE EXCHANGE/ AUTOVON, PLUS EXTENSION		SIGNATURE

TO: (Forward direct to addressee listed in publication) Commander, U.S. Army Aviation and Missile Command ATTN: AMSAM-MMC-MA-NP Redstone Arsenal, AL 35898				FROM: (Activity and location) (Include ZIP Code)			DATE	
PART II – REPAIR PARTS AND SPECIAL TOOL LISTS AND SUPPLY CATALOGS/SUPPLY MANUALS								
PUBLICATION NUMBER				DATE		TITLE		
PAGE NO.	COLM NO.	LINE NO.	NATIONAL STOCK NUMBER	REFERENCE NO.	FIGURE NO.	ITEM NO.	TOTAL NO. OF MAJOR ITEMS SUPPORTED	RECOMMENDED ACTION
PART III – REMARKS (Any general remarks or recommendations, or suggestions for improvement of publications and blank forms. Additional blank sheets may be used if more space is needed.)								
TYPED NAME, GRADE OR TITLE				TELEPHONE EXCHANGE/AUTOVON, PLUS EXTENSION			SIGNATURE	

The Metric System and Equivalents

Linear Measure

1 centimeter = 10 millimeters = .39 inch
 1 decimeter = 10 centimeters = 3.94 inches
 1 meter = 10 decimeters = 39.37 inches
 1 dekameter = 10 meters = 32.8 feet
 1 hectometer = 10 dekameters = 328.08 feet
 1 kilometer = 10 hectometers = 3,280.8 feet

Weights

1 centigram = 10 milligrams = .15 grain
 1 decigram = 10 centigrams = 1.54 grains
 1 gram = 10 decigrams = .035 ounce
 1 decagram = 10 grams = .35 ounce
 1 hectogram = 10 decagrams = 3.52 ounces
 1 kilogram = 10 hectograms = 2.2 pounds
 1 quintal = 100 kilograms = 220.46 pounds
 1 metric ton = 10 quintals = 1.1 short tons

Liquid Measure

1 centiliter = 10 milliliters = .34 fl. ounce
 1 deciliter = 10 centiliters = 3.38 fl. ounces
 1 liter = 10 deciliters = 33.81 fl. ounces
 1 dekaliter = 10 liters = 2.64 gallons
 1 hectoliter = 10 dekaliters = 26.42 gallons
 1 kiloliter = 10 hectoliters = 264.18 gallons

Square Measure

1 sq. centimeter = 100 sq. millimeters = .155 sq. inch
 1 sq. decimeter = 100 sq. centimeters = 15.5 sq. inches
 1 sq. meter (centare) = 100 sq. decimeters = 10.76 sq. feet
 1 sq. dekameter (are) = 100 sq. meters = 1,076.4 sq. feet
 1 sq. hectometer (hectare) = 100 sq. dekameters = 2.47 acres
 1 sq. kilometer = 100 sq. hectometers = .386 sq. mile

Cubic Measure

1 cu. centimeter = 1000 cu. millimeters = .06 cu. inch
 1 cu. decimeter = 1000 cu. centimeters = 61.02 cu. inches
 1 cu. meter = 1000 cu. decimeters = 35.31 cu. feet

Approximate Conversion Factors

<i>To change</i>	<i>To</i>	<i>Multiply by</i>	<i>To change</i>	<i>To</i>	<i>Multiply by</i>
inches	centimeters	2.540	ounce-inches	Newton-meters	.007062
feet	meters	.305	centimeters	inches	.394
yards	meters	.914	meters	feet	3.280
miles	kilometers	1.609	meters	yards	1.094
square inches	square centimeters	6.451	kilometers	miles	.621
square feet	square meters	.093	square centimeters	square inches	.155
square yards	square meters	.836	square meters	square feet	10.764
square miles	square kilometers	2.590	square meters	square yards	1.196
acres	square hectometers	.405	square kilometers	square miles	.386
cubic feet	cubic meters	.028	square hectometers	acres	2.471
cubic yards	cubic meters	.765	cubic meters	cubic feet	35.315
fluid ounces	milliliters	29.573	cubic meters	cubic yards	1.308
pints	liters	.473	milliliters	fluid ounces	.034
quarts	liters	.946	liters	pints	2.113
gallons	liters	3.785	liters	quarts	1.057
ounces	grams	28.349	liters	gallons	.264
pounds	kilograms	.454	grams	ounces	.035
short tons	metric tons	.907	kilograms	pounds	2.205
pound-feet	Newton-meters	1.356	metric tons	short tons	1.102
pound-inches	Newton-meters	.11296			

Temperature (Exact)

F	Fahrenheit temperature	5/9 (after subtracting 32)	Celsius temperature	°C
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